Gambits Engineering: The Historical Foundations

Brad Pierce

4 November 2026
https://doi.org/10.5281/zenodo.23768293

Abstract

This monograph presents historical foundations relevant to gambits engineering, an emerging discipline defined by five questions about capability, requirements, interaction, change and judgment. It explains arguments, mechanisms, examples and criticisms across a wide range of fields, with extended treatments of connected works and close accounts of relevant article passages. The sources include technical constructions, experiments, institutional practices, philosophical proposals and disputes over their interpretation. Reading routes and indexes support study across these different approaches.

Keywords: gambits engineering; historical foundations; systems science; organization; engineering practice; coordination; human-AI systems

Sources and citations

The volume brings together book and article accounts, selected passages, scholarly reconstructions, abstracts and institutional descriptions. Attribution and citations identify the material supporting each account.

The aim is an informative compendium across a wide range of relevant fields. Historical and technical claims remain attached to the sources and circumstances in which they were made.

Bracketed keys identify works in the References section at the end. Related editions or works are listed separately when their identities differ. Printed page numbers, PDF pages and named sections are distinguished where known. The source guide locates each work within the volume.

AI Assistance and Editorial Responsibility

This monograph was developed by Brad Pierce with substantial assistance from an AI model. He established the research objectives and editorial direction and retains responsibility for the claims, source attributions, and decisions about publication.

© 2026 Brad Pierce. All rights reserved.

Contents

S00 The questions and the historical foundations 6

Part 1 Connected systems accounts and their interlocutors 10

S17 Tektology: relations, selection, and the work of organization 11

S18 General systems traditions and distinct programmes 21

S36 Campbell: information, constraint and intellectual reception 29

S19 Jantsch: maintenance, transformation and connected descriptions 31

S20 Gharajedaghi: purposeful organizations and the work of redesign 36

S21 Miller: process allocation, adjustment and purpose-relative costs 42

S22 Beer and viable systems: autonomy, coordination and contested purposes 47

S23 Le Moigne: purposeful modeling and organized activity 54

S24 Capra and Luisi: living organization, meaning and integrative design 59

S25 Homeokinetics, coordination dynamics and systems philosophy 65

S26 Rosen through Louie and Siekmann: organization, repair and realization 69

Part 2 Physical realization, regulation, and information 72

S01 Physical dynamics and passive behavior: mechanisms and operating conditions 73

S02 Constraints and error prevention: triggers, barriers and mistake-proofing 76

S04 Interconnection and dissipativity: what the formal relation preserves 80

S03 Ashby, regulation and models: distinguishable possibilities and constructed capacities 84

S05 Robustness and control: disturbances, criteria and mechanisms 93

S109 Electrical power networks and coordinated operation 97

S54 Stability, maneuverability and recovery 99

S06 Information, coding and production coordination 105

S33 Information acquisition, experiments and dual control 107

S74 Diagnostic testing and the choice of antimicrobial treatment 111

S35 Conditioning, regularization and inverse operators 116

S55 Numerical fidelity and computational cost 120

S75 Algorithm selection, adaptive sorting and computing only what is needed 125

S37 Morphology, embodiment and physical computation 130

S110 Materials processing, degradation and failure 134

S46 General physical principles and cross-domain proposals 135

Part 3 Living, ecological, and flow arrangements 139

S28 Development, plasticity and constructed environments 140

S40 Biological organization, closure and autopoiesis 144

S50 Biological circuits, cellular organization and intervention 148

S43 Ecological resilience and environmental arrangements 154

S107 Chemical process arrangements and integration 159

S47 Networks, flows and competing information measures 160

Part 4 Technical architecture, integration, and assurance 165
S07 Modularity, transactions and platforms: the work of arranging interdependence 166
S111 Manufacturing variation, quality and experimental improvement 176
S08 Form, context and overlapping use: Alexander's design argument 177
S108 Structures, ground conditions and construction sequences 185
S10 Coordination avoidance: the invariant, the operations, and the merge 186
S11 Convergence and self-stabilization: what local rules can establish 191
S12 Integration and expertise across boundaries 195
S84 Specialization across abstraction boundaries 197
S112 Requirements, integration and verification in systems engineering 202
S29 Leveson: constraints, feedback and the organization of safety 203
S30 Reliability, fault tolerance, and what fault experiments establish 211
S57 Preserving essential service under failure 216
S31 Chaos engineering and organizational learning 220
S42 Representations, standards and knowledge infrastructures 228
S39 Lafitte, Simondon and technical objects 234
Part 5 Organizations, institutions, and sustaining work 242
S09 Multilevel coordination: arranging local decisions to meet an overall requirement 243
S13 Information, routines and organizations: Stinchcombe's connected account 249
S51 Organizational silence, dissent and learning 256
S106 Difficult conversations under emotional and relational pressure 261
S14 Dependencies, work design and complexity: changing the demand for coordination 262
S103 Recruiting, selection and forming working relationships 268
S85 Dependencies, bottlenecks and the flow of project work 269
S86 Forecasts, commitments and governing uncertain projects 271
S98 Goal-setting: choosing aims, directing effort and revising commitments 273
S99 Procrastination and beginning intended action 279
S94 Working backward from outcomes 280
S95 Challenging plans before action 284
S53 Absorbing new resources and capabilities 288
S104 Consultancy and the client's capacity to act 294
S58 Transfer and persistence of organizational capability 296
S15 Institutions, messages and invested forms: what people can rely on 300
S59 Expectations, public signals and reflexive feedback 305
S16 Command, shared intent and retained control 311
S27 Voge, information costs and polycellular organization 316
S38 Logistics and sustaining coordinated operation 321
S113 Maneuver warfare and disruption of coordinated action 326
S48 Burden, autonomy, power and evaluation 327
S96 Negotiation: preparation, bargaining and agreements 330

S97 Building, maintaining and renewing personal and professional networks 334
S91 Acquiring, maintaining and contesting power 338
S92 Being strategic at work 343
S115 Charisma, follower recognition and continuing authority 348
S93 Deception, cunning and misdirection 349
Part 6 Choice, commitment, and exchange 354
S32 Antifragility and responses to variability 355
S34 Optionality, flexibility and commitment 360
S89 Constructing portfolios through relationships among investments 362
S90 Position sizing, rebalancing and investment rules over time 364
S61 Late-mover advantage and strategic waiting 366
S62 Learning before commitment 371
S87 Discovering valuable problems and choosing what to build 377
S88 Organizing product development and evolution 379
S100 Founding and developing technology ventures 381
S76 Classical selling models and accounts of buying 383
S77 Questions, diagnosis and commitments in selling 386
S78 Teaching, reframing and buyer indecision 389
S79 Adapting sales encounters and assessing their results 391
S80 Organizing sales work and creating customer value 394
S102 Marketing and the organization of market relationships 397
Part 7 Learning, practice, and continuity 398
S41 Shared practice, skill, environments and cultural coordination 399
S71 Designing practice and developing skill 407
S72 Challenge, difficulty and accelerated learning 413
S101 Fear, avoidance and learning to approach 420
S73 Brief therapy: goals, exceptions and practical change 421
S56 Shared language and collaborative memory 426
S60 External memory and continuity of work 431
S105 Forgetting, discarding and revising retained knowledge 437
Part 8 Inquiry, invention, and collective action 438
S44 Problem structuring, design and systems methods 439
S63 Systematic accounts of methods for generating ideas 446
S81 Systematic invention through contradictions and available resources 450
S64 Invention and scientific discovery 454
S82 Association, bisociation and the construction of new ideas 459
S65 Literary and visual creation through constraint transformation and estrangement 466
S66 Generating performance in music theater and choreography 470
S114 Public speaking, rhetoric and audience judgment 474
S67 Incubation attention and bodily activity 475

S83 Serendipity and the cultivation of unexpected discovery 478
S52 Generating alternatives and organizing exploration 483
S68 Collaborative generation and material methods 489
S69 Computer-supported and human–AI co-creation 493
S70 Learning and practising creative methods 497
S45 Praxeology, economy of knowledge and philosophies of action 500
S49 Stigmergy, emergent patterns and distributed collective action 503
The historical accounts 506
Source specific terminology guide 507
Source guide 508
Subject index 545
Author and institution index 575
References 628

S00 The questions and the historical foundations

This volume develops historical foundations relevant to gambits engineering. The definition, gambit description and five questions below are reproduced from What Is Gambits Engineering? [GE-INTRO]. That paper introduces the field; the accounts in this volume explain relevant material from a wide range of fields.

Gambits engineering is an emerging discipline concerned with how organization and patterns of action enable and limit accomplishment, what it takes to establish and sustain them, and what consequences they produce. Its practical aim is to help people and AI recognize and consciously apply effective engineering moves in particular circumstances.

These moves can change connections, boundaries, responsibilities, or dependencies. They can also change timing, sequences, feedback, commitments, and responses to other participants. They concern machines, software, processes, institutions, and systems combining different kinds of participants.

A gambit is a coherent approach to pursuing a purpose by using or changing the conditions of a situation. It may combine preparation, changes to organization, and coordinated actions, with later choices depending on what happens. Understanding a gambit means understanding how its parts work together, what conditions they depend on, and when to adapt or abandon the approach.

Five defining questions

1. How does the way things are organized affect what they can and cannot do, and how can we use that understanding to achieve particular purposes under real conditions?
2. What does it take to achieve something and keep it going, and how does organizing things differently change those demands and who bears them?
3. How do interactions produce the results we see, and what can those involved actually do to influence what happens?
4. How do things come to be organized as they are, what makes them persist, break down, or change, and how can we work with those processes when building or changing something?
5. How can we judge whether one way of organizing things is better than another, when people want different things and experience different consequences?

Why these questions matter

Engineering choices can change the conditions of accomplishment itself. A well-chosen interface can contain the disruption caused by a later change. A different allocation of responsibilities can permit independent action. A staged commitment can preserve a future choice. A bounded experiment can expose a dependency whose consequences would otherwise remain concealed.

These examples raise several questions together. What capability has been gained? Which knowledge, resources, or continuing activities make it possible? Where has responsibility moved? What happens when circumstances change? Who or what experiences the consequences?

Those affected can include people, other living beings, and their environments. The discipline examines consequences and competing judgments about them, while keeping the evidence, assumptions, and values involved explicit.

What this historical volume provides

The practical ambition is to develop systematic knowledge and a teachable repertoire of gambits. Historical arguments, methods, examples and criticisms can inform that work. This volume makes selected foundations available for study. It preserves the different questions, definitions and conditions of the source accounts, including connections and syntheses advanced by their authors.

The selection follows the questions of gambits engineering. An article may contribute a relevant section rather than its entire argument. Connected book accounts receive extended treatment.

Our own potential advances belong to the later planned Gambits Engineering: A New Synthesis. Application to systems with AI participants is planned for Gambits Engineering: Implications for Agentic AI.

How to study the foundations

Part 1 Connected systems accounts and their interlocutors

For a Tektology-first route, begin with S17 on how combining activities changes what they can accomplish and how the effort is distributed. S18 sets out distinct systems traditions, and S36 follows how readers assessed Jeremy Campbell's account of information and constraint. In S22, Beer asks how an organization can retain local autonomy while coordinating the whole. See Part 2 for the Ashby-first route.

Part 2 Physical realization, regulation, and information

To begin with how systems regulate their behavior, read S03 on Ashby, then S17 on Tektology and S22 on viable systems.

S74 examines when diagnostic tests change whether clinicians start, switch or stop antibiotics.

S75 shows how the input and the result you need can determine whether to adapt the sort or skip a full sort altogether.

To see what a test establishes, begin with S110 on how specimen preparation and service conditions bear on material failure. S111 follows an experiment on molding defects from the first tests through the use of the revised settings in production. S112 examines how engineers check interfaces and establish that an assembled system meets its requirements.

Part 3 Living, ecological, and flow arrangements

S28 follows how organisms develop and alter their environments. S40 compares accounts of how living systems maintain and reproduce their organization. S50 shows how cellular circuits can be built to remember a state or produce an oscillation. S43 asks what enables ecosystems to persist through disturbance.

S47 traces a proposal for quantifying food-web robustness, a critique of its mathematical argument, and a later, narrower conjecture.

For a process retrofit, use S107 on redesigning heat recovery while reusing equipment. The construction and grid-restoration route appears under Part 4.

Part 4 Technical architecture, integration, and assurance

S10 asks when local updates can be combined safely without coordination. S11 asks when distributed systems converge or recover after disruption.

Start with S30 on what deliberately introduced faults can reveal, then read S31 on how teams use those results to change their systems and practices.

S39 uses examples such as fins that cool and stiffen a cylinder head to show how a machine's functions come to work together and depend on its surroundings.

S84 uses exokernels and later examples to show how engineers specialize software across layers.

To study change in stages, read S108 on what supports a structure as construction proceeds and S109 on how operators restore parts of a power network in sequence. For redesign within an existing plant, S107 examines how engineers plan heat-recovery improvements while reusing parts of an existing equipment network.

Part 5 Organizations, institutions, and sustaining work

S12 asks what an organization still needs to know when it buys components from others. S14 shows how changing the design of work can reduce the need for coordination, and S38 examines how logistics keeps an operation supplied. S48 asks who bears the costs, who provides care, and who has a say.

To plan and govern a project, start with S85 on dependencies, bottlenecks and buffers, then read S86 on forecasting outcomes, deciding what to fund, and overseeing work after approval. S94 starts from a desired outcome and works back to what it requires. S95 takes the resulting plan as something to challenge before action.

To compare sources of influence, start with how people acquire and contest power in S91. S114 asks what a speaker can do on a particular occasion and how an audience judges it. S115 examines how followers accept a leader's authority, how staff help sustain it, and what happens when a successor claims it.

Start with S92 on how people pursue strategic aims through everyday conversations, decisions and working relationships. S93 examines situations in which misleading someone is a deliberate move. S113 examines military approaches to disrupting an opponent's ability to act in concert.

For working relationships, begin with S96 on reaching an agreement and S97 on building and renewing ties that can outlast a single exchange. S103 examines how candidates and employers assess one another. S106 considers how to raise a difficult issue once people are working together.

To follow what happens after advice is given, read S104 on what a consultancy engagement leaves a client able to do. S105 takes up a related problem in continuing work. What does it take to revise or withdraw a procedure when people still use it?

To follow an intention into action, begin with choosing and revising aims in S98. S99 asks what makes it easier to begin something one already intends to do. S101 gives a closer look at avoidance and what can be learned by approaching a feared situation.

Part 6 Choice, commitment, and exchange

To study decisions before commitment, start with S32 on when variability helps or harms and S34 on the value of keeping options open. S61 follows the incentives to wait for others to learn, and S62 examines how to gain evidence while a decision can still change.

For investment decisions, read S89 on what holdings contribute together, including how two assets can repeat the same exposure. S90 takes up the decisions that follow allocation, including sizing positions and deciding when to rebalance.

To decide what to build, start with S87 on finding worthwhile problems and S88 on organizing product development so choices can still change. S100 follows how the founder's work changes as a venture develops. S102 broadens the view from choosing a product to understanding the market relationships that sustain it.

Start with S76 on how early selling models explained a buyer's decision. S77 shows how sellers use questions to clarify a need and agree on a next step. S78 examines how sellers reframe a buyer's problem and what they do when the buyer fears choosing badly. S79 looks at how sellers adapt during an encounter and how researchers measure the results. S80 follows the work of shaping an offer and carrying it into delivery.

Part 7 Learning, practice, and continuity

Begin with S41 on how learners recognize the structure of a problem and how a tutor can carry part of the task. S71 examines when to practise parts of a skill, when to return to the whole, and when to give feedback. S72 asks how difficulty, recall and spacing can help learning last beyond the session.

S73 shows how brief therapy turns a client's goals and moments when the problem eases into practical next steps. It also examines how studies compare the results.

For continuity across people and time, read S56 on how people build a shared language, then S60 on how external records help people resume work. S105 asks what happens when knowledge or procedures need to be revised or withdrawn.

Part 8 Inquiry, invention, and collective action

S44 examines how people investigate a situation when they disagree about the problem or what should be achieved.

Start with S63 for ways to choose among methods for generating ideas. S64 follows analogy into invention and scientific discovery. S65 shows how writers and artists use constraints to produce unexpected work. S66 examines how performers improvise. S114 considers how a speaker addresses an audience and how the audience judges the speech. S67 asks what happens when people take a break from a problem.

For work in groups, S52 asks how turn-taking and criticism affect the ideas people produce. S68 shows how teams use cards and shared workspaces to organize ideas, S69 examines what software and AI contribute to creative partnerships, and S70 looks at what people learn through creative exercises.

S81 shows how inventors tackle conflicting requirements and find new uses for resources already at hand. S82 shows how putting a problem in a different frame or connecting ideas from separate fields can suggest a new design or hypothesis. S83 follows what people do after an unexpected encounter gives them an idea worth pursuing.

For different ways of organizing action, compare S45 on preparation and habit with S49 on how traces of earlier activity guide what others do next.

The contents, subject index, author and institution index, and section identifiers S01–S115 provide routes into the volume. The source guide locates each cited work and links it to the shared references.

Part 1 Connected systems accounts and their interlocutors

The extended accounts in this part develop the source authors' connected arguments about organization.

S17 Tektology: relations, selection, and the work of organization

Relations and organizational change

Alexander Bogdanov's *Tektology* asks how things are joined, sustained, separated and reorganized. Its proposed subject extends from work and social cooperation to organisms, machines and scientific thought. Bogdanov examines how an arrangement changes what its elements can accomplish, how an environment assists or frustrates that accomplishment, and how organization persists. [TKT-W001; W017]

Bogdanov's revised 1927 energy discussion describes transformations as not fully reversible. [TKT-W002, p. 236]

Organization is a relation between activities

Bogdanov's starting question is practical: what difference does combining elements make to their capacity to act? Cooperation supplies an accessible case. Two people can sometimes perform a task together that neither can perform separately. A resistance that exceeds either person's effective effort can be overcome when their efforts are appropriately joined. This is more informative than the slogan that a whole exceeds its parts. The relevant comparison concerns what the parts can accomplish separately and what their particular combination can accomplish against a specified resistance. Bogdanov explicitly locates the difference in the relation between their activities and the resistances they encounter. [TKT-W001, part I, ch. II, A §1]

The qualification about the combination is essential. Workers may impede one another, or expend effort in opposed directions. Bogdanov illustrates the latter possibility with people pulling a rope against each other: much of the effort disappears from the resultant available to move the whole arrangement. Even successful cooperation retains some internal interference. His organized, disorganized, and intermediate combinations therefore describe differences in effective relations, rather than a division between flawless organizations and everything else. [TKT-W001, part I, ch. II, A §§1–2]

One of his examples also prevents an easy identification of a collective gain with a gain for every participant. A mother carrying an infant enables both to change location, while the mother's own burden increases. The organizational effect depends on which activities and whose situation are being counted. The example's illustrative coefficients should not be mistaken for measured results. Its substantive point is the difference between the capacity of a combined arrangement and the distribution of effort within it. [TKT-W001, part I, ch. II, A §1]

An activity becomes a resistance when it opposes another activity. Bogdanov makes this change of standpoint explicit through conflict: what is an effort from one participant's position is an obstacle from the other's. The same description cannot silently assign the interests of both to one side. His broad term “activities–resistances” is intended to keep their relationship in view. Some of the accompanying physical explanations belong to the science of his period; the relational distinction does not require adopting those explanations as current physics. [TKT-W001, part I, ch. II, A §2]

The elements of an analysis are similarly relative to the task. A society may be considered through enterprises or people; an organism through cells; a theoretical construction through concepts. An element may in turn be examined as a complex of further elements when the question requires it. This is a method of selecting an analytical scale. It explains why a description of an organization's overall result can leave an important internal burden invisible: the description may have treated the bearer of that burden as an undivided element. [TKT-W001, part I, ch. II, A §2]

Organization is also relative to the activity and environment under consideration. Bogdanov's factory can be effectively arranged for technical production while containing serious social antagonisms. A machine can resist the forces for which it was designed and remain vulnerable to quite different forces. Complexity alone does not make a complex well organized. These passages restrict what can be inferred from a successful performance: the arrangement works in some respect, against some resistance, under some surrounding conditions. [TKT-W001, part I, ch. II, A §3]

Bogdanov relates organization to an activity, the resistance it encounters and the unit whose performance is being considered. The same structure can facilitate one activity while obstructing another.

Specialized experience and the organization of knowledge

Bogdanov presents the fragmentation of knowledge as an organizational problem as well as an intellectual one. Specialization permits increasingly detailed work, but differentiated fields develop their own objects, techniques, and language. Their distance can obscure relations that might otherwise make experience in one field usable in another. Conversely, a resemblance between words can suggest a relationship that the actual activities do not support. His argument for a general organizational science includes the ambition to recover transferable methods without relying on verbal resemblance alone. [TKT-W001, part I, ch. I, discussion of scientific specialization and unity of methods]

Bogdanov's examples of methods crossing disciplinary boundaries provide reasons to investigate relations between different bodies of experience. His later discussion of analogy, considered below, is relevant because it asks for further observations when a pattern admits more than one explanation. [TKT-W001, part I, chs. I–II and appendix, “The doctrine of analogies”]

Bogdanov includes the arrangement of concepts and accumulated experience among the things that people organize. A scientific classification, a technical method, and the communication needed for cooperation can become part of the conditions under which subsequent work occurs. His explanation does not allow their usefulness to be inferred merely from their generality: the relation to the activity still has to be specified.

Connecting through a suitable intermediary

Bogdanov calls the joining or interaction of complexes *conjugation*. A connection need not consist in simply putting two previously separate things next to each other. His account of chain linkage asks what actually enables their activities to connect. In a chain, successive links can have different connecting features: A shares something relevant with B, while B connects to C through another feature. The existence of a long chain consequently does not require every participant to resemble every other participant or even to know that the others exist. His social and cognitive illustrations are intended to disclose the connecting relations, not merely to collect cases of superficial similarity. [TKT-W001, part I, ch. III, A, “Chain linkage”]

Difference can support a connection when the different capacities meet in a common undertaking. Bogdanov considers a theorist and a practitioner. Their contrast is useful only if the activities are directed compatibly and one can carry forward the effort where the other stops. Merely giving two people complementary labels establishes neither cooperation nor the connecting relation. The relevant commonality concerns the undertaking and the point at which their activities meet. [TKT-W001, part I, ch. III, A, “Chain linkage”]

Ingression names the introduction of an intermediary through which a connection becomes possible. His clear social example concerns two people who need to agree on a plan but do not share a language. A translator has a different relevant commonality with each: one language with the first person, another

with the second. The intermediary is therefore not just an additional person in the group. Its particular relations to both sides enable an exchange that the initial pair could not carry out directly. [TKT-W001, part I, ch. III, A, “Ingression”]

The practical interest of this account lies in the specificity of the connection. A useful intermediary must meet the conditions on both sides. In Bogdanov's vocabulary, identifying something as a link does not by itself describe how it links. Nor does the general mechanism imply that adding connections always improves an arrangement. An additional connection may expose a complex to destructive activities, or join participants whose activities interfere. What needs explanation is the effect of the particular linkage upon the particular activities being considered. [TKT-W001, part I, ch. III, A]

Separation, conflicting interests, and the choice of a whole

The book treats separation as an organizational event in its own right. Bogdanov's *disingression* concerns the disruption or cancellation of connections. An apparent destruction at one level can reduce destructive interference at another. An organization already consumed by internal conflict may divide into groups that function more effectively after separation. In his description, some internal oppositions have been replaced by relations between now distinct complexes. Whether the new situation is preferable depends on those relations and on subsequent conditions; separation does not acquire a favorable meaning merely by being described as reorganization. [TKT-W001, part I, ch. III, A, “Disingression”]

Bogdanov makes the asymmetry particularly explicit through divorce and amputation. The same break can relieve one party and devastate another; an amputation can preserve an organism while destroying the separated part. Their function is to demonstrate why the fate of one element does not settle the fate of the combination, or vice versa. A single description such as liberation, destruction, or organizational improvement can conceal the standpoint from which it was made. [TKT-W001, part I, ch. III, A, “Disingression”]

He accordingly asks for whom and for what a change is better. He also proposes considering a disputed arrangement within a larger social complex. The latter move describes his method and social orientation; it does not, by itself, settle the ethical priority of the larger whole. A family, organization, class, or society can have interests that differ from those of its members. [TKT-W001, part I, ch. III, A, “Disingression”]

Bogdanov’s joining and separation can remove recurring friction, enable a blocked activity or redistribute demands among participants. His questions about who benefits and which larger complex is considered distinguish these consequences.

Selection can be performed by an arrangement

Bogdanov's use of *selection* is deliberately broader than biological natural selection. He removes reproduction and heredity from the general schema because those are not properties of every complex to which he wishes to apply it. A pendulum's succession of positions is among his illustrations: the positions are not a simultaneously existing population of organisms. Selection, at this level of abstraction, concerns what persists or is eliminated under specified conditions. Its proposed universality belongs to his program of a general organizational science. [TKT-W001, part I, ch. III, B, “Conservative selection”]

The sieve provides a particularly concrete technical case. Bogdanov distinguishes direct sorting by a person's successive interventions from an arrangement in which material differences and the device's resistances do the discriminating. Smaller particles can pass through an opening that stops larger ones. The distinction between the particles is thereby made effective in their different trajectories. A worker still prepares, moves, and uses the apparatus; the account does not describe a process without activity. It explains how a relation between particle dimensions, openings, and motion can perform a discriminating

operation without a fresh individual judgment about each particle. [TKT-W001, part I, ch. III, B, “Conservative selection”]

His adjacent gold-washing example changes the relevant relation. Water carries away some material while heavier gold remains. In this account, a useful result depends on a material difference that the process can actually exploit. Calling both operations selection should not erase the difference between aperture-based separation and separation in a moving fluid. Bogdanov also invokes fractional distillation as a staged separation through differences in response to heating. [TKT-W001, part I, ch. III, B, “Conservative selection”]

The same chapter extends the sieve image to admission conditions, organizational rules, and examinations. These social examples change the criterion of relevance: what is admitted is appropriate to the selecting party's purposes. A hiring condition can select workers suited to an employer, which is a different assertion from selecting an arrangement beneficial to everyone affected. The earlier distinction between activities and resistances therefore continues to matter. Selection records a relationship to a criterion and surrounding conditions; the term does not certify the criterion's justice or the selected outcome's general superiority. [TKT-W001, part I, ch. III, B, “Conservative selection”]

Selection also enters the search for an arrangement. Bogdanov describes difficult invention and discovery as involving the formation and rejection of many combinations before an acceptable one is found. This is distinct from the later use of a successful discriminating mechanism. The cost of finding a workable arrangement and the activity required to operate it are both present in his examples, even though he does not offer a general calculation comparing them. A description of the successful device alone would omit that history of trial. [TKT-W001, part I, ch. III, B, “Conservative selection”]

Finally, he does not leave selection as a wholly separate class of forces alongside joining and separation. Assimilating something from an environment joins it to one complex and requires detaching it from relations in which it previously participated. He therefore presents selection's regulating mechanism as a combination of forming mechanisms. This is an explicit connection within Bogdanov's theory. It helps explain why his sieve and his social examples concern both persistence and changing relations. [TKT-W001, part I, ch. III, B, conclusion]

A passage from Bogdanov's 1921 *Essays on Universal Organizational Science* makes the intended extension especially plain. Its section on the regulating mechanism describes lawful preservation or elimination as the initial schema of selection. It drops “natural” from the name because the proposed organizational science does not treat the natural–artificial distinction as fundamental, and explicitly extends the schema to complexes, systems, connections, and boundaries. This passage supports an account of the breadth Bogdanov claimed for selection. [GL-W097, II.1]

A sieve selects a property, not a purpose

Bogdanov later returns to the sieve to show a limit that the successful technical example can obscure. A person wants flour with useful material properties, but the sieve discriminates particle size. Under the ordinary preparation conditions he describes, size is associated closely enough with the desired material for the operation to serve its purpose. Size represents the other properties in this procedure. The mechanism does not directly recognize all the qualities that make the material useful. Bogdanov calls this *representative selection*. [TKT-W001, part II, ch. VII, discussion of representative selection]

Alter the mixture and the distinction becomes decisive. Fine sand can pass through alongside flour. Harmful material of an appropriate size can do so as well. The sieve can perform its size discrimination correctly while the intended selection fails. Bogdanov's point is therefore stronger than saying that a device occasionally breaks down. The device's operative distinction and the user's desired distinction can

cease to correspond. Their earlier correspondence depended on the preparation of the input and the surrounding circumstances. [TKT-W001, part II, ch. VII]

His gold-processing discussion makes the same issue cumulative. A further separation can remove an impurity that the preceding operation could not distinguish, while itself leaving other substances undistinguished. This illustrates the limitations of successive practical criteria. The chemical procedures are historical illustrations, rather than instructions offered here for their present use. Bogdanov explicitly separates the task's criterion from the property upon which the available apparatus can act. [TKT-W001, part II, ch. VII]

Stability depends on exposure as well as strength

Bogdanov's law of relative least resistance is often easier to understand through its qualifications than through a bare weakest-link formula. Different parts of a structure face different loads. Even links of a hanging chain do not face identical demands when the upper links also carry the weight of those below. The relevant comparison is consequently between the resistance of a part and the activity directed against it, rather than between the parts' absolute strengths alone. A change in loading or a temporary loss of resistance can change the decisive location. [TKT-W001, part I, ch. IV, discussion of relative resistances]

Bogdanov gives this principle a very broad reach. The examples establish why a locally insufficient resistance can matter to an interconnected whole; they do not constitute a proof that the smallest component capacity determines every conceivable system's performance. The consequences of a local rupture still depend on the surrounding organizational conditions. In particular, the simple chain illustration does not describe every parallel arrangement or every way of preserving a function after a part fails. [TKT-W001, part I, ch. IV]

For practical preparation, Bogdanov distinguishes two situations. In one, the kind and magnitude of the relevant demands can be determined with enough regularity to place suitable resistance where it is needed. In the other, the future demands are substantially undetermined. Designing a machine for specified operating conditions and preparing a child for an uncertain social life are his contrasting examples. These are not two entirely separate worlds: an arrangement can face relatively determinate demands in one respect and uncertain demands in another. [TKT-W001, part I, ch. IV, “Practical tasks”]

Importantly, an activity can change the circumstances of the problem. Building a shelter or cultivating land can make aspects of temperature, moisture, or food supply less uncertain for the person acting. Bogdanov presents these changes as a partial movement from the second kind of problem toward the first. The environment is therefore not merely an outside disturbance that the system must absorb in its original form. It can be worked upon, with the resulting arrangement changing what the system subsequently has to withstand or obtain. [TKT-W001, part I, ch. IV, “Practical tasks”]

His contrast between urban and unfamiliar surroundings adds a cognitive example. Street names, house numbers, and clocks reduce the demands of ordinary orientation in a city. A person accustomed to those supports may face a much harder problem in unfamiliar terrain without them, and may need knowledge and instruments that were unnecessary to the familiar task. The source locates part of the difference in the organized surroundings. It does not reduce the city dweller's performance to an individual endowment considered apart from that setting. [TKT-W001, part I, ch. IV, “Practical tasks”]

Bogdanov uses the same discussion to criticize educational preparation that continues to answer an earlier social situation. He argues that training for obedience can become ill suited to circumstances requiring initiative, and that established educational subjects can consume effort after their former practical role has changed. These are his historically and politically situated judgments about education.

Their relevance here lies in the explicit comparison between the preparation supplied and the demands actually encountered. [TKT-W001, part I, ch. IV, “Practical tasks”]

Differentiation produces both capacity and coordination problems

In the second part of *Tektology*, differentiation is not simply an increase in the number of different parts. Different activities develop under different conditions and at different rates. Complementary relations can arise: what one part supplies enables another to continue. But differences can also deepen into misalignment. Bogdanov's social discussion of the division of labor therefore includes productive specialization, market conflict, disproportions between branches of production, and losses when enterprises are destroyed. His explanation is framed by his account of class society and production. [TKT-W001, part II, ch. V, discussion of system contradictions]

The problem of recovering mutual understanding receives a more intimate illustration. Bogdanov describes a historically conventional household in which a husband occupied with public affairs and a wife occupied with domestic work become increasingly unable to understand each other's experiences. More communication might restore a relation that specialization has weakened. Yet the remedy requires time and energy. If the participants have none to spare, the attempt can impair their other activities, create fatigue and new failures, and make the situation worse. The organizational difficulty he identifies is that an activity intended to repair coordination also consumes resources already committed elsewhere. [TKT-W001, part II, ch. V, “The mechanism of removing contradictions”]

Bogdanov also describes a successful alternative. An exchange of experience can change how the specialized activities themselves are carried out: one participant may gain practical calculation, another a wider view of possible action. In that case, communication need not merely subtract time from otherwise unchanged functions. It can improve their conduct. The text presents possible courses of development, not a measured rule specifying when one will occur. The account's useful detail lies precisely in keeping both possibilities: mutual understanding is neither free nor necessarily an unproductive overhead. [TKT-W001, part II, ch. V]

Egression: concentrating action through unequal influence

Bogdanov distinguishes the concentration of activities through a more influential complex from the mere existence of a connection. He calls this *egression*. The relationship is unequal in its effects: one complex influences the development or movement of another more than the latter influences it in return. His illustrations span physical systems and human leadership. In the social case, the issue concerns the organization of activity through command, example, or other unequal influence. [TKT-W001, part II, ch. VI, §§1–2]

A hierarchy of commanders supplies a direct example. Smaller groups are coordinated through local commanders, whose activities are in turn connected through commanders at another level. This chain can concentrate the action of many participants without requiring the highest commander to deal directly with each person. Bogdanov's powers-of-ten illustration shows the arithmetic possibility of extending such a chain under an assumed span of direct coordination. It does not demonstrate that an actual hierarchy can communicate, understand, or execute every necessary action at those scales. [TKT-W001, part II, ch. VI, §2]

He immediately gives reasons why the concentrating capacity has limits. As a chain lengthens, the most distant participants may be only weakly affected by the nominal center. Apparent authority during a stable situation need not provide effective guidance during rapid development or disintegration. The ruler who seems powerful may be unable to secure obedience or to improve the condition of remote

subjects. The limitation concerns the actual relation through intervening links, rather than the authority represented in a title or diagram. [TKT-W001, part II, ch. VI, §2]

There is also a problem of divergence between organizers and those expected to execute their directions. Their circumstances and experience can become different enough that mutual understanding deteriorates. Bogdanov describes unsuitable orders, unanticipated situations in which obedience supplies no answer, arbitrary acts, and antagonistic responses. These are costs of a particular development of the organization, rather than exceptions that can be removed from an account of concentration. His chapter thus contains both the capacity enabled by hierarchical connections and mechanisms that can undermine that capacity. [TKT-W001, part II, ch. VI, §2]

Degression: protection and fixation without command

Degression concerns a different relation. A relatively stable formation fixes or protects a more plastic, vulnerable activity. Bogdanov introduces the issue through the advantages and vulnerabilities of plasticity: flexible relations allow new combinations and adaptation, but can also be more readily disrupted. The skeletal illustration is literal in some cases, yet he explicitly warns against treating all such formations as mechanically hard shells. The protective relation concerns particular destructive influences. [TKT-W001, part II, ch. VI, §3]

For example, his account includes a cloud of ink that interferes with a predator's visually directed action. The proposed protection is temporary and concerns a particular way an attack is guided. This helps explain why the term's scope extends beyond rigid structures. The shared question is what stabilizes or protects the relevant activity against an identified influence, not what objects happen to look like skeletons. [TKT-W001, part II, ch. VI, §3]

Words and other symbols receive an extended treatment within the same discussion. A word is comparatively stable beside the shifting, elaborate experiences associated with it. In Bogdanov's account, that stability helps fix experience, make it communicable, and preserve it collectively beyond an individual's life. He explicitly resists interpreting the word as a superior center that determines all the changes in its associated representations. The symbol's fixing function is why he places it under degression, rather than egression. His particular account of the labor origins of language draws on Ludwig Noiré. [TKT-W001, part II, ch. VI, §3]

Protection can also become restraint. In Bogdanov's developmental argument, the protected activity can grow or change faster than its stabilizing formation. A child's clothing and a dwelling with a growing population make the difference visible: something that previously accommodated activity can become restrictive without undergoing a matching change itself. The account at issue here is the possibility of a changing relation between an activity and the arrangement that fixes or protects it. [TKT-W001, part II, ch. VI, §4]

Egression and degression can coexist while referring to different activities. Bogdanov's ship is directed in its movement by the crew and captain, while its hull contains and protects captain, crew, passengers, and cargo. One relation does not have to be redescribed as the other. His seven diagrams similarly distinguish connection, concentration, and fixation. Their lines indicate linkages, not directional arrows; their filled and unfilled circles mark his comparison of organizational levels. He warns that their spatial appearance is schematic: a dark circle drawn at the center need not describe an internal physical skeleton. Reading the diagrams as ordinary organizational charts would therefore miss his stated meaning. [TKT-W001, part II, ch. VI, §5]

Reorganization and the meaning of a crisis

Bogdanov's treatment of crises examines changes of organizational form. What counts as such a change depends on the features of the form being studied. A broad description of an organism's persistence can treat many changes as phases within one continuing life; a more detailed description of its functions recognizes further transitions. This is not simply a terminological preference. It makes the scale and purpose of an analysis relevant to the changes that the analysis can detect. [TKT-W001, part III, chapter on crises, §1]

He distinguishes connecting crises, which break an existing boundary and establish new relations, from separating crises, which establish boundaries through the disruption of connections. The abstract contrast is simpler than an actual course of reorganization: real processes combine both kinds of change. The label “crisis” also does not require that the process be rapid on an everyday human timescale. [TKT-W001, part III, chapter on crises, §2]

In a later discussion Bogdanov distinguishes quantitative accumulation within a form from structural improvement through reduced internal interference or reduced fruitless expenditure. These can move in opposite directions. His use of Ringelmann's reported cooperation results illustrates his concern with the relationship between group size and effective result, and he acknowledges that the kind of work and mode of coordination matter. Bogdanov also proposes that moral, aesthetic, and cognitive judgments can ultimately be understood as organizational evaluations. [TKT-W001, part III, discussion of structural progress and regress, §§2–3]

Economic interdependence and the limits of a plan

An earlier appendix on labor and workers' needs makes explicit a burden that descriptions of final output can miss. Bogdanov treats complex work as involving the coordination and mutual checking of many activities, including attention, deliberation, initiative, and regulation. In his physiological explanation these processes require expenditure and replacement of nervous energy. He acknowledges that the relevant multipliers cannot then be measured directly. The passage is evidence that his account includes the worker's continuing support and replenishment; it is not current neuroscientific evidence for a quantitative conversion from skill to energy use. [TKT-W001, part I, appendix, “Labor and the needs of the worker”]

His proposed social calculation consequently turns toward the consumption requirements for sustaining different kinds of labor, rather than claiming to have measured a physiological coefficient. The proposal links the organization of work to the material conditions of maintaining the people who perform it. It is also bound to his theory of value and intended planned economy. [TKT-W001, part I, appendix, “Labor and the needs of the worker”]

An appendix on a unified economic plan shows Bogdanov applying the organizational perspective to linked production. His starting point is the maintenance of people, including workers of different qualifications, people unable to work, and the preparation of future workers. He then traces the goods required for that maintenance back through the materials, fuel, tools, and further productive means needed to supply them. The purpose of the chain is to make the functional dependencies of production explicit. It is not merely an aggregate output target. [TKT-W001, part I, appendix, “A unified economic plan”]

His schematic calculation includes substitutable inputs, such as different fuels, rather than assuming that every input has one irreplaceable form. He compares the resulting requirements with existing productive capacities. If capacities are deficient, the proposed recovery sequence gives special attention to the branches whose deficiencies constrain the rest. Expanding transport fully in advance can be wasteful, in

his example, if other branches cannot yet make use of that carrying capacity. This is Bogdanov's proposed reconstruction procedure under the scheme's assumptions. [TKT-W001, part I, appendix, “A unified economic plan”]

The qualifications within the proposal are substantial. Immediate survival or strategic demands can alter the recovery sequence. Trade requires surpluses and changes the linked requirements. Most strikingly, Bogdanov says that completely accounting for and regulating the vast number of small peasant households is inconceivable, while also rejecting the treatment of them as simply external to the state economy. Food, fuel, labor, and the households' own needs make their relations consequential to the plan. His stated confidence in the problem's tektological solvability should therefore be read alongside the practical limits he acknowledges, rather than used to erase them. [TKT-W001, part I, appendix, “A unified economic plan”]

Analogy requires evidence about a mechanism

The appendix reviewing Michel Petrovitch's *Mécanismes communs aux phénomènes disparates* is relevant to how such a wide-ranging organizational program handles comparison. In Bogdanov's account of that book, a pattern in the course of a phenomenon does not uniquely identify the mechanism producing it. Similar movements can result from forces governed differently. Additional observations are needed to reject mechanisms that do not fit. This restriction matters wherever an argument moves from a familiar pattern to an explanation of how an arrangement works. [TKT-W001, part I, appendix, “The doctrine of analogies”; TKT-W003]

Bogdanov’s review describes comparisons across mechanical, electrical, physiological, and social phenomena, including an effort to match the roles that elements play in different processes. Bogdanov finds heuristic value in these comparisons while criticizing some of the economic explanations. An analogy can suggest a mechanism; further evidence is needed to establish the explanation. [TKT-W001, part I, appendix, “The doctrine of analogies”; TKT-W003]

Rispoli's interpretation of environment and collective knowledge

Giulia Rispoli's study selects a dimension that a treatment confined to devices or organizational charts would miss. She interprets Bogdanov's environment as both a source of conditions upon activity and something changed through organisms' and people's activities. It is neither an empty space awaiting action nor an independently fixed set of determinants. Her discussion of the differing surroundings of a plant's parts and of relations among organisms emphasizes that the relevant environment depends on the relation under examination. Her comparisons with later ecological and systems ideas are the historian's interpretation. [GL-W022, pp. 131–134]

Rispoli then relates this account to knowledge produced through collective work. On her reading, shared practical activity precedes and supports the verbal exchange of experience; skills, tools, practices, and the material setting belong to the production and transmission of knowledge. Her discussion of Proletkult places the proposal within Bogdanov's political and cultural undertaking, including workers' organization of their own cultural activity. [GL-W022, pp. 134–136]

Joining, selection, differentiation and collective knowledge

The relevant account in *Tektology* is extensive because its subjects are connected within the author's project. Joining and separation explain changing relations; selection explains persistence through particular conditions; relative resistance makes loading and exposure consequential; differentiation gives rise to both capacities and conflicts; concentration and fixation describe distinct organizational functions.

The economic and cognitive discussions are applications within that project, with their own purposes and limits.

For a reader studying gambits engineering, the most concrete learning lies in the distinctions that the examples make available: what an arrangement accomplishes and who supplies the effort; what property a selector discriminates and what purpose that property represents; what a center can actually influence and what authority it nominally possesses; what protects an activity and what eventually restricts it.

S18 General systems traditions and distinct programmes

A field of programmes, models and questions

General systems writing contains several different enterprises: identifying recurrent mathematical relations, describing organization across levels, explaining maintenance and development, and arranging inquiry across disciplines.

The International Society for the Systems Sciences describes its founding purposes as identifying isomorphies across fields, encouraging useful model transfers, reducing duplicated theoretical effort and improving communication between specialists. These are substantive proposals for organizing intellectual work. Researchers might reuse an appropriate formal result instead of independently rebuilding it in each discipline. [SG-W006, main body]

Rapoport's account presents general systems theory as a programme or direction of inquiry rather than one particular scientific theory. Its integrating purpose concerns concepts and research methods and permits distinct domain theories. [SG-W037, opening]

Boulding's *The Skeleton of Science* proposes two routes. One starts with phenomena that recur across disciplines and asks whether their relations support common models. The other arranges systems by increasing organizational complexity: static arrangements, dynamic mechanisms, control systems, self-maintaining systems, and further biological, symbolic and social levels. The hierarchy is a research scaffold. A model that captures a lower level may leave out precisely what matters at a higher one. An organization cannot be adequately explained as a thermostat merely because both involve feedback. [SG-W036, article]

This warning is consequential for economy claims. A reduction in the variables of a model is useful only if the description still answers the question. Boulding's article seeks intermediate generality between isolated specialisms and contentless universality. Its historical judgments about biological and social levels are presented as parts of the 1956 programme, rather than current empirical conclusions. [SG-W036]

Bertalanffy: what formal recurrence can and cannot show

The 1950 article *An Outline of General System Theory* develops a connected argument about interacting parts, differentiation, control, maintenance and finality. Bertalanffy's ambition is to find principles applicable to systems as systems, while preserving the special laws and problems of particular fields. [W072, printed 134–165]

He offers several reasons why similar mathematical forms recur. There are relatively restricted mathematical schemes available for solution; some system conditions themselves recur; and formally analogous relations can occur among physically different variables. These possibilities require discrimination. Writing two processes with the same equation does not show that their constituent things, causes or measurements are identical. A proposed mapping must preserve the relevant relations. Otherwise, apparent generality may come from a convenient notation or a superficial analogy. [W072, 137–142]

His treatment of aggregate variables supplies a further condition. Replacing a multitude of particulars with a smaller number of state quantities can make analysis possible, but the quantities must be adequate for the process under study. An aggregate description needs empirical and theoretical warrant. [W072, 141–146]

The finite system of differential equations illustrates interactions among changing quantities. The quadratic growth form is not necessarily sigmoid for arbitrary coefficients and initial conditions; the characteristic-root discussion contains a sign error; and the allometry discussion gives inconsistent forms for a constant and its inequality. [W072, 145–146, 151–152]

Interaction, specialization and the loss of restoration

Wholeness in the article means that the behavior of interacting components differs from what their isolated behavior would imply. Analysis of components remains useful, but it must be joined to an account of their relations. It says that removing a relation can change the phenomenon being explained, so that a parts inventory is insufficient. [W072, 146–150]

Progressive segregation and mechanization describe a change in organization. Components become more differentiated and their activities more specifically assigned. Specialization can support effective differentiated performance. At the same time, a component that once could participate in restoring several functions may lose that flexibility. Greater fixed assignment can make replacement or regulation of a damaged whole more difficult. The article thus discusses a gain accompanied by a possible loss. [W072, 147–150]

A differentiated whole may perform a task well because its parts no longer have to reproduce every function. That arrangement can also depend more heavily on particular parts continuing to work. Bertalanffy's examples connect organization with specialization and restoration. The functions and disturbances being compared must remain explicit. [W072, 147–150]

His discussion of a leading part also separates control from the energy of performance. A small change can initiate or direct a much larger response because the responding process draws energy from elsewhere. The control signal need not supply all the energy released. This gives a concrete meaning to an energetically small determining influence. His nested levels similarly concern organization; a hierarchy in this sense need not be a hierarchy of human authority. [W072, 150–151]

Open maintenance and conditional equifinality

Bertalanffy's open-system account concerns maintenance through material exchange. A living organization can retain an approximately stable pattern while its constituents are continually replaced. Its persistence is an ongoing process involving import, transformation and export. A steady state is therefore different from an unchanging collection of material pieces. In this discussion, “closed” refers to absence of material exchange; it should not silently be strengthened to complete energetic isolation. [W072, 155–158]

Equifinality names arrival at a similar final condition from different initial conditions or through different routes. The important mechanism is the system's organization and continuing processes under appropriate conditions. The endpoint need not encode a unique history. This helps explain why observing the maintained condition does not by itself reveal the complete starting arrangement. [W072, 157–159]

The article's biological qualifications matter: impaired conditions can prevent restoration, and the relevant rates and resources remain necessary. More generally, the claim needs a stable regime, an appropriate basin and maintained parameters. Openness alone does not exclude multiple stable outcomes. Nor is every materially closed system incapable of convergence under its constraints. The account retained here is conditional equifinality, not the blanket opposition between inevitable closed-system history dependence and inevitable open-system convergence. [W072, 157–159]

The account of entropy likewise retains exchange across the boundary. A local increase in organization is not a violation of the second law when the surrounding flows are included. The article extends minimum-

entropy-production reasoning beyond its applicable regimes. Its stellar illustration also involves exchanges with the surroundings. [W072, 161–162]

Different senses of finality

Bertalanffy distinguishes several phenomena often grouped under purpose. Static fitness concerns an arrangement suited to a function. Dynamic finality concerns movement toward a condition. A machine can have structure that guides its operation; feedback can correct departures. An open organization can maintain or recover a state through continuing exchange. Human foresight and purposive language add a different issue. These descriptions should not all become claims that the future causes the present. [W072, 154–155, 159–160]

This distinction preserves significant information for gambits engineering. A suitable arrangement, a correcting loop and an anticipating actor may each affect how performance is achieved. Its final discussion leaves ultimate reduction of the sciences open and rejects treating domain-specific inquiry as dispensable. Generality is pursued alongside the autonomy of particular problems. [W072, 164–165]

Bateson: restraints and levels of choice

Bateson's 1967 *Cybernetic Explanation*, explains events by examining alternatives that restraints exclude. A jigsaw piece is constrained by neighboring shapes, the boundary and the picture. The same features act as clues for a solver and restraints in an explanation of what can fit. The account shifts attention from an isolated object to the relations that make some possibilities more likely than others. [SG-W009, PDF 2–3]

His discussion of mapping concerns relations preserved through transformations. A formal model or simulation can be informative when its relevant correspondences hold. It is not enough to substitute one vocabulary for another. His analogy with logical proof describes the form of cybernetic explanation. [SG-W009, PDF 3–4]

Energy is explicitly retained. In a communication sequence, the response commonly uses the respondent's own energy: a neuronal response or an activated circuit need not be powered entirely by the triggering signal. The process still obeys conservation, and supply can limit what happens. A telephone network can also run out of available routes before exhausting its energy supply; alternative pathways are a different limiting dimension. [SG-W009, PDF 5]

Feedback constrains what happens after a disturbance has traveled around a circuit. The effect depends on gain, timing, thresholds and other characteristics. A circuit is causally closed in the sense of returning influence, while remaining open to energy and outside events. Thus the word “closed” here identifies a causal path, rather than the material-exchange boundary used in Bertalanffy's discussion. Keeping those meanings explicit prevents an apparent contradiction created solely by terminology. [SG-W009, PDF 5–6]

The distinction between levels of choice is especially useful. Once a person chooses a familiar word, its spelling can be available as a pattern; each letter need not be independently selected by eliminating every wrong letter. A higher-level choice can remove lower-level choices from the immediate task. The explanation retains the already established pattern. [SG-W009, PDF 6–7]

Redundancy also depends on the observer's prior knowledge. A person who knows a pattern can infer something about one portion from another. Someone who first has to discover the pattern faces a different task, so the apparent location and availability of information change with the explanatory boundary. Bateson's discussion of educated and initially uninformed observers makes that dependence concrete. [SG-W009, PDF 8–10]

Foerster: the boundary of self-organization

In the selected passages of *On Self-Organizing Systems and Their Environments*, Foerster challenges an account that attributes increased order to an isolated system while neglecting its environment. What counts as self-organization depends on the chosen boundary and on the resources exchanged across it. His opening provocation should be read with that argument, rather than as a literal denial of every process called self-organization. [W080, 2003 reprint 1–3, 6]

The magnetic-cube illustration is concrete. The components have selective properties built into them, and the surrounding process supplies opportunities for interaction. The resulting arrangement is not explained by undifferentiated matter spontaneously acquiring every relevant discrimination. Component design and environmental conditions belong in the explanation. [W080, 11–13, discussion 19]

Maruyama: amplification can generate a history

Maruyama's 1963 *The Second Cybernetics* distinguishes deviation-counteracting from deviation-amplifying mutual causal processes. Both involve reciprocal influence. Counteraction can preserve a pattern; amplification can build differences and produce new patterns. “Second” here names a different kind of process, not automatically the later programme of observer-centered second-order cybernetics. [SG-W061, PDF 1–2]

His economic discussion attributes to Myrdal a circular account of cumulative advantage and disadvantage. Poor health, low working efficiency, low income and exclusion can reinforce one another. Institutions such as transport, education and insurance can affect the direction of the process. [SG-W061, PDF 2–3]

The city illustration separates the initial event from subsequent development. On an initially homogeneous plain, a farm may attract other farms, a tool shop and further services. Increased activity supports more activity. The eventual location need not have had a decisive prior geographical advantage. A historian who seeks only a large initial cause may overlook the cumulative process. The story is an explanatory illustration. [SG-W061, PDF 3–4]

Small initial differences can have large consequences even in deterministic processes. Difficult inverse reconstruction does not imply absence of forward causation. Maruyama's development per initial investment differs from efficiency per total resource input; amplification does not overturn the second law. [SG-W061, PDF 3–7]

The developmental models make the account more specific. Local rules and ordered phases can generate elaborate patterns without storing a separate instruction for every eventual cell. Their operation nevertheless retains a rule set, an interpreting process, an initial state, spatial relations, occupancy and time-dependent stages. [SG-W061, PDF 8–11]

Maruyama also warns that apparent bilateral traffic need not establish mutual causal dependence. A producer may have alternative suppliers or customers; a third activity may drive both sides. An account of dependence needs the substitution possibilities and boundary, rather than arrows showing transactions alone. This qualification is directly relevant to identifying which relations actually determine an organization's performance. [SG-W061, PDF 11–12]

Loop signs describe directional effects, not moral values. An amplifying process can be destructive, while a stable pattern can preserve a harmful condition. Negative feedback can oscillate under particular delays and gains; mixed networks depend on strengths and operating regimes as well as signs. The proposed inventing machine combines search, amplification and counteraction, but is a proposal rather than a demonstrated result. [SG-W061, PDF 13–16]

The 1988 author retrospective is a separate source. Maruyama reports that the popular article omitted mathematics from earlier work, describes repeated rejection, and explains that “second” meant a different kind rather than a serial stage. He gives heterogeneity an increasingly explicit positive role. These are valuable statements of the author's interpretation and intellectual development. [RGC-W037, retrospective]

Qian's knowledge architecture and its later reconstruction

The selected reprint of Qian Xuesen's 1986 speech treats departments of science as perspectives on a connected whole and distinguishes basic science, technical science and engineering, with philosophical relations across the arrangement. This is an account of how inquiry and application can be organized. [WME-W027, 2021 reprint, discussion of the organization of science and technology]

The institutional historical article credited to the Chinese Academy of Sciences reconstructs an expansion from “1+3” to “1+11” fields and emphasizes their connections. Its relevance is the changing architecture through which disciplines and practical work were related. [WME-W028, 2010 main body]

Lu Mingsen’s 2004 reconstruction and Yu Jingyuan’s 2001 account explain the open-complex-giant-systems programme developed in the 1990 Qian–Yu–Dai article and Yu’s 1992 work. [W035; W036; RGC-W039, 17–23; RGC-W040, 10–18]

Lu describes a movement from systems engineering toward problems whose scale, heterogeneity and human participation make a purely decompositional treatment inadequate. The proposed response integrates experts, data, computing, and qualitative and quantitative inquiry. Seminars, broad expertise and organized collective work remain part of the process. [RGC-W039, article]

Metasynthesis retains people, machines and revision

Yu's 2001 account makes the method operationally clearer. Scientific theory, experience and expert judgment first contribute provisional hypotheses, plans or conjectures. Machines and networks handle suitable logical, numerical, storage and communication tasks. Human-led comparison and successive revision connect qualitative judgments with quantitative work and micro descriptions with macro questions. The method is intended to increase the capacity of inquiry through a deliberately composed human–machine arrangement. [RGC-W040, printed 16–17]

His expert, machine and knowledge systems are mutually connected. Experts and machines carry knowledge; they do not make it unnecessary. Collection, storage, retrieval, transmission, analysis and the production of further knowledge remain activities of the arrangement. Decomposition is still used, followed by integration. The programme's opposition to an inadequate reductionism therefore should not be rewritten as a prohibition on analyzing parts. [RGC-W040, 17]

The symbolic “1+1>2” expresses an aspiration to emergent advantage. It is not a measured inequality in one defined unit of knowledge, cost or performance. Yu's claim that repeated comparison yields a scientific conclusion describes the intended process; iteration by itself cannot guarantee correctness without adequate evidence. His division of capabilities between people and computers is a 2001 position, not a proved permanent limit on computation. [RGC-W040, 16–17]

The 2007 *Research on Meta-synthetic Methodology and Systemics* preview provides important self-qualification. Its printed opening asks about necessary and sufficient conditions for “1+1>2” as an unresolved theoretical problem. The programme is described as embryonic. [W078, preview 8–12]

The authors describe practical difficulties in a project involving multiple institutions. A central group could not simply command all participants, and software working inside individual groups did not

automatically interoperate. Funders, researchers seeking innovation and practical users brought different requirements. Even the same experts could reach different conclusions under a different convener or purpose. Knowledge integration consequently includes coordination of institutions, interfaces and purposes, not merely aggregation of statements. [W078, preview 4–10 and prefatory material]

The preview's iterative procedure retains changing goals, thresholds, feedback and limited relative optimization. Its network recommendations require substitutable capabilities and surviving connections; decentralization alone does not prove robustness. Its proposed multi-agent arrangements still include membership management, interaction mechanisms and standards. Security has managerial and social aspects, while an apparently friendly interface can require substantial development and knowledge of users. [W078, preview 9–12]

WSR: physical conditions, organized work and human acceptance

The opening of Gu and Tang's 2002 WSR chapter introduces 物理—事理—人理, commonly rendered Wuli–Shili–Renli. The orientation combines attention to physical and technical conditions, the organization of activities, and human understanding and interests. The authors map historical influences across cultures. The 1997 article below develops a practical application. [WME-W026, printed 16–21; RGC-W041]

Gu and colleagues' article on a commercial-facilities standards framework makes the three aspects concrete. Technical standards must respect actual equipment, material and economic capabilities. They should also support improvement rather than freeze poor performance. Organizing the standards into a coherent framework is another problem: standards need suitable levels and relations to other systems. Human acceptance is equally operative because experts, managers and users must understand, negotiate and implement the result. [RGC-W041, printed 134–137]

The reported workflow begins with literature and expert inquiry, pools results within a multidisciplinary project group, extends inquiry through a wider survey, and returns to group and expert consideration with computer analysis and display. Brainstorming generates alternatives; Delphi treatment and clustering help interpret opinions, including differences associated with respondents' backgrounds. The editable computer knowledge base preserves the framework and survey results for later retrieval, revision and supplementation. Reusability is achieved through an organized maintenance process. [RGC-W041, 135–136, fig. 1]

The reported compromise combines a three-and-a-half-level structure, broad inclusion of related standards and selective detail at the lowest level. The resulting framework contains eight systems, sixty-two specialties, forty related standards and sixty-seven entries. The authors report official appraisal in 1996. [RGC-W041, 136–137]

The article's CATWOE presentation also requires care: it announces six elements but initially lists five, omitting the ownership element subsequently used for leadership opinion; its transformation description is narrower than the general term. [RGC-W041, 135–137]

Other accounts of relations, identity and integration

Dostal's *There are Systems and Systems*, in its 2025 revision of the 2023 proceedings paper, connects entities through activities and treats entities as emerging from interwoven activities. Aims, processes, structures, governance and environmental flows belong in the account. “Tapping” makes an output's actual reception and use a condition of an effective interface: producing something does not ensure that another activity can employ it. Opposed sequences of design and persistence describe the encounter between intended change and existing organization. [SG-W004, 5–14]

Dostal's conceptual synthesis advances practical hypotheses about organizational arrangements. Its information-field and healing ontology extends the argument into speculative claims beyond the organizational examples. [SG-W004, 16–19]

The selected Schwarz excerpt connects action procedures to models and then to underlying beliefs and values. A procedure's practical success, a model's validity and a claim about truth are related questions with different warrants. The relevance is the dependence of organized action on conceptual and evaluative commitments. [SG-W013]

The two Principia Cybernetica pages on metasystem transition describe copies or variants brought into an arrangement with an additional control mechanism. Mere aggregation is insufficient. Control over an activity becomes an activity at another level, and the controller need not be spatially detached from what it controls. Its broader evolutionary progression is a theoretical proposal. [SG-W032, both pages]

Bennett's *Systematics and System Theories* treats a system through qualitatively differentiated terms and their mutual relevance. Compresence and the functional identity of the whole matter alongside pairwise interaction. Successive orders of terms provide ways to organize qualitative judgment. This is a vocabulary for examining kinds of relatedness. [SG-W033, body]

Moreira's reconstruction of Bunge supplies a different explanatory demand through composition, environment, structure and mechanism. A list of components and their bonds can leave novelty unexplained unless the processes that produce and sustain it are specified. The company illustration places activities and environmental interaction inside the description. [SG-W043, composition, environment, structure and mechanism]

Volk and Bloom's metapatterns article proposes a provisional vocabulary of recurrent forms and functions. Borders combine separation with regulated exchange; tubes, webs and centers organize different relations; holons and clonons, triggers and cycles identify further recurring possibilities. Convergence under similar conditions is part of the rationale, with differences acknowledged between cultural and genetic processes. The extension to educational templates is an assumption the authors say has not been rigorously tested. [SG-W044, PDF 1–11, 13 opening]

Wu's letter to pansystems friends links generalized relations, observation and control, parts and wholes, transformation and optimization. Its unusually pertinent qualification is that generalized language should not become an all-encompassing slogan: useful application still requires knowledge of the field in question. [SG-W049, letter, part 1]

Maps and organizational qualifications

Zhang Hao's 2022 *Management Science Research Models and Methods*, second edition, distinguishes the "old" and "new" triads and sets out multiple schools, including mathematical systems work, operations research, dynamics, complex evolution, dissipative structures, synergetics, catastrophe approaches and systems management. It also names grey systems, pansystems and matter-element analysis. [SG-W005, preview 3–13]

The University of Shanghai for Science and Technology's 2021 account similarly names hypercycles, living and social systems, pansystems and grey systems among the branches of systems science. Its principles supply an institutional teaching presentation of the field. [SG-W007]

Kast and Rosenzweig's selected organizational passages bring the issue back to explicit boundaries and contingencies. Organizational survival, consequences for society and outcomes for participants are different criteria. A configuration useful under one set of conditions need not suit another, and a contrived organization should not be identified exactly with a biological organism. The account calls for

specifying levels, relevant relations and conditions rather than relying on systems language alone. [SG-W056, printed 447–448, 455–457]

S36 Campbell: information, constraint and intellectual reception

Information, constraint and reception

Jeremy Campbell's *Grammatical Man* connects information, entropy, language, life and the possibilities created by constraint. The foreword and afterword in Juan José Utrilla's 1989 translation, *El hombre gramatical*, examine how constraints enable variety. [WME-W029]

Constraint can enable variety

The foreword and afterword treat rules and constraints as enabling a rich range of outcomes. An arrangement of rules can make organized possibilities available. The afterword relates Aristotle, form and matter to the question of how form directs what material can do, and discusses costs associated with informational ordering. [WME-W029, foreword, pp. 9–10, and selected afterword, pp. 387–395]

Sless: lucid communication theory, disputed unification

David Sless's review provides the most substantial scholarly criticism among the reviews discussed here. He praises Campbell's accessible explanation of ideas surrounding mathematical communication theory, while objecting that the actual mathematics is omitted. For Sless, this limits the reader's ability to assess the theory and its applications. [SB-W018, printed pp. 186–188]

Sless describes the book as progressing from communication toward symbols, genes, perception, knowledge and society. He argues that Campbell's enthusiasm underplays reservations, including caution from information theory's founders. His objection is not simply that many disciplines appear in one volume; it is that the breadth can give a stronger impression of established unification than the argument warrants.

The review also has limits of its own. Sless expressly declines a competent assessment of the biological-evolution material. That disclaimer leaves a major part of the proposed connection unevaluated by this reviewer. His assertion restricting Shannon theory to finite systems is too broad to adopt as a technical conclusion.

Sless's treatment of perception is more discriminating than a blanket rejection. He praises the explanation of J. J. Gibson's ideas and objects that R. L. Gregory's contrasting position receives insufficient development. [SB-W018]

His closing discussion concerns the extension to social theory and the afterword's link between Aristotle and DNA. He criticizes the favorable social and political implications attached to the language of information and variety. The review consequently distinguishes explanatory clarity in a particular unit from acceptance of the entire cross-disciplinary progression.

Simpson: compact instruction and an unclear connection

Fred Simpson's review describes a book full of interesting perspectives whose overall connection he finds frustratingly unclear. His positive response centers particularly on DNA as a compact, reliable yet flexible instruction scheme. In his account, rules constrain transmission while permitting diversity. Simpson emphasizes the relation between a compact inherited scheme, repeated development and a range of possible outcomes. It remains Simpson's reconstruction of Campbell's biological discussion. [SB-W019, printed pp. 40–41; issue PDF pp. 42–43]

Simpson also reports a progression through entropy, information, language, genetics, the brain and Aristotle. He uses broad equations among information, order and the opposite of randomness to describe both the appeal and difficulty of the book. [SB-W019]

Simpson suggests that the book implies a grand design, while noting that Campbell does not quite say this. It is the reviewer's interpretation of the narrative. [SB-W019]

The review ends favorably despite its frustration: Simpson describes returning to sections and finding the perspectives rewarding. That response shows that unclear integration and useful individual material can coexist in a reader's experience. It is a historical reader response to Campbell's book.

Mair: enthusiastic reception and the difficulty of redundancy-poor text

Victor H. Mair's short notice praises the breadth and informative character of *Grammatical Man*. He associates the difficulty of reading Classical Chinese with ambiguity and low redundancy. His notice applies an informational perspective to a task requiring interpretation. [SB-W020, B28; collection PDF p. 45]

Mair also describes being reassured by the book's broad account of forces opposed to chance. His comments about molecules and the fate of the universe convey the book's rhetorical impact on this reader.

Kirkus: a favorable account that still acknowledges speculation

The Kirkus review identifies themes of uncertainty, redundancy, contextual constraint and interaction across communication, genetics, perception and linguistics. It presents Campbell's synthesis favorably, while recognizing speculative elements and weak evidence in parts of the wider discussion. It therefore supplies both a topic map and a contemporaneous response to the book's explanatory ambition. It is a trade review, with a different purpose from a scholarly adjudication of the underlying fields. [WME-W030; WME-W029, review]

S19 Jantsch: maintenance, transformation and connected descriptions

Evolution and changing organization

Erich Jantsch's *The Self-Organizing Universe* offers an extensive account connecting physical organization, biological development, ecological relations and human meaning. Its central concern is evolution as a process that can change not only its products but also the ways in which further change occurs. [WME-W001, 1980, foreword and chs. 1–4]

Organization can persist through continuous activity

Jantsch begins with a distinction between arrangements that persist as relatively conservative structures and processes that maintain a dynamic organization through exchange. A dissipative structure is not simply a static shape that happens to use energy. Its organized regime depends on continued interactions, including the import of usable resources and the export of products and entropy. What persists can be a pattern of activity even while constituent materials change. [WME-W001, ch. 2, 21–41]

His chemical discussion identifies openness, departure from equilibrium and nonlinear reaction organization, including autocatalytic or cross-catalytic steps. An autocatalytic participant contributes to the production of further participants of its kind; cross-catalysis closes that relation through intermediates. These are specified process relations, rather than an unexplained appeal to a whole acting on its parts. The account describes conditions for the models under discussion. [WME-W001, 31–35]

The maintained organization continues to produce entropy. Jantsch's emphasis is on flows and rates rather than treating internal order as a violation of thermodynamics. A locally persistent regime requires continuing environmental support. If that support fails, the organization can degenerate. “Self” in this account does not mean independence from materials, energy or surroundings. [WME-W001, 31–35, 203]

He distinguishes several aspects of a system: its environmental relations, internal state, logical organization, function, spatial and temporal structure, and total dynamics. These are different questions about the same organized activity. A flow diagram can describe how transformations connect, while a spatial picture describes where they occur. Neither by itself supplies the full account of what makes the organization persist. [WME-W001, 32–35, table 1]

The book uses autopoiesis broadly in this process account. Jantsch extends the term from chemical models and living cells toward institutions. His physical examples give specified ways of understanding organized dynamics. His later social extensions require additional interpretation rather than following as direct experimental deductions from a chemical reaction. [WME-W001, chs. 2–4]

Stability and transformation are different accomplishments

Within a maintained regime, disturbances can be absorbed. A sufficiently amplified fluctuation can instead lead to a different regime. Jantsch relates structure, function and fluctuation: the organization constrains which changes persist, while changes can eventually modify the organization itself. Maintenance and transformation therefore form connected but different aspects of his evolutionary account. [WME-W001, 41–48]

The distinction between microscopic and macroscopic descriptions is substantive. A macroscopic analysis can identify instability thresholds and possible resulting regimes. A microscopic account follows the formation, size and spread of fluctuations. Crossing a parameter threshold need not mean that any tiny local disturbance actually reorganizes a finite system. Spatial extent, diffusion, packing and coupling affect

whether a fluctuation survives or is damped. Jantsch's presentation of nucleation makes the relation between local events and whole-system change conditional. [WME-W001, 44–48, figs. 10–11]

His chemical examples allow several possible branches after a transition. The selected path depends on the process's history as well as the possibilities available under its conditions. A model's available outcomes and a prediction of the particular outcome are different achievements. Jantsch repeatedly presents such models as aids to understanding; the book's city and population models also have simplifying assumptions. [WME-W001, 47–57, 65–72]

Reorganization can require work. The book's hysteresis discussion associates different forward and return paths with the work invested in changing structure. Once the conditions retreat, the organization need not simply trace an identical reversible course. Calling this memory identifies a dependence on prior change. [WME-W001, 48–49, figs. 11–12]

Jantsch connects an established regime with economy of operation and a transition with expenditure on restructuring. That is a useful distinction between using an arrangement and making another arrangement possible. His novelty/confirmation and entropy-production diagrams organize the conceptual story; they are not calibrated accounts in one unit of lifecycle cost. [WME-W001, 42–53, figs. 12–14]

The book's schematic contrasts between maximum and minimum entropy production extend beyond the conditions under which those principles hold. They do not establish an optimizing law across physical, biological and social regimes. [WME-W001, chs. 2–3]

Co-evolution changes the surrounding possibilities

Jantsch's macro–micro narrative makes organisms and their surroundings participants in a connected history. The environment supplies conditions for local processes, while the accumulated effects of those processes alter the environment encountered by later participants. Adaptation to a fixed setting is therefore only one part of the account. The rules and opportunities of further evolution can also change. [WME-W001, chs. 4–8]

The value of this claim lies partly in its causal direction. A feature that appears economical inside one organism may be possible because other processes now supply what it no longer produces. The account of reduced internal biosynthetic capability, illustrated through vitamin supply, locates a corresponding environmental dependence. Its relevant point is that missing internal production and independence are different things. [WME-W001, ch. 7, 133–138]

The architect's plan provides an especially accessible illustration. A plan sent to a distant building site is effective only if local conditions permit realization: materials must be available or obtainable, skills must exist, and infrastructure and environmental relations must support construction. Sending the plan does not transport the completed building or create skilled masons. Jantsch uses the example in explaining the transfer of process programmes rather than all the material conditions of their execution. [WME-W001, 133]

He also discusses the other side of adaptation. Acquired ecological information and an established way of life can increase present capability while narrowing later possibilities. A relation that once opened opportunities can become a dependency. Autonomy and flexibility do not therefore increase together in every step of the evolutionary narrative. The book's changing levels and criteria prevent its account from being reduced to monotonic improvement on one scale. [WME-W001, 141–142]

The biological and cosmic chapters range widely. Their original dates, evolutionary reconstructions, particle-physics models and neurobiological frameworks are historical premises of the 1980 account. [WME-W001, chs. 5–9, 11]

A record becomes useful through selective activation

Jantsch distinguishes ongoing processes from relatively conservative storage. DNA, books, tools and representations can preserve possibilities, yet preserved form becomes effective only through its use in an active context. His term “epigenealogical” describes the relation between stored information and its selective retrieval in living, changing processes. The book's diagram makes both the record and its interpreting context visible. [WME-W001, ch. 11, 195–202, fig. 38]

The musical score illustrates the point. Its marks preserve an ordered possibility, while a performance involves learned conventions, interpretation and present activity. A book likewise does not make its knowledge available independently of readers and their purposes. Selection changes which stored possibilities become operative. Storage can support reuse without executing itself. [WME-W001, 202; ch. 18, 286–296]

This is more than a claim that a container holds information. Jantsch places novelty and confirmation in relation: prior patterns can guide activity, while a changed context can activate them differently. His evolutionary learning process has no permanently fixed school-like target. Existing histories and traditions channel what can happen, but a channeled process is not the same as a sequence directed toward a single externally supplied goal. [WME-W001, 202]

The account also retains implementation work in cultural creation. Vision must acquire rational form, and form must be interpreted and realized through learned and manual capabilities. A notation or representation can make an intention transportable and repeatable while depending on the skills and materials of performance. [WME-W001, ch. 18]

Interaction, communication, symbiosis and fusion

Jantsch distinguishes several kinds of relation rather than treating all coupling as one thing. Interaction names exchange with an environment not organized as a comparable participant at the selected level. Communication involves one organized process evoking activity in another. Symbiosis involves the mutual use of products or services, while fusion describes the limiting case in which the former participants lose their separate autonomous organization. These distinctions concern the processes and autonomy retained, not merely the number of connecting arrows. [WME-W001, 203–207, fig. 39]

In his communication account, the recipient must be capable of the activity evoked. A description of a sunset works through a listener's experience and capacity to form meaning; it does not transfer the sunset itself. A stimulating signal also does not provide all the recipient's energy. Jantsch explicitly retains each system's responsibility for its own structural problems and energy flow. [WME-W001, 203–205]

Dialogue can in turn reorganize knowledge rather than merely exchange already completed packages. The claim is that mutual stimulation can change the participants' cognitive domains. That remains a conceptual account of generative communication. [WME-W001, 205–206]

Symbiosis introduces a different dependency. Participants use one another's transformation products or services. Each can relinquish some individual autonomy while participating in a larger organization with capabilities at another level. The lower-level processes do not necessarily vanish. This is why Jantsch distinguishes a partially autonomous relation from complete fusion. His own account also allows excessive mutual dependence, in which the relation narrows possibilities without producing the anticipated novelty. [WME-W001, 206–207]

His application of “co-operative” to whole ecological processes needs its stated viewpoint. A predator–prey relation may contribute to an ecosystem's pattern while destroying particular prey. Organization at the larger level does not imply benefit to each individual. Jantsch explicitly recognizes that affected

participants might not experience the relation as cooperation. The qualification matters whenever larger-system persistence is used as a criterion. [WME-W001, 135–136, 206]

Multiple levels retain different tests and responsibilities

The book's account of emergent levels includes distinct criteria: energetic penetration, error correction, variety, flexibility, consistency and connectedness. These are not interchangeable numerical forms of one variable. Jantsch describes higher-level openness supported by more settled lower-level activities, but that construction does not make the lower levels inert or eliminate their conditions of existence. [WME-W001, ch. 12, 218–230, figs. 41–43]

His time- and space-binding account relates stored experience, present interactions and anticipation. A present organization can employ records of the past and images of a possible future. [WME-W001, ch. 13, 231–238, fig. 44 and table 5]

Stratified autonomy is his alternative to a picture in which information merely travels upward and orders travel downward. Each organized level can maintain some direct relation with a wider environment. A cell's surroundings, in his illustration, include more than adjacent cells. In the proposed social extension, local participants retain contact with their own circumstances while a broader level performs synthesis. The practical issue is the distribution of initiative and integration. [WME-W001, ch. 14, 239–250]

Jantsch's formulation of decentralized initiative and centralized synthesis therefore retains a management role. The manager as catalyst is an assigned activity, not an absence of organization. He also argues that an institutional pattern can suppress novelty at precisely the levels where broader changes need consideration. These are the author's organizational proposals and criticisms. Jantsch quotes Mesarović in discussing control hierarchies and relative frequencies. [WME-W001, 248–250]

The argument about intermediate organizational levels supplies a further distinction. A choice between very large and very small units can omit forms that connect local capability with wider provision. His machinery illustration, attributed to Schumacher, describes technical development reinforcing an expected disappearance of small producers. It is a historical reported example of the relation between technological options and organizational scale. [WME-W001, 249]

Planning can change what counts as satisfactory

Jantsch separates tactical planning for an already recognized output from strategic, policy and value questions. At the tactical level, efficient realization within a relatively stable structure can be a reasonable aim. At the strategic level, alternatives are imagined, examined and prepared. Policy places the organization's activity in a wider social context, while values influence which futures deserve pursuit. The planning task changes as the level changes. [WME-W001, ch. 16, 266–274, table 6]

It follows within his account that a more open planning process can deliberately introduce options, uncertainty and complexity. Minimizing immediate effort or maximizing short-term profit does not settle whether a new social function should be developed or a harmful one discontinued. He distinguishes the survival of a particular organization from continuity of the social processes it serves. The book thus contains a substantive challenge to using organizational survival as the only criterion. [WME-W001, 266–267]

The evolutionary proposal is not simply enthusiasm for destructive discontinuity. Jantsch recognizes that large transitions can damage the biological and cultural substrate on which further life depends. His proposed “gliding” evolution imagines partially overlapping changes and plural paths instead of requiring an entire civilization to change through a single rupture. [WME-W001, ch. 15, 251–262]

Part IV explicitly offers perspectives rather than solutions to every problem. Its ethical account places several levels of obligation and experience in tension. The passage on Antigone illustrates conflict between personal moral commitment and the law of the larger social order, not a method that automatically reconciles them. Jantsch's wish to ground an evolutionary ethics is a normative and philosophical proposal. Biological or social survival does not by itself establish a moral duty. [WME-W001, ch. 16, 263–274]

The technology discussion also limits a direct transfer from ecology to human economy. Exosomatic tools, restructuring of environments and increased energy use change the conditions of the comparison. A human arrangement can become more capable by expanding its material dependencies. [WME-W001, ch. 17, 275–285]

Connected descriptions without a supermodel

Near the end, Jantsch explicitly rejects one supermodel combining all autonomous levels into a single description. He favors an account of the web of relations among descriptions. His contrast between macroscopic and microscopic approaches returns here: descriptions may be complementary because they answer different questions, rather than waiting to be replaced by one all-purpose vocabulary. [WME-W001, ch. 19, especially 303]

Jantsch brings maintenance, transformation, environmental capability, stored forms, communication and changing purposes into a connected evolutionary account. The epilogue extends his inquiry to meaning. [WME-W001, ch. 19 and epilogue 307–311]

The earlier dissipative-structures excerpt

An excerpt from Jantsch's earlier *Design for Evolution* (1975) proposes the extension from physical to human systems as a hypothesis. Its examples concern a freeway changing urban activity, a newcomer stimulating a bureaucratic office, and a revolution followed by new structures and restrictions. In each, an initial alteration is followed by reorganization and a more established regime. [RGC-W001, excerpt and linked figure]

The freeway example is particularly informative because the new regime can become congested again. Opening capacity changes activity rather than permanently removing every source of congestion. The office example likewise makes the surrounding condition decisive: colleagues can absorb and exclude the newcomer, or existing tensions can permit a wider change. These are illustrative mechanisms and conditional stories, not controlled evaluations of road building or office reform. [RGC-W001]

The revolution example connects initial participation with the later consolidation of power and narrowing of permitted contributions. Jantsch uses entropy, power and rigidity in explaining that trajectory. Those social correspondences are not measured thermodynamic equalities. The examples help identify the proposed sequence and the retained costs of restructuring. [RGC-W001]

Greene's later reconstruction and intellectual reception

Herman Greene's 2018 review reads Jantsch from an ecozoic perspective and traces a connection with Brian Swimme and Thomas Berry. He explains the book through process, self-organization, exchange and co-evolution, and relates those terms to differentiation, interiority and communion in the later writers. This is substantive evidence of a sympathetic intellectual reception. [WME-W002, 363–372]

Greene quotes and interprets Swimme and Berry as well as Jantsch. Their expanded use of autopoiesis and subjectivity belongs to Greene's account of the relationship among these authors; he points to *The Universe Story* in support. [WME-W002, especially 363–364, 370–372]

S20 Gharajedaghi: purposeful organizations and the work of redesign

A connected organizational account

Jamshid Gharajedaghi's *Systems Thinking: Managing Chaos and Complexity*, third edition, connects holistic inquiry, operational dynamics, purposeful social activity and interactive design. His organizational synthesis examines how interdependence, decision criteria, modularity, competence, incentives and governance affect performance. [WME-W003, 2011, preface]

Purpose and choice change the description

In 1984, Gharajedaghi and Russell Ackoff argued that treating a social system as a machine or an organism was an inadequate guide to decisions about it. [RGC-W002]

Gharajedaghi distinguishes passive mechanical systems, organisms treated as having a unified purpose, and social systems composed of purposeful participants. In the last category, participants can choose ends as well as means. A procedure can consequently fail even when it is technically feasible: people may reject its purpose, interpret it differently or pursue another interest. The relevant organizational problem includes choice, rather than only transmitting commands to interchangeable components. [WME-W003, ch. 1, 9–22; ch. 2, 29–37]

Control, influence and context are correspondingly different relations. An actor's ability or authority to determine something differs from an ability to affect it, and both differ from conditions that the actor must accommodate. The boundary is relative to the actors and their capabilities. Describing a supplier, regulator or customer as part of an environment does not make its choices disappear. [WME-W003, 29–37]

The book also distinguishes information about what, knowledge of how and understanding of why. Shared instructions and shared reasons are therefore different accomplishments. A member who understands only an immediate instruction can still require repeated direction when circumstances change. The account of purposeful actors makes the development of relevant understanding an organizational activity in its own right. [WME-W003, ch. 2]

Gharajedaghi's criticism of biological and cybernetic models, including Beer, is part of his argument. His own later foundation diagram acknowledges Beer alongside Ackoff, Forrester and Boulding. [WME-W003, chs. 1, 4; diagram preceding ch. 5]

Inquiry joins structure, function, process and context

The book rejects a simple one-to-one inference from structure to outcome. The same implement can perform different functions depending on its use and setting. Conversely, one function can be realized through different structures. Process explains how an arrangement produces an outcome, while context supplies enabling and limiting conditions. Gharajedaghi therefore proposes iterative inquiry among structure, function, process and context. [WME-W003, ch. 5, 89–93]

Iteration involves examining each perspective and revisiting its relations with the others. His expanding view places the focal activity within progressively wider surroundings until further enlargement ceases to add useful insight. The practical claim is that inquiry should expose assumptions and missing relations. Gharajedaghi proposes the four perspectives as mutually exclusive and collectively exhaustive. [WME-W003, 92–93, fig. 5.4]

His five social dimensions—wealth, power, knowledge, beauty and values—likewise organize the inquiry. Generation and distribution, rational and emotional commitments, and present and future effects can conflict. The dimensions direct attention toward matters that a purely financial account can omit. [WME-W003, chs. 2, 5]

Multidimensionality seeks alternatives beyond a forced choice between two unsatisfactory extremes. An arrangement might improve both production and distribution, or permit local discretion together with broader consistency. That is a design ambition and a question to investigate. The book itself recognizes difficult differences in worldviews and the need to protect minorities. [WME-W003, 38–55, 103–108]

Shared criteria can support local decisions

The contrast between two managers makes the proposed mechanism concrete. A manager can promise autonomy while retaining unspoken preferences; unexpected decisions then provoke intervention, and a subordinate starts seeking permission for everything. Another manager repeatedly explains why decisions are made. As the subordinate learns the relevant criteria, locally proposed action can become acceptable without detailed direction on every occasion. The illustration concerns shared reasons and judgments, not the announcement of decentralization alone. [WME-W003, 98–99]

Gharajedaghi proposes nested learning and design cells that connect a manager, board, superior and direct reports. Their task is to develop common understanding of purpose, cultural defaults and decision criteria. This institutionalizes the conversations rather than expecting every individual to acquire an identical understanding through private contact with the leader. [WME-W003, 99–100, fig. 5.8]

Effective criteria must leave a degree of freedom. If they prescribe every decision, they cease to support meaningful choice. They also need internal consistency, explicit assumptions and expected consequences, and a basis for collective commitment. A production cost target and a sales revenue target can create structural conflict if each rewards conduct that makes the other's task harder. The work of aligning the criteria precedes any claim that local decisions will fit together. [WME-W003, 100–101]

Consensus is described as an agreement to act, rather than majority rule or a requirement that every person privately agree. Leadership remains necessary when discussion does not resolve the issue. The leader can formulate an explicit default, or parties can agree on an experiment with performance criteria and conditions that would count against their positions. The process retains authority, design of tests and judgment about timing. [WME-W003, 101]

Culture and learning reproduce the arrangement

Shared images and default assumptions help reproduce an organization even when individual members change. Improvement within an existing rule and learning that questions the rule are different activities. Gharajedaghi's second-order learning asks how the organization can examine the frames and criteria through which it identifies a problem. Training in an established process is therefore insufficient when that process itself produces the difficulty. [WME-W003, ch. 3, 57–68]

Information-mediated integration requires people who can interpret and use what is communicated. Knowledge can be shared without disappearing from its original holder. Gharajedaghi also draws physical analogies involving entropy, quantum phenomena and attractors; his organizational arguments can be distinguished from those extrapolations. [WME-W003, chs. 2–3]

Development is distinguished from growth. A larger organization need not be more able or willing to serve its members and environment. Differentiation can increase specialized capability while making integration more demanding. Gharajedaghi's concern is the relationship between these changes, rather

than an assumption that scale, output or specialization alone measures development. His sweeping historical classifications remain his interpretations. [WME-W003, ch. 4, 69–85]

Operational models make delays and incentives visible

The stock-and-flow models describe accumulations, rates, feedback and delays. A policy can improve a local indicator while worsening the condition that later drives the same indicator. Modeling is intended to make those interactions available for discussion and testing. The book explicitly makes the usefulness of a model depend on the validity of its underlying equations and assumptions. [WME-W003, ch. 6, 109–131]

The telephone-company example joins several causes. Demand had consumed spare lines previously retained for failures. A repair could then appropriate another active line, briefly satisfying one customer while generating another complaint. Different departments controlled interdependent activities, records deteriorated as more people worked on the plant, and volume-based rewards encouraged activity and overtime. The problem was not described as a shortage of worker effort alone. [WME-W003, 128–129, fig. 6.24]

The reported redesign groups related work in neighborhood units, changes rewards toward results and customer satisfaction, gives managers authority to limit new installations when plant condition and slack require it, and directs expansion toward the limiting parts of the system. Capacity, maintenance, records, local responsibility and incentives change together. The author reports resolution within three months. [WME-W003, 128–129]

The account is particularly informative about spare capacity. Eliminating an apparently idle reserve can increase later interdependence and repair traffic. Its value depends on the disturbances and demand it supports. The book also proposes preserving assumptions and expectations in a read-only log so later experience can be compared with what was expected. [WME-W003, 128–131]

Modularity groups dependencies and maintains competence

Gharajedaghi defines a module through strong internal interdependence and relatively weak external dependence. A platform supplies a predesigned relation through which a class of modules interacts with the rest of the arrangement. The aim is to integrate related activities inside a unit while making its external interactions simpler and more explicit. [WME-W003, ch. 7, 138–140]

The proposed multidimensional architecture distinguishes inputs, markets and outputs. Input modules hold resources and competencies, including knowledge and shared services, and must renew them through research and education. Market modules identify needs and opportunities. Output modules combine external opportunities with internal capabilities in products, projects or programmes of differing duration. Their authority over work and money need not include ownership of all the capital they use. [WME-W003, 139–140, fig. 7.4]

Capability maintenance is therefore deliberately separated from the life of a particular project. The book does not propose that every temporary team independently own every specialist resource. Members of a knowledge bank also need design competence in addition to functional expertise. A simpler interface for an output group can depend on continuing work elsewhere in the architecture. [WME-W003, 139–140]

Cross-functional water-system design

The custom water-purification design example contrasts sequential passage through functional hierarchies with a cross-functional module. In the original reported process, chemical, process, electrical and mechanical engineering each passed work up and down their own chains before transferring it

onward. Costing and customer negotiation followed. The author estimates six to eight months and roughly $500,000 for the prior process. [WME-W003, 140–141]

The proposed experiment formed a learning cell of four junior engineers, a cost accountant and a sales engineer. Each taught the others the essential concerns and variables of the work. This preparation took three months, on the members' own time, with a promised $10,000 salary increase for each participant if the shared learning and project were successful. The team then met the customer together, rather than relying on a distant serial interpretation of requirements. [WME-W003, 141]

The book reports a completed design in six weeks at $120,000 and says the experiment was repeated twice with similar results. Shared understanding and concurrent contact with the customer replaced repeated hierarchical handoffs. [WME-W003, 141]

Design and realization are separate tasks

Idealized design is constrained by technical feasibility and operational viability, while being capable of learning and adaptation. It changes the available alternatives instead of only optimizing the inherited arrangement. [WME-W003, 133–150]

Gharajedaghi describes realization as potentially the most resource-intensive phase. It involves preparation, monitoring of constraints, trust and ownership of the change, following the inquiry and design process. [WME-W003, 144–150]

Measurement and reward are part of the arrangement being designed. Local financial centers can pull against one another, while incompatible present incentives can burden future members. The proposed social calculus incorporates chosen coefficients for need, scarcity and strategic importance. Those choices require continuing revision. They are normative and managerial decisions, not discovered physical constants. [WME-W003, 150–157]

Service, authority and internal markets

The multidimensional design combines horizontal customer–supplier relations with vertical authority. Hiring, promotion and formal accountability do not disappear because units also purchase services. The internal market is intended to make demand and service performance visible, while the authority relation addresses matters it cannot settle. Gharajedaghi presents the two relations as complementary. [WME-W003, ch. 9, 181–207, especially 198]

Centralization requires a specific justification: indivisible functions, common measurement and information arrangements, scale effects or development of a core technology. The benefits must outweigh the disadvantages for the whole. Shared-service provision should not be confused with control, because mixing them can compromise both useful service and legitimate oversight. The control work remains assigned elsewhere. Some redundancy is also deliberately retained for flexibility. [WME-W003, 191, 195]

In his 1984 article on multidimensional modular design, Gharajedaghi argues that narrow specialization leaves an organization with too few ways to respond when conditions change. His proposal builds some redundancy into a modular design so that units can adapt and redefine their functions. [RGC-W003]

Inside–outside purchasing choices are intended to discipline internal providers, but a higher authority can override a choice by accepting the corresponding cost or loss. The account therefore retains governance even where price-like relations organize demand. Its broad assertions that an internal market eliminates bureaucratization are proposals for expected behavior, not universal effects established by the framework's definition. [WME-W003, 198]

The COM/Energy design supplies a particularly concrete cost distinction. Buying an external service does not make corporate fixed costs vanish if the organization still incurs them. The design requires the purchasing unit to bear a proportional internal charge until those costs can actually be eliminated. The comparison is between outside supply plus retained costs and the relevant internal alternative, rather than an allocated-cost figure that moves when the accounting changes. [WME-W003, ch. 13, 309–310]

Measurement itself also costs something. Gharajedaghi says its cost should not exceed the information's value and warns that overhead allocations can be mistaken for causal costs, inducing harmful cuts. The value-added coefficients of his performance centers reflect chosen strategy. These passages specify both a practical accounting issue and the limits of treating a managerial measure as a neutral description. [WME-W003, 202–206]

The cases have different evidential status

“Mess formulation” gathers interacting difficulties into an intelligible pattern. The proposed process searches, maps and tells a story to support recognition and action. It is presented as early warning rather than prediction. The utility named XYZ is expressly fictitious, and its exaggerated scenario is a design aid. [WME-W003, ch. 8, 159–179]

The Part IV cases are selected consultant/client design synopses. Oneida is described as an ongoing fifth iteration; the Butterworth account condenses a much longer design after six iterations, acknowledging losses or misrepresentations in compression. Their detail is valuable for understanding intended arrangements. It does not convert all the proposed activities into implemented and sustained outcomes. [WME-W003, introductions to chs. 10–11, 211, 245]

Oneida's design retains dedicated expertise, boards and membership cells, together with support for meetings, material preparation, minutes and facilitation. Consensus is explicitly time-consuming. Self-reliance differs from doing everything internally: outside contractors can participate while training local counterparts, and outsourcing should avoid excessive vulnerability. Multiple employment possibilities may be desirable redundancy. The account shows how participation and local capability require supporting organization. [WME-W003, ch. 10, 211–243]

The Butterworth health-care design combines community modules with shared specialist, capital-intensive capabilities. It recognizes risk, quality and rights, and the limits of reversible experimentation. Core knowledge is maintained beyond temporary projects. The professional as learner, designer and practitioner is an integrative role that can broaden the competence required of a person. These are features of a proposed organizational design. [WME-W003, ch. 11, 245–279]

The book explicitly associates integration there with cooperation, attention from the president and cultural transformation. Flexible learning can sometimes be less immediately efficient than a perfected routine. A decision to preserve that learning has a purpose different from minimizing all overhead. [WME-W003, 273–279]

The Marriott design combines regional autonomy with shared components, core knowledge, training and maintenance. Initial subsidies may be needed. Policy teams establish criteria; where policy is absent, a local decision can become material for later learning. This is an account of how discretion and common capability are intended to fit together. [WME-W003, ch. 12, 281–289]

COM/Energy distinguishes a regulated core from entrepreneurial platforms and retains technical expertise, alliances, planning and learning networks. Its expectation of automatic or self-administered coordination is conditional on those continuing networks. Confidential supporting reports are not reproduced in the book. [WME-W003, ch. 13, 291–315]

Carrier is even more clearly bounded. The chapter says that implementation was not attempted because corporate circumstances and strategy changed. The architecture is a tentative idealized design produced by a team, not a realized success case. Its content is still informative: purchased components leave a requirement for whole-system design capability, and make, buy and alliance choices each retain relevant knowledge in different relations of control or influence. [WME-W003, ch. 14, 317–332]

The Carrier proposal further includes cross-disciplinary training, technical partnerships, models, shared design capability and policy teams. Revenue shares are proposed strategic allocations informed by benchmarks, alternatives, history and negotiation. [WME-W003, 321, 328–332, fig. 14.5]

S21 Miller: process allocation, adjustment and purpose-relative costs

Processes, structures and adjustment

James Grier Miller's living systems framework connects material and informational flows with structures, functions, regulation and costs. Miller asks how allocating processes to structures changes dependencies, communication, control, vulnerability and the effectiveness of adjustment.

Miller develops these concepts in the *Basic Concepts* excerpt from *Living Systems* and in his 1970 paper *The Nature of Living Systems*. Later author accounts and scholarly reconstructions develop and assess the framework. [RGC-W004; WME-W004; SG-W008]

Matter-energy, markers and meaning

Miller treats living systems as concrete organizations existing in physical space and time. They process matter-energy and information, but information travels through physical markers. A message's pattern, its material carrier and its meaning to a recipient are related aspects, rather than one interchangeable quantity. Communication requires some physical realization and a system capable of receiving and interpreting it. [SG-W008, printed 1–9; WME-W004, §§1–4]

The distinction prevents a particular omission. A brief signal can influence a large activity without supplying all its energy, but the receiver still requires energy and the relevant capability. Information processing also involves channels, transformations, memory and other activities. Describing a message in informational terms does not remove those processes from the system's resource requirements. [SG-W008, 6–9; WME-W004, §§3–4]

Miller expressly separates Shannon information from meaning. Measures defined over alternatives and probabilities do not automatically measure significance, usefulness or agreement. Miller also connects information with negative entropy and develops speculative optimizing extensions. [WME-W004, §§3, 13]

Formal identities also require compatible dimensions, units or specified transformations. A curve drawn in one coordinate system can look different after a change of scale, while superficially similar curves can concern different relations. Miller's cross-level ambition therefore needs more than a visual resemblance. An asserted identity requires the measurement relation as well as a shared name for the process. [SG-W008, 23–25; WME-W004, §§1–4, 8]

Structure, process and history differ

Structure describes spatial organization at a selected instant; process describes changes in matter-energy or information through time. History includes changes in the structure and functioning of the system. Its explanatory question includes how the organization changes, not only how flows pass through an unchanged diagram. [WME-W004, §§5–8]

The distinction between a component and a subsystem is equally important. A component is an identifiable structural unit. A subsystem is identified by the process it carries out and can span several components. One component can perform several processes, and several components can jointly perform one. The mapping between structures and functions is many-to-many. Counting departments, persons or physical parts is therefore different from identifying the activities they perform. [WME-W004, §11]

The discussion of organizational roles makes this concrete. A role specifies demands and activities, but the characteristics and needs of the person filling it also affect what happens. Adjustments can involve compromises between the member and the larger system. A formal position does not by itself establish

who actually leads or decides. Miller's distinction asks the investigator to examine the process rather than infer it directly from a title or shape. [WME-W004, §11]

He also distinguishes a system level from a decision echelon. Levels concern the organization of cells, organs, organisms, groups and further forms. Echelons are steps within a decider's organization, at which different decisions may be made. They need not coincide with levels, and some deciders are not arranged in echelons. [WME-W004, §§8–9]

Cross-level comparison requires both identities and "disidentities." Similar functions can coexist with differences in mobility, duration, transferability of processes, boundaries and other characteristics. Miller's programme seeks quantitative similarities and relevant differences together. [WME-W004, §8.2]

Allocating a function changes dependencies

Miller's allocation patterns include localization in one component, combination with other processes, lateral distribution across components, upward or downward provision within a hierarchy, and outward provision by other systems. The allocation determines which parts depend on which other parts. Two systems can carry out similar functions while distributing the work differently. That difference belongs in an explanation of their performance. [SG-W008, 28–34; WME-W004, §11]

The framework explicitly proposes studying effectiveness and costs under alternative allocations. Distance over which material or messages must move, the number of participating components, difficulties of integration and duplicated capabilities can all matter. These are presented as candidate general principles and research questions. [WME-W004, §11]

Redundancy supplies an important countervailing consideration. A duplicate component capable of the same process can maintain function when another is disabled. Whether that possibility improves survival or just adds an unnecessary cost depends on the failures, capacities and other conditions. Miller offers a proposed relationship between allocation and vulnerability. [WME-W004, §11]

Outward provision is especially relevant. A function absent inside a system may still be carried out by a provider with which it exchanges resources or services. The absence of internal production does not mean the life function is unnecessary. It creates a relation to another system, potentially with exchange costs, asymmetric dependence or parasitism. [WME-W004, §11 and dispersion discussion]

The framework also describes differentiation or "shred-out," in which increasingly specialized components take on processes previously combined. It supplies a way to ask how specialization changes structure without assuming that the function itself has newly come into existence. More differentiated parts can still depend on communication and integration. [WME-W004, §§8, 11]

Critical processes and the decider condition

The earlier architecture names nineteen critical subsystems. They include the boundary and reproducer, matter-energy ingestion, conversion, production, distribution, storage and output, and information transduction, channels, decoding, association, memory, deciding, encoding and output. The list identifies required kinds of activity in the framework. [SG-W008, subsystem table; WME-W004, §11]

A system can lack an internal instance of a critical process if another system effectively supplies it. Miller makes an exception for deciding: an independently existing living system must retain its own decider. If that function is wholly transferred upward, downward or outward, he treats the focal unit's independent system identity differently. This is a substantive condition of his theory. [WME-W004, §11.8]

An artifact can participate in a critical function without itself becoming a living system. Living and nonliving components can be included in a larger organization. Thus the capability to store, filter or

transmit something is not sufficient on its own to establish living-system status. The decider requirement and living-system scope limit transfer to passive devices or decentralized formal arrangements that are described under different definitions. [WME-W004, §11]

This distinction also shows where an organizational decision remains. Miller permits variations in the internal distribution of deciding but requires an account of that process. Describing an organization as decentralized does not answer who settles conflicts, establishes comparison values or changes the allocation. Those activities must be located rather than assumed absent. [WME-W004, §§9, 11, 13]

Adjustment can change more than the immediate response

Adjustment maintains relevant variables within a range despite stress. Negative feedback uses information about actual performance to compare an outcome with a desired condition and alter activity. Timing matters: feedback arriving after the effective opportunity for correction can fail its purpose. A label saying “feedback” does not supply adequate speed, accuracy or action capacity. [SG-W008, 41–44; WME-W004, §13]

Miller's adjustment repertoire also includes learning or reorganization, changing the environment, withdrawal and changing targets. Not every difficulty is answered by exerting more force within an unchanged structure. A system can alter its relations or the condition it seeks. The range of alternatives makes its selected purpose and boundaries essential to evaluating the result. [WME-W004, §§12–13]

Equifinality is correspondingly bounded. Different starting conditions or routes may lead to a similar condition, but only within relevant ranges and with essential inputs available. The text explicitly notes histories and conditions that alter or destroy an organization. [WME-W004, §13.2.4]

Costs are tied to purposes and time

Miller's cost discussion is unusually direct. Adjustment uses material resources, energy, information and time; social systems also use money, which he discusses as information conveyed through markers. Any of these can be scarce. Costs differ between processes and can be immediate or delayed, short-term or long-term. The framework therefore explicitly reaches beyond the visible effort of the current act. [SG-W008, 45–46, §13.2.4; WME-W004, §13.2.5]

Efficiency is defined in relation to success at a known purpose and the costs involved. An operation can be active without being useful in the larger system. One must know what it is expected to achieve before judging whether its resource use is effective. Miller uses programme budgeting to illustrate separate accounts for the components or programmes carrying out distinct activities. [WME-W004, §§13.2.4–13.2.5]

Different performances and resource dimensions can require different comparisons. Miller's claim that economic decisions improve performance and decrease costs expresses the account's optimizing orientation. [SG-W008, 45–46; WME-W004, §13.2.5]

Information itself has a place on both sides of the decision. Relevant information can improve selection among adjustments, but obtaining it costs resources. Reliability, distortion, accuracy, whether the inquiry concerns an act or a policy, and how it is acquired affect its value. The account thus includes information acquisition as work. More messages or more measurement are not automatically an improvement if they do not help the selected performance enough to justify their cost. [WME-W004, §13.2.5]

These passages connect allocation, dependence, regulation, information and performance within Miller's own account.

Hammond and Wilby: a substantive reconstruction with objections

Hammond and Wilby's 2006 article reconstructs Miller's effort to connect behavioral science, cybernetics and an organismic hierarchy through material and informational processes. It traces his research institutions and quantitative ambitions, while examining the implications of his categories. The article is adapted from a chapter of Hammond's 2003 book. [RGC-W005, 429–435]

The decider is a central issue in their account. They describe its role in forming intentions, coordinating activity and allocating power, and ask how that model works as the levels become socially more complex. The existence of a category called deciding does not by itself settle the relation between individual purposes and collective control. They preserve the tension between integrating members and supporting their needs. [RGC-W005, living-systems section]

They also report Boulding's criticism: an essentially physiological template can be too elaborate for lower levels and insufficiently complex for higher ones. This criticism is mediated through Hammond and Wilby. It challenges the adequacy of the cross-level template while leaving open whether particular relations or questions are useful. Their discussion of measurement limits is likewise a critical interpretation. [RGC-W005, concluding living-systems discussion and reflections]

Ruscoe: a schema does not supply its own instruments

Gordon Ruscoe's response to Miller's editorial shares the aspiration to inquiry beyond artificial disciplinary boundaries but questions the proposed strategy. He distinguishes a conceptual system, an empirical theory and a research paradigm. A classificatory arrangement can contain useful concepts without itself supplying empirically testable statements and the instruments needed to examine them. [WME-W005, printed 11–12]

His operative objection is the chain from process to variable to indicator. The investigator needs reasons for selecting a variable, connecting it to the process and deciding that an indicator measures it. Those inferential steps require methodological justification beyond placing the study within a living-systems box. A formally similar label is insufficient to establish a valid cross-level measurement. [WME-W005, 12]

Ruscoe also distinguishes generating a fruitful empirical question from retrospectively fitting completed research into a schema. Both can be useful, but the second does not prove the first. His letter is a methodological challenge to the warrants for the research programme. [WME-W005, 11–12 and editorial note]

Later applications and the status of their evidence

Miller and Miller's *Applications of Living Systems Theory* (1992) reports that the 1978 book contained 173 cross-level hypotheses and discusses studies of overload, communication networks and conflicting commands. Some model applications are explicitly described as compatible with the framework rather than derived from it. Completed and proposed studies remain distinguishable. [RGC-W006]

The overload account reports measurements at several levels, from neural preparations to individuals, small groups and laboratory organizations. It proposes a response in which increasing input initially increases output, then exceeds capacity and can produce deterioration. The later account also stresses compatible units and dimensions for a formal cross-level comparison. [RGC-W006, basic-research section]

The conflicting-command game offers a more specific organizational example. Individuals or groups receive instructions from several nominal superiors, and the account reports slower deciding as commands differ. That result belongs to the reported experimental setting. The biased-network study

similarly concerns specified communication probabilities and reported predictions. [RGC-W006, cross-level tests]

The organizational application describes a three-year study of forty-one Army battalions with more than five thousand participants. It reports relationships between information-processing characteristics and effectiveness, including meaning, lag, volume, cost and distortion. The same account initially lists objective process data but later says only subjective data were collected in that study. [RGC-W006, Army and industrial-project sections]

The later proposed industrial study includes objective measures, computerized responses, voluntary location badges and expert-system analysis. These are plans for a study, not results. [RGC-W006]

The Swanson–Miller EOLSS chapter introduces a later eight-level, twenty-subsystem form. It distinguishes conceptual and concrete systems and retains limitations on measuring meaning. This differs from the earlier seven-level, nineteen-subsystem architecture. [RGC-W007]

Swanson's 2007 proceedings abstract for *Living Systems Theory and Entity-Systems Theory* raises questions about dispersed subsystems, conceptual systems and emergence. [RGC-W008, abstract]

S22 Beer and viable systems: autonomy, coordination and contested purposes

Viability and organizational functions

Stafford Beer's viable system model concerns an organization's continuing existence in a changing environment. It locates activities among local units, coordination, resource allocation, audit, attention to future conditions, and the articulation of identity and policy.

In *The Heart of Enterprise*, Beer develops the viable system model. John Brocklesby, Stephen Cummings and John Davies's 1991 reconstruction draws on *Heart*, *Brain of the Firm* and *Diagnosing the System*. J. P. van Gigch's book review and Michael Jackson's critique offer distinct judgments. Beer's later methodological account and Raul Espejo's 2022 development extend the discussion. [W101; RGC-W009; RGC-W010; RGC-W011; RGC-W012; UX-W007]

The Italian Viable Systems Approach, whose authors acknowledge Beer, appears below. [SG-W014]

Variety is relative to the distinctions and purpose being considered

The reconstruction begins with the problem of maintaining a separate existence while circumstances change, including circumstances not foreseen when an organization was designed. It distinguishes responding within established operating norms from reconsidering those norms. A procedure can correct deviations toward a target while leaving untouched the question of whether that target or procedure remains appropriate. Brocklesby and colleagues use this contrast to explain why viability is more demanding than repeating a previously successful routine. [RGC-W009, printed pp. 3–5]

Their account of variety concerns distinguishable states relevant to a system's purposes. A producer's relevant circumstances include customers, suppliers, regulations, and conditions bearing on what it is trying to do. Different operational activities encounter different local environments, which may overlap. The choice of system and purpose therefore enters the identification of what differences matter. [RGC-W009, pp. 5–7]

Ashby's variety counts distinguishable possibilities within a specified set, as explained in S03. Brocklesby and colleagues use "absorbing local variety" to describe how local circumstances relate to an organization's capacities for action. [W001; RGC-W009, pp. 5–7]

Brocklesby and colleagues describe two ways of changing the relation between demands and response capacity. Amplification increases what a system can bring to bear; attenuation reduces the variety with which it must contend. Their examples include improving a response capability and avoiding an exposure, rather than merely collecting more data. The usefulness of either depends on what the system seeks to accomplish. Avoiding a demand by abandoning the relevant purpose would describe a different accomplishment. [RGC-W009, pp. 6–7]

The original *Heart* passage: ignorance is also a filter

The selected original passage of *The Heart of Enterprise* gives an unusually sharp qualification to the language of attenuation. Beer argues that the variety reaching an operation or management can be reduced through ignorance: the operation recognizes only some environmental features, while a manager recognizes only what the manager's existing model makes perceptible. Such reduction can produce an apparently manageable world without competent management. He explicitly distinguishes this from designing effective amplifiers and attenuators. The existence of filtering is therefore not evidence that the right distinctions have been preserved. [W101, pp. 96–97]

His first principle of organization calls for designing the relationships among managerial, operational, and environmental varieties with attention to damage to people and cost. The passage also criticizes the automation of an inherited information arrangement whose relevant design questions have not been addressed. A computerized filter can preserve an inadequate selection while losing the flexibility of the earlier manual system. A capacity to process more data does not decide what distinctions the organization must be able to act upon. [W101, pp. 97–98]

Beer then considers amplification of managerial influence through meetings, communications, staff, and the development of policies. Their appropriateness depends on the variety of the situation; a managerial style is not assigned a constant effectiveness independent of the circumstances. This selected passage describes design judgments and their possible costs. It also directs attention away from an immediate argument about what a manager wants to know and toward the design of regulation. [W101, pp. 98–99]

System 1: put local response where local knowledge exists

In the 1991 reconstruction, System 1 consists of the operational activities through which the system in focus does what it exists to do. The system in focus is the entity and activity boundary selected for the analysis. For an automobile producer, different model-producing units can be System 1 activities; for a university considered through teaching and research, academic units can occupy that role. These are examples of identifying primary activities from a stated purpose, rather than deriving them from the appearance of an organization chart. [RGC-W009, pp. 10–11]

The operational units are expected to have local management and enough autonomy to deal with their own circumstances. Senior university officers, in the authors' example, cannot keep abreast of every academic field in the way specialists can through journals, conferences, and professional communities. Moving all the relevant decisions upward would move work away from the people who encounter and understand those distinctions. Their account predicts overloaded higher management, weak sensing of local conditions, and inadequate responses. It connects autonomy to the location of knowledge and action. [RGC-W009, pp. 11, 14–16]

“Local” need not mean geographically small. An academic field's environment can be international. Nor does the model's use of an autonomous unit imply a unit without dependence on other people or resources. The reconstruction's criterion concerns potential independent existence and management of its activities, within the chosen account. [RGC-W009, pp. 10–11]

The boundary can also change the classification. A university computer-support service may be a supporting activity when the whole university is the focus. If the focus is a support system whose purpose is to provide that service, computer support can become a primary operational activity, with the rest of the university now part of its environment. The authors use this example to warn against treating corporate boundaries and departmental names as the only possible analytical divisions. A change of focus must remain visible, because it changes what a label such as System 1 refers to. [RGC-W009, pp. 10–11]

Systems 2, 3, and 3*: autonomy keeps coordination and accountability

Autonomous operational units can interfere with one another. They may seek the same scarce resource, create incompatible standards, or withhold something whose shared use matters to the whole. The reconstruction gives System 2 the coordination function that reduces harmful interference or oscillation. This is not a second layer that performs every local task. It concerns relations among tasks whose individual pursuit can create trouble for one another. Scheduling, shared guidance, and the communication necessary for decentralized action are among the relevant activities. [RGC-W009, pp. 9, 12–13]

System 3 concerns internal cohesion and the present conduct of the organization. Resource bargaining connects what operational units undertake with what they receive and with the expectations under which they act. Allocation is one outcome of that bargain, rather than the whole relation. The reconstruction also gives this function a role in implementing policy and establishing criteria of accountability. Coordinating potentially conflicting activities and deciding how the organization uses its resources are consequently related but distinguishable responsibilities. [RGC-W009, p. 13]

The audit function, System 3*, examines operational performance rather than relying entirely on routine representations reaching management. The authors use “audit” more broadly than an occasional accounting review: it includes ways of questioning whether activities are being performed within the agreed guidelines. Their discussion includes both checking what has occurred and a continuing capacity for examination. [RGC-W009, pp. 13, 15]

In the reconstruction's critical case, System 3 tries to occupy the local management role, generating transmissions and interventions until it becomes overloaded and loses capacity for its own tasks. The proposed alternative retains coordination, resource relationships, and audit while allowing operational managers to address the circumstances for which they hold better knowledge. [RGC-W009, pp. 14–16]

Training, induction, and shared guidance are important parts of the alternative. People can respond without seeking a senior decision at every step when they understand the broad expectations. Yet those expectations should leave room for judgment in circumstances that cannot be exhaustively anticipated. A highly elaborate rule network providing a supposed answer to every possible local situation would defeat the use of local knowledge that the account seeks to preserve. These passages locate an enabling arrangement and the continuing competence needed to use it together. [RGC-W009, pp. 15–16]

The guidelines must also be open to challenge. An operational unit may learn that an established way of acting no longer suits its environment, and the communication with coordinating functions must allow proposed changes. The authors further retain an accessible direct connection between operational and higher management for exceptional situations. Being free from constant interference is not the same as being unable to obtain guidance. Their account includes the social possibility of making contact, not just the existence of a line on a diagram. [RGC-W009, p. 16]

Systems 4 and 5: operating now while considering what should change

The reconstruction distinguishes System 3's present, internal concern from System 4's attention to the wider environment and future. Intelligence here can involve research, development, planning, and interpretations of emerging conditions. Its purpose is not simply to produce a larger information store. It must bring the implications of those conditions into a relation with the organization's actual capacity and conduct. [RGC-W009, p. 18]

The authors' automobile example gives this relationship an ordinary managerial form. Market research may favor producing six models, while current operations favor four because of the resources and reorganization required. Both positions bear on the undertaking, and neither alone determines the appropriate decision. A claim about future demand has to meet a claim about present productive capability. [RGC-W009, p. 18]

System 5 concerns identity and policy: what purposes hold the undertaking together and how competing interpretations are resolved. In the example, a policy about a broad range of family cars at a reasonable price admits disagreement over both range and price. Policy is therefore not a substitute for every operational decision, but neither is it a decorative statement unrelated to them. The reconstruction gives it a role in balancing the present and future concerns represented by Systems 3 and 4, and in providing the context for guidelines used elsewhere. [RGC-W009, pp. 18–19, 21]

The performance discussion makes a related distinction among actuality, capability, and potentiality. Actuality concerns what is achieved under existing resources and constraints; capability concerns what could be achieved with those resources and constraints; potentiality concerns possibilities through developing resources and removing constraints. These are different comparisons, rather than three names for a current output measure. The source proposes ratios among them. Their explanatory relevance is the distinction between doing better with what is already available and changing the conditions of what becomes possible. [RGC-W009, p. 17]

A communication link must carry usable distinctions

In Brocklesby and colleagues' account, having each of the five functions present is insufficient if the relationships among them do not work. Channels must be established and maintained. Messages must arrive in a form that recipients can understand. They use *transducer* for the means by which information is carried across a boundary in usable form. The term does not make interpretation disappear; it identifies a function whose adequacy must be examined. [RGC-W009, p. 19]

Their computer-system example concerns departments whose specialists express different requirements in different technical languages. A message's transport does not show that the receiving group can interpret its significance. The authors make this issue explicit through a passage in which Beer distinguishes a translator's linguistic competence from grasp of the intended possible states. They acknowledge that perfect conveyance is an ideal rarely achieved in practice. Thus the reconstructed model retains both communication capacity and the understanding required at the boundary. [RGC-W009, p. 19]

Recursion is a claim about organization at selected boundaries

The model is recursive in the reconstruction's account: an operational unit can itself exhibit the functions needed for viability, and the system in focus can be an operational unit within a wider system. The five labels are therefore not five fixed departments at the top of a corporation. The local operational unit has its own management and relevant internal relationships. Repeating the logical organization is how the model describes relations among selected levels. [RGC-W009, pp. 10–11, 21–23]

The selection of levels remains an analytical act. The authors' automobile example permits different wider systems to be drawn, and they acknowledge interpretive choices in higher-level boundaries. The activity, environment, capacities, and relationships claimed for it still have to be identified. [RGC-W009, pp. 21–24]

The reconstruction itself ends with a warning against the attraction of a tidy diagram. A model does not provide a complete picture of people and their interactions, and its appearance does not make participants committed to changing an organization. The authors offer it as a language and framework for examination, sometimes alongside other models. It also preserves the work of application: empirical understanding, judgment, and participation are not supplied by the diagram alone. [RGC-W009, p. 24]

Van Gigch's review: an explanation still needs practical mastery

J. P. van Gigch's 1981 review of *The Heart of Enterprise* reconstructs the five functions and their variety relationships while raising objections about how the account can be put to work. He finds important insights in the book but considers the presentation difficult to master. His criticism is more precise than saying that the model has no relevance: an uninitiated reader may not acquire the knowledge and practice needed for application simply by reading the book. He also questions the presentation of rules as maxims, principles, theorems, axioms, and laws without the mathematical derivations he expected in that volume. [RGC-W011, pp. 128–129]

The review further distinguishes identifying obvious mismatches from calculating calibrated entropy exchanges in a real organization. Van Gigch describes practical measurement of the relevant entropy as extremely difficult, including attempts to compare mental effort across degrees of mechanization. The review leaves the reader with both an account of what the book illuminates and a concrete reason why conceptual clarity is insufficient for competent implementation. [RGC-W011, p. 129]

Jackson's dispute: structural constraints or an interpretive instrument?

Michael Jackson's 1992 research note rejects a simple reading in which an analyst finds an objective organization and rearranges it to fit an idealized chart. He then distinguishes two more serious interpretations. His preferred structuralist reading allows observers to select identity and purpose, while maintaining that the organization pursuing that purpose faces real structural constraints. On this view, the model seeks underlying relationships that explain operational difficulties; the analyst's task is more than facilitating whichever account participants already favor. [RGC-W010, pp. 561–562]

Jackson contrasts this with the interpretive use he attributes to Harnden and Espejo, where a model supports conversations among different perspectives about emergent organizational forms. The dispute becomes sharp when a cybernetic diagnosis conflicts with participants' beliefs. Jackson asks which should guide action and argues that the existence of cybernetic constraints cannot be settled merely by choosing a preferred conversation. [RGC-W010, pp. 563–564]

The note is relevant to gambits engineering because it makes the status of an organizational explanation consequential. A claim about a constraint, a proposal for collective interpretation, and a participant's agreement are different grounds for action. Jackson argues for one reading. [RGC-W010, pp. 561–564]

Beer on modeling, identity and informational economy

The reproduction of Beer's account of the VSM's provenance and methodology distinguishes literary analogy from more precisely specified mappings. Beer argues that a model selects relationships and invariances, and that the context and boundaries of its application must be nominated. His account of internal regulation, attenuation, and amplification remains connected to that modeling program. The reproduction also lists applications, while acknowledging the absence of a systematic archive and reliance on records and recollection at hand. [RGC-W012, pp. 4–8, 16–17]

Within that account, Beer uses “intrinsic control” for a relation in which a common influence delimits environmental circumstances and evokes a response converging on an adaptive outcome. The described ordering need not be understood only as a rule imposed by a separately identified manager. This is his explanation through Sommerhoff and Ashby, whose positions are mediated through this particular passage. [RGC-W012, p. 5]

Identity is also explicitly difficult in the later text. Beer asks who “we” are when a manager asks what business an organization is in: shareholders, employees, managers, customers, and other interested people can offer different answers. His account recognizes an observer's imputation of purpose and reports disagreement about the purpose of a health system. The source therefore makes the choice of an organizational identity visible rather than treating it as an automatically shared fact. [RGC-W012, p. 10]

Appendix 1 gives a concrete connection between informational economy and lost explanatory power. Beer describes combining the outputs of three shifts into one total. The resulting representation handles fewer distinctions, but calculations within it cannot recover which shift produced what. He treats this as a choice to give up some modeling power for economy in recording and handling data, acceptable only when deliberate and remembered as a limitation. The appendix also distinguishes this known

simplification from aspects of a large system that the modeler has failed to recognize at all. This extract is reproduced from *The World, the Flesh and the Metal*. [RGC-W012, appendix 1, pp. 13–14]

Espejo's later account: residual variety depends on local organization

Raul Espejo's 2022 article develops an explicitly retrospective account of Cybersyn and his subsequent work. One relevant distinction concerns information systems and organizational transformation. He reports that some later clients possessed substantial computing resources and programming capability but sought help understanding their organization and developing meaningful performance indicators. The difficulty in his account was not merely a lack of software. It concerned how activities, purposes, and communications were understood and arranged. [UX-W007, “Real-time management and variety engineering”]

The article's central illustration concerns residual variety. Local self-organization and self-regulation handle much of what occurs before every matter reaches a separate managerial decision. The organization then faces a remaining range of circumstances, and management a further residual. Espejo explicitly calls the lines used to show these reductions impressionistic. They are not measured ratios demonstrating that a particular number of units of effort or information has been eliminated. His argument concerns the response demands that reach particular relationships after other activities have already occurred. [UX-W007, fig. 3 and accompanying discussion]

His account retains the means through which such an arrangement operates. Information filters, communication devices, skills, learning, collaboration, and wider cultural or regulatory conditions remain relevant. Reducing the variety reaching a manager can depend on activities taking place among people and within the surrounding institutions. His political and ethical preferences about how these relations should be formed remain part of his argument. [UX-W007, “Real-time management and variety engineering”]

Espejo also distinguishes two descriptions. A black-box description examines relations among an organization, its environment, and regulation under an assumed agreement about purpose. An operational description examines the interactions through which people construct the organization, its purposes, and coordinated action. He argues that the descriptions are complementary. The second description matters because the agreement presupposed in the first is itself something people have to produce and sustain. [UX-W007, figs. 4–5 and accompanying discussion]

The operational relationships include performance, cohesion, policy, inclusion, and organizational citizenship. Policy involves conversations between present operations and those attending to future circumstances. Inclusion concerns to whom policy makers are accountable; wider normative contexts also enter the account. Espejo contrasts authoritarian and more mutually respectful forms of cohesion and raises questions of power, ownership, and stakeholder participation. These are explicit concerns in this later formulation. [UX-W007, fig. 5 and accompanying numbered relationships]

His discussion of VIPLAN adds a methodological example. Naming the transformation, suppliers, actors, customers, owners, and interveners is a way of making the selected organizational description explicit. Later methodological development attends to communications and interactions through which problematic situations are addressed. The article traces this development through his own projects and collaborations. They supply the source's own account of how governance and meaning enter its organizational method. [UX-W007, “From the Viplan method to the Viplan methodology”]

A distinct Italian programme: structure, interaction, and compatibility

Sergio Barile, Jaqueline Pels, Francesco Polese, and Marialuisa Saviano introduce the Italian Viable Systems Approach as a methodological framework for business and social phenomena, including marketing. They acknowledge Beer as part of its background while drawing on a broader set of systems traditions. Their text explicitly describes the approach as a meta-level orientation, rather than a ready set of operative models for each problem. [SG-W014, printed pp. 62–63]

Their structure–system distinction separates a description of components and organized relations from the dynamics of their actual interaction. The account also identifies decision-making and operations, selective openness to resources, and dependence on other actors. The relevant context includes actors whose resources and influence bear on the undertaking's viability. Their broader propositions about all viable systems are premises of their approach. [SG-W014, pp. 63–65]

Consonance and resonance make one selected distinction particularly concrete. Consonance refers to potential compatibility among entities that might interact. Resonance concerns the harmonious interaction that actually develops. A favorable structural possibility is consequently distinguished from its realization in activity. The source relates this distinction to selecting relationships and engaging actors who control relevant resources. [SG-W014, pp. 65–66]

Viability, identity and organizational purpose

The central VSM account is connected: local units respond using local knowledge; coordination addresses interference; cohesion and resource bargaining relate activities to the undertaking; audit supplies examination; intelligence brings changing wider conditions into view; policy addresses identity and direction. Channels and interpretation connect these functions.

These sources ask whose identity an organization preserves, how purposes are agreed, and what its measurements support. They locate decisions and responsibilities among participants with different information, capacities and authority.

S23 Le Moigne: purposeful modeling and organized activity

A theory of modeling and its changing layers

Jean-Louis Le Moigne's *La théorie du système général: Théorie de la modélisation* develops a connected account of contextual purpose, representation, organized processes, memory, coordination and adaptation. The relevant material concerns both how a model can economize inquiry and how a modeled arrangement can change the demands on its participants.

The 2006 presentation of the 1994 edition includes earlier layers, later rereadings, a postface and appendices. [UX-W027]

The 2006 preface positions the enterprise as a theory of systemic modeling rather than a universal ontology declaring what every natural system really is. The model is constructed for a project in a context. Ends, means, representations and reasoning can recursively revise one another. Finalization therefore includes the ongoing formation of purposes, rather than merely execution toward an eternally fixed goal. Choosing those purposes remains an epistemic and ethical responsibility. [UX-W027, PDF 13–15, 35–47]

The book does not present this as a finished formal calculus. It acknowledges incomplete tools for organization and memory, borrows extensively and permits plural methods. Its polemics challenge the dominance of analytical science. Classical methods remain available where appropriate; the objection is to their exclusive authority over every modeling task. [UX-W027, PDF 27, 35–47, 63]

A representation is selected for a question

Le Moigne's precepts concern relevance to intentions, attention to a larger context, purposes and deliberate aggregation. A model includes selected traits and relations, rather than reproducing its object in every respect. Complexity consequently concerns the representation and its question, not simply a count of variables possessed by an object independently of every observer. The choice of what to aggregate can simplify inquiry or conceal what makes the question difficult. [UX-W027, PDF 51–69]

His “General System” is a modeling artifact, distinct from the aspiration to describe a single universal System-Universe. Context, project, activity, structure and changing identity enter the definition of the modeled object. His triangulation asks what it is, what it does and what it becomes. The point is to keep several aspects in relation, rather than infer function solely from a parts list or treat one instantaneous picture as the complete history. [UX-W027, PDF 70–94]

Systemography then constructs a particular model with reference to this general modeling device and to the observed object. Empirical correspondence is partial and purpose-dependent. Internal consistency and usefulness are different questions: a coherent formal construction can fail to address the intended situation. Conversely, a limited model can be useful without being a complete causal reconstruction. [UX-W027, PDF 95–110]

The observer and the representation-making process can themselves become objects of modeling. Intentions, values and disciplinary conventions affect what is selected and what a measure is taken to mean. Le Moigne's discussion of economic indicators, for example, refuses to let one aggregate stand for every dimension of well-being. That is an argument for explicit purposes and interpretive accountability. [UX-W027, PDF 107–110]

Functional use can require less internal knowledge

The calculator illustration challenges the claim that useful understanding always requires dissecting every internal component. A user can interpret operations, improve use, carry out some upkeep or connect a calculator with another device through knowledge of its behavior. Complete electronics knowledge is not required for each such task. This is a specified economy in what a particular user must know for a particular operation. [UX-W027, PDF 58]

A black-box description can support use while leaving the internal physical relations needed for diagnosis unspecified. Its adequacy depends on the task. [UX-W027, PDF 58–63]

Le Moigne economizes modeling by choosing relatively weakly connected regions while retaining their relevant environmental relations. [UX-W027, PDF 105–106]

Processors make transport and transformation explicit

The process framework distinguishes changes in time, space and form: storage, transport and transformation. Matter, energy and information can be processed, while a processor can itself become the object of another process. The vocabulary permits recursion. The modeler selects the processes needed for the question. [UX-W027, PDF 111–125]

Networks of processors represent activities and their relations. A processor can be multifunctional, and the boundaries of a selected subnetwork must retain its connections to surroundings. The network is an account of organized transformations, not merely a picture of boxes. A matrix showing instantaneous connections cannot by itself replace a description of delays or the sequence in which connections are used. [UX-W027, PDF 126–149]

Le Moigne's ideal neutral connector is a modeling convention. Actual transport requires activity, which can be represented by its own processor. Elsewhere he expressly treats a real communication channel as a processor. The distinction between a schematic relation and the machinery that realizes it is part of the book's account. [UX-W027, PDF 134–136, 256]

Transduction similarly requires specific energy. Events must be turned into signals and signals into events through actual operations. Le Moigne proposes distinguishing maintenance or logistical processors to represent the provision and upkeep of processors themselves. In a simplified diagram maintenance can be aggregated into the operating subsystem, an explicit representational choice. [UX-W027, PDF 164, 171 note 2, 196]

A separate decision box is not always required

The book's nine levels move through passive, active, regulated, informed, deciding, memorizing, coordinating, imagining and self-finalizing descriptions. They are a pedagogical modeling construction. Information attributed to a modeled object also differs from the modeler's own knowledge of it. [UX-W027, PDF 150–173]

The Watt-governor discussion makes a more concrete claim. A predictable algorithmic operation can be embodied in operating structure rather than retained in an additional artificial decision processor. A separate decision box may be a useful representation but is not always necessary. [UX-W027, PDF 162]

When the relevant logic is not completely specified, the account instead retains heuristics, memory of earlier events and a question of when to stop. The child's ability to continue spelling without knowing where to finish illustrates the missing termination judgment. Decisions can require history and purpose, rather than only a transformation of the present input. Le Moigne acknowledges that imagination and self-finalization are only preparatorily represented by this architecture. [UX-W027, PDF 162, 168–171]

The book considers differentiated projects and nonhierarchical possibilities while admitting limits to its formal tools. Its categories help pose who coordinates what, using which information and capabilities. [UX-W027, PDF 164–171]

Programmes reduce description by retaining a rule

The trajectory discussion replaces a large sequence of state pictures with an initial picture and a schedule of parameter changes. An animation need not store every frame independently if a representation and programme specify how the parts move. The economy lies in the selected description and its reusable rule. [UX-W027, PDF 176–190, especially 182]

Organization is then described through structure together with memorized programmes. Memory here is a modeling hypothesis and need not correspond to a literal genome or a distinct physical memory organ. The description can preserve a functional relation across different material realizations. [UX-W027, PDF 191–196]

Common programmes, codes and representations can simplify a network of decision communications. Instead of requiring each participant to obtain a separately tailored instruction through a complicated chain, participants can access a shared resource. Le Moigne describes a common representation of operations and a pool of available programmes as possible supports for coordination. [UX-W027, PDF 203–206]

He immediately preserves a crucial distinction: storing a programme does not make it execute. Memory and deciding are different processes. A code must be established and remain common, representations must be maintained, participants need access, and current purposes must guide selection. [UX-W027, PDF 203–204]

The book also recognizes disagreement about both means and ends. A shared store is not a proof that participants share values or accept the same action. Coalition and conflict questions remain. The interpretation of common memory as more democratic or more economical is attributed to the sources discussed, rather than established as a measured general effect. [UX-W027, PDF 199–208]

Stability can come from absorption, correction or anticipation

Stability is relative to selected purposes and criteria. Le Moigne distinguishes maintaining behavior through an existing structure, changing programmes, modifying the structure, and changing purposes or environmental relations. Treating all of these as the same feedback operation obscures the different activities involved. [UX-W027, PDF 217–237]

He explicitly corrects an overbroad claim for feedback by presenting three routes to regulation. The first absorbs or decouples disturbances through buffers, shields, insulation or similar arrangements. The second corrects deviations through a returning loop. The third anticipates and compensates for disturbances. The modeler is advised to consider these alternatives rather than automatically installing a feedback account. [UX-W027, PDF 223–224]

Shells and buffers attenuate disturbances before they reach the operating process, reducing the occasions for a corrective response. [UX-W027, PDF 223–224]

Adaptation can require stored resources and spare capacity. Le Moigne describes stable regions in which a trial response can be attempted, evaluated and retained or rejected. If almost everything changes faster than the trial, its consequences become difficult to learn. Excessive coupling can remove those regions and consume the slack needed for new connections. Stability and learning are therefore connected under specified timing conditions, rather than simple opposites. [UX-W027, PDF 225–226]

Le Moigne distinguishes effectiveness relative to purposes from input/output efficiency: maintaining a stable outcome or reducing an input can serve different purposes. [UX-W027, PDF 228–234]

Sparse links can require complicated programmes

Le Moigne's differentiation–coordination discussion identifies a concrete tension. A very sparse tree uses few links but can require elaborate indirect sequences to accomplish interactions for which a direct connection is absent. A complete mesh supplies more possible routes but can exceed cognitive and relational capacity. The economical link diagram and the economical programme need not coincide. [UX-W027, PDF 260–277, especially 268]

The proposed design task is consequently to organize the relation between network and programme. New connections can enlarge what coordinators can do, while too many overwhelm them. Missing links can force fragile detours. Resources, purposes, communication capacities and timing must be specified. [UX-W027, PDF 268]

Variety is not necessarily absorbed only at the top. Local arrangements can deal with distinctions before they reach a wider coordinator, and the rate of relevant change matters as well as the number of possibilities. As the Ashby account in S03 explains, not every state distinction requires an individually commanded response, and mere possession of alternatives is not sufficient for appropriate regulation. [UX-W027, PDF 269–271; S03]

The warning about using human beings to absorb the unmodeled residue is particularly direct. An apparently complete technical arrangement can depend on a person's continuing cognitive and relational work to cope with what it omits. Naming a coordination subsystem does not specify that capacity or establish that the job can be done at the required rate. [UX-W027, PDF 270]

His discussions of Simon, Galbraith and Voge are attributed connections within this framework. They help explain conditional assembly, information demand and autonomy with solidarity. [UX-W027, PDF 270–277]

Formal definitions and quantitative claims

Some formal definitions in the book are loose. A bijection alone does not establish an isomorphism preserving the relevant structure, and the printed descriptions of homomorphism and polymorphism contain inconsistent mapping language. [UX-W027, PDF 101]

The Voge passage also places two different quantities near one another: a proposed connection count and a number of possible configurations. They cannot serve as lower and upper bounds on the same measure without further definitions. Similarly, the toy performance expression whose normalized contact fraction peaks halfway does not establish universally optimal fifty-percent connectivity. [UX-W027, PDF 147–148, 215]

The book acknowledges the absence of sufficiently general metrics and rejects one scalar measure of organization. Its entropy, variety, information, value and energy analogies concern different quantities. Schematic curves do not establish that every noise input improves organization. [UX-W027, PDF 215, 240–259, especially 247, 251]

The chapters on evolution and creative time retain changing agendas, historical codes and values. Le Moigne labels important claims as hypotheses and leaves finalization unfinished. The concluding contrast between design, analysis and simulation concerns different intentional uses of models. [UX-W027, PDF 278–301]

Morin through Le Moigne

Le Moigne explicitly reconstructs organization through capacities to transform, produce, relate and maintain identity, drawing on Morin's *La Méthode*, volume 1. He connects producing with self-producing and describes active organization in relation to surroundings. [RGC-W028 through UX-W027, PDF 39–42, 209–214]

The second volume is related through autonomy that requires dependence and through auto-eco-reorganization. Self-maintenance involves environmental relations; greater organization does not imply escape from them. [RGC-W029 through UX-W027, PDF 209–214, 276–290]

Critics: accessibility and the need for internal mechanisms

Motulsky's 1978 review praises a documented and coherent synthesis while finding the entry difficult. Examples are insufficient for an introductory reader, parts resist isolation, and the exposition makes nonlinear demands. The review concerns the 1977 first edition, rather than the later 2006 additions. [RGC-W024, 217–218]

Braudeau's critical discussion addresses a different issue. For the physical delimitation and explanation of the water–soil system, he objects to an anti-Cartesian or black-box orientation that can leave internal mechanisms inadequately specified. He retains triangulation and operating, information and pilot models while recasting the description around a structural representative volume. The account shows how a model useful for one purpose may omit physical relations necessary for another. [RGC-W026, §3.1, printed pp. 16–25/PDF pp. 18–27]

Braudeau's stronger assertion that a black-box approach prevents an account of organization must be read alongside Le Moigne's explicit internal networks, programmes and permission to use classical methods where appropriate. [RGC-W026; UX-W027, PDF 63, 126–149, 191–216]

The author's later retrospective

Le Moigne's 2015 chronicle derives from a revised 2014 oral account, with an editorial introduction by Dominique Genelot. It describes the formation and later development of the modeling programme. As autobiography, it identifies how the author understood his own questions. [RGC-W025]

The recollection of industrial optimization is especially pertinent. Fragile estimates entered elaborate models, while a single profit criterion organized the calculation. Le Moigne's concern was both the reliability of inputs and the choice of what was being optimized. Sophisticated computation cannot by itself decide whether omitted factors are negligible or whether one objective captures the purposes of the organization's members. The account is a criticism of the problem formulation. [RGC-W025, PDF 2]

He later distinguishes the published model from the situated, tentative process of constructing it. Merely replacing the word “model” with “modeling” can leave that work unexamined. His retrospective relates practical action, epistemic justification and ethical reflection, and treats algorithmic methods as possible heuristics within a wider inquiry. The testimony supports an account of his methodological intention while leaving model adequacy to the purposes, evidence and criticism relevant to each application. [RGC-W025, PDF 5–6]

S24 Capra and Luisi: living organization, meaning and integrative design

A broad synthesis with several kinds of claim

Fritjof Capra and Pier Luigi Luisi's *The Systems View of Life* connects biological organization, cognition, social relationships and ecology. Its concluding design material brings these themes into questions of energy, materials, buildings and institutions. The breadth is substantive: the authors explain relationships among self-maintenance, embodied capability, meaning and wider dependencies. [WME-W006; SG-W011, introduction]

The authors describe scientific models as limited, approximate and subject to empirical examination. Their historical narrative about a movement from mechanism toward systems also contains philosophical, ethical and political commitments. A scientific model, a history of ideas and a proposed response to ecological interdependence have different warrants. The book itself says ecological connections do not logically entail ethical norms, while proposing a psychological route through identification with a wider world. [WME-W006, introduction, especially PDF 44; SG-W011]

Relations change what components can do

The organismic account distinguishes organization from an added nonphysical life force. Properties arise through the way components are related and active together. Objects, quantities and structures remain important, even though a parts list alone can omit the relations responsible for a capability. The account is therefore not accurately described as rejecting every form of analysis. [WME-W006, chs. 1, 4, especially PDF 117–142]

The hemoglobin illustration gives the claim a mechanism: the arrangement of subunits changes their binding behavior, so the assembled organization is not described adequately by assigning each isolated subunit an unchanged role. The authors use this to discuss emergence and downward causation without requiring an extra mystical force. Their discussion leaves philosophical differences between stronger and weaker accounts of emergence open. [WME-W006, ch. 8, PDF 245–250]

Repeated patterns can also economize specification. Symmetry permits the description of a repeated arrangement without independently specifying every instance. Homochirality narrows combinatorial variants in molecular arrangements. These are particular ways in which organized constraints alter possibilities. [WME-W006, PDF 265–266]

The mathematical chapter supplies attractors, basins, sensitivity, bifurcations and generative rules as a repertoire for inquiry. It expressly distinguishes a mathematical framework from an empirical scientific explanation: the framework needs a suitable application. The treatment is qualitative and sometimes loose—iteration need not be nonlinear, and a particular logistic-map parameter does not demonstrate chaos for all parameters. [WME-W006, ch. 6, PDF 167–203]

Autopoiesis maintains an organization through exchange

The biological account centers on a network of reactions that produces components, including the boundary supporting those reactions. Organization can persist while particular components and structural details change. The maintained cell is an ongoing production process rather than a fixed collection of matter. Nutrients and energy enter, waste leaves, and the network's activity sustains the conditions for further activity. [WME-W006, ch. 7, PDF 205–230]

Operational closure concerns the organization of processes. It is not thermodynamic independence. The identity of the self-maintaining organization still depends on resources and conditions outside it. A

history of structural coupling also affects the responses available: the present arrangement embodies a history rather than starting without constraints at each encounter. [WME-W006, ch. 7]

The book distinguishes self-organization, autopoiesis and autopoiesis associated with cognition. An organized pattern can form without satisfying the self-producing conditions, and an artificial self-producing structure can raise further questions about whether it counts as alive. The text explicitly reports a necessary-but-not-sufficient qualification to autopoiesis, although it subsequently returns to broader equivalence language in the biological domain. [WME-W006, PDF 219–221, 258]

Reproduction is also distinguished from the identity of an already living individual. Its importance for continuation of lineages does not mean that every living individual must reproduce. The framework asks first how a unit maintains the organization capable of participating in reproduction. Whether a larger aggregate itself satisfies the same criteria remains a further question. [WME-W006, PDF 221]

Assembly, maintenance and historical origin differ

The molecular chapters distinguish thermodynamically favored assembly, energy-throughput-dependent dynamics and kinetically constrained pathways. Confinement, binding, catalysis and signaling can alter what happens. Their description depends on what lies inside the selected boundary: enzymes treated as internal rules in one account can be externally supplied preparations in another. [WME-W006, ch. 8, PDF 231–279]

The authors explicitly acknowledge manual additions and micromanipulation in reconstitution experiments. Such experiments can show that a prepared collection performs an activity under laboratory conditions. They do not establish that the same collection spontaneously assembled without supplied components, nor that it describes the actual historical path by which life originated. The origin-of-life chapter openly presents that historical problem as unsolved. [WME-W006, PDF 243; ch. 10, PDF 329–367]

Minimal cells are consequently relative to the functions and environment being specified. A prepared vesicle may depend on externally supplied enzymes, ribosomes or nutrients. Some replicating core/shell systems discussed in the book are expressly not living. [WME-W006, PDF 350–361]

The proposed explanation for unusual molecular entrapment in vesicles is likewise labeled speculative. Similar power-law appearances do not establish one causal mechanism or complete metabolism. The book also acknowledges small tested subsets in protein research. [WME-W006, PDF 357–366]

Genetic instructions retain a reading apparatus

The chapters on evolution connect gene and protein networks, regulatory feedback, epigenetic processes, symbiosis, structural constraints and historical contingency. One particularly relevant distinction is between stored sequence and the system that uses it. DNA instructions require a protein-reading apparatus and a cellular setting. The text's compact-disc/player analogy makes the relation accessible: a record does not generate the activity independently of the machinery that interprets it. [WME-W006, ch. 9, especially PDF 314–316]

Patrick Bateson's guest essay also distinguishes broad developmental epigenetics from molecular silencing and warns that observed effects across generations can wash out. Such observations are not automatically proof of enduring evolutionary genetic change. The guest's qualifications remain distinct from stronger surrounding rhetoric. [WME-W006, guest essay PDF 303–310]

The human-evolution chapter then proposes connections among tools, hands, brains, language, dependency and social relations. Large brains have metabolic and birth costs in the account; capability is

not described as an unqualified benefit. Merrill's criticism of this historical sequence and its chronology is discussed below. [WME-W006, ch. 11, PDF 368–386]

Cognition is broader than consciousness in this account

The cognition chapters define mind through embodied process and interaction rather than an isolated store of representations. A living system's structure and history condition how it perceives and acts. "Cognition" is used broadly for that relation; consciousness and reflective experience are more specific questions. The authors even disagree explicitly about applying "mind" to organisms without brains. [WME-W006, ch. 12, PDF 387–427, especially 391]

Michel Bitbol's guest essay proposes that phenomenal experience is a methodological prerequisite for science and that functional third-person descriptions do not explain its sheer presence. He favors constraints among first-, second- and third-person inquiry. This is an identifiable philosophical position within the book. The text describes its clinical examples as hints and acknowledges a lack of evidence for a stronger surrounding hypothesis. [WME-W006, PDF 410–416]

Interpreting a signal can depend on embodied capability and history without establishing every metaphysical conclusion discussed later. Spiritual transformation, religion and scientific explanation are treated in different registers. The recorded disagreement with a purely material emergence account further prevents presenting the book as a single undisputed doctrine of consciousness. [WME-W006, ch. 13, especially PDF 449–450]

Social organization adds meaning, rules and power

The main integrative architecture relates form or organization, matter or embodiment, and process; meaning adds a further dimension for human action. A social network is not explained solely by its topology. Its connections involve language, cultural constraints, purposes and power. The authors explicitly say that understanding the chemistry of a cellular network does not supply the material mechanisms of a social one. Domain knowledge remains necessary. [WME-W006, ch. 14, PDF 466–484]

This is also where they qualify the transfer of autopoiesis. Higher-order cases are described as unclear and speculative. A social system whose members are living cells or organisms is not thereby proved to be an autopoietic network in its own right. The book reconstructs Maturana and Varela's restriction of the term to cellular production and their wider use of operational closure. [WME-W006, PDF 482–483]

Luhmann's alternative, reconstructed by the authors, treats communications as producing further communications and sustaining social roles and boundaries. Such boundaries are cultural and symbolic, rather than necessarily a membrane in physical space. Rules can be broken; molecules do not choose whether physical relations apply. [WME-W006, PDF 483–491]

The social-theory reconstructions are also mediated. Giddens and Habermas enter through other interpreters, while the power account involving John Kenneth Galbraith is distinct from Jay Galbraith's organizational information-processing work. [WME-W006, ch. 14]

Designed rules and emergent work both matter

Organizations contain designed structures—formal rules, routines and assigned authority—and emergent networks through which people learn and solve practical difficulties. The book treats both as necessary. Formal stability can support reliable activity, while communities of practice retain tacit knowledge and adaptive capability. Work-to-rule illustrates how much functioning can depend on activity omitted from formal instructions. [WME-W006, PDF 498–507]

Informal relations are not automatically flexible or beneficial. The authors' old-boys-network example supplies a counterexample within their own discussion. Nor is a crisis guaranteed to improve an organization: it can lead to breakdown as well as breakthrough. Creating conditions for participation and meaningful response is a proposed way to support change. [WME-W006, PDF 502–505]

The account retains relationships, communication, skills, learning, formal rules and power. The authors argue that meaningful impulses can require less change effort than coercive direction. The workshops described elsewhere similarly require funding, substantial participation, teaching, practical upkeep and dialogue; positive participant reports are different from controlled outcome evaluations. [WME-W006, chs. 13–14]

Ecology supplies dependencies and tolerance limits

Ecological persistence involves energy flows, cycling, decomposition, overlapping functions and continuing interactions. Redundancy can help preserve activity when one contributor is disrupted, while disturbances outside tolerance limits can collapse it. Diversity has the proposed strategic value within a connected community; assembling diverse parts without suitable relations can also generate friction. [WME-W006, ch. 16, PDF 541–574]

Stephan Harding's Daisyworld guest essay illustrates how temperature regulation can arise through feedback between organisms, growth and albedo without foresight or a coordinating committee. The simulated regulation is a property of the specified model. Later variants reportedly retain resilience over a range and then collapse when available habitat capacity is exhausted. [WME-W006, guest essay PDF 259–263]

The book is explicit that ordinary ecosystem autopoiesis remains open, particularly the requirement of a self-made boundary. Its argument about Gaia is a proposed higher-order case. It also states that ecology does not itself supply justice, democracy or other human values. Those commitments require human judgment alongside ecological understanding. [WME-W006, PDF 550–560]

Ecological education is itself organized work. The integrated curriculum described requires difficult scheduling, summer preparation and coordination among teachers, parents and administrators. Integration changes the pattern of effort rather than making preparation disappear. [WME-W006, PDF 568]

Integrative design can change several loads together

The closing engineering account reconstructs Amory Lovins and colleagues' integrative design. The proposed method brings architects, engineers and designers together early to consider relationships among subsystems and seek several benefits from a change. The objective differs from improving isolated components under unchanged assumptions about the loads they must serve. [WME-W006, ch. 18, PDF 660–670]

The building example gives a causal sequence. Better windows, lighting and office equipment reduce the peak cooling demand. A cooling system can then be renovated at a smaller required capacity, helping offset other retrofit costs. The authors report the Empire State Building project as cutting peak cooling load by a third and saving $17 million in cooling-system capital. The onsite window work and installed equipment remain part of realization. [WME-W006, PDF 667]

The piping example changes the physical resistance of the route. Replacing long, narrow, sharply turning pipes with shorter, wider and straighter routes reduces pumping demand. Smaller pumps, motors and electrical systems can then become possible. The text reports at least eighty-six percent electricity savings

in the cited loop and lower associated capital requirements. The result depends on the loop's flows and installation constraints. [WME-W006, PDF 667–668]

The lightweight-car account describes another cascade: lower mass and altered aerodynamic demand can reduce loads on other systems. It also retains the conditions of material production, propulsion and supporting infrastructure. The book acknowledges energy for separating hydrogen and says a more efficient vehicle cannot alone resolve all urban, social and environmental problems associated with car dependence. Its forecasts, prices and transition dates remain historical proposals rather than current recommendations. [WME-W006, PDF 662–666]

The smart-grid proposal retains substantial coordination machinery: communication, forecasting, geographical and technological diversity, storage, metering and governance. Its simulations concern specified combinations and conditions. [WME-W006, PDF 668–674]

Cascades and service cycles retain providers

The coffee-farm cluster connects several transformations: residues support mushrooms, further residues feed animals or worms, manure enters biogas production, and remaining material fertilizes soil. The arrangement tries to make one activity's output useful to another. It depends on organisms, water, a digester, management, labor and appropriate flows. Calling it a cascade specifies a productive relation; calling it “zero waste” remains a design aspiration rather than thermodynamic independence. [WME-W006, PDF 689–692, fig. 18.2]

The distinction between biological and technical nutrients similarly depends on preserving material quality and avoiding contamination. Repeated use requires suitable separation and processing. A service-and-flow arrangement in which a manufacturer retains ownership can create reasons to recover components, but also needs collection, disassembly, reverse logistics and an enduring provider. Changing ownership does not execute those activities. [WME-W006, PDF 693–695]

Passive building features—orientation, shape, insulation and windows—can reduce heating, cooling or lighting loads. Their performance depends on climate, use, materials and realization. The relevant information is how an arrangement changes the physical loads and the work assigned to active equipment. [WME-W006, PDF 695–696]

The agricultural and ecological proposals also retain local knowledge, collective work and sometimes greater labor intensity. More jobs can be part of the authors' social objective, while lower material throughput is another objective. They are not identical with minimizing all work. The concluding call for cooperation among government, business and civil society identifies institutional work still to be done. [WME-W006, PDF 683–688, 701–702]

Institutional purpose and health claims need their own warrants

Marjorie Kelly's guest essay connects ownership purpose, membership, governance, capital and networks. Its wind, forest, employee, foundation and credit-union illustrations are reported examples of generative ownership and an agenda for further research. [WME-W006, guest essay PDF 636–640]

The health chapter relates care to context, prevention and social conditions. Fabrizio Benedetti's guest essay supplies a narrower distinction: improvement in a placebo group can include natural course and bias, whereas a specific context-induced response needs a different inference. Expectation, learning, reward and anxiety can involve several mechanisms. [WME-W006, ch. 15, especially guest essay PDF 520–525]

Luisi on life, cognition and autopoiesis

Luisi's 2016 *What Is Life? A Systemic Approach*, based on a 2015 conference, explicitly combines biological structure, environment and cognition. The organism's capabilities are specific to its surroundings; cognition is used without requiring an anthropomorphic mind. Outside nutrients and energy sustain the process. [RGC-W013, article and figures]

The paper extends its account to social life and environmental responsibility. These are the author's broader interpretations. [RGC-W013]

The abstract of Bitbol and Luisi's 2004 *Autopoiesis with or without Cognition* proposes a necessary-but-not-sufficient relation and distinguishes metabolism from environmental interaction. The later book reconstructs the proposal. Luisi's 2022 *Reflections on Autopoiesis* abstract raises questions about cognition and molecular versus wider autopoiesis. [RGC-W016; WME-W008]

Reception includes specific criticism

Johny Van Aerde's review reconstructs the book's historical and systemic organization, with particular attention to its biological, cognitive, social and ecological connections. It is an enthusiastic professional response. [WME-W007]

Kübler, Merrill and Anway's collaborative review values the breadth and connected themes while criticizing uneven depth, inaccessible chemistry, schematic mathematics and dated disciplinary material. They specifically contrast the book's account of contingency with a more linear human-evolution narrative, question how examples were selected, and ask for testing and continued revision. [RGC-W014, review section PDF 120–135]

Merrill's separate 24 July 2014 blog review expands the anthropology objection, challenging the proposed links among bipedalism, tools, family arrangements and cultural chronology, as well as reuse of an older narrative. The blog and collaborative review are related critical voices. [RGC-W015; RGC-W014]

S25 Homeokinetics, coordination dynamics and systems philosophy

Distinct questions about organized persistence

The sources in this section pursue different projects. Iberall and Soodak propose a physical account of nested complex dynamics. Kelso develops coordination dynamics with empirically identified variables and task-dependent meaning. Gell-Mann distinguishes kinds of complexity. An abridgment of Laszlo's systems philosophy and reconstructions by Huchingson and Leaning introduce a wider ontological proposal.

Homeokinetics: long actions within nested dynamics

The *Primer on Homeokinetics* describes complex systems through a separation of time scales: a long internal action or "factory day" differs from relatively rapid local exchanges among components. Horizontal interactions and vertical relations among nested levels are both needed in the proposed account. Describing a system requires identifying those physical processes at an appropriate observational scale. [WME-W012, printed 2–8]

The authors distinguish a mathematical framework from the physical and engineering path that connects it with an actual system. Discrete descriptions and continuum approximations also have different conditions. Their declared physical reductionism does not mean simply announcing that biology reduces to chemistry and chemistry to physics; the work lies in specifying the intervening organization, exchanges and scales. The primer supplies a programme and examples. [WME-W012, 2–10]

The supplier analogy makes one relevant claim concrete. A factory could not operate within a compact time and place if every subcomponent at every level had to be constructed anew. Previously produced components arrive from other processes. Local operation employs a history and a surrounding organization of supply. The example explains why the time and capability represented by a supplied component cannot be attributed solely to the current act inside the factory. [WME-W012, 8]

Formation, a long maintenance phase, and degradation are separated. The action sustaining a familiar structure may differ from what assembled it or what eventually dissolves it. Low-energy catalytic changes can switch among more energetic action modes. In that account, a signal negotiates a transition while the active process retains the energy and material needed for performance. A small switch is not a source of the whole activity's energy. [WME-W012, 10–15]

The grandfather clock is an especially concrete example. A falling weight supplies gravitational energy, the escapement couples to pendular motion, gearing produces further time scales, and periodic winding replenishes the store. Oiling and a suitable suspension remain necessary. The authors examine transient changes after winding as well as sustained timekeeping. [WME-W012, 11–15]

The oil-can oscillator and other illustrations similarly involve specific couplings and replenishment. The primer is interested in how relatively autonomous motions become coordinated while retaining different scales. The proposed physical explanation requires the forces, energy source and stability conditions. [WME-W012, 11–16]

Its house illustration extends the maintenance distinction accessibly. A house may look like a completed solid object, but repair and replacement require continuing flows of materials, costs and action from its first use. The maintenance engineer sees a persistent flow process where an owner may initially see a finished construction. [WME-W012, 21]

The proposed social physics remains unfinished

The social passages extend physical language to population, trade, political organization and war. The authors intend condensed and dispersed population phases as literal physical explanations, not merely decorative analogies, and propose civilizational time scales. That intention does not establish the corresponding equations, measurements or historical validity. [WME-W012, 16–23]

The primer's final qualification is unusually explicit. Its account of values and command-control is far from complete or adequately determined, and the hoped-for ultimate neurochemical reduction may be a false hope. [WME-W012, 23]

The abstract of Iberall's 1985 *Outlining Social Physics for Modern Societies* proposes flows of matter, energy, momentum, population and value in trade. The later primer develops his nested physical architecture. [WME-W013, abstract; WME-W012]

Kelso: coordination with continuing autonomy

Kelso's 2008 *An Essay on Understanding the Mind* engages sympathetically with Iberall but develops a distinct account. Coordination concerns how several processes form a collective pattern while retaining their own tendencies. Its functional information is meaningful in relation to a task and the activity organized, rather than a free-standing count of messages. Relevant order variables and control parameters must be identified empirically. [RGC-W017, 180–190]

The extended Haken–Kelso–Bunz relation includes a difference between the intrinsic frequencies of the participating oscillators, nonlinear coupling and noise. The relative phase can then exhibit regimes that the original symmetric equation did not contain. Kelso expressly warns against assigning the extension's results to the unchanged original equation: symmetry breaking changes fixed points and bifurcation structure, and is crucial to the metastable behavior being discussed. [RGC-W017, 188 and notes 2–3]

Fully locked coordination and complete independence are limiting cases. A metastable regime contains tendencies both to bind together and to act independently. Components can linger in a coordinated relation and then depart, rather than remaining permanently locked in one asymptotically stable pattern. The physical components and their coupling remain necessary. [RGC-W017, 187–193]

Kelso proposes dwell and escape times to describe these tendencies. Dwell time concerns the persistence of a collective tendency; escape time concerns expression of the elements' autonomy. Their ratio is a candidate description of the quality of metastability. The measures require operational definitions, and any transition threshold must be worked out for the particular system. [RGC-W017, 192–193, notes 5–6]

The coordination issue is therefore not solved by maximizing either synchronization or independence. Too little coupling can prevent a useful collective activity; too much locking can remove flexibility. Which relation is satisfactory depends on the task. The essay's comparison across behavioral and neural examples is a programme of identifiable relationships. [RGC-W017, 190–202]

Switching also need not require a localized physical switch. A distributed pattern can lose stability and another pattern become possible through the dynamics of interaction. The explanation retains oscillators, coupling, changing parameters and energy. Absence of one command location does not mean absence of determining processes or continuing work. [RGC-W017, 199–202]

The essay relates stability to reported neural measurements, including BOLD changes as coordinated patterns lose stability. The broader extensions to mind, social activity and meditation remain tentative essay proposals. [RGC-W017, 199, note 10, concluding discussion]

Gell-Mann: a short account can be hard to find and use

What Is Complexity? distinguishes information measures from time and space required for computation. Algorithmic information content concerns the shortest programme producing a specified string and halting. An incompressible random string can have high content by that definition, yet random gibberish does not capture every ordinary meaning of complexity. Gell-Mann therefore distinguishes a concise description of regularities from the complete specification of the individual entity. [WME-W014, 16–17]

His effective complexity concerns those regularities. The description depends on coarse-graining, prior knowledge, language, coding and the reference machine. Two apparently different measures can answer different questions. A long string containing a very simple repeated relation can have much mutual algorithmic information while its organizing regularity has a brief description. The amount of shared pattern and the length of the rule describing it are not identical. [WME-W014, 16–17]

The adaptive-schema account then connects observation and activity. A system identifies regularities and compresses them into a schema used for description, prediction or prescribed behavior. Comparison with further data and the consequences of action affect competition among schemata. This process can make mistakes: a pattern can be imagined where it is absent, or an actual relation ignored. The account does not treat every acquired rule as correct merely because it is compact. [WME-W014, 17–18]

Prediction and discovery add separate demands. A simple rule with specified boundary conditions can require lengthy computation to produce a detailed prediction. Finding a short rule from data can be difficult even when using it later is straightforward. Gell-Mann relates these questions to logical depth and crypticity, while saying that a more precise mathematical account will appear elsewhere. [WME-W014, 18–19]

Gell-Mann separates a description from the work of discovering it and computing its predictions. Compactness of a rule can coexist with difficult discovery or lengthy execution. [WME-W014]

Laszlo's patterned flows and a mediated larger architecture

The anonymous abridgment of Laszlo's *The Systems View of the World* presents enduring structures through patterned flows. A structure channels later activity, while that activity sustains or changes the organization. Human action belongs within wider ecological and social relations rather than an isolated world of self-sufficient individuals. [SG-W038]

Huchingson's article and Leaning's thesis reconstruct the foundational philosophy of Laszlo's *Introduction to Systems Philosophy*, distinct from the popular *The Systems View of the World*. [RGC-W018; RGC-W019; RGC-W020; RGC-W021]

Huchingson reconstructs a connected ontology of natural and cognitive systems: ordered wholeness, adaptive stabilization, self-organization and nested relations. Systems can contain other systems, and an enduring identity is formed through relations to an environment. Greater organization permits alternative programmes of action and can support modifying surroundings. The account is a philosophical architecture, not merely a list of examples. [RGC-W020, discussion of reality and environment]

He also reconstructs Laszlo's connection between adaptation, persistence, experience and value. Cognitive systems can pursue imagined conditions, while knowledge arises through observer–world interaction. These are proposed ontological and epistemological relations. Survival or satisfaction does not automatically settle a human ethical choice. [RGC-W020, environment, values and causality sections]

The reconstructions include methodological criticism

Huchingson's comparison with Whitehead preserves differences. Whitehead's actual occasions are not simply Laszlo's nested composite systems. He asks whether an abstract grid is being imposed on experience, whether the primordial matrix coheres with the universal system categories, and how emergence addresses differences in agency. In particular, a matrix that generates systems while not itself being a system raises a problem for the ontology as presented. [RGC-W020]

The article's later theological reconciliation and information-processing extensions belong to Huchingson. [RGC-W020, critical comparison, conclusion and notes]

Leaning's selected section 3.4.3 reconstructs the same broad programme while questioning its grounds for validation. Presupposing that the world is systematically ordered does not establish that a particular general model adequately describes it. Combining correspondence-based confirmation with theory-laden scientific paradigms also creates methodological tensions that require explanation. The objection concerns justification and empirical productivity. [RGC-W021, §3.4.3, printed pp. 89–92/PDF pp. 92–95]

S26 Rosen through Louie and Siekmann: organization, repair and realization

Metabolism and repair through Louie and Siekmann

Robert Rosen's relational biology asks what organized functional relations distinguish living activity. The abstract of his 1958 *A Relational Theory of Biological Systems* identifies metabolism and repair. A. H. Louie's affirmative 2017 chapter and Ivo Siekmann's critical 2017 preprint reconstruct later formulations as well as this earlier programme. [W102; W104; W103]

Louie develops the architecture of *Life Itself* and *Anticipatory Systems*. Rosen was his doctoral supervisor, and Louie continues the intellectual programme. Siekmann offers mathematical and modeling criticism. [RGC-W022; RGC-W023; W104; W103]

Relations specify what processes produce and maintain

Relational biology abstracts from particular material realizations to examine organized dependencies. It asks what different mechanisms can share in their functional organization. That is more specific than saying everything is connected. A process that transforms material, a process that makes the transforming machinery and a process that maintains that productive capability have different places in the account. Louie also respects molecular biology's successful explanations; his target is the claim that they exhaust every question. [W104, 2–5, 15–24]

Louie's diagrams distinguish a flow from the processor constraining or producing it. For a mapping f from A to B, an input a is transformed to an output b by f. He associates the input with material cause, the processor with efficient cause, and the output with final cause in this formal vocabulary. These terms organize the explanation; “final cause” at an output node does not by itself establish foresight or a conscious purpose. [W104, 15–16]

Two processes can then relate in different ways. In material entailment, the output of one becomes material for the other: one transformation supplies what another processes. In functional entailment, what one produces is the processor of another transformation. The distinction identifies a relation between making something and making the capability that makes something. Ordinary chaining of outputs is different from production of the processors themselves. [W104, 17–18]

For gambits engineering, this is a substantive account of continuing capability. A momentary transformation does not explain how the machinery that performs it remains available. Treating a process as a box with an output can omit the production and replacement activity that sustains that box. Relational biology explicitly puts that activity into the model.

Metabolism and repair address different continuities

In Siekmann's reconstruction, metabolism is represented by a map f: A → B. It can be interpreted as an enzyme transforming inputs while remaining unchanged during the particular operation. Real enzymes can nevertheless degrade. A repair component Φ produces or replaces the transforming component f, so sustained operation requires more than a currently available metabolic map. [W103, 6–7]

The notation places Φ's output among mappings from A to B. This expresses production of a functional component, not merely production of another quantity of the original metabolic output. The repair component also needs maintenance, which appears to create a regress: a repairer for the repairer, then another repairer for that repairer. The proposed architecture seeks to close the productive relations rather than add an indefinitely external succession of components. [W103, 6–9]

The environmental boundary remains open to resources. Components can replace one another while drawing on matter and energy supplied from outside. Internal production of efficient causes does not mean that every material cause, energy source or surrounding condition is produced within the organism. Both expositions retain the difference between organizational closure and thermodynamic independence. [W103, 7–9; W104, 15–24]

A closed path and full closure differ

Louie describes hierarchical cycles in which two or more efficient causes enter a closed path. This requires the relevant functional composition, not simply any return arrow in a diagram. A loop of material transformations and a cycle in which processes produce other processors are different relations. [W104, 18–20]

Having such a path among some processes is also different from a claim that every efficient cause relevant to the system is internally entailed. The stronger closure condition demands accounting for all the efficient causes within the selected model. Choosing the system and its modeled functions remains consequential. Neither the circle's appearance nor its name supplies the omitted physical realization, environmental resources or scope. [W104, 20–25; W103, 8–10]

This makes repair organization a positive explanatory proposal even where broader conclusions remain disputed. A system can potentially persist longer than individual components if its organization replaces them in time. To establish an actual instance, the components, their transformations, degradation and resource conditions still need identification. [W103, 7–10]

Siekmann's replication-map objection

For the simple metabolism–repair architecture, a further replication map β is supposed to repair Φ. The operative condition is $\beta(f) = \Phi$. Siekmann argues that one can choose a map satisfying that relation and assign it a parameter b^* in an associated family of maps. He does not object that writing such a family is mathematically impossible. [W103, 8–9]

His objection concerns what follows. Constructing or parameterizing the map does not show that the natural system produces the corresponding physical process. In particular, deriving the family through evaluation maps does not establish the material identification claimed for it. The distinction is between an available mathematical construction and a realized biological organization. [W103, 8–10]

Siekmann therefore proposes treating closure to efficient causation as a postulate or hypothesis to test in concrete biological systems. This preserves the substantive idea of mutually sustained repair. It disputes the claim that the desired realization follows from the abstract architecture alone. [W103, 9–10]

The biological identification is explicitly postulated

Louie's affirmative account distinguishes necessary and sufficient conditions for life. A property can include all living systems while also including nonliving ones. Collecting several necessary conditions can narrow the class; declaring their joint sufficiency adds a biological identification that needs to be seen explicitly. The chapter explains this movement before giving its final taxonomy. [W104, 8–11]

Near the conclusion, the “Postulate of Life” identifies an organism with a natural system realizing an (M,R)-system. The ensuing equivalence between life and closure to efficient causation is presented as the “Fundamental Theorem of Relational Biology.” Its formal derivation incorporates that stated postulate. The biological identification enters as a postulate in the derivation, rather than as an independent empirical finding. [W104, 25–27]

Louie's generalization to a union of interacting organisms includes an ecosystem as itself an organism “in some sense.” That is a qualified extension within his framework. [W104, 26]

Anticipation is not the same as life or impredicativity

Through Louie, an anticipatory system contains an internal predictive model of itself and its environment and acts in the present in accordance with predictions. The antecedent action is the defining feature. A future event need not physically cause the current one; the current model and its use are part of the present organization. [W104, 11–12; RGC-W023 through W104]

The chapter distinguishes the class of living systems, the class of anticipatory systems and the wider class it calls impredicative. It gives a socioeconomic system as a nonliving anticipatory example. Anticipation is therefore not sufficient for life even in this affirmative account. Nor should every automatic response be called anticipation simply because its eventual consequence can be described. [W104, 11–15]

Impredicativity concerns self-referencing definition and the relational dependencies represented. It is not a synonym for an incomprehensible system or a moral ranking of complexity. Louie warns about the misleading status connotations of “simple” and “complex.” The distinction between classes does not itself specify which transitions a dynamical system can undergo. [W104, 12–15]

Modeling relations and purpose-bound mechanisms

Siekmann explains the category-theoretic modeling relation using algebraic curves and associated algebraic objects. The example shows how mathematical structures and relationships can be related, rather than reducing category theory to arrows between unnamed boxes. A meaningful analogy depends on the structures and relations preserved. [W103, 10–14]

His later argument distinguishes a model's purpose from an exhaustive reconstruction of nature. Applied models make simplifying assumptions to investigate selected behavior. Successful predictions can fail to distinguish several mechanisms, and congruence between a model and every causal feature of the natural system may be unverifiable. The model can still provide insight into the selected aspect. [W103, 17–24]

This leads to a disagreement about the word “mechanistic.” Siekmann argues that a relational model can be a kind of mechanistic model if it specifies how the represented interactions produce the selected behavior. That interpretation does not assert that the whole natural system is nothing but the modeled mechanism. Refining a question can require revising assumptions and adding detail, while the resulting model remains partial. [W103, 18, 23]

The choice to abstract internal material structure is consequently one modeling decision, not proof that such structure never matters. A network can leave interpretation, parameter identification or material feasibility underdetermined. Kinetic and physical constraints can help answer those questions. This is a substantive modeling objection alongside the replication-map argument. [W103, 20–26]

Siekmann points toward compositional reaction networks, biochemical realizations and closure-of-constraints approaches. In his reconstruction they illustrate ways of retaining organized relations while giving components material and kinetic content. They are proposals for further operationalization. [W103, 23–26]

The affirmative and critical accounts thus leave a clearly specified literature question: how does a materially realized organization produce and replace the processes on which its continued operation depends? The sources provide both a rich architecture for asking it and explicit disagreements about which conclusions follow formally and which require biological evidence.

Part 2 Physical realization, regulation, and information

These accounts describe specified physical behaviors, control problems and representations.

S01 Physical dynamics and passive behavior: mechanisms and operating conditions

Material arrangement and behavior

Material arrangement can supply a walking cycle, selective binding, attenuated vibration or a period of stellar support. The accounts below explain these mechanisms using different meanings of stability, function and passivity.

McGeer: walking as a natural dynamic mode

Tad McGeer's 1990 *Passive Dynamic Walking* examines a class of two-legged machines that, once started on a shallow downward slope, can settle into a steady gait without active gait control. Gravity supplies the energy through descent. The selected opening distinguishes the problem of dynamic walking from maintaining static equilibrium throughout the motion, and distinguishes several ways active controllers can contribute. The example is informative because the detailed walking cycle need not be imposed as a succession of commanded joint positions. Geometry, mass distribution and the contact dynamics participate in generating it. [W053, abstract and §§1–2]

McGeer's discussion does not simply divide machines into controlled and uncontrolled. The comparison includes feedback toward end-of-step conditions, tracking a specified trajectory, and feedforward schedules supplemented by corrections to the initial conditions of later steps. These locate different tasks for the controller. They help explain why a gait can exploit natural motion without an active controller continuously prescribing its whole trajectory. [W053, §2]

The source reports a test machine as support for the passive effect, while its opening also proposes combining the dynamics with active energy input for a wider range of locomotion. The shallow slope and suitable machine design remain part of the supported accomplishment. A downhill walker is not an equal-task demonstration against every powered walker on level ground or irregular terrain.

Ikemata and Sano: a knee stop can participate in the gait

Yoshito Ikemata and Akihito Sano's review explains passive walking as a way to study the mechanics of locomotion. The first page describes walking down a gentle slope without motors, sensors or a controller, with gravity as the energy source. Its account of McGeer's knee-equipped walker is mechanically specific: paired inner and outer legs constrain the demonstration to a sagittal plane; a stop prevents the knee bending backward; a curved foot also contributes to the arrangement. During swing the knee can bend and extend naturally, then reach the approximately straight configuration needed for contact and support. [W010, printed p. 2]

The stop therefore has a particular job in a sequence of motions. It permits some movement and restricts another, instead of supplying a general intelligent assessment of every possible footfall. [W010, printed p. 2]

Moon: level-ground walking retains energy input and torque compensation

Jae-Sung Moon's 2021 paper examines a torso-equipped biped on level ground through a model and simulations. Its opening explicitly identifies the need to replace energy lost when the swing leg contacts the ground. The model divides a step into take-off, passive swing and contact phases. It treats motion in a two-dimensional sagittal plane, neglects scuffing in the model without knees, and assumes a completely inelastic contact without slip or rebound. These assumptions define the analyzed limit cycles. [W066, pp. 608–610, introduction and §II]

The passive swing is partial in a precise sense. Compensating torques can cancel the generalized torque applied to the leg coordinates while retaining a torque for the torso. When the torso is vertical, the relevant inputs can all be zero for that interval. With a torso deviation, corrective input remains. The absence of generalized forcing on selected coordinates consequently does not mean every actuator is inactive or that there is no energetic cost anywhere in the system. The arrangement separates functions through the coupled mechanics and chosen inputs. [W066, §II.3]

The simulations show how exploiting a natural trajectory can coexist with energy injection and stabilization. Their energy levels, collision assumptions and torso compensation specify how the modeled walker moves beyond downhill passive walking.

Lehn: molecular architecture stores distinctions that binding can read

Jean-Marie Lehn's 1987 Nobel lecture defines supramolecular chemistry around intermolecular bonds and functions of associated chemical species. In the selected recognition discussion, merely binding is insufficient: recognition concerns binding with selection of a substrate and potentially a specified function. The architecture of binding sites, their number and arrangement, and surrounding molecular structure contribute to which association forms and how it persists. Lehn explicitly interprets this as information stored in architecture and read through formation and dissociation of the association. [GL-W009, printed pp. 446–447]

His complementarity includes energetic as well as geometric features. A lock-and-key picture based only on shape would omit part of the account. Thermodynamic and kinetic selectivity matter, and the balance of rigidity and flexibility affects what the receptor can do. A rigidly arranged structure can favor recognition, while exchange, regulation and cooperative changes can need built-in flexibility. The selected discussion thus explains both the capacity provided by a structure and the conditions under which that capacity changes. [GL-W009, p. 447]

Near the lecture's end, Lehn describes designed self-assembly as elaborating molecular components that spontaneously organize into a desired supramolecular architecture. The chemist selects and prepares the components and conditions. His molecular-device and “chemionics” discussion proposes combinations for detection, storage and signal processing. [GL-W009, pp. 483–484]

Rahimi and colleagues: tuning works within a changing frequency relation

Fatemeh Rahimi, Reza Aghayari and Bijan Samali review tuned mass dampers for structural vibration control. In the selected discussion, a secondary mass and its tuned dynamic relationship with a structure can reduce a targeted response. The review also emphasizes a limitation of a single tuned mass damper: a change in structural frequency or damper parameters can impair effectiveness. The absence of a continuously computing feedback controller does not make the response independent of the frequencies and damping for which it was designed. [GL-W017, selected single-/multiple-TMD discussion]

The reviewed multiple-damper approach distributes natural frequencies over a range. The authors report studies in which this reduces sensitivity to uncertainty and can improve performance relative to an optimal single damper of the same total mass. Those comparisons belong to the studies and simplified forcing/model conditions summarized by the review. Widening a range, changing placement or introducing active/semi-active components creates different arrangements with different demands. [GL-W017, discussion of Xu/Igusa, Yamaguchi/Harnpornchai and subsequent studies]

NASA's stellar explanation: persistence has a fuel-dependent history

NASA and the Night Sky Network's public explanation of stellar life cycles describes hydrostatic support as pressure opposing gravity. During the main sequence, nuclear reactions provide interior heat and radiation. Before the main sequence, contraction supplies heat before the core supports those reactions. Exhaustion of available core fuel changes what can continue; later phases depend on stellar mass, and eventual loss of sufficient support can lead to collapse. Approximate balance is a phase sustained under changing material conditions, not an assertion of permanent immobility. [GL-W088, institutional explanation]

The account explains naturally occurring organization through opposing effects, supply and changing regime, without an external operator continuously choosing its successive states.

Mechanical responses shaped by geometry

Bertoldi, Vitelli, Christensen and van Hecke describe a collapsible shell called the buckliball. Its design takes a cue from the Hoberman Twist-O, a toy whose hinged struts let it collapse in one continuous motion. The buckliball uses a patterned elastic shell. Under pneumatic actuation, the shell buckles and folds into a smaller shape. The shell's geometry guides the buckling into the fold the designers want. [GL-W065, fig. 4]

Ling and colleagues' abstract for *A survey on synthesis of compliant constant force/torque mechanisms* proposes using engineered mechanical response to avoid complicated force-control systems. Nonlinear varying stiffness creates design problems addressed through kinetostatic modeling and optimization. [GL-W073, abstract/highlights]

Van Gerven and Stankiewicz organized process intensification around structure, energy, synergy and time. A later account they coauthored explains what some of those design choices aim to achieve. One aim is to make the conditions encountered by different molecules more uniform. Another is to increase the area over which a driving force acts. The framework applies these questions at scales from molecules to whole processing units. [GL-W074] [SMR-W001, fig. 2 and accompanying text]

S02 Constraints and error prevention: triggers, barriers and mistake-proofing

What the arrangement changes

The selected sources concern different ways of changing opportunities for error, exposure or harmful response. Some mechanisms trigger an action; some block an exposure; others remove a hazardous inventory or make an abnormality visible.

Evans and Madrzykowski: the thermal trigger has a response time

David Evans and Daniel Madrzykowski's 1981 NBS report investigates the thermal response of selected fusible-link sprinklers. The opening explains a physical process: hot combustion products transfer heat to a thermal element; added energy first raises its temperature and then contributes to the phase change in its binding material. Sufficient fusion permits rupture of the link and activation. A rated activation temperature therefore does not alone determine when a sprinkler operates in a developing fire. Heat must reach the element, and the element has a thermal response. [W023, printed pp. 1–2]

The report describes sudden immersion in a hot gas flow as a way to characterize that response. Its exploratory apparatus uses a heat gun, tube, mixing/deflection arrangement and instrumented element. The selected methods state that flow turbulence, known to influence heat transfer, was not measured. Instrumentation itself required attention to mechanical robustness and heat conduction through attached wires. These details are useful because a thermal trigger is not an abstract, instantaneous if-temperature-then-release rule independent of its physical realization. The observed response comes from a specified exposure and measurement arrangement. [W023, pp. 2–3]

The abstract distinguishes two response-characterization methods, one explicitly accounting for phase change, and compares predictions for a selected mattress-fire exposure. Thermal sensing, release, available water and successful suppression remain separate parts of the whole undertaking.

Parmelee: a historical specification for automatic release

Henry Parmelee's 1874 patent specification describes an attachment to distributing pipes in which heat automatically opens a valve and permits discharge. The selected description includes a perforated sprinkler chamber, a valve held closed by a spring-and-cap arrangement, and fusible supporting material. It also describes a shield intended to protect that material from a flow of water so heat can act on it. The shield makes an interaction within the protective arrangement visible: water from elsewhere can impede the heating needed for another release. [W024, US154076A, description]

The specification also proposes a main-flow arrangement and alarm associated with opening outlets. Thus the automatic trigger belongs to a larger material system of pipes, pressure, distribution and indication. [W024, description; application 24 June 1874, patent 11 August 1874]

Nakajo: mistake prevention retains prospective analysis

Takeshi Nakajo's 2004 quality-engineering article discusses changing work objects, equipment and procedures to prevent trouble associated with human error. The section describes the practical core of mistake-proofing as improving the work method to suit human characteristics rather than depending solely on changing the person. Objects' shapes and colors, equipment, instructions and the sequence of work can alter what is easy to do or what errors can occur. The article treats these as interventions in the arrangement of work, not simply reminders to be more careful. [W063, §§2.1–2.2, printed pp. 627–628]

The next step is demanding: identify where errors could arise. Nakajo's selected FMEA procedure chooses work for analysis, represents the sequence, divides it into suitable units, identifies possible error modes, evaluates their occurrence and consequences, and plans measures for higher risks. He emphasizes that usefulness depends strongly on identifying appropriate error modes. Units that are too broad obscure what happens; units that are unnecessarily fine waste analysis time. His proposed object-focused cycle asks what object is handled and how its state is changed or recognized. [W063, p. 628]

The same page distinguishes likelihood, seriousness and the effectiveness of measures preventing consequences from spreading. A preventive arrangement can reduce dependence on repeated faultless performance while requiring good decomposition and knowledge during its design.

Amyotte and Khan: removing a hazard is hazard-specific

Paul Amyotte and Faisal Khan's 2021 synthesis distinguishes inherent safety, passive engineered safety, active engineered safety and procedural safety. Eliminating or reducing a hazard at its source is different from adding a device that prevents or mitigates its effects when called upon. Their stairs example explains the distinction: handrails and careful stair use retain a fall opportunity that a one-story arrangement removes. But the authors immediately complicate the favorable example. Changing an existing building is harder than selecting its design initially, and a location subject to flooding poses other questions about a single story. [W064, §§1.2–1.3]

The same argument is explicitly hazard-specific. Removing a flammability hazard through nitrogen can introduce an asphyxiation hazard. In an operational storage example summarized from other work, smaller inventory and greater distance from a community accompany higher pressure. The authors report an improved overall risk profile for that case, while making the competing changes visible. [W064, §1.2 and §6]

The operational section rejects the idea that early-design emphasis makes existing plants irrelevant to inherently safer design. Modifications can create opportunities, and the discussion retains continuing risk analysis and updating from operational information such as near misses. A design choice can reduce a recurring hazard while leaving a need to understand changed conditions and performance. [W064, §6]

NIOSH: minimal user action is a design aim, not the end of evaluation

The NIOSH *Hierarchy of Controls* explanation orders elimination, substitution, engineering controls, administrative controls and personal protective equipment. It explains the preference for the first three partly through reduced reliance on significant continuing human interaction. Engineering-control characteristics include blocking a hazard near its source, integrating protection in the original equipment, limiting interference with the control and requiring minimal user action without making the job harder. These are concrete design aims; they are more informative than the claim that a barrier is simply “safer.” [GL-W059, 2024 institutional page, engineering-controls discussion]

The page also distinguishes upfront and operating costs. Engineering controls can cost more initially, while their longer-term cost can be lower, especially across several workers. Substitution requires examining the new risks and interactions of the substitute. Training and regular evaluation remain explicit responsibilities, including checking whether controls continue to reduce exposure. This is an institutional prevention framework and explanation of its rationale. [GL-W059, substitution, cost and train/evaluate passages]

JIPM: local workers make and understand a karakuri arrangement

The Japan Institute of Plant Maintenance's primary explanation of *karakuri kaizen* starts with workers identifying difficulties at the site. They combine an available power source with simple mechanisms such as levers, links, cams, gears or inclined paths to make an improvement. Gravity, springs, human force and power borrowed from existing machinery are among the sources considered. Reusing a motor's motion or an existing movement is different from claiming that the device operates without energy. The relevant change can be in how an available source performs the task. [UX-W036, definition and method sections]

The explanation includes handling heavy items, presenting parts, moving supplies and reducing awkward work. It also explicitly says realizing an improvement requires understanding the work, the equipment's structure and characteristics, and improvement methods. It treats that effort as a way to develop people as well as devices. The claim that locally made equipment can be repaired locally connects the possible operating benefit to retained skill. [UX-W036, application and personnel/repair sections]

A JIPM technical excerpt illustrates the combination of power sources and mechanisms. Its page 16 describes using an object's weight, human movement or a nearby machine's drive with transmission mechanisms. A gravity-fed roller path moves a part toward a later process; another arrangement borrows one motor's rotation for several motions. [UX-W037, p. 16]

Toyota: stopping changes the work of watching

Toyota's own explanation of its production system gives *jidoka* a particular role. A machine can detect an abnormality and stop, or an operator can pull a stop cord; an andon indication brings attention to the exception. Toyota describes this as allowing people to respond when an abnormality occurs instead of continually watching equipment. The arrangement combines detection, stopping and a means of calling for a response. An indicator alone is not the entire mechanism. [GL-W080, jidoka and andon discussion]

The same company account insists on continuing development of human skills and technologies, with people able to think independently and implement kaizen. Its separate just-in-time pillar concerns linking production and making the needed quantity at the needed time. [GL-W080, two-pillars and human-skill passages]

FDA: the interface includes more than a display

The selected FDA guidance defines the device-user system through users, use environments and user interfaces. Its interface includes packaging, labeling, training materials, physical controls, displays, alarms and operating logic. A task is an interaction or set of interactions toward a desired result. This broad scope prevents an account of error prevention from treating a neat screen or fewer instructions as a complete intervention. Setup, handling and the circumstances of use can affect what a person is actually able to do. [GL-W075, guidance p. 4 and companion explanation]

The guidance's account of use error distinguishes a result differing from what was expected from a failure caused solely by the device. That distinction gives investigation a more precise object than simply blaming an operator. The selected explanation relates design to actual user and environmental characteristics, and calls for confirming that efforts to minimize use-related risk have succeeded. [GL-W075, definitions and overview]

Kabanov, Kupitz and Goetzmann: a grace period retains later judgment

The 1992 IAEA Bulletin article on advanced reactors describes passive features as relying on naturally available or stored sources of motive power, such as natural circulation, with several levels of passivity.

Some features are actively initiated and then operate passively. The selected discussion also explains a grace period during which automatic responses allow operators time to assess the event and plant state before deciding on action. Delaying immediate intervention is different from eliminating the need for later understanding. [GL-W081, printed p. 34]

The article retains demonstration requirements for new features, potentially including full-scale testing for innovative designs. Its following page labels specified probability targets as design goals that had not been shown achievable for each design. The article provides design reasoning about simplification, accessible energy and operator time. [GL-W081, pp. 34–35]

Source inspection and the removal of hazards

The Shingo Institute discussion relates source inspection and mistake-proofing to Shigeo Shingo's work: changing the conditions producing a defect and detecting errors at source can prevent defects. Shingo develops the approach in *Zero Quality Control: Source Inspection and the Poka-yoke System*. [W038, through the Shingo Institute]

Amyotte and Khan's specialist discussion above develops the source-reduction idea associated with Trevor Kletz's *What you don't have can't leak*, published in 1978 following a 1977 lecture. [W039, through Amyotte and Khan]

Nakajo and Hitoshi Kume drew five principles from about 700 examples at mechanical assembly lines. The first three change the chance that an error occurs. Eliminate the operation that permits the error, substitute a method less prone to it, or make the correct action easier. The other two address what happens if the error occurs. Detect it so it can be acted on, or limit its effects. [W076]

S04 Interconnection and dissipativity: what the formal relation preserves

From an arrangement to an admissible behavior

The selected control and network sources make the structure of an arrangement mathematically consequential. They explain different questions: how a governor responds dynamically; what an energy-accounting inequality establishes; how components share a behavior; and when decomposing a network calculation saves computation. The formal words “control,” “passive,” “interconnection” and “decomposition” retain their source-specific meanings here.

Maxwell: stronger correction can destabilize the motion

James Clerk Maxwell's 1868 *On Governors* opens by distinguishing devices that increase resistance with speed from governors whose continuing adjustment can restore a normal speed within working limits. That difference concerns the dynamics of the correction, rather than simply the presence of a speed-responsive part. Maxwell then divides disturbances into growth or decay, with or without oscillation. A useful governor must reject growing departures, including oscillations whose amplitudes increase. The intended direction of an adjustment does not settle that question. [W069, opening, printed pp. 270–272]

He describes a limit at which making the regulation faster leads instead to dancing and jerking motion. The example associated with Fleeming Jenkin's governor gives the mathematical problem a mechanical setting: adjustment parameters that appear to increase governing power can change the character of the coupled motion. Its pertinent lesson remains specific: a regulator's nominal corrective action and the stability of the whole process are different claims. [W069, pp. 271–272]

Willems: stored quantity cannot increase beyond supplied quantity

Jan Willems's 1972 dissipativity theory specifies a dynamical system, an input/output-dependent supply rate, and a nonnegative storage function of state. Dissipativity requires the increase of that storage over every admitted trajectory and time interval to be no greater than the accumulated supply over the interval. Written schematically, $S(x1) - S(x0) \leq \int w\,dt$. The storage function and supply rate must be identified together with the system and its allowed inputs. [W014, §3, definition 2, printed p. 327]

For a small arithmetic illustration, suppose the initial storage is 5 units and the integrated supply is 3 units over the chosen interval. The inequality permits terminal storage no greater than 8 units. It does not determine whether the actual value is 8,6 or another nonnegative value consistent with the trajectory. If the chosen supply has physical energy units, those units must be used consistently.

The universal quantification also matters. Checking one favorable trajectory is insufficient to establish the required inequality for all admitted inputs, states and intervals. Nor is nonnegative net supply on a closed state-space cycle alone the definition: Willems explicitly says that cyclic condition is inadequate by itself for dissipativity as defined. A proposed connection among components consequently needs more than an appealing energy-flow drawing or a single observed balance. [W014, definition 2 and following discussion]

Available storage is a limit on extraction

Willems defines available storage by asking how much can be extracted from a given initial state over all allowed future motions and horizons. It is a supremum of the negative accumulated supply. Including the zero-duration choice makes this quantity nonnegative. Theorem 1 states, in the developed setting, that finite available storage for every state is equivalent to dissipativity, that available storage itself is a

possible storage function, and that it is no larger than any nonnegative storage function satisfying the inequality. [W014, definition 3 and theorem 1, pp. 327–328]

The proof's idea is useful to understand. From one state, first follow an allowed path to a second state, then extract what is available there. This combined procedure cannot exceed the best extraction possible from the first state. That comparison gives the dissipation inequality for available storage. Conversely, an existing nonnegative storage function bounds what can be extracted along any such path. The two directions connect the input/output extraction question to the storage bound. [W014, theorem 1 proof]

The article expressly warns that available storage need not equal actual stored energy. Not every internally stored quantity must be accessible through the external terminals under the admitted operations. Treating extraction capacity as the physical total would therefore overstate what the formal result supplies. The distinction is relevant whenever a selected interface is said to reveal all resources inside an arrangement. Likewise, an energy inequality alone does not establish that a chosen gait or a desired asymptotic equilibrium will occur.

Willems 1997: a controller can be another physical system

In *On Interconnections, Control, and Feedback*, Willems questions the universal use of a sensor-to-computation-to-actuator signal-flow picture. His door and door-closing mechanism share a physical interaction: force and angle are related through both mechanical systems. Assigning force as cause on one side and effect on the other may be convenient for a conventional controller description, but it is not the only way to describe their interconnection. The paper retains intelligent feedback as useful while proposing a more general framework. [UX-W017, §III, pp. 328–329]

In the behavioral approach, a system's behavior is the set of trajectories compatible with its laws over a time axis and signal space. When two systems share signals, their interconnected behavior consists of trajectories satisfying both sets of laws: an intersection of behaviors. Interconnecting only some terminals can be represented by leaving the others free in the appropriate component. Internal connection variables can later be suppressed through the distinction between manifest and latent variables. The relationship supplied by the interconnection is part of the model, rather than something inferred from an unexplained arrow between boxes. [UX-W017, §IV, pp. 329–330]

This viewpoint lets a mechanical attachment count as a controller without requiring it first to sense, represent and compute in the manner of a digital controller. It does not make every physical attachment adequate. The admissible component behavior and the joint constraint must actually yield the required motions. An intersection can remove possible trajectories while also removing behavior needed for the intended task.

An unrestricted mathematical controller can make the problem trivial

The 1997 paper itself demonstrates why controller admissibility matters. If arbitrary additional restrictions are permitted in a linear behavioral setting, an interconnection can trivially force a small stable behavior. That formal construction can look stronger than familiar control results because the permitted controller class is so broad. Willems calls attention to the objection and turns to regular interconnection and implementation. A mathematical subset selected by imposing arbitrary laws is not automatically a physical controller that can be connected through the allowed means. [UX-W017, §§VI–VIII, especially p. 332]

Controller admissibility determines which variables can be constrained and which interconnections can be physically realized. The system assumptions and permitted means of connection are therefore part of Willems’s control problem.

Van der Schaft: composition preserves an energetic structure

Arjan van der Schaft's December 2024 preprint on nonlinear port-Hamiltonian systems describes physical modeling through energy storage, dissipation and power-conserving interconnection. In its selected composition account, components connected through an appropriate power-conserving relation yield another port-Hamiltonian system. The combined stored energy is the sum of component Hamiltonians, with the component dissipation relations retained. The relation at the interface is what permits this accounting; arbitrary coupling is not assumed to have the same property. [GL-W069, §1.1]

Combining the power-preservation relation, storage definition and dissipation condition gives its key inequality: the rate of increase of stored energy is no greater than supplied port power. If the Hamiltonian is bounded below, the system is passive with respect to that supply rate after a suitable storage normalization. “Passive” here is an energetic input/output property. It does not mean that no controller is present or that no external energy arrives. [GL-W069, §1.2, eq. 9]

The preprint also distinguishes zero-input analysis from steady operation with nonzero inflows and outflows, such as a power network. The relevant equilibrium and storage comparison must change accordingly; shifted passivity addresses that different question under further stated conditions. Its treatment of power ports and energy ports also retains different variable roles in later sections. [GL-W069, §§1.2–3]

Van der Schaft and Jeltsema trace port-based modeling to the bond graphs pioneered by Henry Paynter. Each bond carries two quantities whose product is power. Voltage and current serve this purpose in an electrical connection, and force and velocity in a mechanical connection. Using power at the connection gives the model a way to represent exchanges between components from different physical domains. [GL-W070, §1.1]

Willems gives a circuit example with two resistors, a capacitor and an inductor. He writes the laws of the components and connectors, then identifies the variables at the terminals that are joined. To describe what can be observed from outside, he retains the voltage difference across the external terminals and the current entering the circuit. The internal quantities still have to satisfy the component and connection laws. [UX-W019, fig. 8 and accompanying text]

Wu: separating subnetworks retains an interconnection solve

Felix Wu's 1975 *Solution of Large-Scale Networks by Tearing* develops a network solution related to Kron's diakoptics. The network is separated into subnetworks, those are solved locally, and an interconnection-level calculation restores the constraints across the cut. The method retains the Kirchhoff relations rather than treating the cut parts as independent complete answers. The selected explanation distinguishes subnetwork computation from the work of joining their results. [SG-W023, selected opening and algorithm discussion]

Wu describes a possible parallel implementation in which satellite computers perform subnetwork work while a central computer performs an interconnection step. The algorithm can also run serially. Parallel execution and total arithmetic work are thus different comparisons. The selected computational discussion compares node-voltage solutions and counts multiplications, accounting for sparse matrices so operations on zero entries are omitted. Matrix ordering and fill-in during elimination materially affect the count. [SG-W023, algorithm remarks and §IV, printed pp. 14–16]

Its parameterized example reports one range in which tearing takes fewer multiplications and a smaller range, $3 \le p \le 9$, in which it takes more. Another stated class of networks also gives no computational saving for the chosen tearing arrangement. The explanation for a possible advantage concerns fill-in relative to the comparator, not decomposition as an intrinsically better operation. The positive and adverse cases

nevertheless preserve the source's explicit conclusion that neither approach is absolutely superior on this counted task. [SG-W023, §IV, printed pp. 16–19]

The local solves can become easier while an interconnection solve and ordering effects remain. Wu supplies a concrete comparison in one technical problem, including an unfavorable result.

S03 Ashby, regulation and models: distinguishable possibilities and constructed capacities

Behavior, variety and regulation

Ashby's *An Introduction to Cybernetics* develops a language for possible behavior, then variety and communication, then regulation and the construction of regulators. The book first asks what system is being described and what distinctions and couplings its description preserves. Its later accounts of regulation rely on those choices. Its final chapters ask how an arrangement can undertake repeated regulation and where its capacities come from. [W001, parts I–III]

A system is a selected state description

Ashby's pendulum example begins with a failed description. Record the angular position at fixed intervals and it may appear that a position has two different successors. Release the pendulum from ten degrees and its next position differs from the next position when it passes through ten degrees while already moving. The problem is not resolved by insisting that the physical object is the same. Angular position alone omits a distinction that matters to its subsequent behavior. Adding angular velocity supplies the relevant state description for the model under consideration; adding an unrelated variable such as the bob's mass to the original description does not solve that particular omission. [W001, §3/11]

In this usage, a system is a chosen list of variables, with states formed from their possible values. The material object supplies many possible variables; the scientific account selects those relevant to a given interest. This does not license arbitrary simplification. The selected description must support the transition properties claimed for it. If an observed successor depends on an omitted variable, a deterministic account on the reduced state set can fail. Ashby's construction makes finding an adequate description part of investigation, rather than something supplied by pointing at an object and naming it "the system." [W001, §3/11]

His transformations distinguish a state from the rule under which it changes. A closed transformation sends states in the selected set to states within that set. A single-valued transformation assigns one successor to each state. These are formal properties of the description, not claims that the object is thermodynamically isolated, physically self-sufficient or free of environmental dependence. An input can select among transformations; changing an input then changes the transition applied to the state. Omitting relevant inputs can make the reduced transition uncertain. [W001, chs. 2–4]

Ashby abstracts from material substrate so that the same pattern of transitions can describe differently constituted mechanisms. The behavioral description selects the properties relevant to a particular question. [W001, §§1/2–1/5, 3/11]

Coupling supplies a relation that a parts list lacks

Ashby considers coupling two machines while retaining their individual transition laws. To do so, each must affect the other through the inputs admitted by those laws. The coupling specifies how one machine's state determines an input of the other. In his elementary construction, both update on a common time scale; the next joint state is determined from the two present states and the selected coupling. The whole can then be described on pairs of component states. A different coupling can produce different joint behavior even when the component transformations are unchanged. [W001, §§4/6–4/10]

The restriction to input and output is consequential. Forcing two objects together so that their mechanisms are damaged would also connect them in ordinary language, but would not preserve the

component laws used in this construction. Likewise, silently changing update timing would change the model being analyzed. Thus “the parts interact” is too incomplete to explain the whole. The manner and conditions of interaction belong to the explanation. The usefulness of an arrangement lies in those actual relations, rather than in a count or inventory of its parts. [W001, §§4/6–4/7]

The book also warns against reading a whole dynamics from a favorable loop label. In its two-variable example, each next value is half the other's present value. Each variable has a positive immediate effect on the other, yet repeated updates converge toward zero. Calling the relation positive feedback is insufficient to prove instability. Conversely, an undesirable state can be stable. Ashby uses a continuing fire as an example: persistence and return after disturbance are not automatically achievements wanted by the observer. The state, disturbance and transition must remain specified before stability becomes a useful judgment. [W001, §§5/9–5/11]

Black-box knowledge does not identify a unique interior

For Ashby, the experimenter and the black box are coupled: the experimenter's actions change inputs, the box produces observable results, and those results influence subsequent actions. A protocol can support a canonical behavioral description under the assumed determinacy and accessible tests. It does not follow that every relevant transition can be repeatedly tested. A state may become inaccessible after the process passes through it. His examples include an irreversible explosion and an organism that no longer has its original untrained condition. Replacing the specimen introduces an additional assumption about comparability; it is not literally returning the same object to its former state. [W001, §§6/1–6/6]

Even sufficiently described behavior need not determine a unique interior. Ashby's relay networks can realize the same required conducting behavior with different arrangements of contacts. The same object can also have different connection diagrams when studied mechanically and electrically. An insulator can connect two parts mechanically while separating them electrically. Which relation is being investigated therefore matters to the diagram and to any inference about independence. A successful interface description is knowledge of selected behavior, not a transparent picture of every mechanism responsible for it. [W001, §6/7]

Partial knowledge can be complete for a selected operation

Ashby's parity example gives a precise sense to useful incompleteness. One need not know the exact product of every pair of integers to know whether their product is odd or even. Odd times odd is odd; the other parity combinations give an even product. This little account omits almost everything about numerical magnitude, yet gives a determinate answer to the selected parity question. By contrast, a list of a few exact products is incomplete even within multiplication when it leaves further requested cases unanswered. Partiality and adequacy must therefore be judged with respect to what the account is intended to do. [W001, §6/13]

For a dynamic system, grouping detailed states gives a useful simpler machine only if the grouped transitions remain consistent. Suppose two detailed states receive the same reduced label but move to different reduced labels under the same input. The reduced current label then does not determine the next one. Ashby explicitly contrasts suitable and unsuitable groupings. His homomorphic simplification preserves the operative structure required for the reduced behavior; arbitrary lumping need not. The fewer distinctions are justified by what remains predictable or usable, rather than by their small number alone. [W001, §§6/12–6/14]

This supplies his response to practical largeness. A system is very large relative to an observer when that observer cannot obtain all observations, carry out all calculations or exercise all control with the available resources. More variables or finer discrimination can make the same material object larger in this sense.

The mathematical arguments do not fail because their sets are large, provided their assumptions still hold; their application to a material case remains a separate question. Deliberately chosen aspects can be enough for a practical purpose. Their sufficiency must be established for that purpose, not asserted for every omitted detail. [W001, §§4/17–4/18, 6/14]

What variety counts

In *An Introduction to Cybernetics*, W. Ross Ashby gives variety a specified meaning. Consider a sequence containing twelve occurrences of the letters a, b, and c. If order is ignored and only the distinct letters are counted, its variety is three. Counting occurrences, distinguishing types, and retaining an ordered sequence are different descriptions. A numerical statement about variety therefore needs the set of possibilities and the distinctions that define its elements. It is not a count whose object can be left unspecified. [W001, §7/6]

Ashby's semaphore example makes the observer consequential. Two arms with eight positions each can form 64 combinations when their positions are independently set and the arms are individually distinguishable. At a distance, the observer may be unable to distinguish which arm occupies which position. Some combinations then become indistinguishable, leaving 36 distinguishable positions in his example. The physical apparatus has not acquired a different number of arm settings merely because the observer changed. The set of distinctions available in that observation has changed. [W001, §7/6]

This is a useful prerequisite for reading an organizational account that speaks of what a manager or operational unit can distinguish. It directs attention to the relevant possibilities and the means of telling them apart. It does not establish that someone understands a situation merely because a large quantity of data is available.

Counts and bits are different numerical forms

Ashby uses either the number of distinct elements or its base-two logarithm, with the context specifying which. In the logarithmic form, the unit is the bit. Eight possibilities have a logarithmic variety of three bits; two have one bit. When the distinctions combine independently, the number of combinations multiplies, while these logarithmic quantities add. This explains why a report of “no variety” can mean one remaining possibility: the count is one and its base-two logarithm is zero. Confusing the count with its logarithm changes the meaning of a calculation. [W001, §7/7]

The set need not be a collection of simultaneous states

A machine is in one state at a time, so why speak of a set of its states? Ashby gives several reasons. A description may concern a population of machines or replicates, each currently in a different state. It may concern one machine's states at different times. Or it may concern the possibilities an observer considers because the machine's actual state is unknown. Those are different grounds for forming a set. A calculation should preserve which ground applies. [W001, §7/23]

This distinction prevents a common misreading. A decrease in the possibilities consistent with an observer's description does not necessarily describe the physical removal of machines or a reduction in the work being performed. Likewise, a collection of observations over time should not be described as though all the recorded states simultaneously existed within one object. Ashby explicitly develops the actual in relation to the possible; that analytical choice is part of the explanation.

How an arrangement can remove distinctions

Ashby examines a single-valued transformation in which A changes to B, B to C, and C to C. A set initially containing all three possibilities has only two distinct images after one application, B and C. After another application, only C remains. The reason is specific: more than one distinct input can have the same output, while each input has only one output under the transformation. A one-to-one mapping on the inputs actually present would retain their number of distinctions. [W001, §7/22]

This result concerns the stated transformation and set. Losing a distinction need not be beneficial. Ashby's coffee-message illustration shows a case in which convergence toward a common temperature destroys the intended distinction among messages. A person may want an arrangement to remove an unwanted effect or to preserve a useful difference; identifying which is being considered requires the task as well as the count. [W001, §§7/22–7/24]

The response must be appropriate, as well as available

Ashby's introductory regulation game separates disturbances, responses, and outcomes. One player chooses a row of a table; the responding player, knowing that choice, chooses a column. The selected cell determines the outcome. A desired outcome can always be obtained only when an appropriate entry is available for every relevant row and the responder selects it. Merely having many columns does not establish that they contain what is needed. The arrangement of the table matters. [W001, §§11/3–11/4]

For a particular family of tables, Ashby derives a bound. He assumes that a column contains no repeated outcome, and the responding strategy selects one entry from each row. If there are r rows and c available columns, the selected outcomes cannot have fewer than r/c distinct values. To keep selecting the same outcome, a strategy would need a different column each time; the available columns limit how far it can do so. The bound is a necessary limit under those conditions, rather than a sufficient recipe for constructing every desired regulator. Ashby states the corresponding logarithmic form under the same conditions. [W001, §§11/5–11/7]

This is the setting of his concise statement that response variety can reduce outcome variety. In an organizational application, the set of relevant disturbances, available responses, distinctions, and acceptable outcomes must still be identified. A slogan about matching complexity does not supply those identifications or prove their adequacy. A local arrangement can change what distinctions reach another role, while the capacities and purposes of that arrangement remain substantive questions.

Constraint compares an actual set with alternatives

Ashby's constraint is a relation between sets: fewer possibilities occur under one condition than under an explicit comparison condition. His traffic signal has three binary lamps, so independent settings would permit eight combinations. The signal uses four. The excluded combinations establish a constraint in that set of simultaneous lamp settings. Each lamp can still be lit or unlit; the restriction concerns which combinations occur together. The presence of constraint must therefore be attached to the joint possibilities under discussion. It cannot be inferred merely from knowing the individual component ranges. [W001, §§7/8–7/11]

He uses an assembled chair to explain why a thing can act as a unit. Four disconnected rigid legs would each have six degrees of freedom of movement; once rigidly joined as a chair, the assembly has six. Knowing the locations of some parts now restricts those of the others. This is his account of coherence as a constraint on their relative possibilities. [W001, §7/16]

The book connects constraint to prediction and compact description. An aircraft cannot occupy an arbitrary new position at each instant; continuity and characteristics of its motion restrict possibilities.

Prediction can exploit such restrictions. Similarly, a long protocol showing consistent machine behavior can be represented more compactly by its transformation and actual inputs. The transformation does not supply the inputs, and a compressed description is not by itself a cheaper physical realization. Ashby is explaining why regularity makes a useful account possible. [W001, §§7/18–7/20]

His learning examples make an environmental condition explicit. An association can be learned only if the observed sequence has relevant regularity; an unchanged maze offers a learnable pattern that an arbitrarily changing maze would not. These are arguments about what supplies the regularity being exploited. [W001, §7/21]

A smaller channel can require a longer process

Ashby's account of transmission distinguishes what can pass in one step from what a sequence can convey. A channel with two distinguishable values per step can form many distinguishable sequences. Ten binary positions permit up to 1,024 combinations when independent. The restricted alphabet does not impose the same bound on a sequence of ten positions as on one position. A receiver capable of retaining the relevant history can therefore distinguish more possibilities over time than the instantaneous channel alphabet suggests. [W001, §§8/13–8/14]

This is a conditional trade between capacity and duration. It requires usable distinctions, appropriate coding and a receiver/process able to accumulate them. A blocked channel gains no information merely by waiting longer, and distinctions lost before transmission do not reappear by extending the record. Ashby's analysis of delay also retains the intervening stages: if a source affects a receiver only through a chain of parts, some effects reach it later. Redrawing the chain as a single source does not make the delayed internal path instantaneous. The abstraction can change which boundary is described without changing the events. [W001, §§8/14–8/15]

The material environment can participate in such a channel. In his multiplication example, someone unable to complete the task mentally uses pencil and paper. Activity passes from the brain through arm and pencil to marks, then through light and retina back to the brain. This is communication through an external medium between parts of a continuing process. What counts as a channel depends on the behavioral relationship being studied, rather than on whether an anatomical nerve joins the two endpoints. [W001, §9/15]

Redundancy is relative to the message and its realization

Ashby returns to the traffic lights to ask whether the four required signals could be encoded by two binary lamps rather than three. Four combinations suffice to distinguish stop, prepare to go, go and prepare to stop. Under his stated condition that lamps are expensive, this reduces the lamp cost to two-thirds while preserving those signaling distinctions. [W001, §9/16]

The very same lights supply a different redundancy if examined as a clock-driven sequence. When the four signals always repeat in the same order, the initial phase determines later signals in the sequence. The relevant alternatives are now possible sequences, not simultaneous lamp combinations. The book uses the difference to reject an unqualified question about whether “the lights” are redundant: the same physical object provides different sets with different answers. [W001, §9/16]

Regulation preserves a desired range, not constancy alone

Ashby's regulation account specifies essential variables and acceptable values. A water bath is not good merely because it stays constant: it must remain in the desired temperature range despite the disturbances considered. In his communication description, a successful barrier prevents differences

among disturbances from producing unacceptable differences in those variables. This does not mean that every part of the system must be deprived of information. The active regulator may need distinctions that its successful action prevents from reaching the protected variables. [W001, §§10/6–10/7]

The tortoise's shell and the fencer illustrate two different ways of obtaining protection. A passive barrier can reduce the effect of blows on protected tissues. Skilled counteraction instead obtains information and varies action to meet the threat. Ashby groups both within the broader regulatory question; he does not define regulation to require a separate active controller in every case. The shell's success remains relative to what it resists, while the fencer's depends on available perception and action. [W001, §10/7]

The response table discussed above represents one family of regulatory limits. The physical or organizational case must supply the table and its assumptions. Ashby explicitly allows the given environment and mechanism to block some disturbance variety before an additional regulatory response acts. In some cases that baseline blocking is enough; in others it leaves work for a responsive regulator. Thus an unconditional equation equating all environmental variety with a required controller size would lose an important part of his own account. Counting relevant disturbances, the preexisting transformation, response distinctions and acceptable outcomes all matter. [W001, §§11/9–11/11]

The book also states where its normative task stops. At the opening of chapter 12, essential variables and their acceptable states are assumed to be given by outside considerations. The problem developed within the book is how to achieve them despite disturbances. The formal machinery does not derive which human purposes should be adopted or who should bear the burden of protecting them. [W001, §12/1]

Sensing, action and timing are separate restrictions

Ashby distinguishes restrictions on receiving information from restrictions on acting. A driver who cannot see through the windscreen and a rudder that is stuck are different failures of the regulatory path. More information does not supply missing actuation; more possible actions do not identify which one is needed. He also distinguishes knowledge that a disturbance has occurred from knowledge of the system's present state: a signalman can know there is fog while being unable to determine where trains are. These are different required distinctions. [W001, §12/2]

Timing introduces a further condition. The elementary response game gives the regulator time to respond before the outcome is determined. Some regulations can use a warning; others learn of a disturbance through its effects. The thermostat in his water-bath example does not see the cold flask approaching. It receives information after the temperature begins to fall. Calling both processes regulation does not remove the difference in the route or time available for acting. [W001, §§12/3–12/4]

His argument against a perfectly error-controlled regulator is made under a particular static response framing: the response is a determinate function of the present outcome. If that outcome supplies all the disturbance information, holding it exactly constant removes the distinctions available through that channel. Allowing a small deviation can make preventing a much larger deviation possible. A dynamic controller with memory retains additional state that this immediate response table does not represent. [W001, §§12/5–12/7]

An adaptive arrangement must find and retain an acceptable state

Ashby's hunt-and-stick construction changes what is selected. A regulating component is arranged so that it keeps changing under a “bad” input and can remain at equilibrium under a “good” one. It is coupled to the other component so that an acceptable condition supplies the good input. Any joint equilibrium then has the required property. The regulator vetoes unacceptable equilibria rather than explicitly specifying every detail of the acceptable configuration in advance. [W001, §12/14]

That statement about equilibria is not by itself a guarantee that every trajectory reaches one. The text recognizes trapping in an undesirable stable region and discusses richer transition possibilities. Accessibility of an acceptable state, the absence of other closed classes and the actual coupling remain necessary questions. Enlarging a list of regulatory states does not help if none of the available transitions can obtain the required condition. [W001, §§12/11–12/14]

The homeostat makes the mechanism concrete. Resistor settings, relays and a source of successive values allow changes while the system remains unacceptable; acceptable behavior blocks further changes. Viewed without access to the supplied table of values, the process can appear Markovian. Viewed with those determinants included, a different state description becomes available. Neither description removes the physical switching and supplied values. The explanation locates the variation being selected and the mechanism that retains a satisfactory condition. [W001, §12/15]

His telephone selector gives a simpler determinate example. It tests lines in order and stops at a disengaged line. The sequence supplies the selection without requiring a precomputed answer identifying the first free line. Yet it must encounter an available line if it is to make a successful connection; the construction does not make a free line exist when all are busy. Repeated local testing can replace one kind of prior specification while retaining a criterion, apparatus, time and feasible alternatives. [W001, §12/17]

A reusable rule and a complete response sequence differ

Chapter 13 asks what happens when similar classes of disturbance recur over a long period. If three disturbances can appear independently on 400 occasions, there are 3^400 possible disturbance sequences. A matching response sequence can have correspondingly many alternatives. Yet a machine with the right input-response rule can generate the appropriate response on each occasion without someone specifying the whole sequence in advance. The rule stays compact because the recurring relationship is reused; the actual disturbance supplies which response is needed each time. [W001, §§13/9–13/10]

This is Ashby's own transition from a regulator's behavior to its construction. The responsible entity can undertake the recurring responses directly or build an arrangement that undertakes them. The small rule does not contain all future disturbance events. It supplies a response relation that operates when events occur. His later mechanics example explicitly retains maintenance. [W001, §§13/10, 14/5]

Design becomes selection among possible machines. In the book's deliberately broad communication account, components, couplings or input specifications distinguish one resulting machine from alternatives. Selection can occur in stages and can be supplemented by other selectors or by processes unrelated to the main system. Ashby distinguishes an abstract amount of selection from how long a search takes. Successive group divisions can find an item much faster than one-at-a-time examination if the required group-membership tests are actually available. His huge-universe example expressly assumes those tests possible; the counting does not construct a detector capable of making them. [W001, §§13/11–13/20]

Amplification retains the source of supplementation

Ashby's power-amplifier discussion states what is being supplemented. The crane driver's small movement controls a much larger flow of energy, supplied elsewhere. From the designer's account, the output does not exceed total supplied energy; from the user's immediate task, the ability to control the larger output is consequential. Both descriptions can be correct because they account for different boundaries. His analogy for regulatory amplification depends on keeping that distinction visible. [W001, §§14/1–14/3]

One example makes the borrowed capacity explicit: someone unable to correct hundreds of mains-voltage disturbances per second can select a suitable stabilizer from a catalogue. Choosing the product uses the manufacturer's design capacities and the apparatus's continuing response. The person's choice and the ensuing regulation have different alternative sets and time scales. The example demonstrates the logical possibility of obtaining much more regulation than the chooser can perform directly, while openly acknowledging the manufacturer's contribution. [W001, §14/5]

The water-bath example compares an assumed quantity of selection involved in making a thermostat with repeated corrections over a year. Ashby presents a large amplification on that informational account and then explicitly says there need not be amplification in every case. The practical comparison includes the cost, in some sense, of making the machine against doing the job personally. The units used for selection do not automatically measure financial expense, energy, skill or maintenance. Those further quantities require the particular comparison. [W001, §14/5]

His twenty mechanics and two thousand rooms are especially relevant to the work retained by automation. The alternative to manually regulating every room is to make and maintain machines that regulate them. Supplementation comes from other contributors to design and manufacture and from the environment, which supplies current temperature and humidity directly to conditioners rather than through the mechanics. The mechanics become regulators of the regulatory machinery. Ashby also says these relationships need not form one simple hierarchy. [W001, §14/5]

Conant and Ashby: what the “good regulator” result optimizes

Roger Conant and Ashby's 1970 article asks a different formal question. Given regulatory events R, events S in the reguland, and an outcome determined by the two, it defines an optimal regulator for its theorem by minimal outcome entropy. The distribution of S and the joint outcome mapping are given; conditional probabilities of regulatory events given S are varied. Its simplest optimal regulator associates each relevant S event with one R event, a mapping from S to R. “Model” has that mapped-event sense here. It does not initially mean a predictive computer simulation stored inside a controller. [W050, PDF pp. 8–9]

The proof's operative move is to eliminate unnecessary conditional randomization. Where two actions available for one S event would generate different outcomes, shifting probability can lower outcome entropy, contradicting optimality in the conditions used. Where different actions give the same optimal outcome, one can choose a single action without changing optimality. Repeating this selection yields a deterministic conditional mapping. The paper's simplifying uniqueness assumption on the optimal outcome distribution is explicit. The argument establishes a mapping within that optimization. [W050, PDF pp. 8–9]

The authors themselves state that equally optimal regulators may retain unnecessary complexity. Thus their theorem does not literally establish the strongest reading of the title, that every successful regulator must have the simplest model relation. Moreover, minimum entropy is not a complete valuation of outcomes: one constant bad outcome can have zero entropy. [W050, PDF pp. 2–4, 8–9]

Virgo and colleagues: a model can be an attributed interpretation

Nathaniel Virgo, Martin Biehl, Manuel Baltieri and Matteo Capucci's 2025 paper explicitly challenges the inference from Conant and Ashby to every good regulator having a nontrivial internal model. Their new theorem uses deterministic discrete-time coupled machines, a chosen agent/environment boundary and a good set of joint states. A regulating set is nonempty, lies within the good set and is forward-closed: starting in it, the coupled evolution stays in it. Existence of that set establishes regulation from appropriate joint initial states, not recovery from every arbitrary disturbance. [RGC-W036, §§2–2.1]

They then associate an agent state with the set of environment states paired with it in the regulating set. This can be interpreted as a possibilistic belief: it identifies what remains possible, without assigning Bayesian probabilities. Their consistency condition allows forgetting, so the newly attributed possibilities can be broader than the tightest update consistent with the action and observation. The lemma identifies this consistency with forward closure; the theorem adds nonemptiness and inclusion within the chosen good set. The model interpretation is constructed from the dynamics and the observer's choices. It is not an extra causal component that must already have been designed inside the agent. [RGC-W036, definitions 3.1–3.3, lemma 3.2, theorem 3.4]

Their doorstop example makes the qualification concrete. A singleton agent state and singleton action and sensor spaces admit a constant interpretation even if the environment and selected good set are complicated. The complicated description can belong to the observer's account, while the doorstop has no corresponding rich internal state. The authors therefore reject the inference that effective regulator design must always begin by building an explicit model. They distinguish such a trivial interpretation from cases with rich ongoing interaction, and leave more precise measures of that distinction for further work. Their suggested relation to the free-energy principle is qualitative and explicitly unresolved. [RGC-W036, §3.2 and discussion]

The deterministic model maps an environment state and action to the next environment state and sensor value. Its proposed stochastic extension is left for further work. [RGC-W036, definitions 2.2–2.3, §2, fig. 1]

Powers: retain the controlled perception and changing reference

William Powers's 2009 primary summary of perceptual control theory gives another account of what repeated feedback can accomplish. The driver compares a perceived relation between car and road with a reference for how that relation should be. Steering changes as the discrepancy changes. A crosswind can require varying steering even while the car's visible path remains nearly constant. The repeated correction can therefore be active precisely where an observer sees little change in the controlled result. This is the mechanism Powers uses to challenge a description of behavior as a fixed sequence of responses computed in advance. [SG-W027, steps 6–9]

When the driver takes an exit, the reference changes. The lower process need not oppose every departure from the earlier lane position; it acts to bring perception toward the new reference. Powers's hierarchy gives higher systems the role of varying reference perceptions for lower systems, which retain the work of adjusting actions against immediate disturbances. His explanation separates specifying a desired result from specifying every muscle movement. It is especially relevant to how an arrangement can leave detailed action to an ongoing lower process while retaining a higher-level goal. [SG-W027, steps 8–9, 11]

Powers's reference concerns perception. Matching perception to a reference does not by itself establish correspondence to every external fact.

S05 Robustness and control: disturbances, criteria and mechanisms

The scope of a favorable response

The accounts in this section ask different questions about maintaining or recovering performance. Some characterize exact adaptation in a biochemical model, some optimize protection against declared uncertainty, and others reduce controller updates or describe organized physiological responses.

Yi and colleagues: integral feedback in a chemotaxis model

Tau-Mu Yi, Yun Huang, Melvin Simon and John Doyle reconstruct the robust perfect adaptation of the Barkai–Leibler bacterial chemotaxis model through integral feedback. The pertinent variable is approximately total receptor methylation. Its rate of change can be expressed as a methylation contribution minus a term proportional to receptor-complex activity. At a stable steady state that rate is zero, so the activity returns to a value set by the ratio of those rates. If the relevant rates remain independent of ligand level, a changed ligand input does not change that steady-state activity. This is the selected structural reason for adaptation in their model. [W012, pp. 4650–4651]

Integral feedback is more specific than feedback in general. The accumulated error supplies a state that continues to influence the regulated process. A constant present error and a controller whose state retains accumulated history are different situations. The authors' description explicitly requires stability of the closed loop; an algebraic steady-state relation alone does not establish that the dynamics actually approach it. [W012, integral-control derivation and fig. 2]

They also identify four biochemical assumptions needed for the reconstruction: the demethylating action concerns active receptors; specified kinetic parameters and rate ratios are independent of the relevant methylation/occupancy distinctions; activity of the unmethylated form is negligible; and bound CheR does not depend on ligand level. Relaxing these assumptions can spoil exact adaptation. The authors inspect biological evidence for their relevance rather than hiding them behind the word robust. [W012, pp. 4650–4651, assumptions and discussion]

Yi and colleagues reconstruct the model introduced by Barkai and Leibler in *Robustness in simple biochemical networks*. [W033, through W012]

Cannon: constancy can require ongoing cooperation

Walter Cannon's 1929 *Organization for Physiological Homeostasis* defines homeostasis for an open living system whose automatic adjustments keep disturbances of internal conditions within relatively narrow limits. He deliberately distinguishes this from using equilibrium in the simple closed physicochemical sense. His choice of “homeo” also permits variation rather than rigid identity. The selected explanation concerns coordinated physiological reactions, including organs and material supplies, rather than an immobile substance that stays correct without activity. [W062, pp. 399–404]

That distinction makes the apparent absence of change informative only when its means are understood. The selected postulates describe agencies acting or ready to act, stronger resistance to a disturbance, and simultaneous or successive cooperation among factors. Cannon calls these propositions tentative and conditional, acknowledges pathological failure and substantial gaps in knowledge, and distinguishes their status from a strict equilibrium principle. His expectation that further governing agencies will be found is a research position, not proof that every instance has already been mechanistically explained. [W062, pp. 424–426]

Sterling and Ramsay/Woods: anticipation does not settle the terminology

Peter Sterling's allostasis account challenges a version of homeostasis centered on preserving a constant internal milieu through feedback error correction. He argues that regulation instead adjusts conditions to support survival and reproduction, using prediction and coordinated effectors. In the selected description, prior knowledge and current sensing help anticipate required resources; experience updates what can serve as prior knowledge later. Anticipation therefore retains acquired information, sensory input and action. [GL-W028, selected abstract and predictive-coordination discussion]

Sterling's broader article connects that argument to efficiency, behavior and proposed medical implications. These are his model and proposals. A response need not wait for a large harmful departure if relevant demands can be anticipated and effectors prepared. The adequacy of those predictions remains consequential.

Douglas Ramsay and Stephen Woods dispute the idea that anticipation alone requires abandoning homeostasis. Their selected 2014 discussion argues that contemporary homeostatic models can already accommodate anticipatory responses and errors of prediction. They also criticize using homeostasis for virtually any return of any variable to a former value, because that stretching can obscure its physiological scope. The disagreement is about concepts and mechanisms, rather than a simple contest between a modern predictive model and a uniformly static predecessor. [GL-W029, introduction and Scope of Homeostasis]

Their suggested distinction turns attention to underlying effector loops. They discuss persistent overcorrection, continuing effector activity after an initiating disturbance is gone, and competition among effectors as candidates for an allostatic explanation. They explicitly call their suggestions speculative and open to revision. Observing a changed temperature or other regulated value alone is therefore insufficient to decide which account applies. [GL-W029, Scope of Homeostasis and conclusions]

Carlson and Doyle: optimization can expose an unanticipated fragility

Carlson and Doyle's highly optimized tolerance account uses a percolation template to distinguish structured design from random configurations near criticality. Occupied clusters are separated by vacant barriers; a spark damages the cluster it reaches. Optimizing expected yield requires a distribution of spark locations and a set of constraints. In the selected example, more protective cuts are placed where hits are more probable, producing smaller clusters there and larger clusters in less frequently hit regions. A high-performing arrangement is specialized to the risks it was designed to face. [W026, preprint pp. 1–3]

This specialization explains a vulnerability as well as a benefit. Change the hit distribution or place a defect in a barrier and damage can spread differently. The source explicitly considers a tree occupying a location that should be vacant in a separating barrier: a small structural defect lets fire pass into neighboring regions. Adding protection against such flaws or multiple sparks can be possible at a cost in yield. The favorable performance and the sensitivity to what was omitted are consequences of the same specified design comparison. [W026, p. 3]

The authors emphatically say the simple model is not a realistic representation of a particular system and is only remotely related to forest management. Their broader engineering/biological comparisons are their interpretive extension. The model therefore offers a concrete mechanism and a caution about inference from a power-law distribution. Self-organized criticality and optimized tolerance remain different explanations in the source's comparison. [W026, pp. 3–4]

Bertsimas and Sim: robustness is chosen against a stated uncertainty class

Dimitris Bertsimas and Melvyn Sim's robust discrete-optimization framework introduces a tunable degree of conservatism. Instead of requiring an exact distribution for every uncertain coefficient, it uses nominal values, permitted deviations and a declared uncertainty structure. Their selected mixed-integer formulation treats uncertain constraint coefficients as bounded, symmetric and independent, while uncertain costs have a specified range. Unknown distributional detail consequently does not mean all assumptions about uncertainty have been removed. [W060, §§1–2.1]

Protection parameters control how conservative a solution is relative to the uncertainty being represented. A stronger protection requirement can change the feasible choices or objective value; choosing the budget is part of the modeled decision. The article gives tractability results for specified problem classes and transformations. Its relevance is the explicit separation of information that can be dispensed with from bounds, structure and computation that remain. [W060, selected robust-formulation discussion]

Tabuada: an event trigger still has to be detected

Paulo Tabuada's 2007 control-task paper studies execution of a stabilizing control task when plant state makes another update necessary, instead of at a fixed periodic rate. Between updates, an input is held constant; the difference between the last sampled state and current state grows into a measurement error. A trigger limits that error relative to a state-dependent bound so that an appropriate Lyapunov function continues to decrease. The draft separately accounts for time to sense, compute and update actuators: with nonzero delay, execution must start early enough to keep the required bound. [TA-W013, draft PDF pp. 3–4]

Its positive lower bound on inter-execution times is conditional on input-to-state stability with respect to measurement errors and specified Lipschitz/Lyapunov assumptions, for initial states in the stated compact set and sufficiently small implementation delay. Those conditions make the schedule feasible within the model rather than allowing update times to accumulate arbitrarily closely. The discussion of co-schedulability then retains other processor tasks and scheduling assumptions; fewer control executions are useful only in an arrangement that still meets those obligations. [TA-W013, theorem 3.1 and co-scheduling discussion]

The practical discussion is especially revealing. A trigger based on a difference between current and previous control-law values could require repeatedly computing the very control law whose executions one wants to avoid. The measurement-error version instead allows a simpler comparison that can be implemented in separate hardware. The activity is reduced or relocated on a specified resource account; monitoring has not vanished. [TA-W013, §7.2]

Anta and Tabuada: compute the next occasion from current information

Adolfo Anta and Tabuada's 2008 self-triggered account addresses the continuous-monitoring demand of event triggering. When the current state is measured or estimated for a control update, that information and the model can also be used to determine when the controller should execute again. The next occasion is thus computed at the current one. The selected introduction frames this as a trade between resources allotted to control and achieved performance, including processor and communication demands. [TA-W014, introduction]

Their developed conditions concern state-dependent homogeneous and polynomial systems, with the relevant dynamics, desired performance and current measurement explicitly involved. Continuous trigger

tests may be avoided while predictor/model computation and a correct representation of the operating conditions remain. [TA-W014, introduction and §§2–3]

Self-triggering predictive control and feasibility

Patent CN116627039B presents a self-triggering predictive-control proposal by Wang Mengzhi, Zhao Chengcheng, Yang Zeyu and Che Xin. Its abstract constructs a continuous-time networked model, solves a finite-horizon optimization, constructs a candidate feasible sequence at a later time and uses a cost-function deviation to select an update occasion. This gives the claimed reduction in updates a substantial computational setting: optimization and feasibility reasoning are part of the proposal, rather than work removed by the label self-triggering. [TA-W039, abstract and claims]

The patent record dates the application to 2023 and publication to 2026. Reduced computation is a proposed effect of the control scheme, rather than a measured deployment result.

Shraim: reacting to each fluctuation can add variation

Mustafa Shraim's author report of an educational catapult exercise distinguishes reacting to ordinary variation as though every observation signals a new special cause from failing to investigate a genuine special cause. Its three scenarios run the process without adjustment, ask volunteers to adjust toward a target, and then repeat after collaborative improvements to the operating method. The report says informal repeated adjustment produced the most variation, while standardizing how the catapult was held and launched reduced it. A slip-of-hand observation in the first stage was retained and identified, rather than erased as ordinary noise. [W095, 2019 author report]

The example concerns the response to the type of variation and a changed work method. Its relevance is the bounded observation that more corrective activity can worsen a process when its basis is wrong, while deliberate work on the process can improve it.

Doyle and Csete: constraints that permit flexibility

John Doyle and Marie Csete explicitly connect engineering, biology and neuroscience through architecture. In the selected clothing example, component constraints, outfit requirements and protocols jointly determine which combinations are useful. A wardrobe can permit many interchangeable outfits while excluding an even larger set of arbitrary heaps. The common restrictions enable a kind of flexibility by making components compatible. [UX-W042, Architecture as Constraints That Deconstrain]

They also distinguish minimal function under ideal conditions from robustness to broader conditions. Paper can be made into a garment, but that does not give it the endurance and protection of a layered textile architecture. Their discussion retains hidden technologies, layered relationships and control over different time scales. Even familiar outfit protocols do not automatically guarantee fit or suitability: people still try things on and choose for weather and activity. [UX-W042, Layered Architectures Simplified and following discussion]

Further sources on constraints and adaptation

Nagumo's 1942 account of solution curves remaining within a region is developed in S54. [W044]

S109 Electrical power networks and coordinated operation

Scheduled exchange and physical flow

On 4 November 2006, E.ON Netz took the two circuits of the 380 kV Conneforde–Diele line out of service at 21:38 and 21:39. The outage was arranged to let a ship pass under the crossing of the Ems. The operators' analyses before the outage indicated that their grids could accommodate the change. An agreed schedule specifies exchanges between systems, but actual power flows through the connected network. The UCTE investigation reports that flow on a different line, Landesbergen–Wehrendorf, rose by more than 600 MW after the switching and exceeded 1,200 MW. [T11-W001]

Landesbergen–Wehrendorf linked the E.ON Netz and RWE control areas. As loading rose again after 22:02, E.ON Netz dispatchers looked for a switching action to relieve it. At 22:10:11 they coupled two busbars at Landesbergen, joining parts of the substation they expected would redistribute the flow away from the loaded line. The investigation's subsequent simulation estimated that this action instead raised current on the line by 67 A. At 22:10:13, the distance protection at the Wehrendorf end automatically disconnected it. Figure 3 shows the changes in power flow before the trip. Figure 5 puts the manual switch and the automatic trip on the same timeline. [T11-W001]

The report traces further protective trips as the surviving lines took on the changed flows. By about 22:10:29, the network had separated into western, northeastern and southeastern areas operating apart. Frequency fell in the western area, which had lost imports, and rose in the northeastern area, which had lost exports. The switching intended to relieve one line had been a dispatch decision. The trips and the initial frequency responses unfolded through the equipment and the resulting balance of generation and demand in each area. Times are those reported by UCTE. [T11-W001]

Separate responsibilities in a second regime

The joint United States and Canada report on the 2003 blackout gives a different example of the division of work between a local operator and a regional reliability coordinator. It found that FirstEnergy's failed control computers and alarm system helped leave its operators unaware of deteriorating conditions. MISO, the reliability coordinator, had its own limitations. It lacked correct data for the state estimator, the tool it used to infer the condition of the wider network from the measurements available to it. The task force called for stronger real-time tools and backup capabilities for both control areas and reliability coordinators. [T11-W002]

Energizing a network in stages

顾雪平 and colleagues describe restoration beginning with a source that can start without outside electricity. It energizes a path to equipment that requires auxiliary power to start. Operators can then restore additional paths and admit load in steps. A carefully chosen load can help stabilize an energized island, while cold load pickup can cause a sudden increase in demand and constrain how much can be reconnected at once. In the parallel approach reviewed by the authors, operators restore regions separately and then synchronize them. The authors discuss how available starting sources, voltage and frequency control within each region, and the ability to synchronize across boundaries bear on the choice of regions. [T11-W003]

Reserves before a disturbance

The Balance Management Task Force's 2003 account distinguishes responses by how they are activated and what they accomplish. In England and Wales it describes a primary response delivered within 30 seconds to arrest a sharp fall in frequency, and a secondary response delivered from half a minute to 30 minutes to restore frequency. Its reserve tables separately describe fast reserve from hydro generation, which the operator could instruct manually. The first services respond on the timescale of the frequency disturbance. The fast-reserve entry describes capacity the operator could call on. [T11-W004]

S54 Stability, maneuverability and recovery

Engineers can change both how a system moves and how deviations are corrected. Geometry, actuator response, sensory information, permitted operating regions and arrangements for handling failure enter those choices. The historical accounts below develop particular ways of allowing useful motion while maintaining a task, detecting an approaching limit or preparing a different response when ordinary operation fails.

Associated foundations appear in S03 on essential variables and regulation; S01 and S37 on locomotion and body–controller relations; S25 on coordination and metastability; and S43 on ecological resilience and viability. The accounts here add aircraft design and testing, experimental movement analysis, mathematical confinement and decision procedures, and the organization of response near capacity limits.

Walker: a maneuverable aircraft requires a supporting control arrangement

Laurence Walker's 1997 test-pilot account describes the X-36 as a 28-percent-scale demonstrator of a tailless fighter configuration. Removing conventional tail surfaces promised reductions in weight, drag and cost, alongside a different problem of control. Canards, split ailerons, flaps and thrust vectoring supplied the means of changing motion. The control system offered different gains for landing, ordinary airborne operation and the demonstration of fighter response. The aircraft was divergent in pitch and yaw, so favorable handling depended on the controlled aircraft and its supporting arrangements. [SM-W001, pp. 1–5]

Moving the pilot to the ground removed familiar visual, sound and bodily cues. Walker describes compensating through displays, downlinked audio and the cockpit layout. The head-up display was matched geometrically to the camera image. An air-relative velocity symbol was intended to reveal developing divergence or sideslip early, because the remote pilot could not feel lateral acceleration or normal loading. That information still required calibration: after the first landing revealed an error in the predicted airflow near the sensor, the team used a different velocity reference for subsequent landings until the correction was incorporated. [SM-W001, pp. 3–5, 7–8]

Even handling test cards became an engineering problem. Looking down could permit a substantial attitude change without a bodily sensation warning the pilot. A tray immediately below the main display let him remove completed cards while continuing to watch the aircraft's attitude. The team also shielded the pilot from distracting activity and separated his communication channel from engineers' discussions. These arrangements kept information and attention available for controlling a responsive aircraft. [SM-W001, p. 6]

The small demonstrator used a single flight-control channel, accepting less redundancy to reduce cost. A speed restriction was chosen to reduce the danger of structural failure following control failure, so that an onboard parachute could recover the intact aircraft. The team rehearsed procedures and tested the data links through towing, lifting the aircraft and carrying it beneath a helicopter. Qualification, operating restrictions and a recovery device supported a configuration whose ordinary controlled behavior could otherwise conceal its dependence on the link and control system. [SM-W001, pp. 1–2, 6–7]

On the second flight, the data link weakened and the aircraft entered its programmed lost-link mode. Its nearest steering point lay behind it, initially taking it farther from the ground station before it returned toward a designated orbit. Walker reports bank angles beyond those previously flown and a response more aggressive than the team's simulation had shown. Control was regained and the aircraft landed. Subsequent testing located a temperature-sensitive amplifier problem: satisfactory behavior at the tested

low and high temperatures had concealed a failure between them. The event exposed a gap in the supporting tests while exercising a prepared autonomous recovery function. [SM-W001, pp. 8–9]

Testing also changed the control design. In the second phase, revised control laws and better estimates of aerodynamic behavior improved stability margins and produced Level 1 handling ratings in all axes. The report's rolling tests compare time to change bank angle against programme goals, with and without thrust vectoring. Walker reports strong agility alongside the improved controlled handling. His account relates the reported performance to successive configurations, measurements and corrections. [SM-W001, pp. 9–11, figs. 5–6]

A later X-36 account: reconfiguration after an effector failure

Christian Gelzer's 2014 NASA history describes the completed 1997 programme of thirty-one flights, extending beyond the twenty-two-flight stage reported by Walker. It also reports two separate RESTORE research flights in December 1998. In those flights, neural-network reconfiguration software operated alongside the original control laws. Simulated control-surface failures were introduced in flight, and the software compensated sufficiently for the aircraft to continue being flown safely. The reported intervention changed the allocation of control after an effector became unavailable. [SM-W002, Project Summary; Accomplishments]

Kuznetsov and colleagues: small commands can expose a critical control limit

I. P. Kuznetsov, A. A. Parshin, L. V. Khaletsky and V. Yu. Shitov's 2014 article develops actuator requirements for a prospective maneuverable aircraft with an aerodynamically unstable configuration. Their object is the coupled aircraft, flight-control system and actuator. Dead zones, friction, backlash and other nonlinearities can change how an actuator responds even to small commands. Persistent oscillations in that coupled system can then exceed angle-of-attack or normal-load limits. Selecting a control law requires attention to the physical transfer of its commands into movement. [SM-W007, pp. 2–4]

Their model separates a linear airframe-and-control-law part from an actuator response that varies with input amplitude. The unstable airframe changes the gain problem: the authors identify a lower as well as an upper stability boundary. They compare families of actuator frequency responses with the response required by the remaining loop. An approximate construction estimates the amplitude and frequency of possible self-oscillation. Requirements on aircraft pitch and normal loading are carried backward into permissible actuator-input oscillation, taking the tighter restriction where the criteria differ. [SM-W007, pp. 4–13, figs. 1–3]

The worked comparisons concern small-input behavior. A linear approximation can fit an actuator over one range while missing the effect of a hydraulic dead zone or mechanical play at much smaller inputs. The article compares calculated and experimental frequency responses for the RPD-1B actuator and shows why its small-signal behavior can fail the specified whole-loop requirements. At these small inputs, a command can be too small to produce the assumed proportional response. [SM-W007, pp. 13–17, figs. 4–6]

The proposed improvements alter the component itself. Reducing initial dead zones, changing the early slope of hydraulic response and controlling backlash improve the relevant transfer characteristics. Comparisons with SPM-6B responses and calculations of play in the transmission and feedback-sensor coupling make these interventions concrete. A later design uses a direct-control multichannel electrohydraulic amplifier and a linear electric motor to improve low-input behavior. The authors report actuator development and response measurements; their evidence concerns the component and modeled loop requirements. [SM-W007, pp. 17–23, figs. 7–11]

Scholz and Schöner: preserve the task without fixing every component

John Scholz and Gregor Schöner's 1999 article asks which variables a movement-control system stabilizes. Their elementary geometric example uses an arm with three angular coordinates and a hand position described by two coordinates. Several combinations of angles can place the hand at the same location. Movement among those combinations changes the parts while preserving the chosen result. They call the corresponding family of configurations an uncontrolled manifold, using control operationally to mean selective stabilization relative to that task variable. [SM-W004, pp. 289–292, fig. 1]

Their procedure turns that idea into a test. Investigators choose a candidate task variable and a geometric mapping from body configuration to its value. A local linear approximation separates configuration changes that leave the variable unchanged from changes that alter it. Deviations across repeated trials are projected into those two sets of directions. The comparison adjusts for their different numbers of dimensions. Greater variability in directions preserving the task variable than in directions disturbing it supports the hypothesis of preferential stabilization within this model. [SM-W004, pp. 291, 294–295]

The experiment analyzed nine participants rising from a chair under normal conditions, with stiff ski boots restricting ankle movement, and with a narrow support beneath the feet. A sagittal-plane model connected segment angles with center-of-mass, head and hand positions. Trials were aligned by movement phase, and ten with consistent rising durations were selected for analysis. The center-of-mass hypothesis showed substantial variation among configurations that preserved its position, with less variation in directions that changed it. Body segments could therefore vary across successful performances while exhibiting an organized relation to the global task. [SM-W004, pp. 293–299, figs. 7–8]

The head-position analysis gives a less obvious result. Horizontal and vertical head positions could show similar absolute variability, yet their relation to segment configuration differed. Near extension, some angle changes have little effect on head height. A stable-looking height can consequently reflect geometric insensitivity, while a similarly variable horizontal position exhibits preferential stabilization. Combining both coordinates into one outcome can hide the difference. The mapping from component motion to task outcome is part of what the method investigates. [SM-W004, pp. 300–304, fig. 12]

The experimental restrictions changed the inferred structure less broadly than the authors had expected. Their explanation concerns a chosen model, candidate task variables and fluctuations at corresponding movement phases. It offers a procedure for asking what is stabilized and what variation remains compatible with it. Kelso's coordination and metastability account in S25 supplies a neighboring treatment of collective patterns and continuing autonomy. [SM-W004, pp. 302–304]

Huang and Ahmed: maneuverability and balance under uncertain targets

Helen Huang and Alaa Ahmed's 2011 experiment combined a forward-lean task with a lateral balance demand. Eleven young adults used their center of pressure to move a cursor toward a target. A narrow beam beneath a balance board reduced the sideways base of support. In one block, the target could unexpectedly shift forward or backward during the movement. Reaching the target and keeping the board level imposed related demands, while the target changes tested how rapidly participants could redirect their movement. [SM-W003, pp. 2, 8–10, fig. 1]

The authors gave maneuverability and stability operational measures. Mean time to reverse the forward–backward motion after a backward target jump declined from 0.395 seconds in early successful trials to 0.371 seconds in later ones. A separate comparison held final target distance constant: the lateral stability margin, calculated from beam width and sideways center-of-pressure excursion, declined from 1.77 centimeters in late trials of the unchanging-target block to 1.64 centimeters in late no-jump trials within

the uncertain block. Smaller margins were associated with a lower probability of keeping the board level. [SM-W003, pp. 2–5, figs. 2–4, table 1]

Eight of eleven participants exhibited the described more-maneuverable, less-stable pattern. Others did not show that combination. During earlier practice on a wide support, responses became faster without the corresponding loss of margin. The wide-support work always preceded the narrow-support work, and the principal movement estimates used successful trials. The reported tradeoff therefore belongs to the task comparisons, support conditions and individual responses the study actually examined. [SM-W003, pp. 3–6, 8–10, fig. 5]

The experiment also tested a proposed explanation. The authors expected reduced muscle coactivation to accompany increased maneuverability. Surface recordings did not show a statistically significant uncertainty-phase change in the measured muscle activity or coactivation that matched the behavioral shift. Narrow support did increase some muscle activity and lateral-pair coactivation relative to wide support, but that was a different comparison. The expected within-task coactivation mechanism remained unsupported by these measurements, despite the observed change in movement strategy. [SM-W003, pp. 3–4, 6–9, figs. 6–8]

Nagumo: having an admissible continuation and confining every solution

Mitio Nagumo's 1942 article asks when solution curves of an ordinary differential equation remain in a chosen region as the independent variable advances. He distinguishes two properties. A set can allow at least one short continuation from each of its points while a stronger property requires every solution beginning there to remain inside for as long as it stays in the specified ambient domain. The difference matters when the assumptions do not supply a unique solution. His term for the stronger property, attributed to Hukuhara, is a set that is majorant to the right. [W044, §§1–2, pp. 551–552]

For the weaker local property, Nagumo gives an approximate-step criterion. Within the specified relatively closed set, with a continuous vector field, one can find an arbitrarily nearby later point whose secant slope approximates the field at the starting point to an arbitrarily small prescribed tolerance. His proof constructs chains of increasingly accurate steps and obtains an integral curve through a limiting process. The existence of those local steps supports the existence of a continuation remaining in the set. [W044, §2, pp. 551–553]

His treatment of the stronger property uses an auxiliary scalar function called a subintegral. Such a function does not increase along any solution. For the relatively closed ambient domains with local admissibility in both directions that he specifies, confinement is characterized locally by a suitable subintegral that is zero on the selected set and positive outside it. Starting at zero and remaining nonincreasing precludes reaching a positive outside value. The function has the stated continuity and Lipschitz properties in the state coordinates; the local neighborhood and ambient-domain conditions belong to the characterization. [W044, §§3–4, pp. 553–556]

The final comparison results use a scalar bounding equation and a scalar function of the vector system to describe further regions of confinement. The resulting regions permit motion within them; their coordinates need not remain at an equilibrium. [W044, §5, pp. 557–559]

Bonneuil and Saint-Pierre: compute which choices preserve a route to a target

Noël Bonneuil and Patrick Saint-Pierre's 1998 paper develops a decision model from Fredrik Barth's account of fishing captains. Captains can search for new opportunities or follow known ones; catches, running costs and recruitment relations affect their prospects. The authors use this reported setting to

investigate how a changing situation permits some strategies and rules out others. Their contribution is a formal and numerical model drawing on the anthropological description. [SM-W006, §§1–2.1, pp. 43–45]

The state description includes financial loss in a worst-case scenario, expected catch at known locations and the probability that no better catch will be discovered. The modeled control is the population's mean probability of exploring. A captain chooses an individual action that contributes to that mean; a lone captain does not directly set the collective quantity. Search, exploitation and the choices of others consequently enter the decision problem together. The authors supply a catch-distribution model and a target specifying an acceptable terminal situation. [SM-W006, §2.2, pp. 45–49]

Their winning domain consists of states from which at least one modeled trajectory can reach the target while respecting constraints. A viability kernel instead concerns the possibility of continuing within a constraint set indefinitely. For the target-reaching reduction under their stated assumptions, including an empty original viability kernel, the authors add zero velocity at the target so that reaching it permits remaining there. Computing the modified kernel then supplies the required domain. [SM-W006, §§3.1–3.2, pp. 50–54]

The numerical procedure gives the region an operational construction. Start with the constraint set and remove states that have no permissible next step into the currently retained set. Repeat that elimination, then refine the discretization under the stated compactness, convexity and regularity assumptions. The fishing model poses an applicability problem at zero expected known-site catch, where the required compact-image condition fails. The authors work first above a positive cutoff, establish the needed properties there and prove convergence of the resulting domains as the cutoff approaches zero. The computation requires this repair to the model's boundary behavior. [SM-W006, §§4.1–4.3, pp. 55–60]

Knowing the domain still leaves a decision rule to obtain. The authors approximate the correspondence of viable controls and examine a selected feedback rule. In their numerical interpretation, less favorable initial financial circumstances can require more exploration. Underestimating the probability of finding no better catch can postpone necessary action. Their reachable-set calculation also includes destinations outside the winning domain: arriving somewhere does not ensure that the target remains attainable from there. The displayed approximate feedback depends on how closely it follows a selection of the exact rule. [SM-W006, §§4.4–5, pp. 61–64, figs. 5–7]

Cargo-aircraft criteria: protection must retain a usable response

The Civil Aviation Administration of China's 2020 trial standard for high-risk cargo fixed-wing unmanned aircraft provides prescribed design and demonstration criteria. Its scope is primarily regional cargo operations under high-risk conditions. The stability provisions address flight-control augmentation, required weight and center-of-gravity combinations, and degraded control-system states. Transitions among flight states and modes are to be smooth and convergent, with damping assessed relative to the intended flight path. The standard calls for relevant flight-test evidence alongside calculations and simulation. [SM-W009, HY.1–2, HY.171]

Its envelope-protection clause requires limits appropriate to the flight phase and maneuver while retaining response to intentional dynamic maneuvers within those limits. Damping and overshoot must suit the relevant motion. When several protections or internal control restrictions operate together, they must avoid residual oscillation, adverse coupling or unfavorable priority effects. The boundaries and their priorities must be declared. Specifying separate limits therefore leaves the interaction among their control actions to be designed. [SM-W009, HY.334]

The emergency capability has another stated purpose: reducing the consequences of critical failures for third parties. It can involve flight termination, emergency recovery or their combination. The clauses address protection against unintended operation, power for the function and, where applicable, the

dimensions of an expected recovery site. Emergency response can consequently mean ending ordinary flight. [SM-W009, HY.1365]

Communication and procedures remain part of that capability. The standard requires information about data-link range and integrity, addresses latency, and calls for a declared lost-link strategy with an attempt at autonomous reacquisition. The flight manual is to describe normal, abnormal and emergency procedures, including recovery capabilities. These are specified functions, information and tests for the covered aircraft class; the standard is a normative account of what a design should provide. [SM-W009, HY.1585–1587, HY.1707–1711]

Woods: prepare the capacity to respond near saturation

David Woods's 2018 theory of graceful extensibility concerns how systems serving human purposes sustain adaptation as resources remain finite and circumstances continue to change. He organizes the argument into ten proto-theorems about saturation, interdependent adaptive units and changing constraints. He explicitly leaves their formal status and explanatory testing for further development. Within that proposed synthesis, graceful extensibility is the ability to extend response capacity when surprise challenges the boundary of previously adequate behavior. [SM-W008, §§1.2–1.4, 4]

Woods distinguishes base capacity for familiar variations from the potential to mobilize or generate a response near saturation. His construct, capacity for maneuver, concerns the relation between demands and capability available to meet them. Being poised to adapt does not require continually changing plans. It includes recognizing when the current approach still fits, maintaining it when appropriate and revising it when circumstances cease to fit. Preparations made before disruption can supply response options that are difficult to create during a fast cascade. [SM-W008, §2.2.1]

The preparations have costs and can be removed by ordinary performance pressures. Reserves or relationships useful during an unusual event may look idle when assessed only through routine operations. Conversely, a large, poorly targeted investment after a failure can require resources disproportionate to its contribution and become unsustainable. Woods argues for targeted reserves and arrangements that rebalance ordinary and extended capacities. His discussion makes placement, readiness and the ability to deploy resources consequential alongside their amount. [SM-W008, §§2.4.1–2.4.2]

Neighboring units can extend or constrict a unit's response. A local attempt to meet one objective can increase another unit's difficulty; waiting for serial authorization can consume the time in which coordinated action is needed. Woods describes initiative and reciprocity as parts of a response: recognize that a plan no longer fits, act without waiting for a new tasking, and donate limited effort to a neighbor in trouble. Donation imposes an immediate cost while future assistance remains uncertain. Initiative also needs coordination if it is to avoid fragmenting the response. [SM-W008, §§2.3.1–2.3.3]

His reported transportation-firm case during Hurricane Sandy, drawn from Deary's study, makes the reorganization concrete. The firm changed priorities, relaxed usual cost-control processes for safety, formed weather-impact teams and temporary local command centers, increased horizontal coordination and moved authority toward points of action. Those changes sacrificed some ordinary economic and process criteria while making different responses possible. [SM-W008, §2.4.3]

S06 Information, coding and production coordination

Shannon: recover a selected message from alternatives

Claude Shannon's 1948 *A Mathematical Theory of Communication* begins with a particular engineering problem: reproduce at one point a message selected at another. A system must be prepared for the possible selections, rather than only the message that happens to be sent. Its source, transmitter, channel, receiver and destination have different roles. The transmitter makes a suitable signal from the message; the receiver reconstructs the message from that signal. [W025, introduction]

The paper expressly sets semantic aspects aside for this engineering problem. That choice permits analysis of the alternatives and the transmission process without requiring the theory to establish what a recipient understands, believes or values. For equally likely alternatives, a logarithm of their number gives a convenient measure; base two expresses it in bits. Four equally likely messages correspond to two bits in that counting sense. [W025, introduction; arithmetic illustration]

The opening also makes coding a material and procedural operation. A telephone transmitter converts sound pressure into current; a telegraph code supplies a pattern of dots, dashes and spaces; more elaborate systems sample, quantize, encode and interleave. The abstract channel is one part of the process. The source's framework explains how differently constituted communication arrangements can be analyzed under common specified questions. [W025, introduction, communication-system description]

Reliable transmission is a capacity result with a coding obligation

In the selected noisy discrete-channel result, Shannon relates a source's entropy rate to channel capacity. Below the relevant capacity, suitable coding can make the frequency of errors or remaining equivocation arbitrarily small under the model. The result is stronger than saying every bit must travel perfectly in isolation. A receiver can use a suitably organized longer signal to recover what was sent despite particular corruptions. The construction depends on the channel/noise model and the rate being considered. [W025, theorem 11]

Shannon's following discussion is unusually important for interpreting that accomplishment. The proof does not generally supply a straightforward practical method for constructing the ideal code. Appropriate redundancy must combat the particular noise structure, and approaching the ideal generally introduces delay. A longer sample gives the receiver a better basis for deciding which original signal caused the received one. The selected passage thus presents reliable transmission together with the work of coding, decoding and waiting that makes it possible. [W025, §14]

Redundancy serves two distinct roles. Statistical redundancy in a source can permit more compact representation; suitably retained or introduced redundancy can also help recover a message when the channel is noisy. Shannon's English-language example concerns contextual constraints assisting reconstruction, rather than establishing that all repetition should be removed. An arrangement that optimizes one objective can change the other. The desired rate, tolerated error and available delay must remain specified before a coding improvement can be assessed. [W025, §14]

Exchanging information to coordinate production

In a 1999 conference paper, Stanley Reiter compares two tractor makers. One buys a gear train designed specifically for its tractors. The maker and its supplier exchange information and negotiate so that the gear trains arrive when the assembly schedule requires them. The other makes the gear train in a

separate unit of the same firm. Reiter observes that this unit and the tractor plant still face the same scheduling problem and may settle it by the same exchange of information. He contrasts this with divisions of an integrated petrochemical company exchanging intermediate products at transfer prices set to outside market prices. The examples distinguish the channel by which production decisions are coordinated from the ownership of the units making them. [DP-W001]

S33 Information acquisition, experiments and dual control

Information for a specified decision

The selected sources concern the information needed to estimate, decide or control. They also describe situations in which information is not a free input: obtaining it can require an experiment, a control action or a different distribution of knowledge among decision makers.

Fisher: reduction and the criterion of sufficiency

In the definitions and §§2–4 of his 1922 paper, Fisher separates the population characteristics to be estimated from statistics calculated from a sample. This distinction addresses a practical ambiguity: a sample mean is an observed quantity, while the population parameter is what the statistic is meant to illuminate. He organizes statistical work around specification, estimation and distribution. Specification identifies a mathematical form for the hypothetical population; estimation uses sample data to estimate its parameters; distribution concerns how the resulting statistics vary across samples. [W058, definitions and §§2–4, printed pp. 309–317]

His criterion of sufficiency asks whether any other statistic calculated from the same sample can supply additional information about the parameter being estimated. The relevant reduction is consequently relative to a parameter and a statistical setting. It does not mean that the reduced statistic preserves every conceivable use of the observations. The observations may be compressed without losing the information specified by the criterion.

Fisher's separate criteria prevent several ideas from being treated as synonyms. Consistency, efficiency and sufficiency address different properties. An estimate can be evaluated for sampling behavior without thereby establishing that it contains all the relevant sample information. The historical discussion does not justify declaring every maximum-likelihood estimate sufficient. [W058]

The criterion concerns the statistical information retained for a specified model and parameter. Acquisition, computation, storage, explanation and use remain separate resource questions. A new model or question can require the original observations.

Blackwell's 1951 comparison: attainable losses

In *Comparison of Experiments*, Blackwell represents an experiment by a set of probability measures corresponding to possible states or distributions. An observation is used to choose an action. Each action has a loss in each state, and a decision procedure produces a vector of expected losses. Comparing experiments therefore concerns what decision consequences can be attained using their observations, rather than the raw number of observations or their apparent descriptive richness. [W059, §§1–3, printed pp. 93–94]

The construction allows convexification and randomized mixtures of procedures. Its first theorem permits attention to closed convex action sets in the subsequent treatment. Randomized decision capability is part of the mathematical comparison; forbidding it changes the admissible procedure class and can change attainable losses. Blackwell presents equivalent decision criteria for the comparison, including formulations involving expected losses with state weights and worst-state loss. Their meaning depends on the specified class of decision problems.

The standard-experiment construction is especially relevant to information reduction. Blackwell associates an observation with a vector of normalized densities and shows that the corresponding standard experiment has the same attainable loss set in the stated setting. A decision rule on the reduced

representation can be composed with the transformation of the original observation; the converse uses conditional expectation and the convex action set. Thus the claim is not merely that a summary looks informative. It specifies a decision equivalence and the machinery supporting it. [W059, §§4–5, especially theorem 3, printed pp. 94–96]

Blackwell's 1951 summary distinguishes sufficiency-based comparison from the general converse and reports a converse for dichotomies. His later *Equivalent Comparisons of Experiments* appeared in 1953. [W029; W059]

Börgers: a readable finite reconstruction of dominance

Börgers's December 2023 notes provide an operative scholarly reconstruction in a finite-state, finite-outcome setting. An experiment P is a Markov matrix: each row gives the distribution of its possible observations in one state. An action is represented by its losses in the states. A decision rule maps observations into actions, and the resulting risk vector gives expected loss state by state. This makes the comparison inspectable in terms of matrices and feasible decisions. [GL-W024, PDF pp. 1–3]

The central garbling condition is PM = Q, where M is a state-independent Markov matrix. It says that the observations of Q can be generated by taking observations of P and applying an additional randomized transformation that does not require knowledge of the true state. A decision maker with P can therefore reproduce the decision procedure available with Q: first carry out the transformation, then apply the Q-based rule. The transformation does not create hidden access to the very state the decision maker is trying to infer. [GL-W024, proposition 1 and explanation, PDF pp. 3–6]

The notes relate this condition to attainable risk vectors and to decision-value comparisons. Their second proposition gives equivalent conditions expressed through decision problems and their values. This gives an explicit bridge from a relation between observation systems to what an actor can achieve.

The notes develop the Blackwell tradition, including Blackwell and Girshick, in a finite setting with convex action possibilities and randomized procedures. [GL-W024, PDF pp. 1–9]

Brooks, Frankel and Kamenica: information already held can change the comparison

The introduction and early definitions of the March 2024 manuscript distinguish an experiment from a signal. An experiment specifies how data relate to the truth; two separately specified experiments do not determine the joint informational content of their observations. Their use of a signal adds the joint relation to other information. The difference matters when information overlaps or becomes useful only in combination. [W071, introduction and §§2–3.2]

Their newspaper example makes the issue concrete without requiring a theorem proof. If a reader already receives one newspaper, another copy of it need not add as much as a different newspaper. A third publication can also alter the comparison because its reporting overlaps with one source or complements another. An ordering established for sources in isolation need not answer which additional source is more valuable for a reader with existing information.

The authors study comparisons robust both to the decision maker's preferences and to other information the decision maker may possess. Their announced reveal-or-refine condition requires that an observation from the favored signal either reveal the truth or refine what the other signal conveys. [W071]

Information value depends on the knowledge already in place. Two sources can be substitutable or complementary in ways not captured by their separate state–observation distributions.

Saldi and Yüksel: who is allowed to know what

The introduction to Saldi and Yüksel's review presents information structures as restrictions on which observations each decision maker can use. Their example has several agents, each choosing an action from local information, possibly with an additional independent randomization device. A team cost depends on the underlying state and the agents' actions. The admissible policies must respect the local information rather than draw on an omniscient observer's knowledge. [TA-W004, introduction]

Their proposed probability-theoretic and topological approach studies the distributions induced over states, measurements and actions—strategic measures—and the geometry of feasible decision or control policies. The introduction distinguishes questions of existence, approximation, convexity and numerical bounds. An apparently favorable solution obtained by relaxing the information restrictions need not be implementable by the original agents.

These information structures specify local knowledge, admissible action and a team cost, making an information boundary part of the problem formulation. [TA-W004]

Wittenmark: an action can both control and probe

Wittenmark's *Adaptive Dual Control* explains why immediate control and learning about a process can be linked. His stochastic formulation specifies a process model, admissible causal controllers and a loss over a horizon. The model structure and the driving process distribution are assumed known, while some parameters are unknown or changing. The history of inputs and outputs supports parameter estimation. The chosen control signal can affect not only the next output but also the quality of future estimates. [AFC-W012, pp. 1–4]

A certainty-equivalent controller inserts current parameter estimates as though they were the true values in a controller designed for known parameters. A cautious controller additionally responds to uncertainty in those estimates. In the one-step formulation, caution reduces controller gain when parameter uncertainty is high. This can be sensible protection against an unreliable estimate, yet it can also weaken the excitation that supplies information about the process.

Wittenmark describes a resulting turn-off possibility: smaller control signals yield less information; uncertainty can increase; control signals are then reduced further. His integrator example distinguishes the certainty-equivalent controller's difficulty near a zero estimated gain from the cautious controller's tendency toward diminishing action as uncertainty rises. [AFC-W012, pp. 4–6]

Dual control deliberately takes the informational effect of action into account over a longer horizon. An action can have a current cost while improving the basis for later actions. The chapter's summary stresses that the exact optimal problem is computationally intractable in general practice and motivates simpler suboptimal controllers with active probing features. Adding a term that rewards information is not automatically enough: for the intended dual effect it must depend on the control signal being selected and on the quality of the estimates. [AFC-W012, summary and dual-control discussion]

Bayesian design: a limited but explicit decision framing

The abstract of Chaloner and Verdinelli's review frames experimental design as a decision problem in which the chosen design maximizes expected utility from the experiment. It establishes a relevant orientation: the experiment is selected for an anticipated decision or inferential purpose rather than because more data are inherently better. [AFC-W016]

Statistical and decision criteria

The accounts above preserve distinct achievements: a parameter-specific sufficiency criterion; attainable-loss comparisons; a finite garbling reconstruction; a qualification involving correlated prior information; admissible local knowledge; and the informational effect of control action. Each gives a different way of making information operationally precise.

S74 Diagnostic testing and the choice of antimicrobial treatment

Making susceptibility a dependable measurement

Antimicrobial susceptibility testing connects a laboratory observation to a choice among treatments. William Kirby's 1985 retrospective described why the convenience of placing antibiotic disks on an inoculated plate had initially exceeded the comparability of the results. Laboratories varied the medium, inoculum, disk content and interpretation of the boundary around a disk. A clear zone could therefore look precise while reflecting different procedures. Kirby recalled the work that led to the standardized method published in 1966 and its later collaborative evaluation and adoption. [TD-W010]

A disk creates a concentration gradient as the drug diffuses into the medium. The inhibition zone can be related to the concentration required to inhibit growth, with zone diameter inversely related to the logarithm of the minimum inhibitory concentration. The relationship becomes interpretable when the procedure is controlled. Standardization made the convenient measurement useful across laboratories: the question was no longer merely whether a visible zone appeared, but what a zone obtained under specified conditions indicated about susceptibility. [TD-W010]

From growth measurement to a clinical category

The 2025 recommendations of the Comité de l'antibiogramme de la Société française de microbiologie show how much of that interpretive structure remains in a modern standard. Medium, inoculum and test conditions are specified so that a measurement can be compared with an applicable breakpoint. A heavier inoculum, for example, can produce a smaller zone. Without control of that feature, an apparent change in susceptibility may instead be a change in how the test was performed. The recommendations join the measurement procedure to the conditions under which its result has meaning. [TD-W011]

The raw diameter or minimum inhibitory concentration is then interpreted for an organism–drug combination. The categories distinguish susceptibility at standard dosage, susceptibility at increased dosage or exposure, and resistance. The French recommendations discuss increased dosage in relation to EUCAST's increased-exposure terminology, which includes how drug reaches the site of infection. A categorical result therefore already incorporates assumptions about achievable exposure. It is not a context-free property of a drug considered in isolation from the organism and treatment conditions. [TD-W011]

Deciding whether to prescribe

Jochen Cals and colleagues' 2009 trial examined a different decision: whether adults with lower respiratory tract infection should receive an antibiotic in primary care. Twenty practices and forty general practitioners participated, with 431 patients. A factorial design tested point-of-care C-reactive protein measurement and enhanced communication skills as separate interventions. CRP is a marker of the host's inflammatory response. The approximately three-minute measurement supplied evidence for interpreting the presentation; it did not identify an organism or determine which antibiotic would inhibit it. [TD-W012]

The test was introduced with interpretation training and a run-in period. The communication intervention taught eleven tasks addressing such matters as the patient's concerns, expectations and understanding of the illness's likely course. In the consultation, a clinician could establish why the patient had come, discuss the symptoms and test result, explain what recovery might involve, and reach a prescribing decision. The trial consequently tested changes to the use of information as well as the availability of information. [TD-W012]

Antibiotic prescribing at the first consultation was 31% among patients in the CRP-testing comparison and 53% without CRP testing. The corresponding communication comparison was 27% versus 54%. At twenty-eight days, the proportions receiving antibiotics remained lower: 45% versus 58% for CRP and 38% versus 63% for communication. Recovery and satisfaction did not differ detectably between the groups; median recovery took twenty-two days. The combined-intervention cell had particularly low prescribing, but the interaction was not statistically significant, so the study's factorial main effects provide the clearer account of what was established. [TD-W012]

Smedemark and colleagues' 2022 Cochrane review placed biomarker-guided prescribing in a larger primary-care evidence base. Twelve CRP trials included 10,218 participants, of whom 2,335 were children; one procalcitonin trial included 317 participants. CRP-guided care reduced initial prescribing, with a risk ratio of 0.77 and a 95% confidence interval from 0.69 to 0.86. The review rated that evidence moderate certainty and found substantial variation across trials. Prescribing within twenty-eight days was also reduced, with a risk ratio of 0.81, supported by high-certainty evidence. [TD-W013]

Recovery changed little. Deaths were too few to provide a precise assessment of mortality, for which the evidence was low certainty. These outcomes describe different consequences of the intervention: fewer prescriptions, the course of recovery, and uncommon serious events. The review's primary-care procalcitonin evidence should also be distinguished from inpatient studies using repeated procalcitonin measurements to stop treatment already underway. [TD-W013]

Pathogen testing and ordinary prescribing practice

A rapid streptococcal antigen test gives a more targeted answer than a general inflammatory marker. Anna Maria Bonet-Esteve and colleagues' 2021 report studied its introduction for children aged three to fifteen with suspected streptococcal infection. Twenty-seven pediatricians participated; 557 children were eligible and 519 were followed. Adding the rapid test to the ordinary clinical pathway reduced antibiotic prescribing from 88.5% in the control group to 45.5% in the testing group. All positive tests and relatively few negative tests led to prescriptions. The test altered the treatment decision by distinguishing children within a clinically similar presenting group. [TD-W015]

The investigators also followed adherence by telephone. Among those prescribed antibiotics, adherence was 68% in the control group and 62.9% in the test group, a nonsignificant difference. A test that made prescribing more selective did not thereby establish improved completion of the prescribed course. The two outcomes concern different parts of the treatment process: deciding who receives an antibiotic and what happens after that decision. [TD-W015]

Sydenham and colleagues' 2021 register study examined how testing and prescribing were associated in ordinary general practice. It covered 783,252 antibiotic courses in nine Danish municipalities from 2013 through 2017. Diagnostic tests were associated with 43.4% of courses, using a window extending seven days before and after a prescription. CRP use increased over the period, while overall point-of-care testing remained stable and the mixture of tests changed. The study records practice patterns at scale. Because the association window included tests after prescribing, it does not measure the proportion of prescriptions preceded by a diagnostic test, and its observational design does not by itself identify what testing caused. [TD-W014]

When a rapid panel leaves prescribing unchanged

Alastair Hay and colleagues' 2026 primary-care trial supplies an informative contrasting result. A rapid molecular panel tested for twenty-three respiratory targets, nineteen viruses and four atypical bacteria, in about forty-five minutes. The study enrolled 552 patients across sixteen practices, with 276 in each group. Participants had to be willing to wait. Clinicians received information about the panel and

presentations, but the intervention did not include the decision and communication training used in the earlier CRP study. Typical respiratory bacteria were outside the panel's target set. [TD-W016]

Antibiotics were prescribed to 124 patients in each group, 45% of both. The odds ratio was 1.00, with a 95% confidence interval from 0.71 to 1.41. Prespecified subgroup analyses showed less prescribing when a virus was found, offset by more prescribing when no target was detected. A negative result on a panel dominated by viral targets did not establish a bacterial cause. The authors discussed how clinicians might interpret that absence of detection, while the randomized overall result remained no reduction in prescribing. Symptom severity met the study's noninferiority criterion. The trial demonstrates why a faster and more detailed result has to be assessed as part of the clinical decision in which it is used. [TD-W016]

Correcting treatment after a positive blood culture

The RAPIDS-GN trial addressed patients whose blood cultures had already become positive for Gram-negative bacilli. Antibiotic treatment could then need correction: a resistant organism might require escalation, or a susceptible organism might permit a narrower drug. Banerjee and colleagues' report first appeared in 2020. Five hundred patients were randomized; the modified intention-to-treat analysis included 448, with 222 assigned to rapid testing and 226 to standard testing. This was a study of what happened after a positive culture, rather than of withholding initial treatment while waiting for a culture to grow. [TD-W017]

Both groups received antimicrobial stewardship review. A physician or pharmacist was paged when the Gram stain, identification or susceptibility result became available, reviewed the record, and telephoned the treating service when a treatment modification appeared appropriate. Scenario-based recommendations had been developed for the stewardship clinicians. The treating team retained the decision to accept them. In the rapid arm, organism identification and phenotypic susceptibility testing were performed on the rapid platform in addition to ordinary testing, and results entered the medical record. Thus the route from laboratory to medication included a named recipient, review and communication in both groups. [TD-W017]

Median time to identification was 2.7 hours with rapid testing and 11.7 hours with standard testing; susceptibility results took 13.5 and 44.9 hours respectively. The primary outcome, time from randomization to the first antibiotic modification, was 8.6 versus 14.9 hours. For modification of Gram-negative-active antibiotics specifically, the medians were 17.3 versus 42.1 hours. These clinical intervals include more than an instrument's running time. They describe when results became available and when treatment, as recorded in medication administration, actually changed. [TD-W017]

Among patients with resistant organisms, escalation occurred at a median 18.4 hours in the rapid arm and 61.7 hours in the standard arm. The trial did not detect significant differences in clinical outcomes. Its most direct demonstrated achievement was earlier adjustment within this culture-to-review-to-treatment pathway. That finding remains substantial even though faster adjustment and an improvement in mortality are different outcomes. [TD-W017]

The investigators' secondary analysis, first published in 2024, examined 386 patients with monomicrobial, on-panel organisms who had received empiric therapy. It classified treatment as undertreatment, optimal narrow-spectrum treatment or overtreatment at the initial zero-to-two-hour interval and again at twenty-four to twenty-six and forty-eight to fifty-two hours. The clearest impact occurred with resistant *Escherichia coli*, *Klebsiella* and *Proteus*, for which rapid results supported earlier escalation. Broad-spectrum treatment often persisted even when an isolate was susceptible to narrower options. [TD-W018]

For *Acinetobacter*, *Citrobacter*, *Enterobacter*, *Serratia* and *Pseudomonas*, the analysis found little effect on treatment, although the groups were smaller. The authors proposed that clinicians might treat escalation as more urgent than narrowing therapy, or hesitate to remove broader coverage under diagnostic uncertainty. Those were explanations they offered for the observed response. The analysis did not review patient-level clinical details to establish every reason for a choice. It identifies differences in the parent trial's treatment trajectories; it was not a separately tested rule for selecting patients prospectively. [TD-W018]

Laboratory speed and patient outcomes

Anton-Vazquez and colleagues' 2021 Cochrane review examined rapid susceptibility testing in bloodstream infection. Its six randomized trials included 1,638 participants and covered phenotypic and genotypic approaches producing results within eight hours of a positive culture. Individual trials reported earlier treatment changes, but the mortality estimate was imprecise: risk ratio 1.10, with a 95% confidence interval from 0.82 to 1.46, supported by low-certainty evidence. The review did not establish either a mortality reduction or the absence of a clinically meaningful effect. [TD-W019]

The review distinguishes the test's turnaround, time to appropriate treatment, and outcomes such as mortality and discharge. Empiric treatment was already active against the pathogen for many participants, so an earlier result did not always rescue an inactive regimen. Studies also differed in how results reached clinicians and how stewardship was organized. Those differences form part of the intervention being evaluated, rather than merely background to a laboratory comparison. [TD-W019]

Designing the report that reaches the clinician

西村翔's 2024 review develops susceptibility testing as a contribution to antimicrobial stewardship across specimen collection, analysis and reporting. A resistance marker can answer a useful question without supplying a complete susceptibility profile. This is especially consequential for Gram-negative bacteria with several possible resistance mechanisms. Phenotypic testing and molecular detection therefore provide different information, whose interpretation depends on what was actually measured. [TD-W020]

Selective or cascade reporting changes which tested options routinely appear on the clinical report. A laboratory can make a narrower set of appropriate drugs immediately visible while keeping additional results available on request or under specified circumstances. The policy requires communication with clinicians so that an omitted drug is not mistaken for an unperformed test or an unusable treatment. In the author's hospital account, restrictions on routine fluoroquinolone reporting were discussed, tried and then implemented in July 2018, with exceptions for specified organisms. Use fell by roughly half in the reported before-and-after experience. This was an implementation observation, rather than a randomized estimate of the effect of reporting alone. [TD-W020]

The 2016 expert consensus by 王辉, 宁永忠, 陈宏斌 and colleagues sets out the construction of a bacterial susceptibility report. It begins before the display of drug names: the laboratory must consider whether the isolate is clinically significant and recognize intrinsic resistance. Panels combine standard expectations with local needs and consultation. The consensus's required reporting group is a reporting category, not a declaration that every listed agent is the clinician's first choice. [TD-W021]

The report separates smear observations, culture findings and susceptibility results. Measurements such as minimum inhibitory concentrations or zone diameters are accompanied by interpretation under a stated standard; comments can explain a consequential finding. Urgent partial results should be communicated with their timing clear, and unexpected resistance may require confirmation. These details

let the recipient distinguish a preliminary observation from a completed interpretation and act on important information before every part of the laboratory process is finished. [TD-W021]

Deciding when enough treatment has been given

The PROGRESS trial, first published in 2020, used repeated procalcitonin measurements to inform stopping antibiotics in sepsis. Of 266 randomized patients, ten withdrew permission to use their data, leaving 125 in the procalcitonin group and 131 receiving standard care. Patients had already begun treatment and were randomized within twenty-four hours. The intervention measured procalcitonin at baseline and reassessed from day five, with daily measurements when the stopping criterion had not been reached. [TD-W023]

The Methods specified stopping when procalcitonin had fallen by at least 80% from baseline or was below 0.5 micrograms per liter, subject to the patient's condition. Instability, including persisting fever or vasopressor requirements, could justify continuing treatment despite the marker. Thirteen patients who reached the biochemical criterion were considered unstable; ninety-six patients complied with the stopping advice. The protocol therefore coupled a repeated measurement with a clinical exception, rather than making a single laboratory value an unconditional instruction. [TD-W023]

Median antibiotic duration was five days with procalcitonin guidance and ten days with standard care. The primary 180-day composite counted new infections by *Clostridioides difficile* or multidrug-resistant organisms, or death associated with either kind of infection present at baseline. It occurred in nine of 125 and twenty of 131 patients respectively. Twenty-eight-day mortality was also lower, nineteen versus thirty-seven deaths, while the 180-day mortality comparison was not statistically significant. The study involved hospital wards and excluded circumstances requiring prolonged treatment and specified patient groups. Its result concerns a defined stopping strategy after treatment had begun, with its measured outcomes and follow-up periods. [TD-W023]

Organizing a result-to-action pathway

Chen and colleagues' 2026 narrative review proposes an explicit framework for making rapid results actionable. Its four clocks distinguish analytical turnaround, the wider laboratory interval, the point at which a result becomes clinically actionable, and the delay from result to action. A short assay does not uniquely determine any of the other three. A result may be technically available while the clinician still lacks the interpretation, patient information or response arrangement needed to use it. [TD-W022]

The authors organize actionability around five domains: result type and certainty, patient risk, probability of resistance, capacity to respond, and the quality of stewardship. They propose notification, interpretive comments, patient-specific recommendations, documentation, audit and feedback as connected components. This is their organizing clinical framework, not a validated numerical score. Its practical emphasis is on a result reaching someone who can interpret it, choose an action and record the response. S33 develops earlier accounts of information and decisions; these clinical histories show the particular measurements, reporting forms and treatment pathways through which information was used. [TD-W022]

S35 Conditioning, regularization and inverse operators

Several different ways of making a calculation manageable

The mathematical sources in this section explain distinct relations between data, a problem, its representation and a computational procedure. Conditioning describes sensitivity of the problem. Numerical stability concerns error introduced by an algorithm. Equilibration changes scaling; regularization constrains or penalizes possible answers; learned operators use prepared data and models to approximate a map.

Higham: a condition number belongs to a problem and its data

Higham's exposition defines a condition number as a measure of how sensitive a solution is to small perturbations of input data. The definition depends on the function or implicitly defined problem, the input, a norm and the choice between relative and absolute perturbations. A claim that a problem is well conditioned is therefore incomplete without the relevant data and measurement convention. The same physical task can be represented by different mathematical questions, and a condition number answers the question actually specified. [W089, author exposition, 19 March 2020]

For a nonsingular square matrix, the familiar quantity $\kappa(A) = \|A\| \, \|A^{-1}\|$ measures sensitivity associated with inversion in the stated norm. Higham relates it to linear-system sensitivity and distinguishes whether A alone or both A and b are treated as data. Large sensitivity describes what small perturbations can do, rather than asserting that every perturbation realizes the worst case. His diagrams and examples distinguish that potential amplification from the particular response to a given change.

The resource implication appears explicitly in the article: calculating a condition number exactly can be as expensive as computing the function itself, motivating estimates and bounds. Nor does a large but finite condition number imply mathematical discontinuity. [W089]

Higham: numerical stability belongs to an algorithm

Higham's separate account of numerical stability asks how errors introduced while executing an algorithm affect its result. Forward error measures departure from the intended answer. Backward error asks how much the input would have to change for the computed answer to be exact. A backward-stable procedure returns the exact answer to a nearby problem, where the scale of nearby is judged relative to the arithmetic. Forward stability instead evaluates the resulting error in relation to conditioning. Mixed stability allows small changes to both input and output. [W090, author exposition, 4 August 2020]

These distinctions explain why an algorithm can behave well while its answer remains sensitive. A small computational perturbation can be amplified by an ill-conditioned problem. Conversely, inspecting an answer's apparent plausibility does not establish a small backward error. Higham's discussion also makes the data boundary explicit: in a linear system, are the matrix, the right-hand side or both uncertain? In a nonlinear formula, are numerical coefficients fixed constants or uncertain data?

The rank-one outer-product example shows why the error model matters. Rounding its many entries independently need not produce a matrix that is exactly rank one. Requiring the computed result to be representable by perturbing only the original two vectors can therefore be too restrictive. The stability claim therefore depends on which input perturbations the error model permits. [W090]

Diamond and Boyd: equilibration without reading individual matrix entries

Diamond and Boyd study row and column scaling that equalizes the corresponding norms. Their matrix-free method is relevant when a solver can multiply by a matrix and its transpose without convenient access to all its entries. They formulate a convex optimization problem and use a stochastic projected-gradient method with an unbiased gradient estimate. The approach replaces a demand for explicit matrix access with operations available through the computational interface. [MIO-W003, introduction and selected §§3.1–3.3]

Their condition-number result concerns a particular upper bound and, in the square nonsingular setting, its minimization over diagonal scalings. It is not a theorem that every diagonal scaling lowers the exact condition number or accelerates every solver. The optimization also uses a penalty to keep the scaling problem controlled. That penalty belongs to the scaling optimization; it should not be confused with regularization of the original inverse problem.

The selected LSQR example shows how the authors preserve the original task. They solve a scaled system, $(DAE)z = Db$, and recover the original coordinates by $x = Ez$. They assess accuracy using the original residual $\|Ax - b\|$, rather than only the residual in scaled coordinates. A small transformed residual is not allowed to redefine success without translation back to the starting system. [MIO-W003, selected §6.1]

Their experiment also counts preparation. Each equilibration iteration and each LSQR iteration involves one multiplication by A and one by its transpose, making their comparison in total matrix multiplications intelligible. They compare different amounts of equilibration before applying the solver. The reported improvement concerns the constructed numerical setup. [MIO-W003, selected numerical discussion, §7 and conclusion]

Hansen: regularization suppresses unstable detail by changing the reconstruction

Hansen's selected teaching slides explain why directly using small singular values can greatly amplify noisy components of data. Truncated singular-value decomposition removes selected components rather than using the complete inverse expansion. The truncation parameter should respond to where data coefficients become dominated by noise, not simply to a singular-value cutoff considered independently of the data. [MIO-W002, slides 1–8]

The slides make the tradeoff explicit: smaller covariance is purchased with bias. Truncation can reduce the effect of noise, but its abrupt cutoff can introduce artifacts; computing a full singular-value decomposition can also be infeasible for large problems. The convenient formula therefore includes both an approximation choice and a computational prerequisite.

Tikhonov regularization replaces residual minimization by a balance between data fit and a penalty. In the standard form shown, the target is to minimize $\|Ax - b\|^2 + \lambda^2\|x\|^2$. A large λ can oversmooth the answer; a small λ can allow noise to dominate. The selected material treats the smoothing norm and the balancing parameter as choices that affect the result. The regularization term encodes a preference or restriction on acceptable solutions, rather than generating information absent from the measurements. [MIO-W002, slides 10–14; formulas and tradeoffs]

Reducing a solver's burden on an equivalent system differs from accepting a biased or restricted reconstruction.

Nelsen and Yang: learning a family of inverse maps

Nelsen and Yang's 2025 survey distinguishes end-to-end inverse learning from learning selected components inside a conventional inversion procedure. In the first approach, observed data are mapped

directly to an estimated unknown or to a posterior distribution. In the second, learned priors, regularizers or other components enter an algorithm that still carries out inversion. Those arrangements locate the learned information differently. [MIO-W001, selected introduction, §§2.2, 4.1, 4.4 and 5.2]

Their mathematical setup writes data as $y = G(u) + \eta$, with G the forward map, u the unknown and η noise. The true inverse, when it exists on the specified admissible domain, is defined on the range of G. A learned estimator can be evaluated on a larger observation space, including noisy data outside that range. The survey emphasizes that identifying an admissible domain on which the forward map is injective and its inverse continuous can itself require difficult analysis. The training procedure does not make that domain question disappear. [MIO-W001, §4.1]

The selected formulation trains using pairs of unknowns and noisy observations, drawn under a specified distribution and noise model, and minimizes a selected loss over a hypothesis class. Consequently the training distribution, noise and metric are part of what the learned map is prepared to do. Approximation existence, successful optimization and reliable practical use are distinct matters. An expressive architecture is not by itself evidence that an actual training run has recovered the desired inverse.

Their discussion of electrical impedance tomography retains severe sensitivity: the cited stability relation can be logarithmic in the noise. Restrictions to suitable finite-dimensional sets can improve the stability statement, with constants and dimension dependence that matter. These results are reported through the survey. [MIO-W001, selected §4.3, including equation 4.24]

Amortization and the retained preparation

For posterior learning, Nelsen and Yang describe approximating a data-to-posterior map so that a family of inverse problems can be addressed without retraining for every observation. They explicitly acknowledge potentially high upfront training and data-generation costs. Repeated fast evaluations can amortize that preparation. [MIO-W001, §4.4]

One described approach uses a hypernetwork to map the data to parameters of a second network representing transport from prior to posterior. The survey notes that this training task is difficult in practice and that higher accuracy may require subsequent fine-tuning or using the learned model as a preconditioner for a fixed observation. The learned map can thus serve either as an approximate answer or as preparation for an observation-specific calculation.

IPCC: tuning retains physical and observational constraints

The selected IPCC assessment passage on model tuning describes a different modeling practice. Parameters for unresolved processes are not treated as arbitrary controls. Their acceptable ranges are constrained by mathematical consistency, physical considerations such as energy conservation and observational evidence. Component modeling groups make initial choices; assembling the components into an Earth system model requires refinement against predefined tuning targets. [GL-W089, chapter 1, §1.5.3.2]

Meeting predefined tuning targets precedes release for model comparisons. The resulting Earth system model combines governing constraints with parameters refined against observed behavior. [GL-W089]

Sensitivity, preparation and reconstruction

Higham supplies definitions needed to describe sensitivity and computational error accurately. Diamond and Boyd provide a particular preparation method with an original-task residual and counted setup operations. Hansen explains restricted reconstruction and its bias–noise tradeoff. Nelsen and Yang survey

reusable learned maps and retained priors, training and accuracy work. The IPCC assessment describes constrained component tuning.

S55 Numerical fidelity and computational cost

A numerical computation can use different amounts of arithmetic precision in different places. The accounts below explain how programmers and numerical analysts select those places, check the returned result and pay for correction or monitoring. Their tasks range from solving a specified linear system to accepting a measured change in an application's output. The required answer and the means of judging it enter the procedure alongside the available arithmetic.

S35 develops conditioning, stability and preparation of numerical problems. Here the focus is the allocation of fidelity during computation: what work can become cheaper, what must remain accurate and what happens when the initial allocation proves insufficient.

Buttari and colleagues: a cheap factorization with more accurate correction

Buttari and colleagues divide the work of solving a dense linear system, Ax = b, between two arithmetic formats. They convert the matrix and right-hand side to single precision, factor the converted matrix and obtain an initial solution. They preserve the original double-precision matrix. Each subsequent correction begins by calculating the residual, b − Ax, using that original matrix in double precision. The existing single-precision factors are then reused to solve for a correction, which is added to the solution in double precision. The analogous procedure uses Cholesky factorization for a symmetric positive-definite system. [CG-W001, §1.1, Algorithms 2–3]

The procedure places the cubic-cost factorization in the faster arithmetic while retaining more accurate residual calculation and solution updates, whose cost per refinement step is quadratic. It also retains two matrix representations and conversions between them. Single precision therefore describes a particular part of the computation; it does not replace the accuracy required of the returned answer. The paper's error analysis makes the result conditional on a sufficiently well-conditioned matrix relative to the correction arithmetic and an adequately stable correction solver. Low-precision errors must still permit refinement to make progress. [CG-W001, §1.1 and Appendix A]

Their implementation comparisons show why the size and architecture matter. On the conventional processors tested, the reported gains reach approximately 1.8 for the nonsymmetric solver and 1.5 for the positive-definite solver when problems are sufficiently large. For small systems, even a few correction steps can outweigh the cheaper factorization. The Cell results use a different comparison: the mixed-precision solver's achieved throughput is compared with the processor's double-precision peak, giving the reported factors of seven and eleven. The experiments provide implementation evidence for this division of arithmetic work; the graph labels and processor configurations specify what was compared. [CG-W001, §1.2, Table 2 and Figures 1–8]

Carson and Higham: correction accuracy and convergence are different requirements

Carson and Higham make three arithmetic roles independently adjustable: factorization, storage of data and solution, and residual calculation. Their general analysis also distinguishes the effective accuracy with which a correction equation is solved. This fourth parameter is useful when a low-precision factorization supports a more accurate iterative correction. An algorithm can thus store its answer in single precision, factor in half precision and compute residuals in double precision. Their conditions distinguish the rate at which errors decrease from the accuracy eventually attainable. More accurate residuals can lower the final forward-error level without repairing a correction process that fails to converge. [CG-W002, Algorithm 1.1, §§2–4 and 7]

Their GMRES variant uses the low-precision factors as a preconditioner, rather than relying on their triangular solves alone to supply each correction. This can extend the class of systems for which the solution reaches working-precision accuracy. The procedure still needs the original matrix and accurate applications of the preconditioned operator. Half precision's narrow numerical range also requires attention to scaling; when the initial solve overflows, the authors describe starting refinement from zero instead. [CG-W002, §§6 and 8, Algorithm 8.1]

The experiments expose a limit that a count of outer refinement steps would miss. For one generated 100-by-100 system, three outer steps contain 91, 92 and 92 GMRES iterations. A different family, with one small singular value, reaches the required error level with seven GMRES iterations in total despite a very large condition number. Every inner iteration performs additional accurate work. The authors accordingly distinguish successful numerical correction from an economical implementation: enough inner iterations can erase the benefit of a cheaper factorization. These tests use MATLAB, including emulated half precision, to examine error and iteration behavior. The paper's potential speed advantage remains conditional on hardware and correction cost. [CG-W002, §§10–11, Figures 10.7 and 10.9]

Precimonious: search against an explicit acceptance test

Rubio-González and colleagues turn precision allocation into a search over program variants. The programmer supplies a program, test inputs and an accuracy requirement. Precimonious identifies floating-point variables whose types can change and, where alternatives are supplied, calls that can use another mathematical-function implementation. It generates candidate programs, compiles and runs them, and retains changes only when the results pass the accuracy and exception checks and the program satisfies the performance requirement. Their modified delta-debugging procedure tests groups of changes and progressively refines them. It seeks a locally minimal retained set of higher-precision variables rather than exhaustively finding the best possible allocation. [CG-W003, §§2–3, Figures 3–4]

The acceptance test compares a candidate's output with the original program's output on the supplied inputs. This provides a practical engineering contract, but its reference is the original implementation rather than an independently established mathematical answer. The authors' additional summation example makes the distinction consequential: truncating an infinite series and accumulating roundoff introduce different errors. Arithmetic precision can also change the number of iterations before a program stops, so apparently changing only a variable type can change the amount of work actually performed. [CG-W003, §§3.4 and 4.3]

Their experiments connect precision choices to execution cost. In the arc-length program, the reported speedup reaches 41.7% at two selected error thresholds. Several other programs improve less, while three produce no accepted faster configuration at the tested thresholds. The conversion instructions introduced between formats can outweigh savings elsewhere. A more permissive accuracy threshold consequently need not lead this search to a faster program. The FFT case with the strictest tested threshold explores 663 configurations and takes 116 minutes and 51 seconds. At publication, the suggested changes also required manual mapping back into source code, although transformed intermediate code was generated automatically. [CG-W003, §4.2, Tables 1–3; §5]

The authors test the limits of their input selection directly. They tune the NAS benchmarks using Class A inputs and subsequently try Classes B and C. Several allocations that pass the original test fail these new cases; for the EP benchmark, failures remain even at the most relaxed tested threshold. Its EP performance figure also has a specific scope: the 33.2% headline concerns the Gaussian-deviate phase that was tuned, while the reported improvement for the whole program averages 14%. [CG-W003, §4.2, Table 4 and footnote 6]

FPTuner: allocate precision under a conservative expression-wide bound

Chiang and colleagues require a precision allocation to meet an error target throughout a declared input domain. Their object is a real-valued expression with a fixed computational structure, not an arbitrary program. They model rounding at operations and conversions, use symbolic Taylor expansions to relate these errors to the expression's output, and obtain conservative bounds over the input intervals. The allocation is then an optimization problem: maximize a weighted count of lower-precision operations while satisfying the error constraint. [CG-W004, §§2–4]

The procedure includes a further check. Its allocation search initially uses first-order error terms. FPTaylor then checks the generated allocation with higher-order effects included. If the candidate violates the target, the procedure can add an allowance for those effects and repeat the search. The authors report that the candidates in their experiments passed this check immediately. Thus the fast search approximation and the conservative acceptance check perform different jobs. Bounds on conversion error are included, and conversion counts can also be constrained. Groups of operations can be assigned the same precision to support vectorization. [CG-W004, §§4.1–5, Figure 4]

Their measurements show why a rigorous accuracy constraint does not make the cost objective exact. The objective's weighted operator count is a guide to allocation, not a direct measurement of execution time or energy. On a Jetson TK1 board, the authors measure single/double-precision versions using current and voltage instrumentation. Four selected benchmarks save energy on average when moved from all-double to one mixed allocation, but other configurations consume more energy or take longer. In the jetEngine expression, mixing formats increases register pressure and causes constants to be spilled and reloaded. The executed instructions defeat an apparent saving visible in the arithmetic allocation alone. [CG-W004, §6.3, Table 5 and Figures 7–8]

A separate timing comparison replaces some software-implemented quadruple-precision work with hardware-supported double precision. Those results have a different baseline from the Jetson comparison, where both formats have hardware support. The all-double version is also insufficient for the stricter error contracts used in the quadruple/double study; its faster time cannot be treated as an accuracy-matched alternative. Tuning itself can take seconds or, in one reported case, twenty minutes, followed by the final verification. [CG-W004, §§6.1–6.3 and 7.1]

The scope of the conservative result remains explicit. FPTuner treats straight-line expressions over bounded input domains. The paper does not implement general treatment of branches and loops: rounding can change a branch decision, and bounding the resulting difference is a separate problem. The contribution is a developed allocation and checking procedure for its stated expression model, accompanied by measurements that reveal when the intended resource gain survives compilation. [CG-W004, §§7.2–7.4]

Green: calibrate the effect on the output and restore accuracy when needed

Baek and Chilimbi's Green system supports two deliberate approximations: replace an expensive function with a supplied alternative, or stop a loop before its original end. The programmer supplies an output-quality measure, a target and calibration inputs. Calibration relates the approximation to both its output effect and its performance and energy costs. Generated code then selects among function versions according to input ranges or determines when a loop can stop. In the adaptive loop policy, it also observes whether further iterations continue to improve the chosen quality measure. [CG-W005, §§2–3, Figures 1–3]

The reference computation remains part of the operating arrangement. During sampled recalibration, Green records the quality that an early termination would have produced and then continues the precise

computation so that the difference can be assessed. It can increase the work retained, select a more accurate function version, or relax an approximation that is more accurate than the target requires. For a function input range where no available approximation meets the criterion, it uses the precise function. These comparisons and changes have a sampling cost. [CG-W005, §§2.2.3 and 3.2, Figures 3 and 5–7]

Local acceptance also has to survive combination. In the option-pricing benchmark, the locally selected exponential approximation and logarithm approximation do not together meet the application's target. Green changes the logarithm choice from one polynomial approximation to a more accurate one. The resulting combined version has less than 0.8% loss under the chosen output measure, with reported performance and energy improvements of 28.5% and 28%, respectively. The authors retain the application-level comparison rather than assuming that acceptable individual substitutions add up to an acceptable program. [CG-W005, §§3.4 and 4.4, Figures 8 and 23–24]

The search-engine experiment specifies a different output criterion. It counts queries whose ordered top results differ from the precise backend's results; it does not measure human judgments of relevance. Tests use a production index and query logs, with crawling and index updates disabled. One adaptive configuration improves throughput by 21% and reduces joules per query by 14%, while 0.27% of queries produce a different ordered result. Whole-system energy is measured from the main power cable. [CG-W005, §§4.1–4.3, Figures 10–12]

An intentionally inaccurate calibration model exposes the time required for restoration. The chosen target allows 2% differing queries, but the initial cutoff typically produces about 10%. After each 10,000 queries, Green runs the next 100 precisely and uses their comparison to increase the retained search work. In this experiment it reaches the target after 180,000 queries. More frequent monitoring could accelerate adaptation while consuming more resources. The paper's claims about statistical quality rest on calibration and observed target tracking; this experiment makes the delay visible rather than establishing a bound on every query. [CG-W005, §4.3, Figure 14]

EnerJ: protect the part that must remain precise

Sampson and colleagues distinguish permission to approximate from the requirement to preserve precise computation. EnerJ extends Java with types that mark data as approximate or precise. Approximate values can use less reliable storage or operations, and approximate objects can select a supplied alternative algorithm. Unannotated data remains precise. The type system prevents approximate information from flowing into precise state without an explicit endorsement by the programmer. [CG-W006, §§2.1–2.5]

That protection includes indirect routes. An approximate condition cannot silently determine a branch that changes precise state. Array indices and lengths remain precise, and the implementation does not approximate pointers. An endorsement permits a selected approximate value to cross this boundary. It does not repair or recompute the value; the programmer accepts responsibility for how the receiving computation uses it. A checksum, for example, can be calculated precisely over data whose earlier production was approximate. Precisely encoding or storing that data preserves the selected value rather than proving it was the right value. [CG-W006, §§2.2, 2.4, 2.6 and 5.1]

The paper states a noninterference property for a core language without endorsements: changing approximate values does not change its precise state. This is an isolation property. Approximate values themselves have no quantitative accuracy guarantee. The conference article develops the language rules, states the properties and describes the implemented checker. [CG-W006, §3]

Their resource study uses a simulated approximation-aware architecture. It combines reduced floating-point mantissas, less reliable operations and storage, and reduced memory refresh. The energy model counts instruction execution and SRAM and DRAM storage; instruction fetch and decode retain their cost.

Its parameters combine earlier measurements with values explicitly identified as estimates. The model omits implementation and switching overheads, including voltage-scaling latency. The article reports potential CPU/memory energy savings ranging from 7% to 38%, not measurements from constructed approximate hardware. [CG-W006, §§4–5 and 6.1, Table 2 and Figure 4]

Application quality varies independently of that range. Means over twenty simulated runs show relatively small losses under mild approximation, while FFT and SOR lose substantial fidelity at stronger settings. The programmer also has work beyond adding declarations. For the barcode reader, the authors changed an out-of-bounds image-coordinate response to return a white pixel, then endorsed approximate coordinates where they became array indices. This adaptation made the computation more tolerant of errors; it was not performed automatically by the type system. EnerJ's developed contribution is the explicit protected boundary, with application-specific approximation and acceptance still supplied around it. [CG-W006, §§6.2–6.3, Figure 5]

The mandatory and optional task portions in S57 address another way of reducing computational work: stopping a refinement or choosing a shorter procedure while still delivering an acceptable result. That account concerns task execution and scheduling; the procedures here specify arithmetic roles, output comparisons and protected computation.

S75 Algorithm selection, adaptive sorting and computing only what is needed

Choosing a method for an instance

Lars Kotthoff's 2012 survey organizes algorithm selection around problem instances, descriptive features, a set of candidate algorithms and a performance measure. Its account of the earlier Rice framework makes the selection problem explicit: map information about an instance to a method expected to perform well on it. The desired result might be a short running time, a high-quality solution within a budget, or success on a difficult search problem. An algorithm's favorable average over one collection of instances need not make it the best choice for every member. [TD-W024]

The survey distinguishes choosing one algorithm before execution from scheduling a portfolio and from responding to observations during execution. A static selector might use instance features measured in advance. A schedule can allocate time among solvers, while an online approach obtains further information as work proceeds. These arrangements offer different ways to use uneven performance. They also require decisions about which features to compute, how expensive those computations may be, and when to use a default instead. [TD-W024]

Feature computation is itself work that can be budgeted. A predictor that needs an expensive search to describe an instance may consume the time that its choice was supposed to save. Kotthoff discusses selection and feature costs as part of assessing the system, along with the chosen algorithm's performance. The engineering question is consequently concrete: what can be learned economically about this instance, and which decision can still be changed by learning it? The histories of numerical preparation and precision in S35 and S55 concern other ways in which the representation and conduct of a computation affect its performance. [TD-W024]

Different kinds of partial order

Vladimir Estivill-Castro and Derick Wood's 1992 survey shows why “nearly sorted” needs a specified meaning. Straight insertion sort maintains an ordered prefix and inserts each new item by moving larger predecessors aside. Each such shift removes one inversion: a pair whose earlier item is greater than its later item. If there are I inversions among n items, the work is O(n + I). A sequence with only O(n) inversions can therefore be sorted in linear time by this method. The relevant quantity is the number of inverted pairs, rather than simply how many items look displaced. [TD-W025]

Consider two explanatory inputs. The sequence [2, 1, 4, 3, 6, 5, 8, 7] has four inversions and four adjacent descents. Straight insertion needs four shifts of larger predecessors. The sequence [5, 6, 7, 8, 1, 2, 3, 4] has only one adjacent descent, but sixteen inversions: every item in the first half exceeds every item in the second. It consists of two ascending runs, which a merge can combine with linear work. Generalizing the second construction to two equal sorted halves gives $n^2/4$ inversions even though only two runs need merging. [TD-W025]

A linear scan can find adjacent descents and run boundaries. That does not make it a linear-time count of all inverted pairs. Different measures expose different opportunities: few inversions suit ordinary insertion, while a few long runs favor merging. The survey develops adaptive sorting through explicit measures of disorder and bounds relative to the number of inputs compatible with a given amount of disorder. It reports Mannila's formalization and examples of algorithms designed to meet such bounds. [TD-W025]

This analysis does not contradict the worst-case comparison bound for unrestricted sorting. If a comparison procedure must distinguish all n! orders of distinct keys, its decision tree needs enough

leaves to represent them, requiring Ω(n log n) comparisons in the worst case. A restricted class can contain fewer possibilities, and an adaptive algorithm can do less work on favorable inputs while retaining a larger worst case elsewhere. Sortedness is itself recognizable in linear time by checking neighbors. The useful distinction is between the task's worst case and the work justified by the particular order already present. [TD-W025, TD-W026]

Using the key representation

MIT's 2011 linear-time sorting notes contrast comparison sorting with methods that inspect keys as data. Counting sort assumes integer keys in a range of size K. It allocates positions or buckets for those keys, scans the n records to count or distribute them, and visits the key range in order. The time is Θ(n + K); a record-carrying bucket implementation also uses Θ(n + K) space. If keys represent ages from zero through one hundred, K is small. If two keys are zero and a trillion, the same direct range allocation is unattractive despite the small number of records. [TD-W026]

Least-significant-digit radix sort instead represents a key by d digits in base b and performs a stable counting pass for each digit, from least to most significant. Stability preserves the order established by earlier passes among records sharing the present digit. For an explanatory example, sorting 21, 12, 11 by units gives 21, 11, 12, retaining the order of the two records ending in one; a stable tens pass then gives 11, 12, 21. The bound is Θ(d(n + b)). It becomes linear in n under suitable restrictions on d and b. A small key range, a bounded digit representation and a favorable probability distribution are different assumptions, each supporting different methods. [TD-W026]

Making radix sorting practical

Peter McIlroy, Keith Bostic and M. Douglas McIlroy's 1993 engineering study treats string sorting as a problem of using the bytes already present in keys. A most-significant-byte distribution forms piles by the next byte and recursively sorts each unfinished pile. Strings sharing a prefix then advance together to the next position; a completed string need not continue through later characters. This avoids repeatedly rediscovering a long common prefix through whole-string comparisons. [TD-W027]

The bookkeeping matters. Clearing or scanning every possible byte pile can dominate when only a few piles are occupied, so the authors examine ways to track the occupied range or piles. Small piles can be finished by an insertion method rather than paying the full distribution overhead again. Their “American flag” method counts pile sizes, computes their destination ranges, and cycles misplaced string descriptors into the appropriate ranges in place. Once a descriptor is placed, the displaced descriptor supplies the next item to move. This yields an in-place distribution without requiring a separate full array of descriptors, although the permutation is not stable. [TD-W027]

Recursion order also affects storage: leaving the largest pile until last limits the stack needed for the remaining work. The article combines these details with tests on actual workloads and attention to the sorting interface. Its performance gains belong to those implementations and data. The enduring substance of the account is visible in the connected construction: use a key's next byte, group a shared prefix, avoid empty-pile work, move descriptors through counted ranges, and switch methods where a smaller routine is preferable. [TD-W027]

Sampling and equal-key blocks

Jon Bentley and M. Douglas McIlroy's 1993 account of a library sort began with inputs that defeated existing implementations. Organ-pipe patterns and arrays containing only zeros and ones exposed quadratic behavior. The replacement was developed for a general interface in which a caller supplies the comparison operation and records may be expensive to exchange. Comparisons, swaps and partition

bookkeeping therefore had different costs from those in an abstract count of unit-time operations. [TD-W028]

Pivot sampling varied with size: a middle element for small inputs, a median of three for larger ones, and a pseudomedian assembled from nine samples for still larger ones. The authors chose thresholds empirically rather than presenting a single sampling rule as costless at every size. Their partitioning also gathered equal keys so that, after rearrangement, the middle block contained items equal to the pivot. Only the less-than and greater-than blocks required recursive sorting. With an all-equal input, one partitioning scan could finish the problem. [TD-W028]

For an explanatory zero/one input, choosing zero as pivot collects the zeros into the equal block and leaves only the ones on the greater side. Sorting need not repeatedly partition the zeros as if they still had unknown relative ranks. The authors' split-end partitioning implements this treatment of equals while supporting the general library interface. The paper supplies an engineering account of robustness across troublesome inputs; it should not be conflated with later introspective sorting methods that add a worst-case fallback. [TD-W028]

Choosing which runs to merge

Natural mergesort discovers ordered runs, but it still must decide which adjacent runs to combine and in what order. A merge of runs of lengths a and b costs in proportion to a + b. Repeatedly merging a long run with each small arrival can scan its contents many times. A suitable merge tree limits how often an item travels through successive merges while preserving the adjacency needed for stable sorting. This is the problem addressed by J. Ian Munro and Sebastian Wild's 2018 peeksort and powersort algorithms. [TD-W029]

Peeksort finds the run around the middle of an interval and recursively organizes work around the resulting boundaries. Powersort processes runs from left to right and assigns a power to the boundary between neighboring runs. The power records the first binary subdivision depth that separates their normalized midpoints. It represents where the boundary belongs in a nearly balanced tree without requiring the entire run sequence in advance. When a new boundary has a shallower power than a pending one, the deeper work is merged first. A stack retains the unfinished adjacent pieces. [TD-W029]

For run lengths L_1 through L_r totaling n, the authors use the entropy $H = \Sigma(L_i/n)\log_2(n/L_i)$. It reflects the distribution of run lengths: one long run and several tiny ones need not require the merge work of many equally sized runs. Their powersort analysis bounds total merge cost M by nH + 2n. The additive linear term makes the result nearly optimal in this cost model. The measured runtime also depends on the implementation of a merge, memory traffic and comparisons. [TD-W029]

Vincent Jugé's 2020 account of adaptive ShiversSort develops another economical policy. It uses levels derived from the integer parts of the base-two logarithms of run lengths and examines only the top three runs on its stack when deciding the next merge. Run-size comparisons determine whether an older pair should be combined before the newest run is allowed to remain pending. This small local view is used to control the larger merge structure. The paper separates the merge policy from preparation of short runs and from optimizations inside a merge. [TD-W030]

Number the pending runs from the stack's top, so R_1 is newest, and let each level be the floor of $\log_2$ of its length. If either of the top two levels is at least the third, the policy merges R_2 with R_3. Otherwise, if the top level is at least the second, it merges R_1 with R_2; otherwise it admits another run. At the end it merges the remaining stack. For an explanatory stack with lengths 8, 4, 4 from bottom to top, the levels are 3, 2, 2. The two top runs merge to length eight, after which the two eights merge. The older eight is not scanned once for each arriving four. [TD-W030]

Jugé states an nH + O(n) merge-cost bound for the adaptive policy and compares it with earlier guarantees. A better coefficient in an entropy bound concerns accumulated merge work, not an identical percentage reduction in execution time on every machine or input. The account shows how a compact rule operating on a few stack entries can be the subject of a substantial global guarantee. It also preserves a distinction between the historically described TimSort merge policies and later implementations adopting powersort. [TD-W030]

Adaptation during the work

The sorting design notes shipped with CPython 3.14.0, released in October 2025, connect run discovery with the work of sorting itself. The implementation detects natural runs, normalizes descending runs, extends short runs with stable binary insertion, and uses powersort to organize their merges. Its treatment of equal-key subruns allows normalization while preserving stability. These are details of this frozen release, which incorporates developments after the original introduction of TimSort. [TD-W031]

For example, [3, 2, 1, 3, 4, 5, 0] begins with a descending stretch. Reversing its first three items produces [1, 2, 3], after which the scan can extend the ordered run across the following 3, 4, 5. Recognizing order and acting on it are interleaved. There need not be a separate, completed diagnosis of the whole input before sorting begins. If a run is too short for efficient merging, binary insertion brings it to the desired working length. [TD-W031]

During a merge, galloping responds to a sustained winning streak by one run. Exponential search followed by binary search finds a block that can be copied together rather than comparing each item against the other run's next item. The threshold adapts so that useful streaks encourage further galloping and unsuccessful attempts discourage it. The implementation therefore uses information at several scales: runs in the input, their lengths and merge positions, and temporary dominance during a particular merge. [TD-W031]

A rank without a full order

Manuel Blum, Robert Floyd, Vaughan Pratt, Ronald Rivest and Robert Tarjan's 1973 selection paper proved that the item of a specified rank can be found with a worst-case linear number of comparisons. Their PICK construction sorts fixed-size columns, selects representatives, recursively finds a representative of controlled rank, and uses it to partition the original candidates. The representative is chosen so that a guaranteed fraction can be discarded. If the sought rank lies above the discarded lower part, its rank is adjusted within the remaining part; if below, the upper part becomes irrelevant. [TD-W032]

The original construction develops its guarantee through column sizes and representative positions, including a basic analysis with columns of twenty-one. Its result is about finding a rank, not arranging all items in order or necessarily supplying the fastest practical routine for a tiny requested subset. Once a cutoff rank is known, the relevant side of a partition can be retained without sorting the other side. Ordering the retained items is a further task whose size may be much smaller than n. [TD-W032]

Keeping a small extreme subset

CPython 3.14's heapq implementation gives a practical streaming alternative for a small requested k. To retain the largest k items, a min-heap of k candidates exposes the smallest retained item at its root. Each new item is compared with that boundary. If it cannot enter the retained set, it is discarded; if it can, it replaces the root and the heap is repaired. After the scan, only k candidates remain. The worst-case work is O(n log k), with O(k) retained storage, followed by O(k log k) if the retained set is sorted for presentation. For a fixed ten candidates, the scan is linear in n. [TD-W031]

The release's nlargest and nsmallest functions include useful boundary cases: a request for one uses an extremum operation, and a request at least as large as a known input size can use a full sort. The functions return their retained results in order. Their heap entries also carry an order field, preserving a defined treatment of equal keys. A custom routine that only needs the set can leave its heap unordered; it should not assume that a library function returning ordered results omitted the final sort. [TD-W031]

Local top-k results also admit a simple merge. Under a consistent total ordering, an item excluded from a partition's top k already has at least k items ahead of it in that partition, so it cannot enter the global top k. The global answer is therefore contained in the union of the local answers; another top-k selection over that union suffices. Equal keys require a shared tie rule if exact membership matters. This is a direct consequence of the retained-set rule, illustrated by disjoint candidate partitions, rather than a separate empirical performance claim. Scanning a filesystem can involve additional work to compute directory totals or identify shared files; selecting the largest measured candidates is the part explained by this algorithm. [TD-W031]

S37 Morphology, embodiment and physical computation

The body changes the control problem

The sources in this section examine how a body's shape, materials and dynamics affect behavior, perception and control. They disagree about when such contributions should be called computation.

Hoffmann and Müller: a design tradeoff, not an automatic transfer

Hoffmann and Müller's 2014 paper considers a design space extending from relatively simple bodies with model-based control to complex, compliant bodies with distributed or model-free schemes. Conventional rigid-robot engineering often simplifies the body dynamics that a controller must manage. A soft or highly compliant body supplies richer dynamics that can help produce useful behavior, but can also be difficult to model faithfully. The contribution of morphology therefore cannot be evaluated solely by counting fewer instructions in an otherwise unchanged controller. [W007, relevant definitions and body–control discussion]

The paper distinguishes exploiting physical properties to simplify control from performing a computation. It also asks whether a model is required and how difficult that model is to construct. A body well suited to a task can make the task easier, while its high dimensionality and nonlinearity make accurate modeling harder. The proposed tradeoff includes the design and knowledge problem.

Their criticism of proposed morphological-computation measures is particularly pertinent. A measure that rewards little influence by the agent's actions can favor a system that is difficult to control. A measure that rewards strong natural dynamics need not favor the body that makes the intended task easy. The authors distinguish these properties from engineering usefulness. Physical dynamics can resist a controller as well as assist it, and the desired balance depends on the task and environment. [W007]

Müller and Hoffmann: control, perception and computation are distinct contributions

The 2017 article examines characteristic cases including passive walking, self-stabilization and gecko adhesion, task-shaped visual sensing and physical reservoir computing. Its classification separates morphology that facilitates control, morphology that facilitates perception and morphology that performs computation under a defensible account of physical computation. A useful foot or leg does not necessarily calculate an output merely because its properties reduce a controller's difficulty. [W008, characteristic cases and conceptual analysis]

In a passive walker, the movement can result from the physical dynamics without an abstract computational level. In a sensing arrangement, geometry or the distribution of receptors can make task-relevant distinctions available before later processing. These mechanisms deserve explanation in their own terms. Calling them all offloaded computation can obscure what the arrangement actually does and what sensing or action remains.

Reservoir examples offer a more specific computational use of rich dynamics. Inputs perturb a physical system, and a trained readout extracts an output useful for a task. The article discusses a tensegrity body used to classify terrain, as well as arrangements with feedback and a readout producing a periodic command pattern. The trained readout, input coupling and interpretation of the output remain parts of these examples. [W008, physical-reservoir discussion]

Miłkowski: when physical assistance counts as computation

Miłkowski's abstract and opening passages argue that physical implementation matters to computation generally, and that morphology facilitating control need not itself constitute computation. A special label should not suggest that ordinary physical computing is independent of its material substrate. This challenges the language of outsourcing computation to a body. [W054, abstract and opening]

Liu and colleagues: behavior generation and morphology-based intelligence

The opening sections of Liu, Guo, Sun and Zhang's 2023 review place body, brain and environment in interaction. They organize a broader research area around behavior generation through morphology, learning-based morphology control and learning-based morphology optimization. The early sections are especially useful because they distinguish resembling an organism from actually exploiting a body's dynamics in control. A biomimetic shape can enable a new capability while still requiring a complex controller. [W011, printed pp. 1132–1137, §§1–2.2.2]

Their account of morphology uses shape, material and dynamic properties to affect behavior. The examples include passive walking and physical interaction that produces movement, with an emphasis on implicit physical processes rather than merely wearable or implanted digital devices. The review's vocabulary is broad: it often calls this contribution morphological computation. That choice is reported as the review's usage, alongside the stricter distinctions in the preceding papers.

The review also keeps learning and morphology in a two-way relation. Behavior can gather samples that support learning; learning can alter control; control can alter morphology; and optimization can change the designed body. These are different routes, not a single completed method. The wider device literature is mediated through the review. [W011, conclusion, printed pp. 1148–1149]

Chen and colleagues: controllability depends on the arrangement

Chen, Yu, Zhang and Su analyze vibration controllability in an underactuated flexible manipulator. Their example separates two questions. First they examine state controllability of the corresponding rigid system under different actuator placements, disregarding elastic deformation. Then they investigate structural vibration controllability of the flexible system. The operative variables include both joint configuration and where actuation is supplied. [W009, model, analysis and conclusion, printed pp. 391–398]

Their conclusion makes controllability conditional on configuration, placement and the vibration-controllability matrix. It also uses a minimum singular value and a modal-reachability indicator to distinguish configurations and relative difficulty. Thus fewer actuators or a flexible link do not by themselves determine whether a particular motion can be controlled effectively. The arrangement matters within the approximated model.

The paper's simulation supports comparisons within its planar manipulator example and associated generalization. The informative contribution is the explicit dependence of control capability on how physical degrees of freedom and inputs are connected. [W009]

Montúfar, Ghazi-Zahedi and Ay: capacity for embodied behavior

A Theory of Cheap Control in Embodied Systems provides a more formal meaning of cheap control than a general appeal to simplicity. Its main finite, reactive sensorimotor-loop model has fixed body and world channels. A policy maps sensor states to action distributions; the induced behavior also depends on the embodiment. Distinct policies can consequently be equivalent at the level of behavior. A controller need

not represent every possible policy distinction when those distinctions have no effect on the specified embodied behavior. [UX-W041, methods, lemma 1 and main theorem]

The authors exploit this reduction to give a sufficient approximation result for a model class of conditional restricted Boltzmann machines. The main theorem bounds a sufficient number of hidden units using the relevant sensory support and the dimension of embodied behavior. It establishes representational capacity to approximate the specified behavior, rather than a universally exact minimum controller size. The fixed channels, finite variables and reactive policy setting are part of the result.

The experimental treatment shows why estimating the relevant support is itself a substantive task. The authors sample a target behavior, estimate frequencies and prune the data before estimating support and a behavior dimension. They acknowledge that support estimation is a research problem in its own right and state that their pruning choice preceded the reported experimental outcomes. The controller-capacity calculation therefore relies on empirical estimates and a declared treatment of sampling noise. [UX-W041, experimental support estimation and discussion]

The result supplies an explanation of how an embodiment can restrict the behavior space a controller must cover. Representational capacity does not establish how quickly a learning algorithm can find the corresponding policy. [UX-W041]

Langer and Ay, 2021: a controller measure need not measure behavioral importance

In their 2021 study, Langer and Ay use simple simulated agents moving on a racetrack without touching its walls. Changing sensor length changes the relation between the body and environment. They optimize policies and analyze information flows among environment, sensors, controller and actions. This provides a controlled setting for asking how the complexity of a task is distributed across interacting parts. [UX-W022, abstract, §1.1 and §4 discussion]

Their discussion points out that the integrated-information quantity in a controller can be high even when that quantity has no behavioral relevance. Information flowing into and out of the controller must also be considered. An internal measure cannot, by itself, establish what the controller contributes to the task. The optimization can converge to different local optima depending on its starting distribution, yielding more than one reasonably successful strategy.

The discussion separates directly optimizing a policy from learning an adequate world model. A result about a prepared policy does not settle how the agent would acquire the knowledge needed to behave that way. This limitation motivates the later study. [UX-W022]

Langer and Ay, 2024: learning can temporarily require more controller complexity

The later paper makes the acquisition problem explicit. Again using simulated racetrack agents and information-theoretic measures, it adapts an em-algorithm that alternates improvement of goal-directed behavior with updating an internal world model. It compares agents whose world model is supplied with agents that must learn one, and includes split-controller arrangements that cannot integrate information in the same way. [UX-W015, selected v3 introduction, learning results and conclusion]

In the reported internal-world-model agents, successful learning is associated with initially higher controller complexity, followed by a decrease as useful body–environment interaction increases. Agents supplied with an ideal world model do not face the same prediction-learning demand. The comparison locates a possible earlier burden behind an apparently simple eventual controller.

The split internal-world-model agents predict the next sensory state less well and, on average, perform only marginally better than random movement, with relatively few successful agents. The authors take

this as support for the importance of integrated controller processing during world-model learning. They also recognize alternative explanations associated with the simplicity of the task, including the possibility that a single wall-avoidance problem demands little once an appropriate policy is available. [UX-W015, selected PDF pp. 14–18]

The paper supplies evidence about this sensorimotor loop, prediction task and learning scheme. [UX-W015; UX-W022]

Wang and colleagues: responsiveness is not sufficient for their physical-computing definition

The selected opening of Wang, Zhou, Kahak and Li's 2026 Perspective defines a physical-computing kernel in which encoded inputs evolve through physical interactions into decoded outputs, with the possibility of programming the input–output evolution. It distinguishes analog and algorithmic forms and names analog oscillators, physical reservoirs and physical algorithmic computing as strategies for soft robots. The paper explicitly treats its framework as a proposed definition rather than universal agreement. [W098, published PDF pp. 1–2]

A responsive material can change shape or properties under heat, light, magnetic fields or other stimuli. Such a response can help adaptation or simplify operation while failing to satisfy the paper's computing definition. The distinction gives a concrete test of terminology: material responsiveness alone does not establish encoding, decoding and programmability. [W098]

Xie and colleagues: count persistence as well as task achievement

The energy-accounting discussion in Xie and colleagues' article argues that task scores alone can conceal important embodied costs. Their reserve-based formulation tracks an internal energy stock, expenditure during a time step and energy intake. It treats remaining above an admissible energy boundary as a persistence condition. Energy is the accounted physical quantity; viability is a condition assessed through that account, rather than a synonym for energy. [W099, selected *Energy as a Common Currency* section]

The source points to search, movement, docking, processing, conversion and storage as possible demands associated with obtaining energy. Maintenance and persistence costs can also distinguish agents that achieve the same immediate task. Environment structure can change intake or dissipation. The proposal is consequently a call for explicit comparable accounting.

In their illustrations and discussion of open challenges, the authors expressly avoid claiming one universal energetic model for organisms and robots. [W099]

Co-design of morphology and control

Rosendo, von Atzigen and Iida built two-module robots and tested how closely they could move toward a target. Their joint search varied the orientation and offset of the modules along with the amplitude and angular offsets of their motion. In a second set of trials, they froze the body design at several points in the search and used the remaining time to tune only the controller. Controller tuning improved several of the poorer body designs. For the same design time, continuing to search both body and controller produced the higher best scores later in the trials. [W034, figs. 2–5]

Mechanical assistance, perception and control

These sources provide detailed accounts of mechanical assistance, task-shaped perception, physical computation, controller-capacity bounds, learning and physical accounting.

S110 Materials processing, degradation and failure

How a surface flaw enters the fracture question

A. A. Griffith introduces his 1921 paper through a question about filing, grinding and polishing machine parts subjected to repeated loads. He examines stresses near scratches and develops an energetic account of extending a flaw. The argument connects a change in the crack with changes in the energy of the stressed body and the cost of creating new surfaces. His controlled specimens make aspects of that relation observable. They are distinct from the machine parts that motivated the inquiry. [T12-W001]

Properties of a particular processed material

Chen, Zhang and Ritchie tested a silicon carbide ceramic whose grain boundaries were initially glassy. In the as-processed material, measured fracture toughness was 6.2 MPa√m at 25 °C and 4.3 MPa√m at 1300 °C. After annealing for 85 hours at 1300 °C, the corresponding values were 7.2 and 5.2 MPa√m. Microscopy showed that most grain-boundary films had crystallized. In the authors' account of fatigue, cracks extend between grains while intact grains bridging the crack behind its tip oppose its advance. The processing comparison comes from fracture-toughness measurements. The crack-growth observations come from the cyclic tests. [T12-W002]

Service conditions and specimens

中村眞実 examines stainless-steel weld metal in a high-pressure hydrogen setting. The dissertation describes a pre-cooler used in a 70 MPa demonstration station and gives a 77 MPa condition for hydrogen in the cooling pipe. Its earlier discussion treats other hydrogen charging and tensile tests with specified weld metals and treatments. Fracture surfaces and reduction of area supply observations under those conditions. The component establishes an operating context, while a test specimen provides a particular material, exposure and loading arrangement. [T12-W003]

Motooka and Kiuchi compare load-controlled fatigue crack growth for refractory materials in boiling nitric acid and in air at room temperature. The stress-intensity range provides the axis of comparison. They motivate the study through reprocessing equipment. The two test settings differ in both environment and temperature, so the comparison describes behavior under two combined sets of conditions. [T12-W004]

S46 General physical principles and cross-domain proposals

Physical and explanatory quantities

The sources propose physical accounts and connections across physical, biological and cognitive domains. Physical free energy, variational free energy, entropy production, assembly index and reduced descriptive dimension concern different objects and operations.

Prigogine: thermodynamic openness and internal production

Prigogine's Nobel lecture connects thermodynamics, kinetics, spatial and temporal structure, fluctuations and microscopic irreversibility. Its physical Helmholtz free energy is $F = E - TS$, using energy, temperature and entropy. This quantity belongs to the thermodynamic account; it is not introduced as a generic cost function for organizations or as the variational quantity in a later cognitive theory. [SG-W045, lecture opening, equation 1.1]

The entropy balance distinguishes exchange with the environment from nonnegative internal production. An open system can maintain or increase internal organization while exchanging matter and energy. That does not suspend the entropy balance over an appropriate boundary. A visually ordered structure is not enough to determine the whole balance, and an internal decrease cannot be treated as a global disappearance of entropy.

The expression of production through fluxes and forces uses local-equilibrium assumptions. The minimum-entropy-production result requires narrower conditions, including a near-equilibrium linear regime with appropriate constant coefficients. The lecture does not justify extending that result to every open steady state or every self-organizing process. These distinctions are central to what the physical source can contribute. [SG-W045, opening thermodynamic development]

Structure can appear while entropy production increases

The Bénard-convection discussion provides an explicit countercondition to the idea that self-organization always minimizes production. Beyond a threshold, the system develops macroscopic flow structure while entropy production increases. The order of the pattern and the direction of the production change therefore answer different questions. [SG-W045, convection example, PDF p. 5]

Near equilibrium, thermodynamic stability arguments can use a restricted relation between fluctuations and restoring behavior. Farther away, kinetics and the sign of excess-production terms become important. An autocatalytic reaction can supply a necessary feature for instability in the setup being discussed without being sufficient for it. Naming positive feedback does not establish that a transition occurs or that its result is beneficial. [SG-W045, PDF pp. 6–9]

The relevant lesson is source-specific: the operating regime and reaction mechanism determine which result applies.

The Brusselator retains maintained reservoirs and reaction conditions

Prigogine's chemical-clock model makes its supporting arrangement explicit. In the stated simplified reaction scheme, external species are held constant and rate constants are normalized. The homogeneous stationary concentrations are $X = A$ and $Y = B/A$. Under the model's conditions, the threshold $B > 1 + A^2$ marks instability of that branch and the emergence of oscillatory behavior. [SG-W045, selected connected chemical-instability development]

These equations do not describe an isolated chemical device sustaining itself without resources. Maintaining the external species is part of the model. Adding diffusion introduces possibilities of spatial pattern, with system size, shape and boundary conditions affecting what occurs. A repeating chemical pattern can be generated through the arrangement and kinetics without a central agent specifying every oscillation, while its material and energy provision remain.

The lecture also distinguishes behavior along a branch from selection among branches near a bifurcation. Fluctuations and history can matter to which continuation occurs. Its discussion of relative fluctuations retains the conditions behind large-number approximations; nonlinear reaction behavior and critical correlations need not follow the simplest independent-particle picture. [SG-W045, PDF pp. 10–15]

Irreversibility is an argument with restrictions and open questions

The later lecture examines the relation between reversible microscopic descriptions and irreversible macroscopic behavior. Prigogine discusses alternatives involving representation and coarse-graining, then develops his own operator-based approach. The proposed transformation is conditional on properties of the dynamics and is not claimed to be possible for every periodic motion. [SG-W045, PDF pp. 16–21]

One apparent printed sign error in equation 7.8 conflicts with the nonnegative quantity defined earlier. The conclusion retains open questions and a proposal for relating levels of temporality. [SG-W045]

Bruers, Maes and Netočný: the variable and reversal symmetry matter

The selected 2007 preprint studies entropy-production principles in linear electrical circuits through Langevin descriptions and dynamical fluctuation theory. It relates minimum and maximum principles to the close-to-equilibrium structure of fluctuations, including the parity of the relevant observable under time reversal. The choice between a current-like odd variable and an even variable is not an interchangeable labeling choice. [W079, selected setup and §§2.2–3.3]

The discussion explains a known counterexample and qualifications concerning unequal temperatures. Its conclusion questions exact extension beyond the linear regime and argues that dynamics, not thermodynamics alone, must justify further generalizations. These are results and interpretations within the selected circuit analysis. [W079, selected conclusion]

Bejan and Lorente: flow access is their proposed general principle

Bejan and Lorente's selected constructal account proposes that finite flow systems persisting over time evolve configurations offering greater access to their currents. It treats branching flow arrangements across animate, inanimate and engineered settings as instances of a design phenomenon. The authors distinguish the proposal from a fixed final optimum or an inevitable end configuration. [W082, selected §§1–4]

Their engine-and-brake discussion also resists reducing every case to maximizing or minimizing one entropy-production quantity. Flow can involve useful delivery and dissipation in different parts of an arrangement. Their proposed generality therefore differs from the claim that nature always chooses the least resistance.

Bejan and Lorente extend their proposed law through scaling arguments. [W082, selected passages and conclusion]

Friston: variational free energy concerns a model of observations

The opening and Box 1 of Friston's 2010 review present variational free energy in relation to a generative model and an approximate density over hidden causes. The formulation provides a tractable bound related to the surprise of observations, with perception and action interpreted through changes affecting that bound. Its energy and entropy terms are defined in that probabilistic construction. They are not simply the E and TS of the thermodynamic equation above. [W052, opening and Box 1]

Biehl and colleagues, and the response: a scoped mathematical dispute

Biehl, Pollock and Kanai examine particular arguments connecting stochastic dynamics, Markov blankets and an inference interpretation. The selected critique identifies additional assumptions needed for some mathematical steps and gives counterexamples to a free-energy lemma taken at face value. It also questions the relation between a variational density and an ergodic conditional density, and notes that changing what parameterizes a density changes its interpretation. [W055, selected overview, observation discussion and lemma counterexample]

Their critique targets the mathematical arguments linking dynamics with inference, while leaving the wider free-energy programme open.

Friston, Da Costa and Parr's selected response distinguishes conditions on coupling, exact and approximate inference and flow on a manifold of conditional expectations. In their explanation, an average internal flow can be interpreted on a statistical manifold; an average flow on that manifold differs from each fluctuating internal trajectory. They defend the broader programme while addressing the assumptions highlighted by the critics. [W056, selected §§1–2 and 4–6]

The dispute concerns which assumptions support a Bayesian interpretation.

Assembly theory: a construction measure with specified building blocks

Sharma and colleagues define an assembly space through elementary building blocks and permitted recursive joining operations. Previously assembled motifs remain available for reuse. The assembly index is the length of a shortest permitted construction path for an object; copy number adds information about how often a complex object occurs. The proposal links construction history, reusable structure and selection rather than describing complexity only by visual intricacy. [W084, selected printed pp. 321–325 and 327]

The measure depends on justified building blocks and allowed operations. A construction count is not automatically a person's cognitive burden or the full physical cost of making a molecule. The selected paper proposes empirical access, including molecular measurements.

Jaeger's selected critique distinguishes what assembly theory may characterize from what would be required to explain Darwinian evolution. [W085, selected discussion]

Constructor information: physical possibility includes alternatives

Deutsch and Marletto's selected opening frames information through possible and impossible physical tasks. A constructor can perform a task repeatedly while retaining the ability to do so; a substrate undergoes the relevant transformation. The account emphasizes counterfactual alternatives: an actual state carries information in relation to other states it could have occupied. [W091, selected §§1–2, printed pp. 1–6]

The authors seek physical principles accounting for properties such as interoperability of information media. They explicitly present general principles as conjectures. The task-based formulation characterizes which repeatable operations are physically possible. [W091]

Wilson and Flack: a coarser description can have a specific explanatory role

Wilson's renormalization-group lecture examines rescaling, fixed points and universality near critical conditions. Certain large-scale observables can become insensitive to some microscopic details under the relevant limiting description. The condition of being near the critical temperature and the class of system remain important. Universality is not an assertion that all physical systems behave alike or that their detailed design can always be ignored. [GL-W025, selected introduction and PDF pp. 10–11]

The account supplies an asymptotic physical explanation. Describing fewer relevant variables is different from showing that the actual system uses less energy or needs fewer controls.

Flack discusses endogenous coarse-graining: organisms or collectives can respond to relatively slow regularities, with feedback between aggregated descriptions and lower-level activity. The Santa Fe Institute's reconstruction presents this as collective compression and a possible account of downward influence. [TA-W041, through the Santa Fe Institute]

Multitime cognition and System 0

Taniguchi and colleagues' selected quad-process proposal relates embodied dynamics, individual cognitive processes and language-based collective intelligence across time scales. It calls the collective symbolic layer System 3 and seeks to place it alongside Systems 0, 1 and 2. The account connects several research traditions and a philosophical interpretation of multiple rhythms of duration. [W051, selected §§2.2–2.4 and 3.1]

Collective predictive coding remains a hypothesis at this scope.

Haken, catastrophe theory and Bios: proposed transfers retain their prerequisites

Haken's *Can We Apply Synergetics to the Human Sciences?* and his related 2017 article develop the conceptual account discussed in the shared-practice section. [W083, selected pp. 58–66; RGC-W038]

Golubitsky's selected opening to catastrophe theory distinguishes mathematical exposition from the difficulty of applying its assumptions. Classification of qualitative change in a mathematical setting does not automatically validate a social or biological application. Heuristics and proofs are kept distinct. [SG-W046, printed pp. 352–354]

Sabelli's *Bios Theory of Innovation* opens by relating recurrence and novelty within a proposed account of creative processes. The opening defers the empirical foundations of the proposal. [WME-W015, opening pp. 1–3]

Physical measures and cross-domain proposals

The sources supply physical balances, operating regimes, construction measures, interpretive disputes and proposed connections across domains.

Part 3 Living, ecological, and flow arrangements

These accounts address development, production, ecological persistence and representations of flows.

S28 Development, plasticity and constructed environments

Arrangements have developmental and evolutionary histories

These selected sources explain how stable responses, variable responses and environmental modifications arise or persist. Development within an individual, evolutionary change across generations and immediate ecological effects remain different processes.

Waddington: buffered development is conditional

Waddington's 1942 article proposes canalization: a developmental reaction can produce a fairly definite outcome despite minor variation in the conditions along its course. The account includes tissue differentiation and the comparative constancy of wild-type outcomes. Such buffering is not absolute. Large abnormalities, late disturbances and pathological conditions can defeat it. [W032, 563–564]

He distinguishes an environmental influence that selects a developmental path from an influence participating in the processes that maintain it. These are conceptual alternatives rather than a rigid description of every real organ. Depending on how the alternatives are specified, the same influence can appear as a switch or as a modification within a path. The organism's developmental organization is part of the explanation of a response to the environment. [W032, 564]

The ostrich-callosity illustration supports a proposed evolutionary argument. If an environmental stimulus initially elicits a useful response, selection can favor the genetically influenced capacity to respond appropriately. If a relatively definite result is advantageous across variation in stimulus, canalization can develop. A different trigger, including a genetic one, can later activate the path. This is not inheritance of an acquired bodily change by assumption; it is a proposed sequence involving selection on developmental responsiveness. [W032, 563–565]

Timing and prior capability remain necessary. A replacement trigger needs to act before the environmental stimulus has already done the work for it. The developmental path must also be available and able to produce the response. Waddington's “normal” or “optimum” outcome belongs to this historical selection argument. [W032, 565]

Niche construction can leave a usable environmental legacy

Odling-Smee and colleagues' selected practical guide distinguishes alteration of the environment from subsequent evolution in response to the altered conditions. Nests, burrows, webs, decomposition and changes to nutrient or other material flows can modify what later organisms encounter. Ecological inheritance names a legacy that persists and affects descendants' conditions, rather than genetic transmission alone. [TA-W020, PDF 3–8]

The distinction makes persistence and boundaries important. A constructed feature can decay, be counteracted or affect other organisms as well as the constructor. It can alter population dynamics without causing the evolutionary response attributed to it. Conversely, a lasting environmental change can modify selection after the original constructor has gone. The passages describe resource types, signatures and durability as matters to investigate. [TA-W020, PDF 3–8, 13]

The authors advocate recognizing niche construction as an evolutionary process in its own right. That is their theoretical position, not a settled conclusion established by the mere existence of environmental modification. Construction can also reduce fitness or produce effects negligible for the evolutionary question. A changed environment is not necessarily a better one for every participant. [TA-W020]

Outsourcing memory: a specified trade-off model

Lee, Flack and Krakauer's selected revised model balances noisy present observations against memory that can become outdated. Their environment has a specified binary structure and persistence time. Longer memory can improve an estimate during a stable period while making adjustment after a switch slower. A useful memory duration consequently depends on the modeled environment and sensing conditions. [TA-W019, revised v2, model and results 1–2]

In the slow-environment regime examined, the authors derive a sublinear relation between optimal memory duration and environmental persistence under stated approximations. The result concerns their model and objective. Their separate metabolic estimate adds assumptions relating neural tissue, memory burden, size and lifespan. It is a model-based estimate. [TA-W019, results 2–3]

Active stabilization adds another option. Making the environment more persistent can reduce the new information required for adaptation. Their examples include persistent trails, caches, writing and maps. The relevant mechanism is that a maintained external regularity changes what an agent must continually estimate or encode internally. The construction and maintenance of that regularity remain activities. [TA-W019, result 4]

The model also exposes a distribution problem. Stabilizers pay for changing conditions, while other agents can enjoy the more predictable environment. Free riders can therefore exploit the benefit. The authors examine conditions involving monopolization and combine assumed costs of memory complexity, stabilization and sensing precision with adaptation error. The relative weights matter: changing them changes which strategy is favored. [TA-W019, result 4, equations 15–18]

Their local adaptive argument uses perturbative and convexity approximations. The discussion connects the model to institutions and social regularities as hypotheses about outsourced memory. [TA-W019]

The niche-construction dispute is about implications

Scott-Phillips and colleagues' selected critical appraisal is an adversarial collaboration between an advocate and skeptics. The authors agree that organisms modify environments and that those effects can be consequential. They disagree about whether this requires a fundamental reformulation of evolutionary theory, how to define an evolutionary process and what explanatory role construction should receive. It is not a dispute over whether burrows or altered selection environments exist. [TA-W021, introduction and paired accounts]

Their dairy-farming example makes the distinction concrete. The standard account treats the changed nutritional environment as a source of selection on variation in lactose persistence. The niche-construction account emphasizes how the cultural practice generates and coevolves with that selective setting. The paired explanations and summary identify differences in emphasis and causal framing. [TA-W021, case study and summary table]

For gambits engineering, the informative point is the source's distinction between a demonstrated environmental feedback and a stronger claim about the reconstruction of theory. An arrangement's practical consequences can be examined while the wider theoretical interpretation remains contested.

Michod: group performance can constrain its members

Michod's selected 2003 account asks how a group becomes a new evolutionary individual with heritable variation in fitness at the group level. Cooperation can benefit the group while imposing costs on participating cells. That creates potential conflict between levels. A conflict mediator changes the

opportunity for selection within or between groups, rather than merely announcing that the cells have a common interest. [TA-W023, PDF 1–2]

The simple model uses cooperating and defecting alleles and a life cycle from propagule to adult group and further propagules. Group fitness depends on adult size and functionality; cell replication can increase one while impairing the other. These assumptions make the level of the measure explicit. A faster-replicating cell and a more effective multicellular group need not be the same success. [TA-W023, PDF 2]

Propagule size, reproductive specialization, mutation processes and policing provide different possible mediators. A small propagule can reduce within-group conflict but also directly reduce adult group size under the model's conditions. Germ–soma specialization constrains reproductive opportunity for many cells while supporting the larger organism's functioning. Policing can require cooperating cells to spend time and energy monitoring defection. Conflict control is an activity or developmental constraint, not an automatically costless outcome of membership. [TA-W023, PDF 5]

Michod also notes that a mediator can restrict later evolvability, illustrated by a nonreplicative soma in Volvox. Greater current individuality is therefore not a guarantee of unlimited future potential. Cooperation and conflict in multicellularity also frame the abstract of Michod and Roze's earlier 2001 article. [TA-W023, PDF 2, 5; TA-W022]

Costs of responsiveness require a careful comparison

Van Buskirk and Steiner distinguish selection on the phenotype expressed in an environment from costs associated with the capacity for a plastic or canalized response. The mechanisms making a range of responses possible can have consequences even when a particular organism does not express the whole range. Plasticity and canalization can each be targets of cost comparisons; neither is presumed to be the cost-free baseline. [TA-W037, abstract and results/discussion]

Their meta-analysis reports twenty-seven studies and usually mild measured costs, with no simple general predominance of plasticity over canalization costs. Weak detected differences do not establish zero absolute costs. Shared costs, statistical detection and environmental stress remain relevant explanations in the account. [TA-W037]

The critical discussion in Auld, Agrawal and Relyea's article identifies another problem: correlation between trait values and plasticity can make estimated costs imprecise or biased. Bias can itself be environment-specific. They call for re-examining existing estimates, rather than concluding that all past estimates must be false. The passages also distinguish limits involving the range of developmental responses and the reliability of environmental information. [TA-W038, discussion of costs, bias and limits]

Its significant contribution here is a warning about what a comparison actually measures. The cost of producing one phenotype, the cost of retaining a response capacity and the failure to achieve an appropriate response are related but distinguishable questions. [TA-W038]

Developmental resources extend beyond the conventional organism

Griffiths and Gray's selected introductory argument rejects a division in which genes form one fundamental kind of developmental cause and everything else forms another. Genes, cytoplasm and features of a learning environment are among interacting resources. Their proposed developmental unit comprises the resources producing outcomes stably replicated in a lineage, potentially extending beyond the conventional body boundary. [SG-W054, PDF 1–3, printed 277–278]

The parity claim is about reasoning and explanatory status, not equal numerical contributions from every resource or the absence of genetic specificity. Reproducing a developmental process still requires the relevant resources to be present and related appropriately. It identifies a broader boundary for asking how a recurring outcome is produced. [SG-W054]

S40 Biological organization, closure and autopoiesis

Several meanings of closure

The selected sources concern systems that produce, constrain or maintain conditions of their own activity. The term closure has different meanings in the sources: dependence among constraints, a production organization, closure of a reaction set and mathematical conditions for self-maintenance. None of these means that an organism or chemical system needs no external matter or energy.

Montévil and Mossio: constraints and the processes they shape

Montévil and Mossio distinguish thermodynamically open processes from constraints acting on them. A constraint is a contingent structure or dynamics that changes the process while relevant aspects of the constraint remain conserved over the time scale considered. Its presence alone is insufficient: the effect must differ from the unconstrained case. Their vascular-system example concerns oxygen carried through a structured transport system rather than simply diffusing in its absence. [W013; GL-W026, 2015 article, §§2–3, definition 1]

The time-scale qualification matters. Conservation of a relevant property during a constrained process does not mean the structure lasts forever or never needs repair. A relatively persistent arrangement can shape many faster events while its longer-term production and maintenance depend on other processes. This is how the framework avoids treating every constraint as a permanently external boundary condition.

The paper's example of an inclined plane clarifies the difference. The plane constrains a sliding object, but that sliding does not normally produce or maintain the plane. Biological organization, in their proposed account, involves constraints whose continued existence depends on processes constrained by others in the organization. The system uses constrained thermodynamic flows in maintaining the constraints themselves. [W013, §§3–5]

Dependence, boundary and closure are specified rather than inferred from a cycle

The authors ask whether any circular chain is enough and answer by defining dependence more carefully. Their framework distinguishes direct dependence from an indirect contribution through a longer chain of processes. Strict closure requires the relevant mutual dependence among constraints, rather than merely reciprocal benefit in a broad environment. Constraints outside the closed set and specified boundary conditions can still be necessary. Closure is consequently contextual, while thermodynamic openness continues underneath it. [W013; GL-W026, §§4–6]

The upper and lower boundaries pose an explicit problem for the framework. Descriptions of actual organisms are incomplete. The authors propose considering maximally closed systems within the available description, then examining how interacting systems can retain distinguishable organization. Their hunting-and-cooking example treats some collective dependencies as indirect and warns that loose cooperation need not satisfy strict closure. A team doing mutually useful work is therefore not automatically a biological closed organization under their definitions.

The authors state that their purpose is to characterize general features rather than supply a model capturing all the complexity of real organisms. Their later discussion retains the relation between stability and variation. [GL-W026, §§5–7]

The abstract of Mossio, Montévil and Longo's 2016 *Theoretical principles for biology: organization* proposes organization as a principle for making biological objects intelligible and relates it to variation. [GL-W091, abstract]

Corti: causal conditions do not by themselves settle what ought to happen

The selected part of Corti's critique examines a counterfactual argument for biological normativity. In the organizational account as he reconstructs it, a constraint must perform its role or the mutually maintained system will cease to exist. A further inference assigns what the constraint ought to do. Corti questions whether the causal premise is sufficient for that normative conclusion. [W047, counterfactual argument]

His comparison with a simple dependence relation makes the issue concrete. A condition can be necessary for an outcome without thereby acquiring a normative obligation to produce it. He then asks whether making the conditions internally generated and mutually dependent changes that inferential step. He accepts that closure can mark an important descriptive difference while disputing that the difference, on its own, establishes intrinsic norms.

Dittrich and Speroni di Fenizio: chemical closure is a property of a reaction set

Chemical Organisation Theory begins with molecules and reaction rules. A closed set contains the products generated by reactions available within it. This is a different meaning from a physical seal around a container. The theory also distinguishes semi-self-maintenance, in which consumed types can be produced, from the stronger quantitative condition needed to maintain amounts. [W077, author preprint dated 1 December 2006, §§2.2–2.3]

The authors' small reversible example is instructive. Two molecular types produce one another, but both also decay. Having a route that produces each type is not sufficient to keep either at positive concentration. A diagram of reciprocal production can therefore conceal net depletion. Their self-maintenance condition uses the stoichiometric matrix and the existence of a compatible flux vector for which the relevant net production rates are nonnegative. The stipulated positivity of available reactions is part of that definition.

A chemical organization combines closure and self-maintenance in this formal sense. Its existence identifies a structural possibility under the reaction rules and flux conditions. It does not, without further kinetic information, establish that actual concentrations will approach a stable persistent state. An available feasible flux is different from the flux generated by a particular reaction-rate law and initial condition. [W077, definitions 5–11]

The later selected theorem relates fixed points of the specified dynamics to organizations under a mapping from concentrations to present molecular types. Threshold and reaction-positivity assumptions enter the construction. The direction reported is that the appropriately mapped stationary state is an organization, not that every formally feasible organization is a stable attractor. [W077, §3.2, definition 41 and theorem 42]

Varela, Maturana and Uribe: a production organization that realizes its boundary

The early sections of Varela, Maturana and Uribe's 1974 paper characterize an autopoietic system through a network of component-producing processes. The components participate in generating the network that produces them and in realizing the system as a unity in a domain. Its boundary is consequently part of an ongoing production organization, rather than merely a wall placed around a collection of parts. [W081, printed pp. 187–191, §§1–7]

The distinction between organization and a particular material realization allows turnover without making the system an unchanging stock of molecules. The selected model and figures illustrate the proposed organization. Their role is conceptual demonstration in a specified construction. Reproduction is also not identical to the maintenance of such an organization; a system can be described in terms of its organization without making reproduction the definition of each operation.

Marletto: a recipe requires an executing and maintained constructor

Marletto's selected constructor-theoretic argument asks what tasks the laws of physics permit, including accurate construction and reproduction under laws that do not contain a ready-made design for the task. Possibility is defined through arbitrarily good approximate constructors short of perfection. This differs from asking how probable a task is under observed evolution, or how much a particular implementation costs. [W092, §§2–3.1 and selected conclusions]

Her car-factory example separates a task-specific recipe from a more general vehicle that carries out its constituent operations. The recipe can be decomposed into subtasks permitted by the laws, with a modular programme specifying their organization. The task-specific information must be instantiated and preserved; the executing machinery must also remain capable of carrying out the operations. A stored recipe is therefore part of an architecture, rather than an alternative to physical execution.

Marletto connects accurate constructors, replication and maintenance in an architecture that joins a reusable recipe to machinery capable of executing it. [W092]

Hopfield: specificity can require an additional driven path

Hopfield's 1974 kinetic-proofreading paper asks how biosynthetic processes can discriminate more accurately than a simple equilibrium recognition scheme would suggest. In the model, correct and incorrect substrates differ in their dissociation behavior. An additional reaction stage, driven away from equilibrium, creates an effective delay before incorporation. The more rapidly dissociating incorrect complex has more opportunity to leave before the final product forms. [TA-W017, printed pp. 4135–4137, reaction schemes 1–10]

The driving is essential in his analysis. Within the equilibrium constraint, merely adding the intermediate does not supply the desired improvement. Coupling to an energy source, such as phosphate hydrolysis, changes the available kinetic organization. The effective delay is a physical reaction mechanism, not a decision by an external inspector. Reactions that appear superfluous in a simpler description can serve a discrimination function.

Hopfield relates the mechanism to reported biosynthetic examples. The model shows why fidelity can retain additional pathways and free-energy supply. [TA-W017]

Galstyan and colleagues: spatial proofreading relocates part of the work

The selected spatial-proofreading account uses gradients and spatial separation to create a discrimination opportunity. A bound complex must persist long enough during movement or diffusion to reach the region where product formation occurs. Different dissociation rates can then yield different success probabilities. The arrangement supplies a spatially organized delay rather than requiring the same local chemical architecture as the classical scheme. [TA-W018, selected model and gradient-maintenance passages]

The authors make the boundary of energy accounting explicit. An initial reported power calculation concerns the additional cost caused by proofreading's tendency to homogenize a gradient; it excludes the baseline cost of creating that gradient. Appendix B separately examines the continuing power needed to

counter diffusion even without the proofreading enzyme. [TA-W018, energy-cost discussion and appendix B]

The more realistic model describes a membrane-associated kinase and a distributed phosphatase maintaining an active-substrate gradient. At low enough enzyme activity, the gradient can persist and support proofreading. If activity overwhelms the maintaining mechanism, the gradient flattens and proofreading is lost. The selected parameter examples use experimentally constrained ranges, but the study remains a conceptual model rather than a demonstration that every proposed biological implementation uses the mechanism. [TA-W018, gradient-maintenance model]

Spatial organization can alter where discrimination work occurs, while successful use still depends on maintaining the spatial difference.

Synthetic Notch: a selected modular sensing–response account

The summary and introductory material from Morsut and colleagues describe synthetic Notch receptors with replaced extracellular sensing and intracellular transcriptional modules. Binding an external ligand triggers release of a regulator, connecting a selected input to a selected cellular response. Their account reports pathways that do not share common signaling intermediates and can be combined to integrate cues, including Boolean response programmes. [GL-W060]

The engineered sensing interface uses receptor processing, transcriptional machinery and cellular resources to execute its response. [GL-W060]

Hypercycles: cooperative organization with vulnerabilities

The selected 2016 PLOS Topic Page defines a hypercycle as a cyclic autocatalytic arrangement of self-replicating molecular types, each catalyzing a successor. Its historical motivation concerns an error threshold: imperfect copying can prevent a long sequence from retaining the information needed for more accurate replication. Coupled shorter replicators were proposed as a way of organizing a larger informational capacity. [SG-W047]

The review retains difficulties in the traditional ordinary-differential-equation model, including parasites and limits to stable cycle size. It also distinguishes an experimentally reported cooperative ribozyme network from a hypercycle in the strict sense. [SG-W047]

Maintained organization, productive closure and fidelity

The constraint framework describes a conserved causal role and mutual maintenance; chemical organization adds reaction-set and feasible-flux conditions; autopoiesis concerns component production and realized unity; constructor theory examines task possibility and recipes; proofreading makes fidelity and supporting work concrete.

S50 Biological circuits, cellular organization and intervention

From components to cellular behavior

Biological circuits provide concrete cases of organization producing a capability: a temporary signal can leave a persistent expression state, a regulatory loop can generate repeated pulses, and the allocation of cellular machinery can change what an introduced construct achieves. Systems biology examines such behavior through molecular interactions, measurements and models. Synthetic biology also constructs circuits to investigate and use it. These traditions contain pertinent historical foundations for asking how a capability is produced, what sustains it, and what an intervention changes.

The accounts below move from experimentally investigated circuits to metabolic constraints, whole-cell modeling and selected recent computational approaches. Their objects include living cells, mathematical representations of cells, and the conduct of biological research. The experiments show particular realized behaviors; the models and reviews explain the questions, assumptions and methods through which those behaviors can be investigated.

Kitano on structure, dynamics, control and design

Hiroaki Kitano's 2002 overview identifies four aspects of system-level understanding: structure, dynamics, control and design. A list of components supplies a different kind of knowledge from a network of interactions, and an interaction diagram supplies a different kind from the behavior of that network over time. His road-map analogy makes the distinction tangible: the map helps locate possible routes, while explaining traffic requires information about movement and interaction. Control asks how a state can be changed or stabilized; design asks how systems with desired properties can be constructed. [BC-W001, p. 1662]

The research cycle he describes makes a model's assumptions available for testing. Competing explanations can guide the choice of an experiment that distinguishes them, and disagreement with observation can require revision of the model. The needed measurements depend on that purpose: broad coverage, adequate temporal sampling and simultaneous observation of different molecular properties can matter differently. Kitano's account thus joins modeling to the organization of experimental work. He presents expanded control and design capabilities as ambitions supported by continuing advances in measurement, analysis and software. [BC-W001, pp. 1663–1664]

Alon on response time and persistence detection

Uri Alon's 2007 review explains selected regulatory patterns through their dynamics and experimental investigation. Negative autoregulation occurs when a transcription factor represses production of itself. A strong promoter permits rapid initial accumulation; as the repressor approaches its threshold, it reduces further production. Relative to simple regulation reaching the same final concentration, the circuit can respond faster. The steady concentration and the route taken to reach it are separate properties. Alon also discusses reduced variation between cells and the possibility of noise amplification when feedback contains a long delay. [BC-W002, pp. 450–451]

A coherent feedforward circuit gives a different example. X activates Y, and both activate Z. When activation of Z requires both X and Y (an AND rule), it waits for Y to accumulate sufficiently. A brief input can disappear before that happens, whereas a persistent input permits expression. Turning X off can stop new Z production promptly even though Y remains. Alon describes experimental tests in the arabinose system and contrasts OR-like regulation in flagella, where either input can suffice and the delayed

transition changes. Timing therefore depends on the regulatory signs, the input rule and molecular kinetics as well as the wiring. [BC-W002, pp. 452–453]

Gardner, Cantor and Collins construct cellular memory

Gardner, Cantor and Collins constructed a toggle from two genes whose protein products repress one another's promoters. When one side dominates, it suppresses production of its opponent and thereby sustains its own predominance. A temporary inducer weakens the active repressor, allowing the opposing side to accumulate and establish the other state. After the inducer is removed, mutual inhibition can retain the selected expression state. Their experiment thus separates the signal used to change state from the continuing gene expression that maintains it. [BC-W003, pp. 339–340]

The two arrows are only part of the design. Their model requires sufficiently cooperative repression and appropriately balanced effective synthesis rates. Six constructs shared the same basic configuration, but one had only one stable expression state. In one construct, the two culture states persisted for twenty-two further hours without inducer, under the reported growth conditions. These particulars explain what has to be arranged for a temporary input to leave a persistent cellular response. [BC-W003, box 1 and figs. 2–4]

Elowitz and Leibler construct an oscillator

Elowitz and Leibler's 2000 repressilator connects three transcriptional repressors in a cycle: LacI represses tetR, TetR represses cI, and CI represses lacI. Their design model permitted either convergence to a steady state or sustained oscillation, depending on expression, repression and decay parameters. To favor oscillation, the construction used tightly repressible promoters and degradation tags that shortened repressor lifetimes. Cellular protein destruction participates in producing the temporal behavior. [BC-W004, pp. 335–337]

A separate fluorescent reporter allowed the researchers to follow individual cell lineages. They observed oscillations lasting longer than a cell-division cycle and substantial variation in amplitude and period, including skipped or delayed cycles. At least forty percent of cells met their oscillation criterion in each of three movies. A population average would conceal important aspects of this behavior as cells lost synchrony. The experiment supplied a working synthetic oscillator whose irregularity was itself a subject for further investigation. [BC-W004, pp. 335–338]

Adaptation and a programmable sensing interface

The chemotaxis account in S05 supplies another relation between organization and response. Yi and colleagues reconstruct exact adaptation through integral feedback: an internal state associated approximately with total receptor methylation changes with the difference between a methylation contribution and activity-dependent demethylation. At a stable steady state, its rate of change is zero. Under the specified biochemical assumptions, activity returns to a value independent of the changed ligand level. The continuing receptor-modification reactions support that response. [W012]

S05 explains the four biochemical assumptions and the stability requirement in detail. This is a response that returns an activity measure after a changed input, whereas the toggle experiment selects and retains an alternative expression state. The distinction gives a reader two concrete behaviors to investigate without using persistence as a single undifferentiated outcome. Their mathematical descriptions identify different states and different conditions for the reported result. [W012; BC-W003]

The selected synthetic Notch account in S40 describes replaced extracellular sensing and intracellular transcriptional modules. Ligand binding can release a regulator and connect a chosen external cue to a

chosen transcriptional response. The sensing interface is engineered while receptor processing and cellular execution remain involved. S40 develops the sensing interface and combined-cue response programmes. This example places an intervention at the connection between a signal and an existing cellular apparatus. [GL-W060]

Ceroni and colleagues measure the host's remaining capacity

Ceroni, Algar, Stan and Ellis examine the burden imposed by introduced gene-expression constructs. Their capacity monitor uses a gene integrated into the bacterial chromosome that produces a fluorescent protein without an inducing signal. Its production rate provides an indirect assay of expression capacity remaining as another construct draws on cellular resources. Construct output, the monitor's output and cellular growth can then be compared. An introduced function enters a cell that already has demands on its expression machinery. [BC-W006, supplementary figs. 1–3]

Their translational resource model represents ribosomes moving along transcripts and returning to a shared pool. Stronger recruitment and slower elongation can change how much machinery is occupied. In the selected supplementary analysis, changing coding-sequence length or ribosome-binding strength changes both predicted output and remaining capacity. Tests with shorter antibody-fragment constructs captured important trends, while the model overestimated burden in some conditions. The comparison makes the object of improvement explicit: obtaining a desired product while retaining more host capacity. Its diagnostic is a measured reporter response supported by a specified model, rather than a direct inventory of every resource in the cell. [BC-W006, supplementary note 4 and figs. 9, 13–15]

Orth, Thiele and Palsson on feasible metabolic flux

Orth, Thiele and Palsson's 2010 primer explains flux balance analysis through a stoichiometric matrix S and a vector of reaction rates v. Each matrix column records what a reaction consumes and produces. At steady state, Sv = 0 balances the represented metabolites. Upper and lower reaction bounds restrict the allowable rates, including exchange with the environment. These conditions define a feasible set that can contain many flux distributions without requiring a kinetic rate constant for every reaction. [BC-W005, box 1 and figs. 1–2]

Selecting a growth or production objective adds another commitment. Linear programming chooses a feasible distribution maximizing or minimizing a specified linear combination of fluxes. Biomass production can be represented by a reaction consuming measured proportions of precursor materials. The primer illustrates environmental intervention by changing oxygen-uptake bounds, and discusses alternative optima when several pathways attain the same objective. Standard FBA describes steady-state fluxes; it does not determine metabolite concentrations or automatically include regulatory dynamics. Feasibility, the selected objective and the additional biology needed for prediction remain distinct elements of the method. [BC-W005, From constraints to optimizing a phenotype; The many uses of flux balance analysis]

Karr and colleagues integrate a whole-cell model

Karr and colleagues' 2012 model of Mycoplasma genitalium divided cellular function among twenty-eight process models. Different mathematics served different processes, including flux balance for metabolism and stochastic descriptions for degradation. Integration used shared cellular variables and an approximation of process independence over short intervals. Submodels advanced using values determined at the preceding step. The model's architecture therefore included an explicit computational means of reconnecting processes initially developed separately. [BC-W007, pp. 389–391, fig. 1]

The study also used discrepancies to direct experiments. In its examination of twelve gene-disruption strains, two-thirds of predicted growth rates were consistent with the measurements. The lpdA disruption was particularly informative: a colorimetric assay initially failed to detect growth, while an alternative doubling-time measurement indicated viability. The authors interpreted the combined evidence as severe but noncritical impairment of growth. The authors investigated such disagreements to refine parameters and functional assignments. Their whole-cell construction supplies a historical example of connecting heterogeneous representations to experimental correction. [BC-W007, Model-Driven Biological Discovery]

Maintaining processes and realizing their relations

The cellular-resource account has a useful neighboring question in the Rosen material in S26: how are the processes on which operation depends themselves produced and repaired? Louie's reconstruction distinguishes a transformation's output from production of the machinery performing that transformation. Siekmann examines the proposed closure of these productive relations and disputes what can be inferred about their biological realization from the abstract maps. The disagreement concerns an identified step between a formal construction and the organized material process it is intended to describe. [W104; W103]

S40 develops other source-specific accounts of maintained organization. Chemical organization uses closure and feasible-flux conditions for reaction sets. Hopfield's kinetic proofreading adds a driven intermediate stage that gives an incorrectly bound substrate another opportunity to dissociate. The spatial-proofreading account instead examines gradients and movement, including the work needed to maintain the spatial difference. These accounts locate particular requirements inside their mechanisms: productive closure, sustained chemical driving, or a maintained gradient. Their full expositions and qualifications remain in S40. [W077; TA-W017; TA-W018]

Development adds a further historical route. Waddington's canalization account in S28 concerns a fairly definite developmental outcome despite minor variations along its course. His argument distinguishes a trigger selecting a path from conditions participating in its maintenance. The niche-construction accounts consider changes to environments that can outlast the activity producing them. Those sources direct attention beyond an instantaneous input-output relation to the history and persistence of the conditions under which a biological response occurs. [W032; TA-W020]

Zuidberg Dos Martires and colleagues on representing reaction conditions

Zuidberg Dos Martires and colleagues' 2024 position paper argues that knowledge representation and automated reasoning can help connect molecular knowledge to system-level questions. Explicitly listing every combination of a protein's site states can generate an enormous microstate space; representing only whole components can omit distinctions that observations resolve. The authors favor descriptions separating reactions from the site-specific conditions on which they depend. The representation preserves locally meaningful detail without beginning by enumerating every complete configuration. [BC-W011, §§3–4]

Their signaling illustrations distinguish deterministic Boolean simulation from probabilistic reasoning. Adding probabilities can express uncertainty about reactions but introduces a harder inference problem. They propose knowledge compilation and related reasoning methods to address it, with model structure supplied by biological knowledge and parameters learned from data. Their claim that this integration is necessary for precision medicine is their position; the quantitative programme remains a proposed direction alongside the worked illustrations. S42 provides the separate Darwiche–Marquis account of choosing a compiled representation for the operations it must support. [BC-W011, §5; TA-W008]

Soltani and colleagues distinguish an association graph from an assembly

Soltani and colleagues' 2026 methodological review distinguishes several meanings of an edge in a protein-interaction network. Direct binding, recovery in the same purified complex, co-expression and shared localization supply different evidence. Proteins can be associated without touching, and interactions observed at different times can enter the same static graph. A dense cluster can therefore fail to describe a physically possible assembly. [BC-W010, §§2, 4]

Their proposed response brings cellular conditions and molecular interfaces into detection and validation. Compatible locations and binding surfaces qualify a candidate; a plausible pairwise interface still does not show that every proposed subunit can assemble simultaneously. They also identify a risk of circular evaluation when functional annotations help generate a prediction and are reused to validate it. Their recommendations concern evidence integration and comparable tests. [BC-W010, §§4.5–4.6, 6, 8, 11]

Li and colleagues on learned flows and biological trajectories

The flow-matching survey by Zihao Li and colleagues describes learning a vector field that transports a simple probability distribution toward a distribution of biological data. In the continuous formulation, generation follows a learned ordinary differential equation from an initial sample. The flow parameter orders the generation process; its biological interpretation depends on the modeling task. Geometric choices can matter when ordinary straight-line interpolation passes through states unsuitable for the represented data. [BC-W009, Background; Flow Matching Methods]

Selected applications concern cell trajectories and spatial transcriptomics. The review discusses reconstruction from sparse, noisy observations and methods that retain aspects of expression geometry or tissue context. A generated path links modeled observations under learned assumptions; testing a biological transition requires evidence appropriate to that interpretation. The survey's evaluation discussion identifies heterogeneous tasks, datasets and constraints as obstacles to comparable assessment. These passages explain choices in building a computational representation, including the space in which movement is modeled and the conditions by which generated samples are judged. [BC-W009, Dynamic Cell Trajectory Prediction; Cellular Microenvironments from Spatial Transcriptomics; Evaluation Tasks and Datasets]

Li and colleagues on cellular predictors and research agents

Krinos Li and colleagues' 2025 virtual-cell survey separates direct cellular predictors, called oracles, from agents organizing scientific tasks. Their cellular-modeling discussion includes representation, perturbation prediction and gene-regulation inference. The agent discussion concerns planning, tool use and the division of research work. A predictor of cellular response and an agent arranging an analysis occupy different places in the resulting research process. [BC-W008, §§2–3]

The distinction makes their unresolved questions concrete. Predictors must generalize to unfamiliar cellular contexts and make uncertainty intelligible. Agents must sustain coherent work across tool calls and experimental stages, with suitable evaluation of the completed task. In its single-agent architectures, one model manages the workflow, sometimes switching roles through prompts. Multi-agent architectures divide responsibilities among specialized planners, analyzers or executors that coordinate through dialogue or shared memory. Its taxonomy and surveyed examples concern selected approaches to virtual-cell research; the evaluations it calls for remain part of establishing what these approaches can reliably accomplish. [BC-W008, §3.1 and Conclusions and Future Directions]

Reading routes through the biological foundations

For a reader interested in building a particular response, the switch and oscillator accounts supply constructions to study alongside Alon's timing mechanisms. The capacity-monitor account adds a way to examine effects on the host. The metabolic primer makes environmental supply and chosen objectives explicit, while the whole-cell model supplies a worked architecture for relating several processes and confronting their combined predictions with observations. The recent accounts address the representation of reaction conditions, evidence for assemblies, learned biological representations and the conduct of research with AI.

The neighboring sections retain further detail: S05 on adaptation and robustness; S26 on production, repair and realization; S28 on development and environmental legacies; S40 on closure, proofreading and synthetic Notch; S42 on representations and reasoning; and S47 on linked flow accounts. These routes allow the particular intervention or modeling question to determine which historical account the reader studies next.

S43 Ecological resilience and environmental arrangements

Persistence and an acceptable outcome are different questions

The selected sources explain how ecological and environmental arrangements persist, how their behavior can be modeled and how physical or biological processes are used in practice. The accounts distinguish persistence from constancy and from desirability.

Holling: resilience is not rapid return to equilibrium

Holling's 1973 argument begins with two different views of ecological behavior: whether relationships persist at all, and how constant the numbers of organisms remain. A system designed to operate narrowly around a target may be evaluated by fluctuations and correction speed. An ecological system repeatedly encountering external changes can require a different question: can its relationships continue despite changes in variables and parameters? [GL-W067, selected introduction and synthesis]

His synthesis distinguishes resilience, concerning persistence while absorbing change, from stability, concerning return toward an equilibrium after a temporary disturbance. Under these definitions a system can fluctuate substantially and remain resilient. A relatively constant population can be vulnerable to an unusual event outside the range it has experienced. The definitions are not interchangeable labels for generally good performance. [GL-W067, IIASA edition, printed pp. 17–18, PDF pp. 19–20]

Holling reports historical examples and model interpretations. His management discussion contrasts maintaining a predictable yield with preserving a domain within which the system can persist. Interventions that narrow variability can also change the conditions determining vulnerability. The argument concerns those conditions.

Persistence of a harmful state remains persistence under the definition.

Response diversity includes differing reactions, not just a species count

The Stockholm Resilience Centre synthesis distinguishes redundancy in a function from diversity in how contributors respond to disturbances. If several entities perform a similar function but fail under different conditions, their differing responses can help sustain it. Merely counting more entities does not show that they supply that property. [GL-W068, selected response-diversity and redundancy account]

The synthesis also recognizes management demands and costs. Diversity has to be understood in relation to the function, disturbances and boundaries considered.

Cury and colleagues: remain within constraints rather than maximize one target

Cury, Mullon, Garcia and Shannon present viability theory as a modeling approach for ecosystem-based fisheries. Instead of choosing only a single quantity to maximize, the account asks which trajectories can remain within a set of constraints over time. The viability kernel identifies states from which admissible evolution is possible under the model and available controls. A state can satisfy today's limits yet fail to belong to that set if no permissible continuation can keep the constraints satisfied. [UX-W018, viability principle and modeling tools]

The source makes operational specification necessary. Relevant states, dynamics, indicators, reference values and available actions must be described, with data and computation adequate to the problem. In its fisheries discussion, agreement about avoiding collapse, extinction or displacement may be easier than

agreement about one ideal ecosystem target. The constraints still embody choices and can conflict. The southern Benguela application is presented as an initial attempt, not a completed universal solution.

A viable trajectory satisfies the modeled constraints; unmodeled disturbances can change whether those constraints remain attainable. Cury and colleagues apply Aubin's viability theory to the fisheries problem. [UX-W018]

FAO precaution: objectives, limits and the ability to observe them

The selected FAO fisheries guidance distinguishes desired targets from undesirable outcomes that operational constraints should avoid. Such limits may concern stock, age or spatial distribution, ecosystem effects and bycatch. The document also relates these choices to what can be measured and to the time needed to adjust. Approaching a target too quickly can overshoot a limit. [GL-W055, selected chapter 3, paragraphs 19–44]

Data sufficiency, uncertainty, community alternatives, reversibility and wider effects remain relevant. A rule can be intelligible while its monitoring and adjustment still require institutions and skill. This is context-specific prescriptive guidance.

Beven: predictive agreement can leave explanatory alternatives

Beven's equifinality essay questions the assumption that calibration identifies one uniquely correct environmental model. Multiple structures or parameterizations can fit available observations acceptably. Imperfect measurements, model inadequacy, scale and incomplete knowledge can leave several explanations or predictive representations in play. A good fit to selected data is therefore not identical to identifying the system's causal structure or parameters. [GL-W015, selected §2, PDF pp. 3–5]

Beven distinguishes a response adequate for a task and a uniquely determined explanation. Additional observations can constrain alternatives, but their relevance and the criterion of acceptability matter. The source proposes a more serious treatment of multiple acceptable models and developments of GLUE, while leaving research questions open.

Agroecology: a framework and a limited performance account

The FAO ten-elements framework brings diversity, synergies, efficiency, recycling and resilience together with knowledge, human values, culture, governance and economic relations. Its relevance is the breadth of the transition being described. Resource circulation or biological complementarity is not presented in isolation from people who learn, govern and maintain the arrangement. [GL-W010]

The institutional summary of Mouratiadou and colleagues' 2024 review reports varied socioeconomic outcomes, including positive economic or productivity results alongside greater labor requirements and costs in some human-capital outcomes. The practice and metric make a difference. A favorable farm-income result does not imply less labor under every arrangement. [GL-W011]

The reported percentages classify heterogeneous outcomes; they are not the size of a farm-level causal effect. Temporal aspects, magnitudes, success conditions and system-level transitions remain questions in the summary. The useful source information is the coexistence of benefits, costs and metric-dependent results.

Forest adaptation: compare strategies while continuing observation

The Adaptive Silviculture for Climate Change programme describes experimental comparison across resistance, resilience, transition and a no-action control. The source relates management choices to

uncertain future conditions and financial, social and ecological tradeoffs. Its sites include continuing observations of tree survival, growth, regeneration and wildlife. [GL-W054, selected programme orientation]

This is the programme's own description of intended comparison and monitoring. The informative point is the combination of alternative arrangements and an observation programme, with their ongoing demands retained.

The selected FAO forest-operation guidance makes local terrain, soil, equipment and training relevant to extraction. Planning needs operational control and adaptation when conditions differ from the plan. A suitable pattern on paper is not sufficient if the actual equipment, soil or operator cannot realize it. [GL-W085]

Controlled traffic: geometry must match equipment and remain usable

The 2019 USDA requirements describe concentrating repeated machinery travel into consistent lanes to reduce compaction outside them. This requires compatible equipment widths and records of the arrangement. Guidance is needed when lanes cannot be seen, and ruts require attention. The prepared spatial pattern changes where repeated disturbance occurs rather than abolishing machinery movement. [GL-W084, implementation PDF, printed p. 4]

Their relevance lies in the conditions that sustain the arrangement: matching geometry, reliable positioning and repair. A lane drawn once does not ensure that later machinery stays within it or that the land remains suitable indefinitely.

Constructed wetlands: the treatment medium still has to be established and maintained

The EPA manual addresses construction, startup, operation and maintenance of wetlands used for wastewater treatment. A functioning arrangement uses water, plants, microorganisms and its physical setting. Lower continuing energy or less continuous on-site labor does not mean no construction, startup, land requirement, flow-distribution problem or expertise. [GL-W076, selected chapter 6, §§6.1–6.2]

The separate 2012 facilities guidance describes cases in which a wetland can provide hydraulic retention together with a treatment habitat. It also calls for experienced design. Those are specific complementary roles and conditional cost claims. Periodic maintenance remains, and a treatment function must still be evaluated against the required water outcome.

Passive mine-water treatment: chemistry and flow set the conditions

The BLM technical-note overview presents passive treatment as an alternative for removing metals from acid mine drainage, with selection depending on chemistry and discharge flow. Its description of lower power and maintenance demands is scoped to suitable passive arrangements. Different systems or combinations address different effluents; the word passive is not itself a performance specification. [GL-W078]

Monitoring, inspection and potentially substantial periodic work remain, together with institutional financing.

Sensor-based sorting: reject material before further processing

The abstract of Robben and Wotruba's review identifies sensor-based ore sorting as a way to separate material before later processing stages. The relevant proposal is that an earlier discrimination can reduce

the amount entering costly downstream treatment. The sensing and rejection machinery become part of the arrangement. [GL-W077]

Rejecting material can also discard something valuable, and the value of avoiding later processing depends on detection and recovery performance.

Hurdle technology: combinations can change microbial responses

Leistner's review discusses combining preservation factors in relation to microbial homeostasis, metabolic exhaustion and stress responses. The account examines the combined effects of preservation conditions rather than relying on one factor alone for discrimination or suppression. The account also retains the possibility that stress responses complicate the intended preservation effect. [GL-W079]

Consolidated bioprocessing: integration creates a compatibility question

The selected 2005 review summary describes integrating enzyme production, hydrolysis and fermentation in cellulosic-biomass processing. Its two engineering routes concern developing suitable organisms or adding the required capabilities to organisms already useful for another stage. Combining functions can avoid some separations, but the functions still have to coexist under compatible biological conditions. [GL-W086]

This was a research prospect at the source's date, not evidence of current commercial economy. Feedstock pretreatment, biological resources and wider process compatibility are not accounted for merely by naming the integrated step.

Compartmentalisation: a biological boundary includes pathways and oversight

The 2018 animal-health code defines compartmentalisation through functional separation and a managed biosecurity arrangement, rather than geographic distance alone. Infrastructure, supplies, transport and other pathways matter to the boundary, and surveillance and responsible authority remain necessary. An enclosure on a map is not enough to establish the functional separation described. [GL-W056, historical chapter 4.4, selected articles 4.4.1–4.4.3]

The source is a historical standard. Its relevance is a specified support boundary and the records, monitoring and relationships required to maintain it.

The selected dairy-production welfare chapter addresses systems and outcome-based welfare criteria. It prevents productivity or labor saving from being treated as a complete welfare measure. Outcomes depend on conditions and thresholds, and a production arrangement must be examined in relation to the animals' welfare rather than only its output. The account follows the 2018 code. [GL-W057, selected articles 7.11.3–7.11.4]

Recreation and flood protection: intended benefits do not end the lifecycle

The IFAS planning account treats nature-based recreation as serving several benefits and stakeholder criteria. Planning requires identifying what outcomes matter and monitoring whether an arrangement provides them. A common physical setting can be used differently by different people, so an administrative count or the mere presence of a facility need not capture its value. This is a practice framework. [GL-W039, planning framework]

The US Army Corps of Engineers programme material places flood and coastal-storm risk management within operation, maintenance, repair, replacement and rehabilitation. The separate Engineering With

Nature rationale proposes using natural processes within that broader undertaking. The programme statement and funding rationale do not demonstrate the effectiveness of any one feature. [GL-W090]

Persistence, uncertainty and environmental design

The ecological accounts distinguish persistence from return to equilibrium and allow several models to fit the available observations. The practice accounts explain how material flows, spatial arrangements and biological processes are organized for particular environmental purposes.

S107 Chemical process arrangements and integration

Targeting heat recovery before choosing equipment

Truls Gundersen's primer describes a history from process development through unit operations to attention to interactions between operations. Heat recovery gives a concrete reason to study those interactions. The network with minimum utility use, fewest pieces of equipment or lowest total annual cost can differ because these criteria value different features of the design. The primer describes how a more detailed distillation model can change the estimated opportunities for heat integration. The model supplies temperatures and duties on which the estimate depends. [T09-W001]

In the first of their 1978 papers, Bodo Linnhoff and John Flower use a problem table that divides the heat exchange problem into temperature intervals. Its utility requirements can be calculated before specifying the network. The authors' first four-stream example shows how subnetworks respecting those requirements can be joined by moving duties from paired heaters and coolers to new process exchangers. Neighboring units can be merged where the resulting connections allow it. Their five-unit alternative in Fig. 6 gives up some heat recovery and has a higher overall cost in that example. The number of units, heat recovery and annual cost give different descriptions of the same alternatives. [T09-W005]

Evolving a feasible network

The second paper begins with a feasible design, preferably one with maximum heat recovery, and examines changes that promote properties such as low cost or easier start-up. In case 4SP2, the problem table specifies permissible subnetwork utility loads for the recovery objective. The authors form new matches from paired utility duties and merge units. Their later alternatives show roles for both a cyclic topology and parallel stream splitting. The cycle can serve the recovery objective, while splitting can allow a reduction in unit count. The designs and comparisons are worked calculations for stated streams. [T09-W002]

Ryu and Maravelias use a discrete temperature grid to incorporate streams with variable temperatures and flow rates in an optimization model. The utility-targeting objective concerns the cost of hot and cold utilities. A network objective adds the annualized costs of exchanger area and units. In one solution, stream s6 splits at 290 K to exchange heat with s5 and s4. A later split at 344 K supplies matches with s1 and s3. The split is part of a computed configuration whose heat balances and admissible matches are expressed on the temperature grid. [T09-W003]

Changing a crude-oil preheat train

Li and Chang's retrofit design starts with an existing crude-oil preheat train and restrictions on reusing equipment. Their fired-heater target moves from 200 to 166.2 million Btu per hour, a reduction of 33.8. They distinguish the process pinch from a utility pinch associated with steam generation. The specified minimum temperature difference is 70 °F between process streams, 35 °F against steam and 30 °F against cooling water. A proposed match must be judged against the appropriate difference. [T09-W004]

The authors prioritize large cross-pinch duties and propose new connections that retain parts of the old network. For the proposed retrofit, Table III projects a fired-duty reduction of 26.08 million Btu per hour, about 77 percent of the target, and recovery of an additional 18.6 million Btu per hour for generating 120-psig steam. [T09-W004]

S47 Networks, flows and competing information measures

The sources examine support, connectivity and statistical dependence through accounts of maintained capacities, linked budgets, alternative paths and predictive relations.

MuSIASEM: maintained funds and linked flow budgets

Giampietro, Mayumi and Ramos-Martin distinguish funds, whose identity is maintained during the accounting interval, from flows transformed during that interval. People, capital and land provide different kinds of fund; their continued availability requires maintenance and renewal. A rate of energy use per hour of human activity is consequently a relation between a flow and a fund, rather than an account of energy alone. Their account explains how descriptions at household, production and whole-society levels can constrain one another. What is feasible at one scale cannot be chosen independently of the other accounts. [SG-W017]

The authors call this a “Sudoku effect”: linked totals and component allocations restrict the combinations that can consistently be entered elsewhere. Human time allocated to production is unavailable for other activities, while production must support the activities and maintained funds outside its immediate boundary. A locally attractive intensity or productivity figure can therefore conceal an allocation that cannot reproduce the larger system. This is a consistency argument, not a claim that the equations select one uniquely desirable future. The distinction matters when an apparent saving in one production compartment depends on maintenance or supporting work located in another. [SG-W017, accepted manuscript §§1–2]

Categories are chosen before calculation. The narrative about what a system is and does determines which funds, flows and levels enter the representation. The authors expressly allow this narrative to fail: a fund may be unable to handle its proposed flow, cannot be maintained throughout the interval, or requires flows inconsistent with boundary conditions. They treat such failures as reasons to revise the representation. A fixed taxonomy is especially problematic for a system becoming something different, rather than merely changing its quantities within unchanged categories. [SG-W017]

The paper extends the account toward ecological constraints through land-use funds and densities of energy and material flows. The paper distinguishes areas whose flows are controlled by human activity from areas represented through ecosystem self-organization, while acknowledging that it cannot fully develop that application there. The model constrains linked budgets. [SG-W017]

Emergy: proposed accounting for paid and unpaid support

Campbell, Brandt-Williams and Meisch’s West Virginia report proposes accounting for support that market payments can leave invisible. Its selected methodological pages define emergy as energy of one kind previously used, directly and indirectly, to produce a product or service. Transformity supplies a conversion factor between that accumulated supporting energy and a resulting quantity. The intended account includes ecological inputs as well as purchased inputs, so the production boundary extends beyond the immediate enterprise and its monetary accounts. [SG-W018, preface; §§1–2.2]

The proposed common basis depends on the specified transformations, system boundary, input data and treatment of co-products. An accounting convention does not, merely by assigning comparable numbers, demonstrate an equivalence among energy, knowledge and economic value. Allocation of shared inputs determines how co-products enter the supporting-energy account. The report itself calls its state account a first step because important subsystems remain outside the analysis. The report was published in 2005. [SG-W018]

Q-analysis: connectivity at different dimensions

Smirnov, Kurkin and Hramov describe a Python package for Q-analysis and applications including network physiology. In the definitions, a simplex represents a set of incident elements. Two simplices are q-near when they share a q-dimensional face; q-connectivity extends the relation through a chain of such adjacencies. Increasing q makes the required overlap more demanding. The first structure vector records the number of q-connected components at successive dimensions. Connectivity is therefore a graded property of the chosen complex, rather than a single yes-or-no attribute of a graph. [SG-W022, selected definitions and discussion]

This representation can distinguish arrangements that simpler summaries obscure. A large complex connected through single shared vertices can fragment when the analysis requires shared edges or higher-dimensional faces. But the choice of incidence data and the construction of simplices determine what those differences describe. A clique extracted from pairwise observations does not itself demonstrate a causal interaction that requires the entire group. The authors characterize the core framework as exploratory and descriptive, useful for comparison and hypothesis generation. [SG-W022]

Reconstructability analysis: a simpler predictive distribution

Shervais, Kramer, Westaway, Cox and Zwick explain reconstructability analysis through decompositions of categorical probability distributions. Suppose A, B, C and D are inputs and Z is an output. A structure written ABCD:ABZ preserves the joint input distribution and the relation of A and B to Z, while not retaining every dependence in the saturated distribution ABCDZ. Maximum-entropy composition of the selected projections yields a reconstructed distribution. Its conditional distribution can then be used to predict Z. The question is whether a simpler specified structure retains useful information about that output. [SG-W053]

The authors express predictive information as uncertainty reduction, for example $H(Z) - H(Z|AB)$, relative to a stated independence model. Normalizing by $H(Z)$ gives a percentage uncertainty reduction when that denominator is appropriate. This is a defined statistical comparison, not a measurement of energy saved or an automatic causal interpretation of a gene–gene relation. “Independence” also depends on whether the model is directed toward an output or treats all variables symmetrically. The model’s statement must be recovered before its simplicity or information score is interpreted. [SG-W053]

The method passages distinguish loopless feature-selection structures from models with multiple predicting components and loops. Loopless calculations are simpler and faster; loops can require iterative computation. Exhaustive searches over candidate structures can become prohibitive. Thus the representation may reduce a prediction problem while leaving model search, estimation and validation as substantial work. [SG-W053]

Ulanowicz and colleagues, 2009: what the flow identities say

Ulanowicz, Goerner, Lietaer and Gomez begin with nonnegative directed flows T_{ij} and total throughput T. Normalizing gives probabilities $p_{ij} = T_{ij}/T$, together with sender and receiver marginals. Their joint entropy H describes uncertainty in the joint flow distribution; mutual information X describes statistical dependence between origin and destination. The remainder $\Psi = H - X$ is the sum of the two conditional entropies, $H(\text{origin}|\text{destination}) + H(\text{destination}|\text{origin})$. It is not one conventional conditional entropy. These quantities require consistent support, a common logarithm base and the usual zero-probability conventions. [WME-W025, printed pp. 27–29]

Multiplying by throughput yields the paper’s extensive measures $C = TH$, $A = TX$ and $\Phi = T\Psi$, with $C = A + \Phi$. The authors call A ascendency and interpret it in terms of organized constraint or efficiency; Φ is

associated with overhead and reserve. The identity is an accounting decomposition of observed flows. An information-weighted throughput is not automatically a thermodynamic conversion efficiency, a stock of spare equipment, or a measured ability to recover after a disturbance. Those interpretations require further evidence about the system represented. [WME-W025]

The distinction between averages and individual events also matters. Independence maximizes expected joint entropy at fixed marginals. It does not maximize the surprisal of every possible joint event. An individual term $\log[p_{ij}/(p_{i\cdot}p_{\cdot j})]$ can be negative, although the average mutual information is nonnegative. Statistical symmetry between origin and destination is not a causal version of Newton's third law. The paper's discussion of absence, constraint and autocatalysis supplies an interpretation of the equations; these broader philosophical assertions are not all consequences of the entropy identities. [WME-W025, pp. 28–29; WME-W024]

The food-web example and the conditions on fallback

The 2009 article uses a simplified network involving prawns, turtles, fish, snakes and alligators. It isolates a small set of flows from a larger web and deliberately makes extreme changes for illustration. Concentrating modeled flow along fewer paths can increase the indicated throughput while removing an alternative route that would matter if fish failed. The example explains why a present flow arrangement and its apparent efficiency can fail to reveal a dependence on a particular intermediary. It is a didactic counterfactual, not an observed disturbance experiment. [WME-W025, p. 30]

The account retains the inputs, conversions and dissipation that sustain the organisms. Its reserve measure is expressly not a palpable stock. An alternative path is useful only if the organisms or facilities on it remain viable, have capacity, can perform a substitutable function, and can act within the disturbance's time scale. Statistical diversity alone does not establish those properties. The authors also discuss autocatalytic concentration that entrains resources and can starve outsiders, so increased internal organization need not benefit everything affected by the network. [WME-W025; RGC-W042]

The proposed optimum and the critical preprint

The article proposes a family of fitness functions of relative ascendency $\alpha = A/C$, with a positive chosen parameter β. In natural-log notation, the normalized form is $F = -e\,\alpha^{\beta}\ln(\alpha^{\beta})$; robustness also includes throughput. The maximum at $\alpha = \exp(-1/\beta)$ follows from that chosen curve. It is not a theorem identifying the fraction that maximizes actual survival in every network. The article acknowledges limited data, a heuristic balance, a nonuniversal β, an arbitrary bound used in its illustration, and sensitivity to resolution. It leaves transfer from ecology to economics open. [WME-W025, pp. 31–35]

Trancón y Widemann and Bogner's critical preprint targets the formal presentation. The preprint was submitted in 2012 and posted in 2016; rejection without review is the authors' report, not evidence of peer-reviewed acceptance. The critics identify the two-conditional-entropy issue, problems with maximum-surprisal reasoning, ambiguity between source and receiver, and confusion between intensive information measures and throughput-scaled quantities. They expressly do not infer that the ecological observations or philosophical possibilities should be rejected wholesale. [WME-W024]

One consequential problem is the original use of dimensional A and Φ as exponents in transformed connectivity and effective-size measures. The intensive X and Ψ, with compatible logarithm conventions, are required for that construction. A further numerical example does not produce the printed relative ascendency from its stated connectivity and size. Nonnegative conditional entropy also does not establish that an entire network is connected: disjoint deterministic branches can satisfy the same inequality. [WME-W024; WME-W025, pp. 31–32]

The critical source itself requires discrimination. Its conversion between bits and nats is printed in the wrong direction: one bit is ln(2) nats, about 0.693, while one nat is about 1.443 bits. Loose statements about the convexity of logarithms also do not replace the relevant entropy argument. Local derivatives with respect to independently varied flows present another limit: real interventions may have to preserve mass balance and satisfy capacity, timing and material constraints. A sensitivity formula is not a feasible redesign procedure or a convergence proof. [WME-W024; WME-W025, eq. 18]

Goerner, Lietaer and Ulanowicz: the economic extension

The related economic article argues that development depends on the organization and diversity of flows, not simply on aggregate economic throughput. It discusses enterprises at different scales, human capital, concentration, local circulation and the possibility that autocatalytic growth drains resources from surrounding participants. This is the authors' proposed economic synthesis. [RGC-W042, pp. 76–81]

The transfer assumes that a flow framework can apply across matter, energy, information and money. The article does not supply a new measured economic network dataset with out-of-sample survival validation. Its comparisons of local spending and its historical interpretations of trade, banking, food control and empires remain attributed claims. Equity, human capability and the triple bottom line also involve normative commitments not derived from $C = A + \Phi$. [RGC-W042]

There are additional quantitative discontinuities. A footnote identifies sustainability with capacity C, rather than the earlier throughput-scaled fitness function. C alone does not have that function's claimed interior optimum in relative ascendency. The article specifies natural logarithms while repeating numerical information values from the bit-based example without the corresponding conversion, and a redrawn flow differs while the stated total remains unchanged. Calling mutual information "efficiency" and overhead "resilience" does not by itself measure energy efficiency or recovery. These defects limit the promise of a single established economic sustainability metric while leaving the conceptual argument available for study. [RGC-W042, p. 78 and fig. 2]

Ulanowicz, 2020: a narrower balance conjecture

The 2020 article changes notation: A and Φ denote intensive quantities, and $H = A + \Phi$. A reader carrying over the 2009 extensive meanings would misread the argument. The article reports 17 articulated networks clustering near relative ascendency 40 percent, selects an optimum of 0.401 with $\beta = 1.0914$, and explicitly expects optima in other domains to differ. This is a narrower empirical conjecture associated with a chosen fitness curve, not retrospective confirmation of a universal 46 percent rule. [RGC-W043, pp. 1–6]

Plotting networks on $F(\alpha)$ does not supply an independent measurement of their survival. Resolution, boundaries and which surviving systems entered the dataset matter. An oyster-reef example adds a hypothetical flow along a specified route; it is not an observed intervention. The printed derivative also loses a minus sign relative to the fitness function. Recommendations to adjust flows according to sensitivities still require a feasible physical intervention and do not establish dynamic convergence or real recovery. The article's metaphysical claims about indeterminacy and life exceed what the entropy identities demonstrate. [RGC-W043]

These related articles and their criticism put explicit quantities behind a consequential structural proposal. [WME-W024; WME-W025; RGC-W042; RGC-W043]

Double contingency: expectations about expectations

In the introduction and §9.1 of his 2021 chapter, Leydesdorff reconstructs the problem of an actor expecting how another actor will respond, while that other actor also forms expectations. An observed event and expectations about possible events have different roles: an expectation can orient a present selection before the anticipated event occurs. His chapter develops its own anticipatory modeling vocabulary from this reconstruction. [WME-W021, pp. 175–193]

Coordinated communication involves expectations about others, which differ from the existence of a network edge or an information total. For Leydesdorff's related simulation account, see S49. [WME-W021, introduction and double-contingency exposition]

Funds, flows, connectivity and expectations

The distinct units here remain explicit: time and maintained funds, supporting-energy accounts, incidence structures, categorical distributions, flow-weighted information, and expectations in a social model. Each source account identifies what its representation includes and what further interpretation it needs.

Part 4 Technical architecture, integration, and assurance

These accounts concern interfaces, design, distributed computation, safety, technical objects and assurance.

S07 Modularity, transactions and platforms: the work of arranging interdependence

Technology and organizations

Carliss Baldwin's *Design Rules, Volume 2: How Technology Shapes Organizations* develops a connected account of how technical arrangements affect organizations, economic relationships and opportunities for innovation. It begins with tasks and transfers, moves through the design of interfaces and transaction boundaries, and then examines complementary investments, production bottlenecks, platforms and software communities. [UX-W040, chs. 1–4, 17]

The relevant economy is often a reduction in the communication or negotiation required for particular work. But the book also explains why organizational ties remain necessary, why integrated production can outperform dispersed production, why firms may need knowledge beyond what they make, and how a focal firm's favorable results can involve costs borne elsewhere. [UX-W040, chs. 2–3, 5–7, 11–12, 14–17]

Parnas: choosing what a module hides

David Parnas asks which criteria should govern decomposition of a system into modules. In the selected sections of his 1972 article, the important alternative is between dividing a program according to successive processing stages and dividing it according to design decisions that other parts should not have to know. A module is assigned responsibility for a decision liable to change; its interface allows other modules to use the relevant service without depending on the concealed implementation. The resulting economy concerns the spread of design knowledge and the reach of subsequent changes. It does not follow merely from giving different groups different pieces of a flowchart. [W004, “The Criteria, ” “Conclusion”]

Parnas's comparison uses a keyword-in-context indexing program. A decomposition corresponding closely to processing stages can expose choices about representations and intermediate structures to several modules. The alternative arranges responsibility so that such choices can change with less disturbance elsewhere. Even an interface can reveal too much: requiring a particular ordering of circular shifts, for example, commits users to an implementation-related decision that they may not need. The question is what another module must be entitled to assume. Once an unnecessary assumption becomes part of the interface, changes propagate through that assumption. [W004, decomposition comparison and “The Criteria”]

He keeps two further issues separate. A hierarchical relation among modules does not establish that the modules were decomposed well; hierarchy and effective information hiding are distinct properties. Implementation efficiency also requires attention. A conceptually attractive interface can impose costs if realized in an unsuitable way. These programming examples therefore support an account of limiting particular dependencies and change propagation. [W004, “Efficiency and Implementation, ” “Hierarchical Structure”]

From technical recipes to networks of tasks and transfers

Baldwin starts from technology as knowledge of a demonstrated way to serve a human purpose. A technical recipe specifies actions and their relationships. Its implementation requires agents capable of performing the tasks and the transfers between them. Transfers can concern material, energy or information. Depicting only transformation tasks while omitting the actions needed to move things between them would leave out part of the recipe. This representation directs attention to what must actually happen for a result to be obtained. [UX-W040, chs. 1–2]

The task network can be represented in a design structure matrix. Tasks occupy corresponding rows and columns; entries record declared dependencies. Densely connected blocks and the connections between blocks give a more informative picture than an undifferentiated list of stages. A sequence, a reciprocal design exchange and a group linked by shared rules impose different requirements on the people carrying out the work. The matrix is a selected representation of those requirements. It does not itself establish that every consequential dependency has been found. [UX-W040, ch. 2]

The DSM community's tutorial supplies the elementary reading conventions. Parallel activities have no dependency of the represented kind between them. A sequential relation orders activities; reciprocal relations indicate coupling and potentially iteration. Clustering and sequencing can help identify groups that should be managed together or expose iterations concealed by a simple forward schedule. The tutorial also warns that transposed matrix conventions are both used. A reader must establish whether a marked cell means that its row receives from its column or the reverse before interpreting a diagram. [SG-W021, main tutorial]

Independence in a dependency matrix is relative to the relationship represented. A matrix of information requirements can leave material dependencies, shared-resource contention, and the strength and frequency of interactions unspecified. Those relations require their own description. [SG-W021, matrix conventions and parallel/sequential/coupled relations]

Baldwin's example of a disk-drive producer and a laptop producer makes the distinction concrete. Their manufacturing processes can be substantially separate while their designs remain highly interdependent. Questions about commands, timing, connections and compatibility may require repeated exchanges before either design is settled. An answer can depend on another uncertain design choice. Such transfers are not equivalent to handing over a finished, specified object. Leaving out an important exchange may produce a drive that does not work inside the computer. [UX-W040, pp. 24–25]

The technical architecture in this account says what depends on what. The organizational architecture assigns the tasks to people or teams and supplies the ties through which they coordinate. Communication, colocation, common employment and dispute-resolution arrangements can make required transfers and adaptations possible. They are costly to establish and place continuing demands on participants. Baldwin's argument for placing ties where they are needed therefore begins with concrete technical relationships. [UX-W040, pp. 33–34]

Design rules replace selected lateral dependencies

Modularization changes the network rather than accepting every dependency as fixed. In the disk-drive example, an agreed standard settles some interoperability questions before the two designs proceed. Each team can consult that standard in place of repeatedly negotiating the same decisions with the other team. Lateral dependencies are replaced by dependencies on rules. The rule-making work becomes a distinct part of the architecture, and each participating design remains bound by its requirements. [UX-W040, p. 25]

Information hiding follows from this allocation of responsibility. A supplier need not disclose every internal decision if the delivered component satisfies an adequate interface. Other participants can rely on what the interface specifies. That reliance is conditional: a previously hidden interaction, an incomplete specification or a changed technical possibility can make consultation necessary again. Baldwin explicitly draws the disk-drive modularization as incomplete. Standards can lag technology, and vendors may prefer to leave some features unsettled. Standard-setting organizations themselves can experience delay and opportunistic obstruction. [UX-W040, pp. 25–26]

Some questions can be answered once through a design rule and reused in several designs. In Baldwin's modular architecture, rule dependencies connect the modules, which are brought together and tested as a system. [UX-W040, chs. 2, 7]

This explains why modularization can support change without making change costless. A designer may explore an internal alternative while preserving the external commitments. If the alternative alters a commitment or reveals an unrepresented dependency, its effects can extend beyond the module. The architecture has supplied a conditional area for independent action. It has not licensed arbitrary changes to every part of the technical recipe. [UX-W040, chs. 2, 7, 14]

Transfers, transactions and the work avoided inside a zone

Baldwin distinguishes a transfer from a transaction. The latter adds work of defining what is exchanged, counting or otherwise measuring it, and arranging compensation. These are mundane transaction costs, distinct from costs associated with opportunism. Even an exchange completed quickly can depend on extensive supporting infrastructure: product identification, records, payment systems and conventions about ownership. Treating the exchange as frictionless would hide the very work that the account seeks to locate. [UX-W040, ch. 2]

Thin crossing points are places where relatively few, well-specified transfers connect otherwise dense parts of the task network. They make the added work of transactions more manageable. A complicated, contingent design discussion is a thick crossing point: defining every requested answer, checking its completeness and pricing it separately would be difficult. Rules can make a crossing thinner, but transactions remain possible at thicker crossings when contracts and relationships supply the necessary support. The claim is comparative cost, not a prohibition against trading across complicated dependencies. [UX-W040, pp. 24–28]

A transaction-free zone permits transfers without making each one an individually defined, counted and compensated exchange. A kitchen can contain numerous contingent actions among workers without a separate sale every time a tool or ingredient changes hands. A supermarket allows shoppers to move goods into and out of their carts before the boundary transaction at checkout. A library's borrowing procedure defines and counts the book and specifies consequences of nonreturn. These examples identify the location and form of transaction work; they do not imply that the interior of a zone lacks all records, controls, transfers or costs. [UX-W040, pp. 27–28]

Baldwin deliberately uses a broad organizational vocabulary. A productive zone can be maintained by a business corporation, a commons organization or an open collaborative project. Unified governance, ownership and hierarchical authority are distinguishable arrangements rather than interchangeable words for organization. A corporation can protect a zone and establish its boundary relationships, but the ability to work productively without pricing every internal action is not confined to that legal form. [UX-W040, ch. 2]

The 2006 working paper by Baldwin and Kim Clark develops the placement of transactions through the same task-network question. Its selected operative passages explain why dense internal transfers need not all be standardized, counted and separately paid for. Its discussion of pinching a network concerns the redesign required to create manageable transaction interfaces. Incomplete interfaces preserve consultation, and the creation of designs and tests incurs setup costs. [UX-W008, pp. 4–6, 13–18, 35–41, 44]

Baldwin illustrates a thin crossing with a medieval smithy and kitchen. Each team makes many transfers of materials and information among its own members, while a completed pot hook passes from one team to the other. The smith and cook can agree on what counts as a usable hook, count the pieces and settle payment without tracking each contribution within the smithy. Baldwin contrasts this with buying a

partly finished implement, which would require more knowledge to specify the transfer and complete the work. [UX-W009, §4.2]

Complementarity: valuable together need not mean inseparable in every respect

The book next asks why different technologies favor different organizational relationships. Complementarity concerns how the value of one input or action changes with another. Strongly complementary inputs can yield little value separately while producing a valuable result together. Where each actor must incur costs before the joint result exists, dispersed control can expose participants to bargaining problems or leave necessary investments unmade. Consistent decision-making and a capacity to settle disputes can then have substantial value. [UX-W040, ch. 3]

Yet economic complementarity and technical dependency are not identical. Baldwin illustrates the distinction with a recipe. Knowing that the ingredients are valuable together does not tell a cook how their quantities must be adjusted. Too much of one ingredient can ruin the result. Fine tolerances and causal relationships in a technical recipe contain information that the coarse statement of complementarity does not express. The book explicitly preserves this difference rather than translating every engineering dependency into a useful economic complementarity relation. [UX-W040, pp. 51–52]

The spectrum of complementarity supplies Baldwin's explanation of opposing organizational incentives. Dense interdependence, noncontractible effort and cospecialized investment can favor integration. Dispersed knowledge, modularity and opportunities for decentralized exploration can favor autonomy. Between complete independence and unified control lie relationships supported by prices, contracts, platforms, integrators and negotiation. The balance can change when technical designs or strategies change. The book's centripetal and centrifugal language denotes economic incentives. [UX-W040, ch. 3]

Distributed modular complementarity names a more demanding situation than the mere existence of separate firms. Actors must be able to capture value from their contributions; distributed governance must compare favorably with integration; and costs must be aligned with the value available to the actors who incur them. The mathematical appendix distinguishes simultaneous and sequential investment. An actor who can benefit only after another has invested faces a different decision from an actor choosing simultaneously without assurance of the other's participation. [UX-W040, pp. 66–70]

Cost misalignment is particularly consequential for gambits engineering. A system may offer attractive joint value while the participant required to perform one task lacks a sufficient return. The missing investment is then compatible with individually rational action. Joint desirability alone does not cause the task to be performed. [UX-W040, p. 63 and ch. 3 appendix]

Integration and separation also have transition costs. An incumbent cannot necessarily obtain the advantages of a different arrangement simply by redrawing a boundary. Modularizing an integrated product, spinning off a unit, combining separate operations and replacing legacy systems all require changes whose rewards may be uncertain. The book's account of an equilibrium among organizations therefore does not imply an immediate or effortless path to it. [UX-W040, pp. 67–69]

Value structure: finding what constrains a result and what captures its value

Baldwin's value structure analysis follows the way a technology generates value and then asks who can capture it. A production bottleneck limits throughput. A strategic bottleneck gives an actor leverage over value appropriation. These need not be the same location. A component can become inexpensive to manufacture while control of its design, standard or other indispensable attribute remains strategically valuable. Conversely, doing essential productive work does not automatically secure a favorable share of the result. [UX-W040, ch. 4]

Baldwin's 2023 retrospective explains why she turned from detailed design structure matrices to maps of functional components. Building a matrix meant tracing dependencies that could be known only to the people doing the work, often through interviews. Maps organized around what each component does let her trace contributions and strategic bottlenecks across organizations without cataloguing every task. [UX-W038, §§3, 5.2]

Value structure maps provide a selective description of such relationships. Equations permit more specific reasoning when enough is known about the technology and the valuation. Baldwin explicitly says that radical uncertainty limits their use and that the book employs them to derive principles rather than to estimate specific numerical values or probabilities. [UX-W040, ch. 4; p. 105]

The perspective also allows designs and organizational arrangements to alter the location of a bottleneck. A participant may seek an alternative supplier, make an interface more open, acquire control of a scarce complement or prevent another party from becoming indispensable. These are strategies for changing the distribution of dependence and returns. Their success for a focal actor is a separate issue from the costs of the whole productive network. [UX-W040, chs. 4, 9–13]

Flow production: every necessary step still matters

The book's first extended organizational paradigm is flow production. When each step is necessary, output is constrained by the step with the lowest capacity. Increasing capacity elsewhere can leave throughput unchanged. Finding the current bottleneck, improving it and then discovering the new limiting step is therefore a continuing managerial task. Detailed knowledge of the process is required: a factory can contain many potential bottlenecks whose capacities are not evident from an organization chart. [UX-W040, ch. 5, especially pp. 103–108]

The formal comparison is bounded. With a fixed set of capacities and every step required, adding another necessary step cannot increase their minimum. In the stochastic treatment, the adverse uncertainty comparison uses the relevant conditional mean-preserving-spread assumptions. A casual claim that any process with a larger variance must have lower throughput is stronger than this argument. The capacities, their relationships and the definition of the productive sequence matter. [UX-W040, ch. 5 and appendix]

Systematic management addresses this structure through measurement, scheduling, buffers, changes in task assignments and the coordination of complementary investments. Production is not improved by an isolated machine in the abstract; the change must fit the surrounding sequence. A faster machine can merely move the queue. Baldwin's historical accounts of inside contracting show how disputes among departments could obstruct a system-level change even when its proponents expected better production. Unified governance can allow action without renegotiating every affected relationship in real time. [UX-W040, ch. 5]

The book does not make unified governance logically indispensable. It states that sufficiently close information sharing and an ability to act quickly on bottlenecks can also be sustained by long-term relational contracts, although building those relationships takes time. There are also offsetting costs when integration weakens incentives or suppresses useful autonomy. The relevant organizational advantage is a capacity to coordinate the specific coupled process. [UX-W040, p. 115]

Ford, General Motors and Toyota provide different arrangements of flow production. In Baldwin's account, GM combines product platforms with decentralized operational responsibility and corporate coordination. Its smaller divisions also serve as places for developing managers who understand whole businesses. This is a form of retained organizational knowledge. [UX-W040, ch. 6, especially pp. 134–140]

Toyota's low-inventory arrangements expose problems that buffers might otherwise conceal. Making the arrangement work requires diagnosis, improvement, training and relations with suppliers. Local signals

can govern the movement of work, but recurring signals are still coordination, and a production problem still has to be understood and repaired. The source's account cannot be reduced to removing inventory while leaving every surrounding capability unchanged. [UX-W040, ch. 6]

Worker experience is part of this history. The book describes tensions arising when management treats workers' local knowledge and initiative as dispensable, and it contrasts arrangements that develop such capabilities. The later article explicitly describes classical Taylor–Ford tasks as physically arduous, repetitive and detached from the completed product. Its account of authority depends on workers accepting and acting within the arrangement. A scheme that appears economical from a managerial description can impose burdens that undermine that acceptance. [UX-W040, ch. 6; UX-W001, “Synchronized flow production, ” including notes 59 and 63]

Platforms: the ability to refuse an option changes the problem

A platform supports alternatives that can be selected without requiring every one to succeed. In Baldwin's model, the platform exists apart from individual options, and an option can be exercised separately from the others. This is a different value structure from a flow process in which failure of a necessary step prevents the final output. A proposed new feature can be rejected while an adequate fallback remains. [UX-W040, ch. 7, especially pp. 149–156]

That separate exercisability supplies the limited formal reason for a modular option portfolio's advantage. For realized additive increments v_1, …, v_n and a zero incremental fallback, the positive part of their sum cannot exceed the sum of their separate positive parts: $\max(\Sigma v_i, 0) \leq \Sigma \max(v_i, 0)$. The integral choice accepts or rejects the combined result; the separate choices can reject a losing increment without discarding a winning one. It assumes the same component realizations and feasible separate choices; it does not price the interfaces, experiments, maintenance or other costs required to create those choices. [UX-W040, ch. 7 appendix A, especially pp. 175–176]

The weak inequality can be an equality. A strict gain requires relevant outcomes in which accepting some components and rejecting others improves on the combined choice. Modularity by name does not establish that circumstance. Nor does the ability to reject an option protect every participant from its cost. The book explicitly observes that the provider may suffer harm even when a user can reject a poor outcome. [UX-W040, pp. 156–158 and appendix A]

The chapter also advances broader claims about risk, modularity and increasing differences. Its general P4(b) argument moves from separate monotonicities to a multiplicative value representation without establishing the additional structure required. This leaves a gap in the argument for the unrestricted claim. [UX-W040, pp. 176–182; governing notes to ch. 7]

The substantive cost question remains open in the source account. Rule design, testing, buffers and experimentation can be expensive. Gross option value and the net saving after interface and experimentation costs are different comparisons. A technical change may also alter the fallback or couple choices previously treated as separate. These conditions explain why an option argument does not license unlimited fragmentation of actual organizations. [UX-W040, chs. 7–8]

Semiconductor production and the continuing core of integration

The semiconductor history gives the platform account a demanding setting. Components and designs can support decentralized innovation downstream while fabrication remains tightly coupled, capital intensive and dependent on specialized knowledge. Standards, manufacturing discipline and inspection help sustain the interfaces on which other participants rely. Distributed innovation at one level does not imply that every underlying process has become loosely coupled. [UX-W040, ch. 8]

Baldwin's discussion of replication practices and supplier relationships likewise retains coordination. Preserving a successful process across sites can require disciplined control of differences rather than spontaneous local substitution. A stable interface rests on work that prevents or detects incompatibility. [UX-W040, ch. 8]

The IBM PC and subsequent standards-based platforms extend the argument from engineering compatibility to value capture. Opening a system to complementary suppliers can expand the available alternatives while leaving the sponsor vulnerable to parties controlling strategic bottlenecks. Baldwin's Wintel discussion follows that distinction. The creation of a productive ecosystem and the appropriation of its returns need not favor the same actor. [UX-W040, chs. 9–10]

Dell: inventory and finance across an organizational boundary

Dell is an especially revealing case because the book gives the operational and financial boundary together. In the historical arrangement described, orders entered an information system; machines were scheduled when the required parts were available; nearby logistics centers supplied components to a tightly timed assembly process. Dell's internal flow was actively measured and improved. The apparent flexibility of configuring a computer for an individual order rested on a substantial shared logistics and information arrangement. [UX-W040, ch. 11, pp. 292–294]

The components in the logistics centers belonged to suppliers until they crossed Dell's receiving boundary. Suppliers were required to hold inventory and faced consequences when stocks fell below the agreed standard. They also bore devaluation exposure and waited for payment. Baldwin states directly that Dell's supply-chain practices made Dell efficient without thereby making its suppliers efficient. Dell's small reported inventory therefore did not mean that the productive network had stopped holding inventory. [UX-W040, pp. 292–293]

Delayed payment also affected the financing of operations. The book describes a negative cash-conversion cycle in which Dell collected from customers before paying suppliers. Its historical numerical illustration combines receivable days, inventory days and payable days to show the direction of the financing relationship. Suppliers consequently financed part of Dell’s activity. [UX-W040, pp. 294–297]

This arrangement carried vulnerabilities as well as advantages. Suppliers could be constrained by demand changes and inventory requirements; Dell's cash position could come under pressure when purchases and orders declined. Multiple suppliers helped protect against some disruptions, but strategic sole-source dependencies remained. The chapter follows competitive performance over time rather than treating a favorable configuration as permanent. [UX-W040, ch. 11, especially pp. 294–300]

Baldwin describes geographically dispersed modular production as a flow process coordinated end to end by lead firms and assembly providers. Her pandemic discussion follows bottlenecks and cascading disturbances, while leaving the eventual organizational consequences open. [UX-W040, ch. 12, especially p. 325]

Different platforms retain different work

Baldwin distinguishes standards-based product platforms, logistics platforms, transaction platforms and communication platforms. A product interface supports compatible complements. A logistics platform coordinates material flows. A transaction platform supports matching and exchange. A communication platform connects participants and information. Calling all four platforms does not make their channels, governance problems or exposure to disruption identical. [UX-W040, chs. 9–13]

The exchange-platform cases also distinguish relatively asynchronous development from tightly coupled real-time operation. Many outside contributors can develop complements while a core service must still

execute reliably and consistently when used. Decisions over access, architecture and complementary markets affect who can participate and who captures value. A distributed perimeter can coexist with concentrated control of strategically important functions. [UX-W040, ch. 13]

These accounts connect technology and strategy without making organizational change technologically automatic. Interfaces can be opened or restricted; bottlenecks can be controlled, bypassed or displaced; and institutional arrangements affect which choices are available. Baldwin's later conclusion treats technology and organization as developing together. The direction of change cannot be read from a single partition diagram alone. [UX-W040, chs. 13, 17]

Software: modular before addition can become interdependent afterward

Software makes the distinction between an available option and an installed dependency especially clear. A feature may initially be developed separately and evaluated against a working codebase. Once incorporated, other code can come to depend on it. Baldwin states that a codebase may be modular ex ante without being modular ex post. The ability to add a feature one at a time does not guarantee that it can later be removed without consequences. [UX-W040, p. 369]

Formal interfaces and dependency tracking can preserve some separability, but they require time and effort. A running program also differs from the process of designing it. Development can involve trial, rejection and revision; executing a working system involves carrying out the resulting technical recipe. The book uses this distinction to explain why the organization of design should not simply be inferred from a mass-production sequence. [UX-W040, ch. 14]

Information hiding here concerns reliance on interfaces and internal decisions. It is distinct from making source code secret. Open access to code can allow inspection, learning and collaboration while useful boundaries between responsibilities remain. Conversely, visible source code alone does not supply a well-maintained interface or an organization capable of resolving incompatible changes. [UX-W040, chs. 14–16]

Open collaboration: participation, incentives and maintenance

The book's account of open-source collaboration relies on identifiable conditions. Developers can obtain value as users; tasks can be divided and their contributions evaluated; contributors are not rivals in the relevant benefit; and communication costs are low enough for collaboration to be worthwhile. Under such conditions, sharing useful work can create substantial gains without making every contribution an individually compensated transaction. [UX-W040, ch. 15]

Baldwin expressly says that the necessary conditions are not jointly sufficient for a project to form and endure. Community formation, leadership, conflict resolution and continued usefulness still matter. Technologies that lack divisible, value-separable tasks or require large complementary investments may not support the same arrangement. Features with positive aggregate value can also remain missing when no participant can recover its own contribution cost. A group can benefit from a result that nobody has sufficient individual reason to produce. [UX-W040, pp. 402–404]

This explains why the transaction-free character of collaboration is not simply equivalent to the absence of economic problems. Trying to price every internal action can reintroduce the definition, measurement and bargaining costs the zone avoids. Yet refraining from such transactions can leave some needed work unfunded. The book treats this as a specific organizational difficulty, not as evidence that contributors' labor has no cost. [UX-W040, ch. 15]

Corporate participation changes the arrangement again. Contributing an improvement upstream can reduce the burden of maintaining a private fork and allow repairs to be shared. But corporations can also

become dependent on components maintained by very small groups. Baldwin describes developers who unexpectedly become unpaid maintainers for demanding users, and code that loses support when those developers stop. Widely reused software need not have a large, durable maintenance organization behind it. [UX-W040, ch. 16, especially pp. 443–445]

The book's reported vulnerability and contribution-concentration studies illustrate this concern: reuse can spread the value of development while the responsibility and compensation for continuing care remain unresolved. [UX-W040, pp. 445–446]

Mirroring technical dependencies in organizational ties

The mirroring hypothesis proposes correspondence between technical dependencies and organizational ties. Colfer and Baldwin's 2016 review argues that such an arrangement can conserve scarce cognitive resources by placing communication and cooperation where technical work requires them. It reviews 142 studies grouped into industries, firms and open collaborative projects. [SB-W005, abstract/introduction and selected methods/results, pp. 717–723]

Partial mirroring is one important exception. Firms can keep knowledge boundaries broader than operational boundaries, knowing about components they do not make. That knowledge can support integration and adaptation as technologies change. Relational contracts can also support strong dependencies across firm boundaries, while a close-knit organization can deliberately develop modular technical components internally. These patterns challenge any inference that a module interface must eliminate broader technical understanding or that ownership must always align with a technical partition. [SB-W005, selected firm studies and discussion, pp. 721–723, 726–730]

Digital tools furnish further modes of coordination in open collaborative projects. Work can be divided among dispersed participants while shared code, records and other tools preserve information needed to coordinate it. The 2016 article's open-project studies, mostly software, were not supportive of classical mirroring as a general description. The authors' response is to characterize boundary conditions and alternative mechanisms. [SB-W005, pp. 726–730]

The sample and methods constrain the results. Information industries, especially software, are heavily represented; process, extractive and service industries are less well represented. Selection was not random. The classifications are deliberately coarse and partly subjective. In the normative studies, successes substantially outnumber failures. The fact that the reviewed partial-mirroring firms performed well does not show that failed adopters do not exist: researchers seeking best practices can disproportionately choose successful firms. [SB-W005, selected methods and limitations, pp. 717–719, 722–723, 729–730]

Baldwin reuses the mirroring research in the later book and article. [SB-W005, introduction, selected methods/results and limitations; UX-W040, ch. 2; UX-W001, “Organizational ties and the mirroring hypothesis”]

The later article: job shops, flow and platforms as representative arrangements

A theory of technology and organizations provides a later presentation of Baldwin's framework. The article organizes its exposition around job shops, synchronized flow processes and platforms, using task structure matrices to show their different patterns of transfer and coordination. [UX-W001, “Technological paradigms” and subsequent sections]

The job shop adds an instructive case to the book-centered account above. Work arrives in varying quantities and can follow different sequences, with revisits to earlier tasks. Workers assume different roles, and a coordinator redeploys resources as loads change. This versatility supports variety and

responsiveness. Its costs include idle capacity, changing imbalances and repeated adjustment. Resolving today's imbalance need not yield a permanent improvement when tomorrow's job mix differs. Market size and transportation costs help determine whether narrow specialization is viable. [UX-W001, “Job shops”]

The article's flow matrices distinguish a central coordinator from backward local signaling. In the latter, a downstream station signals readiness to its upstream neighbor, potentially through an empty container. Forward material transfers still occur. The conditional self-coordination claim presumes the initial stock and a compatible productive sequence; it does not mean that signals, replenishment or diagnosis have disappeared. The discussion also retains the need to find bottlenecks, rebalance tasks and ensure that workers can perform assigned work within the cycle. [UX-W001, “Synchronized flow production, ” fig. 7]

Its platform matrix separately displays rules, modules and system integration. Providers can work without direct lateral exchanges only insofar as the architecture actually makes their tasks separate and they follow common rules. There may be many integrators, including users. The article's comparison table describes platform efficiency as variable, a useful qualification against reading the sequence of paradigms as an automatic climb toward greater economy. [UX-W001, “Platform organizations, and ecosystems, ” fig. 8 and table 2]

The article finally discusses AI using its contemporary 2025 setting. It distinguishes the organizational work of developing and using systems from the unobserved internal steps of a model, and acknowledges that tasks and transfers may not be an adequate metaphor for those internal processes. This theoretical discussion marks the author's stated limit on extending the representation. [UX-W001, AI discussion and conclusion]

The valuation and scope carried by this account

Baldwin asks how technologies and organizations create and capture value. Her notes define efficiency in terms of sustainable profit relative to investment, rather than simply lowest cost. This business criterion differs from physical resource use, human effort and welfare. [UX-W040, ch. 1 note 42; ch. 17 note 4; chs. 11, 16]

Baldwin explains why design rules can settle recurring questions, why some transfers are better left inside a transaction-free zone, how complementary investments favor coordination, and how separately exercisable options create value. [UX-W040, chs. 2–7, 11–17; UX-W008; SB-W005; UX-W001]

S111 Manufacturing variation, quality and experimental improvement

From a chart signal to an investigation

In his 1931 book, Walter Shewhart asks how fluctuations in the fraction defective might tell an investigator when to look for an assignable cause. Limits derived from previous observations give a basis for examining later points. In the example behind Fig. 6, he reports that the proportion of observations falling within the limits rose from 68 to 84 percent as causes were removed. The statistic counts points within the limits. Shewhart adds that even a point inside them can have an assignable cause. A chart supplies a signal for inquiry while the investigation establishes what changed. [T13-W001]

The NIST/SEMATECH handbook distinguishes the available specification width from the spread and centering of the process. A process can have room within the limits if centered yet put more pieces beyond one specification because its mean lies near that limit. The handbook treats capability estimates in the context of stable behavior and gives assumptions for converting them to expected reject fractions. Sampling uncertainty also affects the estimate. [T13-W002]

Designing for measurement and improvement

仁科健 treats manufacturing as a recipient of the designer’s work. Design choices affect both ease of production and the ease with which an abnormal result can be detected. His exposition considers materials, equipment, operators, methods, measurement and surroundings. Finding a possible cause in the data leads to a further question about the technical relationship, for which he discusses experimental design. [T13-W003]

Sharma, Sahni and Sharma examine a measurement system whose burst test consumes the piece measured. Repeating a test under the same settings requires other pieces judged comparable, so variation within a batch enters the assessment of repeatability. The authors use a crossed gauge study as one part of evaluating the equipment and discuss examining unusual observations and repeating measurements under the same settings. Their case concerns the quality of the measurement system. [T13-W004]

Following a molding experiment into production

สุรศักดิ์ ชุบไธสง and ระพี กาญจนะ studied silver streaking on injection-molded telephone part 1220402. They used a fractional factorial screen on eight candidate settings and then a central composite design to examine four factors. The combination they took forward used 97 MPa injection pressure, 240 tons clamping force, a nozzle temperature of 237 °C and injection speed of 30 mm/s. Screening, confirmation and later production supplied different observations. [T13-W005]

Their part-specific comparison records 1,092 affected pieces out of 32,357 before the change, 663 out of 42,878 during confirmation production and 723 out of 106,665 in later mass production. Those counts give about 33,748, 15,462 and 6,778 pieces with silver streaks per million. The observations concern one defect on one part across different production periods. The authors also call for monitoring other defect modes when applying the settings. [T13-W005]

S08 Form, context and overlapping use: Alexander's design argument

What a form does in a specified context

Christopher Alexander's *Notes on the Synthesis of Form* asks how a designer can organize more interacting requirements than unaided thought can easily manage. Its relevant argument connects the arrangement of a physical form, the conditions around it, practical correction, the representation of requirements and the invention of a design. [TA-W028, chs. 1–9]

Alexander's 1971 paperback preface revisits the 1964 argument. His separate *A City Is Not a Tree* develops a different account of overlapping relationships. [TA-W028, preface; TA-W029]

Alexander divides an ensemble into form and context. Form is the part the designer can change for the question at hand; context supplies the conditions with which it must fit. Fit belongs to their relationship. A property that works well in one situation need not work well in another, and the boundary can be drawn differently for another inquiry. The current inquiry nevertheless has to establish that boundary before judging the result. A kettle's ease of handling, for example, concerns the kettle in relation to human use, rather than an isolated geometrical property. [TA-W028, ch. 2]

Misfit makes the relationship easier to discuss. A particular difficulty can be noticed without possessing a complete positive description of the ideal form. Something burns a hand, obstructs movement, costs too much to maintain or fails under the conditions in which it is used. Alexander proposes organizing such identified failures and their interactions. He does not thereby claim that a finite list exhausts everything that might go wrong. Later chapters explicitly describe the list as a temporary, partly personal statement of the designer's view. [TA-W028, ch. 2; pp. 101–102]

This makes the choice of context consequential. A design may fit a market demand while fitting daily use poorly, or satisfy one participant's requirements while leaving another's out of view. The book's concern with what is included in an ensemble is therefore part of the relevant knowledge, not a preliminary detail that can be discarded once a graph has been drawn. A successful operation on the chosen representation still needs to be related back to the world it represents. [TA-W028, chs. 2, 5, 8]

Physical arrangement can supply several functions

Alexander describes houses whose basic arrangements neglect sun, wind and the separation of incompatible activities. Fans, lamps, heating, ventilation or acoustic insulation are then added to handle problems individually. His criticism is that solving the problems separately can conceal the physical organization they imply together. The example proposes a contrast between arranging the form to fit several requirements and compensating afterward through peripheral devices. [TA-W028, pp. 29–30]

His Mousgoum hut example makes the combined-function account more concrete. Ribs reinforce the hut, guide rainwater and provide footholds during building. The footholds remain available when repair is needed. Construction access is built into the lasting form rather than supplied through disposable scaffolding. In Alexander's description, the same arrangement serves construction, use and maintenance, while siting responds to terrain and erosion. [TA-W028, pp. 30–31]

Practical knowledge is retained in that account. The person who builds also encounters the requirements of upkeep, and recurrent work can affect the initial form. Materials, environmental stresses and patterns of daily life constrain the result. The reported grouping of huts also reflects a particular social order. Its physical coherence does not, by itself, endorse the social relations described. [TA-W028, pp. 30–32]

The one-hole kettle supplies a later example of changed requirements permitting a different component arrangement. Filling, pouring, warning when boiling and avoiding a falling lid had previously been met

through several parts. Alexander argues that the requirements can instead have one physical center: a wide spout. But that possibility depends on noncorrosive metals and cheap, available descaler making a large cleaning-access opening less necessary. A part disappears from the kettle only because materials and maintenance provisions have changed. [TA-W028, pp. 128–129]

The example describes how a specific set of requirements changes when supporting conditions change. Manufacturing, supply and cleaning remain in the ensemble. [TA-W028, pp. 128–129]

Local correction, tradition and the rate of change

The book contrasts what it calls unselfconscious and selfconscious design. These are the author's explanatory constructs, drawn sharply to examine mechanisms; the account is not a comprehensive anthropology of traditional and professional societies. In the first construct, users, makers and repairers remain close to the form and its failures. A noticed difficulty can lead quickly to a practical alteration. Continuing contact with materials and use preserves knowledge that need not first be translated into an elaborate verbal theory. [TA-W028, chs. 3–4]

Direct correction alone is insufficient in Alexander's explanation. If every change spreads through many relationships, a local improvement can reopen other failures. Tradition supplies resistance to needless alteration, keeping much of the arrangement stable while an urgent difficulty is corrected. Prompt response and restraint work together: the first detects and repairs a local misfit; the second limits the disturbance that the repair creates elsewhere. Relative independence among the relevant subsystems is important because it allows improvements to persist. [TA-W028, pp. 50–53]

The account also has a timescale condition. Adjustment has to proceed faster than the context changes. Otherwise a form can remain behind a moving target even though people continue correcting failures. A feedback arrangement that eventually settles under a fixed context does not thereby maintain fit under arbitrarily rapid social or material change. Alexander makes this condition explicit before claiming the advantage of local correction. [TA-W028, p. 51]

His model of linked binary misfit variables gives a simplified way to picture the issue. A variable indicates fit or misfit for a chosen requirement, and couplings allow change in one to disturb another. A highly coupled system can require a long search before all the relevant difficulties are resolved together. If correction can proceed in more restricted groups, gains can be retained locally. [TA-W028, ch. 3]

The numerical light-switching illustration conflates an individual waiting time with the time for an entire collection of independent lights to be off. The latter depends on the joint state of the collection. The illustration therefore does not establish its stated whole-system timing comparison. [TA-W028, chs. 3–4]

Professional design changes these relationships. Specialization can separate the designer from use and repair; feedback can be delayed; a professional culture can reward novelty; and contexts can change rapidly. Alexander argues that the designer must then work through an internal picture of the problem. If that picture organizes the requirements badly, the eventual form may preserve the picture's simplifications rather than the dependencies of actual use. [TA-W028, ch. 5]

The criticism is especially directed at broad verbal headings. Economics, acoustics and safety may be convenient compartments for thought, while a concrete feature participates in several of them. Such categories can hide relationships across their borders and make the wrong combination look natural. [TA-W028, ch. 5]

The governing notes widen the cost question. Maintenance, depreciation and social overheads can be missed when a narrow concept of economics stands in for the cost of a form. The book explicitly locates relevant costs in activities beyond the purchase or construction of the object. [TA-W028, p. 206, ch. 5 note 22]

Embedded adjustment still requires knowledge and invention

One of the book's notes discusses making an ensemble more self-organizing. Governors, automatic regulation of dams or production lines, administrative arrangements and potentially physical urban organization are proposed as ways to take some burden off repeated selfconscious control. This is an unusually direct statement of the issue relevant to gambits engineering: an arrangement can make some adaptations occur through the ensemble's own operation. [TA-W028, pp. 207–208, ch. 6 note 6]

The same note immediately restricts the proposal. Such devices apply in special, limited situations, and designing them can demand an even greater grasp of the ensemble. Unfamiliar circumstances still require inventiveness. Alexander also credits conceptual thought with the ability to develop possibilities that would not emerge through ordinary local correction alone. [TA-W028, pp. 207–208]

Its main text explains why general design is not a solved selection procedure. Automatic selection would require an adequate way to generate alternatives and to express all criteria in a symbolism that permits testing them. For general physical design, Alexander says these conditions are not available. Imagination, experience and encounters with the world remain necessary. His mathematical proposal is intended to help organize the problem confronting that inventive activity. [TA-W028, pp. 74–75]

Requirements, interactions and a program for thought

Alexander represents the selected requirements by a set M and their interactions by links L. Binary classification concerns whether a stated kind of misfit is present; it does not require the underlying physical quantity itself to have only two values. A quantitative property can be judged against an acceptable condition. The procedure is generally satisficing: it seeks to remove specified difficulties, rather than discovering a unique global optimum over every conceivable human aim. [TA-W028, ch. 8]

M is explicitly incomplete. The designer draws on observed failures and anticipated ways a new design might fail. Another designer can identify different issues or give different issues priority. Alexander describes the proposed structure as organizing a person's view of a problem. The method does not make that view objectively complete, ethically neutral or automatically shared by all affected people. [TA-W028, pp. 101–102]

The links also require judgment. Correlations among existing products can express temporary habits rather than necessary relations. Alexander distinguishes conflicts that seem nearly definitional, conflicts related to physical processes and associations arising from contingent current choices. A cheap kettle's hot handle might reflect the materials then used, not an unavoidable conflict between low price and safe handling. Treating that association as necessary could freeze a past design limitation into the new program. [TA-W028, pp. 107–108]

He proposes causal interpretations rather than blind reliance on correlations in a sample of existing forms. Those interpretations are still relative to the domain of possibilities the designer can conceive and to the adequacy of the model used. This is why representing a requirement network already consumes knowledge and why a mathematically neat partition cannot substitute for checking the chosen interactions against the design situation. [TA-W028, pp. 107–110]

The graph is then analyzed into groups whose internal interactions are dense relative to their external interactions. The hierarchical decomposition becomes a program: a way to concentrate on connected parts of the problem and then combine their implications. It is meant to direct attention toward substantive functional relationships rather than familiar headings. The tree represents the proposed organization of the design inquiry; it is not yet a completed physical artifact. [TA-W028, chs. 6, 8–9]

Constructive diagrams connect the analysis to form. A diagram expresses both a physical relationship and the functional forces it is meant to resolve. The designer invents and combines these diagrams, moving between proposed arrangements and the requirements. Conflict among diagrams may require revision, and a promising physical idea can expose a mistaken conception of the context. Synthesis remains a creative, revisable activity. [TA-W028, chs. 7, 9]

The book's notes explicitly say that a functional program can admit many forms. Even if the analysis usefully identifies a related group of requirements, it does not uniquely deduce the arrangement that must realize them. The kettle example shows an inventive recognition of a physical center under changed supporting conditions. [TA-W028, ch. 7 note 11, p. 210; pp. 128–129]

What the formal appendix contributes

The mathematical appendix states a restricted problem: decompose binary stochastic variables under specified assumptions so that information transfer among subsystems is small. It introduces a constructed distribution, simplifies the marginal probabilities and works through a weak-correlation treatment. At the retained second order, a graph-cut expression becomes relevant. The later normalization compares partitions of different types against a random-graph reference. These are mathematical choices within the proposed approximation, not direct measurements of construction cost or cognitive effort. [TA-W028, appendix 2, pp. 174–192]

The approximation requires several conditions. Higher-order interactions must be absent or sufficiently small for the approximation; marginal scope has been deliberately simplified; and the graph must represent the relationships relevant to the question. The notes explicitly distinguish the artificial state distribution from the actual frequency of good and bad forms among real design alternatives. A favorable graph score therefore cannot be read as the probability that a village, building or device will work. [TA-W028, ch. 8; appendix 2; p. 215, appendix 2 note 8]

The entropy expressions on pp. 184 and 186 have an inconsistent sign. The stated random-cut variance omits a finite-population factor; within a fixed graph that factor does not alter the partition ranking in the ordinary nondegenerate case, but it matters to literal variance calibration. The all-singleton boundary of the displayed normalized expression also needs a special convention. [TA-W028, appendix 2, especially pp. 184–190]

The proposed search uses hill climbing. Reaching a stable local arrangement under such a procedure is not a proof that no better partition exists. Nor does a favorable partition at one level guarantee the best set of finer partitions. The connected design account must therefore remain distinguishable from a globally optimal decomposition algorithm. [TA-W028, pp. 190–192; SB-W011, main argument and selected appendices]

An especially important missing step is acknowledged by Alexander himself. The same design question could be described with another set of variables. He says that a secure foundation would require an invariance result showing that the implied physical components survive such a change of representation, but he has not proved it. Without that result, the procedure cannot guarantee representation-independent discovery of the physical constituents of a successful form. [TA-W028, p. 215, ch. 9 note 8]

The village example: several requirements meet in an arrangement

The worked example describes an Indian village of roughly 600 people through 141 requirements. It groups them into a program and develops diagrams concerning cattle and fuel, agricultural production and distribution, communal life, and domestic life. The example shows the passage from a set of stated

needs to proposed physical combinations. It is a worked design proposal, not a report that an implemented village achieved the desired outcomes. [TA-W028, appendix 1]

Water collection, irrigation, roads and protection are treated together. An elevated bund can hold water and support a route protected from floods; the collector and distribution arrangements are related to slope and fields. The design combines functions whose separate treatment could require disconnected works. But water availability, sluices, maintenance, land organization and cooperation remain conditions of operation. Drawing a multipurpose feature does not supply those conditions by itself. [TA-W028, appendix 1, especially pp. 152–165]

The cattle and fuel proposals likewise connect material flows. The diagrams consider keeping cattle, handling manure and using the resulting resources in relation to domestic and agricultural needs. Their relevance lies in the attempt to arrange several recurring tasks together. [TA-W028, appendix 1, cattle/fuel and agricultural diagrams]

The domestic compound combines security, water distribution, shared activities and possibilities of change. In the proposed arrangement, a surrounding wall also carries water, and nearby washing, trees, sitting places and paths are placed in relation to recurrent activity. The text argues that a protected shared area could support women's movement and cottage work differently from secluded individual houses. These are the author's design hypotheses concerning social effects; their success is not established by their appearance on the diagram. [TA-W028, pp. 170–172]

Some requirements also expose unresolved institutional and normative questions. The program includes personal landholding, cooperative cultivation, maintenance responsibility and local rivalries. It includes the social fact of caste separation alongside reforming aims. A graph can register interactions among these statements without settling who should have authority, which arrangements are just or how incompatible demands should be changed. Alexander's proposed physical synthesis does not demonstrate that the relevant ownership, political and social agreements have been achieved. [TA-W028, appendix 1, especially pp. 162–163, 170–172]

The 1971 preface changes the emphasis

In the later paperback preface, Alexander gives priority to constructive diagrams, now called patterns. He says the elaborate formal procedure for discovering independent diagrams is unnecessary. Designers can develop them from experience by examining interacting forces and then combining and improving the resulting patterns. This is a shift in the author's emphasis, stated years after the book's original argument. [TA-W028, 1971 preface]

He objects strongly to a study of design methods detached from practice and warns that blindly following a method will not make someone a better designer. He nevertheless retains a condition of dense internal and relatively weak external interactions for the patterns he recommends. The preface directs attention back to useful physical relationships and the practice of shaping forms. [TA-W028, 1971 preface]

Alexander’s later preface retains diagrams and interacting requirements while rejecting the necessity of the original algorithm. It qualifies the earlier method without abandoning the design problem developed in the book. [TA-W028, 1971 preface and chs. 6–9]

A City Is Not a Tree: preserving overlapping systems of use

The separate essay *A City Is Not a Tree* examines the simplification of urban relationships into nested or disjoint units. Alexander amended the 1965 article for its 1966 reprint. A tree here is a structure of sets, not a botanical analogy: units can be contained within larger units or remain separate, but cannot

partially overlap in the way Alexander wants to preserve. A semilattice permits overlapping units while also including their common part in the collection. [TA-W029, pp. 1–5]

The drugstore, newsrack and traffic-light example explains what makes an urban unit. People waiting at a red light notice headlines and may buy a newspaper. The rack, sidewalk and light provide a relatively stable physical receptacle for changing flows of people, papers, money and electricity. Their coherence arises from an activity that crosses the boundaries of isolated objects. The traffic light need not have been designed for newspaper sales to participate in the system Alexander describes. [TA-W029, p. 3]

This is a precise reason why drawing a city as a set of single-purpose containers can miss something important. One location can participate in several activities, and one person's relationships can cross several administrative or spatial divisions. The essay's networks of friends, catchment areas, pedestrian movement and institutional uses show how actual groupings need not respect a single neighborhood tree. Their intersection may support a useful combination rather than represent an error to be removed. [TA-W029, pp. 5–14]

Alexander's urban-plan examples argue that imposed separations can break such relationships. The relevant object is the living system that uses the built receptacle. Organizing for visual simplicity or administrative convenience can move costs onto users whose activities cross the new boundaries. [TA-W029, pp. 5–14]

The essay is not a demand to mix everything. Separation of children from fast traffic can be appropriate; the problem is prescribing separation without examining the relevant functions. Alexander also distinguishes formal organizational hierarchy from the more changeable patterns of influence he reports through Banfield. [TA-W029, discussion of pedestrian/traffic systems and organizational influence, pp. 11–14]

Most importantly, he admits that he cannot yet show a satisfactory plan for the desired modern city. Overlap for its own sake can produce chaos. The right relationships depend on changing functions, so copying the appearance of a historical city will not necessarily reproduce useful organization. The Nicholson painting and its multiple groupings supply an image for thought, not empirical proof that a corresponding urban design will perform well. [TA-W029, pp. 16–17]

Alexander reports psychological grouping experiments. The substantive urban challenge can be stated without them: a representation that excludes partial overlaps may destroy relationships that matter to use. The essay leaves the task of discovering and realizing the appropriate overlaps unfinished. [TA-W029, pp. 14–17]

The book proposes a hierarchy for organizing a chosen design problem; the essay attacks an imposed urban tree that severs significant overlaps. Their shared set-theoretic vocabulary serves these different constructions. The sequence contains both a decomposition programme and a challenge to disjoint urban organization. [TA-W028, chs. 6–9; TA-W029]

Habraken: levels of form and the distribution of control

N. John Habraken's *The Uses of Levels* asks a related but distinct question: which parties control decisions at different levels of the built environment? The text is the 2002 reissue of a 1988 keynote. It distinguishes urban structure, building or support, infill and furniture. Lower-level changes can vary within higher-level provisions. But a diagram of physical levels does not identify control until responsibilities have also been assigned. [GL-W021, selected primary text, pp. 9–14, 16–17]

Design control and control during use require separate descriptions. Professionals might design a building whose subsequent management belongs to an outside organization or a collective of users. A resident might control only furniture, or might also control infill. Sites-and-services and core-house

arrangements distribute the work differently again. Habraken explicitly says that his model does not identify one universally superior strategy. Circumstances affect which division of responsibility is suitable. [GL-W021, pp. 9–11]

Technical systems can cross levels. In his wall and sewage examples, an individual, a collective and a municipality may control different portions of a system, and the distributions need not be identical for the two technologies. This makes the responsibility question more precise than a choice between central control and user freedom in the abstract. The relevant parties and the actual system have to be identified together. [GL-W021, pp. 11–12]

The support/infill proposal allows repetitive shared support while users choose and change individual layouts. Habraken argues that separating infill can support industrial production, installation efficiencies and continued improvement of the infill systems. Variety at the dwelling level need not require a separately invented support for each dwelling. [GL-W021, pp. 12–14]

Professional expertise remains essential in this account. Users need clearly defined responsibilities, and professionals need skill in organizing parties and coordinating their relationships across levels. Habraken says that misunderstandings of levels and poor coordination can cause problems regardless of the project's budget. More freedom of local choice therefore does not mean less need for understanding how the shared provisions work. [GL-W021, p. 16]

Habraken's synopsis of *The Structure of the Ordinary*, edited by Jonathan Teicher, distinguishes orders of form, territorial control and shared understanding, extending the subject beyond physical geometry. [GL-W092, author synopsis]

Gabriel: attempting to recover the decomposition

Richard Gabriel's 2023 essay reconstructs the early design method through attempts to recover the published village decomposition and through the HIDECS program history. Gabriel reports difficulties with the interaction data, notation, goodness measures and the relation between the presented decomposition and the programs. His investigation makes the reproducibility of the worked formal example a substantive issue rather than treating the printed tree as a self-validating result. [SB-W011, pp. 1–25; selected appendices D–H]

He reports fifty asymmetries in the printed interaction table: a listed relation in one direction is not reciprocally listed where symmetry is intended. His initial inclination was to add the missing reciprocal relation. He later concluded from the program material that the input treatment instead removed such inconsistent relations. Which repair is made changes the graph that the algorithm receives. These are Gabriel's reported findings and historical reconstruction. [SB-W011, main reconstruction; appendices B, F]

The search procedure supplies a second challenge. A succession of favorable binary splits can fail to give the best finer-level partition. Gabriel describes how later program variants addressed holistic grouping and allowed overlaps, and he reports that his attempts did not reproduce the exact published decomposition. His account therefore separates several procedures that a casual reading might compress into one successful computer method. [SB-W011, main argument, especially §§1.5–1.9; selected appendices D, G, H]

Gabriel identifies difficulties in reconstructing the published decomposition, obtaining a global partition through local search, and relating the published account to the program variants. Notation problems in his alternative formulas also leave their numerical rankings uncertain. [SB-W011, main argument and appendix D]

Steadman: intentional craft and the biological analogy

Philip Steadman's 2020 retrospective explains how his research into biological analogy developed partly through criticism of Alexander. He reads the early design account as depending on an analogy between the adaptation of organisms and the gradual adjustment of vernacular forms, while explicitly recognizing that this is his interpretation rather than Alexander's own description. He argues that organisms are not simply molded by environmental forces and that craftspeople deliberately introduce changes. Cultural variation is not equivalent to blind genetic mutation. [SB-W012, retrospective essay]

For gambits engineering, the intentionality objection matters because inherited forms can embody judgment and practical understanding. Treating a working tradition as an automatic product of environmental selection can conceal who noticed the failure, proposed a change and maintained the result. Steadman contrasts an ambition to generate an optimal solution automatically with scientific knowledge and simulations that help designers compare ranges of choices. His retrospective favors the latter use of research. [SB-W012, discussion of Alexander and architectural science]

Alexander's text complicates the strongest version of the criticism. It expressly reserves invention for designers, acknowledges the flexibility of conceptual thought and allows multiple forms for a program. Steadman's essay retrospectively examines his own intellectual work. [SB-W012; TA-W028, pp. 74–75, 207–208, 210]

Zhao and Jiang: design agency and implementation institutions

Zhao Botao and Jiang Hong's 2024 article interprets Alexander's urban-design work across a later trajectory. Their relevant criticism is that an approach centered on constructive design agency can underestimate the regulations, finance and institutional arrangements that condition implementation. A proposed spatial process does not secure those arrangements simply by offering a coherent design logic. [SB-W013, §§1–2 and later discussion]

In their interpretation of Alexander's 1987 work, Zhao and Jiang resist describing his later development as an abandonment of constructive reasoning in favor of mysticism. Their account emphasizes design agency alongside the institutional conditions of realization. [SB-W013]

Form, correction and overlapping use

The relevant material teaches several distinct things in detail: a form can combine construction, use and maintenance functions; prompt local correction depends on knowledge, restraint and time; a graph can organize selected interactions without exhausting a problem; and a built environment can support overlapping uses that a disjoint plan conceals. Habraken separately shows how physical provisions and control responsibilities can be distinguished. [TA-W028; TA-W029; GL-W021]

S108 Structures, ground conditions and construction sequences

What supports a slab while it is raised

The National Bureau of Standards investigation of the L'Ambiance Plaza collapse describes construction by lifting slabs. Floor slabs were cast near ground level, post-tensioned, raised by hydraulic jacks and secured to columns at the required elevations. During lifting, the jacking equipment and its connection to the slab carry loads that the final connection is intended to carry after attachment. The investigating report describes equipment, construction sequence and witness accounts as evidence from which to reconstruct the condition at collapse. [T10-W001]

How movement changes with excavation stage

刘征 and 罗志琪 describe a large semi-top-down excavation near an existing 48-storey building. The staged work gave them reason to observe deformation while support conditions were changing. At point CX14 they report 38.9 mm at completion of the base slab and 48.4 mm at the later construction stage identified in Fig. 9. Most of the other points had final cumulative maximum displacements of 30 to 38 mm. The point, direction and construction stage matter to interpreting the values. The later reading concerns a different condition of the excavation and its supporting structure. [T10-W002]

Changing support during continued use

The French engineering account of reinforcing ground below existing structures distinguishes treating the ground from transferring support through underpinning. Its Fig. 32 example uses temporary jacks and then sliding supports, followed by readjustments. The same account describes micropile arrangements for places where access is restricted. A transverse support can pass through a load-bearing wall and transfer its load to two vertical micropiles. What carries the load changes during the intervention. The example describes a sequence adapted to an existing structure and its ground conditions. [T10-W003]

The Federal Highway Administration's bridge preservation guide connects inspection to decisions about maintenance and rehabilitation. It treats severity and the quantity of affected elements as relevant to the choice of action. Deterioration or loss of support can prompt more frequent monitoring, restrictions or closure. Its condition descriptions are institutional guidance for making those decisions. The excavation study instead provides measurements taken while a particular construction sequence was under way. [T10-W004]

S10 Coordination avoidance: the invariant, the operations, and the merge

When local transactions compose safely

Peter Bailis and colleagues ask when database transactions can execute without blocking coordination while preserving stated application invariants. Their framework makes the answer depend on the combination of an invariant, a transaction set, and a merge operation. The result concerns a declared database model. [W005, §§3–4]

The model represents a database state through versions of data items. A transaction operates on a logical replica, a partial view of versions, and changes to committed state are subsequently shared through merge. A logical replica is not necessarily a separate physical server. In the initial model, merge is set union: the resulting state includes the versions present on either side. The authors assume merge is associative, commutative, and idempotent for their discussion. Those algebraic properties describe how merges combine; they do not determine what application states are acceptable. [W005, §3]

Four requirements for coordination avoidance

An invariant is a predicate that says whether a state is acceptable. The initial state must satisfy it. Global validity requires replica states to remain valid, including states reached by merging independently produced changes. Checking a transaction's immediate local result is therefore only part of what has to be established. [W005, §3, definitions 1 and 4]

Transactional availability is deliberately nontrivial. Under the stated access conditions, a transaction eventually commits except when its own abort operation or a violation of a declared invariant over its replica justifies aborting. Always refusing transactions is not an available solution in this definition. Convergence also rules out permanently isolated valid histories as a substitute for sharing state: without new writes and indefinite communication delay, servers eventually agree on versions of items they both store. [W005, §3, definitions 2–3]

Coordination-freedom concerns the progress of concurrently executing transactions. A transaction's progress depends on the versions it reads, rather than blocking synchronization with work performed for another transaction. This does not mean there is no communication. The model explicitly retains subsequent propagation and merge; an implementation may also communicate to provide a chosen durability level. Removing synchronous coordination between concurrent decisions is a more specific claim than eliminating every kind of network work. [W005, §3, definition 5 and discussion]

Reachability and the common ancestor

The key property, invariant confluence, is defined over reachable states. A state is reachable in the relevant sense when transactions and merges can produce it through a history whose intermediate states satisfy the invariant. The criterion considers two such states with a common ancestor and asks whether their merge is also valid. Both reachability and common ancestry matter. The definition does not require arbitrary states that happen individually to be valid to merge safely, even when the permitted transactions could never have produced their divergent histories. [W005, §4.1, definition 6]

The source states a necessary-and-sufficient result: within its model, global invariant validity can coexist with coordination-freedom, transactional availability, and convergence exactly when the transaction set is invariant-confluent with respect to the invariant. The paper provides a proof sketch and directs the reader to a technical report for the full proof. [W005, §4.2, theorem 1]

Explanatory illustration: a property that survives union

The following small example illustrates the paper's model and distinction. It is an explanatory construction for this monograph. Suppose the state is a set of records, and the only permitted transactions add records with nonnegative ages. Let the invariant require every stored age to be nonnegative. Start from the empty valid state. One history adds Alice, aged 30; another adds Bob, aged 40. Each result is valid, and their union is valid because each record in the union came from a side that met the same requirement. [W005, §§3–5, model and attribute constraints]

The reasoning extends over the specified add-only transaction set: none of its operations introduces a negative-age record, and taking unions does not introduce a record absent from both inputs. This explains why the property in this example is preserved through the permitted divergent histories and merge. It does not say anything yet about unique identifiers, correct billing, or other requirements that were not included in the invariant.

Change the invariant and the answer changes

Now give the records identifiers, retaining set union as merge. Let the application invariant require that no two different people share an identifier. Two histories diverge from the same empty ancestor. On one replica, a transaction sees identifier 7 unused and inserts Alice with identifier 7. On the other, a transaction sees it unused and inserts Bob with identifier 7. Each local state satisfies uniqueness. Their merged state contains two different records with the same identifier and violates it. The local commit checks have not made the independent decisions composable. [W005, §§3–5, uniqueness discussion; explanatory illustration]

Idempotence does not repair this outcome. Unioning the already merged set with itself changes nothing. Alice's record and Bob's record are distinct elements; the duplicated application identifier does not make them the same version for set union. A stable, repeatable merge can therefore preserve an unacceptable result. The example shows why an algebraic merge property and an application invariant must not be substituted for each other.

Under the theorem's combination of obligations, the non-confluent pair prevents the unrestricted coordination-free conclusion for these operations and this invariant. One cannot save all the obligations simply by refusing to merge forever or by aborting every otherwise locally valid transaction. A different allocation operation, invariant, or representation would pose a different question and needs its own analysis. The paper's distinction between choosing a specific identifier and choosing some identifier is important for precisely this reason: uniqueness is not an unconditional verdict that every conceivable identifier-allocation design requires the same coordination. [W005, §§4.1–5]

Correctness still depends on what was specified

The authors explicitly limit the framework to the invariants and operations supplied. If those requirements are incomplete or wrong, satisfying them need not satisfy the application's actual purposes. Specifying the requirements is itself substantive work. The framework can identify a possibility or an impossibility under its assumptions; it does not select every implementation, determine the relevance of every business rule, or demonstrate a net lifetime saving. [W005, §4.3]

Whether the local decisions compose safely depends on the allowed histories and the invariant under the chosen merge. The worked example exposes the temptation to infer application validity from idempotence alone. The answer depends on the specified database model and requirement.

What the coordination comparison actually compares

Bailis and colleagues motivate their criterion by distinguishing serializability from application correctness. If each transaction is correct in isolation, an execution equivalent to a serial order supplies a powerful general guarantee. But not every interaction that conflicts at the read/write level compromises the application's selected invariant. Their proposal asks which combinations actually need synchronous coordination, rather than paying that cost for every general-purpose conflict. The distinction raises the specification from low-level interference to the application property that must remain true. [W005, §2]

Their measurements illustrate why that distinction can matter. An eight-server prototype comparison contrasts an optimized two-phase-locking implementation with execution without coordination. The latter avoids the locking bottleneck for its specified operations; distributed coordinated transactions suffer a much larger throughput drop as their scope expands. These are particular implementation and workload comparisons, supplemented by network-delay calculations using earlier trace studies. The paper explicitly notes that rare coordinating operations need not be a bottleneck and that disjoint read/write sets can allow a serializable database to perform similarly. The favorable case is therefore informative without establishing that removing coordination always improves the application's overall performance. [W005, §2, figs. 1–2]

The theorem and these measurements do different work. The criterion characterizes a possibility under the declared semantics; the benchmark measures particular costs under particular execution arrangements. Neither permits a correctness requirement to be silently discarded because avoiding it runs faster. Conversely, using a stronger general guarantee than the application needs can impose work that a more specific analysis could avoid.

Changing identifier allocation changes the problem

The article's uniqueness discussion distinguishes “give this record this particular identifier” from “give this record some unique identifier.” In the latter case, replicas with known identities can allocate from disjoint portions of a namespace. Each then uses a space other replicas do not use; merging locally allocated identifiers does not introduce the collision caused by two replicas freely choosing the same value. The arrangement changes the allowed allocation operation and relies on membership information. The unrestricted-insertion counterexample above consequently does not refute this different design. [W005, §5.1]

This is a concrete example of prior structure changing a recurring interaction. The separated namespaces embody a constraint on which local choice can occur. Their suitability still depends on the actual identifier requirements and reliable handling of membership; a design requiring one particular shared identifier poses another task. The paper does not establish that choosing any identifier is an acceptable substitute in every application. Its result is strongest when the changed semantics remain explicit.

Referential integrity depends on the allowed updates

For a foreign-key relationship, an inserted record requires a corresponding referenced record. In the article's non-destructive merge account, if every locally valid inserted reference has its target, union cannot make that target disappear. That argument explains a positive result for the specified insertions. Arbitrary deletion changes the situation: one history can add an employee to a department while another deletes the department. Each local view can meet its immediate requirements, yet their combined effect can leave a dangling reference. The condition is not simply “foreign keys are safe” or “foreign keys require coordination.” It depends on what updates and merge semantics are allowed. [W005, §5.1]

The source also considers cascading deletion and suitable propagation of related effects as different operations. Their treatment depends on the modeled semantics, not merely on attaching the word cascade to an implementation. It preserves the paper's reason for distinguishing the cases: validity can concern the visibility and persistence of related records as well as the immediate value in one field. Coordination requirements cannot be read from the invariant's name alone. [W005, §5.1]

Borrego and colleagues: a losing operation can become ineffective

Dina Borrego, Nuno Preguiça, Elisa Gonzalez Boix and Carla Ferreira's 2025 No-Op approach maintains convergence and invariants through explicit conflict-resolution policies. In their music-data example, concurrently removing an artist and adding an album referring to that artist can threaten referential integrity. An update-wins policy can render the artist removal ineffective; a remove-wins policy can instead suppress the album addition. Both arrangements make a choice about which effect survives. They do not preserve both unrestricted effects while somehow making the conflict disappear. [W027, §§2–2.1]

The mechanism records operation calls and their causal dependencies in an execution graph. Concurrent calls remain unordered by causal edges. A winning call issues a block against an incompatible concurrent call, which is marked as a No-Op. When the other replica receives the relevant call, it can make the same determination. Replicas can therefore initially differ and subsequently agree after the required information is delivered. In the paper's update-wins trace, an artist removal that looked effective locally is later recognized as a No-Op. Immediate local appearance and eventual effective state must be distinguished. [W027, §2.1]

Their correctness argument assumes that every replica receives all updates, updates arrive in causal order, and a resolution policy exists for every conflicting pair. The claimed safe serializations and convergence retain those prerequisites. The workflow also retains programmers' responsibility for identifying conflicts and selecting policies. Verification of invariant preservation does not check that a chosen priority matches the programmer's intended meaning. A user-priority rule, for example, makes whose operation wins an explicit part of the arrangement. This remains a substantive semantic decision even when a formal check succeeds. [W027, §§2.2, 3.1]

The account is relevant alongside invariant confluence because it makes a changed obligation visible. Suppressing a losing effect is a particular concurrency semantics, not a counterexample to a theorem holding fixed the original transactions and its availability/validity obligations. Comparing the sources requires retaining what each permits to be discarded or made ineffective.

Bailis's dissertation: specification and mixed execution

In the conclusion of his 2015 doctoral dissertation, Bailis describes four limitations of coordination avoidance. The invariant-confluence analysis has effectively assumed that program text and constraints are known in advance. A restricted safe language could permit some dynamic programming, but its usefulness still depends on admissible logic and complexity. This is an explicit condition on what has to be available before the runtime advantage can be established. [W057, ch. 8, printed pp. 185–186]

The same passage emphasizes complete correctness specifications. An invariant absent from the specification is treated as irrelevant to correctness. The programmer therefore bears a burden that does not disappear when the selected invariant is formally preserved. Bailis also describes the analysis and construction of algorithms as primarily manual, using proofs, rudimentary static analysis and design principles. [W057, printed p. 186]

Finally, the passage identifies mixed execution as an unresolved practical issue. Always coordinating whenever the criterion does not hold can waste opportunities; realizing the benefit may require combining coordinated and coordination-free semantics within one runtime. [W057, §8.2]

Sato: files and prior transactions supply state

Ryo Sato's 1992 paper offers a state-space representation of business transaction systems, drawing its general system definitions from Mesarovic and Takahara. The selected opening pages treat operations such as orders, purchasing and bookkeeping as discrete events that can occur in parallel. Files retain accumulated past transactions, while internally set events also contribute to the state governing subsequent response. The proposed model therefore does not identify a business system only with a stream of current requests: its stored history changes how an incoming event can be handled. [SG-W035, printed pp. 184–187]

Sato's data-model definitions distinguish file types, attributes, keys and actual file contents, then formulate integrity conditions, including references between files. They supply a language for saying what information and constraints a transaction system contains. [SG-W035, definitions and prose, printed pp. 185–188]

S11 Convergence and self-stabilization: what local rules can establish

Two questions about distributed behavior

Edsger Dijkstra's self-stabilization paper and the replicated-data-type report by Marc Shapiro, Nuno Preguiça, Carlos Baquero and Marek Zawirski examine different global properties obtained through constrained local operations. Dijkstra asks whether a distributed state can enter a legitimate set from an arbitrary initial state and remain legitimate. The CRDT report asks when replicas that receive the same updates give equivalent answers despite asynchronous exchange. [W006; W061]

The permitted local actions replace some continuing global direction. Dijkstra specifies a scheduler and local interfaces. The CRDT report specifies update/merge properties and eventual delivery.

Dijkstra: recovery differs from preserving a good initial state

In his 1974 two-page paper, Dijkstra begins with synchronization as preservation of legitimate global states. A system can be designed so that correctly admitted steps preserve legitimacy once established. That leaves another problem: an illegitimate state reached through an error may persist indefinitely. Self-stabilization adds recovery from any initial state admitted by the finite-machine model. Thus a local rule that keeps an already correct execution correct has not yet met the stronger requirement. [W006, p. 643]

His model places finite-state machines at nodes of a connected graph, with each able to inspect its own state and those of its neighbors. A privilege is a Boolean condition on that available state. When the privilege holds, its associated move is enabled. A central daemon chooses one present privilege, that machine changes state, and the process repeats. The central daemon models arbitrary relative speeds through nondeterministic choice of successive moves. Replacing it with a distributed daemon is expressly outside this short paper. “Distributed control” therefore does not mean the model has no global scheduling assumption. [W006, p. 643]

Legitimacy is a criterion on the joint state, although a node makes its move from local information. Dijkstra requires legitimate states to have at least one enabled privilege and to remain legitimate under every permitted next move. He also imposes conditions connecting the privileges and legitimate states. Self-stabilization then requires that some privilege is always present and that every allowed succession of daemon choices reaches a legitimate state after finitely many moves, regardless of the initial state. The quantification over allowed choices matters: one fortunate simulation is not the result being asserted. [W006, p. 643]

This is a recovery guarantee under fixed rules and scheduling semantics. Arbitrary initial state does not mean a continuing external agent can corrupt state at every step while the system is guaranteed to remain legitimate. Nor does an arbitrary corruption of the machine's transition rules fit the same model merely because its register contents were allowed to start arbitrarily.

The ring constructions retain an asymmetry

Dijkstra supplies three ring constructions and says their full proofs and much of the discovery process are omitted from the article. In the K-state construction, the distinguished bottom machine advances its state cyclically when it matches its predecessor. Other machines copy their predecessor when they differ. The legitimate states selected for the ring have exactly one privilege present. The source's sufficient state-space relation is K greater than N for N+1 machines under the central daemon. The distinction of the bottom machine is consequential: this is not a construction of wholly identical nodes each executing an identical rule. [W006, pp. 643–644 and note 1]

A small legitimate trace helps explain the mechanism without purporting to prove recovery. Take three machines, four state values, and write the states in bottom-to-top order. Starting at (0,0,0), only the bottom machine's equality condition is enabled. It advances to 1, giving (1,0,0). The next machine now differs from its predecessor and copies it, giving (1,1,0). The third then copies, giving (1,1,1), where the bottom is again enabled. The single privilege has moved through the ring. This displayed trace illustrates behavior within the legitimate set; it does not establish the universal finite recovery assertion from all states or all daemon choices. [W006, K-state construction; explanatory instance]

The three-state and four-state constructions use different neighbor conditions, with special endpoint roles. Their presence shows that the short paper is offering concrete ways of meeting a global criterion with locally determined moves, rather than only naming self-stabilization. The constructions distinguish between closure of the legitimate set and convergence into it, together with the precise local-information and scheduling conditions used to demonstrate the possibility. [W006, p. 644]

For gambits engineering, the relevant material is consequently more specific than a slogan about self-repair. State and permitted relations can be arranged so that recovery need not require a separate controller to inspect the entire state and issue each corrective instruction. The construction nevertheless retains a selected notion of legitimacy, special rules, sufficient state space and an admitted order of moves.

Shapiro and colleagues: equal histories must give equivalent answers

The 2011 INRIA report defines eventual convergence through both safety and liveness. Replicas with the same causal history must have equivalent abstract states, meaning their query operations give the same values. Updates must also eventually reach the causal histories of the other replicas. Identical answers obtained only by permanently ignoring some updates would not satisfy the combined condition. Nor does temporary disagreement before propagation automatically violate it. The property concerns histories and eventual communication in an asynchronous system. [W061, §§2.1–2.3]

The surrounding model allows network partitions and later recovery. A process can crash and recover while its memory survives. Byzantine behavior is excluded. These are substantial assumptions about what the arrangement must tolerate. [W061, §2]

The report gives two sufficient routes to convergence. State-based replication sends state, while operation-based replication sends operations. They place different requirements on the payload and communication. [W061, §§2.2–2.3, 3.1]

State-based merging retains order through the payload

For a state-based convergent replicated data type, payload states form a join semilattice: any two states have a least state above both in the specified partial order. Merge returns that least upper bound. Local updates advance monotonically in the order. This supplies a way to retain all relevant progress while combining states. The order concerns the representation, rather than necessarily an increase in every visible application quantity. A type that supports a decreasing visible value may need a representation whose accumulated information still advances. [W061, §2.3.1]

Least-upper-bound merge is associative, commutative and idempotent. Changing the order of combining states, grouping several combinations, or receiving the same state again does not change the abstract combined result. These properties relieve the communication channel of some demands: individual messages can be duplicated, reordered or lost if sufficiently new state eventually reaches every replica, directly or through other merges. They do not remove the eventual-delivery requirement. A permanently missing update cannot be inferred from these algebraic properties. [W061, §2.3.1, proposition 2.1]

The proof's core comparison is between replicas with the same accumulated histories. Their respective merges contain the same updates, and commutativity of the least upper bound gives equivalent abstract states. Liveness supplies the eventual arrival needed to obtain those histories. The proof is therefore informative about how the obligations fit together. [W061, proposition 2.1]

A convergent scalar can still be the wrong counter

The report's counter example shows why satisfying a convergence pattern is not enough to meet an arbitrary type specification. A single integer with merge by maximum and local increments forms a monotonic semilattice. Yet if two replicas start at zero and each increments to one, their maximum is one. If the desired counter should reflect both increments, that result loses one of them. The replicas can agree perfectly on an answer that fails the intended count. [W061, §3.1.2]

The state-based grow-only counter instead gives each source replica an entry in a vector. A replica increments its own entry; merge takes the componentwise maximum; query sums the entries. For two replicas, independent local states (1,0) and (0,1) merge to (1,1), and the query returns two. Receiving the same vector again makes no further change. Source-specific components preserve distinctions that a single maximum discarded. The explanation depends on the specified source identities and ownership of increments; arbitrary writers changing another source's entry would be a different operation set. [W061, §3.1.2, specification 6; two-replica explanatory instance]

This version also assumes the payload does not overflow and the set of replicas is known. It is a precise example of using a richer maintained representation to simplify combination. The vector retains additional state. The report states that its portfolio specifications favor clarity over efficiency and notes that equivalent representations can use less space. Choosing an actual representation and its maintenance still involves work beyond recognizing the semilattice property. [W061, §3 introduction and §3.1.2]

Operation-based replication has another channel obligation

For an operation-based commutative replicated data type, operations not ordered by the required delivery relation must commute. Their permitted execution orders then give equivalent abstract states. The report assumes reliable broadcast that delivers operations in the order specified by the type. For the studied types, causal delivery is sufficient; some types can use weaker order if more pairs commute. The state-based tolerance of repeated state messages is therefore not a license to apply arbitrary operation messages twice without checking their semantics. [W061, §2.3.2]

Suitable operation algebra and delivery order replace a need to agree on a single order for all concurrent operations. The surrounding implementation still has to ensure that updates reach replicas and obey the chosen constraints.

Immediate updates, storage and later synchronization

The report itself identifies costs beyond foreground synchronization. Some proposed designs grow without bound, and garbage collection can require a weaker form of synchronization. It treats that synchronization as off the critical path and not essential to liveness, rather than claiming the storage issue absent. It also discusses the need for synchronization for some less common operations as a direction beyond the main study. [W061, introduction]

The preceding coordination-avoidance account gives the separate application-validity question a concrete form. Idempotent union can preserve repeated-merge behavior while violating a unique-identifier invariant. Here, similarly, an integer maximum can converge while failing the intended total count.

Dijkstra's recovery property remains distinct again: agreement among replicas says nothing by itself about recovery from arbitrary corrupt initial states.

S12 Integration and expertise across boundaries

A supplied component does not specify the whole obligation

The sources address boundaries between firms and suppliers, applications and communication layers, flight crews and automation, and internet design under changing requirements.

Brusoni, Prencipe and Pavitt: knowing can extend beyond making

Stefano Brusoni, Andrea Prencipe and Keith Pavitt distinguish a firm's production boundaries from its knowledge boundaries. In the selected discussion of complex products such as aircraft engines, the systems integrator can outsource detailed design and manufacture while retaining concept design and integration capabilities. Those capabilities coordinate suppliers' research, design and manufacturing and handle interdependencies at the product level. Buying the component consequently does not settle whether the buyer can also dispense with technological knowledge about it. [W068, abstract and discussion, pp. 617–618]

Their account gives two reasons overlapping knowledge can matter. A rapid advance in one technology can create an imbalance among component performances; an innovative opportunity at a specialized supplier can also make a different system architecture worth exploring. In either case, the assembler needs capabilities to recognize and work with the opportunity, including temporarily tighter links with outside knowledge. The authors call for cognitive overlap between assembler and supplier, rather than treating perfect separability as the definition of every useful supply relationship. [W068, pp. 617–618]

They describe this as a loosely coupled network organization: differentiated production and knowledge activities remain connected in ways that can change as technologies and products change. The selected conclusion also identifies limits. The empirical framework should be extended across sectors, and variables such as appropriability, past research investments and internal coalitions can affect make-or-buy decisions without being captured by their technological categories. [W068, pp. 617–618]

The source's substantive contribution is the explicit distinction between what is performed elsewhere and what must still be understood to integrate and change a complex product.

Saltzer, Reed and Clark: the endpoint must check the actual application result

Jerome Saltzer, David Reed and David Clark's *End-to-End Arguments in System Design* asks where a function belongs among distributed-system modules. Its careful-file-transfer example identifies several places a transfer can go wrong: storage, programs, memory and transmission all participate. Making a communication subsystem highly reliable addresses only some of those opportunities. It cannot by itself check the complete application result, including the copy actually written at the destination. [GL-W030, author reprint pp. 1–3]

The alternative procedure retains an end-to-end checksum, comparison and retry/commit plan. After transfer, the destination reads its stored copy back, computes a checksum, and returns it for comparison with the original. Agreement permits the application to commit; disagreement can prompt another attempt. The proposed effectiveness depends on sufficiently rare faults, an appropriate checksum and a correct endpoint procedure. Repeated failures also indicate a need to repair the system, rather than retrying indefinitely while declaring reliability achieved. The detailed sequence shows what knowledge and action remain at the endpoint. [GL-W030, careful-file-transfer example]

The reported gateway incident makes the boundary problem concrete. Per-hop checksums protected transmissions but not data during internal copying in a gateway. A transient copying fault could therefore

alter data between those protected portions. Application programmers' inference that the network already guaranteed the whole required transfer was the mistake the account exposes. [GL-W030, pp. 3–4, A too-real example]

The argument does not make all lower-level reliability useless. Lower-level checks can substantially improve performance by reducing endpoint retries. The choice depends on error rates, relative costs and the application's obligation; an extremely unreliable lower level can make a theoretically correct end-to-end arrangement impractical. The selected source distinguishes completeness of the correctness function from helpful partial support. [GL-W030, pp. 3–4 and performance discussion]

Flight-deck automation: responsibility includes monitoring and procedure revision

The recommendations of the PARC/CAST Flight Deck Automation Working Group frame the operational goal as flight-path management, rather than merely correct use of automated equipment. Their selected policy recommendation retains pilots' responsibility for that goal. The procedure-design recommendation emphasizes monitoring, cross-verification and allocating tasks between the pilot flying and pilot monitoring. These are identified parts of the supported operational arrangement, not residual activities that become unnecessary when automation is introduced. [GL-W083, recommendations 8–9]

The same passage calls for procedures to be updated using operational experience and research, and for investigation of when and why procedures are not followed. Retained expertise therefore includes learning from the interaction of people, procedures and equipment over time.

Clark and Blumenthal: stakeholders can change the architecture's problem

David Clark and Marjory Blumenthal's September 2000 working paper revisits end-to-end arguments under a wider range of internet purposes. The institutional abstract identifies new stakeholders, including service providers and government interests, changed user motivations, and a tension between trustworthy overall operation and limited trust in individual users. An architecture chosen for one set of assumptions can therefore face a different task as its users and institutions change. This is the authors' proposed problem. [GL-W095, working-paper abstract]

Their stated concern is that accommodating emerging requirements could compromise support for new and unanticipated applications, one of the original architecture's valued features. That places benefits to one purpose beside possible restrictions on another.

S84 Specialization across abstraction boundaries

An operating system can offer the same operations to many applications while fixing choices that matter differently to each of them. A file-copying program, a web server and a database can use familiar interfaces yet benefit from different arrangements of memory, storage and communication. The architectures examined here change which choices an application can make, which implementations it can replace and how those choices are protected. Their histories include both exposing lower-level resources and retaining interfaces while changing the work performed behind them.

Moving management into a replaceable library

Dawson Engler, Frans Kaashoek and James O'Toole's 1995 exokernel design separates protection of resources from their management. A small privileged kernel tracks ownership and controls access to physical resources. Library operating systems implement higher-level abstractions, such as virtual memory, interprocess communication and files. An application can use a library supplying familiar services or a specialized implementation suited to its task. Replacing a library need not require replacing the kernel. [XA-W001]

The authors identify several restrictions imposed by a fixed privileged implementation. A general replacement policy can be poorly matched to a database's predictable access pattern. Hiding faults or timing information can obstruct an application's own management. A new abstraction can require awkward emulation through an existing one. Their proposed change gives application software control of implementation decisions that would otherwise belong to the operating system. It extends the question of function placement developed in S12 to resource management. [XA-W001]

A secure binding separates authorization from subsequent use. When access is established, the system can examine the relevant ownership and permissions; later accesses can be checked through simpler hardware or software mechanisms. The kernel need not understand every higher-level purpose for which an authorized resource will be used. The design also exposes allocation and resource names so that an application can choose specific resources using information about their properties and availability. [XA-W001]

Control remains bounded by the kernel's power to allocate and reclaim. Visible revocation lets a library choose which resource to surrender and adjust its own state. An abort protocol allows the kernel to reclaim resources when an application refuses to cooperate. Allocation denial and revocation can enforce quotas or other policies among competing applications. Moving management into a library therefore changes the application's available decisions while preserving an authority that arbitrates access. [XA-W001]

Sharing disk data without prescribing one filesystem

The later Xok/ExOS system makes protected sharing a concrete design problem. A library filesystem needs control of disk layout, caching and requests, but several libraries may access the same files. Partitioning the disk among applications would prevent useful sharing. XN, Xok's storage substrate, instead protects blocks and metadata while allowing library filesystems to interpret and arrange them. [XA-W002]

Its buffer-cache registry records the association between a disk block and the physical page holding a cached copy. The data remain in application-managed pages. A library can inspect the registry to discover what is cached, but mapping a page requires corresponding access rights to the disk block. This arrangement permits reuse of cached data without giving every participant authority to modify it.

Libraries retain choices about replacement, prefetching and the division of their memory between cached files and other uses. [XA-W002]

Protection also concerns how metadata establishes ownership. XN uses deterministic metadata-interpreting functions to check whether a proposed update correctly tracks the blocks controlled by a filesystem object. It enforces ordering rules needed to prevent a crash from creating unauthorized access. For example, a persistent pointer must not expose uninitialized metadata, and a freed block must not be reused while old persistent pointers still refer to it. Libraries choose operations and schedules consistent with those rules. [XA-W002]

C-FFS supplies Unix filesystem behavior on this substrate, including access-control mappings and invariants such as unique names within a directory. ExOS supplies ordinary Unix abstractions through libraries. In the 1997 implementation, some shared tables were not yet fully protected; the ordinary-application experiments included extra calls to estimate the cost of checking writes to them. Those experiments showed comparable performance with conventional systems, including a 41-second workload total on ExOS/C-FFS against 51 seconds on OpenBSD/C-FFS. The specialized applications discussed next were measured without the newly developed XN substrate. [XA-W002]

Copying files without touching their contents

XCP uses the exposed storage and cache interfaces to change how copying is scheduled. Given a list of files, it first enumerates and sorts their disk blocks. It can then issue large asynchronous reads in an order suited to disk access, rather than follow an application-level sequence of small reads and writes. Concurrent instances can have their disk schedules merged by the driver. [XA-W002]

While reads proceed, XCP creates destination files and allocates their blocks. As data arrive, it constructs large writes using the buffer-cache entries. The disk controller transfers data into and out of the cache through direct memory access; the CPU does not copy or examine the contents. Allocation, reading and writing overlap, and cached pages become the connection between the two sides of the copy. [XA-W002]

The authors report XCP running three times faster than cp using Unix interfaces on the same Xok/ExOS system. For resident files, avoiding CPU data movement contributes to the gain. For files on disk, large contiguous transfers and a schedule with fewer seeks also matter. The library filesystem makes it possible to combine operations around the copying task instead of expressing the entire task through the usual file-operation sequence. [XA-W002]

Coordinating an HTTP response across storage and networking

Cheetah applies application knowledge across another boundary. An HTTP server receives a request, locates a document and sends it to a client. The XIO library and Cheetah implementation allow storage and TCP handling to be organized together around this task. Their optimizations concern the path from cached or stored document to transmitted response. [XA-W002]

Cheetah merges the file cache with the TCP retransmission pool. It transmits document data directly from cached pages, using precomputed checksums stored with the files. If a retransmission is required, the same cached data can supply it. Separate copies for file caching and retransmission become unnecessary. The arrangement joins knowledge of the stored object to the communication protocol’s requirements. [XA-W002]

Knowledge of the request also affects packet scheduling. Cheetah delays acknowledgment of a client’s request when it expects to include that acknowledgment with the response, reducing separate control packets. This matters particularly for small documents, where one packet is a substantial part of the

exchange. On disk, it groups files included by an HTML document near that document when possible, making related requests cheaper when the cache does not absorb most of the workload. [XA-W002]

The evaluation compares conventional OpenBSD servers, a base socket server on OpenBSD, that server using socket interfaces on Xok, and Cheetah's specialized path. For documents of one kilobyte or less, Cheetah obtained approximately four times the throughput of the Xok socket-based implementation and eight times the best performance the authors achieved on OpenBSD. For larger documents, Cheetah delivered more than 29.3 megabytes per second with the CPU idle over 30 percent of the time, whereas the socket implementation reached 16.5 megabytes per second at full CPU utilization. The machines used 200 MHz Pentium Pro processors; the server's available network bandwidth came from three 100 Mbit/s Ethernet connections. [XA-W002]

The external HTTP/TCP exchange remains recognizable while internal operations are coordinated differently. The paper reports that even its default socket and filesystem implementations benefited from changes such as packet merging and cached pointers to file blocks. Further specialization then used knowledge of the complete response. In this history, retaining a familiar interface and exposing another route for application-specific control were both productive choices. [XA-W002]

What the exokernel builders learned

The authors' later account distinguishes fast primitive operations from control over expensive work. Aegis had been heavily tuned for microbenchmarks; Xok had not. They nevertheless obtained strong application results by changing I/O and other substantial operations. They also describe a practical development advantage: a library can be edited, compiled, relinked and debugged without the repeated kernel reboot cycle. Another library operating system can remain available while the experimental one is developed. [XA-W002]

Designing a suitable protected interface still took repeated iterations. It had to expose enough control while preserving the invariants needed for sharing. Self-paging libraries also required careful treatment of circumstances in which the software responsible for obtaining memory itself needed memory. These were engineering problems encountered while constructing the systems, alongside the improvements their architecture enabled. [XA-W002]

The project's distribution notes describe a system suitable for library-OS and application development, with restricted disk and Ethernet controller support, bugs and unimplemented features. They also distinguish source-history events and available snapshots. The release account presents the state of a developing system; the performance papers present particular implementations and experiments. [XA-W008]

Reusing a kernel by introducing an adaptation layer

田崎創's 2017 account of Linux Kernel Library begins with a reuse problem. Accessing a filesystem created under another operating system requires an implementation of its semantics. Reimplementing a large, evolving filesystem is substantial work; booting a complete guest operating system is another possible route. LKL makes existing Linux kernel functionality available as a library that a surrounding program can use. [XA-W005]

Its adaptation layer occupies the place normally used to accommodate a processor architecture. Instead of adapting Linux solely to a different CPU, it mediates between the kernel code and the surrounding execution environment. Time, scheduling and memory are supplied through that environment, and virtio supports transfers to external devices. This makes the same core functionality usable through several hosts. [XA-W005]

Applications can call the LKL system-call interface, use an interposition library that redirects calls, or follow a musl-based compilation route. The interfaces above and below the reused kernel make that kernel useful in a new setting. Here additional abstraction serves reuse: the design does not seek its benefit simply by removing intermediate components. [XA-W005]

The author examines the implementation with netperf against ordinary Linux 4.9.0, using two PCs connected through Intel X540-T2 10 Gbit/s Ethernet interfaces. Large TCP transfers exploit segmentation and checksum offload, limiting the reported throughput reduction to about ten percent. Smaller transfers lack the same batching advantage. The reported UDP deficit reaches 47 percent at a 1024-byte payload, while much larger payloads approach ordinary Linux performance. These measurements connect the reused functionality to the actual transfer sizes and processing path. [XA-W005]

Keeping familiar calls while selecting the machinery underneath

Unikraft organizes a specialized system from selectable micro-libraries. Its 2021 account describes components with explicit dependencies and build configurations, so that the final image includes the functionality required for the application. Scheduling can be omitted for a suitable single-threaded or run-to-completion task. Locking primitives can similarly disappear when the configured execution circumstances do not require them. [XA-W003]

The memory-allocation design preserves a familiar external interface while opening choices underneath it. A POSIX-facing allocation interface supports existing application code. The internal ukalloc interface directs requests to an allocator backend, and a caller using that internal interface specifies which backend to use. Backends have their own memory regions and allocation functions. Boot code and application code can therefore use different allocators without requiring every ordinary allocation call to expose that choice. [XA-W003]

Application support uses native builds and compatibility components. A syscall shim connects libc’s required operations to implementations available in the image, allowing calls to become ordinary function calls. The paper reports 146 implemented system calls and additional work for glibc-specific compatibility. Its application evaluation uses manual ports of nginx and Redis, selected configurations and single-core virtual machines. Those workloads ran 70–170 percent faster than the Linux VM comparisons reported there; allocator choices and security-mitigation settings affected the differences. [XA-W003]

Startup measurements distinguish the guest from the virtual-machine monitor. A guest’s initialization can take far less time than the complete operation of launching it. The paper reports total startup of approximately three milliseconds with Solo5 and Firecracker, including their monitor overhead. Selecting and initializing fewer components contributes to the result. These are choices about the constructed system and its launch path as well as about application instructions. [XA-W003]

Lefeuvre and colleagues’ developer overview places this work within earlier attempts to make unikernels useful. They identify restricted application support, unfamiliar tooling, insufficient modularity and missing security mechanisms as obstacles. They also question the inference that a small image alone provides strong security. Their proposed development direction joins specialized execution to support for applications, actual protection mechanisms and familiar operational tools. [XA-W007]

Shortening a known path inside Linux

Unikernel Linux takes a different starting point: retaining Linux and its ecosystem while allowing one application to be linked with the kernel. Raza and colleagues’ base model replaces system-call instructions with procedure calls but preserves much of the existing execution machinery. Application

and kernel code retain different stack and execution conventions. Ordinary user processes can run alongside the optimized application. [XA-W004]

The optimized application runs with supervisor privilege. Other processes are not protected from it in the manner that separate Xok applications are protected from one another; the UKL account uses external isolation where work requires that protection. Within this arrangement, developers can select further changes, including bypassing portions of entry and exit processing and choosing how stacks are handled. Preserving execution conventions supplies a working starting point from which more specific changes can be investigated. [XA-W004]

Redis supplies the developed application case. Profiling shows its read and write operations passing through generic dispatch before reaching TCP routines. Since the relevant Redis operations use TCP, a shortcut can invoke tcp_recvmsg and tcp_sendmsg directly. The authors report modifying about ten lines of Redis to use it. This changes the underlying path, rather than only the instruction used to enter that path. [XA-W004]

In the reported bare-metal experiment, the base model improved throughput by 1.6 percent over Linux. The bypass/return configuration improved it by twelve percent; adding the shortcut improved it by 26 percent. The 99th-percentile latency changed from 3.26 milliseconds for Linux to 2.54 milliseconds for the shortcut configuration. The comparison used matching kernel and library versions, with the specified security mitigations disabled on both systems. Its separate virtual-machine experiments produced different, smaller gains. [XA-W004]

The authors' performance-counter analysis follows instructions, cycles, branches and caches. The shortcut configuration executes fewer instructions and branches and makes fewer cache accesses; the reduction in cycles is greater than the reduction in instructions. They interpret the changed path and improved cache behavior as important to the result. The experiment therefore develops the engineering beyond replacing a system-call instruction: it follows how application knowledge changes the work that remains in the execution path. [XA-W004]

Changing a working platform with its maintainers

UKL's implementation coordinates changes to Linux, glibc and application code. Configuration-specific changes keep the base kernel patch relatively small and allow normal configurations to remain available. The authors describe months of discussion with kernel developers at Red Hat and an October 2022 request for comments to the Linux kernel mailing list. Their comparison of patches records the files and subsystems affected, including the maintainers whose review and approval would be required. [XA-W004]

Richard Jones's 2023 project account describes further work before another community submission: separating patches, preparing useful commit messages and checking Linux standards and tests. Demonstrating a useful configuration and obtaining its acceptance into a shared platform are separate parts of this history. The participating team could build and measure a coordinated optimization while still needing review of changes intended for wider incorporation. [XA-W006]

This account gives the organizational questions developed in S07 a particular engineering setting. Decisions about an experimental image, changes to an application and acceptance into a maintained common platform involve different participants and commitments. In Jones's account, the subsequent work is to prepare the changes for review and demonstrate their usefulness with additional workloads and collaborators. [XA-W006]

S112 Requirements, integration and verification in systems engineering

Requirements as an activity

Jeanneret Medina, Baudet and Bonnal look back on the development of requirements engineering as an area of work. In their account, the early 1990s conferences mark the emergence of an identifiable research current. They point to Pohl's distinction among specification, representation and agreement, and contrast that account with standards that enumerate analysis, verification, validation, communication, documentation and management. [T14-W001]

NASA's systems engineering handbook describes technical requirements definition as an iterative process. Baselined stakeholder expectations and a concept of operations supply inputs. Requirements are elaborated from the product to lower levels and address relationships with other products, operators and maintainers. The handbook's example of a specified force with a tolerance makes a condition checkable. The specification gives the test a criterion. The observed result comes from carrying out the test. [T14-W002]

A specified unit at an interface

The 1999 Mars Climate Orbiter investigation traces a numerical mismatch through an interface. The specification required impulse in newton-seconds. A ground software application wrote values in pound-force seconds, and the downstream navigation process treated them as the required metric values. The unit conversion is about 4.45. The report identifies the application, exchanged file and consumer, and records responsibilities on the contractor and JPL sides as they existed in the project. The case shows the practical distance between establishing an interface requirement and producing effective evidence that exchanged values meet it. [T14-W003]

Change and acceptance evidence

Ågren and colleagues' interviews at two automotive manufacturers address the visibility of requirements change. A current baseline tells a reader what the requirements now say. The history of changes shows which requirements moved and gives people examining the consequences a way to trace the change. The study's proposed improvements come from interviews and a second round of inquiry. [T14-W004]

The transport prototype acceptance report gives an example of evidence at a specified configuration. A script describes the expected result, and its steps are repeated at least 15 times. The test team observes displays or physical events and keeps logs for inspection after the run. Step pass rates concern those scripts and the configuration they exercised. The report distinguishes these acceptance checks from the later field evaluation. [T14-W005]

S29 Leveson: constraints, feedback and the organization of safety

The connected safety account

Nancy Leveson's *Engineering a Safer World* connects physical design, automated and human control, organizational responsibilities, regulation and change throughout a system's life. Its relevant contribution is an extensive account of preserving acceptable behavior while arranging the information, action and support that preservation requires. The connection is the author's applied synthesis. [UX-W021, 2012]

Leveson develops the framework in *Engineering a Safer World*, her 2004 accident-model article and a related 2009 formulation. [UX-W021; UX-W020; SB-W017]

Reliable components can participate in unsafe interactions

The book distinguishes component reliability from system safety. A component can perform its specified function while its interaction with others violates a system constraint. A specification can also be wrong for the surrounding physical situation. Searching only for a failed part can therefore miss the conditions that made an otherwise expected action hazardous. [UX-W021, chs. 1–3]

An event sequence records what happened but can leave out what should have happened, why a control was missing, or which organizational process failed to maintain an assumption. Leveson's proposed account starts with constraints on acceptable behavior and asks how they are established and preserved. Physical barriers, software requirements, operating procedures, management and regulation can each contribute. [UX-W021, chs. 2–4]

The selected chemical-reactor example in the earlier article makes the interaction concrete. Cooling must be available when catalyst flow begins. Software following a locally stated instruction can still be unsafe if the instruction omits the relation between those operations. Executing the code as written is different from satisfying the necessary system-level condition. The example is Leveson's reconstruction. [UX-W020, 7–15]

Safety is also relative to losses judged unacceptable and to the environment and use considered. Several constraints can conflict. A train door, for example, should ordinarily remain closed while moving yet permit evacuation under different conditions. A useful safety account has to state those conditions and competing obligations. The vocabulary of constraints does not make stakeholder judgment unnecessary. [UX-W021, chs. 1, 6–9]

Actual feedback differs from evidence that a command was issued

The control account distinguishes goals, action capacity, information about the controlled process and the assumptions through which a controller interprets that information. A command can be delayed, fail to execute or achieve only part of its intended effect. Information that power was applied to a valve is not proof that the valve reached its required position. The feedback must concern what the relevant constraint actually requires. [UX-W020, 13–16; UX-W021, ch. 4]

Direct physical coupling can supply information that a remote interface must reconstruct. Vibration, sound, temperature and the behavior of a mechanically connected gauge can reveal both the process and whether the display remains connected to it. Moving control to a remote electronic interface can improve capability while removing those incidental cues. Designers then have to decide what to represent and how the operator will obtain it under abnormal as well as routine conditions. [UX-W020, 14–15]

Delays occur in actuators, process response, measurement and communication. A controller's interpretation can be internally consistent but late or wrong about the actual state. The book's process-model analysis asks what each participant believes or assumes and how feedback changes that account. [UX-W021, chs. 4–5]

Leveson invokes Conant and Ashby to state a general requirement for controllers to contain a process model. The practical method can make hidden assumptions visible, but the universal inference needs the qualifications attached to the original result: its optimal-response premises do not establish that every successful passive constraint contains a rich internal representation. An observer's model of a controller is also different from a stored representation inside it. The original arguments are treated separately in the Ashby section. [UX-W020, 15–16; UX-W021, ch. 4]

Responsibilities can fall into gaps or overlap unsafely

The development and operating control structures relate organizations as well as physical devices. Assumptions, requirements, reports, maintenance findings and changes must cross their interfaces. A nominal chart can omit who actually supplies or receives relevant information. Conversely, two controllers can each assume that the other has responsibility, or act incompatibly on the same process. [UX-W021, chs. 4–5; UX-W020, 17–25]

Overlap is therefore not always a remedy. The book's boundary-responsibility examples ask which controller handles the case at the edge of two domains and how their actions are coordinated. Removing an overlap can also leave a gap. The issue is the actual relation among responsibilities, knowledge, timing and action, rather than maximum integration or maximum separation in the abstract. [UX-W021, chs. 4, 8]

Asynchronous evolution names a further difficulty: one part changes while related parts retain assumptions about its earlier behavior. Software, equipment, procedures, training, contracts and regulation can change on different schedules. A previously acceptable arrangement can consequently become unsafe without one dramatic component breakdown. Maintenance of the relationships is part of safety work. [UX-W021, chs. 4, 12; UX-W020]

The friendly-fire chapter illustrates differences in local information, schedules, service expectations, liaison and interpretation. It reconstructs testimony and reports, preserving disagreements among sources. An unexplained identification-system failure prevents describing the whole event categorically as an accident without any component failure. The example's relevant contribution is how different local actions and understandings fit into a wider control structure. [UX-W021, ch. 5]

STPA begins with losses, hazards and a control structure

Systems-Theoretic Process Analysis, STPA, first identifies the losses and hazards to be considered, relevant constraints and the control structure. Its four unsafe-control-action categories concern an action not supplied when needed, an action supplied unsafely, wrong timing or order, and an action stopped too soon or sustained too long. Timing and absence can therefore be analyzed alongside the occurrence of an explicit wrong action. [UX-W021, chs. 6–8]

The next task develops causal scenarios. Analysts examine controller assumptions, feedback, actuators, the controlled process, communications and environmental conditions that can lead to an unsafe action or prevent a necessary one from taking effect. The objective is to make design requirements and potential failures of their enforcement visible before relying on operation to reveal them. [UX-W021, ch. 8]

The analysis can begin early, with a coarse design, and be refined as decisions add detail. A new feature can remove one hazard while creating another. The model is revised with the design, rather than treated

as a document made once after the architecture has already been fixed. This gives analysis an active role in generating and comparing alternatives. [UX-W021, chs. 7–9]

Completeness remains bounded. Examining every listed control action does not prove that the list, the chosen boundary or the causal scenarios contain everything relevant in the real system. Some losses and interactions can have been omitted from the initial model. The book's comparisons with other analyses are author reports. [UX-W021, chs. 6–8]

A physical constraint can change the necessary response chain

The high-energy door example starts with a requirement that opening the door must not expose a person to power. A design can detect door opening and command a separate interruption, requiring sensors, logic, action and timing. Alternatively, the physical circuit can pass through the door so opening it breaks that circuit. In the modeled example, the arrangement directly removes the identified exposure condition. Its physical integrity and other hazards still need consideration. [UX-W021, 224–225]

Adding an indicator offers another alternative: the operator can see whether power is present. That adds information but also a component that can fail and a display whose safe interpretation must be designed. The book explicitly says that each solution can introduce drawbacks and that the rationale for the choice should be recorded for later certification, reuse and maintenance. [UX-W021, 224–225]

The important distinction is between changing the set of possible physical interactions and asking a participant to detect and correct the same dangerous interaction. Either may be appropriate under particular obligations. Leveson's design process keeps the alternatives and their support requirements available for comparison, rather than identifying safety with a larger number of active checks. [UX-W021, chs. 8–9]

The robot example: a solution changes the problem

The mobile robot's manipulator can destabilize its base. A sufficiently heavy base is a candidate physical solution, but added mass can worsen collision damage or impede manual movement in an emergency. A wider base can improve stability while preventing movement through doors and crowded spaces. These competing requirements constrain an apparently simple solution. [UX-W021, 257]

Deployable stabilizer legs offer another option. They must be extended before arm movement and retracted for travel. The change introduces additional hazards, synchronization requirements and feedback. An assurance that a motor was energized cannot replace evidence that the legs reached the required position; an obstruction could prevent full deployment. [UX-W021, 258–262]

The analysis can then change the decomposition. Combining the arm and leg controllers can reduce the communications needed to keep their assumptions consistent. This is an explicit example in which integration is considered to manage a strong coupling. [UX-W021, 261–262, figs. 9.3–9.5]

The example begins with a hazard, examines physical alternatives and environmental constraints, analyzes hazards introduced by a candidate solution, and revises the control architecture. Design and safety analysis inform one another through this sequence. [UX-W021, ch. 9]

Automation changes cognitive and contingency demands

Automation can remove routine actions while leaving harder supervision, diagnosis and recovery tasks. Operators need opportunities to learn, practice and understand unusual conditions. An arrangement that performs ordinary work smoothly can still make an abnormal event difficult to recognize or handle.

Leveson therefore treats training, skill retention and useful feedback as design obligations, rather than assuming fewer routine interventions settle the comparison. [UX-W021, ch. 9]

Display design supplies a more specific economy. If an operator needs to know whether a variable exceeds a limit, a dial with a reference can reduce the extra processing required to compare an absolute number with a remembered threshold. Patterned displays can support recognition, while sequentially requested screens can require remembering one portion while seeking another. The claimed benefit concerns the actual decision task. [UX-W021, 304–305]

The book also preserves a trade-off: an effortlessly interpreted display may do less to develop the knowledge and thinking needed in abnormal conditions. It advocates involving operators, understanding the reasons behind established conventions, and evaluating the display in the work it supports. The design aim is appropriate information use with retained capability, not merely minimum momentary mental effort. [UX-W021, 304–305]

Intent specifications preserve why as well as what

The intent-specification approach arranges information along intent, part–whole and refinement relations. It makes the rationale for a requirement available together with the requirement and its realization. A decision that appears arbitrary to a later engineer can have depended on an environmental assumption or a rejected alternative. Losing the reason can make maintenance and change hazardous. [UX-W021, ch. 10]

Refinement is not presented as an irreversible waterfall. Later findings can change earlier assumptions and choices. A specification can provide links among purpose, behavior, architecture, implementation and operation so that a revision can be examined for consequences elsewhere. The organization of information supports review and reuse; it still requires people who understand the domain and can judge the implications. [UX-W021, ch. 10]

Black-box behavioral specifications can serve as an interface between an integrator and contractors. The boundary hides selected implementation details while making required behavior and assumptions explicit. Reusing a component does not establish that its environmental assumptions remain satisfied in a new system. Those assumptions need ownership and revalidation; more documentation without such relationships is insufficient. [UX-W021, ch. 10]

The chapter's architecture-screening example uses severity and mitigatibility as comparative design information. Its ordinal scales are not calibrated probabilities or a complete quantitative risk model. Similarly, a promising operational alternative can shift demands to landing precision, infrastructure, communications or other supporting capabilities. Specifying an interface does not eliminate those external conditions. [UX-W021, ch. 10]

CAST reconstructs a wider control failure

Causal Analysis based on Systems Theory, CAST, reconstructs events and then examines the control structure, constraints, context and decisions around them. It seeks explanations beyond a final error label. Information available somewhere in an organization may not have been observable to the relevant participant through the actual task and interface. An account must follow those relations rather than equate existence of data with usable knowledge. [UX-W021, ch. 11]

Citichem is a fictional teaching scenario. The chemical-overflow and Überlingen accounts are reconstructions from reports. Their practical value lies in making responsibilities and missing controls available for examination. Recommendation selection, priority and implementation remain judgment-laden. [UX-W021, ch. 11]

In the earlier 2004 article, Leveson reports completeness and analysis-effort advantages from informal evaluations and accident analyses. These evaluations compare the analyses rather than measuring a controlled resource intervention. [UX-W020, 26–28]

Designed constraints must survive operation and change

Operational safety work includes review of planned change, detection of unplanned migration, audits, incident and problem reports, and feedback about the result. A constraint can be correctly designed and subsequently defeated by altered procedures, incentives, equipment or resources. The book asks who will notice and respond when an assumption no longer holds. [UX-W021, ch. 12]

The reporting example changes a small part of the work system. Instead of requiring a separate cumbersome report, a worker contacting those who can fix a problem can copy the safety function into an existing email. This can reduce the act of reporting. Acknowledgment, analysis, action and communication back to the reporter remain necessary; the extra copy is not the resolution. [UX-W021, 405]

Reporting incentives affect the data obtained. Fear of punishment, lack of confidence that anything will happen, confidentiality and time requirements can suppress reports. More reports can therefore indicate a changed reporting environment rather than a higher underlying hazard rate. The book also distinguishes a just culture from blanket immunity for gross negligence or deliberate disregard. The organizational purpose is useful learning and accountability in context. [UX-W021, chs. 12–13]

Generic indicators are not assumed to reveal every system's safety condition. Leading indicators should relate to the actual constraints and processes being maintained. Data collection has to be connected with interpretation and response. Training also needs to preserve reasons and assumptions, rather than only the memorized steps of a procedure. [UX-W021, ch. 12]

Authority, resources and insight require actual bearers

Safety responsibilities must be matched with authority, resources and feedback. They occur at several levels and can differ by level. Leveson describes working groups as a means of preserving communication while maintaining review independence. Oversight can fail by becoming too entangled with the project or by withdrawing so far that it loses relevant understanding. [UX-W021, ch. 13, especially 435]

The book's warning about a transition from oversight to insight is concrete. A management-by-objectives label can conceal a reduction in personnel and budgets without anyone assuming critical tasks. Local interpretation of objectives needs competence and an actual process for satisfying the constraint. [UX-W021, 435]

Leveson argues for redesigns that improve safety and cost together, rather than treating their conflict as inevitable. Her account also addresses conflicting objectives, limited resources and negotiation. The related 2009 article seeks to reduce avoidable trade-offs. [UX-W021, chs. 8–13; SB-W017]

SUBSAFE illustrates a bounded, maintained institution

The SUBSAFE account concerns watertight integrity and recovery from flooding. Other hazards, including fire, weapons and reactor safety, have other arrangements. Its assurance system includes material and fabrication evidence, design reviews, tests, calibration, maintenance, audits, training and reentry into controlled work processes. It is an extensive institutional architecture. [UX-W021, ch. 14]

The historical record reported for certified submarines is bounded by that purpose. The book's San Francisco collision account includes death and injuries even though the hull and recovery features performed. [UX-W021, ch. 14]

The example is informative because the design and operation are connected. An assurance claim depends on knowing materials, changes and test conditions and on continuing to enforce them. Certification is not a one-time substitute for maintenance or knowledge. The related 2009 source also identifies unusually favorable institutional resources, limiting simple transfer to a different setting. [UX-W021, ch. 14; SB-W017]

Interfaces can lose end-to-end ownership

The satellite appendix traces a flight-constant entry through several organizations, formats and tests. A test can use a default rather than the actual flight data while participants assume the relevant assurance has occurred elsewhere. Many reviews do not compensate automatically for an unowned end-to-end process. [UX-W021, appendix B, 469–493]

Mergers, restructuring and staff departures also remove knowledge of why a process was designed as it was. The appendix describes lost history and rationale, no clearly identified process owner and misunderstandings about the coverage of testing. A filter was retained for consistency after a relevant need had changed. Some higher-level explanations remain speculative because the reports lacked data. [UX-W021, appendix B, especially 491–493]

The Walkerton appendix follows physical contamination controls together with monitoring, laboratory feedback, training and institutional change. The book explicitly distinguishes privatization itself from implementing a change without preserving notification and oversight. [UX-W021, appendix C, especially 514]

The system-dynamics appendix supplies another kind of account: qualitative reinforcing and balancing loops with delays. These diagrams identify hypotheses about migration and pressure. Signs alone do not establish exponential growth, equilibrium or instability without rate functions, parameters and conditions. [UX-W021, appendix D]

Related frameworks and a separate 2009 formulation

La Porte's reconstruction of Perrow explains why interactive complexity and tight coupling can create conflicting demands for centralization and decentralization. A choice about authority depends on both dimensions. [UX-W004, through La Porte, printed 136]

Rosa’s 2005 symposium introduction compares normal-accident and high-reliability frameworks and calls for evidence from cases with and without accidents. Considering both challenges explanations built only from the selection of disasters. [UX-W016, abstract]

Leveson, Dulac, Marais and Carroll's 2009 article argues for examining concrete couplings and hazards rather than treating an industry label as sufficient explanation. Local decisions can be appropriate where coupling is reduced or where participants possess relevant knowledge and prepared responses. The account retains stakeholder judgment and goals in conflict. It is a related formulation by the same author group, with overlapping examples. [SB-W017]

SOHO: differences between analysts’ reconstructions

Johnson and Holloway's 2003 SOHO analysis uses the same official report for two analysts. They identify useful systemic emphases but differ in model granularity and detailed classification. The comparison asks how analysts choose an appropriate representation, handle time and connect findings to recommendations. A framework does not mechanically determine all those judgments. [SB-W014]

They also acknowledge Leveson's response that constraints were underemphasized in their application. The criticism is therefore part of a methodological exchange, not a verdict based on an uncontested application. Their early concern about practical guidance should not be projected unchanged onto the later detailed 2012 book, which addresses temporal processes, actual versus nominal structures and operational change. [SB-W014; UX-W021]

A controlled method comparison reports a trade-off

Abdulkhaleq and Wagner's 2015 study compared STPA, failure-mode-and-effects analysis and fault-tree analysis. Twenty-one students worked in three groups on three systems, with methods rotated. The results concern the requirements found and the effort of these participants, not observed future accidents. [SB-W015]

Reported mean recall was 0.443 for STPA, 0.326 for FMEA and 0.231 for FTA. Mean recorded time was 116, 94 and 88 minutes respectively. The method finding more requirements also required more time in this setting. [SB-W015, results and table 6]

The recall denominator was the union of reported requirements, not an independently known complete gold standard. Coverage also used a taxonomy partly drawn from Leveson. Group-level reports, novice participants, small sample and task ordering limit generalization. The paper's coverage table and prose disagree, as do its abstract and body about one applicability comparison. These inconsistencies make the corresponding coverage and applicability comparisons ambiguous. [SB-W015]

An application supplies conditional simulation evidence

Liu and colleagues' 2020 differential-braking article models seven unsafe control actions and simulates selected scenarios using a cited aircraft model. It demonstrates how the approach is applied to specified control timing and state assumptions. It retains power, speed and torque sensing, position information, human response and the model's conditions. [SB-W016]

Its five-second threshold belongs to the stated dimensions, initial conditions and disturbances. It is not an operational rule or flight-validated universal limit. [SB-W016]

Information and coordinated pathways in selected practice sources

The National Commission on Forensic Science's 2015 policy-development abstract asks which information belongs in a forensic assessment. A fact can support a conclusion about a suspect while being irrelevant to the inferential connection between an examined pattern and the requested scientific opinion. The proposed context management therefore concerns the arrangement and timing of information access, not simply less information in every task. Deciding relevance itself requires expertise. [GL-W063]

The document is an abstract of a developing proposal, not a final demonstrated intervention. Its reference to cognitive and contextual bias identifies the problem and intended principle. [GL-W063]

Rotter and colleagues' 2025 clinical-pathways review, reports fifty-eight studies and searches through July 2024. Stand-alone multidisciplinary pathways probably reduce in-hospital complications and length of stay, with moderate certainty; mortality, readmission and cost conclusions remain uncertain. Multifaceted implementations have less conclusive results. [GL-W058]

These findings are outcome-specific. A shorter stay does not establish every patient benefit or identify one coordination mechanism as the cause. [GL-W058]

The selected NHS England guidance, updated 23 July 2026, describes longer care packages joining prevention, personalized goals, team competence and continuing review, with funding arrangements intended to support them. The worked follow-up example illustrates planned work over time rather than episodic repair alone. It is an implementation design under a specific institutional arrangement, not comparative outcome evidence. [GL-W064, §§1.1–1.6 and worked example]

The forensic proposal addresses bias in evaluating evidence; the clinical-pathways review compares patient and hospital outcomes; the dental guidance organizes prevention and follow-up. They describe different activities within the conduct and assessment of care.

S30 Reliability, fault tolerance, and what fault experiments establish

Reliability under disruption

A system can contain unreliable components and still provide a useful service. Explaining how requires more than counting spare components or naming a recovery mechanism. The relevant literature specifies relationships among components, assumptions about failures, the service to be preserved, and ways of checking what happens under disruption.

Fault injection is especially informative because it intervenes in a system rather than waiting for a naturally occurring failure. It can reveal a weakness, exercise a recovery path, or test a bounded proposition about a model. The meaning of its result depends on what was changed, when the change occurred, the workload that exercised it, the observations collected, and the criterion of success. These dependencies are explicit in the sources considered below.

Von Neumann: reliability through an arrangement of fallible elements

The reconstruction of John von Neumann's lectures develops the possibility of reliable computation using unreliable elements. A majority arrangement is a concrete part of the discussion. Its inputs are intended to represent the same state, while their errors are assumed independent in the relevant calculation. Agreement among several representations can then help preserve a correct result when an individual representation is wrong. The majority element can itself fail, so its error must also enter the account; counting additional representations alone does not settle the output's reliability. The source later explicitly calls a constant error probability independent of time and previous inputs unrealistic. Actual failure probabilities can vary with input, history, environment, and element. [W043, §§7–8, especially reconstructed pp. 23–24; §11.3]

Stott and colleagues: a modeled improvement can come from counting assumptions

James Stott and colleagues examine a problem in transferring common-cause failure models and generic data from nuclear applications to aerospace. Redundant components can be affected together; their failures therefore cannot always be treated as independent events. The paper's selected methodological discussion goes further than this familiar warning. It asks what the recorded failures and testing arrangements actually allow an analyst to estimate. A test interval, an individual test episode, and a demand on the system are not automatically the same observational unit. The number of opportunities to detect failures changes how observed counts relate to the quantities in a model. [W065, §§3–5]

Their comparison distinguishes testing all members of a group in one episode from testing them in separate, staggered episodes. In the latter scheme, finding a failed component can lead to additional tests of the others. Those procedures affect the challenges and observations represented by the data. Using factors estimated under one interpretation with a model that assumes another can therefore alter the calculated risk without establishing a corresponding alteration in the physical system. The authors specifically criticize using mixed interpretations of generic data within one risk assessment or trade study. An alternative can receive a lower calculated risk because of the assumption assigned to it, then appear to earn that advantage as an analytical result. [W065, §§3–5]

The paper does not conclude that the actual testing schedule can never affect common-cause risk. It gives a possible real mechanism: separating test episodes may separate operators and thereby reduce some shared influences. Its point is that this causal difference is not captured merely by switching the parametric counting convention. Quantifying that benefit requires analysis of the relevant causes and coupling factors. The authors' recommendations concern the NASA modeling problem they address in

2010. The substantive lesson supplied by this source is how an apparent advantage of an arrangement can arise from the interpretation of evidence rather than an established change in the arrangement's failure behavior. [W065, §§5–6]

Hsueh, Tsai, and Iyer: the experiment includes a workload and an observer

Mei-Chen Hsueh, Timothy Tsai, and Ravishankar Iyer's 1997 survey explains fault injection as an experimental approach to dependability evaluation. Its relevance extends beyond the act of introducing a fault. Their account distinguishes simulation during conception and design, injection into a prototype or operational system, and measurement of naturally occurring behavior in operation. Each supplies different evidence. A simulation can explore a design before implementation but embodies simplified assumptions. A physical or software intervention exercises an implemented system but represents the faults that the experimenter can emulate. Operational measurement observes real events and workloads but can require long collection periods and face changing field conditions. [AFC-W003, pp. 75–76]

The survey's experimental architecture makes these differences concrete. A target system runs a workload drawn from a workload generator and library. A fault injector applies changes selected from a fault library, including choices of type, location, and time. A controller organizes the experiment; monitoring and data collection record behavior; analysis interprets the observations. These are distinct functions even when some share a machine or program. The result is thus produced by a target-plus-experiment arrangement. The injector alone does not determine whether the relevant behavior was exercised or whether its consequences were observed. [AFC-W003, p. 76, fig. 1]

An introduced fault can produce an internal error without causing failure of the service being judged. The workload may not exercise a changed location. An internal error may be detected, contained, or masked before it affects the chosen result. Conversely, an incorrect result may escape a detector. The survey describes a Messaline device that senses whether an injected fault is activated and produces an error, and Ferrari experiments in which detection depended on fault type and on how frequently the affected routines were exercised. These details explain why the mere number of injections is an inadequate account of the experiment's outcome. [AFC-W003, pp. 77–80]

Selecting an injection technique also selects what can be reached and how the experiment may interfere with the system. Pin-level hardware methods can control particular physical intervention points and offer high time resolution, but do not literally reproduce every fault inside a chip. Contactless physical methods can reach internal behavior while providing less precise control over the exact time and location. Software methods offer access to applications and operating-system state without requiring the same specialized hardware, yet cannot reach every hardware location. Their added execution and instrumentation can disturb the workload or change the software structure. These are substantive differences in the relationship between intervention and object. [AFC-W003, pp. 76–79]

Timing matters to the observation as well as the intervention. A method with inadequate time resolution may miss rapid propagation even if it is adequate for a longer-lived memory error. The survey discusses combining software injection with hardware monitoring to address such cases, while retaining the extra cost and constraints of the monitoring equipment. Compile-time modification and runtime triggering likewise provide different control: placing a change in a program image differs from invoking an injector when a timer expires, a specified instruction is reached, or an exception occurs. The useful technique depends on the fault behavior and question being investigated. [AFC-W003, pp. 79–80]

The reported tool comparisons show why reach is consequential. In the cited MARS experiments, pin-level injection, heavy-ion radiation, and electromagnetic interference exercised different error-detection mechanisms. The methods were complementary in the surveyed results. Evidence about one chosen family of interventions cannot automatically establish coverage of error behavior that the family does not

produce. The survey also reports undetected incorrect outcomes in selected Xception tests; that observation remains attached to the tested processor units and setup rather than becoming a general percentage for all computers. [AFC-W003, pp. 78, 81]

Finally, the survey expressly distinguishes fault-injection findings from operational measures such as mean time between failures and availability. Choosing and injecting faults does not determine how often those faults occur in service. A collection of tests can investigate behavior conditional on specified interventions without providing the occurrence rates and representative exposure needed for an operational frequency estimate. A high pass rate across chosen injections therefore cannot simply be relabeled high field availability. The distinction is relevant even when the injections and observations are perfectly executed. [AFC-W003, pp. 75–76]

Alvaro, Rosen, and Hellerstein: choosing faults from an outcome's dependencies

Peter Alvaro, Joshua Rosen, and Joseph Hellerstein address a difficulty that arises when a service can fail through combinations of disruptions. Selecting individual faults at random may quickly reveal shallow bugs while rarely reaching a particular sequence of message losses and crashes. Their lineage-driven fault injection, or LDFI, reasons backward from a successful outcome. It asks which events and messages supported that outcome and what combinations of allowed faults could remove all of the known support. This is a method for choosing informative candidate experiments. [AFC-W010, §§1–2]

The paper's historical Kafka replication example shows why a combination matters. In the described beta-version protocol, a leader becomes separated from other replicas. Those replicas are removed from the in-sync set; the remaining leader acknowledges a write without disseminating it, then crashes. The acknowledged write is lost. A nominal count of replicas does not describe the sequence that defeats the stated stable-write requirement. [AFC-W010, §1.1]

LDFI starts from the computational lineage of a good outcome: the sequence and dependencies through which it was obtained. Its MOLLY prototype turns information about uncertain communications into a logical problem for a satisfiability solver. A solution identifies a candidate combination of failures that could defeat the currently known derivations of the outcome. The system then performs a concrete execution with that candidate. This forward step is essential. A program may have a recovery path or alternative derivation that was absent from the earlier successful execution, so removing its known support need not actually make it fail. [AFC-W010, §1.2; appendix B]

The procedure alternates between those steps. A further successful execution can reveal additional support that the next candidate must defeat. An execution that violates the chosen assertion supplies a counterexample. When the specified search completes without such a counterexample, the authors give an assurance for the declared configuration and bounded model. Their soundness account depends on confirming candidate violations through execution rather than publishing every candidate generated by the solver as an actual bug. The completeness argument concerns coverage within its formal setting. [AFC-W010, §1.2; appendix B]

The bounds define the proposition being checked. The program, topology, inputs, and assertions are supplied. A logical end-of-time bound limits the execution length; an earlier end-of-finite-failures bound specifies when message loss ceases, leaving time for the program to respond; a separate bound limits permanent node crashes. The paper's example of six global transitions, message loss permitted during the first four, and at most one crash is an example of a particular failure specification. Byzantine behavior and crash recovery are outside the stated failure model. [AFC-W010, §2.1.1]

The abstraction of time has an additional effect. MOLLY evaluates the programs in a synchronous simulation with a deterministic ordering for successfully delivered messages. This focuses the analysis on faults and reduces the execution space, while potentially missing bugs that depend on asynchronous

ordering and timing. The paper explicitly says that this sacrifices completeness in the more general model. Similarly, the assurance relies on internal determinism: repeating the same faults must reproduce the relevant outcome lineage. Merely using the tool on an internally nondeterministic protocol does not confer the same completeness guarantee. [AFC-W010, §§2.1, 7]

Language and evidence requirements also remain. The method needs identifiable outputs, sufficiently fine-grained lineage, and control over the relevant communications. The prototype uses Dedalus programs to provide these facilities. Where a derivation depends on a fact's absence, the lineage analysis must account for why that fact did not occur. Appendix B describes different treatments of this issue, including an explicitly incomplete option that ignores such provenance. The assurance depends on these analysis conditions. [AFC-W010, §2.2; appendix B]

The evaluation is informative both for favorable cases and for a case that resists a blanket efficiency claim. In the reported Kafka model comparison, MOLLY found a counterexample in 38 concrete executions and 3.74 seconds; the random strategy averaged 1,183.12 executions and 133.30 seconds over 25 runs. For the three-phase-commit model, however, MOLLY took 55 executions and 9.60 seconds, while the random strategy averaged 40.60 executions and 6.24 seconds. These are source-reported measurements for the studied configurations. They show why fewer irrelevant candidate executions can be valuable, while also showing that the analysis is not uniformly faster for every example. [AFC-W010, §5, fig. 12]

The paper also distinguishes finding a counterexample from ruling out counterexamples within a bounded space. A random strategy may discover one quickly, yet an absence claim would require coverage it has not thereby provided. The reported examples of bug-free models compare MOLLY's executed tests with the much larger set of possible failure combinations under the chosen parameters. Their relevance is to the work required for that specified verification task. [AFC-W010, §5, figs. 11–13]

To see the boundary concretely, change the proposed application from the given inputs and message-loss/crash model to a new topology with recovery after a crash. The previously completed analysis no longer establishes the new proposition. The difference follows from the declared scope of the paper's model.

La Porte: efficiency and vulnerability in interdependent infrastructures

Todd La Porte's selected chapter passage addresses large technical systems whose components and organizations are connected across locations. The examples include mutual dependence among electricity, telecommunications, transport, and water services. A disruption is not confined by the organizational ownership of the first affected component. He describes tight coupling as a source of efficiency in ordinary operation that can also increase vulnerability. Defensive management, stockpiling, and relationships with other organizations can then consume some of the efficiency the coupling was intended to provide. These are propositions in his organizational account of infrastructure protection. [UX-W005, pp. 135–137]

The same passage reports Charles Perrow's argument about conflicting demands for authority: tightly coupled operation can call for centralized coordination, while unexpected problems can require discretion among those closest to them. La Porte's surrounding discussion also makes the response depend on the character of disruption. Relatively predictable hazards support routinized preparation; purposive, changing threats can require other capabilities. Maintaining several modes of readiness creates organizational and budgetary demands that may be difficult to sustain after a conspicuous event has passed. [UX-W005, pp. 136–138]

An official practice account: resilience retains resources and responsibility

The 2020 US interagency *Sound Practices to Strengthen Operational Resilience* supplies a further kind of source: an official practice formulation. The cyber-risk table describes governance, identification, protection, detection, response, recovery, and third-party risk management. It identifies data, personnel, devices, facilities, systems, and external providers as part of the dependencies of critical operations. Its account preserves a declared tolerance for disruption rather than promising uninterrupted performance in every respect. [SG-W029, selected body and table 1]

The table's detail is relevant because protection does not remove ongoing activity. It calls for training, maintenance, configuration management, monitoring, tested backups, exercised response and recovery processes, and independent oversight. Lessons from incidents are to inform later activities. Outsourcing likewise does not remove the firm's responsibility for risks associated with a critical operation: the formulation retains due diligence, contracts that allocate responsibilities, monitoring, and checks of a provider's ability to continue or restore service. These prescriptions describe organizational work that remains around the technical arrangements. [SG-W029, table 1]

Fault construction, observation and recovery

The selected sources offer different answers to different questions: a conditional construction from fallible elements, an analysis of common-cause modeling, a survey of experimental apparatus, a lineage-guided search method, and organizational or official accounts of resilience.

S31 follows the subsequent work of analyzing a discovered weakness, changing an arrangement and retaining the change through chaos engineering and organizational learning.

S57 Preserving essential service under failure

Insufficient capacity need not remove every useful function at once. The accounts here develop ways to retain an acceptable computation or service while relinquishing some work. Scheduling, application design, feedback control and incident response determine what survives. A further problem arises when the changed operation is difficult for its human supervisors to recognize.

S30 treats reliability and fault handling. The following sources concentrate on deliberately reduced service: how the reduction is prepared, selected and operated, and what its recipients lose.

Liu and colleagues: finish the mandatory computation

Jane Liu and colleagues' 1991 account of imprecise computation divides a time-critical task into mandatory and optional work. The mandatory part produces an acceptable result; optional execution improves its quality. A scheduler can stop the optional work when its time is exhausted and still return something useful. This requires a program designed to produce such a result. An arbitrary program interrupted before completion does not acquire that property merely because a deadline has arrived. [CG-W013, pp. 58–59]

The authors describe several ways to construct the task. In the milestone method, a computation records intermediate results with indicators of their error. Further execution improves the result, and the indicators let a recipient assess an early answer. A sieve method instead omits complete steps. Their radar example can omit an update to an estimate of background noise and reuse the earlier estimate. A third method provides two versions: a precise primary computation and a shorter, less accurate alternative. The alternatives make different forms of partial work useful. A half-executed sieve step or an unfinished alternate version may contribute nothing to the returned result. [CG-W013, p. 59]

Each construction has an implementation cost. Milestones require recording intermediate values. Sieve methods complicate scheduling when the optional steps must finish as units. Multiple versions need storage and a decision about which version can finish. The scheduler also needs bounds on mandatory execution. If all mandatory portions cannot fit within the available time, discarding optional portions cannot by itself make the task set feasible. The authors make mandatory feasibility a separate test before pursuing better quality. [CG-W013, pp. 59–61]

Their basic model includes ready times, deadlines, processing requirements, importance weights and precedence relations. Additional optional execution is assumed not to increase the result's error. Several possible objectives remain: minimizing total error, limiting the greatest error, reducing the number of discarded optional tasks, or controlling delay. These choices produce different scheduling problems. Finishing by the deadline is consequently only one part of the specified service; the quality measure and the distribution of optional work also matter. [CG-W013, pp. 59–61]

The later queueing discussion distinguishes responsive service from a service commitment made on admission. A responsive scheme can reduce the service assigned to waiting tasks as congestion develops. It can clear an overload sooner, while changing what an already waiting task will receive. A guaranteed-service scheme decides on arrival which service a task will receive and preserves that decision. Tasks admitted before overload retain their promised full service, at the expense of slower clearance of the backlog. The authors develop these choices through queue models with specified arrival and service assumptions and identify further comparisons to be studied. [CG-W013, pp. 66–67]

Klein and colleagues: regulate optional application work

Cristian Klein and colleagues' 2014 Brownout system adjusts a cloud application's demand by omitting optional response components. A parameter called the dimmer controls the probability that an optional component will execute. Individual requests sample that probability, while a feedback controller changes the parameter from measured response times. The application can thereby reduce the work required per request when its workload grows or its assigned processing capacity declines. [CG-W014, §§2–3]

The application must expose useful divisions of work. In their auction-site implementation, the researchers added a simple recommendation feature that could be omitted while the main response remained available. A script reads a small cached file containing the dimmer, decides whether to generate recommendations and reports response time to a local controller. The controller writes a replacement dimmer file and renames it atomically. This implementation makes the response path, measurement and actuation concrete: optionality is prepared in the code before the controller can use it. [CG-W014, §4]

Their bulletin-board implementation has two optional components, comments and recommendations. It first samples whether to include comments and then, if comments are included, samples again whether to add recommendations. With the same dimmer value for both independent trials, the probability of including comments is the dimmer value, while the probability of including both is its square. The researchers chose that dependency between features. It determines which combinations users can receive and how a controller setting changes the work performed. [CG-W014, §4]

The controller uses a simplified relationship between the dimmer and the maximum response time observed in a sampling window. It estimates changing gain online and applies feedback with protection against accumulated control action beyond the dimmer's limits. Tuning changes the balance between fast response and sensitivity to noise or model error. If response time remains excessive with all optional work disabled, the requested operating point is infeasible: the controller has no further optional work to relinquish. [CG-W014, §3]

The experiments run an application stack inside a virtual machine on one physical machine, with the CPU as the bottleneck. Tests change processor resources, concurrent users, or both, and measure the time to receive a complete HTTP response. The controller reacts after observing the change, so transient violations remain. More conservative tuning recovers more slowly while reducing aggressive overshoot. The authors compare settings using a constructed revenue measure that awards one unit for a request and an additional half-unit for a recommendation. Their favorable comparisons concern that valuation and the tested workloads. The measurement was not customer sales. Extending the approach to applications distributed over several machines remained future work. [CG-W014, §§5, 7]

Meza and colleagues: maintain emergency feature controls

Justin Meza and colleagues' 2023 account of Defcon begins with an operational choice at Meta: reduce the work needed to provide a product by temporarily changing some of its features. Sending a message can remain available while an online-status indicator is omitted. A comment service can return comments while replacing model-based ranking with chronological order. Explicit switches, called knobs, select these changes in the affected control flow. The reductions alter what people receive while allowing the core request to proceed. [CG-W015, §§2.2–2.4, 3.2, pp. 609–612]

Product engineers identify candidate features and connect them to a shared configuration system. A testing framework measures resource savings and changes in user behavior. Operators use a portal showing those results and a command-line interface to activate groups of knobs. Knobs can span source files and binaries, identify an owner and belong to a degradation level. Keeping these definitions useful requires product teams to maintain them as their software changes. [CG-W015, §§3.1–3.3, pp. 611–612]

The actuation path is deliberately kept dependent on few other systems so that it remains usable during partial failures. Server changes can apply in seconds. Client changes can prevent requests from reaching servers, but configuration must first reach the devices. A push mechanism takes around thirty minutes across the client population. Periodic full configuration pulls require bandwidth and server resources; an emergency mechanism pulls selected configuration fields with manual intervention. The location of the switch therefore affects both the work saved and the time needed to obtain that saving. [CG-W015, §3.2, p. 611]

Ordinary operation is Level 4, with degradation knobs disabled. Levels 3, 2 and 1 were designed around target demand reductions of 5, 10 and 20 percent, with progressively greater product impact. These targets reflect Meta's failure domains and expected disruptions. Responders inspect demand, tested savings and product effects, activate an appropriate level and continue monitoring. The reported October 2022 demand surge prompted successive activation of two levels, followed by withdrawal when the surge passed. This is a reported production response; the savings targets guide its preparation rather than guarantee a fixed reduction in every event. [CG-W015, §1, pp. 607–608; §3.5, pp. 613–614]

Savings cannot be inferred solely from the code that a knob skips. Tests trace changes into dependent services, including services without knobs of their own. Region-drain experiments also show that traffic relocation and cold caches change the comparison. Small controlled tests have a concurrent comparison group; larger exercises can instead use a modeled baseline built from forecast and backcast signals. Those estimates and the directly observed comparisons perform different evidential jobs in the account. [CG-W015, §§4.1–4.5, pp. 614–618]

Users bear some of the cost. One Level 1 product test reduced app usage time by 11 percent; combined tests increased reports of unexpected behavior. Alongside such outcomes, operators need current measurements, regression investigations, rehearsals and training. The paper describes continuing engineering and organizational work to keep the emergency option available. The existence of a switch is only the beginning of that preparation. [CG-W015, table 2, pp. 616–618; §5, pp. 619–620]

Sarter and Woods: recognize what the changed mode will do

Nadine Sarter and David Woods's 1995 article develops the human demands of supervising automation with many operating modes. A mode changes how a system responds to input. Transitions can follow an environmental event, a protection function or an earlier command without a new direct instruction. A supervisor needs to know which mode is active, which is armed for later activation and what conditions will cause a transition. A status indication may supply only part of that knowledge. [CG-W016, pp. 5–9]

Their account includes a part-task simulator study involving twenty airline pilots. The scenario introduced abnormal, time-critical events, including loss of the glide-slope signal during final approach. About half the pilots did not understand what that loss meant for the flight-management system; some had difficulty detecting the problem during the simulation. Only five knew when to expect the indication that go-around mode was available. The difficulty involved both recognizing a changed condition and anticipating which action would next become possible. [CG-W016, pp. 11–12]

The aborted-takeoff probe showed another distinction between knowledge and action. Sixty-five percent of participants were unaware that the autothrottle system controlled thrust in that situation. Only four of twenty managed the automation completely correctly, and one of those explained that he was following a procedure rather than understanding the active automation. Some pilots could describe relevant settings afterward without having used that knowledge during the event. The authors call attention to knowledge that remains inert when the situation requires it. [CG-W016, pp. 11–12]

Their proposed responses include feedback that makes present and future behavior more apparent and training that activates knowledge in context. An interaction history or an additional sensory channel

might help, but added information can also burden attention. Requiring consent before every change could keep people involved while making repeated checks automatic or interrupting other work. The article presents these as design directions requiring investigation. Its study explains a specific difficulty in operating changed automation; its proposals address the information and practice needed to use the remaining capability. [CG-W016, pp. 13–16]

S54 develops other accounts of control, reconfiguration and recovery. The present accounts concentrate on a selected reduction of service and on the knowledge required to operate when behavior changes.

S31 Chaos engineering and organizational learning

Connecting experiments with changes in practice

An experiment can identify an unexpected dependency without changing it. An organization can observe repeated failures without learning how to prevent them. The material selected in this section concerns the practices that connect intervention, observation, interpretation, and subsequent work.

The practitioner principles: make a claim about observable behavior

The *Principles of Chaos Engineering* formulation defines a discipline of experimentation intended to build confidence in a system's ability to withstand turbulent production conditions. Its problem is not limited to a failed component. Interactions among functioning services can produce unexpected outcomes, while unusual events can expose weaknesses in fallback settings, retries, capacity, and dependencies. These are examples of the systemic uncertainty that the formulation proposes to investigate. [AFC-W004, definition and opening discussion]

The proposed experiment begins with a measurable output that indicates ordinary behavior, then states a hypothesis that this behavior will continue in a control and an experimental group. The experimental change represents an event such as a server crash or severed connection. Investigators try to disprove the hypothesis by looking for a difference in the measured behavior. The formulation's "steady state" therefore concerns a selected observational description over a period, rather than an internal state of each component. Throughput, error rates, and latency percentiles are among its suggested outputs. [AFC-W004, "Chaos in practice"; "Build a hypothesis around steady state behavior"]

The distinction between output and internal attribute is consequential. The experiment is directed toward whether the service continues to work under the chosen conditions. Merely confirming that a disturbance was introduced does not answer that question. Conversely, finding a weakness supplies a target for further improvement. In this formulation, discovery is followed by the opportunity to address the weakness; a disturbance by itself is not a repair. The underlying choice of measurable output remains part of what the experiment asks. [AFC-W004, "Chaos in practice"; advanced principles]

The advanced principles retain the surrounding engineering work. They recommend choosing realistic events by likely occurrence or potential impact, including non-failure changes such as traffic surges. They prefer production experiments because environment and traffic affect behavior. They also call for automation of orchestration and analysis, partly because repeated manual experiments consume labor. At the same time, they explicitly recognize possible short-term customer harm and assign responsibility for containing the experiment's effects. [AFC-W004, advanced principles]

Brier: connect stated probabilities with subsequent events

Glenn Brier's 1950 article begins with a practical objection to forecast verification: a scoring system can reward statements that serve the score better than the forecast's users. A forecaster may learn to report something other than the probability they judge credible. Brier investigates a verification method for probability forecasts, asking how its incentives affect what the forecaster says. His categories must be mutually exclusive and exhaustive, with their assigned probabilities summing to one. The record therefore specifies both what outcomes count and how much confidence was assigned to them. [CG-W021, p. 1]

For each forecasting occasion, his score sums the squared differences between every category's assigned probability and its observed indicator, one for the category that occurred and zero for the others. It then

averages those sums over the forecasting occasions. A perfect forecast scores zero; certainty in a category that does not occur scores two. His rain/no-rain example counts the errors for both categories. The article's normalization consequently gives a range from zero to two. [CG-W021, pp. 1–2, equations 1–2]

The worked example links ten forecasts with the events that subsequently occurred. Their score is 0.19. A forecaster assigning the sample's rain frequency, 0.3, on every occasion would score 0.42; using a climatological rain probability of 0.2 would score 0.44. The sample frequency is a benchmark calculated after observing the period, whereas a working forecaster must estimate such probabilities from available information. Brier uses the comparison to distinguish recognition of an average departure from climatology from discrimination among individual occasions. Both can contribute information. [CG-W021, p. 2, table 1 and equations 3–4]

Brier also groups a larger series of eighty-five forecasts by issued rain probability and compares those groups with observed rain frequencies. The relationship shows where the forecaster's confidence needs adjustment. He says that this relationship alone does not establish usefulness: how often different probabilities are issued also matters. His closing account reports several months of experimental six-hour precipitation forecasts at the Weather Bureau. Some initial criticism concerned misunderstanding the method or disappointment at how little confidence the forecasts justified; he also leaves substantive questions about the scoring method for further investigation. [CG-W021, pp. 2–3, table 2 and conclusion]

Nosek and colleagues: preserve the plan and make changes inspectable

Brian Nosek and colleagues' 2018 account concerns the relationship between research questions, analysis choices and observed results. They distinguish predictions tested by observations from explanations generated after seeing them. Both activities contribute to inquiry. The difficulty arises when an explanation discovered in the data is presented as though the same data had tested an advance prediction. Their account joins reasoning biases with research incentives and the many decisions available when analyzing a dataset. [CG-W023, "Mental Constraints on Distinguishing Predictions and Postdictions"; "Standard Tools of Statistical Inference Assume Prediction"]

Their proposed preregistration mechanism deposits the questions, design and analysis plan in an independent registry before outcomes are observed. The registry preserves the record and makes it discoverable, sometimes after an embargo. A later report can then be compared with the plan, distinguishing planned tests, amendments and newly generated questions. The authors describe the case for preregistration as theoretically strong and supported by some correlational evidence, while explicitly saying that sufficient experimental evidence establishing superiority for reproducibility was not yet available. [CG-W023, "Preregistration Distinguishes Prediction and Postdiction"]

The practical examples retain the work required when research departs from its plan. A study may recruit fewer participants than intended or reveal violated statistical assumptions. Researchers can report amendments, their reasons and what outcomes had already been seen. Other possibilities include staged registration, blinded datasets, decision trees and standing procedures. These methods have conditions: an earlier stage may reveal the outcome intended for a later test, and a decision tree can formalize a biased stopping rule. Recording that rule makes it inspectable without improving its statistical properties. [CG-W023, challenges 1–2]

For existing datasets, the relevant separation concerns information access. Data may already exist while remaining unseen by the analyst. Conversely, earlier summaries or knowledge of correlated variables can influence a nominally new analysis. The authors therefore ask who has seen what and what has been communicated. They also describe splitting a dataset: exploration develops a prediction using one part, and registration precedes testing on a sealed part. Fast experimental workflows can use a related

sequence, treating initial experiments as exploration and then repeating a selected experiment with its design and analysis recorded in advance. [CG-W023, challenges 3–5 and 7]

The record must also extend beyond the successful attempt. In their example of a research program producing one positive result in twenty tries, registering every attempt does not prevent misleading selection if only the positive result is reported. Preserved, accessible registrations and outcomes permit examination of that wider history. The authors distinguish ordinary registration from Registered Reports, which add methodological review and in-principle publication acceptance before results are known. They also retain a further difficulty: selective narrative emphasis can overstate a subset of results even when the planned quantitative outputs are all reported. [CG-W023, challenges 6 and 9; “Advancing the Motive for Preregistration”]

Van den Akker and colleagues: compare expected benefits with published results

Olmo van den Akker and colleagues tested several expected benefits in an article first published in 2023. They compared 193 published psychology studies earning preregistration prizes or badges with 193 related non-preregistered studies published in the same year. Registered Reports were excluded. Controls were selected from Web of Science’s related-publication lists, using shared references to match subject matter. This compared observed publication groups whose researchers had chosen their own practices. [CG-W024, pp. 5427–5428]

The comparison found almost the same proportion of positive results, 0.69 and 0.68. It did not support the expected reductions in effect sizes or gross statistical reporting inconsistencies. Preregistered studies more often reported power analyses, 55 percent against 23 percent, and had larger median samples, 216 against 116. The authors also disclosed a change to their planned positive-result analysis after recognizing an asymmetry in how hypotheses had been coded; they reported both versions, each giving an unsupported predicted difference. [CG-W024, p. 5430]

Assessing the results required choices about what counted as a hypothesis test. In preregistered studies, reported results were matched to recorded hypotheses. For controls, the researchers extracted reported statistical results while excluding identified exploratory tests and checks. They acknowledge that some remaining results might not have been intended as hypothesis tests. They also discuss self-selection into preregistration and the difficulty of making causal claims from the comparison. Their evidence supports particular differences in reported planning and sample size, while leaving the expected reductions in the selected bias indicators unsupported. [CG-W024, pp. 5429–5432]

Microsoft's guidance: test a complete flow against its targets

The Microsoft guidance distinguishes three levels of activity. Fault injection introduces a fault into a component, dependency, or system path. Chaos testing investigates a particular resilience hypothesis under disruption. Chaos engineering incorporates such testing into culture, engineering practice, and the development cycle. These definitions place a technical intervention within an experiment and the experiment within continuing organizational work. [AFC-W005, definitions]

Its full-flow account explains why component recovery is insufficient as the sole criterion. A component may recover within its target while the dependent service still takes too long to become usable. Detection, response, and the recovery of other dependencies matter to the observed result. Similarly, responsibility for testing depends on what a team controls in infrastructure, platform, and software services. A managed service can move some work to a provider while leaving the workload's interactions, fallback behavior, throttling response, and boundary assumptions to be examined. The source calls attention to these differing responsibilities rather than treating a purchased service as the end of the reliability question. [AFC-W005, scope by service type; end-to-end targets]

The guidance also keeps preparation and revision explicit. It recommends scenarios based on critical flows and failure modes, measured baselines, experiments with clear hypotheses, containment, and the ability to stop and restore. It favors beginning in lower-risk environments before proceeding to production with safeguards. Results feed new tests, recovery-plan changes, and remediation work; architectural changes can make earlier assumptions obsolete. [AFC-W005, critical flows and failure scenarios; fault injection and chaos engineering]

Monperrus: improvement must occur in the software or its engineering process

Martin Monperrus's *Principles of Antifragile Software*, in the 2017 paper, asks how an error can become more than something to survive. His account distinguishes static fault tolerance from an adaptive mechanism that learns when errors occur. Static tolerance can preserve a result without becoming more capable afterward. In his proposed use of "antifragility," learning and improvement are the distinguishing conditions. [AFC-W007, §§3.1–3.2]

His distinction between state repair and behavioral repair adds detail. A state repair changes the current program state, such as a data structure. Behavioral repair changes the program's subsequent conduct through a patch. The latter can leave a change in the code after the triggering failure, which is why he treats it as particularly relevant to improvement through encountered faults. The paper refers to existing repair research as examples. [AFC-W007, §3.2]

Fault self-injection in production enters his argument through its effect on the engineering ecosystem. Repeated disruption exercises recovery behavior, draws engineering attention to it, and creates opportunities to learn from observations. He explicitly says that the injected fault alone is not what improves the system: design principles, the engineers' thinking, and other surrounding practices matter. He also requires a balance between the dependability lost through introduced failures and gains from software improvements. Measuring that tradeoff is a problem for the proposed engineering program. [AFC-W007, §3.3]

The process discussion supplies two specific mechanisms. First, a regression test that reproduces a discovered bug can remain in the test suite after repair. The accumulated suite can support later changes by making some regressions easier to detect. Monperrus relates testing, refactoring, and frequent deployment through this retained aid. [AFC-W007, §4.1]

Second, he discusses dependency on indispensable individuals. Moving people among projects can distribute knowledge and reduce the consequences of losing one person, while producing immediate learning costs, delays, and frustration. Reducing concentrated key-person dependency can require repeated smaller costs. [AFC-W007, §4.2]

The broader relation between process and product remains expressly hypothetical. Drawing on Conway's law, Monperrus proposes that processes embodying his error-oriented principles may be better at producing software with the corresponding properties. His mention of Ariel Tseitlin's *The Antifragile Organization* identifies a related organizational argument. [AFC-W007, §4.3; AFC-W019, through Monperrus]

Edmondson and Cannon: seeing failure is a different competency from analyzing it

The HBS *Working Knowledge* excerpt by Amy Edmondson and Mark Cannon explains why awareness of a failure may leave its causes unchanged. An error count can show that something is wrong without showing what to correct. Their examples include discussion forums in medicine and military after-action reviews, but the operative point is the work of inquiry: people need to disclose relevant knowledge, hear

different accounts, and examine the event deeply enough to alter their understanding. [AFC-W015, opening analysis discussion]

The authors identify organizational and personal obstacles to that inquiry. Revisiting a failure can threaten confidence and invite defensive explanation. Reward systems can favor rapid, decisive action over patient investigation. Habitual attribution can explain away a large failure as outside interference or a small one as an irrelevant fluke. The result can be repeated experience without corresponding learning. Their reported telecommunications case illustrates such rationalization. [AFC-W015, barriers to analysis]

Their proposed supports are correspondingly substantial. They describe forums, resources, incentives, technical expertise, diverse perspectives, and skill in managing group inquiry. Such arrangements are meant to make interpretations contestable and the findings usable elsewhere in the organization. A reported hospital example extends analysis to near misses and small errors, with formal studies, clinical teams, and logs that preserve information for further examination. These details matter to gambits engineering because learning depends on organized work and retained records, rather than on failure's occurrence alone. [AFC-W015, organizational forums and selected hospital example]

The excerpt expressly distinguishes identifying failures, analyzing them, and deliberate experimentation as relatively independent competencies, rather than a required three-step sequence. Different competencies can be inhibited by different social and technical conditions. Their treatment supplies reasons to investigate how an organization notices, understands, and acts on evidence, without making failure a necessary or sufficient cause of improvement. [AFC-W015, concluding distinction among competencies]

The practice guide: experiment design and its supporting organization

The China Academy of Information and Communications Technology's 2021 *Chaos Engineering Practice Guide* (《混沌工程实践指南》) presents experimentation as a core activity supported by strategy, people, culture, risk management, and assessment. It also describes subsequent improvements to architecture and development or operations procedures as extensions of the practice. This organization of the guide makes the relationship between an experiment and its surrounding institution explicit. [AFC-W014, printed pp. 9, 23–28]

The design account begins with a hypothesis about a possible stability weakness, informed by user feedback, test records, and understanding of the architecture. Probability, scope of effect, and seriousness enter the proposed prioritization. The system needs to be performing a business scenario for the experiment's effects to be observed; an oversimplified scenario can leave important tasks unexercised. The guide describes production traffic branches or canary versions as possible ways of limiting exposure, and suggests replaying production variables or using representative simulated data when working outside production. These are alternative arrangements for the experimental situation, each retaining work to make the scenario relevant. [AFC-W014, printed pp. 9–10]

Measurement is designed alongside the intervention. The selected passage recommends observing extremes as well as averages, supporting tasks as well as final outputs, and designated stopping indicators. Its categories include time, throughput, failures, resource use, and business feedback such as user retries or error reports. The choice of metrics is to depend on the object, scenario, and monitoring facilities. This connects what can be concluded to what the experiment is equipped to observe. The guide does not treat the introduction of a disturbance as a sufficient account of the resulting service condition. [AFC-W014, printed pp. 10–11]

Cost limits the proposed coverage. The guide says that simulating every disturbance is impractical and therefore requires selection. It proposes relevance to the hypothesis, frequency, representativeness, and

significant impact as selection considerations. Fixed and random injection are suited to different situations in its account: repeated automation is more suitable for comparatively contained, short-lived disturbances with lower recovery cost, whereas wider, longer, or more costly cases may call for designed exercises at specified times. It further distinguishes single from combined disturbances, with combinations able to expose cascading behavior missed by individual factors. [AFC-W014, printed pp. 11–13]

The support discussion keeps possible losses visible. Production experiments can cause failure; a system with inadequate observability makes defects harder to identify and locate; a weak initial stability basis can make repeated exercises disruptive and demoralizing. The guide discusses containment and lower-risk environments as possible responses. Its culture account calls for exposing rather than hiding defects and for habits developed through practice. [AFC-W014, printed pp. 23–24]

Return on investment is explicitly difficult to assess, partly because a prevented failure is less conspicuous than a realized incident. The guide proposes separate assessment of adoption, capabilities, and benefits, with the framework adapted to the organization's people, business, and architecture. Selected benefit indicators concern response efficiency, repeated defects, monitoring and detection, transparency, and experimental efficiency. Its architectural and procedural extensions ask whether discovered problems are actually resolved, including whether recurring lower-level errors warrant changes in development or operations discipline. The framework proposes what to examine when assessing benefits. [AFC-W014, printed pp. 24–28]

Owotogbe and colleagues: what a review can tell us about the practice

Joshua Owotogbe and colleagues' v2 review manuscript, dated 19 June 2025, brings together 96 selected sources: 49 academic sources and 47 grey-literature sources. It reports on a chaos-centered body of material covering January 2016 through April 2024. [AFC-W011, §§3, 6, 8]

Its selection and analysis procedure helps define that scope. Searches covered four academic databases and grey literature, followed by eligibility screening and quality assessment. The eligibility requirements included English and accessible full text, while exclusions concerned short items and material outside the chosen chaos-engineering focus, among other conditions. The authors used thematic coding with review and reconciliation between the first two authors. [AFC-W011, §§3.2–3.5, 8.3]

The practice account distinguishes clear objectives, small initial experiments, automation, monitoring, stakeholder participation, and documentation. Its associated problems include misaligned experiments, inadequate planning, insufficient tools, weak collaboration, and neglected containment or restoration mechanisms. The tables record sources associated with those themes. [AFC-W011, §6.3]

Its evaluation discussion is also broader than a single technical metric. Numerical observations include response behavior, throughput, resource use, errors, availability, and recovery. Other forms of evaluation concern user experience, business consequences, monitoring gaps, and team communication or decision-making. [AFC-W011, §6.4]

The review's discussion retains substantial work around the experiment: environments, telemetry, configurations, specialized knowledge, repeated analysis, and coordination among teams. Its authors identify limited empirical research on how teams acquire the relevant skills, deal with cultural barriers, or manage resource constraints. They also acknowledge that the taxonomy has not received formal external validation and that alternative terminology may leave relevant material outside the corpus. [AFC-W011, §§8.2–8.3]

Two narrower engineering accounts

The Volcengine developer-community account separates whether a fault was injected from whether a business metric changed. It describes cases in which an injection succeeds while service metrics remain normal because protective arrangements work, and proposes fault-specific verification indicators alongside service observations. Its workflow also separates execution, result analysis, and subsequent optimization. A further passage describes a Bayesian structural time-series model for estimating what the observed series might have been without injection; this modeled counterfactual is not a simultaneous observation of both alternatives. [AFC-W020]

A selected SRE passage from Huawei Cloud's 2023 whitepaper uses 反脆弱 in connection with controlled fault-injection exercises intended to improve reliability and recovery. It places this within organization, culture, clear processes, and an operations platform. The selected responsibilities include planning, exercise design, observation, emergency restoration, review, accumulated scenarios, and revisions to procedures. [AFC-W021, §5.2.3.1, printed p. 35]

Verhulst: where the system boundary places improvement

Eric Verhulst proposes assured reliability and resilience levels distinguished by how components meet a contract in the presence of faults. A component may itself be a system within a larger system. He distinguishes preserving service through redundancy and repair from a wider process that updates the system, and describes aviation through a boundary including operating conditions, stakeholders, feedback, and authorities. If a technical artifact changes only through surrounding maintenance and institutional activity, the source account concerns that larger arrangement. Verhulst's proposed levels are normative, and his industry-wide characterization remains his claim. [AFC-W018, pp. 844–849]

Caulkin's account of Seddon: some recurring work is created by earlier failure

Simon Caulkin's journalistic account of John Seddon's "failure demand" concerns a different relationship between failure and subsequent activity. A customer calls again because something was not done, was done incorrectly, or did not resolve the original need. The repeat contact is work that the service's own earlier conduct helped create. Caulkin contrasts this with the demand the service exists to meet. The distinction concerns the history and purpose of a request, rather than treating every arrival as an independent unit of useful output. [SG-W020]

His selected insurance example describes agents constrained by detailed contracts, repeated transactions to resolve one case, and a redesign intended to let an agent complete the case with appropriate access to expertise. A second example contrasts apparently successful project and activity indicators with whether distressed customers actually return to normal. These reported cases make the argument more informative than a general call for efficiency: the proposed change concerns the organization of the work that generates later demand. Caulkin does not identify the organizations. [SG-W020]

The relevant proposed reduction in work depends on meeting the original purpose more successfully. Refusing access, leaving a problem unresolved, or reclassifying a legitimate need out of the count would not establish the mechanism Caulkin describes. His argument is about reducing preventable return work through service design. [SG-W020]

How experiments enter continuing practice

The accounts in this section describe several different places where a consequence may be retained: a revised program, a regression test, an experimental record, shared expertise, an altered procedure, or a service arrangement that avoids repeated corrective requests.

S42 Representations, standards and knowledge infrastructures

Making something available differs from making it usable

Physical records, symbolic systems, programs, classifications, standards and assurance arrangements make information available for action. Their design shapes which operations a person can perform and which distinctions a representation preserves.

Pattee: symbols require physical realization and interpretation

Pattee's selected discussion relates dynamical laws, constraints, memory and symbolic control. He argues against accounts that detach symbols from physics and against accounts that assume real-time sensorimotor dynamics explain every aspect of life. A relatively persistent memory structure can affect a time-dependent process through physical mechanisms that read or respond to it. The symbolic description and the dynamics are distinguishable, but the connection needs an implementation. [UX-W012, discussion of laws, constraints and memory]

This account does not classify every useful physical constraint as a symbol. A structure can constrain motion without serving as a record interpreted within a symbolic-control system. Pattee's discussion of flexible mechanical constraints, switches and molecular mechanisms asks how stored alternatives can enter dynamic operation. The relevant material is the connection between a comparatively rate-independent record and rate-dependent execution, not a free-standing informational cause outside physical processes.

Pattee's epistemic-cut argument relates descriptions to operations and connects symbolic control with his account of biological origins. [UX-W012]

Etxeberria and Moreno's reconstruction of Pattee's hierarchical-control account [UX-W014] distinguishes structural hierarchies from control hierarchies and discusses constraints, records and symbols in that setting. Its usefulness is the distinction between assembling levels of material organization and specifying how a record controls a process. A higher descriptive level does not become a controller merely by being drawn above another level. [UX-W013, selected manuscript pp. 2–4 and concluding caveat]

Darwiche and Marquis: choose a representation for the operations required

A Knowledge Compilation Map examines propositional theories compiled in advance into representations suited to later reasoning. Darwiche and Marquis compare languages along two dimensions: succinctness and the queries or transformations they support in polynomial time. A language can be convenient for one operation and unsuitable for another. The choice is therefore application-dependent rather than a contest to find one universally smallest or fastest representation. [TA-W008, printed pp. 229–236]

Their framework treats several classes as restrictions on directed acyclic representations in negation normal form. Conditions such as decomposability, determinism and smoothness impose different structures. Those restrictions affect which operations are tractable and how large a representation may need to be. A more concise language can leave an operation hard; a less concise compiled form can make repeated queries manageable.

The authors propose identifying the needed operations before choosing a target language. The relevant saving is not simply deleting information. An equivalent representation can reorganize the information so that a particular query is easier. Conversely, having a small equivalent representation in principle does not establish that it can be constructed efficiently from the original input. Compilation and representation size remain separate obligations.

Jones, Gomard and Sestoft: move known computation to an earlier stage

The opening of *Partial Evaluation and Automatic Program Generation* explains specialization using a program with some inputs already known. A partial evaluator produces a residual program for the inputs still unknown. When given those remaining inputs, the specialized program is intended to return the result the original would have returned from the complete input. The relation is semantic, while its efficiency depends on what can actually be evaluated in advance. [TA-W009, selected printed pp. 1–4 and staging passages pp. 6–9]

The authors' power-function example is concrete. Knowing the exponent in advance can allow recursive control and exponent-dependent work to be unfolded or precomputed. Knowing the base instead, while leaving the exponent unknown, need not yield a significant speedup in the same example. Advance information has to match the structure of the computation; any known input is not equally useful.

Their term binding-time engineering concerns changing when subcomputations occur. It retains the earlier computation, the specialized code and the circumstances in which the preparation remains valid. Correct specialization is not by itself proof of faster execution or lower total cost. [TA-W009]

Necula and Rahul: a smaller oracle can guide a trusted checker

The selected 2001 oracle-based checking paper describes a variant of proof-carrying code. Trusted inference rules are represented as a logic programme, and a nondeterministic interpreter checks the property. A supplied bit stream resolves choices that would otherwise require search. The oracle guides the checker; it does not replace the rules that determine whether the proposed steps are acceptable. [TA-W011, PDF pp. 1–3]

The article contrasts this with carrying an explicit proof syntax for every inference. Its proposal can exploit directed or simple checking procedures without charging the full proof-size overhead of a very general representation. In the reported type-safety examples for code compiled from Java, the authors report oracle sizes averaging twelve percent of code size and a factor-of-thirty reduction relative to an earlier proof representation.

The retained trusted components are decisive: the policy, verification-condition generation, inference rules and checker must be appropriate. Cryptographic evidence of who supplied code differs from evidence about what the code does. A compact oracle can reduce transmitted proof information while verification still requires execution of the trusted procedure. Safety remains relative to the specified property, not unrestricted harmlessness. [TA-W011]

Necula introduced *Proof-Carrying Code* in 1997; the account above follows the later paper. [TA-W010]

Edmonds: greedy choice depends on an independence structure

Edmonds considers finite linear-objective combinatorial optimization and asks why certain problems admit efficient algorithms. In a matroidal problem, feasible independent sets satisfy a structure that permits a greedy procedure to select weighted elements while preserving feasibility. A simple sequence of choices can be globally justified under these conditions, rather than relying on a general belief that local decisions are good. [TA-W012, printed pp. 127–130]

The representation of the feasible family matters: the source explicitly asks how membership is determined. An independence test or oracle is part of the algorithmic setting. Arbitrary constraints do not necessarily have the exchange property behind the characterization.

The selected characterization explains how an independence structure supports a computational method. [TA-W012]

Liskov and Wing: an interface must preserve behavior over time

Liskov and Wing's selected account of behavioral subtyping asks when an object of a subtype can be used while reasoning with the supertype's specification. Matching method names or signatures is insufficient. Invariants and the allowed evolution of state matter. The source includes ways of preserving history properties directly through constraints or indirectly by explaining additional methods through operations already available at the supertype. [W087, selected §§1–2 and 5.2–5.3]

Aliasing makes the issue substantive. The same mutable object can be accessible through more than one name, with one user reasoning through the supertype and another invoking subtype methods. Effects of the additional method can later be visible through the first name. A contract that considers only an isolated call can therefore miss behavior that undermines the first user's reasoning. [W087, printed pp. 1811–1815]

The stack-and-bag discussion explains how an additional operation can preserve or violate what remains fixed in the abstract history. The account thus concerns substitution under a stated behavioral contract, not an absence of knowledge about implementation. The contract itself must specify the relevant guarantees and be justified against the implementation. Liveness and certain cross-object safety properties lie outside the stated scope. [W087, selected printed pp. 1822–1829 and 1836–1837]

Tonet and da Paz: knowledge sharing continues after initial transmission

Tonet and da Paz propose four phases of workplace knowledge sharing: initiation, implementation, support and incorporation. Their account attends to the source, recipient, knowledge and context rather than treating sharing as the delivery of a detachable message. Support after initial transfer matters when the recipient encounters difficulties using what was received. [W088, selected four-phase model and support discussion]

The sequence helps distinguish access from incorporation into practice. A proposal can fail because participants lack the relevant understanding, opportunities or contextual support, even after information has been supplied. The model offers questions and guidance for examining the process.

Buckland: stored things are not identical to knowledge

The conceptual discussion in Buckland's *Information as thing* distinguishes information as a process, knowledge and evidence treated as a thing. Storage and retrieval systems directly handle objects or representations: documents, records, data and other things regarded as informative. Calling them information does not guarantee that a user has learned from them or that their content is true. What counts as informative is situational and can depend on agreement about evidence. [GL-W020, situational information and the topography of information-related activities]

His topography of information-related activities distinguishes the constraints of museums, archives, printed libraries and coded-data systems. Selection, representation, identification, location and physical access are not the same task in each setting. A convenient representation can support one activity without preserving every physical or evidential property of its subject. The account is conceptual classification.

Potts: conventions and context both contribute to meaning

Potts's exposition distinguishes conventional linguistic content from context-dependent interpretation. Indexicals require a setting to determine what is referred to; implicatures involve what a speaker conveys beyond a literal formulation. Compositional rules can support interpretation without eliminating the situation in which an utterance is used. [GL-W014, selected body on semantics and pragmatics]

This supplies a substantive qualification to any picture in which a message is wholly self-contained. The recipient's context and conventions can carry information not repeated in the words.

Star and Lampland: standards coordinate while retaining local work

The selected pages of *Reckoning with Standards* examine standards as arrangements embedded in activities, rather than neutral descriptions floating above them. A standardized category, measure or procedure can enable work across settings while fitting some local circumstances better than others. The work of interpreting, fitting and maintaining the standard does not vanish when the written form is shared. [TA-W025, pp. 3, 6–7 and 10–11]

The chapter's account directs attention to classifications and their consequences for those who must use them. Uniformity can facilitate coordination and also obscure variation or make accommodations necessary.

Star and Ruhleder followed the Worm Community System, built so geographically dispersed geneticists could share information. Some laboratories needed to obtain and configure windowing software and learn how to reach a remote machine before they could use it. The instructions assumed that users already knew how to obtain help with those steps. Even with favorable feedback on the prototype, many biologists struggled to get to routine use. During the study, many respondents turned to the World Wide Web for information exchange. [TA-W024, case study and discussion of signing on]

Archives: context is part of description

The introduction to *Records in Contexts–Conceptual Model*, version 1.0, frames archival description through records and their relations to contexts. It supplies a descriptive model for recording relationships rather than only a flat list of isolated documents. The relevant contribution is an explicit structure for representing provenance and context. [GL-W023, 2023 version 1.0, selected §§1.1–1.2]

Traceability: a documented chain still carries uncertainty

The CNAS 2021 guidance defines metrological traceability through a measurement result's relation to a reference by a documented chain of calibrations, with uncertainty contributions retained. The account concerns traceability requirements and definitions. [TA-W030, cover, foreword and printed pp. 2–3]

The NIST FAQ and policy make the responsibilities explicit. The provider supports a traceability claim; the user assesses its validity. NIST's provision of its own services does not certify every other measurement claimed to be traceable to it. Traceability also does not guarantee fitness for a purpose: the uncertainty may still be too large for the intended use. A certificate cannot substitute for the user's accuracy requirement. [TA-W031; GL-W101, policy §3.2 and FAQ]

NIST attributes its formal definition to VIM. [TA-W031; GL-W101]

Role-based access: permissions are prepared through assignments and constraints

Sandhu, Ferraiolo and Kuhn's 2000 NIST model organizes permissions through roles rather than treating every authorization only as an isolated individual grant. The proposal provides levels of RBAC capability, including hierarchical and constrained forms. The four levels describe the capabilities distinguished in this 2000 proposal. [TA-W032, selected PDF pp. 1–3 and 8]

The selected constrained model distinguishes restrictions on assigning users to roles from restrictions on activating roles in sessions. Separation of duty seeks to prevent one person from possessing excessive

combined authority. It retains the possibility of deliberate collusion among multiple people. Administrators must still assign roles, maintain permissions and enforce the runtime rules.

Internal control: separation, authorization and review remain different functions

The selected 2001 GAO exposure draft discusses segregation of duties, authorization, records and review. Dividing incompatible responsibilities can make some errors or fraud harder for one actor to carry out unnoticed, while authorization specifies who may act and documentation supports later examination. The source also recognizes that collusion can undermine separation. [TA-W033, historical draft, printed pp. 38–45]

This is historical prescriptive guidance. A formal division of duties does not establish that staff actually perform them, that records are complete or that review is competent. The arrangement's operation remains part of what has to be assessed.

The archived AU319 standard distinguishes control activities and monitoring in a financial-statement-audit setting. Monitoring concerns whether controls continue to operate appropriately and whether needed changes are made. [GL-W044, selected §§.21–.22 and monitoring paragraphs]

The IFAC guidance similarly addresses evaluating and improving controls as part of organizational activity. Its overview and appendix do not equate a passed reporting control with effective operational control. [GL-W048]

Zero trust: the boundary is not an unconditional authorization

The abstract of *Zero Trust Architecture* describes an approach that does not assign implicit trust merely from network location or ownership. The relevant shift is toward explicit decisions about access to resources. [GL-W045]

Journalism verification: source and content are separate objects of checking

The selected UNESCO training module sets out work with a source, image, metadata and location evidence. Establishing who supplied an image differs from establishing what it depicts, where it was made or whether its surrounding claim is accurate. A verification workflow distributes those questions across different evidence rather than expecting a single cue to answer all of them. [GL-W047, selected module 6, PDF pp. 100 and 102–103]

The EXIF discussion retains a specific limitation: metadata may be absent, altered or otherwise insufficient. The handbook is a professional curriculum. Interpretation and source assessment remain work performed within the workflow.

Mapping and classification: the useful omission is relative to a purpose

The USGS cartographic orientation treats generalization as a way to make geographic information useful at a selected scale and purpose. Selection and simplification can remove visual or informational detail that a particular map cannot usefully present. The textbook warning makes the condition clear: what is omitted may be important for a different task. [GL-W066, institutional orientation and selected sample passage]

The UBC guide to the Brian Deer Classification System describes a specialized original scope and local adaptations at different institutions. A classification can fit a collection's communities and relationships better when it can be adapted. The same flexibility can create differences among collections that matter for cross-collection use. [GL-W072]

The guide traces the classification system through earlier accounts and describes the work of maintaining categories suited to a local setting.

Shaw and the proof-theory entry: explicit levels of description

Shaw's selected chapter opening distinguishes source, data, generative, structure and meta levels in knowledge and expert systems, drawing on Klir and applying the account to personal constructs. Observations, a model that generates them, a structure organizing that model and questions about the modeling arrangement are not interchangeable. The framework helps identify which level a proposed knowledge representation is meant to address. [SG-W034, selected PDF pp. 1–5]

Shaw reconstructs and applies Klir's epistemological hierarchies. [SG-W034, selected chapter passages]

Rathjen and Sieg's selected proof-theory reconstruction examines formal proof, foundations, reverse mathematics and feasibility. It helps distinguish formal derivability from questions about the resources or strength needed to establish a result. [GL-W087]

Using representations in practice

The section provides source-specific explanations of preparation, representation, interpretation and assurance. Their guarantees differ: semantic equivalence, behavioral contracts, formal tractability, measurement traceability, archival context and professional procedure answer different questions.

S39 Lafitte, Simondon and technical objects

Technical operation, its conditions and its human setting

Simondon's *On the Mode of Existence of Technical Objects* examines technical genesis, functioning and culture. Four scholarly articles reconstruct Lafitte's programme. Archaeology and conservation accounts address human dependence on things and the preservation of material objects.

Simondon: understand an object through its genesis

Simondon begins by arguing that technical objects are poorly understood when culture sees them only as useful tools, alien machinery or imagined robots. They embody human invention and natural processes. To understand them requires technical knowledge: neither owning a machine, operating it habitually nor knowing an isolated scientific law supplies the whole account. His proposed cultural integration of technical knowledge is normative as well as explanatory. [W020, introduction, pp. 15–21]

The first part favors the genesis of a technical type over classification only by use. Objects performing the same named function can have very different internal organizations and histories. A later object is not better in every setting. Simondon's comparison of engines notes that an older design can be more repairable, autonomous and tolerant of imperfect conditions in a particular use. The history is not simply a list of newer devices replacing worse ones. [W020, part I, printed pp. 25–29]

Concretization is his central account of technical development. In an abstract arrangement, parts are conceived as separately complete organs, each with a dedicated function, and must then be made compatible. Their residual interactions can create burdens or conflicts. A more concrete arrangement develops a mutually supporting organization in which a structure participates in several compatible functions. The change is not merely cosmetic simplification or a smaller number of components.

Cooling fins that also reinforce an engine

The engine cylinder-head example makes the mechanism precise. Fins exchange heat with external air, while the same ribbed structure stiffens the head. Because ribbing supplies rigidity, the wall can be thinner than an equally rigid unreinforced shell; the thinner wall can also improve heat exchange. In Simondon's account, structural and thermal functions converge rather than simply competing for a compromise thickness. [W020, printed pp. 27–28; PDF pp. 39–40]

The best location of ribs for strength need not coincide with the best position for airflow and cooling. Simondon therefore describes designs that remain mixed or incompletely concrete.

Simondon connects internal coherence with material, construction-work and operating-energy economies in particular developments. An air-cooled design still needs heat transfer to its surroundings; a water-cooled system can retain useful thermal buffering even while adding pumps and other dependencies. Similarly, removing a manual starting action can create dependence on a battery. The examples are historically situated comparisons, not timeless rankings of all engines. [W020, printed pp. 29–33]

Technical progress can reorganize the functions themselves

Simondon distinguishes small improvements within an existing organization from a more substantial reorganization that changes the relation among functions. A previously troublesome effect can sometimes become a functional resource. The vacuum-tube sequence illustrates how electrode arrangement, feedback and electric fields are reorganized rather than handled as entirely separate problems. Changing

relative geometry can perform work otherwise assigned to an added specialized electrode, while another change can require an additional component. Concretization therefore has no general rule that fewer parts are always better. [W020, printed pp. 32–39]

The Crookes-to-Coolidge example concerns the separation of roles previously assigned to gas that was needed for one effect while obstructing another. The later arrangement retains a heated cathode, vacuum, fields, cooling and precise construction. Likewise Simondon's account of the diesel engine treats compression heat as useful while retaining more exact fuel-injection requirements. A structural gain can be accompanied by a narrower tolerance for some other condition. [W020, printed pp. 38–47]

Minor improvements can be useful without resolving an underlying antagonism. The rotating-anode discussion adds motion to spread heating while preserving a concentrated source. It also adds a new dependency: actual rotation must occur. An energized motor circuit is not the same fact as a rotating target, and a protective interlock can itself fail. [W020, printed pp. 39–43]

He treats complete concretization as a limiting condition, not an achieved property of every mature machine. Full knowledge of all relevant physical interactions is unavailable in practice, and experiments remain necessary.

Hypertely: high performance can narrow the usable setting

Simondon uses hypertely for overadaptation to determinate conditions of production, use or milieu. A highly specialized object may perform well locally while being less capable of functioning elsewhere. Dependence on another technical system is also relevant: a device fitted to one supply network inherits its constraints. Technical success in a narrow environment is consequently different from broad autonomy. [W020, printed pp. 53–58]

Simondon distinguishes adaptation to a given environment from creating a technical–geographical relation that helps sustain the object. This leads to the associated milieu.

The Guimbal turbine: a mutually enabling environment

Simondon's Guimbal example places a turbine and a compact generator within a water conduit. Water supplies the driving energy and removes heat. Pressurized oil lubricates, insulates the windings, transfers heat toward the casing and prevents water entering through the shaft seals. The small generator becomes possible because the surrounding arrangement supplies the heat removal needed at that size. The environment is not an incidental container; its operation participates in making the architecture viable. [W020, printed pp. 57–62; PDF pp. 69–70]

Simondon calls this reciprocal operational relation self-conditioning. A generator designed to depend on water cooling cannot deliver the same power in air.

The associated milieu is partly constituted through natural processes and partly through the technical operating regime. Inventive anticipation represents a future compatible whole during design. Simondon extends this relation philosophically to life and thought.

An ensemble can require separation rather than a common milieu

The audiometer discussion identifies the technical boundary by the conditions of functioning, rather than the manufacturer's enclosure. A supply, oscillators, an amplifier, headphones or a speaker, and the acoustic room can all matter. Stable frequencies and intensities require suitable temperature, voltage and noise conditions. In a relevant configuration, the room belongs to the complete apparatus even though it is outside the instrument's box. [W020, printed pp. 63–66]

Simondon distinguishes merely stabilizing a power supply from feedback that regulates a particular variable. If the supply does not respond to the oscillator's relevant deviation, it cannot be described as regulating that deviation. He also identifies beneficial and harmful common dependencies. Shared oscillator drift can cancel in a beat-frequency arrangement, while interactions with an amplifier can interfere with the intended behavior.

An ensemble can therefore require separate supplies, shielding and high-impedance connections. A common associated milieu is useful for a technical individual only when the coupled functions support one another appropriately. Making every component share a medium is not his criterion for a well-organized ensemble. The source's own example includes deliberate insulation against unwanted interaction. [W020, PDF pp. 75–77]

This is also why element, individual and ensemble are not simply names for small, medium and large objects. Their distinction concerns relations of functioning and the associated milieu.

Small objects can carry a large technical history

Simondon describes technical inheritance through elements, individuals and ensembles. An element can carry properties developed in a much larger production setting. A spring, adze or needle embodies metallurgy, forging, treatment and skilled fabrication, even if the finished item looks simple. Its visible shape does not reveal all the knowledge and organized work needed to give it the relevant properties. [W020, printed pp. 67–76]

Simondon distinguishes synchronization of technical developments from complete historical determination. He asks the reader to consider the ensemble that manufactured an object's elements.

The discussion of tools and machine operation likewise retains cleaning, lubrication, sharpening, waste removal, belt adjustment and maintenance. The historical displacement of the human tool-bearing individual does not remove humans from technical ensembles. Their relation changes, and Simondon assigns importance to coordination, invention, interpretation and upkeep. [W020, printed pp. 77–81 and 129–140]

Technical knowledge is neither mere habit nor a detached formula

The second part considers different modes of acquiring technical knowledge. Apprenticeship can transmit implicit practical skill; reflective engineering connects operation with more explicit scientific understanding. Simondon criticizes social rankings that separate learned culture from technical work. Some categorical historical and psychological claims in the discussion remain source idealizations rather than established findings about all children, adults or trades. [W020, printed pp. 103–127]

The encyclopedia supplies a concrete example of portable technical representation. Schematics communicate selected structure without reproducing every detail of an image or practical encounter. Simondon also warns that a simultaneous inventory can hide the inventions, failures and crises through which technical objects became possible.

His ambition for information theory is broad: it would mediate among sciences, techniques and learning. [W020, printed pp. 114–127]

Timing, signals and the energy that an operation uses

In discussing the governor and load changes, Simondon distinguishes reacting after speed has changed from receiving a separate signal about a disturbance soon enough to act. A faster information path can

matter because waiting for the output deviation can allow an undesirable event. [W020, printed pp. 142–147]

The book distinguishes efficiency in energy conversion from fidelity in signal transmission. A narrow resonance useful for energy transfer can be poor for preserving a broader range of informational variation. Low carrier energy also encounters noise. Simondon discusses thermal and shot noise and the importance of bandwidth and signal conditions. [W020, printed pp. 145–150]

Metastability and relaxation explain how a small trigger redirects stored potential energy: the signal does not supply all the energy released. Synchronizing oscillators require compatible phase and frequency conditions. [W020, printed pp. 151–156]

Openness, memory and the historical machine claims

Simondon does not equate increasing fixed automation with greater technical perfection. He values a margin of indeterminacy through which a machine can receive information and enter relations with other machines. This can include earlier programming and configuration rather than a human supplying every instruction at the instant of action. [W020, introduction and printed pp. 135–163]

His contrasts between machine memory and lived human memory, and his claims concerning future-oriented invention, are historically situated philosophical positions from the 1958 work. His criticism of the homeostat concerns a particular device and problem.

The culturally important claim is that technical understanding cannot be replaced merely by ownership or command. Both an owner and a worker can be separated from the machine's operation. Simondon's preference for a responsible technical relation is normative; it does not settle every conflict over property or labor.

The philosophical third part: a proposed genesis, with plural values

The third part explicitly advances a hypothetical genetic interpretation of technical and other modes of thought. Simondon describes an original unity differentiating into technical and religious phases, with aesthetic and philosophical mediation. These phases are not presented simply as a necessary chronological succession. Their role is to situate technicity within a wider human–world relation, rather than make technology the answer to every question. [W020, part III, printed pp. 167–190]

The relevant aesthetic examples depend on functioning and setting: a sail in wind, a pylon in a valley or a communications installation can be understood differently when their operation is known. Cosmetic appearance can obscure that relation. Yet aesthetic, ethical and technical judgments remain distinguishable. Simondon also warns that a premature apparent reconciliation can stop inquiry too soon. [W020, printed pp. 191–210]

Failure has an explanatory role. A suction pump's limit cannot necessarily be removed by more precise construction; an adequate account must incorporate pressure, material and environmental conditions. This is a source example of revising the conceptual problem rather than merely improving execution. [W020, printed pp. 211–216]

Simondon explicitly separates the value of ease from the value of efficiency. At printed p. 217 he denies an analytic link between them even while regarding their combination as desirable. An easy action is not automatically resource-efficient. Practical experience informs a plurality of values. The later chapters develop political, ethical and cultural proposals about mediation and responsibility. [W020]

The material mold and the continued act of construction

The conclusion returns to a technical operation obscured by descriptions that name only a worker, an intention and a result. In the clay-and-mold example, the worker prepares malleable clay and a material mold, brings them together and applies pressure. The coupled physical system then effects the taking of form. The mold is an actual prepared object, not an abstract shape floating independently of matter. [W020, printed pp. 247–253; PDF p. 261]

Simondon calls for knowledge of the operation itself, beyond the customary gesture that brings it about. The worker prepares the clay and mold, while their physical interaction effects the taking of form.

The following pages reject a once-for-all view of construction. Manufactured units realize the inventive schema imperfectly and require adjustment, repair and sometimes running-in. Maintenance continues construction. His claim that serviceable objects do not deteriorate is explained through replacement and repair, not literal immunity to wear. A technical object can carry knowledge and still require a competent recipient to interpret, maintain and adapt it. [W020, conclusion, PDF pp. 267–269]

The proposed response includes channels for technical information and worker participation following actual operations rather than only the social hierarchy. The conclusion also distinguishes a productivity number from the technical schema: different arrangements can produce the same output figure while functioning differently.

Lafitte through Sandrone, Vaccari and Lawler

Sandrone, Vaccari and Lawler reconstruct three connected dimensions of Lafitte's thought: how to study machines, what sort of organized bodies machines are, and how human life and machinery develop in relation. They distinguish the art of conceiving, building and ensuring operation from descriptive history or mechanography and from a proposed science of machines, mechanology. Physical events occur within machines, but a list of those events does not exhaust the organization of the machine as an object of inquiry. [UX-W023, especially PDF pp. 2–6]

Their account identifies passive, active and reflex or reflective machines. Passive cases such as columns or rafts can do useful work through resistance or support without an internally cycling transformation. Active machines transform, transmit or transport energy. Reflex machines regulate operation through sensitivity to their surroundings. These are Lafitte's categories as reconstructed by the authors. A passive object is not useless, and an active or reflex object is not automatically more economical. [UX-W023, PDF pp. 6–11]

The wider proposed organology includes natural and artificial organized bodies. It asks about relationships among machines, organisms and human organization without reducing them to a single physics formula. Human creation matters to the existence of machines, but their operating properties and lineages are not arbitrary consequences of an inventor's intention. The article relates this to ethical and political questions about living with machines. It expressly presents an early programme and germs of arguments, rather than a finished unified theory. [UX-W023, PDF pp. 9–14]

Sandrone, Vaccari and Lawler's account reconstructs Lafitte's *Réflexions sur la science des machines* (1932). [W074, through UX-W023]

Guffroy and Bontems: a lineage is a historical problem

Guffroy and Bontems reconstruct the genealogy of mechanology while asking how a technical lineage can be identified. Their account distinguishes the 1932 work, Simondon's 1958 thesis, his mediated encounter with Lafitte through Jean Le Moyne in 1968, and later colloquia and continuators. The dates

prevent an easy assertion that Lafitte directly generated the argument of the 1958 book. Similarities and later acknowledgment do not establish that earlier influence. [UX-W024, printed pp. 255–280]

The article also treats the organization of machine science as unfinished. What counts as a lineage, branch or generation depends partly on the scale and criterion of comparison. Data collection, processing and representation need more stable methods. Its discussion of genetic diagrams includes abandonment, resurgence, saturation, branching and hybridization, rather than inevitable progress along a single line. Institutional failure is not taken to settle the worth of the questions, but neither is methodological ambition treated as a completed synthesis.

The authors report criticism of attempts to unite naturalist and formalist approaches in other scholarship. They also distinguish classification by function from classification through functioning in comparing Lafitte and Simondon. This distinction concerns functioning, while Sandrone and colleagues emphasize intrinsic organization. [UX-W024]

Bontems: several aspects can coexist in the same object

Bontems's article reconstructs Simondon's classification and develops it through later material. It explains concretization through structural and functional synergy, while retaining hypertely and the associated-milieu criterion for element, individual and ensemble. The article notes ambiguities in early examples and the need to distinguish the 1958 text from later lectures and ethical work. [UX-W025, printed pp. 184–190]

The later passive, active and informational classification can identify different performance aspects within one object. Structural integrity, energy transformation and fidelity of information all matter, with their own supports: resistance to wear, energy supply, sensors, filtering or cooling. Bontems proposes a matrix crossing types and levels; it is his analytical proposal rather than a verbatim table from Simondon. [UX-W025, printed pp. 190–196; matrix, PDF p. 12]

His ChemCam example describes separate structural and thermal validation, optical and electronic validation, and later combination and compatibility checking. The point is not that separate tests make integration unnecessary. Different aspects must first be examined and then checked together. He also qualifies the idea that infrastructure, energy and communication technologies necessarily form one beneficial synergy. Network participation can impose dependencies and constraints.

A programme prepared earlier can reduce instructions supplied during action. Bontems connects stored programmes with later adaptation. [UX-W025]

Marratt: resonance without a simple direct descent

Marratt offers an architectural and historiographic reading of the relation between Lafitte and Simondon. It explores convergences, differences and multiple technical temporalities, including networks as encounters among natural structure, human need and invention. Its operative body recognizes the mediated later encounter, even where an abbreviated introductory account might invite a simpler genealogy. This is an interpretive account, not a finished formal mechanology. [UX-W026, printed pp. 203–214]

Marratt's paired quotation and translation differ: the quotation says that machines cannot be defined any better than living beings, while the translation suggests that living beings define machines best. The latter wording changes the claim. [UX-W026, note 11 and associated body]

Sigaut and related accounts of mechanology

Part IV of Sigaut's manuscript reconstructs a debate about machines, technology and human activity. It questions whether renaming a science of machines supplies a genuinely new undertaking, and it distinguishes the machine phenomenon from an account exhausted by physical sciences. It also locates proposals extending inquiry beyond objects toward human activity. [RGC-W035, PDF pp. 25–30]

The abstract of Hayward and Thibault's *Ethics in Jacques Lafitte's Mechanology* objects to a precursor-to-cybernetics reading that overlooks Lafitte's ethical and social purpose. [RGC-W033, abstract]

Sigaut comments on Lafitte's 1933 *Sur la science des machines*. [RGC-W034, through RGC-W035]

Hodder: mutual dependencies and an unfinished integration

The conclusions of Hodder's 2011 article describe human–thing entanglement through interdependent changes in material and social practices. His pottery discussion relates changes to houses, hearths, bricks and cooking, rather than explaining an artifact lineage solely by transmission or reproductive success. A trait's fittingness concerns how it works within those mutual dependencies. [GL-W012, printed pp. 172–173]

Hodder explicitly acknowledges that his attempted integration has not fully brought the different theories together. He still argues for the value of a partial integrative account.

In *Where Are We Heading?*, Hodder asks where the wheel ends as an object of historical explanation. A wagon wheel needs an axle and a frame, as well as a source of motion. Making the wheel calls for tools and carpentry skills. Drawing the wagon with cattle calls for a yoke and trained animals. Where wheels serve road transport, road building and upkeep enter the account as well. Hodder argues that different answers about where the object ends lead to different accounts of its origin and use. [GL-W096, pp. 54–55]

Conservation: preserving information can constrain intervention

The Canadian Association for Conservation of Cultural Property defines conservation around safeguarding culturally significant qualities with the least appropriate intervention. Its statement includes examination, documentation, prevention, preservation, restoration and reconstruction. Preventive action can be directed at an object's surroundings rather than repeatedly treating the object itself. [GL-W099, definition]

The source explains why more intervention can be worse: treatment can erase information about manufacture, use and prior history. Earlier repairs or wear may themselves be significant. Treatment materials should avoid future harm and, as far as possible, remain removable; documentation should distinguish an addition from original material. The degree of intervention involves trained judgment and consultation with the custodian. This is a professional purpose and set of principles.

The Canadian Conservation Institute's selected painting guidance gives more concrete material examples. An unstable support or inherent material problem can persist after a treatment. A controlled microenvironment can reduce environmental variation, while ongoing delamination can still require professional intervention. The account therefore distinguishes shielding an object from repairing every internal defect. [GL-W100, inherent vice and microenvironments]

In the documented *Buffalo Stampede* example, consolidation did not remove the painting's sensitivity to humidity because reactive glue remained in its structure. An enclosed frame was added to moderate the local environment. In *Charlottetown Revisited*, the account describes adding supportive inserts and a

protective back rather than removing the painting from its original strainer. These documented professional decisions retain historical material and improve support. [GL-W100, selected examples]

Part 5 Organizations, institutions, and sustaining work

These sources examine the organization of decisions, knowledge, authority, supply and burdens, with different purposes and criteria of evaluation.

S09 Multilevel coordination: arranging local decisions to meet an overall requirement

Local decisions and overall performance

Mihajlo Mesarović, D. Macko and Yasuhiko Takahara's *Theory of Hierarchical, Multilevel, Systems* concerns descriptions of complex systems, local decision problems and overall performance. Its publisher preview sets out a programme for arranging local decisions and higher-level intervention to meet a larger requirement without making the higher level solve the entire problem again. [W096, publisher preview; W097, pp. 23–26; SB-W008, pp. 3–5]

The 1969 coordination paper develops the formal mechanism. Richardson's 1975 conference paper, Findeisen and colleagues' 1980 book, and Abraham, Tribolet and Winter's 2013 article examine practical control, modeling and organizational difficulties. [W097; SB-W006; SB-W007; SB-W008]

Strata, layers and echelons ask different questions

Richardson describes three distinct hierarchy concepts in the 1970 programme. Strata concern levels of description or abstraction. Layers concern decision complexity. Echelons concern priority of action and organizational coordination. A coarser representation, a more complex decision and a superior organizational office are consequently different things. They may be related in an application without becoming interchangeable meanings of higher level. [W097, p. 24]

The distinction helps recover what has been delegated. A detailed model and an aggregate model can describe the same system differently. A local regulator can act within a setting selected by another decision process. An organization can assign intervention rights among offices. Calling all three hierarchical would conceal whether the relevant economy concerns representation, the division of computation or the allocation of authority. The sources deliberately provide several descriptions for these different relationships. [W097, pp. 24–26; SB-W007, hierarchy framework]

Richardson also describes the programme as combining heuristics for decomposing and representing large problems, mathematical theory, and methods for improving behavior. Satisfactory performance, rather than exact optimization, is included in his reconstruction. [W097, pp. 24–25]

The 1969 paper: local success has to map into overall success

In *Two Coordination Principles and Their Application in Large Scale Systems Control*, the same three authors distinguish an overall control problem, local control problems and a coordination problem. Local units directly influence the controlled process. The higher or supremal unit acts indirectly by influencing those units. Its task is not defined simply as being above them: it has to make their decisions compatible with the overall requirement. [SB-W008, pp. 3–5]

The formal setup considers families of local problems indexed by intervention parameters. A transformation maps a joint solution of the local problems into a candidate solution of the overall problem. The crucial condition is that every relevant joint solution under the chosen intervention maps into an overall solution. Without such a condition, separately satisfactory local decisions need not constitute a satisfactory whole. A decomposition supplies multiple problems; it does not automatically supply their compatibility. [SB-W008, pp. 4–5]

The law of consistency constrains the coordinator's own problem. Solving that problem should be sufficient for the joint local results to meet the overall requirement. One trivial way to ensure this is to

make the coordinator solve the complete original problem. The authors explicitly recognize why that misses the intended advantage: the art is to find a simpler coordination problem while retaining consistency. [SB-W008, p. 5]

Coordinability adds an existence question. It concerns whether some intervention in the specified family can produce the required relation between local solutions and the overall solution. That differs from saying what follows if a suitable intervention is found. The paper itself distinguishes a principle's applicability from the existence of a parameter that meets its matching condition. [SB-W008, pp. 5, 8]

Prediction and balance retain different work

Interaction prediction begins by specifying anticipated interface inputs for the local units. Each local controller optimizes with those predictions fixed. When their controls are applied to the connected process, actual interface inputs result. The coordinator compares actual inputs with predictions and changes the predictions to reduce the discrepancy. The architecture has separated local decisions by supplying information about their interfaces; it has retained a problem of making that information consistent with actual joint operation. [SB-W008, pp. 6–8]

Interaction balance instead modifies local performance functions. The local problems then optimize over their controls and desired interface inputs. Actual operation supplies interfaces that may differ from those desired. The higher level changes its intervention parameters to bring the desired and actual values into agreement. Local autonomy in this construction is conditional on the objective modifications and on the interfaces supplied by the connected system. [SB-W008, pp. 7–8]

The authors allow a mismatch measure to be minimized when exact zero error is not attainable. Such a formulation is a further problem definition. [SB-W008, p. 8]

What the conditional optimality argument actually uses

Theorem I in the 1969 paper supplies sufficient conditions for applying interaction balance. For every relevant intervention and feasible overall control, global cost must be expressible through an order-preserving aggregation of the modified local costs evaluated at actual interfaces. Local units must attain minima over the stated local controls and freely chosen interface inputs. If their chosen interfaces then match those arising from the joint controls, the resulting overall control minimizes global cost. [SB-W008, pp. 9–10; appendix A]

The proof explains why all these premises matter. Each local minimizing pair is no worse in its local cost than the corresponding pair associated with an arbitrary feasible overall alternative. Order preservation carries these componentwise comparisons into a comparison of global cost. Matching allows the locally selected pairs to be interpreted as the actual interconnected result. The conclusion concerns that declared objective and feasible set, not every property a real system might be expected to have. [SB-W008, appendix A, proof of theorem I]

Theorem II provides an additive special case. Modifications can be introduced into local objectives while summing to zero at feasible interconnections. The overall cost is then preserved by those modifications, allowing the first theorem's reasoning to apply. The important feature is the verified relation among objectives, rather than an assumption that any collection of individually optimized goals will combine well. [SB-W008, p. 10; appendix A]

Existence results add further hypotheses about attained and unique local minima, convexity, lower semicontinuity, compactness and the relevant dual relation. These hypotheses cannot be suppressed when moving from the conditional matching argument to a claim that matching can be achieved. The

paper does not provide a general convergence algorithm, a response-time guarantee or a measured reduction in total computation. [SB-W008, theorems III–V and appendices]

In Example III of the 1969 paper, the printed parameter makes one local quadratic unbounded below despite the claimed minimum. Theorem V's compactness reasoning also requires appropriate space and attainment assumptions beyond an unrestricted normed-space reading. [SB-W008, example III, proof of theorem V and appendix B]

Four questions consequently remain separate in this account: whether matched local solutions satisfy the global criterion; whether a matching intervention exists; whether an actual procedure reaches it; and whether the procedure does so reliably and soon enough for the real application. An answer to an earlier question does not answer the later ones. Correct specification of the real objective and constraints is another obligation throughout. [SB-W008; SB-W006, ch. 1]

Findeisen and colleagues: control must work on an operating system

The introductory chapter of *Control and Coordination in Hierarchical Systems*, by W. Findeisen, F. N. Bailey, M. Brdys, K. Malinowski, P. Tatjewski and A. Wozniak, develops the practical coordination problem. [SB-W006, introduction and ch. 1, pp. 4–74]

Its introduction identifies reasons to divide decision responsibility: limited individual capability, spatial separation, communication cost and delay, distorted information, and private local information. For control, it also insists that the system is operating under disturbances and that current information can be used to improve decisions. Models, real constraints, measurement and response horizons are therefore part of the subject from the beginning. [SB-W006, pp. 1–4]

The chapter distinguishes multilayer organization of different control tasks from multilevel decomposition and coordination of related local problems. Different horizons and kinds of information can justify different decision processes. Treating all units as copies of one static optimizer would hide what a faster regulator, a slower planning process or a model-updating function actually does. The architecture has to respect the process and the time available for action. [SB-W006, §§1.2–1.4]

The formulation of global performance has a consequential assumption: it is an order-preserving function of local performances. The authors say that this excludes gamelike situations among decision makers from their discussion. Their coordination results therefore cannot be carried directly into an organization with unresolved conflicting aims simply because its chart contains local units and a center. The required objective relationship has to be supplied, not inferred from the word hierarchy. [SB-W006, p. 36]

Direct coordination illustrates the feasibility problem. The coordinator prescribes outputs and allocates resources; local units then try to achieve those demands. Some demands may be impossible under local equations or constraints, and disturbance or model uncertainty can change the feasible region. Staying inside a known safe region can preserve feasibility but may require conservative decisions and substantial knowledge. Issuing a target is not proof that a local unit can produce it. [SB-W006, pp. 36–38]

The chapter also distinguishes solving local models offline from coordinating people or controllers acting on a real process. An optimization result computed from models becomes open-loop control if applied without correcting it from actual system behavior. In real interaction, separate decision makers can cause instability while each is attempting to do well locally. Even a mathematically convergent coordination scheme needs an appropriate relation to the process operating between its updates. [SB-W006, pp. 40–41]

An adverse case: decomposition can enlarge the central problem

The ammonia-plant example directly challenges an automatic saving from decomposition. The plant contains feed-forward and recycle relationships, including heat flows that contribute to its operation. Cutting it into subprocesses while representing all relevant cross-boundary quantities would require roughly forty coordination variables. Yet the original overall optimization concerned only twenty-two adjustable control variables. The result would be a larger coordination problem plus the added local problems. The authors call that version more complex and expensive than direct treatment. [SB-W006, pp. 42–45]

Their reported response uses engineering understanding and operating experience. Only five quantities were retained as essential coordination variables for an approximate solution; others were related to them, treated as nearly constant or judged less relevant to the objective. The reduction is not supplied by the drawing of the boundary. It depends on knowing which couplings can be compressed or neglected for the application. [SB-W006, pp. 44–45]

This motivates subcoordination: using fewer variables than a strict interface description would require. Its economy is bought through an approximation whose adequacy must be examined. Knowing a model exactly enough to encode a perfect low-dimensional prescription would be a different situation from controlling a process with model–reality differences. The source itself cautions against making that exact-knowledge assumption. [SB-W006, pp. 45–47]

The authors' plant example provides significant adverse evidence within their account: hierarchical decomposition can increase work, and successful simplification can require more retained system knowledge than the abstract partition suggests. [SB-W006, pp. 43–47]

Dynamic control, information and model revision

The introductory chapter extends coordination to dynamic problems with states, time functions and finite horizons. Solving a steady-state problem repeatedly is not automatically an adequate dynamic controller. Actions influence later conditions, and the time spent measuring, communicating and computing can matter to the result. Decomposing a horizon or using a shorter horizon changes the problem unless its consequences are controlled. [SB-W006, ch. 1, dynamic control and information-structure discussions, pp. 49–63]

The information structure specifies who knows what and when. Local decision makers may have measurements unavailable to the coordinator, while the center may possess information about interactions unavailable locally. Feedback can help with model error, but measurements, transmission and interpretation are retained activities. A formally convenient hierarchy is therefore only part of the arrangement needed for actual control. [SB-W006, §1.4]

Richardson: building and using integrated models

Richardson's 1975 paper reports on the Multilevel Regionalized World Modeling Project and related work on Lake Erie pollution. Richardson is an insider describing a continuing research programme. His paper connects the earlier framework to later modelling practice and explicitly says that technical and organizational skills developed together. [W097, pp. 23–25]

His Lake Erie discussion uses strata to organize submodels associated with different disciplines. Partial decomposability permits some development and validation before final integration. The qualification is important: in crisis situations, couplings can become stronger. A decomposition useful under ordinary conditions may not preserve the same independence when the situation changes. Integration remains a task even when subgroups can make progress separately. [W097, pp. 25–27]

The regionalized world model differentiates regions by relevant economic and political conditions rather than treating the globe as one uniform aggregate. Trade, migration and other flows connect the regions. Within regions, causal, organizational and normative descriptions address different aspects of behavior. Values and beliefs affect the selection among satisfactory policies; allowing users to supply norms interactively is one response to the difficulty of simulating them completely. [W097, pp. 28–30]

This makes retained judgment concrete. Model builders must decide which regional differences matter, how decisions are represented and what belongs in the scenarios. The paper's food-policy example joins separately developed submodels whose detailed documentation is much larger than the conference exposition. Their existence and integration are reported facts about the project. The scenario curves remain historical model outputs under their stated policy assumptions. [W097, pp. 30–31 and figs. 10–12]

Richardson's agenda is unusually explicit about limits. He argues that difficulty quantifying social and value questions does not justify ignoring them. He emphasizes judgment and design in training, cooperation across disciplines and an institutional environment that supports integrated work. A framework for modelling cannot itself create the professional relationships and durable organizations required to use it. [W097, pp. 32–33]

He also questions whether fitting historical data is sufficient validation for long-term models when parameters and structures are changed to explore alternatives. Claims to decision makers should remain conservative relative to actual utility. Institutional pressures can favor short horizons even where a model asks long-term questions. [W097, pp. 33–35]

Abraham, Tribolet and Winter: transformation as well as operation

Abraham, Tribolet and Winter's 2013 article applies distinctions among strata, layers and echelons to enterprise architecture management. It positions architecture management across these dimensions and proposes interlinked feedback loops for transformation. [SB-W007, hierarchy framework and reference 6]

The authors distinguish a functional understanding of an enterprise from an understanding of its construction. Principles can restrict design freedom and help coordinate planned change, but the relevant perspective is broader than a single place in an organization chart. An architectural model can describe different strata while transformation decisions and intervention responsibilities occupy other dimensions. [SB-W007, introduction and hierarchy discussion]

Their feedback model distinguishes routine optimization within an existing understanding, planned transformation that changes models and goals, and improvisation under severe unpredictable disturbance. The latter can require making sense of a new situation before returning to planned projects and routine operation. The categories are part of the authors' conceptual proposal. [SB-W007, feedback-loop model and fig. 4]

The model also makes continued activity visible. An enterprise must keep operating while it undertakes a transformation, and useful experience must enter its representations if it is to handle a formerly unfamiliar exception routinely later. Models, goals and practices do not update simply because a new diagram is available. The Portuguese Air Force case serves as an illustrative application in the article. [SB-W007, feedback and application sections]

The article’s assertion that observability is universally necessary for controllability is not a general control-theoretic result. Its enterprise-transformation proposal joins observation, modeling and intervention, whereas Mesarović and colleagues’ 1970 book also addresses dynamic control. [SB-W007; W096, contents; W097, p. 24]

Interventions, objectives and interface compatibility

These coordination accounts connect local and overall problems through interventions, objectives, interface compatibility, measurement and revision. [SB-W008; SB-W006; W097; SB-W007]

S13 Information, routines and organizations: Stinchcombe's connected account

The task determines which news matters

Arthur Stinchcombe's *Information and Organizations* develops a connected sociological explanation of organizational structures through the information needed to handle uncertainty. The relevant unit is often a lasting part of an organization engaged with a particular task, rather than either a whole legal firm or an isolated individual decision. Different tasks expose people to different uncertainties, sources of news and opportunities for action. An arrangement that makes useful information available to competent decision makers can consequently have a different structure from an arrangement serving another task. [UX-W003, chs. 1, 10]

Stinchcombe asks where early, useful news about a task's constraints appears and what organizational structure makes a timely response possible. A department may need detailed information about an immediate production difficulty while corporate investment decisions require a different kind of summary over a longer period. Combining all information indiscriminately would obscure those differences. The question concerns what someone needs to know in order to do a particular job, with authority and practical capacity to act on it. [UX-W003, chs. 1, 3–4]

Uncertainty does not mean that no action is possible until everything is known. The book follows staged commitments and differences in the reliability of information. An exploration report, a first successful well and a more developed reserve estimate carry different degrees of uncertainty in its oil-industry example. A knowledgeable recipient can use them differently. Treating all three as equally established would damage the information system rather than simplify it usefully. [UX-W003, ch. 1, especially pp. 13–15]

Trustworthy information therefore need not be error-free. It must avoid routinely misleading users about how uncertain its conclusions are. Evidence and qualifying information allow a user to go behind an assertion when that is necessary. Stable matters need not be reinvestigated at every use, while unsettled matters continue to receive scrutiny. The resulting economy depends on expert care and an intelligible assessment of uncertainty, not on an assumption that all communicated conclusions are certain. [UX-W003, ch. 1]

Incentives can corrupt this arrangement. A person punished for overestimates but rarely exposed for missed opportunities may bias an estimate downward. A reporting system can look appropriately safeguarded while serving another interest. Stinchcombe treats the causal explanation of such arrangements as a problem: identifying an apparent useful consequence does not prove that it caused the structure or that the structure actually performs as advertised. [UX-W003, ch. 1]

Prepared routines economize on new decisions

The book distinguishes complexity inside a routine from discretionary decisions needed to choose or change it. A complicated sequence can be performed with little new decision-making if its relevant choices have been settled in advance. Stinchcombe uses the analogy of a computer program: once an appropriate routine is called, much of what follows is prescribed. The existence of many internal steps does not imply that a worker must invent each step anew. [UX-W003, ch. 2]

The selection of the routine can remain a substantial decision. Choosing a product mix, for example, can call several different production routines. The units then repeat the prescribed operations for that mix. Preparing, testing and improving those routines is different work again. The economy in recurrent choice

is therefore relative to a prepared set of options, dependable inputs and an appropriate selection process. [UX-W003, ch. 2]

The registrar example exposes the surrounding intelligence. A university's clerical routines can be highly standardized while professors decide course content, class schedules and grades, departments decide requirements, and students choose courses within the arrangement. The clerk's task can be simple because the system obtains complex decisions through other participants. Looking only at the clerical procedure would mistake a local reduction in discretion for a reduction of intelligence throughout the university. [UX-W003, ch. 2]

Craft skill includes a repertoire of routines and the discretion to choose among them. Professional principles can guide unfamiliar cases; experience can also allow someone to revise the repertoire itself. These are not interchangeable capacities. Stinchcombe uses occupational distinctions to ask who acquires, preserves and supplies the relevant knowledge, rather than assuming that a formal procedure carries all the competence needed to apply it. [UX-W003, ch. 2]

Fordist production, in his account, combines extensive advance programming with relatively few new decisions per unit produced. It nevertheless retains managerial choices, maintenance, toolmaking and engineering. Its favorable conditions include stable demand, a standardized product, relatively stable technology and standardized, available inputs in a suitable environment. If those conditions vary unpredictably, more discretion must be supplied locally or variability must be buffered elsewhere. [UX-W003, ch. 2, especially pp. 54–72]

That distinction changes the comparison between skilled and routinized work. The relevant question is not whether one worker uses more discretion in the abstract. It is how much preparation, selection, adjustment and repair are required to produce a specified quality under the given conditions. An apparent local saving can depend on training, machinery, management or another occupation's work. Stinchcombe also treats occupational jurisdictions and management ideology as influences on which arrangements are chosen. Existing division of labor is not assumed to be the uniquely efficient one. [UX-W003, ch. 2]

The book's displayed ratios of routinized decisions and skill help state its conceptual distinctions, but they do not provide an operationally validated universal scale of work. Decisions differ in significance, and design or debugging cannot simply be inferred from a count of repetitive actions. The impressionistic wage–age illustration and dated industrial figures remain the author's historical material. [UX-W003, ch. 2]

Managers construct different information systems

Chapter 3 treats managers partly as builders and maintainers of information systems. Such systems are shaped by the decisions for which the information will be used. A cost-comparison account, a search for production improvements, a maintenance record and a commissioning procedure do not require the same categories, detail, timing or authority. Information flows laterally as well as upward through a hierarchy. [UX-W003, ch. 3]

The chapter identifies several dimensions of variation. Information can improve through experience, serve one locality or several users, require supplementary innovation, prioritize flexible categories or comparability, support audit, need reorganization from another activity, and require different approval procedures. The differences explain why a common store of data does not automatically become a useful system for every purpose. The material must be organized so that the relevant recipient can interpret and act on it. [UX-W003, ch. 3, pp. 76–99]

Design and construction documents supply a particularly concrete example. Near the end of a construction project, a builder making a change needs to know what work to perform. An operator or maintainer needs to know the final configuration: as-built drawings, technical data and parts lists. The project record and the operating record overlap, but they answer different questions. [UX-W003, ch. 3]

Stinchcombe says that much design information has to be extensively reorganized for operations. Spare-parts procurement may require supplementary information about vendors that the construction project could obtain readily but does not itself need. The project can regard final documentation as marginal paperwork even though it is crucial to the organization inheriting the plant. The difficulty is partly responsibility and motivation, not merely storage capacity. [UX-W003, ch. 3]

Commissioning and steady operation also differ. Testing, approval and the authority to accept a plant occur under one set of conditions; ongoing operation and maintenance occur under another. A completed physical asset therefore does not deliver its own usable organizational knowledge. The source's Statoil-based discussion has an explicit empirical limit: the organization studied was not yet operating. Some claims about its eventual information needs concern anticipated use rather than observed long-term performance. [UX-W003, ch. 3]

Preparation and handover connect physical design with continuing organizational knowledge. A designer may settle a physical relation in advance, while another group needs accurate records, intelligible categories and authority to maintain it. [UX-W003, ch. 3]

Divisionalization combines local integration and corporate abstraction

Stinchcombe reconstructs Chandler's divisionalization argument in terms of information and decisions. A division can integrate detailed product and market decisions while the general office works with more abstract measures for allocating investment. This is not decentralization understood as simply giving every local unit more freedom. Some decisions become more tightly integrated within a division while others are summarized for a higher-level purpose. [UX-W003, ch. 4]

The time horizon matters. Product-market conditions can require detailed, relatively fast response. The returns to major investments develop over a longer period. Information appropriate to those slower decisions can be abstracted over time without requiring corporate executives to make every operating choice. The economy is selective: an abstraction has to retain what its recipient needs while allowing detailed expertise to remain where operating decisions are made. [UX-W003, ch. 4]

The DuPont, General Motors and Sears discussions concern different causal problems. DuPont supplies a relatively detailed account of diagnosing and changing organizational arrangements. GM raises the general office's investment-coordination role. Sears involves purchasing and information about the quantities wanted in different markets. Stinchcombe does not treat these as interchangeable examples of one beneficial decentralization slogan. [UX-W003, ch. 4]

He also asks how individuals, committees, concepts and authority actually produced the structural changes. A functional explanation has to connect useful consequences with a causal process. People may need a theory of what is going wrong before they can reorganize, and existing experience can make some proposals intelligible while others remain difficult. An organization chart appearing after the change does not itself explain the transition. [UX-W003, ch. 4]

The chapter acknowledges gaps, including what Chandler establishes about the general office's superiority in investment decisions. Its displayed relation between divisionalization and several graded conditions leaves the numerical function unspecified. It is a proposal about explanatory relationships rather than a ready quantitative prediction of the optimal corporate structure. [UX-W003, ch. 4]

Invention becomes useful through an operating organization

The innovation chapter distinguishes a technical invention from an operating concern capable of producing and supplying its benefits. Technical theory, investment, production, markets, the division of benefits and personnel all enter the problem. An attractive device or process still needs a social vehicle through which it can be made, financed, introduced and maintained. [UX-W003, ch. 5]

Learning curves in the chapter concern accumulated cost-reducing improvements to equipment, procedures and skill. An innovator can gain an advantage by developing those capabilities earlier, but followers' learning, input costs and other conditions can alter it. Converting an area under unit-cost curves indexed by time into money requires production-rate and pricing assumptions. [UX-W003, ch. 5]

Stinchcombe's replacement-board example shows the organizational work behind an apparently effortless improvement. A board can be designed to fit an existing computer and software arrangement so that users need not rewrite programs, change their own routines or learn a new system. From the user's perspective the innovation can be close to a zero-change event. The supplier's perspective is different. [UX-W003, ch. 5]

Engineers collect information about failures and limitations from maintenance and software work, design modifications and test that the interfaces remain compatible and that new defects have not been introduced. Investment and production decisions also remain. Stable interfaces are part of the supplier's market reasoning because forcing a client to reorganize can make switching suppliers attractive. The small burden visible to the client thus depends on a substantial supplier organization. [UX-W003, ch. 5]

The example preserves a specific distinction between reducing a user's need to adapt and eliminating adaptation work altogether. It also gives continuing service relations an informational role: maintenance brings knowledge of problems back to the producer and provides a route for improvements to reach users. [UX-W003, ch. 5]

Incentives and the distribution of gains affect whether an improvement is introduced. A technical reduction in cost is not sufficient if the parties needed to make it happen cannot obtain a worthwhile return or expect to lose valued rights. The chapter's criticism of technological utopianism concerns precisely the omission of the organization, training, investment and power surrounding a technology. [UX-W003, ch. 5]

The author does not infer that bureaucracy necessarily prevents innovation. An innovative division can require dependable routines as well as rapid adaptive information links. The issue is whether the structure can support the different kinds of work, not whether the organization has escaped every procedure. [UX-W003, ch. 5]

A contract can constitute hierarchy across firm boundaries

Chapter 6 examines contracts as organizational arrangements. A legal firm boundary does not necessarily mark the end of hierarchical coordination. A contract can establish authority, incentives, procedures, dispute resolution and a method of pricing changes. These functions can support integrated work across legally separate organizations. [UX-W003, ch. 6]

Stinchcombe identifies five forms of hierarchical content. Command structures allocate authority. Incentives guide discretionary action. Operating procedures specify actions involving both contractor and client. Dispute-resolution arrangements reduce reliance on fresh market bargaining or court proceedings. Pricing provisions, including cost-related methods, govern variations in performance. Together these can create relationships resembling an internal hierarchy, despite separate ownership. [UX-W003, ch. 6, pp. 194–236]

The reasons include uncertainty over specifications, costs and measurable performance. A client may not yet know precisely what it wants because development is unfinished. Engineering errors and execution inefficiency may be difficult to distinguish in an overrun. A rigid original description can then be inadequate for the work that must actually be done. Adaptive authority and agreed procedures permit changes without reconstructing the relationship through a new market transaction each time. [UX-W003, ch. 6]

Such adaptation retains knowledge and bargaining. The client still needs enough understanding to specify aims, recognize problems, evaluate information and participate in changes. A supplier's discretion has to be related to authority, compensation and accountability. The chapter distinguishes constituting an organizational arrangement through a contract from making its continuing information operations work. Outsourcing does not remove those functions by changing the legal location of the people performing them. [UX-W003, ch. 6]

The book reuses and develops material from the author's related work. The relevant knowledge is the concrete organization of adaptive cooperation and the distinction between legal and functional boundaries. [UX-W003, ch. 6 and notes]

Personnel information can economize on judgment and preserve privilege

The labor-market chapter asks how employers and others know about skill. Direct observation of future performance is difficult. Educational records, tenure, experience, peer judgments and other certificates can substitute for repeated complete assessment. Such devices are information arrangements: people rely on knowledge produced and maintained elsewhere. [UX-W003, ch. 7]

The arrangements have requirements that can conflict. Information should be relevant, competent, honest, auditable and comparable, yet the procedures that support one quality can obstruct another. A standardized certificate can support comparison while losing local detail. A judgment from someone who knows the work can carry expertise while creating opportunities for reciprocal favoritism or exclusion. [UX-W003, ch. 7]

Access to opportunities for producing a credible record is itself uneven. Occupational groups and institutions can control training, certification and recruitment, creating segmented labor markets and preserving advantages. An information system useful to those making personnel decisions can still reproduce unjust distributions of opportunity. Stinchcombe explicitly distinguishes functional explanation from moral justification and acknowledges that his explanation does not fully account for discriminatory allocation. [UX-W003, ch. 7]

The chapter’s national, gender and class generalizations are historical interpretations. Its wage illustration also has an arithmetic problem: a thirty-percent premium requires a reciprocal cost comparison to determine the corresponding labor hours, rather than a direct thirty-percent reduction. Production rates, other costs and causal estimates enter any wider monetary comparison. [UX-W003, ch. 7]

Chapter 8 gives cultural interpretation and political organization a larger role. Incentives and standardized work categories do not uniquely determine perceived interests, collective action or citizenship. People interpret fairness, authority and careers through histories and institutions. Stinchcombe's information is therefore not simply a quantity of signals: its meaning, purposes and use can be contested. [UX-W003, ch. 8]

These chapters qualify any inference that a structure's persistence proves a universally favorable economy. An arrangement can survive because it serves the interests of people with the power to

maintain it. The criterion of whose work is reduced and whose purposes are served remains within the author's explanation of the institution. [UX-W003, chs. 1, 7–8, 10]

University space: local knowledge, reserves and credible promises

University administration supplies a case where managers cannot readily judge the detailed work being performed. Research space is specialized, demand changes with grants and projects, and people know more about their own future needs than central administrators do. A simple aggregate measure cannot capture every relevant difference between laboratories, offices and equipment. [UX-W003, ch. 9]

Individual researchers may retain space to insure against future expansion or uncertainty. A common reserve can reduce that incentive only if space is usable for different claims and people trust that giving it up now will not destroy their future access. Stinchcombe's discussion connects a larger liquid reserve with less need for private hoarding, conditional on a credible return guarantee and the physical suitability of the space. [UX-W003, ch. 9]

Physical flexibility matters alongside administrative promises. Movable partitions can help with some reallocation, while drains, water and exhaust requirements make laboratory space harder to treat as interchangeable. The arrangement thus includes a material problem and an authority problem. A central claim that all space is available does not make every room useful for every research activity. [UX-W003, ch. 9]

The proposed strategies are not reported as universally tested remedies. They show how the distribution of knowledge, uncertainty and responsibility can produce reserve holdings that look redundant from one viewpoint but protect important local activity from another. Changing the rules without sustaining trust or usable capacity may leave the underlying demand for insurance intact. [UX-W003, ch. 9]

The connected argument and its own limits

The conclusion joins these accounts through lasting task-related substructures, prepared routines, skill and the handling of useful news. Different parts of an organization can embody different rationalities because their tasks, uncertainties and response tempos differ. Abstraction can allow one part to rely on another, while expertise and correction maintain the adequacy of that reliance. The connections are Stinchcombe's own organizational explanation. [UX-W003, ch. 10]

He also names limits. The book explains structures more than particular individual errors, and its cultural treatment is incomplete. Functional accounts need causal pathways, and the historical evidence is uneven. A principle of decoupling or a category of uncertainty must be mapped to a specific situation rather than assumed to determine its structure on its own. [UX-W003, chs. 1, 4, 7–8, 10]

Later formalization through du Gay and Lopdrup-Hjorth

Paul du Gay and Thomas Lopdrup-Hjorth's 2024 *Organizing States* provides a selected reconstruction of Stinchcombe's later work on formality. In their account, useful formalization requires abstraction, communication and correction. People can act through an established representation without independently checking every detail because others have checked, and continue to check, what the abstraction allows them to rely on. [SB-W001, “Formality and Formal Organization”]

This explains a particular economy of reliance. An abstraction must preserve something useful, be communicable to its users and remain open to correction when its relation to practice fails. A formal document by itself does not establish those conditions. The reconstruction therefore differs from treating formality as either inherently obstructive or automatically trustworthy. Its relevant mechanism includes the organization that maintains the abstraction. [SB-W001, selected section]

Du Gay and Lopdrup-Hjorth draw on Stinchcombe's *When Formality Works: Authority and Abstraction in Law and Organizations* (2001). [SB-W002, through SB-W001]

Zhou and Zhao: historical and institutional organization

Zhou Xueguang and Zhao Wei's 2009 review of research on Chinese organizational phenomena discusses Stinchcombe's 1965 argument about organizations and social structure. They emphasize the resources, legitimacy, behavioral patterns and opportunity structures supplied by an organization's institutional environment. [SB-W003, pp. 1–6]

The authors also recover historical influences on organizational structures and the role of formal organizations in stratification. Formal organizations help shape access to education, employment and other positions, rather than merely responding to external social categories. Their discussion places information and organizational arrangements within legal, cultural and political conditions. [SB-W003, pp. 5–6]

Their introduction distinguishes studying organizations from explicitly using organizational analysis. Economic, political and organizational models can illuminate different aspects of a decision process, but one does not automatically replace the others. This is the authors' methodological argument in a Chinese research setting. [SB-W003, pp. 3–4]

Expertise, routines and organizational relationships

The selected book material explains more than the presence of information in organizations. It shows who supplies expert inputs to a routine, how documents must change between construction and operation, which details a corporate abstraction can omit, what supplier work makes an upgrade easy, and how contracts sustain adaptation across legal boundaries. The personnel and university accounts also show that incentives, privilege, material differences and credible commitments affect whether a seemingly economical arrangement works for particular participants. [UX-W003, chs. 2–9]

S51 Organizational silence, dissent and learning

An organization can possess information that never becomes available to the people deciding what to do. It can also receive a warning and interpret it through assumptions that make the warning seem unimportant. The historical accounts in this section examine the organization of voice, learning, commitment and the interpretation of danger. They explain how reporting channels, authority, shared expectations and ordinary work practices affect what can be questioned and corrected.

Stinchcombe's account in S13 already connects useful information with tasks, incentives and recipients' ability to act. Beer's treatment in S22 explains how ignorance can filter the world into an apparently manageable form, while operational examination supplements routine reporting. The sources developed here examine more closely why people withhold concerns, how disagreement becomes usable, and why some decisions resist revision. The relevant engineering interventions concern the conduct of inquiry and the institutional relations that support it.

Hirschman: exit and voice are different responses to deterioration

Albert Hirschman's 1970 book begins with organizations and firms that can deteriorate and potentially recover. Dissatisfied customers or members can leave, withdrawing purchases or participation. They can also attempt to change the objectionable condition. Hirschman calls the latter response voice. It ranges from a direct complaint to collective organization, appeals to higher authority and public protest. A person deciding whether to remain and speak faces a different problem from one deciding whether another supplier or organization offers a better alternative. Voice requires some prospect that effort can influence what happens. [OI-W007, opening; ch. 3]

The distinction gives deteriorating performance two possible routes back to management. A loss of customers or members can create pressure through reduced revenue or participation. Complaints and organized demands can convey more specific information about what is wrong and what might repair it. These routes need not reinforce each other. Easy departure can remove the people most willing to object, leaving managers with a quieter organization whose surviving members appear more satisfied. Whether exit supplies an effective corrective signal depends on how sensitive the organization is to the loss and whether those responsible can recognize and respond to it. [OI-W007, chs. 3–4]

Hirschman's discussion of Nigerian railways makes that dependency concrete. Dissatisfied shippers could turn to road transport. This alternative relieved their immediate difficulty, but it could also weaken the constituency pressing for railway improvement. Public financial backing could cushion the railway against the consequences of lost business. Competition would then provide an escape for dissatisfied users without necessarily producing the expected recovery inside the railway. The argument depends on both relations: access to a substitute and the organization's weak response to departure. [OI-W007, ch. 4]

Who leaves matters as well as how many leave. People particularly sensitive to quality may notice deterioration early and have strong reasons to complain. If they are also especially capable of finding an alternative, their departure can remove disproportionately valuable pressure for correction. The organization loses potential advocates for improvement along with their custom. Hirschman's analysis therefore asks about the composition of those who leave, the experience of those who remain and the relation between an exit signal and the specific knowledge that voice could have supplied. [OI-W007, chs. 3–4]

Voice itself has costs. A complaint can require time, coordination, information and confidence that somebody will respond. A person may wait for others to bear those costs. Repeated, ineffective protest can discourage further effort; excessive protest can also impede an organization's attempts to recover. Hirschman treats channels for complaints and the organization of collective action as capacities that can

develop. A judgment that voice is presently ineffective need not describe its permanent potential. Choosing departure can terminate the effort through which a more effective form of influence might have emerged. [OI-W007, ch. 3]

Loyalty changes the relation between these responses. Attachment can postpone exit long enough for someone to attempt repair, or to invent a way of influencing an organization that initially seems unresponsive. Yet an available exit can make voice more effective by supplying bargaining power. Hirschman distinguishes the willingness to speak from the ability to make speaking count. A credible possibility of leaving can strengthen a complaint even while an easy alternative reduces the incentive to make that complaint in the first place. [OI-W007, ch. 7, pp. 76–84]

Remaining is consequently ambiguous. A member may stay out of attachment, hope that others will intervene, lack an acceptable alternative, or fail to recognize deterioration. Management may also impose entry costs or penalties that obstruct both departure and opposition. Hirschman's selected discussion distinguishes such restraints from loyalty that sustains active efforts at repair. Continued membership can therefore conceal quite different responses to deterioration. [OI-W007, ch. 7, pp. 91–93]

Argyris: questioning the assumptions that govern correction

Chris Argyris's 1976 article distinguishes correcting action within accepted goals from questioning the goals, policies and assumptions that govern it. He calls these single-loop and double-loop learning. An organization can encourage improvements while excluding inquiry into its fundamental design. He also distinguishes what people say guides their conduct from the theory of action inferred from their observable behavior. A professed commitment to openness can coexist with unilateral control and avoidance of threatening questions. [OI-W003, pp. 365–368]

His proposed alternative combines advocacy with inquiry: people state their reasoning, make it open to challenge and share control with those relevant to the problem. Changing an organization chart or teaching the vocabulary does not, in his account, produce that behavior automatically. Existing practices can make candor risky and recreate the old pattern inside a nominally different structure. The article presents an explanatory and prescriptive argument drawing on earlier studies; its discussion acknowledges incomplete evidence and the difficulty of testing the transition. [OI-W003, pp. 369–373]

Staw: responsibility for an earlier choice can affect the next allocation

Barry Staw's 1976 experiment examines a particular way that an unfavorable outcome can strengthen a commitment. An investor can respond to disappointment by changing an interpretation, but can also commit additional resources in an attempt to justify or rescue the earlier decision. Staw asks whether personal responsibility for the initial choice changes the response to subsequent performance information. The question concerns the interaction between responsibility and outcomes, rather than persistence considered in isolation. [OI-W005, pp. 27–30]

Participants took the role of a financial vice-president allocating research and development funds between two divisions of a hypothetical company. In the high-responsibility condition, they made an initial choice and explained it. In the low-responsibility condition, they inherited an allocation attributed to another officer. Favorable or unfavorable financial outcomes were then presented independently of that manipulation. Participants subsequently divided a new pool of hypothetical funds between the divisions. This design separated having made the prior choice from learning that its consequences were disappointing. [OI-W005, pp. 30–35]

The largest allocation to the previously selected division occurred when participants had personally chosen it and then received unfavorable results. The corresponding response to negative outcomes was

not significant among those who inherited the decision. Nor did responsibility significantly change allocation following favorable results. These conditional comparisons are central to the finding. The experiment supports a combined effect of responsibility and outcome valence, rather than a claim that any unfavorable news necessarily increases investment or that taking responsibility always impedes correction. [OI-W005, pp. 36–39]

The analysis included 239 undergraduate business students, after one participant was excluded because the initial choice could not be made. The experiment measured one subsequent allocation of hypothetical funds in a stylized financial task. [OI-W005, pp. 30–39]

Staw interprets the result through self-justification and discusses competing accounts of how people infer their own attitudes. He also distinguishes maintaining a coherent image of oneself from demonstrating to other people that an earlier costly decision was defensible. The latter could matter especially where status and evaluation depend on the appearance of competence. Those motives were not separately manipulated in this experiment. They are parts of the explanatory discussion that follows the observed interaction. [OI-W005, pp. 40–42]

Morrison and Milliken: how silence becomes an organizational property

Elizabeth Wolfe Morrison and Frances J. Milliken's 2000 article asks how withholding concerns about organizational problems can become a collective practice. Their object is more than the reluctance of an isolated employee. People can come to share an expectation that speaking is dangerous or useless, even when they possess different information and have never explicitly agreed to remain silent. The article develops a conceptual explanation linking managerial beliefs, organizational arrangements and employees' interpretation of their experience. [OI-W001, pp. 706–710]

The proposed mechanism begins partly with managers' fear of negative feedback and their assumptions about employees. If managers regard employees as self-interested, believe that management knows best, and interpret agreement as a sign of organizational health, they have reasons to retain decisions centrally and restrict the channels through which objections reach them. Informal conduct matters alongside formal design. Dismissing an unwelcome suggestion, soliciting advice mainly from agreeable people, or reacting defensively to adverse news can teach employees what happens to those who speak. Middle managers may withhold subordinate concerns because they anticipate the reactions of people above them, even when they do not personally share the senior managers' assumptions. [OI-W001, pp. 708–714]

Employees also learn from one another. They discuss encounters with managers, observe the treatment of colleagues and infer the local rules. Morrison and Milliken explain how these interactions can produce a shared climate of silence. An unfavorable career event following an objection may be interpreted as retaliation even when the two events were unrelated. This is an illustration of their proposed sensemaking process: beliefs about the danger of speaking can spread beyond the people who directly encountered a discouraging response. Silence itself supplies a further cue, since the absence of objections can make an unwritten prohibition seem established. [OI-W001, pp. 714–717]

The consequences in their model include impoverished information, fewer challenges to prevailing interpretations and weaker correction of errors. Management can mistake acquiescence for agreement. Employees who withdraw psychologically may then appear to confirm the managerial suspicion that workers lack commitment, reinforcing the conditions that discouraged voice. The authors nevertheless distinguish useful participation from unlimited input, which can overload and delay decisions. [OI-W001, pp. 719–721]

Edmondson: interpersonal risk and the observable work of learning

Amy Edmondson's 1999 field study treats learning as observable activity within teams: asking for help, seeking feedback, discussing errors, sharing information and experimenting. These activities can reveal uncertainty or failure to colleagues and supervisors. Someone may recognize that an inquiry would help the task and still avoid it because of the expected personal consequences. The paper examines team psychological safety, a shared belief that interpersonal risk-taking is safe, as a condition associated with carrying out that work. [OI-W004, pp. 350–355]

Psychological safety in this account differs from confidence that the team has the ability to succeed. A team can think itself competent while making it difficult to admit a mistake. It also differs from cohesion or an atmosphere in which every view must be agreeable. The survey asks about the treatment of mistakes, the ability to bring up difficult issues, asking for help and the acceptance of differences. The construct concerns the expected interpersonal consequences of participating, while team efficacy concerns confidence in collective capability. [OI-W004, pp. 351–355, appendix]

The study covered 51 work teams in one office-furniture manufacturing company. Team-member surveys were complemented by interviews, observation and measures from people receiving the teams' work. Psychological safety was positively associated with learning behavior, including measures supplied by outside observers. The analysis also examined whether learning could help account for the relation between safety and performance. Because the principal measurements were cross-sectional, those relationships do not demonstrate that an intervention increasing safety would cause learning or performance to improve. [OI-W004, pp. 358–368, 378–379]

The qualitative cases show what learning activity could look like. Members of the Stain Team described responding to reports of defects by working on the defects instead of attacking the person who reported them. They also sometimes sought another person's opinion without first announcing which defect they suspected. This practice made an additional inspection less dependent on the team's initial interpretation. [OI-W004, pp. 371, 375–377]

Resources and leadership remain part of the account. Time, staffing and access to support affect the opportunity to learn, while coaching and reactions to problems affect what team members expect from speaking. Some teams worked around material constraints; teams with apparently similar formal support could nevertheless differ in their learning behavior. Edmondson investigates those relations together. Learning also consumes time, and its value depends partly on uncertainty and task demands. The detailed failure-analysis account by Edmondson and Cannon in S31 develops a separate contribution concerning the work after a failure has been recognized. [OI-W004, pp. 354, 376–379; AFC-W015]

Vaughan: warning evidence is interpreted within a history of work

Diane Vaughan's 1997 article examines how earlier policy decisions and ordinary engineering work contributed to the Challenger launch decision. It condenses her larger historical reconstruction, drawing on archival material and interviews. Her argument challenges accounts that locate the explanation chiefly in the final teleconference, the moral character of particular managers or a simple opposition between engineers concerned with safety and managers concerned with schedules. She traces how production concerns, bureaucratic accountability and the inherited technical culture affected both groups. [OI-W017, pp. 80–82]

The history of the booster joints matters to that explanation. Damage to O-rings was initially an anomaly relative to design expectations. Engineers analyzed the damage and accepted explanations under which a second O-ring would preserve the required seal. Subsequent decisions could rely on that analysis and on the precedent of flights that returned successfully. Vaughan describes a gradual extension of acceptable

risk: individual decisions could appear reasonable within the established assessment process while cumulatively accepting conditions that had originally been anomalous. Her term normalization of deviance concerns this development of official risk judgments, including judgments made by working engineers. [OI-W017, p. 85]

She distinguishes patterns in the evidence that helped sustain those interpretations. A danger signal followed by successful flights could suggest that a repair had worked. An unclear or seemingly unlikely event could remain a weak signal. Repeated erosion could become routine when engineers believed its occurrence confirmed their understanding of the joint. The same recorded observation could therefore have a different significance to someone reconstructing the disaster than to someone assessing the next flight within the accumulated history. Vaughan's explanation concerns the interpretation and sequence of information, as well as whether information was transmitted. [OI-W017, pp. 86–87]

The eve-of-launch discussion introduced circumstances that did not fit the routine. Predicted temperatures were below those of earlier launches, the contractor's engineering presentation was being handled by teleconference, and its engineers recommended against launching. Preparing charts under time pressure divided the work among specialists. Some participants assembled the final recommendation without examining the other charts as a complete set. The group did not collectively review all those charts before transmitting them. These details connect organizational divisions of work and timing to the evidence available in the decision itself. [OI-W017, pp. 91–92]

Within Vaughan's reconstruction, the presentation contained signals that weakened the recommendation as others interpreted it. Charts reused from earlier launch recommendations looked familiar. Damage at a warm launch complicated a simple temperature argument, while some statements about redundancy appeared inconsistent with the proposed launch limit. A culture requiring quantitative support for engineering recommendations made the engineers' observational concerns difficult to defend in the expected form. Hierarchy and the separation of participants among locations also affected who spoke and what they knew about support for their position. The warning was voiced, but the way it was assembled, challenged and interpreted reduced its force. [OI-W017, pp. 92–94]

The article's practical discussion addresses policy, culture and signals. Vaughan argues that leaders need to examine how goals and resource decisions change the conditions of hazardous work, including consequences appearing after the original decision makers have moved on. She also asks how an organization can preserve coordination through rules while recognizing a situation for which familiar rules are inappropriate. Her warning about organizational change includes the loss of memory, disrupted working relations and new opportunities for error during a transition. [OI-W017, pp. 96–99]

The concrete problem of preserving the reasons behind a decision also appears in Leveson's safety account in S29 and in the treatment of records and traceability in S42. The sources here add detailed accounts of voice, interpretation and the personal and collective consequences of questioning a course of action. S52 turns to the related work of generating and testing alternatives, including situations in which people are willing to discuss a decision but have considered only a narrow range of possible responses.

S106 Difficult conversations under emotional and relational pressure

Deciding who should have the conversation

The Acas guide treats preparation as a decision about who has the authority and information to address a problem. Its timekeeping scenario follows a sales manager who has kept a record of an employee's late arrivals and long lunches. The manager begins by asking if anything is wrong. The employee says all is well. When the manager produces the record, the employee disputes its accuracy and asks why he is being singled out. As he leaves, he mentions that his wife has been unwell. A few days later he says he will meet again only with a union representative and is considering a grievance. [T08-W001]

Acas uses this instructional scenario to ask whether the manager had raised the issue earlier, told the employee about the monitoring or noticed a change in his behavior. The guide credits the opening question and the use of specific evidence. The scenario ends with the terms of a second meeting in dispute. [T08-W001]

Reasons for avoiding an exchange

King, Edlington and Williams interviewed 20 clinical supervisors in Australia about avoiding difficult conversations. Participants spoke of feared consequences, dislike of confrontation and limited confidence. The authors used convergent interviews to examine explanations across participants. The organizational work in S51 gives a broader account of the risks that can accompany speaking up. [T08-W002]

Changing the next turn

初鹿野 and 岩田 examine an interaction in which one person speaks about another while addressing a third. In their second case, Fujii addresses Doi about the reliability of Egawa's account. Egawa enters the exchange before the selected addressee. The analysis follows gaze, address and the timing of overlapping talk to explain why the entry matters. By taking a turn before Doi's response, Egawa alters the immediate course of the exchange. The authors interpret that entry as a way of checking the continuation of teasing. [T08-W003]

S14 Dependencies, work design and complexity: changing the demand for coordination

The question each arrangement answers

The sources in this section examine how dependencies, uncertainty and the allocation of work affect coordination. Some redesign tasks so that fewer exceptions arise across boundaries. Others increase the capacity to handle information, distribute expertise or compare mechanisms for managing a dependency. Their criteria differ. Fewer messages, less delay, lower learning cost and broader control capability are not interchangeable results.

Galbraith: reduce information demand or increase processing capacity

Jay Galbraith's 1974 article starts with a specified level of performance. When a task is well understood, much activity can be planned before execution. When uncertainty is greater, decisions about resources, schedules and priorities have to be made while work proceeds. More information then has to be processed among the relevant decision makers. Organizational design concerns the arrangements that permit the interdependent work to continue under those conditions. [W002, pp. 28–29]

Rules and programs precoordinate recurring situations. A worker can recognize the situation and apply an agreed response without communicating with everyone affected each time. Exceptions can be referred upward until they reach a level with a sufficiently broad perspective. Goals allow local discretion about behavior, but Galbraith's qualification is explicit: the tasks, budgets and completion dates still have to remain within the assumptions of the plan. New products, technologies or markets can invalidate the subgoals and generate renewed exceptions. [W002, p. 29]

The hierarchy has finite capacity. Galbraith distinguishes reducing the need to process information from increasing the ability to process it. Slack can absorb variation: longer lead times, inventories, budget allowances or other margins make fewer events demand immediate joint decisions. But slack has costs. Customers can wait longer, capital can be tied up and performance can be relaxed. Lower information demand obtained by accepting a less demanding output is not automatically an improvement at unchanged performance. [W002, pp. 30–31]

Self-contained tasks provide another demand-reduction strategy. Instead of grouping all specialists by resource or occupation, an organization can give a product, region, project or customer group much of what it needs to produce its output. Fewer conflicts over shared priorities then require cross-group scheduling. The tradeoff is reduced specialization, duplicated equipment or lost scale economies. Galbraith's example has groups do their own programming when each group needs too little work to sustain a separate full-time specialist. Some shared resources can remain centralized, so self-containment has degrees. [W002, pp. 31–32]

Vertical information systems take another route. They support more frequent replanning through clerks, computers, devices and decision capacity. Formalized language can allow more information to be carried in a given number of symbols under the model's assumptions. Yet more frequent reports can overload the recipient. Where relevant knowledge is qualitative and ambiguous, moving a decision toward the people who hold it can be more useful than repeatedly transmitting it upward. [W002, p. 32]

Lateral relationships include direct contact, liaison roles, task forces, teams and integrating roles. They can place joint decisions below the top of the hierarchy while preserving information about effects across departments. Nor does an informal relationship necessarily arise spontaneously across language, cultural or physical barriers. Galbraith treats its design as part of the task. [W002, pp. 32–35]

The article reports associations between uncertainty and integrative roles, including an adapted comparison of successful firms. These are reported cases and associations. Its concluding possibility of contracts, coalitions and influence over the environment also qualifies any claim that the four internal strategies exhaust all ways to reduce uncertainty. [W002, pp. 35–36]

De Sitter and colleagues: redesign production before allocating control

L. Ulbo de Sitter, J. Friso den Hertog and Ben Dankbaar's 1997 article proposes integral organizational renewal. Its starting point is the architecture of functionally differentiated relationships. The authors object to designing separate social and technical systems and only afterward attempting to optimize their combination. Their proposed relations are simultaneously technical, social and normative. [W003, pp. 503–506]

They distinguish the likelihood that disturbances arise through an arrangement from its sensitivity to those disturbances. Production design can reduce the variation that has to be dealt with; control design supplies opportunities to respond effectively to what remains. Their controllability concerns the ability to formulate and achieve a range of objectives. It is not identical to the technical reachability definition in every other control theory or a guarantee of one specified outcome under every disturbance. [W003, pp. 505–510]

Parallelization divides incoming work into relatively separate flows, preferably around product–market combinations. Segmentation groups highly interdependent operations within a flow so that fewer interfaces require cross-boundary management. The aim is to avoid sending each order through many specialized departments with repeated scheduling and priority conflicts. The proposed economy comes from changing the pattern of dependencies, before designing a larger apparatus to handle the old pattern. [W003, pp. 510–512]

The examples preserve residual sharing. At St. Antonius, the three product flows retain common preparatory and final work. The insurance example groups work around regions and customers while distinguishing risk acceptance and claims. These arrangements are not wholly independent systems. Decisions about what to combine, what to separate and what to leave shared remain part of the design. [W003, p. 512, table I]

Complete-task groups retain flexible equipment, multiskilled work, internal coordination, process control, boundary control and responsibility for improvement. Their greater local scope requires capabilities. Training, reliable and timely information, memory, experience and learning remain necessary. A group can ask for external help. [W003, pp. 512–514]

The authors propose designing production structure before control structure, moving from overall flows through segments to groups, and selecting process technology in relation to those requirements. Control allocation then works upward from decisions that can be handled locally. The article also says that actual redesign is iterative and can take years. Leadership, design expertise, a shared language, participation and resources are implementation conditions rather than incidental additions to a finished chart. [W003, pp. 515–524]

The article uses schematic functions to relate control opportunities to required variation and reports that architecture can resolve up to eighty percent of control questions. It does not supply calibrated general prediction formulas for these claims. Its account explains how changing production relations can alter control demands. [W003, pp. 506, 509–512]

A commercially successful change toward a more customized product mix overloaded the former workflow. The redesign introduced four self-managing teams while retaining inspection, scheduling,

calculation and information systems. The authors report improved delivery reliability and productivity. [W003, appendix B, pp. 530–531]

The authors themselves discuss conflict, failed consensus, mistrust, participation used badly, mergers and exhausted funds. They warn that adoption and satisfaction do not establish validity or superiority and note the lack of adequate comparative evaluations. Their design scheme is a tool for developing a local arrangement. It is not a guarantee that interests align or that the implementation succeeds. [W003, pp. 525–529]

Simon: stable intermediate forms and conditional simplification

Herbert Simon's *The Architecture of Complexity* connects assembly, dynamic interaction, description and problem solving. Hierarchy means nested subsystems and need not mean an authority relationship. Interaction strength matters more than simple physical distance. [W015, pp. 467–469]

The watchmaker parable concerns interruption. If partial work falls apart whenever an interruption occurs, completing a complex product can require repeatedly starting over. Stable intermediate subassemblies allow progress to be retained. The illustrative comparison shows why the availability of intermediate stable forms can affect the path to complexity. Its numerical ratio is a toy result under its specified probabilities and assembly assumptions, not an estimate of biological evolution or manufacturing productivity. [W015, pp. 470–473]

The associated search argument also requires recognizable partial success. A solver must be able to identify useful components or intermediate states and preserve them. Arbitrarily partitioning a difficult problem does not ensure that local correctness can be recognized independently or that components will combine. Simon's account of experience and reproduction retains information about solutions already found. [W015, pp. 471–473]

He explicitly preserves energetic conditions in biological discussion. Organisms are not energetically closed, and intermediate forms may be stable only while an external source of free energy continues. Classical thermodynamic considerations alone do not provide the direction and rate of evolution described by the assembly argument. [W015, p. 471 and governing notes]

Near-decomposability is another conditional property. Interactions within subsystems are strong relative to those among subsystems. Short-run behavior can then be approximately independent, while long-run behavior depends on more aggregate relationships. Simon's heat example uses a building insulated from its surroundings, rooms with relatively resistant walls and cubicles with weaker partitions. Temperatures equilibrate within rooms before the building reaches its overall equilibrium. [W015, pp. 474–475]

After the within-room transients have subsided, a smaller set of room-level measurements can suffice for the relevant dynamic description. The simplification depends on the timescale and approximation tolerated. Simon calls near-decomposability a strong property and qualifies his economic illustration with a consumption subsystem broadly connected to others. A hierarchy on paper therefore does not establish the required separation of coupling strengths or timescales. [W015, pp. 474–477]

The description argument uses actual redundancy. Repeated subpatterns can be named and reused, and a generating law can sometimes replace an explicit account of every successive state. Simon's array illustration replaces sixty-four displayed symbols with a thirty-five-symbol description under its stated notation. Constructing the notation, interpreting it and retaining adequate information remain conditions of use. [W015, pp. 477–479]

An entirely unredundant object may be its own shortest description. Simon also raises the observation problem: perhaps hierarchical things are more understandable partly because less hierarchical aspects escape attention. His distinction between a desired state and a process that produces it further preserves

invention and search. Recognizing an acceptable result is different from possessing a recipe for obtaining it. [W015, pp. 478–482]

Simon develops three mechanisms: stable intermediate assemblies retain progress after interruption; weak intersystem coupling permits approximate local treatment; and redundancy permits a reusable description. Each addresses a different part of the complexity problem. [W015]

Malone and Crowston: the dependency and the mechanism

Thomas Malone and Kevin Crowston define coordination as managing dependencies between activities. Their 1994 article develops a vocabulary for identifying those dependencies across human, computational and other settings. The definition includes competition, conflicting interests and political processes insofar as they manage interdependent activities. It does not assume that all participants cooperate toward one agreed good. [WME-W011, pp. 90–92]

Different dependencies require different work. Activities may compete for a shared resource. A consumer may need a producer's result, which introduces prerequisite, transfer and usability issues. Activities may need simultaneous occurrence, or a task may depend on subtasks. The authors describe the list as nonexhaustive. What looks like one object dependency becomes more precise when the activities that require or change that object are identified. [WME-W011, pp. 92–97]

Transfer is not just communication. Moving a material output can require transportation, storage and handling. Timing production closely can reduce storage, while an inventory can buffer timing differences. A standard can make an output usable without renewed joint design, while participative design can address requirements that were not settled in advance. [WME-W011, pp. 94–95]

The activity description is selective. A whole division can be treated as one activity for one question and individual stations as activities for another. The authors deny that there is a single correct granularity. Evaluation criteria also have to be identified: speed, accuracy, cost and the treatment of conflicting goals can lead to different assessments. More coordination is not always worth its cost. [WME-W011, pp. 100–101]

The survey reports related models comparing production, coordination and vulnerability costs. A central allocator can reduce some communication while creating exposure to a central failure. Decentralized bidding requires announcements, bids, awards and estimates. Dedicated product hierarchies can retain unused capacity. These comparisons are reported through Malone and Crowston’s review under the stated model assumptions. [WME-W011, pp. 98–100]

Cross-domain transfer needs additional care. Human motivation, incentives, emotion, reputation and the capacities of the communication system can change a proposed mechanism's effects. The article distinguishes varying a model's parameters from adding behavioral qualifications to a simple baseline. Shared terminology is an invitation to investigate the comparison, not evidence that people and processors are identical. [WME-W011, pp. 91, 100–101]

The discussion of cheaper coordination is especially consequential. Lower unit cost can substitute technology for human activity, increase the quantity of coordination used, and enable more coordination-intensive structures. The authors describe a case where removing a layer of managers was followed by the creation of roughly as many specialist staff positions, some devoted to developing the new systems. Their interpretation concerns reallocation toward more complex analysis. [WME-W011, pp. 102–103]

Tools also retain contribution and interpretation problems. A shared database can be technically accessible while experts have incentives to withhold useful knowledge. Rules for filtering messages, making commitments, choosing tasks and maintaining standards still require organization. Distributed

computer mechanisms likewise incur resource bookkeeping and message overhead. [WME-W011, pp. 105–109]

The article ends with a research programme concerning dependency types, the composition of processes and comparison across design spaces. The source offers a connected vocabulary and partial results, with questions of generality still open. [WME-W011, pp. 110–114]

Garicano: specialize learning and pay for communication

Luis Garicano's production model examines how people can specialize in knowledge. A worker who encounters problems need not learn to solve every possible problem personally if unresolved cases can reach someone with the relevant knowledge. Specialized learning can broaden the knowledge usable by a productive organization while reducing what each person needs to acquire. Escalation, matching and communication then have costs. [W030, selected introduction and model, pp. 875–880]

The setup specifies a problem distribution and explicit learning and communication costs. Those assumptions determine which problems are routine enough for production workers and which justify specialized assistance. The relevant claim is conditional organization of knowledge. [W030, §II]

Radner: processor count and delay are joint criteria

Roy Radner's *The Organization of Decentralized Information Processing* models programmed networks of limited processors. Associative operations allow portions of an information-processing task to be carried out and combined. A network can then be assessed through both the number of processors and the delay before a result is available. A reduction in one measure need not settle the other. [TA-W040, pp. 1109–1115]

The model uses prescribed programs and timing, identical limited processors and a fixed amount of environmental data. Communication has no additional cost in that formulation. The selected account consequently cannot be used as a whole organizational cost analysis that includes every communication and learning burden. Information-processing decentralization is also distinct from decentralized authority. [TA-W040, introduction and §2]

Distributed control has a setup phase

Debbie Harrison, Kristin Munksgaard and Frans Prenkert propose distributed control for coordinating dissimilar but complementary activities across actors. Their selected discussion distinguishes the intensive interaction needed to establish the arrangement from thinner, potentially digitally mediated later interaction. Actors can achieve partial local goals while participating in a wider interdependent activity structure. [UX-W002, abstract and “A new coordination principle”]

The distinction makes setup substantive. Establishing relationships, specifying interfaces and finding workable accommodations precedes the apparent ease of later coordination. The circular-economy and blockchain discussions illustrate a conceptual proposal. [UX-W002]

Alter: work systems include customers and continuing adaptations

Steven Alter's 2019 account defines a work system through participants or machines performing processes and activities with information, technology and other resources to produce products or services for customers. The framework treats these elements in relation rather than forcing every situation into separately designed social and technical systems. Its nine-element representation also includes the surrounding environment, infrastructure and strategy. [SG-W015, selected pp. 1–3, 5–13]

The lifecycle includes planned projects and unplanned adaptations or workarounds. A specification can fail to describe useful work, and compliance with it can obstruct the actual purpose. Partial and total automation change participants and activities without making the rest of the work system disappear. The article's emphasis on products, services and customers also makes it possible to distinguish a provider's local efficiency from the customer's total costs or ability to obtain value. [SG-W015, selected lifecycle, workaround and service discussions, including p. 19]

Alter presents a descriptive and design framework with methods and local hypotheses. [SG-W015]

Sociotechnical groups in citizenship services

Almir Cordeiro Júnior and Mônica Bianco apply a sociotechnical work-group perspective to conflict-resolution services in integrated citizenship centers in Vitória. The selected abstract and final discussion extend the subject beyond factory flows. Semi-autonomous service work remains dependent on technical knowledge, relationships among people and learning. The service's social interaction is part of the work rather than interference that can simply be removed from a schedule. [W028, abstract and final discussion]

This organizational analysis asks what a work group must be capable of doing in a service whose cases and interpersonal requirements cannot be treated as identical manufactured units. [W028]

Joseph and Sengul: design problems are distinguishable

John Joseph and Metin Sengul's specialist review distinguishes configuration, control, channelization and coordination. The selected sections treat organization design as more than drawing formal reporting lines. Choices about the arrangement of units, the guidance of behavior, the channels through which information and attention move, and the management of interdependence answer different questions. Their integrative argument retains contingent findings and differences in assumptions. [W046, “Contemporary Research, ” table 1, “State of Research, ” “Organization Design as a Theory, ” conclusion]

Sternieri: autonomy and shared dependencies

Armando Sternieri's 2026 *Dependency Restructuring* distinguishes changes in activities from changes in dependency and coordination relationships. The abstract proposes that local autonomy can rise while reliance on common capabilities also rises. More freedom at a visible operating point does not by itself establish less dependence throughout the arrangement. [UX-W011, abstract and illustrative cases]

Sternieri presents the cases as illustrations rather than tests. [UX-W011, abstract and illustrative cases]

Production relations, local control and shared capability

The selected explanations identify different places where work can change: recurring exceptions, production interfaces, retained partial assemblies, descriptive redundancy, shared resources, specialist learning and the processing of a fixed computation. They also identify different costs and failure conditions.

S103 Recruiting, selection and forming working relationships

Reaching a candidate pool

The French study coordinated by Yannick Fondeur and Guillemette de Larquier distinguishes reaching candidates from assessing them after they come into view. Its Table 1 reproduces an analysis by de Larquier and Marchal of the 2005 Dares *Offre d'emploi et recrutement* survey. In 35 percent of the reported cases, the establishment examined one application per vacancy. In that group, personal and professional networks and contacts with former staff together accounted for 49.8 percent of the channels reported to have brought in the most candidates. Press or internet advertising accounted for 4.3 percent. In the group with more than ten applicants per vacancy, the corresponding shares were 7.2 percent for networks and 31.2 percent for advertising. Those percentages describe recruitment cases in each group, classified by the channel that brought in the most candidates. [T05-W001]

The authors distinguish a channel that brings a candidate already known or recommended to the employer from a channel that invites a wider field to apply. In their retail fieldwork, some employers placed a notice inside the shop or advertised only in the local press to limit the response or reach people living nearby. Even within advertising, the place where an employer announces a vacancy can serve as an initial filter. S13 discusses other ways access to occupational opportunities is organized. [T05-W001]

Assessment goes in both directions

Wilhelmy and Kleinmann illustrate reciprocal selection through a publishing house. A potential candidate sees the vacancy, checks the publisher's site and reads an employer portal. Inside the publishing house, staff identify job requirements and choose ways of assessing them. The candidate's own inquiry also continues during the contact. The authors discuss the tension between a standardized interview, which gives applicants a more comparable assessment, and room for the two parties to learn about each other. [T05-W002]

What a validity estimate describes

In their 1998 review, Schmidt and Hunter distinguish the validity of a single predictor from that of a combination. In Table 1, their estimated correlation with job performance is 0.51 for the structured interview alone and 0.63 for a combination of the interview and a general mental ability test. The second value estimates the performance of the pair. [T05-W003]

Sackett and colleagues revisited the adjustments behind estimates of predictive validity. When people are hired partly on the basis of a test, their scores can have less spread than the scores of the full applicant pool. A study of hired workers can then understate the relationship between test scores and later performance. A correction needs a defensible estimate of how much variation was lost. The authors question applying a correction from studies with evidence of restriction to other studies where the selection process could have been different. [T05-W004]

Their reanalysis of job-specific knowledge tests gives a concrete example. In the 59 studies of such tests assembled by Dye and colleagues, the observed mean correlation with job performance was 0.31. Correcting for unreliability in the performance measure raised it to 0.40. Dye and colleagues also assumed that selection directly on the test retained half the applicants, yielding an estimate of 0.62 after the additional adjustment. Sackett and colleagues retained 0.40 as a conservative estimate from that evidence. The selection ratio had been assumed, and the original analysis did not identify which studies followed applicants into the job and which tested workers already employed. The difference between 0.62 and 0.40 is therefore a difference in the treatment of the same body of studies. [T05-W004]

S85 Dependencies, bottlenecks and the flow of project work

Project schedules express dependencies, but they also allocate opportunities to work. A job can be technologically ready while the crew or equipment it needs is occupied elsewhere. A task can have room to move without delaying completion, yet moving it can change inventory, employment or contractual costs. The following accounts develop network calculations, constrained resources and the placement of protective time.

Planning the network and choosing a completion date

Kelley and Walker's 1959 critical-path account distinguishes planning from scheduling. Planning defines the jobs and their technological order; scheduling supplies durations, costs and timing. Choosing what counts as a job is itself consequential. A delivery, an authorization or a safety requirement may need representation because subsequent work depends on it. The useful level of detail follows the purpose of the analysis, rather than a requirement to divide every project into equally sized tasks. [MG-W001]

The network permits forward and backward calculations. Earliest event times follow from predecessor completion; latest permissible times follow backward from the required completion. Their difference locates room for displacement. Critical work has no such room under the calculated schedule, so a delay there propagates to completion unless another change compensates for it. Float makes some timing choices available without changing that completion date. It is information about a schedule, rather than a reason to treat every noncritical task as unimportant. [MG-W001]

The authors also ask what completion time should be purchased. Expediting a job changes its duration and cost, while a project's completion can have costs beyond the jobs themselves. Their parametric linear-programming formulation develops a family of time/cost choices. The calculation therefore supports comparison among completion dates and combinations of expedited jobs, rather than treating the fastest technically attainable schedule as the only objective. [MG-W001]

Manpower and equipment enter the original account explicitly. A technologically feasible schedule can demand resources that are unavailable. Adding precedence relations to serialize jobs sharing a resource is one response, but the chosen order can exclude alternatives. With three jobs and one crew there are six possible orders; fixing one order hides the other five. The authors' practical resource-leveling discussion instead examines workforce demand and shifts eligible jobs within their float, including separate totals for different crafts. A precedence diagram and a workforce curve expose different constraints on the same work. [MG-W001]

Buffers, releases and operating priorities

In Sood's account of critical-chain project management, resource contentions are resolved before the constraining sequence is identified. Protective time is organized into a project buffer and feeding buffers, while the release of projects is related to constrained-resource availability. This changes what a task estimate and a local delay mean operationally: attention is directed to protection of the project and its critical chain, rather than every activity carrying an independent allowance that managers try to preserve. [MG-W002]

Buffer management continues after the plan is drawn. Progress and consumption of protective time provide signals for priority decisions. A task's position in relation to the threatened buffer can matter more than its place in a static list of individual due dates. Work release and daily priorities thus form part of the arrangement; adding buffers to a schedule without changing those decisions would leave much of the reported method unexplained. [MG-W002]

Sood describes a construction implementation at Shea Homes in which a shared portal supplied subcontractors with priorities for the following day. The portal joined the scheduling model to the people deciding what work to undertake. His reported cases illustrate that operational connection. They are implementation accounts supplied by a vendor executive, and the useful historical detail lies in how release, protection and coordination were organized. [MG-W002]

Where protective time is placed

諏訪晴彦, 森田大輔 and 三道弘明 distinguish the total amount of slack from its distribution. Their 2009 model begins with a resource-feasible baseline schedule, with fixed estimated processing times and constant resource availability. It then inserts dummy activities whose durations represent protective time. The sum of these durations is the chosen allowance K; allocating that sum is a further decision. [MG-W003]

The construction places a dummy before the first critical activity and after critical activities, including a terminal dummy. Critical work and affected successors are shifted to the right to accommodate these insertions. Later critical start times incorporate preceding dummy durations; ordinary activities start as early as their technological and resource predecessors permit in the changed schedule. Distinguishing the terminal dummy also separates the last real job's completion from the project's buffered due date. [MG-W003]

The objective concerns displacement from announced start times. Starting earlier can create inventory or other preparation costs; starting later can extend employment or disturb subcontracting arrangements. The model assigns early and late deviations their own weights. Buffer placement is judged through expected weighted displacement costs, so two schedules with the same K can have different consequences. The authors consider allocation rules based on position and successor relationships as well as uniform allocation. [MG-W003]

This formulation makes a concrete choice visible: how much protection to provide and where to provide it are separate scheduling decisions. A buffer's position changes which jobs move and which relationships absorb variation. The operating signals in Sood's account and the displacement-cost model address different parts of managing protective time. Work dependencies and coordination mechanisms receive broader treatment in S14.

S86 Forecasts, commitments and governing uncertain projects

Authorizing a project involves more than estimating its tasks. Forecasts shape the commitment, people have reasons to present particular forecasts, and governing bodies decide whether the work still serves its purpose. Exploratory projects add another difficulty: the investigation can change what the project is supposed to accomplish.

Forecasting from completed projects

Flyvbjerg's reference-class forecasting turns attention from the internal story of a proposed project to outcomes of comparable completed projects. The procedure identifies a reference class, establishes the distribution of its outcomes and locates the proposed project in relation to that distribution. The comparison must be appropriate to the kind of project and to the decision stage: an early estimate and an estimate made after detailed design answer different forecasting questions. [MG-W004]

The outcome distribution permits a decision about tolerated overrun probability. A budget based near the middle of a distribution accepts a different chance of overrun from one based at a higher percentile. Flyvbjerg illustrates the connection with a £100 million road estimate and a 32% uplift, producing £132 million. The uplift belongs to a specified reference class and risk tolerance, rather than being a general contingency percentage to attach to every undertaking. [MG-W004]

His Edinburgh Tram Line 2 account concerns an independent review before construction. It shows how an outside-view assessment can enter an authorization process. Its historical role is the review of an estimate at that stage, not a retrospective demonstration of the line's eventual cost. Independence matters because the people assembling a project's internal case may also be committed to getting it approved. [MG-W004]

Flyvbjerg distinguishes optimism bias from strategic misrepresentation. Better information can correct a sincere but unrealistic estimate; it has a different relationship to a forecast deliberately made attractive to secure approval. His proposals therefore concern incentives, scrutiny and accountability as well as the statistical procedure. The distinction explains why installing a forecasting technique and making an institution want an accurate forecast are separate interventions. [MG-W004]

The Heilmeier Catechism in research funding

George H. Heilmeier recalled that he read and commented on each DARPA order before deciding whether to sign it. The director’s signature was the last step before funding. He described how staff critiqued one another’s proposals and agreed to use a common set of questions. Asked how many projects survived that scrutiny, Heilmeier said the more telling question was how many had changed during the discussions. [HCT-W001, pp. 180–181]

DARPA describes using the “catechism” to assess proposed research programs. It publishes the following eight questions. [HCT-W002]

1. What are you trying to do? Articulate your objectives using absolutely no jargon.

2. How is it done today, and what are the limits of current practice?

3. What is new in your approach and why do you think it will be successful?

4. Who cares? If you are successful, what difference will it make?

5. What are the risks?

6. How much will it cost?

7. How long will it take?

8. What are the mid-term and final “exams” to check for success?

Governing the undertaking beyond its initial approval

Too and Weaver place project management within a governance framework linking project outputs to organizational strategy. The governing board supplies direction and remains accountable for governing the organization. Project managers exercise delegated authority to carry out work. Delegating a task does not make the governing responsibility disappear, while retaining accountability does not require the board to manage each operational decision. [MG-W005]

Their framework connects four elements: portfolio management, sponsorship, a project management office and project/program support. Portfolio management selects undertakings that support strategy and reconsiders those that cease to do so. Sponsorship maintains the connection between an undertaking and the organization that expects its benefits. The other elements supply oversight information and support for delivery. The authors present this as a conceptual construction drawn from management and governance literature. [MG-W005]

Continuing review changes the meaning of authorization. A project approved under one strategy or expectation need not retain its place indefinitely. Governing information must make it possible to consider the undertaking's continuing contribution, rather than merely whether its original schedule is being followed. Too and Weaver's account thus treats delivering the specified output and selecting the right continuing commitments as connected but different responsibilities. [MG-W005]

Organizing exploration before the project is fully specified

Lenfle examines innovation projects for which a customer, goal or decomposition of work is not yet settled. The uncertainty extends beyond the time needed to carry out known activities. Investigation can alter the potential uses, technical questions and criteria by which an opportunity should be judged. A detailed development plan presupposes answers that this kind of project is trying to obtain. [MG-W006]

He gives the undertaking a provisional research program. It organizes related technical and use questions without pretending that the final product is already known. Tests supply occasions for learning and coordination: a deadline can require a team to produce knowledge rather than a completed feature. Exploring uses and technical possibilities together also avoids a sequence in which a technical answer arrives after the question about its value has changed. [MG-W006]

Legitimacy and evaluation are part of the construction. Without a recognized undertaking and criteria appropriate to exploration, investigations can appear to be disconnected studies competing for resources. The program gives them a common provisional purpose while allowing that purpose to be revised. The Usinor history developed in S62 supplies a related account of exploratory learning. Here Lenfle's contribution is the problem of organizing and evaluating such work when the usual specification-led project representation is premature. [MG-W006]

S98 Goal-setting: choosing aims, directing effort and revising commitments

A goal can specify what an engineer is trying to construct, what an athlete attends to during performance, or what a manager rewards. It can also express an aim that someone accepts, revises or relinquishes. These uses bring different problems into view: constructing an adequate target, sustaining effort, learning how to act, recognizing a useful opportunity, and deciding whether pursuit should continue. Historical accounts of goal-setting address each of these problems through different models, procedures and kinds of evidence.

Constructing and adopting aims

桑原哲郎 and 岸野哲's 1977 account begins before the target is fixed. An engineering problem does not always arrive with a satisfactory objective function awaiting optimization. The target itself must be constructed through inquiry into the system and its environment. They present this inquiry as a way of discovering causal regularities rather than restricting the goal to improvements under conditions already fixed. Their proposed reasoning follows effects from the system into its surroundings and the responses that return to it. Understanding those relations can change what it would be purposeful to accomplish. In this account, abstraction is part of discovering the target, rather than merely simplifying a target already supplied. [GS-W02]

The authors distinguish a creative phase and three execution phases. The creative phase abstracts the object, considers its wider relations and makes a judgment about purpose. The first execution phase introduces concrete conditions and asks whether the proposed account is effective; inadequacy can require returning to abstraction. The second works toward consistency between system and environment, including conditions left unexplained earlier. The third introduces economic and feasibility conditions and establishes the functions to be realized and the constraints under which conventional engineering design can proceed. Their table and diagrams distinguish phases by the determinacy of the target and the kind of judgment being made. The result is a proposed architecture for reasoning about an engineering objective, with revision built into its construction. [GS-W02]

Sheldon and Elliot's 1999 self-concordance model asks why people pursue the personal goals they have adopted. A person may act because the activity is enjoyable, because its importance is personally accepted, because failure would provoke guilt or self-reproach, or because other people demand it. Personally accepting an unpleasant but valued task differs from enjoying it, and both differ from acting under pressure. Self-concordance concerns these reasons for pursuit. It is therefore possible to specify the same outward achievement while having substantially different relationships to that achievement. [GS-W04]

Their model connects concordance with sustained effort and goal attainment. It then connects attainment with well-being through experiences of autonomy, competence and relatedness. A further relationship matters: concordance is expected to modify what attainment provides. Achieving a personally concordant goal should yield more need-satisfying experience than achieving a goal pursued for less personally accepted reasons. The account thus concerns both persistence toward an end and the experience of having reached it. [GS-W04]

In the first study, 169 students listed ten personal goals and rated their reasons for pursuing them. Repeated assessments over a semester followed effort and progress. The concordance index added intrinsic and identified reasons and subtracted external and introjected reasons. The reported associations were consistent with effort mediating attainment. The model begins with selected goals and commitment; its explanation concerns differences among adopted pursuits rather than the prior deliberation through which every possible end might be compared. [GS-W04]

Harald Seelig's operational account makes the component reasons especially tangible through exercise. Interest in the activity, acceptance of its value, self-reproach for failing to exercise and pressure from others can be examined separately. He explains that the difference index is a simplified global measure: inspecting the component profile gives more detail than the resulting score alone. The profile preserves distinctions between different combinations of reasons that a single total can conceal. [GS-W05]

Д. А. Леонтьев's account of self-regulation also gives the adoption of goals an active role. An aim may be supplied externally, but accepting it still involves the actor's selective judgment and responsibility. He contrasts the absence of personal criteria for selection with rigidity that makes an accepted goal impossible to relinquish. These are different problems: being unable to choose among demands and being unable to revise a choice once made. His discussion places both within the structure of activity and its correction. [GS-W10]

Directing attention, effort and learning

In their 2002 retrospective, Locke and Latham explain four pathways through which goals affect task performance. Goals direct attention toward relevant activities, mobilize effort, prolong persistence and prompt the use or acquisition of task knowledge. Specificity reduces ambiguity about what counts as the required performance; difficulty supplies challenge. Knowledge, commitment and information about progress affect how those processes operate. The account explains why merely announcing a target can differ from giving someone a target that is understood, valued and connected to workable action. [GS-W01]

A particularly consequential distinction concerns performance and learning goals. A performance goal specifies the score or result to achieve. A learning goal concerns discovering or acquiring strategies for the task. When a task is unfamiliar and complex, a demanding score can draw effort toward immediate performance while the performer still lacks an effective method. The relevant work may instead be systematic strategy acquisition. Locke and Latham's discussion therefore distinguishes applying available knowledge from learning what action the task requires. Their account of commitment likewise includes the goal's importance and the person's perceived capability, rather than treating difficulty alone as sufficient motivation. [GS-W01]

Díaz-Ocejo and Mora-Merida's review distinguishes outcome, performance and process goals in sport. Winning a race is an outcome goal. Improving one's own previous time is a performance goal. Directing attention to a technical action is a process goal. The distinctions concern the content of the target and what the athlete can attend to or influence during execution. A process goal specifies an action to attend to, whereas a learning goal specifies knowledge or a strategy to acquire. [GS-W16; GS-W01]

The review reports more consistent support for goal-setting interventions in motor performance than for a uniform ranking of specificity, difficulty or proximity. Comparisons differ in who sets the target and how its difficulty is determined. Its discussion of proximity also raises flexibility: a commitment to a number of training sessions during a week permits adjustments that a rigid daily requirement may prevent when injury or another interruption occurs. The practical question is how the target organizes training under changing conditions, as well as how challenging it sounds when first stated. [GS-W16]

Frutuoso's sport-practice review develops a concrete coach–athlete account. Written goals are coupled with a strategy, a deadline, periodic evaluation and feedback. Goals are developed with the athlete, and the target actually pursued is compared with the one formally stated. Competition and training targets are distinguished; simultaneous aims need priorities and compatibility. Revision is part of the proposed practice. The inventory he describes elicits athletes' perceptions of goal-setting, while his recommended package brings formulation, execution and evaluation together as a coaching practice. [GS-W17]

From a desired future to situated action

Oettingen, Pak and Schnetter's 2001 fantasy realization account distinguishes imagining a desired future from relating it to an obstructing present. Indulging in the future elaborates its attractive aspects; dwelling on the present elaborates difficulties. Mental contrasting brings the desired future and the present obstacle into relation. The proposed effect is expectancy-dependent commitment: determination strengthens when success appears feasible and weakens when it appears unlikely. The procedure is consequently concerned with selective commitment, rather than simply making every wish more motivating. [GS-W06]

Their first experiment involved 136 female students who named an important interpersonal problem, rated the importance and likelihood of a happy outcome, and listed positive aspects of that future and obstacles in the present. Contrasting instructions alternated elaboration of future and obstacle; comparison instructions elaborated one side alone. Immediate planning and perceived responsibility served as commitment indicators. The authors' general discussion reports the predicted expectancy-dependent pattern across their four experiments. Commitment here means determination to pursue the future, rather than a binary choice or a ranking of competing ultimate values. [GS-W06]

Gollwitzer and Sheeran's account of implementation intentions begins with a different difficulty: a person can intend an outcome yet fail to initiate the necessary action. Forgetting, missing an opportunity, initial reluctance and distraction pose different obstacles. Constructing an implementation intention involves identifying an instrumental response, choosing a feasible situation in which to perform it, and linking situation and response in an if–then plan. The selected response must serve the goal; the cue must be one the person can actually encounter and recognize. [GS-W07]

Their explanation has two parts. Specifying the situation makes its cue more accessible, and linking that cue to the response facilitates initiation without renewed deliberation at every encounter. Deliberate preparation is intended to make later action less dependent on a fresh decision. In a coupon-collection example discussed by Gollwitzer and Sheeran, location cues are connected with collecting the coupon; the account examines cue recognition and subsequent action, rather than intention alone. Their discussion also considers the strength and activation of the underlying goal. A pursuit already proceeding easily need not gain in the same way as one hindered by an initiation problem. [GS-W07]

Wang, Wang and Gai's 2021 meta-analysis evaluates mental contrasting with implementation intentions, or MCII, as a combined intervention. A desired future is related to an obstacle, and an obstacle-linked response plan is constructed. Included comparisons encompass MCII against no treatment and adding MCII to an intervention against that intervention without MCII. The review analyzes 24 independent samples or effect sizes from 21 reports, with 15,907 participants. Its random-effects estimate for goal attainment is g = 0.336: an average advantage of about one third of a standard deviation over the included controls. The 95% confidence interval of 0.229–0.443 expresses uncertainty around this pooled estimate. Heterogeneity I^2 = 59.321% indicates substantial variation among study estimates beyond sampling error. [GS-W08]

The review's publication-bias diagnostics give different indications. Its trim-and-fill procedure estimates how the pooled result would change if apparently missing studies were included, yielding an adjusted g of 0.242. This is a sensitivity estimate based on that procedure's assumptions. Its pooled comparison evaluates the specified intervention against the included controls. Most comparisons do not isolate whether the combination outperforms each component separately. The delivery-mode analysis also has a different basis: it excludes two large online-course samples totaling 13,909 participants. Differences among delivery modes in that reduced collection are comparisons across studies, rather than randomized assignments to delivery modes within the full participant sample. [GS-W08]

Feedback and the people who manage goals

野上真 and 古川久敬's 2010 review examines managerial action across goal formulation, execution and achievement appraisal. It distinguishes communicating or emphasizing demands, providing support through advice and feedback, and incorporating performers' views into decisions. These activities can occur at different points in the work. A manager can consult someone about the target, advise on the method, discuss progress during execution or involve the performer in appraisal. Each changes a different part of the goal-setting process. [GS-W03]

The review considers learning, satisfaction and organizational commitment alongside productivity and motivation. Advice about performing an unfamiliar task supplies something different from a report of the distance still remaining to a numerical target. Participation in devising a strategy also differs from participation in choosing the number. The authors find more developed evidence for support than for achievement demands and identify combinations across phases as a research question. Their summary table collects reported positive findings; their proposed complementarity among managerial actions is an interpretation and direction for research. [GS-W03]

Locke and Latham's discussion of participation similarly distinguishes a reasoned assignment from a terse command. Their account of the Erez–Latham collaboration examines what information and rationale accompany assignment, as well as whether the performer participated in setting the goal. Participation in developing task strategies can supply useful knowledge even when choosing the target is not the decisive motivational difference. Progress feedback gives the performer a basis for altering effort or strategy. The comparison therefore involves the interaction and information surrounding the target, as well as its nominal source. [GS-W01]

Леонтьев separates receiving feedback, comparing the actual state with a desired criterion, and correcting action. Information can be available without being received; an error can be recognized without producing a change of conduct. His conceptual structure distinguishes failure to begin acting from failure to notice a discrepancy and from failure to respond after recognizing it. Correction may concern the route being taken or the goal itself. These distinctions make it possible to describe several different failures that would otherwise appear simply as failure to attain a target. [GS-W10]

In Frutuoso's coaching account, periodic evaluation and feedback return attention to the goal actually being pursued. A formal agreement between coach and athlete does not by itself establish that training is organized around that agreement. Comparing stated and pursued aims, reviewing progress and revising the plan give the target an ongoing role in the working relationship. The accounts of participation and organizational strategy in S92 develop related questions about interpreting objectives and taking part in decisions. [GS-W17]

Revising, relinquishing and replacing goals

Wrosch and colleagues' 2003 account addresses self-regulation when a goal becomes unattainable. Disengagement has two components: reducing effort and relinquishing commitment. Stopping an activity while remaining intensely committed to its outcome differs from releasing the commitment itself. Reengagement has its own components: identifying alternative goals, investing them with value and beginning their pursuit. The distinction permits someone to develop a new pursuit while the old commitment is still being relinquished. [GS-W09]

Participants rated how readily they could reduce effort and release commitment to an unattainable goal, and how readily they could find, value and pursue alternatives. The authors consider lost opportunities, changed life circumstances and competing demands on limited time and energy. These contexts give the problem a practical form: continuing to pursue one aim can interfere with pursuits that remain available,

while the absence of an alternative can make disengagement harder. Their framework distinguishes capacities for releasing and replacing goals rather than requiring a fixed sequence in which one must be completed before the other begins. [GS-W09]

The three samples comprised 115 undergraduates, 120 younger and older adults, and 45 parents of children with cancer or medically healthy children. Associations connected goal-adjustment capacities with aspects of subjective well-being. The authors identify the cross-sectional design and possible cohort differences as constraints on causal interpretation. The studies relate adjustment capacities to well-being; they do not supply a test for deciding that a difficult undertaking is objectively unattainable. [GS-W09]

Леонтьев's discussion of flexibility distinguishes changing a route from changing the desired destination. Recognizing a discrepancy can prompt either response. A goal that cannot be reconsidered can organize repeated action around a criterion that circumstances no longer support, while indiscriminate acceptance of new demands creates a different difficulty. His conceptual account places revision within self-regulation itself. The backward-planning accounts in S94 examine how means are derived from intended results; these adjustment accounts concern the possibility that the intended result will also change. [GS-W10]

Organizational objectives and the debate over targets

Doran's 1981 formulation of SMART concerns making managerial objectives usable and communicable. Specific identifies an area for improvement. Measurable calls for quantity or an indicator of progress. Assignable identifies who will do the work. Realistic relates the intended results to available resources. Time-related specifies when results are to be achieved. The original A therefore concerns assignment of responsibility. His illustrative objective calls for developing and implementing an inventory system by December 31 of a designated year to reduce inventory costs by $1 million. The work is to take no more than 200 work hours and $15,000 in initial out-of-pocket expenditure. The example connects the intended result with a deadline and specified resources. [GS-W11]

Doran also allows abstract objectives and does not require every objective to satisfy every criterion or be quantified. Pairing the objective with an action plan matters to his account: writing an intelligible result and organizing its accomplishment are connected managerial tasks. He discusses managers' anxiety about commitments involving variables they cannot control and senior managers' role in communicating objectives. The proposal addresses formulation and coordination. [GS-W11]

Ordóñez, Schweitzer, Galinsky and Bazerman's 2009 critique asks what targets cause people to attend to and what selected performance measures leave out. They discuss a proofreading experiment in which explicit grammar or content targets are contrasted with an instruction to do one's best. Concentration on a specified dimension can leave the other dimension unattended. Their discussion of multiple goals likewise raises competition between quantity and quality: adding a second objective changes the demands rather than automatically resolving the allocation of attention. [GS-W12]

The critics also distinguish using dishonest means to attain a goal from falsely reporting that it has been attained. A target can organize conduct leading to a result, and a reward can depend on a report about that result; the two create different opportunities for misconduct. Corporate incidents illustrate their argument for examining risk and conduct alongside output. In their subsequent rejoinder, they distinguish using case histories to identify consequential questions from using them to prove causality and argue for studies that permit and measure broader consequences. [GS-W12; GS-W14]

Locke and Latham's reply contests inference from anecdotes, selective citation and the characterization of earlier research. Their stated position is that proponents had already described many of the pitfalls presented as neglected. The disagreement thus concerns both evidential standards and what the existing theory actually asserts. The proponents' earlier retrospective develops conditions involving knowledge,

commitment, rationale, participation and feedback; the critics ask whether investigations adequately capture consequences beyond the chosen performance score. [GS-W13; GS-W01; GS-W14]

文鹏, 任晓雅 and 陈诚's 2017 review separates target attributes, the setting process and target content. Difficulty and specificity concern attributes; participation and rewards concern how goals operate organizationally; result-oriented and process-oriented formulations concern content. The authors also distinguish violation of ethical norms from violation of an organization's internal rules. Conduct can serve an organization's immediate interests and still violate broader moral expectations. That distinction prevents organizational compliance from silently becoming the definition of ethical behavior. [GS-W15]

The review develops three proposed explanations. Goal shielding concentrates attention on an important target while other goals receive less attention. Organizational targets can thereby signal what is valued and where effort should go. Moral disengagement concerns justifying misconduct: treating the result as overriding other considerations can make deception seem acceptable, or an imposed target can encourage an actor to shift responsibility to the person who set it. The depletion account treats self-control as a limited resource. Demanding goals require effort and increase cognitive load; reduced self-control can then make conduct less consistent with internal standards. [GS-W15]

The review also examines studies involving unverifiable performance reports, nearly attained targets and moral justification. It records inconsistent findings and limited organizational-field evidence. The contrast between result-oriented and process-oriented content is presented as theoretically proposed, with direct tests still lacking in that comparison. The organizational debate extends from how an objective is written to what conduct is observed, what outcome is rewarded and what counts as an adequate explanation of the result. [GS-W15]

S99 Procrastination and beginning intended action

When delay becomes procrastination

Piers Steel's 2007 review defines procrastination through the relationship between an intention, a delay and its expected consequences. A person voluntarily puts off an intended action while expecting the postponement to leave them worse off. The definition makes the prior intention consequential. Evidence that an action started late leaves open whether the person changed their mind, lost an opportunity or chose to wait because the task had become more favorable later. Steel discusses task aversiveness and the delay before a reward as different influences on the appeal of acting now. The histories of intentional waiting in S61 and S62 concern decisions in which postponement can serve a purpose. [T01-W001]

Recognizing delay in ordinary behavior

Svartdal and colleagues asked how behavioral delay might be examined alongside a questionnaire. Among the routines they considered was bringing lunch from home versus buying it. Participants who generally brought lunch and those who generally bought it had mean Irrational Procrastination Scale scores of 2.71 and 2.99 respectively. The contrast associates a reported routine with a score. Classifying a particular postponement would require further evidence about what the person intended to prepare and what happened to that intention. [T01-W002]

Changing willingness to start

童廷婷, 白幼玲 and 冯廷勇 investigated positive reappraisal among participants selected for high procrastination. Their report describes two groups of 34 and a week of experience sampling. They assessed willingness to execute the task, how aversive it seemed and the usefulness of the result. At post-test, the positive-reappraisal group reported greater willingness to begin the task and less aversion to it than the comparison group. In the authors' mediation analysis, reduced aversion accounted for part of the relationship they studied, while increased perceived usefulness did not mediate it. [T01-W003]

Rozental and colleagues' review pooled 21 comparisons from 12 studies, involving 718 participants. Against inactive controls, the pooled post-treatment difference on procrastination measures was 0.34 standard deviations in favor of treatment, with a 95 percent confidence interval from 0.11 to 0.56. Effects varied considerably among comparisons. Most studies assessed the outcome at the end of treatment. The cue-linked plans discussed in S98 specify when to initiate an action, while the reappraisal study above asks how the task feels when a person considers doing it. [T01-W004]

S94 Working backward from outcomes

A proposition to be proved, a future to be examined and a performance to be demonstrated each give planning a possible starting point. The following accounts show how authors derive intermediate requirements, construct pathways and decide what evidence would establish an intended result.

S62 examines reverse business calculations and staged assumptions; S85 explains backward calculations in project scheduling.

Starting with the proposition to be established

Stephanie Dick's 2015 history follows mathematical heuristics into the construction of the Logic Theory Machine. The program began with a proposition to be proved and sought subproblems from which it could be deduced. It continued generating such subproblems until it could connect the chain to axioms; reversing a successful chain supplied the proof. A search could also fail to find the connection. Dick places Pólya's working-backward heuristic within this history alongside other aids to discovery. Implementing the procedure on JOHNNIAC required representations and operations adapted to the machine. Her account makes the intended result an active part of the search: it determines which intermediate propositions would be useful. [PM-W001]

Regressing requirements and protecting goals

Richard Waldinger's Achieving Several Goals Simultaneously, issued as a technical report in 1975, addresses plans in which satisfying one goal can interfere with another. His procedure first constructs a plan for one goal, then modifies it to achieve the second while preserving required relations. Regression passes a desired relation backward across an action and asks what must have been true before the action for that relation to hold afterward. For an assignment X receives t, the desired relation P(X) becomes P(t) before the assignment. The substitution supplies a requirement that can be examined earlier in the plan. [PM-W002]

The three-block example shows why this operation matters. The target tower has A on B and B on C. Starting with separate blocks, the planner first puts A on B. A one-block-at-a-time arm cannot then put B on C while maintaining that arrangement. Passing the requirement that B be on C backward across the first action permits the planner to put B on C before placing A on B. Protection points identify relations that must hold at particular positions. Skeleton models retain selected facts and use regression to infer others as the plan changes. The action descriptions and protected relations make the modification inspectable. [PM-W002]

Backcasting, participation and the choice of a future

Jaco Quist and Philip Vergragt's 2006 account traces energy backcasting as an alternative to extrapolative forecasting and follows its extension into participatory sustainability work. Their proposed framework includes problem orientation, future visions, backcasting, elaboration of follow-up agendas, and embedding and implementation. Participation and iteration can change across these stages. In their historical reconstruction, Robinson's earlier policy-analysis purpose and later stakeholder-oriented applications give backcasting different practical settings. [PM-W003]

Their SusHouse food-centre scenario illustrates how a vision enters further assessment. A neighbourhood centre for food preparation appeared attractive, but calculations using contemporary restaurant energy data suggested that energy demand might increase. The authors called for better data, and the workshop scenarios underwent environmental, economic and consumer assessment. The desirable image initiated

questions that subsequent investigation could alter. Their two food cases also had different follow-up histories: producing a vision and embedding subsequent activity were distinguishable achievements. [PM-W003]

Emilie Mutombo, Tom Bauler and Grégoire Wallenborn's 2007 comparative report separates constructing a final image from constructing an evolution toward it. In their analytical grid, putting the final image first gives a backcasting order; putting the evolution first gives a forecasting order. Objectives, measures and monitoring remain further activities, with actual exercises often combining or overlapping the simplified stages. [PM-W004]

The report also examines how a destination is chosen. A vague image can accommodate apparently compatible principles; specifying it more precisely exposes different priorities and values. In its World Water Vision account, polarized conceptions remained unresolved and produced two reports. The staff report nevertheless supported initiatives by actors who adopted its vision. Participation could thus produce material for action without establishing one shared destination. The authors' sequencing distinctions sit alongside this practical history of agreement and disagreement. [PM-W004]

Sebastian Bollien's 2021 institutional account begins with a concrete future image and asks which events must precede it. Its example is peace negotiation between hostile parties: the backward questions concern conditions and timing that would make talks possible. Examining those requirements can lead to revising the image or moving it in time. The destination participates in the inquiry and can change as its prerequisites are examined. [PM-W012]

David Bengston, Lynne Westphal and Michael Dockry's 2020 Backcasting Wheel places a preferred future at the centre of a diagram. Dimensions of success lead outward to signposts, opportunities and obstacles, and concrete management actions. Participants score importance and timing, and subsequent analysis organizes the results into action plans. The outward movement reaches earlier enabling actions, whereas a Futures Wheel explores later consequences of a change. [PM-W013]

Their Forest Service case involved two units preparing to share leadership and a headquarters. Mixed groups developed signposts, obstacles, opportunities and actions, scored them, and assigned leaders to urgent tasks. Participants later described the exercise as useful. Turnover interrupted continuity, leaving the eventual effect on the reorganization unresolved. The case records how an aspiration was translated into actions and assignments within a participatory exercise. [PM-W013]

Yusuke Kishita, Mattias Höjer and Jaco Quist's 2024 comparison distinguishes participatory, goal-oriented and scenario-design approaches to backcasting. Implementation and stakeholder engagement receive a different emphasis from constructing goal-fulfilling future images. Their comparison retains those differences while developing the authors' own design-framework questions about when, which, how and what. Backcasting continues to encompass different objects of design and different arrangements for involving participants. [PM-W014]

A backward tree and a forward storyline

Haruna Wada and colleagues' 2013 scenario-construction method starts with a target and builds a logic tree of events and actions that could lead to it. Key events divide the material into subscenarios. The writers then develop storylines forward from the present and turn them into scenario text. References and logical jumps are represented explicitly. [PM-W005]

Their worked target is an undesirable collapse of manufacturing in 2050. The tree separates inability to produce from inability to sell. When the resource branch is developed forward, substitution of aluminium for copper wiring introduces a possibility absent from the initial backward tree, one that could interrupt the collapse story. Backward tree construction and forward elaboration therefore perform different work

in the authors' procedure. Selecting key events and judging the tree's adequacy remain part of scenario construction; the paper discusses AND/OR operators as a subsequent improvement. [PM-W005]

Designing evidence of learning

Grant Wiggins and Jay McTighe's Understanding by Design (1998) orders curriculum decisions around desired learning, acceptable evidence and then instruction. Their hypothetical teacher Bob James wants fifth-grade students to use nutrition knowledge to plan balanced diets. He specifies a three-day camp menu and an analysis of a hypothetical family's diet as performances through which students can demonstrate that understanding. Quizzes supply evidence about prerequisite knowledge. [PM-W006]

These choices guide the teaching resources and activities: interpreting nutrition labels, scaling recipes, consulting a visiting nutritionist, and using a brochure, video and selected textbook passages. Assessment includes evidence gathered during learning through observations, dialogue and other tasks. The example changes the textbook's role from an obligatory sequence to one resource among several. The authors acknowledge Tyler's earlier objectives-based logic and supply filters, templates and design questions for putting such decisions into practice. [PM-W006]

Wiggins and McTighe's 2011 guide distinguishes acquiring knowledge, making meaning and transferring learning. Evidence is chosen in relation to the intended result. Using knowledge independently in a new situation calls for a different demonstration from recalling facts. Their nutrition materials sequence analysis of a hypothetical family's diet, feedback on misunderstandings and independent preparation of a camp menu. Instruction and practice prepare that performance. The GRASPS prompts make task design explicit through goal, role, audience, situation, product or performance, and standards. These are tools for designing a unit and inspecting its coherence. [PM-W007]

Wang Lingling's 2023 account applies backward unit design to fifth-grade texts about Venice, Dutch pastures and the pyramids. Intended learning is expressed through performances such as explaining a place and preparing a travel guide. In a proposed tour-guide competition, groups choose locations, develop guide materials and presentations, and are judged on features of content and delivery. Further assessment concerns participation and cooperation. Wang's exposition constructs the performance activity before separately discussing assessment evidence. The resulting account makes the product, audience and criteria visible within her formulation of unit teaching. [PM-W008]

Planning direction and predictions of completion

Jessica Wiese, Roger Buehler and Dale Griffin's 2016 experiments distinguish hours spent working from the calendar date on which a task is completed. Their four studies compare backward, forward and unspecified planning directions. In the fourth, adults nominated a multiday project with a deadline within two weeks, planned its steps and predicted when they would start and finish and how many work hours they would need. A follow-up asked what had happened. [PM-W009]

The initial analysed sample contained 187 participants; 125 reported completed projects and supplied the completion-bias analysis. Backward planners showed less average optimistic bias in predicted completion dates. The intervention shifted predictions without a corresponding reduction in actual completion time. Predicted work hours and starting-time bias had different results. The authors discuss attention to obstacles, additional steps and planning insights as possible processes, with variable support across their studies. [PM-W009]

Jooyoung Park, Fang-Chi Lu and William Hedgcock's 2017 paper reports comparisons of chronological and reverse-chronological planning for personal goals, treating motivation, expectancy, perceived pressure and performance as distinct outcomes. Their 2018 corrigendum revises effect sizes, intervals

and portions of Study 3's statistical reporting while stating that the interpretations stand; some corrected comparisons are marginal or nonsignificant. Their reported goal-pursuit findings concern different outcomes from Wiese and colleagues' completion-time predictions. [PM-W010] [PM-W029]

Transforming computational planning problems

Allen Ren, Brian Ichter and Anirudha Majumdar's 2024 computational study examines graph paths, array transformations and block manipulation. A narrow passage near a goal can make searching from that end attractive. Simply asking their model to plan backward, however, performs worse. They instead flip the problem, plan forward in the transformed representation and convert the solution back. Directed edges must be reversed, and actions must have usable inverses for the relevant transformation. [PM-W011]

The experiments generate candidate plans, use model self-verification and apply an external task verifier. Combining ordinary and flipped planning improves success in their tested settings. Transforming the problem and checking the returned plan are substantial parts of the procedure. [PM-W011]

S95 Challenging plans before action

A plan can depend on assumptions its proponents have not examined. Organized opposition, an assumed failure or an opponent's next action can make those assumptions available for discussion. The historical accounts below describe what participants are asked to do and what the resulting arguments, reasons and decisions reveal.

S44 examines approaches to organizing inquiry; S86 discusses independent forecast review and project governance.

Exposing assumptions behind rival plans

David McDonald, Gabriele Bammer and Peter Deane's Research Integration Using Dialogue Methods (2009) reconstructs strategic assumption surfacing and testing through group formation, assumption surfacing and rating, debate, and attempted accommodation. Groups develop their positions and identify assumptions that trouble them in rival positions. The procedure makes the premises sustaining competing courses of action available for examination. [PM-W015]

Their Winterton example concerns an agency divided between finding business opportunities and recruiting cooperatives to pursue them, or helping people form cooperatives and then find opportunities. The groups identified different stakeholders and assumptions. Debate did not reconcile their positions, but participants agreed to gather information and conduct limited experiments. The reported action preserved the unresolved issue while making further investigation possible. [PM-W015]

The Pfeiffer Library's 1998 robotics exercise provides a teachable implementation adapted from Mason and Mitroff. Participants receive the same automation proposal and form pro and con subgroups. They identify stakeholders, surface assumptions and rate their importance and certainty before presenting positions and debating damaging disagreements. Mixed groups subsequently propose integrative plans, retaining unresolved issues where agreement remains impossible. [PM-W016]

The stipulated Elmire case combines an attractive financial calculation for robot conversion with possible job losses, strikes, employee commitments and community reactions. It supplies participants with a situation in which assumptions about other people's responses matter to the proposed investment. The exercise shows how those assumptions enter discussion and revision within the training design. [PM-W016]

Criticism, counterproposals and consensus

George Wright and Gene Rowe's 2011 review reconstructs the procedures compared by Schweiger and colleagues. In dialectical inquiry, subgroups develop opposing alternatives and debate their assumptions and recommendations. In devil's advocacy, one subgroup offers a proposal and another critically probes its elements and recommendations. The returned material consequently differs: a rival proposal supplies another course of action to debate, while an organized critique identifies weaknesses in the proposal under examination. They contrast these conflict procedures with discussion directed toward consensus, without an assigned adversarial role. Their account also cautions that disagreement can turn toward personalities rather than factual issues. [PM-W028]

Charles Schwenk's 1990 meta-analysis reports an overall advantage for devil's advocacy over an expert-only approach. It reports no demonstrated superiority of dialectical inquiry over expert advice on relatively ill-structured tasks and no established winner between the two conflict methods. These reported conclusions address comparisons earlier reviews had interpreted differently. The task

qualification and comparator are part of the result: an advantage over expert-only advice answers a different question from an advantage over another organized-conflict procedure. [PM-W017]

Jerry Fjermestad's 2019 review introduction frames a disagreement between organizational case and field studies and laboratory experiments. Active and passive participation, leadership and communication technology distinguish the settings. Groups whose participants merely receive an argument have a different experience from groups that construct and defend it. His framing also questions whether results from face-to-face work will carry into distributed computer-mediated teams. These distinctions direct attention to the organization of the comparison as well as the technique's name. [PM-W018]

How participants interpret opposition

Charlan Nemeth, Keith Brown and John Rogers' 2001 experiment used a vacation-scheduling dispute in computer-mediated groups. The opposing participant delivered the same scripted arguments, while the conditions changed whether others understood her as expressing a minority belief or performing a devil's-advocate role. After the discussion, participants generated possible solutions. [PM-W019]

The final usable pool was 94 female undergraduates, with smaller samples for individual measures. Perceived authentic dissent produced more alternatives and a stronger combined quantity/quality score than the consistent-role and control conditions. Mean quality alone did not differ significantly between those conditions. The manipulation thus concerns how participants interpreted the same opposition, while the separate measures distinguish producing more alternatives from improving their average quality. [PM-W019]

Explaining an assumed failure

Deborah Mitchell, J. Edward Russo and Nancy Pennington's 1989 experiments varied both the temporal setting of an event and the certainty of its outcome. Participants explained professional and personal successes or failures. The first experiment enrolled 114 MBA students; the second combined 76 MBA students with 32 paid students, with time recorded for the latter. [PM-W020]

In the dinner-party scenario, the same background was paired with success or failure presented as certain or possible and situated in the past or future. Certainty affected the number and episodic character of reasons more consistently than temporal framing. [PM-W020]

Gary Klein's 2007 project premortem briefs a team on its plan and stipulates that the project has failed. Members independently write reasons for the failure, then contribute different reasons around the group. The manager considers changes in response. The exercise gives reservations a concrete occasion for expression before implementation. [PM-W021]

Klein's software example concerns algorithms that would run too slowly on field laptops. A previously unmentioned shortcut was substituted after the concern surfaced. In his account, the change emerged from examining an assumed failure and identifying a specific technical reservation. [PM-W021]

Beth Veinott, Gary Klein and Sterling Wiggins' 2010 study assigned 178 students to evaluate an H1N1 university response plan using a baseline, critique, pros/cons, cons-only or premortem condition. Participants rated confidence before criticism, after pooling concerns and after proposing improvements. The premortem condition showed a larger reduction in confidence followed by a larger recovery after improvements were considered. [PM-W022]

The paper reports confidence and subjective understanding; its overall understanding test was nonsignificant. Participants reviewed a supplied plan and did not implement it. [PM-W022]

Whose knowledge enters the review

Guillermo Wippold and colleagues' 2025 implementation premortem brings potential recipients into examining local conditions. In the Fade to Fitness example, recipients were recruited beyond the barbershop customers involved in shaping the original programme. After an overview and a stipulated failure, participants generated reasons individually, discussed responses and prioritized concerns. Transcripts were analysed through an implementation framework. [PM-W023]

The authors illustrate how limited shop space could lead to planning delivery during non-peak hours. The article explains that procedure and its contextual example, directing readers to a separate report for empirical findings. [PM-W023]

Giovanni Cunico, Nici Zimmermann and Nuno Videira's study, first published online in 2023, addresses homogeneous groups whose disagreements may circulate only knowledge already present. Their New Devil's Advocate is a prepared facilitator who temporarily represents absent stakeholders through documented information. Other facilitators maintain the process. A role-plot connects a stakeholder, evidence, a possible intervention and relevant model variables. [PM-W027]

In their UK housing-retrofit application, the first workshop left supply-chain and homeowner perspectives poorly represented. The team gathered information through three expert interviews, literature and earlier research, and prepared 45 inputs for two advocates. The role-plot paired each issue with a stakeholder, a way to raise it and possible model variables. One prepared intervention asked how household reluctance to borrow could affect renovation policy, connecting that concern to mortgage refinancing and the willingness to take loans. [PM-W027]

The resulting map incorporated relationships concerning installer skills, retrofit quality and performance, as well as household behaviour and perceived monthly savings. The authors traced 20 of the 45 prepared inputs into the map by comparing the role-plot, workshop transcripts and model changes. Their appendix shows how an input could be reformulated during modelling: household compliance with a retrofit measure became a link from behavioural compliance to measure performance. [PM-W027]

The two advocates participated very differently. The performer experienced in group model building supplied nearly all their interventions; the other, despite substantial topic knowledge, reported difficulty entering the discussion while trying to formulate contributions in the modelling language. The experienced performer occasionally returned to facilitation or joined a decision about model content. In one such instance, the facilitator asked the regular participants again whether they agreed to include the proposed variable. Transcripts and feedback thus reveal both how missing perspectives entered the model and how participants retained authority over its construction. [PM-W027]

Critical review and adversary perspectives

The UK Ministry of Defence's Red Teaming Handbook (2021) distinguishes critical review by a red team from adoption of an adversary or stakeholder perspective by a red cell. Its assumptions review identifies explicit and implicit assumptions in an estimate or plan, examines their support, asks when they might fail and considers what failure would change. Judgments about support and caveats are recorded for subsequent use. [PM-W024]

Argument mapping starts at a recommendation and traces supporting deductions, assumptions and assertions backward. Missing links and unsupported steps are marked. Reviewers then examine the chain forward to determine the consequences of flaws and the information needed to repair them. Red-cell work has a different information arrangement: separation from friendly planning and withholding privileged knowledge of friendly intentions help participants reason from the represented actor's position. The handbook makes actor expertise part of that prescription. [PM-W024]

Héloïse Fayet and Amélie Férey's 2023 institutional account distinguishes red cells modelling an opponent's reactions, red teams questioning assumptions and defence foresight constructing fictional futures. Short brain games bring officials and external experts into roles, alongside more developed games and scenario efforts. The activities give participants different jobs and work across different time horizons. [PM-W025]

The authors also examine promotion incentives, hierarchy and personnel rotation. Critical perspectives have to be voiced, retained and used within an institution whose career and staffing practices influence those possibilities. Their account places the design of a session alongside the organization in which its results must circulate. [PM-W025]

Examining a course of action through a game

Masahiro Kurita's 2025 review compares traditional analytical wargaming with experiments that use repeated games as synthetic data. Human decisions, adjudication and consequences make the course of play responsive to participants. His review examines the tension between simplifying scenarios to isolate a variable and preserving the rich setting and interaction considered necessary for realistic choices. [PM-W026]

His account of the International Crisis Wargame describes teams of four to six acting as a fictional nuclear-armed state's national security council. They choose responses to two separate crises. Repeated sessions vary whether their own nuclear command arrangements are vulnerable to cyberattack and whether they can attack the opponent's arrangements. The principal material for analysis is the teams' response plans. Play ends with the response decision, without a further exchange of actions between competing sides. [PM-W026]

SIGNAL uses a different design. Players representing three fictional states pursue economic, military and diplomatic objectives over five rounds. They negotiate and signal intentions, then take actions whose consequences are governed by rules. A player can depart from an earlier signal. This produces a sequence of interacting choices, rather than only a team's response to a stipulated crisis. Kurita uses the two examples to show that experimental wargaming includes substantially different forms of play and data. [PM-W026]

The disagreement about immersion concerns those concrete choices. Fictional settings can help separate the variable under study from participants' knowledge of real countries; removing subsequent interaction also simplifies comparison. Critics question whether such designs elicit decisions resembling those of actual policymakers. Kurita reports little empirical evidence establishing how much immersion is needed or which design features produce it. He treats the appropriate degree of simplification as an unresolved question for traditional analytical games as well as experimental ones. [PM-W026]

S53 Absorbing new resources and capabilities

Acquiring equipment, finance or outside knowledge can leave substantial work between the acquisition and a useful result. People may have to learn unfamiliar operations, establish relationships, develop supporting services or change how decisions are made. The historical accounts below examine that work and the order in which it develops. Their interventions include organizing training, combining investments, distributing knowledge, adapting institutions and managing the pace of expenditure.

Baldwin's account in S07 explains complementary investments and transition costs; S14 develops changes in work design and information processing. Miller's overload discussion in S21 and the moving-constraint account in S44 address limits on increasing input. The sources here examine how the receiving capabilities themselves develop, including situations in which beginning an activity helps create the conditions for its further expansion.

Rosenstein-Rodan: training and investment beyond a single enterprise

P. N. Rosenstein-Rodan's 1943 proposal for industrialization in Eastern and South-Eastern Europe treats training as an investment problem. A firm paying to prepare workers may lose them to another employer. The benefit of their acquired skills can exceed the return available to the firm that bears the training cost. He therefore considers training from the standpoint of the group of industries being created. Transport and housing enter the provision for bringing people into industrial work, alongside instruction itself. His proposed large investment unit would include this preparation among its capital costs. [AB-W003, pp. 204–205]

Complementarity also concerns the market for what new enterprises produce. In his hypothetical example, workers employed by one large shoe factory will spend their wages on many products, leaving the factory unable to rely on their demand alone. A group of industries producing the goods those workers require can generate demand for one another's output. The argument for coordinated investment thus includes relations among production, income and consumption. It asks why a project that looks unattractive by itself might contribute to a more valuable combined undertaking. [AB-W003, pp. 205–207]

Rosenstein-Rodan also questions a different proposal: build transport and power facilities and expect other industries to follow automatically. The region's existing transport provision and its intended productive activities affect what infrastructure is needed, where and in what quantity. His coordinated industrial programme is a proposed response to a particular reconstruction problem; the paper explicitly leaves detailed institutional implementation for later work. Training and complementary demand receive explanations, while the task of establishing the proposed investment organization remains to be worked out. [AB-W003, pp. 208–209]

Hirschman: developing the ability to invest

In *The Strategy of Economic Development* (1958), Albert Hirschman distinguishes available savings and investment opportunities from the ability to connect them. Recognizing a project, obtaining cooperation and carrying it through require abilities that cannot be assumed to appear whenever finance becomes available. He argues that these abilities develop partly through practice in existing productive activities. An expanding sector can train people and create experience useful for further expansion. This makes the ability to undertake projects part of the development process itself. [AB-W004, pp. 35–38]

His schematic account allows the limiting factor to change. Initially, available or mobilizable savings may exceed what the existing productive sector can turn into investments. As that sector expands, its investing

abilities can increase until finance becomes the tighter restriction. Hirschman cautions that this transition need not occur at one neat date: skills and savings move imperfectly between sectors, so different activities can encounter the change at different times. He prefers investigating this developing ability to treating every difficulty as an obstacle to an assumed normal state in which all available finance would be absorbed. [AB-W004, pp. 36–39]

This interpretation also changes the role assigned to external capital. In his first situation, an outside investment can matter through skills, organization and ways of conducting an enterprise, as well as the money supplied. In a later situation, domestic managerial and entrepreneurial abilities may already outrun available savings, making finance itself more important. These are distinctions within his proposed account of development, with different implications for what a capital transfer must actually provide. [AB-W004, pp. 38–39]

Hirschman: choosing a sequence that elicits a response

Hirschman's investment discussion compares expanding infrastructure before productive activity with allowing productive activity to advance far enough to create pressure for additional infrastructure. More transport or power capacity can reduce costs and attract investment. Conversely, existing producers facing costly shortages can press authorities to expand those services. The first route invites an activity through improved conditions; the second attempts to elicit action through a difficulty people already experience. His comparison depends on entrepreneurial response and on the responsiveness of the authorities responsible for infrastructure. [AB-W004, pp. 86–93, figs. 3–4]

His diagram separates the resources needed for a static least-cost combination from the decisions through which a country might reach a larger productive scale. He assumes, for that comparison, that infrastructure and directly productive activity cannot both be expanded at once. Infrastructure built ahead of demand supplies spare capacity whose value depends on subsequent use. Production expanded first can make the need for an additional road or power supply concrete enough to generate a decision that previously attracted little support. Those assumptions and induced responses, rather than a preference for imbalance as such, do the explanatory work. [AB-W004, pp. 86–90, fig. 3]

There is a technical minimum. Access and some supporting services must exist before an activity can operate at all. Prolonged neglect can also make expansion self-defeating: another factory may claim scarce services at the expense of existing firms, leaving little gain in aggregate productivity. The absence of an absolute barrier to building the new factory can conceal that displacement. Hirschman's discussion therefore distinguishes a shortage capable of provoking a useful response from one that continues to damage the productive system. [AB-W004, pp. 94–97]

He also allows experience to change the comparison. Investors who expect infrastructure providers to respond, and providers who anticipate growing demand, may act before an actual shortage or excess appears. Under those conditions, his alternative inducement curves permit coordinated expansion to sustain further investment. The same account that explains the possible usefulness of an uneven sequence also explains how learned expectations can reduce reliance on the difficulties that initially elicited action. [AB-W004, pp. 91–93, figs. 4a–4b]

Dierickx and Cool: expenditure flows and accumulated capabilities

Ingemar Dierickx and Karel Cool's 1989 article distinguishes resources that can be traded from assets that develop through a history of activity. Trust, firm-specific knowledge and a reputation for reliable quality are among their examples. Current expenditure can purchase inputs to that history without immediately producing its accumulated result. Their stock-and-flow description makes the distinction

explicit: additions build an asset stock, while erosion diminishes it. A current research budget and accumulated technical know-how therefore describe different quantities. [AB-W002, pp. 1505–1506]

They call one possible property of accumulation time compression diseconomies. Increasing expenditure over a shorter period may not reproduce what a longer process would have developed. Their examples include crash research programmes and attempts to compress education. They also identify interconnectedness: building one asset can depend on another stock already in place. A service network, for example, can provide customer knowledge that contributes to technological development. These are proposed properties to investigate in particular accumulation processes, and the authors explicitly say that a process need not possess all of them. [AB-W002, pp. 1507–1509]

An accumulated advantage also needs maintenance and can be displaced by a different approach. Their Canon–Xerox illustration describes design capability substituting for a service network on which an earlier competitive position depended. The article's question is the sustainability of competitive advantage. Its conceptual examples explain why buying more inputs, accelerating expenditure, preserving an existing asset and developing a substitute are different strategic actions. [AB-W002, pp. 1508–1510]

Cohen and Levinthal: prior knowledge and the organization of reception

Wesley Cohen and Daniel Levinthal's 1990 account names a firm's ability to recognize useful external knowledge, assimilate it and apply it commercially as absorptive capacity. Prior related knowledge helps people understand and connect new information. Their argument joins existing knowledge to effort and practice: information can be available without being intelligible or usable to its recipients. Research, technical training and experience in production can contribute to this capacity through different activities. [AB-W001, pp. 128–131]

Organizational reception involves more than employing several knowledgeable individuals. Someone encountering an outside idea must connect it with people who can evaluate or use it internally. A specialist at the boundary may translate unfamiliar material, but recipients still need enough related understanding to act on what is communicated. Where technical change is rapid and uncertain, the authors argue for a broader distribution of people able to notice relevant developments. Shared understanding supports communication, while excessive uniformity can restrict the different associations through which something new is recognized. [AB-W001, pp. 131–135]

They also explain why purchasing expertise may leave integration work unresolved. An external consultant or newly hired specialist can bring knowledge without possessing the firm's understanding of its own needs, routines and complementary capabilities. Maintaining some internal expertise helps participants interpret outside work and connect it with those particulars. Similarly, participation in a joint research venture leaves a firm with internal work if it is to use the venture's output. The authors distinguish knowledge related to ordinary activity, for which capacity may develop as a byproduct, from unfamiliar domains requiring deliberate preparation. [AB-W001, pp. 135, 148–150]

Their empirical discussion examines an indirect implication of this account: how learning conditions affect firms' research expenditure. It combines reported characteristics of technical knowledge with manufacturing business-unit data, rather than directly observing a completed absorption process. The R&D evidence reported in the article derives from their earlier work. Greater relevance of basic rather than applied science was associated with more research spending, as predicted when less targeted knowledge requires more internal effort to absorb. These spending patterns do not by themselves establish the causal effectiveness of the organizational practices described. Their closing discussion retains the difficulty of valuing indirect benefits when deciding how much to invest. [AB-W001, pp. 139–150]

Noblet and Simon: making organizational practices observable

Jean-Pierre Noblet and Eric Simon's 2008 study attempts to make absorptive capacity examinable through what firms do. They interview leaders of ten innovative companies, largely small and medium enterprises, using two appointments separated by two to four weeks and about thirty hours of interviews in total. Their literature-derived grid distinguishes acquiring, assimilating, transforming and exploiting knowledge. Indicators include alliances, shared understanding, research activity, changed routines and management support. They analyze meaningful passages in the interviews, with interpretation checked by another researcher. [AB-W007, pp. 9–12]

The reported practices give those categories substance. One leader describes joint research followed by a report and discussion that can change working practices. Others identify employees who scan external information or help colleagues find an outside application when their own work is blocked. The body of the paper emphasizes managerial openness, contact with final customers and trust. These concern willingness to reconsider an activity, opportunities to learn what a product must do, and relationships through which people exchange knowledge. [AB-W007, pp. 12–16]

Measurement itself becomes a practical issue. Interviewed leaders challenge patent counts as an adequate indicator for their firms because confidentiality, retained know-how and filing costs affect whether they patent. Noblet and Simon offer an exploratory operationalization based on leaders' accounts and call for further development of the instrument. Their practices and proposed indicators help identify what an investigation could observe, while the small selected sample and self-reports limit conclusions about their effects on growth or innovation. [AB-W007, pp. 13–16, appendices 1–2]

Picoli and Takahashi: similar equipment, different incorporation into work

Florindo Rhaoni Picoli and Adriana Takahashi's 2016 study compares two public schools in Curitiba that received computer laboratories through Paraná Digital. Both received twenty computers in 2007 and had access to training and online manuals. The researchers collected interviews, documents and nonparticipant observations in 2013, reconstructing implementation since the programme began. They deliberately selected schools with contrasting levels of use and broadly comparable staffing, enrollment and location. [AB-W008, pp. 8–12]

At Professora Luiza Ross, earlier experience with a school computer laboratory helped staff approach the new equipment, although its Linux system differed from the software many already knew. Formal meetings and training were joined by informal exchanges during breaks. Staff discussed successful and unsuccessful experiences, tried the technology in practical situations and adapted how they worked. The authors associate these interactions with the transformation and use of knowledge obtained through the programme. [AB-W008, pp. 9–11]

At Santo Agostinho, staff also had some personal computer experience and access to the programme's support. The researchers nevertheless found little exchange about classroom use or change in teaching routines. The reported use was student internet research outside classes, without pedagogical guidance. The source's comparative coding locates the difference beyond initial introduction and understanding, in the sharing, experimentation and incorporation of knowledge into work. The knowledge-sharing account in S42 gives a neighboring treatment of support after transfer. [AB-W008, pp. 11–16; W088]

Picoli and Takahashi distinguish their observed knowledge-flow processes from a stronger claim about organization-wide learning. They say the latter would require deeper investigation of changes in culture, values and practice. Prior laboratory experience, space and other institutional differences also remain part of the comparison. The evidence develops an account of how people came to use supplied

technology; computer-use totals and staff reports do not themselves measure improvements in students' learning. [AB-W008, pp. 13–17]

Zhang and colleagues: imported technology and delayed innovation effort

Zhang Jie, Chen Zhiyuan, Wu Shufeng and Sun Wenhao's 2020 article investigates the relation between foreign technology introduction and firms' own innovation effort. Its firm-level analysis uses the 2008–2014 National Enterprise Innovation Survey database. The principal outcomes are spending on science and technology activity per science and technology worker, including a measure net of government funding. Technology-introduction expenditure is excluded from those outcomes. The study therefore examines changes in further innovation inputs rather than directly measuring inventions, productivity or operational mastery. [AB-W009, pp. 93–95]

The authors address reverse causation using an instrument based on government catalogues encouraging technology and product imports, matched to industries and lagged to allow for implementation. Their instrumental-variable estimates do not show a statistically significant contemporaneous increase in the spending outcomes. More consistent positive estimates appear for outcomes two years ahead. They interpret the delay through the work of learning, digestion and assimilation following a technology purchase. The estimates rely on the policy instrument's assumptions, while the learning process remains the authors' explanation of the observed delay. [AB-W009, pp. 94–98, tables 2–3]

Their theoretical model and subsequent comparisons also distinguish firms by position relative to industry leaders. The more consistent positive spending responses occur in the middle portions of their innovation-gap distribution. The reported tables are less uniform than a simple middle-only rule: one measure in the most distant group has weakly significant positive estimates. The model and the data comparisons make the initial capabilities and competitive setting relevant to the question, while the estimated delay and group boundaries remain properties of this investigation. [AB-W009, pp. 98–103, table 4]

Polterovich: institutions can change with acquired capabilities

Victor Polterovich's 2009 account of building a national innovation system argues for intermediate institutions that help relax constraints on further development. His technological borrowing includes choosing a technology, obtaining rights, adapting it to local conditions, constructing prototypes and testing. The receiving activity can itself modify or improve what was acquired. Domestic scientific work remains important for adaptation, so borrowing and new knowledge creation are not mutually exclusive phases in his argument. [AB-W010, “Problem statement: seeking a strategy of intermediate institutions”; “Absorptive and innovative capabilities of a country”; “Transition to innovative development”]

A proposed institutional sequence changes as firms become able to do more. Polterovich reports the establishment of Korea's KIET in 1976 to support learning in semiconductor technologies and coordinate research. In the historical account he cites, the growth of corporate research led to a reorientation of that institution eight years later toward basic research supporting innovation. He uses the example to explain why an organization built to compensate for a missing capability can require a different role once that capability develops elsewhere. This is his report from the cited development literature. [AB-W010, “Main agents of technology imitation: the state and large firms”]

His account also examines interdependent investment. Refineries can hesitate over costly modernization while domestic equipment producers hesitate over supplying machinery without assured demand; downstream production may require changes as well. He proposes repeated interaction among public authorities, business, workers and consumers to coordinate expectations and investments. Indicative planning is an intermediate institution in this argument, capable of changing or receding as other

relations become adequate. His prescriptions concern catching-up development and its institutional conditions, and he expressly rejects a universal recipe for dividing resources between basic research and borrowing-based modernization. [AB-W010, “Escaping the trap: increasing absorptive capability through interactive planning”; “Transition to innovative development”; “Situation in Russia: approaches to modernization”]

Collier and colleagues: investing resource revenues requires an investment process

Paul Collier, Frederick van der Ploeg, Michael Spence and Anthony Venables examine the use of resource revenues in capital-scarce economies. Their 2009 account distinguishes a high return on small additions to existing capital from the returns obtainable when investment expands rapidly. A spending increase can run into construction, skills and other domestic supply constraints. Prices then rise and other activities can be displaced. Some equipment can be imported, but productive, human and managerial capabilities are not all obtainable through an international purchase. [AB-W005, section 4.1]

The response they discuss includes changing the supply conditions. Investment in infrastructure and skills can support private production; administrative barriers, equipment-import delays and earlier underinvestment can otherwise restrict the response to new demand. They also distinguish technical and informational limitations from incentives that favor diversion or poorly chosen projects. A larger budget does not settle which of these difficulties a spending system faces. [AB-W005, sections 4.1, 4.5]

Their account separates ownership of the funds, control of aggregate spending and choice of individual projects. Transfers to citizens, public spending, domestic lending and foreign-asset accumulation allocate those functions differently. Passing project decisions to private actors can use their information and incentives, but the argument still requires channels that connect savers with productive investments. Their discussion of public expenditure separately examines honesty and efficiency, scrutiny before and after spending, and enforcement through authority, citizens, peers and professional conduct. [AB-W005, sections 4.2–4.5, tables 1–2]

Revenue volatility adds a different timing problem. A buffer of liquid foreign assets can protect spending from abrupt revenue changes, yet a fund large enough to stabilize everything can postpone valuable domestic consumption and investment. The authors consider allowing investment flows to vary while the accumulated capital stock changes less sharply. They also discuss designing construction-sector employment and financial commitments to withstand contractions. The capacity to manage an investment programme includes its response to lower revenues as well as to an initial expansion. [AB-W005, sections 5.1–5.3]

Their conclusion about a liquidity fund is consequently conditional on its role in the investment process. It can buy time for adjustment. It can also become counterproductive if saving overseas substitutes for developing the ability to invest effectively at home. Their proposed task is to restructure that process for a higher and more variable level of investment, with buffering as one instrument in the adjustment. [AB-W005, section 5.3]

S104 Consultancy and the client's capacity to act

Inquiring into an engagement together

Lambrechts and colleagues describe Edgar Schein's process consultation as joint inquiry. A consultant and client consider an interaction while the participants can still examine how it unfolds. That work differs in purpose from a consultant's report produced after collecting facts from others. The authors develop Schein's account through their emphasis on relationships and on choices participants can author together. Their reconstruction gives process consultation a place among different ways of organizing a commission. [T06-W001]

Loizos Heracleous distinguishes interventions that ask a client to explain an episode from those that supply a concept or suggest an action. In his account of working with NASA, he describes a two-hour workshop on how to assess the value of the agency's relationships with other organizations. The senior sponsor introduced the workshop and left. Heracleous supplied a framework for discussing the relationships, and participants used it to consider what criteria their agency could adopt. They did not finish choosing the criteria within the time allotted. [T06-W002]

The sponsor later expressed dissatisfaction with how little had been settled. The division leader and other participants told Heracleous that the framework had helped them think through the relationships and that they intended to use it in subsequent internal discussions. Heracleous decided against a further intervention. His account records the sponsor's judgment and the participants' intended next step. [T06-W002]

Work on both sides of the commission

Isabel Boni-Le Goff examines management consultants in France through the demands of client service and the career pressures inside consulting firms. Consultants move across assignments and present work to clients while doing other work for colleagues and supervisors. The expectation of advancement or departure creates a second audience for their efforts. The account distinguishes what a client receives from the professional judgments governing a consultant's career. [T06-W003]

Francesco Ciampi asks what knowledge is created during consulting and which organization can use it. His inquiry includes interviews with 15 medium and large Italian firms in 2015. Client learning and a consultancy's ability to reuse what it learned are separate outcomes in this account. S53 addresses a related question about a recipient's capacity to absorb outside knowledge. [T06-W004]

Following an intervention into operations

In Bloom and colleagues' field experiment in Indian textile firms, treatment and comparison plants both received a month of diagnostic consulting. The treatment plants received a further four months of intensive help putting recommended practices into use. One practice was to classify and record weaving defects daily. The authors illustrate how the resulting quality index could flag a faulty loom for discussion at the next day's quality meeting, where it had previously been possible for such a problem to persist for weeks. [T06-W005]

After the implementation phase, the authors estimated that the treatment reduced the weighted quality-defect index by about 43 percent relative to the comparison, based on the intervention coefficient in Table II. They also reported lower yarn inventories and higher weaving output. The defect estimate is for the intervention package. Daily logging and review are among the practices the authors use to explain how it could improve quality. [T06-W005]

The authors also tracked practices at five other plants owned by treatment firms. By August 2010, adoption at those plants had risen by 17.5 percentage points. The authors report that firm directors copied practices from the plants that received the intervention. The researchers returned in 2011 to collect longer-run performance and management data. The trial's comparison establishes an operational result, and the account of copying shows a specific use of the practices elsewhere within the firms. [T06-W005]

S58 Transfer and persistence of organizational capability

A practice working in one place may be difficult to establish elsewhere. These accounts examine the transfer process, the work of reproducing an operating organization and the dependence of an individual's performance on the setting. They ask what the receiving unit can do and what support makes that possible. S53 develops investment in receiving capacity; the material here follows particular attempts to transfer or reproduce capability.

Szulanski: transferring a practice through actual use

Gabriel Szulanski's 1996 study treated the transfer of a best practice as the recreation of coordinating relationships among resources. A practice could already work in one unit while remaining difficult to establish in another unit of the same company. He divided this process into initiation, implementation, ramp-up, and integration. The decision to transfer was followed by exchanges of resources and preparation, then by actual use and attempts to reach satisfactory performance, and finally by efforts to sustain routine use. A signed decision or a completed installation therefore did not exhaust the transfer process. [CG-W017, pp. 28–30]

Szulanski distinguished a transfer's cost from the difficulty that people experienced while carrying it out. Costs could be hard to interpret without knowing what the same transfer would have cost in the absence of problems. Moreover, two equally expensive problems might place very different demands on an organization. One could be handled through established procedures; another could require senior managers and improvised coordination. His alternative was to examine the transfer's "eventfulness": the nonroutine problems that called for special attention. A company with established ways of handling an otherwise demanding transfer might experience it as relatively uneventful. The measure consequently concerned difficulty in relation to the organization undertaking the transfer. [CG-W017, pp. 29–30]

The four stages made that difficulty more specific. Before the decision to transfer, participants had to identify a need, recognize a promising practice, and investigate its suitability. During implementation they had to establish communication, prepare resources, and make adaptations. During ramp-up, actual use exposed problems that preparations had missed. The questionnaire asked, for example, whether trained people left, replacements were poorly qualified, changes introduced further problems, or support teams disbanded because other work intervened. Integration concerned maintaining the practice after satisfactory performance had first been achieved. Temporary workarounds could become habitual, unresolved problems could be displaced by daily pressures, or the recipient could reconsider adoption. These questionnaire items made visible several ways in which a transfer could remain incomplete after an apparently successful start. [CG-W017, pp. 28–30, 41–42]

The survey collected 271 questionnaires concerning 122 transfers of 38 practices in eight companies. Responses came from source units, recipient units, and third parties. The practices required coordinated work by multiple people; they included software-development procedures, drawing standards, appraisal, and activity-based costing. Szulanski measured difficulty partly through the problems participants encountered at each stage. His measures of the recipient's ability to absorb knowledge included existing technical and managerial competence, clear responsibilities, and knowing whom to approach for help. The relationship between units was measured through the difficulty of communication and each side's readiness to collaborate. These were organizational conditions of learning, rather than a count of documents delivered. [CG-W017, pp. 32–34, 41–43]

The strongest barriers in the survey analysis were limited recipient absorptive capacity, uncertainty about why the practice worked, and a laborious relationship between source and recipient. Motivational barriers were less prominent in this sample. The first analysis used 87 complete observations; excluding

an outcome measure increased the usable set to 142 and preserved the ordering of the three leading barriers. These were associations among survey measures, with transfers reported four to eight months after first use. Aborted transfers were difficult to include, and the design could not establish that changing a particular barrier would cause a corresponding improvement. Szulanski accordingly presented investment in learning capacity, relationships, and understanding practices as an alternative to relying only on incentives. [CG-W017, pp. 35–38]

The study's unexpected result for retentive capacity also concerned timing. Its coefficient had the opposite sign from the one anticipated. Szulanski suggested that, only four to eight months after first use, some established routines might still preserve the old practice rather than the new one. Replacing them could itself be difficult. He offered this as a possible interpretation of a survey result, rather than evidence that retaining knowledge generally harms transfer. In interpreting the leading barriers, he instead stressed the recipient's prior ability to understand and use the practice, uncertainty about its workings, and access to help from the source. [CG-W017, pp. 36–38]

Winter and Szulanski: reproduce the capability to operate

Winter and Szulanski called a transfer broad in scope when it created or substantially changed the recipient organization's context. Recreating an outlet capable of producing a service locally required more than relabeling goods received from a central supplier. It involved a collection of routines, people, equipment, supplies, and procedures. The central organization therefore needed knowledge of the features customers valued, the processes producing those features, the resources needed by those processes, and the environments where the combination could work. Their account distinguished copying the visible result from developing the capability to establish and maintain its production elsewhere. [CG-W018, pp. 732–733]

In *Replication as Strategy* (2001), Sidney Winter and Gabriel Szulanski distinguished a working example, or template, from the features that were both worth reproducing and reproducible elsewhere. Their term for knowledge of those features, how to create them, and suitable environments was the "Arrow core." A successful outlet might also contain irrelevant details, a manager's unrepeatable talents, or advantages peculiar to its location. Its success alone did not identify which elements a new outlet needed. The authors described learning through successive attempts, followed by a decision to stabilize a business model for wider replication. [CG-W018, pp. 731–736]

The template supplied something that a description alone could not: a functioning example to consult when a difficulty arose. Yet it could embody many things that a replicator should not reproduce. Winter and Szulanski offered a thought experiment about a successful local business whose profits might depend on an unusually capable manager or on local access to inexpensive skilled workers. They also used the example of a diner whose apple pie might travel well while its proprietor's decorating tastes would not. Their question was which parts of a successful combination could be reproduced, at what cost, and in which other environments. This was the purpose of distinguishing the Arrow core from the actual outlet that served as a template. [CG-W018, pp. 733–735]

Their Banc One example concerned acquired banks being converted to common computer systems and operating procedures. A recently converted sister bank, selected for comparable systems and market conditions, supplied a working reference. Staff could visit it, receive training, and obtain temporary specialist assistance. A separate model bank provided a place to try new systems and procedures without disturbing an operating bank. The central organization's job included selecting examples, training staff, and coordinating the conversion, rather than merely distributing software and instructions. [CG-W018, pp. 737–738]

Banc One Systems Corporation coordinated the conversion. In the authors' reported history, departmental work initially proceeded with limited coordination and tacit procedures. The later creation of conversion management brought documentation, meetings across application groups and explicit attention to dependencies. Problems encountered during conversions became checkpoints and tasks. The organization doing the transfer developed alongside the transfers it carried out. [CG-W018, pp. 737–739]

The authors also examined combining selected features from different outlets. A proposed outlet might borrow one element from a St. Louis design and another from a San Francisco innovation. They argued that such a combination could introduce interactions absent from either original. The same employees might, for example, have to carry out both activities, creating conflicts over schedules and workloads. A mix of individually successful parts was consequently a different proposition from copying an operating whole. The authors allowed that new combinations could produce favorable interactions as well as problems. Their hypothesis favored guidance by a working template during exploitation; it did not establish that every combination or adaptation would fail. [CG-W018, pp. 735–736]

The authors proposed that using a working template, and initially copying it closely, could improve replication once early exploration had produced a satisfactory model. They also described the costs of this approach: copying irrelevant features, delaying improvement, and reproducing a design poorly suited to its new environment. Their precision-versus-adaptation argument was a set of hypotheses, illustrated by reported organizational experience. The Banc One account described 1970–1995 and did not establish a controlled comparison of replication methods. [CG-W018, pp. 735–739, endnotes 2, 7–8]

Huckman and Pisano: performance that depends on the setting

Robert Huckman and Gary Pisano's *The Effect of Organizational Context on Individual Performance* (2003) examined whether a skilled person's performance could be carried unchanged between organizations. Their setting was coronary artery bypass graft surgery in Pennsylvania in 1994–1995. Many surgeons worked at more than one hospital during the same period, while nurses, technicians, anesthesiologists, and other resources were supplied by the hospitals. This allowed the researchers to compare a surgeon's results across organizations without relying solely on a move from one employer to another many years apart. The underlying records covered more than 38,000 procedures by 203 surgeons at 43 hospitals. [CG-W019, pp. 1–10]

The principal regressions treated death in hospital following an individual procedure as the outcome. They included patient characteristics and clinical conditions, calendar-quarter effects, and measures of prior surgeon and hospital quality. The key comparison divided a surgeon's case volume in the preceding quarter into procedures at the hospital where the current operation occurred and procedures at other hospitals. Using prior-quarter volume was intended to reduce reverse-causality concerns. A specification with hospital fixed effects checked whether the relationship survived a different way of accounting for stable hospital differences. The study did not randomly allocate surgeons or patients between hospitals. [CG-W019, pp. 11–18]

Higher volume at the current hospital was associated with a lower probability of death; volume at other hospitals had a much smaller, statistically insignificant association. In the first hospital-specific specification, a one-standard-deviation increase in local volume corresponded to a 0.34-percentage-point reduction relative to a predicted mortality probability of 1.8 percent. The corresponding estimate for volume elsewhere was 0.01 percentage points. The local and elsewhere coefficients differed at the 3 percent significance level in the first four specifications, including the hospital-fixed-effects comparison. These results concerned where a surgeon had recently performed procedures, rather than treating all of that surgeon's experience as interchangeable. [CG-W019, pp. 17–18]

Huckman and Pisano considered two explanations for this hospital specificity. Familiarity with a hospital's people and resources could improve coordination. Alternatively, a surgeon accounting for a large share of the hospital's business might obtain preferential access to its resources. To investigate the latter explanation they added the surgeon's share of the hospital's procedure volume. Local volume retained its association with mortality, but the statistical distinction between local volume and volume elsewhere weakened to the 13 percent level. The share measure was only a proxy for influence and was correlated with local volume. The authors accordingly described the separation of familiarity and influence as suggestive. The data did not identify whether particular employees, team structures, or operating routines were responsible. [CG-W019, pp. 18–22]

Their staffing implications followed from this limited portability of measured performance. Hiring a highly skilled individual might not reproduce the results that person achieved with another organization's complementary resources. The study also raised a question about dividing skilled workers among projects or units. Its evidence remained specific to cardiac surgery in this period. It could not distinguish learning to work together from a stable compatibility between a surgeon and a hospital. The authors' estimates of the benefits from consolidating split practice were thought experiments under simplifying assumptions, rather than observed results from a consolidation program. [CG-W019, pp. 4–6, 19–22]

S07 and S12 develop neighboring accounts of organizational knowledge and retained expertise. The studies here concern the work of recreating a practice and the conditions under which demonstrated performance can be sustained in another setting.

S15 Institutions, messages and invested forms: what people can rely on

Hayek: acting without knowing every cause

F. A. Hayek's *The Use of Knowledge in Society* asks how an economic arrangement can make use of knowledge dispersed among people. The selected argument distinguishes scientific knowledge from knowledge of particular circumstances of time and place. Relevant knowledge is not already collected in one mind. The organizational question is how decisions can use it when needs, supplies and opportunities change. [W016, §§I–V]

Prices can communicate a relevant change without conveying its full history. In the tin example, people can economize on a scarcer input without each learning whether its scarcity arose from a new use or a lost supply. Changes in their demand affect other people and substitutes. The proposed economy concerns what a participant must know in order to adjust. [W016, §§IV–V, pp. 523–526]

Hayek explicitly resists the idea that a developed production apparatus makes important economic decisions necessary only at long intervals. Local adjustments continue. The price signal complements knowledge of available capacities and opportunities rather than replacing that knowledge with a complete centrally supplied instruction. [W016]

Ostrom: contingent commitment retains monitoring

The selected passage from Elinor Ostrom's *Governing the Commons* concerns contingent commitments, observation and graduated sanctions. Participants' willingness to continue following rules depends partly on knowing that others generally do so. Monitoring makes behavior visible, while graduated responses allow the institution to address violations without treating every departure as a reason to abandon the whole arrangement. [W049, p. 186]

Local governance sustains cooperation by distributing responsibility for rules, monitoring, judging exceptions and applying graduated responses. [W049, p. 186]

Hurwicz: prescribed rules and feasible behavior differ

Leonid Hurwicz's prize lecture *But Who Will Guard the Guardians?* connects mechanism design with information, feasible conduct and enforcement. Its selected opening sections examine information decentralization and limits of message exchange, while later selected sections distinguish the actions permitted by a rule from the actions people can actually take. A mechanism analyzed only over its prescribed legal strategies can omit profitable violations. [TA-W001, selected pp. 280–286]

The lecture calls the larger set the true strategy domain. Illegal actions can still have consequences. Enforcement matters because those actions are feasible; if violations were impossible, the enforcement question would have a different form. An equilibrium among permitted actions therefore need not be self-enforcing when a participant can move outside that set. The relevant institution includes the arrangements that affect the attractiveness and consequences of those departures. [TA-W001, p. 285]

Hurwicz's discussion of genuine implementation is correspondingly broad. His social-security illustration includes procedures for determining eligibility, calculating payments, collecting funds and authorizing the institutions that do the work. A desired allocation written in legislation does not perform those activities itself. Money, information and administrative capacity are part of implementation in this sense. The lecture distinguishes that meaning from narrower technical uses of the same term. [TA-W001, p. 286]

The argument is not that enforcement is universally impossible because guardians would require an infinite chain of guardians. The lecture uses that challenge to investigate how enforcement can be organized. [TA-W001]

Hurwicz and Reiter: informational requirements are design variables

The introduction to Hurwicz and Stanley Reiter's *Designing Economic Mechanisms* treats the mechanism itself as a design variable. Given agents, an environment class and a goal, alternative decentralized arrangements can be compared by their informational requirements. Privacy means that an agent cannot simply observe another's private circumstances: information not directly observed has to arrive through an appropriate communication process. [TA-W002, introduction, pp. 1–6]

The introduction distinguishes realizing a goal when agents do not use information strategically from implementing it when they do. It also distinguishes recognizing an equilibrium from finding one. In a verification scenario, agents can respond to an announced candidate; unanimity of appropriate confirmations verifies it. That does not establish a practical process that discovers the candidate or reaches it dynamically within a deadline. [TA-W002, pp. 2–3]

The size of a message space supplies one way to compare mechanisms' information requirements. It is not a count of all resources involved in an institution. The introduction separately lists observation precision, communication, computation, losses from strategic behavior and enforcement. It describes cost comparison in a vectorial sense, and expressly says that computation is not treated formally in the book, although discussed in places. A smaller message space cannot therefore be presented as a proved minimum of every operating cost. [TA-W002, pp. 3–4]

Groves and Radner: the value of a restricted message exchange

Theodore Groves and Roy Radner's *Allocation of Resources in a Team* studies decisions made after limited communication rather than assuming an adjustment process has run to completion. Enterprise managers observe local production information, and a resource manager observes available supply. The model asks which decision rules maximize expected total output under alternative information structures. A common objective is assumed; inducing truthful messages and obedience through incentives is set aside. [TA-W003, pp. 415–420]

In the selected quadratic model, a single price-and-demand exchange can have the same expected payoff as specified more informative one-stage messages. Theorem 2 compares a declared class: a resource message goes to enterprise managers, replies return to the resource manager, and communication among enterprises is prohibited. Additional information can have zero marginal value for the model's criterion even though the alternative message is more informative. That is the particular informational economy being described. [TA-W003, pp. 435–437]

The premises remain substantial. Quadratic production, the probability structure, the common expected-output objective and the timing and direction of messages matter. The introductory simple model assumes independent local variations and corresponding similarities among enterprises. The source explicitly distinguishes these results from the full-information comparator and observes that the optimal decisions need not maximize each enterprise's profit at a common price. It also leaves the optimal choice of information-processing technology outside the analysis. [TA-W003, pp. 416, 418–420]

Smith: exclusion as a limited-information interface

Henry Smith's *Property as the Law of Things* proposes an architectural account of property through information costs. Specifying every possible relationship among every pair of people and every use of

resources would be difficult. A thing-based exclusion boundary can let outsiders respect a relatively simple instruction without learning all the uses and decisions inside it. [TA-W005, pp. 1691–1694, 1699–1705]

His Blackacre example makes the mechanism concrete. An owner can choose among many uses without coordinating every choice with every neighbor. The law need not enumerate them all. Some activities, however, cause effects outside the boundary, such as nuisance, and require more detailed governance. Exclusion supplies a baseline; it does not settle every interaction. The ends still concern people's interests in using things. [TA-W005, p. 1704]

Different audiences need different information. Strangers can use a standardized boundary message. Potential transactors may need formal information about the interest offered. Insiders need richer arrangements for their own interactions. Notice, records and governance remain work. Smith also treats water and information as cases where interactions can make simple compartmentalization less adequate, rather than claiming all resources are naturally separable. [TA-W005, selected pp. 1699–1705, 1709–1711]

The account explicitly draws on modularity arguments associated with Simon and Baldwin; those are the author's connections. Enforcement, competing uses and distributional judgments are not eliminated by the informational convenience of a boundary. [TA-W005]

Merrill: information savings do not exhaust ownership

Thomas Merrill's response *Property as Modularity* agrees with much of the information-cost account. He finds modularity helpful for explaining why a property system presents different levels of complexity to strangers, transactors and insiders. A person outside the arrangement can rely on a relatively simple message while intricate relationships remain inside. [TA-W006, pp. 151–155]

His reservation is that a structural description can understate incentives and residual authority. Ownership includes the capacity to rearrange uses and claim the returns from doing so. Exclusion does more than reduce outsiders' information requirements: it can support initiative and continuing management. The critique therefore asks what the modular picture leaves less visible, rather than denying every informational advantage. [TA-W006, especially pp. 151–152]

The response is a reasoned criticism of explanatory emphasis. Smith's and Merrill's work has a partly shared authorship history. [TA-W005; TA-W006]

Diaz-Bone: invested forms extend coordination but retain interpretation

Rainer Diaz-Bone's chapter reconstructs the economics of convention. It treats coordination as occurring in equipped situations that include material arrangements, categories, standards and conventions of evaluation. People do not arrive at each encounter without shared forms, but neither do those forms mechanically determine one interpretation in every situation. [GL-W031, chapter discussion]

Through Eymard-Duvernay and Thévenot, Diaz-Bone describes investment in cognitive as well as material forms. Organizations invest in means of recognizing relevant information, and those forms can extend coordination across time and space. A classification, label or standard can make future recognition and comparison possible without repeating all the original work of agreement. [GL-W031, investment in forms]

Plurality is integral to the reconstruction. Different conventions support different evaluations, and competent actors judge appropriateness, switch conventions, criticize one another or construct compromises. The approach rejects a single best method of coordination for all production, distribution

and consumption situations. Its treatment of law also includes enacted interpretation rather than treating law solely as an external constraint. [GL-W031, conventions, plural evaluation and law]

Diaz-Bone reconstructs the investment-in-forms argument of Laurent Thévenot's 1984 *Rules and implements: investment in forms*. [GL-W032, through GL-W031]

The abstract of Thévenot's *Organized Complexity* presents the firm as an arrangement compromising among modes of coordination and evaluation. Plural requirements enter the description of the firm itself. [GL-W033, abstract]

Rules and standards: when specification work is done

Louis Kaplow's *Rules Versus Standards* distinguishes giving legal content in advance from determining it later. Its repository abstract also separates that timing question from complexity and identifies promulgation, learning and enforcement costs. The relevant point is that moving a decision into an advance rule can change when and by whom interpretive work is performed; rule and simplicity are not synonyms. [GL-W043, abstract]

Lehavi's chapter excerpt develops the front-end/back-end tradeoff between advance specification and later interpretation or enforcement. [GL-W043; GL-W094, chapter excerpt]

A reusable provision can make later decisions easier. Its value depends on how often it can be reused and which differences between cases require fresh interpretation. [GL-W043; GL-W094]

Scharpf: a broker can reduce consultations without eliminating politics

Fritz Scharpf's *The Joint-Decision Trap Revisited* examines mixed-motive bargaining, vetoes and the status quo in historical German and European arrangements. A compulsory negotiation system differs from one where parties can simply act separately if talks fail. Once an agreed rule binds them, a veto can prevent others from correcting it when circumstances change. Agreement and the absence of agreement therefore have consequences that a simple voluntary-exchange picture can miss. [SG-W025, selected body]

The central-proposer comparison identifies a possible organizational economy. With N participants and a salient concerns for each, the displayed all-channel comparison counts $N(N-1)a^2$ impact examinations. A central agent's consultations are represented by Na. For fixed a, the first expression is quadratic, despite the text's exponential adjective. These are idealized consultation counts, not a measurement of the full cost of bargaining or a general proof of administrative efficiency. [SG-W025, printed pp. 850–851]

The broker still needs knowledge. The account describes extensive consultation with governments, interests and experts to identify constraints and develop proposals preferred to the status quo. The agent must be accepted as an honest broker, and a suitable mutually acceptable solution must exist. Its knowledge-gathering and design work have not vanished merely because every pair of participants need not perform the same comparison. [SG-W025, pp. 850–851]

Scharpf expressly says that the broker cannot solve an empty solution space. Preference diversity, transaction costs, political dynamics and compulsory commitments continue to limit agreement. His discussion qualifies pessimism about joint decisions rather than replacing it with an unrestricted case for centralization. [SG-W025, pp. 2–8, 16–17]

Signals, commitments and institutional boundaries

These selected sources explain several forms of reliance: a scarcity signal, a monitored commitment, a restricted message, a property boundary, an invested classification, an advance rule and a broker's proposal. Their criteria and institutional premises remain different.

Across the accounts, the reader can recover which information may be omitted by a particular participant and where observation, interpretation, adjustment or enforcement must continue. [W016; W049; TA-W001; TA-W002; TA-W003; TA-W005; TA-W006; GL-W031; SG-W025]

S59 Expectations, public signals and reflexive feedback

An announcement can matter because of what it reveals, because people can see that others have received it, or because it changes the consequences of acting alone. Prices, votes and other observable actions can themselves become information. The following accounts examine these processes through communication, investment decisions and the interaction between financial markets and institutions. They ask what participants can observe, what they infer, how they are evaluated, and how their responses change subsequent events.

Chwe: knowing that others received the message

Michael Suk-Young Chwe's 1998 article starts with two coworkers who usually leave the bus together for a drink. On a crowded journey they become separated. A mutual acquaintance calls to invite them off at another stop. Each would enjoy joining the acquaintance, but mainly wants the other's company. Both may hear the invitation and nevertheless remain on the bus: neither is sure that the other heard it and will get off. The message has reached both people without supplying the assurance each needs to act. [EF-W003, pp. 49–50]

Knowing that the other heard is not always enough. One coworker may see the other look up but remain unsure whether the other has noticed that the invitation was mutually received. Chwe uses this progression to explain common knowledge: everyone knows something, everyone knows that everyone knows it, and the corresponding further levels hold. In the example, mutually observed eye contact supplies an immediate way to establish the relevant shared awareness. The communication problem includes what people know about one another's reception of the message. [EF-W003, pp. 49–50]

The dependence on joint participation matters. Chwe changes the example so that one coworker's partner is calling from the street. That coworker will get off regardless of what the other does. The remaining coworker then needs to know that the invitation was heard, but need not obtain the same assurance about mutual awareness. The original difficulty came from each wanting to participate only if the other participated. [EF-W003, p. 50]

Chwe applies this argument to the physical organization of public encounters. In inward-facing circles, people can observe others receiving a communication and reacting to it. His discussion of circular seating at Fort Worth's city hall emphasizes the audience's visual contact with one another as well as with the council. His reconstruction of revolutionary festivals similarly distinguishes the circle's symbolic meaning from the opportunity it gives spectators to see one another. Removing partitions or arranging sight lines can change this opportunity even when the spoken content remains the same. [EF-W003, pp. 58–59]

Publicity also enters his account of advertising. A buyer may value a product more when others use it, as with compatible computers or goods consumed together. Advertising on a conspicuously popular programme can inform viewers that many other viewers have encountered the product. Chwe compares advertising for 119 brands over three representative months of a network year. The eighteen brands classified as social goods appeared before larger audiences and paid substantially more per viewer than the other brands. The comparison is suggestive: the available brands were not a representative sample, classification was partly judgmental, and the social group was dominated by products aimed at men, particularly beer. [EF-W003, pp. 59–63, tables 1–2]

Chwe also addresses the apparent implausibility of indefinitely repeated beliefs about beliefs. People need not consciously recite an infinite chain. They can rely on familiar situations, recursive recognition or practical heuristics for judging shared awareness. He explicitly distinguishes such supports from a failsafe communication procedure. Common knowledge helps explain coordinated action without guaranteeing it. [EF-W003, p. 66]

Informational cascades: more followers need not mean more evidence

Sushil Bikhchandani, David Hirshleifer and Ivo Welch's 1992 article examines a different reason for doing what others do. Participants decide sequentially whether to adopt an action. They observe predecessors' choices, but not their private information. Each also receives a private signal about whether adoption is worthwhile. The model's payoffs do not require a taste for conformity or a reward for agreeing with other people. Earlier actions matter because they may reveal information. [EF-W005, pp. 994–996]

In the article's elementary example, adoption has a known cost and an uncertain benefit, with two possible values. A signal is more likely to favor adoption when adoption is valuable. The first person's choice reveals that person's signal. The second combines this information with a new private signal, using a random tie-break when the two indications balance. After two preceding adoptions, the third person can find adoption preferable even after receiving an unfavorable signal. After two rejections the corresponding reversal occurs. With one adoption and one rejection, the third person's own signal again determines the choice. [EF-W005, pp. 996–997, table 1]

The consequential step is the loss of information in subsequent actions. Once someone chooses the same action regardless of the private signal received, observing that choice tells later participants nothing about that signal. Later people can then face the same informational balance and make the same choice. Many identical decisions may therefore rest on the information conveyed by only a few early decisions. They are not necessarily many independent confirmations of the action's merit. [EF-W005, pp. 998–1001]

The authors compare this with a setting in which previous signals remain observable. An individual may still choose against a private signal, but the signal enters the common information pool. Enough contrary signals can eventually reverse behavior. In the actions-only setting, signals arriving after the cascade begins remain hidden behind identical choices. The difference concerns what an observed action preserves about the reasoning or evidence behind it. [EF-W005, p. 999]

This also explains the fragility of a cascade. A public disclosure less informative than one person's private signal can make a long-established cascade unravel, because it need only counter the limited information that originally started the process. The crowd's size has not continually deepened its evidential basis. The article distinguishes disclosure before a cascade from disclosure afterward. An early noisy disclosure can change predecessors' choices in a way that reduces what later people learn from those choices; after a cascade has begun, new public information can add evidence without destroying information that later actions were still transmitting. These are results about the specified sequence and information structure. [EF-W005, pp. 1004–1007]

Scharfstein and Stein: the reputation gained by following others

David Scharfstein and Jeremy Stein's 1990 model makes evaluation by other people part of the decision. Two managers consider investing in a cost-saving technology, with the first deciding before the second. Each receives a signal about the investment's eventual profitability. The outcome later becomes observable even if neither invests. Investment is available at a fixed price, so the model isolates reputation and inference without also determining a market-clearing asset price. [EF-W004, pp. 465–468]

Managers differ in their ability to obtain useful signals, but initially neither they nor the labor market know which managers have that ability. In the central case, capable managers receive correlated informative signals; other managers receive independent noise. An evaluator can therefore learn about ability from both the investment's eventual success and the similarity of the managers' decisions. Two managers making the same unsuccessful choice may have encountered an adverse circumstance that neither could have foreseen. A manager making an isolated unsuccessful choice has less of that explanation available. The authors call this sharing the blame. [EF-W004, pp. 466–469]

The second manager consequently has a reason to imitate beyond learning whether the investment will be profitable. Agreement can make that manager appear to have received the kind of information a capable colleague received. In the authors' herding equilibrium, the second manager follows the first even when a private signal points elsewhere. The information possessed by both firms need not be reflected in investment. Reputation affects future compensation, and maximizing that compensation can conflict with making the best present investment. [EF-W004, pp. 469–473]

This is a model of incentives within an evaluation system. Its managers need no explicit agreement to suppress private information. Nor does it require evaluators to be unaware of the incentive to imitate: managers' attempts to influence the inference about their ability occur within the equilibrium being analyzed. The definition of ability, what evaluators observe, the sequence of decisions and the dependence between informative signals all enter the original construction. [EF-W004, pp. 468–473]

The paper also develops countervailing incentives. Giving managers a material stake in profits can make sufficiently costly imitation unattractive. Rewarding relative standing can make merely matching another manager less appealing, because imitation prevents becoming the top-ranked performer. A broader conception of ability can reward finding a useful alternative rather than forecasting which of the same two choices will succeed. These changes affect what a manager gains by preserving a distinct judgment. [EF-W004, pp. 475–476]

The authors apply the argument to committee procedure. Sequential public voting may allow later members to echo the first speaker instead of contributing their information. Simultaneous written votes are one proposed response, although they can reduce valuable exchanges of ideas. Another is to let those with stronger reputational concerns speak first; under the conditions discussed, this can mean hearing younger members before established senior members. [EF-W004, pp. 477–478]

A later exchange clarifies what counts as herding

Marco Ottaviani and Peter Sørensen's comment challenges the original claim that correlated prediction errors are necessary for reputational herding. They connect reputational imitation with statistical learning from predecessors' choices. An observed decision can change both the probability assigned to a favorable state and the reputational payoff associated with acting in that state. Relaxing a restriction on prior probabilities allows cases in which a later manager disregards private information even without the original form of conditional signal correlation. Their account gives learning and evaluation joint roles in the resulting decision. [EF-W009, opening argument; EF-W010, pp. 705–706]

In their 2000 reply, Scharfstein and Stein accept that result under the comment's broader definition, but distinguish it from a manager always following a predecessor. A sufficiently favorable prior can make even the first manager ignore a contrary signal. After an informative first action, the second manager can face similarly favorable odds and invest regardless of a new signal. In that case, the second manager could make the same choice even if evaluators could not observe the predecessor's action and could not tell that copying had occurred. The reply reserves the stronger sharing-the-blame interpretation for the mechanism in which visible agreement itself improves the evaluation. The exchange separates sometimes ignoring information, always copying another person, and benefiting from being seen to agree. [EF-W010, pp. 705–706]

Orléan: the resale price changes what it is rational to anticipate

André Orléan's 1988 article, *L'auto-référence dans la théorie keynésienne de la spéculation*, reconstructs Keynes's distinction between estimating an asset's productive returns and anticipating its resale price. A person intending to hold an asset indefinitely has to consider its future income. A person expecting to sell also needs to consider the price another buyer will pay. The two valuations can be algebraically

compatible while depending on different activities of inquiry: examining productive conditions, or anticipating other market participants' judgments. [EF-W006, pp. 229–231]

Orléan emphasizes the institutional role of negotiability. The possibility of selling a financial claim does not make the corresponding physical investment withdrawable by the community as a whole. It does, however, expose the individual holder to changing market quotations. In the example he draws from Keynes, someone can judge an investment's long-term value to be thirty while expecting the market to value it at twenty in three months. Paying twenty-five now need not be attractive merely because the long-term assessment is higher. Expectations about a prospective transaction affect the sensible present action. [EF-W006, pp. 231–233]

Resources also affect whether a person can maintain a long-term position. Someone exposed to an unexpected need to liquidate cannot simply disregard an adverse current quotation, however confident that person is about eventual productive returns. Orléan uses this dependence to explain why attention to short-term market judgment can follow from the organization of a liquid market, rather than from a fixed personal disposition to speculate. In his interpretation, the same participant can be drawn into market-oriented anticipation by the conditions under which commitments must be financed and revised. [EF-W006, pp. 233–234]

Self-reference enters when the object being anticipated is produced by participants' anticipations. Each tries to anticipate the average judgment while recognizing that others are doing likewise. Orléan's discussion of Keynes's beauty contest gives this its recursive form: success depends on anticipating how other people anticipate the collective selection. A configuration of expectations can be validated by the price that those expectations help produce. Such validation answers a different question from whether investment is serving long-term productive requirements. [EF-W006, pp. 235–238]

Orléan presents this as a substantive reading of Keynes and a challenge to the organization of financial inquiry. He also finds formal resources in rational-expectations models that permit self-validating price paths in addition to a fundamental-value solution. His conclusion asks how conventions, institutions and forms of information beyond private competitive arbitrage can be understood. [EF-W006, pp. 238–241]

Morris and Shin: a public signal can coordinate a common error

Stephen Morris and Hyun Song Shin's 2002 model gives public information two roles. It informs each participant about an underlying state, and it provides something each expects others to use. Participants want their actions to suit the state, but also want to be close to the actions of other participants. Each observes a shared noisy signal and a separate private signal. Their choices are made together rather than in the sequential chain used in an informational cascade. [EF-W008, pp. 1521–1526]

The equilibrium action places more weight on the public signal than a forecast concerned only with the underlying state would place on it. The reason is strategic: the common signal helps predict what others will do as well as what the state is. Higher-order expectations pull toward the information known to be publicly shared. The same property gives any error in that signal an unusually wide influence. Participants can all respond rationally within the model while reinforcing a common informational error. [EF-W008, pp. 1525–1530]

The welfare result depends on what is being evaluated. The authors construct the rewards for matching others as a zero-sum competition. Those rewards cancel when individual payoffs are added, leaving the average squared distance between actions and the underlying state as the social loss. Individually useful coordination therefore receives more weight in decisions than it receives in this welfare criterion. With a sufficiently strong coordination motive and sufficiently accurate private information, an improvement in a still-noisy public signal can worsen that criterion. [EF-W008, pp. 1524, 1528–1529]

Other cases in the paper preserve the value of information. When the shared signal is the only information available, greater accuracy improves welfare. Greater precision of private signals improves welfare in the central model. The authors also examine what changes when participants want to differentiate their actions rather than match them; the tendency to overweight public information then changes. Their disclosure problem concerns the interaction between signal precision, incentives and the stated objective, including the practical choice between early provisional information and later more accurate information. [EF-W008, pp. 1522–1523, 1529–1532]

Bacchetta and van Wincoop: finding a scapegoat in the price

Philippe Bacchetta and Eric van Wincoop's 2004 article asks why market participants repeatedly change the importance they assign to particular macroeconomic indicators when explaining exchange rates. Its scapegoat is an observable economic variable given excessive weight through inference from the exchange rate. The model includes observed money supply and output, unobserved currency trades, and uncertainty about the persistence of economic conditions. Investors receive different private signals about the relevant persistence parameters. [EF-W001, pp. 114–116]

An exchange-rate movement is therefore ambiguous. It may reflect unobserved trading, or it may reflect information other investors possess about how persistent a change in money or output will be. Each investor rationally uses the price as a public signal alongside private information. The current price is partly an outcome of participants' expectations and partly evidence from which they estimate what those expectations contain. [EF-W001, pp. 115–116]

Consider the article's case in which money supply is unusually high and the exchange rate rises because of unobserved currency trades. An investor seeing a higher rate than expected can infer that other investors have information suggesting greater monetary persistence. Even investors whose own information gives little reason for that inference can revise their assessment because they do not know everyone else's signals. In the model, the private-signal errors average to zero, yet inference from the price produces a market-wide mistaken estimate. Everyone can think others possess supporting information that no one actually possesses. [EF-W001, pp. 116–117]

The resulting weight affects behavior and the exchange rate. Confusion amplifies the influence of the hidden trading shock and can increase both the actual and perceived importance of the visible monetary variable. Output can become the scapegoat under the corresponding conditions. The model also permits reduced weights, and its average weight can be correct across shocks while particular realizations produce substantial errors. Its result concerns the changing relation between observable conditions, private estimates and a price used as evidence. [EF-W001, pp. 117–118]

The short model limits information differences to one period and rules out bubbles in its later common-information setting. Its authors leave a fuller dynamic account to subsequent work. The operative explanation here is dispersed information and rational signal extraction; an observed indicator becomes influential because of what investors infer from the price, and that inference feeds into the price itself. [EF-W001, pp. 115, 118]

Soros: responses can change the conditions being interpreted

George Soros distinguishes two functions of participants' thinking. The cognitive function attempts to understand conditions; the manipulative function attempts to influence events. In his account of reflexivity, these functions interfere because participants act within the reality they are trying to understand. Their interpretations influence conduct, while conduct changes the situation that will subsequently be interpreted. His 2013 symposium remarks distinguish corrective feedback from

reinforcing feedback and place interactions between authorities and markets alongside the more dramatic case of financial bubbles. [EF-W002, opening conceptual argument]

His 2009 lecture on financial markets gives a concrete credit mechanism. Easier credit encourages transactions and raises real-estate values. Defaults become less frequent, recorded loan performance improves, and lenders relax their standards. The increased apparent safety of lending is partly produced by the credit expansion whose safety is being judged. Treating collateral value as independent of credit availability misses this dependence. When the process reverses, forced liquidation can depress property values and weaken the collateral supporting remaining loans. [EF-W007, pp. 1–3]

In this account, a price is more than an inaccurate description of an independently fixed condition. Credit terms, transactions and collateral provide routes by which valuation affects what can actually be financed. Soros also discusses equity-financed expansion and the conglomerate boom: elevated expectations permit acquisitions that improve reported earnings, until performance can no longer sustain expectations. The credit example supplies the clearest operative sequence, while the conglomerate case illustrates his wider claim that different financing arrangements can transmit the interaction. [EF-W007, pp. 1–3]

Soros describes a boom as a real trend reinforcing a misconception about that trend. Tests can interrupt it; surviving a test can strengthen both confidence and the trend. He includes a period in which doubts grow while activity continues, followed by reversal and reinforcement in the other direction. He also says that corrective feedback can stop a reinforcing process before it completes this sequence. The account does not predict in advance whether a particular test will be survived. Soros acknowledges that he expected the 1997–1998 crisis to end a longer credit expansion and was wrong. [EF-W007, pp. 2, 4, 6]

Intervention is part of the feedback in his historical interpretation. Authorities rescue failing institutions and support activity; participants then interpret the resulting survival as evidence about the system. Soros argues that repeated successful rescues can reinforce confidence that markets are self-correcting even when official action was what prevented collapse. His proposed responses include varying margin and capital requirements and paying attention to credit availability, rather than treating control of the money supply as sufficient. He also distinguishes emergency support during a contraction from restrictions intended to limit a later expansion. [EF-W007, pp. 6–9]

The 2013 remarks extend his analysis to institutional rules. In his account of the euro, common treatment of government bonds in bank-capital and central-bank collateral arrangements encouraged purchases of higher-yielding debt. Converging borrowing costs contributed to divergent national economic trajectories; later recognition of default risk changed the pressures on indebted countries and on the authorities. The rules, interpretations and financing conditions developed together. This is Soros's application of reflexivity to a particular institutional history, where public authorities are participants rather than outside observers. [EF-W002, euro-crisis discussion]

Related accounts

S15 develops the communication of dispersed knowledge through prices and institutional messages. S49 examines distributed action and expectations about others; S51 explains organizational silence and its reinforcing beliefs. The studies of common ground and collaborative memory in S56 concern how particular participants establish and retain shared understanding.

S16 Command, shared intent and retained control

Acting together under uncertainty

The relevant military literature asks how people can act together when no commander can know or decide every detail. Shared intent, training, delegated decisions, reporting, lateral relationships and intervention are different parts of the answer. The sources also ask when the organization itself should change.

Clausewitz: apparent simplicity and the resistances of execution

The selected chapter on friction in *On War* describes why an organization that looks simple on paper becomes difficult to operate. It consists of people rather than one perfectly integrated mechanical part. Small delays, fatigue, uncertainty, weather and mistakes accumulate. An instruction can be intelligible while its execution still encounters circumstances absent from the diagram. [MIL-W036, book I, ch. VII]

The journey illustration makes the distinction accessible: expected transport can be unavailable, roads can be worse than anticipated and darkness can compound the delay. The point is the interaction of ordinary obstacles, not an unexplained mystical force. Clausewitz emphasizes experienced judgment and warns against expecting unattainable precision. Forceful will can also damage the organization while overcoming obstacles. [MIL-W036, book I, ch. VII]

Sunzi: disposition and collective potential

The selected *Art of War* passage, traditionally attributed to Sunzi, directs attention toward disposition and collective situational potential, associated with 勢, shi, rather than demanding the whole result from isolated individuals. Giles's 1910 translation and commentary use the image of logs and stones: shape, position and slope affect whether they remain still or move. The arrangement changes what effort can accomplish. [MIL-W027, ch. V, §§21–23 and commentary]

The received text and commentaries contain variant readings and different explanations of the passage. Across these interpretations, disposition and collective potential connect the configuration of circumstances with the effort required. [MIL-W027]

Implicit communication is prepared, not empty

The *Command and Control*, MCDP 6, passage says that shared experience and a common outlook can permit implicit communication. People can interpret a brief indication through an established background rather than requiring every detail to be stated. The economy concerns explicit message traffic; interactive feedback remains necessary. It is a doctrinal explanation. [MIL-W001, selected printed 2-32 passage]

MCDP 6 was published in 1996 and amended by Change 1 in 2018. [MIL-W001]

The selected 2019 Army ADP 6-0 passages likewise tie delegation to competence, resources, shared understanding and trust. A commander can make fewer detailed decisions when subordinates understand the purpose and can act appropriately within it. Responsibility is redistributed; it is not enough to remove an approval step while leaving people without the capability or means to carry out the task. [MIL-W002, selected paras.1-26–1-41, 4-48–4-49]

Boyd: orientation grows through interaction

John Boyd's *Organic Design for Command and Control* emphasizes implicit orientation and the relationships that make initiative coherent with a larger purpose. In the selected slides, orientation is shaped by heritage, culture, previous experience and unfolding circumstances; it also changes through interaction. It is a process of cross-reference, comparison and rejection, not merely a fixed instruction stored once. [MIL-W008, slides 7–23]

The proposed preparation exposes people with different abilities to varied situations in which they orient to one another as well as to events. Repeated shared experience can build similar expectations and trust. Boyd then argues that lower-level initiative can follow higher-level intent with less explicit communication. The source thus places substantial developmental work before the apparent ease of coordination. [MIL-W008, especially slides 13–18]

The warning against inward-looking formal arrangements is part of this argument. A procedure can become an obstacle if adherence cuts people off from changes in their environment. Yet the briefing's appeals to Gödel, Heisenberg and thermodynamics do not establish formal transfers of those results into organizations. Its organizational proposals are interpreted as a conceptual briefing, not a theorem derived from physics or logic. [MIL-W008, slides 20–23]

Logistics can require more than common intent

Christopher Paparone and George Topic's 2014 article uses James Thompson's dependency distinctions to challenge a simple application of unity of command to the logistics enterprise. Pooled contributions can be coordinated partly by keeping activities from interfering. Sequential dependencies add requirements for order and transfer. Reciprocal dependencies mean that the output of one activity becomes another's input and vice versa. [MIL-W004, main argument]

The authors argue that reciprocal logistics relationships require continuing coordination across organizational boundaries. A common operational picture, lateral communication and trusted relationships support mutual adjustment. A single commander's statement of intent cannot specify every relevant interaction in a network shaped by industries, appropriations, transport, raw materials and political arrangements. This is a specific conceptual diagnosis of retained coordination work. [MIL-W004]

Paparone and Topic reconstruct Thompson's 1967 *Organizations in Action* to explain these dependencies and challenge enterprise-level central regulation. [MIL-W005, through MIL-W004]

Van Creveld through later accounts, and Smith's criticism

An MCDP 6 epigraph quotes Martin van Creveld's *Command in War*: a commander can increase information-processing capacity or redesign tasks and organization to reduce the information required. [MIL-W006, through MIL-W001, ch. 2 epigraph]

Christopher Smith's *Network Centric Warfare, Command, and the Nature of War* supplies a separate selected reconstruction of van Creveld's qualifications about technology and command. Smith criticizes narrow definitions of command and assertions of network advantage detached from context. More information or connectivity does not, in his argument, automatically resolve the human and organizational character of command. [MIL-W007, pp. 12–15, 64–68]

Power to the Edge: self-synchronization has conditions

David Alberts and Richard Hayes's *Power to the Edge* proposes conditions under which participants can synchronize their activity without detailed central direction. The selected passage lists consistent

understanding of intent, good information and shared awareness, competence and trust. It explicitly retains command through congruent intent, authoritative allocation of resources and rules that guide action without dictating every detail. [MIL-W009, pp. 27–33]

The text also says that self-synchronization is not assumed appropriate when its conditions are absent. Its organizational proposal therefore differs from the claim that installing a network creates collective control. Responsibilities for information, architecture, preparation and resources persist, including in the selected later discussion. [MIL-W009, selected pp. 186–187, 208–210]

A passage in Alberts and Hayes's *Understanding Command and Control* distinguishes understanding from prediction. The SAS-085 overview develops the command-approach framework discussed below. [MIL-W010; MIL-W011]

SAS-085: changing the command arrangement is itself work

The SAS-085 executive overview distinguishes decision rights, interaction patterns and information distribution as dimensions of a command approach. They can vary separately. Giving more people access to information does not, by itself, specify who may decide or which relationships carry coordination. The approach has to be described across those dimensions. [MIL-W011, selected pp. 9–15]

Agility includes recognizing when an arrangement no longer fits and being able to change it. That requires diagnosis, an alternative and a feasible transition, not simply permanent maximum decentralization. The group reports retrospective and simulation support. [MIL-W011]

The 2016 report on the 2014 Västmanland forest fire is a civil-emergency application of this programme. Its institutional abstract describes an initial mismatch in command arrangements and a case-analysis template. It demonstrates the scope of an application claim beyond combat. [MIL-W033, abstract and metadata]

Bousquet: a history of scientific images is not completed transformation

In the abstract of *Chaoplexic warfare*, Bousquet relates military organization to scientific accounts of mechanism, thermodynamics, cybernetics, chaos and complexity. These provide the historical categories for his interpretation of changing images of order. [MIL-W018, abstract]

In Josefin Svensson's interview about *The Scientific Way of Warfare*, Bousquet describes scientific images of order and adaptation alongside a continuing tension between decentralizing ambitions and the hierarchical organization of state military institutions. [MIL-W019, through MIL-W020]

Stahel: doctrine, culture and historical practice do not coincide

David Stahel's 2025 article examines the origins and application of Auftragstaktik and the problems of importing it into another institutional culture. A compact definition can hide differences in command level, available communications, personalities and organizational expectations. Stahel rejects a uniformly understood and practiced historical ideal. [MIL-W021, introduction and historical discussion]

Through Stephan Leistenschneider, he describes contested origins and early debates. Through Marco Sigg, he reports uneven practice across levels and periods. Better communications could permit close direction at lower levels, while higher-level autonomy could extend beyond the spirit of orders through rivalry and ambition. Initiative was not uniformly beneficial, and a culture emphasizing boldness had costs as well as capacities. [MIL-W021, reconstruction of historical studies]

Stahel also distinguishes the social and ethical purposes of later professional institutions from a purely functional reading of earlier military effectiveness. A historical technique cannot simply be imported as though its surrounding contract, authority and values were unchanged. [MIL-W021, conclusion]

Stahel reconstructs the mission-command histories developed in Sigg's *Der Unterführer als Feldherr im Taschenformat* and Leistenschneider's *Auftragstaktik im preußisch-deutschen Heer*. [MIL-W022; MIL-W023, through MIL-W021]

The supported historical qualification is more informative than a binary choice between always delegating and always directing. Stahel's account leaves an arrangement responsive to the operational circumstances and to the institution that can sustain it. [MIL-W021]

French, German and allied statements retain boundaries

The selected French FT-05 passages distinguish command by detailed orders from command by objectives as complementary tendencies. Information collection does not remove uncertainty, and waiting for exhaustive information can itself obstruct decisions. Delegation can place execution where appropriate information can be acquired and used, while acute circumstances can lead to more centralized decisions. [MIL-W024, pp. 27–28]

Its subsidiarity passage retains systematic reporting and the commander's responsibility while warning against unnecessary intrusion into subordinate conduct. Local freedom exists within a wider relation of intent, reports and accountability. These are the source's doctrinal prescriptions. [MIL-W024, p. 38]

The selected German *Agile Fehlerkultur* passages similarly describe bounded independence under *Führen mit Auftrag*, with resources, training, responsibility and an acceptance of some errors. Nor does a modern discussion of *Innere Führung* establish that its social and legal assumptions are identical to every historical use of Auftragstaktik. [MIL-W025]

The selected 2022 allied land-doctrine passage makes trust earned and delegation conditional on capability, common training and professional understanding. Multinational cooperation can make that common background difficult. Sharing a doctrinal label cannot be assumed to supply the relationships and interpretation needed among organizations. The text is doctrinal prescription. [GL-W036, p. 43]

Command by veto: a prepared default with continuing oversight

Alastair Cooper and Felicity Petrie's *Mission Command at Sea and Command by Veto* describes delegation within defined limits, with the delegating commander present physically or on a communication circuit. A trained delegate can proceed on a stated intention unless the commander modifies or vetoes it. Routine approval exchanges can therefore be reduced while the commander remains informed and able to intervene. [MIL-W034, definition and comparative argument]

The mechanism retains shared interpretation, competent delegates, explicit limits and attention. Authority may be delegated while accountability remains with the commander. The ability to act before a veto is useful only within the defined scope and appropriate timing. Silence in an unrelated, unobserved or misunderstood situation is not the same arrangement as the authors' command-by-veto construct. [MIL-W034]

The article connects changing communication and information capacities with changing command practice. Better information can support either more local action or a more active higher headquarters. It explicitly treats naval development on its own terms rather than reducing it to an identical land-doctrine history. [MIL-W034, conclusion and notes]

Duus: the bill for discretion is paid in preparation

Anthony Duus's *Paying the Mission Command Bill* argues that trust rests on education, training, shared experience and execution. A hands-off posture without initial guidance, resources and appropriate supervision can leave a subordinate to discover the commander's intention only after expensive divergence. The relevant preparation is institutional as well as personal. [MIL-W035, selected introduction and trust/education sections]

Duus treats detailed and guided command as available within a spectrum, with circumstances and the relationship between people affecting the appropriate choice. His criticism is that terminology can hide the work needed to make discretion useful. This is a professional argument with historical illustrations. [MIL-W035]

Automation can remove interactions that were doing useful work

The selected *Re-Envisioning Command and Control* passage proposes that repetition, communication in several forms, group ideation and reformulation can build understanding and trust. Efforts to automate apparently repetitive activity may remove some of the interactions through which that shared understanding develops. An activity can transmit content and also contribute to the relationship needed for later coordination. [MIL-W029, arXiv v2, introduction and framing]

The paper's human–AI organization is a research vision. [MIL-W029]

Civil incident organization and public accountability

FEMA's 2017 *National Incident Management System* supplies a selected civil example of modular organization. Functions can be activated according to incident size and complexity; management responsibility is delegated, and unused elements can be demobilized. The arrangement changes the organization with the task rather than requiring every position to operate at every scale. [GL-W046, pp. 30, 90 (printed p. 81)]

Manageable span and modular expansion remain prescriptive principles. They depend on communications, experience and the relationships through which functions cooperate. [GL-W046]

Kevin Krupski's *Who's the Boss?* introduces another limit to importing an organizational model. He questions the wholesale transfer of a private-market principal–agent picture into civil–military relations, where public values and multiple principals matter. The selected discussion presents cultivated political and professional relationships as relevant to how accountability operates. [GL-W053, p. 28]

Krupski reconstructs Feaver's monitoring argument while asking which purposes and constituencies an accountable command arrangement must serve. [GL-W053]

Common understanding, local decisions and accountability

The selected sources offer substantive accounts of preparing common understanding, distributing decisions, preserving lateral coordination, monitoring defaults and adapting command relationships. They also expose the work of education, trust, resources, communication and accountability that a short instruction or a sparse organization chart can conceal.

S27 Voge, information costs and polycellular organization

An original article and a chain of development

Jean Voge's 1985 *Crise, information et communication dans l'économie américaine* connects organizational architecture with communication, selection, regulation and costs. The abstract of his 1983 article identifies the related information-economy inquiry, while later accounts interpret his 1997 *Le complexe de Babel*. [UX-W031; W094, abstract; RGC-W031, through UX-W028 and RGC-W030]

Hubert Landier's *Polycellular Human Organisation* develops this line of thought organizationally. Le Moigne's book review, the French Senate's discussion and Villiers's information-economy argument document other routes of development. [UX-W030; UX-W028; UX-W029; RGC-W030]

Voge: information work can offset productive gains

Voge's 1985 article divides material production from informational activity and asks how gains within material production can coexist with weak overall productivity. His proposed explanation includes rising organizational information and communication costs. Technical improvements to processing alone need not settle the issue if the organization creates increasing quantities of work to select, route and use information. The article concerns the arrangement of activity as well as its technical tools. [UX-W031, printed 39–47]

The historical macroeconomic argument uses a particular classification. In the notes Voge explicitly assumes, following Jonscher, that government information services such as education, research, documentation, regulation and administration should be treated with material final demand because material goods and services are their principal beneficiaries. That decision affects the resulting account. It is an assumption about classification, not an observed natural boundary dividing all beneficial information from overhead. [UX-W031, 46, notes]

The article's statistics, explanation of the early-1980s recovery and prediction of its duration are historical claims. The selected contribution is an explicit hypothesis about productive work and organizational demands, with its classification visible. [UX-W031, 39–46]

Branching and meshed communications differ

Voge contrasts a branching communication structure with a cellular mesh. In the first, a downward message follows prescribed branches, while a message traveling upward follows a limited route through intermediaries. Selection is concentrated, and upper points can become bottlenecks. In the mesh, alternative routes and selective choices can exist at several points. The distinction is about where a choice of path is available. [UX-W031, 44–45]

His telephone comparison concerns particular quantities. He contrasts the number of switchable contacts needed for a caller to reach others, scaling linearly with subscriber count in the described pyramid and potentially logarithmically in the cellular arrangement. He separately contrasts operating costs as call volume grows. These are different measures. The comparison depends on switch design, ports, capacity, blocking, reliability and traffic. [UX-W031, 45]

The proposed organizational transfer decentralizes selection and supplies feedback at several levels. Voge favors interwoven serial and parallel processes and interactive dialogue instead of treating an organization solely as the execution channel for leaders' ideas. This is a connected architectural proposal. A communication diagram alone does not establish whether its human participants share purposes, possess relevant skills or can bear the assigned work. [UX-W031, 45–47]

The thermodynamic correspondence is a stronger claim

Voge relates his economic account to thermodynamic models in which dissipative terms depend quadratically on forces or flows and useful power can peak at a particular point. He transfers the correspondence to informational production and a fifty-percent maximum. These are the author's analogies and proposed generalizations. Entropy production, dissipated power and monetary value are not treated here as interchangeable physical quantities. [UX-W031, 43–44]

Voge's extension from a thermodynamic analogy to universal minimization of entropy production requires premises beyond the comparison. The architectural account instead specifies where information work occurs and how it can offset productive gains. [UX-W031, 43–47]

Landier's book: ideal types and actual informal work

Landier's 2024 book revises material from the distinct 1987 *L'entreprise polycellulaire* through a later transdisciplinary framework. [UX-W030, prefatory material, PDF 13]

The pyramid and polycellular organization function as ideal types. Real organizations already contain informal relations, local improvisation and support work missing from their official diagrams. Landier's account of a strict pyramid says that employees often make it function by departing from impractical instructions. Each level can hide local accommodations from another because an intervention that appears rational above can be disruptive below. The actual coordination is richer than the formal reporting lines. [UX-W030, PDF 14–32, 51–59]

The proposed change therefore concerns both organization and representation. Calling an organization pyramidal can obscure the initiatives that sustain it; renaming it a network does not necessarily change its practices. Restructuring also encounters job losses, learning requirements, uncertainty and resistance tied to status or power. [UX-W030, opening chapters and PDF 57–59]

Landier draws the Voge argument directly from the 1985 article. His discussion of overflowing inboxes and intermediaries gives a contemporary-to-the-book interpretation of the same information-burden concern. The repeated thermodynamic and fifty-percent formulations remain dependent reception of that argument, not independent physical demonstrations. [UX-W030, PDF 57–59]

Proliferation, multiple routes and conditional autonomy

The polycellular proposal has three connected principles. Units develop around a common project rather than simply filling a fixed chart. Communication has multiple possible paths. Local units retain operational autonomy conditional on effectiveness in relation to the shared project. The conditions distinguish the proposal from an arbitrary aggregate of independent teams. [UX-W030, PDF 60–64]

Multiple routes can permit communication after some failures, but require resources and appropriate surviving paths. Survival depends on node failures, loads and shared dependencies. The benefit must be specified against the disturbances and loads at issue. [UX-W030, PDF 62–63, fig. 9]

The autonomy principle explicitly retains central and intermediate responsibility. Management affirms the common project, monitors effectiveness and can intervene, reorganize or eliminate an ineffective unit. Apart from this corrective responsibility, it is meant to leave internal arrangements to the unit. Thus the proposal reallocates decisions. [UX-W030, PDF 64]

Development also involves initiative from the center and the periphery. A local unit can encounter possibilities unknown above, while broader management must be able to recognize and connect them. The intended organization depends on responses in both directions. [UX-W030, PDF 65–70]

Local knowledge substitutes for some directives

Salespeople, technicians, buyers and production workers encounter different parts of an organization's environment. Their knowledge can be lost or delayed when every appropriate response must await a distant directive. Landier proposes local initiative because the person at the point of encounter can have relevant information unavailable to management. The argument concerns a particular distribution of knowledge and decision opportunity. [UX-W030, PDF 99–100]

The substitution increases other demands. Local members must know the organization's purposes and relate their actions to them. Internal communication, shared standards, selection and evaluation replace some detailed instructions. Landier also says the balance differs between activities such as accounting control and research. [UX-W030, PDF 99–102]

Common codes require work. When one level's information becomes unintelligible at another, the proposed responses include adopting a code, developing a compromise or negotiating a new relation. Surveys, audits, training and decoding support that adjustment. Disruption can destroy the organization or provoke intervention from outside it. [UX-W030, PDF 88–103, especially 94]

Leadership and arbitration remain functions

Landier explicitly rejects the inference that local autonomy automatically gives an optimal outcome. Apparently equal relations can conceal age, status or other informal hierarchies, and conflicts can require legitimate arbitration. Participants can also have personal interests inconsistent with the common project, while relevant information is not automatically transparent. Removing an official supervisor does not remove these questions. [UX-W030, PDF 81–82]

His team-climate account associates better conditions with available leadership, opportunities for peer exchange and information situating the team in the wider organization. Worse conditions include inaccessible managers, absent exchange and deficient information. The passage supplies the author's practical account of retained support. [UX-W030, PDF 82–83]

Networks themselves also have several possible characters. They predate the Internet and can support secrecy, clientelism or coercion as well as cooperation. Technology can enable a relation without ensuring its purpose or quality. The book's political or organizational forecasts are proposals and interpretations. [UX-W030, PDF 73–87]

A common project is made and maintained

Landier criticizes a corporate-project exercise that ends in slogans and a brochure. A management statement is not automatically the project of all participants. Bringing it to life requires taking seriously the perspectives of individuals, teams, establishments and the organization, together with capital providers, customers, suppliers and public authorities. Participatory development permits outcomes management cannot fully specify in advance. [UX-W030, PDF 112–123]

The process can take years as an identity develops. It is iterative across levels: a site can formulate a project before a corporate document exists, and local work can contribute to the broader formulation. The relation is therefore not simply a complete goal flowing downward through a chart. Negotiation, mediation, labor representation and continuing revision are part of the intended arrangement. [UX-W030, PDF 112–123]

The expressions A = f(C) and C = f(A) represent a mutual dependence between autonomy and coherence. Appropriate initiative requires room to act, while continued autonomy depends on effectiveness and

relation to the common purpose. The functions, units and empirical optimum are not specified. The notation states a conceptual dependency. [UX-W030, PDF 118–119]

The individual is also more than an organizational component. Family, personal existence and other commitments have purposes that cannot be absorbed without remainder into a corporate aim. Landier makes manipulation and sacrifice of participants ethical problems. His biological analogy is expressly limited by the status of human beings as subjects. [UX-W030, PDF 70–72, 117–118]

Institutional boundaries do not contain every capability

Outsourcing can assign a function to a complementary organization rather than develop it internally. Landier's account retains the provider's viability and the reciprocal conditions of continued exchange. A client can weaken a supplier through its terms, but doing so can damage the wider productive relation on which the client depends. The local disappearance of a function is therefore not a disappearance of its provision. [UX-W030, PDF 69–70, 85–86]

Cross-firm cooperation has explicit conditions: transaction costs, continuity, confidentiality, trust, shared concepts and tools, and an appropriate institutional setting. Legal and public institutions participate in the supporting arrangement. A nominally independent network of suppliers still requires negotiation and may contain severe asymmetries. The book's expected gains are conditional organizational proposals. [UX-W030, PDF 85–86]

The time discussion identifies waiting, repeated reporting and conflicts among horizons as effects of architecture. Managers must bridge several time scales, and plans need adjustment. Its Toyota account retains stocks, signals, foreman regulation and learning between upstream and downstream activities. [UX-W030, PDF 104–111]

The broader societal discussion includes externalities and a reported generalized travel-time calculation that counts acquisition, earning and maintenance as well as movement. The particular historical calculation is mediated. Its relevance is the stated boundary of the account. Landier's planetary and human-care commitments remain normative positions. [UX-W030, PDF 124–139, especially 136]

Le Moigne's review of Babel

The review of *Le complexe de Babel* praises its transdisciplinary reach and recommends the chapter on diminishing returns to information and human capital. Le Moigne emphasizes cooperative production of meaning rather than sheer proliferation of signs. He also objects that Voge gives some contemporary management fashions too much credit. [UX-W028]

The Senate section adds supporting institutions

The pertinent section of the French Senate's 1998 report reproduces the contrast between pyramids and networks and relates it to the Internet. Its conclusion also retains trust, education, market and state regulation, public participation in infrastructure and substantial intangible investment. Technology, a public, enterprises and a culture must come together for a medium to develop. The report therefore does not explain a communication society by the network alone. [UX-W029, section 2]

Its expectations about democracy, peace, technological progress and European opportunity are historical institutional forecasts. [UX-W029]

Villiers: the fifty-percent result has assumptions

Villiers's professional essay reuses Babel to examine information employment and productivity. It supposes a proportional relation Q/L = kXi, where Xi is information employment as a share of total employment, and assumes average wages grow proportionally with output per worker. With the chosen division between informational costs and material benefit, the account arrives at a maximum. These are explicit modeling premises. [RGC-W030, information-economy discussion and notes]

The treatment of final informational consumption affects the result. Changing what counts as an intermediate informational cost changes the claimed maximum. Aggregate output, per-worker normalization and wage coefficients also require clarification before numerical reuse. [RGC-W030]

S38 Logistics and sustaining coordinated operation

Sustaining operation is a continuing activity

These selected accounts examine the provision, movement, maintenance and information that allow other activities to continue.

MCDP 4: visibility requires interpretation

The 2023 edition of Marine Corps *Logistics* separates demand visibility from supply visibility. The first concerns what forces need in relation to an operation; the second concerns the resources available, including equipment, units, infrastructure and supplies. The document emphasizes that data must be processed and analyzed to create shared understanding. A current snapshot is not equivalent to knowing whether anticipated demands can be met. [MIL-W003, printed pp. 2-6–2-8]

Its sustainability discussion places self-sufficiency and dependency on a spectrum. A force carrying or producing everything it needs can have autonomy over a specified period. An outsourced arrangement can reduce its footprint, but requires access to external providers. The doctrine expects combinations, chosen in relation to mission and context, rather than pure self-sufficiency or pure outsourcing. The external supply relationship remains part of the account even when the supported unit is smaller.

Simplicity is likewise described as a designed and maintained property. Clear command relationships, standardized interoperable procedures, fewer required actions and practiced methods can make execution more manageable. The document does not describe the logistical undertaking as intrinsically simple: planners still estimate needs using usage, equipment, personnel and mission information. It seeks actionable concepts without instructions for every eventuality. [MIL-W003]

Push and pull distribute anticipation differently

In the selected distribution passage, a push system forecasts requirements and supplies them on a predetermined schedule. Its advantages include regularity and predictability; its limitations include the time required to adjust when requirements change. Forethought, planning and coordination remain necessary. A pull system responds to requirements submitted by the receiving unit. It can reduce provision of unneeded supplies, but the receiver must identify and communicate its requirements. [MIL-W003, printed pp. 2-26–2-27, figure 2-6]

The figure explicitly assigns anticipation to the provider under push and to the receiver under pull. Pull is therefore not an absence of anticipation. It depends on developed procedures, reliable communications and available packages. The doctrine also recommends combinations: routine requirements can favor scheduled support while unusual requirements call for requests.

The logistics snowball discussion addresses inflated requirements and habitual extras. Repeated across a force, such demands burden the system and can divert support away from more urgent needs. The proposed response includes disciplined requirement generation, training, trust and good habits. Emerging digital mapping or machine learning is presented as a possible improvement to visibility and responsiveness. [MIL-W003]

The selected education and training pages retain the institutional investment behind manageable execution. People learn how the system works, how to request support, how occupational fields interact and how logistics limits plans. Training introduces friction and uncertainty and develops initiative. The foreword identifies the document as guidance and distinguishes fictional scenarios. [MIL-W003, printed pp. 4-12–4-14]

Thorpe: define sustaining functions so they are actually assigned

In the opening of *Pure Logistics*, Thorpe argues that the definition of logistics determines a division of labor. If that division is incomplete, functions may be omitted; if it overlaps carelessly, effort may be duplicated. His example is care of the wounded: merely possessing medical personnel does not ensure that their numbers, equipment and employment have been estimated for the proposed operation. Assigning a function makes it part of planning. [MIL-W016, 1917 work in the 1997 edition, printed pp. 9–13]

This historical argument is relevant because it treats classification as consequential for preparation. It does not suggest that a label performs the function. The care still needs people, resources and an estimate related to the planned activity. The selected later organizational passage also debates where intelligence belongs: information serves strategy, tactics and logistics, and must be systematically supplied even when its bureau is located elsewhere. Functional use and administrative location need not coincide. [MIL-W016, printed p. 45]

Thorpe's broader organizational proposals are prescriptive historical arguments. Later favorable editorial comments in the edition are distinguishable from Thorpe's own text.

Eccles through a doctrinal attribution

The MCDP introduction draws on Eccles's *Logistics in the National Defense* to connect a national economy with combat operations. It treats logistics as transforming resources and productive capacity into the material means for operations. [MIL-W015, through MIL-W003, introduction]

Kress: intrinsic versatility and structural flexibility

Kress's 2000 article distinguishes versatility of assets from flexibility supplied by their deployment in a logistics network. Intrinsic flexibility concerns how many relevant demands an asset can serve and how many resources a support asset can handle. The palletized-load example separates a truck from a particular load, allowing a vehicle to switch among resource types. General-purpose maintenance capability supplies another kind of versatility, conditional on staff qualification, equipment and spares. [MIL-W013, printed pp. 44–49]

The intrinsic-flexibility index counts associations between assets and essential demand units. An essential demand unit pairs a supported unit with a resource it can actually consume. The association is not established by a generic name: a maintenance unit supports a particular vehicle type only if it has the capability to do so. The worked examples show how a change in transport or maintenance versatility changes the index. The index measures those defined associations, rather than the complete cost or effectiveness of every asset.

Structural flexibility concerns where resources are held and how they can flow. Kress represents source, intermediate and destination nodes, with edge capacities affected by transport and reliability. Because meaningful lateral flow at a command level is largely excluded in the model, the graph is essentially a tree. This is a substantive assumption. A result for that structure should not be presented as a theorem about an arbitrary network with extensive peer redistribution. [MIL-W013, printed pp. 45–47]

Holding supplies higher in a tree can postpone commitment to a particular subordinate unit. It preserves alternative destinations, provided reliable transport can deliver them in time. Concentration is therefore coupled to flow capacity and lead time. The paper's sufficient comparison of two systems sharing a graph explicitly includes the requirement that the relevant edge capacities not be worse in the more concentrated system. Rearward inventory without usable delivery is not the claimed mechanism.

The flexibility comparison can reverse

Kress's structural-flexibility measure combines resource sufficiency and timeliness, with stated weighting across command levels. In its worked comparison, posture B is often favored, but not under every supply and demand condition. For one parameter combination the reported indices are 0.574 for B and 0.353 for A. With relatively abundant resources and lower variation in demand, the ordering reverses: the reported comparison gives 0.795 for A and 0.768 for B. [MIL-W013, printed pp. 52–53]

The paper explains the reversal. When both arrangements are highly likely to have enough supplies, the extra benefit associated with concentration becomes less important, while the time to deliver dominates the index. A resource-positioning advantage depends on which constraint binds.

These numbers belong to a constructed model with specified demand, timeliness and weighting assumptions. [MIL-W013]

Identifying and locating a convoy

Kress illustrates convoy tracking by combining two kinds of data. Radio-frequency tags associate a convoy with its cargo. GPS receivers locate its vehicles. A map can then show where the convoy is and how it is moving alongside its destination and load. Tracking the vehicles answers where the convoy is. Identifying its load lets the logistician see which supplies are on the move. [MIL-W012, example 5.13]

RAND: sense and response retain prediction and command

The selected RAND report combines prediction, responsiveness and command-and-control capabilities. Its future-work discussion calls for continuous monitoring of resources and process performance, estimates of changing campaign needs, assessment of plan feasibility and review of infrastructure configurations. The report places logistics information early in planning so proposed operations can be supportable. [MIL-W017, summary, printed pp. 9–10 and 79–81]

The selected passages require weighing support alternatives against time, capability, risk and cost. Near-real-time sensing is intended to alert decision makers to deviations before mission capability is affected. Metrics should concern outcomes rather than only outputs. The apparent responsiveness is consequently supported by maintained information, planning, thresholds, configuration and authority. It is a program analysis and proposal.

Shared standards and a shared system

The official account describes joint support using common technical standards, interfaces, containers and pallets. Its examples contrast incompatible equipment interfaces with arrangements that can be combined across services. A further passage describes several services using a joint logistics information system during an exercise. The relevant mechanism is compatibility prepared in advance, together with shared information during coordinated operation. [MIL-W026, sections 统一标准 and 共用系统]

The account reports implementation and exercises from an official organizational perspective. Common standards and systems require design, implementation and maintenance.

Fox: threat, environment and one's own organization

Fox's conceptual essay divides contested-logistics challenges among the threat, the environment and the supported organization's own arrangements. It argues that the categories must be addressed together and calls for attention to transport possibilities, protection, deception and masking. The support system's

exposure is thus part of its usefulness: a technically efficient movement route can still be vulnerable. [MIL-W030, framing and conceptual support-boundary discussion]

The essay questions whether reducing or removing a movement demand could reduce opportunities for interference, while calling for theory as well as technology. This is a conceptual proposal with a hypothetical future vignette.

Sherbrooke: multi-echelon recoverable-item inventory

Sherbrooke's METRIC model compares where to hold spare parts in a system of bases and a depot. When an item fails, the base repairs it if it can. Otherwise the item goes to the depot for repair and the base requests a replacement from the depot. For costly parts with low demand, the model starts repair and replenishment one item at a time. It leaves out transfers between bases. [MIL-W031]

Sherbrooke's program compares stocking plans by the number of requests expected to remain unfilled at a given inventory investment. It can evaluate an existing distribution of spares as well as compare alternatives, so the decision concerns where to stock a part as well as how many to buy. [MIL-W032] [MIL-W031]

A queueing benchmark for pooling

Van Leeuwaarden, Mathijsen and Zwart's review explains resource pooling using an M/M/s queue. Jobs arrive as a Poisson process, service times are independent exponential variables, servers share a common service capability, and waiting occurs in an unbounded buffer in arrival order. Separate queues can leave one server idle while another has waiting jobs. A common queue removes that particular mismatch under the model's assumptions. [TA-W016, printed pp. 403–411, especially p. 405]

The quality-and-efficiency-driven regime adds capacity of square-root order above offered load as the system grows. Utilization can approach one while the probability of delay approaches a value strictly between zero and one, with mean delay decreasing at the specified scale. Proposition 2.1 states the asymptotic relation for the benchmark setting. Pooling therefore does not mean that every job avoids waiting, nor does high utilization alone establish good service.

The pooling result depends on the model's arrival, service, capacity and limiting assumptions. [TA-W016, introduction and proposition]

IMO: integrate navigation information while retaining integrity work

The IMO e-navigation framework concerns harmonized collection, integration, exchange, presentation and analysis of marine information aboard ship and ashore. Its five solutions address bridge design, standardized automated reporting, equipment and information integrity, graphical integration, and communication of vessel-traffic services. These are distinguishable components of an intended reduction in administrative burden and improvement of navigation services. [GL-W082, definition and implementation-plan sections]

The plan remains subject to review as user needs, technologies and lessons change. The source states intended benefits rather than measured outcomes.

Positioning supplies and pooling capacity

The sources explain sustaining operation through assigned responsibilities, versatile assets, retained alternatives, delivery capacity, compatible interfaces, shared information and controlled pooling. Kress's

ranking reversal shows how the advantage of a resource position depends on supply, demand and delivery time; the doctrine distributes anticipation across organizational roles.

S113 Maneuver warfare and disruption of coordinated action

Firepower and the scale of operations

В. К. Триандафиллов's 1929 work examines changes in weapons and in the composition and movement of forces. In his account of cavalry, the capacity to move and the ability to fight mounted are different problems. The spread of automatic weapons changes what can be done in a mounted engagement and shifts the importance of the fire weapons a formation can bring. His argument is framed by the forces and operations of his period. [T15-W001]

The Marine Corps account of maneuver warfare

The 1997 Marine Corps publication *Warfighting* describes maneuver and attrition as styles along a spectrum. Its account of maneuver emphasizes identifying and exploiting gaps and vulnerabilities, moving rapidly enough to exploit a changing situation and giving subordinates room to act under uncertainty. The discussion still gives combat power a place. Disruption is an intended effect on the opponent's capacity to respond coherently. S16 explains the separate command question of how shared intent can support subordinate decisions. [T15-W002]

Cole Petersen examines the difficulty of carrying the maneuver and attrition distinction into strategy. His account traces part of the debate through Liddell Hart's discussion of an indirect approach. A method of fighting can describe how a force acts in a particular engagement, while a strategic account has to explain the relationship of those actions to a wider object. Petersen disputes treating the two labels as complete competing strategies. [T15-W004]

Competing accounts of a campaign

A Command and General Staff College study asks how to interpret the coalition campaign in the Gulf War. It contrasts an explanation centered on AirLand Battle maneuver precepts with one centered on firepower and attrition. The study concludes that the coalition used maneuver-warfare precepts in defeating Iraqi forces. The author's judgment concerns the campaign as interpreted through AirLand Battle doctrine. [T15-W003]

The Defense Department's 1992 report to Congress documents how different aircraft could supply parts of a single engagement. In the northern Persian Gulf, U.S. SH-60B helicopters provided electronic surveillance and radar capability, while British Lynx helicopters brought radar-guided missiles. The report describes this combination in operations against Iraqi vessels and distinguishes it from helicopter patrols for mines. [T15-W005]

Florian Waitl's Army University Press study describes a different operation at the opening of the air campaign. Early on 17 January 1991, Task Force Normandy combined eight Army AH-64 Apache attack helicopters with Air Force MH-53J Pave Low helicopters. It destroyed two Iraqi air-defense command and control sites. In Waitl's account, the strikes opened a corridor through the air-defense network for coalition aircraft. The chapter then traces the shift in targeting from the air campaign to preparations for the ground campaign, including strikes on communications, logistics and armored units. Waitl describes what destroying the sites enabled. The staff-college study's judgment about maneuver and attrition concerns the campaign as a whole. [T15-W006]

S48 Burden, autonomy, power and evaluation

These sources examine effort, authority, care, autonomy and alienation, asking whose purposes and experiences enter judgments of an economical arrangement.

Knowledge and power inside hierarchies

Dosi, Marengo and Virgillito treat hierarchy as an arrangement of knowledge and power as well as a means of organizing production. In this historical and interpretive framework, authority can shape who knows what, whose decisions count, and which choices remain available to others. The organization of knowledge is therefore not necessarily independent of the organization of control. Making a worker's local task simpler can coincide with relocating understanding and discretion into managerial routines. Productivity alone does not describe that redistribution. [W093, §§1–3]

Their comparison of Taylorism and Toyotism concerns different ways of using and organizing workers' knowledge, rather than a simple contrast between an organization possessing knowledge and one lacking it. Participation in improvement and discretion over execution need to be read alongside the surrounding authority relations. The consequences of local simplification depend on who acquires the corresponding knowledge and power. [W093]

Tainter: complexity as a costly problem-solving investment

Tainter's selected introduction, definitions and discussion of problem-solving costs present social complexity as an investment made in response to problems. More elaborate organization can provide benefits, but it must also be supported. As solutions accumulate, the resources required to sustain institutions and specialized activities can become consequential to sustainability. In this argument a successful response to an immediate difficulty can create continuing obligations rather than simply remove a problem from the system. [UX-W006, printed pp. 91–94]

The relevant material is historical and organizational reasoning about returns and support requirements. Tainter expressly cautions about transferring ecological generalizations to societies. [UX-W006]

Work and care are different evaluative concerns

Murphy and Parry examine household sustainability through work and care. Counting domestic activities as unpaid work can reveal effort hidden by monetary accounts, but the authors argue that this description can also lose personal and relational aspects of the activity. Their selected care discussion reconstructs scholarship in which maintaining people and their world is an ongoing relational activity. Who undertakes care, and whether adequate care occurs, are consequential questions in addition to the number of hours recorded. [GL-W006, care section]

Their integration discussion consequently does not equate a more equal division of household work with every possible improvement in care. Technology can alter expectations, standards and relations as well as the duration of a task. Its relevance is the more discriminating question: which work changes, whose work it becomes, and which quality or relation the proposed improvement is intended to preserve. Equal allocation and better care remain distinct normative goals. [GL-W006]

Administrative burden: three kinds of cost

Moynihan, Herd and Harvey distinguish learning, psychological and compliance costs in citizen–state interactions. Learning includes discovering that a programme exists, determining eligibility and

understanding its procedures. Compliance concerns the actions required to meet those procedures, such as supplying information or satisfying process requirements. Psychological costs concern experiences such as stress, stigma or loss of autonomy associated with the interaction. These categories distinguish burdens that an account of administrative processing time alone can miss. [GL-W018, introduction and table 1, printed pp. 46–47]

The allocation of these costs between citizen and state can be political. A procedure that is inexpensive for an agency may require the applicant to assemble information, interpret requirements or repeatedly establish eligibility. Conversely, reducing citizen effort can require the agency to perform additional work. The authors also recognize that procedures can protect legitimate public values. “Minimum burden” is therefore not automatically the institution’s sole objective. [GL-W018]

What the administrative-burden review covers

Halling and Bækgaard’s selected systematic-review passages distinguish government actions, citizens’ experienced burdens and subsequent outcomes. These are links in an explanatory account, not interchangeable observations. A formal requirement can exist without affecting everyone in the same way; an experienced difficulty does not by itself identify which state action caused it; an outcome can have several causes. The review identifies a larger research agenda while flagging missing causal links and questions about legitimacy. [GL-W019]

Its corpus comprises 119 English-language studies from 2012 to February 2023 within the administrative-burden framework. The authors note the predominance of Western settings. [GL-W019]

Relational autonomy: dependence does not settle the question

The selected Stanford Encyclopedia account distinguishes causal from constitutive relational theories of autonomy. On a causal account, relationships and social conditions can develop or undermine a person’s capacity for autonomy. Support from parents, teachers or friends can enable agency; oppressive socialization can impede it. On a constitutive account, certain interpersonal or social conditions belong to what autonomy itself consists in, rather than merely explaining how it arose. The distinction changes what would count as an adequate explanation of a person’s autonomy. [GL-W040]

The selected discussion of stronger substantive and social-relational views also considers severely constraining relationships in which routine aspects of a person’s life remain under another’s control. Subjective endorsement alone need not settle the autonomy question for those theories. Other positions resist making every preferred way of life subject to the same substantive test. The entry presents a controversy, not a single established metric. Dependence on a person, institution or technical aid cannot simply be read as absence of autonomy. The quality and structure of the relationship matter within the competing accounts. [GL-W040, selected lines 246–260]

Alienation: technology and its social employment

Leopold’s selected reconstruction separates objectification from alienation and distinguishes a criticism of technology’s social employment from a rejection of technology as such. Producing an object does not, merely because the product exists outside its maker, establish an objectionable separation. A theory of alienation must identify the relationship that has become problematic. Similarly, a machine’s technical capability does not determine the social relationship in which people use it. [GL-W049, objectification/fetishism discussion]

The entry’s selected later discussion raises problems of measurement and of connecting a diagnosis of alienation to a practicable account of unalienated production. This prevents a short circuit from “less

effort" or "more automation" to emancipation. Whether an arrangement improves control over activity, its products or its social meaning is a further question within the theory being discussed. [GL-W049, §§6.2–6.3]

Peripheral case: coordinated roles in a traditional account

China's National Health Commission's 2017 institutional description presents traditional Chinese medicine as treating a person's condition holistically and tailoring formulations to that condition. Its selected account assigns differentiated roles within a formulation: a chief ingredient, an ancillary ingredient, an ingredient intended to reduce adverse effects, and one intended to coordinate the others. The source is relevant here as a public description of how a tradition conceptualizes an organized combination, rather than as evidence that every ingredient has the proposed effect. [GL-W061]

The NCCIH public evidence orientation distinguishes modalities and outcomes instead of validating traditional Chinese medicine as a unitary whole. It reports that some physical or psychological approaches may help particular outcomes, while results for herbal products vary; it also identifies contamination, substitution and other safety problems. The relevant lesson within this source is that an account of a coordinated formulation must remain distinguishable from evidence about specific interventions and harms. [GL-W062, selected effectiveness, safety and definition sections]

Effort, care, autonomy and power

These source accounts leave their evaluative objects distinct. Task effort, allocation of knowledge, political control, experienced burden, adequacy of care, autonomy and alienation cannot be substituted for one another merely by calling an arrangement efficient.

S96 Negotiation: preparation, bargaining and agreements

Negotiation can change the proposal under discussion, the alternatives available outside it and the parties' understanding of what they want. Reaching an agreement also creates questions about representation, implementation and continuing relationships. The accounts developed here examine these activities through practitioner methods, training exercises, reviews and experiments. Institutional bargaining and brokers have their developed account in S15; deception during negotiation is discussed in S93.

Preparation and the choice to negotiate

Алла Стремовская's 2012 discussion makes strategy depend on the importance of substantive results and of the relationship with the other party. Avoidance, accommodation, competition and cooperation consequently have different purposes. Her account includes the possibility that someone already satisfied with their position has little to gain by entering a negotiation. A presidential candidate leading in the polls, for example, may decline an opponent's request for additional debates. Preparation includes identifying other parties' strategies and adjusting one's own conduct as the exchange develops. The article brings together earlier writers' approaches and examples. [NN-W007]

Aníbal Sierralta Ríos places commercial negotiation within institutions and their histories. People represent firms, states and other organizations whose policies, market positions and experience shape what their representatives can pursue. His account of negotiator education recalls the OEA–Brazil initiative CICOM and its contribution to training in Latin America. Preparation in this setting concerns the organization represented as well as the person speaking: institutional experience and commitments enter the encounter through the negotiator. His opening discussion gives these collective circumstances a place alongside the individual's conduct. [NN-W008]

Fisher and Ury: interests, options, standards and alternatives

Roger Fisher and William Ury's Getting to Yes, first published in 1981, develops principled negotiation through four connected practices: separating interpersonal difficulties from the substantive problem, investigating interests behind positions, generating possible agreements and discussing standards for choosing among them. The parties can remain concerned about their relationship while examining a disputed question on its merits. Their method asks negotiators to discover what a demand is meant to accomplish, rather than treating the demand as the only possible expression of the interest. [NN-W001]

Their orange illustration makes the distinction concrete. Dividing the fruit appears to require a compromise until the parties discover that one wants the flesh to eat and the other wants the peel for baking. The uses differ, so an exchange can satisfy more of both parties' interests than equal division. Their interpretation of the Sinai settlement develops a larger example. They describe Egypt's concern with sovereignty and Israel's concern with security: returning territory while restricting military deployment addresses different interests within what initially appears to be an incompatible territorial demand. This is the authors' explanation of the settlement. [NN-W001]

Generating an option and judging it are distinct activities in their method. The discussion of objective criteria asks the parties to examine standards and the reasons for selecting them. In a disagreement about an insurance settlement, the relevant question can become the cost of replacing the destroyed car with a comparable one. In a disagreement about foundations, construction requirements provide a different kind of standard. The proposed change is in what the negotiators argue about: they can examine a basis for the decision instead of merely repeating how far they are willing to concede. Competing standards can themselves require discussion. [NN-W001]

The book's best alternative to a negotiated agreement, or BATNA, is developed through three operations. First, list possible actions if agreement fails. Second, improve promising possibilities into practical alternatives. Third, select the best available option. Thinking about a job in another city is a possibility; obtaining an offer there makes it an alternative against which a current offer can be assessed. The book similarly describes turning possible union action into operational decisions, such as obtaining authorization for a strike. Negotiating power in this account includes work performed outside the exchange. [NN-W001]

The resulting alternative supplies a basis for accepting or rejecting an offer. Fisher and Ury also ask negotiators to investigate the other party's alternatives, including whether that party overestimates them. If both sides have attractive alternatives, discovering that no agreement would improve their positions can itself bring the negotiation to a satisfactory close. Their treatment therefore joins the creation of possible agreements to the development of a feasible course of action without agreement. [NN-W001]

James White credits the book's search for mutual gains but argues that it neglects distributive bargaining, where an additional benefit to one party imposes a cost on the other. His labor-contract example distinguishes changing an insurance provider to obtain better services at lower cost from bargaining over wages that reduce the profits available to shareholders. Discovering a mutually beneficial adjustment does not, in his account, eliminate the subsequent dispute over distribution. [NN-W009, pp. 115–116]

White also questions the distinction between objective criteria and persuasive rationalizations. A seller may obtain different prices from buyers with different needs, while numerical tables for pain and suffering can give subjective judgments an appearance of objectivity. He regards standards as useful arguments, but disputes the suggestion that they generally remove the influence of bargaining power. His objection concerns the scope claimed for the method, alongside his appreciation of its practical techniques. [NN-W009, pp. 116–117]

First offers and the unfolding exchange

John Oesch and Adam Galinsky's review manuscript examines four connected questions: who opens, how extreme the opening is, how it affects the final agreement and how participants subsequently evaluate the outcome. An opening can convey information about a bargaining position, signal an intention or establish a reference point. Their discussion of resistance to anchoring concerns the recipient's attention. Considering the opponent's alternatives or reservation point, or concentrating on one's own target, can direct judgment away from the initial proposal. They also discuss the possibility that an implausibly extreme offer undermines agreement. [NN-W004]

Wolfram Lipp, Remigiusz Smolinski and Peter Kesting's review of 119 articles distinguishes a first settlement proposal from anchoring, the judgment process that it may activate. A proposal can contain qualitative conditions as well as a numerical amount. Their process model connects antecedents and offer characteristics to the counterpart's response and to subsequent interaction. What the recipient knows and does consequently belongs within the explanation of an opening's effect. Economic results and subjective evaluations occupy different places among the outcomes considered. The review develops comparisons and open research questions across this literature. [NN-W003]

张真 and 皇甫刚 organize influences on integrative negotiation into contextual and process factors. Their contextual category includes cultural values, social motives and emotions already present when formal negotiation begins. Process factors concern the frequency and sequence of strategies and their use at different stages of the exchange. This distinction separates initial circumstances from conduct that unfolds between the parties, allowing a question about a tactic's use to include when it occurs and what precedes or follows it. [NN-W011]

苏勇, 程骏骏 and 吴展's review of cooperative negotiation argues that much of the research has remained dominated by linear and static models. They propose developing dynamic and relational perspectives on the mechanisms of cooperation. Their criticism concerns the way a research setting represents negotiation over time and among people in relationships. Their proposed direction gives the evolving exchange a different emphasis from a comparison of isolated strategies or average outcomes. [NN-W010]

Erica Boothby, Gus Cooney and Maurice Schweitzer's review likewise emphasizes interactions, evolving conversations and real settings. Its relationship discussion treats relational outcomes as part of what negotiation produces. Its cultural discussion includes adaptation and flexible exchange of information alongside knowledge of the setting. The authors also examine how simplified role-play paradigms shape available evidence. These accounts distinguish a settlement's economic terms from whether agreement is reached and from what happens to the relationship between the participants. [NN-W005]

Discovering trades and making an agreement operate

The Organisation internationale du Travail's 2010 training manual makes differences in preference tangible through an exchange between Mariama and Abdoulaye. Mariama has a green sweet and two orange sweets, dislikes green and particularly wants yellow. Abdoulaye has a yellow sweet and two brown sweets, dislikes yellow and particularly wants green. Both also value variety. Exchanging the unwanted green and yellow sweets benefits both; exchanging an orange for a brown creates a further opportunity. The exercise asks participants to discover interests that a simple division of the stock would conceal. [NN-W002]

The manual carries the exchange into the work of establishing an agreement. Participants assemble acceptable options, prepare a draft and check whether the parties understand it in the same way. Responsibilities then require specification: who will do what, where, when and how. Evaluation, follow-up and procedures for difficulties make the agreement an operating arrangement. Partial, trial and time-limited agreements appear among the possibilities when a final settlement is difficult. Its labor-dispute setting also includes consultation with constituencies and assistance from third parties. [NN-W002]

Maurice Schweitzer and colleagues' handbook chapter distinguishes opportunity negotiations from disputes arising after loss or conflict. Preparation includes recognizing the setting, identifying issues, gathering information and considering alternatives and the counterpart. Its account of logrolling develops exchanges across issues valued differently by the parties. Bringing a start date into a salary discussion, for example, introduces a further dimension on which preferences can differ. Its phases extend through closing and subsequent behavior, placing continuing relationships and implementation within the subject. [NN-W006]

The handbook also distinguishes teams negotiating against one another from negotiation among several parties. A team can bring several participants to the interpretation and exchange of information. Several parties introduce additional questions about coalitions and whose agreement is required. Economic gains and relational outcomes remain separate matters in the chapter's assessment of group negotiation. These distinctions connect preparation and exchange to the people entitled to participate and to the rules under which a decision can be made. [NN-W006]

Decision rules and procedural fairness

Bianca Beersma and Carsten De Dreu's experiments assign participants to represent a bakery, flower shop and grocer planning a joint market. They negotiate design, temperature and rental-cost allocation with private profit schedules. Their first experiment involves 90 students; the second involves 141. The instructions vary social motives, and the second experiment also varies the decision rule. Unanimity

requires all three parties to agree. Majority rule allows two to settle, making it possible to impose an agreement on the third participant. [NN-W025]

Under majority rule, prosocial groups report greater procedural fairness than egoistic groups; the corresponding difference does not appear under unanimity. Joint agreement points and participants' fairness ratings are distinct outcomes. The experimental construction makes the mechanism of exclusion concrete: the minority's consent has a different role under the two rules, while motive instructions change what the participants are encouraged to pursue. The available coalition is a feature of the task, distinct from what every group actually chooses to do. These experiments give a particular institutional form to the broader bargaining and veto questions developed in S15. [NN-W025]

S97 Building, maintaining and renewing personal and professional networks

A relationship can provide companionship, belonging, advice or practical assistance. Its development involves encounters and communication over time, while changes of workplace, residence or activity can alter the opportunities for contact. The accounts here examine how people form relationships, sustain them and reconnect after an interval. Friendship and mutual support are purposes in their own right alongside professional access and advancement.

Encounters and the development of a friendship

Claire Bidart describes sociability through repeated encounters that acquire a shared history. As a relationship develops, its participants cease to be interchangeable: knowing this person involves a history that another person does not simply replace. Her discussion distinguishes a network centered on an individual from the network of a collective. Specifying which relations are included is part of describing the network. This account gives particular relationships and their development a place within a picture that might otherwise appear to consist of nodes and connections alone. [NN-W018]

Jeffrey Hall examines friendship formation among 355 recently relocated adults and begins a nine-week longitudinal study with 112 first-year students. Participants identify acquaintances, report time and activities together and describe their closeness. Shared work or study does not carry the same association with closeness as chosen social time: leisure and everyday interaction help distinguish developing friendship from merely being present in the same workplace or classroom. The studies examine both the amount of time together and what people do during it. [NN-W013]

In the relocated-adult study, the logistic models put the probability of the casual-friend to friend transition above one half at approximately 164 hours together, and the friend to good/best-friend transition above one half at approximately 219 hours. These retrospective modeled estimates concern different relationship categories among the sampled adults, distinct from the students' longitudinal observations. They describe observed relationships between time and closeness rather than durations that guarantee friendship. [NN-W013]

岡田涼's review brings friendship research together with achievement goals, social goals, social cognition and self-determination accounts. His proposed model distinguishes motivation, conduct toward a friend and the relationship that develops through their interaction. Environmental conditions, including parent–child relations and classroom climate, and individual characteristics influence motivation for friendship. Motivation in turn encourages actions such as initiating conversation, helping and disclosing personal thoughts. The model places these actions between motivation and the closeness, satisfaction, trust and affection characterizing a friendship. [NN-W016, pp. 578–580, Figure 1]

The resulting relationship feeds back into later motivation while also contributing to adjustment and mental health. 岡田 distinguishes possessing social skills from being motivated to use them: someone may know how to help or disclose appropriately without taking the initiative to engage. His discussion therefore follows the emergence of interaction, not just the qualities of an already established friendship. The review proposes this explanatory framework using findings primarily about schoolchildren and university students; it identifies developmental differences and tests of motivational intervention as questions for further research. [NN-W016, pp. 580–582]

Maintenance, motives and reciprocity

Debra Oswald, Eddie Clark and Cheryl Kelly identify positivity, supportiveness, openness and interaction as dimensions of friendship maintenance. Across three studies they develop and confirm a measure and examine the interdependence of friends' behavior. Their findings connect both partners' reported conduct with dyad-level satisfaction and commitment. Maintenance in this account belongs within a reciprocal relationship: what one person does and what their friend does contribute to the relationship-level picture. The dimensions describe a family of maintenance behaviors rather than a count of how many contacts a person possesses. [NN-W014]

本田周二 studies relatively less intimate friendships, asking how motivation relates to communication, responses to conflict, satisfaction and mental health. Participants nominate a less-close friend of the same sex and answer questions about that relationship. Of 247 surveyed students, 211 remain in the analysis after exclusions. The study distinguishes motives arising from enjoyment or personal importance from motives involving obligation, anxiety or internal pressure. It also records communication through different media and responses to actual interpersonal conflict. [NN-W015]

In 本田's analyses, introjected motivation is associated with avoidance and self-concession in conflict and with lower relationship satisfaction. More autonomous motivation has few associations with the measured communication behaviors. The finding gives a particular difficulty in maintenance an explanatory home: continuing a relationship can involve pressure to retain it, and that pressure relates to how disagreements are handled. These are associations in the students' reported relationships. The reasons for continuing contact and the frequency or form of communication remain distinct parts of the investigation. [NN-W015]

李松, 刘峻君, 鲍秀琴 and 陈旭 review interpersonal gratitude in relationship maintenance. They distinguish buffering a relationship against threats from strengthening it after the threat has subsided. Their account also includes maladaptive maintenance, in which preserving a relationship conflicts with a person's wishes or social norms. They propose self- and other-evaluation as explanatory links and call for investigation at the dyadic level. Gratitude thus enters their account in both repair and further development of a connection, with the value of preserving the connection itself remaining part of the question. [NN-W017]

Professional contacts inside and outside an organization

Hans-Georg Wolff and Klaus Moser's 2006 measurement work crosses two distinctions: contacts inside versus outside the organization, and building, maintaining versus using contacts. The resulting six subscales concern conduct, rather than network size alone. Three studies examine the proposed structure and validity. This formulation separates meeting a person from continuing the relationship and from seeking help through it, while keeping the organizational boundary explicit. Their later longitudinal investigation uses this measurement tradition to examine career outcomes. [NN-W019]

In the 2009 study, examples of the measured conduct include using company events to make contacts, catching up with colleagues in other departments about their work and seeking confidential business advice through those contacts. Building an internal relationship, maintaining it and making use of it consequently appear as different activities. External contacts receive the corresponding distinction. The three-wave analysis includes 235 participants using available repeated observations, a different number from the respondents present at the final wave. [NN-W020]

Networking is related to salary level and salary growth in this study. Career satisfaction is related to networking concurrently, with no corresponding satisfaction-growth effect. The different results matter

to the interpretation of professional success: income trajectories and satisfaction trajectories are separate outcomes. The study follows behavior and careers over time; its account does not reduce professional relationships to attendance at events or treat every favorable career measure as the same result. [NN-W020]

Tiziana Casciaro, Francesca Gino and Maryam Kouchaki distinguish deliberately instrumental professional networking from personal support and spontaneously formed relationships. Their experiments examine feelings of moral contamination, while a law-firm survey relates networking experiences to networking frequency and billable work. In that setting, relationships with clients and colleagues are consequential for work allocation. The study makes the experience of forming an instrumental tie part of the account of whether people engage in professional networking. [NN-W021]

The 2024 correction records an independent reanalysis of the available data for Studies 2–4; Study 1's original data were unavailable. In Study 3, eight participants began the survey twice but completed it only once, and the analyses used their complete entries. The correction also clarifies the office-location controls in models predicting billable hours; supplementary controls did not change the key findings. In Study 4, correcting a coding error involving two participants changes the interaction between networking content and power on cleansing-product preference to p = .072. [NN-W022]

Reconnecting after losing touch

Daniel Levin, Jorge Walter and Keith Murnighan investigate the usefulness of dormant ties for work-project advice. They ask 224 Executive MBA participants to reconnect with two contacts after at least three years without communication, one formerly strong and one formerly weak. Participants select people who might help with an important current project, locate them, reopen communication and seek their advice. Ultimately, 129 executives complete both surveys. The comparison concerns the usefulness and features of these exchanges relative to current ties. [NN-W012]

Reconnected strong ties combine novel knowledge with retained trust and shared perspective in the authors' findings; dormant weak ties are also useful. A past relationship can retain something relevant to a new exchange while the participants have acquired different knowledge during their separation. The authors also examine contact selection in a further sample by asking participants to rank possible contacts and try lower-ranked choices. Their account describes purposeful selection and project-related reconnection. Its demonstrated outcome is useful renewed exchange, rather than sustained restoration of a close friendship. [NN-W012]

Mentorship and support communities

Lillian Eby and colleagues' meta-analysis combines 116 independent samples comparing mentored and non-mentored individuals across youth, academic and workplace settings. Their definition involves a more experienced person taking an interest in and encouraging someone less experienced or disadvantaged. The review considers behavioral, attitudinal, relational, motivational, health and career outcomes. Associations are generally favorable and small, with differences across settings and outcome categories. [NN-W023]

Mentorship includes attention and encouragement extended to another person rather than only access sought for oneself. Its developmental and supportive outcomes differ from the salary and networking-frequency measures in the professional-contact studies. Sponsorship through advocacy and allocation of opportunities has its related account in S91, while S92 examines enrolling support for strategic work. [NN-W023]

Giovania Maesima, Lívia Gomes and Daniela Levandowski review 25 studies of internet use and interpersonal relationships. Their account describes online communication as part of existing relationships and as an entrance to new ones. Shared activity and support communities give the practices a concrete setting. Among the populations considered are parents of autistic children and people affected by cancer; exchanges can provide practical knowledge and a sense of belonging. These examples are the review's accounts of the studies it examines. [NN-W024]

The review challenges a simple division between online and offline relationships by examining particular media practices and communities. Communication can continue a connection across separation, support participation in a group or help a new relationship develop. The relevant conduct concerns what participants do together and what they exchange, as well as the medium available to them. Alongside Hall's relocation and university transitions, these accounts show relationships developing and continuing as the conditions for encounter change. Encounters and the pursuit of unexpected connections receive their complementary account in S83. [NN-W024]

S91 Acquiring, maintaining and contesting power

Getting a proposal approved, controlling a resource and holding an office present different problems of influence. An office supplies certain rights; practical dependence can give someone influence well beyond those rights. Support may have to be assembled before a decision, renewed after it or protected against a rival. The accounts developed here examine how people acquire room to act, obtain cooperation, establish dependence and sustain positions of control.

Political skill and willingness to act

Liam Maher, Aqsa Ejaz, Chi Lan Nguyen and Gerald Ferris distinguish the ability to act politically from the willingness to do so. Their review describes political skill through four connected capacities: understanding other people and situations, influencing interpersonal encounters, building networks, and appearing sincere. Social astuteness helps someone recognize an opportunity; relationships make resources accessible; interpersonal influence concerns using the understanding effectively in an encounter. Apparent sincerity concerns how the conduct is received. The same overt attempt to please a superior may be accepted as appropriate or dismissed as transparent flattery. [PWR-W001, pp. 6–10]

The distinction matters for interpreting a repertoire of influence tactics. Knowing a tactic's name does not establish an ability to perform it successfully, and possessing that ability does not establish a willingness to incur the relational or reputational risks of using it. The review follows political skill into relationships, reputation and evaluations by others, including supervisors and colleagues. These outcomes explain much of its relevance to careers: an employee's work is performed and assessed among people whose cooperation, information and judgments affect what becomes possible. [PWR-W001, pp. 5–10]

Dependence outside the hierarchy

Michel Crozier's investigation of French factories gives practical dependence a concrete organizational form. Elaborate rules specify work, rights and relations between occupational groups. Seniority governs movement within a category, while recruitment from outside limits promotion between categories. Security and impersonal procedures reduce the personal discretion of supervisors. Yet machine breakdowns interrupt the otherwise regulated activity, and repairs depend on the maintenance workers. Production workers consequently experience dependence on people who are not their formal superiors. [PWR-W002]

Crozier asks why breakdowns become so consequential in these particular factories. When much of working life is specified beforehand, the events escaping the rules acquire exceptional importance. The maintenance workers control the response to one such uncertainty. Foremen object to the relationship but cannot resolve it through their own technical competence. Their position in the official hierarchy does not supply mastery of the practical problem on which production depends. [PWR-W002]

The maintenance group's position also has protection higher in the organization. An operations engineer possesses practical knowledge that formally superior engineers need, while his career prospects remain subordinate to theirs. Crozier interprets his support for the maintenance workers as protecting his own advantage. The resulting relationship runs across ranks: practical competence, dependence and backing reinforce a pattern that the official chart does not describe. Baldwin's strategic bottlenecks in S07 and Stinchcombe's occupational authority in S13 provide neighboring accounts of dependence and knowledge; Crozier develops the relations among the people occupying these particular positions. [PWR-W002]

Robert Greene's *The 48 Laws of Power* presents dependence as a deliberate objective. Its numbered prescriptions recommend keeping others in need of one's contribution and avoiding teaching them enough to dispense with one's services. Other prescriptions concern allowing a superior to feel secure, protecting reputation, controlling disclosure and taking credit for other people's work. These recommendations direct attention to different objects of political action: the distribution of capability, the attribution of achievement and others' beliefs about an actor. Greene treats preserving dependence and managing recognition as conduct to cultivate in the pursuit of power. [PWR-W004, laws 1, 3–7, 11]

Assembling support before a decision

Jeffrey Pfeffer's *Power Play* begins with the difficulty of implementing an agenda among people who have competing purposes. Discretion over money, equipment, space or information can supply a means of recruiting support. Access to influential people can also be valuable. His account treats favors and resource distribution as ways of creating reciprocal support, alongside the substantive case for a proposed change. [PWR-W003]

Laura Esserman's efforts to change breast-cancer care illustrate several avenues of support operating over time. At the University of California, San Francisco, she sought more integrated care, easier enrollment in clinical trials and better collection of treatment information. When local changes met resistance, she continued building scientific and clinical visibility and relationships beyond the immediate organization. She also maintained relationships with patients and former patients, some of whom had substantial resources and connections. In Pfeffer's account, these activities kept adding advocates while parts of the program remained obstructed. [PWR-W003]

The account makes the construction of support part of implementing the clinical agenda. A relationship established outside the immediate dispute could become useful within it. Continuing on several fronts meant that resistance to one part did not stop every means of advancing. Pfeffer's reported case supplies this history of persistence and relationship building; the proposal itself was accompanied by efforts to acquire the practical capacity to carry it through. [PWR-W003]

大嶋玲未 and 廣川佳子 examine preparatory influence in new proposals within large Japanese firms. Their 2020 study draws on interviews with seven managers, analyzed using the KJ method. They identify understanding the informal decision process, preparing the surrounding conditions and 根回し, or advance approaches to relevant participants. The managers needed to know who actually mattered, what concerns would arise and which parts of the approval process could be influenced through prior discussion. [PWR-W005, pp. 61–64]

The actions are specific. A manager might tell a key person about an impending proposal and ask which points mattered. Participants could develop a compromise before a formal confrontation, exchange concessions or speak beforehand to someone likely to obstruct the discussion. An intermediary could approach a person more effectively than the proposer could. Table 2 includes an account of asking a vice-president to supplement a presentation to the president where additional support was needed. The sequence of approaches and the choice of messenger form part of the work. [PWR-W005, Table 2]

Formal procedures continue alongside these activities. The managers' preparation sought both timely approval and cooperation once an approved proposal had to be implemented. The authors describe functions of coordinating intentions, avoiding stalled decisions and making subsequent collaboration easier. Their evidence concerns proposal situations in hierarchical firms. They also observe that short, single interviews may make improper political conduct difficult to disclose. [PWR-W005, pp. 68–69]

Opponents, superiors and subordinates

Pfeffer describes incorporating opponents into an institution as one way of changing a conflict. In his university example, a committee, modest resources and an official role gave critics a stake in the organization they had been challenging. Their attention came to include sustaining the committee itself. Another example concerns helping rivals obtain attractive positions elsewhere: Willie Brown supported assembly rivals' moves to other elected offices. A departure could remove local opposition while giving the departing person an opportunity worth accepting. These cases concern changes in participants' positions and interests, as well as arguments addressed to them. [PWR-W003]

Fong Keng-Highberger, Zhiyu Feng, Kai Chi Yam, Xiao-Ping Chen and Hu Li study a different configuration: a middle manager situated between a superior and subordinates. Their account distinguishes coercive power over subordinates from relational empowerment supplied by a senior manager. They propose that Machiavellian middle managers can cultivate close upward relationships while using abusive supervision downward. The two relations concern different dependencies and different sources of power. [PWR-W006, §§2–3]

The proposed coercive mechanism involves the expectation of punishment. A subordinate may comply to avoid abuse, making the withholding of an undesirable consequence valuable. Upward closeness supplies a different route, through the superior's relational support and empowerment. Perceived threats to the existing hierarchy enter the authors' explanation of why both strategies may intensify. In the laboratory manipulation, subordinates challenged the middle manager's usefulness and threatened to report the manager to the senior manager. [PWR-W006, §§3–4]

The research combines a laboratory study with 338 students in Singapore, a multiwave and multisource field study in China, and a two-wave study of 716 managers from the United States and United Kingdom. The results did not support every proposed relationship uniformly. The association between Machiavellianism and abusive supervision appeared in the laboratory and US/UK study but not in the China field study. The authors' general discussion preserves that inconsistency alongside the broader support for their model. Laboratory responses to a manipulated threat and reports of relationships at work are distinct parts of the evidence. [PWR-W006, §§4–7]

Positions that control access

Alexander Beresford's account of gatekeeper politics examines the African National Congress through party documents, interviews and observations around its 2012 centenary policy conference. Party or public positions can place their occupants in control of access to resources and opportunities. Granting or denying that access helps form relations of support and dependence. Rivals consequently compete over the positions through which distribution is controlled. [PWR-W010]

Beresford follows this relation into patronage networks, accumulation and factional struggles. Access is important to aspiring elites, but also to ordinary people seeking opportunities amid inequality, unemployment and poverty. The relationships are therefore sustained by several kinds of need. His account does not restrict patronage to kinship or inherited traditional authority: loyalties can run through modern party organization and become volatile as alliances and positions change. Holding the gate and fighting over who holds it are connected parts of the political process he describes. [PWR-W010]

This gives organizational reform a difficult setting. Integrity measures, political education and membership procedures address conduct within the party, while access to office and resources continues to support the relationships under examination. Beresford questions whether disciplinary measures alone can resolve that problem. He places gatekeeping within a broader political economy of patronage

and accumulation, explicitly resisting an interpretation of it as a uniquely African deviation. [PWR-W010, conclusion]

В. П. Пугачев's 2006 discussion of public human-resource management uses a narrower conception of micropolitics. He identifies sustained action to obtain, use and preserve power or resource-distribution positions for personal interests conflicting with official organizational purposes. Scarce resources, weak accountability, areas left to discretion and mismatches between individual and organizational interests help explain the opportunities for such action in his diagnosis. [PWR-W008]

Pugachev also recognizes defensive motives. Employees who experience injustice or neglect may seek protection for themselves and those close to them. His account follows increasing suspicion into additional preparation, safeguards and monitoring, with energy diverted into protecting positions and countering other people's moves. This adverse conception of micropolitics differs from the broader treatment of informal coordination in the proposal study. The difference concerns which activities the authors include in the category, as well as how they evaluate them. [PWR-W008]

Ezequiel Alpuche de la Cruz, José Luis Bernal López and Cesaire Chiatchoua examine organizational power through property rights, contracts and hierarchy. In their institutional account, ownership supports appropriation of the benefits attached to rights, while contracts organize exchanges and the allocation of resources among competing uses. Control is consequently embedded in legal and organizational forms as well as exercised through personal relationships. Their discussion connects these forms with bounded rationality, transaction costs and control of uncertainty. S15 develops the neighboring history of property and residual authority in greater detail. [PWR-W012]

Historical networks and room to maneuver

John Padgett and Christopher Ansell reconstruct the rise of the Medici through the marriage, economic and patronage networks of early fifteenth-century Florence. Cosimo's position linked relations that did not coincide neatly with a single unified faction. Their interpretation of robust action centers on multivocality: the same conduct could have coherent but different meanings for different participants. Ambiguity could preserve options when others sought to narrow them. [PWR-W011, pp. 1262–1264]

The historical development of the position is central to their explanation. They argue that Cosimo did not design the centralized party from the outset. Earlier oligarchic efforts to consolidate control helped produce the network configuration around the Medici. The political potential of that configuration became apparent through subsequent events. The authors therefore describe a process in which actors learned to use relationships formed partly through their opponents' actions. [PWR-W011, pp. 1286–1287]

Cosimo's restrained public role and multiple identities also enter their account of control. Different participants could interpret his actions through their particular relations with him, while he maintained discretion and an appearance above the immediate conflict of interests. The authors' explanation combines the relational position with its historical formation and the style of action it made possible. It does not depend on Cosimo having specified the eventual outcome in advance. [PWR-W011, conclusion]

Personality, conduct and responses within the organization

Cameron Anderson, Daron Sharps, Christopher Soto and Oliver John investigate whether disagreeable people attain more power at work. Their preregistered longitudinal studies connect personality measured before entry to the labor market with power assessed about fourteen years later. In Study 1, the 457 participants reported workplace power, control over subordinates and organizational rank. Disagreeableness did not predict greater power overall. [PWR-W007]

The authors' examination of conduct helps interpret the result. Dominant-aggressive behavior was associated with greater power, while less communal and generous conduct was associated with less power. The opposing associations could offset one another at the level of personality. This concerns a different predictor and time horizon from a particular upward or downward influence tactic. The study examines attained workplace power rather than treating a single successful confrontation as the outcome. [PWR-W007]

Juliana Mansur and Filipe Sobral examine employees' experience of organizational politics in three private firms based in Rio de Janeiro. Their 2011 study analyzed 157 valid questionnaire responses. It related perceived politics to satisfaction, reported performance evaluations, neglect and intentions to leave. The authors found adverse relationships with satisfaction and performance and positive relationships with neglect and exit intentions. [PWR-W009, §§3–4]

Their discussion situates these findings in a setting associated with jeitinho, while the observations concern the particular firms and respondents studied. The measured outcomes show how employees experience and respond to a political workplace. The Japanese proposal study, by contrast, develops managers' accounts of using informal preparation to secure decisions and subsequent cooperation. Their objects of inquiry and outcomes differ: one follows employees' perceptions and work responses, the other the conduct of a proposal through an approval process. [PWR-W009, §5; PWR-W005]

S92 Being strategic at work

Being strategic at work can concern the course someone pursues, the understanding brought to a situation, the attention obtained for a problem or the opportunity to participate in deciding what happens. These questions arise in implementing an established direction as well as proposing a different one. The accounts below examine the work through which direction is formed, interpreted and pursued, and the organizational conditions under which a contribution becomes possible or recognized.

Intention and a course of action

Henry Mintzberg and James Waters distinguish intended strategies from patterns realized through action. Their 1985 account places deliberate and emergent strategies along a continuum. An umbrella strategy sets boundaries while allowing others to determine details. Leaders can specify the kinds of activity they want without possessing the information or control needed to settle every action. A process strategy instead influences who makes decisions and the organizational setting in which they work. [ST-W001, pp. 263–265]

Their National Film Board example concerns Norman McLaren's sustained production of experimental films. His individual course could appear emergent from the organization's perspective. The authors question whether he needed an advance intention to continue for decades: a sequence of worthwhile projects could develop into such a course. Their architectural-firm example similarly distinguishes the partners' broad preferences from the performing-arts commissions clients supplied. Both examples make the actor and level of intention consequential to the interpretation of a strategic pattern. [ST-W001, pp. 265–269]

Carrying a direction through everyday work

Linda Rouleau and Julia Balogun's 2008 working paper examines practical knowledge in middle managers' strategic sensemaking. At Radio-Canada, Mary entered a technical directorate without its established technical credentials. She acknowledged that she was there to organize the work, listened to technicians' difficulties and related the proposed changes to problems they experienced. The account gives conversation a specific setting: what could be said credibly depended on knowledge of the people and their work. [ST-W003, pp. 15–18]

Mary also prepared production-meeting reports and discussed them with colleagues and employees. These records helped consolidate compromises between program and technical production and could subsequently inform disputes. Visits to television sets supplied further opportunities to learn and establish relationships. Rouleau and Balogun describe enacting conversations and enrolling networks through semantic and sociocultural knowledge. Their vignette locates strategic implementation in continuing interpretation, documentation and relationships, alongside formal appointment and technical expertise. [ST-W003, pp. 15–18]

Christophe Torset's 2005 conceptual account questions a simple progression from reflection through decision to action. Ideas can remain active without producing a discrete decision; decisions about different questions can affect one another; action can contribute to subsequent understanding. His proposed conception of strategic reflection brings these activities into the same inquiry rather than treating thought as an isolated preliminary stage. [ST-W005, §1]

Torset also treats strategic reflection as an object and an instrument of research. Attention to how ideas arise and circulate can bring contributors into view who disappear when inquiry follows only formal decision makers. He includes analysis and planning practices within this account of meaning creation. His

argument changes what a researcher follows through an organization: the development of an idea and its practical influence may cross several decisions and levels. [ST-W005, §3]

Obtaining attention for an issue

Jane Dutton, Susan Ashford, Regina O'Neill and Katherine Lawrence examine 82 issue-selling accounts from 42 individuals. Their concern is how people influence attention to developments with organizational significance. Presentation can make an issue intelligible through a business case; bundling connects it to valued goals or other concerns. One reported proposal for additional intensive-care space required work with physicians and data specialists on trends, demand and financial considerations. [ST-W002, pp. 716–724]

Involvement and timing extend the process beyond presenting an argument. Earlier conversations can build support before an opportunity to act, while persistence can include setting an issue aside and returning later. The authors distinguish relational, normative and strategic contextual knowledge: knowing the people, acceptable ways of proceeding and organizational priorities affects the conduct of selling. Preparation also appears in unsuccessful accounts. These retrospective accounts describe how participants sought consideration for an issue and understood the outcome. S51 develops the conditions of voice, and S91 examines the construction of support and dependence. [ST-W002, pp. 724–731]

Participation and access to direction

Saku Mantere and Eero Vaara study strategy work in twelve organizations, drawing on interviews with 301 people across hierarchical levels and additional organizational material. Their six discourses concern how strategy is understood, who is entitled to define it and what others are expected to do. They associate mystification, disciplining and technologization with restricted participation, and self-actualization, dialogization and concretization with wider participation. These tendencies coexist in several of the organizations. [ST-W004, pp. 344–347, Tables 2–3]

Mystification gives strategy a special status accessible to a privileged few. Visions and missions arrive from senior management, while others are expected to accept their authority and implement them. In the telecommunications firm of their second vignette, access to strategy documents depended on rank. A manager described learning the organization's intended direction only after promotion. Another employee contrasted earlier experience of participating in planning with a new employer's small inner circle and annual presentation. The authors connect restricted information and unquestionable direction with the expectation that employees lack either the standing or the knowledge to contribute. [ST-W004, pp. 347–349]

Disciplining makes hierarchy and obedience central to strategy work. In a retail chain, military language cast managers as commanders and operating personnel as followers; sanctions for noncompliance formed part of that understanding. A governmental office used an educational version: managers appeared as teachers or parents, and attendance at strategy discussions was an obligation. Being present in a discussion could thus accompany a subordinate role rather than an opportunity to help determine direction. [ST-W004, pp. 349–350]

Technologization places a particular system in command of what can be discussed and measured. An insurance firm adopted a balanced scorecard throughout its strategy process. Senior managers welcomed its greater controllability, but some employees found organization-level specifications difficult to use in departmental work. A team leader could not find a measure that would make excessive workload visible. The authors describe tools becoming control devices: employees were responsible for supplying measures within the system while having little opportunity to question its starting points. [ST-W004, pp. 350–351]

The participation-promoting discourses give people different roles. Self-actualization concerns finding meaning and defining objectives through one's own involvement. In an expert governmental office, people interpreted shared goals in relation to their work, and an employee with an idea could seek an appointment with the chief executive. Dialogization instead emphasizes organized exchange between levels and groups. In a financial organization, management supplied broad plans, teams developed action plans and sent them upward, and targets were discussed before being finalized. Senior management retained its role in setting direction while others had a recognized part in developing it. [ST-W004, pp. 351–352]

Concretization makes strategy intelligible through the continuing specification of activities, rules and procedures. In a municipal health authority, practitioners helped craft guidelines for problems encountered in their own work. Specifications were developed at different levels rather than simply arriving from the center. The distinction from disciplining lies in the roles that the rules make possible: the people doing the work can help construct its practical direction. Mantere and Vaara also describe concretization as a demand voiced within organizations where strategy remained remote and obscure. Their analysis follows how these understandings legitimize or restrict participation through meetings, information access, systems and everyday interaction. [ST-W004, pp. 352–355]

Roles connecting decisions and implementation

李炜文, 高岩 and 李晓彤's 2022 review examines 103 articles published between 1986 and 2020. It organizes middle managers' roles around knowledge management, relationship connections, participation, autonomous decisions and implementation. Knowledge and relationships supply routes through which managers influence decisions, while implementation entails interpreting aims and coordinating their realization. The review brings decision and execution into the same account of middle-management work. [ST-W006, research method and Table 2]

The review distinguishes benefits attributed to participation from a contrary explanation of managerial conduct. Studies it discusses associate middle managers' involvement with correcting senior decision makers' cognitive biases, improving decision quality, aligning objectives and enabling strategic change. It also reports Natarajan and colleagues' finding that managers may make unnecessary resource allocations to demonstrate their management ability and increase personal income. Such activity can reduce operating efficiency. The disagreement concerns what managers do with their position and the consequences of that conduct, rather than whether they take part in decisions at all. [ST-W006, decision participation]

Implementation requires work on both interpretation and activity. The reviewers describe managers developing shared understandings with peers and making strategic aims meaningful in particular settings. They also discuss dividing complex objectives into tasks that employees can understand and carry out. Their reconstruction of Beatty and Lee's study of computer-aided systems in companies in Britain and Canada emphasizes managers solving employees' difficulties through communication, encouragement and control of the implementation process. [ST-W006, strategic implementation]

Emotions enter this account through managers' own commitment and through their responses to other employees. The review describes resistance arising when middle managers question the legitimacy of a major change, alongside accounts of managers responding to employees' negative emotions during implementation. It identifies institutional differences, cultural settings and digital forms of work as questions for further investigation. S86 examines project governance, and S88 follows the continuing responsibilities of product development. [ST-W006, strategic implementation and discussion]

谷口正一郎's 2021 study reconstructs an employee innovation program at Company X, a food manufacturer, through interviews with eight people across functions and ranks. Introduced by a new president in 2011, the program asked employees to report initiatives outside their ordinary duties to directors. Selected initiatives could receive awards and become new businesses or formal company practices. Employees initially differed over what innovation meant for their work. Some associated it with patented products and could see little opportunity for someone in sales to contribute. The president's explanation of an award-winning example helped broaden their understanding of what could qualify. [ST-W008, pp. 26–28, 31–32]

Participation developed through changes to the program and its relationship to ordinary management. One interviewee recalled that it had not initially been required of everyone, but that roughly one initiative per person was encouraged from the second year. Participation became part of appraisal, and a manager reported that subordinates' failure to participate could affect the manager's own evaluation. Employees who preferred routine work consequently experienced an obligation to contribute. A dedicated budget made initiatives possible that otherwise lacked funding; an employee also described approved funding as a commitment to carry the work through. [ST-W008, pp. 32–33]

These arrangements changed supervisors' responsibilities. One interviewee described managers helping teams finish routine work earlier or reduce it so that employees could devote time to innovation. Other departments' assistance and senior managers' attention to individual initiatives made support visible. The program office coordinated the scheme, training and adjustments, while department heads retained appraisal authority. Its office therefore worked with personnel planners on the appraisal system rather than directly assigning employees' evaluations. [ST-W008, pp. 30–34, note 4]

The case also follows difficulty sustaining involvement. Employees could run out of ideas, struggle to combine initiatives with operating targets or feel that another department had better opportunities to win recognition. Adjustments emphasized the reasoning and experience behind an attempt: what need the employee had identified, what hypothesis had been tried and what happened. Reporting formats asked for that explanation, and interviewees described being encouraged to try even when an initiative failed. The author's process model connects these responses with renewed participation. [ST-W008, Table 5, Figure 2, pp. 34–35]

Developing an initiative brought employees into contact with the work of other departments: pricing required finance, volumes required product management and logistics, and an idea could require training staff in stores. Interviewees described improved internal communication, greater visibility for their abilities and links between recognition, promotion and transfers. Some also described reorganizing routine duties to create time for further initiatives. The account follows participation through changes in working relationships, career opportunities and the organization of everyday work. [ST-W008, pp. 35–36]

What judgments and measures observe

许志行 and 曹骞's 2019 review distinguishes attention to strategic activities from attention to kinds of thinking. Its landscape includes management-oriented accounts and game-theoretic reasoning. The reviewers show how conceptions of strategic thinking shape the instruments used to recognize it. An attribute scale begins with proposed characteristics, such as a systems perspective or attention to the future, and turns them into questionnaire items. Differences over the characteristics therefore become differences between the scales. [ST-W007, §§III–V]

The cognitive logic approach described in the review is also a questionnaire. It distinguishes ways of gathering information, through intuition or sensation, from ways of evaluating it, through logic or feeling.

Their combinations produce four proposed modes of thought; the instrument treats strategic thinking as their interaction. Written-text analysis takes another route, examining semantic features of an author's work to estimate tendencies such as risk-taking and foresight. Experimental approaches instead observe choices in structured tasks or games, including how participants recognize similarities and respond to risks and rewards. [ST-W007, §V]

The reviewers' objections are specific to these approaches. The proposed division into four cognitive modes remains disputed. Text analysis depends on the quantity and quality of available writing. Experiments can narrow strategic thinking to reasoning within a game. Widely used attribute scales inherit disagreements over what the capability consists of, making findings difficult to accumulate and compare. The reviewers also question scales developed without sufficiently rigorous item refinement and reliability and validity evaluation. Naming desirable characteristics does not by itself establish a sound measure. [ST-W007, §V]

Their criticism of attribute lists also concerns explanation. They criticize one comprehensive model for simply packaging characteristics collected from earlier authors. They also question whether existing attribute definitions distinguish strategic thinking sufficiently from other thought. The reviewers connect concept definition, attributes, instruments and research on effects: a change in what the concept means changes what subsequent measurements can tell us. Their treatment separates individual from organizational influences and asks how they interact over time. S71 discusses practice and skill development. [ST-W007, §V and prospects]

Stefan Kühl's 2017 interview with Martin Claßen supplies a different diagnostic perspective on managerial judgments. Kühl describes pressure on newly appointed managers to display change even where continuity may be appropriate. He distinguishes formal hierarchy from influence arising through expertise, customer access and informal communication. His argument directs attention to organizational expectations that make some conduct visible as an achievement. [ST-W009]

Kühl's account also places the possibility of retaining a course alongside the expectation of intervention. The workplace questions consequently concern what is being pursued, who can contribute to it and how that contribution is recognized. The historical accounts give these questions different practical settings: a sustained individual activity, the interpretation of an established direction, a concern seeking attention and an institution organizing participation.

S115 Charisma, follower recognition and continuing authority

Recognition and the work of a continuing staff

In the posthumously published *Wirtschaft und Gesellschaft*, Max Weber examines what happens as a relationship sustained by recognition of an exceptional leader has to continue in ordinary conditions. The administrative staff acquires interests in recruitment, economic opportunities and the regular conduct of its work. People initially joined through devotion may later occupy positions with more stable claims to resources or authority. Weber describes ways in which those claims and procedures can change the relationship through which authority is recognized. [T17-W001]

François Chazel distinguishes two layers of Weber's analysis. In his reading, the question of succession is also a question of the basis on which a successor can be recognized. An expectation that authority passes within a lineage differs from attaching it to an office with its own recognized conditions. Chazel connects these variants to different aspects of everyday continuation and objectivation in Weber's writings. The distinctions are part of Chazel's textual reconstruction. [T17-W002]

Continuation and disputed succession

Claudia Jetter compares Joseph Smith Jr. and Ralph Waldo Emerson through what each made of continuing revelation. In her account, Smith gave church-wide revelation a place in a hierarchy. Emerson treated access to revelation as a capacity individuals could exercise and rejected an exclusive institution as its center. Jetter uses the contrast to explain why their accounts of revelation supported different kinds of religious organization. [T17-W003]

Joseph F. Smith Jr.'s later polemic locates succession in the authority of the Twelve Apostles and their recognition by the church. He cites votes in August and October 1844 at which, in his account, the assembled members sustained the Twelve as the presiding quorum. He treats those votes as confirmation of the quorum's right to preside and disputes the authority of those who later ordained Joseph Smith III. [T17-W005]

Community of Christ's institutional history recounts the recognition of a different successor. It describes Joseph Smith III, the founder's son, as coming to believe he had a divine calling, accepting leadership and being ordained president at Amboy, Illinois, on 6 April 1860. It also describes the *Herald*, first published in 1860, as a forum where leaders and members exchanged accounts of their missions and discussed the church's affairs. The ordination and the periodical are two concrete acts in that institution's account of its reorganization. [T17-W006]

Behaviors that observers rate

Antonakis, Fenley and Liechti examine whether speakers and managers can learn behaviors judged charismatic. One study randomly assigned 34 managers and reassessed them after three months. In a second, 41 MBA participants gave a speech and delivered it again six weeks after instruction, with ratings by independent assessors. The authors report increases in ratings of charisma and discuss how those ratings relate to judgments of who looks like a leader or emerges as one. The outcomes are behavior and leadership impressions under specified measures. Continuing authority, in Weber's account, concerns the relationships and arrangements through which people keep recognizing a claim. [T17-W004]

S93 Deception, cunning and misdirection

A concealed intention, a misleading statement and a sporting feint present different problems. Writers have asked how an opponent interprets appearances, how people sustain deceptive practices together, how an audience reconstructs what it has seen and which forms of confidence may legitimately be invoked. Their accounts include strategic prescriptions, exemplary narratives, organizational interpretations, experiments and legal distinctions.

Assessment, appearances and military strength

In 孫子兵法, assessment precedes the instructions to present misleading appearances. 始計 compares rulers, commanders, conditions and discipline. 兵勢 develops the combinations of straightforward and extraordinary operations; 虛實 connects concealment of one's disposition and inducement of the enemy with the concentration and dispersal of forces. An opponent obliged to prepare in many places must distribute defenses, while a commander who conceals the intended point of engagement can concentrate. 軍爭 adds the management of direct and roundabout routes. Preparation, display and deployment therefore work together in the text's strategic prescriptions. [DC-W01]

Lisa Raphals reconstructs a disagreement within early Chinese thought over deception and military strength. Her Sunzi combines direct and indirect operations; she objects to interpretations that make Chinese strategy uniformly indirect or equate its deliberate preparation with effortless action. Her reading of 荀子 distinguishes morally constructive artifice from deception. In 議兵, Linwu praises positional advantage and dissimulation, whereas Xunzi makes popular attachment and orderly government the basis of military strength. He argues that deception cannot break the army of a humane ruler. Raphals also identifies substantial agreement on military preparations. The disagreement concerns the sources of strength, deception's acceptability and its efficacy, within a textual dialogue whose teaching rhetoric shapes the presentation. [DC-W02]

Stratagem collections and princely dissimulation

Frontinus's Strategemata arranges examples for commanders seeking usable precedents. Its preface distinguishes the broad conduct of strategy from particular ingenious stratagems and organizes material around situations before battle, in battle and in siege warfare. Book I.1 tells how Ventidius used an intermediary known to pass information to the Parthians. He displayed fear of the route he wanted them to take and confidence about the route he feared. In Frontinus's telling, the enemy chose the slower crossing, allowing reinforcements to arrive. The intermediary's reporting makes the expression of preference consequential: the intended recipient hears it as intelligence about what Ventidius wants to avoid. Frontinus presents the sequence as an instructive example. [DC-W03]

Chapter XVIII of Machiavelli's Il Principe places dissimulation within the problem of preserving princely rule under changing circumstances. The fox recognizes traps and the lion frightens wolves; neither capacity substitutes for the other. Machiavelli argues that promises may cease to bind a ruler when the conditions that occasioned them disappear. He distinguishes possessing admired qualities from appearing to possess them and remaining able to act otherwise when necessity requires. Alexander VI illustrates the contrast: Machiavelli describes him as exceptionally forceful in his assurances and oaths, exceptionally unreliable in observing them, and nevertheless successful in deceiving people. The example supports his argument that an appearance of fidelity can coexist with conduct that violates it. He also

argues that people judge largely by what they see, while few can test the reality behind appearances. Public recognition and adaptability therefore enter his account of maintaining princely power. [DC-W04]

Practical intelligence and political counsel

Marcel Meulder's 1975 review presents Détienne and Vernant's account of mētis as practical, resourceful intelligence operating amid unstable circumstances. It enables people to overcome another's strength and find ways around natural obstacles. Its reach includes human crafts as well as deliberate deception. Navigation and chariot guidance connect skilled direction with the construction of ships and chariots; in Meulder's reconstruction, the pilot must know the available passages and anticipate chance and the opportune moment. The relevant intelligence deals with changing, multiple and ambiguous circumstances. Meulder places this practical intelligence against its devaluation in Plato and praises the book for challenging an excessively Platonic picture of Greek antiquity. His review thus gives cunning a place in human skill and practical judgment beyond the production of false beliefs. [DC-W05]

Olga Torres reads al-Harawi's manual as political and military counsel associated with Ayyubid state-building. Its chapter outline places officials, counsel and envoys alongside campaigns and sieges. Torres describes advice to receive an enemy envoy warmly, seek counsel and encourage dissatisfaction with the envoy's own patron. She connects this with an attempted inducement of the Green Knight, who refused. The recommended approach and the reported response reveal different aspects of the account: a prescription for cultivating disaffection and a historical reconstruction in which the intended recipient resisted. Torres also discusses uncertainty over whether the manual's intended patron was Saladin or his son. [DC-W06]

Michel Pretalli's introduction to Ruse et magie de l'Antiquité à nos jours follows an earlier inquiry into military stratagems. He recalls that project's definition of military ruse through ingenious artifice, mistaken representations and surprise, then emphasizes its moral ambiguity. An ingenious escape from a desperate situation can appear honorable in one telling and dishonorable or diabolical in another. Moving into magic changes the setting and the meanings attached to secrecy and extraordinary effects. His historical project brings these practices into conversation while investigating the judgments made about them. [DC-W21]

Strategic assessment and reflexive control

Richards Heuer applies cognitive psychology to intelligence assessment. Reinforcing an existing interpretation, he argues, is often easier than replacing it: discrepant information can be absorbed into a settled diagnosis, while organizational commitment to earlier reports makes revision harder. He also examines the planner's vulnerability. Familiarity with a planned message can make its meaning seem clearer than it will be to an opponent. Greater suspicion is not his sole answer to counterdeception. Salient stories may encourage analysts to overestimate deception's prevalence, and more collection or alertness may leave interpretation unchanged. He recommends cognitive aids and a counterdeception staff to give the analytic problem sustained attention. [DC-W07]

Georgy Smolyan explains reflexive control as influencing a decision by supplying the grounds on which another participant chooses. He opens with Lefebvre's illustration of Brer Rabbit pleading with Brer Fox to do anything except throw him into the thorn bush. The plea presents a destination as something Rabbit fears. In Smolyan's illustration, that apparent aversion supplies Fox with a reason to choose the very action Rabbit seeks. The influence works through the grounds of the opponent's choice rather than an overt command. Smolyan connects this with concealed direction that preserves a participant's conviction that the choice is free. His conceptual account concerns manipulation of the decision, including what the

target regards as desirable or threatening. Reflexive control in this account differs from the market reflexivity discussed in S59. [DC-W08]

João Ricardo da Cunha Croce Lopes's professional exposition distinguishes demonstrations of force, false information, effects on the decision process and effects on decision timing. Demonstrations include overt threats and readiness displays; false information includes concealed dispositions and apparent strengths. A timing intervention can seek to delay the opponent by directing attention elsewhere. The repertoire he reconstructs through military writers, including Ionov, therefore reaches beyond lying. Pressure and concealment enter alongside attempts to shape interpretation. These distinctions form part of Lopes's interpretation of reflexive control and its military uses. [DC-W09]

Hayden Peake's review describes Whaley's STRATAGEM as a way of reasoning about deception operations, supported by a collection of surprise cases. He distinguishes its discussion of history, theory, ethics and counterdeception from its summaries of 115 instances of wartime surprise, divided into strategic and tactical cases. Peake reports Whaley's objection to a signals-obscured-by-noise explanation of deception: the objection concerns its failure to predict deception, whereas Whaley proposes looking for clues that point to deception itself. Peake presents this as a way of thinking and studying operations, rather than a step-by-step recipe for assessing or conducting them, and notes Whaley's acknowledgment of chance. His appraisal also raises a problem within the case collection: he reports that one celebrated ruse attributed to Richard Meinertzhagen was fabricated, while accepting the principles it illustrates. Peake thereby distinguishes the standing of an illustrative story from that of the principles associated with it. [DC-W10]

Lying within organizations

Sarah Jenkins and Rick Delbridge relate lying to organizational contexts and social relations. Coercion, compensation and condoning describe different pressures or permissions; cooperative, caring and conflictual relations help explain why employees comply, cope or challenge. Their reconstruction of Sallaz's CallCo study supplies a concrete sequence. Short call-handling targets, a concealed depot contact and promises of same-day redelivery helped staff end calls without resolving customers' problems. Their account distinguishes tolerated coping from an explicit order to lie. Targets, working relationships and the practical handling of calls enter the authors' relational explanation together. [DC-W11]

David Shulman separates notorious criminal fraud, familiar workplace deception and forms easily overlooked in nonprofits. Misleading donors, favoring clients and diverting budgets to preferred causes differ from the crimes that dominate public discussion. His exploratory explanation connects resource pressure, moral commitment, small size, autonomy and limited career paths with opportunities and inducements to mislead. A worker may defend an organization's mission while hiding how its work is carried out. His essay challenges the inference that benevolent purposes make deception unlikely, while warning that American sector divisions and scandal reporting do not establish worldwide rates. [DC-W12]

David Arellano Gault examines corruption as an organizational process. A small lie that passes unnoticed may require further lies to sustain it, while colleagues and authorities can draw others into maintaining the account. Rationalizing language can recast bribery as ordinary assistance to getting work done. Repeated action, shared justification and socialization support his explanation of persistence. His conceptual interpretation concerns how corruption becomes normalized, beyond the identification of an individual's moral defect. Deception enters this account through practices and justifications that participants sustain together. [DC-W13]

Emotion and misleading truths in negotiation

Joseph Gaspar and Maurice Schweitzer explain deception decisions through present and anticipated emotion, as well as incentives and information asymmetry. Present feelings may arise from the negotiation itself or unrelated events. Their emotion-deception model proposes that these feelings affect both what negotiators think about and which opportunities they notice. Anticipated consequences, including guilt, can influence the decision before it is made. Their discussion of incidental anger proposes a path through reduced trust to defensive deception. Their review also questions the dominance of one-shot role-play studies: later discovery, retaliation and continuing relationships can change what an account of negotiation must explain. [DC-W14]

Todd Rogers and colleagues distinguish paltering, the active use of true statements to create a false impression, from false assertions and silence. In Study 4B, buyers with private knowledge about rezoning negotiated with sellers online. Buyers instructed to palter could answer a question about intended use by invoking their history of residential development. The authors report more mistaken seller beliefs in that condition. Buyers obtained more surplus among completed deals; including impasses removed the profit difference. Sellers also judged discovered paltering more harshly than buyers judged themselves. The study connects an assigned response, the inference it encouraged and measured outcomes in an incentivized role-play negotiation. [DC-W15]

Misdirection and sporting feints

Gustav Kuhn and colleagues distinguish misdirection affecting perception, memory and reasoning. Attention control is only part of their taxonomy. Separating a method from its apparent effect can make the earlier action difficult to reconstruct. Incidental actions, convincing apparent shuffles and retrospective descriptions can shape what an audience remembers. The spectator must nevertheless see enough of the effect to experience the impossibility; preventing observation altogether would defeat the performance. Their psychologically based taxonomy links craft practices and research through the relation between a concealed method, the visible event and its later reconstruction. [DC-W16]

Mustafa Alhaj Ahmad Alaboud and colleagues investigate whether a head fake's frequency changes its effect. Participants identify pass direction while a pictured player's gaze agrees or conflicts with it; blocks vary the proportion of deceptive trials. The experiments use static and apparent-motion displays. Conflicting gaze interferes with identifying the pass, and the effect is reduced when deceptive displays become frequent. Immediate trial history also enters the analysis, with differing strength across experiments. The measured result concerns identifying a displayed pass under these conditions. [DC-W17]

Plausible requests and security procedures

Fatima Salahdine and Naima Kaabouch describe security attacks that combine a plausible social situation with a technical or physical action. In pretexting, an invented scenario gives a request for information a reason. In tailgating, a request to hold a door substitutes social cooperation for a credential check. Baiting supplies an apparently desirable object carrying the technical compromise. These sequences differ in what the participant is asked to accept and do. The survey also classifies attacks by contact and medium, organizing a security repertoire that crosses social, physical and technical activity. [DC-W18]

Ruses and protected confidence

Article 37 of Additional Protocol I distinguishes perfidy from permissible ruses of war. Its prohibition addresses killing, injuring or capturing an adversary through betrayal of confidence in protection under

the applicable law. Feigned surrender is one example. Paragraph 2 permits ruses that mislead or induce recklessness without breaching applicable rules or invoking legally protected confidence; camouflage, decoys, mock operations and misinformation are examples. The operative boundary concerns the confidence invited, the intention to betray it and the prohibited result. The legal classification gives a different question from whether the misleading conduct achieves its purpose. [DC-W19]

吉田靖之 examines how rules developed around visual identification bear on contemporary naval deception. His discussion distinguishes navigation lights, an apparent ship identity and a claim to protected legal status. Lights can indicate movement or vessel characteristics without necessarily asserting immunity from attack. His treatment of false flags distinguishes conduct before an attack from conduct during one and identifies specially protected categories that cannot simply be impersonated as an ordinary ruse. Signals and appearances acquire legal significance through the status they claim and the circumstances in which they are used. His analysis makes those distinctions within the naval rules and manuals he discusses. [DC-W20]

The accounts place judgment in different settings: preparing an enemy's interpretation, sustaining an organization's explanation, reconstructing an apparent impossibility and invoking a protected status. Their practical questions involve what others can observe, what they infer, how they respond and which expectations the participants are entitled to rely on.

Part 6 Choice, commitment, and exchange

These accounts examine how people decide what to undertake under uncertainty, when they commit resources, and how they organize exchange and market relationships.

S32 Antifragility and responses to variability

The response to variation is the object of inquiry

The sources considered here ask how variation in an input changes a specified outcome. Taleb and Douady formulate sensitivity to changes in distributions and nonlinear exposure. Axenie and colleagues organize technical and biological applications across several scales. De Florio proposes a scheme for choosing between provisioned resources and more elaborate adaptive behavior. Danchin, Binder and Noria investigate biological maintenance and molecular flexibility.

The relevant distinction is more exacting than survival after disturbance. A system can withstand a change without benefiting from greater variability. It can also improve a selected outcome while imposing losses elsewhere, or benefit over one range and fail beyond it. The sources' definitions and examples therefore require the reader to identify an input, an outcome, a range of variation and a boundary.

Taleb and Douady: exposure beyond a threshold

In the 2012 preprint of *Mathematical Definition, Mapping, and Detection of (Anti)Fragility*, Taleb and Douady begin with the sensitivity of an object or outcome to environmental variability. Their coffee-cup example separates small ordinary disturbances from events large enough to break the cup. The accumulated effect of many harmless movements need not resemble that of one sufficiently large movement. A bridge, building or balance sheet can likewise tolerate a range of stresses and then encounter a sharply adverse region. Their interest is the shape of the exposure, including the threshold, rather than merely the average magnitude of an input. [AFC-W001, 2012 preprint, pp. 1–4]

The paper distinguishes this exposure from a person's attitude toward a gamble. A cup need not possess a utility function for its breaking response to be nonlinear. Taleb and Douady criticize definitions of risk centered on subjective preferences and seek measures applicable to objects with distributions of outcomes. This is their position in a methodological debate; it does not make economic preferences irrelevant to the choice of an outcome or to the distribution of its consequences. Their mathematical development concerns the sensitivity of the selected exposure.

Their initial formal construction uses a reference level, conventionally denoted Ω, a lower stress threshold K, and a one-parameter family of probability densities. The scale parameter is related to a left semi-absolute deviation: the expected magnitude of departures on one side of the reference. A finite first semi-moment is required for that construction. It is not simply an assumption that every distribution has finite variance. They require the relation between the family parameter and this dispersion measure to be continuous and strictly increasing over the specified range, allowing the dispersion measure to parameterize the family. The tail sensitivity then concerns the change in the expected adverse exposure below K as this dispersion changes. [AFC-W001, pp. 5–8]

The restrictions matter. A result for movement within a specified family is not automatically a result for every possible replacement distribution. Nor does a semi-deviation identify every feature of a tail. For nonsmooth threshold effects, the authors also supply a finite-difference version rather than requiring the derivative to exist everywhere. Later two-sided definitions and regularity qualifications extend the treatment beyond the introductory left-tail picture. [AFC-W001, pp. 12–14]

Nonlinear transformation and uncertainty about a model

Taleb and Douady distinguish the distribution of an input from the transformation through which an object inherits exposure to that input. The same disturbance can have different consequences for

different arrangements. Their analysis connects nonlinear transformations with sensitivity to the tails and develops a heuristic for detecting an unfavorable response to additional dispersion. The aim is practical detection even when a confidently specified probability model is unavailable. The authors apply this reasoning to stress testing, forecasting and hidden exposure. [AFC-W001, pp. 2–4 and subsequent transfer-function development]

The account of model error follows the threshold example. If an arrangement is engineered to tolerate a specified range but fails outside it, uncertainty about the stressor distribution can increase the chance of reaching the failure region. Treating the estimated distribution as exact may conceal this sensitivity. Their argument asks what happens to exposure when uncertainty itself changes. It is consequently possible to be robust to familiar disturbances while remaining fragile to error in the assumed range or distribution.

For gambits engineering, the relevant information in this paper is the specified mapping from stress to outcome and the consequences of changing that mapping or its domain. A favorable response over a local range can turn adverse beyond it.

Axenie and colleagues: three scales of response

The 2024 Perspective by Axenie and colleagues treats antifragility as benefit derived from variability in environmental perturbations. It asks how the response should be described in technical and natural systems and emphasizes the need to specify the scale. The article proposes three technical categories—intrinsic, inherited and induced—with ecological, evolutionary and interventional counterparts in its biological discussion. This is the authors' taxonomy. Its applications are reconstructed from cited work. [AFC-W008, pp. 1–2, figures 1–2]

Intrinsic antifragility concerns the response associated with the system's internal dynamics. The authors discuss the shape of the input–output relation and its second-order effects. Inherited antifragility brings in external signals that modulate those internal dynamics. Induced antifragility adds a closed loop: a controller or driving signal guides the system toward a desired region of operation. The categories therefore distinguish sources of response capacity. A favorable curve obtained under active control cannot be described as a property supplied solely by passive internal structure.

Time is also explicit in their account. Technical systems may be designed around known constraints and specifications; heterogeneous natural systems involve entities that differ in speed, strength, proximity and other characteristics. The relevant time scale can alter which response is being measured. The measured response depends on its metric, time scale and environment. [AFC-W008, pp. 2–3]

Curvature, payoff direction and the meaning of improvement

Axenie and colleagues use Jensen's inequality to explain why a nonlinear response to variable inputs can differ from the response to the mean input. For an integrable convex response, the expected response is at least the response evaluated at the expected input. Strict improvement requires appropriate strictness and nondegeneracy conditions; the article's displayed strict signs should not be read as excluding equality cases. The direction reverses for a concave response. [AFC-W008, p. 3, equations 1–2]

The meaning of the response remains essential. Increasing the expectation of a benefit differs from increasing the expectation of a cost. A convex damage function makes variability more harmful, even though the same inequality applies. Likewise a desired effect on one population can be an adverse outcome for another. The article's treatment examples illustrate the dependence on a response curve and dose distribution. [AFC-W008, pp. 3–5, figure 4]

In the evolutionary discussion, the authors distinguish the fate of individuals from changes in a heterogeneous population. Some individuals can die while the surviving population becomes better fitted

to fluctuating conditions. That boundary prevents a population-level claim from being reported as an individual benefit. [AFC-W008, pp. 3–4]

Induced response retains a controller

In the technical discussion of induced antifragility, Axenie and colleagues describe combinations of time-scale separation, redundant overcompensation and variable-structure control. A controller chooses signals intended to move the dynamics into an advantageous operational region. The robot illustration measures such matters as tracking and the speed of reaching desired dynamics under uncertainty. The favorable behavior belongs to a specified closed-loop system, including the control law. [AFC-W008, pp. 4–5, figure 5]

The traffic applications are described in terms of response to disruptions, including their amplitude and timing, and comparisons with particular baseline controllers. Machine-learning approaches are also reported. The intervention changes how the system responds, and the claimed benefit is assessed against an operational objective and alternatives.

De Florio: fidelity and the cost of adaptation

De Florio's preliminary 2014 paper begins from fidelity: correspondence between figures of interest in two communicating domains. A physical quantity and its computer representation provide one example. Real time and a system's representation of time provide another. The relevant engineering problem is not only whether the device continues running, but whether its representations and behavior remain sufficiently faithful for the application. The paper develops an algebraic treatment of the correspondence and a behavioral treatment of the changing relation between system and environment. [AFC-W009, arXiv version, pp. 1–5]

One design approach anticipates conditions and provisions resources through worst-case analysis, redundancy or diversity. De Florio calls attention to its simplicity and low nonfunctional overhead. Unfavorable events can be masked within the provisioned range without extensive runtime reconfiguration. The corresponding assumption is that the system and its intended environment are sufficiently known and stable. Reserved resources are part of this arrangement, even when the runtime behavior is simple.

The behavioral approach allows more elaborate awareness, planning and adaptation as conditions drift. Such capabilities can be necessary in open cyber-physical settings, but they introduce overhead and can complicate the very timing analyses on which guarantees depend. De Florio therefore does not equate more adaptation with a costless improvement. He explicitly frames a tradeoff between the advantages and costs of the approaches. [AFC-W009, pp. 4–5]

His proposed scheme monitors fidelity and environmental conditions, maintains a comparatively simple model while conditions remain safe, and invokes more complex reactive or proactive models when identity is at risk. Actions can include self-reconfiguration or entering and leaving relationships with neighboring systems. The listed relationships include improving one's own position at a neighbor's expense and mutual assistance.

The final learning step measures attempted solutions, ranks them against past solutions, retains conclusions and updates models. That stored experience is the basis for the conjecture that repeated challenges might improve future response. Monitoring, comparison, memory, model revision and implementation remain active functions. The title's expression, *Antifragility = Elasticity + Resilience + Machine Learning*, is a conceptual proposal, not a measured additive identity. The paper presents a

preliminary scheme and conjecture without a simulation result demonstrating effectiveness. [AFC-W009, pp. 5–7]

Danchin, Binder and Noria: maintaining a cell rather than merely reproducing it

Danchin, Binder and Noria approach antifragility through tinkering, molecular flexibility and the persistence of living systems. Their account distinguishes stationary life from reproduction. Their thought experiment asks what functions would be needed to construct a long-lived organism. A cell that is not multiplying still needs a maintained envelope, metabolism, energy supply, import and export, sensing and the replacement of components. Dormancy and continued low-level metabolism are distinct possibilities. [AFC-W013, introduction and §§2.1–2.3]

The paper describes salvage and turnover as ways of reducing dependence on fresh environmental supply. It also distinguishes functions whose redundancy or low demand permits gradual deterioration from those required at the front line of maintenance. Molecular chaperones are important in its account because temperature changes and ageing can both disrupt folding. Repair and refolding cost energy; when they fail, degradation and replacement must occur. The authors discuss degradation machinery that consumes energy even when the underlying chemical breakdown is energetically favorable. Selection and orderly processing, rather than the mere release of energy, are part of the biological work being described.

The relevant observation about long proteins is deliberately puzzling. Their production can be vulnerable to transcription or translation errors, yet the authors identify large proteins in important maintenance and information-processing machinery. They discuss flexible regions and report experiments from other sources as evidence relevant to possible functional advantages. A molecule can have a vulnerability in its production and a different advantage in its operation.

Flexibility, ageing and the boundary of biological benefit

In their discussion of flexible molecules, Danchin and colleagues consider how modifications need not always be damage. Some modifications participate in regulation or protection, and time-dependent structural changes may alter interactions with control mediators. A protein's changing state can, in their proposed account, help time a later response. Cooperation among imperfect molecular clocks is another possibility they discuss. The question is whether some molecules can acquire useful functional properties after synthesis, including under challenging conditions. [AFC-W013, §§3–4]

The authors present aspects of the antifragile interpretation of protein ageing as proposals and questions for investigation. Their account contains reported biological examples, explanatory conjectures and an agenda for further study. Expansion or adaptation of a lineage can involve turnover, destruction and replacement of organisms or components. [AFC-W013, §5]

Flexibility has a material carrier; retained information requires continued physical processes; and maintenance distinguishes functions by their timing, replaceability and importance.

Related book and risk-analysis critique

Taleb also develops antifragility in *Antifragile: Things That Gain from Disorder*. [AFC-W002, publication description]

In his abstract, Terje Aven argues that antifragility contributes to risk analysis by emphasizing how risk and performance develop over time. Some exposure to variation, uncertainty and error can support improvement at later stages, making the time horizon part of the appraisal of risk. He relates this argument to established risk-management ideas. [AFC-W006, abstract]

Reading across these accounts

Taleb and Douady analyze sensitivity to variability; Axenie and colleagues distinguish sources and scales of response; De Florio considers provision, monitoring and learning; Danchin and colleagues examine maintenance and flexible molecular processes.

S34 Optionality, flexibility and commitment

Arrangements that preserve a later choice

The real-options passages and the scholarly reconstruction of Katsenelinboigen concern decisions whose consequences cannot all be specified in advance. An intermediate arrangement can preserve a future action, improve a position or avoid premature commitment.

Smit and Trigeorgis: cash flows, flexibility and strategic interaction

The opening of *Strategic Investment: Real Options and Games* links corporate finance and strategic planning. Smit and Trigeorgis distinguish appraisal of projected cash flows from the additional value that managerial flexibility and strategic interaction can bring to growth opportunities. Their framework treats those opportunities as a package of real options actively managed by a firm, with competitors and new technologies capable of changing their value. It is therefore an account of contingent choices in a competitive setting, not simply a favorable valuation of uncertainty. [AFC-W017, chapter 1, selected opening, printed pp. 3–7]

Their diagram relates project appraisal, planning of growth opportunities and competitive strategy to an expanded valuation. Discounted cash-flow analysis contributes one part; real-option reasoning considers adaptive capability; game-theoretic industrial organization addresses the interaction with competitors. The point of linking the approaches is that a firm's action need not leave the strategic environment unchanged. Waiting, expanding or committing can affect, and be affected by, other actors.

The selected deferral passage uses partly irreversible capacity investment to explain why timing can matter. A plant that becomes unattractive after prices fall may not be saleable at its original cost. Exercising the investment opportunity destroys the possibility of waiting for further information before making that expenditure. In such a setting, a project just at a conventional zero-net-present-value cutoff need not be worth undertaking immediately: the authors describe a premium associated with giving up the option to defer. [AFC-W017, flexibility/deferral passage]

Their analogy to a call option separates the opportunity from its exercise. Holding the right to acquire an asset at a specified cost differs from already owning the asset. The passage also states a qualification: postponement to expiration in the analogy assumes no opportunity costs of waiting or dividend-like benefits of holding the asset. Those foregone benefits are not erased by calling waiting flexible. Similarly, the book's opening places competitors' moves in the problem.

An irreversible commitment changes the future choice set. Its value depends on cash flows, the exercise cost, information revealed over time and the opportunities lost by delay. The object is the firm's strategic investment valuation under specified circumstances.

Katsenelinboigen through a scholarly reconstruction

The selected portions of the *Filosofia* encyclopedia entry reconstruct Katsenelinboigen's theory of predispositioning. The account distinguishes material, positional and relational parameters. Material resources form part of an arrangement; position concerns relations such as proximity or centrality; relational parameters in its biological and social examples include relations among entities. A predisposition is an intermediate arrangement whose connections are partly formed, rather than a complete program linking an initial state to a determined outcome. [WME-W016, selected definition and parameter sections]

Chess is the principal explanatory model in the account. A forced combination can be evaluated through a specified sequence toward an outcome, while positional play must assess a state without such a complete chain. Material can be sacrificed to improve a position whose later use remains uncertain. The relevant question is not only how many pieces remain, but what their arrangement allows.

The entry also distinguishes valuations tied to a particular initial position, goal, program and rules from more general valuations based primarily on the rules. Even the latter are not literally conditionless; the reconstruction explains their partial or qualified unconditionality. A broadly useful piece value and its value in a particular position answer different questions. The theory's strategic valuation is meant to support action when the eventual sequence cannot yet be specified. [WME-W016, selected valuation and style sections]

The account describes programming, predispositioning and randomness as different methods of relating stages of development. Programming supplies complete, consistent links; predispositioning prepares partly connected possibilities. The evaluation of an intermediate position is itself the difficult work. The entry's discussion of the calculus of predisposition emphasizes decomposing material and positional parameters and assigning weights; it also retains the role of experience and subjectivity when unique situations lack repeatable frequency data. [WME-W016, selected calculus and subjectivity sections]

A positional sacrifice is a judgment under uncertainty: it prepares possibilities whose eventual use cannot yet be specified. The resulting position can fail to produce the anticipated advantage.

Valuing a future choice and preparing a position

Smit and Trigeorgis supply an attributed argument about valuation, irreversibility, timing and strategic interaction. The Katsenelinboigen reconstruction supplies an account of preparing and evaluating an intermediate position without a complete program. These accounts can help a reader locate questions about retained choice without turning real options and predisposition into synonyms.

S89 Constructing portfolios through relationships among investments

A portfolio is more than a list of investments. Holdings can move together, duplicate an exposure or supply cash at times when liabilities must be met. The following constructions make relationships among investments part of the decision: covariance in portfolio selection, factor exposure in risk allocation and timing in asset–liability immunization.

Expected returns and joint variability

Markowitz's 1952 account begins by separating maximizing expected return from choosing a combination of investments. Expected return alone can favor concentration in the holding with the greatest expected return. Diversification requires another consideration. His mean–variance approach represents the portfolio through both its expected return and the variability produced by the holdings together. [MG-W014]

If x_i is the share assigned to asset i and μ_i its expected return, portfolio expected return is the sum of x_i μ_i. Portfolio variance contains terms x_i x_j σ_ij, where σ_ij is the covariance of returns on assets i and j. The cross terms express how holdings vary together. Adding names therefore does not automatically make risk disappear: correlated movements remain in the combination even when the number of holdings grows. [MG-W014]

With nonnegative weights summing to one and specified probability beliefs, feasible combinations occupy an attainable region of expected return and variance. An efficient combination has no attainable competitor offering a higher expected return without greater variance, or lower variance without less expected return. The construction separates identifying these combinations from the investor's choice among them. It also separates the calculation from obtaining credible expectations and covariances, which the paper leaves as further work. [MG-W014]

When another asset repeats the same exposure

Roncalli and Weisang's 2012 working paper examines the distinction between risk contributions attached to named assets and contributions attached to underlying factors. Their motivating example starts with four assets and introduces a fifth that duplicates one of them. The economic exposure available has not become more diverse merely because the same investment appears under another label. Yet allocating equal risk contributions across the asset list can change the resulting allocation. [MG-W020]

The example tests what the allocation rule treats as its units. Dividing risk equally among five asset labels is a different instruction from dividing it among four underlying exposures. The authors turn toward factor risk contributions to address this dependence on the nominated universe. Their motivating argument gives a reason to inspect exposure beneath asset counts; the particular factors and their representation remain part of specifying the allocation problem. [MG-W020]

Matching assets to liabilities

Redington's 1952 treatment of life-office valuation asks how assets relate to the payments the institution must make. Immunization concerns the effect of a general interest-rate change on that relationship. The starting calculation values asset proceeds and liability outgo consistently. Equal present values establish a balance at the chosen rate; matching first-order sensitivity makes a small change in that rate have no first-order effect on their difference. [MG-W015]

The second-order condition then matters. Positive second-order behavior of the asset-minus-liability value supplies local protection around the valuation rate. The timing of interest receipts enters the

calculation alongside repayment of principal, so matching asset maturity dates alone does not reproduce the construction. Redington distinguishes the term of an investment from the mean discounted time of all the proceeds it supplies. [MG-W015]

His numerical illustration considers a unit liability payable in ten years at a valuation rate of 2.5%. A single coupon-bearing security can need a maturity of 11.485 years to give the asset proceeds the appropriate mean term. Earlier interest receipts shorten that mean relative to the final maturity. The example shows why a longer-maturing security can perform a matching role that its maturity date, viewed alone, would conceal. [MG-W015]

This is a local construction for a small uniform rate change under the stated valuation setting. Exact matching of cash flows is a different construction. The relationship also changes as time passes, rates change or new business enters, requiring reconsideration of the asset arrangement. Redington's objective is meeting obligations; Markowitz's efficient combinations use expected return and variance. Their different criteria explain why a portfolio can be well constructed for one purpose without settling another. [MG-W015]

S90 Position sizing, rebalancing and investment rules over time

Investment rules act repeatedly. A fraction of current capital determines a changing monetary stake; market movements alter portfolio weights; a trigger determines when those weights are restored. The accounts here develop these temporal arrangements and then consider uncertainty about the return distribution used to choose an allocation.

Sizing for repeated participation

Kelly's 1956 paper attaches a value to information through a betting construction. A noisy channel supplies information about outcomes, giving the bettor an advantage without certainty. Betting all capital can maximize expected wealth in the simple favorable game, yet a single loss eliminates the capital required to continue. Repeated participation changes the significance of a favorable expected payoff. [MG-W016]

In his even-money example, q is the probability of winning and p the probability of losing. Betting a fraction ℓ of current capital multiplies wealth by $1 + \ell$ after a win and $1 - \ell$ after a loss. Over many independent repetitions, the logarithmic growth rate is $q \log(1 + \ell) + p \log(1 - \ell)$. Maximizing this expression gives $\ell = q - p$. Both the multiplication and the objective matter: expected wealth and long-run logarithmic growth need not favor the same fraction. [MG-W016]

The monetary stake adapts even when the chosen fraction stays constant. It rises after gains and falls after losses because it is applied to current wealth. Kelly connects the resulting maximum growth to information rate in his specified channel model. The known probabilities, odds and nonterminating comparison define the result; the investor's valuation of a finite-horizon outcome is another question, which Kelly explicitly recognizes. [MG-W016]

Maintaining weights rather than holding quantities fixed

Willenbrock distinguishes a constant-weight portfolio, restored through rebalancing, from a buy-and-hold portfolio whose weights drift. After relative gains, an asset occupies more of the portfolio; after relative losses, less. Restoring weights sells some of the relative gainers and buys some of the relative losers. Maintaining the allocation is an ongoing rule of action, rather than a property secured by the initial purchase. [MG-W017]

His diversification-return analysis distinguishes arithmetic returns from compounded, or geometric, returns. It compares portfolio geometric return with the weighted geometric returns of the component investments. The difference uses that specified baseline. In his 2000–2009 illustration, a portfolio initially split equally between the S&P 500 total-return index and a long-Treasury total-return index is rebalanced annually. Its geometric return is 4.44%, against a 3.32% weighted component baseline, a difference of 1.12 percentage points. [MG-W017]

Willenbrock argues that rebalancing is operative in this increment and examines why attributing the entire effect to a reduction in variance is misleading. Buy-and-hold also has a changing composition: its successful holdings become more influential without corrective trades. The decomposition and the historical example explain these different mechanisms; the stated difference is not a comparison establishing that rebalancing always beats buy-and-hold. [MG-W017]

Deciding when to rebalance

陆靖昶 and 朱坤 distinguish calendar rules from tolerance-band rules. A calendar rule checks an elapsed period; a band rule checks how far actual weights have moved from targets. The observation controlling action differs. A portfolio can reach a calendar date with little drift, or cross a band long before that date. Defining target weights is therefore only one part of specifying the allocation process. [MG-W018]

Their practitioner account relates rebalancing to maintaining intended exposure and discusses frequency, turnover and market movement. Repeated restoration during a trend and restoration amid reversals need not produce the same comparison. The method consists of observing actual weights, testing the chosen trigger, trading when called for and continuing to monitor drift. More frequent trading is not itself a definition of better maintenance. [MG-W018]

Allocating when the distribution is uncertain

Li's 2023 Wasserstein-Kelly formulation addresses uncertainty in the distribution used for growth optimization. Historical observations supply an empirical distribution, but treating that distribution as exact can make a chosen allocation sensitive to sampling and misspecification. The formulation constructs a neighborhood of distributions around the empirical one and chooses an allocation against the worst expected log growth within that neighborhood. [MG-W019]

This changes the optimization problem rather than merely replacing one return forecast with another. The size and form of the permitted departures, together with restrictions on possible returns, determine which adverse distributions enter the comparison. Support assumptions matter because historical observations alone may not include outcomes that the investment can experience. The selected formulation explains how distributional uncertainty enters the decision alongside the growth objective. [MG-W019]

The temporal rules developed here specify different objects: a fraction of current wealth, a target allocation and trading trigger, or a set of distributions over which an allocation is evaluated. Each account makes its object and objective explicit. Financial allocation complements the histories of retained choice and investment timing in S34 and S61, where the decisions concern options and commitments of a different kind.

S61 Late-mover advantage and strategic waiting

Entering later can mean arriving after someone else has paid to discover what works. A follower may encounter a proven technology, informed customers, trained workers and an established market. Delaying an investment can also leave a choice open until better information arrives. These possibilities raise different questions about what becomes available with time, who produces the knowledge and whether the opportunity remains open to the actor who waits.

The investment and organizational literatures examine these questions through accounts of pioneering and imitation, irreversible commitments, observable investment decisions and shared experimentation. The treatment of optionality and flexibility elsewhere concerns the choices an actor preserves; the accounts here develop the timing of investment and the relationships between pioneers and followers. Absorptive capacity provides a related explanation of how firms acquire and use external knowledge. [AFC-W017; AB-W001]

What followers can obtain from pioneers

Marvin Lieberman and David Montgomery's 1988 *First-mover advantages* organizes the benefits of entering early around technological leadership, control of scarce assets and buyer switching costs. Its account of first-mover disadvantages explains why pioneering can also create opportunities for followers. A pioneer may pay for research, the discovery of a viable market and the education of buyers, while later entrants use knowledge and demand that those expenditures helped create. The follower does not necessarily reproduce the pioneer's sequence of unsuccessful trials. [SW-W001]

The authors separate several channels through which this happens. Technical knowledge can spread through published information, imitation and the movement of personnel. Infrastructure and buyer education can serve more than the firm that financed them. A pioneer's marketing may establish that a category of product is useful, making the task of a subsequent seller easier. How much the pioneer retains depends on the ability to appropriate these gains. Patents, secrecy, cumulative learning and specialized complementary assets can give an early entrant advantages that a follower cannot simply purchase. [SW-W001]

Waiting can also resolve uncertainty about the product and the market. An early entrant must choose a design before knowing which characteristics customers will value or which technical approach will prevail. A follower can observe that experience and concentrate on a more clearly defined target. The opportunity is particularly visible when an industry moves from competing designs toward a dominant design. Early entry may help a firm shape that outcome; later entry may let it avoid commitments to designs that disappear. In Lieberman and Montgomery's analysis, those are different possible consequences of the same period of uncertainty. [SW-W001]

Technological change and incumbent inertia provide two further routes. A new technology may make an established firm's knowledge and equipment less valuable while allowing an entrant to start with the newer approach. An incumbent may rationally continue harvesting an existing investment or hesitate to displace its own profitable products. Organizational routines can also make a new opportunity difficult to perceive or pursue. These explanations differ from a follower merely copying a finished invention: they concern the relationship between past success and the firm's ability to change direction. [SW-W001]

Complementary assets can decide which firm captures an innovation's returns. Lieberman and Montgomery discuss EMI's computed-tomography scanner and Pilkington's float-glass process as contrasting illustrations of the ability to retain the benefits of pioneering. Manufacturing, marketing, distribution and the organization of licensing can matter alongside the invention itself. Their question is consequently more precise than whether being first is good. It concerns which advantage an early

position creates, how durable it is and whether a later entrant possesses assets that make the innovation easier to exploit. [SW-W001]

Entry order and the resources of the entrant

In their 1998 retrospective, Lieberman and Montgomery give greater emphasis to the two-way relationship between entry timing and resources. Entering early can build customer recognition, experience and privileged access to resources. But a firm's resources also influence when it can enter and whether it is better suited to pioneering or following. An organization with strong manufacturing and distribution may be able to exploit an opportunity that another firm first developed. A firm without the knowledge needed to enter early may arrive late without having chosen delay as a strategy. [SW-W002]

This distinction changes how evidence about entry order is interpreted. Successful early entrants may have possessed advantages before entering, rather than receiving all of them from being first. Conversely, comparing surviving pioneers with surviving followers can hide pioneers that failed. The retrospective discusses research distinguishing inventors, product pioneers and market pioneers, and shows why the category being defined matters. The first supplier of a technology need not be the first firm to develop the market in which a later product succeeds. [SW-W002]

The authors also locate some resources outside the entrant's own organization. Customers' beliefs, habits and switching costs can preserve an early firm's position, while standards and network effects can make a market more difficult to enter after it has developed. Their resource-based interpretation therefore includes both what a follower can acquire from earlier activity and what that earlier activity may make harder to obtain. [SW-W002]

A profitable investment that is worth postponing

Robert Pindyck's 1991 *Irreversibility, Uncertainty, and Investment* explains why an investment with positive net present value can nevertheless be worth postponing. His opening factory example makes one widget a year forever at no operating cost. Construction costs $800. The widget sells for $100 initially; a year later its price will either rise to $150 or fall to $50, with equal probabilities, and then stay there. At a 10 percent discount rate, immediate construction has a net present value of $300. Waiting a year and building only after the favorable price change is worth about $386. [SW-W004]

The comparison includes the production forgone while waiting. The gain comes from retaining the ability to reject the unfavorable investment. Immediate construction exchanges that choice for an operating factory. Pindyck treats the opportunity given up as part of the cost of committing now. In this example, the difference of about $86 is the value of waiting after allowing for delayed production and discounting. A positive conventional net present value does not by itself establish that immediate construction is the best available action. [SW-W004]

Irreversibility gives the later choice its force. The investor cannot undo construction and recover the full outlay when the price falls. The opportunity to invest must also remain available next year. If construction could be reversed without loss, or if the opportunity would disappear unless taken immediately, the alternatives would change. Pindyck's example makes the information arrive through the passage of time. His separate account of research and demonstration projects, discussed in Learning before commitment, concerns information that early expenditure itself produces. [SW-W004]

Learning from another investor's action

Christophe Chamley and Douglas Gale's 1994 model investigates delay when investment decisions reveal information. Each agent knows whether it has an investment opportunity, while the distribution of

opportunities contains information about the state of the economy and hence the attractiveness of investment. An agent observes how many others invest and can use that evidence when deciding whether to exercise its own opportunity. Waiting can improve its information without requiring it to undertake the first investment. [SW-W005]

The informational relationship is isolated from ordinary product-market competition. Another agent's investment does not directly reduce the investor's payoff by taking a customer or increasing supply. Its action matters because it is informative. That separation makes it possible to examine a pure informational externality: an investment can provide useful evidence to other decision makers, while the investor does not capture all the value of that evidence. [SW-W005]

The timing of information is endogenous. There is no independent announcement that automatically supplies everything the agents want to learn. Information appears through actions, so an agent's wish to observe others affects the very process from which it hopes to learn. The model produces costly delay and the possibility that investment collapses. An absence of investment can itself discourage further investment, rather than providing a neutral pause in an otherwise unchanged sequence. [SW-W005]

Chamley and Gale also examine the interval between opportunities to revise decisions. Faster reaction does not automatically improve the collective result: their analysis shows how a rapid succession of responses can bring investment to an early halt. It explains a mechanism by which individually useful waiting can reduce the supply of informative actions. [SW-W005]

Experimentation as a shared source of knowledge

Patrick Bolton and Christopher Harris's 1999 *Strategic Experimentation* studies actors who can actively generate information while also learning from one another. Each chooses how much effort to allocate between a safe activity with a known return and a risky activity whose underlying quality is uncertain. The risky activity can be good or bad. Its noisy results gradually reveal which state holds. The players observe one another's actions and outcomes, so an experiment undertaken by one player informs the others. [SW-W006]

The players share a changing belief about the risky activity. When they are sufficiently optimistic, they devote their effort to it; when sufficiently pessimistic, they abandon it in favor of the safe return. Between those regions, the symmetric equilibrium divides effort between the two activities. Compared with a team that jointly values all the experimental information, independent players have an incentive to rely on the experiments of others. Each receives only part of the total benefit created by its own information-producing effort. [SW-W006]

Bolton and Harris identify an encouragement effect alongside free-riding. A player's experiment may produce favorable evidence that induces others to experiment in the future. Their subsequent observations then benefit the original experimenter. An action can therefore be attractive partly because it changes the future willingness of other players to supply information. This feedback can sustain experimentation that would be missed by considering only the immediate return and the information generated by one isolated experiment. [SW-W006]

The balance is more complicated than a fixed amount of experimentation divided among more beneficiaries. The equilibrium allocation changes as beliefs change; an additional player changes both the opportunity to free-ride and the prospect of receiving future observations. Bolton and Harris show that individual equilibrium payoffs rise with the number of players, while aggregate experimentation need not vary monotonically. Even an arbitrarily large group does not in general attain the payoff available with full information. These results preserve both the gains from shared experience and the incentives governing its production. [SW-W006]

When competition accelerates commitment

Shinsuke Kamoto's 2006 study of market entry under uncertainty examines the competing pressure to enter before a rival. Two firms face irreversible investment and uncertain demand. The first entrant initially serves the market alone. After the second enters, the first retains an advantage through its ability to choose output first. The model makes the attraction of a first position part of the timing problem rather than assuming that each firm can preserve its investment opportunity independently of its competitor. [SW-W007]

The option to wait, familiar from a single-investor calculation, now interacts with preemption. Waiting for a more favorable demand level can expose a firm to being overtaken. The first-mover advantage can bring entry forward relative to the timing that would be chosen without that race. Kamoto's analysis also considers demand jumps and the risk that demand falls soon after an irreversible outlay. A firm can win the early position while still undertaking the investment at a moment that exposes it to an unfavorable subsequent state. [SW-W007]

This model gives the timing question a different structure from Chamley and Gale's informational externality. There, other investments are informative without directly changing the agent's payoff through competition. Here, who enters first changes the competitive relationship itself. The choice preserved by waiting depends on whether the actor will still be able to obtain the position it wants. [SW-W007; SW-W005]

How latecomers acquire the knowledge to compete

Momoko Kawakami's 2018 study follows MediaTek and MStar into systems-on-chip for LCD televisions. The transition from analog cathode-ray-tube sets to digital LCD televisions shifted many integration tasks into semiconductor components. A chip supplier could supply a larger part of a television's functioning system, rather than one narrowly separated part. Kawakami examines how these later entrants acquired both component knowledge and knowledge of the finished television, allowing them to supply what she calls problem-solving core components. [SW-W008]

Her interview-based account identifies five routes for acquiring component knowledge. The firms transferred technologies and integration experience from their existing businesses, bought intellectual property from specialist suppliers, learned through commercial integration with established chip vendors, obtained technical information through television manufacturers and worked directly with manufacturers on tasks such as picture adjustment. Some transfers were paid and formal; others passed through working relationships and informal circulation of technical material. The possibility of learning from earlier suppliers was realized through identifiable channels. [SW-W008]

The existing businesses supplied different starting points. MediaTek brought knowledge from optical-disc-drive chips, while MStar drew on LCD-monitor chips. Suppliers of complementary components sometimes shared integration knowledge to secure a place for their own products in a recommended design. Over time, the Taiwanese firms could replace some of those components with their own developments. Such exchanges connected a supplier's immediate interest in selling its product with the recipient's accumulation of technical competence. [SW-W008]

Product-level knowledge developed through helping customers solve problems. Newer television manufacturers faced shortages of chips, limited technical support, differing regional specifications and the need to combine panels and software rapidly. MediaTek and MStar offered extensive support and reference designs, including engineers working near or within important customers' development activities. Solving a customer's problem exposed the supplier to requirements and adjustment knowledge

that it could use in later designs. Better support helped it obtain further customers, whose problems opened further learning opportunities. [SW-W008]

Kawakami's history explains more than access to an already developed technology. It describes the accumulation of an ability to integrate components and satisfy product requirements. The later entrants' opportunities included purchasable knowledge and transfers from established firms, but their advance also drew on prior experience and continuing engineering with customers. [SW-W008]

Testing latecomer advantage at the regional level

Dou Lichen and Li Guoping's 2004 analysis of technological innovation and diffusion examines latecomer advantage across 29 regions of China during 1991–2002. Their question is whether a region's distance from the technological frontier supplies an opportunity for faster productivity growth. The empirical analysis uses labor productivity to represent technological levels, patents to represent innovation, investment per worker to represent absorptive conditions and foreign direct investment as a route for external technological diffusion. [SW-W009]

The authors investigate a nonlinear relationship between technological distance and the benefit obtainable from diffusion. A gap creates room for catching up, while the ability to receive and use incoming knowledge affects whether that room becomes productive growth. Their regional comparisons find stronger evidence of effective diffusion among more developed regions; the corresponding relationships are weaker or statistically unconvincing for less developed groups. Being further behind is therefore a different property from possessing an effective route for using knowledge developed elsewhere. [SW-W009]

This is an observational regional study, with proxy measures for innovation and absorption. It adds an empirical account of the conditions under which a latecomer position can become useful. The fuller organizational treatment of recognizing, assimilating and applying external knowledge appears in Absorbing new resources and capabilities. [SW-W009; AB-W001]

S62 Learning before commitment

An organization can begin learning before it decides exactly what to build or whether to undertake a large investment. It can test a technical principle, investigate customers, develop several compatible designs or build the ability to use a technology whose final form is still changing. These activities alter what a later decision can be based on. Their histories concern the organization of investigation as well as the timing of commitment.

The accounts below develop sequential investment, discovery-driven planning, set-based engineering and software-investment management. Information acquisition, experiments and dual control treats actions that serve both operational and informational purposes. Generating alternatives and organizing exploration examines the production and evaluation of alternatives. Here the emphasis is on what is learned before a particular choice is closed, how the knowledge is retained and how subsequent decisions use it.

Information produced by early expenditure

In his 1991 discussion of irreversible investment, Robert Pindyck distinguishes waiting for information from spending to obtain it. His exposition of Kevin Roberts and Martin Weitzman's work on research, development and exploration considers projects that reveal more about their value as successive stages are undertaken. The investor can stop after an unfavorable discovery instead of completing the entire sequence. An initial expenditure can therefore be worthwhile even when a calculation that assumes unconditional completion gives the project a negative net present value. [SW-W004]

The informative activity changes the alternatives available at the next decision. A drug-development program, an aircraft prototype or a demonstration plant can disclose something that the mere passage of time would not reveal. In the sequential-learning formulation Pindyck discusses, the project's underlying value is fixed but unknown. Learning requires investment. Deferring every stage also defers the acquisition of the information that could justify, redirect or end the project. [SW-W004]

Pindyck also describes a market study costing $50 in a version of his widget example in which the factory opportunity is available only in the current year and costs $1,200. The study reveals whether the future selling price will be high or low, allowing the investor to condition construction on the result. Its value lies in changing the construction decision. This gives the early expenditure a different purpose from buying a smaller amount of production capacity: it purchases a better-informed choice about the larger outlay. [SW-W004]

Planning around assumptions that can be tested

Rita Gunther McGrath and Ian MacMillan's 1995 *Discovery-Driven Planning* addresses ventures whose plans contain many assumptions that cannot yet be treated as facts. Their method connects four working documents: a reverse income statement, a specification of the operations needed to deliver the business, a list of important assumptions and a milestone plan. Together these identify what would have to be true for the venture to justify its investment and when those propositions can be tested. [LC-W002]

The reverse income statement starts with the profit needed to make the venture worthwhile and works backward to required revenue and allowable cost. The operations specification then asks what activities, capacity and resources would be needed to produce that result. Putting those two descriptions beside one another exposes assumptions about prices, market size, production, distribution and the time needed to reach scale. The assumption list makes the uncertain parts explicit so that a detailed forecast does not quietly turn them into established knowledge. [LC-W002]

The milestone plan orders expenditures around opportunities to learn. A milestone is useful when reaching it tests an assumption that matters to continuation. McGrath and MacMillan use Kao's entry into floppy disks as a retrospective worked illustration, assembled from public information. They show how prototype production, technical testing with prospective buyers, subcontracted production and sales, a pilot plant and full-scale production could form a sequence. Each stage addresses a different question about performance, demand or the production economics required by the intended business. [LC-W002]

The worked plan does not require all of these questions to be settled before any activity begins. It gives each uncertain proposition a place in the sequence before the commitments that depend on it. Evidence from a technical test can alter an operations estimate; a sales result can change the volume assumed in the reverse income statement. The documents are meant to be revised together, so that a newly learned fact changes the plan rather than remaining an isolated observation. [LC-W002]

McGrath and MacMillan assign responsibility for maintaining the assumptions and updating the linked documents. This makes the procedure more than a list of risks written at the start. A changing account of what is known governs the next commitment. [LC-W002]

Keeping compatible designs open

Allen Ward, Jeffrey Liker, John Cristiano and Durward Sobek's 1995 account of Toyota describes an engineering process in which a broader early exploration can support a faster later convergence. A point-based process selects a particular design and then revises it as problems appear. Because subsystem decisions interact, revising one part can invalidate another team's work. The alternative they call set-based concurrent engineering maintains sets of possibilities, communicates the ranges other teams can work within and narrows those ranges as knowledge improves. [LC-W003]

The authors compare coordination to finding a meeting time. Asking each person for a single preferred time can lead to repeated proposals and rejections. Asking for each person's available times reveals an intersection that can satisfy everyone. In engineering, the corresponding sets describe acceptable dimensions, performance, manufacturing conditions and other requirements. The task is to find combinations that remain compatible across the system while investigating the choices that have not yet been settled. [LC-W003]

Their account links five activities: describing possibilities at both system and subsystem levels; investigating several possibilities in parallel; narrowing them as knowledge accumulates; deriving feasible specifications from that knowledge; and preserving the commitments once the selected set has been settled. Parallel work is supported by information about allowable variation. A team does not need to know one final design to establish that an entire family of proposed shapes would be difficult to manufacture. [LC-W003]

The Nippondenso example concerns advanced component development that begins before a particular Toyota vehicle program. Engineers explore ideas and partial prototypes, combine promising features and progressively reduce the candidates. The reported sequence reaches three designs and five prototypes in its fourth year and a chosen design in the fifth year of an eight-year cycle. Compatible component families and interchangeable parts allow the development to serve several vehicles and customers. The early work creates knowledge that a later vehicle program can draw on. [LC-W003]

Toyota's lessons-learned books give that knowledge a form that can outlast the project. They describe manufacturing limits and relationships between design choices and outcomes, rather than merely recording the sequence of changes that produced the last vehicle. A body designer can use them to recognize troublesome shapes before asking manufacturing engineers to evaluate a completed design. What is retained is a map of workable possibilities, available for a future design that may occupy a different part of that map. [LC-W003]

Ward and colleagues also explain how progressively narrower communications can make independent work more reliable. A later message refines an earlier range instead of replacing one promised point with an incompatible point. Suppliers know how much variation remains and can judge how far to advance their own work. The authors develop these ideas from observed practices and present the relationship to Toyota's development performance as a hypothesis. [LC-W003]

Moving investigation ahead of design completion

Jeffrey Liker and James Morgan's 2006 account of lean product development explains the timing of Toyota's early investigation in greater detail. In the body-development comparison they describe, engineering at other manufacturers often begins in earnest after the stylists' clay model is frozen. At Toyota, substantial investigation of the product and its manufacturing process begins during an earlier study period. Cross-functional teams produce study drawings and examine alternative solutions while the shape and detailed design still have room to change. [LC-W006]

The intended result is system compatibility before individual designs are completed. Structural and manufacturing questions can influence the developing product while alternatives remain available, instead of arriving as demands to revise finished work. The authors connect this front-loading to a more stable execution phase. Their account places delayed selection within an active engineering process: investigation starts earlier even though some final decisions are taken later. [LC-W006]

A cheap test before expensive redesign

Brian Kennedy, Durward Sobek and Michael Kennedy's study, first published online in 2013, distinguishes purposeful experimental iterations from rework caused by premature decisions. In their Teledyne TapTone case, engineers used carpenter's clamps costing $14 on an existing machine to vary rigidity and investigate its effect. A few hours of testing helped establish a relationship between structural stiffness and performance before detailed redesign. The resulting knowledge supported a smaller, less rigid and less costly design than an untested assumption would have suggested. [LC-W011]

The case also describes customer input on cleanable rounded stainless-steel legs and metric components before drawings and tooling fixed the design. The authors advocate expressing ranges and tradeoffs that teams can use to coordinate decisions. The clamp experiment is a concrete instance of learning a design relationship with inexpensive equipment; it did not require first producing a complete candidate machine. [LC-W011]

Representing preferences and feasible ranges

Haruo Ishikawa and Naoko Sasaki's 2018 treatment gives set-based design a computational form for simultaneous structural and control design. They represent both required performance and design variables as ranges with preferences within those ranges. A preference value indicates which portions are more desirable, from acceptable boundary regions to the most desired region. The method propagates the design-variable ranges into predicted performance ranges and compares them with the requirements. [LC-W007]

When a predicted range includes unacceptable performance, the method narrows the design-variable ranges. It evaluates candidate subranges using both desirability and robustness, then seeks common ranges that satisfy the multiple performance requirements together. In this formulation, choosing acceptable ranges and expressing preferences are distinct operations. Designers can retain a region of acceptable choices while still indicating that some parts of it are more attractive. [LC-W007]

The worked example is the controlled rotation of a stepped rectangular arm containing holes. Its six design variables include structural dimensions, the number of holes and three controller gains. Four performance requirements concern rise time, overshoot, settling time and mass. The authors compare two preference patterns over the same initial variable ranges: an equal preference throughout each range, and a triangular pattern favoring the midpoint. [LC-W007]

The different preferences produce different retained ranges while meeting the specified performance targets in the example. The initial arm-width range of 0.12–0.18 metres narrows to 0.135–0.150 metres in the first case and 0.150–0.165 metres in the second. The selected ranges of controller gains also differ. Thus the result includes a family of structural and control choices shaped by both the performance limits and the designer's stated preferences, rather than a single supposedly inevitable optimum. [LC-W007]

Investing in a technological position

Rita Gunther McGrath's 1997 paper distinguishes expenditure to establish a technological position from the later decision to commercialize it. An early research investment can build knowledge and a right or practical ability to pursue a business if conditions become favorable. Commercialization is a further exercise of that opportunity. The organization may instead license the result, redirect it or stop. The initial investment and the final business are therefore different decisions with different information available at their respective times. [LC-W004]

McGrath separates uncertainty that the investor's work can reduce from uncertainty that evolves outside the project. A technical question about whether a design can function may require experiments. An input price or a regulatory decision may become clearer through other events. Some external conditions may also be influenced by strategic action. These distinctions affect whether an organization gains more by waiting, investigating or taking action that changes the conditions under which it might later commercialize. [LC-W004]

Her pollution-control illustration separates the smaller investment needed to establish a technological position from the much larger outlay required for commercialization. The attraction of the later business depends partly on regulation. Initial work can create a position from which the firm can respond if the market develops, without paying the entire commercialization cost at once. Sequential milestones allow the project to absorb information and change direction as this process unfolds. [LC-W004]

The paper develops a real-options logic for these decisions rather than treating the research investment as a miniature version of the eventual business. The relevant opportunity can be created or improved by the investor's own activity. Its value depends on what that activity makes possible and on the decisions that remain available afterward. [LC-W004]

Connecting tests to decisions that can still change

Ian MacMillan, Alexander van Putten, Rita McGrath and James Thompson's 2006 account of real-options discipline connects discovery-driven planning to the management of uncertain technology investments. Their argument requires actual freedom to make a later decision. Dividing an already unavoidable expenditure into accounting stages does not create the same opportunity as funding an investigation whose result can lead to continuation, alteration or abandonment. [LC-W005]

The procedure uses the proposed business, its operating requirements and its assumptions to identify what must be learned before further resources are committed. Its BioBarrier illustration concerns a proposed hospital hand-washing product. Assumptions about the market, production capacity and selling costs influence whether the business could meet its financial requirements. Simulation is used to examine

uncertain outcomes and identify assumptions that deserve attention; milestone tests then address the assumptions that materially affect the next decision. [LC-W005]

The illustrative business makes the connection between analysis and action inspectable. A simulation does not itself improve the product or reveal customer behavior. It helps identify which unknowns matter to the choice of another expenditure. The managerial procedure then links an investigation to the decision it can inform. This extends the earlier discovery-driven account by making the freedom to stop or change the investment part of its real-options interpretation. [LC-W005]

Exploration before a customer has specified the product

Sylvain Lenfle and Christophe Midler's 2002 study at Usinor concerns an upstream supplier developing an innovative offer before a customer has defined a particular product. Their hydroforming case involves shaping metal tubes through pressure and deformation, with possible uses across several markets. The project must discover both what the process can accomplish and which applications are worth pursuing. A conventional development plan with one agreed object and a delivery date does not yet describe the work. [LC-W009]

The authors organize the effort as a portfolio of studies and experiments. Research on forming possibilities, process capability and crash behavior can interact with particular customer applications. Bringing these investigations into a common project makes their combined significance visible and can justify shared experimental equipment. A customer study can redirect technical research, while technical knowledge can reveal an application that was not apparent at the start. [LC-W009]

They distinguish two kinds of achievement within this exploration. A study may produce a sale with little new technical knowledge, or it may fail to produce immediate revenue while supplying knowledge important to several later applications. Their proposed evaluation therefore follows both commercial results and the growth of knowledge across the portfolio. An unsuccessful application can still establish a useful boundary of the process; a successful part can serve as evidence about a broader field of possible uses. [LC-W009]

Tests play a central coordinating role because no finished product yet provides an agreed schedule. Experiments, simulations and presentations to customers challenge the project's starting assumptions and can redirect the exploration. The research program is a provisional organization of what to investigate. It changes as the team learns, rather than requiring a complete specification of the discovery before work begins. [LC-W009]

Lenfle and Midler also describe a masked urgency in this upstream work. A new material or process must be validated in time to enter a customer's own design sequence. A vehicle may seem six or seven years from the market, while the practical choice to include or exclude the innovation comes within two or three years. Learning has to begin early enough to reach that choice. The late market launch does not mean that the opportunity to influence the design remains open until then. [LC-W009]

Making flexibility exercisable in software investment

Philip Weihs's 2008 *Steuerung von Flexibilität* develops a framework for managing software investments with real options. Its selected project-management and case-study chapters follow investment beyond the initial appraisal. The framework separates initialization from use, includes a continuing search for new options and assigns roles to the investment decision maker, an adviser on real options and the project manager. The decision maker owns the relevant choices and commissions the work; the project manager implements the projects through which the investment is realized. [LC-W010]

The distinction matters when the investment is larger than any single delivery project. A first project may establish a platform or capability from which later projects can be undertaken. Those later choices require identification and management during use, not merely a statement of their possible value at the original approval. Weihs permits ranges or qualitative assessments when a numerical option value cannot be established, with further valuation as the parameters become clearer. [LC-W010]

One retrospective case concerns a configuration-management application intended to organize software, plans and documentation, support version control and enable reuse. The project began in 2004, stopped late that year, resumed with changed requirements in 2005 and was then put into service. Weihs's analysis found that the needs actually extended across configuration management, project management and customer relationship management. Treating all of them as additions to one project had obscured what the first investment should contain. [LC-W010]

The thesis proposes two alternative interpretations of that history. One would define the larger investment in business-process support, implement a clearly bounded initial application and retain the others as possible subsequent expansions. The other would explicitly recognize a right to interrupt or abandon the narrower project as its requirements became clearer. These are retrospective redesigns of the investment decision; the case does not measure savings from a framework implemented at its start. [LC-W010]

The comparison across the thesis's cases connects options to practical authority. A project consciously organized in small increments offered more scope for adjustment than projects with fixed long-term plans. Decisions requiring approval through several levels, including a senior body meeting only twice a year, could not be exercised promptly when conditions changed. The ability to choose later consequently depended on who could decide and when, as well as on the technical form of the software. [LC-W010]

A contemporary model of enterprise AI pilots

Gaurav Tewari's 2026 preprint *Pilot Early, Commit Late* formulates an enterprise AI adoption problem with three choices: deploy immediately, wait or conduct a pilot before deciding whether to deploy. The two-date model separates the external technological frontier from the organization's ability to use it. Deployment produces current operating value and learning by doing. A pilot costs money and produces no full operating return in the first period, but increases capability available for the later decision. Waiting preserves the later choice without creating that internal capability in the model. [LC-W008]

The installed system can capture a specified fraction of subsequent technological improvement. That fraction represents the economic effects of modularity, portability and the ability to upgrade without another sunk implementation cost. Immediate deployment can therefore remain attractive when it captures future progress well, produces substantial current returns or teaches the organization a great deal through production use. The model makes these advantages comparable with the flexibility of choosing a later technology. [LC-W008]

Piloting is preferable to pure waiting when its cost is no greater than the expected increase it creates in the value of the later deployment choice. The increase can come from making adoption profitable in more future states or making profitable adoption more valuable. In this formulation a demonstration with little transferable organizational learning contributes little to that option, however polished the demonstration may be. [LC-W008]

The analysis also derives conditions under which faster expected technological progress makes immediate deployment less attractive relative to waiting or piloting. A low ability to capture future improvements favors that result; strong production learning or a large present operating benefit favors earlier deployment. These are comparisons within a theoretical model that omits strategic competition and financing constraints. [LC-W008]

S87 Discovering valuable problems and choosing what to build

Choosing what to build involves representations of needs, technical possibilities and a firm's offering mix. A customer can report a present difficulty without imagining a new product, while a technically attractive concept can fit poorly with other products. The accounts here develop ways of representing product decisions and finding users whose experience contains information about emerging needs.

Representing the product decision

Krishnan and Ulrich organize product-development literature around decisions. Their 2001 review compares the concerns of marketing, organizational research, engineering design and operations management. A product can appear as a bundle of attributes, an organizational undertaking, an interacting assembly or a flow of development activities. These representations make different relationships and measures prominent. Stating the decision helps identify which representation supplies useful information. [MG-W008]

The review distinguishes decisions within a development project from decisions that establish its setting. Concept development, architecture, variants, shared components, prototyping and launch occur within a context of product strategy, project selection and development organization. Choices about an individual component can therefore depend on a product-line decision that the local design problem takes as given. Optimizing one project does not settle which projects should exist. [MG-W008]

Their launch-sequencing discussion illustrates the connection. An earlier introduction can earn profit sooner, yet a new offering can displace sales of another product. Likewise, offering variants and sharing components can change both market coverage and development work. Their review locates these relationships across research traditions, rather than reducing product-development performance to minimum lead time. Its decision map helps a reader distinguish the object being chosen, the objective used to compare alternatives and the related decisions outside the immediate project. [MG-W008]

Finding needs ahead of the market

Von Hippel's lead users have two characteristics: they encounter needs ahead of a significant market trend and expect substantial benefit from solving those needs. Those conditions make them informative in different ways. Being ahead provides experience with a problem others may face later; expecting a large benefit supplies a reason to search for or construct a solution. An early purchaser is not necessarily a lead user, since buying a new offering does not itself establish either condition. [MG-W007]

The inquiry begins with a consequential trend and seeks users who are already confronting its demands. It then examines their needs, adaptations and solutions, before assessing which findings transfer to the intended market. Von Hippel argues that typical users can be constrained by their experience of familiar products and attributes. Lead-user inquiry instead finds people for whom an emerging problem is already practical experience. [MG-W007]

The search can extend beyond the producer's existing customers and industry. Users facing demanding conditions elsewhere may already have addressed a related need: the paper discusses such connections as lightweight materials in aerospace and automotive uses, and controls for commercial buildings and homes. Transfer is a further question, however. A demanding user's solution provides something to inspect and evaluate, rather than a substitute for understanding the target users. [MG-W007]

The IBM component-insertion machine illustrates a user-developed solution moving into supply. In von Hippel's account, the user developed the design and supplied drawings with an order to a manufacturer. Permission to sell the machine more widely followed. The sequence matters: an experienced user's work

supplied a concrete design before the supplier treated it as a general commercial offering. Information about need and information about a possible solution had been produced together in use. [MG-W007]

Lead-user inquiry therefore includes both discovery and evaluation. Finding an inventive user can reveal a possibility; assessing its applicability determines whether that possibility belongs in a proposed product. Product-line decisions remain another level of choice. Krishnan and Ulrich's review concerns how those decisions relate, while von Hippel supplies a particular route for obtaining information about needs and solutions. The histories of inventive and associative methods elsewhere develop further ways of generating concepts.

S88 Organizing product development and evolution

Product management continues through development, introduction, use, revision and withdrawal. Its work includes deciding what information is needed before a commitment, which aspects of a design can remain changeable, and how an existing offering fits a changing portfolio. The following accounts examine processes and continuing responsibilities across that span.

Stages of work and gates of commitment

Smolnik and Bergmann's 2020 review traces the evolution of the Stage-Gate process. Stages gather and analyze information; gates decide whether further resources should be committed. Discovery and screening lead through scoping and a business case toward development, validation and launch, with post-launch review. The distinction between work and authorization explains how a process can organize inquiry while reserving a decision about proceeding. [MG-W009]

The review also describes adaptations involving iteration and agile work. Iterating within a stage and deciding whether to continue the larger undertaking need not be the same event. That distinction allows repeated development cycles to coexist with broader investment decisions. The authors discuss the evidence available for hybrids, including its limitations for some physical-product applications. Their historical review presents a changing family of processes rather than a single unaltered sequence. [MG-W009]

Making changes less expensive

Thomke and Reinertsen define development flexibility through the economic cost of modifying a product under development. Its value depends on what may change and when. Improving a forecast can reduce surprises; changing a process or technology can make a later revision cheaper. Their account considers both routes and explains why a design strategy should reflect the cost of changing its underlying technology. [MG-W010]

Progressive freezing makes commitments piece by piece. A requirement or design feature can be approved when sufficient information is available, while other features remain open. Qualifying a backup material early similarly prepares an alternative before it becomes urgently needed. These arrangements require identifying which changes have consequential downstream effects and which permissions can be granted independently. [MG-W010]

The Boeing 777 illustration makes authorization granular. A part drawing could have six or more release levels, permitting such different downstream activities as purchasing material, designing tools or machining them. Releasing a drawing was therefore a statement about what someone could safely begin, rather than a declaration that every detail was final. The authors contrast this with one release that freezes the entire part at its procurement lead time. [MG-W010]

Their tooling-shop discussion also changes the criterion used to manage a supporting activity. Minimizing the unit cost of a tool can delay an iteration whose timing matters to the larger program. Delivery timing brings that relationship into the measure. Their survey of 391 integrated-circuit projects compares development using technologies with different change costs; respondents were not randomly assigned to those technologies. The comparison supports their investigation of flexibility and development strategy within that observed setting. [MG-W010]

Actual processes and long-lived platforms

Jetter, Albar and Sperry compare prescribed processes with development practice through literature and high-technology cases. Their measurement-tool company maintains a platform over years while adapting interchangeable heads or modules. Several projects can therefore contribute to the evolution of an offering whose life exceeds that of any one project. Platform continuity and project closure are different time horizons. [MG-W011]

The case research describes departures from standard processes, including agile software development and decisions concentrated in the owner of a small firm. Its second phase includes twelve project cases and interviews concerning project management offices. These qualitative accounts show why a standard process may be modified and how monitoring is carried out. They also distinguish disciplined management of projects from choosing a successful product mix: following a process supplies information about execution without by itself determining whether the right offerings were selected. [MG-W011]

Strategy throughout the product's life

Bauch's 2020 process navigator distinguishes product, product-portfolio and enterprise strategy. A product moves through definition, development, launch, growth and improvement toward withdrawal, while its business plan and strategy continue alongside that work. The roadmap communicates a time dimension across functions. It connects intended changes with the continuing offering rather than serving only as a schedule for an initial release. [MG-W012]

The portfolio perspective compares the current and intended offering mix. Introducing one product and phasing out another must be coordinated with product-level plans and enterprise direction. Bauch also identifies supporting work, such as troubleshooting and assisting sales, that competes for attention with strategic work. His practitioner framework makes these continuing processes visible as product-management responsibilities. [MG-W012]

Springer and Miler investigate software product-management challenges through a literature review, ten interviews and an 89-person survey. Reported difficulties include establishing customer value, changing priorities, technical debt, organizational silos and balancing reactive with proactive work. Their categories distinguish core product decisions from participation in related work and orchestration across functions. The product manager's responsibility can consequently extend across activities over which that person has no direct command. [MG-W013]

The study reports perceived frequency and severity. These reports describe difficulties experienced in practice, including continuing maintenance and withdrawal as well as development. Together with the lifecycle frameworks, they give substance to product management after delivery: the offering, its dependencies and the organization supporting it continue to change. S62 develops learning before commitment, and S80 examines the connection between sales work and customer value. [MG-W013]

S100 Founding and developing technology ventures

Who founded the firm and when

Arnold Cooper's 1971 account gives a semiconductor company that nearly failed with its first founders. A later group brought management and capital and got the enterprise operating. The episode makes the founding date a question about what event counts: organizing the original company or assembling the group that made it viable. In another firm, an early founder took up product development before a later founder took responsibility for management and marketing. The histories of the team and the legal organization can therefore begin at different dates. [T02-W001]

Cooper also describes the previous employer as an incubator in a specific sense. Prospective founders encounter one another through their work and learn what their future colleagues can do before forming a venture. In S83 the Nairobi incubator is a program that deliberately connects entrepreneurs with contacts. Cooper's example concerns an employing organization that can become a source of founding teams. [T02-W001]

The changing work of a founder

榊原清則 and colleagues asked founders of Japanese technology ventures which duties they performed as the business developed. The report describes a pattern of founders covering many functions at the beginning and gradually concentrating on line functions. The same broad shift appears among firms aiming for an initial public offering, although the report finds a stronger tendency for the chief executive to combine several functions in that group. The study compares reported allocations of duties across stages and groups of respondents. [T02-W002]

Sarasvathy distinguishes deciding on an objective and finding means to reach it from starting with means already at hand and considering which ends those means make possible. A conventional marketing sequence begins with a market and a selected target. Her example of a business developing through contingent problem solving illustrates the second direction of choice. It explains a proposed logic of founding and the place of opportunity in it. [T02-W003]

Which ventures enter a survival estimate

Arcuri, Russo and Gandolfi matched the Italian register of innovative start-ups with financial records. Their analysis covers 9,171 firms from 2013 to 2021 and excludes firms founded after 2020 and known exits such as mergers or acquisitions. They measure survival from foundation and identify failure from the cessation of annual reporting, with non-failing firms observed through the study cutoff. The three registration criteria they examine concern research spending, staff qualifications and ownership of industrial property. Firms could satisfy more than one. [T02-W004]

Table 4 puts estimated five-year survival for the whole sample just under 90 percent. The curves in Fig. 2 compare firms that met each criterion with the other registered ventures in the sample. Those meeting the research-spending criterion had lower estimated survival in that comparison. The curves went the other way for staff qualifications and for ownership of industrial property. The authors then used a model that included firm characteristics and regional, sector and year differences. In their further models accounting for selection into the criteria, the results for staff qualifications and industrial property stayed positive, while the association for research spending was less clear once the other criteria were included. These are comparisons within the register of innovative ventures. [T02-W004]

The Kauffman Firm Survey uses a different frame. Its baseline report records 4,928 completed surveys and a weighted response rate of 43 percent, with an oversample of high-technology firms. A firm founded in 2004 that closed before the sampling frame was constructed could be absent even though the survey describes a cohort of new firms. The report identifies this interval explicitly. Each survey therefore offers evidence about a defined population of entrants and survivors, while the founding episodes in Cooper describe events before such a count is possible. [T02-W005]

S76 Classical selling models and accounts of buying

Attention, interest, desire, action and satisfaction

AIDAS describes five successive objects of attention in a selling encounter: attention, interest, desire, action and satisfaction. In Pramod Kumar Nayak's teaching account, the seller first establishes rapport and a receptive audience, then develops the relevance of the offering. A presentation can attract attention without producing interest, and interest can remain weaker than a desire to obtain the benefit. Objections and the buyer's concerns enter the work of developing that desire. The seller must then recognize an appropriate point to seek a purchase, rather than assuming that wanting something automatically produces action. [SL-W003]

The Sales and Distribution Management course illustrates the sequence through insurance and family security. The concern for a family's future gives a proposed policy its relevance; the presentation connects the policy to that concern, addresses questions and seeks the purchase. Satisfaction extends the account beyond closing. Delivery, reassurance and subsequent service bear on whether the experience fulfils what the buyer expected. AIDAS consequently gives follow-through a place within the selling sequence rather than treating the order as its sole endpoint. [SL-W004]

What the early formulations meant by a completed sale

Akinori Iwamoto's historical investigation questions the familiar attribution of AIDA to E. St. Elmo Lewis in 1898. He compares dated formulations and their later retelling, including Strong's accounts, rather than treating repetition of an attribution as evidence for it. His reconstruction distinguishes Dukesmith's connected selling argument in 1903–1904, Sheldon's discussion of conviction, resolve and action, and Brown's 1904 ordering of the familiar four terms. Differences among these formulations matter because they concern what completes a sale. [SL-W005] Iwamoto argues that Sheldon can be regarded as the originator and that Sheldon and Dukesmith contributed decisively to formulation; this is the historical conclusion advanced by his reconstruction. [SL-W005]

Conviction that an offering is suitable, a resolve to buy and an actual purchase are different accomplishments. An account that ends with conviction may leave the transition to action unexplained; adding another mental state does not by itself identify the practical step that completes the transaction. Iwamoto uses the early texts to examine this distinction as part of the history of the formula. His argument contests an inherited origin narrative while preserving the differences among the formulations that narrative had grouped together. [SL-W005]

One of the Sheldon discussions reproduced by Iwamoto concerns the retail customer who arrives knowing what is wanted. Attention, interest, desire and resolve may already have developed before the encounter. The seller can then assist completion instead of recreating each state through a fresh presentation. This example qualifies a mechanical use of the familiar sequence: where the encounter begins depends on what the customer has already done. [SL-W005]

The right set of circumstances

The Right Set of Circumstances account directs attention to a seller's use of the situation. Suitable circumstances, stimuli and appeals are expected to secure attention and interest and produce the desired response. Nayak presents it as a seller-oriented situation–response formulation. The practical object is the selling setting and the action taken within it, rather than only the sequence of mental states named by AIDAS. [SL-W003]

The university-course exposition similarly makes the seller's creation or use of circumstances central. Choosing an appeal, arranging the occasion and presenting it to a particular prospect belong to the proposed influence process. The formulation is a teaching account of what the seller should attend to; it does not provide a calibrated prediction of how a given buyer will respond. [SL-W004]

Need, product, source and satisfaction in the Buying-Formula

The Buying-Formula treats a purchase as a set of connections on the buyer's side. A need or problem is connected to a product or service, then to a particular trade name or source, purchase and satisfaction. Recognizing a kind of solution differs from accepting a particular supplier. A buyer may want the product category while remaining unconvinced that this source will supply it satisfactorily. Nayak's teaching diagram makes both connections explicit. [SL-W003]

At the product/service connection and again at the trade-name connection, the account places adequacy alongside a pleasant anticipation of satisfaction. The proposed purchase must answer the problem, and thinking about it must carry an agreeable expectation. The seller's problem-solving work therefore concerns more than making the brand familiar: it may concern whether the product will do the job, whether the supplier can be relied upon, or what experience the buyer anticipates. Satisfaction after purchase feeds the continuing relationship and later buying. The corresponding university-course exposition develops these links as a guide to understanding buying. [SL-W003; SL-W004]

Drive, cues and reinforcement in the Behavioral Equation

Nayak's Behavioral Equation connects a buying response with learning. Its printed expression is B = P × D × K × V: B is the buyer's response, P predisposition or habit, D drive or motivation, K incentive potential, and V the intensity of cues. Incentive potential concerns the product's value or potential satisfaction. A familiar response can thus be supported by habit while still depending on motivation and the attractiveness of the anticipated result. [SL-W003]

The account distinguishes cues that trigger a decision from cues that influence a choice without initiating one. Product colour, weight and price can provide product cues; advertising and conversation can provide informational cues. The categories concern what a cue does in the decision: a price can also trigger action. A rewarding result reinforces the response and can strengthen the predisposition to respond when a cue appears again. [SL-W003]

Multiplication makes a strong claim inside the formulation: if drive is zero, the proposed response is zero even when habit is strong. The slides use that implication to explain why predisposition alone is insufficient. They supply meanings and a learning account rather than fitted coefficients or a probability scale for estimating purchases. [SL-W003]

Confidence, evidence and participation in the customer's work

The chapter of 销售管理实务, edited by 安贺新, compares several ways of organizing a selling argument. GME concerns the seller's own confidence in the goods, the enterprise and personal capability. FABE separates features, advantages, benefits and evidence, making support for a benefit claim a distinct part of the presentation. The six steps printed under DIPDA move through definition, identification, proof, acceptance, desire and action. The related discussion connects discovering a need, showing the offering's relevance, demonstrating it and obtaining acceptance. [SL-W006]

The chapter also distinguishes attending to customer needs in order to sell a fixed product from participating more broadly in the customer's work. Its consultative comparison gives the seller roles as team member, adviser and partner. Two-way information, knowledge of the customer's business,

participation in decisions and subsequent service become part of the relationship. These distinctions appear in the third edition of a work first published in 2009. [SL-W006]

The Changan illustration brings a service adviser and technician into continued contact with the customer. They visit, inspect the vehicle and answer questions; the chapter reports renewed intentions following that contact. The pairing joins technical examination to the service relationship, extending the encounter beyond an initial product argument. [SL-W006]

S77 Questions, diagnosis and commitments in selling

SPIN questions and the development of an explicit need

SPIN places questions inside an encounter that includes preliminary contact, investigation, demonstration of capability and obtaining commitment. In Huthwaite's exposition, situation questions establish relevant background; problem questions elicit difficulties and dissatisfaction; implication questions explore their consequences; and need-payoff questions invite the buyer to articulate the value of a resolution. The next question responds to what has been learned, rather than merely moving to the next letter. [SL-W008]

A difficulty can constitute an implied need without yet becoming a buyer's explicit desire for a solution. Implication questions develop why the difficulty matters, while need-payoff questions help the buyer express what improvement would be worth. Capability can then be demonstrated in relation to that expressed value. Huthwaite treats the questioning pattern as a logical framework rather than a fixed interview script. Relevant preparation also reduces the need to ask buyers for background that a seller could already know. [SL-W008]

Chaker and colleagues' reconstruction of Neil Rackham's 1988 SPIN Selling preserves a distinction about the scale of the sale. In a smaller sale, responding to an implied need may be sufficient; a larger commitment may require developing the seriousness of the problem and the value of acting before presenting a solution. This is the Rackham account reported in the review, alongside Huthwaite's current exposition of the method. [SL-W001; SL-W008]

An advance, a continuation and the work of implementation

The Urgo account first distinguishes a feature, an advantage and a benefit. A feature names a technical or functional capability; an advantage describes its generic usefulness; a benefit connects that capability to the particular customer's expressed problem and valued result. Its illustrative wound-care conversation links recovery time with care costs, staff pressure and the customer's operating targets. The seller's earlier questions establish why these consequences matter to that customer, so the eventual benefit statement answers a developed need rather than reciting product properties. [SL-W009]

Huthwaite's Urgo Medical account distinguishes a concrete advance from a pleasant continuation. An advance commits resources to an agreed action that moves an opportunity forward. Access to a board presentation, a trial with an agreed reporting date, and an order differ from an invitation to send another proposal or meet for lunch. In the report's coding, orders and advances count as successful call outcomes; continuations and no sale do not. An advance can therefore be successful under this scheme without being a completed sale. [SL-W009]

The implementation joined training to coaching and common records. Sellers and managers needed a shared way to recognize what had happened during a call; recorded outcomes then supplied material for coaching and subsequent calls. Tablets, marketing involvement and management support belonged to the same arrangement. The report describes 104 calls by sixteen sellers, with a nested set of 48 recently coached calls. Its comparison concerns this implementation package and selected calls, rather than an isolated experiment on the four SPIN question types. [SL-W009]

Sandler: agreements, pain, investment and the decision process

David Mattson's Sandler account connects rapport and equal business stature with an up-front agreement, exploration of pain, investment, decision arrangements, matched fulfilment and post-sale confirmation. Up-front agreements establish the purpose, conduct and possible outcomes of the present

conversation. They can be renewed as an opportunity develops. Investment includes time and other resources as well as money; decision arrangements concern who will decide, how and when. [SL-W010]

Mattson's budget example concerns a seller who knows that money should be discussed but regards asking about it as rude. The seller postpones the issue until after presenting and receives an indefinite response. His distinction among attitude, behaviour and technique explains why knowing the appropriate question does not ensure that it will be asked. Beliefs about the conversation affect enacted behaviour, and a learned technique can remain unused. [SL-W010]

The presentation follows qualification and is matched to what has been established. Post-sale work confirms the decision and next steps, including the possibility of buyer remorse. Chaker and colleagues' reconstruction of Sandler's earlier rules similarly places qualification, pain, money and decision authority before matching the offering. The review's account and Mattson's exposition together show both the qualification logic and the work around its use. [SL-W001; SL-W010]

Consultation and the creation or revision of a buying vision

Ben Taylor distinguishes working with a buyer whose challenge remains unclear from responding to a request already framed around a solution. Consultation in the first situation develops understanding of the problem before an offering can be matched to it. The historical title Consultative Selling, by Mack Hanan, James J. Cribbin and Herman C. Heiser, dates from 1970 [SL-W029]; the later accounts discussed here give the operative comparisons between consultation and other selling approaches. [SL-W021]

In Chaker and colleagues' reconstruction of Michael Bosworth's Solution Selling, the seller locates the buyer within a buying process. Latent or acknowledged pain without a vision calls for creating a vision; an existing vision calls for reconsidering or re-engineering it. The resulting vision is commercially directed toward the seller's product or service. The account therefore joins inquiry into the buyer's difficulties with an attempt to shape how an eventual solution is understood. [SL-W001]

Pain chains, sponsorship and coordinated evaluation

The current Solution Selling exposition connects preparation with a pain chain: how a business problem affects different people. Preparation identifies relevant participants and makes an opening inquiry more useful. Subsequent work includes vision creation or revision, sponsorship and access to power, collaboration planning, evaluation criteria, proof of value and negotiation. A user's recognized problem does not by itself supply the authority or arrangements needed to proceed. [SL-W021; SL-W027]

The collaboration plan coordinates actions and evaluation so that proof can be judged against criteria the participants have accepted. Sponsorship connects a conversation with the people able to authorize a commitment. Evaluation and negotiation then concern a proposal whose intended value and assessment have already been discussed. Cooper and Winters' separate account of sales and project coordination develops how such conversations can span people and meetings within the work of defining a deliverable solution. [SL-W027; SL-W022]

From control to consensus: related qualification formulations

The Eades account reconstructed by Chaker and colleagues treats Solution Selling as a philosophy, map, methodology and management system aligned with buying. Situational fluency joins knowledge of the buyer's job and problems to knowledge of the seller's relevant capabilities. Preparation is consequently part of the capacity to ask useful questions and interpret an answer. Its Pain × Power × Vision × Value × Control = Sale formulation expresses qualification conditions; it is not an estimated probability equation.

Control in that account concerns facilitating a process within which buyers direct their choices. [SL-W001]

Richardson's current PPVVC account instead names Pain, Power, Vision, Value and Consensus. It asks whether the issue is acknowledged, authority is engaged, a desired outcome is articulated, value is measurable and the buying group agrees. A compelling problem and offering can coexist with missing authority or agreement. Those missing conditions focus subsequent conversations and coaching. Consensus in this current account should therefore be understood in its own terms alongside the earlier Control formulation. [SL-W027]

S78 Teaching, reframing and buyer indecision

The sequence of commercial teaching

The current Challenger exposition organizes commercial teaching into six connected steps. The warmer demonstrates informed understanding of the customer's setting. The reframe then makes an unfamiliar problem or opportunity visible. Rational drowning supplies a quantified account of the consequences of continuing as before; emotional impact makes those consequences recognizable in a situation the customer can imagine inhabiting. The seller is developing a reason to reconsider the existing course before asking the buyer to consider a product. [SL-W020]

The fifth step, a new way, establishes the value of a different approach. Only the sixth, your solution, introduces the supplier's differentiated offering as a way to realize it. Moving directly from a problem to the product would omit the intervening argument. Challenger describes an adaptable choreography, supported by teaching, tailoring and taking control, rather than a script to recite regardless of the buyer's answers. [SL-W020]

Producing an insight and tailoring it across a buying group

Commercial teaching relies on an organizational supply of insight as well as a seller's conduct. Challenger's exposition connects knowledge of the customer's business with message development, marketing, coaching and adaptation across a buying group. Tailoring concerns different stakeholders' drivers, even when the underlying commercial insight is shared. The seller need not invent a new economic argument in each meeting. Taking control includes discussing money and commitments, while constructive tension asks the buyer to reconsider a familiar course. Stakeholder tailoring connects that challenge to what matters to each participant. [SL-W020]

The provider's Metrie illustration concerns molding and the basis on which builders and distributors consider an offering. Prepared messages shift attention from purchase price toward what the offer contributes to their work. Implementation joins this changed conversation to organizational support. The case reports use of the approach, while the seller-profile research reconstructed by Chaker and colleagues concerns a different object: manager assessments across 44 attributes and their association with performance. Observing a profile associated with high performers differs from testing the effect of training sellers to enact it. [SL-W020; SL-W001]

Trust-based selling and the dispute about relationships

Avila and Inks organize trust-based selling as initiating, developing and enhancing a relationship. Initiation includes qualification and relevant information; development concerns value and commitment; enhancement includes reliable delivery and further opportunities. Competence and dependable performance belong to trust alongside interpersonal conduct. [SL-W024]

Their distinctions among need-satisfaction, problem-solving and consultation also preserve different tasks. A seller may match an expressed need, compare alternative solutions, or work on the buyer's strategic goals. Relationship selling in this account is consequently more substantial than personal amiability. Their criticism of Challenger credits its value focus and marketing coordination while questioning whether inexperienced sellers are equipped to teach or challenge before understanding the buyer's situation. The dispute concerns the preparation and knowledge that make an encounter credible. [SL-W024]

JOLT and the fear of making the wrong decision

Matt Dixon and Ted McKenna's JOLT exposition distinguishes reluctance to change from fear of choosing badly. Strengthening the case against the status quo need not settle anxiety about making the wrong decision. In their account, another urgent argument or another menu of options can deepen indecision instead of resolving it. [SL-W025]

Judging indecision means investigating uncertainty about the choice and its outcome. Offering a recommendation supplies a reasoned path through tradeoffs. Limiting exploration directs further investigation toward information that can actually improve the decision. Taking risk off the table supplies an appropriate safety net, such as a trial, opt-out or smaller initial deployment. The authors connect these actions to realistic expectations about implementation, so that reassurance does not depend on an implausibly optimistic promise. [SL-W025]

A recommendation carries some of the responsibility for choosing; a smaller deployment makes the first commitment less final. These illustrations explain how the authors intend to address hesitation after a conceptual case for buying already exists. The four activities concern the buyer's ability to move forward, rather than simply adding more reasons why the offering is attractive. [SL-W025]

S79 Adapting sales encounters and assessing their results

Prepared responses and room to adapt

Frédéric Buchet begins with a technical engineer entering sales who asks for effective techniques, buyer types and convincing tricks. Buchet separates tricks, reported best practices and organized methods. A reusable reply can keep a novice from being disconcerted, but a recognizable stock answer can also invite distrust. Transferring a star seller's practice requires attention to whether the example is representative, whether conduct was actually observed, and whether it can be organized into something teachable. [SL-W023]

CNZ organizes objection handling through investigating the objection, reformulating it constructively and then giving a targeted argument. The first two actions change what the answer should address. PLAISIR places contact, expression of needs, clarification, an argued offering, use of opportunities, purchase and reassurance within a longer encounter. A method provides a common vocabulary for instruction and coaching, while leaving a practical question about how much discretion its implementation permits. [SL-W023]

Jolson's prepared/adaptive continuum, reconstructed in Chaker and colleagues' review, makes room between a memorized presentation and unrestricted improvisation. A seller can prepare usable modules and combine them for the encounter. Structure supplies a repertoire; adaptation concerns selecting and connecting it in response to the situation. The proposal gives preparation a role in flexibility instead of treating the two as opposites. [SL-W001]

Buchet also distinguishes implementations that give sellers discretion from those that require near-verbatim use. A common method can help a manager and seller discuss the same part of an encounter, but the organization still decides how tightly the conduct must follow it. The method's teachable structure and the latitude available to its user are therefore separate features of the arrangement he describes. [SL-W023]

Information, strategy, communication and adjustment

Bigné, Küster and Torán describe adaptive selling through collecting information, formulating a strategy, transmitting a message, evaluating and adjusting it. A response can change within an encounter or across encounters. Their account distinguishes confidence in the ability to adapt from enacted changes in selling behaviour. A salesperson can endorse flexibility without reporting the same conduct as another who actually changes an approach. [SL-W012]

The authors connect this process with market orientation: generating information, sharing it and coordinating a response. Their scale-development work with 119 sellers is distinct from the focal study of thirty sellers in eleven paint and varnish firms. Reported associations with market orientation differ for the broader adaptation measure and a single adaptation item. The distinction makes the organization's information practices and the particular measure part of interpreting what adaptation means in the study. [SL-W012]

Different meanings of adaptation and different measures of success

Chaker and colleagues' review separates adaptive selling understood as behaviour, ability, mindset and trait. Their scholarly corpus contains 188 articles; a separate practitioner review compares 71 resources. The review shows how the broad label can cover different objects and how diverse performance

measures complicate an unconditional claim that adaptation improves results. Its integrative framework is the authors' proposed organization of that literature. [SL-W001]

McGowan's review develops requirements that recur across selling approaches: qualification, listening, needs, product knowledge, adaptation, relationships and digital capabilities. Listening includes several forms of communication, and relevant networks extend within both buyer and seller firms. These connections make failure a concern of preparation and organizational access as well as individual conversation. The review of 104 articles identifies common requirements rather than establishing trial-based winners among branded systems. [SL-W011]

Verbeke, Dietz and Verwaal's meta-analysis brings together 268 studies, 292 samples, 79,747 sellers and 4,317 firms over 1982–2008. Knowledge and adaptability are positively associated with performance, while role ambiguity is negatively associated. The interpretation depends partly on whether performance is self-rated, manager-rated or measured objectively. Their knowledge-broker interpretation concerns sellers' use of knowledge across relationships. Pooled model coefficients describe associations in that analysis, rather than percentages of revenue obtained by adopting a technique. [SL-W018]

The Sales Agility Code, as reconstructed in Chaker and colleagues' practitioner review, organizes action as assessing the situation, choosing a path and executing while monitoring impact and changing circumstances. Vazzana and Sottosanti Doyle name consultative, disruptive, competitive and financial strategies. The account concerns this repertoire and the choice among its paths; proprietary performance descriptions remain claims reported by the review. [SL-W001]

A verbal prompt at the point of ordering

A field experiment at two Austrian fast-food branches examined a much smaller intervention: a suggestion when a customer had ordered a main dish but no side. The procedure first required attention to the order already given. Only an eligible transaction received the assigned prompt. Successive groups of ten transactions rotated among no prompt, a potato-salad suggestion, a fries suggestion and a salad-or-fries choice. The intervention was therefore a particular utterance at a particular point in ordering, rather than a general instruction to be more persuasive. [SL-W026]

Across 2,160 transactions, side-dish purchases were approximately 61 percent under the control condition and 78 percent across the pooled prompting conditions. The forced-choice version did not significantly outperform the simpler suggestions. An added verbal device could thus fail to improve on a useful prompt. The measured result was purchase of a side dish; the authors' discussion of fit with the main dish and unfamiliarity concerns interpretation of those choices. [SL-W026]

Suggestive selling within a meal

Lagueux and Rajaobelina place suggestive selling within a meal and service encounter. Menu information, the occasion and the diner's involvement affect what may be pertinent. The encounter can involve several diners rather than a single buyer facing a single seller. Companions' choices and experiences make the situation polyadic: one suggestion can enter a group conversation and influence more than its immediate recipient. [SL-W014] A business meal and a social occasion may permit different amounts of interaction with a server, while an informative menu can answer questions before the server intervenes. [SL-W014]

Their conceptual model also includes the server's circumstances and workload. A relevant suggestion depends on an opportunity to notice, interpret and address what is happening during service. Bill size and customer satisfaction remain different proposed consequences. The model develops possible relations around the activity, complementing the narrower transaction outcome examined in the prompt experiment. [SL-W014]

Cross-selling and differences among retail settings

Vieira, Pires and Galeano studied cross-selling in three settings: 49 clothing sellers, 57 pharmacy staff across thirteen stores and 27 wholesale respondents. Their measures include beliefs about training, selling conduct, organizational feedback and self-reported firm performance. Co-creation here concerns whether a firm listens and acts on seller feedback, giving frontline information a place in the organization's response. [SL-W013]

An item about directing buyers toward what the seller thinks best has a negative association in clothing and a positive one in pharmacy. Those different results concern the sampled settings and measured constructs. The authors' cold-remedy-and-tissues illustration shows what an additional related purchase might look like; the study's performance measures concern selling and the firm. Beliefs about training likewise differ from an experiment that assigns training and measures its effects. [SL-W013]
Understanding needs is positively associated with cross-selling in both the clothing and pharmacy accounts. The authors interpret the contrasting selling postures through their different settings, alongside the contribution of management's response to frontline information. [SL-W013]

S80 Organizing sales work and creating customer value

From seven selling steps to continuing organizational functions

Moncrief and Marshall contrast the traditional seven selling steps with a changed organization of sales work. Prospecting, preapproach, approach, presentation, handling objections, closing and follow-up organize a familiar sequence. Their evolving functions include customer retention and relinquishment, knowledge and database management, relationship development, marketing the product, problem solving, adding value and relationship maintenance. These functions need not occur once each in a fixed order. [SL-W002]

Their argument follows changes in who performs an activity and what supports it. Prospecting can be supplied by another unit; preparation can use shared account records rather than an individual's index cards; a team can work with several members of a buying centre. Problem solving and continuing service can span meetings and participants. The resulting account concerns an ongoing customer relationship and division of work, rather than simply renaming stages in a single conversation. [SL-W002]

Situational, functional and process accounts of sales management

Руденко distinguishes situational, functional and process approaches to sales management. A situational approach attends to the particular conditions under which selling takes place. A functional approach makes responsibilities and management activities visible. A process approach follows connected work across those functions. Each directs attention toward a different object of management, from an encountered condition to an assigned responsibility or a sequence that crosses departmental boundaries. [SL-W016]

In the functional account, planning, organization, motivation and control organize management responsibilities. The process account instead follows market information through connected transformations and supporting activities toward results and satisfied needs. A situational response may solve an immediate issue without specifying that continuing whole. Rudenko's comparison consequently concerns how a sales operation is managed, with conversation techniques located within a larger set of linked activities. [SL-W016]

Understanding a customer's business and shaping a valuable offer

Terho, Haas, Eggert and Ulaga conceptualize value-based selling through understanding the customer's business model, crafting a value proposition and communicating customer value. Understanding includes earning logic and operating facts, not just requested specifications. Crafting uses that knowledge to shape an offering; communication makes the proposed difference credible through calculations, comparisons, references or tests. Their qualitative investigation draws on eleven purposefully selected experienced firms, senior informants and 176 transcript pages. [SL-W019]

The truck illustration begins with a request for twenty vehicles. Investigation of materials, distances, time and maintenance requirements supports a proposal for eighteen. The seller changes the proposed fleet by understanding the work it must perform, rather than merely reducing the price of the requested twenty. Capital and operating implications become part of explaining value, alongside the reported continuing relationship with the customer. [SL-W019]

The dairy illustration follows washing frequency and the use of water, energy, chemicals and time. Information about the process and relevant plant experience supports an estimate of a saving opportunity. Tests or a savings-sharing arrangement can connect that estimate to a commercial proposal.

Information access, calculation and the form of commitment thus remain together in the account. The authors distinguish their explicit concern with customer business value from related constructs such as adaptation, consultation and partnering. [SL-W019]

The scope of becoming a solution provider

Evers and colleagues' solution-provider report joins the scope of an offering to its realization, selling and management. A product/service combination can involve a different allocation of responsibility rather than a new label on an existing bundle. The three compass categories direct attention to how a chosen solution scope is made workable across these activities. [SL-W007] Its distinction among solution scopes moves from support for a customer's process steps to taking responsibility for subprocesses and, at the broadest scope, results. The report treats these as different commitments, rather than assuming that the largest scope is always preferable. [SL-W007]

Their comparison of consultative, insight and value selling is one classification of the approaches, including their placement of SPIN and Challenger. The Siemens ShareNet discussion gives shared lessons and knowledge a supporting role across solution work. Making such information available concerns the organization behind the seller, as well as the skills brought to an encounter. [SL-W007] In the ShareNet illustration, project participants review the work and its results, record success and failure factors, and make them searchable by other departments before a similar project begins. The practical connection is between a completed project's interpretation and preparation elsewhere in the firm. [SL-W007]

Proposal approval and the handoff into project delivery

Cooper and Winters connect selling services with defining and delivering the promised work. Their six iterative stages move through opportunity, initial need and vision, a detailed feasible roadmap, a formal solution, contract and execution. Pursuit strategy, opportunity validation, internal proposal review and project kickoff supply four checkpoints. Continuing opportunity strategy, project oversight and account planning accompany the stages. [SL-W022]

Internal proposal review brings delivery authority into the decision before an external promise. The project manager can withhold a signature where feasibility is unacceptable; discussion can also lead management to accept an identified risk explicitly. The arrangement makes the connection between a commercial commitment and its execution a decision to be taken, rather than a handoff assumed to work by itself. [SL-W022]

Their e-commerce contrast explains why involvement can differ with scope. Helping define an entire venture, its business and its systems requires interdisciplinary development of a feasible vision. A later bounded catalogue assignment can be much more specified. The nine-box conversation they discuss can span several people and meetings: it is a guide to developing understanding and agreement, rather than a dialogue that must fit in one call. [SL-W022]

Ordering, settlement and a wholesaler's selling time

粟島浩二's industrial-parts wholesaler case connects changes in ordering and delivery with changes in settlement and sales work. Customers could order around the clock and receive flexible small deliveries. Automated, consolidated settlement and quicker collection reduced routine administrative demands on the sales function. Released attention could then be directed to new customers. The arrangement depended on the commercial and financial process as well as the convenience of delivery. [SL-W017]

Specialization in power components also changed who could become a customer. Firms that might otherwise have been competitors could buy from the wholesaler, with goods shipped directly to their

own buyers. The case thus concerns both the handling of existing transactions and a changed role within a trading network. The author goes on to propose further links involving research and development, quality coordination and incentives; these proposals extend the argument beyond the reported company arrangement. [SL-W017]

The buyer's progress and the seller's process

西川功一's DSPM and cloud-CRM account records the buyer's stage separately from the seller's process. Completing a seller task does not establish that the buyer has progressed. Keeping the two axes distinct makes a mismatch visible and supplies a question about what action might help the buyer move forward. [SL-W015]

The procedure then follows an action with observation and revision. A candidate response is tried, subsequent buyer progress examined and the process standard adjusted. Modifiable CRM fields and tests let frontline participants revise the arrangement without waiting weeks for an information-technology change. The system supports the trial and the record; the sales procedure concerns what participants do with the discrepancy and what they learn from the next response. [SL-W015]

Paired customer exploration and the use of prototypes

山中正樹's exploratory study concerns the Japan business of a precision-instrument company of US origin. Its business-development group separates exploration responsibility from short-term order targets while bringing technical and sales participants together in customer work. The study interviewed two technical-sales staff, two sellers and one marketer between December 2016 and February 2017. This company is distinct from the industrial-parts wholesaler discussed above. [SL-W028]

Paired preparation develops hypothetical needs from market and technical knowledge, including understanding of the customer's customer. During an encounter, one participant can speak while the other notices, supplements and interprets. Subsequent internal discussion develops a shared account. Yamanaka distinguishes credibility obtained by competent responses to explicit needs from the trust that supports deeper dialogue. Pairing and debriefing help connect what was noticed to what can be explored next. [SL-W028]

Minimum prototypes are then used by customers. Use can reveal needs that were not specified beforehand; the firm can revise the offering, and a customized product may develop into a standard one. Yamanaka's interpretation connects exploration responsibility, complementary attention, discussion and actual customer use. The reported contrast with less successful solo work forms part of this exploratory case and the author's proposed explanation. [SL-W028] The contrast includes technical staff working alone who discuss technology without establishing the same exchange; joint interpretation within the organization is part of the proposed account, as well as pairing during the visit. [SL-W028]

S102 Marketing and the organization of market relationships

What the seller takes to be the market

Wendell Smith's 1956 article distinguishes product differentiation and market segmentation as alternative approaches to marketing. Differentiation seeks preference for a particular offering. Segmentation begins from differences in demand and asks which offerings or efforts fit the resulting groups. The distinction directs attention to a decision made before a selling encounter: whether to shape preference around an offering or to organize the offering and effort around differences already apparent in the market. [T04-W001]

小原博's 2010 essay follows changes in Japanese accounts of sales management, marketing management and market management. In his discussion of 光澤's two books, the earlier work moves from the formation of marketing management through management of processes and strategy. The later work sets aside space for what lies beyond marketing management. 小原's account locates continuity in the turn toward 市場管理, or market management. [T04-W002]

Exchange as a continuing relationship

Christian Grönroos argues that the conventional marketing mix leaves much of service and industrial exchange unexplained. His relationship account draws attention to continuing interaction and to contributions made by participants beyond the immediate buyer and seller. [T04-W003]

Robert Salle and Jean-Paul Valla's Lyon Acier case in Håkansson's 1982 volume shows what particular terms and adaptations can mean. Lyon Acier obtained refractory bricks from three suppliers within a larger pool and kept a reserve stock. Field, the British supplier, adjusted the alumina content to fit local working practices. When one of its early oven vaults failed after about 60 percent of its normal service life, Field sent a technical engineer and sales director and replaced the vault at its expense. It accepted Lyon Acier's usual payment terms. The buyer reported preferring Field when prices were equal. The authors also describe the bearing of the exchange rate on Field's position. [T04-W005]

In the same study, Lyon Acier accounted for little of a German supplier's sales of oil coke. The supplier provided a standard product and required shorter payment terms. The French company's large size provided little advantage in that exchange. Salle and Valla use the comparison to explain the different prospects for negotiation and adaptation across the buyer's purchases. The report gives the buyer's interview perspective and the authors' interpretation of the terms and practices they found. [T04-W005]

Part 7 Learning, practice, and continuity

These accounts examine how people acquire and change capabilities, and how practices, language, and records help carry them forward.

S41 Shared practice, skill, environments and cultural coordination

Organized action has purposes as well as demands

These sources examine how people act with environments, learned skills, conventions, other people and material things. Their purposes include artistic performance, education, cultural participation, religious practice and therapy. The value of these activities is not exhausted by minimizing effort.

Kirsh and Maglio: an action can change what is easy to know

Kirsh and Maglio distinguish pragmatic actions, which advance a physical task, from epistemic actions, whose principal function is to improve cognition. Their Tetris account considers translations and rotations that can look unnecessary when judged only as moves toward a final board position. An action can instead expose information that is difficult to compute mentally, reduce a memory demand or make an error less likely. The extra movement can be part of a better problem-solving process. [W041, selected manuscript pp. 2–5 and 40–43]

The definition separates the apparent physical distance to a goal from the informational work needed to act. An external intermediate state can make a distinction visible. A reminder or partially organized work area can similarly change the memory or search task. Its informative contribution is that an account of action can miss its purpose if it considers only direct physical progress.

Maglio and Kirsh's 1996 *Epistemic Action Increases With Skill* reports a relation between skill and actions such as backtracking, qualifying the supposition that expertise always eliminates such movements. [GL-W027, abstract and introductory and result passages]

Todd and Gigerenzer describe a shortcut for predicting the winner of a Wimbledon match. If you recognize only one of the two players, predict that player will win. They explain why recognition can be informative even for tennis novices. In a setting where successful players are more often discussed, knowing a name provides a clue about performance. [W040, pp. 167–168]

The other authors discussed here draw on Gibson's *The Theory of Affordances*. [W042, through the accounts below]

Ecological interfaces: display functional relations rather than isolated readings

Vicente and Rasmussen tested two displays for a simulation of a system of tanks, pipes and heaters. Both showed physical settings and target values. One also showed higher-level information about how the system was functioning. After viewing each sequence for 25 to 30 seconds, participants recalled values and diagnosed the system state. Diagnosis was more accurate with the display that included functional information, chiefly among participants with theoretical expertise. The authors report better recall of functional variables with the richer display on trials showing normal operation or a fault. More accurate recall of those variables was associated with more accurate diagnosis. [TA-W026, §VII]

Shi and Zhang's 2004 review explains an abstraction hierarchy and the distinction among skill-, rule- and knowledge-based behavior. A work domain must be represented in its complexity, while its information must also be communicated to an operator. Means–ends links can show how a lower-level variable relates to a higher-level purpose, rather than merely arranging readings by part–whole location. [TA-W027, printed pp. 65–67]

Shi and Zhang also review studies of fault detection and navigation. They report advantages for displays that include functional information in demanding tasks, with normal-operation completion times sometimes similar across displays. [TA-W027]

The review also addresses training, self-explanation and individual differences. The review explicitly limits generalization from DURESS studies. Its assertions about fixed cognitive styles are not adopted as settled current psychology. A useful display still depends on the information pathway and the operator's acquired understanding; fewer visible numbers or a cleaner appearance alone do not establish the effect. [TA-W027]

Sweller: solving a problem and acquiring a schema can compete

The analysis in Sweller's 1988 article distinguishes successful immediate problem solving from learning a reusable schema. His explanation concerns means–ends search: repeatedly comparing a current state with a goal and seeking moves that reduce the difference can consume processing capacity without directing attention to the relations needed for later generalization. Effort spent reaching an answer is not automatically effort spent learning how to solve a class of problems. [TA-W035, printed pp. 257–259, 273–274 and 284]

The paper discusses domain-specific knowledge rather than a general superiority in memory or search. Reported expert–novice contrasts concern meaningful configurations, not simply random items. Its computational model and educational examples support a proposal for changing instructional tasks, including worked examples or less specific goals under appropriate conditions.

Kalyuga, Ayres, Chandler and Sweller's review of the expertise-reversal effect adds a material qualification. Guidance that helps a novice can become redundant or interfere once a learner has acquired the relevant schemas. Instructional support has to be assessed in relation to existing knowledge, rather than assigned one permanent benefit. [TA-W036, printed pp. 23 and 29]

Wood, Bruner and Ross: a tutor carries part of the task

The selected discussion of tutoring specifies six functions: recruitment, reduction of degrees of freedom, direction maintenance, marking critical features, frustration control and demonstration. The tutor can temporarily handle parts of a task that the learner cannot yet manage, leaving a component within reach. That arrangement explains the scaffold metaphor more precisely than simply saying assistance makes learning easier. [GL-W016, printed pp. 97–99]

Direction maintenance is not the same as repeated instruction. A learner can remain attached to a simpler success and need support to attempt the next step. Marking a critical feature also requires interpreting the discrepancy between the learner's production and what would satisfy the task. Demonstration is more than doing the task alongside the learner; it supplies a modeled solution relevant to the learner's attempt.

The source is a small tutoring study with proposed instructional functions. The tutor's knowledge, attention and judgment remain active. A reduced task for the learner is a distribution of work in this account, not evidence that the rest of the task has ceased to exist. [GL-W016]

Billett and Orth: participation is not guaranteed by an available setting

Billett's abstract frames workplace learning through both workplace affordances and individual engagement. Opportunities offered by a workplace do not compel a person to learn, and participation can depend on how access and activities are distributed. [GL-W052]

The abstract and implications discussion in Orth, van der Kamp and Button's article situate adaptive learning across a coach–athlete system. In their constraints-led interpretation, coaching helps shape conditions under which an athlete explores and adapts. The coach's design, observation and adjustment remain part of the arrangement. [GL-W071]

Ferrer i Cancho and Solé: a model with competing effort proxies

The language article models a binary association matrix linking signals and referents. It assigns a speaker-side effort proxy using signal entropy and a hearer-side proxy using uncertainty about the referent conditional on a signal. Its stated objective combines the two with a weight. A single signal can be economical under one proxy while leaving the hearer unable to distinguish referents; a large differentiating vocabulary places a different demand on the speaker. [WME-W010, equation 9 and model definition]

The authors optimize the associations and report a transition between regimes, with a Zipf-like frequency pattern near a transition in their model. The proposition is not merely that language is short. It concerns a balance between the modeled demands of producing signals and distinguishing what they refer to, with assumptions including uniformly treated referents and the permitted association structure.

The effort variables are proxies, not direct measurements of all human effort or physical energy. The paper's governing objective in equation 9 specifies the tradeoff used in its model. [WME-W010]

Kirby and colleagues: learnability and expressivity are different pressures

Kirby, Tamariz, Cornish and Smith examine cultural transmission and communication using models and artificial-language experiments. Their claim is that compositional structure can arise when a language must be both learnable by new users and sufficiently expressive for communication. Compressibility alone can favor degenerate systems that merge distinctions. Expressivity alone can preserve holistic forms that work for established users without being easy for new learners. [TA-W034, selected PDF pp. 2–3 and 8–9]

The experiment contrasts chains introducing fresh pairs of learners with closed groups in which the same pair continues. Training, communication and transmission occur in both, but the introduction of naive learners changes the learning pressure. The design uses sixty students and twelve meanings. Its initial holistic languages and controlled tasks are part of the evidence, not incidental details to be discarded when generalizing.

The selected discussion explicitly treats the appearance of structure as contingent rather than inevitable. Their relevant contribution is a specified account of how communication and transmission can impose different demands on a shared system. Underlying natural-language examples reported in the discussion remain mediated. [TA-W034, selected PDF p. 13]

Nagy: formulas support reperformance without fixing an exact text

Nagy's discussion of oral poetry examines composition in performance and the relation between text and reperformance. In the oral-poetic account, formulas and conventions can support a performance as it is made. It is misleading to assume that the performer's task is always verbatim preservation of one prior written text. The identity and success of a reperformance belong to the practice under discussion. [GL-W013, §§1.3–2]

The selected account is relevant because a shared repertoire can organize action without an explicit complete script for every event. It does not follow that the repertoire is merely compressed text or that its only value is reducing memory effort.

Goodman through Giovannelli: removing distinctions can change the work

The selected scholarly reconstruction of Goodman's aesthetics examines work identity and notation. A score can specify properties a performance must possess to count as an instance of a work. Other arts give different significance to production history and authenticity. The relevant question is what distinctions are constitutive of the work, rather than how much descriptive detail can be removed without any stated criterion. [WME-W009, selected main-text paragraphs 14–15 and 46–48]

This is a useful limit on an economy interpretation. A convenient simplification may change what is being performed or preserved, instead of producing the same work with less effort.

Musical ensemble: common frames support skilled mutual adjustment

Keller, Novembre and Loehr's chapter distinguishes musical dimensions and explains an underlying metric frame. Rhythmic patterns can be coordinated relative to temporal units organized into beats and bars, while pitch relations create another aspect of joint organization. A shared framework can help performers locate their own and others' parts without making those parts identical. [GL-W001, manuscript PDF pp. 2–3]

The discussion retains representation of self, others and joint outcomes. Skilled performance involves anticipation and adjustment as well as common structure. Expressive timing and the desired musical result matter; strict mechanical simultaneity is not a universal success criterion.

Vesper and colleagues' joint-action review similarly considers shared information and cultural coordination while calling for a more overarching framework. Conventions can help align expectations, but their presence does not eliminate contextual adaptation or prove that every joint action relies on one mechanism. This is a review-level account with its own open framework question. [GL-W002]

Bitner: the physical service setting is part of action

Bitner's selected *Servicescapes* passages distinguish spatial layout and functionality from signs, symbols and artifacts. The arrangement of equipment, furniture and movement routes can affect whether people can carry out a service task. Signs can communicate where to go or what conduct is expected. Customers and employees may occupy the same production setting, so an environment can shape more than one participant's experience. [GL-W005, opening framework and printed pp. 65–66]

These are conceptual propositions supported by reported prior work. [GL-W005]

Lucas: a behavior setting includes a maintained pattern

Lucas's field guide reconstructs behavior-settings theory and provides an operative method. A room or physical milieu alone is insufficient. A setting includes a standing pattern of behavior in relation to a bounded physical and temporal environment. The pattern can persist as different people enter and leave: a shop open for business is more than the walls of the shop. People create and maintain the norms and material conditions to which they subsequently conform. [SG-W026]

The method also examines interdependence within a proposed setting and between neighboring ones. Structural and dynamic tests help determine the boundary rather than accepting an architectural label. The gallery application and K21 method illustrate the observational work involved. Identifying a stable-looking setting requires attention to what people actually do, how parts depend on one another and where the pattern stops.

Lucas's method provides a way to describe and investigate recurring behavior settings, addressing their boundaries through observation and interpretation. [SG-W026]

Maller and Hyysalo: practices are performed, reproduced and contested

Maller's selected practice-theory discussion relates materials, competences and meanings. A practice as an entity is distinguishable from its particular performances. The breakfast example places an apparently simple activity within interwoven practices and material arrangements. Knowing how to act is not represented solely as information stored in one person's head; it is connected to equipment, routines and understandings reproduced through performance. [GL-W034, practice and materiality discussion]

Reproduction is an accomplishment. Changing a material element can alter a practice, but competence, meaning and other practices can sustain or obstruct the change. Maller's account also notes differences among materialist and practice approaches over agency, which are not resolved by grouping them under one heading.

Hyysalo's two-page review of that book supplies a separate critical reconstruction. It questions the adequacy of illustrative evidence and the abstraction involved in accounts of how practice elements are transmitted and connected. This is a useful challenge to an attractive diagram: naming elements and links is not yet a complete empirical account of how a practice changes. [GL-W035, printed pp. 41–42]

Li, Holstein and Wedekind: material resistance participates in knowing

The selected qualitative study of craft draws on observation, interviews with twenty studio potters and an eighteen-month participation context. It treats knowing as something enacted through relations among bodies, clay, tools and surroundings. Resistance, unexpected encounters and changing material behavior can alter how a practitioner senses and acts. A learned hand is not working on an inert object whose behavior is already completely known. [GL-W038, selected methods and interpretive findings]

The relevant contribution is a detailed account of situated adjustment and the material conditions of learning. The authors' language of material agency belongs to their interpretive framework. The study retains practice and encounter rather than assuming that knowledge can be fully transferred as a detachable verbal instruction.

Public cultural guidance: knowledge has custodians and conditions of use

The Assembly of Alaska Native Educators' *Guidelines for Respecting Cultural Knowledge* addresses researchers, language specialists and community organizations in its own setting. The selected pages call for informed consent, accurate representation, protection of cultural integrity and attention to community expertise. They distinguish public knowledge from sensitive information whose distribution has not been authorized. [GL-W003, selected preface and printed pp. 15–20]

The recommendations also concern local review, accurate interpretation, language and community repositories. Their relevance is that circulation of knowledge involves relationships, rights and responsibilities, not only an efficient transmission channel. The document is not represented as the view of every Indigenous people.

Armstrong and colleagues' ethnoecological discussion concerning Gitxsan and nłeʔkepmx territories makes a related but distinct argument. It locates stewardship in laws, institutions, beliefs and relational views of land. Its account explicitly resists treating reciprocity merely as a useful management tool: reciprocity can be a moral or legal obligation within the described relationships. Reducing the account to a technique for lowering management effort would change its stated purpose. The authors also criticize extractive use of knowledge. [GL-W004, selected discussion]

Xunzi through Wang: ritual and custom are not the same category

Wang Weiwei's scholarly abstract discusses the roles of ritual propriety and custom in Xunzi's thought. The relevant contrast concerns different ways that human conduct is shaped through learned forms and habitual social practice. [GL-W008]

礼义 and 习俗 have distinct normative and cultural roles in the interpretation.

Ritual and ascetic practice can involve deliberately sustained effort

The Stockholm University field orientation presents asceticism and ritual self-practices as subjects of religious-studies inquiry. The relevant account concerns practices through which people cultivate or transform themselves in relation to religious aims. Reducing effort need not be their purpose; disciplined effort can be constitutive of the activity. [GL-W037]

The abstract and ritual-components discussion in Boyer and Liénard's article explicitly resist treating ritual as one scientific mechanism. Their proposed components occur in different combinations. The coalitional discussion concerns signals of affiliation, commitment and cohesion; costly initiation or training can be meaningful as a signal because commitment is difficult to establish in advance. It is an account of a proposed mechanism within some practices. [GL-W042]

An action that appears repetitive or costly under an external task criterion can have a different meaning within the practice.

Assistance includes services and participation around an object

The WHO framework describes assistive technology as more than a product. Its five-part approach brings people together with products, provision, personnel and policy. A device's usefulness depends on selection, fitting, learning, maintenance, access and the wider service environment. A product can enable participation while still requiring continuing support. [GL-W041, 2024 framework and GATE five-P account]

Its relevance lies in the explicit support work that an apparently simple aid can depend on.

Personal constructs: interpreting a situation is itself an organized activity

Gaines's introduction presents Personal Construct Psychology through systems of constructs by which people anticipate and interpret events. It identifies research and elicitation resources, including repertory grids. [WME-W017]

The George Kelly Society FAQ supplies a field-native description, including the distinction from claims that a person simply creates reality. Its account of repertory-grid elicitation retains the respondent's distinctions and the interviewer's influence. A structured grid can make an interpretive system discussable, but eliciting it requires skill and can introduce bias. [WME-W018]

Repertory-grid elicitation makes assumptions explicit in a representation that can be discussed and examined.

Kriz: influencing conditions differs from prescribing a result

The selected translated chapter by Kriz proposes a person-centered systems account in which patterns arise through interaction. The environment supplies conditions but does not uniquely prescribe the resulting pattern. His family example asks how patterns once appropriate to one stage can become rigid when circumstances change. The therapist's role, in the argument, is to influence conditions under which

change becomes possible rather than dictate a new interaction in every detail. [WME-W022, translated chapter, pattern formation and family interaction]

IFS: protective parts and internal relationships

The IFS Institute outline describes parts, a leading Self, protective patterns and contextual constraints. It proposes that parts can take on different roles and that extreme beliefs or emotions can be described as burdens. The stated goal involves changing relations within the model rather than simply suppressing a troublesome part. These are the model's own terms and assumptions. Its therapeutic use of burden is not the monograph's accounting of material or informational demands. [WME-W023, selected public outline]

The outline also lists external unsupportive conditions and practitioner mistakes, including proceeding before a system is ready or mistaking a part's activity for the Self. Its account consequently retains interpretation, timing and the practitioner within the process. This is the institute's public model description.

Bowen: apparent stability can leave a problem unresolved

The Bowen Center's public theory presentation describes a triangle as a three-person emotional relationship in which tension can shift among relationships. It explicitly says that distributing tension can stabilize a system without resolving the underlying issue. The outsider position and changes in alliance can matter to the people involved even if the overall pattern continues. [SG-W028, Introduction to the Eight Concepts; Triangles]

Its emotional-cutoff account similarly distinguishes apparent improvement through reduced contact from resolution. The public presentation proposes that unresolved patterns can recur in later relationships. These are Bowen-theory descriptions, not universal facts established by the institutional page.

The useful relevant distinction is persistence of a pattern versus benefit to each participant or resolution of its causes. [SG-W028]

Haken: order parameters summarize and influence collective patterns

Haken's 2017 conceptual article presents synergetics through control parameters, instability, collective order parameters and circular causality. The parts exchange matter, energy or information. Under appropriate conditions, a small number of collective variables can describe a pattern and constrain how the parts move. [RGC-W038, article]

The examples include coordinated movements, common rhythms and pattern recognition. A low-dimensional description is not automatically a low-energy implementation. It is a claim about relevant collective variables within a model or observed organization. Applying it elsewhere requires identifying those variables, checking disturbance responses and the relation between slow collective change and faster component dynamics.

The pattern-completion example makes retained preparation particularly visible. Stored or learned prototypes and network dynamics allow an incomplete input to develop toward a complete pattern. Storage, learning, sensing, transmission and physical implementation remain. Haken's use of a landscape to discuss intention is an interpretation, not a direct physical energy measurement. [RGC-W038, PDF pp. 7–10]

His discussion of embodied aesthetics goes on to proposed relations among perception, action, beauty and therapeutic conditions. Group consciousness and mind–body identifications are philosophical or speculative extensions, not experimentally proved consequences of the network. His conclusion explicitly

warns that the apparent generality of order parameters can be seductive and requires care. [RGC-W038, PDF pp. 9–12]

Environments, skills and shared practices

The accounts offer concrete ways to examine action organized by surroundings, skills, representations, social relations and cultural forms.

S71 Designing practice and developing skill

Practice tasks and performance tasks

Seventeen advanced pianists were given the same three measures of Shostakovich to learn. They chose how to practise and returned about a day later to perform the passage. Robert Duke, Amy Simmons and Carla Davis Cash examined the recorded practice sessions alongside independent rankings of those later performances. Their 2009 study made the organization of practice visible: people could spend their time on the same notes while doing substantially different work. [LP-W006]

The three highest-ranked pianists dealt with errors in a distinctive way. They identified the precise place and source of a mistake, worked on its repair and changed tempo systematically. Slowing down meant slowing enough to play the passage correctly; speeding up followed what the player could already control. A successful correction was then repeated until it remained secure in subsequent attempts. Diagnosis, a changed attempt and stabilization formed a connected sequence. [LP-W006]

These sessions also included musical inflection early in learning. The players were preparing an expressive performance, not postponing interpretation until after mechanical accuracy. Silent examination of the score, singing, annotations and changes in playing could participate in that preparation. The later rankings included musical quality as well as correct pitches and rhythms. That matters when interpreting what the best practice sessions accomplished. [LP-W006]

Total practice time and the number of attempts did not significantly predict the retention rankings in this sample. The top performers had not simply learned the excerpt more quickly. Their handling of errors distinguished them, particularly the combination of precise diagnosis, controlled tempo changes and repetition that secured a repair. The researchers observed these strategies in the pianists' chosen practice; they did not assign a strategy to each pianist experimentally. [LP-W006]

K. Anders Ericsson, Ralf Krampe and Clemens Tesch-Römer's 1993 account of deliberate practice distinguishes improvement-directed training from work and play. Occupational performance must deliver a result with the capabilities already available. Play may be organized around enjoyment. Deliberate practice instead constructs an activity to improve a particular aspect of performance. A practice task can therefore differ markedly from the activity for which someone is preparing. [LP-W001]

Their baseball example concerns opportunities that ordinary performance supplies only sparsely. A game gives a batter particular pitches at particular moments, amid other demands. Training can concentrate attempts on the kind of pitch or response that needs improvement. Selecting, repeating and adjusting those attempts changes what the player gets to learn from. The quantity of relevant opportunities is organized deliberately rather than left to the frequency with which the situation occurs in competition. [LP-W001]

In the authors' account, a teacher or coach identifies a suitable task and helps the learner understand what to attempt. The learner then has opportunities to repeat it, obtain informative feedback and revise the attempt. As capability changes, the task can change too. Solitary practice by an advanced music student can belong to this relationship: what looks like independent repetition may carry forward tasks designed through individualized teaching. The existing tutoring account in S41 explains other ways support can organize what a learner is able to attempt. [LP-W001]

The 1993 paper connects this training account with expert development, including studies of musicians and their accumulated practice. It emphasizes sustained concentration and efforts directed at improvement, rather than treating enjoyable absorption as the defining condition of practice. Its definition of the activity, its studies of attained performance and its larger explanation of expertise

became objects of later debate. Those questions return below, after several accounts of how practice itself can be arranged. [LP-W001]

Building an action in stages

Pyotr Galperin's 1959 account asks how a mental action can be formed through instruction. He distinguishes the form in which an action is carried out from its degree of generalization, abbreviation and mastery. An action performed with objects can become more economical while still depending on those objects. Removing a visible step is therefore a different change from developing an action that can be performed mentally. [LP-W009]

The orienting basis specifies what the learner attends to in deciding how to act and whether the action is correct. Galperin distinguishes orientation built around a model, orientation supplied by complete instructions for particular tasks, and orientation that teaches a method of analysing a class of tasks. The third approach changes what instruction supplies: the learner is helped to establish relevant conditions in a new case, rather than receiving only another ready-made procedure. [LP-W009]

His account follows an action through material or materialized execution, overt speech, speech addressed to oneself and increasingly internal execution. Speech can make operations explicit after the learner ceases to rely directly on the objects. Subsequent abbreviation changes how the action proceeds without requiring every earlier operation to be expressed each time. The instructional sequence concerns the form and organization of activity as well as its speed. [LP-W009]

The counting example shows why withdrawing a support and teaching an unsupported action are different instructional events. A child counts two groups of objects and is then asked to determine their combined number when the objects are covered or no longer visible. Initially, the child may try to reproduce the objects in imagination and count those images. Galperin's discussion treats acting without the objects as something that may need to be taught, rather than a result guaranteed by successful visible counting. His larger project concerns how an action is formed and transformed through these changing supports. [LP-W009]

Practising parts and returning to the whole

Judith Jiménez-Díaz and María Morera-Castro's 2018 review distinguishes several choices involved in designing motor practice: physical or mental execution, the composition of the skill, variability, distribution of attempts and their ordering. These choices concern different properties of a session. Imagining a movement changes the form of execution; separating its components changes what is practised together; changing the order of tasks changes what must be selected or reconstructed between attempts. [LP-W015]

Their account of part practice separates fragmentation, progressive segmentation and simplification. In a swimming example, the learner works on kicking, arm movements and breathing separately, then performs them together. Progressive segmentation joins successively learned units: learn one group of dance steps, learn a second, combine them, then incorporate a third. Fragmentation separates coordinated components; segmentation rebuilds a sequence through successive combinations. Both accounts include returning to the activity whose performance is intended. [LP-W015]

Simplification can preserve the overall action while easing one of its demands. Hitting a stationary ball on a tee changes the difficulty of batting without requiring the learner to practise only an isolated body movement. Mental rehearsal provides another choice: imagining the execution of a skill without carrying out its full movement. The review discusses physiological and cognitive explanations for such rehearsal and organizes decisions about practice around the learner and task. Its distinctions describe alternatives

that can be combined within an instructional design, rather than one standard way to break every skill into pieces. [LP-W015]

Building a route through a performance

Roger Chaffin and Gabriela Imreh's 1997 study follows Imreh's learning and relearning of the Presto from Bach's Italian Concerto. Recordings, verbal reports and analysis of practice starts and stops make the development of one concert pianist's memory unusually visible. The account asks how a performer can use conceptual knowledge of a piece while also relying on highly developed motor memory. [LP-W007]

The piece's formal structure organized practice. Boundaries between musical segments supplied places to begin and end attempts, and local difficulties received concentrated attention within that larger structure. Learning the notes was joined to identifying where the performer was in the piece and how one passage related to another. A repeated theme could require a different continuation at a different point in the performance; remembering the motor sequence alone did not describe the whole problem. [LP-W007]

The pianist identified conceptual, technical and expressive features that could function as performance cues. A fingering decision, a dynamic change or a musical interpretation could become a point of deliberate attention. Later work increased the speed with which the conceptual organization could bring the relevant feature to mind during playing. The resulting route through the music supported expression and control as well as the recall of a sequence. [LP-W007]

The study also examined memory after a long interruption, including written reconstruction and recall at the piano. Formal organization remained relevant to what could be recovered. Chaffin and Imreh describe a representation that helps elicit motor performance, keep track of location and retrieve important features when they are needed. Their case complements the rehearsal and transmission accounts in S66 by showing in detail how one performer develops access to a demanding performance. [LP-W007]

Who chooses when feedback arrives

Suzete Chiviacowsky and Gabriele Wulf's 2002 experiment gives control over feedback a precise form. Thirty adults learned a four-key timing task. Some could ask for information about a completed attempt. Each participant in a comparison group received feedback on the same numbered attempts as a corresponding participant who chose it. The paired schedule held the occurrence of feedback and no-feedback trials together while changing who controlled those occurrences. [LP-W034]

The self-controlled group performed better on a delayed transfer test involving a changed version of the timing task. The participants tended to request feedback after attempts they thought had gone well. Questionnaire responses and comparisons of errors on feedback and no-feedback trials supported that pattern. Requests were serving partly to confirm a successful attempt, rather than simply to obtain correction after an obvious failure. [LP-W034]

This made the relation between an attempt and the subsequent information important. A schedule could be identical in numbered trials yet fit one learner's experience differently from another's. Chiviacowsky and Wulf proposed that self-controlled requests were more closely aligned with the learner's needs. Their reported advantage concerned delayed transfer; the experiment's value also lies in showing how the selection of feedback can be studied separately from simply supplying more of it. [LP-W034]

Chiviacowsky, Camila Neves, Liliane Locatelli and Cibelly Oliveira's 2005 study examined forty children, approximately ten years old, using a timed four-key sequence and self-controlled or paired externally scheduled feedback. Its results separated relative timing, the proportions within the sequence, from absolute timing, its overall duration. On the relative-timing retention measure, the externally scheduled

group performed better. The absolute-timing comparison did not show the same significant difference. [LP-W013]

The authors considered children's ability to recognize good and bad attempts when discussing why feedback control might work differently in this setting. Selecting when to ask for information itself requires a judgment about the attempt. Their developmental interpretation concerns that judgment and the demands of the task. It is a separate account from the adult study, with its own population, assessment and finding. [LP-W013]

Grouping attempts and feedback

Satoshi Shibuya and Masanobu Ito's 1999 experiment combined two scheduling choices. Participants practised timing tasks in blocked or random order and received feedback summarizing one, three or five attempts. Blocking repeated the same task together; random practice switched among tasks. Changing the summary size altered how many attempts preceded a presentation of their results. The experiment could therefore examine whether the effect of one choice depended on the other. [LP-W014]

For relative timing in the retention test, the random-practice group with three-attempt summaries performed best. Among the blocked groups, five-attempt summaries produced the best relative-timing accuracy. Relative timing and overall duration were assessed separately. The pattern concerned the combination of practice order and feedback summary, rather than establishing one summary size as preferable independently of the practice arrangement. [LP-W014]

The authors interpreted these findings through the information-processing demands created by switching tasks and retaining information about recent attempts. A longer summary changed the demands under which the learner interpreted feedback; switching among tasks changed them again. Their discussion asks how these arrangements interact in forming the timing structure of an action. [LP-W014]

Ivan Wallan Tertuliano, Orlando Pereira de Souza Júnior, Antônio Sabino da Silva Filho and Umberto Cesar Corrêa's 2008 volleyball study used a larger motor activity. In 144 children, it combined four practice arrangements with feedback on every attempt or on approximately a third of attempts. The feedback concerned movement quality, and the study distinguished the score achieved by the serve from the pattern of movement used to achieve it. A later adaptation phase changed the task after the stabilization phase. [LP-W012]

The authors did not establish learning in the experiment and consequently did not infer a successful combination of schedules from the group comparisons. Their design nevertheless states the practical question concretely: organize where serves are directed, choose when to provide information about movement, then examine both the result and the developing movement pattern when conditions change. The intended skill, the information supplied and the measures used to assess adaptation were separate parts of that question. [LP-W012]

Relating practice to expertise

Brooke Macnamara, David Hambrick and Frederick Oswald's 2014 meta-analysis related accumulated practice to attained performance across music, games, sport, education and professional activity. It combined studies of structured activity intended to improve performance. The reported overall correlation was .35, with substantial differences across domains and ways of measuring practice and performance. The review treated practice as important while arguing that it left considerable variation among people unexplained. [LP-W002]

Those comparisons concern differences between people: whether participants reporting more practice tend to attain higher performance. They do not assign a percentage of one person's skill to a cause. The review's domain and measurement analyses also show why a single correlation conceals consequential choices about the activity counted and the result being explained. Retrospective reports, recorded practice and different performance assessments contribute different kinds of information to the pooled comparison. [LP-W002]

The authors' broad structured-practice category became central to disagreement about what their estimates could test. A measure of accumulated activity can include practice that meets some descriptions of deliberate practice while failing stricter requirements advanced by its proponents. The dispute therefore concerns the identity of the activity as well as the size of its association with performance. [LP-W002]

Friedrich Platz, Reinhard Kopiez, Andreas Lehmann and Anna Wolf's 2014 music meta-analysis organized its pool around accumulated task-relevant practice and objectively assessed musical achievement. Its final pool contained thirteen studies and 788 participants. It reported a correlation of .61, with a confidence interval from .54 to .67. The authors argued for comparisons in which the predictors and performance outcomes were sufficiently alike to support aggregation. [LP-W008]

Their category of task-relevant practice included deliberate practice but was not identical to the strict teacher-guided category subsequently defended by Ericsson. The music analysis supplies a developed example of how defining the domain, predictor and outcome changes a research question. Its larger association and the broader review's smaller one describe differently assembled evidence, with the authors' own reasons for their choices. [LP-W008]

Ericsson and Kyle Harwell's 2019 criticism returns to what counts as expertise and what counts as deliberate practice. They distinguish reproducibly superior performance on representative tasks from some educational or professional proxies. They also require evidence that the measured activity fits the original account of improvement-directed, individualized training. Their argument challenges the inclusion of studies whose performance measures or practice categories do not meet those criteria. [LP-W003]

Their more restrictive reanalysis reports different practice-performance associations, including an adjustment for measurement error. They regard these changes as evidence that broader pools can underestimate the relationship they intend to study. They also recommend examining concentration, analysis and problem solving during practice. In their account, duration alone cannot represent the quality of a training activity whose very definition includes how it is organized. [LP-W003]

This position joins an empirical question to a methodological one. Which activity has been measured, and is the performance assessment appropriate to the domain's expertise? Ericsson and Harwell argue that answers are required before a correlation can bear on their theory. Their proposed criteria specify the body of evidence to which they believe the theory applies. [LP-W003]

Macnamara and Hambrick's 2020 commentary disputes how the terms have been used across publications. They compare earlier descriptions with later distinctions between purposeful and deliberate practice, and object to relabelling activities previously discussed under the latter heading. Their examples include the treatment of solitary practice and research on word-game performance. The criticism concerns whether changes in terminology alter what evidence is accepted as a test of a theory. [LP-W004]

For these authors, cumulative research depends on definitions that permit a finding to count against an explanation as well as for it. If an activity can be accepted under one description and excluded under another, the relation between prediction and evidence becomes unstable. Their commentary develops that objection rather than introducing another training intervention. It argues for practice as an

important contributor to expertise whose explanatory reach can be investigated alongside other factors. [LP-W004]

Ericsson's 2020 reply defends continuity in the original definition. He specifies a task that the learner understands, opportunities for independent attempts, actionable feedback, repetition and individualized guidance from a teacher or coach. He calls activity satisfying the first four requirements purposeful practice; deliberate practice also includes the fifth. On this interpretation, broadly structured self-study is not automatically evidence about the narrower training category. [LP-W005]

The reply also explains why solitary activity and teacher-guided training need not be opposites. An advanced student can work alone on exercises established through individual instruction. What matters to Ericsson is the relation of those exercises to the student's present performance and subsequent guidance. His answer defends a narrower training category, while Macnamara and Hambrick question whether its usage has remained stable across the research being debated. [LP-W005]

What is carried into the next task

Geneviève Oléron's 1955 review begins with an operational definition of transfer. Compare learning task B after learning task A with learning B without that preceding experience. An improvement, an impairment or no measured difference supplies the corresponding positive, negative or null transfer result. The comparison asks about the effect of prior activity on a later one, rather than inferring transfer simply because the activities look alike. [LP-W011]

Oléron also emphasizes that an organized activity can combine movement, perception, memory and intellectual operations. Describing two tasks by one conspicuous surface feature may therefore miss what the learner is carrying between them. The analogy experiments developed in S64 give another concrete setting in which recognizing and using a relevant relation matters. The retention, transfer and scheduling accounts in S72 follow how different practice conditions affect what becomes available after the immediate attempt has ended. [LP-W011]

S72 Challenge, difficulty and accelerated learning

Learning that outlasts a good performance

Training can make someone perform well while the training conditions remain in place. Richard Schmidt and Robert Bjork's 1992 account asks what that performance reveals about the capability that will remain when those conditions change. They compare findings from motor and verbal learning and argue that practice should be judged in relation to the later performance for which it prepares a person. A successful attempt during instruction and a successful attempt after a delay answer different questions. [LP-W030]

Their comparison includes arrangements that make acquisition look less successful while improving later performance. Varying or switching tasks, reducing some forms of assistance and distributing activity can change what the learner must do during practice. The authors interpret such findings through the processes required by the eventual task. Their training argument begins with that criterion performance and asks whether practice exercises the processes it will require. [LP-W030]

This is an ambitious historical integration across experimental traditions. Schmidt and Bjork seek common principles in results that had often been considered within separate research paradigms. Their proposed account changes the role of the practice session: its immediate score becomes one observation among others, while retention and transfer become central tests of its value as training. [LP-W030]

Nicholas Soderstrom and Robert Bjork's 2015 review develops the distinction between performance and learning through a larger literature. They use learning to concern relatively durable changes that support later retention and transfer, while performance is the behavior observed under present conditions. A temporary aid can improve current execution without producing the same improvement when removed. Conversely, a condition that initially disrupts fluent execution can prepare better later performance. [LP-W019]

The distinction also affects learners' decisions. A fluent response can seem to show that material is mastered; a difficult attempt can seem to show that a method is failing. The review examines cases in which these judgments follow immediate performance more closely than later results. Its verbal- and motor-learning accounts explain why judging a learning procedure requires specifying what should become possible afterward and how that capability will be assessed. The schema-acquisition and guidance studies in S41 develop related instructional questions through their own accounts. [LP-W019]

Effort and the learning edge

Mihaly Csikszentmihalyi and Judith LeFevre's 1989 study examined the quality of experience in work and leisure. Seventy-eight workers supplied repeated reports during a week, allowing experiences to be related to perceived challenge, perceived skill and the activity being undertaken. Their analysis distinguished situations with high challenge and high skill from those with little of either. Absorption and engagement were examined in relation to what a person was doing and felt capable of doing. [LP-W031]

Work and leisure did not simply divide experience into burdensome and rewarding states. The authors found patterns in which work supplied challenges and opportunities for skill that could be associated with a richer experience, even while participants might prefer to be doing something else. Their subject was the organization and felt quality of experience. That provides the setting for later arguments about flow without making experience-sampling reports measurements of how quickly a skill was acquired. [LP-W031]

Phillip Armour's 2006 essay The Learning Edge brings competence and task difficulty into a practical argument about software work. His diagrams place routine activity toward boredom, demands far beyond

capability toward anxiety, and a zone of competence between them. The learning edge lies near the boundary of what the person can presently do. As competence grows, an unchanged task can move from challenge into routine; the edge moves with the learner. [LP-W018]

Armour argues that organizations should help people reach and remain near this moving edge. He makes a stronger claim about accelerated learning at its outer boundary: stress can be pushed far enough to force development, provided it does not cause the learner to shut down. Marine recruit training and Outward Bound supply his illustrations of people placed in demanding situations. The essay treats discomfort as potentially productive and challenges managers to attend to learning rather than viewing software work only as the delivery of already mastered activity. [LP-W018]

His account is a practitioner's argument, supported by these illustrations and a proposed relation among challenge, competence and experience. Its useful practical question is how an organization supplies tasks beyond comfortable repetition while keeping a person able to engage with them. The instructional role of support is developed in S41; the interpersonal risks of exposing uncertainty are examined in S51. [LP-W018]

Elizabeth Ligon Bjork and Robert Bjork's 2011 account of desirable difficulties concerns demands that can improve later learning even when they impede immediate performance. Their distinction between retrieval strength and storage strength separates current accessibility from the more enduring entrenchment of learning. Material can be easy to bring to mind now because it has just been seen, while that accessibility gives an unreliable impression of what will be available after an interval. [LP-W028]

Generating an answer, attempting recall, spacing encounters and switching among examples change the activity that must be performed. The authors explain their desirability through what those activities can contribute to learning, rather than through unpleasantness itself. Their chapter also addresses judgments made during study: familiar material can invite confidence without requiring the learner to reproduce or use it. The experiments below give particular procedures and outcomes behind several of these recommendations. [LP-W028]

Matching challenge to the learner

Mark Guadagnoli and Timothy Lee's 2004 challenge-point framework distinguishes nominal task difficulty from functional difficulty. Nominal difficulty concerns the task's demands. Functional difficulty also depends on the person attempting it and the conditions under which it is attempted. The same task can thus supply a different challenge at different stages of a learner's development. [LP-W021]

Their golf example makes the distinction tangible. A pitch shot over water can be a substantial problem for a beginner and a familiar task for a skilled golfer. Keeping the physical shot unchanged does not keep the functional difficulty unchanged. Feedback and practice conditions further alter what information is available and what the performer must process. The account concerns a relation among task, skill and the organization of practice. [LP-W021]

The proposed framework distinguishes potential information from information the learner can interpret and use. Increasing difficulty can supply more opportunity to learn, until processing demands exceed what that learner can manage. Their diagrams consequently distinguish the functional difficulty associated with the best current practice performance from the difficulty associated with the greatest potential learning benefit. A session chosen solely to produce the cleanest immediate performance can occupy a different point on those proposed relations. [LP-W021]

As skill increases, the framework predicts that suitable functional difficulty should increase too. The golfer can attempt a more demanding task or practise under conditions that require more selection and interpretation. Guadagnoli and Lee use this argument to interpret findings about task order and feedback,

including why their effects can depend on task complexity. Their challenge point is a theoretical account of interpretable information and learning, rather than a scale for measuring emotional distress. [LP-W021]

Learning by trying to remember

A memory test can change the memory it examines. In Henry Roediger and Jeffrey Karpicke's 2006 experiments, students learned prose passages and then either studied them again or tried to recall them. The recall attempts supplied no corrective feedback. A later test compared what remained available after different delays. Replacing another encounter with the text by an attempt to produce it changed the subsequent course of remembering. [LP-W022]

In their first experiment, additional study produced better recall five minutes later: 81 percent of the scored idea units, compared with 75 percent after an initial recall test. Two days later the order reversed. Students who had previously attempted recall remembered 68 percent, compared with 54 percent after additional study. After a week, the corresponding figures were 56 and 42 percent. The apparent winner depended on when the outcome was measured. [LP-W022]

The second experiment arranged four learning periods in three ways: four periods of study; three periods of study followed by one recall test; or one period of study followed by three tests. Students wrote what they could remember rather than reproducing exact wording or a required order. Repeated testing provided further attempts to retrieve the passage without showing it again. It therefore changed both what students did and how often they encountered the original text. [LP-W022]

Again, repeated study led on the five-minute test, whereas repeated testing led after a week. The students who studied throughout read their passage approximately fourteen times, compared with about three and a half readings in the repeated-test condition. Nevertheless, after a week the repeated-test group recalled 61 percent of the idea units and the repeated-study group 40 percent. More exposure to the passage had produced stronger immediate performance but less durable recall in this comparison. [LP-W022]

Students' expectations pointed in the other direction. Those who repeatedly studied were more confident that they would remember the material a week later. Repeated access to a readable passage could feel reassuring while a closed-text attempt exposed what was difficult to produce. The experiment thus joined a learning procedure to a problem of judgment: the experience that makes study feel successful need not identify the procedure that leaves more available later. [LP-W022]

Wang Zhixun and Zhang Jinkun's 2017 review compares explanations of retrieval-practice effects. Additional exposure, effort during retrieval, elaboration and changes in episodic context make different claims about why attempting recall can benefit subsequent remembering. Their discussion moves from descriptions of an observed effect to processes that might produce it. In particular, a difficult attempt and a successful learning outcome do not by themselves identify the mechanism connecting them. [LP-W017]

The episodic-context account they discuss links an event with the context encoded when it was encountered. A later retrieval attempt uses present cues to reconstruct the earlier context. Successful retrieval then updates the representation so that elements of the old and new contexts can participate in subsequent remembering. On this account, repeated retrieval changes the cues and search conditions through which the material can later be found. [LP-W017]

Their review also examines how a retrieval cue can become more diagnostic: it can direct a search toward a target while excluding competing items. It compares this kind of account with explanations emphasizing semantic elaboration or the effort involved in retrieval. Wang and Zhang develop the integration of

encoding and retrieval as an explanatory direction. The review's contribution is this comparison of proposed processes, beyond merely repeating the recommendation to test oneself. [LP-W017]

Returning after an interval

In the selected experiments from Hermann Ebbinghaus's 1885 work on memory, learning continued through returns to material on successive days. He learned series of syllables to the point of a first successful reproduction, waited twenty-four hours and relearned them to that criterion. The number of repetitions required on later days made the persistence of previous work observable even when the material could no longer simply be reproduced at once. [LP-W027]

The chapter considers series of different lengths and stanzas from Don Juan. These materials posed different learning demands, but the repeated procedure made later savings comparable with earlier effort for the material being learned. Each new occasion began with whatever remained from the previous occasions and continued until the specified reproduction criterion was reached. Learning was being examined as a history of returns, not only as the progress made within one sitting. [LP-W027]

The number of repetitions needed fell over successive days. Ebbinghaus examined the relation between repetitions performed and later savings, while keeping the outcome criterion stable. His self-experimental procedure made the effort of relearning a measure of persistence. It could reveal retained effects of earlier work that a simple present-versus-absent memory judgment would miss. [LP-W027]

He also distinguished the observed series from an estimate of the isolated contribution of a particular day's repetitions. To calculate that contribution, one would need to know how much relearning would have been required had those repetitions not occurred. Subtracting consecutive daily totals would not supply that missing comparison. Ebbinghaus set aside uncertain assumptions about it and reported the relations actually observed. [LP-W027]

Nicholas Cepeda, Edward Vul, Doug Rohrer, John Wixted and Harold Pashler's 2008 study varied two intervals separately: the gap between learning occasions and the later delay before the final test. Its 1,354 participants contributed to twenty-six combinations of study gap and test delay, using learned facts. A spacing schedule was therefore evaluated against a specified future occasion for remembering. [LP-W023]

The procedure can be followed on two clocks. Learn the facts, wait the selected gap and study them again; then wait the separately selected retention interval before testing what can be recalled. Altering the first interval changes when relearning occurs. Altering the second changes how long that learning must remain useful. The design examines their relationship rather than treating all delayed tests as equivalent. [LP-W023]

The best-performing gap changed with the intended retention interval. Cepeda and colleagues described the resulting surface as a temporal ridgeline: effective spacing depended on the horizon over which memory was being assessed. Their empirical account provides a way to investigate a scheduling decision that a simple instruction to spread practice out leaves unspecified. [LP-W023]

Relearning to a criterion

Katherine Rawson and John Dunlosky's 2012 review asks when practice testing can support learning that is both durable and efficient. Its question includes the time or effort needed to obtain a later result. Correct recall during an initial session is one possible stopping criterion, but it leaves open what should happen at later encounters with the same material. The authors examine a schedule that combines initial retrieval with returns to successful recall on subsequent occasions. [LP-W029]

This changes the unit in which practice is planned. Instead of deciding only how many additional repetitions to perform before ending the first session, the learner plans several sessions in which the material must again be produced correctly. The review's accounts connect initial learning, later relearning and the final retention measure. Durability and efficiency are evaluated together, so a schedule can be considered in relation to both what is remembered and what it took to establish that memory. [LP-W029]

Julian Roelle and Tobias Richter's 2025 review develops successive relearning as the combination of retrieval practice with distributed returns to a mastery criterion. Their example begins with an explanation of a biological mechanism learned until it can be given correctly. The learner then practises the explanation again after two, nine and sixteen days, each time relearning until correct. The return is an attempt to reconstruct the explanation, followed by whatever additional learning is needed to meet the criterion. [LP-W016]

A mathematical procedure can be organized similarly. Having applied it successfully in the first learning phase, the learner returns on several later occasions and continues until successful application is again possible. These examples distinguish a return to a criterion from a return for a fixed number of minutes. The later session responds to what the learner can then produce. [LP-W016]

Roelle and Richter also distinguish learning to recognize similarities within a category from learning to discriminate between categories. Blocking can keep related examples together; interleaving can put contrasting examples close enough to compare. Their review uses these distinctions to explain why retrieval, spacing, successive relearning and interleaving are separate design choices, even when more than one occurs within a learning programme. [LP-W016]

Mixing what must be distinguished

Nate Kornell and Robert Bjork's 2008 experiments asked participants to learn painters' styles from examples. In blocked study, examples by the same artist appeared together. In interleaved study, paintings by different artists were mixed. A later test used previously unseen paintings and asked who had painted them. Success required using what had been learned about a style to classify a new instance. [LP-W024]

Interleaving improved classification in the reported comparisons. Yet participants tended to judge blocked study more effective. Seeing several works by one artist in succession could make the style seem easier to grasp while that sequence was present. The later task required distinguishing among possible artists, which made comparisons across styles important. The authors discussed improved discrimination as an explanation for the interleaving benefit. [LP-W024]

The manipulation also clarifies a difference between interleaving and temporal spacing. Alternating artists increases the interval between examples by one artist, but it fills that interval with potentially informative examples from others. Kornell and Bjork explicitly distinguish their arrangement from simply waiting before showing another example of the same category. Both the time between related examples and the material placed between them belong to the procedure. [LP-W024]

Stanisław Czyż, Aleksandra Wójcik and Petra Solarská's 2024 review concerns contextual interference and motor transfer. It included forty-two studies, with thirty-four entering its quantitative analysis of delayed transfer. The review separated that outcome from immediate transfer and examined blocked and random practice in relation to performance on a changed task after an interval. Its average result favored random practice, with differences across settings and age groups. [LP-W032]

The review gives a broader empirical setting for motor scheduling experiments such as those in S71. It keeps transfer delay, the transfer task and the practice setting within the comparison, rather than pooling every later score as the same kind of result. Laboratory and applied conditions can organize substantially

different learning situations. The authors' synthesis makes those distinctions part of interpreting the estimated effect. [LP-W032]

Problem solving and productive failure

Manu Kapur's 2008 study placed 309 eleventh-grade students in three-person groups working on Newtonian-kinematics problems in a computer-supported collaborative setting. The students had already completed the relevant curricular unit. Groups were assigned more or less structured problems. Car-accident scenarios supplied situations requiring the students to interpret events, propose approaches and assess their attempts. The manipulation concerned problem structuredness after that prior instruction. [LP-W025]

The ill-structured groups struggled during their collaborative work, but the article reported better later individual near and far transfer. Their discussions provided occasions to generate and examine approaches without following the same well-specified path through the problem. Kapur investigated whether the apparently unsuccessful group activity could prepare resources for a later individual task. Initial solution quality and subsequent learning were deliberately examined at different points. [LP-W025]

The term productive failure gives that difference a practical force. A group's difficulty in reaching a satisfactory answer need not exhaust what its members have gained from trying. In this study, that argument is tied to the structure of the problems, the prior curriculum and the later tests. It concerns students working on physical problems; S31 develops separate histories of how organizations identify and learn from failures. [LP-W025]

Tanmay Sinha and Kapur's 2021 meta-analysis examines another explicit comparison: problem solving followed by instruction, against instruction followed by problem solving. The review included fifty-three studies and 166 comparisons. Its overall result favored problem solving first, and it examined how closely interventions followed productive-failure design principles. The sequence includes subsequent instruction as part of the learning arrangement. [LP-W026]

In this research family, learners' generated approaches can supply material on which instruction builds. Comparing an incomplete or unsuccessful attempt with a developed solution can make relevant features and distinctions available for discussion. The instructional phase is consequently more than a reward supplied after struggle: its connection to the preceding attempts is part of the proposed design. Sinha and Kapur examine implementation along with conceptual knowledge and transfer outcomes. [LP-W026]

Sinha and Kapur also report comparisons favoring instruction first, particularly among younger learners in second through fifth grades and in work on domain-general skills. Their analysis considers grade level, intervention duration and the experimental or quasi-experimental design alongside fidelity to productive-failure principles. It develops the question of when a sequence works, including how instruction makes use of the attempts that precede it. [LP-W026]

Learning techniques in classrooms

John Dunlosky, Katherine Rawson, Elizabeth Marsh, Mitchell Nathan and Daniel Willingham's 2013 review evaluates ten learning techniques through several dimensions: learning materials, conditions of learning, learner characteristics and the tasks used to assess the result. A technique's utility is considered in relation to the breadth of evidence and the situations in which it can be used. This supplies an evaluation framework as well as findings about particular techniques. [LP-W020]

Practice testing and distributed practice receive high utility assessments in that review. Testing includes low-stakes attempts to retrieve information, including self-testing; it need not mean another

consequential examination. Distributed practice concerns how learning opportunities are placed over time. The review's practical question is how these activities can be incorporated into studying and teaching, with evidence about retention beyond the initial encounter. [LP-W020]

Its interleaving discussion also concerns selecting what to do. A blocked run of one kind of problem can repeatedly cue the same procedure. A mixed set may require identifying the kind of problem before choosing a method. Thus changing the order can change the operation practised, even when the set contains the same problems. The review develops this explanation alongside the state of the evidence, while S70 examines the distinct content and outcomes of training in creative methods. [LP-W020]

Sterre Ruitenburg, Mieke Goos, Guus Lambert, Paul Kirschner, Halszka Jarodzka and Gino Camp's 2026 classroom meta-analysis examines retrieval practice, distributed practice and interleaving in school settings. It includes fifty-three studies, 233 effect sizes and 7,303 participants. The reported average effects favor the techniques in the selected comparisons. Its analysis also retains information about the alternative activity against which a technique was compared. [LP-W033]

For spacing, a comparison with massed practice differs from a comparison with practice that is already distributed to some degree. The ratio of study interval to retention interval and the form of the later test further distinguish the reported outcomes. For interleaving, the analysis examines whether feedback is immediate, delayed or absent. These are concrete differences in classroom arrangements, rather than interchangeable instances of a technique's name. [LP-W033]

The classroom review brings the scheduling questions back to activities conducted within teaching. A pupil can be asked to retrieve, to return after an interval or to choose among different problems, while the teacher also determines feedback and assessment. Its findings develop the study of these arrangements in their actual instructional settings. They extend a history in which what is practised, when it is practised and what later performance should show remain central questions. [LP-W033]

S101 Fear, avoidance and learning to approach

What an avoidance procedure observes

Krypotos and colleagues distinguish arrangements in which a designated response prevents an aversive event from arrangements in which withholding a response prevents it. The act can precede an event or terminate one already under way, depending on the experimental contingency. Unsignaled avoidance schedules give researchers a way to ask whether a warning cue is necessary for the behavior under study. The procedure specifies the available action, the timing of the aversive event and how the two depend on one another. [T03-W001]

Hofmann and Hay's account asks what function avoidance serves in its setting. Avoiding an actual threat can protect the person. In other circumstances, a protective action can keep the person from finding out that the feared outcome would have failed to occur even without it. Their discussion brings both functions into clinical assessment. The costs of silence discussed in S51 arise in organizational settings where an adverse response can be a real possibility. [T03-W002]

What is learned during approach

Craske and colleagues organize exposure around a prediction about a feared event. Their treatment illustrations describe eliciting the prediction, arranging the exposure, observing what occurs and revisiting the expectation. Reduction in distress within the exercise and a change in expectation are different observations. Their account gives varied contexts and retrieval cues a role because the person may need to draw on the learning in circumstances different from those of the treatment. The cases illustrate the procedure and its rationale. [T03-W003]

Shiban, Pauli and Mühlberger compared exposure in one virtual context with exposure in several contexts in a study of spider phobia. The subsequent renewal test asked whether the feared response returned in another context. The authors reported less renewal with the multiple-context arrangement under their test conditions. This result concerns transfer between specified experimental settings. Practical functioning in everyday settings requires its own observations. [T03-W005]

Magdalena Sich's 2024 dissertation asked 62 patients receiving inpatient or day treatment to rate how willing they would be to undertake exposure in virtual reality or in person. They viewed example videos of the two formats and rated them before and after an experience with the virtual medium. They rated their willingness to undertake virtual exposure higher at both points. In the separate forced choice between formats, 40 initially preferred virtual reality and 38 did afterward. Eleven switched to in-person exposure and nine switched the other way. The study measured willingness and preference for the format. [T03-W004]

S73 Brief therapy: goals, exceptions and practical change

When attempted solutions sustain a problem

In the Palo Alto model of brief therapy, an effort to remove a difficulty can become part of what keeps it going. Michael Rohrbaugh and Varda Shoham's 2001 account calls attention to the interaction between a complaint and the attempted solutions surrounding it. A partner presses for discussion because the other withdraws; the other withdraws further under pressure; the first responds by pressing harder. The complaint need not be imaginary, and neither participant need intend the result. The repeated response to it supplies a pattern that the therapist can try to interrupt. [TD-W001]

The therapist first asks what actually happens, what the participants have tried, and what follows those attempts. A complaint such as poor communication becomes a sequence of observable actions. The intervention is then fitted to that sequence: reducing pursuit, changing the circumstances of a conversation, or asking someone to observe a pattern instead of immediately trying to correct it. The next encounter examines whether the response changed. The point of reconstructing the sequence is to locate an intervention, rather than to require a complete explanation of the person's history before anything useful can be attempted. [TD-W001]

Rohrbaugh and Shoham describe other ironic processes, including reassurance that invites further requests for reassurance and efforts to force sleep that increase wakefulness. Their account distinguishes a complaint's origins from the transactions maintaining it. It also makes the therapist's manner of proposing a change consequential. An instruction that encounters opposition may need to acknowledge the client's position or take an indirect form. The same complaint label can conceal different interaction patterns, so the proposed interruption depends on the attempted solution and the response it evokes. [TD-W001]

Brevity in this tradition did not mean that an arbitrary smaller allowance was always better. The review reports a small experiment comparing limits of five and ten sessions, with thirteen and fourteen cases respectively. Although the clinic's treatment averaged about six sessions, the five-session condition fared worse. An average describes what happened across cases; a cap changes the conditions under which each case can proceed. The authors' discussion of this experiment is particularly useful for understanding why a brief approach can seek economy without treating the shortest permitted treatment as its objective. [TD-W001]

Developing solutions from exceptions

Steve de Shazer and his colleagues' 1986 paper described a further change of emphasis at the Brief Family Therapy Center. Instead of organizing every intervention around the attempted solution that maintained a complaint, they increasingly developed solutions from clients' goals and from exceptions to the complaint. Their distinction between an explanatory match and a practical fit was deliberate. A useful intervention might fit what the client needed to accomplish without reproducing a detailed model of the problem. The therapist therefore sought descriptions of what would be different, what was already happening on better occasions, and what the client could recognize as progress. [TD-W002]

One family was asked to observe, between visits, what each person did that the others wanted to continue. This assignment changed what the family was looking for. At the next meeting, particular acts worth preserving could be described and discussed, rather than only another inventory of failures. The therapists then asked how the family had helped those events occur and how they might encourage more of them. In the authors' account, the family invented its eventual solution between the first and second sessions; the later work helped it sustain and extend the useful changes. Observation was connected to

action through the selection of concrete events and the participants' account of their own contribution. [TD-W002]

Another task introduced a contrast into parents' efforts to get homework done. Coin-toss days determined when they would refrain from mentioning it, giving them occasions to notice what happened without their usual prompting. If the child did the homework on those days, the next conversation could examine a successful difference already present in the family's experience. The instruction was more specific than a general demand to be positive: alter a recurring response, observe the result, and use the result to decide what to continue. [TD-W002]

The first-session format connected the complaint, exceptions, goals and possible solutions. Concrete questions established who did what, when the difficulty occurred, and how anyone would know that it had improved. After a consultation break, the team delivered a message combining acknowledgments of the client's resources with a task or clue directed toward useful change. Even an initially vague complaint could lead to an observable goal. Asking what a close friend would notice when the person felt more certain of who they were made a broad aspiration answerable in terms of everyday conduct. [TD-W002]

At a subsequent session, the therapists elicited what was better, distinguished new changes from previously overlooked exceptions, and asked how the client had brought them about. A useful difference became material for the next action. When nothing had improved, the search shifted toward what was preventing deterioration or what small difference might still be amplified. When things were worse, the authors described questions about the course of that worsening and signs of an approaching improvement. These were the center's reported conversational practices, rather than results establishing the effectiveness of each question separately. [TD-W002]

Continuation also depended on the developing case. The center usually placed the second session a week after the first, then lengthened intervals when the client reported improvement: two weeks, followed by three weeks to a month. When the team thought the change might be sufficient, it asked the client whether another meeting was needed and when. The center reported about 1,600 cases from 1978 to 1983, averaging six sessions. Follow-up was based on self-report, and only twenty-eight of fifty-six clients in a separate first-task project were reached. Those service observations accompanied the description of an evolving practice; they were not a randomized demonstration that six sessions constituted a minimum effective dose. [TD-W002]

How the conversation is built

The Solution Focused Brief Therapy Association's 2025 treatment manual develops the conversation itself as a sequence of listening, selecting and building. The therapist listens for the client's descriptions, selects an expression that can be developed toward the preferred future, and asks a question that builds on it. The client's words provide more than rapport. They supply the distinctions through which a change will be recognized and make the next question responsive to this person's situation. A preferred future becomes increasingly specific as the therapist asks who would notice, what they would see, and what would happen next. [TD-W003]

In the manual's mother–daughter example, a successful conversation about a science project supplies a concrete instance of better communication. The useful detail is that the mother listened without simultaneously doing something else. Following that detail reveals an interaction the participants can recognize and potentially repeat. The therapist can ask how it happened, what the daughter noticed, and what the mother did that made a difference. The exception thus supplies actions and circumstances for a goal that would otherwise remain an abstract wish to communicate better. [TD-W003]

Scaling questions develop another kind of specificity. A rating locates the client's current position relative to a desired state, after which the therapist asks what makes that position possible and what a small

further improvement would look like. The work lies in the descriptions attached to the numbers. Asking why the current rating is not lower can elicit resources already in use; asking what the next increment would reveal can identify an attainable difference. The scale is a conversational device for constructing and noticing change, rather than a substitute for every outcome measure. [TD-W003]

The manual's EARS sequence links eliciting a useful change, amplifying its details, reinforcing the client's contribution, and starting the process again. If the person reports no progress, coping questions can explore how they have managed to continue under difficult conditions. A report of worsening calls first for acknowledgment, followed by questions fitted to what has happened. The decision about another meeting remains joint. This later manual gives a developed account of the approach in 2025, including how its procedures adapt during an encounter, without imposing a fixed session count. [TD-W003]

What outcome comparisons measure

Wallace Gingerich and Lance Peterson's 2013 systematic qualitative review included forty-three controlled outcome studies. Thirty-two reported statistically significant favorable outcomes, ten reported favorable trends, and one reported no improvement. Their inclusion procedure accepted studies using at least one of ten identified solution-focused techniques. The resulting body of evidence therefore encompassed varied interventions, settings and comparators. A comparison with no treatment or usual services answered a different question from a comparison with another active psychotherapy. [TD-W004]

Only three studies in that review directly compared treatment duration. One comparison involved experienced therapists and trainees, making experience relevant to the interpretation of the difference. The review's broad conclusion that the approach was promising and often brief did not arise from forty-three independent dose comparisons. Nor does a nonsignificant difference between therapies establish equivalence in the statistical sense. Reading the review by comparator and outcome makes it possible to distinguish evidence of beneficial change from evidence about the contacts required to obtain it. [TD-W004]

Contacts, elapsed time and lasting improvement

The Helsinki Psychotherapy Study gives these distinctions concrete form. Paul Knekt and colleagues randomized 326 patients with mood or anxiety disorders among solution-focused therapy, short-term psychodynamic psychotherapy and long-term psychodynamic psychotherapy. The first report appeared in 2007, with the journal issue following in 2008. The solution-focused arm allowed up to twelve sessions over eight months; the short psychodynamic treatment provided twenty weekly sessions over about five to six months; the long treatment met two or three times weekly for approximately three years. These were three different treatment schedules, rather than equal-duration packages with interchangeable labels. [TD-W005]

Among those who started the assigned therapies, mean assigned contacts were 9.8, 18.5 and 232 respectively. Their mean elapsed treatment periods were 7.5, 5.7 and 31.3 months. The solution-focused treatment used fewer contacts than the short psychodynamic treatment while extending over a longer calendar interval. When auxiliary psychotherapy was included, mean total contacts rose to 29.9, 46.9 and 235. Counting only scheduled visits would have missed some of the treatment received during follow-up. [TD-W005]

The short treatments produced improvement earlier, while the long psychodynamic treatment achieved greater symptom improvement by the three-year assessment. No statistically significant difference was found between the two short therapies. The history is consequently intelligible at several scales: the pace of early relief, the number of meetings, the period over which they occurred, additional care, and

symptoms years later. Choosing a single one of these quantities as the meaning of “faster” would discard a substantial part of what this study measured. [TD-W005]

A brief intervention in primary care

Cooper and colleagues' 2024 primary-care trial tested three weekly solution-focused sessions lasting twenty to thirty minutes, alongside usual care. The comparison group received usual care and problem-focused assessment contacts matched in frequency and duration. The intervention was consequently tested against more than an absence of professional attention. It was not a comparison with cognitive behavioral therapy. The brief schedule gave the researchers a defined intervention to examine, with goals and exceptions developed into actions and progress reviewed across the three weeks. [TD-W006]

Eighty people were randomized. The body of the report gives sixty-nine baseline participants and sixty-eight completing the intervention period. Mean depression scores on the PHQ-9 fell from 18.17 to 9.71 in the solution-focused group, compared with 17.76 to 18.94 in the comparison group; anxiety and well-being measures also favored the intervention. The assessment horizon was three weeks. The trial supports improvement under this short primary-care arrangement, while its design does not identify the fewest sessions that would have worked or establish how long the improvement persisted. [TD-W006]

Constructing alternatives to harmful coping

邹晓伟 and colleagues' 2021 analysis of solution-focused work with nonsuicidal self-injury begins with the function of a person's coping behavior. An attempt to regulate distress may pursue a comprehensible aim through a harmful means. The proposed therapeutic work recognizes the aim, explores occasions when the person coped differently, and develops safer actions that can serve a similar purpose. Preferred-future and scaling questions make the alternative specific: what would be happening at a slightly better point, and what has already helped the person reach the present one? [TD-W007]

The authors connect conversation with practical aids such as a written repertoire of coping actions, reminders and crisis cards, together with rehearsal of alternatives. The repertoire must be available when the situation arises, rather than remaining an insight confined to the interview. This is a conceptual clinical account drawing on a limited research base, including some combined interventions. Its contribution here is the connected proposal: identify what the action is doing for the person, find an existing exception, construct an accessible substitute, and prepare to use it. [TD-W007]

From a teacher's case knowledge to a workable response

黒沢幸子 and colleagues' 2015 eleven-step consultation model applies solution-focused work to teachers seeking help with cases. The consultant first establishes the setting and history, then explores resources in the case, exceptions, earlier helpful and unhelpful actions, and the teacher's own capacities. The resource steps allow flexibility in order. The conversation then revisits the teacher's need, specifies a small positive goal, considers helpful actions, offers affirmative feedback, and brings in the consultant's professional proposal. The teacher's knowledge of the classroom and the consultant's expertise are both used. [TD-W008]

Closure includes an explicit test of the proposed response: does the teacher judge it feasible and useful? If either answer is unfavorable, the proposal needs alteration. This prevents the consultation from ending merely because the expert has supplied an intelligible recommendation. What the teacher can carry out in the actual setting becomes part of the construction of that recommendation. The eleven steps thus connect description, resource discovery, a revised goal and a workable next action. [TD-W008]

Twenty-seven teachers brought two cases each to the study, with the order of solution-focused and problem-exploration consultations randomized. Fifty-one consultations were usable for analysis, twenty-six solution-focused and twenty-five problem-exploration, all conducted by one consultant. Immediate ratings favored the solution-focused format on such dimensions as usefulness, reduced difficulty and action planning. These were evaluations of the consultation and the teacher's readiness to act, rather than measurements of children's subsequent recovery. [TD-W008]

The shape of the evidence

Michele Orr, Sharlette Ryder and Julie Considine's 2026 scoping review of adult community mental-health services illustrates why the shape of a literature matters alongside its size. Twenty included papers did not represent twenty independent trials: fourteen arose from the Helsinki study. The review maps a range of applications and reports uneven specification of the intervention across studies. It directs attention to which populations and comparisons have actually been examined. Treatment benefit, efficient use of contacts, and a procedure that reliably chooses among modalities for a particular person remain distinct empirical questions. The earlier accounts of feedback and practice in S71 and S72 provide neighboring histories of change through activity and review; the therapeutic studies above retain their own interventions and outcomes. [TD-W009]

S56 Shared language and collaborative memory

A short expression can depend on a history of joint work. The following accounts examine how people establish a reference together, learn what another person knows, reuse local descriptions and remember an exchange. Their experiments also expose occasions when agreement, familiar wording and successful action come apart.

S41 treats language, learnability and shared repertoires; S42 develops context, interpretation and records. The sources here follow the interaction through which particular partners make an expression usable, and the consequences when participation, membership or memory changes.

Clark and Wilkes-Gibbs: a reference is completed together

Herbert Clark and Deanna Wilkes-Gibbs's 1986 account treats referring as an activity that the speaker and listener complete together. A speaker presents a description or invites the other person to supply one. The listener can accept it, reject it or hold judgment while seeking more information. A problem can lead either person to repair the wording, add information or offer a different way of seeing the object. The resulting description is considered again. Their model follows this cycle through to mutual acceptance, including cases where an apparent acceptance is followed by another check. [CG-W025, pp. 7–9, 16–25, table 3]

The distinction among ways of changing a description matters. Adding that a person is standing up can supplement an earlier description of someone holding a book. Replacing an awkward description of an upside-down hat with a person pointing left proposes another perspective. In the recorded exchange, the original speaker subsequently uses the listener's replacement. The listener's contribution changes what both can say next. An acknowledgment can also accept the description so far while postponing a final decision about which object it identifies. [CG-W025, pp. 21–25]

Their experiment used eight pairs of students, with a screen preventing each person from seeing the other's twelve unfamiliar tangram figures. One person's ordering supplied the target; the other rearranged the cards through conversation. The same figures appeared in different orders across six trials. The experimenter identified mismatches after each trial, and the overall error rate was two percent. Transcripts recorded hesitations, interruptions, acknowledgments and repairs alongside the descriptions themselves. These features made it possible to examine how an exchange reached completion. [CG-W025, pp. 10–11, fig. 1]

The director's average word count per figure fell from about forty-one on the first trial to eight on the sixth. Speaking turns fell from 3.7 to about one. Requests for expansion and confirmation became less frequent, as did the speaker's repairs and follow-up checks. The authors explain the change through previously accepted perspectives: an elaborate description of a person skating can later support a compact reference to the ice skater. Their proposed economy concerns the work of both participants through the whole exchange. A brief initial formulation that requires extensive repair can demand more joint work than a somewhat longer one that is readily understood. [CG-W025, pp. 12–17, 21–28, figs. 2–3]

In their discussion, acceptance is judged against the purpose of the conversation. Directions intended to help someone reach a house require a different level of understanding from mentioning an address at a party. They also describe pressures that can make acceptance misleading: a listener may avoid requesting clarification to spare embarrassment, avoid offending the speaker or keep the exchange moving. The authors distinguish the participants' acceptance of an utterance from perfect comprehension, and allow an earlier acceptance to be reopened when a problem becomes apparent. [CG-W025, pp. 26, 33–35]

Wegner: remembering an item and remembering where to find it

Daniel Wegner's 1987 chapter develops a transactive memory system from individual memories and the communication connecting them. His distinction among an item, its label and its location explains how people can retrieve information they have not retained personally. Someone may remember the label of a needed document and who has it. Another person can therefore serve as a location for knowledge. The resulting system includes differences in what its members know, together with knowledge of those differences. [CG-W028, pp. 186–191]

Retrieval can require a transaction that changes the initial cue. Wegner describes a requester asking a colleague for information; the colleague cannot immediately supply it but remembers having given it to the requester on a previous occasion. That reminder lets the requester locate the document. The information is recovered through their combined exchange. At encoding, participants can likewise negotiate what an item should be called, who should retain it and where it will be available. These arrangements distribute responsibility as well as content. [CG-W028, pp. 190–192]

Wegner reports a study involving twenty dating couples who attempted to remember sixty-four items from different knowledge domains. Participants judged their own and their partners' relative expertise. One partner studied an item set for a minute before passing it to the other for thirty seconds, making the difference in exposure apparent to both. Recall varied with personal expertise and this circumstantial responsibility. When a person had the longer exposure, recall also depended on whether that person believed the partner possessed relevant expertise. Wegner interprets this as an individual's memory responding to what the other member of the pair is expected to remember. An inaccurate understanding of the partner's expertise can leave both relying on storage that has not occurred. [CG-W028, pp. 192–194, table 9.1]

Labels can also change the information being retained. In Wegner's illustrative account, calling an ambiguous sighting a UFO supplies an organizing interpretation for later discussion. Recollection can come to reflect that discussion rather than the original observations. Retrieval can drift from its initial question, while bringing together separately held details can produce an interpretation that neither person initially possessed. These possibilities are part of his account of how the transactions transform information. [CG-W028, pp. 195–197]

The arrangement can be lost. When a group dissolves, a remembered location may point to a person who is no longer available. Private expressions can remain intelligible to former participants but convey little to someone outside the relationship. Wegner describes this as a loss of access and organization even when individuals retain portions of the information. [CG-W028, pp. 198–199]

Schober and Clark: hearing an agreement and participating in it

Michael Schober and Herbert Clark's 1989 study examined the difference between participating in such an agreement and hearing it develop. Ten pairs arranged twelve of sixteen unfamiliar figures across six trials. Forty other students later attempted the same arrangement while listening to recordings. Some heard the conversation from its beginning; others entered at the third trial. Half of each group could pause the recording. Listeners present from the beginning placed 88 percent of figures correctly, compared with 68 percent for those entering later. The original conversational partners nevertheless achieved 99 percent accuracy. Hearing every word had not supplied the same opportunity to establish what an expression meant. Pausing produced no statistically significant accuracy advantage. [CG-W009, pp. 214–220, figs. 1–3]

A second experiment addressed the possibility that recordings themselves caused the disadvantage. An overhearer now listened in the room, separated by barriers and presented as a coder so that the speakers

would not address that person. The principal analysis used ten triples after four exclusions, including two groups whose original matchers continued making unusual numbers of errors. Matchers achieved 98 percent accuracy and overhearers 85 percent. Including those two error-prone groups reduced matcher accuracy to 95 percent but preserved the difference. [CG-W009, pp. 222–223, fig. 4]

The difference was not simply a slower average response by overhearers. Their placements were spread much more widely around the points when the partners agreed to proceed. When overhearers placed a figure after that point, errors were more frequent. The authors' explanation concerns participation in grounding: a listener can request another description or propose a different perspective until understanding is adequate for the task. An overhearer must work with the perspective and stopping point chosen by the other two. Their experiment distinguishes access to the conversation from influence over how its references become established. [CG-W009, pp. 224–230, figs. 5–7]

Brennan and Clark: provisional agreements with particular partners

Susan Brennan and Herbert Clark's 1996 experiments examine what people establish when they agree on a way of referring to something. Their conceptual pact is a temporary agreement about how to conceptualize a referent. One person can propose a description, the other can question it or offer a substitute, and both can subsequently use the expression they settle on. In one recorded exchange, a listener did not recognize a proposed shoe name and initially wondered whether it named a dog. An explanation established an acceptable shoe description. In another, the listener supplied the fish identification that both subsequently used. [CG-W007, pp. 1484, 1487]

The first experiment separated this history from the information needed to distinguish an object at the present moment. Forty-eight students, working in pairs, arranged picture cards without seeing their partners' cards. An initial set contained only one shoe; another introduced competing shoes, requiring a more specific description. When the original set returned, the specific expression could remain useful as an established precedent even though a simpler term would now distinguish the target. Directors retained the same content words on that return 69 percent of the time after four trials with competitors, compared with 35 percent after one. The number of previous uses mattered, beyond simply repeating the most recent formulation. [CG-W007, pp. 1484–1487, table 2]

The pact could also change. In the second experiment, partners alternated the speaking and matching roles, and either could use an established description. Expressions became simpler, and markers of uncertainty became less frequent as an agreement developed. A third experiment introduced a different listener after four matching trials. Across fifty participants in ten continuing-partner groups and ten partner-switch groups, directors retained the previous description on the first subsequent trial 48 percent of the time with their existing partner, compared with 18 percent with a new partner. Some initially offered an elaborated version of the old description; over further trials they accommodated to the new listener. The authors describe agreements that participants can revise, rather than names fixed once and retained indefinitely. [CG-W007, pp. 1488–1492, table 5]

Their appendix preserves an informative exception. Two partners repeatedly used different terms for the same dog when their roles changed. The authors describe them as understanding each other's terms while maintaining different perspectives. Agreement sufficient to perform the task could therefore coexist with a failure to converge on one expression. [CG-W007, pp. 1492–1493, appendix]

Mitsuda and colleagues: compare the layouts people believe they share

Koh Mitsuda, Ryuichiro Higashinaka, Yuhei Oga and Tetsuya Kinebuchi's 2021 study records a shared task as the participants negotiate and act. Each person sees a private screen containing the same collection of shapes, initially arranged differently. The pair chooses what to make through text chat and

moves shapes toward a common layout. The shapes can be moved but cannot be rotated, resized or deleted. The task therefore lets the researchers compare the conversation with the participants' changing arrangements, including occasions when verbal agreement and the resulting objects diverge. [CG-W026, pp. 1697–1698, fig. 1]

One pair discusses making a face, then agrees on Pinocchio. When a participant discovers that a triangle cannot be rotated, they decide to make the face tilted instead. They continue discussing the positions of its parts and eventually express satisfaction. Their final layouts nevertheless differ. A name for the intended whole and repeated assent have supported the interaction while leaving a consequential difference unresolved. [CG-W026, pp. 1698–1699, table 2, fig. 2]

The report describes a corpus of 984 dialogues involving 287 participants. To examine final outcomes, the authors developed seven layout categories and had a separate annotator classify two hundred sessions. Exact matches accounted for nineteen percent. Another six percent matched after allowing a translation of the whole layout, which the task instructions had not required participants to eliminate. Twenty-eight percent were wholly mismatched. Other outcomes included swapped shapes, a reflected arrangement and a difference in the overall scale. These categories give the researchers a way to describe what the apparent agreement had preserved and what it had left unsettled. [CG-W026, pp. 1698–1699, tables 1, 3]

They also introduce a distance based on the relationships between pairs of shapes in the two layouts. Following its changes over the conversation distinguishes arrangements that converge early, those that converge later and those that fail to converge. Inspection of unsuccessful exchanges reveals different interpretations of an oblique orientation and failures to check locations. In this preliminary analysis, common ground is made observable through the specified layout task and its operations. The researchers can compare what participants say with what they actually construct. [CG-W026, p. 1700, fig. 3]

Naming a whole and describing its parts

The human-dialogue analysis reported by 齋藤結, 光田航, 東中竜一郎 and 南泰浩 in 2023 examines how names enter the construction of a shared layout. Their annotation distinguishes a name for one shape, a name for several shapes forming the intended goal, and names for other groups of shapes. A separate classification describes what kind of thing the name invokes, including ordinary names already associated with the displayed shapes or building icons. This separates naming the intended whole from identifying a component. [CG-W027, pp. 1985–1986, §§2–3]

Their example begins with a proposal to arrange two circles like the wheels of a car. The other participant recognizes a truck. The conversation can then describe a quadrilateral as its passenger compartment, another part as its front and a particular orientation for those parts. The shared proposed object supplies relationships that the participants can use in subsequent instructions. Naming here participates in choosing and describing what to build. [CG-W027, p. 1985, fig. 1]

In the comparison of recorded dialogues, the first name for the overall goal appeared at a mean of 5.7 turns in successful dialogues and 8.1 turns in unsuccessful ones. The authors associate this earlier agreement with subsequent discussion of constituent parts. Their closer classification also preserves different routes. Many dialogues succeeded without naming an overall goal, using familiar names for the shapes instead. Conversely, repeated proposals for a goal in the middle of a dialogue could accompany participants talking past one another or losing track of which idea was under discussion. A goal agreed near the end could leave insufficient time for the remaining construction. The recorded examples and timing comparisons make the progression of the dialogue consequential alongside the availability of names. [CG-W027, pp. 1986–1987, §§5.1–5.2, figs. 4–5]

Nault and colleagues: remembered words, remembered meaning and familiarity

Daniel Nault, Rohit Voleti, Matthew Nicastro and Kevin Munhall's 2023 experiments extend the referential task to delayed memory for what was said. Participants discussed images while arranging them to match a partner's screen. In a later individual test, they saw the same images and were asked to recall the most efficient descriptions used in the conversation. The images served as cues, and participants had not been told during the original interaction that their memory would later be tested. The study compares retained wording with an estimate of retained meaning. [CG-W011, Experiment 1, Procedure and Dependent Measures]

The first experiment analyzed twenty-four participants in twelve acquainted pairs; twenty-three completed the later recall task. They matched photographs of ordinary objects across two rounds, with three trials in each round and a change of roles between rounds. Delayed recall occurred seven to fourteen days later. A second experiment analyzed forty-eight participants, half in friendship pairs and half paired with a stranger, using unfamiliar faces whose visual similarity could be more systematically controlled. Recall in that experiment was supervised individually over a video call. Both experiments also included immediate recognition of previously seen images, a separate task on which performance was near ceiling. [CG-W011, Experiments 1–2, Methods and Recognition Memory]

The local common-ground measure was the director's normalized reduction in words between the first and third descriptions of an image. For later wording, the researchers measured word overlap with the final description, allowing specified spelling and word-root variants. For meaning, they used sentence-model estimates of semantic similarity between that final description and the typed recollection. These measures let them ask whether the efficiency developed during the interaction predicted two different aspects of later recall. [CG-W011, Experiment 1, Dependent Measures]

In both experiments, greater local shortening predicted more overlap with the original words at follow-up. It did not significantly predict the estimated semantic similarity. The authors also found that shorter final descriptions were easier to recall. They explicitly identify the need to separate that explanation from a contribution of common-ground formation itself. Their account therefore preserves an association with remembered wording while leaving the proposed relationship with richer remembered meaning unresolved. [CG-W011, Experiments 1–2, Recall Memory; Experiment 1, n. 1]

An additional control addressed a basic difficulty with image-cued recall. A picture can elicit similar words from people who never discussed it together. Twelve additional participants in each experiment described the images without the preceding conversation. The recalled descriptions from conversational participants overlapped their own original interaction more strongly than these independently generated descriptions did. The authors used that comparison as evidence for conversation-specific recollection, while recognizing that some raw overlap came from the images themselves. [CG-W011, Experiments 1–2, Control Experiment]

Friendship supplied a different pattern from local shortening. Friends described the faces with fewer words overall, but their descriptions did not shorten at a significantly different rate from those of strangers. They also showed no significant recall advantage in the principal mixed-model analyses. Secondary analyses that considered average description lengths favored strangers. The authors distinguish efficiency associated with prior familiarity from the change achieved during the task. They also argue that constrained picture descriptions may give a poor view of the richer semantic memories formed in ordinary conversation. Their results leave room for that further investigation while retaining the observed difference between concise communication and later recall. [CG-W011, Experiment 2, Description Lengths and Recall Memory; General Discussion]

S60 External memory and continuity of work

How can someone continue an enquiry, a design or a collective undertaking after an interruption, or after the people who began it have left? A finished document may preserve the answer while losing the route to it. Meeting minutes may preserve the conversation while scattering one decision across many files. A conclusion may remain in a collection after the reasons for accepting it have changed. Historical approaches to external memory addressed these problems by changing what was recorded and how a later participant could work with it.

Records and archival context are treated in S42; collaborative remembering and knowing who knows what are treated in S56. The accounts here develop external working structures, recorded deliberation, the retrieval of project experience and the maintenance of a changing position. Their concrete objects range from a trail through documents to a design-review history and a program's reasons for believing a proposition.

Bush and the preservation of an enquiry

Vannevar Bush's *As We May Think* (1945) proposed the memex, a personal apparatus holding books, records and communications on microfilm. A user could bring material onto viewing screens, add notes and retrieve an item by its code. Bush's distinctive operation was associative indexing: connecting items so that calling up one could lead directly to another. Ordinary classification placed an item under a heading; a trail recorded a particular route through material that might belong to many different classifications. The same item could participate in several trails. [CM-W001, §§6–8]

The construction of a trail was itself intellectual work. Bush described a reader investigating bows and arrows, moving from an encyclopedia to historical accounts and material about the properties of wood and other materials. The example illustrated an enquiry, rather than a search for one already formulated answer. The reader selected passages, linked them, added comments and inserted an analysis of his own. A trail could branch into a subsidiary line of investigation and then return to the main route. Its organization recorded connections that the reader had found useful, including connections that no cataloguer would have anticipated.

Bush made continuation tangible by returning to this reader years later. A conversation revived the old interest. Instead of reconstructing the investigation from memory, the reader called up the trail and followed its steps. He could reproduce the trail and give it to his friend, who could insert it into another memex and connect it to further material. What travelled was consequently more than a list of references: it included the sequence of consultation and the reader's annotations. Another person could examine the path, depart from it or extend it.

The memex was a proposal, and Bush described its mechanisms through projected uses. Within that proposal, however, the preservation of unfinished intellectual work was quite explicit. He envisaged the inheritance of the scaffolding of an enquiry as well as its completed products. A private working path could become a usable contribution to someone else's work without first being rewritten as a conventional publication.

Engelbart and a working structure that can be revisited

Douglas C. Engelbart's *Augmenting Human Intellect: A Conceptual Framework* (1962) considered the person together with language, artifacts, methods and training. His concern was the organization of the activities through which someone understood a problem and developed a solution. External symbols belonged inside that activity: rearranging them could change what the participant could see, compare and

attempt next. A finished report was one possible product of a more extensive working structure. [CM-W002, Introduction; II.A; III.A–B]

Engelbart's own experiment with cards gave this distinction practical detail. He placed relatively small units of thought on separate cards, including source references and page numbers where appropriate, and used descriptors to help sort them. While preparing a memorandum he distinguished material concerning its specifications, its organization and its content. Within these groups he kept considerations, tentative possibilities and currently accepted product statements. A reader of the eventual memorandum would see the resulting exposition; the writer's working collection also contained the questions, alternatives and reasons through which that exposition had been developed.

One difficulty appeared when a product statement changed. A new consideration might suggest a revision, but the author might no longer remember why the earlier wording had been adopted. Being able to recover the other considerations linked to that statement would help him assess the revision. Engelbart also wanted to try several organizations of the same statements without making several copies of every card. Recorded associative trails could preserve these trial organizations so that an earlier structure could be reconstructed when needed. His card experiment exposed both the usefulness of these relations and the difficulty of managing them with the equipment available to him.

The report's computer-supported demonstration developed these possibilities through the imaginary worker Joe. Statements could be linked into an argument rather than confined to one linear sequence. A participant could inspect supporting reasoning, follow implications or change the view of a structure without rewriting all its contents. When an investigation had to be put aside, its assumptions, objectives, attempts and intermediate results could be left in an organized form. Later information or a new technique could then provide an occasion to revisit it. The retained structure gave the returning worker a place from which to proceed.

Engelbart extended this demonstration to collaboration. Several participants could examine a common working structure, explore alternatives and attach objections where they applied. Their separate contributions could enter the shared reasoning at the relevant statement, rather than waiting for a final report to circulate. These were capabilities illustrated in his proposed augmentation environment. The account made the coordination of tools and working methods central: storing symbols was useful because participants could manipulate their relationships while reasoning, revising and communicating.

Doyle and conclusions whose support changes

Jon Doyle's *A Truth Maintenance System* (1979) addressed continuity inside a reasoning program. A program could derive conclusions from assumptions and then encounter information requiring an assumption to change. Simply adding the new information left an important question unanswered: which earlier conclusions should still govern the program's reasoning? Doyle recorded reasons for potential beliefs and used their dependencies to maintain the current set. [CM-W003, §§1–2.1, 3.2, 4]

The system represented a potential belief by a node. A node could have several justifications. In a support-list justification, one list contained nodes that had to be currently believed and another contained nodes that had to be outside the current set of beliefs. This second list allowed a default to depend on the absence of a defeating belief. A node was *in* when it had an acceptable supporting justification and *out* when it did not. Doyle explicitly distinguished these statuses from true and false: they described the program's present support for a belief.

His scheduling example showed the operation. A program tentatively placed a meeting on Wednesday because it had no reason to reject that day. A rule connected Wednesday with a meeting time of 1 p.m. New scheduling information then supplied a reason against Wednesday. The system could trace the dependencies: the Wednesday assumption lost its support, and the conclusion about 1 p.m. lost the

support it had obtained from that assumption. The statements and their justifications remained represented, while their current status changed. A later change in the defeating reason could permit an assumption to become current again.

The maintenance procedure examined the changed node and its repercussions, seeking well-founded support rather than allowing beliefs to sustain one another through a circular argument. The problem-solving program supplied the statements, inference steps and declarations of contradiction; the maintenance system operated on the recorded justification relationships. A node could stay in if another valid justification survived the loss of one route of support. Thus a change did not require indiscriminately discarding everything recorded after an earlier choice.

Doyle also described dependency-directed backtracking. When a contradiction was marked, the procedure traced its supporting argument to the assumptions involved. It recorded the incompatible combination as a *nogood* and revised an assumption with a reason tied to that incompatibility. In a second meeting example, a proposed time and room conflicted with an existing booking. Trying another room preserved the time preference until another contradiction required the time itself to change. The retained nogood helped the program avoid simply returning to the rejected combination. The example explained a revision mechanism; the implementation's arbitrary selection of an assumption to reject did not establish a generally good policy for choosing among revisions.

Conklin and Begeman on recording deliberation

Jeff Conklin and Michael L. Begeman's gIBIS system (1988) organized design discussion around issues, positions and arguments. An issue posed a question; a position proposed a response; arguments supported or objected to a position. Other links connected related questions and proposals. The arrangement preserved rejected possibilities and the reasons discussed around them, alongside proposals that participants continued to support. It offered a later reader an entry into the deliberation behind a design. [CM-W004, §§2–3, 5.1–5.7]

The interface combined a graph browser, a structured index and a window displaying the selected item's contents. Selecting an issue allowed the user to create a responding position with the appropriate link already established. Selecting a position made supporting or objecting arguments available as subsequent operations. These typed operations gave a contribution a place in the discussion as it was written. Nodes also carried attributes such as author and subject. Queries could retrieve, for example, nodes by a particular author, after a date and containing specified text, while showing the matches in the graph and index.

For a reader entering an unfamiliar network, gIBIS distinguished a node's primary connection from additional links. Hiding the additional links produced a simpler hierarchy. A NEXT operation followed the corresponding sequence in the structured index, while the graph showed where the current item belonged. Further passes could expose the cross-connections. Aggregation could also combine an issue, its positions and arguments into a larger unit. These facilities addressed a specific reading problem: a network could retain many relations while still providing a manageable first route through them.

The authors reported use by 32 people creating 33 issue groups; by February 1988 the collection contained 2,091 nodes. Some groups were largely individual workspaces, while others supported structured communication. Users reported that the framework helped them notice incomplete reasoning and make assumptions explicit. The authors also observed cases in which readers understood individual nodes but could not follow the larger line of thought. Their discussion connected this fragmentation to hurried, minimal entries and to the loss of the unfolding context supplied by continuous prose. Preserving discrete contributions and preserving an intelligible argument were distinguishable accomplishments within the same working system.

Issue resolution required its own representation. An issue-position-argument network could expose the alternatives without indicating which position a group had adopted. A recently added facility marked an aggregated discussion as resolved and recorded which position had been selected. Conklin and Begeman further noted that a resolution might need an explanation beyond a flag: it could combine earlier options or abandon an assumption they had shared. This observation concerned the meaning of a decision record. The selected position, its rationale and the discussion preceding it were related records with different jobs.

Ackerman and a memory that grows through questions

Mark S. Ackerman's Answer Garden field study (1994) examined a different route into organizational memory. A user first navigated a diagnostic question tree to seek a stored answer. When the database could not answer the question, the system could forward it to a suitable human expert. The reply returned to the user, and an expert could add a commonly needed answer to the database. Repeated enquiries could therefore leave a record available to later users, while the system continued to provide access to people. [CM-W005]

The field-study version anonymized questions to reduce the social cost of exposing a gap in one's knowledge. Its initial database contained 98 nodes covering roughly 225 questions and answers about the X Window System. The study involved software engineers in a research group and a university extension setting over three months. Of 59 potential users, 35 used the system, accounting for 194 sessions. A little over half the sessions resulted in mail being sent. The stored collection and the expert route were both active parts of the service.

Ackerman combined usage records with interviews about particular incidents. Users valued finding an answer promptly and finding out promptly whether the database contained one. They also came to treat access to the human network as a natural part of the system. Experts answered questions and the database grew. An unanswered question was consequently useful input into the development of the memory: it could reveal an omission and bring an expert's response into a reusable form.

The study also explained why the form of that response mattered. Some experts saw their answers as a public representation of their professional competence and wrote formal, general explanations. Some users wanted a shorter answer suited to their immediate level of understanding. An answer could be technically sound yet poorly matched to the particular enquiry. Users' reluctance to bother experts also persisted in some interactions. Ackerman distinguished possible concerns about professional authority and about calling on volunteer labour, rather than assuming that anonymity had removed every social feature of asking for help.

Answer Garden's achieved combination was specific: it connected a searchable body of answers with an expertise network and allowed new questions to enlarge the recorded material. The three-month field study observed this combination in use.

Matta and colleagues on project experience and design reasons

Nada Matta, Myriam Ribière and Olivier Corby's *Définition d'un modèle de mémoire de projet* (1999) proposed a model for making design-project experience accessible to future designers. It organized the memory into project characteristics and problem resolution. The first part described the setting in which work had occurred: requirements and constraints, methods, tasks, participants and roles, and results such as prototypes, software, tests and technical documents. A later designer could examine these characteristics when deciding whether earlier experience was relevant. [CM-W006, pp. 13–27]

The problem-resolution part retained the choices made during design. For a problem it recorded participants, methods, candidate solutions, rejected solutions and the arguments for rejecting them, as well as the selected solution and the arguments attached to it. The distinction between results and their rationale was operational. Finding an earlier product could suggest a solution; finding why one candidate had been rejected could explain why an apparently attractive alternative had not been pursued. The model linked such descriptions to project resources in their existing forms.

The authors described two access scenarios. A designer could start with the task currently being performed, retrieve comparable earlier projects and inspect their context, organization and results. Alternatively, the designer could start with a type of problem and retrieve earlier instances, then examine the methods and options considered in those instances. The two routes supplied different ways into the same project experience. Neither required a future user to know in advance the filename or chronological location of the relevant record.

The report presented an initial validation with industrial collaborators and access scenarios for the model; implementation support using intranet techniques and XML was planned. Its developed contribution was the organization of project experience around the later designer's activity and immediate need. The project description supplied context for judging relevance, while the problem account made the reasoning around alternatives accessible with the result.

Kudo on retrieving a decision across successive reviews

In his 2011 dissertation on knowledge use in information-system development and operation, Yutaka Kudo developed an account of a design-review management system built around meeting minutes. A single meeting could address many design items, while one item could recur in several meetings. Keeping one file per meeting preserved the meetings but dispersed the history of a design item. The system retained the familiar minute-taking practice while adding a way to retrieve that history directly. [CM-W007, Chapter 2]

The record format supplied the connection. Discussion headings corresponded to items in the review plan; each item paired a conclusion with its reason; reviewed design documents could be attached. The database separated project, meeting, discussion-item, attendee and related-document records. Heading numbers and heading text were stored separately so that the same design topic could be matched across minutes. The resulting relationships supported both an overview of review coverage and a detailed enquiry into a particular decision.

The cross-meeting search extracted the relevant conclusions and reasons and displayed them in date order, together with the meeting titles and participants. A link opened the full source minutes when the reader needed the surrounding discussion. Kudo's example followed changes to a client's computer specification. The display brought successive specifications together with the reasons for them, including a change in what processing the client needed to perform. The reader could see the development of the decision without opening each meeting file merely to locate its relevant paragraph.

The coverage view counted reviews of each design item and participation by responsible people or knowledgeable participants. It also provided a separate manager-approval field. A project manager could mark an item as sufficiently reviewed, with the system recording the approver and time. The display therefore distinguished the existence of recorded discussions from a manager's judgment that the review was complete. Kudo reported a practice of bringing the retrieved history into the next meeting so that participants could begin with the earlier conclusions and reasons in view.

The system entered service in August 1999. Kudo reported that by June 2011 it held 100,724 minutes for 2,469 projects across six divisions. These counts documented sustained operational use. His evaluation also compared the work of locating a topic's history in a project with 350 minute records. The manual

estimate was 130 seconds, based on opening 13 matching files at ten seconds each; the system's retrieval and ordered display took under four seconds. The comparison excluded reading and understanding the discussion. It measured the assembly of the relevant history for inspection, a distinct operation the system made much quicker.

In this account, the review heading carried a design concern across meetings, while conclusions, reasons and source links made its development inspectable. Participants could consult the assembled sequence instead of depending on their own recollection. The record remained connected to the meetings and people from which it came, and the team could resume its discussion with that history already assembled.

S105 Forgetting, discarding and revising retained knowledge

Recall after an instruction to forget

Robert Bjork's 1972 chapter distinguishes instructions given during a sequence, after a set, or item by item. In the latter arrangement the cue can coincide with presentation or follow it. He separates two questions about the effect of such a cue. One concerns how much of the material designated for remembering is subsequently recalled. The other concerns how much of the material designated for forgetting remains accessible on recall or recognition tests. An instruction can reduce interference with what is to be remembered while leaving some of the other material accessible. [T07-W001]

Kluge and Gronau examine retrieval cues in organizations. Their proposal concerns an old routine that still comes to mind because familiar tools, records or circumstances cue it. Changing those cues while introducing conditions for the new routine is a way to change which practice is prompted in everyday work. [T07-W002]

Written procedures and work that people continue to do

Martin de Holan and Phillips studied seven Cuban hotels through visits, documents and 78 interviews. Five hotels were associated with alliances and two were comparison properties. The authors distinguish failure to retain a new practice, decay of an established practice, and deliberately abandoning an adopted arrangement. Their Withwind account describes manuals and training specifying new procedures while managers repeatedly intervened to keep everyday practice aligned with them. The manager's example of kitchen staff substituting ingredients illustrates how an old way of working could remain available. [T07-W003]

At Caribbean Hotel, management adopted another hotel's structure and introduced corresponding operating and communication procedures. The manager reported that procedures went unused unless checked by the same level that had imposed them. Meetings were taking time without solving the problems. The organization changed its structure and routines. The authors discuss this as abandonment of an innovation. Their separate comparison of food and beverage costs per guest in three hotels concerns deterioration of an established practice. They relate it to turnover and dependence on supervision. Those hotels were the subset that could supply complete cost records. [T07-W003]

Withdrawing support

The truth-maintenance system described in S60 concerns another kind of retention. Doyle's 1979 memo explains how the current support for a proposition can change while the stored statement and its reasons remain available. The system can withdraw a conclusion without losing the record needed to reconsider it. The object being revised is the current set of supported conclusions. [CM-W003]

Part 8 Inquiry, invention, and collective action

These accounts examine how people frame problems, develop possibilities, create and perform work, and coordinate through shared practices or traces of earlier action.

S44 Problem structuring, design and systems methods

Framing problems and organizing inquiry

The sources here propose ways to frame problems, make models, organize discussion, design arrangements and examine consequences. Some sources contain their own extensive methodological syntheses.

Checkland: use declared activity models to question a situation

Checkland's plenary summary describes the development of Soft Systems Methodology through action research in management situations. Its contrast is with a systems-engineering approach that takes objectives as given and systematically compares means. In the situations he describes, people act purposefully but do not necessarily share one objective or interpretation. SSM becomes a learning system rather than a procedure for optimizing an already agreed task. [SG-W001, sections I–VII]

Its purposeful-activity models are built according to declared worldviews. Several models can structure debate about a messy situation, helping people find accommodations that permit action. A model is therefore not simply asserted to be the uniquely true system hidden in the world. Connecting an activity model to information support asks what people would need to know to carry out that activity.

Checkland's 1999 retrospective summarizes his methodological development. [SG-W001]

Ulrich: the boundary determines which facts and values count

Ulrich's mini-primer defines boundary critique as examining the judgments that make some observations and values relevant while excluding or subordinating others. Its twelve-question framework, also developed in the Ulrich–Reynolds chapter, distinguishes sources of motivation, control, knowledge and legitimacy. It asks about an actual arrangement and what ought to be the case, including the difference between people involved in a decision and people affected by it. [SG-W002]

The contribution is not a command to include everything. It is a way to make the selection discussable and expose a claim that disguises its boundary as necessary or neutral. Relevant expertise does not remove normative choices, while a person affected by a choice can question those choices without possessing all of an expert's knowledge. The primer presents this as useful in cooperative inquiry and in critical challenges where agreement is absent.

The source explicitly does not offer a self-contained method that settles every issue. [SG-W002]

Vickers: appreciation includes standards of fact and value

Hudson's notes reconstruct Checkland's account of Vickers's appreciative systems. In this account, people respond to events and ideas through standards of fact and value, selecting what seems relevant and judging what relationships to maintain or change. Action can modify the situation, while experience can modify the standards through which later situations are interpreted. A situation is not first perceived without any selection and only afterward assigned a value. [SG-W010, text]

Hudson draws on Checkland's notes on Vickers, adding his own explanation.

The separate 1978 interview transcript supplies Vickers's own short discussion of people as agents and observers. It situates systems inquiry in the relationships people participate in, rather than only a detached description. [SG-W012]

Shchedrovitsky: distinguish thinking, communication and practical action

The chapter by Georgy Petrovich Shchedrovitsky presents a thought-activity schema with three relatively autonomous belts. Practical collective thought-action concerns what people do together; thought-communication is the polyphonic exchange through which positions and meanings are articulated; schematic thinking works with organized representations. Understanding and reflection connect the belts. They are not three interchangeable names for the same process. [WME-W019, chapter and schemes]

The account of the 1980 game involving Meerovich makes this distinction concrete. A participant could share communication and collective action with other architects while insisting that his individual thinking differed. Separate boards made that difference visible. A common conversation is not evidence that all participants possess the same conceptual content. The scheme preserves individual and group representations rather than imposing identity by drawing one group box.

A schematic sign can retain a long chain of intellectual operations, but it still needs interpretation. The chapter describes reconstructing meaning, making it explicit through material boards or paper and learning the conventions through which a scheme can be used. Concision at the level of a diagram does not eliminate the history or competence needed to understand it.

The middle communication belt can accommodate conflict, questioning and multiple positions in a way different from formally checking a symbolic derivation. The author's concern is the work of problematization and coordination among heterogeneous practices. He also warns about isolating a belt: formal thought can become detached symbols, communication empty talk, and action mere repetition. These are his methodological evaluations. [WME-W019]

The organizational-activity game supplies a proposed operating process. Facilitated encounters expose incompatible practices, create opportunities for reflection, develop new schemes and lead toward programmes that participants enact and revise. It retains organizers, participant effort, conflict over days, shared material representations, learned norms, goal setting, testing and later institutionalization.

The chapter's conclusion also limits what its schema represents. It is an ideal theoretical basis for deriving models, rather than a model of one concrete real system. Its synthesis of thinking, communication and action is the author's philosophical and methodological proposal. The article was first published in the 1986 yearbook issued in 1987 and reprinted in Избранные труды in 1995. [WME-W019]

Pyotr Shchedrovitsky's later thought-activity-ontology account develops a distinct formulation from Georgy Shchedrovitsky's original essay. [SG-W024]

Midgley and Lindhult: systemic innovation includes inquiry into purpose

Midgley and Lindhult's article distinguishes several uses of systemic innovation: complementary technologies, enabling governance, sustainability transitions, collaborative value networks and systemic thinking-and-action practice. The authors favor the last as a way to engage with the others. This is their conceptual synthesis. [WME-W020, article, sections 1–8]

Their account relates boundaries, purposes and values in both directions. Values affect whom and what a study considers; established boundaries can also constrain which values count as legitimate. A system can be used as a tool for thinking without denying that the world has material consequences beyond the model. The authors' assumptions about interconnectedness, partial knowledge and action's consequences are philosophical starting points, not proved equations that determine every ethical judgment.

The critique applies to expertise and participation alike. Expert modeling can impose a client's purposes without examining affected interests. Participation can preserve bias, coercion, deception or groupthink. Dialogue is therefore not a sufficient condition of good outcomes. The article examines conflict, stereotypes, marginalization and the possibility that a ritual of participation conceals exclusion. [WME-W020]

The figures help articulate these relations: differing concerns, overlapping boundaries, conflict shaped by stereotypes and nested boundaries that marginalize people or issues. They are conceptual diagrams, not calibrated causal models. Broadening a boundary can also exceed available resources or collide with wider institutions.

Their examples include reframing in a water company, an imposed supplier-margin reduction followed by failure, and the marginalization of an innovation unit. These are author-reported vignettes. They illustrate why an apparently local improvement can fail within wider relations.

The article changes methodological questions across values, resource and risk distribution, delivery, scaling and governance, with a return to purpose when needed. It does not provide a universal algorithm for selecting the correct method. Internal expertise can sometimes replace an outside facilitator, while facilitation and knowledge work remain. The authors explicitly acknowledge no guarantee of success, call for evaluation and retain adaptation to context. [WME-W020]

Ackoff: idealized design, realization and decision rules

Ackoff's guide distinguishes removing disliked deficiencies from designing a desired future. Its idealized design is not a claim to create a perfect organization unconstrained by reality. The guide separates idealization from means, resources, implementation and controls, and asks how a design can learn and improve. Eliminating an unwanted condition need not produce the desired one. [SG-W040, guide]

The organizational questions make retained responsibilities visible: which services to provide internally or acquire, how to assure quality, how to change facilities and staffing, and how to respond when the situation changes. A stated ideal is therefore an input to further work rather than a complete plan of construction.

Its final decision procedure is especially important. Practical agreement can permit action without agreement on every principle. If participants cannot agree on alternatives, the guide first proposes a test they agree to accept. If a test cannot be agreed or completed in time, the responsible manager declares a position. Agreement among the other participants can override that position; if they fail to agree, the manager's position enters the design. Authority remains in the procedure. [SG-W040, final decision pages]

The guide also seeks comments from stakeholders not involved in preparation and asks for explanations when suggestions are not incorporated. These are prescriptive procedures.

Inquiring organizations: different ways of warranting a claim

Courtney, Croasdell and Paradice reconstruct Churchman's five inquirers and apply them to learning organizations. The Leibnizian account relies on derivation from axioms; the Lockean emphasizes observation and agreement; the Kantian combines models with data; the Hegelian develops knowledge through opposed interpretations; the Singerian treats measures and the inquiry arrangement as revisable. The different guarantors are the point of the comparison, rather than five names for a universally reliable search. [SG-W039, selected main review and table]

In the Singerian discussion, persistent disagreement can motivate sweeping in new variables or revising measures. That expands the inquiry rather than forcing agreement under a fixed representation. Yet the authors explicitly warn that a community's knowledge generation can drift toward its own illusion without monitoring. Their IT-support discussion retains debate, justification and checks on inquiry.

Courtney and colleagues reconstruct and apply Churchman's inquiring-systems account. [SG-W039]

Gigamapping, SODA and Interactive Management

Sevaldson's Gigamapping account proposes rich, large mappings to explore complex relations across disciplinary boundaries. Creating the map is a way to investigate and communicate. [SG-W016, selected account]

Ackermann and Eden's selected SODA opening uses cognitive mapping to represent participants' construals and means–ends relationships. The central example helps organize an issue through connected concepts. A link records an understood relation in the inquiry; it is not thereby a demonstrated causal law. Map construction, merging, facilitation and political feasibility remain part of the method. [SG-W051]

The Warfield archive's Interactive Management exhibit is more concrete about the support arrangement. Sessions can require several days from selected participants, trained facilitators, a room with extensive display space and computational support. Structured questions and consensus methods organize contributions. The sponsoring organization helps identify the participants, so a room designed to hear every voice does not itself determine whose voice was invited. [SG-W060, main exhibit]

The Warfield archive reconstructs a structured group-inquiry process. Facilitators help participants develop and organize their judgments into a problem representation.

Meadows: leverage depends on the structure and purpose

Meadows's essay distinguishes parameters from stocks, flows, delays, feedback, information, rules, self-organization, goals and paradigms. The stock–flow explanation makes a useful initial point: a stock changes through rates of inflow and outflow, and an intervention's timing can matter as much as an isolated numerical setting. The essay's ranking is a heuristic account of leverage, not a universal theorem. [W107, selected printed pp. 1–6 and 12–18]

Information-flow changes can expose consequences that actors previously did not see, while rules govern incentives and permitted action. The examples are reported anecdotes or interpretations. Changing a parameter is not always weak: the essay itself contains qualifications to that kind of ranking.

The selected goal and paradigm passages concern changing what a system is organized to do and the beliefs through which it is interpreted. Such a change may be consequential, but it changes the comparison. A new purpose cannot be counted as supplying the same outcome with fewer resources unless that equivalence is separately established. [W107]

The selected introduction and table of Crielaard and colleagues' 2026 review of twenty-nine studies discuss systems archetypes in public-health applications. Recurring feedback patterns can help formulate a hypothesis about a problem, but a causal-loop diagram does not itself validate an intervention. [W106]

Schwarz: a three-plane metamodel and six cycles

Schwarz's 2005 formulation proposes a metamodel with physical, information and existential planes, associated with objects, relations and wholes. It distinguishes cycles supplying persistence—material circulation, homeostasis and self-reference—from morphogenesis, autopoiesis and autogenesis as change

processes. The metamodel organizes his proposed interpretation of viable systems and informs his intervention discussion. [SG-W003, central diagram and dynamics, PDF pp. 5–9]

Schwarz offers an authorial synthesis with ontological commitments. Its necessity across physical, biological, social and cognitive systems is asserted rather than established through comparative tests. [SG-W003]

Suh: functional independence and probability of success

Suh's chapter introduces functional requirements, design parameters and process variables in distinct domains. Its independence axiom concerns satisfying the selected functional requirements without undesired coupling. Some designs can preserve independence directly, while others require a sequence that avoids interactions. The account concerns the relation among stated requirements and the parameters used to realize them. [SG-W052, chapter discussion of domains and design axioms]

The information axiom evaluates success using a probability-based information content. That quantity is not automatically knowledge possessed by an actor, sensing demand or total lifecycle cost. Functional independence is also different from statistical independence: multiplying probabilities requires assumptions about their joint relation.

Suh's chapter introduces the axiomatic-design approach associated with his 1990 *The Principles of Design*. [W037, through SG-W052]

Altshuller and Shapiro, and a later QFD–function-analysis–TRIZ proposal

The selected 1956 article organizes inventive work through analytic and operative stages, examining a technical difficulty and seeking changes or transfers that can overcome it. Its examples and stated limits provide an account of organizing search rather than treating invention as an unexplained inspiration. Later formulas of TRIZ ideality are not retroactively attributed to this early article. [W022, analytic and operative stages]

León-Rovira and Aguayo's later proposed integration assigns different tasks to QFD, function analysis and TRIZ. QFD helps identify what customers want and how existing functions relate to those wants; the authors explicitly say it is not a problem-solving tool. Mapping the relationships is nontrivial and requires experienced designers. The proposal therefore retains expertise inside a structured method. [SG-W057, integration of QFD, function analysis and TRIZ]

Their account reports problems when a team tries to define new functional parameters before establishing the existing relationships. The ideality expression combining benefits, costs and harms is a heuristic, not a universal commensurate measure. A ratio cannot establish comparable quantities or remove a conflicting requirement simply by being written down. The source's tool boundaries and reported inappropriate use matter as much as its promised integration.

Theory of Constraints: a bottleneck can move

The practitioner account presents five focusing steps: identify a constraint, exploit it, subordinate other activity, elevate it and repeat when the constraint changes. The method asks why improving a nonbinding part may fail to improve overall throughput, and why relieving one constraint can make another important. Buffers and the distinction between investment and operating expense are part of the account. [SG-W019, five focusing steps and accounting distinctions]

Single-constraint rules have difficulties in product-mix problems with multiple constraints; ratio heuristics can then give misleading priorities. [SG-W019, practitioner account and abstract-level counterpoint]

Robust Decision Making: inspect vulnerabilities across possible futures

RAND's account describes an iterative process of decision framing, evaluating strategies across futures, vulnerability analysis, tradeoff analysis and developing new futures or strategies. It seeks strategies that perform acceptably across a range rather than relying on one predicted future. A vulnerability summary can help stakeholders understand where a strategy fails. [SG-W031, main body and FAQ]

The futures are simulations, not automatically probabilities. Model construction, computation, storage, training and stakeholder judgments remain.

The EPTA method orientation addresses technology assessment through examination of consequences and options in relation to public decision making. The Collingridge dilemma concerns limited early knowledge versus costly later change. Constructive assessment is a response to that difficulty, not a guarantee that anticipatory knowledge can be made complete. [SG-W030]

Dori, DSRP, grey prediction and Extenics: representations with stated limits

Dori's selected Object-Process Methodology opening relates structure and behavior in a common representation. Its bread-baking example connects objects, states and transforming processes rather than leaving a static parts list separate from action. Representational coherence can help an inquiry, but does not ensure that the model corresponds to the world. [SG-W058]

Cabrera and Cabrera's selected DSRP primer organizes questions through distinctions, systems, relationships and perspectives. These operations supply a vocabulary for making a model explicit and examining its implications. The primer advances universal claims and strong comparisons with other approaches. Its proposed ratio between model and reality does not specify operational resource quantities. [SG-W041]

Yang and colleagues' selected grey-prediction account uses satellite observations of vegetation recovery after a landslide. Classification and observation feed a particular modeling method with its own assumptions. [SG-W048]

Cai, Yang and He's publisher abstract describes Extenics as formalizing and transforming descriptions of contradictions or incompatible requirements. The application discussion develops this orientation. Changing a goal or constraint is different from satisfying the original task. [SG-W050, abstract and application discussion]

Brand: different rates of change can retain different functions

Brand's selected pace-layering essay proposes that faster-changing layers explore or respond while slower layers preserve continuity and impose constraints. Their interaction, rather than uniform speed, is the explanatory idea. The forest and social-layer examples are conceptual illustrations, not proof that every durable system has one fixed hierarchy. [SG-W059, pace-layering discussion]

Slow structures can also obstruct needed change, and fast or slow does not constitute a moral ranking. The proposal supplies a way to ask about timing and retained memory.

Purposes, expertise and the organization of inquiry

The sources explain substantial operations: making a boundary explicit, separating kinds of activity, modeling a requirement, inviting criticism, guiding a search or comparing a strategy across futures. They also retain the people and institutions that choose purposes, supply expertise, facilitate disagreement, build representations and test consequences.

S63 Systematic accounts of methods for generating ideas

Accounts of creativity methods organize different kinds of knowledge. Some compare named techniques, some examine their instructions and some relate procedures to stages of innovation or conditions of use. A method may specify both an operation on ideas and an arrangement of participants, time and materials. The accounts below make those parts available for study and show several ways authors have classified them.

Kirjavainen and Hölttä-Otto and the parts of a method

Senni Kirjavainen and Katja Hölttä-Otto analyse the instructions of eighty-eight documented idea-generation methods. They identify twenty-five mechanisms used as coding categories and distinguish implementation mechanisms from idea-promoting mechanisms. The first group concerns how an activity is organized; the second concerns sources of ideas and operations during generation. Several mechanisms can occur in one method, and the same mechanism can recur under different method names. [CRE-W001, §§3.3–4.2, tables 1–3]

Their examples begin with actual instructions. Asking participants to draw possible solutions specifies a visual modality. Providing a card with an action prompt can supply stimulation and invite modification of an existing idea. Asking for many possibilities specifies a quantity aim. These are separable features even when one exercise includes all of them. A method's name does not by itself reveal the combination. [CRE-W001, table 1]

The idea-promoting categories include analogy, association, building on other people's contributions, incubation and stimulation. They also include operations such as combining, modifying, framing, classifying, evaluating and selecting ideas. Analogy and association have different definitions in the coding: one concerns a pertinent relationship between domains, while the other can follow a word, image or idea without requiring that relationship. Suspending judgment appears as one component within this larger account. [CRE-W001, table 3]

The implementation categories distinguish individual and team participation, given or open duration, written and spoken expression, physical activity, visualization and thought. Facilitation and special tools are also recorded. A method designed for a team can contain individual phases. The arrangement of participation is therefore more specific than simply calling an exercise collective. [CRE-W001, table 2]

The authors refine their categories through coding and comparison. A second coder examines fifteen percent of the instructions, disagreements lead to revised guidance and the definitions are used in the remaining analysis. Their framework gives a way to inspect and compare the operations present in documented methods. It also makes the organization of a session visible as part of what a method specifies. [CRE-W001, §§3.3–4.2]

Geschka and Zirm and connected procedural families

Horst Geschka and Andrea Zirm distinguish free association, structured association, configuration, confrontation and imagination. Their descriptions connect these families to ways of carrying out an exercise. In free association, an initial contribution becomes a source for another. Structured association introduces successive directions of thought. Configuration organizes combinations of features. Confrontation introduces material from outside the immediate problem, while imagination uses internally developed images and perspectives. [CRE-W002, §§4.1–4.5]

Their rotating-sheet procedure replaces a single stream of speech with parallel writing. A sheet has three columns, and each participant enters an idea in each column. After roughly five minutes, the sheet passes

to another participant, who develops ideas beneath the earlier entries. The page supplies both material to respond to and a position for recording a continuation. A contribution can be developed without first winning an opportunity to speak. [CRE-W002, §4.1]

In a circulating-card variant, participants put one idea on each card and pass cards in a common direction. During this phase they do not speak. Cards received from a neighbor can stimulate new ideas. The later phases group contributions by possible solution direction, remove duplicates, clarify understanding and make an initial selection through adhesive dots. The exercise includes both the circulation that generates material and a way to organize what has accumulated. [CRE-W002, §4.1, fig. 4-2]

Confrontation techniques use a different sequence. First the problem is clarified and participants briefly express the familiar solutions that occur to them. Only then is unrelated material introduced. A stimulus word is examined for its properties, structure and function; those features can be carried back to the problem. The intended departure is supported by an actual intervening object of attention. [CRE-W002, §4.4]

Visual confrontation can proceed through images, and one described variant combines them with written circulation. Participants inspect picture cards and record ideas before circulating their contributions. Other imagination exercises ask the participant to identify with a problem or picture it in another way. These procedures specify a route between an initial task, a change in attention and a return to possible solutions. [CRE-W002, §§4.4–4.5]

The chapter's families consequently contain different operations and different session arrangements. A confrontation technique may use a technical principle as stimulus; a written technique may circulate a visual or verbal contribution. The descriptions show how procedures can be combined while preserving the particular role of each step. S52 develops experimental accounts of group interaction and production blocking, and S68 develops other material and collaborative methods.

Takahashi and four purposes of creative techniques

Makoto Takahashi's classification, presented by the Japan Creativity Society, distinguishes divergent, convergent, integrative and attitude methods. Divergent methods produce facts or ideas. Convergent methods bring that material together. Integrative methods are characterized by repeated movement between divergence and convergence. The classification therefore gives organizing material a place inside creative activity. [CRE-W004]

The fourth category concerns the creator's attitude. Such a method is intended to cultivate a disposition rather than directly solve the current problem. Its immediate purpose differs from that of a technique for producing a particular collection of alternatives. Readiness to participate and operations on a present problem can be addressed through different activities. [CRE-W004]

The distinction also changes how a procedure can be understood over time. An integrative method includes recurring transitions, while an attitude method can concern capacities brought to later work. Takahashi's short account provides a classification by those purposes. The particular methods described elsewhere give concrete forms to generating, organizing and developing material. [CRE-W004]

Yao and Chu and methods across an innovation process

Yao Wei and Chu Zhaowei review systematic innovation methods as a subject wider than TRIZ. They distinguish methods drawing on human inventive experience from those abstracted from natural phenomena. They also follow the use of methods across identifying an opportunity, formulating a problem, developing a response and putting it into use. Their account treats a method's place in the process as part of its practical significance. [CRE-W003, §§1–2]

The integration examples assign different jobs to different methods. Quality function deployment can help articulate requirements; TRIZ can address a technical contradiction; other analyses can examine implementation or failure. Joining methods in this way concerns the output of one activity becoming useful to the next. The important question is what contribution each method makes within the connected process. [CRE-W003, discussion of integrated application]

Computer assistance enters that account through knowledge and retrieval. Case-based assistance can draw on earlier experience within a domain, while inventive principles seek connections across domains. The authors' proposed AI engineer is a prospective development of such assistance. Their review also separates learning about a method from using it effectively in an enterprise. Training, application and practical outcome are examined as different parts of its development and assessment. [CRE-W003, §§3–4]

Aznar and what an exercise is intended to accomplish

Guy Aznar distinguishes imaginative expression, artistic creation, the production of ideas and innovation. His account of imaginative expression includes collective drawing, storytelling and imagined journeys within a permissive group. The immediate point is participation and the development of expression. A drawing need not become an exhibited artwork for the activity to serve that purpose. [CRE-W005, pp. 26–28]

Producing ideas for a stated problem has another direction in his account. An idea must become both new and fitted to a situation. The work includes departures from an existing organization and a return toward a possible response. Aznar describes this process through tension and transformation, rather than identify it solely with a relaxed expressive state. [CRE-W005, pp. 28–30]

The distinction helps make the task of a session explicit. An expressive exercise and an inventive working session can share operations while pursuing different accomplishments. Aznar presents artistic creation as responding chiefly to an internal demand and idea production as responding to an external problem. His characterization is one historical account of those purposes. S66 develops collaborative artistic practices in which teachers, performers and shared material have substantial roles. [CRE-W005, pp. 27–30]

Leopoldino and colleagues and a comparative inventory

Kleidson Daniel Medeiros Leopoldino and colleagues map fifty-six publications and report 120 creativity techniques. Their detailed comparison treats nine frequently cited techniques. It distinguishes individual and collective application, the type of thinking, session time, operating criteria, material resources and the approach assigned to each technique. The table makes several different aspects of a technique available for comparison. [CRE-W006, §§3–4, tables 1–2]

Their operating criteria are exploration, combination, transformation and validation. The resulting descriptions compare such matters as whether the session is time-bounded, whether materials are needed and whether work is organized around several kinds of operation. These dimensions concern how techniques are described in the selected literature. The frequency with which a technique is cited is another part of the review, separate from the comparative description. [CRE-W006, §4]

The article gives a route into a wider literature and an inspectable comparison of its selected techniques. The reported inventory and the detailed table have different scopes: naming 120 techniques does not supply 120 complete procedures. Its useful contribution includes showing which features a comparative account can make explicit, alongside the instruction-level decomposition developed by Kirjavainen and Hölttä-Otto. [CRE-W006, §§4–5]

Acar and colleagues and different kinds of constraint

Oguz Acar, Murat Tarakci and Daan van Knippenberg review empirical research on constraints in creativity and innovation. Their taxonomy distinguishes input, process and output constraints. Restricted time or materials limit inputs; a prescribed procedure governs the process; a specification defines what an outcome must contain or achieve. These restrictions act on different parts of the work. [CRE-W045, taxonomy of constraints]

The authors also distinguish motivational, cognitive and social routes through which constraints may affect activity. A restriction can alter willingness to experiment, the information attended to or relationships among participants. Classifying a constraint and explaining its effect are therefore different steps in their review. A time limit and a rule for exchanging ideas can both constrain a session while changing different conditions of action. [CRE-W045, underlying mechanisms]

The evidence discussed includes arguments that resources enable experimentation and findings in which pressure or restriction redirects effort. The framework relates these apparently different results to the object of a restriction and to proposed mediating processes. It also attends to the situation in which the restriction operates, rather than assign every constraint the same effect. [CRE-W045, input constraints]

These systematic accounts provide several ways to examine a method: its instructions, procedural family, immediate purpose, place in innovation and conditions of use. The developed practices in S64–S70 give those distinctions substantial cases. The classifications remain useful as guides to what a particular account explains.

S81 Systematic invention through contradictions and available resources

From an inventive situation to a conflict model

Altshuller's ARIZ-85-V begins by changing the description of a difficulty. An inventive situation can contain many inconveniences and possible ambitions. The mini-problem asks for a useful effect to be obtained, or a harmful effect removed, while retaining or simplifying the existing system. This formulation directs attention away from immediately adding a device and toward what, precisely, the system must do differently. The mature 1985 procedure develops the earlier inventive-problem tradition discussed in S44. [NC-W001]

The next task is to identify a conflicting pair and state two opposed conditions of an element. Improving one desired result under one condition can worsen another. The system's principal function determines which condition to retain as the starting point. Intensifying that condition makes the conflict more explicit. The X-element names the change required to resolve it; the name does not commit the solver to another component. A change in a substance's state can fulfil the requirement. [NC-W001]

In the radio-telescope example, conducting rods provide lightning protection but disturb reception. Reducing their presence helps reception while weakening protection. Because reception is the principal function, the worked analysis retains the condition that avoids interference and intensifies it to the absence of the conductor. The question becomes how protection can be obtained where the interfering rod is no longer present. [NC-W001]

This is a provisional model to be checked against the original situation. The procedure repeatedly returns the solver to the initial requirements rather than treating the first formulation as final. Choosing a conflict and intensifying it supplies a more precise problem; it has not yet supplied a physical means of solving it. [NC-W001]

The place and time of the contradiction

The operating zone locates the conflict, and operating time distinguishes when the opposed requirements must hold. In the telescope analysis, the zone is the air column where a lightning conductor would have been. Air is already present and becomes a candidate resource. The ideal-result direction asks for protection without the obstruction that had accompanied it. Altshuller uses ideality to direct the search toward the required effect with less added complication. [NC-W001]

The opposed requirements can now be stated at a different scale. The column should conduct when a lightning discharge develops and remain nonconducting during ordinary reception. These conditions need not apply at the same time. At the particle level, free charges are wanted during discharge and neutral molecules otherwise. The reformulation separates a general demand for a transparent protector into a particular place, occasion and material behaviour. [NC-W001]

A resource inventory considers substances and fields in and around the system, including their states and possible changes. The fact that air has been identified does not yet explain how it will acquire the required charges. The next part of the worked analysis must make the behaviour physically attainable. Altshuller's ideal-result formulation is a direction of inquiry, not a claim that an ideal answer is always obtainable; the program also permits outside substances or fields when existing resources are insufficient. [NC-W001]

Turning a physical requirement into a design

The little-people representation develops the particle requirement by asking what must differ between the selected air column and the surrounding air. In technical terms, the selected gas must ionize more readily. Combining air with emptiness suggests reduced pressure. That change makes a resource already implicated in the problem behave differently under the developing field. [NC-W001]

The technical scheme then places gas in a sealed dielectric tube with pressure selected for discharge as the lightning field develops. Altshuller cites author's certificate 177497 in connection with this answer. The ending of the example is therefore more specific than a proposal to use an ionized column: it includes the pressure change and the construction that maintains it. The source presents a worked design and cited invention. [NC-W001]

Later stages distinguish a physical answer from its technical realization. A composite problem may need to be divided, the selected conflict reconsidered, or the system placed within a larger one. Checks ask what contradiction has actually been removed, how the result can be controlled, and whether it works over repeated cycles. An unsatisfactory answer sends the solver back to the initial mini-problem. Other uses and an examination of the route to the answer follow. [NC-W001]

The thermal-protective suit is a separate example of moving to a supersystem. Liquid oxygen removes heat while evaporating and warming; it subsequently supplies respiration. Cooling and breathing become connected functions within a larger arrangement. This example changes the boundary of the technical problem, whereas the telescope example develops the material behaviour required within a selected zone. [NC-W001]

Standards, physical effects and other inventive principles form part of the program's wider repertoire. The worked analysis shows why choosing among tools is embedded in reformulation, development and checking. A diagram of successive stages would miss the returns and changes of direction that the instructions make conditional on the emerging answer. [NC-W001]

Making a new dependency visible

Goldenberg and Mazursky's Creativity in Product Innovation organizes invention through representations of products and their relations. Their attribute-dependency procedure distinguishes components from variables. Connecting a chair leg to a backrest concerns components; making the length of one vary with the length of the other concerns a dependency between attributes. The latter representation can expose a relation that a component list alone would not show. [NC-W002]

Their forecasting matrix places manufacturer-controlled internal variables in columns, with internal and external variables in rows. Existing dependencies are marked. Selected empty cells then become possible changes: what would follow if these variables were made dependent? The direction of a proposed dependency matters because it specifies what is to change with what. [NC-W002]

A candidate relation is developed into a feasible construction and examined for a useful consequence, market benefit or new segment. The representation supplies possibilities that still require technical and commercial interpretation. It is a way of generating a different product concept, rather than simply ranking completed concepts already supplied by an inventor. [NC-W002]

Their ice-coated antenna-pole example has a different conflict from Altshuller's lightning problem. A transportable pole is light enough to carry but insufficiently strong under ice loading. Installed freezing conditions allow strength to depend on a resource that is absent during transport: a roughened pole collects an ice coating while the antenna acquires its load. The source of the difficulty becomes part of the

proposed support. Lightness in transport and strength in service are separated by circumstances and time. [NC-W002]

Reassigning a function to what is already there

Replacement temporarily removes a component from a product or process representation while retaining its function. The unassigned function prompts a search among components already available. Goldenberg and Mazursky illustrate the procedure with rejected butter patties that had been melted slowly in a steam-jacketed vat. Steam provides heat indirectly through the vessel. [NC-W002]

Removing the steam component leaves heating as a function that must still be supplied. The search is constrained by what else is present and by the properties needed for the task. Hot milk already used upstream can carry the function: it contacts and melts the butter directly. The resulting mixture can return to the production process. [NC-W002]

The intermediate steps explain the change. The inventors do not merely substitute the word milk for steam; they change the heat-transfer relation and the role of an existing process material. The butter and vessel remain intelligible parts of the revised arrangement, while a separate steam system is removed. [NC-W002]

The authors report faster melting and reduced time and energy costs. The case's contribution is the connected reassignment from a removed component to an available one, followed through to the changed process. [NC-W002]

Six informational stages in USIT

中川徹's six-box USIT account distinguishes information produced during problem solving from a flowchart of named techniques. Its boxes contain a user's concrete problem, an appropriately defined problem, understanding of the present and ideal systems, ideas for a new system, a solution concept, and the user's concrete solution. The difference between an idea fragment and a technically coherent concept is explicit. [NC-W003]

Analysis develops objects, attributes and functions alongside relations in space and time. The present system is examined in relation to desired behaviour and properties. Operators then help generate nuclei of change. Technical knowledge is needed to combine those fragments into a concept that can answer the problem, rather than leaving the process at a collection of suggestions. [NC-W003]

Problem definition and concept construction form interfaces between practical work and the thinking process. Nakagawa identifies the conceptual solution as USIT's output; concrete realization belongs to the user's practical activity. He credits Sickafus's earlier development and explains his own reorganization of TRIZ operators. His argument for a route that does not depend on first finding an external analogy concerns analysis of one's own problem and construction from its results. [NC-W003]

What application and training studies report

Goldenberg and Mazursky's Study 3 compared methods with 120 participants divided among six method-by-product groups, using baby ointment or mattresses. Template instruction lasted about four hours, compared with about two hours for lateral thinking or random stimulation. A thirty-minute task produced 277 ideas, which three senior marketers rated blind for originality and investment value. Template ideas received higher ratings, including comparisons of selected leading ideas. Training time was therefore one difference in this design alongside the methods. [NC-W002]

Study 4 changed that comparison. Eighteen template trainees, nineteen HIT trainees and eighteen untrained participants generated drinking-glass ideas. Both trained groups received four hours of instruction. A different panel of three marketers rated the 82 ideas, again favouring the template condition. The two studies concern originality and value judgments of generated ideas, with the second addressing the earlier training-time difference. [NC-W002]

Birdi's review chapter treats creativity training as a family of interventions whose tasks, delivery and setting matter. Systematic cognitive instruction, realistic practice, and support for evaluation as well as generation recur in his account. In his later description of the TRIZ workplace evaluation he coauthored, trainees showed stronger subsequent idea generation, while support for implementation change was more limited and included self-reported impact. The chapter distinguishes developing ideas from securing their use in an organization. [NC-W004]

Dumas, Schmidt and Alexander's study of 44 engineering graduate students separates fluency from originality before and after TRIZ instruction. The reported post-instruction pattern contains fewer ideas but greater originality. Relational reasoning is the strongest reported predictor of originality, with antinomous reasoning implicated in the TRIZ-supported work. Their distinction among similarity, anomaly, incompatibility and opposition gives several kinds of relation a place in understanding the task. [NC-W005]

Spreafico and Russo classify more than two hundred published industrial applications by purpose, design activity, tools, combinations and output maturity. Contradictions and inventive principles predominate among reported tools, with QFD, FMEA and patent tools among their partners. The survey describes how tools are selected and combined in published engineering cases rather than treating TRIZ as a single intervention used identically everywhere. [NC-W006]

Their output distinctions range from a physical-effect hint to a concept with behaviour and structure, a detailed design and a market-ready product. Fewer than a quarter of the surveyed cases are classified as market-ready. The classification identifies what the reported application produced: a promising principle and a finished product are different endpoints, both of which may be useful in an engineering history. [NC-W006]

姚威、储昭卫 and 胡顺顺 examine 148 usable project reports from 216 collected in a supported innovation-method program. Two experienced researchers code tool uses and concepts, while expert ratings provide separate benefit measures. Productivity, quality and reliability are prominent aims in these reports of engineering problems. The method makes actual uses and the form of the resulting proposals available for comparison. [NC-W007]

Generating a concept and adopting it as a solution are counted separately. Contradiction and principle use is common, with a lower concepts-per-adopted-measure ratio than some other tools in the reports. This describes the relation between concepts and adopted measures in the selected field material. Alongside the published-case survey and the training studies, it gives a distinct view of how a proposed method enters engineering work. [NC-W007]

S64 Invention and scientific discovery

An inventive task can begin with a difficulty in an existing design, an unexplained result or a representation that makes some possibilities hard to see. The accounts in this section follow transformations of a technical problem, the recognition of a mathematical relationship, the use of analogy and the construction of new terms in a computational search. They make the work between encountering a problem and obtaining a candidate solution available for close examination.

Bush and transformations of the technical problem

G. Ya. Bush's Methods of Technical Creativity, published in 1972, classifies inventive methods along several dimensions. A method can be general or specialized, simple or complex, and directed toward a particular stage of creative work. His stages include preparation, forming a conception, searching for a solution and realizing it. The classification gives problem formulation and implementation a place alongside generating ideas. Bush describes heuristic methods as recommendations or incomplete procedures whose use requires judgment. [CRE-W007, §3.1]

Inversion is one of his developed families. It includes changing an object's orientation, reversing an action, substituting a physical state, changing the course of a process in time, or turning an unwanted influence toward a useful purpose. The operation begins by identifying something in the situation that has been taken for granted. A specific reversal can then be considered. The question is not simply whether an object can be turned upside down, but which relation, property or sequence might usefully be changed. [CRE-W007, §3.3]

Bush's turbine-shaft example gives this change of direction a concrete form. He recounts the difficulty of obtaining stable operation at high rotational speed by shortening, thickening or strengthening a shaft. In his account of de Laval's experiment, a deliberately flexible shaft supplies an alternative to pursuing greater rigidity. The example belongs to Bush's explanation of antithesis: examine a response opposed to the one that has organized the unsuccessful attempts. The technical function provides the reason to reconsider the conventional remedy. [CRE-W007, §3.3]

His complex methods distinguish several kinds of combination. Established elements can be combined with new ones, new elements with one another, or established elements in a new relationship. Integration can bring functions together within an object. Agglutination attaches something to a principal object, potentially without redesigning the connected parts and potentially only for a time. Bush's computing example joins peripheral devices to a principal machine as their functions are required. The combination changes what the larger arrangement can do. [CRE-W007, §3.4]

An intermediate element can also make a combination possible and then be removed. Bush describes an assembly method in which a film helps position components for a printed-circuit board. After the connections have been made, the intermediate film is removed. The temporary support participates in making the final object without remaining one of its permanent parts. This example explains why an inventive method may concern the sequence of construction as much as the finished geometry. [CRE-W007, §3.4]

Bush's account supplies several different questions to put to a technical difficulty. What has been held fixed? Which property could be reversed? What functions could be joined? Could an intermediate object or operation make a desired connection possible? His examples give those questions specific operations and technical settings. S44 develops the earlier account of inventive creativity by Altshuller and Shapiro; Bush's selected method families provide a further substantial treatment of technical invention.

Poincaré on recognizing a fruitful combination

Henri Poincaré's 1908 chapter on mathematical invention begins with a problem of selection. New combinations of familiar mathematical objects are easy to describe in the abstract; most would have little interest. Invention, in his account, includes discerning the rare connections that reveal an unsuspected relationship between facts previously treated as separate. A connection across distant areas can be fruitful, but mere distance does not make it so. [CRE-W008, Livre I, §III]

He illustrates the work through the history of his first memoir on Fuchsian functions. For about a fortnight he made sustained attempts without obtaining the result he sought. One night, after taking black coffee contrary to his habit, he experienced a crowd of ideas while unable to sleep. He describes two ideas forming a stable combination. The next morning brought work that could be written out. The episode appears within a longer sequence of deliberate attempts and subsequent mathematical development. [CRE-W008, Livre I, §III]

A later episode occurs during a geological excursion. Boarding an omnibus, Poincaré suddenly recognizes a relationship between transformations used in defining Fuchsian functions and those of non-Euclidean geometry. He does not verify it on the spot. He continues the conversation and later checks the result at leisure. Other episodes include unsuccessful work, a stay by the sea and an interruption for military service. The chapter gives mathematical invention a temporal texture: directed effort, interruption, recognition and renewed deliberate work. [CRE-W008, Livre I, §III]

Selection raises a further question for him. The inventor cannot inspect every possible combination one after another. Poincaré proposes that mathematical sensibility admits some combinations into attention because of their promise of order, elegance or useful relation. His aesthetic explanation concerns the recognition of fruitful structure. It does not equate mathematical invention with generating arbitrary novelties and later checking all of them. [CRE-W008, Livre I, §III]

He also distinguishes the felt certainty of an arrival from a completed demonstration. Sudden ideas can be mistaken. Conscious work is required to formulate and verify them, as well as to prepare the problem that precedes them. The chapter joins a first-person history to a proposed explanation of why some combinations attract attention. Its description of interrupted work gives a historical setting for the later experimental accounts of incubation in S67. [CRE-W008, Livre I, §III]

Gick and Holyoak and the analogy that must be noticed

Mary Gick and Keith Holyoak's 1980 experiments use two stories with different objects but a corresponding problem structure. A general wants to capture a fortress. Roads leading toward it are mined: small groups can pass safely, while a large force would trigger destruction. The army is divided, approaches along different roads and converges at the fortress. The separate passages preserve the surroundings while concentrating the force where it is needed. [CRE-W009, pp. 308–312]

The medical problem concerns a tumour that cannot safely be treated by sending a destructive intensity of radiation through healthy tissue along a single path. Several low-intensity rays can approach from different directions and combine at the target. Mapping the general's solution requires relating troops to rays, separate roads to different paths and convergent attack to combined treatment. The analogy concerns organized relations among the elements. [CRE-W009, pp. 308–312]

The fourth experiment separates knowing the source story from using it. Twenty-seven participants learn the critical story among other stories. Twelve receive a hint that a previous story may help solve the radiation problem. Eleven of those twelve produce the complete dispersion solution. Among the fifteen without the hint, three produce complete or partial dispersion solutions, and the authors examine whether those participants actually noticed the analogy. [CRE-W009, pp. 340–343]

The source can therefore be available in memory while its relevance goes unrecognized. Gick and Holyoak's procedure brings retrieval and mapping into the problem-solving account. A participant must find a useful earlier situation, identify the relations that matter and use them to construct a response in the target setting. Learning an instructive example and recognizing when to apply it are distinguishable accomplishments. [CRE-W009, pp. 340–343]

Dunbar and Blanchette and analogies used at work

Kevin Dunbar and Isabelle Blanchette study analogy during scientists' laboratory meetings. Their initial analysis covers sixteen meetings in four laboratories and finds more than ninety-nine analogies. Most comparisons stay within biology or immunology. A useful analogy need not cross into a remote intellectual area: a familiar experiment can supply the comparison needed to understand why another experiment behaved unexpectedly. [CRE-W010, discussion of analogy in laboratory meetings]

A near comparison can isolate a consequential difference. Two experiments may share a gene, proteins and temperature while differing in incubation time. Drawing attention to that difference can help scientists repair an experimental problem. The authors therefore give superficial similarity a useful role when the immediate goal is troubleshooting. The relation between the comparison and the goal matters more than a general preference for distant analogies. [CRE-W010]

When the goal changes to formulating a hypothesis, the comparisons also change. Scientists draw on underlying structural relationships, including similarities between genetic sequences, to suggest a possible common function. The paper reports that structural analogies were a minority of the analogies observed, but more than eighty percent of those structural analogies were used to formulate hypotheses. What scientists compare is responsive to what they are trying to accomplish. [CRE-W010]

The authors connect these observations to laboratory tasks in which participants generate analogies rather than merely remember or choose a supplied one. Constructing a comparison, recognizing a reminder and using a comparison in a discussion are different activities. Their combined observational and experimental approach makes that distinction visible. It gives analogy a place within continuing research, where an anomaly, a colleague's suggestion and the design of another experiment can redirect the conversation. [CRE-W010]

Klahr and Simon and connected spaces of discovery

David Klahr and Herbert Simon's 2001 review examines historical studies, laboratory tasks, naturalistic observations and computational models of discovery. Their account describes a researcher searching among hypotheses, experiments, representations and instruments. Work in one space changes what is worth trying in another. The initial problem is not always a fully specified question waiting for a single correct answer. [CRE-W011, pp. 75–78]

In the device-learning studies they discuss, participants investigate an unfamiliar control on a simulated rocket ship. They must propose an account of what the control does and choose manipulations that reveal its function. Some give greater emphasis to developing explanations; others concentrate on experiments. The task makes the coordination of the two searches observable. An intervention is informative in relation to a question, and its outcome can create a new question. [CRE-W011, discussion of dual search]

The review also gives surprise an active place. A result that contradicts expectations can become the beginning of an inquiry into a phenomenon. The Curies' encounter with unexpectedly strong radioactivity in pitchblende is one of its historical examples. Their attempt to obtain a substance helps reveal a problem about what else is present. The review follows this reversal between hypotheses motivating experiments and phenomena motivating hypotheses. [CRE-W011, discussion of surprise]

Representations and instruments can join the search as well. A different way to describe a problem can make a new analogy or experiment possible. The review's contribution is an account of discovery in which generating, testing and reformulating are connected activities. It complements the treatment of experiments and dual control in S33 by explaining how the possible objects of investigation themselves change. [CRE-W011, discussion of multiple search spaces]

BACON.5 and a discovered relation used as new data

Pat Langley, Gary Bradshaw and Herbert Simon's 1981 BACON.5 paper describes a program that seeks regularities in numerical and nominal data. Its user specifies independent terms, values to consider and dependent variables. The program applies heuristics for constancy, trends and combinations of terms. Its ideal-gas example shows how these operations build on one another. [CRE-W012, pp. 121–122]

The program first holds temperature and amount of gas fixed while pressure varies. Volume decreases as pressure increases, suggesting that the product of pressure and volume should be examined. That product is found to be constant for the setting. Changing temperature produces different constant products. Relating those products to temperature suggests a ratio. Changing the amount of gas leads to a further relation, ultimately expressing the ideal-gas law through a constant value of pV/(nT). [CRE-W012, p. 121]

Each step supplies a term that can be used at the next level. The product pV is a hypothesis about the lower-level observations and becomes data for investigating its relationship to temperature. The program reapplies similar heuristics to the resulting descriptions. A useful representation is produced during the search instead of being supplied only as the final name for what was found. [CRE-W012, pp. 121–122]

The discovered pattern also changes data collection. Once the program expects a lower-level relation, later settings need not repeat the entire earlier variation. A selected observation can be used to estimate the relevant constant. The paper contrasts this procedure with a complete factorial search. Representation and control of investigation work together: finding a relation can reduce the observations required to investigate a higher-level relation. [CRE-W012, pp. 121–122]

The authors present BACON.5 as a computational way to perform part of discovery, including demonstrations on known physical laws. Their aim is to exhibit productive heuristics, not to reconstruct every action taken by the historical discoverers. The worked example makes the program's inputs, transformations and use of intermediate results explicit. [CRE-W012, Introduction]

Bonnardel and Lubart and constraints that make an analogy useful

Nathalie Bonnardel and Todd Lubart describe creative design through an interaction between analogy and constraint management. Analogy can expand the field of ideas. Constraints help direct attention, evaluate a candidate and progressively determine what a satisfactory solution requires. Some constraints are prescribed; others are inferred from the consequences of those already accepted. A requirement for a lightweight but resistant aerospace component raises further questions about material and manufacture. [CRE-W013, §2.4]

Their field observations give this account a physical example. Designers need a reflector larger than those they normally make, but it must fit beneath a rocket fairing during transport. An umbrella suggests a structure that can be compact for launch and deployed later. Another designer draws on the movement of curtains in the office to propose a different solution. Ordinary objects become sources of invention when a designer recognizes a relation to a specific difficulty. [CRE-W013, §3.1]

The transport constraint makes deployability relevant. The analogy then supplies something concrete to examine and develop. In the authors' model, analogy, association and management of constraints continue to interact as representations and evaluations change. Their account gives a developed

alternative to treating creative work as an unrestricted generation phase followed by an entirely separate judgment phase. [CRE-W013, §2.4]

Across these histories, the work can concern what is varied, what is compared, what becomes a new term and what counts as a promising relation. The technical transformations, mathematical memoir, experiments and observational cases give those questions different forms. Their detailed operations can be studied alongside the accounts of form in S08 and problem structuring in S44.

S82 Association, bisociation and the construction of new ideas

Accounts and measurement of creative connection

Two frames for one situation

Koestler's The Act of Creation distinguishes association within an accustomed frame from apprehending a situation through two self-consistent but habitually incompatible frames. A matrix is a pattern of possible activities; its code supplies governing rules; strategy is the particular course taken within those rules. His chess comparison separates the rules that define a game from the many ways a player may proceed. Bisociation brings frames into an intersection that ordinary conduct within either frame would not supply. [NC-W008]

The distinction gives surprise a structure. In the opening jokes, a situation that appears intelligible under one social code is reinterpreted through another. A reply can be reasonable within one frame while overturning the expectation established by the other. The connection is more specific than the occurrence of two thoughts: a situation is made to belong to two organized patterns of meaning or action. [NC-W008]

Koestler deliberately extends this account across humour, scientific discovery and artistic creation. The domains differ in what happens when frames meet, but his theoretical ambition is to describe creative activity through their encounters. Matrix, code and strategy preserve room for established organization as well as a course that departs from habitual expectations. Learning a domain's code and discovering an unexpected intersection are consequently distinguishable activities within his account. [NC-W008]

His printing-press discussion connects practices of making impressions with the use of a press. The example makes a relationship between established activities, rather than a wholly unprecedented component, central to the proposed creative episode. Koestler presents a historical reconstruction of this connection. It explains his argument without requiring his account of Gutenberg's private reflections to be taken as an independently authenticated discovery record. [NC-W008]

The opening of his discussion of false inspirations also separates the experience of an idea arriving from a connection that survives development. An apparent intersection can be suggestive before its implications have been worked through. Elaboration therefore belongs within the theory's own account of creation. The sudden interpretation is part of a longer activity in which its usefulness, coherence or expressive possibilities become clearer. [NC-W008]

A concert contract seen through two financial problems

Chabrillat, Lecoutre and Lièvre follow a managerial innovation at the concert venue they call Le Cabestan. A changed grant rule replaces a fixed subsidy with an amount calculated as a percentage of actual expenditure. At the same time, reopening the venue requires renewed relations with private concert coproducers. The manager initially faces these as separate financial and commercial problems. [NC-W019]

Revised coproduction agreements allow the venue to pay some production expenses directly while maintaining the agreed financial result and profit sharing with a partner. This reassignment can support concessions and negotiations needed to rebuild the commercial relationship. Expenses are not simply renamed; who pays and records them changes under the contract. [NC-W019]

A tense conversation with the financial manager connects that arrangement to the grant problem. Expenditure now borne by the venue affects the base used to calculate its public subsidy. A practice developed in the private coproduction relationship thus becomes consequential under the public-

financing rule. The connected operation depends on both the contractual allocation and the particular subsidy basis. [NC-W019]

The authors trace partner responses, contract versions and stabilization of the practice. Their concurrent observation and consulting involvement from February 2014 to March 2015 supplied meetings, interviews and documents as the activity developed. They interpret the case through matrices of public financing and private coproduction, with stages of an unresolved problem, a creative clash and validation in practice. Expertise in both domains and the longer work of stabilization are part of that interpretation. [NC-W019]

Associative routes and practical requirements

Mednick describes creative thinking as combining associative elements in ways that meet specified requirements or prove useful. Remoteness contributes to his account of creative quality, but an unusual conjunction is not useful solely because it is remote. Serendipitous contiguity, similarity and mediation provide different routes by which elements can be brought together. His account of steep and flat associative hierarchies concerns how readily common and more remote responses become available. Requisite knowledge remains important to what can be connected. [NC-W010]

The refrigerator example keeps practical requirements beside the association. Preventing frost is different from reacting after it has accumulated, but a proposal never to open the door defeats the refrigerator's purpose. A counter responding to door openings suggests a different way of controlling defrosting. The useful connection concerns an event associated with frost accumulation and an operation that can be controlled, while retaining the appliance's ordinary use. [NC-W010]

The Remote Associates Test turns mediation into a research task. A participant finds an associate connecting three cues; in Mednick's example, rat, blue and cottage can each be joined with cheese. A candidate that fits one cue must be tested against the others. The task gives associative theory an operational instrument, with a specific kind of successful answer. [NC-W010]

What an association test measures

Wu, Huang, Chen and Chen's systematic review retained 172 studies from 256 screened records covering 2000–2019. The review shows RAT tasks being used for different research purposes and with differing relations among cues. Producing candidate associates can involve divergent search; finding one that satisfies all cues involves convergence. Vocabulary, retrieval and overcoming a prominent wrong answer can also influence the result. A test score therefore needs to be interpreted through the task and version that produced it. [NC-W011]

王硕阳、林崇德 and 孙汉银 distinguish what is connected from the efficiency with which a connection is retrieved. Language-based co-occurrence differs from relations built through experience, function or shared features. Their account develops partly independent associative abilities and different roles in problems that do or do not require restructuring. A person's facility with conventional verbal relations need not capture every way that relations can support a new problem representation. [NC-W012]

西田勇樹、服部雅史 and 織田涼 compare a compound-word RAT with insight problems, alternative uses, vocabulary and cognitive reflection. Their version requires a character that makes compounds with each cue. After vocabulary was controlled, their first experiment did not find the expected association with the combined insight or creativity tasks. The result concerns those tasks and the contribution vocabulary makes to answering them. [NC-W013]

Their second experiment retained 68 of 91 recruited participants and found a relation to making fewer typical cognitive-reflection errors. The relation with the correct-answer score did not persist in the same way after vocabulary control. The authors resist a binary verdict that RAT either is or is not insight: insight tasks themselves need not correlate closely. Their discussion gives inhibiting a salient but wrong response a visible role alongside vocabulary and associative search. [NC-W013]

Constructing and interpreting ideas

Making the familiar strange and returning to the problem

Gordon's Synectics seeks to make deliberate work possible around activities often described only as intuition. Making the strange familiar is the analytic effort to understand a problem. Making the familiar strange changes its habitual representation, allowing a group to approach it through a different experience or image. The departure is followed by a return to the technical problem; an excursion that never returns would leave the original task unanswered. [NC-W009]

Personal analogy asks participants to identify with an element or process. Direct analogy draws on another phenomenon. Symbolic analogy compresses a problem into an evocative expression or image, while fantasy analogy imagines a desired fulfilment. Gordon treats these mechanisms as ways to encourage useful psychological states, not simply four labels for any unusual comparison. The mechanism chosen affects how the group leaves its accustomed account of the problem. [NC-W009]

Appendix II joins the initial problem, analytic understanding, excursions through the mechanisms, integration with the problem, a new viewpoint and a solution or research target. Some targets require a model that tests a principle; others require work on materials. The return to a technical viewpoint can change or reverse part of the image that prompted it. What is retained is a relation useful to construction, rather than a demand that a product imitate every feature of the analogy. [NC-W009]

From an imagined stitch to a space-suit closure

The space-suit session concerns an opening that must close strongly and reliably. Familiar fasteners give the group an established set of possibilities but do not settle the task. Insect and stitching imagery opens a different way of thinking about joining the sides. The group's responses move between imagined conduct and the load the closure must bear. [NC-W009]

The discussion develops steel stitching into overlapping springs embedded in the rubber lips of the opening. A wire passed through the springs can pull the sides together. This proposal gives the image an arrangement of materials and connections: the opposed lips carry the springs, the springs overlap, and the wire supplies the joining action. [NC-W009]

The load requirement brings the excursion back to the suit. The mechanism must answer the difficulty that made an alternative to ordinary fastening worth seeking. Gordon's account ends with a proposed construction, so the significant result is the route from an unfamiliar image to a technically describable closure. The session's ingenuity lies in the successive interpretations and their return to the original requirement. [NC-W009]

A roof that changes its exposed colour

A second session asks for a roof that is light in hot weather and dark in cold weather. Changes in animal appearance supply possible analogies, but the time scale matters. A seasonal coat does not give the rapid response wanted for the roof. Chromatophores suggest coloured elements whose exposure can change instead. [NC-W009]

The technical viewpoint transforms that suggestion. White elements embedded in a black roof would expand into view when heated, changing what the surface exposes. The construction does not copy the animal unchanged; it selects a relation between temperature, element behaviour and visible colour, and reverses part of the analogy for the building's purpose. [NC-W009]

Gordon presents a research target for materials and construction. The group has obtained something more definite than an animal metaphor: a proposed response and arrangement whose realization can be investigated. His account thus connects imaginative departure with a specification for further technical work. [NC-W009]

A form before its purpose

Finke's constrained imagery task asks participants to combine three selected object parts into a form before deciding what useful object it will become. Size, material and some relations can change while specified geometric constraints are retained. The result is a preinventive structure, open to interpretation rather than a drawing of a finished purpose already chosen. [NC-W015]

Supplying an interpretation category after form generation changes what the participant is doing. A structure first assembled without that category can subsequently be explored for functions that were not its initial rationale. Broad functional categories also leave different interpretive possibilities from narrow object categories. Finke reports especially creative inventions under conditions that delayed the category, judged through originality and practicality ratings. [NC-W015]

The operation is construction followed by a search for meaning under constraints. A part can acquire a role not apparent when it was selected, and its relation to another part can suggest a function for the whole. Interpretation then supports refinement, expansion or the generation of a different structure. Delaying a category in this task preserves a particular set of possibilities; it does not imply that every purpose should be postponed in every design problem. [NC-W015]

Geneplore makes the return paths explicit. Generation produces a preinventive structure; exploration interprets it under product constraints, and the interpretation can lead back to further generation. Constraints act on both activities. The process distinguishes a suggestive form from a developed invention while giving the former an active role in producing the latter. [NC-W015]

Comparing taught methods

Azadyekta compared brainstorming, forced association and synectics with a control condition in 160 elementary pupils. Eight weeks of training were assessed with pre- and post-training Torrance pictorial tests. All three trained groups improved on the reported measure relative to the control pattern, while the overall differences among the trained methods were not statistically significant. This is a comparison of taught activities and task scores in those conditions. It supplies a different kind of evidence from Gordon's accounts of technical sessions and preserves the distinction between improving on a measure and establishing a general ranking among invention methods. [NC-W018]

Exploring a space and changing what it permits

Boden's 1997 account distinguishes what is new to an individual from what is new in recorded history. A person can produce an idea without copying it even when someone else has already produced it. That distinction separates a psychological question about the individual's creative activity from a historical question about priority. [NC-W014]

The paper organizes its discussion under combinational creativity and exploratory-transformational creativity. Exploration works within a conceptual space of permitted structures and operations.

Transformation changes that space so that something previously excluded becomes possible. The distinction concerns the representation that defines what can be generated, rather than only whether the resulting item looks unfamiliar. [NC-W014]

Computer examples make the representation consequential. Permitted structures, generative operations and changes to them affect a program's productions; Copycat illustrates flexible representations. Evaluation remains a separate issue involving values and historical or cultural context. Boden's account therefore connects producing possibilities to the conditions under which they can count as valuable, without treating a generator's output count as an adequate assessment of creativity. [NC-W014]

A combinatorial account of discovery

Simonton proposes a combinatorial account of scientific discovery and invention that includes activities Boden distinguishes as exploration and transformation. His formalization concerns an individual scientist's subjective episode. It defines p as the initial probability or response strength of an idea, u as its final utility and v as prior knowledge of that utility, each ranging from zero to one. The proposed creativity measure is $c = (1 - p)u(1 - v)$. [NC-W016]

The expression assigns its maximum to a useful idea that initially had low probability and whose usefulness was little known beforehand. An unlikely candidate is insufficient if it proves useless; a useful result whose utility was already known has a different status in the model. The extreme combinations distinguish such situations as expertise, blind problem solving, lucky guesses and perseveration. They expose the difference between producing a candidate and knowing that it will work. [NC-W016]

The formalization is Simonton's proposed way of characterizing creative episodes, with its variables and interpretation kept together. His discussion of optimization and no-free-lunch reasoning depends on the averaging assumptions of that argument. His broader account and Boden's conceptual-space account remain distinct theoretical positions. [NC-W016]

Running an imagined combination

Fauconnier and Turner describe mental spaces as partial representations constructed for local understanding. Counterparts can be mapped between input spaces, a generic space captures shared structure, and a blend selectively projects elements from the inputs. Composition forms relations, completion recruits background frames, and elaboration runs the imagined situation. Emergent structure in the blend can then inform the understanding of an input. [NC-W017]

Their monk walks up a path on one day and down on another, with arbitrary changes in pace and stops. The question is whether a location is occupied at the same time of day on both journeys. Imagine the journeys occurring simultaneously, with two monks moving in opposite directions on the same path. They must meet. Mapping the meeting back to the separate days supplies the shared time and location. [NC-W017]

The inference depends on selective combination: the path and time of day are shared in the imagined scene, while the two journeys remain represented by distinct travellers. Running that scene makes a relation apparent that was harder to see in the separate accounts. The model explains the representational work of the inference rather than merely naming combination as its cause. [NC-W017]

Literature-based discovery and computational bridges

Finding a hypothesis between disconnected literatures

Swanson and Smalheiser ask what may follow from complementary literatures that do not cite one another. One literature connects A with B; another connects B with C. Reading them together can suggest an A–C hypothesis that neither states. The absence of citation concerns the public literature and does not establish that no person has privately made the connection. Nor do the two links alone prove the proposed hypothesis. [NC-W020]

ARROWSMITH begins with provisional A and C literatures, collects MEDLINE title files, filters shared-word candidates and displays titles and citations for guided reading. The expert interprets whether a common term marks a substantive connection. The representation and filtering make a large set of possible links examinable; interpretation identifies which might be scientifically complementary. [NC-W020]

The authors recount the fish-oil and Raynaud connection through literatures involving blood viscosity and report later corroboration by others. Their retrospective account preserves the difference between proposing a relation from public knowledge and testing it subsequently. A separate migraine/magnesium history supplies a benchmark for later computational work on recovering bridge terms. [NC-W020]

Representing and ranking bridges

Kötter and Berthold's information networks represent heterogeneous concepts, relations and pieces of evidence. A network can accommodate incompletely typed information, with weights expressing confidence in evidence. The representation provides relations that can be explored across domains without requiring every link to be treated as a causal assertion. [NC-W021]

Their bisociative patterns are distinct. A bridging concept links dense groups; a bridging graph connects domains through several elements or returns through another domain; structurally similar disconnected subgraphs supply a comparison without an existing connecting edge. The last pattern is especially different from finding a shared word: the relation lies in how structures correspond. [NC-W021]

Juršič and colleagues begin with two known literatures, represent documents as term vectors and identify outliers classified toward the other domain. These documents may contain ideas atypical of their labelled field and closer to the other. Frequency, similarity and outlier-related heuristics then rank candidate terms. Selected relations can be turned into a weighted network with pruning options. [NC-W022]

Their retrospective migraine/magnesium experiment uses 8,058 pre-1988 titles and 43 historical bridging terms. All tested rankings already put terms shared by both domains ahead of other terms, which gives the baseline a substantial advantage. Improvements in ranking concern recovery of known bridges, not a percentage of correct new hypotheses. The constructed network supports examination of possible relations after that retrieval step. [NC-W022]

Topics as sources of design inspiration

Ahmed and Fuge use topics, represented by collections of words, rather than isolated bridging words. Their data contain 3,918 OpenIDEO ideas across fourteen challenges. Topic proportions represent each idea; a classifier identifies ideas that resemble another challenge more than their assigned one; topics prominent in these outliers receive higher bisociation scores. Topic co-occurrence and proportion relations supply network edges. Both the classifier and the bridging stage affect what the resulting network proposes. [NC-W023]

The approach seeks inspirational frames across several domains without requiring a designer to nominate only two literatures in advance. In the city-vibrancy example, a gardening and urban-farming topic prompts an idea for a community garden on vacant land, growing and selling produce. A paired response to a common topic centres on city, government and economic activity. The example shows a change in the proposed idea, not construction of a garden by the network. [NC-W023]

One experiment compares perceived promise of topics. Another asks workers to generate paired ideas from bridging and common topics, with other workers judging the results in two challenges. Novelty generally benefits, while quality findings are mixed and depend on the challenge and prompt. The authors thus connect a computational representation, a source of inspiration and an observed judgment of resulting ideas, with each stage retaining its own outcome. [NC-W023]

S65 Literary and visual creation through constraint transformation and estrangement

A word, a sentence, an image or a way of attending can become material for deliberate transformation. The practices examined here include changing perception, suspending familiar control during writing, substituting words through a rule, following associations and assembling images taken from different settings. Their artistic purposes give the operations a particular direction. What is being generated may be a text, an image, a new relation to familiar material or a starting point for further work.

Rimbaud and the cultivation of perception

Arthur Rimbaud's letter to Paul Demeny, written on 15 May 1871, places the transformation of the poet within the work of poetry. He calls for knowing and cultivating the self and for a sustained, reasoned disordering of the senses. The ambition is to reach what has not yet been apprehended and find a language capable of expressing it. The letter addresses a way of becoming a poet as well as the making of individual poems. [CRE-W014]

The programme joins experience to form. If perception reaches something unfamiliar, inherited ways of writing may fail to express it. Rimbaud's demand for new forms follows from that ambition. The poet is not simply asked to vary the surface of conventional verse while leaving the conditions of perception untouched. The creator's habits of sensing, interpreting and speaking become part of what is to be changed. [CRE-W014]

The letter also imagines a future of poetic work beyond its present participants and conventions, including women's future place as poets. Its rhetoric is expansive and extreme, and its practical object is an artistic life and language. The text's historical force lies partly in making those capacities a matter of deliberate cultivation. Written in 1871, the letter first appeared complete in print in 1912. [CRE-W014]

Breton and the procedure of automatic writing

André Breton's 1924 Manifesto of Surrealism includes explicit instructions for automatic writing. The writer first obtains materials and chooses a setting favorable to concentration. Breton asks for a receptive state, a departure from judgments about one's own or other people's talent, and rapid writing without a preconceived subject. The pace is intended to keep the writer from holding back and rereading what has just been written. [CRE-W015, “Composition surréaliste écrite”]

The sequence changes when ordinary control begins to return. If the writing falls silent or its continuation becomes too clear, Breton recommends breaking the line of development. After a word whose origin seems suspect, the writer can impose a fixed letter as the beginning of the next word. The arbitrary letter interferes with the expected continuation. A procedure intended to allow a flow therefore includes an intervention for restarting it. [CRE-W015, “Composition surréaliste écrite”]

Breton's account is concerned with the relationship between conscious supervision and expression. Preparing the place and materials, writing quickly, suspending rereading and disrupting an overly controlled continuation belong to one exercise. Removing immediate judgment does not mean that no conditions have been established. The conditions are directed toward changing what can emerge before the familiar act of approval or correction takes over. [CRE-W015]

The manifesto gives this practice its surrealist purpose and explanation. Breton treats the activity as a means of encountering thought and expression that ordinary literary monitoring can inhibit. The instructions make that claim available through an actual way of working. They also show how a writer can detect a change in the exercise and alter the procedure in response. [CRE-W015]

Lescure and a rule that produces an unexpected text

Jean Lescure presented the S+7 procedure at an Oulipo meeting on 13 February 1961. It begins with an existing text and a chosen dictionary. Each noun is replaced by the seventh noun following it in that dictionary. The source supplies much of the grammatical frame, while the dictionary supplies the replacement vocabulary. The writer does not choose each new noun because it seems the best expression of an intended message. [CRE-W016]

The Oulipo account demonstrates the operation through a transformation of Baudelaire's L'Étranger into L'Étreinte. The question-and-answer pattern remains recognizable, but the successive nouns disturb what a reader expects the exchange to concern. The result invites another reading of an inherited form. Both the starting text and the dictionary matter: changing either changes what the counting rule encounters. [CRE-W016, example]

Once those inputs and the rule are fixed, the substitutions are determined. Their effect can be surprising without arising from a random draw. Creative decisions have partly moved to the choice of material and operation. The produced wording then becomes something to read, interpret and potentially develop. Cage's chance-selection procedures in S66 place the source of variation differently. [CRE-W016]

The extensions show the procedure itself becoming material for invention. Raymond Queneau applies related substitutions to adjectives, verbs and other grammatical categories. Another rule admits a replacement only if metrical or rhyme requirements are met. A simple instruction can thus be combined with further constraints that govern which candidate is accepted. [CRE-W016, Extensions]

Other variants alter the displacement number, change the kind of dictionary or construct reciprocal texts. In Jacques Jouet's example, one part is the S−7 of the other, while the other is its S+7. The two parts share a syntactic organization but exchange related vocabulary through the rule. The account develops a family of operations with specified inputs and relations, rather than an instruction to be novel in general. [CRE-W016, Extensions]

Rodari and the word that sets other things moving

Gianni Rodari opens Grammar of Fantasy, published in 1973, with the image of a stone entering a pond. It produces movements at several depths, disturbing things that had seemed separate. He uses the image to describe what can happen when a word enters the mind. Sounds, meanings, memories, images and earlier experiences become active together. The mind also intervenes, connecting, accepting and rejecting what arises. [CRE-W017, chapters 1–2]

His example begins with sasso, a stone. One path follows initial sounds; another follows rhyme; another follows meanings associated with stone and other materials. These are different ways to continue from the same word. Personal memory supplies another path. Rodari recalls a place, a friend, conversations and musical instruments. A word's usefulness as a starting point can therefore depend on the person who encounters it and the experiences available to that person. [CRE-W017, chapter 2]

A later association joins a brick to a song. Rodari develops the conjunction into a musical house whose parts can be sounded, then into a musical village. The image is not complete when the first unexpected pair appears. It can be expanded by asking what follows from the relation and letting remembered music, social experience and further images change the development. Rodari's account follows that movement closely enough to show a story becoming possible. [CRE-W017, chapter 2]

He then changes the form of the starting material. The letters S A S S O are written vertically, each supplying the initial of another word. One result suggests seven geese on a swing. A further attempt produces seven hundred lawyers playing seven hundred ocarinas. The second attempt develops material

already active in the exercise: seven becomes seven hundred, and geese helps prompt ocarinas. The formal positions remain the same while the material continues to accumulate associations. [CRE-W017, chapter 2]

The chapter supplies a repeatable operation without expecting the first output to be valuable. An initial word can be followed through sound and memory, split into letters, rebuilt as a sentence and developed into an image or story. Rodari distinguishes the educational purpose of such activity from using an imaginative task solely to rank children. His exercise is intended to make the capacities available to the participants themselves. [CRE-W017, chapter 2]

Álvarez Rodríguez and the writing workshop

Maritza Isabel Álvarez Rodríguez describes adapting Rodari's techniques for a university course in children's literature. Students are asked to produce texts through the techniques, rather than encounter them only as a theoretical topic. The workshop supplies prompts, a coordinator and other participants. The resulting activity joins the beginning of a text to reading, discussion and further work on what has been written. [CRE-W018]

Her account stresses an experimental relationship to language. Students attend to sound, images, oral expression and alternative ways of reading everyday experience. The prompt initiates activity, but revision and rewriting also belong to the workshop. The coordinator helps participants sustain writing and makes it possible to discuss the material they produce. [CRE-W018, characteristics of creative writing and workshop adaptation]

The article reports the course experience and argues that the arrangement helped students approach writing with less fear and develop expression. Its practical contribution includes this passage from a named technique to a teaching situation. Reading, speaking, generating and revising are organized around the production of texts for children. S70 develops other accounts of how creative methods are taught and assessed. [CRE-W018, conclusions and recommendations]

Pérez Jiménez and the operations of image making

Mauricio Pérez Jiménez distinguishes the material equipment of image production from the rules, decisions and intentions through which it is used. A camera or digital tool supplies possibilities, but a procedure determines what is done with them. Acquiring a technique also changes ways of seeing: isolating a detail, framing a subject, attending to a relation or imagining a transformation. [CRE-W019, pp. 72–75]

His discussion develops operations such as changing apparent size, combining events separated in time or space, reversing movement and altering its speed. A small detail can be emphasized until it stands for a larger event. Something motionless can be given movement, or an action can be slowed enough to invite a different perception of it. The artistic choice concerns both what is represented and how the technical operation transforms it. [CRE-W019, pp. 72–75]

The distinction between equipment and operation also makes room for purposes and attitudes. Identical equipment can support different work because the maker selects, combines and interprets differently. Pérez Jiménez explains the image as the result of a sequence involving an idea, material means and a developing representation. His account gives technically mediated creation an operative vocabulary rather than treating a new device as a creative method by itself. [CRE-W019, pp. 72–75]

Adamowicz and the selection and assembly of images

Elza Adamowicz's Surrealist Collage in Text and Image begins with the 1921 Paris exhibition of Max Ernst's work. Her historical account identifies materials taken from encyclopaedic plates, anatomical treatises, engineering diagrams, commercial catalogues and advertisements. Printed elements and photographs are cut and joined in new combinations. Words and inscriptions can participate in the work as well. [CRE-W046, pp. 1–3]

These operations move material out of the contexts in which it ordinarily has a recognizable use. A technical illustration can acquire a poetic role through its placement beside another image. Selection, combination and accompanying language alter what the familiar element can suggest. The material remains recognizable enough for its new relationship to have force. [CRE-W046, pp. 2–4]

Adamowicz examines the interpretations offered by Breton and Louis Aragon. Breton emphasizes the encounter of disparate realities and the disorientation it can produce. Aragon gives a different emphasis to the diversion of an image into a new role and to the relationship between literal and figurative meanings. Their interpretations are part of the history of what artists understood collage to make possible. [CRE-W046, pp. 3–5]

She also places collage beside automatism. Deliberate cutting and assembly and the flowing production of automatic writing become two significant modes in the development of surrealist practice. Their relationship does not erase the difference between their operations. Adamowicz gives both material process and intellectual interpretation enough detail to explain why joining fragments can concern a transformation of meaning, not merely an altered surface. [CRE-W046, pp. 4–5]

The sources in this section work on perception, words, images and the conditions of expression. Their procedures can be followed through particular changes to material and activity. The related performing practices in S66 extend the account to bodies, partners, scores and rehearsals.

S66 Generating performance in music theater and choreography

A performer may need a way to begin, a partner to respond to, a rule that redirects attention, or material that can be varied in rehearsal. The practices described here organize those conditions in different ways. Theater games make a shared problem available for action and observation. Choreographic work develops movement through images, imitation, discussion and recording. Chance composition specifies operations through which material becomes a score. Each account explains a practice in which generating something and learning how to work with it are closely connected.

Spolin and the shared acting problem

Viola Spolin's Improvisation for the Theater, first published in 1963, organizes learning around games that give actors a concrete problem. Its Point of Concentration is the focus that keeps participants working together. At an early stage it might concern a cup, a rope or a door. Later problems can involve more complex events. Spolin compares the focus to a ball in a game: participants attend to something they can act upon together, rather than continually assess themselves as performers. [CRE-W020, pp. 21–25]

The simplicity of the immediate task is deliberate. Acting involves many simultaneous relationships between body, voice, setting, partners and audience. Spolin divides this complexity into problems that can be experienced. A player can concentrate on the object or action while developing other capacities through participation. The teacher understands how the exercise fits within the larger work, but need not give the player a lecture on every capacity it is intended to develop. The focus supplies enough direction to act and enough openness to respond to what happens. [CRE-W020, pp. 21–25]

Mirror Exercise #1 shows how that arrangement works. Two players face each other. One initiates an ordinary activity such as washing or dressing; the other reflects the movements and expressions as a mirror. They later exchange roles. The task requires attention from head to foot. Spolin asks the observer to notice whether the reflecting player stays with the partner or anticipates a familiar sequence. Knowing what usually happens when someone applies makeup can become an obstacle if it replaces observation of what this person is doing now. [CRE-W020, pp. 60–61]

The initiating player has a different opportunity. Actions can vary in rhythm and inventiveness and can challenge the partner's attention. Spolin also suggests performing the exercise without telling the audience who initiates. Concealing that distinction intensifies the pair's concentration on one another. A small change to the conditions of presentation therefore alters how the exercise is played. Its parts include an activity, differentiated roles, a shared focus and an audience with something specific to observe. [CRE-W020, pp. 60–61]

Side coaching helps sustain the task while the action continues. Spolin recommends short, direct interventions that return attention to the problem: share the stage picture, make contact with an object, continue working. The teacher is responding to what is happening, rather than stopping the experience to deliver an explanation after every difficulty. The intervention must also preserve the nature of the game. A stream of commands directed toward the teacher's approval would replace the shared problem with a different one: satisfying the teacher. [CRE-W020, pp. 28–29]

Evaluation completes the arrangement. Players and observers ask what was actually communicated and whether the concentration was sustained. Spolin recommends questions about the acting problem rather than personal verdicts about a good or bad actor. The teacher joins this inquiry instead of supplying the entire judgment. The audience consequently has an active responsibility: watch closely enough to help the players understand what occurred. Acting, observing, coaching and evaluation become related ways of working on the same problem. [CRE-W020, pp. 25–29]

The discipline of the exercise is compatible with spontaneity in Spolin's account. A player does not have to invent every condition of an encounter before entering it. The task supplies a point of engagement, and its demands change as the other player acts. Her handbook develops this relationship through instructions, coaching suggestions and observations attached to particular exercises. The practical knowledge includes how to keep an exercise alive, not merely its opening rule. [CRE-W020, pp. 21–29, 60–61]

Itani on similar exercises and different explanations

Nobuhiko Itani's historical comparison of Spolin and Konstantin Stanislavski examines what can differ when an exercise looks similar. Both ask actors to work with objects that are not physically present. A performer must give an action enough detail for an absent object to become part of theatrical experience. Itani relates Spolin's space-object exercises to Stanislavski's work on action without objects, then examines the purposes attached to them. [CRE-W023, §§2.4–2.5]

In Itani's interpretation, Spolin emphasizes contact and present physical activity, while Stanislavski gives belief in the truth of the action a different explanatory role. The similarity of an instruction therefore does not settle its place within a teaching practice. Itani also discusses Spolin's resistance to deliberately recalling past emotions. Her body memory describes experience becoming available through present action, rather than a requirement to retrieve and display an earlier feeling. [CRE-W023, §§2.4–2.5]

This comparison gives a history to the relationship among exercise, attention and explanation. The physical action of handling an absent object can be taught within different accounts of what the actor is learning. Itani's analysis preserves both the overlap and the disagreement, including the distinction between avoiding forced emotional recall and excluding past experience altogether. [CRE-W023, §§2.4–2.5]

Waterhouse on making and learning Duo

Elizabeth Waterhouse's Processing Choreography follows William Forsythe's Duo through archival rehearsal videos, interviews and her experience as a dancer. Her account moves between the creation of the piece and the practices through which dancers continued to perform and transmit it. The chronology of the 1996 creation begins with Forsythe developing movement while Jill Johnson and Regina van Berkel observe, imitate and repeat it. Recording is already part of the encounter: he checks that the camera is running as the material takes shape. [CRE-W021, chapter 10, pp. 221–224]

Imagery helps make a movement available to the dancers. A gesture can be described through a steering wheel, a suitcase or another familiar action. The image does not simply name an appearance to reproduce. It gives a direction for making and attending to movement. Waterhouse describes a moment in which Forsythe's steering-wheel image becomes a view over trees for van Berkel, while Johnson imagines something else. Their exchange can proceed with several images in play. What matters in the rehearsal is how those images help the dancers produce and recognize the movement. [CRE-W021, pp. 231–234]

The movement called showerhead receives particularly close attention. Dancers imagine the fingers moving around a circular shower dial. The action begins with an image that seems local to the hand, but its development involves shoulders, torso and both sides of the body. Waterhouse explains the twists and counter-twists through épaulement, the coordinated orientation of the upper body in ballet. The body is not held rigid while a hand draws a circle. The image invites movement that passes through a trained, responsive body. [CRE-W021, pp. 138–146]

Her discussion of Improvisation Technologies gives other examples of this relationship between an image and an operation. A dancer can imagine a line between the fingertips, extend or match lines with the arms, or write in space with different body parts. In the instructional videos, white lines make some of

that imagined geometry visible. The operations begin from knowledge familiar to ballet dancers, including points, lines, circles and planes, then use it to explore unfamiliar movement. Going backward or downward can enlarge a range that habitual movement would leave unused. [CRE-W021, pp. 143–146]

Waterhouse describes these procedures as ways to develop perception and movement rather than routines to reproduce verbatim. Her account of residual response gives the rest of the body an active place in what follows an initiated action. A movement can be deliberately begun and then developed through its physical consequences. Learning involves becoming sensitive to those consequences and coordinating them, as well as acquiring a recognizable shape. [CRE-W021, pp. 143–146]

The office rehearsals show another part of the creative work. Johnson and van Berkel study the recorded material together without Forsythe present. They alternate between watching the small viewing device and getting up to move. They ask where an action begins, which parts of the body participate and what paths they take. The video permits a developing movement to become something that can be inspected repeatedly. Their discussion, bodily attempts and occasional use of a mirror develop its details. [CRE-W021, pp. 234–235]

They also name phrases and make decisions within material that contains variants. Waterhouse recounts how seven phrases from the initial rehearsal became nine sequences as the dancers worked through the movement. Names help the dancers refer to parts and retain them. Forsythe's recorded comments sometimes indicate a preferred version, while other details become settled through the dancers' joint interpretation. Making a piece includes this passage from an improvised event to material that others can remember and perform. [CRE-W021, pp. 233–235]

The wider creation practice permits changes of position, reversals, restrictions and exceptions. Waterhouse's dancers describe an exploratory situation in which neither the final piece nor the precise path toward it is fully known in advance. Their existing skills and shared practices make it possible to continue under those conditions. The detailed Duo history gives this openness a concrete setting: people work with particular movements, images, recordings and one another over time. [CRE-W021, §4.5, pp. 115–117]

Cage and the composition of chance

In Silence, John Cage describes the procedures used to compose Music of Changes and Imaginary Landscape No. 4. The procedure begins with prepared possibilities. Charts provide elements for sounds, durations, dynamics, tempi and the number of events occurring together. Coin operations derived from the I Ching select among them. Three coins tossed six times yield a hexagram that can be related to one of sixty-four positions. The selection is put to work through the contents and organization of the charts. [CRE-W022, pp. 57–59]

The charts have different behaviors. Some are fixed; others change after an element has been used, allowing a new element to take its place. In the sound charts, silence is represented alongside sounds. A decision about how many events can occur together interacts with the durations and sounds selected for those events. The musical result arises through these connected selections. Chance enters a prepared compositional system whose different components have specific jobs. [CRE-W022, pp. 57–59]

The radio piece changes the relationship between selection and sounding material. Numbers on a tuning dial take the place of specified sounds. A selected position may yield a broadcast, static or silence. Cage accepts what occurs there. The chart can determine a setting while the broadcasting environment supplies what that setting produces. His account identifies the operation clearly enough to show how the composition engages a situation beyond the score. [CRE-W022, pp. 58–59]

Music for Piano 21–52 begins with imperfections in paper. Cage prepares a master page containing musical systems, then uses chance operations to determine how many sounds a page will contain. On a transparent sheet, he marks the corresponding number of imperfections. Registering that sheet against the master allows the points to be converted into musical notation. Further operations determine clefs and characteristics of the tones. A material feature that would ordinarily be ignored becomes an input to composition. [CRE-W022, pp. 60–61]

The procedure combines incidental material with deliberate conventions. The paper supplies points, but the master page supplies their musical interpretation. Decisions about notation and playing methods turn those points into something a pianist can use. Chance operations also work within ranges related to performance difficulty. The process is therefore describable as a sequence: prepare the field, select and mark points, map them onto notation, and determine further features through specified operations. [CRE-W022, pp. 60–61]

Cage also states what remains open for performers. The pieces may be played alone or together, with or without other pieces, and with possible overlap. Durations and dynamics remain free within the conditions described. A programmed concert length can still be maintained through advance calculation and a stopwatch. Determination during composition and freedom during performance occupy different places in this account. Cage's stated aim of freeing continuity from personal taste is expressed through these particular arrangements of choice, material and performance. [CRE-W022, pp. 60–61]

Different purposes for teaching improvisation

Eeva Siljamäki and Panagiotis Kanellopoulos review music-education research to ask what improvisation is understood to be for. Their analysis of seventy-seven selected articles identifies five visions and eleven approaches. They examine how those views inform an educator's immediate response to learners. A teaching decision about an improvised sound can depend on whether the aim is expression, musical understanding, stylistic accomplishment or the generation of material. [CRE-W024, Method; Results, research question 2]

One vision encourages an open relationship to sound and questions inherited criteria about what counts as music. Another treats improvisation as a way to cultivate expression and social relationships. A third uses it to learn musical structures and skills. In that setting, manipulating tonal and rhythmic material can help learners understand conventions through use, rather than encounter them only as an explanation. The same broad word, improvisation, can therefore identify activities with different educational directions. [CRE-W024, Results, visions I–III]

A fourth vision emphasizes expertise within a musical tradition. Students learn building blocks and stylistic conventions and acquire the capacity to vary them fluently. A fifth emphasizes invention, giving improvisation a place in discovering material for composition. Faithfulness to a tradition and discovery of unfamiliar material can each organize an exercise, but they do not ask the teacher or learner to attend to precisely the same thing. The review makes those purposes available for examination through its map of the literature. [CRE-W024, Results, visions IV–V]

The performing practices in these accounts include setting a problem, sustaining attention, generating movement, preparing a compositional field and choosing how a learner will participate. S41 develops other aspects of shared skilled practice and ensemble coordination. Here the games, rehearsals and scores show how particular working conditions become part of making the performance itself.

S114 Public speaking, rhetoric and audience judgment

What a speaker can furnish

In the Greek text of the *Rhetoric*, Aristotle distinguishes evidence already available from means of persuasion a speaker can construct. He differentiates the character the speaker conveys, the hearer's disposition and demonstration or apparent demonstration in the speech. This last distinction makes room for an address to affect judgment without settling whether its case is true. The University of Geneva's historical account places instruction in these arts in settings such as advocacy, where the speaker has to address a particular audience about a disputed matter. [T16-W001] [T16-W004]

Preparing speech for an occasion

Quintilian's eleventh book treats suitability as a property of speech in its circumstances. Elaborate language can be unsuitable to a small case, just as a light manner can be wrong for a grave one. The account of delivery treats voice and gesture as parts of the speaker's practice. A written draft can establish the words available to the speaker. How a voice is used or a gesture timed belongs to the performance for which the author offers instruction. [T16-W002]

The received Chinese text of the *Guiguzi* describes opening and closing in persuasive exchange. Its opening chapter considers when to encourage another person to disclose something and when to close or withhold. The interlocutor's words give the speaker material from which to consider intent. The chapter develops its own questions about exchange and disclosure. Its traditional attribution and the dating of the received text require separate historical judgment. [T16-W003]

An address in manuscript and circulation

Helen Dalrymple distinguishes an early draft of the Gettysburg Address from the manuscript prepared for publication in *Autograph Leaves* in 1864. The Library of Congress describes the occasion as the dedication of a military cemetery on 19 November 1863. Edward Everett had been invited to deliver the main oration. Lincoln was asked to follow him with brief remarks formally setting apart the grounds. [T16-W005] [T16-W006]

In the Hay manuscript, the passage that begins by referring to the dedication of part of the battlefield turns to the soldiers who fought there. Lincoln says their actions have already consecrated the ground beyond the speakers' power to add to it. He ends by asking the living to dedicate themselves to the unfinished work. Within the text, the object of dedication shifts from the burial ground to the work left to the living. [T16-W006]

The next day Everett wrote to Lincoln that the short remarks had come closer to the central idea of the occasion than he felt his own long oration had. He singled out their simplicity and fitness to the occasion. His letter records the judgment of the speaker who preceded Lincoln. [T16-W007]

S67 Incubation attention and bodily activity

Setting a problem aside can be part of working on it. The interval may contain rest, an undemanding activity or another demanding task, and the return can concern the same problem or a new one. Walking changes another condition of thinking: what the body is doing while a person generates a response. The studies in this section make these arrangements explicit and examine what changes in the resulting work. Poincaré's account in S64 supplies an earlier history of preparation, interruption and renewed mathematical effort.

Sio and Ormerod and the structure of an incubation interval

Ut Na Sio and Thomas Ormerod's 2009 meta-analysis examines experiments in which people return to a problem after an interval. Their selection contains twenty-nine publications, many reporting several experiments or conditions. The analysis constructs 117 comparisons between an incubation condition and a control condition, adjusting the allocation of shared controls. [CRE-W026, Method]

The review finds a positive average effect of incubation and asks how the arrangement of work relates to it. Preparation matters because the person has already encountered a particular problem before leaving it. The intervening task then occupies the interval in a particular way. A return to that problem is the occasion on which an incubation effect can be observed. [CRE-W026, Results]

Problem type, preparation and the demands of the intervening task differentiate the results. Highly demanding intervening activity tends to be associated with smaller benefits. The relation between low-demand activity and rest varies with the tasks being examined, and preparation time is related to outcomes differently across problem types. The review gives the interval contents and the earlier work a substantive place in the explanation. [CRE-W026, moderator analyses]

Incubation is consequently studied as a sequence, not simply as the passage of time. A task is attempted, an interval changes the person's activity, and performance is examined on return. This structure allows different explanations to be compared and makes it possible to ask which part of the arrangement contributes to an improvement. [CRE-W026, Results and Discussion]

Baird and colleagues and returning to the same material

Benjamin Baird and colleagues investigate the contents of the interval through an unusual-uses task. Participants first generate uses for objects. The intervening conditions include a demanding task, an undemanding task and rest, alongside a no-break comparison. The break lasts twelve minutes in the interval conditions. Participants subsequently work on previously encountered objects and on new ones. [CRE-W025, Method]

The undemanding condition uses a simple response task, while the demanding condition requires maintaining and updating information. Both occupy participants with something other than the creative problem. They nevertheless make different demands on attention. Including new objects in the later task also helps separate an effect on material already encountered from a general improvement in generating responses to anything. [CRE-W025, Method]

The undemanding task leads to greater improvement in uniqueness on the previously encountered problems than the other conditions. The same advantage does not appear for new problems, and the finding is distinct from simply producing more responses. What changes is connected to returning to material on which work has already begun. [CRE-W025, Results]

Participants in the undemanding condition also report more mind wandering, without reporting more explicitly directed thought about the unusual-uses problems. The authors connect that pattern with their incubation result. The study gives a concrete example of an interval that helps creative work while attention is engaged in a modest external task. It distinguishes the condition that was assigned from the mental activity participants later describe. [CRE-W025, Results and Discussion]

Ritter and Dijksterhuis and explanations of what happens during a break

Simone Ritter and Ap Dijksterhuis review several explanations of incubation. A break can allow recovery from fatigue, weaken misleading associations, change a mental set or create an opportunity to encounter a useful environmental cue. Their own argument also gives active unconscious processing a role. These accounts describe different changes that could occur while direct attention is elsewhere. [CRE-W027, “Mechanisms Underlying Incubation Effects”]

The review examines studies that vary whether people expect to return to a problem. In one arrangement, participants receive a task and then perform another activity before making a later response. In a comparison condition, participants believe the original task has ended. Both groups can be distracted, but only one retains an active problem-solving goal. The contrast asks whether having a goal during the interval contributes something beyond distraction itself. [CRE-W027, mechanisms section]

The authors also distinguish generating ideas from selecting among ideas. A period without direct deliberation could affect one of those activities differently from the other. Their discussion of studies with and without misleading cues likewise bears on an explanation through forgetting: an improvement where no misleading cue was supplied requires further explanation. These are ways to give competing accounts different empirical implications. [CRE-W027, mechanisms section]

Ritter and Dijksterhuis favour an active contribution from unconscious processes. Their review makes that position part of a developed argument about goals, association, selection and the contents of the interval. The earlier finding that a break helped is a starting point for those questions. The proposed mechanism is presented through the comparisons on which the authors rely. [CRE-W027]

Oppezzo and Schwartz and thinking while walking

Marily Oppezzo and Daniel Schwartz's fourth experiment separates walking from moving through an outdoor setting. Forty adults are assigned to sitting indoors, walking on a treadmill indoors, being pushed outdoors in a wheelchair or walking outdoors. The moving-wheelchair condition supplies changing outdoor stimulation without requiring the participant to walk. [CRE-W028, Experiment 4]

The task asks participants to generate symbolic analogies for three prompts within five minutes. An appropriate response must express an analogy rather than merely restate the prompt or identify a surface feature. The scoring then distinguishes novelty from quality. High-quality analogies are specific, reach a relatively distant domain and preserve the relevant relational structure. [CRE-W028, Experiment 4, Method]

Walking produces more novel high-quality analogies whether it occurs indoors or outdoors. Ninety-five percent of those who walk produce at least one such analogy, compared with fifty percent of those who sit in the conditions combined. When lower-quality novel analogies are included, outdoor movement also contributes to novelty. The different measures reveal different relationships with bodily activity and surroundings. [CRE-W028, Experiment 4, Results]

The paper's earlier experiments distinguish divergent production from a convergent association task. Its developed account therefore concerns particular kinds of creative response, rather than an undifferentiated improvement in thought. The analogy experiment gives a particularly clear case: a

person receives the same kind of prompt while the body and environment are arranged differently, and the resulting relations among ideas are assessed. [CRE-W028, Experiments 1–4]

These accounts develop several ways of examining activity around a problem. Preparation and return organize work over time; the intervening task changes attention; bodily movement changes a condition under which responses are generated. Their detailed comparisons give substance to the intervals and activities that a brief description of inspiration can leave unexplained.

S83 Serendipity and the cultivation of unexpected discovery

Accident, sagacity and an unexpected finding

Walpole introduced serendipity in his letter to Horace Mann of 28 January 1754 while recounting a heraldic discovery associated with a portrait. His explanation joins accidental encounter to sagacity. The princes' inference about an animal from traces illustrates interpretive ability: what matters is recognizing what an incidental observation may reveal. Walpole explicitly excludes finding what one was already seeking, giving his usage a boundary that later accounts do not all retain. [NC-W024]

Merton places unexpected observations within a larger argument about the functions of empirical research. Research can initiate, reformulate, redirect and clarify theory as well as test a hypothesis. His serendipity pattern concerns an observation that is unexpected, anomalous relative to an existing expectation, and strategic for a broader theoretical question. Strategic value depends partly on what an observer recognizes in a particular discrepancy. [NC-W025]

In the Craftown inquiry, parents reported easier access to adolescent child minders and greater participation in community organizations. Yet the housing community had fewer adolescents relative to young children than the places residents had left. The numerical discrepancy prevented an apparently satisfactory explanation from closing the inquiry. Further interviews suggested that more adolescents were known and trusted in the new community. They were therefore available as usable helpers within the parents' social relations. [NC-W025]

Merton uses the sequence to develop an account of perception shaped by social relationships. The useful observation is not simply that parents made a surprising statement. Their statement, the conflicting demographic comparison and the renewed interviews together support a different interpretation of what availability means in the community. The developed account shows an empirical finding redirecting a theoretical question. [NC-W025]

What was being sought, and what was found?

Yaqub's taxonomy separates the initial intent of inquiry from the eventual role of a result. Walpolian serendipity occurs when targeted search answers a different problem. Mertonian serendipity reaches the intended goal by an unexpected route. Bushian serendipity involves untargeted inquiry serving an existing need, while Stephanian serendipity concerns a solution whose useful problem emerges later. These are different relations between search and use, rather than degrees of good fortune. [NC-W028]

The mechanisms he considers are a separate set of distinctions. Theory-led expectations can make an anomaly conspicuous; observer-led tools and routines can make it visible; an error can expose an unfamiliar effect; and a network can supply information, collaborators or resources. These mechanisms can occur together. Networks can also sustain expectations that make a new interpretation harder to entertain. [NC-W028]

His archival exercise assembled 118 examples to develop ideal types using historical plausibility. The time window matters to the resulting classification. An episode can appear untargeted when preceding years of inquiry are omitted, and one result can acquire several uses much later. His retrospective discussion of Goodyear illustrates how the account changes with the period and purpose considered. The taxonomy therefore asks both what was being sought and when the relation between a result and a problem is recognized. [NC-W028]

Informed attention and freedom of inquiry

Catellin and Loty argue that reducing serendipity to chance conceals the activity of attending to a surprising fact and imagining an interpretation. Knowledge and receptivity, reason and imagination work together in their account. Their broad claim is that every scientific discovery has this character, including the imaginative formation of a hypothesis that later guides a deliberate search. Their neutrino illustration separates an imaginative response to an anomaly from subsequent detection of what the hypothesis predicts. [NC-W029]

Indisciplinarity starts from a person's question and the needs of inquiry rather than a problem's assigned location within a discipline. It draws on disciplined knowledge and self-discipline while resisting conceptual or institutional authority that obstructs investigation. The term consequently concerns the relation between inquiry and its governing frames, rather than merely an organization containing several departments. [NC-W029]

The relation works in both directions. A surprising observation can make departure from an accepted frame necessary, while freedom to reconsider the frame can make an observation surprising in the first place. Their broad argument about discovery differs from Walpole's exclusion of finding a sought result. Preserving those positions separately shows what each author is trying to explain. [NC-W029]

From an encounter to a useful connection

Makri and Blandford interviewed 28 researchers without first prescribing a definition of serendipity. Their model connects a need to something encountered, projects possible value, follows the connection and reflects on what results. Encounter and recognition need not coincide. Perceived usefulness can change during pursuit, and unexpectedness, insight and value can be matters of degree. [NC-W027]

A researcher struggling to frame a thesis on digital music encoding encountered Latour's account of inscriptions during a broad search. Days later, a connection with the findings became apparent. Close reading then helped develop the argument that supplied the thesis's narrative structure. The episode joins exposure, delayed recognition, interpretation and drafting: the useful result is achieved through those later activities as well as the encounter. [NC-W027]

Another researcher first found a website about internal child trafficking, then unexpectedly sat beside its author at a conference lunch. The earlier encounter made the person relevant. Continued discussion and a subsequent meeting helped refine a PhD subject and produced collaboration and further contacts. The chain would be incompletely described as either a fortunate search result or a lucky seating arrangement on its own. [NC-W027]

The two histories give the model different temporal forms. In one, an encountered idea becomes useful after an interval and is developed through writing. In the other, prior information prepares a later personal encounter, which becomes productive through follow-up. The participants' accounts supply the activities through which the authors explain a useful connection taking shape. [NC-W027]

Practices for making one's own luck

Makri, Blandford, Woods, Sharples and Maxwell asked fourteen creative professionals about practices they believed encouraged unexpected useful connections. Reported activities include varying routines, observing, making mental space, relaxing boundaries, drawing on previous experience, looking for patterns and acting on opportunities. The study distinguishes these reported practices from the researchers' proposals for digital support. [NC-W030]

The composer's Fallen Angel episode starts with an intended piece about Lucifer. During a journey from Angel to Highgate, an advertisement showing an eye recalls peacock-feather imagery. The descent and station names connect with the story. The journey subsequently becomes a structure for the work, including film recorded along the route. The piece is performed, and the composer reports strong responses. Preparation, imagery and the encountered route develop into a particular artistic form. [NC-W030]

The authors' design suggestions support activities within such histories. Capture and revisiting can preserve material before its possible use is understood. Exploratory displays can help someone form patterns, while follow-up facilities can support development of an idea or contact. The connection with external memory is practical: something encountered now may matter when another problem or interpretation becomes available later. [NC-W030]

An unexpected model for a library service

Trindade and Vechiato combine questionnaire responses from 51 librarians with one returned diary record. A respondent describes watching television images of Black Friday crowds and adapting the promotional form to a library offer: one-month loans for all categories of user. The service is announced online, and the respondent finds a crowded library at opening. The reported episode connects an observed activity in another setting with a specific change in a lending offer. [NC-W032]

The diary exercise asked for a problem, encounter, trigger and possible discovery. Its returned record instead mainly described benchmarking, sketches and discussions in redesigning communication and online registration. The authors observe that it did not identify a distinct unexpected discovery. The distinction is informative because useful service development can be reported in detail without exhibiting the unexpected trigger that the instrument sought. Their proposed relation between serendipity, creativity and innovation is an interpretation developed alongside those observations. [NC-W032]

Recognizing and developing a possibility in an organization

Busch's systematic review proposes agency, surprise and value as necessary conditions for serendipity. In his model, an unplanned trigger becomes meaningful through association; materialization enacts a valuable possibility. The trigger and the association may occur together or develop over time. The account gives pursuit a role alongside noticing, and treats the process and its related outcome together. [NC-W026]

His potato-washing-machine illustration follows a customer complaint into a different interpretation. Washing potatoes damages ordinary machines. Employees recognize a possible application, and a dirt filter helps develop a product for that use. The complaint changes from an account of misuse into information about an activity a machine could be redesigned to support. Interpretation is followed by a construction that makes the newly perceived use practicable. [NC-W026]

Organizational conditions can support different parts of this development. Interaction across perspectives makes unfamiliar interpretations available. A meaningful but not excessively narrow problem formulation leaves room for them. Permission to express an incomplete idea and access to people or resources can help a possibility be pursued after it is noticed. These conditions concern encounters, interpretation and development through different activities. [NC-W026]

Busch combines sensemaking and event-based theory with a quantum-inspired vocabulary of potentiality and materialization. That vocabulary belongs to his conceptual account of possibilities and their realization. His theory stands alongside the other definitions and histories as an ambitious source contribution, with its own proposed boundary around agency, surprise and value. [NC-W026]

A community in which needs and partners emerge

Busch and Barkema's longitudinal study examines the Nairobi incubator they call T-Hub. Fifty-one months of interviews, observation and documents follow a setting in which entrepreneurs' needs, markets and potential collaborators change. A fixed matching service cannot specify every useful connection in advance. The authors examine how a community can make emerging needs and relationships easier to recognize and develop. [NC-W031]

Cultivating reframing legitimizes changing an idea in response to information. Elevating commitment develops trust and reasons to help other members. Agile platform design adjusts seating, shared places and events. Elevating emerging opportunities adapts partnerships and support as possibilities become clearer. Community-enabling leadership listens and lets members contribute to shaping the community. These named conditions are concrete parts of the authors' process interpretation. [NC-W031]

One reported encounter places a person with relevant health software near a social enterprise communicating with mothers, opening an application in day-care settings. Other accounts concern finding a collaborator before recognizing that one was needed. Introductions, training and user-research support can then help develop the relationship; some histories also involve investment. The shared place supplies encounters, while the surrounding activity supports what follows them. [NC-W031]

The process includes entrepreneurial agency and feedback. Members interpret what a contact may make possible, and the community can change its support as their needs develop. The authors' account therefore concerns both emerging relations and an organization capable of responding to them, rather than the presence of a social space alone. [NC-W031]

Learning to arrange flexible matches

张颖颖 and 王进富 study three orchestration events at 中科创星: project roadshows, an ecosystem-strategy event and industrial matchmaking. Interviews, documents and participation support their account of how an entrepreneurial platform develops alertness to connections between enterprises and resources. Need and value may become clearer during the activity rather than being fully specified beforehand. [NC-W033]

In the roadshow case, staff examine other practices and elicit members' needs. They circulate and interpret possible forms with stakeholders, judge them against investment conditions and the platform's own capabilities, and create an organizational structure and procedures. Search, interpretation, judgment and institutionalization name connected activities through which a possible arrangement becomes an operating one. [NC-W033]

The relations also feed back. Interpretation can redirect the search, and an institutionalized practice shapes later searches. The study's process explains the formation of an organizational capacity to arrange flexible matches. Its events and learning activities provide a different unit of analysis from an individual noticing an unfamiliar item, while keeping the development of a possibility visible. [NC-W033]

Making overlooked possibilities visible

高橋菜奈子 and colleagues compare five virtual bookshelf configurations through gaze measures and the choice of unfamiliar books participants would like to read. Twenty-five people participate across the experiments, with only three taking part in all environments. The configurations change vertical placement, show covers lower down, or use very tall and cylindrical shelves. The arrangement changes what a visitor encounters within the same collection. [NC-W034]

Reversing positions changes the books selected. Showing covers lower down draws attention to an otherwise less-noticed region. Tall and cylindrical arrangements do not distribute attention uniformly. Subject interest and visibility both contribute to selection, so a display feature can change exposure without producing a single even pattern of browsing. [NC-W034]

The measured outcomes are gaze and immediate unfamiliar-book choice. Those observations locate a concrete point at which an interface can affect an encounter: what becomes visible, noticed and selected for possible reading. They complement the histories of interpretation and pursuit by examining how the field of available possibilities is presented. [NC-W034]

S52 Generating alternatives and organizing exploration

Engineering often requires discovering a possible response before its merits can be assessed. The work includes recognizing a problem differently, producing genuinely different candidates, testing them and deciding when to commit. The sources in this section examine how examples, discussion, representations, feedback and organizational learning affect that work. Their procedures range from short individual design tasks to the allocation of resources across an organization.

Several neighboring accounts address different parts of the problem. Alexander's treatment in S08 preserves the inventive work of general design; TRIZ and facilitated problem-structuring methods appear in S44. Ashby's selection and regulatory variety are developed in S03. Baldwin's modular options in S07 and the flexibility accounts in S34 concern retaining choices that can be exercised later. Here the emphasis is on how alternatives arise and how exploration is organized before a choice becomes settled.

Jansson and Smith: an example can constrain designs without reducing their number

David Jansson and Steven Smith's 1991 experiments examine design fixation: adherence to a limited set of ideas that constrains conceptual design. They ask whether an example supplied with a task can restrict the designs people produce, even when they have been asked for many possible solutions. Their procedure compares participants who receive a problem alone with participants who receive the same problem and an illustrative design. Similarity to selected features of that example supplies a way to observe restriction that a simple count of sketches might miss. [OI-W015, pp. 3–4]

In the bicycle-rack experiment, 25 senior mechanical-engineering students were randomly divided between these conditions. The task called for a rack that would be easy to mount, avoid damage and accommodate different bicycles and cars. The example showed a roof-mounted design with suction cups and rails supporting the tires. Students who saw it produced approximately the same number of designs as the control group, but used its characteristic features more often. Suction cups appeared in 54 percent of their designs, compared with 6 percent in the control condition. The quantity of output concealed a substantial difference in the kinds of solution considered. [OI-W015, pp. 5–6, table 1]

Other tasks introduced clearly undesirable features. In the measuring-device task, the example failed to offer continuous adjustment or protection against overflow, both pertinent to the requirements. In a spill-proof cup task, the sample used a straw and mouthpiece despite instructions excluding them, and its construction could leak. The groups shown examples again reproduced more of their features, including defects. The cup experiment also assessed flexibility across approaches and originality relative to other participants' designs; both measures were lower in the example condition. Merely asking for multiple designs did not ensure that the underlying approach changed. [OI-W015, pp. 6–9, tables 2–3]

The fourth experiment involved thirteen professional engineers and a device for taking measurements and samples along the intestinal tract. The example used a cable and control box, with an opening at the front of the device. Engineers exposed to it incorporated those features more often than the comparison group. This small professional sample extended the investigation beyond students, while remaining a brief task with a deliberately limited problem. [OI-W015, pp. 9–10, table 4]

Jansson and Smith retain a difficulty important to the interpretation of their experiment: prior knowledge is indispensable to design. Learning from an earlier design and becoming unable to consider a relevant alternative are different things. They ask about useful features, long-term experience and more complex tasks, leaving the conditions of beneficial and harmful influence open. The work demonstrates a barrier under specified conditions and provides a method for examining it; the discussion does not supply a validated general procedure for eliminating it. [OI-W015, p. 10]

Ritchey: organize combinations through dimensions and consistency judgments

Tom Ritchey's account of general morphological analysis begins by identifying dimensions of a problem and possible conditions within each dimension. A configuration combines one condition from each. The dimensions can concern technical properties, financial arrangements, policy aims or ethical commitments, so the representation does not require every difference to be expressed as a numerical quantity. The first work is therefore defining what matters and specifying meaningful alternatives within each dimension. The account first appeared in 2002 and was revised in 2013. [OI-W014, pp. 1–4]

The resulting morphological field can contain many combinations that nobody would have listed unaided. In Ritchey's reduced teaching example, six dimensions contain four, four, four, three, three and four conditions, producing 2,304 formal configurations. This is a count of combinations inside the defined field. It does not mean that 2,304 policies were implemented, evaluated or found workable. Defining a field makes a particular range of possibilities explicit while retaining the judgment that selected its dimensions and conditions. [OI-W014, pp. 4–5]

Cross-consistency assessment reduces that range. Participants consider pairs of conditions from different dimensions and judge whether they can coexist. A configuration containing a rejected pair can then be excluded. The procedure records the judgments used to obtain the remaining configurations, allowing later inspection of how the result arose. Ritchey distinguishes logical contradiction, empirical incompatibility and normative objections, arguing that judgments about what is possible should not be silently collapsed into judgments about what is desirable. The assessments still require people with relevant knowledge and the ability to make disagreements explicit. [OI-W014, pp. 5–7]

His shelter-policy example concerns Sweden's reconsideration of a civil-defense system after the Cold War. The field includes geographical priorities, functional priorities, crowding, new construction, maintenance and the general philosophy governing provision. A policy of equal shelter quality for everyone conflicts, in the illustrated assessment, with selectively prioritizing some geographical areas. Selecting a condition as an input lets participants inspect which other conditions remain consistent with it. The representation supports investigation of the implications of a commitment, as well as the discovery of previously unlisted combinations. [OI-W014, pp. 4–7, figs. 2–5]

The method's operative result is a set of configurations consistent with recorded judgments. It does not simulate their consequences or resolve political choices through arithmetic. Software helps manage combinations and display the implications of selected conditions, while the definition of the problem and the consistency assessments remain substantive work. Ritchey's claims about traceability and the value of the method are methodological arguments illustrated through applications. They do not amount to a controlled comparison showing that the resulting policies outperform alternatives. [OI-W014, pp. 6–8]

Diehl and Stroebe: giving everyone a chance to speak is a production constraint

Michael Diehl and Wolfgang Stroebe's 1987 study investigates why interacting brainstorming groups can produce fewer ideas than an equivalent number of people working independently. Their comparison is between an actual group and a nominal group constructed by pooling independent contributions and removing duplicates. A four-person nominal group's output is not the output of one person. Holding the session length constant gives both arrangements the same total person-time, making the organization of interaction a possible explanation for the difference. [OI-W012, pp. 497–501]

Four experiments examine proposed explanations including reduced effort when assessment is collective, fear of evaluation and production blocking. Blocking occurs when one person's opportunity to express an idea depends on somebody else finishing. The first experiments varied assessment and evaluation conditions. These changes affected productivity, but did not remove the large difference between actual

and nominal groups. The findings therefore required an explanation beyond attributing all losses to unwillingness to contribute or fear of criticism. [OI-W012, pp. 499–505]

The fourth experiment made the turn-taking restriction explicit. People in separate rooms used microphones and a light display indicating whether another person was speaking. In blocking conditions, they were to wait until the lights permitted them to speak. One condition retained the restriction while preventing participants from hearing others' ideas. Another allowed participants to ignore the lights and speak whenever they had an idea. Individual assessment was retained across the conditions. This arrangement separated the opportunity to speak from exposure to another person's contribution. [OI-W012, pp. 505–506]

The restricted conditions produced fewer ideas even when participants could not hear one another. Removing the restriction brought output close to the independent control. In that experiment, the mean pooled output of four people working independently was 106 nonredundant ideas, compared with about 56 for an actual four-person group. The result supports production blocking as an important mechanism in those short sessions. It does not by itself identify whether a waiting person's difficulty arose from memory, disrupted development of an idea, motivation or some combination of processes. [OI-W012, pp. 506–508, table 5]

The resources in the comparison matter. Giving an interacting group several times longer would increase its total person-time. Moreover, idea counts, the number of good ideas and average quality are different measures; the later experiments concentrated on counts. The authors suggest independent generation followed by discussion and evaluation, but that combined procedure was not the treatment tested in these experiments. Their participants and brief prompts also differ from an established expert team working over months. The paper identifies a cost of organizing generation around one shared speaking channel and supplies controlled ways to examine it. [OI-W012, pp. 501, 507–508]

Nemeth and colleagues: criticism and generation need not be separate stages

Charlan Nemeth, Marie Personnaz, Bernard Personnaz and Jack Goncalo's 2003 working paper tests an instruction frequently associated with brainstorming: participants should avoid criticizing one another's ideas. The authors contrast it with instructions encouraging debate and criticism, and with a minimal condition giving only the task. Their question is whether the attempt to remove evaluation can also remove an interaction that stimulates new thinking. The manipulated object is the instruction governing a discussion, rather than whether every participant agrees with the others. [OI-W008, printed pp. 10–12]

Groups worked on reducing traffic congestion, with one participant recording ideas while the others discussed them. The discussion lasted twenty minutes. Afterward, people separately recorded ideas they had considered but not expressed and new ideas occurring after the discussion. This extended the observation beyond the public record of the meeting. An idea absent from that record could have been withheld, crowded out or developed only afterward; the two individual questions attempted to distinguish those possibilities in the participants' reports. [OI-W008, printed pp. 11–14]

The study used samples in Berkeley and Paris. Debate instructions produced more group ideas than minimal instructions in each sample. A direct comparison between debate and conventional brainstorming instructions was less decisive for ideas spoken during the session. In the combined analysis, debate was superior on the authors' total-production measure, which combined a share of the group output with the individual's subsequent contributions. The pattern therefore depends on which output is counted. It is stronger evidence for attending to discussion and its aftermath than for claiming that every instance of criticism improves every measure of creativity. [OI-W008, printed pp. 13–19]

The authors interpret their findings through the stimulating effects of dissent and competing views. Their procedure encouraged criticism of ideas within a short task; it did not manipulate retaliation by a

superior or the consequences of challenging an established organizational commitment. This makes its relevance different from the silence and voice accounts in S51. Its substantive contribution concerns how the terms of discussion can affect the production of ideas, including ideas that become available after the discussion has ended. [OI-W008, printed pp. 17–19]

Dow and colleagues: make several prototypes before receiving feedback

Steven Dow and colleagues' 2010 journal article compares two ways of distributing the same amount of prototyping and critique. In the serial condition, a designer made one prototype, received feedback and continued to the next. In the parallel condition, a designer first made three alternatives and received feedback on them, then made two more and received another round of feedback. Both conditions produced five prototypes followed by a final design, with equal creation time and the same amount of descriptive critique. Parallel here means developing several alternatives before feedback; it does not mean assigning different prototypes to a team of workers. [OI-W013, pp. 18:5–18:10]

Thirty-three participants completed the task of designing graphic advertisements for the same client. They were nonprofessional designers using a common unfamiliar tool, with assignment balanced for prior experience and gender. The final advertisements were assessed through a live online campaign and by independent client and professional judges. Other raters assessed similarity among each participant's designs. The study thus separated several outcomes: response to the advertisement, judged quality, variation among prototypes and the designer's confidence in performing the task. [OI-W013, pp. 18:5–18:13]

The parallel condition performed better on click-through rate and expert ratings. Exposure must be included in reading the campaign results because the advertising system changed how often advertisements appeared. During the first five days, when aggregate exposure was approximately balanced, the parallel advertisements received 44 clicks from 79,800 impressions and the serial advertisements 26 clicks from 79,658 impressions. The paper also reports higher expert ratings and greater design diversity in the parallel condition. The live response and the expert ratings were distinct criteria and did not simply reproduce one ranking. [OI-W013, pp. 18:11–18:14]

The authors propose several explanations. Seeing alternatives can make the differences among criticisms easier to understand, while attachment to any one prototype may be reduced when several are available. Serial refinement can instead preserve the assumptions of an early design through successive improvements. The experimental comparison changed the schedule of work and feedback as a package. It did not independently establish the causal contribution of each proposed psychological mechanism. Nor did it show that greater diversity necessarily produced higher quality: more experienced participants could produce less diverse work while achieving better results. [OI-W013, pp. 18:14–18:18]

The task was short, inexpensive and allowed consequences to be measured relatively quickly. The authors discuss the need to investigate longer design processes and teams. Maintaining several expensive prototypes could allocate effort differently from the banner-ad task, and the experimenter administering the conditions was not blind to assignment. The equal time and prototype count nevertheless make the demonstrated comparison informative: the observed advantage cannot simply be attributed to the parallel condition receiving more opportunities to build or more feedback. [OI-W013, pp. 18:18–18:19]

Nutt: searching for a result rather than defending a ready-made solution

Paul Nutt's 1999 article examines how organizations establish a direction, identify possible responses and implement a decision. Its cases distinguish defining a desired result from beginning with a solution that somebody already favors. His heat-pump example makes the distinction concrete. A firm spent years trying to make a particular heat-wheel technology work, although its broader opportunity concerned

economical heating. Once that solution supplied the definition of the problem, technical effort went into repairing it rather than examining other ways to obtain the desired result. The example concerns how a commitment organizes subsequent search. [OI-W009, pp. 78–81]

Nutt distinguishes several ways of finding options. Copying a single comparison organization imports a package that may fit a different setting. Integrated benchmarking instead examines several organizations and combines pertinent features. A search can also occur in stages: an initial request for proposals teaches the buyer what suppliers can offer, and a later request uses that knowledge to specify a more informed demand. Designing a custom option is another route, requiring design skill, attention to users and the capacity to put the result into operation. His categories describe different work, rather than identifying all search with asking for a longer list. [OI-W009, pp. 81–85]

The underlying evidence is a retrospectively reconstructed collection of decisions in medium and large organizations in the United States and Canada. Interviews and documents were used to establish the process, with subsequent checks of implementation. Success was assessed through such measures as sustained use and full use after two years. Those measures distinguish adoption from abandonment or partial use; they do not by themselves establish profitability, public benefit or the absence of harmful effects. Managers were not randomly assigned to search methods. [OI-W009, pp. 76–77, 89, note 5]

Nutt associates the development of multiple options with more successful implementation and argues that discovering alternatives can reduce later repair work. The multiple-search category contained nine cases, all sustained and fully used after two years. Additional search requires time, access to comparison cases and the ability to combine what is learned. The article also treats participation and implementation separately, so discovering an attractive response does not complete the task of securing its acceptance and use. [OI-W009, pp. 83–89]

March: learning can consume the differences from which it learns

James March's 1991 article examines exploration and exploitation in organizational learning. Exploration includes experimentation with new alternatives, while exploitation develops and uses established capabilities. Both consume resources, but their returns can arrive at different times and benefit different people or parts of an organization. Procedures for setting targets, rewarding work, allocating spare capacity and choosing investments therefore affect the balance even when nobody explicitly makes a decision called exploration versus exploitation. [OI-W010, p. 71]

March explains why ordinary adaptation can favor existing activities. Greater competence produces more reliable rewards, which make that activity more attractive and support further competence. A less familiar alternative can perform poorly while people learn how to use it, even if its potential is greater. Feedback about established work is often faster, clearer and closer to the people doing it than feedback about exploration. A process that improves short-term performance can consequently narrow the experience from which later possibilities would be discovered. This is the argument developed through the paper's models. [OI-W010, p. 73]

In the mutual-learning model, individuals hold beliefs about an external reality and an organizational code holds accepted beliefs. Individuals can adopt the code's beliefs; the code can learn from a group of individuals stipulated by the model to be more accurate than itself. The model specifies reality so that accuracy can be measured. Individuals do not directly experience that reality or teach one another outside the code. Those simplifications make it possible to study how two learning processes interact, with rates of socialization and code revision treated separately. [OI-W010, pp. 74–75]

Rapid socialization removes differences between individuals and the code. That can improve an individual's knowledge in the short term while reducing the variation from which the code can learn. In the simulations, slower socialization can preserve useful differences long enough for the organization to

absorb them. Mixtures of fast and slow learners can also improve the eventual code relative to a population with a uniform learning rate. The benefits are distributed unevenly: fast learners can acquire knowledge whose development depended disproportionately on people who remained different longer. [OI-W010, pp. 75–77]

Turnover provides another source of variation in the model. A newcomer has less knowledge on average but may hold beliefs that are less redundant with the code. Moderate turnover can therefore improve the code under some learning-rate conditions, while excessive turnover or turnover combined with already slow socialization can undermine the exploitation of what has been learned. When the external reality changes, a population that has fully converged can cease to adapt. Introducing newcomers can prevent that degeneration in the modeled system. [OI-W010, pp. 78–80]

March also distinguishes improvement in average performance from competition to finish first. Reducing variability can make outcomes more reliable while reducing the chance of an unusually good result at the upper end of a distribution. Whether that trade-off matters depends on the competitive objective and the distributions being compared. He uses it to question the assumption that more reliable knowledge necessarily improves every kind of competitive position. The discussion is a model-based argument about the conditions of performance, not an observed comparison of a particular firm's management practices. [OI-W010, pp. 84–85]

The article does not determine one optimal allocation between exploration and exploitation for every organization. Its models explain particular ways that learning rates, recruitment, time horizons and competitive criteria can change the consequences of an allocation. Too much exploration can leave ideas undeveloped and competence insufficient; too little can leave an organization highly practiced at an inferior activity. The work makes those choices and their distribution of costs explicit. It extends the discussion beyond generating candidates during one decision to sustaining the capacity to encounter and develop alternatives over time. [OI-W010, pp. 71, 73, 85]

S68 Collaborative generation and material methods

An observation written on a card can be moved beside another observation, given a new heading and connected to a larger account. A workshop can supply pictures of technologies, invite children to explain their schoolbags to imaginary visitors, or let designers build a proposal from physical pieces. These procedures give participants something to work on together. Their histories show how the preparation of material, the framing of a task and the opportunities for rearrangement enter into creative work.

Kawakita's field cards and the KJ method

In a 2003 interview with Tetsuro Matsuzawa and Yoko Yamada, Jiro Kawakita recalled the fieldwork behind the KJ method. He described transferring observations from a notebook to separate cards, a practice he traced to his field investigations in 1946. A notebook preserves a sequence of recording. Cards allow observations to be read in different sequences and assembled according to relationships that become apparent later. The interview presents this development through Kawakita's recollections of his own research. [CRE-W030, pp. 14–16]

The movable statement retained information about its circumstances: time, place, source and recorder. Rearrangement did not require stripping the observation of its origin. Kawakita emphasized repeated reading of the cards and attention to what they actually said. Statements that seemed close in meaning could be brought together, and a heading could express the message emerging from their association. The heading was preferably a sentence conveying something substantive, rather than a noun that merely named a subject. [CRE-W030, pp. 14–16]

Successive groupings made larger patterns available for consideration. Arranging the material could also expose a gap: an apparent relationship might require information the researcher did not yet possess. The work could therefore lead back to inquiry. Kawakita's account joins the manipulation of records, the interpretation of their relationships and further observation within an ongoing investigation. [CRE-W030, pp. 14–16]

Hiroaki Tanaka's 2013 quick manual specifies a sequence for carrying out KJ work, based on Kawakita's later exposition. It begins by putting one message on each label. Labels are numbered so that they can be related to the underlying data. They are spread out and read before grouping begins. The initial groups are small and arise from similarities of meaning in the statements. A classification prepared in advance would bypass the interpretive work the procedure is intended to support. [CRE-W029, pp. 102–104]

Each small group receives a heading expressing what its labels say together. A label that does not fit can remain on its own. The headings can then be treated as material for another round of grouping, producing larger groups without requiring every statement to enter the same hierarchy. Different colors and the retention of the lower-level labels help participants keep track of these successive operations. The emerging account can be traced through its headings to the statements from which it was developed. [CRE-W029, pp. 103–105]

The groups are next placed spatially to make their relationships intelligible. Positioning comes before an elaborate network of lines; the manual recommends using relations economically rather than connecting everything to everything. A diagram gives the arrangement a visible form, and prose develops an explicit account of what it means. Writing is thus a further operation on the material, requiring the interpreter to state relationships that a suggestive diagram might leave implicit. [CRE-W029, pp. 105–106]

The sequence gives several distinct tasks their own working forms: a label states a message, a heading expresses a group's meaning, a diagram displays relationships, and prose articulates an interpretation. Their combination is what makes Tanaka's account more useful than the instruction to sort observations

into clusters. It explains both how the material is transformed and how earlier statements remain available during that transformation. [CRE-W029, pp. 102–106]

Preparing cards and inventing a workshop situation

Michael Mose Biskjaer, Peter Dalsgaard and Kim Halskov examined creativity methods in design through nine components: process structure, materials, tools, combination, metaphor, analogy, framing, divergence and convergence. Their 2017 analysis grouped these into concrete, conceptual and design-space aspects. A workshop's instructions become more intelligible when considered with the material supplied, the situation participants are asked to imagine and the forms of development expected from them. [CRE-W031]

Their account of the Inspiration Card Workshop begins before participants assemble. Organizers prepare Domain Cards representing knowledge of the setting, often obtained through field studies, and Technology Cards representing technologies or particular applications. Each card carries an image and may include a short text. Selecting familiar technologies can make the workshop easier to enter; selecting unfamiliar or provocative ones can unsettle established expectations. Preparation helps determine the possibilities participants will encounter. [CRE-W031, analysis of Inspiration Card Workshops]

In the workshop, the cards are introduced and then combined to develop design concepts. Participants record emerging ideas on notes and collect them on poster-sized sheets. The main creative phase has room for different ways of working rather than prescribing every operation. Concepts are presented and discussed, but this procedure does not require reducing them to a winning selection. Afterwards, posters can be scanned and distributed. The physical combination of prepared cards, discussion around a developing concept and preservation of its representation occupy different parts of the sequence. [CRE-W031, analysis of Inspiration Card Workshops]

Fictional Inquiry changes the social situation in which participants contribute. In one case described by Biskjaer and colleagues, designers acted as Martians visiting Earth, and primary-school children explained their schoolbags to them. Children could no longer assume that their adult interlocutors already understood an ordinary bag or its use. The fictional visitors gave them a reason to articulate familiar experiences and imagine alternatives while contributing to the design of digitally enhanced schoolbags. [CRE-W031, analysis of Fictional Inquiry]

Another workshop asked participants to help the king of Atlantis understand what humans found extraordinary. Seashells, a scepter and other props supported exploration of interactive aquarium exhibits. The fictional situation supplied roles and objects through which participants could develop ideas; those ideas then had to be related to the real aquarium. The authors describe a method that adapts its narrative to a particular design setting, with the connection back to that setting as an important part of the work. [CRE-W031, analysis of Fictional Inquiry]

Their third example, Extreme Characters, exaggerates the people imagined in a design scenario. A proposed appointment manager might be explored through an unconventional character whose jewelry becomes a temporary stand-in for the device. Touching a ring can suggest an interaction such as moving between appointments. The procedure uses role, appearance, action and props together to provoke a different view of a user group. Biskjaer and colleagues treat such methods as combinations of interacting features, and their examples explain why a label such as analogy alone would leave much of the workshop undescribed. [CRE-W031, analysis of Extreme Characters and discussion]

What analogue tools supply

Dorian Peters, Lian Loke and Naseem Ahmadpour reviewed analogue tools for collaborative ideation. Their review selected 76 tools from research and commercial sources and illustrated a smaller set in the article. They distinguished seven types: methods, prompts, components, concepts, stories, embodiment and construction. A card explaining a method, a question that provokes a response and a physical component to assemble offer different forms of assistance, even when all are described as creativity tools. [CRE-W032, §§4–5]

These distinctions concern what a participant can do with the supplied material. Prompts stimulate responses; components can be combined into a developing proposal; concepts make potentially relevant knowledge available. Stories give participants situations or sequences to develop. The format alone does not determine the activity: two decks of cards may support substantially different kinds of work. The review records features such as a tool's focus, availability and supporting studies alongside its type. [CRE-W032, §§5.1–5.3]

Embodiment and construction extend the account beyond printed cards. Reduced-visibility glasses and reduced-mobility gloves in an inclusive-design toolkit let designers experience particular difficulties while handling a product. Construction materials allow participants to make and discuss a representation. The authors distinguish generic building pieces from LEGO Serious Play's combination of a prescribed method and a purpose-made kit. In each case, the activity depends on the relation between the object and its use. [CRE-W032, §§5.3.5–5.3.6]

The review also draws attention to participation itself. Equipment for simulating an impairment serves a different purpose from tools that enable people with disabilities to participate in design. Peters and colleagues describe examples addressing such participation when discussing gaps and related tools. Their survey makes both material variety and the intended participants visible; it does not rank the 76 tools as though they had all undergone a common effectiveness trial. [CRE-W032, §§5.4.1, 6.2.2]

Working spaces and the continuing design cycle

Scarlett R. Herring, Brett R. Jones and Brian P. Bailey interviewed ten designers about idea generation throughout their work. Seven interviews took place in project spaces, where the researchers could inspect artifacts, and three took place by telephone. The study identified nineteen techniques and placed them within an account of the design cycle, including implementation and evaluation as well as initial concept generation. [CRE-W033, §§3–5]

Their IR3 model describes research, representation and refinement within an encompassing search for inspiration. Research can produce several concepts; sketches or other external representations let a designer inspect and discuss them; refinement can change what further information is needed. These movements need not occur in long, neatly separated phases. A quick sketch may be evaluated and discarded almost immediately. A concept can concern the definition of the problem as well as a potential solution. [CRE-W033, §4.1]

The techniques reported by the designers include seeking examples, consulting others, acting out situations, manipulating concrete stimuli, documenting ideas and criticizing developing proposals. A critique can generate further ideas by drawing attention to a defect that needs a response. Work on a representation can similarly reveal something that was not apparent when an idea remained verbal. The study presents these activities as recurring parts of professional work rather than as exercises confined to a single introductory meeting. [CRE-W033, §5]

Project spaces make this recurrence tangible. Photographs, notes, comparable products and prototypes can remain spread out, visible and available to handle. A designer can return to an earlier representation

while developing another one, and colleagues can refer to the same material. Herring and colleagues use these observations to discuss requirements for computer support, including access to the working environment when a person or dispersed team is elsewhere. Their proposal for support grows out of observed practices and interview accounts. [CRE-W033, discussion]

Collaborative generation for accessible websites

Daniela Satomi Saito and colleagues considered how a shared web environment could support creative collaboration in developing accessible websites. The proposed participants include people with different professional knowledge and users whose experiences matter to accessibility. The chapter examines techniques together with the functions a platform would need to make them available in a collaborative design process. [CRE-W034, pp. 170–203]

The authors consider written circulation of ideas, role-based generation, mind maps and evaluation. Circulating contributions offers a way for a multidisciplinary group to add to and develop proposals. Role-based exercises ask participants to approach the problem from another identity or perspective. A mind map can make requirements and their associations visible for further discussion. The chapter connects these operations to the task of eliciting and developing accessible interfaces, where the knowledge required is distributed among different participants. [CRE-W034, pp. 184–186]

This is a design and instructional proposal for bringing established techniques into a shared environment. Its account explains how participation, representation of requirements and generation of interface ideas could be supported together. The general written-exchange procedures discussed in Systematic accounts of methods for generating ideas provide another route into this material; Saito and colleagues develop their application to a specific collaborative task. [CRE-W034]

S69 Computer-supported and human–AI co-creation

Computer support for creative work includes preserving alternatives, sharing developing representations, exchanging ideas and generating candidates. These activities have different histories. Work on creativity-support interfaces and electronic brainstorming provides a setting for later studies of human–AI collaboration, whose interventions range from an initial suggestion to continuing participation in a shared performance or design.

Shneiderman's creativity-support tools

Ben Shneiderman's 2007 account places exploration at the center of creativity-support design. A person looking for an answer may need to inspect distributions, relationships, gaps and unusual cases before knowing exactly what to ask. Faceted search and dynamic queries allow incremental changes to a view. Rapid, reversible responses help users learn from successive attempts rather than commit to one query and receive a final answer. [CRE-W035, pp. 25–27]

Rich history-keeping supports this extended activity. A record of alternatives lets users compare what they have tried, return to an earlier state and modify it. A promising case can be sent to a colleague, and an investigation can resume without reconstructing its previous path. Shneiderman connects such facilities to both systematic and relatively free exploration. His design proposals concern the working relationship between successive attempts, their representations and the people who may develop them. [CRE-W035, pp. 26–27]

The familiar combination of a low threshold, a high ceiling and wide walls expresses another part of his argument. Beginners should be able to start, experienced users should be able to attempt ambitious work, and the environment should permit different kinds of activity. A tool that supports only a narrow demonstration may be easy to learn while offering little room to develop. Movement among exploration, representation, analysis and communication is therefore part of the design problem. [CRE-W035, design principles]

Collaboration also changes over a project's life. Early exchanges concern problem definition and goals; later ones may involve information, refinement, validation and dissemination. Shneiderman discusses trust, records of contributions and opportunities to expose uncertainty safely. He argues for evaluating creativity-support tools through observation of serious work over extended periods, alongside shorter studies. The article supplies an account of computer support in which a useful intervention can preserve, connect or expose work without generating the central idea itself. [CRE-W035, pp. 26–31]

Electronic brainstorming and time to use others' ideas

Hamit Coşkun's 2011 study varied group size, memory instructions and session length in three electronic-brainstorming experiments. Participants worked at separate computers and completed a typing-speed test before the exercise. Tasks included imagining the consequences of children being born with an extra thumb and proposing improvements to a university. Electronic exchange made other participants' contributions available while people generated their own. [CRE-W036, pp. 91–94]

Across the experiments, larger groups generated more ideas in total and more unique and original ideas. The corresponding individual-output analyses did not show that each participant produced more merely by belonging to a larger group. The results describe an enlarged collective pool alongside a different finding about its individual contributors. Both are useful for understanding what the electronic arrangement accomplished. [CRE-W036, results and general discussion]

The memory instruction asked participants to remember ideas because they would later be tested. It reduced production in a fifteen-minute session but improved production in a twenty-five-minute session. In the longer exercise, participants had more time in which attention to the shared material could contribute to further generation. Coşkun presents this interaction as a way of understanding inconsistent findings about such instructions. The same request can have a different effect when the period available for using it changes. [CRE-W036, experiment 3]

Typing speed affected total and unique production, while the study did not find corresponding effects on originality and feasibility. The experiments thus bring the interface, attention to others' ideas and session duration into the account of the exercise. The earlier discussion of group idea generation in this volume provides the broader background; Coşkun's contribution concerns these particular arrangements for electronic participation. [CRE-W036, general discussion]

COFI and the design of a creative partnership

Jeba Rezwana and Mary Lou Maher developed COFI to describe interaction in human–AI co-creative systems. Their framework considers collaboration style, communication, creative process and creative product. Partners can work on a shared task or divide the work, take turns or act in parallel, and follow a plan or respond spontaneously. These choices describe how participation unfolds, beyond identifying what kind of output the system produces. [CRE-W037, framework]

Communication receives particular attention. A person can intentionally instruct a system through text, voice, gesture or direct manipulation. A system may also collect consequential information that the person gives off without making it an explicit instruction, such as gaze, bodily movement or facial expression. Communication from the AI to the person can provide feedback, reveal a contribution's status or support a response. The framework separates these channels so that the presence of an input facility is not mistaken for a complete two-way exchange. [CRE-W037, communication components]

The creative process and product provide further distinctions. A partner may contribute to generating a new element, evaluating an existing one or changing material already present. A shared dance, drawing or narrative presents different opportunities for timing and response. COFI asks how the contributions relate to the evolving product and to one another, making interaction itself an object of analysis. [CRE-W037, creative-process and product components]

The authors applied the framework to 92 systems across thirteen creative domains. Two coders independently examined a quarter of the systems, discussed differences and established a coding basis for the remainder. Their analysis identifies recurring interaction designs in the described systems. Its contribution is a structured account of creative participation, grounded in those designs, rather than an effectiveness ranking of every possible form of partnership. [CRE-W037, §§4–5]

Suggested beginnings and the stories people write

Anil R. Doshi and Oliver P. Hauser studied access to AI-generated ideas in a short-story task. The 293 writers were asked to produce an eight-sentence story suitable for a teenage and young-adult audience. They were assigned to write without AI assistance, with access to one generated idea, or with access to as many as five. A generated idea consisted of three sentences offered as starting material; the writer composed the story. [CRE-W038, experimental design]

The second phase obtained evaluations of the stories from a separate group of participants. Evaluators first judged the work without being told about its AI assistance. In the condition allowing five ideas, average novelty ratings were 8.1 percent higher and usefulness ratings 9 percent higher than in the

unassisted condition. The study therefore found an improvement in judged individual work under this specific form of assistance. [CRE-W038, results]

The authors also compared stories with one another and with their generated starting ideas. AI-assisted stories were more similar across writers. Access to a pool of suggestions could improve a writer's evaluated result while drawing the collection toward more similar content. These analyses concern different objects: the quality attributed to one story and the relationships among many stories. [CRE-W038, collective-diversity analysis]

The intervention's sequence is important to understanding the result. The model supplied a possible beginning, and a person interpreted and developed it within a tightly bounded writing task. Doshi and Hauser's study makes this early contribution visible as part of the subsequent human work. It also provides a concrete case in which an individual's benefit and the variety of the resulting collection can be assessed separately. [CRE-W038]

Comparing a language model with oblique prompts

Barrett R. Anderson, Jash Hemant Shah and Max Kreminski compared ChatGPT with the Oblique Strategies prompt deck. Thirty-three participants were included in the analysis. Each worked with both forms of assistance on timed tasks involving product improvements and improbable consequences. Examples included making a jigsaw puzzle more engaging and imagining a world in which only people's feet could be seen. [CRE-W039, §3]

The deck offers brief provocations that a participant must interpret in relation to the task. A language model can provide developed candidates and respond to further requests. The experiment gave these two kinds of assistance a common set of ideation tasks and an eight-minute period for each session. It thereby compared an established prompt device with conversational model support, rather than comparing assistance with no intervention at all. [CRE-W039, materials and procedure]

With ChatGPT, participants produced approximately 15 percent more ideas and covered 27 percent more categories in the reported analysis. Their ideas were also more homogeneous across participants. The analyses did not detect the same change in diversity within an individual's collection, and the unique-idea measure did not show a significant improvement. More output and more represented categories can therefore coexist with greater similarity among the collections produced by different people. [CRE-W039, results]

The authors discuss the interpretive distance between an oblique prompt and the current task, compared with the more developed material a model can supply. This is their interpretation of how the forms of assistance may shape ideation. Together with the task and outcome distinctions, it gives the comparison more explanatory value than a single judgment that one tool makes people more creative. [CRE-W039, discussion]

Generating a pool of product concepts

Karan Girotra, Lennart Meincke, Christian Terwiesch and Karl T. Ulrich's 2023 working paper compared student product concepts with concepts generated using GPT-4. The task was to propose a physical product for college students that could retail for less than fifty dollars. The comparison used 200 student ideas from a 2021 course, 100 model-generated ideas and another 100 generated with examples of highly rated student concepts in the prompt. [CRE-W040, approach]

The researchers requested descriptions of forty to eighty words in batches of ten, using GPT-4-0314 with a temperature setting of 0.7. To continue generating a large pool within the available context, they had the model compress its previous ideas into shorter summaries. Those summaries accompanied subsequent

requests for ideas different from the preceding ones. The candidate pool was thus constructed through repeated generation with a representation of what had already been proposed. [CRE-W040, approach]

The example-seeded condition supplied seven highly rated student ideas from a separate set. It made examples part of the intervention while retaining the same product challenge. The paper compares distributions of evaluated concepts as well as their averages, reflecting the innovation-tournament setting in which a large pool is generated and then selected. [CRE-W040, approach and evaluation]

Survey respondents rated purchase intent for the descriptions. Average ratings were higher for both generated pools, and AI-generated ideas occupied 35 of the best-rated forty places in the combined collection. The human ideas received higher novelty ratings. These results describe judgments of proposed concepts, including the promising end of their distribution; the study did not observe the products being launched and purchased. [CRE-W040, results]

This account connects prompt construction, repeated generation, retention of prior candidates and evaluation of a pool. Its particular contribution differs from the story-writing studies: the generated descriptions are themselves the concepts being screened, rather than starting ideas that participants subsequently turn into finished stories. The procedures and assessed outcomes make those differences intelligible. [CRE-W040]

S70 Learning and practising creative methods

A creative procedure can be taught as a sequence of actions, practised in a particular setting and assessed through the work that follows. Training studies examine these connections at several levels. Their accounts range from the effects measured across many programmes to the placement of an exercise within a proposed course, and from a learner's personally meaningful discovery to professional achievement.

What creativity training improves

Ut Na Sio and Hugues Lortie-Forgues reviewed five decades of creativity-training research in 2024. Their principal meta-analysis included 169 studies, 844 effect sizes and 18,486 participants. The authors distinguished cognitive content, such as idea-generation skills, from content directed at motivation, emotion or personality, with a further category for combined programmes. Outcomes included divergent-thinking tasks, problem solving and creative products. [CRE-W041, method and results]

The unadjusted average training effect was 0.53 standard deviations. The review also found evidence consistent with publication bias and applied several corrections. PEESE, trim-and-fill, robust Bayesian meta-analysis and the top-ten-percent approach produced estimates around 0.29 to 0.32 standard deviations. The PET estimate was smaller and statistically uncertain. The authors report a positive average effect under several adjustments while showing how substantially its estimated size depends on accounting for the evidence available. [CRE-W041, publication-bias analyses]

The review examined the contents and duration of training as well as average effects. Cognitive and combined programmes compared favorably with the smaller non-cognitive category; the analysis did not establish a difference between cognitive and combined content. A positive association with duration in the simpler analysis weakened when other characteristics were considered together. These findings give more information than an undifferentiated claim that creativity can be taught. [CRE-W041, moderator analyses]

Sio and Lortie-Forgues also examined study quality and the concentration of outcomes in particular kinds of task. Their account treats measured improvement, the content of instruction and the quality of its evaluation as connected questions. A programme teaching generation skills and assessed with a divergent-thinking task supplies evidence about that accomplishment. Its result is most informative when the reader knows what was practised and what the assessment asked participants to do. [CRE-W041, discussion]

Arranging interventions for children

Patrícia Alves-Oliveira and colleagues' review of children's creativity interventions organized programmes at eight levels: cognitive processing skill, training technique, delivery medium, practice exercise, target, setting, administrator and dimension of creativity. A skill such as flexibility is distinct from a technique used to develop it, and that technique is distinct from its delivery through discussion, programmed instruction or a digital environment. The classification allows these parts of an intervention to be examined together. [CRE-W042, taxonomy]

The reviewed programmes include imaginative exercises, expressive activity, elaboration, cooperative learning and movement. A teacher can give sequential instructions, arrange discussion or ask learners to work together; an activity can use familiar materials or a computer or robotic partner. These are accounts of the programmes assembled in the review. Their arrangement makes the people conducting the

teaching and the circumstances of practice visible alongside the named creative technique. [CRE-W042, results and discussion]

The review distinguishes the dimension assessed from the activity used for teaching. In its classification, assessments concentrated on the person, commonly through tests: approximately 83 percent, compared with 9 percent directed at process and 8 percent at product. An intervention may involve producing something together while its evaluation chiefly measures individual performance on another task. This distinction helps explain what the accumulated evidence describes. [CRE-W042, assessment of creativity]

The authors' eight-level account also allows a method to be adapted without treating every adaptation as an entirely new technique. Delivery, exercises and teaching personnel can change while a programme continues to address a related skill. The review provides a way of describing these arrangements and their diversity across children's interventions. [CRE-W042]

Placing exercises within an online course

Viviane Kuntz and colleagues proposed an arrangement of creative techniques for massive open online courses in 2015. They related particular operations to occasions in a course: welcoming participants, introducing people and the platform, forming groups, establishing plans and working on content. The chapter's procedure is to ask what an activity could contribute at each point, rather than add an isolated creativity exercise to an otherwise unchanged course. [CRE-W043, pp. 217–227]

An opening combinatorial activity asks learners to imagine another form or structure for the subject. Unusual questions can change an introductory exchange. Exploring the positive, negative and interesting aspects of a platform gives participants an activity through which to consider the environment they will use. Profile-based games can help form groups. These proposals connect generating alternatives with the practical work of entering a shared course. Word-based recreation can explore another organization of the course content. [CRE-W043, table 7]

For planning, the chapter brings a desired future into comparison with the current situation and considers inviting someone unfamiliar with the subject to introduce another perspective. For work on content, a mind map can support individual interpretation, inversion can break down and rearrange a logical structure, and unusual analogy can connect material that initially seems unrelated. Each technique is assigned a teaching occasion and an intended contribution. [CRE-W043, tables 4 and 7]

The chapter concludes by proposing a pilot course to examine this arrangement. It is therefore an instructional design and rationale, with concrete adaptations available to inspect. Its historical contribution lies in the proposed connection between established creative operations and the structure of an online course. [CRE-W043, conclusion]

Different levels of creative accomplishment

Agustín Tristán López and Liliana Mendoza González's 2016 review of creativity taxonomies considers the range of accomplishments that education and assessment can address. Their account of the four-C model distinguishes personally meaningful discovery, everyday creativity, professional creativity and historically influential achievement. These levels give a place to work that is creative for the learner without requiring it to transform a field. [CRE-W044, four-C model]

Their examples make the distinctions concrete. A preschool child's drawing can embody a personally significant discovery. A child with some instruction in sculpture may produce an independent work worth displaying. An engineer's product can solve a client's problem and receive recognition within professional practice. A major work in music, art, technology or science can alter a domain much more widely. The

review uses these cases to explain why the expectations appropriate to one achievement need not fit another. [CRE-W044]

The authors also distinguish an unfolding process from a finished product. Observation and recorded sessions can help reconstruct how an activity developed; expert judgment can address qualities of completed work. Choreography, a technical device and a mathematical contribution pose different problems of description and evaluation. The review examines such differences while seeking more explicit accounts of creative qualities. [CRE-W044, assessment and creative-process taxonomies]

These distinctions give learning and practice several legitimate objects. A programme can develop a person's ways of approaching material, improve performance on a task or support a valued product. The studies and proposals in this section specify these accomplishments in different ways, providing historical accounts of what people have tried to teach and how they have recognized a result. [CRE-W041; CRE-W042; CRE-W043; CRE-W044]

S45 Praxeology, economy of knowledge and philosophies of action

Several different accounts of prepared action

The sources here examine effective action, economical representation, division of intellectual work, habit and institutions. Some describe how a favorable situation or prepared arrangement supports an action; others ask how people interpret conduct and its purposes.

Kotarbiński: a favorable position relative to a requirement

Sprawność i błąd includes a chapter on praxeology, the study of effective action. The chapter examines minimizing intervention by finding or creating a favorable situation. A plant requiring moisture is in a positive position with respect to that requirement when its site already supplies the moisture. In a dry site, the gardener must act to provide it. The favorable relation is to a specified property at a later time, not an assertion that the plant is generally independent of conditions. [W019, pp. 100–101]

The example is informative because it identifies the work supplied by the situation. The gardener's reduced action depends on a continuing environmental provision. A condition relevant to one demand can be favorable while other demands remain.

The selected chapter also distinguishes intervention from surveillance or monitoring, alongside potentialization, mechanization and organization as ways of improving action. Reducing the frequency of direct action is different from abandoning readiness to notice and respond. The proposed categories organize possible improvements. [W019, selected chapter, pp. 97–111]

Kotarbiński developed praxeology in *Traktat o dobrej robocie* (1955) and *Sprawność i błąd* (first published in 1956 and subsequently expanded). [W018; W019]

Babbage: distribute the intellectual work of calculation

Babbage's selected chapter on mental labor reports the organization of de Prony's calculation of mathematical tables. It distinguishes people who determine analytical methods, those who translate them into executable arrangements and those carrying out large quantities of elementary arithmetic. The report's interest is not simply that a calculation becomes shorter for everyone. The required knowledge is distributed across roles, with substantial repetitive labor assigned to people who need not know the complete analysis. [W086, chapter XIX, §§183 and 187–193]

The manufacturing analogy retains the intermediate role. An inventor or designer prepares the machinery; operative engineers construct it and understand the relevant processes; other workers use it. Babbage explicitly says that the work and performance of the machines still require supervision by the second group. Less knowledge demanded of an individual operator is not disappearance of the knowledge needed by the undertaking. [W086, §188, first-edition printed p. 158]

The difference-table explanation gives an actual computational mechanism. Successive squares have first differences increasing by a constant second difference. Once appropriate starting values and differences are known, repeated additions can extend the table. The prepared mathematical arrangement changes the later operations required; it does not make their correctness independent of the initial setup and execution. [W086, §§189–193]

Babbage connects the scheme to the possibility of a calculating engine replacing routine arithmetic. In the text that is a proposed development. His account of the French tables is historical author testimony. [W086]

Mach: economical thought represents more than it repeats

In the opening of his 1882 lecture, Mach starts from thought's limited resources in relation to the richness of experience. Scientific concepts and organized representations can let thought handle relationships without repeating every encounter. The relevant economy concerns representing experience and using thought to anticipate it, rather than possessing an exhaustive duplicate of the world. [W021, chapter XIII, selected opening]

The account treats scientific formulae and conceptual organization as intellectual instruments. Its pertinent claim is about what a representation can spare a thinker from separately retaining or rehearsing. The economy of a description and the operation described remain distinct.

Dewey: habit incorporates an environment

The discussion of habit in *Human Nature and Conduct* argues against locating habit wholly inside an isolated person. Walking implicates ground as well as legs, and speech requires air and other people as well as vocal organs. Dewey describes habits as organized skills involving materials, tools and surroundings. An acquired disposition operates in relation to a world that supports and resists it. [GL-W093, selected printed pp. 14–17 and 25–26]

This is more specific than saying that a repeated action becomes automatic. A craft can depend on external implements and arrangements, and moral or social conduct can likewise depend on institutions and other people. The selected account treats habit as a way of using and incorporating an environment. It does not make the environment an inert background or the person irrelevant to action.

The discussion of deliberation examines how habits and impulses enter inquiry when action is unsettled. Considering consequences and possible courses can revise conduct, rather than merely applying a ready-made end to neutral means. Established habits make action possible but can also require change. [GL-W093, selected printed pp. 172–184 and revised-habits passage]

Anderson's selected reconstruction of Dewey's moral philosophy adds the relation between means, ends and social ethics. Ends are examined through the activity and consequences of pursuing them, rather than assumed permanently fixed outside inquiry. The reader can therefore study a philosophy of organized conduct without turning every stable habit into a justified institution. [GL-W007]

Institutional and technical relief

Turner reads Gehlen's *Der Mensch* as an account of human beings whose conduct is only partly directed by instinct. Institutions and habits provide a relatively stable cultural setting, reducing repeated demands for motivation and control. Turner presents Entlastung as relief supplied through a taken-for-granted background that permits more focused foreground activity. [UX-W032, introduction, printed pp. 9–11] [UX-W033, through Turner]

The same reconstruction connects Gehlen's account to a conservative criticism of modernity: persistent reflection can destabilize the institutional background on which action relies. This is a controversial philosophical position as presented by Turner. Stability of a background does not settle who benefits from it or bears its costs.

In *Die Seele im technischen Zeitalter* (1957), Gehlen distinguishes tools that substitute for a bodily capacity, amplify it or relieve people of exerting it. A microscope amplifies sight. A wheeled wagon relieves people of dragging the load themselves. He relates these functions to the ways humans change their surroundings to make action possible. [UX-W039, ch. I, §1]

Spinner gives relief a different setting in his account of scientific institutions. He argues that the separation of inquiry from practice gives researchers room to pursue questions without an immediate obligation to act on the answers or assume responsibility for their use. In his account this separation becomes harder to sustain as research moves into technologies that have to be built, financed and operated. [UX-W034, §§III, V]

Historical time: a persistent structure is not necessarily beneficial

The selected passages of Braudel's *La longue durée* direct attention to structures and temporal scales that differ from a short event sequence. A slowly changing arrangement can condition what is possible over many events, without being immediately visible in an event-centered narrative. The relevant contribution is a perspective on duration and constraint. [GL-W051, printed p. 726 and discussion of structure]

Daniel Little's *Philosophy of History* entry, first published in 2007 and revised in 2025, examines causal mechanisms, contingency and the selectivity of historical explanation as methodological questions. A broad pattern needs evidence about particular processes and contexts. [GL-W050]

Daoism through Hansen: guidance and normative autonomy

Hansen's selected interpretive account distinguishes strands associated with Laozi and Zhuangzi and examines natural possibilities, guidance and normative autonomy. Its relevance is a philosophical question about how action relates to an order or path without every step being imposed through an explicit command. [GL-W098]

Hansen’s discussion of wuwei concerns guidance and normative autonomy. Mechanical passivity and the absence of effort describe different questions from those philosophical distinctions.

Effective action, representation and institutions

The sources provide distinct information about favorable conditions, delegated calculation, economical representation, environmentally supported habits, institutional relief and philosophical action.

S49 Stigmergy, emergent patterns and distributed collective action

This section reports selected explanations of coordination through local interactions, a modified environment, and expectations about others. It also includes an important adverse case: local choices can produce an aggregate pattern unlike the mixture the participants would tolerate. “Emergent” identifies a relation between interactions and a pattern.

Grassé: the product of work stimulates subsequent work

In the selected passages of his 1959 termite article, Grassé describes a mechanism in which construction changes a material product, and that changed product becomes a stimulus for subsequent activity by the same or other workers. The environment is consequently involved in the sequence of coordination. The next act need not be directed by a central representation of the completed nest or an instruction exchanged between each pair of workers. A worker can encounter a physically altered situation and act in relation to it. [W031, printed pp. 62–63, 71–75]

The selected §V also supplies boundaries. Grassé does not treat this construction mechanism as an exhaustive explanation of every social-insect behavior, or eliminate all possible roles for communication. [W031]

Heylighen: a broader trace-mediated proposal

Heylighen’s selected definitions and components generalize stigmergy as indirect coordination mediated by traces left in an environment. An activity changes the medium; a later activity responds to the change. This description can cover more than insect building and need not require every participant to represent the whole collective process. It also makes the medium operationally important: the relevant change must persist sufficiently, be perceivable, and support a response appropriate to the activity. An uninterpretable mark is not a coordination mechanism merely because it is a trace. [W073, §§2–3]

The example and conclusion present Heylighen's broader explanatory proposal. The account acknowledges imperfect use of the available workforce. A trace can guide selection without ensuring that work is evenly distributed, that obsolete information disappears in time, or that all required tasks are covered. [W073]

Multi-robot coordination retains an engineered organization

Gil, Aguilar, Rivas and Dapena’s article proposes an architecture for emergent coordination in multi-robot systems. In the selected architecture and communication passages, robots do not simply follow unsupported local rules. The design retains sensing, knowledge management, collective memory, and both direct and indirect communication. These layers provide the conditions under which local acts can become relevant to a collective task. The account therefore makes the supporting organization visible within a proposal described as emergent. [W067, §§2–4, printed pp. 76–79]

Schelling: tolerable local mixtures can produce segregation

Schelling’s selected opening and model passages distinguish individual neighborhood rules from the collective pattern produced by movements under those rules. A person may tolerate some unlike neighbors without requiring the final population to be segregated. Yet moves made in response to locally unsatisfactory mixtures can change other neighborhoods, trigger further moves, and produce substantial separation. An aggregate pattern can therefore be different from the pattern one would infer by treating

each participant's preference as a direct preference for the final collective outcome. [W070, printed pp. 147, 158–161]

The result depends on the model's population ratios, neighborhood definition, thresholds and movement rules. Schelling explicitly limits quantitative generalization from an artificial board. Even describing the resulting arrangement as undesirable requires a stated social goal; the model does not derive that goal from its updating rules. This is relevant adverse material for a study of structural arrangements: local responses and successful individual adjustment need not establish an acceptable collective outcome. [W070]

Distributed intelligence as a research landscape

JST CRDS's 2026 landscape section on autonomous distributed intelligence maps questions across material, bodily and collective scales, including design theory, prediction, oversight and implementation. [W100, S2.03, selected pp. 1–6]

Double and multi-contingency: order between two agents may not scale

Dittrich, Kron and Banzhaf reconstruct a problem associated with Parsons and Luhmann: agents form expectations about how others will respond, including expectations about others' expectations. The article contrasts shared-culture explanations with the development of expectations through interaction history. The selected abstract and paragraphs 1.1–2.5 introduce their own simulation framework. [SG-W042]

In the dyadic model, two agents alternately display one of a finite set of messages. There is no prior semantic relation among the messages. Each agent's selection is influenced by wanting to meet the other's expectations and wanting the other's response to be predictable. Those two motives can conflict. The model thus specifies a selection problem rather than assuming that "communication" automatically produces a common understanding. Memory and information about the other's response are part of the arrangement that can produce regularities. [SG-W042, §§2.1–2.5]

The authors report that order achieved between two agents often fails to scale to a larger setting and that observation and the proliferation of information matter. That reported finding belongs to their model. The contribution is a specified obstacle to generalization: a mechanism that produces a stable relation in a pair need not organize many participants under changed information conditions. [SG-W042]

Leydesdorff: present decisions using a representation of a future

Leydesdorff's 2005 article relates Rosen's account of anticipation to Dubois's incursive formulations and to social meaning. In the selected introduction, a system can use an internal representation to consider alternatives and alter present behavior. This differs from describing every adjustment as a reaction only to a past disturbance. The selected derivation then proposes a mathematical treatment and distinguishes an observing system from the system under observation. The quality of observation and the difference between interacting observers and interacting observations become explicit modeling choices. [SG-W055, introductory definitions]

The proposal retains the work of constructing, evaluating and updating the model, selecting among possibilities and making a decision. For Leydesdorff's later double-contingency exposition, see S47. [SG-W055; WME-W021]

Environmental traces and expectations in collective action

The material distinguishes a physical stimulus, a readable environmental trace, an engineered memory and communication architecture, a neighborhood updating rule, an institutional research agenda, and formal models of expectations. These descriptions make different commitments.

The historical accounts

The historical accounts examine capabilities produced by physical structures, computational representations, learned practices and institutions. They describe how these capabilities developed, how they were investigated and where their explanations remain disputed.

The source accounts differ in their quantities, mechanisms, aims and evidence. The reading routes and indexes support comparison among them.

Source specific terminology guide

Variety in Ashby counts distinguishable possibilities in a specified set, or uses the base-two logarithm of that count. Actual state, possible states and observed sequences are different objects. See S03.

Selection in Tektology is a deliberately generalized account of persistence and elimination under conditions, not necessarily biological reproduction and heredity. Egression concerns concentrated influence; degression concerns fixation and protection of more plastic activity. See S17.

Autonomy in the reconstructed viable-system model remains related to coordination, resources, audit, intelligence, policy and communication. Relational autonomy in the selected philosophical account asks a different normative question. See S22 and S48.

Invariant confluence concerns merges of valid reachable states with a common ancestor under specified transactions and invariants. Idempotence says repeating a merge adds no further change; it does not establish arbitrary application validity. See S10 and S11.

Lineage in lineage-driven fault injection describes dependencies explaining a known successful outcome, which guide candidate disruptions that must still be checked by execution. See S30.

Concretization in Simondon concerns development toward compatible functional relations within a technical object and its milieu. See S39.

Closure refers to different specified relations in the biological and chemical sources. These relations coexist with exchanges of matter and energy. See S26 and S40.

Resilience in Holling's selected account differs from return-rate stability. In the ecological-information articles an overhead measure receives a related interpretation that requires separate empirical support. See S43 and S47.

Free energy denotes different quantities in the thermodynamic lecture and variational probabilistic account. See S46.

Burden in the administrative account includes learning, psychological and compliance costs in citizen–state interactions. See S48.

Source guide

This guide locates each work in the historical accounts. Section identifiers link to discussions; citation keys link to the shared references.

[AB-W001] Absorptive Capacity: A New Perspective on Learning and Innovation. See S53, S61.

[AB-W002] Asset Stock Accumulation and Sustainability of Competitive Advantage. See S53.

[AB-W003] Problems of Industrialisation of Eastern and South-Eastern Europe. See S53.

[AB-W004] The Strategy of Economic Development. See S53.

[AB-W005] Managing Resource Revenues in Developing Economies. See S53.

[AB-W007] Capacité d'absorption : revue de littérature, opérationnalisation et exploration [Absorptive capacity: literature review, operationalization and exploration]. See S53.

[AB-W008] Capacidade de Absorção, Aprendizagem Organizacional e Mecanismos de Integração Social [Absorptive capacity, organizational learning and social integration mechanisms]. See S53.

[AB-W009] 对外技术引进与中国本土企业自主创新 [Foreign Technology Introduction and Local Firms' Innovation in China]. See S53.

[AB-W010] Проблема формирования национальной инновационной системы [The Problem of Creating a National Innovation System]. See S53.

[AFC-W001] Mathematical Definition, Mapping, and Detection of (Anti)Fragility. See S32.

[AFC-W002] Antifragile: Things That Gain from Disorder. See S32.

[AFC-W003] Fault Injection Techniques and Tools. See S30.

[AFC-W004] Principles of Chaos Engineering. See S31.

[AFC-W005] Architecture strategies for reliability testing. See S31.

[AFC-W006] The concept of antifragility and its implications for the practice of risk analysis. See S32.

[AFC-W007] Principles of Antifragile Software. See S31.

[AFC-W008] Antifragility in complex dynamical systems. See S32.

[AFC-W009] Antifragility = Elasticity + Resilience + Machine Learning: Models and Algorithms for Open System Fidelity. See S32.

[AFC-W010] Lineage-driven Fault Injection. See S30.

[AFC-W011] Chaos Engineering: A Multi-Vocal Literature Review. See S31.

[AFC-W012] Adaptive Dual Control. See S33.

[AFC-W013] Antifragility and Tinkering in Biology (and in Business): Flexibility Provides an Efficient Epigenetic Way to Manage Risk. See S32.

[AFC-W014] 混沌工程实践指南（2021 年） [Chaos Engineering Practice Guide (2021)]. See S31.

[AFC-W015] The Hard Work of Failure Analysis. See S31, S51.

[AFC-W016] Bayesian Experimental Design: A Review. See S33.

[AFC-W017] Strategic Investment: Real Options and Games. See S34, S61.

[AFC-W018] Applying Systems and Safety Engineering Principles for Antifragility. See S31.

[AFC-W019] The Antifragile Organization. See S31.

[AFC-W020] 火山引擎混沌工程之云原生场景实现 [Implementing Volcano Engine Chaos Engineering in Cloud-Native Scenarios]. See S31.

[AFC-W021] 确定性运维白皮书：稳定可靠篇 [Deterministic Operations and Maintenance White Paper: Stability and Reliability]. See S31.

[BC-W001] Systems Biology: A Brief Overview. See S50.

[BC-W002] Network motifs: theory and experimental approaches. See S50.

[BC-W003] Construction of a genetic toggle switch in Escherichia coli. See S50.

[BC-W004] A synthetic oscillatory network of transcriptional regulators. See S50.

[BC-W005] What is flux balance analysis?. See S50.

[BC-W006] Quantifying cellular capacity identifies gene expression designs with reduced burden. See S50.

[BC-W007] A Whole-Cell Computational Model Predicts Phenotype from Genotype. See S50.

[BC-W008] Large Language Models Meet Virtual Cell: A Survey. See S50.

[BC-W009] Flow Matching Meets Biology and Life Science: A Survey. See S50.

[BC-W010] Beyond Network Topology: Biological Evidence Integration and Reproducible Benchmarking for Protein Complex Detection. See S50.

[BC-W011] Automated Reasoning in Systems Biology: a Necessity for Precision Medicine. See S50.

[CG-W001] Mixed Precision Iterative Refinement Techniques for the Solution of Dense Linear Systems. See S55.

[CG-W002] Accelerating the Solution of Linear Systems by Iterative Refinement in Three Precisions. See S55.

[CG-W003] Precimonious: Tuning Assistant for Floating-Point Precision. See S55.

[CG-W004] Rigorous Floating-Point Mixed-Precision Tuning. See S55.

[CG-W005] Green: A Framework for Supporting Energy-Conscious Programming using Controlled Approximation. See S55.

[CG-W006] EnerJ: Approximate Data Types for Safe and General Low-Power Computation. See S55.

[CG-W007] Conceptual pacts and lexical choice in conversation. See S56.

[CG-W009] Understanding by addressees and overhearers. See S56.

[CG-W011] Investigating the influence of local and personal common ground on memory for conversation using an online referential communication task. See S56.

[CG-W013] Algorithms for Scheduling Imprecise Computations. See S57.

[CG-W014] Brownout: Building More Robust Cloud Applications. See S57.

[CG-W015] Defcon: Preventing Overload with Graceful Feature Degradation. See S57.

[CG-W016] How in the World Did We Ever Get into That Mode? Mode Error and Awareness in Supervisory Control. See S57.

[CG-W017] Exploring internal stickiness: Impediments to the transfer of best practice within the firm. See S58.

[CG-W018] Replication as Strategy. See S58.

[CG-W019] The Effect of Organizational Context on Individual Performance. See S58.

[CG-W021] Verification of Forecasts Expressed in Terms of Probability. See S31.

[CG-W023] The preregistration revolution. See S31.

[CG-W024] Preregistration in practice: A comparison of preregistered and non-preregistered studies in psychology. See S31.

[CG-W025] Referring as a collaborative process. See S56.

[CG-W026] 共同図形配置課題における対話の共通基盤構築過程の分析 [Analyzing common-ground construction in dialogue during a collaborative shape-arrangement task]. See S56.

[CG-W027] 共通基盤の構築における名付けの有用性の分析 [An analysis of the usefulness of naming in constructing common ground]. See S56.

[CG-W028] Transactive Memory: A Contemporary Analysis of the Group Mind. See S56.

[CM-W001] As We May Think. See S60.

[CM-W002] Augmenting Human Intellect: A Conceptual Framework. See S60.

[CM-W003] A Truth Maintenance System. See S60, S105.

[CM-W004] gIBIS: A Hypertext Tool for Exploratory Policy Discussion. See S60.

[CM-W005] Augmenting the Organizational Memory: A Field Study of Answer Garden. See S60.

[CM-W006] Définition d'un modèle de mémoire de projet [Definition of a Project Memory Model]. See S60.

[CM-W007] 情報システムの開発及び運用における知識活用手法に関する研究 [Research on Methods for Using Knowledge in the Development and Operation of Information Systems]. See S60.

[CRE-W001] Deconstruction of Idea Generation Methods Into a Framework of Creativity Mechanisms. See S63.

[CRE-W002] Kreativitätstechniken [Creativity Techniques]. See S63.

[CRE-W003] 系统化创新方法研究：理论进展与实践效果评价 [Systematic Innovation Methods: Theoretical Progress and Evaluation of Practical Effects]. See S63.

[CRE-W004] 創造技法の分類とその種類 [Classification and Types of Creativity Techniques]. See S63.

[CRE-W005] Préciser le sens du mot « créativité » [Clarifying the Meaning of Creativity]. See S63.

[CRE-W006] Creativity techniques: a systematic literature review. See S63.

[CRE-W007] Методы технического творчества [Methods of Technical Creativity]. See S64.

[CRE-W008] L'invention mathématique [Mathematical Invention], in Science et méthode. See S64.

[CRE-W009] Analogical Problem Solving. See S64.

[CRE-W010] The in vivo/in vitro approach to cognition: the case of analogy. See S64.

[CRE-W011] What Have Psychologists (And Others) Discovered About the Process of Scientific Discovery?. See S64.

[CRE-W012] BACON.5: The Discovery of Conservation Laws. See S64.

[CRE-W013] La créativité : approches et méthodes en psychologie et en ergonomie [Creativity: Approaches and Methods in Psychology and Ergonomics]. See S64.

[CRE-W014] Lettre à Paul Demeny du 15 mai 1871 [Letter to Paul Demeny, 15 May 1871]. See S65.

[CRE-W015] Manifeste du surréalisme [Manifesto of Surrealism]. See S65.

[CRE-W016] S+7 [S+7 constraint]. See S65.

[CRE-W017] Grammatica della fantasia: Introduzione all'arte di inventare storie [Grammar of Fantasy: An Introduction to the Art of Inventing Stories]. See S65.

[CRE-W018] Escritura creativa. Aplicación de las técnicas de Gianni Rodari [Creative Writing: Applying Gianni Rodari's Techniques]. See S65.

[CRE-W019] Técnicas y procedimientos. Un acercamiento al proceso creativo en el medio tecnográfico [Techniques and Procedures: An Approach to the Creative Process in the Technographic Medium]. See S65.

[CRE-W020] Improvisation for the Theater: A Handbook of Teaching and Directing Techniques. See S66.

[CRE-W021] Processing Choreography: Thinking with William Forsythe's Duo. See S66.

[CRE-W022] Silence: Lectures and Writings. See S66.

[CRE-W023] Theater Games and Stanislavski's System: A Study on the Origins of Viola Spolin's Theatrical Education. See S66.

[CRE-W024] Mapping visions of improvisation pedagogy in music education research. See S66.

[CRE-W025] Inspired by Distraction: Mind Wandering Facilitates Creative Incubation. See S67.

[CRE-W026] Does Incubation Enhance Problem Solving? A Meta-Analytic Review. See S67.

[CRE-W027] Creativity—the unconscious foundations of the incubation period. See S67.

[CRE-W028] Give Your Ideas Some Legs: The Positive Effect of Walking on Creative Thinking. See S67.

[CRE-W029] KJ 法クイックマニュアル [KJ Method Quick Manual]. See S68.

[CRE-W030] KJ 法の原点と核心を語る：川喜田二郎さんインタビュー [Emergence and Essence of the KJ Method: An Interview with Jiro Kawakita]. See S68.

[CRE-W031] Understanding Creativity Methods in Design. See S68.

[CRE-W032] Toolkits, cards and games – a review of analogue tools for collaborative ideation. See S68.

[CRE-W033] Idea Generation Techniques among Creative Professionals. See S68.

[CRE-W034] A web como espaço de criatividade e colaboração para o desenvolvimento de sites acessíveis [The Web as a Space for Creativity and Collaboration in Developing Accessible Websites]. See S68.

[CRE-W035] Creativity Support Tools: Accelerating Discovery and Innovation. See S69.

[CRE-W036] The Effects of Group Size, Memory Instruction, and Session Length on the Creative Performance in Electronic Brainstorming Groups. See S69.

[CRE-W037] Designing Creative AI Partners with COFI: A Framework for Modeling Interaction in Human-AI Co-Creative Systems. See S69.

[CRE-W038] Generative AI enhances individual creativity but reduces the collective diversity of novel content. See S69.

[CRE-W039] Homogenization Effects of Large Language Models on Human Creative Ideation. See S69.

[CRE-W040] Ideas Are Dimes a Dozen: Large Language Models for Idea Generation in Innovation. See S69.

[CRE-W041] The impact of creativity training on creative performance: A meta-analytic review and critical evaluation of 5 decades of creativity training studies. See S70.

[CRE-W042] Creativity Landscapes: Systematic Review Spanning 70 Years of Creativity Interventions for Children. See S70.

[CRE-W043] Técnicas Criativas Aplicáveis em Cursos Online Abertos e Massivos (MOOCS) [Creative Techniques Applicable to MOOCs]. See S70.

[CRE-W044] Taxonomías sobre creatividad [Taxonomies of Creativity]. See S70.

[CRE-W045] Creativity and Innovation Under Constraints: A Cross-Disciplinary Integrative Review. See S63.

[CRE-W046] Surrealist Collage in Text and Image: Dissecting the Exquisite Corpse. See S65.

[DC-W01] 孫子兵法 [The Art of War]. See S93.

[DC-W02] Sunzi versus Xunzi: Two Views of Deception and Indirection. See S93.

[DC-W03] Strategemata [Stratagems]. See S93.

[DC-W04] Il Principe [The Prince]. See S93.

[DC-W05] Marcel Détienne et Jean-Pierre Vernant, Les ruses de l'intelligence, la métis chez les Grecs [Review of The ruses of intelligence]. See S93.

[DC-W06] Los consejos de al-Harawi sobre estratagemas de guerra: un manual político-militar árabe del siglo XII [Al-Harawi's counsels on stratagems of war]. See S93.

[DC-W07] Strategic Deception and Counterdeception: A Cognitive Process Approach. See S93.

[DC-W08] Рефлексивное управление — технология принятия манипулятивных решений [Reflexive control: a technology of manipulative decision-making]. See S93.

[DC-W09] Controle reflexivo russo: teoria militar e aplicações [Russian reflexive control: military theory and applications]. See S93.

[DC-W10] The Intelligence Officer's Bookshelf: review of STRATAGEM: Deception and Surprise in War. See S93.

[DC-W11] Exploring Organizational Deception: Organizational Contexts, Social Relations and Types of Lying. See S93.

[DC-W12] More Lies than Meet the Eyes: Organizational Realities and Deceptions in Nonprofit Organizations. See S93.

[DC-W13] Corrupción como proceso organizacional: comprendiendo la lógica de la desnormalización de la corrupción [Corruption as an organizational process]. See S93.

[DC-W14] The Emotion Deception Model: A Review of Deception in Negotiation and the Role of Emotion in Deception. See S93.

[DC-W15] Artful Paltering: The Risks and Rewards of Using Truthful Statements to Mislead Others. See S93.

[DC-W16] A psychologically-based taxonomy of misdirection. See S93.

[DC-W17] Täuschungshandlungen im Sport: Eine experimentelle Untersuchung zur Wirkung der Häufigkeitsverteilung auf die Blicktäuschung im Basketball [Deceptive actions in sport: frequency and the basketball head fake]. See S93.

[DC-W18] Social Engineering Attacks: A Survey. See S93.

[DC-W19] Protocol Additional to the Geneva Conventions of 12 August 1949, relating to the Protection of Victims of International Armed Conflicts (Protocol I). See S93.

[DC-W20] 今日の海上における欺瞞規則：海戦法規上の背信行為と奇計に関する一考察 [Rules of deception at sea today: perfidy and ruses in naval warfare]. See S93.

[DC-W21] Introduction, Ruse et magie de l'Antiquité à nos jours [Ruse and magic from antiquity to the present]. See S93.

[DP-W001] Coordinating Economic Activity: An Example. See S06.

[EF-W001] A Scapegoat Model of Exchange-Rate Fluctuations. See S59.

[EF-W002] Remarks delivered at the Global Economic Symposium. See S59.

[EF-W003] Culture, Circles, and Commercials: Publicity, Common Knowledge, and Social Coordination. See S59.

[EF-W004] Herd Behavior and Investment. See S59.

[EF-W005] A Theory of Fads, Fashion, Custom, and Cultural Change as Informational Cascades. See S59.

[EF-W006] L'auto-référence dans la théorie keynésienne de la spéculation [Self-reference in the Keynesian theory of speculation]. See S59.

[EF-W007] Financial Markets. See S59.

[EF-W008] Social Value of Public Information. See S59.

[EF-W009] Herd Behavior and Investment: Comment. See S59.

[EF-W010] Herd Behavior and Investment: Reply. See S59.

[GL-W001] Musical Ensemble Performance: Representing Self, Other, and Joint Action Outcomes. See S41.

[GL-W002] Joint Action: Mental Representations, Shared Information and General Mechanisms for Coordinating with Others. See S41.

[GL-W003] Guidelines for Respecting Cultural Knowledge. See S41.

[GL-W004] Ethnoecological perspectives on environmental stewardship: Tenets and basis of reciprocity in Gitxsan and nłeʔkepmx (Nlaka'pamux) Territories. See S41.

[GL-W005] Servicescapes: The Impact of Physical Surroundings on Customers and Employees. See S41.

[GL-W006] Gender, households and sustainability: Disentangling and re-entangling with the help of 'work' and 'care'. See S48.

[GL-W007] Dewey's Moral Philosophy. See S45.

[GL-W008] "礼义化性"与"习俗化性"——论荀子思想中礼与俗的作用 ["Transforming Nature Through Ritual and Rightness" and "Transforming Nature Through Custom": On the Roles of Ritual and Custom in Xunzi's Thought]. See S41.

[GL-W009] Supramolecular Chemistry—Scope and Perspectives: Molecules—Supermolecules—Molecular Devices. See S01.

[GL-W010] The 10 Elements of Agroecology framework. See S43.

[GL-W011] The socio-economic performance of agroecology. A review. See S43.

[GL-W012] Human-thing entanglement: towards an integrated archaeological perspective. See S39.

[GL-W013] Text and reperformance: do you really need a text for your reperformance?. See S41.

[GL-W014] Semantics–pragmatics interactions. See S42.

[GL-W015] A manifesto for the equifinality thesis. See S43.

[GL-W016] The role of tutoring in problem solving. See S41.

[GL-W017] Application of Tuned Mass Dampers for Structural Vibration Control: A State-of-the-art Review. See S01.

[GL-W018] Administrative Burden: Learning, Psychological, and Compliance Costs in Citizen-State Interactions. See S48.

[GL-W019] Administrative Burden in Citizen–State Interactions: A Systematic Literature Review. See S48.

[GL-W020] Information as thing. See S42.

[GL-W021] The Uses of Levels. See S08.

[GL-W022] Sharing in Action: The Systemic Concept of the Environment in Aleksandr Bogdanov. See S17.

[GL-W023] Records in Contexts–Conceptual Model, Version 1.0. See S42.

[GL-W024] Notes on Blackwell Dominance. See S33.

[GL-W025] The Renormalization Group and Critical Phenomena. See S46.

[GL-W026] Biological organisation as closure of constraints. See S40.

[GL-W027] Epistemic Action Increases With Skill. See S41.

[GL-W028] Allostasis: a model of predictive regulation. See S05.

[GL-W029] Clarifying the roles of homeostasis and allostasis in physiological regulation. See S05.

[GL-W030] End-to-End Arguments in System Design. See S12.

[GL-W031] Economics of Convention and its Perspective on Knowledge and Institutions. See S15.

[GL-W032] Rules and implements: investment in forms. See S15.

[GL-W033] Organized Complexity: Conventions of Coordination and the Composition of Economic Arrangements. See S15.

[GL-W034] Understanding health through social practices: performance and materiality in everyday life. See S41.

[GL-W035] Review of The Dynamics of Social Practice: Everyday Life and How It Changes. See S41.

[GL-W036] AJP-3.2 Allied Joint Doctrine for Land Operations, Edition B Version 1. See S16.

[GL-W037] Asceticism and Ritual Self-Practices. See S41.

[GL-W038] Vital materiality and its constitution of knowing in craft practice. See S41.

[GL-W039] Planning for the Many Benefits of Nature-Based Recreation. See S43.

[GL-W040] Feminist Perspectives on Autonomy. See S48.

[GL-W041] Assistive technology. See S41.

[GL-W042] Ingredients of 'rituals' and their cognitive underpinnings. See S41.

[GL-W043] Rules Versus Standards: An Economic Analysis. See S15.

[GL-W044] AU Section 319: Consideration of Internal Control in a Financial Statement Audit. See S42.

[GL-W045] Zero Trust Architecture. See S42.

[GL-W046] National Incident Management System, Third Edition. See S16.

[GL-W047] Journalism, 'Fake News' and Disinformation: Handbook for Journalism Education and Training. See S42.

[GL-W048] Evaluating and Improving Internal Control in Organizations. See S42.

[GL-W049] Alienation. See S48.

[GL-W050] Philosophy of History. See S45.

[GL-W051] Histoire et Sciences sociales: La longue durée [History and the Social Sciences: The Long Duration]. See S45.

[GL-W052] Learning through work: workplace affordances and individual engagement. See S41.

[GL-W053] Who's the Boss? Defining the Civil-Military Relationship in the Twenty-First Century. See S16.

[GL-W054] Adaptive Silviculture for Climate Change (ASCC). See S43.

[GL-W055] Precautionary Approach to Fishery Management. See S43.

[GL-W056] Application of compartmentalisation, Terrestrial Animal Health Code chapter 4.4. See S43.

[GL-W057] Animal welfare and dairy cattle production systems, Terrestrial Animal Health Code chapter 7.11. See S43.

[GL-W058] Clinical pathways for secondary care and the effects on professional practice, patient outcomes, length of stay and hospital costs. See S29.

[GL-W059] Hierarchy of Controls. See S02.

[GL-W060] Engineering Customized Cell Sensing and Response Behaviors Using Synthetic Notch Receptors. See S40, S50.

[GL-W061] What is TCM?. See S48.

[GL-W062] Traditional Chinese Medicine: What You Need To Know. See S48.

[GL-W063] Institutional and Organizational Best Practices: Abstract developed by the Human Factors Subcommittee. See S29.

[GL-W064] NHS dentistry: care pathways guidance. See S29.

[GL-W065] Flexible mechanical metamaterials. See S01.

[GL-W066] Generalization. See S42.

[GL-W067] Resilience and Stability of Ecological Systems. See S43.

[GL-W068] Diversity is essential for resilience to thrive. See S43.

[GL-W069] Port-Hamiltonian nonlinear systems. See S04.

[GL-W070] Port-Hamiltonian Systems Theory: An Introductory Overview. See S04.

[GL-W071] Learning to be adaptive as a distributed process across the coach–athlete system: situating the coach in the constraints-led approach. See S41.

[GL-W072] Brian Deer Classification System. See S42.

[GL-W073] A survey on synthesis of compliant constant force/torque mechanisms. See S01.

[GL-W074] Structure, Energy, Synergy, Time—The Fundamentals of Process Intensification. See S01.

[GL-W075] Applying Human Factors and Usability Engineering to Medical Devices. See S02.

[GL-W076] Constructed Wetlands Treatment of Municipal Wastewaters. See S43.

[GL-W077] Sensor-Based Ore Sorting Technology in Mining—Past, Present and Future. See S43.

[GL-W078] Passive Treatment Systems for Acid Mine Drainage. See S43.

[GL-W079] Basic aspects of food preservation by hurdle technology. See S43.

[GL-W080] Toyota Production System. See S02.

[GL-W081] Advanced reactors: Safety and environmental considerations. See S02.

[GL-W082] E-navigation. See S38.

[GL-W083] Operational Use of Flight Path Management Systems: Final Report of the PARC/CAST Flight Deck Automation Working Group. See S12.

[GL-W084] Controlled traffic farming to reduce compaction (E334A). See S43.

[GL-W085] Guidelines for implementation of forest management operations. See S43.

[GL-W086] Consolidated bioprocessing of cellulosic biomass: an update. See S43.

[GL-W087] Proof Theory. See S42.

[GL-W088] Life Cycles of Sun-like and Massive Stars. See S01.

[GL-W089] Climate Change 2021: The Physical Science Basis, Chapter 1, §1.5.3.2: Model Tuning and Adjustment. See S35.

[GL-W090] Flood and Coastal Storm Risk Management. See S43.

[GL-W091] Theoretical principles for biology: organization. See S40.

[GL-W092] The Structure of the Ordinary: Form and Control in the Built Environment. See S08.

[GL-W093] Human Nature and Conduct: An Introduction to Social Psychology. See S45.

[GL-W094] Rules and standards: an institutional analysis of property. See S15.

[GL-W095] Rethinking the Design of the Internet: End to End Arguments vs. the Brave New World. See S12.

[GL-W096] Where Are We Heading? The Evolution of Humans and Things. See S39.

[GL-W097] Очерки всеобщей организационной науки [Essays on Universal Organizational Science]. See S17.

[GL-W098] Daoism. See S45.

[GL-W099] What is Conservation?. See S39.

[GL-W100] Caring for paintings. See S39.

[GL-W101] Metrological Traceability: Frequently Asked Questions and NIST Policy. See S42.

[GS-W01] Building a Practically Useful Theory of Goal Setting and Task Motivation: A 35-Year Odyssey. See S98.

[GS-W02] 目標設定理論の概念：目標設定理論（第 1 報）. See S98.

[GS-W03] 目標設定に関わる運用方略の効果性に関する研究の概括. See S98.

[GS-W04] Goal Striving, Need Satisfaction, and Longitudinal Well-Being: The Self-Concordance Model. See S98.

[GS-W05] Das Selbstkonkordanz-Modell. See S98.

[GS-W06] Self-Regulation of Goal Setting: Turning Free Fantasies About the Future Into Binding Goals. See S98.

[GS-W07] Implementation Intentions. See S98.

[GS-W08] A Meta-Analysis of the Effects of Mental Contrasting With Implementation Intentions on Goal Attainment. See S98.

[GS-W09] Adaptive Self-Regulation of Unattainable Goals: Goal Disengagement, Goal Reengagement, and Subjective Well-Being. See S98.

[GS-W10] Структура и нарушения саморегуляции деятельности: точка роста культурно-исторической деятельностной психологии. See S98.

[GS-W11] There's a S.M.A.R.T. Way to Write Management's Goals and Objectives. See S98.

[GS-W12] Goals Gone Wild: The Systematic Side Effects of Over-Prescribing Goal Setting. See S98.

[GS-W13] Has Goal Setting Gone Wild, or Have Its Attackers Abandoned Good Scholarship?. See S98.

[GS-W14] On Good Scholarship, Goal Setting, and Scholars Gone Wild. See S98.

[GS-W15] 目标设置对非伦理行为的影响：边界条件与理论基础. See S98.

[GS-W16] Revisión de algunas variables relevantes en el establecimiento de metas deportivas. See S98.

[GS-W17] Estabelecimento de metas como estratégia motivacional. Artigo de revisão. See S98.

[HCT-W001] Interview, January 16, 2007. See S86.

[HCT-W002] The Heilmeier Catechism. See S86.

[LC-W002] Discovery-Driven Planning. See S62.

[LC-W003] The Second Toyota Paradox: How Delaying Decisions Can Make Better Cars Faster. See S62.

[LC-W004] A Real Options Logic for Initiating Technology Positioning Investments. See S62.

[LC-W005] Using Real Options Discipline for Highly Uncertain Technology Investments. See S62.

[LC-W006] The Toyota Way in Services: The Case of Lean Product Development. See S62.

[LC-W007] 選好度を有する範囲概念に基づく多目的同時満足化設計（構造と制御の同時設計への適用）[Multi-Objective Simultaneous Satisficing Design Based on a Set-Based Concept with Preference: Application to Simultaneous Structural and Control Design]. See S62.

[LC-W008] Pilot Early, Commit Late: A Real-Options Model of Enterprise AI Adoption under Rapid Technological Progress. See S62.

[LC-W009] Stratégies d'innovation et organisation de la conception dans les entreprises amont. Enseignements d'une recherche chez Usinor [Innovation Strategies and the Organization of Design in Upstream Firms: Lessons from Research at Usinor]. See S62.

[LC-W010] Steuerung von Flexibilität. Management von Softwareinvestitionen mit Realoptionen [Managing Flexibility: Managing Software Investments with Real Options]. See S62.

[LC-W011] Reducing Rework by Applying Set-Based Practices Early in the Systems Engineering Process. See S62.

[LP-W001] The Role of Deliberate Practice in the Acquisition of Expert Performance. See S71.

[LP-W002] Deliberate Practice and Performance in Music, Games, Sports, Education, and Professions: A Meta-Analysis. See S71.

[LP-W003] Deliberate Practice and Proposed Limits on the Effects of Practice on the Acquisition of Expert Performance: Why the Original Definition Matters and Recommendations for Future Research. See S71.

[LP-W004] Toward a cumulative science of expertise: commentary on Moxley, Ericsson, and Tuffiash (2017). See S71.

[LP-W005] Given that the detailed original criteria for deliberate practice have not changed, could the understanding of this complex concept have improved over time? A response to Macnamara and Hambrick (2020). See S71.

[LP-W006] It's Not How Much; It's How: Characteristics of Practice Behavior and Retention of Performance Skills. See S71.

[LP-W007] "Pulling Teeth and Torture": Musical Memory and Problem Solving. See S71.

[LP-W008] The influence of deliberate practice on musical achievement: a meta-analysis. See S71.

[LP-W009] Развитие исследований по формированию умственных действий [Development of research on the formation of mental actions]. See S71.

[LP-W011] Récents travaux sur le transfert [Recent work on transfer]. See S71.

[LP-W012] Estrutura de prática e freqüência de “feedback” extrínseco na aprendizagem de habilidades motoras [Practice structure and extrinsic feedback frequency in learning motor skills]. See S71.

[LP-W013] Aprendizagem motora em crianças: efeitos da freqüência autocontrolada de conhecimento de resultados [Motor learning in children: effects of self-controlled frequency of knowledge of results]. See S71.

[LP-W014] 運動学習における文脈干渉と要約フィードバックの相互作用 [Interaction of contextual interference and summary feedback in motor learning]. See S71.

[LP-W015] ¿Cómo diseñar la práctica para optimizar el desempeño y aprendizaje motor? Una revisión de literatura [How to design practice to optimize motor performance and learning? A literature review]. See S71.

[LP-W016] Üben aus Sicht der Lernpsychologie: Wie kann Üben das nachhaltige Lernen fördern? [Practice from the perspective of the psychology of learning: How can practice promote lasting learning?]. See S72.

[LP-W017] 提取练习效应的产生机制：编码、提取的整合演变 [The mechanism of the retrieval practice effect: From encoding and retrieval to their integration]. See S72.

[LP-W018] The Learning Edge. See S72.

[LP-W019] Learning Versus Performance: An Integrative Review. See S72.

[LP-W020] Improving Students’ Learning With Effective Learning Techniques: Promising Directions From Cognitive and Educational Psychology. See S72.

[LP-W021] Challenge Point: A Framework for Conceptualizing the Effects of Various Practice Conditions in Motor Learning. See S72.

[LP-W022] Test-Enhanced Learning: Taking Memory Tests Improves Long-Term Retention. See S72.

[LP-W023] Spacing Effects in Learning: A Temporal Ridgeline of Optimal Retention. See S72.

[LP-W024] Learning Concepts and Categories: Is Spacing the “Enemy of Induction”? See S72.

[LP-W025] Productive Failure. See S72.

[LP-W026] When Problem Solving Followed by Instruction Works: Evidence for Productive Failure. See S72.

[LP-W027] Über das Gedächtnis. Untersuchungen zur experimentellen Psychologie [On memory: Investigations in experimental psychology]. See S72.

[LP-W028] Making Things Hard on Yourself, But in a Good Way: Creating Desirable Difficulties to Enhance Learning. See S72.

[LP-W029] When Is Practice Testing Most Effective for Improving the Durability and Efficiency of Student Learning? See S72.

[LP-W030] New Conceptualizations of Practice: Common Principles in Three Paradigms Suggest New Concepts for Training. See S72.

[LP-W031] Optimal Experience in Work and Leisure. See S72.

[LP-W032] The effect of contextual interference on transfer in motor learning: a systematic review and meta-analysis. See S72.

[LP-W033] A Meta-Analysis of the Effectiveness and Boundary Conditions of Retrieval Practice, Distributed Practice, and Interleaved Practice in Authentic School Classrooms. See S72.

[LP-W034] Self-Controlled Feedback: Does It Enhance Learning Because Performers Get Feedback When They Need It? See S71.

[MG-W001] Critical-Path Planning and Scheduling. See S85.

[MG-W002] Taming uncertainty. See S85.

[MG-W003] プロジェクト管理におけるダミーに基づくクリティカルチェーン・スケジューリング [Dummy-Based Critical Chain Scheduling in Project Management]. See S85.

[MG-W004] From Nobel Prize to Project Management: Getting Risks Right. See S86.

[MG-W005] The management of project management: A conceptual framework for project governance. See S86.

[MG-W006] Peut-on gérer l'innovation par projet ? [Can innovation be managed through projects?]. See S86.

[MG-W007] Lead Users: A Source of Novel Product Concepts. See S87.

[MG-W008] Product Development Decisions: A Review of the Literature. See S87.

[MG-W009] Structuring and managing the new product development process – review on the evolution of the Stage-Gate® process. See S88.

[MG-W010] Agile Product Development: Managing Development Flexibility in Uncertain Environments. See S88.

[MG-W011] The Practice of Project Management in Product Development: Insights from the Literature and Cases in High-Tech. See S88.

[MG-W012] Produktmanagement Prozess-Navigator [Product-management process navigator]. See S88.

[MG-W013] A comprehensive overview of software product management challenges. See S88.

[MG-W014] Portfolio Selection. See S89.

[MG-W015] Review of the Principles of Life-office Valuations. See S89.

[MG-W016] A New Interpretation of Information Rate. See S90.

[MG-W017] Diversification Return, Portfolio Rebalancing, and the Commodity Return Puzzle. See S90.

[MG-W018] 广发 FOF 观察第 27 期丨资产配置中的再平衡策略 [FOF observation27: rebalancing strategies in asset allocation]. See S90.

[MG-W019] Wasserstein-Kelly Portfolios: A Robust Data-Driven Solution to Optimize Portfolio Growth. See S90.

[MG-W020] Risk Parity Portfolios with Risk Factors. See S89.

[MIL-W001] Command and Control (MCDP 6). See S16.

[MIL-W002] Mission Command: Command and Control of Army Forces (ADP 6-0). See S16.

[MIL-W003] Logistics (MCDP 4). See S38.

[MIL-W004] Mission command and logistics interdependencies. See S16.

[MIL-W005] Organizations in Action: Social Science Bases of Administrative Theory. See S16.

[MIL-W006] Command in War. See S16.

[MIL-W007] Network Centric Warfare, Command, and the Nature of War. See S16.

[MIL-W008] Organic Design for Command and Control. See S16.

[MIL-W009] Power to the Edge: Command and Control in the Information Age. See S16.

[MIL-W010] Understanding Command and Control. See S16.

[MIL-W011] Task Group SAS-085 Final Report on C2 Agility: Executive Overview. See S16.

[MIL-W012] Operational Logistics: The Art and Science of Sustaining Military Operations. See S38.

[MIL-W013] Flexibility in Operational-Level Logistics. See S38.

[MIL-W014] Operational Logistics: The Art and Science of Sustaining Military Operations.

[MIL-W015] Logistics in the National Defense. See S38.

[MIL-W016] Pure Logistics: The Science of War Preparation. See S38.

[MIL-W017] Sense and Respond Logistics: Integrating Prediction, Responsiveness, and Control Capabilities. See S38.

[MIL-W018] Chaoplexic warfare or the future of military organization. See S16.

[MIL-W019] The Scientific Way of Warfare: Order and Chaos on the Battlefields of Modernity. See S16.

[MIL-W020] How scientific ideas are reflected in practices of war. See S16.

[MIL-W021] Auftragstaktik (Mission Command): The Prussian-German Origins and Application of Mission Command. See S16.

[MIL-W022] Der Unterführer als Feldherr im Taschenformat: Theorie und Praxis der Auftragstaktik im deutschen Heer 1869 bis 1945 [The Subordinate Leader as a Field Commander in Pocket Format: Theory and Practice of Mission Command in the German Army, 1869–1945]. See S16.

[MIL-W023] Auftragstaktik im preußisch-deutschen Heer 1871 bis 1914 [Mission Command in the Prussian-German Army, 1871–1914]. See S16.

[MIL-W024] FT-05: L'exercice du commandement en opérations pour les chefs tactiques [The Exercise of Command in Operations for Tactical Commanders]. See S16.

[MIL-W025] Agile Fehlerkultur – Realität (auch) in der Bundeswehr? [An Agile Error Culture: A Reality in the Bundeswehr Too?]. See S16.

[MIL-W026] 我军加速构建跨军种一体化联合保障模式 [Our Military Accelerates the Development of an Integrated Joint Support Model Across Services]. See S38.

[MIL-W027] 孫子兵法 [The Art of War]. See S16.

[MIL-W028] Трансформация управления — централизация или децентрализация [Transformation of Management: Centralization or Decentralization].

[MIL-W029] Re-Envisioning Command and Control. See S16.

[MIL-W030] Contested Logistics: A Primer. See S38.

[MIL-W031] METRIC: A Multi-Echelon Technique for Recoverable Item Control. See S38.

[MIL-W032] METRIC: A Multi-Echelon Technique for Recoverable Item Control. See S38.

[MIL-W033] A case study of C2 Agility in the 2014 Västmanland forest fire. See S16.

[MIL-W034] Mission Command at Sea and Command by Veto. See S16.

[MIL-W035] Paying the Mission Command Bill (Mission Command response). See S16.

[MIL-W036] Vom Kriege [On War], Buch I, Kapitel VII: Friktion im Kriege [Book I, Chapter VII: Friction in War]. See S16.

[MIO-W001] Operator learning meets inverse problems: A probabilistic perspective. See S35.

[MIO-W002] Introduction to Inverse Problems, Chapter 4: Regularization Methods. See S35.

[MIO-W003] Stochastic Matrix-Free Equilibration. See S35.

[NC-W001] Алгоритм решения изобретательских задач АРИЗ-85-В [Algorithm for solving inventive problems, ARIZ-85-V].. See S81.

[NC-W002] Creativity in Product Innovation. See S81.

[NC-W003] 創造的問題解決の新しいパラダイム—類比思考に頼らない USIT の 6 箱方式 [A new paradigm for creative problem solving: the six-box scheme of USIT without depending on analogical thinking]. See S81.

[NC-W004] Creativity Training. See S81.

[NC-W005] Predicting creative problem solving in engineering design. See S81.

[NC-W006] TRIZ industrial case studies: a critical survey. See S81.

[NC-W007] TRIZ 真的是创新“点金术”吗——对浙江省 TRIZ 应用效果的分析 [Is TRIZ really an innovative “Midas touch”? Application analysis based on Zhejiang Province]. See S81.

[NC-W008] The Act of Creation.. See S82.

[NC-W009] Synectics: The Development of Creative Capacity.. See S82.

[NC-W010] The associative basis of the creative process. See S82.

[NC-W011] A Systematic Review of Creativity-Related Studies Applying the Remote Associates Test From 2000 to 2019. See S82.

[NC-W012] 创造性问题解决中的联想能力及其作用 [The mechanism of associative abilities in creative problem solving]. See S82.

[NC-W013] 遠隔連想テストは何を測るか [What does the Remote Associates Test measure?]. See S82.

[NC-W014] Creativity and Artificial Intelligence. See S82.

[NC-W015] Creative Inference in Imagery and Invention. See S82.

[NC-W016] Scientific Creativity: Discovery and Invention as Combinatorial. See S82.

[NC-W017] Conceptual Integration Networks. See S82.

[NC-W018] Comparison of the Effect Of Three Methods of Creativity Development in Elementary Students in Tehran City. See S82.

[NC-W019] Le processus de bissociation au cœur de la fabrique des idées : le cas d’une innovation managériale [The process of bisociation at the heart of idea generation: the case of a managerial innovation]. See S82.

[NC-W020] Undiscovered Public Knowledge: a Ten-Year Update. See S82.

[NC-W021] From Information Networks to Bisociative Information Networks. See S82.

[NC-W022] Bridging Concept Identification for Constructing Information Networks from Text Documents. See S82.

[NC-W023] Creative exploration using topic-based bisociative networks. See S82.

[NC-W024] Letter to Sir Horace Mann, 28 January 1754.. See S83.

[NC-W025] The Bearing of Empirical Research upon the Development of Social Theory. See S83.

[NC-W026] Towards a Theory of Serendipity: A Systematic Review and Conceptualization. See S83.

[NC-W027] Coming across information serendipitously, Part 1: A process model. See S83.

[NC-W028] Serendipity: Towards a Taxonomy and a Theory. See S83.

[NC-W029] Sérendipité et indisciplinarité [Serendipity and indisciplinarity]. See S83.

[NC-W030] “Making my own luck”: Serendipity strategies and how to support them in digital information environments. See S83.

[NC-W031] Planned Luck: How Incubators Can Facilitate Serendipity for Nascent Entrepreneurs Through Fostering Network Embeddedness. See S83.

[NC-W032] Criatividade e serendipidade na inovação de produtos e serviços em bibliotecas universitárias [Creativity and serendipity in university-library product and service innovation]. See S83.

[NC-W033] 机缘巧合实现：创业平台编配警觉性形成机理——组织间学习视角下的案例研究 [Seizing serendipity: the formation mechanism of entrepreneurial-platform orchestration alertness, a case study from interorganizational learning]. See S83.

[NC-W034] VR 図書館内の 3D 書架におけるブラウジング行動：視線データに着目した探索的分析 [Browsing behavior on 3D bookshelves in a VR library: exploratory analysis focusing on eye-tracking data]. See S83.

[NN-W001] Getting to Yes: Negotiating Agreement Without Giving In. Second edition with Bruce Patton, 1991.. See S96.

[NN-W002] Processus et techniques de négociation : manuel de formation pour les acteurs du dialogue social en Afrique [Negotiation processes and techniques: A training manual for participants in social dialogue in Africa]. Dakar: OIT.. See S96.

[NN-W003] Toward a Process Model of First Offers and Anchoring in Negotiations. Negotiation and Conflict Management Research 16(1). DOI 10.34891/2022.574.. See S96.

[NN-W004] 2003. First Offers in Negotiations: Determinants and Effects. Working paper, SSRN 399722.. See S96.

[NN-W005] 2022/2023. Embracing Complexity: A Review of Negotiation Research. Annual Review of Psychology 74:299–332. DOI 10.1146/annurev-psych-033020-014116.. See S96.

[NN-W006] 2025. Negotiation. In The Handbook of Social Psychology, sixth edition. DOI 10.70400/NUZE7621.. See S96.

[NN-W007] Возможные стратегии ведения переговорного процесса [Possible strategies for conducting negotiations]. Российский внешнеэкономический вестник 8:79–88.. See S96.

[NN-W008] Negociaciones comerciales internacionales: texto y casos. Subsequent expanded second-edition expression, 2014. Fondo Editorial PUCP.. See S96.

[NN-W009] The Pros and Cons of Getting to YES. Journal of Legal Education 34:115–124.. See S96.

[NN-W010] 合作式谈判研究述评与展望 [Review and prospects of cooperative negotiation research]. 外国经济与管理 36(3):65–73.. See S96.

[NN-W011] 影响整合式谈判的情境因素和过程因素 [Contextual and process variables influencing integrative negotiation]. 心理科学进展 15(3):518–523.. See S96.

[NN-W012] 2010/2011. Dormant Ties: The Value of Reconnecting. Organization Science 22(4):923–939. DOI 10.1287/orsc.1100.0576.. See S97.

[NN-W013] How many hours does it take to make a friend? Journal of Social and Personal Relationships 36(4):1278–1296. DOI 10.1177/0265407518761225.. See S97.

[NN-W014] Kelly. 2004. Friendship Maintenance: An Analysis of Individual and Dyad Behaviors. Journal of Social and Clinical Psychology 23(3):413–441. DOI 10.1521/jscp.23.3.413.35460.. See S97.

[NN-W015] 友人関係における動機づけと友人とのコミュニケーションおよび精神的健康との関連 [Friendship motivation, communication and mental health]. 人間生活文化研究 26:572–582. DOI 10.9748/hcs.2016.572.. See S97.

[NN-W016] 親密な友人関係の形成・維持過程の動機づけモデルの構築 [A motivational model of the formation and maintenance of close friendships]. 教育心理学研究 56(4):575–588. DOI 10.5926/jjep1953.56.4_575.. See S97.

[NN-W017] 人际感恩对关系维持的影响 [The impacts of interpersonal gratitude on relationship maintenance]. 心理科学进展 30(11):2586–2594. DOI 10.3724/SP.J.1042.2022.02586.. See S97.

[NN-W018] Étudier les réseaux sociaux [Studying social networks]. Diversité 157:179–184. DOI 10.3406/diver.2009.3090. Related earlier article: Étudier les réseaux. Apports et perspectives pour les sciences sociales, 2008.. See S97.

[NN-W019] Entwicklung und Validierung einer Networkingskala [Development and validation of a networking scale]. Diagnostica 52(4):161–180. DOI 10.1026/0012-1924.52.4.161.. See S97.

[NN-W020] Effects of Networking on Career Success: A Longitudinal Study. Journal of Applied Psychology 94(1):196–206. DOI 10.1037/a0013350.. See S97.

[NN-W021] The Contaminating Effects of Building Instrumental Ties: How Networking Can Make Us Feel Dirty. Administrative Science Quarterly 59(4):705–735. DOI 10.1177/0001839214554990.. See S97.

[NN-W022] Corrigendum to The Contaminating Effects of Building Instrumental Ties: How Networking Can Make Us Feel Dirty. Administrative Science Quarterly 69(2):568. DOI 10.1177/00018392241243394.. See S97.

[NN-W023] Evans, Thomas Ng, and David DuBois. 2008. Does Mentoring Matter? A Multidisciplinary Meta-Analysis Comparing Mentored and Non-Mentored Individuals. Journal of Vocational Behavior 72(2):254–267. DOI 10.1016/j.jvb.2007.04.005.. See S97.

[NN-W024] Internet e Relacionamentos Interpessoais: Uma Revisão Narrativa Crítica da Literatura Científica (2011–2020) [Internet and interpersonal relationships: A critical narrative review]. Interações 42:22–47. DOI 10.31211/interacoes.n42.2022.a2.. See S97.

[NN-W025] De Dreu. 2003. Social Motives in Integrative Negotiation: The Mediating Influence of Procedural Fairness. Social Justice Research 16:217–239.. See S96.

[OI-W001] Organizational Silence: A Barrier to Change and Development in a Pluralistic World. See S51.

[OI-W003] Single-Loop and Double-Loop Models in Research on Decision Making. See S51.

[OI-W004] Psychological Safety and Learning Behavior in Work Teams. See S51.

[OI-W005] Knee-Deep in the Big Muddy: A Study of Escalating Commitment to a Chosen Course of Action. See S51.

[OI-W007] Exit, Voice, and Loyalty: Responses to Decline in Firms, Organizations, and States. See S51.

[OI-W008] The Liberating Role of Conflict in Group Creativity: A Cross Cultural Study. See S52.

[OI-W009] Surprising but True: Half the Decisions in Organizations Fail. See S52.

[OI-W010] Exploration and Exploitation in Organizational Learning. See S52.

[OI-W012] Productivity Loss in Brainstorming Groups: Toward the Solution of a Riddle. See S52.

[OI-W013] Parallel Prototyping Leads to Better Design Results, More Divergence, and Increased Self-Efficacy. See S52.

[OI-W014] General Morphological Analysis: A General Method for Non-Quantified Modelling. See S52.

[OI-W015] Design Fixation. See S52.

[OI-W017] The Trickle-Down Effect: Policy Decisions, Risky Work, and the Challenger Tragedy. See S51.

[PM-W001] Of Models and Machines: Implementing Bounded Rationality. See S94.

[PM-W002] Achieving Several Goals Simultaneously. See S94.

[PM-W003] Past and Future of Backcasting: The Shift to Stakeholder Participation and a Proposal for a Methodological Framework. See S94.

[PM-W004] Méthodes participatives de prospective et de planification pour un développement durable : analyse d'approches et de réalisations [Participatory foresight and planning methods for sustainable development: approaches and applications]. See S94.

[PM-W005] 持続可能社会に向けたバックキャスティング型シナリオ作成手法の提案 [A design-support method for backcasting scenarios for a sustainable society]. See S94.

[PM-W006] Understanding by Design. See S94.

[PM-W007] The Understanding by Design Guide to Creating High-Quality Units. See S94.

[PM-W008] 基于逆向教学设计视野下单元整体教学研究 [Integrated unit teaching from the perspective of backward instructional design]. See S94.

[PM-W009] Backward Planning: Effects of Planning Direction on Predictions of Task Completion Time. See S94.

[PM-W010] Relative Effects of Forward and Backward Planning on Goal Pursuit. See S94.

[PM-W011] Thinking Forward and Backward: Effective Backward Planning with Large Language Models. See S94.

[PM-W012] Methoden zur Strategischen Vorausschau: Backcasting [Methods of strategic foresight: backcasting]. See S94.

[PM-W013] Back from the Future: The Backcasting Wheel for Mapping a Pathway to a Preferred Future. See S94.

[PM-W014] Consolidating Backcasting: A Design Framework towards a Users' Guide. See S94.

[PM-W015] Research Integration Using Dialogue Methods. See S95.

[PM-W016] The Robotics Decision: Solving Strategic Problems. See S95.

[PM-W017] Effects of Devil's Advocacy and Dialectical Inquiry on Decision Making: A Meta-analysis. See S95.

[PM-W018] Structured Conflict Approaches Used in Strategic Decision Making: From Mason's Initial Study to Virtual Teams. See S95.

[PM-W019] Devil's Advocate versus Authentic Dissent: Stimulating Quantity and Quality. See S95.

[PM-W020] Back to the Future: Temporal Perspective in the Explanation of Events. See S95.

[PM-W021] Performing a Project Premortem. See S95.

[PM-W022] Evaluating the Effectiveness of the PreMortem Technique on Plan Confidence. See S95.

[PM-W023] Proactive Planning for Contextual Fit: The Role of the Implementation Premortem. See S95.

[PM-W024] Red Teaming Handbook. See S95.

[PM-W025] « Imaginer au-delà de l'imaginaire ». Red teaming et serious games au service de l'anticipation et de la prospective [Imagining beyond the imaginable: red teaming and serious games for anticipation and foresight]. See S95.

[PM-W026] 国際政治学における分析手法としてのウォーゲームの活用–実験的ウォーゲームの台頭とその活用をめぐる議論を中心として– [Wargaming as an analytical method in international politics: the rise and debate over experimental wargaming]. See S95.

[PM-W027] Playing the New Devil's Advocate Role in Facilitated Modelling Processes to Address Group Homogeneity. See S95.

[PM-W028] Group-based judgmental forecasting: An integration of extant knowledge and the development of priorities for a new research agenda. See S95.

[PM-W029] Corrigendum: Relative Effects of Forward and Backward Planning on Goal Pursuit. See S94.

[PWR-W001] Forty years of political skill and will in organizations: a review, meta-theoretical framework and directions for future research. See S91.

[PWR-W002] Les relations de pouvoir dans un système d'organisation bureaucratique [Power relations in a bureaucratic organizational system]. See S91.

[PWR-W003] Power Play. See S91.

[PWR-W004] The 48 Laws of Power. See S91.

[PWR-W005] 日本企業での新規提案場面における組織内政治の実態と機能 [Organizational politics in new-proposal situations in Japanese companies]. See S91.

[PWR-W006] Middle power plays: How and when Mach middle managers use downward abuse and upward guanxi to gain and maintain power. See S91.

[PWR-W007] People with disagreeable personalities (selfish, combative, and manipulative) do not have an advantage in pursuing power at work. See S91.

[PWR-W008] Микрополитический аспект государственного управления человеческими ресурсами [The micropolitical aspect of public human-resource management]. See S91.

[PWR-W009] Política na terra do "jeitinho": consequências dos comportamentos políticos em organizações no Brasil [Politics in the land of jeitinho: consequences of political behavior in Brazilian organizations]. See S91.

[PWR-W010] Power, patronage, and gatekeeper politics in South Africa. See S91.

[PWR-W011] Robust Action and the Rise of the Medici, 1400–1434. See S91.

[PWR-W012] El poder y la organización: una mirada desde la Nueva Economía Institucional (NEI) [Power and organization: a view from New Institutional Economics]. See S91.

[RGC-W001] The Theory of Dissipative Structures. See S19.

[RGC-W002] Mechanisms, organisms and social systems. See S20.

[RGC-W003] Organizational implications of systems thinking: Multidimensional modular design. See S20.

[RGC-W004] Living Systems. See S21.

[RGC-W005] The life and work of James Grier Miller. See S21.

[RGC-W006] Applications of Living Systems Theory. See S21.

[RGC-W007] Living Systems Theory. See S21.

[RGC-W008] Living Systems Theory and Entity-Systems Theory. See S21.

[RGC-W009] Cybernetics and organisational analysis; Towards a better understanding of Beer's Viable System Model. See S22.

[RGC-W010] The Soul of the Viable System Model. See S22.

[RGC-W011] Review of The Heart of Enterprise. See S22.

[RGC-W012] The Viable System Model: Its Provenance, Development, Methodology and Pathology. See S22.

[RGC-W013] What Is Life? A Systemic Approach. See S24.

[RGC-W014] The Systems View of Life: A Unifying Vision — collaborative book review. See S24.

[RGC-W015] Reviewing The Systems View of Life. See S24.

[RGC-W016] Autopoiesis with or without cognition: defining life at its edge. See S24.

[RGC-W017] An Essay on Understanding the Mind. See S25.

[RGC-W018] Introduction to Systems Philosophy: Toward a New Paradigm of Contemporary Thought. See S25.

[RGC-W019] The Systems View of the World: The Natural Philosophy of the New Developments in the Sciences. See S25.

[RGC-W020] Organization and Process: Systems Philosophy and Whiteheadian Metaphysics. See S25.

[RGC-W021] The validity and validation of mathematical models: Methodological, theoretical, and practical studies with emphasis on the modelling of complex biological systems. See S25.

[RGC-W022] Life Itself: A Comprehensive Inquiry into the Nature, Origin, and Fabrication of Life. See S26.

[RGC-W023] Anticipatory Systems: Philosophical, Mathematical, and Methodological Foundations. See S26.

[RGC-W024] Jean-Louis Le Moigne. La théorie du système général (théorie de la modélisation), 1977 [Jean-Louis Le Moigne, General System Theory: Theory of Modeling, 1977] (book review). See S23.

[RGC-W025] Chronique de La Théorie du Système Général, Théorie de la Modélisation [Chronicle of General System Theory: Theory of Modeling]. See S23.

[RGC-W026] Modélisation systémique du couple eau-sol: la pédologie hydrostructurale [Systemic Modeling of the Water–Soil Coupling: Hydrostructural Pedology]. See S23.

[RGC-W027] Le Moigne Jean-Louis, La théorie du système général. Théorie de la modélisation [Jean-Louis Le Moigne, General System Theory: Theory of Modeling] (book review).

[RGC-W028] La Méthode, tome 1: La Nature de la Nature [Method, Volume 1: The Nature of Nature]. See S23.

[RGC-W029] La Méthode, tome 2: La Vie de la Vie [Method, Volume 2: The Life of Life]. See S23.

[RGC-W030] Productivité et équité, la fin d'un rêve ? [Productivity and Equity: The End of a Dream?]. See S27.

[RGC-W031] Le complexe de Babel: crise ou maîtrise de l'information ? [The Babel Complex: Crisis or Mastery of Information?]. See S27.

[RGC-W032] A comparison and critical assessment of Porat and Rubin's information economy and Wallis and North's transaction sector.

[RGC-W033] Ethics in Jacques Lafitte's Mechanology. See S39.

[RGC-W034] Sur la science des machines [On the Science of Machines]. See S39.

[RGC-W035] Homo Faber: Documents d'un débat oublié, 1907–1941. Partie IV: D'Espinas à Mauss [Homo Faber: Documents of a Forgotten Debate, 1907–1941. Part IV: From Espinas to Mauss]. See S39.

[RGC-W036] A "Good Regulator Theorem" for Embodied Agents. See S03.

[RGC-W037] This Week's Citation Classic: The Second Cybernetics. See S18.

[RGC-W038] What Can Synergetics Contribute to Embodied Aesthetics?. See S41.

[RGC-W039] "开放的复杂巨系统"概念的形成 [Formation of the Concept "Open Complex Giant System"]. See S18.

[RGC-W040] 钱学森的现代科学技术体系与综合集成方法论 [Qian Xuesen's Contemporary System of Science and Technology and Meta-synthesis Methodology]. See S18.

[RGC-W041] 物理-事理-人理系统方法论在建立商业设施与技术装备标准规范体系表结构框架中的应用 [Application of Wuli–Shili–Renli Systems Methodology to Establishing the Structural Framework of a Standards System for Commercial Facilities and Technical Equipment]. See S18.

[RGC-W042] Quantifying economic sustainability: Implications for free-enterprise theory, policy and practice. See S47.

[RGC-W043] Quantifying sustainable balance in ecosystem configurations. See S47.

[SB-W001] Organizing States: The continuing relevance of formal organization within political administration. See S13.

[SB-W002] When Formality Works: Authority and Abstraction in Law and Organizations. See S13.

[SB-W003] 英文文献中的中国组织现象研究 [Research on Chinese Organizational Phenomena in the English-Language Literature]. See S13.

[SB-W004] The Alchemist of Contingency Theory.

[SB-W005] The mirroring hypothesis: theory, evidence, and exceptions. See S07.

[SB-W006] Control and Coordination in Hierarchical Systems. See S09.

[SB-W007] Transformation of multi-level systems – theoretical grounding and consequences for enterprise architecture management. See S09.

[SB-W008] Two Coordination Principles and Their Application in Large Scale Systems Control. See S09.

[SB-W009] A Review of Theory of Hierarchical, Multilevel Systems.

[SB-W010] Modeling the Evolution and Breakdown of Adaptive Systems: A Review.

[SB-W011] Notes on "Notes on the Synthesis of Form": Dawning Insights in Early Christopher Alexander. See S08.

[SB-W012] Biological Analogy and an Architectural Science. See S08.

[SB-W013] 走向都市认同的空间实践——亚历山大城市设计理论的价值、局限与启示 [Spatial Practices Toward Urban Identity: The Value, Limitations and Implications of Alexander's Urban Design Theory]. See S08.

[SB-W014] The ESA/NASA SOHO Mission Interruption: Using the STAMP Accident Analysis Technique for a Software Related Mishap. See S29.

[SB-W015] A Controlled Experiment for the Empirical Evaluation of Safety Analysis Techniques for Safety-Critical Software. See S29.

[SB-W016] 飞机差动刹车纠偏过程的 STAMP/STPA 安全性分析 [STAMP/STPA Safety Analysis of Aircraft Differential-Braking Direction Correction]. See S29.

[SB-W017] Moving Beyond Normal Accidents and High Reliability Organizations: A Systems Approach to Safety in Complex Systems. See S29.

[SB-W018] Review of Grammatical Man. See S36.

[SB-W019] Book Review: Grammatical Man. See S36.

[SB-W020] Short notice of Grammatical Man. See S36.

[SG-W001] Soft Systems Methodology: a 30-year retrospective (plenary-address summary). See S44.

[SG-W002] A Mini-Primer of Critical Systems Heuristics. See S44.

[SG-W003] From Epistemology to Action. See S44.

[SG-W004] There are Systems and Systems: Towards an Integrated General Systems Theory. See S18.

[SG-W005] 管理科学研究模型与方法（第 2 版）, 第 1 章: 系统科学 [Models and Methods of Management Science, Second Edition, Chapter 1: Systems Science]. See S18.

[SG-W006] About ISSS. See S18.

[SG-W007] 学术研究中的系统科学法 [The Systems Science Method in Academic Research]. See S18.

[SG-W008] The Nature of Living Systems: An Exposition of the Basic Concepts in General Systems Theory. See S21.

[SG-W009] Cybernetic Explanation (reprint in Steps to an Ecology of Mind). See S18.

[SG-W010] Appreciative systems. See S44.

[SG-W011] The Systems View of Life: A Unifying Vision. See S24.

[SG-W012] Systems approaches: Sir Geoffrey Vickers. See S44.

[SG-W013] Reality, Validity, Truth. See S18.

[SG-W014] An Introduction to the Viable Systems Approach and Its Contribution to Marketing. See S22.

[SG-W015] Applying Socio-technical Thinking in the Competitive, Agile, Lean, Data-Driven World of Knowledge Work and Smart, Service-Oriented, Customer-Centric Value Creation Ecosystems. See S14.

[SG-W016] What is Gigamapping?. See S44.

[SG-W017] Multi-scale integrated analysis of societal and ecosystem metabolism (MuSIASEM): Theoretical concepts and basic rationale. See S47.

[SG-W018] Environmental Accounting Using Emergy: Evaluation of the State of West Virginia. See S47.

[SG-W019] Theory of Constraints (TOC). See S44.

[SG-W020] Failure demand: what's the big secret?. See S31.

[SG-W021] Introduction to DSM. See S07.

[SG-W022] A Q-analysis package for higher-order interactions analysis in Python and its application in network physiology. See S47.

[SG-W023] Solution of Large-Scale Networks by Tearing. See S04.

[SG-W024] Организационно-деятельностная игра [The Organizational-Activity Game]. See S44.

[SG-W025] The Joint-Decision Trap Revisited. See S15.

[SG-W026] A practitioner's field guide to the behaviour settings method. See S41.

[SG-W027] PCT in 11 Steps. See S03.

[SG-W028] Introduction to the Eight Concepts; Triangles; Emotional Cutoff. See S41.

[SG-W029] Sound Practices to Strengthen Operational Resilience. See S30.

[SG-W030] What is Technology Assessment?. See S44.

[SG-W031] Robust Decision Making. See S44.

[SG-W032] Metasystem Transition Theory; Metasystem Transition. See S18.

[SG-W033] Systematics and System Theories. See S18.

[SG-W034] Epistemological Hierarchies in Knowledge and Expert Systems. See S42.

[SG-W035] A State Space Representation of Business Transaction System. See S10.

[SG-W036] General Systems Theory—The Skeleton of Science. See S18.

[SG-W037] Systems Analysis: I. General Systems Theory. See S18.

[SG-W038] The Systems View of the World: The Natural Philosophy of the New Developments in the Sciences. See S25.

[SG-W039] Inquiring Organizations. See S44.

[SG-W040] A Brief Guide to Interactive Planning and Idealized Design. See S44.

[SG-W041] DSRP Theory: A Primer. See S44.

[SG-W042] On the Scalability of Social Order: Modeling the Problem of Double and Multi Contingency Following Luhmann. See S49.

[SG-W043] O Sistemismo de Bunge como Instrumento Metodológico para Aprendizagem Ativa [Bunge's Systemism as a Methodological Instrument for Active Learning]. See S18.

[SG-W044] The Use of Metapatterns for Research into Complex Systems of Teaching, Learning, and Schooling —Part I: Metapatterns in Nature and Culture. See S18.

[SG-W045] Time, Structure and Fluctuations—Nobel Lecture. See S46.

[SG-W046] An Introduction to Catastrophe Theory and Its Applications. See S46.

[SG-W047] Hypercycle. See S40.

[SG-W048] Chi-Chi Earthquake-caused Landslide: grey prediction model for pioneer vegetation recovery monitored by satellite images. See S44.

[SG-W049] 泛系友人书（1）[Letter to Pansystems Friends, Part 1]. See S18.

[SG-W050] New Development of the Basic Theory of Extenics. See S44.

[SG-W051] Strategic Options Development and Analysis. See S44.

[SG-W052] Introduction to Axiomatic Design. See S44.

[SG-W053] Reconstructability Analysis as a Tool for Identifying Gene-Gene Interactions in Studies of Human Diseases. See S47.

[SG-W054] Developmental Systems and Evolutionary Explanation. See S28.

[SG-W055] Anticipatory Systems and the Processing of Meaning: a Simulation Study Inspired by Luhmann's Theory of Social Systems. See S49.

[SG-W056] General Systems Theory: Applications for Organization and Management. See S18.

[SG-W057] A new Model of the Conceptual Design Process using QFD/FA/TRIZ. See S44.

[SG-W058] Object-Process Methodology for Structure-Behavior Co-Design. See S44.

[SG-W059] Pace Layering: How Complex Systems Learn and Keep Learning. See S44.

[SG-W060] The IM Process. See S44.

[SG-W061] The Second Cybernetics: Deviation-Amplifying Mutual Causal Processes. See S18.

[SL-W001] The past, present, and future of adaptive selling: Toward an integrative framework. See S77, S78, S79.

[SL-W002] The evolution of the seven steps of selling. See S80.

[SL-W003] Theories of Selling. See S76.

[SL-W004] Sales and Distribution Management, MS 508. See S76.

[SL-W005] The Origin of AIDA: Who Invented and Formulated the AIDA model?. See S76.

[SL-W006] 销售管理实务 [Sales Management Practice]. See S76.

[SL-W007] Vom Produkt- zum Lösungsanbieter – Die Transformation zum Solution-Seller meistern [From product provider to solution provider: Mastering the transformation to solution seller]. See S80.

[SL-W008] The SPIN Methodology. See S77.

[SL-W009] SPIN Evidence of return on investment: Seven companies share their SPIN journeys and return on investment evidence.. See S77.

[SL-W010] How To Sell To the Modern Buyer ...The Sandler Way. See S77.

[SL-W011] Sales failure: A review and future research directions. See S79.

[SL-W012] La venta adaptable y la orientación al mercado en el contexto de la venta personal [Adaptive selling and market orientation in personal selling]. See S79.

[SL-W013] Determinantes do desempenho empresarial e das vendas cruzadas no varejo [Determinants of business performance and cross-selling in retail]. See S79.

[SL-W014] Proposition d'un modèle des antécédents et conséquences de la vente suggestive dans le domaine de la restauration [A proposed model of antecedents and consequences of suggestive selling in restaurants]. See S79.

[SL-W015] 法人営業イノベーションとクラウド CRM システムの活用 [B2B sales innovation and the use of cloud CRM systems]. See S80.

[SL-W016] Управление продажами: истоки, сущность, подходы [Sales management: Origins, meaning and approaches]. See S80.

[SL-W017] 卸売業における営業革新—生産財卸売業を事例として [Sales innovation in wholesaling: A case of industrial-goods wholesaling]. See S80.

[SL-W018] Drivers of sales performance: A contemporary meta-analysis. Have salespeople become knowledge brokers?. See S79.

[SL-W019] "It's almost like taking the sales out of selling"—Towards a conceptualization of value-based selling in business markets. See S80.

[SL-W020] What is the Challenger Sales Methodology?. See S78.

[SL-W021] Defining Solution Selling. See S77.

[SL-W022] Selling Services—Coordinating Sales and Project Management. See S77, S80.

[SL-W023] Vendeurs : passez en mode solution [Salespeople: Switch to solution mode]. See S79.

[SL-W024] The Evolution of the Sales Process: Relationship Selling Versus the Challenger Sales. See S78.

[SL-W025] What is the JOLT Effect? Everything You Need To Know. See S78.

[SL-W026] The effectiveness of verbal prompts on sales. See S79.

[SL-W027] Solution Selling® Training for Buyer Alignment and Sales Success.. See S77.

[SL-W028] 生産財における潜在的ニーズ獲得の論理 [A logic of latent-needs acquisition for industrial products]. See S80.

[SL-W029] Consultative Selling. See S77.

[SM-W001] Flight Testing the X-36: The Test Pilot's Perspective. See S54.

[SM-W002] X-36 Tailless Fighter Agility Research Aircraft. See S54.

[SM-W003] Tradeoff between Stability and Maneuverability during Whole-Body Movements. See S54.

[SM-W004] The uncontrolled manifold concept: identifying control variables for a functional task. See S54.

[SM-W006] Domaine de victoire et stratégies viables chez les pêcheurs décrits par l'anthropologue Fredrik Barth [Winning domain and viable strategies among the fishermen described by anthropologist Fredrik Barth]. See S54.

[SM-W007] Формирование требований к динамическим характеристикам и базовым параметрам контуров управления рулевого привода перспективного маневренного самолета [Developing requirements for the dynamic characteristics and basic parameters of flight-control actuator loops for a prospective maneuverable aircraft]. See S54.

[SM-W008] The theory of graceful extensibility: basic rules that govern adaptive systems. See S54.

[SM-W009] 高风险货运固定翼无人机系统适航标准（试行） [Airworthiness standards for high-risk cargo fixed-wing unmanned aircraft systems (trial)]. See S54.

[SMR-W001] Process intensification education contributes to sustainable development goals. Part 1. See S01.

[ST-W001] Of Strategies, Deliberate and Emergent. See S92.

[ST-W002] Moves That Matter: Issue Selling and Organizational Change. See S92.

[ST-W003] Exploring Middle Managers' Strategic Sensemaking Role Through Practical Knowledge. See S92.

[ST-W004] On the Problem of Participation in Strategy: A Critical Discursive Perspective. See S92.

[ST-W005] La réflexion stratégique, objet et outil de recherche pour le management stratégique ? [Strategic reflection: object and research tool for strategic management?]. See S92.

[ST-W006] 战略过程中的中层管理者 [Middle managers in the strategy process]. See S92.

[ST-W007] 企业战略思维研究述评与展望 [Enterprise strategic thinking: a literature review and prospects]. See S92.

[ST-W008] 組織メンバーの創発的行為形成に着目した組織変革のプロセスモデル：M-GTA を用いた X 社の事例分析 [A process model of organizational change focused on members' emergent actions: an M-GTA case analysis of Company X]. See S92.

[ST-W009] Die Dimension Macht: Funktion bei Veränderungen [The dimension of power: its function in change]. See S92.

[SW-W001] First-mover advantages. See S61.

[SW-W002] First-mover (dis)advantages: Retrospective and link with the resource-based view. See S61.

[SW-W004] Irreversibility, Uncertainty, and Investment. See S61, S62.

[SW-W005] Information Revelation and Strategic Delay in a Model of Investment. See S61.

[SW-W006] Strategic Experimentation. See S61.

[SW-W007] 不確実性下における市場参入に対する先発の優位性の影響 [The Effect of First-Mover Advantage on Market Entry under Uncertainty]. See S61.

[SW-W008] 「問題解決型コア部品」ベンダーとしての台湾企業の興隆過程と知識の獲得：液晶テレビ用SoC 事業の事例分析 [The Rise of Taiwanese Firms as Suppliers of Problem-Solving Core Components and Their Acquisition of Knowledge: A Case Study of LCD Television SoCs]. See S61.

[SW-W009] 对"后发优势"的国内实证：基于技术创新扩散视角的分析 [An Empirical Test of Latecomer Advantage within China: An Analysis from the Perspective of Technological Innovation and Diffusion]. See S61.

[T01-W001] The Nature of Procrastination: A Meta-Analytic and Theoretical Review of Quintessential Self-Regulatory Failure. See S99.

[T01-W002] On the Behavioral Side of Procrastination: Exploring Behavioral Delay in Real-Life Settings. See S99.

[T01-W003] 情绪调节改善拖延行为的认知机制：任务厌恶中介作用. See S99.

[T01-W004] Targeting Procrastination Using Psychological Treatments: A Systematic Review and Meta-Analysis. See S99.

[T02-W001] The Founding of Technologically-Based Firms. See S100.

[T02-W002] 日本における技術系ベンチャー企業の経営実態と創業者に関する調査研究. See S100.

[T02-W003] Causation and Effectuation: Toward a Theoretical Shift from Economic Inevitability to Entrepreneurial Contingency. See S100.

[T02-W004] Productivity of innovation: the effect of innovativeness on start-up survival. See S100.

[T02-W005] An Overview of the Kauffman Firm Survey: Results from the 2004–2008 Data. See S100.

[T03-W001] Avoidance learning: a review of theoretical models and recent developments. See S101.

[T03-W002] Rethinking Avoidance: Toward a Balanced Approach to Avoidance in Treating Anxiety Disorders. See S101.

[T03-W003] Maximizing Exposure Therapy: An Inhibitory Learning Approach. See S101.

[T03-W004] Hemmschwelle Expositionstherapie – virtuelle Realität (VR) als Lösungsansatz: Untersuchungen von Automatisierung und Akzeptanz von VR bei Patienten mit Angststörungen. See S101.

[T03-W005] Effect of multiple context exposure on renewal in spider phobia. See S101.

[T04-W001] Product differentiation and market segmentation as alternative marketing strategies. See S102.

[T04-W002] 歴史的視角からのマーケティング―販売管理，マーケティング管理，市場管理―. See S102.

[T04-W003] From marketing mix to relationship marketing: Towards a paradigm shift in marketing. See S102.

[T04-W005] Lyon Acier. See S102.

[T05-W001] Pratiques de recrutement et sélectivité sur le marché du travail. See S103.

[T05-W002] Selektion und Attraktion: Wie Organisationen und Bewerbende sich gegenseitig auswählen und beeinflussen. See S103.

[T05-W003] The validity and utility of selection methods in personnel psychology: Practical and theoretical implications of 85 years of research findings. See S103.

[T05-W004] Revisiting meta-analytic estimates of validity in personnel selection: Addressing systematic overcorrection for restriction of range. See S103.

[T06-W001] Process consultation revisited: Taking a relational practice perspective. See S104.

[T06-W002] Helping at NASA: Guidelines for using process consultation to develop impactful research. See S104.

[T06-W003] Pratiques et identités professionnelles dans le conseil en management en France : entre ethos du service au client et pression du « up or out ». See S104.

[T06-W004] Knowing Through Consulting in Action: Meta-consulting Knowledge Creation Pathways. See S104.

[T06-W005] Does Management Matter? Evidence from India. See S104.

[T07-W001] Theoretical implications of directed forgetting. See S105.

[T07-W002] Intentional Forgetting in Organizations: The Importance of Eliminating Retrieval Cues for Implementing New Routines. See S105.

[T07-W003] Remembrance of things past? The dynamics of organizational forgetting. See S105.

[T08-W001] Challenging conversations and how to manage them. See S106.

[T08-W002] Avoiding Difficult Conversations in the Australian Health Sector. See S106.

[T08-W003] 選ばれていない参加者が発話するとき：もう一人の参加者について言及すること. See S106.

[T09-W001] A Process Integration Primer. See S107.

[T09-W002] Synthesis of heat exchanger networks: II. Evolutionary generation of networks with various criteria of optimality. See S107.

[T09-W003] Simultaneous Process and Heat Exchanger Network Synthesis Using a Discrete Temperature Grid. See S107.

[T09-W004] Retrofitting industrial heat exchanger network based on pinch analysis. See S107.

[T09-W005] Synthesis of heat exchanger networks: I. Systematic generation of energy optimal networks. See S107.

[T10-W001] Investigation of L'Ambiance Plaza Building collapse in Bridgeport, Connecticut. See S108.

[T10-W002] 超大型深基坑半逆作法设计与实践. See S108.

[T10-W003] Renforcement de sol sous ouvrage existant et reprise en sous œuvre [Ground reinforcement below existing structures and underpinning]. See S108.

[T10-W004] FHWA Bridge Preservation Guide. See S108.

[T11-W001] Final Report: System Disturbance on 4 November 2006. See S109.

[T11-W002] Final Report on the August 14, 2003 Blackout in the United States and Canada: Causes and Recommendations. See S109.

[T11-W003] 电力系统黑启动恢复问题的研究评述. See S109.

[T11-W004] Current State of Balance Management in Europe. See S109.

[T12-W001] The phenomena of rupture and flow in solids. See S110.

[T12-W002] Mechanisms of high-temperature fatigue and fracture in silicon carbide ceramics. See S110.

[T12-W003] 高圧水素ガス中におけるオーステナイト系ステンレス鋼溶接金属の SSRT 特性と疲労寿命特性に関する研究 [Slow strain rate and fatigue-life properties of stainless-steel weld metal under high-pressure hydrogen]. See S110.

[T12-W004] Corrosion Fatigue of Refractory Materials in Boiling Nitric Acid. See S110.

[T13-W001] Economic Control of Quality of Manufactured Product. See S111.

[T13-W002] What Is Process Capability? See S111.

[T13-W003] 品質のつくり込みは「ばらつき」への挑戦. See S111.

[T13-W004] Validating a destructive measurement system using Gauge R&R — a case study. See S111.

[T13-W005] การลดของเสียในกระบวนการฉีดพลาสติกของชิ้นส่วนโทรศัพท์ โดยการออกแบบการทดลอง. See S111.

[T14-W001] Cinquante ans d'ingénierie des exigences: un concept suffisamment mûr pour être revisité. See S112.

[T14-W002] NASA Systems Engineering Handbook: Technical Requirements Definition. See S112.

[T14-W003] Mars Climate Orbiter Mishap Investigation Board Phase I Report. See S112.

[T14-W004] The Impact of Requirements on Systems Development Speed: A Multiple-Case Study in Automotive. See S112.

[T14-W005] Sharing Data between Mobile Devices, Connected Vehicles and Infrastructure. See S112.

[T15-W001] Характер операций современных армий. See S113.

[T15-W002] MCDP 1 Warfighting. See S113.

[T15-W003] Desert Storm: Attrition or Maneuver? See S113.

[T15-W004] Clearing the air: Taking manoeuvre and attrition out of strategy. See S113.

[T15-W005] Conduct of the Persian Gulf War: Final Report to Congress. See S113.

[T15-W006] Crossing the Berm: Maneuver Support during Desert Shield and Desert Storm. See S113.

[T16-W001] Τέχνη ῥητορική [Rhetoric]. See S114.

[T16-W002] Quintilian: Institutio Oratoria XI. See S114.

[T16-W003] 《鬼谷子》 [Guiguzi], 捭闔第一 [Opening and closing]. See S114.

[T16-W004] La mise en scène du discours. See S114.

[T16-W005] ‘Four Score and Seven Years Ago…’ Lincoln’s Words Still Inspire 130 Years Later. See S114.

[T16-W006] Gettysburg Address: Hay Copy. See S114.

[T16-W007] Edward Everett to Abraham Lincoln, 20 November 1863. See S114.

[T17-W001] Die Veralltäglichung des Charisma im Verwaltungsstab. See S115.

[T17-W002] La sociologie wébérienne de la domination. See S115.

[T17-W003] Continuing Revelation and Institutionalization: Joseph Smith, Ralph Waldo Emerson and Charismatic Leadership in Antebellum America. See S115.

[T17-W004] Can Charisma Be Taught? Tests of Two Interventions. See S115.

[T17-W005] Origin of the “Reorganized” Church and the Question of Succession. See S115.

[T17-W006] History. See S115.

[TA-W001] But Who Will Guard the Guardians?. See S15.

[TA-W002] Designing Economic Mechanisms. See S15.

[TA-W003] Allocation of Resources in a Team. See S15.

[TA-W004] Geometry of Information Structures, Strategic Measures and associated Control Topologies. See S33.

[TA-W005] Property as the Law of Things. See S15.

[TA-W006] Property as Modularity. See S15.

[TA-W008] A Knowledge Compilation Map. See S42, S50.

[TA-W009] Partial Evaluation and Automatic Program Generation. See S42.

[TA-W010] Proof-Carrying Code. See S42.

[TA-W011] Oracle-Based Checking of Untrusted Software. See S42.

[TA-W012] Matroids and the Greedy Algorithm. See S42.

[TA-W013] Event-Triggered Real-Time Scheduling of Stabilizing Control Tasks. See S05.

[TA-W014] To sample or not to sample: Self-triggered control for nonlinear systems. See S05.

[TA-W016] Economies-of-Scale in Many-Server Queueing Systems: Tutorial and Partial Review of the QED Halfin–Whitt Heavy-Traffic Regime. See S38.

[TA-W017] Kinetic Proofreading: A New Mechanism for Reducing Errors in Biosynthetic Processes Requiring High Specificity. See S40, S50.

[TA-W018] Proofreading through spatial gradients. See S40, S50.

[TA-W019] Outsourcing Memory Through Niche Construction. See S28.

[TA-W020] Niche Construction Theory: A Practical Guide for Ecologists. See S28, S50.

[TA-W021] The Niche Construction Perspective: A Critical Appraisal. See S28.

[TA-W022] Cooperation and conflict in the evolution of multicellularity. See S28.

[TA-W023] Cooperation and Conflict in the Evolution of Complexity. See S28.

[TA-W024] Steps Toward an Ecology of Infrastructure: Design and Access for Large Information Spaces. See S42.

[TA-W025] Reckoning with Standards. See S42.

[TA-W026] Ecological Interface Design: Theoretical Foundations. See S41.

[TA-W027] 生态学界面设计研究综述 [A Review of Research on Ecological Interface Design]. See S41.

[TA-W028] Notes on the Synthesis of Form. See S08.

[TA-W029] A City Is Not a Tree. See S08.

[TA-W030] 测量结果的计量溯源性要求 [Requirements for the Metrological Traceability of Measurement Results] (CNAS-CL01-G002:2021). See S42.

[TA-W031] Metrological Traceability: Frequently Asked Questions and NIST Policy. See S42.

[TA-W032] The NIST Model for Role-Based Access Control: Towards A Unified Standard. See S42.

[TA-W033] Internal Control Management and Evaluation Tool: Exposure Draft (GAO-01-131G). See S42.

[TA-W034] Compression and communication in the cultural evolution of linguistic structure. See S41.

[TA-W035] Cognitive Load During Problem Solving: Effects on Learning. See S41.

[TA-W036] The Expertise Reversal Effect. See S41.

[TA-W037] The fitness costs of developmental canalization and plasticity. See S28.

[TA-W038] Re-evaluating the costs and limits of adaptive phenotypic plasticity. See S28.

[TA-W039] Self-triggering predictive control method for continuous-time networked control system (CN116627039B). See S05.

[TA-W040] The Organization of Decentralized Information Processing. See S14.

[TA-W041] Coarse-graining as a downward causation mechanism. See S46.

[TD-W001] Brief Therapy Based on Interrupting Ironic Processes: The Palo Alto Model. See S73.

[TD-W002] Brief Therapy: Focused Solution Development. See S73.

[TD-W003] Solution Focused Therapy Treatment Manual for Working with Individuals. See S73.

[TD-W004] Effectiveness of Solution-Focused Brief Therapy: A Systematic Qualitative Review of Controlled Outcome Studies. See S73.

[TD-W005] Randomized trial on the effectiveness of long- and short-term psychodynamic psychotherapy and solution-focused therapy on psychiatric symptoms during a 3-year follow-up. See S73.

[TD-W006] Addressing depression and comorbid health conditions through solution-focused brief therapy in an integrated care setting: a randomized clinical trial. See S73.

[TD-W007] 焦点解决短程治疗非自杀性自伤的理念分析 [An analysis of solution-focused brief therapy for nonsuicidal self-injury]. See S73.

[TD-W008] 事例とコンサルティを活かす解決志向ブリーフセラピーのコンサルテーション–11 ステップモデルの効果研究と実践への誘い– [Solution-focused brief-therapy consultation drawing on the case and consultee: an effectiveness study of an eleven-step model and an invitation to practice]. See S73.

[TD-W009] Use of Solution-Focused Brief Therapy Within Adult Community Mental Health: A Scoping Review. See S73.

[TD-W010] [Citation Classic commentary on “Antibiotic susceptibility testing by a standardized single disk method”]. See S74.

[TD-W011] Recommandations 2025 [Recommendations 2025]. See S74.

[TD-W012] Effect of point of care testing for C reactive protein and training in communication skills on antibiotic use in lower respiratory tract infections: cluster randomised trial. See S74.

[TD-W013] Biomarkers as point-of-care tests to guide prescription of antibiotics in people with acute respiratory infections in primary care. See S74.

[TD-W014] Prescribing antibiotics: the use of diagnostic tests in general practice. A register-based study. See S74.

[TD-W015] Implantación de una prueba rápida de infección estreptocócica: ¿su uso también mejora la adherencia antibiótica? [Introducing a rapid test for streptococcal infection: does its use also improve antibiotic adherence?]. See S74.

[TD-W016] Rapid Respiratory Microbiological Point-of-Care Testing and Antibiotic Use in Primary Care. See S74.

[TD-W017] Randomized Trial Evaluating Clinical Impact of RAPid IDentification and Susceptibility Testing for Gram-negative Bacteremia: RAPIDS-GN. See S74.

[TD-W018] Impact of rapid antibiotic susceptibility testing for Gram-negative bacteremia varies by pathogen type and resistance: a secondary analysis of the RAPIDS GN trial. See S74.

[TD-W019] Rapid versus standard antimicrobial susceptibility testing to guide treatment of bloodstream infection. See S74.

[TD-W020] 抗菌薬適正使用に貢献する薬剤感受性検査とは [Susceptibility testing that contributes to appropriate antimicrobial use]. See S74.

[TD-W021] 常见细菌药物敏感性试验报告规范中国专家共识 [Chinese expert consensus on standardized reporting of susceptibility tests for common bacteria]. See S74.

[TD-W022] Rapid Diagnostics and Rapid Antimicrobial Susceptibility Testing for Bloodstream Infections: A Narrative Review and Proposed Clinical Actionability Framework. See S74.

[TD-W023] Procalcitonin to Reduce Long-Term Infection-associated Adverse Events in Sepsis. A Randomized Trial. See S74.

[TD-W024] Algorithm Selection for Combinatorial Search Problems: A survey. See S75.

[TD-W025] A Survey of Adaptive Sorting Algorithms. See S75.

[TD-W026] Lecture 7: Linear-Time Sorting. See S75.

[TD-W027] Engineering Radix Sort. See S75.

[TD-W028] Engineering a Sort Function. See S75.

[TD-W029] Nearly-Optimal Mergesorts: Fast, Practical Sorting Methods That Optimally Adapt to Existing Runs. See S75.

[TD-W030] L'algorithme de tri ShiversSort adaptatif [The adaptive ShiversSort sorting algorithm]. See S75.

[TD-W031] CPython 3.14.0. See S75.

[TD-W032] Time Bounds for Selection. See S75.

[TKT-W001] Тектология: Всеобщая организационная наука [Tektology: Universal Organizational Science]. See S17.

[TKT-W002] Всеобщая организационная наука (Тектология). Часть II [Universal Organizational Science (Tektology), Part II]. See S17.

[TKT-W003] Mécanismes communs aux phénomènes disparates [Mechanisms Common to Disparate Phenomena]. See S17.

[UX-W001] A theory of technology and organizations. See S07.

[UX-W002] Coordinating Activity Interdependencies in the Contemporary Economy: The Principle of Distributed Control. See S14.

[UX-W003] Information and Organizations. See S13.

[UX-W004] Normal Accidents: Living with High-Risk Technologies. See S29.

[UX-W005] Organizational Strategies for Complex System Resilience, Reliability, and Adaptation. See S30.

[UX-W006] Social complexity and sustainability. See S48.

[UX-W007] Cybersyn, big data, variety engineering and governance. See S22.

[UX-W008] Where Do Transactions Come From? A Network Design Perspective on the Theory of the Firm. See S07.

[UX-W009] Where do transactions come from? Modularity, transactions, and the boundaries of firms. See S07.

[UX-W010] Management and information: the complementarity control-autonomy. See S22.

[UX-W011] Dependency Restructuring: How Technological Enactment Changes Organizational Requirement Relations. See S14.

[UX-W012] The Physics of Symbols: Bridging the Epistemic Cut. See S42.

[UX-W013] From complexity to simplicity: nature and symbols. See S42.

[UX-W014] The Physical Basis and Origin of Hierarchical Control. See S42.

[UX-W015] Outsourcing Control requires Control Complexity. See S37.

[UX-W016] Celebrating a Citation Classic—and More: Symposium on Charles Perrow's Normal Accidents. See S29.

[UX-W017] On Interconnections, Control, and Feedback. See S04.

[UX-W018] Viability theory for an ecosystem approach to fisheries. See S43.

[UX-W019] The Behavioral Approach to Open and Interconnected Systems. See S04.

[UX-W020] A New Accident Model for Engineering Safer Systems. See S29.

[UX-W021] Engineering a Safer World: Systems Thinking Applied to Safety. See S29.

[UX-W022] How Morphological Computation Shapes Integrated Information in Embodied Agents. See S37.

[UX-W023] The centrality of the machine in the thought of Jacques Lafitte. See S39.

[UX-W024] La mécanologie : une lignée technologique francophone ? [Mechanology: A Francophone Technological Lineage?]. See S39.

[UX-W025] Sur la classification des objets techniques selon Simondon [On the Classification of Technical Objects According to Simondon]. See S39.

[UX-W026] Mechanological Dynamics in Jacques Lafitte and Gilbert Simondon. See S39.

[UX-W027] La théorie du système général. Théorie de la modélisation [General System Theory: Theory of Modeling]. See S23.

[UX-W028] Le complexe de Babel, crise ou maîtrise de l'information ? [The Babel Complex: Crisis or Mastery of Information?] (book review). See S27.

[UX-W029] Des pyramides du pouvoir aux réseaux de savoirs [From Pyramids of Power to Networks of Knowledge]. Tome 1. See S27.

[UX-W030] Polycellular Human Organisation: For a Transdisciplinary Approach. See S27.

[UX-W031] Crise, information et communication dans l'économie américaine [Crisis, Information and Communication in the American Economy]. See S27.

[UX-W032] The Body & Society: Explorations in Social Theory. See S45.

[UX-W033] Der Mensch: Seine Natur und seine Stellung in der Welt [Man: His Nature and Place in the World]. See S45.

[UX-W034] Erkenntnis und Entlastung – zu Arnold Gehlens Verständnis von Wissenschaft und Technik im Rahmen der Klassischen Wissensordnung [Knowledge and Relief: On Arnold Gehlen's Understanding of Science and Technology Within the Classical Order of Knowledge]. See S45.

[UX-W036] からくり改善とは [What is karakuri kaizen?]. See S02.

[UX-W037] からくり改善のメカニズム ～動力源とからくり機構～ [The Mechanisms of Karakuri Kaizen: Power Sources and Karakuri Mechanisms]. See S02.

[UX-W038] Design rules: past and future. See S07.

[UX-W039] Die Seele im technischen Zeitalter: Sozialpsychologische Probleme in der industriellen Gesellschaft [The Soul in the Technological Age: Social-Psychological Problems in Industrial Society]. See S45.

[UX-W040] Design Rules, Volume 2: How Technology Shapes Organizations. See S07.

[UX-W041] A Theory of Cheap Control in Embodied Systems. See S37.

[UX-W042] Architecture, constraints, and behavior. See S05.

[UX-W043] Contrasting views of complexity and their implications for network-centric infrastructures.

[W001] An Introduction to Cybernetics. See S03.

[W002] Organization Design: An Information Processing View. See S14.

[W003] From Complex Organizations with Simple Jobs to Simple Organizations with Complex Jobs. See S14.

[W004] On the Criteria To Be Used in Decomposing Systems into Modules. See S07.

[W005] Coordination Avoidance in Database Systems. See S10.

[W006] Self-stabilizing Systems in Spite of Distributed Control. See S11.

[W007] Trade-offs in exploiting body morphology for control: from simple bodies and model-based control to complex bodies with model-free distributed control schemes. See S37.

[W008] What Is Morphological Computation? On How the Body Contributes to Cognition and Control. See S37.

[W009] 欠驱动柔性机器人的振动可控性分析 [Analysis of Vibration Controllability in Underactuated Flexible Robots]. See S37.

[W010] 受動歩行の力学的原理と応用 [Mechanical Principles and Applications of Passive Walking]. See S01.

[W011] 基于形态的具身智能研究: 历史回顾与前沿进展 [Morphology-Based Embodied Intelligence Research: Historical Review and Frontier Advances]. See S37.

[W012] Robust perfect adaptation in bacterial chemotaxis through integral feedback control. See S05, S50.

[W013] Biological organisation as closure of constraints. See S40.

[W014] Dissipative Dynamical Systems Part I General Theory. See S04.

[W015] The Architecture of Complexity. See S14.

[W016] The Use of Knowledge in Society. See S15.

[W017] Всеобщая организационная наука (Тектология). Часть I [Universal Organizational Science (Tektology), Part I]. See S17.

[W018] Traktat o dobrej robocie [Praxiology: An Introduction to the Science of Efficient Action]. See S45.

[W019] Sprawność i błąd. Z myślą o dobrej robocie nauczyciela [Efficiency and Error: With Good Teaching Work in Mind]. See S45.

[W020] Du mode d'existence des objets techniques [On the Mode of Existence of Technical Objects]. See S39.

[W021] Die ökonomische Natur der physikalischen Forschung [The Economical Nature of Physical Research]. See S45.

[W022] О психологии изобретательского творчества [On the Psychology of Inventive Creativity]. See S44.

[W023] Characterizing the Thermal Response of Fusible-Link Sprinklers. See S02.

[W024] Improvement in fire-extinguishers. See S02.

[W025] A Mathematical Theory of Communication. See S06.

[W026] Highly Optimized Tolerance Robustness and Power Laws in Complex Systems. See S05.

[W027] Ensuring Convergence and Invariants Without Coordination. See S10.

[W028] Resolução de conflitos em centros integrados de cidadania: uma abordagem à luz da perspectiva sociotécnica de grupos de trabalho [Conflict Resolution in Integrated Citizenship Centers: An Approach Through the Sociotechnical Perspective on Work Groups]. See S14.

[W029] Equivalent Comparisons of Experiments. See S33.

[W030] Hierarchies and the Organization of Knowledge in Production. See S14.

[W031] La reconstruction du nid et les coordinations interindividuelles chez Bellicositermes natalensis et Cubitermes sp. La théorie de la stigmergie [Nest Reconstruction and Interindividual Coordination in Bellicositermes natalensis and Cubitermes sp.: The Theory of Stigmergy]. See S49.

[W032] Canalization of Development and the Inheritance of Acquired Characters. See S28, S50.

[W033] Robustness in simple biochemical networks. See S05.

[W034] The trade-off between morphology and control in the co-optimized design of robots. See S37.

[W035] 一个科学新领域 开放的复杂巨系统及其方法论 [A New Scientific Field: Open Complex Giant Systems and Their Methodology]. See S18.

[W036] 钱学森关于开放的复杂巨系统的研究 [Qian Xuesen's Research on Open Complex Giant Systems]. See S18.

[W037] The Principles of Design. See S44.

[W038] 源流検査とポカヨケ・システム : 不良=0 への挑戦 0QC 方式への展開 [Source Inspection and the Poka-Yoke System: The Challenge of Zero Defects and Development Toward the Zero Quality Control Method]. See S02.

[W039] What you don't have can't leak. See S02.

[W040] Environments That Make Us Smart. See S41.

[W041] On Distinguishing Epistemic from Pragmatic Action. See S41.

[W042] The Theory of Affordances. See S41.

[W043] Probabilistic Logics and the Synthesis of Reliable Organisms from Unreliable Components. See S30.

[W044] Über die Lage der Integralkurven gewöhnlicher Differentialgleichungen [On the Position of the Integral Curves of Ordinary Differential Equations]. See S05, S54.

[W046] Organization Design: Current Insights and Future Research Directions. See S14.

[W047] Organizational normativity and teleology a critique. See S40.

[W048] Body Machinery and Intelligence.

[W049] Governing the Commons The Evolution of Institutions for Collective Action. See S15.

[W050] Every good regulator of a system must be a model of that system. See S03.

[W051] System 0/1/2/3 Quad-process theory for multi-timescale embodied collective cognitive systems. See S46.

[W052] The free-energy principle a unified brain theory. See S46.

[W053] Passive Dynamic Walking. See S01.

[W054] Morphological Computation Nothing but Physical Computation. See S37.

[W055] A Technical Critique of Some Parts of the Free Energy Principle. See S46.

[W056] Some Interesting Observations on the Free Energy Principle. See S46.

[W057] Coordination Avoidance in Distributed Databases. See S10.

[W058] On the Mathematical Foundations of Theoretical Statistics. See S33.

[W059] Comparison of Experiments. See S33.

[W060] Robust discrete optimization and network flows. See S05.

[W061] A comprehensive study of Convergent and Commutative Replicated Data Types. See S11.

[W062] Organization for Physiological Homeostasis. See S05.

[W063] 組織における不適切な人間行動とそのリスク評価 [Inappropriate Human Behavior in Organizations and Its Risk Assessment]. See S02.

[W064] The role of inherently safer design in process safety. See S02.

[W065] Common Cause Failure Modeling: Aerospace Versus Nuclear. See S30.

[W066] 독립적 기능 제어를 이용한 몸통이 부착된 이족 로봇의 자연스럽고 유연한 평지 보행 [Natural and Flexible Level-Ground Walking of a Torso-Equipped Biped Robot Using Independent Function Control]. See S01.

[W067] Coordinación emergente en sistemas multirobots [Emergent Coordination in Multirobot Systems]. See S49.

[W068] Knowledge Specialization, Organizational Coupling, and the Boundaries of the Firm: Why Do Firms Know More Than They Make?. See S12.

[W069] On Governors. See S04.

[W070] Dynamic Models of Segregation. See S49.

[W071] Comparisons of Signals. See S33.

[W072] An Outline of General System Theory. See S18.

[W073] Stigmergy as a universal coordination mechanism I: Definition and components. See S49.

[W074] Réflexions sur la science des machines [Reflections on the Science of Machines]. See S39.

[W075] On theories with a combinatorial definition of equivalence.

[W076] 作業のフールプルーフ化に関する研究 —フールプルーフ化の原理— [A Study of Foolproofing Work: Principles of Foolproofing]. See S02.

[W077] Chemical Organisation Theory. See S40, S50.

[W078] 综合集成方法体系与系统学研究 [Research on the Meta-synthesis Methodological System and Systematology]. See S18.

[W079] On the validity of entropy production principles for linear electrical circuits. See S46.

[W080] On Self-Organizing Systems and Their Environments. See S18.

[W081] Autopoiesis: The organization of living systems, its characterization and a model. See S40.

[W082] The constructal law of design and evolution in nature. See S46.

[W083] Can we apply synergetics to the human sciences?. See S46.

[W084] Assembly theory explains and quantifies selection and evolution. See S46.

[W085] Assembly Theory: What It Does and What It Does Not Do. See S46.

[W086] On the Economy of Machinery and Manufactures. See S45.

[W087] A Behavioral Notion of Subtyping. See S42.

[W088] Um modelo para o compartilhamento de conhecimento no trabalho [A Model for Knowledge Sharing at Work]. See S42, S53.

[W089] What Is a Condition Number?. See S35.

[W090] What is Numerical Stability?. See S35.

[W091] Constructor theory of information. See S46.

[W092] Constructor theory of life. See S40.

[W093] Hierarchies, Knowledge, and Power Inside Organizations. See S48.

[W094] The political economics of complexity: From the information economy to the ‘complexity’ economy. See S27.

[W095] Impact of Process Tampering on Variation (Experiment). See S05.

[W096] Theory of Hierarchical, Multilevel, Systems. See S09.

[W097] Hierarchical models of very large problems, dilemmas, prospects, and an agenda for the future. See S09.

[W098] Embodying physical computing into soft robots. See S37.

[W099] Energy as a Common Currency for Understanding Embodied Evolution. See S37.

[W100] 自律分散知能システム [Autonomous Distributed Intelligent Systems]. See S49.

[W101] The Heart of Enterprise. See S22.

[W102] A relational theory of biological systems. See S26.

[W103] An applied mathematician’s perspective on Rosennean Complexity. See S26, S50.

[W104] Relational Biology. See S26, S50.

[W106] Using Systems Archetypes in Public Health: A Review Connecting System Dynamics Literature and Public Health Applications. See S44.

[W107] Leverage Points: Places to Intervene in a System. See S44.

[WME-W001] The Self-Organizing Universe: Scientific and Human Implications of the Emerging Paradigm of Evolution. See S19.

[WME-W002] Review of Erich Jantsch, The Self-Organizing Universe. See S19.

[WME-W003] Systems Thinking: Managing Chaos and Complexity: A Platform for Designing Business Architecture. See S20.

[WME-W004] Living Systems. See S21.

[WME-W005] A Response to Miller's Editorial. See S21.

[WME-W006] The Systems View of Life: A Unifying Vision. See S24.

[WME-W007] Book review: The Systems View of Life: A Unifying Vision. See S24.

[WME-W008] Reflections on Autopoiesis, the Signature of Life. See S24.

[WME-W009] Goodman's Aesthetics. See S41.

[WME-W010] Least effort and the origins of scaling in human language. See S41.

[WME-W011] The Interdisciplinary Study of Coordination. See S14.

[WME-W012] Primer on Homeokinetics: A Physical Foundation for Complex Systems. See S25.

[WME-W013] Outlining social physics for modern societies—locating culture, economics, and politics: The Enlightenment reconsidered. See S25.

[WME-W014] What is complexity?. See S25.

[WME-W015] Bios Theory of Innovation. See S46.

[WME-W016] Katsenelinboigen, Aron. See S34.

[WME-W017] Introduction to Personal Construct Psychology. See S41.

[WME-W018] Your Questions, Answered — Personal Construct Psychology FAQ. See S41.

[WME-W019] Схема мыследеятельности — системно-структурное строение, смысл и содержание [The Schema of Thought-Activity: Systemic-Structural Organization, Meaning and Content]. See S44.

[WME-W020] A systems perspective on systemic innovation. See S44.

[WME-W021] Subdynamics in Knowledge-Based Systems. See S47.

[WME-W022] Towards a Synthesis: Person-Centered Systems Theory. See S41.

[WME-W023] The Internal Family Systems Model Outline. See S41.

[WME-W024] Comments on “the return of information theory”. See S47.

[WME-W025] Quantifying sustainability: Resilience, efficiency and the return of information theory. See S47.

[WME-W026] 物理—事理—人理系统方法论：一种东方的系统思考 [Wuli–Shili–Renli Systems Methodology: An Eastern Approach to Systems Thinking]. See S18.

[WME-W027] 我对系统学认识的历程 [The Course of My Understanding of Systematology]. See S18.

[WME-W028] 科学技术体系学研究的开创者 [A Pioneer in the Study of the System of Science and Technology]. See S18.

[WME-W029] Grammatical Man: Information, Entropy, Language, and Life. See S36.

[WME-W030] Grammatical Man — review. See S36.

[XA-W001] Exokernel: An Operating System Architecture for Application-Level Resource Management. See S84.

[XA-W002] Application Performance and Flexibility on Exokernel Systems. See S84.

[XA-W003] Unikraft: Fast, Specialized Unikernels the Easy Way. See S84.

[XA-W004] Unikernel Linux (UKL). See S84.

[XA-W005] ライブラリ OS をとりまく現況 [The current landscape of library operating systems]. See S84.

[XA-W006] Unikernel Linux (UKL) moves forward. See S84.

[XA-W007] Unikraft and the Coming of Age of Unikernels. See S84.

[XA-W008] Exopc Exokernel Distribution. See S84.

Subject index

Entries follow the manuscript's substantive headings. Identifiers refer to sections; digital links locate the corresponding heading.

A backward tree and a forward storyline — S94

A brief intervention in primary care — S73

A broad synthesis with several kinds of claim — S24

A cheap test before expensive redesign — S62

A City Is Not a Tree: preserving overlapping systems of use — S08

A closed path and full closure differ — S26

A combinatorial account of discovery — S82

A common project is made and maintained — S27

A communication link must carry usable distinctions — S22

A community in which needs and partners emerge — S83

A concert contract seen through two financial problems — S82

A connected organizational account — S20

A contemporary model of enterprise AI pilots — S62

A contract can constitute hierarchy across firm boundaries — S13

A controlled method comparison reports a trade-off — S29

A convergent scalar can still be the wrong counter — S11

A distinct Italian programme: structure, interaction, and compatibility — S22

A field of programmes, models and questions — S18

A form before its purpose — S82

A later exchange clarifies what counts as herding — S59

A later X-36 account: reconfiguration after an effector failure — S54

A physical constraint can change the necessary response chain — S29

A profitable investment that is worth postponing — S61

A queueing benchmark for pooling — S38

A rank without a full order — S75

A record becomes useful through selective activation — S19

A representation is selected for a question — S23

A reusable rule and a complete response sequence differ — S03

A roof that changes its exposed colour — S82

A separate decision box is not always required — S23

A sieve selects a property, not a purpose — S17

A smaller channel can require a longer process — S03

A specified unit at an interface — S112
A supplied component does not specify the whole obligation — S12
A system is a selected state description — S03
A theory of modeling and its changing layers — S23
A verbal prompt at the point of ordering — S79
Abraham, Tribolet and Winter: transformation as well as operation — S09
Acar and colleagues and different kinds of constraint — S63
Accelerated learning — S72
Accident, sagacity and an unexpected finding — S83
Accounts and measurement of creative connection — S82
Ackerman and a memory that grows through questions — S60
Ackoff: idealized design, realization and decision rules — S44
Acquiring, maintaining and contesting power — S91
Acting together under uncertainty — S16
Action formation — S71
Actual feedback differs from evidence that a command was issued — S29
Actual processes and long-lived platforms — S88
Adamowicz and the selection and assembly of images — S65
Adaptation and a programmable sensing interface — S50
Adaptation during the work — S75
Adapting sales encounters and assessing their results — S79
Adjustment can change more than the immediate response — S21
Administrative burden: three kinds of cost — S48
Agroecology: a framework and a limited performance account — S43
Algorithm selection — S75
Algorithm selection, adaptive sorting and computing only what is needed — S75
Alienation: technology and its social employment — S48
Allocating a function changes dependencies — S21
Allocating when the distribution is uncertain — S90
Alon on response time and persistence detection — S50
Alter: work systems include customers and continuing adaptations — S14
Altshuller and Shapiro, and a later QFD–function-analysis–TRIZ proposal — S44
Alvaro, Rosen, and Hellerstein: choosing faults from an outcome's dependencies — S30
Amortization and the retained preparation — S35
Amplification retains the source of supplementation — S03
Amyotte and Khan: removing a hazard is hazard-specific — S02

An adaptive arrangement must find and retain an acceptable state — S03
An address in manuscript and circulation — S114
An advance, a continuation and the work of implementation — S77
An adverse case: decomposition can enlarge the central problem — S09
An application supplies conditional simulation evidence — S29
An ensemble can require separation rather than a common milieu — S39
An official practice account: resilience retains resources and responsibility — S30
An original article and a chain of development — S27
An unexpected model for a library service — S83
An unrestricted mathematical controller can make the problem trivial — S04
Analogy requires evidence about a mechanism — S17
Anta and Tabuada: compute the next occasion from current information — S05
Anticipation is not the same as life or impredicativity — S26
Archives: context is part of description — S42
Argyris: questioning the assumptions that govern correction — S51
Arrangements have developmental and evolutionary histories — S28
Arrangements that preserve a later choice — S34
Arranging interventions for children — S70
Assembling support before a decision — S91
Assembly theory: a construction measure with specified building blocks — S46
Assembly, maintenance and historical origin differ — S24
Assessment goes in both directions — S103
Assessment, appearances and military strength — S93
Assistance includes services and participation around an object — S41
Association, bisociation and the construction of new ideas — S82
Associative routes and practical requirements — S82
Attempted solutions — S73
Attention, interest, desire, action and satisfaction — S76
Authority, resources and insight require actual bearers — S29
Automation can remove interactions that were doing useful work — S16
Automation changes cognitive and contingency demands — S29
Autopoiesis maintains an organization through exchange — S24
Available storage is a limit on extraction — S04
Axenie and colleagues: three scales of response — S32
Aznar and what an exercise is intended to accomplish — S63
Babbage: distribute the intellectual work of calculation — S45

Bacchetta and van Wincoop: finding a scapegoat in the price — S59
Backcasting, participation and the choice of a future — S94
BACON.5 and a discovered relation used as new data — S64
Bailis's dissertation: specification and mixed execution — S10
Baird and colleagues and returning to the same material — S67
Bateson: restraints and levels of choice — S18
Bayesian design: a limited but explicit decision framing — S33
Beer on modeling, identity and informational economy — S22
Behavior, variety and regulation — S03
Behaviors that observers rate — S115
Being strategic at work — S92
Bejan and Lorente: flow access is their proposed general principle — S46
Bertalanffy: what formal recurrence can and cannot show — S18
Bertsimas and Sim: robustness is chosen against a stated uncertainty class — S05
Beven: predictive agreement can leave explanatory alternatives — S43
Biehl and colleagues, and the response: a scoped mathematical dispute — S46
Billett and Orth: participation is not guaranteed by an available setting — S41
Bitner: the physical service setting is part of action — S41
Black-box knowledge does not identify a unique interior — S03
Blackwell's 1951 comparison: attainable losses — S33
Bonnardel and Lubart and constraints that make an analogy useful — S64
Bonneuil and Saint-Pierre: compute which choices preserve a route to a target — S54
Bontems: several aspects can coexist in the same object — S39
Borrego and colleagues: a losing operation can become ineffective — S10
Bousquet: a history of scientific images is not completed transformation — S16
Bowen: apparent stability can leave a problem unresolved — S41
Boyd: orientation grows through interaction — S16
Branching and meshed communications differ — S27
Brand: different rates of change can retain different functions — S44
Brennan and Clark: provisional agreements with particular partners — S56
Breton and the procedure of automatic writing — S65
Brief therapy: goals, exceptions and practical change — S73
Brier: connect stated probabilities with subsequent events — S31
Brooks, Frankel and Kamenica: information already held can change the comparison — S33
Bruers, Maes and Netočný: the variable and reversal symmetry matter — S46
Brusoni, Prencipe and Pavitt: knowing can extend beyond making — S12

Buckland: stored things are not identical to knowledge — S42
Buffers, releases and operating priorities — S85
Building a route through a performance — S71
Building an action in stages — S71
Building, maintaining and renewing personal and professional networks — S97
Bush and the preservation of an enquiry — S60
Bush and transformations of the technical problem — S64
Buttari and colleagues: a cheap factorization with more accurate correction — S55
Börgers: a readable finite reconstruction of dominance — S33
Cage and the composition of chance — S66
Cannon: constancy can require ongoing cooperation — S05
Cargo-aircraft criteria: protection must retain a usable response — S54
Carlson and Doyle: optimization can expose an unanticipated fragility — S05
Carrying a direction through everyday work — S92
Carson and Higham: correction accuracy and convergence are different requirements — S55
Cascades and service cycles retain providers — S24
CAST reconstructs a wider control failure — S29
Caulkin's account of Seddon: some recurring work is created by earlier failure — S31
Ceroni and colleagues measure the host's remaining capacity — S50
Challenge point — S72
Challenge, difficulty and accelerated learning — S72
Challenging plans before action — S95
Change and acceptance evidence — S112
Change the invariant and the answer changes — S10
Changing a crude-oil preheat train — S107
Changing a working platform with its maintainers — S84
Changing identifier allocation changes the problem — S10
Changing support during continued use — S108
Changing the next turn — S106
Changing willingness to start — S99
Charisma, follower recognition and continuing authority — S115
Checkland: use declared activity models to question a situation — S44
Chemical process arrangements and integration — S107
Chen and colleagues: controllability depends on the arrangement — S37
Choosing a method for an instance — S75
Choosing which runs to merge — S75

Chwe: knowing that others received the message — S59
Civil incident organization and public accountability — S16
Clark and Blumenthal: stakeholders can change the architecture's problem — S12
Clark and Wilkes-Gibbs: a reference is completed together — S56
Classical selling models and accounts of buying — S76
Clausewitz: apparent simplicity and the resistances of execution — S16
Co-design of morphology and control — S37
Co-evolution changes the surrounding possibilities — S19
COFI and the design of a creative partnership — S69
Cognition is broader than consciousness in this account — S24
Cohen and Levinthal: prior knowledge and the organization of reception — S53
Collaborative generation and material methods — S68
Collaborative generation for accessible websites — S68
Collier and colleagues: investing resource revenues requires an investment process — S53
Command by veto: a prepared default with continuing oversight — S16
Common understanding, local decisions and accountability — S16
Comparing a language model with oblique prompts — S69
Comparing taught methods — S82
Compartmentalisation: a biological boundary includes pathways and oversight — S43
Competing accounts of a campaign — S113
Complementarity: valuable together need not mean inseparable in every respect — S07
Computer-supported and human–AI co-creation — S69
Conant and Ashby: what the “good regulator” result optimizes — S03
Confidence, evidence and participation in the customer’s work — S76
Conklin and Begeman on recording deliberation — S60
Connected descriptions without a supermodel — S19
Connecting experiments with changes in practice — S31
Connecting tests to decisions that can still change — S62
Connecting through a suitable intermediary — S17
Conservation: preserving information can constrain intervention — S39
Consolidated bioprocessing: integration creates a compatibility question — S43
Constraint can enable variety — S36
Constraint compares an actual set with alternatives — S03
Constructed wetlands: the treatment medium still has to be established and maintained — S43
Constructing alternatives to harmful coping — S73
Constructing and adopting aims — S98

Constructing and interpreting ideas — S82
Constructing portfolios through relationships among investments — S89
Constructor information: physical possibility includes alternatives — S46
Consultancy and the client's capacity to act — S104
Consultation and the creation or revision of a buying vision — S77
Contacts, elapsed time and lasting improvement — S73
Continuation and disputed succession — S115
Controlled traffic: geometry must match equipment and remain usable — S43
Cooling fins that also reinforce an engine — S39
Coordinating an HTTP response across storage and networking — S84
Copying files without touching their contents — S84
Correcting treatment after a positive blood culture — S74
Correctness still depends on what was specified — S10
Corti: causal conditions do not by themselves settle what ought to happen — S40
Costs are tied to purposes and time — S21
Costs of responsiveness require a careful comparison — S28
Counts and bits are different numerical forms — S03
Coupling supplies a relation that a parts list lacks — S03
Critical processes and the decider condition — S21
Critical review and adversary perspectives — S95
Criticism, counterproposals and consensus — S95
Critics: accessibility and the need for internal mechanisms — S23
Cross-functional water-system design — S20
Cross-selling and differences among retail settings — S79
CRP testing — S74
Culture and learning reproduce the arrangement — S20
Curvature, payoff direction and the meaning of improvement — S32
Cury and colleagues: remain within constraints rather than maximize one target — S43
Danchin, Binder and Noria: maintaining a cell rather than merely reproducing it — S32
Daoism through Hansen: guidance and normative autonomy — S45
Darwiche and Marquis: choose a representation for the operations required — S42
De Florio: fidelity and the cost of adaptation — S32
De Sitter and colleagues: redesign production before allocating control — S14
Deception, cunning and misdirection — S93
Deciding when enough treatment has been given — S74
Deciding when to rebalance — S90

Deciding whether to prescribe — S74
Deciding who should have the conversation — S106
Decision rules and procedural fairness — S96
Degression: protection and fixation without command — S17
Delegation, independent top-k selection — S75
Deliberate practice — S71
Dell: inventory and finance across an organizational boundary — S07
Dependence outside the hierarchy — S91
Dependence, boundary and closure are specified rather than inferred from a cycle — S40
Dependencies, bottlenecks and the flow of project work — S85
Design and realization are separate tasks — S20
Design rules replace selected lateral dependencies — S07
Designed constraints must survive operation and change — S29
Designed rules and emergent work both matter — S24
Designing evidence of learning — S94
Designing for measurement and improvement — S111
Designing practice and developing skill — S71
Designing the report that reaches the clinician — S74
Developing solutions from exceptions — S73
Developmental resources extend beyond the conventional organism — S28
Dewey: habit incorporates an environment — S45
Diagnostic stewardship — S74
Diagnostic testing and the choice of antimicrobial treatment — S74
Diamond and Boyd: equilibration without reading individual matrix entries — S35
Diaz-Bone: invested forms extend coordination but retain interpretation — S15
Diehl and Stroebe: giving everyone a chance to speak is a production constraint — S52
Dierickx and Cool: expenditure flows and accumulated capabilities — S53
Different kinds of partial order — S75
Different levels of creative accomplishment — S70
Different meanings of adaptation and different measures of success — S79
Different platforms retain different work — S07
Different purposes for teaching improvisation — S66
Different senses of finality — S18
Differentiation produces both capacity and coordination problems — S17
Difficult conversations under emotional and relational pressure — S106
Dijkstra: recovery differs from preserving a good initial state — S11

Directing attention, effort and learning — S98
Discovering trades and making an agreement operate — S96
Discovering valuable problems and choosing what to build — S87
Distinct questions about organized persistence — S25
Distributed control has a setup phase — S14
Distributed intelligence as a research landscape — S49
Distributed practice — S72
Dittrich and Speroni di Fenizio: chemical closure is a property of a reaction set — S40
Divisionalization combines local integration and corporate abstraction — S13
Dori, DSRP, grey prediction and Extenics: representations with stated limits — S44
Double and multi-contingency: order between two agents may not scale — S49
Double contingency: expectations about expectations — S47
Dow and colleagues: make several prototypes before receiving feedback — S52
Doyle and conclusions whose support changes — S60
Doyle and Csete: constraints that permit flexibility — S05
Drive, cues and reinforcement in the Behavioral Equation — S76
Dunbar and Blanchette and analogies used at work — S64
Duus: the bill for discretion is paid in preparation — S16
Dynamic control, information and model revision — S09
Eccles through a doctrinal attribution — S38
Ecological interfaces: display functional relations rather than isolated readings — S41
Ecology supplies dependencies and tolerance limits — S24
Economic interdependence and the limits of a plan — S17
Edmonds: greedy choice depends on an independence structure — S42
Edmondson and Cannon: seeing failure is a different competency from analyzing it — S31
Edmondson: interpersonal risk and the observable work of learning — S51
Effective action, representation and institutions — S45
Effort and the learning edge — S72
Egression: concentrating action through unequal influence — S17
Electrical power networks and coordinated operation — S109
Electronic brainstorming and time to use others' ideas — S69
Elowitz and Leibler construct an oscillator — S50
Embedded adjustment still requires knowledge and invention — S08
Emergy: proposed accounting for paid and unpaid support — S47
Emotion and misleading truths in negotiation — S93
Encounters and the development of a friendship — S97

Energizing a network in stages — S109
EnerJ: protect the part that must remain precise — S55
Engelbart and a working structure that can be revisited — S60
Entry order and the resources of the entrant — S61
Environments, skills and shared practices — S41
Espejo's later account: residual variety depends on local organization — S22
Evans and Madrzykowski: the thermal trigger has a response time — S02
Evolution and changing organization — S19
Evolving a feasible network — S107
Examining a course of action through a game — S95
Exceptions, therapeutic — S73
Exchange as a continuing relationship — S102
Exchanging information to coordinate production — S06
Expected returns and joint variability — S89
Experimentation as a shared source of knowledge — S61
Expertise, routines and organizational relationships — S13
Explaining an assumed failure — S95
Explanatory illustration: a property that survives union — S10
Exploration before a customer has specified the product — S62
Exploring a space and changing what it permits — S82
Exposing assumptions behind rival plans — S95
External memory and continuity of work — S60
FAO precaution: objectives, limits and the ability to observe them — S43
FDA: the interface includes more than a display — S02
Fear, avoidance and learning to approach — S101
Feedback and the people who manage goals — S98
Feedback choice — S71
Ferrer i Cancho and Solé: a model with competing effort proxies — S41
Findeisen and colleagues: control must work on an operating system — S09
Finding a hypothesis between disconnected literatures — S82
Finding needs ahead of the market — S87
Firepower and the scale of operations — S113
First offers and the unfolding exchange — S96
Fisher and Ury: interests, options, standards and alternatives — S96
Fisher: reduction and the criterion of sufficiency — S33
Flexibility, ageing and the boundary of biological benefit — S32

Flight-deck automation: responsibility includes monitoring and procedure revision — S12
Flow production: every necessary step still matters — S07
Foerster: the boundary of self-organization — S18
Following a molding experiment into production — S111
Following an intervention into operations — S104
Forecasting from completed projects — S86
Forecasts, commitments and governing uncertain projects — S86
Forest adaptation: compare strategies while continuing observation — S43
Forgetting, discarding and revising retained knowledge — S105
Form, correction and overlapping use — S08
Formal definitions and quantitative claims — S23
Founding and developing technology ventures — S100
Four requirements for coordination avoidance — S10
Fox: threat, environment and one's own organization — S38
FPTuner: allocate precision under a conservative expression-wide bound — S55
Framing problems and organizing inquiry — S44
French, German and allied statements retain boundaries — S16
Friston: variational free energy concerns a model of observations — S46
From a chart signal to an investigation — S111
From a desired future to situated action — S98
From a teacher's case knowledge to a workable response — S73
From an arrangement to an admissible behavior — S04
From an encounter to a useful connection — S83
From an imagined stitch to a space-suit closure — S82
From an inventive situation to a conflict model — S81
From components to cellular behavior — S50
From control to consensus: related qualification formulations — S77
From growth measurement to a clinical category — S74
From seven selling steps to continuing organizational functions — S80
From technical recipes to networks of tasks and transfers — S07
Functional use can require less internal knowledge — S23
Further sources on constraints and adaptation — S05
Gabriel: attempting to recover the decomposition — S08
Galbraith: reduce information demand or increase processing capacity — S14
Galstyan and colleagues: spatial proofreading relocates part of the work — S40
Gardner, Cantor and Collins construct cellular memory — S50

Garicano: specialize learning and pay for communication — S14
Gehlen through Turner: institutional relief — S45
Gell-Mann: a short account can be hard to find and use — S25
Generating a pool of product concepts — S69
Generating performance in music theater and choreography — S66
Genetic instructions retain a reading apparatus — S24
Geschka and Zirm and connected procedural families — S63
Gick and Holyoak and the analogy that must be noticed — S64
Gigamapping, SODA and Interactive Management — S44
Goal-setting: choosing aims, directing effort and revising commitments — S98
Goerner, Lietaer and Ulanowicz: the economic extension — S47
Goodman through Giovannelli: removing distinctions can change the work — S41
Governing the undertaking beyond its initial approval — S86
Grassé: the product of work stimulates subsequent work — S49
Green: calibrate the effect on the output and restore accuracy when needed — S55
Greene's later reconstruction and intellectual reception — S19
Grouping attempts and feedback — S71
Groves and Radner: the value of a restricted message exchange — S15
Guffroy and Bontems: a lineage is a historical problem — S39
Habraken: levels of form and the distribution of control — S08
Haken, catastrophe theory and Bios: proposed transfers retain their prerequisites — S46
Haken: order parameters summarize and influence collective patterns — S41
Hammond and Wilby: a substantive reconstruction with objections — S21
Hansen: regularization suppresses unstable detail by changing the reconstruction — S35
Hayek: acting without knowing every cause — S15
Heylighen: a broader trace-mediated proposal — S49
Higham: a condition number belongs to a problem and its data — S35
Higham: numerical stability belongs to an algorithm — S35
Hirschman: choosing a sequence that elicits a response — S53
Hirschman: developing the ability to invest — S53
Hirschman: exit and voice are different responses to deterioration — S51
Historical networks and room to maneuver — S91
Historical time: a persistent structure is not necessarily beneficial — S45
Hodder: mutual dependencies and an unfinished integration — S39
Hoffmann and Müller: a design tradeoff, not an automatic transfer — S37
Holling: resilience is not rapid return to equilibrium — S43

Homeokinetics: long actions within nested dynamics — S25
Hopfield: specificity can require an additional driven path — S40
How a surface flaw enters the fracture question — S110
How an arrangement can remove distinctions — S03
How experiments enter continuing practice — S31
How latecomers acquire the knowledge to compete — S61
How movement changes with excavation stage — S108
How participants interpret opposition — S95
How the conversation is built — S73
Hsueh, Tsai, and Iyer: the experiment includes a workload and an observer — S30
Huang and Ahmed: maneuverability and balance under uncertain targets — S54
Huckman and Pisano: performance that depends on the setting — S58
Hurdle technology: combinations can change microbial responses — S43
Hurwicz and Reiter: informational requirements are design variables — S15
Hurwicz: prescribed rules and feasible behavior differ — S15
Hypercycles: cooperative organization with vulnerabilities — S40
Hypertely: high performance can narrow the usable setting — S39
IFS: protective parts and internal relationships — S41
Ikemata and Sano: a knee stop can participate in the gait — S01
Immediate updates, storage and later synchronization — S11
IMO: integrate navigation information while retaining integrity work — S38
Implicit communication is prepared, not empty — S16
Incubation attention and bodily activity — S67
Induced response retains a controller — S32
Information and coordinated pathways in selected practice sources — S29
Information for a specified decision — S33
Information produced by early expenditure — S62
Information, constraint and reception — S36
Information, strategy, communication and adjustment — S79
Informational cascades: more followers need not mean more evidence — S59
Informed attention and freedom of inquiry — S83
Inquiring into an engagement together — S104
Inquiring organizations: different ways of warranting a claim — S44
Inquiry joins structure, function, process and context — S20
Institutional boundaries do not contain every capability — S27
Institutional purpose and health claims need their own warrants — S24

Integrative design can change several loads together — S24
Intent specifications preserve why as well as what — S29
Intention and a course of action — S92
Interaction, communication, symbiosis and fusion — S19
Interaction, specialization and the loss of restoration — S18
Interfaces can lose end-to-end ownership — S29
Interleaving — S72
Internal control: separation, authorization and review remain different functions — S42
Interventions, objectives and interface compatibility — S09
Invention and scientific discovery — S64
Invention becomes useful through an operating organization — S13
Investing in a technological position — S62
IPCC: tuning retains physical and observational constraints — S35
Irreversibility is an argument with restrictions and open questions — S46
Itani on similar exercises and different explanations — S66
Jackson's dispute: structural constraints or an interpretive instrument? — S22
Jansson and Smith: an example can constrain designs without reducing their number — S52
JIPM: local workers make and understand a karakuri arrangement — S02
Joining, selection, differentiation and collective knowledge — S17
JOLT and the fear of making the wrong decision — S78
Jones, Gomard and Sestoft: move known computation to an earlier stage — S42
Joseph and Sengul: design problems are distinguishable — S14
Journalism verification: source and content are separate objects of checking — S42
Kabanov, Kupitz and Goetzmann: a grace period retains later judgment — S02
Karr and colleagues integrate a whole-cell model — S50
Katsenelinboigen through a scholarly reconstruction — S34
Kawakita's field cards and the KJ method — S68
Keeping a small extreme subset — S75
Keeping compatible designs open — S62
Keeping familiar calls while selecting the machinery underneath — S84
Kelso: coordination with continuing autonomy — S25
Kirby and colleagues: learnability and expressivity are different pressures — S41
Kirjavainen and Hölttä-Otto and the parts of a method — S63
Kirkus: a favorable account that still acknowledges speculation — S36
Kirsh and Maglio: an action can change what is easy to know — S41
Kitano on structure, dynamics, control and design — S50

Klahr and Simon and connected spaces of discovery — S64

Klein and colleagues: regulate optional application work — S57

Knowledge and power inside hierarchies — S48

Kotarbiński: a favorable position relative to a requirement — S45

Kress: intrinsic versatility and structural flexibility — S38

Kriz: influencing conditions differs from prescribing a result — S41

Kudo on retrieving a decision across successive reviews — S60

Kuznetsov and colleagues: small commands can expose a critical control limit — S54

La Porte: efficiency and vulnerability in interdependent infrastructures — S30

Laboratory speed and patient outcomes — S74

Lafitte through Sandrone, Vaccari and Lawler — S39

Landier's book: ideal types and actual informal work — S27

Langer and Ay, 2021: a controller measure need not measure behavioral importance — S37

Langer and Ay, 2024: learning can temporarily require more controller complexity — S37

Laszlo's patterned flows and a mediated larger architecture — S25

Late-mover advantage and strategic waiting — S61

Later applications and the status of their evidence — S21

Later formalization through du Gay and Lopdrup-Hjorth — S13

Le Moigne's review of Babel — S27

Leadership and arbitration remain functions — S27

Learning and performance — S72

Learning and practising creative methods — S70

Learning before commitment — S62

Learning by trying to remember — S72

Learning from another investor’s action — S61

Learning techniques in classrooms — S72

Learning that outlasts a good performance — S72

Learning to arrange flexible matches — S83

Lehn: molecular architecture stores distinctions that binding can read — S01

Leopoldino and colleagues and a comparative inventory — S63

Lescure and a rule that produces an unexpected text — S65

Leydesdorff: present decisions using a representation of a future — S49

Li and colleagues on cellular predictors and research agents — S50

Li and colleagues on learned flows and biological trajectories — S50

Li, Holstein and Wedekind: material resistance participates in knowing — S41

Liskov and Wing: an interface must preserve behavior over time — S42

Literary and visual creation through constraint transformation and estrangement — S65
Literature-based discovery and computational bridges — S82
Liu and colleagues: behavior generation and morphology-based intelligence — S37
Liu and colleagues: finish the mandatory computation — S57
Local correction, tradition and the rate of change — S08
Local decisions and overall performance — S09
Local knowledge substitutes for some directives — S27
Logistics can require more than common intent — S16
Logistics information and flexibility — S38
Lucas: a behavior setting includes a maintained pattern — S41
Luisi on life, cognition and autopoiesis — S24
Lying within organizations — S93
Mach: economical thought represents more than it repeats — S45
Maintained organization, productive closure and fidelity — S40
Maintaining processes and realizing their relations — S50
Maintaining weights rather than holding quantities fixed — S90
Maintenance, motives and reciprocity — S97
Mair: enthusiastic reception and the difficulty of redundancy-poor text — S36
Making a new dependency visible — S81
Making changes less expensive — S88
Making flexibility exercisable in software investment — S62
Making overlooked possibilities visible — S83
Making radix sorting practical — S75
Making something available differs from making it usable — S42
Making susceptibility a dependable measurement — S74
Making the familiar strange and returning to the problem — S82
Maller and Hyysalo: practices are performed, reproduced and contested — S41
Malone and Crowston: the dependency and the mechanism — S14
Managers construct different information systems — S13
Maneuver warfare and disruption of coordinated action — S113
Manufacturing variation, quality and experimental improvement — S111
Mapping and classification: the useful omission is relative to a purpose — S42
Maps and organizational qualifications — S18
March: learning can consume the differences from which it learns — S52
Marketing and the organization of market relationships — S102
Marletto: a recipe requires an executing and maintained constructor — S40

Marratt: resonance without a simple direct descent — S39
Maruyama: amplification can generate a history — S18
Matching assets to liabilities — S89
Matching challenge to the learner — S72
Material arrangement and behavior — S01
Materials processing, degradation and failure — S110
Matta and colleagues on project experience and design reasons — S60
Matter-energy, markers and meaning — S21
Maxwell: stronger correction can destabilize the motion — S04
MCDP 4: visibility requires interpretation — S38
McGeer: walking as a natural dynamic mode — S01
Meadows: leverage depends on the structure and purpose — S44
Mechanical assistance, perception and control — S37
Mechanical responses shaped by geometry — S01
Mentorship and support communities — S97
Merrill: information savings do not exhaust ownership — S15
Metabolism and repair address different continuities — S26
Metabolism and repair through Louie and Siekmann — S26
Metasynthesis retains people, machines and revision — S18
Meza and colleagues: maintain emergency feature controls — S57
Michod: group performance can constrain its members — S28
Microsoft's guidance: test a complete flow against its targets — S31
Midgley and Lindhult: systemic innovation includes inquiry into purpose — S44
Mirroring technical dependencies in organizational ties — S07
Misdirection and sporting feints — S93
Mitsuda and colleagues: compare the layouts people believe they share — S56
Mixing what must be distinguished — S72
Miłkowski: when physical assistance counts as computation — S37
Modeling relations and purpose-bound mechanisms — S26
Modularity groups dependencies and maintains competence — S20
Monperrus: improvement must occur in the software or its engineering process — S31
Montévil and Mossio: constraints and the processes they shape — S40
Montúfar, Ghazi-Zahedi and Ay: capacity for embodied behavior — S37
Moon: level-ground walking retains energy input and torque compensation — S01
Morin through Le Moigne — S23
Morris and Shin: a public signal can coordinate a common error — S59

Morrison and Milliken: how silence becomes an organizational property — S51
Moving investigation ahead of design completion — S62
Moving management into a replaceable library — S84
Multi-robot coordination retains an engineered organization — S49
Multiple levels retain different tests and responsibilities — S19
Multitime cognition and System 0 — S46
MuSIASEM: maintained funds and linked flow budgets — S47
Musical ensemble: common frames support skilled mutual adjustment — S41
Müller and Hoffmann: control, perception and computation are distinct contributions — S37
Nagumo: having an admissible continuation and confining every solution — S54
Nagy: formulas support reperformance without fixing an exact text — S41
Nakajo: mistake prevention retains prospective analysis — S02
Naming a whole and describing its parts — S56
NASA's stellar explanation: persistence has a fuel-dependent history — S01
Nault and colleagues: remembered words, remembered meaning and familiarity — S56
Necula and Rahul: a smaller oracle can guide a trusted checker — S42
Need, product, source and satisfaction in the Buying-Formula — S76
Negotiation: preparation, bargaining and agreements — S96
Nelsen and Yang: learning a family of inverse maps — S35
Nemeth and colleagues: criticism and generation need not be separate stages — S52
Niche construction can leave a usable environmental legacy — S28
NIOSH: minimal user action is a design aim, not the end of evaluation — S02
Noblet and Simon: making organizational practices observable — S53
Nonlinear transformation and uncertainty about a model — S32
Nosek and colleagues: preserve the plan and make changes inspectable — S31
Nutt: searching for a result rather than defending a ready-made solution — S52
Obtaining attention for an issue — S92
Open collaboration: participation, incentives and maintenance — S07
Open maintenance and conditional equifinality — S18
Openness, memory and the historical machine claims — S39
Operation-based replication has another channel obligation — S11
Operational models make delays and incentives visible — S20
Oppezzo and Schwartz and thinking while walking — S67
Opponents, superiors and subordinates — S91
Ordering, settlement and a wholesaler’s selling time — S80
Organization can persist through continuous activity — S19

Organization is a relation between activities — S17
Organizational objectives and the debate over targets — S98
Organized action has purposes as well as demands — S41
Organizing a result-to-action pathway — S74
Organizing exploration before the project is fully specified — S86
Organizing product development and evolution — S88
Organizing sales work and creating customer value — S80
Orléan: the resale price changes what it is rational to anticipate — S59
Orth, Thiele and Palsson on feasible metabolic flux — S50
Ostrom: contingent commitment retains monitoring — S15
Other accounts of relations, identity and integration — S18
Outsourcing memory: a specified trade-off model — S28
Owotogbe and colleagues: what a review can tell us about the practice — S31
Pain chains, sponsorship and coordinated evaluation — S77
Paired customer exploration and the use of prototypes — S80
Parmelee: a historical specification for automatic release — S02
Parnas: choosing what a module hides — S07
Partial knowledge can be complete for a selected operation — S03
Participation and access to direction — S92
Passive mine-water treatment: chemistry and flow set the conditions — S43
Pathogen testing and ordinary prescribing practice — S74
Pattee: symbols require physical realization and interpretation — S42
Peripheral case: coordinated roles in a traditional account — S48
Persistence and an acceptable outcome are different questions — S43
Persistence, uncertainty and environmental design — S43
Personal constructs: interpreting a situation is itself an organized activity — S41
Personality, conduct and responses within the organization — S91
Personnel information can economize on judgment and preserve privilege — S13
Physical and explanatory quantities — S46
Physical arrangement can supply several functions — S08
Physical measures and cross-domain proposals — S46
Picoli and Takahashi: similar equipment, different incorporation into work — S53
Placing exercises within an online course — S70
Planning around assumptions that can be tested — S62
Planning can change what counts as satisfactory — S19
Planning direction and predictions of completion — S94

Planning the network and choosing a completion date — S85
Platforms: the ability to refuse an option changes the problem — S07
Plausible requests and security procedures — S93
Poincaré on recognizing a fruitful combination — S64
Political skill and willingness to act — S91
Polterovich: institutions can change with acquired capabilities — S53
Position sizing, rebalancing and investment rules over time — S90
Positions that control access — S91
Potts: conventions and context both contribute to meaning — S42
Power to the Edge: self-synchronization has conditions — S16
Powers: retain the controlled perception and changing reference — S03
Powersort — S75
Practical intelligence and political counsel — S93
Practice composition — S71
Practice tasks and performance tasks — S71
Practices for making one's own luck — S83
Practising parts and returning to the whole — S71
Precimonious: search against an explicit acceptance test — S55
Prediction and balance retain different work — S09
Preparation and the choice to negotiate — S96
Prepared responses and room to adapt — S79
Prepared routines economize on new decisions — S13
Preparing cards and inventing a workshop situation — S68
Preparing speech for an occasion — S114
Presortedness — S75
Prigogine: thermodynamic openness and internal production — S46
Problem solving and productive failure — S72
Procalcitonin — S74
Processes, structures and adjustment — S21
Processors make transport and transformation explicit — S23
Procrastination and beginning intended action — S99
Producing an insight and tailoring it across a buying group — S78
Production relations, local control and shared capability — S14
Productive failure — S72
Professional contacts inside and outside an organization — S97
Programmes reduce description by retaining a rule — S23

Proliferation, multiple routes and conditional autonomy — S27
Properties of a particular processed material — S110
Proposal approval and the handoff into project delivery — S80
Public cultural guidance: knowledge has custodians and conditions of use — S41
Public speaking, rhetoric and audience judgment — S114
Purpose and choice change the description — S20
Purposes, expertise and the organization of inquiry — S44
Push and pull distribute anticipation differently — S38
Pérez Jiménez and the operations of image making — S65
Q-analysis: connectivity at different dimensions — S47
Qian's knowledge architecture and its later reconstruction — S18
Questions, diagnosis and commitments in selling — S77
Radix sorting — S75
Radner: processor count and delay are joint criteria — S14
Rahimi and colleagues: tuning works within a changing frequency relation — S01
RAND: sense and response retain prediction and command — S38
Rank selection — S75
Reachability and the common ancestor — S10
Reaching a candidate pool — S103
Reading routes through the biological foundations — S50
Reasons for avoiding an exchange — S106
Reassigning a function to what is already there — S81
Recall after an instruction to forget — S105
Reception includes specific criticism — S24
Recognition and the work of a continuing staff — S115
Recognizing and developing a possibility in an organization — S83
Recognizing delay in ordinary behavior — S99
Reconnecting after losing touch — S97
Reconstructability analysis: a simpler predictive distribution — S47
Recreation and flood protection: intended benefits do not end the lifecycle — S43
Recruiting, selection and forming working relationships — S103
Recursion is a claim about organization at selected boundaries — S22
Redundancy is relative to the message and its realization — S03
Referential integrity depends on the allowed updates — S10
Regressing requirements and protecting goals — S94
Regulation preserves a desired range, not constancy alone — S03

Related accounts — S59
Related book and risk-analysis critique — S32
Related frameworks and a separate 2009 formulation — S29
Relating practice to expertise — S71
Relational autonomy: dependence does not settle the question — S48
Relations and organizational change — S17
Relations change what components can do — S24
Relations specify what processes produce and maintain — S26
Relearning to a criterion — S72
Reliability under disruption — S30
Reliable components can participate in unsafe interactions — S29
Reliable transmission is a capacity result with a coding obligation — S06
Reorganization and the meaning of a crisis — S17
Representing and ranking bridges — S82
Representing preferences and feasible ranges — S62
Representing the product decision — S87
Requirements as an activity — S112
Requirements, integration and verification in systems engineering — S112
Requirements, interactions and a program for thought — S08
Reserves before a disturbance — S109
Response diversity includes differing reactions, not just a species count — S43
Responsibilities can fall into gaps or overlap unsafely — S29
Retrieval practice — S72
Returning after an interval — S72
Reusing a kernel by introducing an adaptation layer — S84
Revising, relinquishing and replacing goals — S98
Richardson: building and using integrated models — S09
Rimbaud and the cultivation of perception — S65
Rispoli's interpretation of environment and collective knowledge — S17
Ritchey: organize combinations through dimensions and consistency judgments — S52
Ritter and Dijksterhuis and explanations of what happens during a break — S67
Ritual and ascetic practice can involve deliberately sustained effort — S41
Robust Decision Making: inspect vulnerabilities across possible futures — S44
Rodari and the word that sets other things moving — S65
Role-based access: permissions are prepared through assignments and constraints — S42
Roles connecting decisions and implementation — S92

Rosenstein-Rodan: training and investment beyond a single enterprise — S53
Rules and standards: when specification work is done — S15
Running an imagined combination — S82
Runs, sorting — S75
Ruscoe: a schema does not supply its own instruments — S21
Ruses and protected confidence — S93
Saldi and Yüksel: who is allowed to know what — S33
Saltzer, Reed and Clark: the endpoint must check the actual application result — S12
Sampling and equal-key blocks — S75
Sandler: agreements, pain, investment and the decision process — S77
Sarter and Woods: recognize what the changed mode will do — S57
SAS-085: changing the command arrangement is itself work — S16
Sato: files and prior transactions supply state — S10
Scharfstein and Stein: the reputation gained by following others — S59
Scharpf: a broker can reduce consultations without eliminating politics — S15
Scheduled exchange and physical flow — S109
Schelling: tolerable local mixtures can produce segregation — S49
Schober and Clark: hearing an agreement and participating in it — S56
Scholz and Schöner: preserve the task without fixing every component — S54
Schwarz: a three-plane metamodel and six cycles — S44
Selection can be performed by an arrangement — S17
Selective reporting — S74
Self-triggering predictive control and feasibility — S05
Semiconductor production and the continuing core of integration — S07
Sensing, action and timing are separate restrictions — S03
Sensitivity, preparation and reconstruction — S35
Sensor-based sorting: reject material before further processing — S43
Separate responsibilities in a second regime — S109
Separation, conflicting interests, and the choice of a whole — S17
Serendipity and the cultivation of unexpected discovery — S83
Service conditions and specimens — S110
Service, authority and internal markets — S20
Several different accounts of prepared action — S45
Several different ways of making a calculation manageable — S35
Several meanings of closure — S40
Shannon: recover a selected message from alternatives — S06

Shapiro and colleagues: equal histories must give equivalent answers — S11
Shared criteria can support local decisions — S20
Shared standards and a shared system — S38
Sharing disk data without prescribing one filesystem — S84
Shaw and the proof-theory entry: explicit levels of description — S42
Shchedrovitsky: distinguish thinking, communication and practical action — S44
Sherbrooke: multi-echelon recoverable-item inventory — S38
Shneiderman's creativity-support tools — S69
Shortening a known path inside Linux — S84
Shraim: reacting to each fluctuation can add variation — S05
Siekmann's replication-map objection — S26
Sigaut and related accounts of mechanology — S39
Signals, commitments and institutional boundaries — S15
Simon: stable intermediate forms and conditional simplification — S14
Simondon: understand an object through its genesis — S39
Simpson: compact instruction and an unclear connection — S36
Sio and Ormerod and the structure of an incubation interval — S67
Situational, functional and process accounts of sales management — S80
Six informational stages in USIT — S81
Sizing for repeated participation — S90
Sless: lucid communication theory, disputed unification — S36
Small objects can carry a large technical history — S39
Smit and Trigeorgis: cash flows, flexibility and strategic interaction — S34
Smith: exclusion as a limited-information interface — S15
Social organization adds meaning, rules and power — S24
Sociotechnical groups in citizenship services — S14
Software: modular before addition can become interdependent afterward — S07
SOHO: differences between analysts' reconstructions — S29
Soltani and colleagues distinguish an association graph from an assembly — S50
Solution-focused brief therapy — S73
Soros: responses can change the conditions being interpreted — S59
Source inspection and the removal of hazards — S02
Spacing — S72
Sparse links can require complicated programmes — S23
Specialization across abstraction boundaries — S84
Specialized experience and the organization of knowledge — S17

SPIN questions and the development of an explicit need — S77
Spolin and the shared acting problem — S66
Stability and transformation are different accomplishments — S19
Stability can come from absorption, correction or anticipation — S23
Stability depends on exposure as well as strength — S17
Stages of work and gates of commitment — S88
Stahel: doctrine, culture and historical practice do not coincide — S16
Star and Lampland: standards coordinate while retaining local work — S42
Starting with the proposition to be established — S94
State-based merging retains order through the payload — S11
Statistical and decision criteria — S33
Staw: responsibility for an earlier choice can affect the next allocation — S51
Steadman: intentional craft and the biological analogy — S08
Sterling and Ramsay/Woods: anticipation does not settle the terminology — S05
Sternieri: autonomy and shared dependencies — S14
Stott and colleagues: a modeled improvement can come from counting assumptions — S30
STPA begins with losses, hazards and a control structure — S29
Strata, layers and echelons ask different questions — S09
Stratagem collections and princely dissimulation — S93
Strategic assessment and reflexive control — S93
Strategy throughout the product's life — S88
Structure can appear while entropy production increases — S46
Structure, process and history differ — S21
Structures, ground conditions and construction sequences — S108
SUBSAFE illustrates a bounded, maintained institution — S29
Successive relearning — S72
Suggested beginnings and the stories people write — S69
Suggestive selling within a meal — S79
Suh: functional independence and probability of success — S44
Sunzi: disposition and collective potential — S16
Susceptibility testing — S74
Sustaining operation is a continuing activity — S38
Sweller: solving a problem and acquiring a schema can compete — S41
Synthetic Notch: a selected modular sensing–response account — S40
System 1: put local response where local knowledge exists — S22
Systematic accounts of methods for generating ideas — S63

Systematic invention through contradictions and available resources — S81
Systems 2, 3, and 3*: autonomy keeps coordination and accountability — S22
Systems 4 and 5: operating now while considering what should change — S22
Szulanski: transferring a practice through actual use — S58
Tabuada: an event trigger still has to be detected — S05
Tainter: complexity as a costly problem-solving investment — S48
Takahashi and four purposes of creative techniques — S63
Taleb and Douady: exposure beyond a threshold — S32
Targeting heat recovery before choosing equipment — S107
Teaching, reframing and buyer indecision — S78
Technical knowledge is neither mere habit nor a detached formula — S39
Technical operation, its conditions and its human setting — S39
Technical progress can reorganize the functions themselves — S39
Testing latecomer advantage at the regional level — S61
The 1969 paper: local success has to map into overall success — S09
The 1971 preface changes the emphasis — S08
The author's later retrospective — S23
The biological identification is explicitly postulated — S26
The body changes the control problem — S37
The Brusselator retains maintained reservoirs and reaction conditions — S46
The buyer’s progress and the seller’s process — S80
The cases have different evidential status — S20
The changing work of a founder — S100
The connected argument and its own limits — S13
The connected safety account — S29
The earlier dissipative-structures excerpt — S19
The flexibility comparison can reverse — S38
The food-web example and the conditions on fallback — S47
The Guimbal turbine: a mutually enabling environment — S39
The Heilmeier Catechism in research funding — S86
The later article: job shops, flow and platforms as representative arrangements — S07
The Marine Corps account of maneuver warfare — S113
The material mold and the continued act of construction — S39
The niche-construction dispute is about implications — S28
The original *Heart* passage: ignorance is also a filter — S22
The philosophical third part: a proposed genesis, with plural values — S39

The place and time of the contradiction — S81
The practice guide: experiment design and its supporting organization — S31
The practitioner principles: make a claim about observable behavior — S31
The proposed optimum and the critical preprint — S47
The proposed social physics remains unfinished — S25
The question each arrangement answers — S14
The reconstructions include methodological criticism — S25
The response must be appropriate, as well as available — S03
The response to variation is the object of inquiry — S32
The right set of circumstances — S76
The ring constructions retain an asymmetry — S11
The robot example: a solution changes the problem — S29
The scope of a favorable response — S05
The scope of becoming a solution provider — S80
The Senate section adds supporting institutions — S27
The sequence of commercial teaching — S78
The set need not be a collection of simultaneous states — S03
The shape of the evidence — S73
The task determines which news matters — S13
The thermodynamic correspondence is a stronger claim — S27
The valuation and scope carried by this account — S07
The village example: several requirements meet in an arrangement — S08
Theory of Constraints: a bottleneck can move — S44
Thorpe: define sustaining functions so they are actually assigned — S38
Timing, signals and the energy that an operation uses — S39
Tonet and da Paz: knowledge sharing continues after initial transmission — S42
Top-k selection — S75
Topics as sources of design inspiration — S82
Toyota: stopping changes the work of watching — S02
Traceability: a documented chain still carries uncertainty — S42
Transfer of learning — S71
Transfers, transactions and the work avoided inside a zone — S07
Transforming computational planning problems — S94
Treatment duration — S73
Trust-based selling and the dispute about relationships — S78
Turning a physical requirement into a design — S81

Two frames for one situation — S82
Two narrower engineering accounts — S31
Two questions about distributed behavior — S11
Ulanowicz and colleagues, 2009: what the flow identities say — S47
Ulanowicz, 2020: a narrower balance conjecture — S47
Ulrich: the boundary determines which facts and values count — S44
Understanding a customer's business and shaping a valuable offer — S80
University space: local knowledge, reserves and credible promises — S13
Using representations in practice — S42
Using the key representation — S75
Value structure: finding what constrains a result and what captures its value — S07
Valuing a future choice and preparing a position — S34
Van Creveld through later accounts, and Smith's criticism — S16
Van den Akker and colleagues: compare expected benefits with published results — S31
Van der Schaft: composition preserves an energetic structure — S04
Van Gigch's review: an explanation still needs practical mastery — S22
Varela, Maturana and Uribe: a production organization that realizes its boundary — S40
Variety is relative to the distinctions and purpose being considered — S22
Vaughan: warning evidence is interpreted within a history of work — S51
Verhulst: where the system boundary places improvement — S31
Viability and organizational functions — S22
Viability, identity and organizational purpose — S22
Vickers: appreciation includes standards of fact and value — S44
Villiers: the fifty-percent result has assumptions — S27
Virgo and colleagues: a model can be an attributed interpretation — S03
Voge: information work can offset productive gains — S27
Von Neumann: reliability through an arrangement of fallible elements — S30
Waddington: buffered development is conditional — S28
Walker: a maneuverable aircraft requires a supporting control arrangement — S54
Wang and colleagues: responsiveness is not sufficient for their physical-computing definition — S37
Waterhouse on making and learning Duo — S66
Wegner: remembering an item and remembering where to find it — S56
What a form does in a specified context — S08
What a speaker can furnish — S114
What a validity estimate describes — S103
What an association test measures — S82

What an avoidance procedure observes — S101
What analogue tools supply — S68
What application and training studies report — S81
What creativity training improves — S70
What followers can obtain from pioneers — S61
What is carried into the next task — S71
What is learned during approach — S101
What judgments and measures observe — S92
What outcome comparisons measure — S73
What supports a slab while it is raised — S108
What the administrative-burden review covers — S48
What the arrangement changes — S02
What the conditional optimality argument actually uses — S09
What the coordination comparison actually compares — S10
What the early formulations meant by a completed sale — S76
What the exokernel builders learned — S84
What the formal appendix contributes — S08
What the seller takes to be the market — S102
What variety counts — S03
What was being sought, and what was found? — S83
When a rapid panel leaves prescribing unchanged — S74
When another asset repeats the same exposure — S89
When attempted solutions sustain a problem — S73
When competition accelerates commitment — S61
When delay becomes procrastination — S99
When local transactions compose safely — S10
Where protective time is placed — S85
Which ventures enter a survival estimate — S100
Who chooses when feedback arrives — S71
Who founded the firm and when — S100
Whose knowledge enters the review — S95
Willems 1997: a controller can be another physical system — S04
Willems: stored quantity cannot increase beyond supplied quantity — S04
Wilson and Flack: a coarser description can have a specific explanatory role — S46
Winter and Szulanski: reproduce the capability to operate — S58
Withdrawing support — S105

Wittenmark: an action can both control and probe — S33
Wood, Bruner and Ross: a tutor carries part of the task — S41
Woods: prepare the capacity to respond near saturation — S54
Work and care are different evaluative concerns — S48
Work on both sides of the commission — S104
Working backward from outcomes — S94
Working spaces and the continuing design cycle — S68
Written procedures and work that people continue to do — S105
WSR: physical conditions, organized work and human acceptance — S18
Wu: separating subnetworks retains an interconnection solve — S04
Xie and colleagues: count persistence as well as task achievement — S37
Xunzi through Wang: ritual and custom are not the same category — S41
Yao and Chu and methods across an innovation process — S63
Yi and colleagues: integral feedback in a chemotaxis model — S05
Zero trust: the boundary is not an unconditional authorization — S42
Zhang and colleagues: imported technology and delayed innovation effort — S53
Zhao and Jiang: design agency and implementation institutions — S08
Zhou and Zhao: historical and institutional organization — S13
Zuidberg Dos Martires and colleagues on representing reaction conditions — S50
Álvarez Rodríguez and the writing workshop — S65

Author and institution index

Entries link to the shared references and to the sections discussing each work.

A. H. Louie — W104 — S26, S50

A. Wozniak — SB-W006 — S09

Abhigya Giri — TD-W018 — S74

Abhishek Sharma — W084 — S46

Abraham Lincoln — T16-W006 — S114

Acas — T08-W001 — S106

Adam D. Galinsky — GS-W12 — S98

Adam D. Galinsky — GS-W14 — S98

Adegboyega Lawal — GL-W058 — S29

Administrative Science Quarterly editors and authors — NN-W022 — S97

Adnan Darwiche — TA-W008 — S42, S50

Adolfo Anta — TA-W014 — S05

Adrian Bejan — W082 — S46

Adrian Sampson — CG-W006 — S55

Adriana Takahashi — AB-W008 — S53

Agnès Alsius — GL-W058 — S29

Agustín Tristán López — CRE-W044 — S70

Ahmed Eid — CG-W015 — S57

Akinori Iwamoto — SL-W005 — S76

Alaa A. Ahmed — SM-W003 — S54

Alain Degenne — RGC-W027

Alan Fekete — W005 — S10

Alana Glassco — OI-W013 — S52

Alaska Native Knowledge Network — GL-W003 — S41

Alastair Cooper — MIL-W034 — S16

Alastair D. Hay — TD-W016 — S74

Albert Chuang-shi Yu — CG-W013 — S57

Albert O. Hirschman — AB-W004, OI-W007 — S51, S53

Alberto Cebrián-García — W048

Aleksandr Bogdanov — GL-W097 — S17

Aleksandra M. Wójcik — LP-W032 — S72

Alessandra Stefane Cândido Elias da Trindade — NC-W032 — S83

Alessandro Giovannelli — WME-W009 — S41

Alex Molnar — TD-W002 — S73
Alexander B. van Putten — LC-W005 — S62
Alexander Beresford — PWR-W010 — S91
Alexander C. DeHaven — CG-W023 — S31
Alexander E. Hramov — SG-W022 — S47
Alexander Frankel — W071 — S33
Alexander Haas — SL-W019 — S80
Alexander Jung — XA-W003 — S84
Alexander Jung — XA-W007 — S84
Alexander Wezel — GL-W011 — S43
Alexandru Stancu — AFC-W008 — S32
Alexey Solovyev — CG-W004 — S55
Alfredo Buttari — CG-W001 — S55
Ali Ghodsi — W005 — S10
Ali Raza — XA-W004 — S84
Allen C. Ward — LC-W003 — S62
Allen Z. Ren — PM-W011 — S94
Almir Cordeiro Júnior — W028 — S14
Alvaro Moreno — UX-W013 — S42
Amanda Ross-White — GL-W058 — S29
Amnon Lehavi — GL-W094 — S15
Amos C. Fox — MIL-W030 — S38
Amy C. Edmondson — AFC-W015 — S31, S51
Amy Edmondson — OI-W004 — S51
Amy L. Simmons — LP-W006 — S71
Amélie Férey — PM-W025 — S95
Ana Paiva — CRE-W042 — S70
Anastasia Antoniadou — TD-W023 — S74
Anatol Rapoport — SG-W037 — S18
Anders Fournaise — TD-W013 — S74
Anderson Simas Frutuoso — GS-W17 — S98
Andre Rosendo — W034 — S37
Andrea Prencipe — W068 — S12
Andrea Zirm — CRE-W002 — S63
Andreas C. Lehmann — LP-W008 — S71
Andreas Eggert — SL-W019 — S80

Andreas Machotta — GL-W058 — S29
Andrew J. Elliot — GS-W04 — S98
Andronikos Spyrou — TD-W023 — S74
Andrzej Stankiewicz — GL-W074 — S01
André Breton — CRE-W015 — S65
André Orléan — EF-W006 — S59
Andrés J. Serrano-Balbontín — W048
Andrés Vaccari — UX-W023 — S39
Angelica Marchetti — GL-W011 — S43
Anil R. Doshi — CRE-W038 — S69
Anirudha Majumdar — PM-W011 — S94
Ann Blandford — NC-W027 — S83
Ann Blandford — NC-W030 — S83
Anna Madison — MIL-W029 — S16
Anna Makina — TD-W023 — S74
Anna Maria Bonet-Esteve — TD-W015 — S74
Anna Wolf — LP-W008 — S71
Anthony Duus — MIL-W035 — S16
Anthony J. Venables — AB-W005 — S53
Antoine Bousquet — MIL-W018, MIL-W019 — S16
Antoine Danchin — AFC-W013 — S32
Antonakis, J., Fenley, M., & Liechti, S. — T17-W004 — S115
Antonia Panagaki — TD-W023 — S74
Antonia Retamal- Cañiz — TD-W015 — S74
Antonie Jetter — MG-W011 — S88
Antonio Possolo — TA-W031 — S42
Antônio Sabino da Silva Filho — LP-W012 — S71
Anurag A. Agrawal — TA-W038 — S28
Ap Dijksterhuis — CRE-W027 — S67
Aqsa Ejaz — PWR-W001 — S91
Arantza Etxeberria — UX-W013 — S42
Arcuri, M. C., Russo, I., & Gandolfi, G. — T02-W004 — S100
Ardalan Kahak — W098 — S37
Ariel Domlyn — PM-W023 — S95
Ariel Tseitlin — AFC-W019 — S31
Arjan van der Schaft — GL-W069, GL-W070 — S04

Armando Sternieri — UX-W011 — S14

Arnold Gehlen — UX-W033, UX-W039 — S45

Arthur Koestler — NC-W008 — S82

Arthur L. Stinchcombe — SB-W002, UX-W003 — S13

Arthur Rimbaud — CRE-W014 — S65

Arthur S. Iberall — WME-W012, WME-W013 — S25

Arvind Murugan — TA-W018 — S40, S50

Asim Abdulkhaleq — SB-W015 — S29

Aske Halling — GL-W019 — S48

Assembly of Alaska Native Educators — GL-W003 — S41

Balance Management Task Force — T11-W004 — S109

Baltasar Trancón y Widemann — WME-W024 — S47

Barbara H. Liskov — W087 — S42

Barrett R. Anderson — CRE-W039 — S69

Barry M. Staw — OI-W005 — S51

Bart Dietz — SL-W018 — S79

Battelle Memorial Institute — T14-W005 — S112

Beersma, Bianca, and Carsten K. W. De Dreu — NN-W025 — S96

Ben Dankbaar — W003 — S14

Ben Shneiderman — CRE-W035 — S69

Ben Taylor — SL-W021 — S77

Benjamin Baird — CRE-W025 — S67

Benjamin Bolival Jr. — BC-W007 — S50

Benjamin Brooks — W071 — S33

Bent Flyvbjerg — MG-W004 — S86

Bernard Lietaer — RGC-W042, WME-W025 — S47

Bernard Motulsky — RGC-W024 — S23

Bernard Personnaz — OI-W008 — S52

Bernhard Ø. Palsson — BC-W005 — S50

Bert Zwart — TA-W016 — S38

Beth Veinott — PM-W022 — S95

Bidart, Claire — NN-W018 — S97

Bijan Samali — GL-W017 — S01

Birger Sevaldson / Systems Oriented Design — SG-W016 — S44

Bjork, R. A. — T07-W001 — S105

Björn J. E. Johansson — MIL-W033 — S16

Björn Wittenmark — AFC-W012 — S33

Blas M. Vinagre — W048

Bloom, N., Eifert, B., Mahajan, A., McKenzie, D., & Roberts, J. — T06-W005 — S104

Board of Governors of the Federal Reserve System — SG-W029 — S30

Bojan Cestnik — NC-W022 — S82

Boni-Le Goff, I. — T06-W003 — S104

Boothby, Erica J., Gus Cooney, and Maurice E. Schweitzer — NN-W005 — S96

Borut Sluban — NC-W022 — S82

Brad Pierce — GE-INTRO — S00

Brett R. Jones — CRE-W033 — S68

Brian A. Nosek — CG-W023 — S31

Brian Ichter — PM-W011 — S94

Brian M. Kennedy — LC-W011 — S62

Brian P. Bailey — CRE-W033 — S68

Brian R. Gaines — WME-W017 — S41

Britt W. J. Mathijsen — TA-W016 — S38

Brooke N. Macnamara — LP-W002 — S71

Brooke N. Macnamara — LP-W004 — S71

Bruno Mazaré — T10-W003 — S108

Bryan S. Turner — UX-W032 — S45

C. A. Goetzmann — GL-W081 — S02

C. H. Waddington — W032 — S28, S50

C. M. Holloway — SB-W014 — S29

C. Renlund — TD-W005 — S73

C. S. Holling — GL-W067 — S43

C. W. Johnson — SB-W014 — S29

Cai Wen — SG-W050 — S44

Cameron Anderson — PWR-W007 — S91

Camila Neves — LP-W013 — S71

Canadian Association for Conservation of Cultural Property — GL-W099 — S39

Carl Llor — TD-W013 — S74

Carl von Clausewitz — MIL-W036 — S16

Carla Davis Cash — LP-W006 — S71

Carla Ferreira — W027 — S10

Carla Xavier — CRE-W042 — S70

Carliss Y. Baldwin — SB-W005, UX-W001, UX-W008, UX-W009, UX-W038, UX-W040 — S07

Carlos Baquero — W061 — S11

Carlotta Langer — UX-W015, UX-W022 — S37

Carsten K. Gomard — TA-W009 — S42

Carsten Wrosch — GS-W09 — S98

Casciaro, Tiziana, Francesca Gino, and Maryam Kouchaki — NN-W021 — S97

Cecily Maller — GL-W034 — S41

Centre de doctrine d'emploi des forces, armée de Terre — MIL-W024 — S16

Cesaire Chiatchoua — PWR-W012 — S91

Chad Hansen — GL-W098 — S45

Chad Nachiappan — CG-W015 — S57

Challenger — SL-W020 — S78

Chang-Hung Chou — SG-W048 — S44

Charlan Jeanne Nemeth — OI-W008 — S52

Charlan Nemeth — PM-W019 — S95

Charles Babbage — W086 — S45

Charles Perrow — UX-W004 — S29

Charles R. Cantor — BC-W003 — S50

Charles R. Ebersole — CG-W023 — S31

Charles R. Schwenk — PM-W017 — S95

Charles S. Carver — GS-W09 — S98

Chazel, F. — T17-W002 — S115

Che Xin — TA-W039 — S05

Chelsey Geralda Armstrong — GL-W004 — S41

Cherilyn Ireton — GL-W047 — S42

Chi Lan Nguyen — PWR-W001 — S91

Chi-Ming Yang — SG-W048 — S44

Chiara Marletto — W091, W092 — S40, S46

Ching-Lin Wu — NC-W011 — S82

Chloe Coulson — GL-W058 — S29

Chris Argyris — OI-W003 — S51

Chris Button — GL-W071 — S41

Chris Metcalfe — TD-W016 — S74

Christian Busch — NC-W026 — S83

Christian Busch — NC-W031 — S83

Christian Gelzer — SM-W002 — S54

Christian Maes — W079 — S46

Christian Mullon — UX-W018 — S43
Christian Schmitz — SL-W007 — S80
Christian Spreafico — NC-W006 — S81
Christian Terwiesch — CRE-W040 — S69
Christina Bogner — WME-W024 — S47
Christophe Chamley — SW-W005 — S61
Christophe Midler — LC-W009 — S62
Christophe Torset — ST-W005 — S92
Christopher Alexander — TA-W028, TA-W029 — S08
Christopher C. Butler — TD-W012 — S74
Christopher Harris — SW-W006 — S61
Christopher J. Soto — PWR-W007 — S91
Christopher K. Ansell — PWR-W011 — S91
Christopher Kelshaw — MIL-W029 — S16
Christopher P. Kempes — W084 — S46
Christopher Plishka — GL-W058 — S29
Christopher Potts — GL-W014 — S42
Christopher R. Paparone — MIL-W004 — S16
Christopher R. Smith — MIL-W007 — S16
Christopher Robben — GL-W077 — S43
Ciampi, F. — T06-W004 — S104
Cibelly Oliveira — LP-W013 — S71
Cindy Rubio-González — CG-W003 — S55
Claude E. Shannon — W025 — S06
Claudia R. Batista — CRE-W034 — S68
Claudia Regina Batista — CRE-W043 — S70
Claus Ebster — SL-W026 — S79
Clemens Tesch-Römer — LP-W001 — S71
Cliff Joslyn — SG-W032 — S18
Cole Petersen — T15-W004 — S113
Colin Eden — SG-W051 — S44
Comité de l’antibiogramme de la Société française de microbiologie — TD-W011 — S74
Community of Christ — T17-W006 — S115
Cooper, A. C. — T02-W001 — S100
Cordula Vesper and colleagues — GL-W002 — S41
Costi Răducanu — XA-W003 — S84

Costin Iancu — CG-W003 — S55
Costin Lupu — XA-W003 — S84
Costin Raiciu — XA-W003 — S84
Costin Raiciu — XA-W007 — S84
Craig C. Sherbrooke — MIL-W031, MIL-W032 — S38
Craske, M. G., et al. — T03-W003 — S101
Cristian Axenie — AFC-W008 — S32
Cristian Banu — XA-W003 — S84
Cristian Klein — CG-W014 — S57
Cuong Nguyen — CG-W003 — S55
Cyril Soldani — XA-W003 — S84
D. D. Clark — GL-W030 — S12
D. E. Campbell — SG-W018 — S47
D. Macko — SB-W008, W096 — S09
D. P. Reed — GL-W030 — S12
Da Chen, Xiao-Feng Zhang and Robert O. Ritchie — T12-W002 — S110
Daan van Knippenberg — CRE-W045 — S63
Dale Griffin — PM-W009 — S94
Dalrymple, H. — T16-W005 — S114
Damian Andrew Tamburri — AFC-W011 — S31
Dan Grossman — CG-W006 — S55
Dana Amiri — SL-W001 — S77, S78, S79
Daniel A. Levinthal — AB-W001 — S53, S61
Daniel Bristot de Oliveira — XA-W004 — S84
Daniel Dinca — XA-W007 — S84
Daniel L. Schwartz — CRE-W028 — S67
Daniel L. Schwartz — OI-W013 — S52
Daniel Little — GL-W050 — S45
Daniel M. Wegner — CG-W028 — S56
Daniel Madrzykowski — W023 — S02
Daniel R. Nault — CG-W011 — S56
Daniel T. Willingham — LP-W020 — S72
Daniel Tabak — SB-W009
Daniela Satomi Saito — CRE-W034 — S68
Danushen Gnanapragasam — CG-W006 — S55
Daron L. Sharps — PWR-W007 — S91

Darío Sandrone — UX-W023 — S39
David Arellano Gault — DC-W13 — S93
David B. Montgomery — SW-W001 — S61
David B. Montgomery — SW-W002 — S61
David B. Paradice — SG-W039 — S44
David Blackwell — W029, W059 — S33
David C. Krakauer — TA-W019 — S28
David D. Clark — GL-W095 — S12
David D. Evans — W023 — S02
David D. Woods — CG-W016, SM-W008 — S54, S57
David Deutsch — W091 — S46
David Evers — SL-W007 — S80
David Fernandez Rivas et al. — SMR-W001 — S01
David Ferraiolo — TA-W032 — S42
David G. Jansson — OI-W015 — S52
David H. Bailey — CG-W003 — S55
David Hirshleifer — EF-W005 — S59
David Hough — CG-W003 — S55
David Ke Hong — CG-W015 — S57
David Kirsh — GL-W027, W041 — S41
David Klahr — CRE-W011 — S64
David L. Alderson — UX-W043
David L. Parnas — W004 — S07
David Leopold — GL-W049 — S48
David M. Shuker — TA-W021 — S28
David Mattson — SL-W010 — S77
David Mazières — XA-W002 — S84
David Mazursky — NC-W002 — S81
David McDonald — PM-W015 — S95
David N. Bengston — PM-W013 — S94
David S. Alberts — MIL-W009, MIL-W010 — S16
David S. Scharfstein — EF-W004, EF-W010 — S59
David Shulman — DC-W12 — S93
David Sless — SB-W018 — S36
David Stahel — MIL-W021 — S16
David T. Croasdell — SG-W039 — S44

David T. Mellor — CG-W023 — S31
David W. George — MIL-W017 — S38
David Wood — GL-W016 — S41
David Z. Hambrick — LP-W002 — S71
David Z. Hambrick — LP-W004 — S71
Davide Russo — NC-W006 — S81
Dawson R. Engler — XA-W001 — S84
Dawson R. Engler — XA-W002 — S84
Dean Keith Simonton — NC-W016 — S82
Deanna Wilkes-Gibbs — CG-W025 — S56
Debbie Harrison — UX-W002 — S14
Debora Hammond and Jennifer Wilby — RGC-W005 — S21
Deborah J. Mitchell — PM-W020 — S95
Deborah Maxwell — NC-W030 — S83
Debra Daly Hartin — GL-W100 — S39
Defense Advanced Research Projects Agency — HCT-W002 — S86
Deividi Pires — SL-W013 — S79
Denis Dumas — NC-W005 — S81
Denis Roze — TA-W022 — S28
Dependency and Structure Modeling community — SG-W021 — S07
Derek Cabrera — SG-W041 — S44
Derek N. Macklin — BC-W007 — S50
Derick Wood — TD-W025 — S75
Desert Storm: Attrition or Maneuver? — T15-W003 — S113
Diane Vaughan — OI-W017 — S51
Diego Lawler — UX-W023 — S39
Dimitra Petrou — TD-W023 — S74
Dimitri Jeltsema — GL-W070 — S04
Dimitris Bertsimas — W060 — S05
Dina Borrego — W027 — S10
Diplomatic Conference, Geneva — DC-W19 — S93
Dmitry Chernyshev — CG-W015 — S57
Dominic Orth — GL-W071 — S41
Don R. Swanson — NC-W020 — S82
Donald Moynihan — GL-W018 — S48
Donald Reinertsen — MG-W010 — S88

Donella H. Meadows — W107 — S44

Dorian Peters — CRE-W032 — S68

Dorte Ejg Jarbøl — TD-W014 — S74

Doug Rohrer — LP-W023 — S72

Douglas C. Engelbart — CM-W002 — S60

Douglas Gale — SW-W005 — S61

Douglas H. Erwin — TA-W020 — S28, S50

Douglas S. Ramsay — GL-W029 — S05

Dov Dori — SG-W058 — S44

Durward K. Sobek II — LC-W003 — S62

Durward K. Sobek II — LC-W011 — S62

Dylan Wong — PM-W023 — S95

Dániel Czégel — W084 — S46

E. Virtala — TD-W005 — S73

Eby, Lillian T., Tammy D. Allen, Sarah C. Evans, Thomas Ng, and David DuBois — NN-W023 — S97

Edgar Morin — RGC-W028, RGC-W029 — S23

Edsger W. Dijkstra — W006 — S11

Edward D. Lee — TA-W019 — S28

Edward Everett — T16-W007 — S114

Edward Vul — LP-W023 — S72

Edwin A. Locke — GS-W01 — S98

Edwin A. Locke — GS-W13 — S98

Eero Vaara — ST-W004 — S92

Eeva Siljamäki — CRE-W024 — S66

Efrosyni Roulia — TD-W023 — S74

Eladio Dapena — W067 — S49

Elam Nunnally — TD-W002 — S73

Elina Drakou — TD-W023 — S74

Elinor Ostrom — W049 — S15

Elisa Gonzalez Boix — W027 — S10

Elisa Maria Pivetta — CRE-W034 — S68

Elisabeth Dostal — SG-W004 — S18

Elizabeth Anderson — GL-W007 — S45

Elizabeth G. McDougal — SL-W001 — S77, S78, S79

Elizabeth J. Marsh — LP-W020 — S72

Elizabeth Ligon Bjork — LP-W028 — S72

Elizabeth Waterhouse — CRE-W021 — S66

Elizabeth Wolfe Morrison — OI-W001 — S51

Ellen Novoseller — MIL-W029 — S16

Elza Adamowicz — CRE-W046 — S65

Emilie J. K. Mutombo — PM-W004 — S94

Emily Brown — TD-W016 — S74

Emily Fortuna — CG-W006 — S55

Emir Kamenica — W071 — S33

Emma Bridgeman — TD-W016 — S74

Enrique Bigné Alcañiz — SL-W012 — S79

Enrique Mancha-Sánchez — W048

Eric B. Munson — XA-W004 — S84

Eric G. Too — MG-W005 — S86

Eric P. Palkovacs — TA-W020 — S28, S50

Eric Schwarz — SG-W003, SG-W013 — S18, S44

Eric Simon — AB-W007 — S53

Eric van Wincoop — EF-W001 — S59

Eric Verhulst — AFC-W018 — S31

Eric von Hippel — MG-W007 — S87

Erich Jantsch — RGC-W001, WME-W001 — S19

Erik Braudeau — RGC-W026 — S23

Erik Lindhult — WME-W020 — S44

Erin Carson — CG-W002 — S55

Ernst Mach — W021 — S45

Ernst Verwaal — SL-W018 — S79

Ervin Laszlo — RGC-W018, RGC-W019, SG-W038 — S25

Eugene A. Rosa — UX-W016 — S29

European Parliamentary Technology Assessment network — SG-W030 — S44

Evangelos J. Giamarellos-Bourboulis — TD-W023 — S74

Evdoxia Kyriazopoulou — TD-W023 — S74

Eve Lipchik — TD-W002 — S73

Ezequiel Alpuche de la Cruz — PWR-W012 — S91

F. A. Hayek — W016 — S15

F. M. Redington — MG-W015 — S89

F. N. Bailey — SB-W006 — S09

Faez Ahmed — NC-W023 — S82

Faisal I. Khan — W064 — S02

Fang-Chi Lu — PM-W010 — S94

Fang-Chi Lu — PM-W029 — S94

Fangzhou Xiao — TA-W018 — S40, S50

FAO — GL-W055, GL-W085 — S43

Fatemeh Rahimi — GL-W017 — S01

Fatima Albar — MG-W011 — S88

Fatima Salahdine and Naima Kaabouch — DC-W18 — S93

FDIC — SG-W029 — S30

Feihao Fang — BC-W009 — S50

Felicity Petrie — MIL-W034 — S16

Felipe Huici — XA-W003 — S84

Felipe Huici — XA-W007 — S84

Felix A. Pollock — W055 — S46

Felix F. Wu — SG-W023 — S04

FEMA — GL-W046 — S16

Fernand Braudel — GL-W051 — S45

Fernando Luiz Vechiato — NC-W032 — S83

FHWA — T10-W004 — S108

Filipe João Bera de Azevedo Sobral — PWR-W009 — S91

Filosofia: An Encyclopedia of Russian Thought — WME-W016 — S34

Fisher, Roger, and William Ury — NN-W001 — S96

Florian L. Waitl — T15-W006 — S113

Florindo Rhaoni Picoli — AB-W008 — S53

Fondeur, Y., de Larquier, G., et al. — T05-W001 — S103

Fong Keng-Highberger — PWR-W006 — S91

Food and Agriculture Organization of the United Nations — GL-W010 — S43

Fran Ackermann — SG-W051 — S44

Frances J. Milliken — OI-W001 — S51

Francesca Ceroni — BC-W006 — S50

Francesco Polese — SG-W014 — S22

Francis Heylighen — SG-W032, W073 — S18, S49

Francisco G. Varela — W081 — S40

Francisco Hernández-Rodriguez — CG-W014 — S57

Francisco Torán Torres — SL-W012 — S79

Francois Richard — CG-W015 — S57

Frank Hark — W065 — S30

Frank Winters — SL-W022 — S77, S80

Frans Prenkert — UX-W002 — S14

François Sigaut — RGC-W035 — S39

Fred Simpson — SB-W019 — S36

Frederick L. Oswald — LP-W002 — S71

Frederick van der Ploeg — AB-W005 — S53

Fremont E. Kast — SG-W056 — S18

Friedrich Platz — LP-W008 — S71

Fritjof Capra — SG-W011, WME-W006 — S24

Fritz W. Scharpf — SG-W025 — S15

Frédéric Buchet — SL-W023 — S79

Fumiya Iida — W034, W099 — S37

G. Spencer Hatfield — W065 — S30

G.A.Swanson and James Grier Miller — RGC-W007 — S21

G.A.Swanson — RGC-W008 — S21

Gabriel Moreno — SL-W001 — S77, S78, S79

Gabriel Szulanski — CG-W017, CG-W018 — S58

Gabriela Imreh — LP-W007 — S71

Gabriele Bammer — PM-W015 — S95

Gabriele Oettingen — GS-W06 — S98

Gabriele Wulf — LP-W034 — S71

Gail Ross — GL-W016 — S41

Ganesh Gopalakrishnan — CG-W004 — S55

Gary A. Klein — PM-W022 — S95

Gary Groot — GL-W058 — S29

Gary Klein — PM-W021 — S95

Gary L. Bradshaw — CRE-W012 — S64

Gary P. Latham — GS-W01 — S98

Gary P. Latham — GS-W13 — S98

Gary P. Pisano — CG-W019 — S58

Garyfallia Poulakou — TD-W023 — S74

Gaulthier Gain — XA-W003 — S84

Gaulthier Gain — XA-W007 — S84

Gaurav Tewari — LC-W008 — S62

Ge Liu — BC-W009 — S50

Geert-Jan Dinant — TD-W012 — S74

Gene Rowe — PM-W028 — S95

Geneviève Oléron — LP-W011 — S71

Geoffrey Vickers (1978 interview) — SG-W012 — S44

George Adamis — TD-W023 — S74

George C. Necula — TA-W010, TA-W011 — S42

George C. Thorpe — MIL-W016 — S38

George H. Heilmeier — HCT-W001 — S86

George Kelly Society — WME-W018 — S41

George L. Topic Jr. — MIL-W004 — S16

George Mason University Warfield archive — SG-W060 — S44

George Soros — EF-W002, EF-W007 — S59

George T. Doran — GS-W11 — S98

George Wright — PM-W028 — S95

Georgios Chrysos — TD-W023 — S74

Gerald Midgley — WME-W020 — S44

Gerald R. Ferris — PWR-W001 — S91

Gerd Gigerenzer — W040 — S41

Ghislain Thibault — RGC-W033 — S39

Giacomo Novembre — GL-W001 — S41

Gianni Rodari — CRE-W017 — S65

Gilbert Simondon — W020 — S39

Gilles Fauconnier — NC-W017 — S82

Gino Camp — LP-W033 — S72

Giovanni Cunico — PM-W027 — S95

Giovanni Dosi — W093 — S48

Giulia Rispoli — GL-W022 — S17

Giuseppe Longo — GL-W091 — S40

Glenn W. Brier — CG-W021 — S31

Gong Xinzhong — RGC-W041 — S18

Gordon C. Ruscoe — WME-W005 — S21

Grace Young — TD-W016 — S74

Grant Wiggins — PM-W006 — S94

Grant Wiggins — PM-W007 — S94

Greg W. Marshall — SL-W002 — S80

Gregor Schöner — SM-W004 — S54

Gregory Bateson — SG-W009 — S18
Gregory E. Miller — GS-W09 — S98
Gregory Nagy — GL-W013 — S41
Gregory R. Ganger — XA-W002 — S84
Griffith, A. A. — T12-W001 — S110
Grégoire Wallenborn — PM-W004 — S94
Grönroos, C. — T04-W003 — S102
Gu Jifa — RGC-W041 — S18
Guang Yang — BC-W008 — S50
Guido Montúfar — UX-W041 — S37
Guillaume Weisang — MG-W020 — S89
Guillermo M. Wippold — PM-W023 — S95
Gundersen, T. — T09-W001 — S107
Guoxia Wang — GS-W08 — S98
Gustav Kuhn, Hugo A. Caffaratti, Robert Teszka and Ronald A. Rensink — DC-W16 — S93
Guus Lambert — LP-W033 — S72
Guy Aznar — CRE-W005 — S63
Guy Hoffman — CRE-W042 — S70
Guy-Bart Stan — BC-W006 — S50
Hall, Jeffrey A — NN-W013 — S97
Halszka Jarodzka — LP-W033 — S72
Hamit Coşkun — CRE-W036 — S69
Han T. J. Smit — AFC-W017 — S34, S61
Hanghang Tong — BC-W009 — S50
Hannah Cornish — TA-W034 — S41
Hannah V. Thornton — TD-W016 — S74
Hans Ragas — CG-W015 — S57
Hans-Jurgen Engelbrecht — RGC-W032
Harald Seelig — GS-W05 — S98
Harold Pashler — LP-W023 — S72
Harri Terho — SL-W019 — S80
Harry G. Barkema — NC-W031 — S83
Harry Markowitz — MG-W014 — S89
Harry Soodak — WME-W012 — S25
Hayden B. Peake — DC-W10 — S93
Hayley Dash — TD-W016 — S74

He Bin — SG-W050 — S44

Headquarters Department of the Army — MIL-W002 — S16

Headquarters United States Marine Corps — MIL-W001, MIL-W003 — S16, S38

Heinz von Foerster — W080 — S18

Helen J. Huang — SM-W003 — S54

Helena Correa Tonet — W088 — S42, S53

Helmut F. Spinner — UX-W034 — S45

Helsinki Psychotherapy Study Group — TD-W005 — S73

Henri Poincaré — CRE-W008 — S64

Henry E. Eccles — MIL-W015 — S38

Henry E. Smith — TA-W005 — S15

Henry L. Roediger III — LP-W022 — S72

Henry Mintzberg — ST-W001 — S92

Henry S. Parmelee — W024 — S02

Herbert A. Simon — CRE-W011 — S64

Herbert A. Simon — CRE-W012 — S64

Herbert A. Simon — W015 — S14

Herbert H. Clark — CG-W007, CG-W009, CG-W025 — S56

Herman C. Heiser — SL-W029 — S77

Herman Greene — WME-W002 — S19

Hermann Ebbinghaus — LP-W027 — S72

Hermann Haken — RGC-W038, W083 — S41, S46

Hermann Wotruba — GL-W077 — S43

Hiroaki Kitano — BC-W001 — S50

Hofmann, S. G., & Hay, A. C. — T03-W002 — S101

Hong Diep Nguyen — CG-W003 — S55

Hope Harvey — GL-W018 — S48

Horace Walpole — NC-W024 — S83

Horst Geschka — CRE-W002 — S63

Howard H. Pattee — UX-W012, UX-W014 — S42

Hsueh-Chih Chen — NC-W011 — S82

Hu Li — PWR-W006 — S91

Hubert Landier — UX-W030 — S27

Hugo Lefeuvre — XA-W003 — S84

Hugo Lefeuvre — XA-W007 — S84

Hugues Lortie-Forgues — CRE-W041 — S70

Humberto Aguayo — SG-W057 — S44
Humberto R. Maturana — W081 — S40
Huthwaite International — SL-W008 — S77
Huthwaite International — SL-W009 — S77
Hyeon-ju Pak — GS-W06 — S98
Hyun Song Shin — EF-W008 — S59
Håkansson, H. (ed.) — T04-W005 — S102
Héctor M. Briceño — XA-W002 — S84
Héctor Sabelli — WME-W015 — S46
Héloïse Fayet — PM-W025 — S95
Ian Briggs — CG-W004 — S55
Ian C. MacMillan — LC-W002 — S62
Ian C. MacMillan — LC-W005 — S62
Ian Hodder — GL-W012, GL-W096 — S39
IFAC Professional Accountants in Business Committee — GL-W048 — S42
IFS Institute — WME-W023 — S41
Ilya Prigogine — SG-W045 — S46
Indika Kumara — AFC-W011 — S31
Ines Thiele — BC-W005 — S50
Ingemar Dierickx — AB-W002 — S53
Insoo Kim Berg — TD-W002 — S73
International Council on Archives, Expert Group on Archival Description — GL-W023 — S42
International Maritime Organization — GL-W082 — S38
International Society for the Systems Sciences — SG-W006 — S18
Inés Küster Boluda — SL-W012 — S79
Inés Tejado — W048
Ioanna Mouratiadou — GL-W011 — S43
Ioannis Anagnostopoulos — TD-W023 — S74
Ion Stoica — W005 — S10
IPCC Working GroupI chapter authors — GL-W089 — S35
Isabella Verdinelli — AFC-W016 — S33
Isabelle Blanchette — CRE-W010 — S64
Ivan Wallan Tertuliano — LP-W012 — S71
Ivo Siekmann — W103 — S26, S50
Ivo Welch — EF-W005 — S59
J. A. Scott Kelso — RGC-W017 — S25

J. Athene Lane — TD-W016 — S74
J. Edward Russo — PM-W020 — S95
J. Friso den Hertog — W003 — S14
J. G. Bennett — SG-W033 — S18
J. H. Saltzer — GL-W030 — S12
J. Ian Munro — TD-W029 — S75
J. Kupitz — GL-W081 — S02
J. L. Kelly Jr. — MG-W016 — S90
J. M. Carlson — W026 — S05
J. P. van Gigch — RGC-W011 — S22
J. Van Buskirk — TA-W037 — S28
Jacek Blazewicz — SG-W047 — S40
Jack A. Goncalo — OI-W008 — S52
Jack Dongarra — CG-W001 — S55
Jack Edmonds — TA-W012 — S42
Jaco Quist — PM-W003 — S94
Jaco Quist — PM-W014 — S94
Jacob Goldenberg — NC-W002 — S81
Jacques Chabrillat — NC-W019 — S82
Jacques Lafitte — RGC-W034, W074 — S39
Jacques Villiers — RGC-W030 — S27
Jade Lafontaine — GL-W004 — S41
Jaime Díaz-Ocejo — GS-W16 — S98
Jakub Kurzak — CG-W001 — S55
Jakub Miler — MG-W013 — S88
James A. Waters — ST-W001 — S92
James Clerk Maxwell — W069 — S04
James D. Thompson — LC-W005 — S62
James D. Thompson — MIL-W005 — S16
James Demmel — CG-W003 — S55
James E. Huchingson — RGC-W020 — S25
James E. Kelley Jr. — MG-W001 — S85
James E. Rosenzweig — SG-W056 — S18
James E. Stott — W065 — S30
James F. Courtney — SG-W039 — S44
James G. March — OI-W010 — S52

James G. Miller — SG-W008 — S21

James Grier Miller and Jessie L. Miller — RGC-W006 — S21

James Grier Miller — RGC-W004, WME-W004 — S21

James J. Collins — BC-W003 — S50

James J. Cribbin — SL-W029 — S77

James J. Gibson — W042 — S41

James M. Morgan — LC-W006 — S62

James O'Toole Jr. — XA-W001 — S84

Jamshid Gharajedaghi and Russell L. Ackoff — RGC-W002 — S20

Jamshid Gharajedaghi — RGC-W003, WME-W003 — S20

Jan C. Willems — UX-W017, UX-W019, W014 — S04

Jan Wieseke — SL-W007 — S80

Jan-Chang Chen — SG-W048 — S44

Jane E. Dutton — ST-W002 — S92

Jane W. S. Liu — CG-W013 — S57

Janeen Loehr — GL-W001 — S41

Janet Kübler — RGC-W014 — S24

Jaqueline Pels — SG-W014 — S22

Jared M. Jacobs — BC-W007 — S50

Jash Hemant Shah — CRE-W039 — S69

Jay McTighe — PM-W006 — S94

Jay McTighe — PM-W007 — S94

Jay O'Toole — SL-W001 — S77, S78, S79

Jay R. Galbraith — W002 — S14

Jayodita C. Sanghvi — BC-W007 — S50

Jean Lagueux — SL-W014 — S79

Jean Lescure — CRE-W016 — S65

Jean Voge — RGC-W031, UX-W031, W094 — S27

Jean-Louis Le Moigne — RGC-W025, UX-W027, UX-W028 — S23, S27

Jean-Marie Lehn — GL-W009 — S01

Jean-Pierre Noblet — AB-W007 — S53

Jeanneret Medina, M., Baudet, C., & Bonnal, P. — T14-W001 — S112

Jeannette M. Wing — W087 — S42

Jeannie Holstein — GL-W038 — S41

Jeba Rezwana — CRE-W037 — S69

Jeff Conklin — CM-W004 — S60

Jeffrey D. Karpicke — LP-W022 — S72

Jeffrey D. Orth — BC-W005 — S50

Jeffrey K. Liker — LC-W003 — S62

Jeffrey K. Liker — LC-W006 — S62

Jeffrey Pfeffer — PWR-W003 — S91

Jeffrey W. Bloom — SG-W044 — S18

Jeffrey West — AFC-W008 — S32

Jelte M. Wicherts — CG-W024 — S31

Jen-Yao Chung — CG-W013 — S57

Jennifer Grenz — GL-W004 — S41

Jennifer Zyp-Loring — GL-W004 — S41

Jens Rasmussen — SB-W010, TA-W026 — S41

Jeremy C. Stein — EF-W004, EF-W010 — S59

Jeremy Campbell — WME-W029, WME-W029-ES1989 — S36

Jerome S. Bruner — GL-W016 — S41

Jerry Fjermestad — PM-W018 — S95

Jessica C. Flack — TA-W019, TA-W041 — S28, S46

Jessica Wiese — PM-W009 — S94

Jesus Ramos-Martin — SG-W017 — S47

Jetter, C. — T17-W003 — S115

Ji Yan — TD-W022 — S74

Jie Ling — GL-W073 — S01

Jillian O'Mara — W106 — S44

Jing-Shu Li — TD-W022 — S74

Jingrui He — BC-W009 — S50

Jiri Trnka — MIL-W033 — S16

Joan Cornuet — MIL-W017 — S38

Jochen W. L. Cals — TD-W012 — S74

Jodi Taylor — TD-W016 — S74

Johan Christensen — GL-W065 — S01

Johan S. H. van Leeuwaarden — TA-W016 — S38

Johannes Jaeger — W085 — S46

John Brocklesby, Stephen Cummings and John Davies — RGC-W009 — S22

John C. Doyle — UX-W042, UX-W043 — S05

John Cage — CRE-W022 — S66

John Carroll — SB-W017 — S29

John Dewey — GL-W093 — S45
John Doyle — W012, W026 — S05, S50
John Dunlosky — LP-W020 — S72
John Dunlosky — LP-W029 — S72
John E. McBride — GL-W086 — S43
John F. Padgett — PWR-W011 — S91
John F. Padgett — SB-W004
John I. Glass — BC-W007 — S50
John J. Cristiano — LC-W003 — S62
John J. Hopfield — TA-W017 — S40, S50
John Jannotti — XA-W002 — S84
John Joseph — W046 — S14
John M. Richardson, Jr. — W097 — S09
John Odling-Smee — TA-W020 — S28, S50
John P. Scholz — SM-W004 — S54
John R. Boyd — MIL-W008 — S16
John R. Hudson — SG-W010 — S44
John Rogers — PM-W019 — S95
John Sweller — TA-W035, TA-W036 — S41
John T. Wixted — LP-W023 — S72
John van der Kamp — GL-W071 — S41
John von Neumann — W043 — S30
Johny Van Aerde — WME-W007 — S24
Jon Doyle — CM-W003 — S60; S105
Jon L. Bentley — TD-W028 — S75
Jonathan Appavoo — XA-W004 — S84
Jonathan Kass — OI-W013 — S52
Jonathan R. Karr — BC-W007 — S50
Jonathan Smallwood — CRE-W025 — S67
Jonathan Teicher (editor) — GL-W092 — S08
Jonathan W. Schooler — CRE-W025 — S67
Jonathan Yu-Meng Li — MG-W019 — S90
Joost Elffers (producer) — PWR-W004 — S91
Jooyoung Park — PM-W010 — S94
Jooyoung Park — PM-W029 — S94
Jose Tribolet — SB-W007 — S09

Josefin Svensson (interviewing Antoine Bousquet) — MIL-W020 — S16

Josep Vidal-Alaball — TD-W015 — S74

Joseph A. Tainter — UX-W006 — S48

Joseph M. Hellerstein — AFC-W010, W005 — S10, S30

Joseph Murphy — GL-W006 — S48

Joseph P. Gaspar and Maurice E. Schweitzer — DC-W14 — S93

Josh R. Auld — TA-W038 — S28

Joshua Owotogbe — AFC-W011 — S31

Joshua Rosen — AFC-W010 — S30

José Aguilar — W067 — S49

José Luis Bernal López — PWR-W012 — S91

José Raeudo Pereira — CRE-W006 — S63

João Ricardo da Cunha Croce Lopes — DC-W09 — S93

Jr-Syu Yang — SG-W048 — S44

Juan Antonio Mora-Merida — GS-W16 — S98

Judit Dorca-Vila — TD-W015 — S74

Judith Jiménez-Díaz — LP-W015 — S71

Judith LeFevre — LP-W031 — S72

Julia Balogun — ST-W003 — S92

Julia W. Y. Kam — CRE-W025 — S67

Julian Roelle — LP-W016 — S72

Juliana Arcoverde Mansur — PWR-W009 — S91

Julie Considine — TD-W009 — S73

Julie Langou — CG-W001 — S55

Julie Posetti (editors) — GL-W047 — S42

Julien Langou — CG-W001 — S55

Jun Wang — W098 — S37

Justin J. Meza — CG-W015 — S57

Jürgen Kriz — WME-W022 — S41

K. Anders Ericsson — LP-W001 — S71

K. Anders Ericsson — LP-W003 — S71

K. Anders Ericsson — LP-W005 — S71

K. L. Ford — GL-W078 — S43

K. Malinowski — SB-W006 — S09

Kabir Husain — TA-W018 — S40, S50

Kai Chi Yam — PWR-W006 — S91

Kaleb McDowell — MIL-W029 — S16
Kamal S. Birdi — NC-W004 — S81
Karan Girotra — CRE-W040 — S69
Karel Cool — AB-W002 — S53
Karel Netočný — W079 — S46
Karen Marais — SB-W017 — S29
Karen Ruhleder — TA-W024 — S42
Karien Stronks — W106 — S44
Karl Friston — W052 — S46
Karl J. Friston — W056 — S46
Karl T. Ulrich — CRE-W040 — S69
Karl T. Ulrich — MG-W008 — S87
Karl-Erik Årzén — CG-W014 — S57
Karoline Schnetter — GS-W06 — S98
Karsten Juhl Jørgensen — TD-W013 — S74
Katherine A. Lawrence — ST-W002 — S92
Katherine A. Rawson — LP-W020 — S72
Katherine A. Rawson — LP-W029 — S72
Kathryn Chaloner — AFC-W016 — S33
Kathy Eastwood — TD-W016 — S74
Katia Bertoldi — GL-W065 — S01
Katja Hölttä-Otto — CRE-W001 — S63
Kaylyn A. Garcia — PM-W023 — S95
Kazutoshi Tanaka — W051 — S46
Keith Beven — GL-W015 — S43
Keith Bostic — TD-W027 — S75
Keith Brown — PM-W019 — S95
Keith J. Holyoak — CRE-W009 — S64
Keith Pavitt — W068 — S12
Kenneth E. Boulding — SG-W036 — S18
Kenneth G. Wilson — GL-W025 — S46
Kenneth Mackenzie — XA-W002 — S84
Kennon M. Sheldon — GS-W04 — S98
Kenny Smith — TA-W034 — S41
Kerenza Hood — TD-W012 — S74
Kevin Crowston — WME-W011 — S14

Kevin Dunbar — CRE-W010 — S64

Kevin G. Munhall — CG-W011 — S56

Kevin Krupski — GL-W053 — S16

Kevin N. Laland — TA-W020, TA-W021 — S28, S50

Keyan Ghazi-Zahedi — UX-W041 — S37

Kim B. Clark — UX-W008 — S07

Kim Halskov — CRE-W031 — S68

Kim J. Vicente — TA-W026 — S41

Kim Sears — GL-W058 — S29

King, C., Edlington, T., & Williams, B. — T08-W002 — S106

Kintan Kamilia — GL-W011 — S43

Kirkus Reviews — WME-W030 — S36

Kleidson Daniel Medeiros Leopoldino — CRE-W006 — S63

Kluge, A., & Gronau, N. — T07-W002 — S105

Kole T. Roybal — GL-W060 — S40, S50

Konstantinos Marousis — TD-W023 — S74

Koushik Sen — CG-W003 — S55

Kozo Mayumi — SG-W017 — S47

Krinos Li — BC-W008 — S50

Kristin B. Munksgaard — UX-W002 — S14

Krypotos, A.-M., Effting, M., Kindt, M., & Beckers, T. — T03-W001 — S101

Kwei-Jay Lin — CG-W013 — S57

Kyle W. Harwell — LP-W003 — S71

L. Kabanov — GL-W081 — S02

L. Ulbo de Sitter — W003 — S14

Laia Font-Ribera — TD-W015 — S74

Lambrechts, F., Grieten, S., Bouwen, R., & Corthouts, F. — T06-W001 — S104

Lan-Lin Peng — SG-W048 — S44

Lance T. Peterson — TD-W004 — S73

Lancelot Da Costa — W056 — S46

Larry Woodman — XA-W004 — S84

Lars Kotthoff — TD-W024 — S75

Laura Cabrera — SG-W041 — S44

Lauren Komarow — TD-W018 — S74

Laurence A. Walker — SM-W001 — S54

Laurent Loty — NC-W029 — S83

Laurent Mathy — XA-W003 — S84

Laurent Mathy — XA-W007 — S84

Laurent Thévenot — GL-W032, GL-W033 — S15

Lee R. Lynd — GL-W086 — S43

Leigh Anne Bakel — GL-W058 — S29

Leigh D. Kinsman — GL-W058 — S29

Lennart Meincke — CRE-W040 — S69

Lenos Trigeorgis — AFC-W017 — S34, S61

Leonardo Morsut — GL-W060 — S40, S50

Leonid Hurwicz — TA-W001, TA-W002 — S15

Leroy Cronin — W084 — S46

Leslie C. Johnson — TD-W006 — S73

Leslie Main Johnson — GL-W004 — S41

Levin, Daniel Z., Jorge Walter, and J. Keith Murnighan — NN-W012 — S97

Li, B.-H., & Chang, C.-T. — T09-W004 — S107

Liam P. Maher — PWR-W001 — S91

Lian Loke — CRE-W032 — S68

Liang Zhu — TD-W016 — S74

Liliana Mendoza González — CRE-W044 — S70

Liliane Locatelli — LP-W013 — S71

Linda C. Schmidt — NC-W005 — S81

Linda Rouleau — ST-W003 — S92

Line Bjørnskov Pedersen — TD-W014 — S74

Linnhoff, B., & Flower, J. R. — T09-W002 — S107

Linnhoff, B., & Flower, J. R. — T09-W005 — S107

Lionel Giles (translator and commentator) — MIL-W027 — S16

Lipp, Wolfram Emanuel, Remigiusz Smolinski, and Peter Kesting — NN-W003 — S96

Lisa D. Ordóñez — GS-W12 — S98

Lisa D. Ordóñez — GS-W14 — S98

Lisa Raphals — DC-W02 — S93

Liu Puhe — RGC-W041 — S18

Loes Crielaard — W106 — S44

Loet Leydesdorff — SG-W055, WME-W021 — S47, S49

Loizos Heracleous — T06-W002 — S104

Lothar Leistner — GL-W079 — S43

Louis Kaplow — GL-W043 — S15

Lova Rajaobelina — SL-W014 — S79

Lu Mingsen / 卢明森 — RGC-W039 — S18

Luan Moreira — SG-W043 — S18

Luc De Raedt — BC-W011 — S50

Luc Hagenaars — W106 — S44

Luca Corti — W047 — S40

Lucas He — BC-W008 — S50

Lucy Yardley — TD-W016 — S74

Ludwig von Bertalanffy — W072 — S18

Luigi Marengo — W093 — S48

Luis Ceze — CG-W006 — S55

Luis Garicano — W030 — S14

Lydia Liaskou-Antoniou — TD-W023 — S74

Lynne Bradshaw — TD-W016 — S74

Lynne J. Shannon — UX-W018 — S43

Lynne M. Westphal — PM-W013 — S94

Lyra J. Colfer — SB-W005 — S07

M. A. Laaksonen — TD-W005 — S73

M. Brdys — SB-W006 — S09

M. D. Mesarovic — SB-W008 — S09

M. Douglas McIlroy — TD-W027 — S75

M. Douglas McIlroy — TD-W028 — S75

M. E. A. Meisch — SG-W018 — S47

M. Frans Kaashoek — XA-W001 — S84

M. Frans Kaashoek — XA-W002 — S84

M. H. A. Newman — W075

M. Kaipainen — TD-W005 — S73

M. Marttunen — TD-W005 — S73

M. S. Leaning — RGC-W021 — S25

M. Välikoski — TD-W005 — S73

Mack Hanan — SL-W029 — S77

Maesima, Giovania Mitie, Lívia Sanseverino Gomes, and Daniela Centenaro Levandowski — NN-W024 — S97

Magoroh Maruyama — RGC-W037, SG-W061 — S18

Mahmoud Elsherif — CG-W024 — S31

Mahyar A. Amouzegar — MIL-W017 — S38

Malene Plejdrup Hansen — TD-W014 — S74

Malvina Lada — TD-W023 — S74
Mandy Lui — TD-W016 — S74
Manu Kapur — LP-W025 — S72
Manu Kapur — LP-W026 — S72
Manuel Baltieri — RGC-W036 — S03
Manuel Blum — TD-W032 — S75
Marc Lecoutre — NC-W019 — S82
Marc Shapiro — W061 — S11
Marcel A. L. M. van Assen — CG-W024 — S31
Marcel Meulder — DC-W05 — S93
Marcin Miłkowski — W054 — S37
Marco Ottaviani — EF-W009 — S59
Marco Sigg — MIL-W022 — S16
Marco von Atzigen — W034 — S37
Marcus Eduardo Costa Souto — CRE-W006 — S63
Marcus Krantz — BC-W011 — S50
Marcus W. Feldman — TA-W020 — S28, S50
Marek Zawirski — W061 — S11
Margaret A. Boden — NC-W014 — S82
Maria das Graças Torres da Paz — W088 — S42, S53
Maria E. Fernandez — PM-W023 — S95
Maria Enrica Virgillito — W093 — S48
Maria Luisa Paracchini — GL-W011 — S43
Maria Souli — TD-W018 — S74
Marialuisa Saviano — SG-W014 — S22
Marie Csete — UX-W042 — S05
Marie Personnaz — OI-W008 — S52
Marily Oppezzo — CRE-W028 — S67
Marina Koupetori — TD-W023 — S74
Mario Giampietro — SG-W017 — S47
Mario Orestes Aguirre González — CRE-W006 — S63
Marisabel Marratt — UX-W026 — S39
Maritza Isabel Álvarez Rodríguez — CRE-W018 — S65
Marjan Bakker — CG-W024 — S31
Marjory Blumenthal — GL-W095 — S12
Mark A. Guadagnoli — LP-W021 — S72

Mark Baranowski — CG-W004 — S55
Mark D. Cannon — AFC-W015 — S31, S51
Mark Fuge — NC-W023 — S82
Mark Hayward — RGC-W033 — S39
Mark Laser — GL-W086 — S43
Mark S. Ackerman — CM-W005 — S60
Mark Turner — NC-W017 — S82
Markus W. Covert — BC-W007 — S50
Mars Climate Orbiter Mishap Investigation Board — T14-W003 — S112
Martha Lampland — TA-W025 — S42
Martin Biehl — RGC-W036, W055 — S03, S46
Martin Bækgaard — GL-W019 — S48
Martin de Holan, P., & Phillips, N. — T07-W003 — S105
Martin Golubitsky — SG-W046 — S46
Martin Monperrus — AFC-W007 — S31
Martin van Creveld — MIL-W006 — S16
Martin van Hecke — GL-W065 — S01
Martin Zwick — SG-W053 — S47
Martina Maggio — CG-W014 — S57
Marty Chaplin — TD-W019 — S74
Marvin B. Lieberman — SW-W001 — S61
Marvin B. Lieberman — SW-W002 — S61
Mary Jo Bitner — GL-W005 — S41
Mary L. Gick — CRE-W009 — S64
Mary Lou Maher — CRE-W037 — S69
Mary Nicolaou — W106 — S44
María Morera-Castro — LP-W015 — S71
Masahiro Suzuki — W051 — S46
Massachusetts Institute of Technology — TD-W026 — S75
Matej Hoffmann — W007, W008 — S37
Matjaž Juršič — NC-W022 — S82
Matt Dixon — SL-W025 — S78
Matteo Capucci — RGC-W036 — S03
Matteo Mossio — GL-W091, W013 — S40
Matteo Saveriano — AFC-W008 — S32
Matthew Boyd — XA-W004 — S84

Matthew Nicastro — CG-W011 — S56
Matthew Ridd — TD-W016 — S74
Matthew Thomson — GL-W060 — S40, S50
Matthias Rief — SL-W007 — S80
Mattias Höjer — PM-W014 — S94
Maurice E. Schweitzer — GS-W12 — S98
Maurice E. Schweitzer — GS-W14 — S98
Mauricio Pérez Jiménez — CRE-W019 — S65
Max H. Bazerman — GS-W12 — S98
Max H. Bazerman — GS-W14 — S98
Max Kreminski — CRE-W039 — S69
Maël Montévil — GL-W091, W013 — S40
Md Nazim Uddin — CG-W015 — S57
Mehrdad Jalali — BC-W010 — S50
Mehrnaz Azadyekta — NC-W018 — S82
Mei-Chen Hsueh — AFC-W003 — S30
Meisam Akbarzadeh — AFC-W008 — S32
Mel Woods — NC-W030 — S83
Melissa Schwarz — OI-W013 — S52
Melvin I. Simon — W012 — S05, S50
Melvyn Sim — W060 — S05
Metin Sengul — W046 — S14
Michael B. Elowitz — BC-W004 — S50
Michael C. Jackson — RGC-W010 — S22
Michael Cooper — SL-W022 — S77, S80
Michael D. Mrazek — CRE-W025 — S67
Michael Diehl — OI-W012 — S52
Michael F. Scheier — GS-W09 — S98
Michael F. Schober — CG-W009 — S56
Michael J. Dockry — PM-W013 — S94
Michael J. Franklin — W005 — S10
Michael J. Rohrbaugh — TD-W001 — S73
Michael K. Buckland — GL-W020 — S42
Michael L. Begeman — CM-W004 — S60
Michael Lachmann — W084 — S46
Michael Mose Biskjaer — CRE-W031 — S68

Michael N. Kennedy — LC-W011 — S62

Michael R. Berthold — NC-W021 — S82

Michael Rathjen — GL-W087 — S42

Michael S. Franklin — CRE-W025 — S67

Michael Spence — AB-W005 — S53

Michael Suk-Young Chwe — EF-W003 — S59

Michail A. Makridis — AFC-W008 — S32

Michel Bitbol — RGC-W016 — S24

Michel Crozier — PWR-W002 — S91

Michel Petrovitch (Mihailo Petrović) — TKT-W003 — S17

Michel Pretalli — DC-W21 — S93

Michele Orr — TD-W009 — S73

Michele Weiner-Davis — TD-W002 — S73

Michelle Y. Merrill — RGC-W014, RGC-W015 — S24

Microsoft — AFC-W005 — S31

Mieke Goos — LP-W033 — S72

Miha Grčar — NC-W022 — S82

Mihajlo D. Mesarović — W096 — S09

Mihaly Csikszentmihalyi — LP-W031 — S72

Mildred L. G. Shaw — SG-W034 — S42

Miltiades Kyprianou — TD-W023 — S74

Miranda Lucas — SG-W026 — S41

Miriam V. Gutschow — BC-W007 — S50

MIT PDOS exokernel project — XA-W008 — S84

Mitchell J. Nathan — LP-W020 — S72

Mitio Nagumo — W044 — S05, S54

Mixue Li — GL-W038 — S41

Mohammed K. Ali — TD-W006 — S73

Moisés Bolívar-Vivas — W048

Monica Tamariz — TA-W034 — S41

Morgan R. Walker — MG-W001 — S85

Moshe Kress — MIL-W012, MIL-W013, MIL-W014 — S38

Motooka, T., & Kiuchi, K. — T12-W004 — S110

Murat Tarakci — CRE-W045 — S63

Murray Gell-Mann — WME-W014 — S25

Mustafa Alhaj Ahmad Alaboud, Yvonne Steggemann, Timo Klein-Soetebier, Wilfried Kunde and Matthias Weigelt — DC-W17 — S93

Mustafa Shraim — W095 — S05

Myriam Ribière — CM-W006 — S60

Mônica de Fátima Bianco — W028 — S14

N. J. Habraken — GL-W092 — S08

N. John Habraken — GL-W021 — S08

Naama Barkai — W033 — S05

Naci Saldi — TA-W004 — S33

Nacyra Assad-Garcia — BC-W007 — S50

Nada Lavrač — NC-W022 — S82

Nada Matta — CM-W006 — S60

Nadine B. Sarter — CG-W016 — S57

Nam P. Suh — SG-W052, W037 — S44

Nancy G. Leveson — UX-W020, UX-W021 — S29

Nancy J. Cox — SG-W053 — S47

Nancy J. Turner — GL-W004 — S41

Nancy Leveson — SB-W017 — S29

Nancy Pennington — PM-W020 — S95

NASA / Night Sky Network — GL-W088 — S01

NASA — T14-W002 — S112

Naseem Ahmadpour — CRE-W032 — S68

Nassim N. Taleb — AFC-W001 — S32

Nassim Nicholas Taleb — AFC-W002 — S32

Natalia Szostak — SG-W047 — S40

Natalie Stoljar — GL-W040 — S48

Nate Kornell — LP-W024 — S72

Nathalie Bonnardel — CRE-W013 — S64

Nathaniel Virgo — RGC-W036 — S03

National Bureau of Standards — T10-W001 — S108

National Commission on Forensic Science — GL-W063 — S29

National Health Commission of China — GL-W061 — S48

NATO Research Task Group SAS-085 — MIL-W011 — S16

NATO — GL-W036 — S16

Nawar N. Chaker — SL-W001 — S77, S78, S79

NCCIH — GL-W062 — S48

Neil D. Jones — TA-W009 — S42

Neil R. Smalheiser — NC-W020 — S82

NHS England — GL-W064 — S29

Niccolò Machiavelli — DC-W04 — S93

Nicholas C. Soderstrom — LP-W019 — S72

Nicholas H. Nelsen — MIO-W001 — S35

Nicholas J. Cepeda — LP-W023 — S72

Nicholas J. Higham — CG-W002, W089, W090 — S35, S55

Nici Zimmermann — PM-W027 — S95

Nicolas Dulac — SB-W017 — S29

Nihat Ay — UX-W015, UX-W022, UX-W041 — S37

Nikita Smirnov — SG-W022 — S47

Nikolaos Alexiou — TD-W023 — S74

Nikolaos Melachroinopoulos — TD-W023 — S74

NIOSH — GL-W059 — S02

Nobuhiko Itani — CRE-W023 — S66

Noel León-Rovira — SG-W057 — S44

Noël Bonneuil — SM-W006 — S54

Nuno Preguiça — W027, W061 — S10, S11

Nuno Videira — PM-W027 — S95

O. Lindfors — TD-W005 — S73

OCC — SG-W029 — S30

Oesch, John, and Adam D. Galinsky — NN-W004 — S96

Oguz A. Acar — CRE-W045 — S63

Ohid Yaqub — NC-W028 — S83

Ole Olsen — TD-W013 — S74

Olga Springer — MG-W013 — S88

Olga Torres — DC-W06 — S93

Oliver Borchert — GL-W045 — S42

Oliver López-Corona — AFC-W008 — S32

Oliver P. Hauser — CRE-W038 — S69

Oliver P. John — PWR-W007 — S91

Olivier Corby — CM-W006 — S60

Olmo R. van den Akker — CG-W024 — S31

Organisation internationale du Travail — NN-W002 — S96

Orion Mowbray — TD-W006 — S73

Orlando Pereira de Souza Júnior — LP-W012 — S71

Orran Krieger — XA-W004 — S84

Oswald, Debra L., Eddie M. Clark, and Cheryl M. Kelly — NN-W014 — S97

Oulipo — CRE-W016 — S65

P. Knekt — TD-W005 — S73

P. M. Binder — AFC-W013 — S32

P. N. Rosenstein-Rodan — AB-W003 — S53

P. Tatjewski — SB-W006 — S09

Pamela Herd — GL-W018 — S48

Panagiotis A. Kanellopoulos — CRE-W024 — S66

Paolo Barberi — GL-W011 — S43

PARC/CAST Flight Deck Automation Working Group — GL-W083 — S12

Parul Sohal — XA-W004 — S84

Pascal Boyer — GL-W042 — S41

Pascal Lièvre — NC-W019 — S82

Paschal Sheeran — GS-W07 — S98

Pat Langley — CRE-W012 — S64

Patricia A. Alexander — NC-W005 — S81

Patricia L. Kramer — SG-W053 — S47

Patrick Bolton — SW-W006 — S61

Patrick Saint-Pierre — SM-W006 — S54

Patrick Weaver — MG-W005 — S86

Patrícia Alves-Oliveira — CRE-W042 — S70

Patrícia Arriaga — CRE-W042 — S70

Paul A. Kirschner — LP-W033 — S72

Paul Ayres — TA-W036 — S41

Paul C. Nutt — OI-W009 — S52

Paul Chandler — TA-W036 — S41

Paul Collier — AB-W005 — S53

Paul du Gay — SB-W001 — S13

Paul E. Griffiths — SG-W054 — S28

Paul Hine — TD-W019 — S74

Paul M. Mitchell — TD-W016 — S74

Paul Maglio — W041 — S41

Paul P. Maglio — GL-W027 — S41

Paul R. Amyotte — W064 — S02

Paul T. Britton — W065 — S30

Paula de Oliveira Ferreira — CRE-W006 — S63

Paulo Tabuada — TA-W013, TA-W014 — S05

PCAOB — GL-W044 — S42

Pedro Zuidberg Dos Martires — BC-W011 — S50

Pei-Zhen Chen — NC-W011 — S82

Per Christian Hansen — MIO-W002 — S35

Pere Roura-Poch — TD-W015 — S74

Peter Alvaro — AFC-W010 — S30

Peter Bailis — W005, W057 — S10

Peter Berggren — MIL-W033 — S16

Peter Checkland — SG-W001 — S44

Peter Dalsgaard — CRE-W031 — S68

Peter Deane — PM-W015 — S95

Peter Dittrich — SG-W042, W077 — S40, S49, S50

Peter E. Keller — GL-W001 — S41

Peter M. Gollwitzer — GS-W07 — S98

Peter M. McIlroy — TD-W027 — S75

Peter M. Todd — W040 — S41

Peter Muir — TD-W016 — S74

Peter Sestoft — TA-W009 — S42

Peter Sterling — GL-W028 — S05

Peter Sørensen — EF-W009 — S59

Petra Solarská — LP-W032 — S72

Phil Woods — GL-W058 — S29

Philip Steadman — SB-W012 — S08

Philip Vergragt — PM-W003 — S94

Philip Weihs — LC-W010 — S62

Philippe Bacchetta — EF-W001 — S59

Philippe M. Cury — UX-W018 — S43

Phillip G. Armour — LP-W018 — S72

Phillip Jesse McGowan — SL-W011 — S79

Pier Luigi Luisi — RGC-W013, RGC-W016, SG-W011, WME-W006, WME-W008 — S24

Pierre Liénard — GL-W042 — S41

Pierre Marquis — TA-W008 — S42, S50

Pierre Olivier — XA-W007 — S84

Pierre-Paul Grassé — W031 — S49

Pietro Speroni di Fenizio — W077 — S40, S50

Piotr Luszczek — CG-W001 — S55

Pramod Kumar Nayak — SL-W003 — S76

Principles of Chaos Engineering contributors — AFC-W004 — S31

Pyotr G. Shchedrovitsky — SG-W024 — S44

Python development team — TD-W031 — S75

Qian Chen — TD-W022 — S74

R. Uribe — W081 — S40

Rachel C. M. Brierley — TD-W016 — S74

Rafael Rivas — W067 — S49

Rainer Diaz-Bone — GL-W031 — S15

Ralf Abraham — SB-W007 — S09

Ralf Th. Krampe — LP-W001 — S71

Ramon A. Avila — SL-W024 — S78

Ramon Ferrer i Cancho — WME-W010 — S41

RAND Corporation — SG-W031 — S44

Randall Anway — RGC-W014 — S24

Raphael Douady — AFC-W001 — S32

Raul Espejo — UX-W007, UX-W010 — S22

Ravi Sandhu — TA-W032 — S42

Ravishankar K. Iyer — AFC-W003 — S30

Rebecca Clarke — TD-W016 — S74

Regina M. O'Neill — ST-W002 — S92

Regina Schrank — SL-W007 — S80

Reinhard Kopiez — LP-W008 — S71

Renato Mancuso — XA-W004 — S84

René Bourrigaud (editor) — RGC-W035 — S39

Reza Aghayari — GL-W017 — S01

Reza Sheibani — BC-W010 — S50

Rhett T. Epler — SL-W001 — S77, S78, S79

Rhys Algar — BC-W006 — S50

Ricard V. Solé — WME-W010 — S41

Richard A. Schmidt — LP-W030 — S72

Richard C. Sperry — MG-W011 — S88

Richard E. Hayes — MIL-W009, MIL-W010 — S16

Richard E. Michod — TA-W022, TA-W023 — S28

Richard Jones — XA-W004 — S84

Richard Jones — XA-W006 — S84

Richard Kuhn — TA-W032 — S42

Richard P. Gabriel — SB-W011 — S08

Richard Schulz — GS-W09 — S98

Richard Waldinger — PM-W002 — S94

Richards J. Heuer, Jr. — DC-W07 — S93

Richardson Sales Performance — SL-W027 — S77

Rick A. Relyea — TA-W038 — S28

Rick Bereit — MIL-W017 — S38

Rikke Vognbjerg Sydenham — TD-W014 — S74

Rita Gunther McGrath — LC-W002 — S62

Rita Gunther McGrath — LC-W004 — S62

Rita Gunther McGrath — LC-W005 — S62

Ritu Banerjee et al., for the Antibacterial Resistance Leadership Group — TD-W017 — S74

Ritu Banerjee — TD-W018 — S74

Rob Phillips — TA-W018 — S40, S50

Robb, A., Reedy, E. J., Ballou, J., DesRoches, D., Potter, F., & Zhao, Z. — T02-W005 — S100

Robert A. Bjork — LP-W019 — S72

Robert A. Bjork — LP-W024 — S72

Robert A. Bjork — LP-W028 — S72

Robert A. Bjork — LP-W030 — S72

Robert A. Duke — LP-W006 — S71

Robert E. Tarjan — TD-W032 — S75

Robert E. Ulanowicz — RGC-W042, RGC-W043, WME-W025 — S47

Robert Greene — PWR-W004 — S91

Robert Grimm — XA-W002 — S84

Robert K. Merton — NC-W025 — S83

Robert L. Watters, Jr. — TA-W031 — S42

Robert Rosen — RGC-W022, RGC-W023, W102 — S26

Robert S. Huckman — CG-W019 — S58

Robert S. Pindyck — SW-W004 — S61, S62

Robert S. Tripp — MIL-W017 — S38

Robert W. Floyd — TD-W032 — S75

Robert W. Ring — W065 — S30

Robert Winter — SB-W007 — S09
Robin Patel — TD-W018 — S74
Rocio Gomez — WME-W025 — S47
Roger Buehler — PM-W009 — S94
Roger C. Conant — W050 — S03
Roger Chaffin — LP-W007 — S71
Rogier M. Hopstaken — TD-W012 — S74
Rohan Das — CG-W015 — S57
Rohit Voleti — CG-W011 — S56
Ronald A. Finke — NC-W015 — S82
Ronald A. Fisher — W058 — S33
Ronald G. McGarvey — MIL-W017 — S38
Ronald L. Rivest — TD-W032 — S75
Ronie Galeano — SL-W013 — S79
Rosane de Fatima Antunes Obregon — CRE-W034 — S68
Roy Radner — TA-W003, TA-W040 — S14, S15
Rozental, A., et al. — T01-W004 — S99
Rune Aabenhus — TD-W013 — S74
Rune Munck Aabenhus — TD-W014 — S74
Russell D. Gray — SG-W054 — S28
Russell Hunt — XA-W002 — S84
Russell L. Ackoff — SG-W040 — S44
Russell M. Gordley — GL-W060 — S40, S50
Ryo Sato — SG-W035 — S10
Ryota Kanai — W055 — S46
Ryu, J., & Maravelias, C. T. — T09-W003 — S107
Răzvan Deaconescu — XA-W003 — S84
Răzvan Deaconescu — XA-W007 — S84
S. L. Brandt-Williams — SG-W018 — S47
S. P. Rahul — TA-W011 — S42
Sabine Valis — SL-W026 — S79
Sackett, P. R., Zhang, C., Berry, C. M., & Lievens, F. — T05-W004 — S103
Saku Mantere — ST-W004 — S92
Sally J. Goerner — RGC-W042, WME-W025 — S47
Sally S. Bruce — TA-W031 — S42
Samantha Abbs — TD-W016 — S74

Sampsa Hyysalo — GL-W035 — S41

Sanjeev Krishna — TD-W019 — S74

Sankaralingam Panneerselvam — CG-W015 — S57

Sara I. Walker — W084 — S46

Sarah B. Doernberg — TD-W018 — S74

Sarah Jenkins and Rick Delbridge — DC-W11 — S93

Sarah Parry — GL-W006 — S48

Sarah Sharples — NC-W030 — S83

Sarasvathy, S. D. — T02-W003 — S100

Sari Tran — CG-W015 — S57

Sarnoff A. Mednick — NC-W010 — S82

Scarlett R. Herring — CRE-W033 — S68

Schmidt, F. L., & Hunter, J. E. — T05-W003 — S103

Schweitzer, Maurice E., Kori L. Krueger, Erica Boothby, and Gus Cooney — NN-W006 — S96

Scott A. Inks — SL-W024 — S78

Scott M. Coyle — GL-W060 — S40, S50

Scott R. Klemmer — OI-W013 — S52

Scott Rose — GL-W045 — S42

Scott Willenbrock — MG-W017 — S90

Sean Connelly — GL-W045 — S42

Sebastian Bollien — PM-W012 — S94

Sebastian Wild — TD-W029 — S75

Semen Kurkin — SG-W022 — S47

Senni Kirjavainen — CRE-W001 — S63

Serdar Yüksel — TA-W004 — S33

Serge M. Garcia — UX-W018 — S43

Sergio Barile — SG-W014 — S22

Sextus Iulius Frontinus — DC-W03 — S93

Shailesh Sood — MG-W002 — S85

Shannon D. Scott — GL-W058 — S29

Sharan Santhanam — XA-W003 — S84

Sharlette Ryder — TD-W009 — S73

Sharma, M., Sahni, S. P., & Sharma, S. — T13-W004 — S111

Shawn K. Westaway — SG-W053 — S47

Shenglong Deng — BC-W008 — S50

Shewhart, W. A. — T13-W001 — S111

Shiban, Y., Pauli, P., & Mühlberger, A. — T03-W005 — S101

Shih-Yuan Huang — NC-W011 — S82

Shingo Murata — W051 — S46

Shuyang Shi — CG-W015 — S57

Sich, M. M. — T03-W004 — S101

Sidney G. Winter — CG-W018 — S58

Sierralta Ríos, Aníbal — NN-W008 — S96

Sima Soltani — BC-W010 — S50

Simon Caulkin — SG-W020 — S31

Simon Kirby — TA-W034 — S41

Simon Kuenzer — XA-W003 — S84

Simon Kuenzer — XA-W007 — S84

Simone M. Ritter — CRE-W027 — S67

Siri Aas Smedemark — TD-W013 — S74

Slava Kalyuga — TA-W036 — S41

Smith, J. F., Jr. — T17-W005 — S115

Smith, W. R. — T04-W001 — S102

Solution Focused Brief Therapy Association Research Committee — TD-W003 — S73

Sonja Wehberg — TD-W014 — S74

Stafford Beer — RGC-W012, W101 — S22

Stanislas Leibler — BC-W004, W033 — S05, S50

Stanislas Noria — AFC-W013 — S32

Stanisław H. Czyż — LP-W032 — S72

Stanley Reiter — DP-W001 — S06

Stanley Reiter — TA-W002 — S15

Steel, P. — T01-W001 — S99

Stefan Kühl, in conversation with Martin Claßen — ST-W009 — S92

Stefan Thomke — MG-W010 — S88

Stefan Wagner — SB-W015 — S29

Stefano Brusoni — W068 — S12

Stephan Leistenschneider — MIL-W023 — S16

Stephanie Dick — PM-W001 — S94

Stephann Makri — NC-W027 — S83

Stephann Makri — NC-W030 — S83

Stephen Billett — GL-W052 — S41

Stephen Boyd — MIO-W003 — S35

Stephen C. Woods — GL-W029 — S05
Stephen Granier — TD-W016 — S74
Stephen Morris — EF-W008 — S59
Stephen Shervais — SG-W053 — S47
Sterling Wiggins — PM-W022 — S95
Sterre K. Ruitenburg — LP-W033 — S72
Steve de Shazer — TD-W002 — S73
Steven Alter — SG-W015 — S14
Steven Diamond — MIO-W003 — S35
Steven M. Smith — OI-W015 — S52
Steven P. Dow — OI-W013 — S52
Stewart Brand — SG-W059 — S44
Stijn Bruers — W079 — S46
Stockholm Resilience Centre — GL-W068 — S43
Stockholm University — GL-W037 — S41
Stuart A. West — TA-W021 — S28
Stuart Mitchell — GL-W045 — S42
Styliani Lagou — TD-W023 — S74
Styliani Symbardi — TD-W023 — S74
Sunzi (traditional attribution) — MIL-W027 — S16
Susan E. Brennan — CG-W007 — S56
Susan J. Ashford — ST-W002 — S92
Susan Leigh Star — TA-W024, TA-W025 — S42
Sushil Bikhchandani — EF-W005 — S59
Suyi Li — W098 — S37
Suzete Chiviacowsky — LP-W013 — S71
Suzete Chiviacowsky — LP-W034 — S71
Svartdal, F., et al. — T01-W002 — S99
Svetlin Manavski — CG-W015 — S57
Sylvain Lenfle — LC-W009 — S62
Sylvain Lenfle — MG-W006 — S86
Sylvie Catellin — NC-W029 — S83
Sylvie Lorente — W082 — S46
Szymon Wasik — SG-W047 — S40
Sénat français — UX-W029 — S27
T. Härkänen — TD-W005 — S73

Tad McGeer — W053 — S01
Tadahiro Taniguchi — W051 — S46
Tadeusz Kotarbiński — W018, W019 — S45
Takato Horii — W051 — S46
Tang Xijin — RGC-W041 — S18
Tanmay Sinha — LP-W026 — S72
Tao Sun — W099 — S37
Tarcisio Vanzin — CRE-W043 — S70
Tau-Mu Yi — W012 — S05, S50
Taylor V. Stein — GL-W039 — S43
Ted McKenna — SL-W025 — S78
Terje Aven — AFC-W006 — S32
The Bowen Center for the Study of the Family — SG-W028 — S41
The Open University (course framing) — SG-W012 — S44
The Pfeiffer Library — PM-W016 — S95
Theodore Groves — TA-W003 — S15
Theologia Gkavogianni — TD-W023 — S74
Thierry Roncalli — MG-W020 — S89
Thomas Bauch — MG-W012 — S88
Thomas C. Ormerod — CRE-W026 — S67
Thomas C. Schelling — W070 — S49
Thomas C. Scott-Phillips — TA-W021 — S28
Thomas E. Dickins — TA-W021 — S28
Thomas Kron — SG-W042 — S49
Thomas Lopdrup-Hjorth — SB-W001 — S13
Thomas Nitsotolis — TD-W023 — S74
Thomas Parr — W056 — S46
Thomas Pinckney — XA-W002 — S84
Thomas Rotter — GL-W058 — S29
Thomas Unger — XA-W004 — S84
Thomas W. Malone — WME-W011 — S14
Thomas W. Merrill — TA-W006 — S15
Thorsten Bergmann — MG-W009 — S88
Thote Gowda — CG-W015 — S57
Tianxin Wei — BC-W009 — S50
Tilman Börgers — GL-W024 — S33

Tim Smolnik — MG-W009 — S88

Timo Spatzier — SL-W007 — S80

Timothy D. Lee — LP-W021 — S72

Timothy J. Schultz — GL-W058 — S29

Timothy K. Tsai — AFC-W003 — S30

Timothy Planche — TD-W019 — S74

Timothy S. Gardner — BC-W003 — S50

Tina Luo — CG-W015 — S57

Tingting Ye — GL-W073 — S01

Tobias Kötter — NC-W021 — S82

Tobias Richter — LP-W016 — S72

Todd Lubart — CRE-W013 — S64

Todd M. La Porte — UX-W005 — S30

Todd Rogers, Richard Zeckhauser, Francesca Gino, Michael I. Norton and Maurice E. Schweitzer — DC-W15 — S93

Tom Bauler — PM-W004 — S94

Tom Ellis — BC-W006 — S50

Tom Ritchey — OI-W014 — S52

Tom Van Gerven — GL-W074 — S01

Tomiwa Ijaware — CG-W015 — S57

Toyota Motor Corporation — GL-W080 — S02

Trevor Kletz — W039 — S02

Trishul M. Chilimbi — CG-W005 — S55

Tsz Keung Wong — CG-W024 — S31

Tyler Volk — SG-W044 — S18

U. K. Steiner — TA-W037 — S28

U.S. Department of Defense — T15-W005 — S113

U.S. Marine Corps — T15-W002 — S113

U.S.–Canada Power System Outage Task Force — T11-W002 — S109

UBC Library Indigenous Librarianship Research Guide — GL-W072 — S42

UCTE — T11-W001 — S109

Udo Wagner — SL-W026 — S79

UK Ministry of Defence, Development, Concepts and Doctrine Centre — PM-W024 — S95

Ulrich Drepper — XA-W004 — S84

Ulrich Ronellenfitsch — GL-W058 — S29

Ulrik Stenz Justesen — TD-W014 — S74

Umberto Cesar Corrêa — LP-W012 — S71
United States General Accounting Office — TA-W033 — S42
Uri Alon — BC-W002 — S50
US Army Corps of Engineers — GL-W090 — S43
US Environmental Protection Agency — GL-W076 — S43
US Food and Drug Administration — GL-W075 — S02
USDA Forest Service Northern Research Station — GL-W054 — S43
USDA Natural Resources Conservation Service — GL-W084 — S43
USGS Center of Excellence for Geospatial Information Science — GL-W066 — S42
Ut Na Sio — CRE-W026 — S67
Ut Na Sio — CRE-W041 — S70
Uttarakhand Open University — SL-W004 — S76
V. Krishnan — MG-W008 — S87
Vahe Galstyan — TA-W018 — S40, S50
Valentin Turchin — SG-W032 — S18
Valter Afonso Vieira — SL-W013 — S79
Vanesa Anton-Vazquez — TD-W019 — S74
Vania Ribas Ulbricht — CRE-W034 — S68
Vania Ribas Ulbricht — CRE-W043 — S70
Vannevar Bush — CM-W001 — S60
Varda Shoham — TD-W001 — S73
Vasileios Apostolopoulos — TD-W023 — S74
Vaughan Pratt — TD-W032 — S75
Victor H. Mair — SB-W020 — S36
Vincent Bontems — UX-W024, UX-W025 — S39
Vincent C. Müller — W007, W008 — S37
Vincent Derkinderen — BC-W011 — S50
Vincent Jugé — TD-W030 — S75
Vincenzo De Florio — AFC-W009 — S32
Vincenzo Vitelli — GL-W065 — S01
Vinicius G. Goecks — MIL-W029 — S16
Viola Spolin — CRE-W020 — S66
Virginia Kolonia — TD-W023 — S74
Viviane Kuntz — CRE-W043 — S70
Vlad Bădoiu — XA-W007 — S84
Vlad-Andrei Bădoiu — XA-W003 — S84

Vladimir Estivill-Castro — TD-W025 — S75

Volker Wedekind — GL-W038 — S41

Vorne / Lean Production — SG-W019 — S44

W. Findeisen — SB-W006 — S09

W. Ross Ashby — W001, W050 — S03, S22

Wallace Gingerich — TD-W002 — S73

Wallace J. Gingerich — TD-W004 — S73

Walter B. Cannon — W062 — S05

Wang Lin — RGC-W041 — S18

Wang Mengzhi — TA-W039 — S05

Weber, M. — T17-W001 — S115

Wei Zhao — CG-W013 — S57

Wei-Fan Chiang — CG-W004 — S55

Wei-Kuan Shih — CG-W013 — S57

Weidong Wang — CG-W015 — S57

Weiye Bao — BC-W008 — S50

Wendell A. Lim — GL-W060 — S40, S50

Wendy Baker — GL-W100 — S39

Werner Dietl — CG-W006 — S55

Werner Ulrich — SG-W002 — S44

Wesley M. Cohen — AB-W001 — S53, S61

White, James J — NN-W009 — S96

Wilfried Sieg — GL-W087 — S42

Willem H. van Zyl — GL-W086 — S43

Willem Verbeke — SL-W018 — S79

Willem-Jan van den Heuvel — AFC-W011 — S31

William C. Moncrief — SL-W002 — S80

William J. J. Gordon — NC-W009 — S82

William Kahan — CG-W003 — S55

William M. Hedgcock — PM-W010 — S94

William M. Hedgcock — PM-W029 — S94

William M. M. Kirby — TD-W010 — S74

William T. Powers — SG-W027 — S03

Wilma Waterlander — W106 — S44

Wolff, Hans-Georg, and Klaus Moser — NN-W019 — S97

Wolff, Hans-Georg, and Klaus Moser — NN-W020 — S97

Wolfgang Banzhaf — SG-W042 — S49

Wolfgang Boos — SL-W007 — S80

Wolfgang Stroebe — OI-W012 — S52

Wolfgang Ulaga — SL-W019 — S80

Woongki Baek — CG-W005 — S55

World Health Organization — GL-W041 — S41

World Organisation for Animal Health (OIE/WOAH) — GL-W056, GL-W057 — S43

Xianglu Xiao — BC-W008 — S50

Xiao Lin — BC-W009 — S50

Xiao-Ping Chen — PWR-W006 — S91

Xiaohui Xiao — GL-W073 — S01

Xiaosong Gai — GS-W08 — S98

Xin Xiong — GL-W060 — S40, S50

Xuying Ning — BC-W009 — S50

Y. Takahara — SB-W008 — S09

Yahya Forghani — BC-W010 — S50

Yang Chunyan — SG-W050 — S44

Yang Zeyu — TA-W039 — S05

Yangmin Li — GL-W073 — S01

Yanru Qu — BC-W009 — S50

Yasuhiko Takahara — W096 — S09

Yasushi Hirai — W051 — S46

Yi Wang — GS-W08 — S98

Yi Yu — CG-W015 — S57

Yi Zhang — W099 — S37

Yi-Ge Qu — TD-W022 — S74

Yohann Guffroy — UX-W024 — S39

Yu Jingyuan / 于景元 — RGC-W040 — S18

Yuanjie Zou — BC-W008 — S50

Yuchuan Zhu — GL-W073 — S01

Yue Xie — W099 — S37

Yun Huang — W012 — S05, S50

Yunan Yang — MIO-W001 — S35

Yusuke Kishita — PM-W014 — S94

Zach W. Cooper — TD-W006 — S73

Zentrum Innere Führung / Bundeswehr — MIL-W025 — S16

Zhao Chengcheng — TA-W039 — S05

Zhao Feng — GL-W073 — S01

Zhe Xu — BC-W009 — S50

Zheng Hui — BC-W008 — S50

Zhichen Zeng — BC-W009 — S50

Zhining Liu — BC-W009 — S50

Zhiyu Feng — PWR-W006 — S91

Zhonghao Zhan — BC-W008 — S50

Zihao Li — BC-W009 — S50

Zijun Zhong — BC-W008 — S50

Ziyang Zhou — W098 — S37

Zoi Alexiou — TD-W023 — S74

Zvonimir Rakamarić — CG-W004 — S55

Ángel Gil — W067 — S49

Ågren, S. M., et al. — T14-W004 — S112

Ștefan Teodorescu — XA-W003 — S84

А. А. Паршин [A. A. Parshin] — SM-W007 — S54

А. Богданов — TKT-W002 — S17

А. В. Вдовин — MIL-W028

Александр Александрович Богданов — TKT-W001, W017 — S17

В. В. Карпович — MIL-W028

В. Е. Севрук — MIL-W028

В. М. Полтерович [Victor Polterovich] — AB-W010 — S53

В. П. Пугачев — PWR-W008 — S91

В. Ю. Шитов [V. Yu. Shitov] — SM-W007 — S54

Г. С. Альтшуллер — NC-W001 — S81

Г. С. Альтшуллер — W022 — S44

Г. Я. Буш [G. Ya. Bush] — CRE-W007 — S64

Георгий Львович Смолян — DC-W08 — S93

Георгий Петрович Щедровицкий (Georgy Petrovich Shchedrovitsky) — WME-W019 — S44

Д. А. Леонтьев — GS-W10 — S98

И. В. Руденко [I. V. Rudenko] — SL-W016 — S80

И. П. Кузнецов [I. P. Kuznetsov] — SM-W007 — S54

Л. В. Халецкий [L. V. Khaletsky] — SM-W007 — S54

П. Я. Гальперин [P. Ya. Galperin] — LP-W009 — S71

Р. Б. Шапиро — W022 — S44

Стремовская, Алла Леонидовна — NN-W007 — S96

Триандафиллов, В. К. — T15-W001 — S113

สุรศักดิ์ ชุบไธสง、ระพี กาญจนะ — T13-W005 — S111

やまだようこ [Yoko Yamada] — CRE-W030 — S68

万爱兰 — TD-W007 — S73

三道弘明 — MG-W003 — S85

上海系统科学研究院 — SG-W007 — S18

中国信息通信研究院云计算与大数据研究所 — AFC-W014 — S31

中国合格评定国家认可委员会 (CNAS) — TA-W030 — S42

中国民用航空局 [Civil Aviation Administration of China] — SM-W009 — S54

中川 徹 — NC-W003 — S81

中村眞実 — T12-W003 — S110

中條武志 (Takeshi Nakajo) — W063 — S02

中條武志 — W076 — S02

久米均 — W076 — S02

于景元 — W035, W036 — S18

仁科健 — T13-W003 — S111

任晓雅 — GS-W15 — S98

伊藤政展 [Masanobu Ito] — LP-W014 — S71

佐々木直子 (Naoko Sasaki) — LC-W007 — S62

佐野明人 — W010 — S01

余跃庆 — W009 — S37

储昭卫 [Chu Zhaowei] — CRE-W003 — S63

储昭卫 — NC-W007 — S81

光田航 — CG-W026, CG-W027 — S56

刘利丹 — TD-W007 — S73

刘华平 — W011 — S37

刘征、罗志琪 — T10-W002 — S108

刘炳琪 — SB-W016 — S29

刘畅 — SB-W016 — S29

初鹿野· 岩田 — T08-W003 — S106

华为云计算技术有限公司 SRE 部 — AFC-W021 — S31

南泰浩 — CG-W027 — S56

古川久敬 — GS-W03 — S98

吉田靖之 — DC-W20 — S93

名倉早都季 — NC-W034 — S83

吴书凤 [Wu Shufeng] — AB-W009 — S53

吴学谋 / Xuemou Wu — SG-W049 — S18

周雪光 — SB-W003 — S13

和田春菜 — PM-W005 — S94

唐锡晋 (Tang Xijin) — WME-W026 — S18

唐锡晋 等 — W078 — S18

嘉本慎介 (Shinsuke Kamoto) — SW-W007 — S61

大嶋玲未 — PWR-W005 — S91

大賀悠平 — CG-W026 — S56

姚威 [Yao Wei] — CRE-W003 — S63

姚威 — NC-W007 — S81

孙富春 — W011 — S37

孙文浩 [Sun Wenhao] — AB-W009 — S53

孙汉银 — NC-W012 — S82

孫子 (traditional attribution) — DC-W01 — S93

宁永忠 — TD-W021 — S74

安贺新（主编） — SL-W006 — S76

小原博 — T04-W002 — S102

山中正樹 [Masaki Yamanaka] — SL-W028 — S80

山崎裕子 — NC-W034 — S83

岡田涼 — NN-W016 — S97

岸野哲 — GS-W02 — S98

川上桃子 (Momoko Kawakami) — SW-W008 — S61

川喜田二郎 [Jiro Kawakita] — CRE-W030 — S68

工藤裕 [Yutaka Kudo] — CM-W007 — S60

廣川佳子 — PWR-W005 — S91

张侃 — TA-W027 — S41

张新钰 — W011 — S37

张杰 [Zhang Jie] — AB-W009 — S53

张浩 [Zhang Hao] — SG-W005 — S18

张添溯 (Zhang Tiansu) — MIL-W026 — S38

张真, 皇甫刚 — NN-W011 — S96

张绪平 — W009 — S37

张锦坤 [Zhang Jinkun] — LP-W017 — S72

张颖颖 — NC-W033 — S83

戴汝为 — W035 — S18

文鹏 — GS-W15 — S98

新郷重夫 (Shigeo Shingo) — W038 — S02

日本プラントメンテナンス協会 [Japan Institute of Plant Maintenance] — UX-W036, UX-W037 — S02

日本創造学会 [Japan Creativity Society] — CRE-W004 — S63

曹骞 — ST-W007 — S92

服部雅史 — NC-W013 — S82

木下裕介 — PM-W005 — S94

本田周二 — NN-W015 — S97

朱坤 — MG-W018 — S90

李俊 — SB-W016 — S29

李国平 (Li Guoping) — SW-W009 — S61

李晓彤 — ST-W006 — S92

李松, 刘峻君, 鲍秀琴, 陈旭 — NN-W017 — S97

李炜文 — ST-W006 — S92

東中竜一郎 — CG-W026, CG-W027 — S56

杵渊哲也 — CG-W026 — S56

松沢哲郎 [Tetsuro Matsuzawa] — CRE-W030 — S68

林崇德 — NC-W012 — S82

栗田真広 — PM-W026 — S95

桑原哲郎 — GS-W02 — S98

梅田靖 — PM-W005 — S94

森俊夫 — TD-W008 — S73

森田大輔 — MG-W003 — S85

榊原清則· 古賀款久· 本庄裕司· 近藤一徳 — T02-W002 — S100

水野有智 — PM-W005 — S94

江泓 — SB-W013 — S08

池俣吉人 — W010 — S01

浅石卓真 — NC-W034 — S83

渋谷賢 [Satoshi Shibuya] — LP-W014 — S71

火山引擎开发者社区 — AFC-W020 — S31

王威威 — GL-W008 — S41

王植洵 [Wang Zhixun] — LP-W017 — S72

王浣尘 — W078 — S18

王玲玲 — PM-W008 — S94

王硕阳 — NC-W012 — S82

王辉 — TD-W021 — S74

王进富 — NC-W033 — S83

田中博晃 [Hiroaki Tanaka] — CRE-W029 — S68

田崎創 — XA-W005 — S84

田洋丽 — TD-W007 — S73

真家美咲 — NC-W034 — S83

石川晴雄 (Haruo Ishikawa) — LC-W007 — S62

石庆馨 — TA-W027 — S41

福重真一 — PM-W005 — S94

科学技術振興機構 研究開発戦略センター (JST CRDS) — W100 — S49

科学时报社 — WME-W028 — S18

窦丽琛 (Dou Lichen) — SW-W009 — S61

童廷婷、白幼玲、冯廷勇 — T01-W003 — S99

粟島浩二 — SL-W017 — S80

織田涼 — NC-W013 — S82

胡剑波 — SB-W016 — S29

胡顺顺 — NC-W007 — S81

苏丽颖 — W009 — S37

苏勇, 程骏骏, 吴展 — NN-W010 — S96

藤村聡 — NC-W034 — S83

西川功一 [Koichi Nishikawa] — SL-W015 — S80

西村翔 — TD-W020 — S74

西田勇樹 — NC-W013 — S82

西野明樹 — TD-W008 — S73

諏訪晴彦 — MG-W003 — S85

许志行 — ST-W007 — S92

谷口正一郎 — ST-W008 — S92

赵伟 — SB-W003 — S13

赵博韬 — SB-W013 — S08

邹晓伟 — TD-W007 — S73

郭迪 — W011 — S37

野上真 — GS-W03 — S98

钱学森 (Qian Xuesen) — WME-W027 — S18

钱学森 — W035 — S18

陆靖昶 — MG-W018 — S90

陈宏斌 — TD-W021 — S74

陈小菁 (Chen Xiaojing) — MIL-W026 — S38

陈志远 [Chen Zhiyuan] — AB-W009 — S53

陈炜 — W009 — S37

陈诚 — GS-W15 — S98

顾基发 (Gu Jifa) — WME-W026 — S18

顾基发 — W078 — S18

顾雪平、白岩松、李少岩、刘艳 — T11-W003 — S109

高岩 — ST-W006 — S92

高橋菜奈子 — NC-W034 — S83

高橋誠 [Makoto Takahashi] — CRE-W004 — S63

鶴田芳映 — TD-W008 — S73

黒沢幸子 — TD-W008 — S73

齋藤結 — CG-W027 — S56

문재성 (Jae-Sung Moon) — W066 — S01

References

References identify original publication dates and distinguish substantive later editions or revisions where necessary. Bracketed title renderings do not necessarily name a published translation.

[W104] A. H. Louie (2017). *Relational Biology*. https://doi.org/10.1007/978-3-319-31737-3_17-1

[W084] Abhishek Sharma; Dániel Czégel; Michael Lachmann; Christopher P. Kempes; Sara I. Walker; Leroy Cronin (2023). *Assembly theory explains and quantifies selection and evolution*. https://doi.org/10.1038/s41586-023-06600-9

[TA-W008] Adnan Darwiche; Pierre Marquis (2002). *A Knowledge Compilation Map*. https://arxiv.org/pdf/1106.1819

[TA-W014] Adolfo Anta; Paulo Tabuada (2008). *To sample or not to sample: Self-triggered control for nonlinear systems*. https://arxiv.org/abs/0806.0709

[W082] Adrian Bejan; Sylvie Lorente (2010). *The constructal law of design and evolution in nature*. https://doi.org/10.1098/rstb.2009.0302

[RGC-W027] Alain Degenne (1978). *Le Moigne Jean-Louis, La théorie du système général. Théorie de la modélisation [Jean-Louis Le Moigne, General System Theory: Theory of Modeling] (book review)*. https://doi.org/10.2307/3320802

[MIL-W034] Alastair Cooper; Felicity Petrie (2025). *Mission Command at Sea and Command by Veto*. https://doi.org/10.61451/210305

[GL-W097] Aleksandr Bogdanov (1921). Очерки всеобщей организационной науки *[Essays on Universal Organizational Science]*. Samara: Государственное издательство [State Publishing House]. https://search.rsl.ru/ru/record/01008733462

[WME-W009] Alessandro Giovannelli (2005). *Goodman's Aesthetics*. Stanford Encyclopedia of Philosophy. First published 2005; substantively revised 2017. https://plato.stanford.edu/entries/goodman-aesthetics/

[W028] Almir Cordeiro Júnior; Mônica de Fátima Bianco (2005). *Resolução de conflitos em centros integrados de cidadania: uma abordagem à luz da perspectiva sociotécnica de grupos de trabalho [Conflict Resolution in Integrated Citizenship Centers: An Approach Through the Sociotechnical Perspective on Work Groups]*. https://doi.org/10.1590/S1679-39512005000200008

[GL-W094] Amnon Lehavi (2013). *Rules and standards: an institutional analysis of property*. https://doi.org/10.1017/CBO9781139548847.004

[MIL-W030] Amos C. Fox (2024). *Contested Logistics: A Primer*. https://www.ausa.org/publications/contested-logistics-primer

[AFC-W015] Amy C. Edmondson; Mark D. Cannon (2005). *The Hard Work of Failure Analysis*. https://www.library.hbs.edu/working-knowledge/the-hard-work-of-failure-analysis

[SG-W037] Anatol Rapoport (1968). *Systems Analysis: I. General Systems Theory*. In David L. Sills (ed.), International Encyclopedia of the Social Sciences, vol. 15, pp. 452–458. Macmillan and The Free Press. Spanish translation: Teoría general de los sistemas, Pensamiento Crítico 47 (December 1970), pp. 76–91. https://www.encyclopedia.com/science-and-technology/computers-and-electrical-engineering/computers-and-computing/systems-analysis

[W034] Andre Rosendo; Marco von Atzigen; Fumiya Iida (2017). *The trade-off between morphology and control in the co-optimized design of robots*. https://doi.org/10.1371/journal.pone.0186107

[MIL-W035] Anthony Duus (2025). *Paying the Mission Command Bill (Mission Command response)*. https://doi.org/10.61451/2103013

[MIL-W018] Antoine Bousquet (2008). *Chaoplexic warfare or the future of military organization*. https://doi.org/10.1111/j.1468-2346.2008.00746.x

[MIL-W019] Antoine Bousquet (2009). *The Scientific Way of Warfare: Order and Chaos on the Battlefields of Modernity*. Revised second edition, 2022. Hurst.

[AFC-W013] Antoine Danchin; P. M. Binder; Stanislas Noria (2011). *Antifragility and Tinkering in Biology (and in Business): Flexibility Provides an Efficient Epigenetic Way to Manage Risk*. https://doi.org/10.3390/genes2040998

[TA-W031] Antonio Possolo; Sally S. Bruce; Robert L. Watters, Jr. (2021). *Metrological Traceability: Frequently Asked Questions and NIST Policy*. NIST Technical Note 2156. National Institute of Standards and Technology. https://doi.org/10.6028/NIST.TN.2156

[UX-W013] Arantza Etxeberria; Alvaro Moreno (2001). *From complexity to simplicity: nature and symbols*. BioSystems, 60(1–3), 149–157. https://doi.org/10.1016/S0303-2647(01)00112-5

[AFC-W019] Ariel Tseitlin (2013). *The Antifragile Organization*. https://doi.org/10.1145/2492007.2492022

[GL-W069] Arjan van der Schaft (2024). *Port-Hamiltonian nonlinear systems*. https://arxiv.org/html/2412.19673v1

[GL-W070] Arjan van der Schaft; Dimitri Jeltsema (2014). *Port-Hamiltonian Systems Theory: An Introductory Overview*. https://doi.org/10.1561/2600000002

[UX-W011] Armando Sternieri (2026). *Dependency Restructuring: How Technological Enactment Changes Organizational Requirement Relations*. Information, 17(10), 1015. https://doi.org/10.3390/info17101015

[GL-W004] Chelsey Geralda Armstrong; Jennifer Grenz; Jennifer Zyp-Loring; Jade Lafontaine; Leslie Main Johnson; Nancy J. Turner (2025). *Ethnoecological perspectives on environmental stewardship: Tenets and basis of reciprocity in Gitxsan and nłe ʔkepmx (Nlaka'pamux) Territories*. https://besjournals.onlinelibrary.wiley.com/doi/full/10.1002/pan3.10641

[UX-W039] Arnold Gehlen (1957). *Die Seele im technischen Zeitalter: Sozialpsychologische Probleme in der industriellen Gesellschaft [The Soul in the Technological Age: Social-Psychological Problems in Industrial Society]*. Rowohlt. https://katalog.bibliothek.kit.edu/bib/665047

[UX-W033] Arnold Gehlen (1940). *Der Mensch: Seine Natur und seine Stellung in der Welt [Man: His Nature and Place in the World]*. Substantially revised fourth edition, 1950. Junker und Dünnhaupt. https://cup.columbia.edu/book/man/9780231052184/

[UX-W003] Arthur L. Stinchcombe (1990). *Information and Organizations*. University of California Press. ISBN 9780520909625. https://publishing.cdlib.org/ucpressebooks/public/book/information-and-organizations.html

[SB-W002] Arthur L. Stinchcombe (2001). *When Formality Works: Authority and Abstraction in Law and Organizations*. https://books.google.com/books?id=TvOmZEXSDvkC

[WME-W013] Arthur S. Iberall (1985). *Outlining social physics for modern societies—locating culture, economics, and politics: The Enlightenment reconsidered*. https://pmc.ncbi.nlm.nih.gov/articles/PMC390594/

[WME-W012] Arthur S. Iberall; Harry Soodak (1998). *Primer on Homeokinetics: A Physical Foundation for Complex Systems*. https://homeokinetics.org/pdf%20files/HomeokineticsPrimer.pdf

[SB-W015] Asim Abdulkhaleq; Stefan Wagner (2015). *A Controlled Experiment for the Empirical Evaluation of Safety Analysis Techniques for Safety-Critical Software*. https://arxiv.org/pdf/1612.00330

[GL-W003] Assembly of Alaska Native Educators; Alaska Native Knowledge Network (2000). *Guidelines for Respecting Cultural Knowledge*. https://www.uaf.edu/ankn/publications/pdfs/Knowledge.pdf

[CM-W002] Douglas C. Engelbart (1962). *Augmenting Human Intellect: A Conceptual Framework*. Stanford Research Institute, October. https://www.dougengelbart.org/pubs/papers/scanned/Doug_Engelbart-AugmentingHumanIntellect.pdf

[CM-W004] Jeff Conklin; Michael L. Begeman (1988). *gIBIS: A Hypertext Tool for Exploratory Policy Discussion*. ACM Transactions on Office Information Systems 6(4), 303–331, October. https://doi.org/10.1145/58566.59297

[CM-W003] Jon Doyle (1979). *A Truth Maintenance System*. MIT Artificial Intelligence Laboratory, AI Memo 521, June. https://hdl.handle.net/1721.1/5733

[CM-W005] Mark S. Ackerman (1994). *Augmenting the Organizational Memory: A Field Study of Answer Garden*. Proceedings of the ACM Conference on Computer-Supported Cooperative Work, 243–252. https://web.eecs.umich.edu/~ackerm/pub/94b12/cscw94.final.pdf

[CM-W006] Nada Matta; Myriam Ribière; Olivier Corby (1999). *Définition d'un modèle de mémoire de projet [Definition of a Project Memory Model]*. INRIA, Research Report RR-3720, June. https://hal.science/inria-00072945

[CM-W001] Vannevar Bush (1945). *As We May Think*. The Atlantic Monthly 176(1), July. https://www.theatlantic.com/magazine/archive/1945/07/as-we-may-think/303881/

[CM-W007] 工藤裕 [Yutaka Kudo] (2011). *情報システムの開発及び運用における知識活用手法に関する研究 [Research on Methods for Using Knowledge in the Development and Operation of Information Systems]*. Doctoral dissertation, Osaka University, July. https://hdl.handle.net/11094/1101

[SG-W005] 张浩 [Zhang Hao] (2022). *管理科学研究模型与方法（第2版）, 第1章: 系统科学 [Models and Methods of Management Science, Second Edition, Chapter 1: Systems Science]*. 清华大学出版社 [Tsinghua University Press]. ISBN 9787302613756. https://www.tup.tsinghua.edu.cn/upload/books/yz/095972-01.pdf

[W048] Blas M. Vinagre; Inés Tejado; Andrés J. Serrano-Balbontín; Enrique Mancha-Sánchez; Moisés Bolívar-Vivas; Alberto Cebrián-García (2026). *Body Machinery and Intelligence*. Jornadas de Automática, 47. https://doi.org/10.17979/ja-cea.2026.47.13876

[WME-W024] Baltasar Trancón y Widemann; Christina Bogner (2016). *Comments on "the return of information theory"*. https://arxiv.org/abs/1602.05056

[W087] Barbara H. Liskov; Jeannette M. Wing (1994). *A Behavioral Notion of Subtyping*. https://doi.org/10.1145/197320.197383

[W071] Benjamin Brooks; Alexander Frankel; Emir Kamenica (2024). *Comparisons of Signals*. Working paper. https://benjaminbrooks.net/downloads/bfk_comparisons.pdf

[RGC-W024] Bernard Motulsky (1978). *Jean-Louis Le Moigne. La théorie du système général (théorie de la modélisation), 1977 [Jean-Louis Le Moigne, General System Theory: Theory of Modeling, 1977] (book review)*. https://www.persee.fr/doc/comin_0382-7798_1978_num_2_3_1078

[SG-W016] Birger Sevaldson / Systems Oriented Design (2012). *What is Gigamapping?*. https://systemsorienteddesign.net/what-is-gigamapping/

[MIL-W033] Björn J. E. Johansson; Jiri Trnka; Peter Berggren (2016). *A case study of C2 Agility in the 2014 Västmanland forest fire*. FOI-R--4259--SE. https://foi.se/en/foi/reports/report-summary.html?reportNo=FOI-R--4259--SE

[AFC-W012] Björn Wittenmark (2002). *Adaptive Dual Control*. In H. Unbehauen (ed.), Control Systems, Robotics and Automation, vol. X. EOLSS Publishers. https://www.eolss.net/sample-chapters/c18/e6-43-15-06.pdf

[SG-W029] Board of Governors of the Federal Reserve System; OCC; FDIC (2020). *Sound Practices to Strengthen Operational Resilience*. https://www.federalreserve.gov/frrs/guidance/interagency-paper-on-sound-practices-to-strengthen-operational-resilience.htm

[WME-W017] Brian R. Gaines (2017). *Introduction to Personal Construct Psychology*. https://pages.cpsc.ucalgary.ca/~gaines/pcp/

[UX-W032] Bryan S. Turner (2008). *The Body & Society: Explorations in Social Theory*. Third edition. SAGE. First edition 1984. 9781412929875. https://api.pageplace.de/preview/DT0400.9781849205412_A23446745/preview-9781849205412_A23446745.pdf

[W032] C. H. Waddington (1942). *Canalization of Development and the Inheritance of Acquired Characters*. https://www.evolocus.com/publications/Waddington1942.pdf

[GL-W067] C. S. Holling (1973). *Resilience and Stability of Ecological Systems*. https://pure.iiasa.ac.at/id/eprint/26/1/RP-73-003.pdf

[SB-W014] C. W. Johnson; C. M. Holloway (2003). *The ESA/NASA SOHO Mission Interruption: Using the STAMP Accident Analysis Technique for a Software Related Mishap*. https://www.dcs.gla.ac.uk/~johnson/papers/STAMP.pdf

[SG-W050] Cai Wen; Yang Chunyan; He Bin (2003). *New Development of the Basic Theory of Extenics*. https://www.engineering.org.cn/sscae/EN/1160102827823522516

[GL-W099] Canadian Association for Conservation of Cultural Property (n.d.). *What is Conservation?*. https://www.cac-accr.ca/conservation/

[GL-W100] Debra Daly Hartin; Wendy Baker (2018). *Caring for paintings*. Canadian Conservation Institute/Canadian Heritage. Catalogue CH 57-4/6-10-2018 E-PDF. ISBN 9780660280080. https://www.canada.ca/en/conservation-institute/services/preventive-conservation/guidelines-collections/paintings.html

[MIL-W036] Carl von Clausewitz (1832). *Vom Kriege [On War], Buch I, Kapitel VII: Friktion im Kriege [Book I, Chapter VII: Friction in War]*. https://www.marxists.org/reference/archive/clausewitz/works/on-war/book1/ch07.htm

[UX-W009] Carliss Y. Baldwin (2008). *Where do transactions come from? Modularity, transactions, and the boundaries of firms*. Industrial and Corporate Change, 17(1), 155–195. https://doi.org/10.1093/icc/dtm036

[UX-W038] Carliss Y. Baldwin (2023). *Design rules: past and future*. Industrial and Corporate Change, 32(1), 11–27. https://doi.org/10.1093/icc/dtac055

[UX-W040] Carliss Y. Baldwin (2024). *Design Rules, Volume 2: How Technology Shapes Organizations*. MIT Press. DOI: 10.7551/mitpress/7374.001.0001. 9780262049337. https://direct.mit.edu/books/oa-monograph/5887/Design-Rules-Volume-2How-Technology-Shapes

[UX-W001] Carliss Y. Baldwin (2025). *A theory of technology and organizations*. Strategic Organization, 24(2) (2026); first published online 11 December 2025. https://doi.org/10.1177/14761270251409566

[UX-W008] Carliss Y. Baldwin; Kim B. Clark (2006). *Where Do Transactions Come From? A Network Design Perspective on the Theory of the Firm*. Harvard Business School Working Paper 06-051, 11 May 2006. https://doi.org/10.2139/ssrn.901790

[UX-W022] Carlotta Langer; Nihat Ay (2021). *How Morphological Computation Shapes Integrated Information in Embodied Agents*. Frontiers in Psychology, 12, 716433. https://doi.org/10.3389/fpsyg.2021.716433

[UX-W015] Carlotta Langer; Nihat Ay (2024). *Outsourcing Control requires Control Complexity*. First published as a preprint in 2022; revised 2024. https://doi.org/10.48550/arXiv.2209.01418

[GL-W034] Cecily Maller (2015). *Understanding health through social practices: performance and materiality in everyday life*. https://onlinelibrary.wiley.com/doi/10.1111/1467-9566.12178

[MIL-W024] Centre de doctrine d'emploi des forces, armée de Terre (2010). *FT-05: L'exercice du commandement en opérations pour les chefs tactiques [The Exercise of Command in Operations for Tactical Commanders]*. https://www.terre.defense.gouv.fr/sites/default/files/ccf/FT-05%20-%20l%20exercice%20du%20commandement%20en%20ope%CC%81rations%20pour%20les%20chefs%20tactiques%20-%20Octobre%202010.pdf

[GL-W098] Chad Hansen (2025). *Daoism*. https://plato.stanford.edu/entries/daoism/index.html

[W086] Charles Babbage (1832). *On the Economy of Machinery and Manufactures*. ISBN 978-1-108-00910-2 (2009 facsimile).
https://rybn.org/human_computers/articles/Babbage_Charles_On_the_Economy_of_Machinery_and_Manufactures.pdf

[UX-W004] Charles Perrow (1984). *Normal Accidents: Living with High-Risk Technologies*.
https://press.princeton.edu/books/paperback/9780691004129/normal-accidents

[GL-W047] Cherilyn Ireton; Julie Posetti (editors) (2018). *Journalism, 'Fake News' and Disinformation: Handbook for Journalism Education and Training*.
https://www.unesco.org/sites/default/files/journalism_fake_news_disinformation_print_friendly_0.pdf

[SG-W048] Chi-Ming Yang; Jan-Chang Chen; Lan-Lin Peng; Jr-Syu Yang; Chang-Hung Chou (2002). *Chi-Chi Earthquake-caused Landslide: grey prediction model for pioneer vegetation recovery monitored by satellite images*. https://ejournal.sinica.edu.tw/bbas/content/2002/1/bot431-08.html

[W092] Chiara Marletto (2015). *Constructor theory of life*. https://doi.org/10.1098/rsif.2014.1226

[TA-W028] Christopher Alexander (1964). *Notes on the Synthesis of Form*. Harvard University Press.
https://monoskop.org/images/f/ff/Alexander_Christopher_Notes_on_the_Synthesis_of_Form.pdf

[TA-W029] Christopher Alexander (1965). *A City Is Not a Tree*.
https://www.patternlanguage.com/archive/cityisnotatree.html

[GL-W014] Christopher Potts (2010). *Semantics–pragmatics interactions*. In Patrick Colm Hogan (ed.), The Cambridge Encyclopedia of the Language Sciences, pp. 758–759. Cambridge University Press.
https://web.stanford.edu/~cgpotts/entries/potts-cup-semantics-pragmatics.html

[MIL-W004] Christopher R. Paparone; George L. Topic Jr. (2014). *Mission command and logistics interdependencies*.
https://www.army.mil/article/136556/mission_command_and_logistics_interdependencies

[MIL-W007] Christopher R. Smith (2010). *Network Centric Warfare, Command, and the Nature of War*. ISBN9780642297211.
https://researchcentre.army.gov.au/sites/default/files/sp318ncwcommandandnatureofwarchristopher_smith.pdf

[GL-W077] Christopher Robben; Hermann Wotruba (2019). *Sensor-Based Ore Sorting Technology in Mining—Past, Present and Future*. https://doi.org/10.3390/min9090523

[W025] Claude E. Shannon (1948). *A Mathematical Theory of Communication*.
https://doi.org/10.1002/j.1538-7305.1948.tb01338.x

[SG-W032] Cliff Joslyn; Francis Heylighen; Valentin Turchin (1997). *Metasystem Transition Theory; Metasystem Transition*. https://pespmc1.vub.ac.be/MSTT.html

[GL-W002] Cordula Vesper and colleagues (2017). *Joint Action: Mental Representations, Shared Information and General Mechanisms for Coordinating with Others*. https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2016.02039/full

[MIL-W032] Craig C. Sherbrooke (1966). *METRIC: A Multi-Echelon Technique for Recoverable Item Control*. RANDRM5078PR. https://www.rand.org/pubs/research_memoranda/RM5078.html

[MIL-W031] Craig C. Sherbrooke (1968). *METRIC: A Multi-Echelon Technique for Recoverable Item Control*. https://doi.org/10.1287/opre.16.1.122

[AFC-W008] Cristian Axenie; Oliver López-Corona; Michail A. Makridis; Meisam Akbarzadeh; Matteo Saveriano; Alexandru Stancu; Jeffrey West (2024). *Antifragility in complex dynamical systems*. https://doi.org/10.1038/s44260-024-00014-y

[SG-W018] D. E. Campbell; S. L. Brandt-Williams; M. E. A. Meisch (2005). *Environmental Accounting Using Emergy: Evaluation of the State of West Virginia*. U.S. Environmental Protection Agency, Office of Research and Development, National Health and Environmental Effects Research Laboratory, Atlantic Ecology Division. EPA/600/R-05/006. https://archive.epa.gov/aed/html/collaboration/web/pdf/wvevaluationposted.pdf

[SB-W009] Daniel Tabak (1975). *A Review of Theory of Hierarchical, Multilevel Systems*. https://doi.org/10.1080/03081077508960831

[UX-W023] Darío Sandrone; Andrés Vaccari; Diego Lawler (2022). *The centrality of the machine in the thought of Jacques Lafitte*. Philosophy & Technology, 35, article 28. https://doi.org/10.1007/s13347-022-00523-5

[W059] David Blackwell (1951). *Comparison of Experiments*. https://doi.org/10.1525/9780520411586-009

[W029] David Blackwell (1953). *Equivalent Comparisons of Experiments*. https://doi.org/10.1214/aoms/1177729032

[GL-W095] David D. Clark; Marjory Blumenthal (2000). *Rethinking the Design of the Internet: End to End Arguments vs. the Brave New World*. https://dspace.mit.edu/entities/publication/9ea63dd3-3384-46b9-a3a7-3c25a0a9ae4e

[W023] David D. Evans; Daniel Madrzykowski (1981). *Characterizing the Thermal Response of Fusible-Link Sprinklers*. NBSIR 81-2329. https://nvlpubs.nist.gov/nistpubs/Legacy/IR/nbsir81-2329.pdf

[W091] David Deutsch; Chiara Marletto (2015). *Constructor theory of information*. https://doi.org/10.1098/rspa.2014.0540

[W041] David Kirsh; Paul Maglio (1994). *On Distinguishing Epistemic from Pragmatic Action*. https://doi.org/10.1207/s15516709cog1804_1

[UX-W043] David L. Alderson; John C. Doyle (2010). *Contrasting views of complexity and their implications for network-centric infrastructures*. IEEE Transactions on Systems, Man, and Cybernetics—Part A: Systems and Humans, 40, 839–852. https://doyle.caltech.edu/2010-2011

[W004] David L. Parnas (1972). *On the Criteria To Be Used in Decomposing Systems into Modules*. https://doi.org/10.1145/361598.361623

[GL-W049] David Leopold (2022). *Alienation*. https://plato.stanford.edu/entries/alienation/

[MIL-W009] David S. Alberts; Richard E. Hayes (2003). *Power to the Edge: Command and Control in the Information Age*. ISBN1893723135. https://universityofleeds.github.io/philtaylorpapers/pmt/exhibits/751/poweredge.pdf

[MIL-W010] David S. Alberts; Richard E. Hayes (2006). *Understanding Command and Control*. https://www.dodccrp.org/files/Alberts_UC2.pdf

[SB-W018] David Sless (1985). *Review of Grammatical Man*. https://www.scienceopen.com/hosted-document?doi=10.1080%2F08109028508628983

[MIL-W021] David Stahel (2025). *Auftragstaktik (Mission Command): The Prussian-German Origins and Application of Mission Command*. https://doi.org/10.61451/210301

[GL-W016] David Wood; Jerome S. Bruner; Gail Ross (1976). *The role of tutoring in problem solving*. https://w.pauldowling.me/tmr/readings/Wood,%20Bruner%20&%20Ross.pdf

[UX-W002] Debbie Harrison; Kristin B. Munksgaard; Frans Prenkert (2022). *Coordinating Activity Interdependencies in the Contemporary Economy: The Principle of Distributed Control*. British Journal of Management, 34(3), 1488–1509 (2023); first published online 28 August 2022. https://doi.org/10.1111/1467-8551.12650

[RGC-W005] Debora Hammond and Jennifer Wilby (2006). *The life and work of James Grier Miller*. https://go.gale.com/ps/i.do?id=GALE%7CA148267709&it=r&p=AONE&u=googlescholar&v=2.1

[SG-W021] Dependency and Structure Modeling community (2019). *Introduction to DSM*. https://dsmweb.org/introduction-to-dsm/

[SG-W041] Derek Cabrera; Laura Cabrera (2022). *DSRP Theory: A Primer*. https://doi.org/10.3390/systems10020026

[W060] Dimitris Bertsimas; Melvyn Sim (2003). *Robust discrete optimization and network flows*. https://doi.org/10.1007/s10107-003-0396-4

[W027] Dina Borrego; Nuno Preguiça; Elisa Gonzalez Boix; Carla Ferreira (2025). *Ensuring Convergence and Invariants Without Coordination*. https://doi.org/10.4230/LIPIcs.ECOOP.2025.4

[GL-W071] Dominic Orth; John van der Kamp; Chris Button (2019). *Learning to be adaptive as a distributed process across the coach–athlete system: situating the coach in the constraints-led approach*. https://research.vu.nl/en/publications/learning-to-be-adaptive-as-a-distributed-process-across-the-coach/

[GL-W018] Donald Moynihan; Pamela Herd; Hope Harvey (2015). *Administrative Burden: Learning, Psychological, and Compliance Costs in Citizen-State Interactions*. https://digitalgovernmenthub.org/wp-content/uploads/2022/07/muu009.pdf

[W107] Donella H. Meadows (1999). *Leverage Points: Places to Intervene in a System*. Sustainability Institute 1999. https://donellameadows.org/wp-content/userfiles/Leverage_Points.pdf

[GL-W029] Douglas S. Ramsay; Stephen C. Woods (2014). *Clarifying the roles of homeostasis and allostasis in physiological regulation*. https://pubmed.ncbi.nlm.nih.gov/24730599/

[SG-W058] Dov Dori (2011). *Object-Process Methodology for Structure-Behavior Co-Design*. In Handbook of Conceptual Modeling, pp. 209–258. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-642-15865-0_7

[RGC-W028] Edgar Morin (1977). *La Méthode, tome 1: La Nature de la Nature [Method, Volume 1: The Nature of Nature]*.

[RGC-W029] Edgar Morin (1980). *La Méthode, tome 2: La Vie de la Vie [Method, Volume 2: The Life of Life]*.

[W006] Edsger W. Dijkstra (1974). *Self-stabilizing Systems in Spite of Distributed Control*. https://doi.org/10.1145/361179.361202

[TA-W019] Edward D. Lee; Jessica C. Flack; David C. Krakauer (2023). *Outsourcing Memory Through Niche Construction*. https://arxiv.org/html/2209.00476v2

[W049] Elinor Ostrom (1990). *Governing the Commons The Evolution of Institutions for Collective Action*. https://www.lowimpact.org/wp-content/uploads/ostrom_1990.pdf

[SG-W004] Elisabeth Dostal (2023). *There are Systems and Systems: Towards an Integrated General Systems Theory*. Proceedings of the 67th Meeting of the International Society for the Systems Sciences (2023); revised 2025. https://journals.isss.org/index.php/jisss/article/download/4163/1282/13850

[GL-W007] Elizabeth Anderson (2018). *Dewey's Moral Philosophy*. https://plato.stanford.edu/archives/spr2021/entries/dewey-moral/

[SG-W013] Eric Schwarz (1995). *Reality, Validity, Truth*. https://web3.isss.org/primer2/schwar.html

[SG-W003] Eric Schwarz (2005). *From Epistemology to Action*. https://www.res-systemica.org/afscet/resSystemica/Paris05/schwarz.pdf

[AFC-W018] Eric Verhulst (2014). *Applying Systems and Safety Engineering Principles for Antifragility*. https://doi.org/10.1016/j.procs.2014.05.500

[RGC-W001] Erich Jantsch (1975). *The Theory of Dissipative Structures*. In Design for Evolution, Chapter 4. https://www.panarchy.org/jantsch/dissipativestructures.html

[WME-W001] Erich Jantsch (1980). *The Self-Organizing Universe: Scientific and Human Implications of the Emerging Paradigm of Evolution*. https://monoskop.org/images/9/9e/166495032-The-Self-Organizing-Universe-by-Erich-Jantsch.pdf

[RGC-W026] Erik Braudeau (2015). *Modélisation systémique du couple eau-sol: la pédologie hydrostructurale [Systemic Modeling of the Water–Soil Coupling: Hydrostructural Pedology]*. https://www.res-systemica.org/afscet/resSystemica/vol13/ResSystemica13-systeme-sol.pdf

[W021] Ernst Mach (1882). *Die ökonomische Natur der physikalischen Forschung [The Economical Nature of Physical Research]*. https://projekt-gutenberg.org/authors/ernst-mach/books/populaer-wissenschaftliche-vorlesungen/chapter/15/

[RGC-W018] Ervin Laszlo (1972). *Introduction to Systems Philosophy: Toward a New Paradigm of Contemporary Thought*. https://www.routledge.com/Introduction-to-Systems-Philosophy-Toward-a-New-Paradigm-of-Contemporary-Thought/Laszlo/p/book/9781003205586

[RGC-W019] Ervin Laszlo (1972). *The Systems View of the World: The Natural Philosophy of the New Developments in the Sciences*. https://obnb.uk/p11338001-the-systems-view-of-the-world-the-natural-philosophy-of-the-new-developments-in-the-sciences

[SG-W038] Ervin Laszlo (1972). *The Systems View of the World: The Natural Philosophy of the New Developments in the Sciences*. https://sciphilos.info/docs_pages/docs_Laszlo_css_copy.html

[UX-W016] Eugene A. Rosa (2005). *Celebrating a Citation Classic—and More: Symposium on Charles Perrow's Normal Accidents*. Organization & Environment, 18(2). https://doi.org/10.1177/1086026604270873

[SG-W030] European Parliamentary Technology Assessment network (n.d.). *What is Technology Assessment?*. https://eptanetwork.org/about/what-is-ta

[W016] F. A. Hayek (1945). *The Use of Knowledge in Society*. https://german.yale.edu/sites/default/files/hayek_-_the_use_of_knowledge_in_society.pdf

[GL-W085] FAO (1998). *Guidelines for implementation of forest management operations*. In Guidelines for the Management of Tropical Forests. 1: The Production of Wood. FAO Forestry Paper 135. Rome: FAO. ISBN 92-5-104123-7. https://www.fao.org/4/w8212e/w8212e08.htm

[GL-W055] FAO (1996). *Precautionary Approach to Fishery Management*. Chapter 3 of Precautionary Approach to Capture Fisheries and Species Introductions. FAO Technical Guidelines for Responsible Fisheries, 2. Rome: FAO. ISBN 92-5-103915-1. https://www.fao.org/4/W3592E/w3592e07.htm

[GL-W017] Fatemeh Rahimi; Reza Aghayari; Bijan Samali (2020). *Application of Tuned Mass Dampers for Structural Vibration Control: A State-of-the-art Review*. https://www.civilejournal.org/index.php/cej/article/download/2261/pdf/6416

[SG-W023] Felix F. Wu (1975). *Solution of Large-Scale Networks by Tearing*. https://digicoll.lib.berkeley.edu/record/140050/files/ERL-m-532.pdf

[GL-W046] FEMA (2017). *National Incident Management System, Third Edition*. https://training.fema.gov/emiweb/is/is700b/handouts/national_incident_management%20system_third%20edition_october_2017.pdf

[GL-W051] Fernand Braudel (1958). *Histoire et Sciences sociales: La longue durée [History and the Social Sciences: The Long Duration]*. https://www.persee.fr/doc/ahess_0395-2649_1958_num_13_4_2781

[WME-W016] Filosofia: An Encyclopedia of Russian Thought (2021). *Katsenelinboigen, Aron*. https://filosofia.dickinson.edu/encyclopedia/katsenelinboigen-aron/

[GL-W010] Food and Agriculture Organization of the United Nations (n.d.). *The 10 Elements of Agroecology framework*. https://www.fao.org/agroecology/overview/the-10-elements-of-agroecology/en

[SG-W051] Fran Ackermann; Colin Eden (2020). *Strategic Options Development and Analysis*. https://strathprints.strath.ac.uk/72174/1/Ackermann_Eden_2020_Strategic_options_development_and_analysis.pdf

[W073] Francis Heylighen (2016). *Stigmergy as a universal coordination mechanism I: Definition and components*. https://doi.org/10.1016/j.cogsys.2015.12.002

[W081] Francisco G. Varela; Humberto R. Maturana; R. Uribe (1974). *Autopoiesis: The organization of living systems, its characterization and a model*. https://doi.org/10.1016/0303-2647(74)90031-8

[RGC-W035] François Sigaut; René Bourrigaud (editor) (n.d.). *Homo Faber: Documents d'un débat oublié, 1907–1941. Partie IV: D'Espinas à Mauss [Homo Faber: Documents of a Forgotten Debate, 1907–1941. Part IV: From Espinas to Mauss]*. Unpublished manuscript, posthumously edited by René Bourrigaud. https://francois-sigaut.com/phocadownload/Inedits/Ouvrages-inedits/homo%20faber-documents-1907-1941.pdf

[SB-W019] Fred Simpson (1989). *Book Review: Grammatical Man*. https://www.narayanagurukula.org/Media/MagArchive/fulltext/1989-2.pdf

[SG-W056] Fremont E. Kast; James E. Rosenzweig (1972). *General Systems Theory: Applications for Organization and Management*. https://doi.org/10.2307/255141

[SG-W011] Fritjof Capra; Pier Luigi Luisi (2014). *The Systems View of Life: A Unifying Vision*. https://www.capracourse.net/wp-content/uploads/2015/12/Introduction.pdf

[WME-W006] Fritjof Capra; Pier Luigi Luisi (2014). *The Systems View of Life: A Unifying Vision*. https://www.capracourse.net/book/

[SG-W025] Fritz W. Scharpf (2006). *The Joint-Decision Trap Revisited*. https://pure.mpg.de/rest/items/item_1233619_4/component/file_1825078/content

[RGC-W008] G.A.Swanson (2007). *Living Systems Theory and Entity-Systems Theory*. https://journals.isss.org/index.php/proceedings51st/article/view/576

[RGC-W007] G.A.Swanson and James Grier Miller (n.d.). *Living Systems Theory*. In Systems Science and Cybernetics, vol. I. Encyclopedia of Life Support Systems (EOLSS). https://www.eolss.net/sample-chapters/c02/e6-46-01-03.pdf

[SG-W012] Geoffrey Vickers (1978 interview); The Open University (course framing) (1978). *Systems approaches: Sir Geoffrey Vickers*. https://www.open.edu/openlearn/mod/oucontent/view.php?id=65611&section=4

[TA-W010] George C. Necula (1997). *Proof-Carrying Code*. https://homes.cs.washington.edu/~mernst/teaching/6.893/readings/necula-popl97.pdf

[TA-W011] George C. Necula; S. P. Rahul (2001). *Oracle-Based Checking of Untrusted Software*. https://people.eecs.berkeley.edu/~necula/Papers/oracle_popl01.pdf

[MIL-W016] George C. Thorpe (1917). *Pure Logistics: The Science of War Preparation*. https://upload.wikimedia.org/wikipedia/commons/6/60/Pure_logistics_%28IA_purelogistics00thor%29.pdf

[WME-W018] George Kelly Society (n.d.). *Your Questions, Answered — Personal Construct Psychology FAQ*. https://www.georgekellysociety.org/faq

[SG-W060] George Mason University Warfield archive (n.d.). *The IM Process*. https://warfield.gmu.edu/exhibits/show/warfield/IM/process

[WME-W020] Gerald Midgley; Erik Lindhult (2021). *A systems perspective on systemic innovation*. https://onlinelibrary.wiley.com/doi/10.1002/sres.2819

[W020] Gilbert Simondon (1958). *Du mode d'existence des objets techniques [On the Mode of Existence of Technical Objects]*. https://s3.amazonaws.com/arena-attachments/2444378/e3006bb0230952ea3d9a3e2d4cf65c1c.pdf

[W093] Giovanni Dosi; Luigi Marengo; Maria Enrica Virgillito (2021). *Hierarchies, Knowledge, and Power Inside Organizations*. https://doi.org/10.1287/stsc.2021.0136

[GL-W022] Giulia Rispoli (2021). *Sharing in Action: The Systemic Concept of the Environment in Aleksandr Bogdanov*. https://iris.unive.it/handle/10278/3761930

[WME-W005] Gordon C. Ruscoe (1984). *A Response to Miller's Editorial*. https://www.coexploration.org/systems/bulletins/General_Systems_Bulletin_Winter_1984_Vol_XIV_No_3.pdf

[SG-W009] Gregory Bateson (1967). *Cybernetic Explanation (reprint in Steps to an Ecology of Mind)*. https://jlombardi.net/pdf/bateson_explanation.pdf

[GL-W013] Gregory Nagy (2021). *Text and reperformance: do you really need a text for your reperformance?*. Classical Inquiries, 24 June 2021. https://classical-inquiries.chs.harvard.edu/text-and-reperformance-do-you-really-need-a-text-for-your-reperformance/

[RGC-W041] Gu Jifa; Wang Lin; Tang Xijin; Liu Puhe; Gong Xinzhong (1997). *物理-事理-人理系统方法论在建立商业设施与技术装备标准规范体系表结构框架中的应用 [Application of Wuli–Shili–Renli Systems Methodology to Establishing the Structural Framework of a Standards System for Commercial Facilities and Technical Equipment]*. https://meta-synthesis.amss.cas.cn/Publication/MSKS_Publications/Journal_papers/Papers_2000/201410/P020141016378061060114.pdf

[UX-W041] Guido Montúfar; Keyan Ghazi-Zahedi; Nihat Ay (2015). *A Theory of Cheap Control in Embodied Systems*. PLOS Computational Biology, 11(9), e1004427. https://doi.org/10.1371/journal.pcbi.1004427

[GL-W019] Aske Halling; Martin Bækgaard (2024). *Administrative Burden in Citizen–State Interactions: A Systematic Literature Review*. Journal of Public Administration Research and Theory, 34(2), 180–195. https://doi.org/10.1093/jopart/muad023

[AFC-W017] Han T. J. Smit; Lenos Trigeorgis (2004). *Strategic Investment: Real Options and Games*. ISBN9780691010397. https://assets.press.princeton.edu/chapters/s7727.pdf

[RGC-W032] Hans-Jurgen Engelbrecht (1997). *A comparison and critical assessment of Porat and Rubin's information economy and Wallis and North's transaction sector*. https://ideas.repec.org/a/eee/iepoli/v9y1997i4p271-290.html

[MIL-W002] Headquarters Department of the Army (2019). *Mission Command: Command and Control of Army Forces (ADP 6-0)*. 31 July 2019. https://irp.fas.org/doddir/army/adp6_0.pdf

[MIL-W001] Headquarters United States Marine Corps (1996). *Command and Control (MCDP 6)*. 4 October 1996; Change 1, 4 April 2018. https://www.marines.mil/Portals/1/Publications/MCDP%206.pdf

[MIL-W003] Headquarters United States Marine Corps (2023). *Logistics (MCDP 4)*. 21 March 2023. Revision of the 1997 publication. https://www.mca-marines.org/wp-content/uploads/MCDP-4-1.pdf

[W080] Heinz von Foerster (1960). *On Self-Organizing Systems and Their Environments*. https://doi.org/10.1007/0-387-21722-3_1

[W088] Helena Correa Tonet; Maria das Graças Torres da Paz (2006). *Um modelo para o compartilhamento de conhecimento no trabalho [A Model for Knowledge Sharing at Work]*. https://doi.org/10.1590/S1415-65552006000200005

[UX-W034] Helmut F. Spinner (1994). *Erkenntnis und Entlastung – zu Arnold Gehlens Verständnis von Wissenschaft und Technik im Rahmen der Klassischen Wissensordnung [Knowledge and Relief: On Arnold Gehlen's Understanding of Science and Technology Within the Classical Order of Knowledge]*. In Zur geisteswissenschaftlichen Bedeutung Arnold Gehlens, pp. 19–56; discussion pp. 58–65. Schriftenreihe der Hochschule Speyer, 113. https://publikationen.bibliothek.kit.edu/318594

[MIL-W015] Henry E. Eccles (1959). *Logistics in the National Defense*. https://www.marines.mil/Portals/1/Publications/FMFRP%2012-14.pdf

[TA-W005] Henry E. Smith (2012). *Property as the Law of Things*. 125 Harvard Law Review 1691–1726. https://harvardlawreview.org/wp-content/uploads/2012/05/vol125_h.smith.pdf

[W024] Henry S. Parmelee (1874). *Improvement in fire-extinguishers*. US154076A. https://patents.google.com/patent/US154076A/en

[W015] Herbert A. Simon (1962). *The Architecture of Complexity*. https://www.andrew.cmu.edu/course/15-440/assets/READINGS/simon-architecture-of-complexity-1962.pdf

[WME-W002] Herman Greene (2018). *Review of Erich Jantsch, The Self-Organizing Universe*. https://ecozoicstudies.org/wp-content/uploads/2022/10/Greene.H.Review-of-Erich-Jantsch-The-Self-Organzing-Universe-Scientific-and-Human-Implications-of-the-Emerging-Paradigm-of-Evolution.EJ5_.2018.pdf

[W083] Hermann Haken (1998). *Can we apply synergetics to the human sciences?*. ISBN 3-11-015619-9. https://api.pageplace.de/preview/DT0400.9783110801194_A19971969/preview-9783110801194_A19971969.pdf

[RGC-W038] Hermann Haken (2017). *What Can Synergetics Contribute to Embodied Aesthetics?*. https://doi.org/10.3390/bs7030061

[UX-W014] Howard H. Pattee (1973). *The Physical Basis and Origin of Hierarchical Control*. In H. H. Pattee (ed.), Hierarchy Theory: The Challenge of Complex Systems, pp. 73–108. George Braziller. https://casci.binghamton.edu/publications/pattee/bibliography.html

[UX-W012] Howard H. Pattee (2001). *The Physics of Symbols: Bridging the Epistemic Cut*. BioSystems, 60, 5–21. https://casci.binghamton.edu/publications/pattee/pattee.html

[UX-W030] Hubert Landier (2024). *Polycellular Human Organisation: For a Transdisciplinary Approach*. ADJURIS International Academic Publisher. 978-606-95862-9-7. https://www.adjuris.ro/books/phor/phor.pdf

[WME-W015] Héctor Sabelli (2008). *Bios Theory of Innovation*. https://innovation.cc/wp-content/uploads/2008_13_3_12_sabelli_bios-theory.pdf

[GL-W012] Ian Hodder (2011). *Human-thing entanglement: towards an integrated archaeological perspective*. https://www.ian-hodder.com/s/human-thing-entanglement-towards-an-integrated-archaeological-perspectivejrai_1674.pdf

[GL-W096] Ian Hodder (2018). *Where Are We Heading? The Evolution of Humans and Things*. https://yalebooks.co.uk/display.asp?K=9780300204094

[GL-W048] IFAC Professional Accountants in Business Committee (2012). *Evaluating and Improving Internal Control in Organizations*. https://www.ifac.org/knowledge-gateway/professional-accountants-business-paib/publications/evaluating-and-improving-internal-control-organizations

[WME-W023] IFS Institute (n.d.). *The Internal Family Systems Model Outline*. https://ifs-institute.com/resources/articles/internal-family-systems-model-outline

[SG-W045] Ilya Prigogine (1977). *Time, Structure and Fluctuations—Nobel Lecture*. https://www.nobelprize.org/uploads/2018/06/prigogine-lecture.pdf

[GL-W023] International Council on Archives, Expert Group on Archival Description (2023). *Records in Contexts–Conceptual Model, Version 1.0*. https://www.ica.org/app/uploads/2023/12/RiC-CM-1.0.pdf

[GL-W082] International Maritime Organization (n.d.). *E-navigation*. https://www.imo.org/en/ourwork/safety/pages/enavigation.aspx

[SG-W006] International Society for the Systems Sciences (n.d.). *About ISSS*. https://www.isss.org/about-isss/

[GL-W011] Ioanna Mouratiadou; Alexander Wezel; Kintan Kamilia; Angelica Marchetti; Maria Luisa Paracchini; Paolo Barberi (2024). *The socio-economic performance of agroecology. A review*. https://publications.jrc.ec.europa.eu/repository/handle/JRC135379

[GL-W089] IPCC Working GroupI chapter authors (2021). *Climate Change 2021: The Physical Science Basis, Chapter 1, §1.5.3.2: Model Tuning and Adjustment*. https://www.ipcc.ch/report/ar6/wg1/chapter/chapter-1/

[W103] Ivo Siekmann (2017). *An applied mathematician's perspective on Rosennean Complexity*. Preprint. LJMU repository eprint7133. https://researchonline.ljmu.ac.uk/id/eprint/7133/1/Siekmann2017.pdf

[SG-W033] J. G. Bennett (1970). *Systematics and System Theories*. https://duversity.org/systematics-and-system-theories/

[GL-W030] J. H. Saltzer; D. P. Reed; D. D. Clark (1984). *End-to-End Arguments in System Design*. https://web.mit.edu/saltzer/www/publications/endtoend/endtoend.pdf

[W026] J. M. Carlson; John Doyle (1998). *Highly Optimized Tolerance Robustness and Power Laws in Complex Systems*. Preprint, 8 December 1998. arXiv:cond-mat/9812127. https://arxiv.org/pdf/cond-mat/9812127

[TA-W037] J. Van Buskirk; U. K. Steiner (2009). *The fitness costs of developmental canalization and plasticity*. https://doi.org/10.1111/j.1420-9101.2009.01685.x

[RGC-W017] J. A. Scott Kelso (2008). *An Essay on Understanding the Mind*. https://doi.org/10.1080/10407410801949297

[RGC-W011] J. P. van Gigch (1981). *Review of The Heart of Enterprise*. https://journals.sagepub.com/doi/pdf/10.3233/HSM-1981-2212

[TA-W012] Jack Edmonds (1971). *Matroids and the Greedy Algorithm*. https://deltadbu.github.io/UCAS_algorithm_course/Lectures/Edmonds1971.pdf

[W074] Jacques Lafitte (1932). *Réflexions sur la science des machines [Reflections on the Science of Machines]*. https://repository.gatech.edu/entities/publication/58b7b43b-88ee-4c06-b9fe-8a992bafa699

[RGC-W034] Jacques Lafitte (1933). *Sur la science des machines [On the Science of Machines]*. https://gallica.bnf.fr/ark:/12148/bpt6k101631q

[RGC-W030] Jacques Villiers (2002). *Productivité et équité, la fin d'un rêve ? [Productivity and Equity: The End of a Dream?]*. https://www.lajauneetlarouge.com/productivite-et-equite-la-fin-dun-reve/

[W069] James Clerk Maxwell (1868). *On Governors*. https://doi.org/10.1098/rspl.1867.0055

[MIL-W005] James D. Thompson (1967). *Organizations in Action: Social Science Bases of Administrative Theory*. https://www.army.mil/article/136556/mission_command_and_logistics_interdependencies

[W065] James E. Stott; Paul T. Britton; Robert W. Ring; Frank Hark; G. Spencer Hatfield (2010). *Common Cause Failure Modeling: Aerospace Versus Nuclear*. NASA NTRS 20100025991. M10-0548. https://ntrs.nasa.gov/api/citations/20100025991/downloads/20100025991.pdf

[RGC-W020] James E. Huchingson (1981). *Organization and Process: Systems Philosophy and Whiteheadian Metaphysics*. https://www.religion-online.org/article/organization-and-process-systems-philosophy-and-whiteheadian-metaphysics/

[SG-W039] James F. Courtney; David T. Croasdell; David B. Paradice (1998). *Inquiring Organizations*. https://www.bauer.uh.edu/parks/fis/inqorg.htm

[SG-W008] James G. Miller (1970). *The Nature of Living Systems: An Exposition of the Basic Concepts in General Systems Theory*. https://files.eric.ed.gov/fulltext/ED039774.pdf

[RGC-W004] James Grier Miller (1978). *Living Systems*. https://openlibrary.org/books/OL4553786M/Living_systems

[WME-W004] James Grier Miller (1978). *Living Systems*. https://www.panarchy.org/miller/livingsystems.html

[RGC-W006] James Grier Miller and Jessie L. Miller (1992). *Applications of Living Systems Theory*. https://web3.isss.org/primer2/PP1/asem05jm.html

[W042] James J. Gibson (1979). *The Theory of Affordances*. Chapter 8 in The Ecological Approach to Visual Perception. Houghton Mifflin, Boston. https://cs.brown.edu/courses/cs137/2017/readings/Gibson-AFF.pdf

[RGC-W003] Jamshid Gharajedaghi (1984). *Organizational implications of systems thinking: Multidimensional modular design*. https://doi.org/10.1016/0377-2217(84)90181-4

[WME-W003] Jamshid Gharajedaghi (1999). *Systems Thinking: Managing Chaos and Complexity: A Platform for Designing Business Architecture*. Third edition, 2011. Butterworth-Heinemann; Morgan Kaufmann (third edition). https://ocw.ui.ac.id/pluginfile.php?file=%2F593%2Fmod_folder%2Fcontent%2F0%2FJamshid+-+Systems+Thinking.pdf&forcedownload=1

[RGC-W002] Jamshid Gharajedaghi and Russell L. Ackoff (1984). *Mechanisms, organisms and social systems*. https://doi.org/10.1002/smj.4250050308

[W014] Jan C. Willems (1972). *Dissipative Dynamical Systems Part I General Theory*. https://doi.org/10.1007/BF00276493

[UX-W017] Jan C. Willems (1997). *On Interconnections, Control, and Feedback*. IEEE Transactions on Automatic Control, 42(3), 326–339. https://homes.esat.kuleuven.be/~sistawww/smc/jwillems/Articles/JournalArticles/1997.4.pdf

[UX-W019] Jan C. Willems (2007). *The Behavioral Approach to Open and Interconnected Systems*. IEEE Control Systems Magazine, 27(6), 46–99. https://ieeexplore.ieee.org/document/4384643/

[RGC-W014] Janet Kübler; Michelle Y. Merrill; Randall Anway (2014). *The Systems View of Life: A Unifying Vision — collaborative book review*. https://test.zqjournal.org/pdfs/ZQ_issue_10.pdf

[W002] Jay R. Galbraith (1974). *Organization Design: An Information Processing View*. https://doi.org/10.1287/inte.4.3.28

[W094] Jean Voge (1983). *The political economics of complexity: From the information economy to the 'complexity' economy*. https://doi.org/10.1016/0167-6245(83)90022-7

[UX-W031] Jean Voge (1985). *Crise, information et communication dans l'économie américaine [Crisis, Information and Communication in the American Economy]*. Futuribles, 90, 39–47. https://www.futuribles.com/wp-content/uploads/related-documents/revue-futuribles-n090.pdf?postId=4510

[RGC-W031] Jean Voge (1997). *Le complexe de Babel: crise ou maîtrise de l'information ? [The Babel Complex: Crisis or Mastery of Information?]*.

[UX-W028] Jean-Louis Le Moigne (2003). *Le complexe de Babel, crise ou maîtrise de l'information ? [The Babel Complex: Crisis or Mastery of Information?] (book review)*. Review of Jean Voge (1997). https://www.intelligence-complexite.org/bibliotheque/note-de-lecture/complexe-babel-crise-ou-maitrise-linformation

[UX-W027] Jean-Louis Le Moigne (1977). *La théorie du système général. Théorie de la modélisation [General System Theory: Theory of Modeling]*. Fourth edition, 1994. Presses Universitaires de France. https://www.intelligence-complexite.org/sites/default/files/media/text/ateliers/0609tsgtm.pdf

[RGC-W025] Jean-Louis Le Moigne (2015). *Chronique de La Théorie du Système Général, Théorie de la Modélisation [Chronicle of General System Theory: Theory of Modeling]*. https://www.intelligence-complexite.org/sites/default/files/media/text/conseil_scient/1505jlmdg.pdf

[GL-W009] Jean-Marie Lehn (1987). *Supramolecular Chemistry—Scope and Perspectives: Molecules—Supermolecules—Molecular Devices*. https://www.nobelprize.org/uploads/2018/06/lehn-lecture.pdf

[SB-W010] Jens Rasmussen (1993). *Modeling the Evolution and Breakdown of Adaptive Systems: A Review*. https://orbit.dtu.dk/en/publications/modeling-the-evolution-and-breakdown-of-adaptive-systems-a-review/

[WME-W029] Jeremy Campbell (1982). *Grammatical Man: Information, Entropy, Language, and Life*. Simon and Schuster. https://books.google.com/books?id=7ItQAAAAMAAJ

[WME-W029-ES1989] Jeremy Campbell (1989). *El hombre gramatical [Grammatical Man]*. Spanish translation by Juan José Utrilla. Fondo de Cultura Económica. Original English work published 1982. ISBN9789681629489.

[TA-W041] Jessica C. Flack (2017). *Coarse-graining as a downward causation mechanism*. https://pmc.ncbi.nlm.nih.gov/articles/PMC5686398/

[GL-W073] Jie Ling; Tingting Ye; Zhao Feng; Yuchuan Zhu; Yangmin Li; Xiaohui Xiao (2022). *A survey on synthesis of compliant constant force/torque mechanisms*. https://doi.org/10.1016/j.mechmachtheory.2022.104970

[TA-W016] Johan S. H. van Leeuwaarden; Britt W. J. Mathijsen; Bert Zwart (2019). *Economies-of-Scale in Many-Server Queueing Systems: Tutorial and Partial Review of the QED Halfin–Whitt Heavy-Traffic Regime*. https://ir.cwi.nl/pub/28868/28868.pdf

[W085] Johannes Jaeger (2024). *Assembly Theory: What It Does and What It Does Not Do*. https://doi.org/10.1007/s00239-024-10163-2

[RGC-W009] John Brocklesby, Stephen Cummings and John Davies (1991). *Cybernetics and organisational analysis; Towards a better understanding of Beer's Viable System Model*. https://ojs.wgtn.ac.nz/somwp/article/view/7161

[UX-W042] John C. Doyle; Marie Csete (2011). *Architecture, constraints, and behavior*. Proceedings of the National Academy of Sciences, 108(Supplement 3), 15624–15630. https://doi.org/10.1073/pnas.1103557108

[GL-W093] John Dewey (1922). *Human Nature and Conduct: An Introduction to Social Psychology*. https://www.gutenberg.org/files/41386/41386-h/41386-h.htm

[SB-W004] John F. Padgett (1992). *The Alchemist of Contingency Theory*. https://doi.org/10.1086/229909

[TA-W017] John J. Hopfield (1974). *Kinetic Proofreading: A New Mechanism for Reducing Errors in Biosynthetic Processes Requiring High Specificity*. https://garcialab.berkeley.edu/courses/papers/Hopfield1974.pdf

[W046] John Joseph; Metin Sengul (2025). *Organization Design: Current Insights and Future Research Directions*. First published online 2024; journal issue 2025. https://doi.org/10.1177/01492063241271242

[W097] John M. Richardson, Jr. (1975). *Hierarchical models of very large problems, dilemmas, prospects, and an agenda for the future*. NTRS19750021774;75N29847. https://ntrs.nasa.gov/api/citations/19750021774/downloads/19750021774.pdf

[TA-W020] John Odling-Smee; Douglas H. Erwin; Eric P. Palkovacs; Marcus W. Feldman; Kevin N. Laland (2013). *Niche Construction Theory: A Practical Guide for Ecologists*. https://lalandlab.wp.st-andrews.ac.uk/files/2015/08/Publication191.pdf

[MIL-W008] John R. Boyd (2014). *Organic Design for Command and Control*. Edited reconstruction, Chet Richards and Chuck Spinney, 2014, of material dated 1987–c. 1991. https://ooda.de/media/john_boyd_-_organic_design_for_command_and_control.pdf

[SG-W010] John R. Hudson (2019). *Appreciative systems*. https://johnrhudson.me.uk/systems/Appreciative_systems.pdf

[TA-W035] John Sweller (1988). *Cognitive Load During Problem Solving: Effects on Learning*. https://www.waddesdonschool.com/wp-content/uploads/2021/02/Cognitive-Load-Theory-Sweller-J-1988.pdf

[W043] John von Neumann (1956). *Probabilistic Logics and the Synthesis of Reliable Organisms from Unreliable Components*. https://static.ias.edu/pitp/archive/2012files/Probabilistic_Logics.pdf

[WME-W007] Johny Van Aerde (2015). *Book review: The Systems View of Life: A Unifying Vision*. Canadian Journal of Physician Leadership, 1(3), Winter. https://cjpl.ca/systems.html

[MIL-W020] Josefin Svensson (interviewing Antoine Bousquet) (2022). *How scientific ideas are reflected in practices of war*. https://www.forsvarshogskolan.se/en/swedish-defence-university/stories/2022-06-10-how-scientific-ideas-are-reflected-in-practices-of-war.html

[UX-W006] Joseph A. Tainter (2006). *Social complexity and sustainability*. Ecological Complexity, 3, 91–103. https://doi.org/10.1016/j.ecocom.2005.07.004

[GL-W006] Joseph Murphy; Sarah Parry (2021). *Gender, households and sustainability: Disentangling and re-entangling with the help of 'work' and 'care'*. https://journals.sagepub.com/doi/10.1177/2514848620948432

[TA-W038] Josh R. Auld; Anurag A. Agrawal; Rick A. Relyea (2010). *Re-evaluating the costs and limits of adaptive phenotypic plasticity*. https://www.biology.pitt.edu/sites/default/files/facilities-images/Auld-Relyea-270.pdf

[AFC-W011] Joshua Owotogbe; Indika Kumara; Willem-Jan van den Heuvel; Damian Andrew Tamburri (2025). *Chaos Engineering: A Multi-Vocal Literature Review*. arXiv:2412.01416v2. https://arxiv.org/abs/2412.01416v2

[W098] Jun Wang; Ziyang Zhou; Ardalan Kahak; Suyi Li (2026). *Embodying physical computing into soft robots*. https://doi.org/10.1038/s41467-026-70866-6

[WME-W022] Jürgen Kriz (n.d.). *Towards a Synthesis: Person-Centered Systems Theory*. English redaction associated with chapter 20, Epilog: Prolegomena zu einem Mehr-Ebenen-Konzept von Therapie [Epilogue: Prolegomena to a Multi-Level Concept of Therapy], in Grundkonzepte der Psychotherapie: Eine Einführung [Fundamental Concepts in Psychotherapy: An Introduction], Munich: Urban & Schwarzenberg, 1985, pp. 298–309; revised second edition, Psychologie Verlags Union, 1989. https://www.jkriz.de/jk-pcst.htm

[MIL-W029] Kaleb McDowell; Ellen Novoseller; Anna Madison; Vinicius G. Goecks; Christopher Kelshaw (2024). *Re-Envisioning Command and Control*. arXiv2402.07946v2. https://arxiv.org/html/2402.07946v2

[W052] Karl Friston (2010). *The free-energy principle a unified brain theory*. https://doi.org/10.1038/nrn2787

[W056] Karl J. Friston; Lancelot Da Costa; Thomas Parr (2021). *Some Interesting Observations on the Free Energy Principle*. https://doi.org/10.3390/e23081076

[AFC-W016] Kathryn Chaloner; Isabella Verdinelli (1995). *Bayesian Experimental Design: A Review*. https://doi.org/10.1214/ss/1177009939

[GL-W065] Katia Bertoldi; Vincenzo Vitelli; Johan Christensen; Martin van Hecke (2017). *Flexible mechanical metamaterials*. https://amolf.nl/publications/flexible-mechanical-metamaterials

[GL-W015] Keith Beven (2006). *A manifesto for the equifinality thesis*. https://mechanism.ucsd.edu/bill/ModelsAndPrediction/papers/beven.ManifestoJH.pdf

[SG-W036] Kenneth E. Boulding (1956). *General Systems Theory—The Skeleton of Science*. https://doi.org/10.1287/mnsc.2.3.197

[GL-W025] Kenneth G. Wilson (1982). *The Renormalization Group and Critical Phenomena*. https://www.nobelprize.org/uploads/2018/06/wilson-lecture-2.pdf

[GL-W053] Kevin Krupski (2023). *Who's the Boss? Defining the Civil-Military Relationship in the Twenty-First Century*. https://www.armyupress.army.mil/Portals/7/military-review/Archives/English/JF-23/Krupski/krupski-defining-civil-military-relationship.pdf

[TA-W026] Kim J. Vicente; Jens Rasmussen (1992). *Ecological Interface Design: Theoretical Foundations*. https://orbit.dtu.dk/en/publications/ecological-interface-design-theoretical-foundations/

[WME-W030] Kirkus Reviews (1982). *Grammatical Man — review*. https://www.kirkusreviews.com/book-reviews/a/jeremy-campbell-2/grammatical-man-information-entropy-language-an/

[GL-W081] L. Kabanov; J. Kupitz; C. A. Goetzmann (1992). *Advanced reactors: Safety and environmental considerations*. https://www.iaea.org/sites/default/files/publications/magazines/bulletin/bull34-2/34202043236.pdf

[W003] L. Ulbo de Sitter; J. Friso den Hertog; Ben Dankbaar (1997). *From Complex Organizations with Simple Jobs to Simple Organizations with Complex Jobs*. https://doi.org/10.1177/001872679705000503

[GL-W032] Laurent Thévenot (1984). *Rules and implements: investment in forms*. https://journals.sagepub.com/doi/10.1177/053901884023001001

[GL-W033] Laurent Thévenot (2001). *Organized Complexity: Conventions of Coordination and the Composition of Economic Arrangements*. https://journals.sagepub.com/doi/10.1177/13684310122225235

[GL-W086] Lee R. Lynd; Willem H. van Zyl; John E. McBride; Mark Laser (2005). *Consolidated bioprocessing of cellulosic biomass: an update*. https://doi.org/10.1016/j.copbio.2005.08.009

[GL-W060] Leonardo Morsut; Kole T. Roybal; Xin Xiong; Russell M. Gordley; Scott M. Coyle; Matthew Thomson; Wendell A. Lim (2016). *Engineering Customized Cell Sensing and Response Behaviors Using Synthetic Notch Receptors*. https://doi.org/10.1016/j.cell.2016.01.012

[TA-W001] Leonid Hurwicz (2007). *But Who Will Guard the Guardians?*. https://www.nobelprize.org/uploads/2018/06/hurwicz_lecture.pdf

[TA-W002] Leonid Hurwicz; Stanley Reiter (2006). *Designing Economic Mechanisms*. https://doi.org/10.1017/CBO9780511754258

[W106] Loes Crielaard; Luc Hagenaars; Jillian O'Mara; Mary Nicolaou; Wilma Waterlander; Karien Stronks (2026). *Using Systems Archetypes in Public Health: A Review Connecting System Dynamics Literature and Public Health Applications*. https://doi.org/10.1002/sdr.70032

[SG-W055] Loet Leydesdorff (2005). *Anticipatory Systems and the Processing of Meaning: a Simulation Study Inspired by Luhmann's Theory of Social Systems*. https://www.jasss.org/8/2/7.html

[WME-W021] Loet Leydesdorff (2021). *Subdynamics in Knowledge-Based Systems*. https://link.springer.com/chapter/10.1007/978-3-030-59951-5_9

[GL-W079] Lothar Leistner (2000). *Basic aspects of food preservation by hurdle technology*. https://www.sciencedirect.com/science/article/pii/S0168160500001616

[GL-W043] Louis Kaplow (1992). *Rules Versus Standards: An Economic Analysis*. https://scholarship.law.duke.edu/dlj/vol42/iss3/2/

[RGC-W039] Lu Mingsen / 卢明森 (2004). *"开放的复杂巨系统"概念的形成 [Formation of the Concept "Open Complex Giant System"]*. https://devp-service.oss-cn-beijing.aliyuncs.com/c1bef5bc349340ef8ae8faa19e39d47e/file_1562114167506.pdf

[SG-W043] Luan Moreira (2020). *O Sistemismo de Bunge como Instrumento Metodológico para Aprendizagem Ativa [Bunge's Systemism as a Methodological Instrument for Active Learning]*. https://doi.org/10.1590/1516-731320200058

[W047] Luca Corti (2023). *Organizational normativity and teleology a critique*. https://doi.org/10.1007/s11229-023-04255-7

[W072] Ludwig von Bertalanffy (1950). *An Outline of General System Theory*. https://doi.org/10.1093/bjps/I.2.134

[W030] Luis Garicano (2000). *Hierarchies and the Organization of Knowledge in Production*. https://doi.org/10.1086/317671

[SB-W005] Lyra J. Colfer; Carliss Y. Baldwin (2016). *The mirroring hypothesis: theory, evidence, and exceptions*. Industrial and Corporate Change, 25(5), 709–738. https://doi.org/10.1093/icc/dtw027

[W075] M. H. A. Newman (1942). *On theories with a combinatorial definition of equivalence*. https://turingarchive.kings.cam.ac.uk/publications-lectures-and-talks-amtb/amt-b-50

[SB-W008] M. D. Mesarovic; D. Macko; Y. Takahara (1969). *Two Coordination Principles and Their Application in Large Scale Systems Control*. https://zbc.uz.zgora.pl/Content/30559/PDF/ifac_28.pdf

[RGC-W021] M. S. Leaning (1980). *The validity and validation of mathematical models: Methodological, theoretical, and practical studies with emphasis on the modelling of complex biological systems*. https://openaccess.city.ac.uk/id/eprint/27445/

[SG-W061] Magoroh Maruyama (1963). *The Second Cybernetics: Deviation-Amplifying Mutual Causal Processes*. https://pespmc1.vub.ac.be/books/Maruyama-SecondCybernetics.pdf

[RGC-W037] Magoroh Maruyama (1988). *This Week's Citation Classic: The Second Cybernetics*. https://garfield.library.upenn.edu/classics1988/A1988L962400001.pdf

[W061] Marc Shapiro; Nuno Preguiça; Carlos Baquero; Marek Zawirski (2011). *A comprehensive study of Convergent and Commutative Replicated Data Types*. inria-00555588v1. https://ericmoritz.github.io/comp-sci-papers/papers/crdt-comprehensive.pdf

[W054] Marcin Miłkowski (2018). *Morphological Computation Nothing but Physical Computation*. https://doi.org/10.3390/e20120942

[MIL-W022] Marco Sigg (2014). *Der Unterführer als Feldherr im Taschenformat: Theorie und Praxis der Auftragstaktik im deutschen Heer 1869 bis 1945 [The Subordinate Leader as a Field Commander in Pocket Format: Theory and Practice of Mission Command in the German Army, 1869–1945]*. https://researchcentre.army.gov.au/library/australian-army-journal-aaj/auftragstaktik-mission-command

[SG-W017] Mario Giampietro; Kozo Mayumi; Jesus Ramos-Martin (2009). *Multi-scale integrated analysis of societal and ecosystem metabolism (MuSIASEM): Theoretical concepts and basic rationale*. https://ddd.uab.cat/pub/artpub/2009/81dc1ab73df1/AAM_Giampietro_et_al_2009.pdf

[UX-W026] Marisabel Marratt (2021). *Mechanological Dynamics in Jacques Lafitte and Gilbert Simondon*. ConCave 2020 proceedings, pp. 203–214. Georgia Institute of Technology, 2021. https://doi.org/10.35090/gatech/93

[RGC-W033] Mark Hayward; Ghislain Thibault (2021). *Ethics in Jacques Lafitte's Mechanology*. https://journals.sagepub.com/doi/10.1177/0263276420981156

[W055] Martin Biehl; Felix A. Pollock; Ryota Kanai (2021). *A Technical Critique of Some Parts of the Free Energy Principle*. https://doi.org/10.3390/e23030293

[SG-W046] Martin Golubitsky (1978). *An Introduction to Catastrophe Theory and Its Applications*. https://www.asc.ohio-state.edu/golubitsky.4/reprintweb-0.5/output/papers/catastrophe_theory.pdf

[AFC-W007] Martin Monperrus (2017). *Principles of Antifragile Software*. https://doi.org/10.1145/3079368.3079412

[MIL-W006] Martin van Creveld (1985). *Command in War*. ISBN0674144406. https://library.marshallfoundation.org/portal/Default/en-US/RecordView/Index/18864

[GL-W005] Mary Jo Bitner (1992). *Servicescapes: The Impact of Physical Surroundings on Customers and Employees*. https://studylib.net/doc/27211687/bitner--1992-

[W007] Matej Hoffmann; Vincent C. Müller (2014). *Trade-offs in exploiting body morphology for control: from simple bodies and model-based control to complex bodies with model-free distributed control schemes*. arXiv preprint, 2014. arXiv:1411.2276v1. https://arxiv.org/html/1411.2276v1

[GL-W091] Matteo Mossio; Maël Montévil; Giuseppe Longo (2016). *Theoretical principles for biology: organization*. https://montevil.org/publications/articles/2016-mml-theoretical-principles-organization/

[W013] Maël Montévil; Matteo Mossio (2015). *Biological organisation as closure of constraints*. https://doi.org/10.1016/j.jtbi.2015.02.029

[AFC-W003] Mei-Chen Hsueh; Timothy K. Tsai; Ravishankar K. Iyer (1997). *Fault Injection Techniques and Tools*. Computer, 30(4), 75–82. https://doi.org/10.1109/2.585157

[RGC-W010] Michael C. Jackson (1992). *The Soul of the Viable System Model*. https://www.researchgate.net/publication/226077715_The_soul_of_the_viable_system_model

[GL-W020] Michael K. Buckland (1991). *Information as thing*. https://escholarship.org/content/qt4x2561mb/qt4x2561mb.pdf

[GL-W087] Michael Rathjen; Wilfried Sieg (2024). *Proof Theory*. https://plato.stanford.edu/entries/proof-theory/

[RGC-W016] Michel Bitbol; Pier Luigi Luisi (2004). *Autopoiesis with or without cognition: defining life at its edge*. https://doi.org/10.1098/rsif.2004.0012

[TKT-W003] Michel Petrovitch (Mihailo Petrović) (1921). *Mécanismes communs aux phénomènes disparates [Mechanisms Common to Disparate Phenomena]*. https://gtmarket.ru/library/basis/5909/5916

[RGC-W015] Michelle Y. Merrill (2014). *Reviewing The Systems View of Life*. https://perplexedprimate.org/2014/07/24/reviewing-the-systems-view-of-life/

[AFC-W005] Microsoft (n.d.). *Architecture strategies for reliability testing*. Azure Well-Architected Framework. https://learn.microsoft.com/en-us/azure/well-architected/reliability/reliability-test

[W096] Mihajlo D. Mesarović; D. Macko; Yasuhiko Takahara (1970). *Theory of Hierarchical, Multilevel, Systems*. Academic Press. Mathematics in Science and Engineering, vol. 68. ISBN 0124915507. https://openlibrary.org/books/OL14827911M/Theory_of_hierarchical_multilevel_systems_by_M.D._Mesarovi%C4%87_D._Macko_and_Y._Takahara.

[SG-W034] Mildred L. G. Shaw (1986). *Epistemological Hierarchies in Knowledge and Expert Systems*. https://gaines.library.uvic.ca/pdf/EpistHierKES86.pdf

[SG-W026] Miranda Lucas (2024). *A practitioner's field guide to the behaviour settings method*. https://pmc.ncbi.nlm.nih.gov/articles/PMC11338572/

[W044] Mitio Nagumo (1942). *Über die Lage der Integralkurven gewöhnlicher Differentialgleichungen [On the Position of the Integral Curves of Ordinary Differential Equations]*. Proceedings of the Physico-Mathematical Society of Japan, Third Series, 24, 551–559. https://doi.org/10.11429/ppmsj1919.24.0_551

[GL-W038] Mixue Li; Jeannie Holstein; Volker Wedekind (2025). *Vital materiality and its constitution of knowing in craft practice*. https://journals.sagepub.com/doi/10.1177/13505076251317965

[MIL-W013] Moshe Kress (2000). *Flexibility in Operational-Level Logistics*. https://doi.org/10.5711/morj.5.1.41

[MIL-W012] Moshe Kress (2016). *Operational Logistics: The Art and Science of Sustaining Military Operations*. 2nd edition. https://doi.org/10.1007/978-3-319-22674-3

[MIL-W014] Moshe Kress (2026). *Operational Logistics: The Art and Science of Sustaining Military Operations*. Announced third edition. Springer, Cham. ISBN9783032301925. https://link.springer.com/book/9783032301925

[WME-W014] Murray Gell-Mann (1995). *What is complexity?*. https://www.its.caltech.edu/~matilde/GellMannComplexity.pdf

[W095] Mustafa Shraim (2019). *Impact of Process Tampering on Variation (Experiment)*. https://deming.org/impact-of-process-tampering-on-variation-experiment-presented-at-engineering-education-conference/?lost_pass=1

[GL-W092] N. J. Habraken; Jonathan Teicher (editor) (1998). *The Structure of the Ordinary: Form and Control in the Built Environment*. https://mitpress.mit.edu/9780262581950/the-structure-of-the-ordinary/

[GL-W021] N. John Habraken (2002). *The Uses of Levels*. https://www.habraken.com/html/downloads/the_uses_of_levels.pdf

[W033] Naama Barkai; Stanislas Leibler (1997). *Robustness in simple biochemical networks*. https://doi.org/10.1038/43199

[TA-W004] Naci Saldi; Serdar Yüksel (2020). *Geometry of Information Structures, Strategic Measures and associated Control Topologies*. arXiv:2010.07377v1. https://arxiv.org/html/2010.07377

[W037] Nam P. Suh (1990). *The Principles of Design*. https://www.axiomaticdesign.com/wp-content/uploads/thesis_sweden96.pdf

[SG-W052] Nam P. Suh (1996). *Introduction to Axiomatic Design*. Chapter 1. Version dated 3 December 1996. https://web.mit.edu/2.882/www/chapter1/chapter1.htm

[UX-W020] Nancy G. Leveson (2004). *A New Accident Model for Engineering Safer Systems*. Safety Science, 42(4), 237–270. https://doi.org/10.1016/S0925-7535(03)00047-X

[UX-W021] Nancy G. Leveson (2012). *Engineering a Safer World: Systems Thinking Applied to Safety*. MIT Press. 9780262016629. https://tile.loc.gov/storage-services/master/gdc/gdcebookspublic/20/20/71/92/47/2020719247/20207192 47.pdf

[SB-W017] Nancy Leveson; Nicolas Dulac; Karen Marais; John Carroll (2009). *Moving Beyond Normal Accidents and High Reliability Organizations: A Systems Approach to Safety in Complex Systems*. https://jori.ncsu.edu/documents/2015/09/daiaoepvdapov.pdf/

[GL-W088] NASA / Night Sky Network (2018). *Life Cycles of Sun-like and Massive Stars*. https://science.nasa.gov/asset/webb/life-cycles-of-sun-like-and-massive-stars/

[AFC-W001] Nassim N. Taleb; Raphael Douady (2012). *Mathematical Definition, Mapping, and Detection of (Anti)Fragility*. arXiv:1208.1189. https://doi.org/10.1080/14697688.2013.800219

[AFC-W002] Nassim Nicholas Taleb (2012). *Antifragile: Things That Gain from Disorder*. Random House. https://www.penguinrandomhouse.com/books/176227/antifragile-by-nassim-nicholas-taleb/

[SG-W047] Natalia Szostak; Szymon Wasik; Jacek Blazewicz (2016). *Hypercycle*. https://doi.org/10.1371/journal.pcbi.1004853

[RGC-W036] Nathaniel Virgo; Martin Biehl; Manuel Baltieri; Matteo Capucci (2025). *A "Good Regulator Theorem" for Embodied Agents*. https://uhra.herts.ac.uk/id/eprint/26230/2/isal.a.874.pdf

[GL-W063] National Commission on Forensic Science (2015). *Institutional and Organizational Best Practices: Abstract developed by the Human Factors Subcommittee*. https://www.justice.gov/sites/default/files/ncfs/pages/attachments/2015/04/16/abstract_human_factors_best_practices.pdf

[GL-W061] National Health Commission of China (2017). *What is TCM?*. https://en.nhc.gov.cn/2017-07/05/c_71908.htm

[GL-W036] NATO (2022). *AJP-3.2 Allied Joint Doctrine for Land Operations, Edition B Version 1*. https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/1085158/20220207-AJP3-2_EDB_Land_Ops.pdf

[MIL-W011] NATO Research Task Group SAS-085 (2014). *Task Group SAS-085 Final Report on C2 Agility: Executive Overview*. STO Technical Report STO-TR-SAS-085, Executive Overview. NATO Science and Technology Organization. https://internationalc2institute.org/s/sas-085_report_overview.pdf

[GL-W062] NCCIH (n.d.). *Traditional Chinese Medicine: What You Need To Know*. https://www.nccih.nih.gov/health/traditional-chinese-medicine-what-you-need-to-know

[TA-W009] Neil D. Jones; Carsten K. Gomard; Peter Sestoft (1993). *Partial Evaluation and Automatic Program Generation*. https://www.cs.utexas.edu/~novak/jonesgomardsestoft.pdf

[GL-W064] NHS England (2026). *NHS dentistry: care pathways guidance*. https://www.england.nhs.uk/long-read/nhs-dentistry-care-pathways-guidance/

[MIO-W001] Nicholas H. Nelsen; Yunan Yang (2025). *Operator learning meets inverse problems: A probabilistic perspective*. arXiv preprint, 27 August 2025. https://doi.org/10.48550/arXiv.2508.20207

[W089] Nicholas J. Higham (2020). *What Is a Condition Number?*. Author publication 19 March 2020. https://nhigham.com/2020/03/19/what-is-a-condition-number/

[W090] Nicholas J. Higham (2020). *What is Numerical Stability?*. Author publication 4 August 2020. https://nhigham.com/2020/08/04/what-is-numerical-stability/

[SG-W022] Nikita Smirnov; Semen Kurkin; Alexander E. Hramov (2025). *A Q-analysis package for higher-order interactions analysis in Python and its application in network physiology*. https://pmc.ncbi.nlm.nih.gov/articles/PMC12605130/

[GL-W059] NIOSH (2024). *Hierarchy of Controls*. https://www.cdc.gov/niosh/topics/hierarchy/default.html

[SG-W057] Noel León-Rovira; Humberto Aguayo (1998). *A new Model of the Conceptual Design Process using QFD/FA/TRIZ*. https://www.metodolog.ru/triz-journal/archives/1998/07/d/index.htm

[GL-W083] PARC/CAST Flight Deck Automation Working Group (2013). *Operational Use of Flight Path Management Systems: Final Report of the PARC/CAST Flight Deck Automation Working Group*. https://www.faa.gov/sites/faa.gov/files/aircraft/air_cert/design_approvals/human_factors/OUFPMS_Report.pdf

[GL-W042] Pascal Boyer; Pierre Liénard (2020). *Ingredients of 'rituals' and their cognitive underpinnings*. https://pmc.ncbi.nlm.nih.gov/articles/PMC7423267/

[SB-W001] Paul du Gay; Thomas Lopdrup-Hjorth (2024). *Organizing States: The continuing relevance of formal organization within political administration*. https://doi.org/10.1177/26317877241235944

[SG-W054] Paul E. Griffiths; Russell D. Gray (1994). *Developmental Systems and Evolutionary Explanation*. https://doi.org/10.2307/2940982

[GL-W027] Paul P. Maglio; David Kirsh (1996). *Epistemic Action Increases With Skill*. https://escholarship.org/uc/item/66t3v8mv

[W064] Paul R. Amyotte; Faisal I. Khan (2021). *The role of inherently safer design in process safety*. https://doi.org/10.1002/cjce.23987

[TA-W013] Paulo Tabuada (2007). *Event-Triggered Real-Time Scheduling of Stabilizing Control Tasks*. Author manuscript. Published article: IEEE Transactions on Automatic Control, 52(9), 1680–1685 (2007). https://doi.org/10.1109/TAC.2007.904277

[GL-W044] PCAOB (n.d.). *AU Section 319: Consideration of Internal Control in a Financial Statement Audit*. Archived consolidated standard incorporating SAS Nos. 55, 78 and 94 and later amendments; superseded for audits of fiscal years beginning on or after 15 December 2010. https://pcaobus.org/oversight/standards/archived-standards/details/AU319

[MIO-W002] Per Christian Hansen (n.d.). *Introduction to Inverse Problems, Chapter 4: Regularization Methods*. Teaching slides. Technical University of Denmark. https://people.compute.dtu.dk/pcha/DIP/chap4.pdf

[AFC-W010] Peter Alvaro; Joshua Rosen; Joseph M. Hellerstein (2015). *Lineage-driven Fault Injection*. https://doi.org/10.1145/2723372.2723711

[W057] Peter Bailis (2015). *Coordination Avoidance in Distributed Databases*. Doctoral dissertation. https://www2.eecs.berkeley.edu/Pubs/TechRpts/2015/EECS-2015-206.pdf

[W005] Peter Bailis; Alan Fekete; Michael J. Franklin; Ali Ghodsi; Joseph M. Hellerstein; Ion Stoica (2014). *Coordination Avoidance in Database Systems*. https://doi.org/10.14778/2735508.2735509

[SG-W001] Peter Checkland (1999). *Soft Systems Methodology: a 30-year retrospective (plenary-address summary)*. https://proceedings.systemdynamics.org/1999/PAPERS/KEYNOTE3.PDF

[W077] Peter Dittrich; Pietro Speroni di Fenizio (2007). *Chemical Organisation Theory*. Bulletin of Mathematical Biology, 69(4), 1199–1231. https://doi.org/10.1007/s11538-006-9130-8

[SG-W042] Peter Dittrich; Thomas Kron; Wolfgang Banzhaf (2003). *On the Scalability of Social Order: Modeling the Problem of Double and Multi Contingency Following Luhmann*. https://www.jasss.org/6/1/3.html

[GL-W001] Peter E. Keller; Giacomo Novembre; Janeen Loehr (2016). *Musical Ensemble Performance: Representing Self, Other, and Joint Action Outcomes*. In Sukhvinder S. Obhi and Emily S. Cross (eds.), Shared Representations: Sensorimotor Foundations of Social Life, pp. 280–310. Cambridge University Press. https://janeenloehr.com/publications/KellerNovembreLoehrInPress.pdf

[W040] Peter M. Todd; Gerd Gigerenzer (2007). *Environments That Make Us Smart*. https://doi.org/10.1111/j.1467-8721.2007.00497.x

[GL-W028] Peter Sterling (2012). *Allostasis: a model of predictive regulation*. https://www.lifeweavings.com/wp-content/uploads/2020/11/Allostasis-amodelofpredictiveregulation.12.pdf

[SB-W012] Philip Steadman (2020). *Biological Analogy and an Architectural Science*. https://www.buildingsandcities.org/insights/research-pathways/biological-analogy-architectural-science.html

[UX-W018] Philippe M. Cury; Christian Mullon; Serge M. Garcia; Lynne J. Shannon (2005). *Viability theory for an ecosystem approach to fisheries*. ICES Journal of Marine Science, 62(3), 577–584. https://doi.org/10.1016/j.icesjms.2004.10.007

[RGC-W013] Pier Luigi Luisi (2016). *What Is Life? A Systemic Approach*. https://doi.org/10.22323/1.246.0074

[WME-W008] Pier Luigi Luisi (2022). *Reflections on Autopoiesis, the Signature of Life*. https://constructivist.info/18/1/020.luisi

[W031] Pierre-Paul Grassé (1959). *La reconstruction du nid et les coordinations interindividuelles chez Bellicositermes natalensis et Cubitermes sp. La théorie de la stigmergie [Nest Reconstruction and Interindividual Coordination in Bellicositermes natalensis and Cubitermes sp.: The Theory of Stigmergy]*. https://doi.org/10.1007/BF02223791

[AFC-W004] Principles of Chaos Engineering contributors (2019). *Principles of Chaos Engineering*. https://principlesofchaos.org/

[SG-W024] Pyotr G. Shchedrovitsky (2019). *Организационно-деятельностная игра [The Organizational-Activity Game]*. https://shchedrovitskiy.com/organizacionno-dejatelnostnaya-igra/

[GL-W031] Rainer Diaz-Bone (2018). *Economics of Convention and its Perspective on Knowledge and Institutions*. https://link.springer.com/chapter/10.1007/978-3-319-75328-7_4

[SB-W007] Ralf Abraham; Jose Tribolet; Robert Winter (2013). *Transformation of multi-level systems – theoretical grounding and consequences for enterprise architecture management*. https://doi.org/10.1007/978-3-642-38117-1_6

[WME-W010] Ramon Ferrer i Cancho; Ricard V. Solé (2003). *Least effort and the origins of scaling in human language*. https://pdodds.w3.uvm.edu/research/papers/others/2003/ferrericancho2003a.pdf

[SG-W031] RAND Corporation (2021). *Robust Decision Making*. In Water Planning for the Uncertain Future: An Interactive Guide to the Use of Methods for Decisionmaking Under Deep Uncertainty (DMDU) for U.S. Bureau of Reclamation Water Resources Planning. RAND Corporation, TL-320-BOR. https://doi.org/10.7249/TL320

[UX-W010] Raul Espejo (1983). *Management and information: the complementarity control-autonomy*. Cybernetics and Systems, 14(1), 85–102. https://doi.org/10.1080/01969728308927719

[UX-W007] Raul Espejo (2022). *Cybersyn, big data, variety engineering and governance*. AI & Society, 37, 1163–1177. https://doi.org/10.1007/s00146-021-01348-0

[TA-W032] Ravi Sandhu; David Ferraiolo; Richard Kuhn (2000). *The NIST Model for Role-Based Access Control: Towards A Unified Standard*. https://tsapps.nist.gov/publication/get_pdf.cfm?pub_id=916402

[TA-W023] Richard E. Michod (2003). *Cooperation and Conflict in the Evolution of Complexity*. https://cdn.aaai.org/Symposia/Spring/2003/SS-03-02/SS03-02-002.pdf

[TA-W022] Richard E. Michod; Denis Roze (2001). *Cooperation and conflict in the evolution of multicellularity*. https://www.nature.com/articles/6888080

[SB-W011] Richard P. Gabriel (2023). *Notes on "Notes on the Synthesis of Form": Dawning Insights in Early Christopher Alexander*. https://doi.org/10.22152/programming-journal.org/2023/7/8

[RGC-W043] Robert E. Ulanowicz (2020). *Quantifying sustainable balance in ecosystem configurations*. https://doi.org/10.1016/j.crsust.2019.09.001

[WME-W025] Robert E. Ulanowicz; Sally J. Goerner; Bernard Lietaer; Rocio Gomez (2009). *Quantifying sustainability: Resilience, efficiency and the return of information theory*. https://doi.org/10.1016/j.ecocom.2008.10.005

[W102] Robert Rosen (1958). *A relational theory of biological systems*. https://doi.org/10.1007/BF02478302

[RGC-W023] Robert Rosen (1985). *Anticipatory Systems: Philosophical, Mathematical, and Methodological Foundations*. https://ahlouie.com/wp-content/uploads/2019/08/louie-relational-biology-2017.pdf

[RGC-W022] Robert Rosen (1991). *Life Itself: A Comprehensive Inquiry into the Nature, Origin, and Fabrication of Life*. https://www.yoono.org/download/lifei.pdf

[MIL-W017] Robert S. Tripp; Mahyar A. Amouzegar; Ronald G. McGarvey; Rick Bereit; David W. George; Joan Cornuet (2006). *Sense and Respond Logistics: Integrating Prediction, Responsiveness, and Control Capabilities*. RANDMG488AF. https://www.rand.org/pubs/monographs/MG488.html

[W050] Roger C. Conant; W. Ross Ashby (1970). *Every good regulator of a system must be a model of that system*. https://doi.org/10.1080/00207727008920220

[W058] Ronald A. Fisher (1922). *On the Mathematical Foundations of Theoretical Statistics*. https://doi.org/10.1098/rsta.1922.0009

[TA-W040] Roy Radner (1993). *The Organization of Decentralized Information Processing*. https://pages.stern.nyu.edu/~rradner/publishedpapers/83OrganizationDecentralizedInfo.pdf

[SG-W040] Russell L. Ackoff (2001). *A Brief Guide to Interactive Planning and Idealized Design*. https://www.ida.liu.se/~steho87/und/htdd01/AckoffGuidetoIdealizedRedesign.pdf

[SG-W035] Ryo Sato (1992). *A State Space Representation of Business Transaction System*. https://www.kurims.kyoto-u.ac.jp/~kyodo/kokyuroku/contents/pdf/0809-14.pdf

[RGC-W042] Sally J. Goerner; Bernard Lietaer; Robert E. Ulanowicz (2009). *Quantifying economic sustainability: Implications for free-enterprise theory, policy and practice*. https://doi.org/10.1016/j.ecolecon.2009.07.018

[GL-W035] Sampsa Hyysalo (2013). *Review of The Dynamics of Social Practice: Everyday Life and How It Changes*. https://www.ntnu.no/ojs/index.php/njsts/article/view/2125/pdf

[GL-W045] Scott Rose; Oliver Borchert; Stuart Mitchell; Sean Connelly (2020). *Zero Trust Architecture*. https://www.nist.gov/publications/zero-trust-architecture

[SG-W014] Sergio Barile; Jaqueline Pels; Francesco Polese; Marialuisa Saviano (2012). *An Introduction to the Viable Systems Approach and Its Contribution to Marketing*. https://www.econstor.eu/bitstream/10419/66007/1/727048856.pdf

[W038] 新郷重夫 (Shigeo Shingo) (1985). *源流検査とポカヨケ・システム : 不良=0 への挑戦 0QC 方式への展開 [Source Inspection and the Poka-Yoke System: The Challenge of Zero Defects and Development Toward the Zero Quality Control Method]*. 日本能率協会. ISBN 4-8207-0226-2. https://ndlsearch.ndl.go.jp/books/R100000002-I000001736438

[SG-W020] Simon Caulkin (2018). *Failure demand: what's the big secret?*. https://beyondcommandandcontrol.com/2018/02/01/failure-demand-whats-the-big-secret/amp/

[TA-W034] Simon Kirby; Monica Tamariz; Hannah Cornish; Kenny Smith (2015). *Compression and communication in the cultural evolution of linguistic structure*. https://dspace.stir.ac.uk/handle/1893/22564

[TA-W036] Slava Kalyuga; Paul Ayres; Paul Chandler; John Sweller (2003). *The Expertise Reversal Effect*. https://www.davidlewisphd.com/courses/EDD8121/readings/2003-Kalyuga_et_al.pdf

[W101] Stafford Beer (1979). *The Heart of Enterprise*. Wiley. ISBN 0-471-27599-9. https://www.scribd.com/document/1019215454/The-Heart-of-Enterprise-Beer-Stafford

[RGC-W012] Stafford Beer (1989). *The Viable System Model: Its Provenance, Development, Methodology and Pathology*. https://www.kybernetik.ch/dwn/Viable_System_Model.pdf

[GL-W050] Daniel Little (2007). *Philosophy of History*. Stanford Encyclopedia of Philosophy. First published 2007; substantively revised 2025. https://plato.stanford.edu/entries/history/

[GL-W040] Natalie Stoljar (2013). *Feminist Perspectives on Autonomy*. Stanford Encyclopedia of Philosophy. First published 2013; substantively revised 2024. https://plato.stanford.edu/entries/feminism-autonomy/

[W068] Stefano Brusoni; Andrea Prencipe; Keith Pavitt (2001). *Knowledge Specialization, Organizational Coupling, and the Boundaries of the Firm: Why Do Firms Know More Than They Make?*. https://doi.org/10.2307/3094825

[MIL-W023] Stephan Leistenschneider (2002). *Auftragstaktik im preußisch-deutschen Heer 1871 bis 1914 [Mission Command in the Prussian-German Army, 1871–1914]*. https://researchcentre.army.gov.au/library/australian-army-journal-aaj/auftragstaktik-mission-command

[GL-W052] Stephen Billett (2001). *Learning through work: workplace affordances and individual engagement*. https://doi.org/10.1108/EUM0000000005548

[SG-W053] Stephen Shervais; Patricia L. Kramer; Shawn K. Westaway; Nancy J. Cox; Martin Zwick (2010). *Reconstructability Analysis as a Tool for Identifying Gene-Gene Interactions in Studies of Human Diseases*.

Statistical Applications in Genetics and Molecular Biology, 9(1), article 18. https://doi.org/10.2202/1544-6115.1516

[SG-W015] Steven Alter (2019). *Applying Socio-technical Thinking in the Competitive, Agile, Lean, Data-Driven World of Knowledge Work and Smart, Service-Oriented, Customer-Centric Value Creation Ecosystems*. https://csimq-journals.rtu.lv/csimq/article/download/csimq.2019-18.01/112/118

[MIO-W003] Steven Diamond; Stephen Boyd (2017). *Stochastic Matrix-Free Equilibration*. Journal of Optimization Theory and Applications, 172(2), 436–454. https://doi.org/10.1007/s10957-016-0990-2

[SG-W059] Stewart Brand (2018). *Pace Layering: How Complex Systems Learn and Keep Learning*. https://longnow.org/ideas/pace-layers/

[W079] Stijn Bruers; Christian Maes; Karel Netočný (2007). *On the validity of entropy production principles for linear electrical circuits*. arXiv:cond-mat/0701035v2. https://arxiv.org/pdf/cond-mat/0701035

[GL-W068] Stockholm Resilience Centre (n.d.). *Diversity is essential for resilience to thrive*. https://www.stockholmresilience.org/research/synthesis-reports/resilience-science-must-knows/the-must-knows/5.-diversity-is-essential-for-resilience-to-thrive.html

[GL-W037] Stockholm University (n.d.). *Asceticism and Ritual Self-Practices*. https://www.su.se/english/research/research-catalogue/research-subjects/6/asceticism-and-ritual-self-practices

[MIL-W027] Sunzi (traditional attribution); Lionel Giles (translator and commentator) (n.d.; translation and commentary 1910). *孫子兵法 [The Art of War]*. Ancient Chinese text, traditionally attributed to Sunzi. Lionel Giles, translation and commentary, 1910. https://www.gutenberg.org/cache/epub/132/pg132-images.html

[TA-W024] Susan Leigh Star; Karen Ruhleder (1996). *Steps Toward an Ecology of Infrastructure: Design and Access for Large Information Spaces*. https://doi.org/10.1287/isre.7.1.111

[TA-W025] Susan Leigh Star; Martha Lampland (2009). *Reckoning with Standards*. https://sociology.ucsd.edu/_files/people/lampland/Star%20and%20Lampland%20Reckoning%20Introduction%20to%20Standards%20and%20their%20Stories.pdf

[UX-W029] Sénat français (1998). *Des pyramides du pouvoir aux réseaux de savoirs [From Pyramids of Power to Networks of Knowledge]. Tome 1*. Tome 1. https://www.senat.fr/rap/r97-331-t1/r97-331-t1104.html

[W053] Tad McGeer (1990). *Passive Dynamic Walking*. https://doi.org/10.1177/027836499000900206

[W051] Tadahiro Taniguchi; Yasushi Hirai; Masahiro Suzuki; Shingo Murata; Takato Horii; Kazutoshi Tanaka (2025). *System 0/1/2/3 Quad-process theory for multi-timescale embodied collective cognitive systems*. Preprint, 2025. arXiv:2503.06138v1. https://arxiv.org/html/2503.06138v1

[W018] Tadeusz Kotarbiński (1955). *Traktat o dobrej robocie [Praxiology: An Introduction to the Science of Efficient Action]*. Zakład Narodowy im. Ossolińskich. https://bc.gbpizs.gov.pl/dlibra/publication/1181/edition/1128/content

[W019] Tadeusz Kotarbiński (1956). *Sprawność i błąd. Z myślą o dobrej robocie nauczyciela [Efficiency and Error: With Good Teaching Work in Mind]*. First edition 1956; expanded editions 1957, 1960 and 1966. The chapter, Zagadnienia prakseologii, czyli nauki o działaniu skutecznym [Problems of Praxeology, or the Science of Effective Action], identifies a lecture delivered on 22 May 1957. https://www.scribd.com/document/1007042028/Tadeusz-Kotarbi%C5%84ski-Sprawno%C5%9B%C4%87-i-B%C5%82%C4%85d

[W012] Tau-Mu Yi; Yun Huang; Melvin I. Simon; John Doyle (2000). *Robust perfect adaptation in bacterial chemotaxis through integral feedback control*. https://doi.org/10.1073/pnas.97.9.4649

[AFC-W006] Terje Aven (2015). *The concept of antifragility and its implications for the practice of risk analysis*. Risk Analysis, 35(3), 476–483. https://doi.org/10.1111/risa.12279

[SG-W028] The Bowen Center for the Study of the Family (n.d.). *Introduction to the Eight Concepts; Triangles; Emotional Cutoff*. https://www.thebowencenter.org/triangles

[TA-W003] Theodore Groves; Roy Radner (1972). *Allocation of Resources in a Team*. https://doi.org/10.1016/0022-0531(72)90131-7

[W070] Thomas C. Schelling (1971). *Dynamic Models of Segregation*. https://doi.org/10.1080/0022250X.1971.9989794

[TA-W021] Thomas C. Scott-Phillips; Kevin N. Laland; David M. Shuker; Thomas E. Dickins; Stuart A. West (2014). *The Niche Construction Perspective: A Critical Appraisal*. https://pmc.ncbi.nlm.nih.gov/articles/PMC4261998/

[GL-W058] Thomas Rotter; Leigh D. Kinsman; Agnès Alsius; Shannon D. Scott; Adegboyega Lawal; Ulrich Ronellenfitsch; Christopher Plishka; Gary Groot; Phil Woods; Chloe Coulson; Leigh Anne Bakel; Kim Sears; Amanda Ross-White; Andreas Machotta; Timothy J. Schultz (2025). *Clinical pathways for secondary care and the effects on professional practice, patient outcomes, length of stay and hospital costs*. https://www.cochrane.org/evidence/CD006632_effects-clinical-pathways-hospitals-patient-outcomes-length-hospital-stay-hospital-costs-and-charges

[WME-W011] Thomas W. Malone; Kevin Crowston (1994). *The Interdisciplinary Study of Coordination*. https://crowston.syr.edu/sites/default/files/acmcs94.pdf

[TA-W006] Thomas W. Merrill (2012). *Property as Modularity*. 125 Harvard Law Review Forum151–163. https://harvardlawreview.org/wp-content/uploads/2012/05/forvol125_merrill.pdf

[GL-W024] Tilman Börgers (2023). *Notes on Blackwell Dominance*. https://websites.umich.edu/~tborgers/BlackwellDominance.pdf

[UX-W005] Todd M. La Porte (2006). *Organizational Strategies for Complex System Resilience, Reliability, and Adaptation*. https://polisci.berkeley.edu/sites/default/files/people/u3825/0521857961c10_p135-154.pdf

[GL-W074] Tom Van Gerven; Andrzej Stankiewicz (2009). *Structure, Energy, Synergy, Time—The Fundamentals of Process Intensification*. https://doi.org/10.1021/ie801501y

[GL-W080] Toyota Motor Corporation (n.d.). *Toyota Production System*. https://global.toyota/en/company/vision-and-philosophy/production-system/

[W039] Trevor Kletz (1978). *What you don't have can't leak*. Chemistry and Industry, 6 May 1978, pp. 287–292. https://www.aiche.org/sites/default/files/cep/20120140-1.pdf

[SG-W044] Tyler Volk; Jeffrey W. Bloom (2007). *The Use of Metapatterns for Research into Complex Systems of Teaching, Learning, and Schooling—Part I: Metapatterns in Nature and Culture*. https://doi.org/10.29173/cmplct8759

[GL-W072] UBC Library Indigenous Librarianship Research Guide (n.d.). *Brian Deer Classification System*. https://guides.library.ubc.ca/Indiglibrarianship/briandeer

[TA-W033] United States General Accounting Office (2001). *Internal Control Management and Evaluation Tool: Exposure Draft (GAO-01-131G)*. https://www.govinfo.gov/content/pkg/GAOREPORTS-GAO-01-131G/html/GAOREPORTS-GAO-01-131G.htm

[GL-W039] Taylor V. Stein (2010). *Planning for the Many Benefits of Nature-Based Recreation*. UF/IFAS Extension, FOR 233. Revised June 2019. https://ask.ifas.ufl.edu/publication/FR295

[GL-W090] US Army Corps of Engineers (n.d.). *Flood and Coastal Storm Risk Management*. https://www.usace.army.mil/Missions/Civil-Works/Flood-and-Coastal-Storm-Risk-Management/

[GL-W078] K. L. Ford (2003). *Passive Treatment Systems for Acid Mine Drainage*. Technical Note 409. BLM/ST/ST-02/001+3596. Bureau of Land Management. https://www.blm.gov/sites/default/files/documents/files/Library_BLMTechnicalNote409.pdf

[GL-W076] US Environmental Protection Agency (2000). *Constructed Wetlands Treatment of Municipal Wastewaters*. Manual. EPA/625/R-99/010. National Risk Management Research Laboratory, U.S. Environmental Protection Agency. https://nepis.epa.gov/Exe/ZyPURL.cgi?Dockey=30004TBD.TXT

[GL-W075] US Food and Drug Administration (2016). *Applying Human Factors and Usability Engineering to Medical Devices*. Guidance for Industry and Food and Drug Administration Staff. U.S. Food and Drug Administration. https://www.fda.gov/files/medical%20devices/published/Applying-Human-Factors-and-Usability-Engineering-to-Medical-Devices---Guidance-for-Industry-and-Food-and-Drug-Administration-Staff.pdf

[GL-W054] USDA Forest Service Northern Research Station (n.d.). *Adaptive Silviculture for Climate Change (ASCC)*. https://research.fs.usda.gov/nrs/partnerships/ascc

[GL-W084] USDA Natural Resources Conservation Service (2019). *Controlled traffic farming to reduce compaction (E334A)*. https://www.nrcs.usda.gov/sites/default/files/2022-11/E334A%20July%202019.pdf

[GL-W066] USGS Center of Excellence for Geospatial Information Science (n.d.). *Generalization*. https://www.usgs.gov/centers/cegis/science/science-topics/generalization

[TA-W018] Vahe Galstyan; Kabir Husain; Fangzhou Xiao; Arvind Murugan; Rob Phillips (2020). *Proofreading through spatial gradients*. https://arxiv.org/html/2005.11615v1

[SB-W020] Victor H. Mair (1989). *Short notice of Grammatical Man*. https://sino-platonic.org/complete/spp014_chinese_book_reviews.pdf

[UX-W025] Vincent Bontems (2016). *Sur la classification des objets techniques selon Simondon [On the Classification of Technical Objects According to Simondon]*. Artefact, 3, 183–198 (issue dated 2015; published 2016). https://doi.org/10.4000/artefact.8270

[W008] Vincent C. Müller; Matej Hoffmann (2017). *What Is Morphological Computation? On How the Body Contributes to Cognition and Control*. https://doi.org/10.1162/ARTL_a_00219

[AFC-W009] Vincenzo De Florio (2014). *Antifragility = Elasticity + Resilience + Machine Learning: Models and Algorithms for Open System Fidelity*. https://doi.org/10.1016/j.procs.2014.05.499

[SG-W019] Vorne / Lean Production (n.d.). *Theory of Constraints (TOC)*. https://www.leanproduction.com/theory-of-constraints/

[W001] W. Ross Ashby (1956). *An Introduction to Cybernetics*. 1956 Chapman and Hall. https://www.ashby.info/Ashby-Introduction-to-Cybernetics.pdf

[SB-W006] W. Findeisen; F. N. Bailey; M. Brdys; K. Malinowski; P. Tatjewski; A. Wozniak (1980). *Control and Coordination in Hierarchical Systems*. https://pure.iiasa.ac.at/id/eprint/1227/1/XB-80-109.pdf

[W062] Walter B. Cannon (1929). *Organization for Physiological Homeostasis*. https://doi.org/10.1152/physrev.1929.9.3.399

[TA-W039] Wang Mengzhi; Zhao Chengcheng; Yang Zeyu; Che Xin (2026). *Self-triggering predictive control method for continuous-time networked control system (CN116627039B)*. https://www.goveda.com/patent/CN-116627039-B

[SG-W002] Werner Ulrich (2005). *A Mini-Primer of Critical Systems Heuristics*. First published 2005; revised 2023. https://www.wulrich.com/csh.html

[SG-W027] William T. Powers (2009). *PCT in 11 Steps*. https://www.iapct.org/pct/pct-in-11-steps/

[GL-W041] World Health Organization (2024). *Assistive technology*. https://www.who.int/news-room/fact-sheets/detail/assistive-technology

[GL-W057] World Organisation for Animal Health (OIE/WOAH) (2018). *Animal welfare and dairy cattle production systems, Terrestrial Animal Health Code chapter 7.11*. https://www.woah.org/fileadmin/Home/eng/Health_standards/tahc/2018/en_chapitre_aw_dairy_cattle.htm

[GL-W056] World Organisation for Animal Health (OIE/WOAH) (2018). *Application of compartmentalisation, Terrestrial Animal Health Code chapter 4.4*. https://www.woah.org/fileadmin/Home/eng/Health_standards/tahc/2018/en_chapitre_application_compartment.htm

[UX-W024] Yohann Guffroy; Vincent Bontems (2018). *La mécanologie : une lignée technologique francophone ? [Mechanology: A Francophone Technological Lineage?]*. Artefact, 8, 255–280. https://doi.org/10.4000/artefact.2218

[RGC-W040] Yu Jingyuan / 于景元 (2001). *钱学森的现代科学技术体系与综合集成方法论 [Qian Xuesen's Contemporary System of Science and Technology and Meta-synthesis Methodology]*. https://devp-service.oss-cn-beijing.aliyuncs.com/c1bef5bc349340ef8ae8faa19e39d47e/file_1562055285973.pdf

[W099] Yue Xie; Yi Zhang; Tao Sun; Fumiya Iida (2026). *Energy as a Common Currency for Understanding Embodied Evolution*. https://doi.org/10.34133/cbsystems.0680

[MIL-W025] Zentrum Innere Führung / Bundeswehr (2021). *Agile Fehlerkultur – Realität (auch) in der Bundeswehr? [An Agile Error Culture: A Reality in the Bundeswehr Too?]*. https://www.bundeswehr.de/de/organisation/zentrum-innere-fuehrung/unsere-angebote/if-zeitschrift-fuer-innere-fuehrung/agile-fehlerkultur-5100954

[W067] Ángel Gil; José Aguilar; Rafael Rivas; Eladio Dapena (2019). *Coordinación emergente en sistemas multirobots [Emergent Coordination in Multirobot Systems]*. https://doi.org/10.18273/revuin.v18n3-2019008

[TKT-W002] А. Богданов (1927). Всеобщая организационная наука (Тектология). Часть *II [Universal Organizational Science (Tektology), Part II]*. 3rd edition, expanded and revised. Книга, Ленинград–Москва. https://bogdanovlibrary.org/wp-content/uploads/2019/01/%D0%91%D0%BE%D0%B3%D0%B4%D0%B0%D0%BD%D0%BE%D0%B2-%D0%90.%D0%90.-%D0%92%D1%81%D0%B5%D0%BE%D0%B1%D1%89%D0%B0%D1%8F-%D0%BE%D1%80%D0%B3%D0%B0%D0%BD%D0%B8%D0%B7%D0%B0%D1%86%D0%B8%D0%BE%D0%BD.pdf

[W017] Александр Александрович Богданов (1912). *Всеобщая организационная наука (Тектология). Часть I [Universal Organizational Science (Tektology), Part I]*. Revised third edition, 1925. https://traumlibrary.ru/book/bogdanov-tektologia-1/bogdanov-tektologia-1.html

[TKT-W001] Александр Александрович Богданов (1912–1917). *Тектология: Всеобщая организационная наука [Tektology: Universal Organizational Science]*. Expanded three-part edition, 1922; revised parts, 1925–1929. https://gtmarket.ru/library/basis/5909

[MIL-W028] В. Е. Севрук; А. В. Вдовин; В. В. Карпович (2021). *Трансформация управления — централизация или децентрализация [Transformation of Management: Centralization or Decentralization]*. https://library.rucont.ru/efd/726278

[W022] Г. С. Альтшуллер; Р. Б. Шапиро (1956). *О психологии изобретательского творчества [On the Psychology of Inventive Creativity]*. https://www.altshuller.ru/triz/triz0.asp

[WME-W019] Георгий Петрович Щедровицкий (Georgy Petrovich Shchedrovitsky) (1987). *Схема* мыследеятельности — системно-структурное строение, смысл и содержание *[The Schema of Thought-Activity: Systemic-Structural Organization, Meaning and Content]*. Системные исследования. Методологические проблемы. Ежегодник 1986 [Systems Research: Methodological Problems. Yearbook 1986]. Moscow: Наука, 1987. Reprinted in Избранные труды [Selected Works], Moscow: Издательство Школы культурной политики, 1995, pp. 281–298. https://priss-laboratory.net.ru/library/library_GP_shema-MD-stroenie-i-soderjanie.html

[SG-W007] 上海系统科学研究院 (2021). *学术研究中的系统科学法 [The Systems Science Method in Academic Research]*. https://sass.usst.edu.cn/2021/1102/c12541a258349/page.htm

[AFC-W014] 中国信息通信研究院云计算与大数据研究所 (2021). *混沌工程实践指南（2021 年）[Chaos Engineering Practice Guide (2021)]*. https://www.caict.ac.cn/kxyj/qwfb/ztbg/202112/P020211223588643401747.pdf

[MIL-W026] 张添溯 (Zhang Tiansu); 陈小菁 (Chen Xiaojing) (2025). *我军加速构建跨军种一体化联合保障模式 [Our Military Accelerates the Development of an Integrated Joint Support Model Across Services]*. 解放军报 [PLA Daily], 15 December 2025. https://www.81.mil.cn/jwtt/16427017.html

[TA-W030] 中国合格评定国家认可委员会 (CNAS) (2021). *测量结果的计量溯源性要求 [Requirements for the Metrological Traceability of Measurement Results] (CNAS-CL01-G002:2021)*. https://www.cnas.org.cn/cms_files/filemanager/328304128/attach/202412/EE1ABB819EAE436B209D6003663618B7.pdf

[W063] 中條武志 (Takeshi Nakajo) (2004). *組織における不適切な人間行動とそのリスク評価 [Inappropriate Human Behavior in Organizations and Its Risk Assessment]*. 日本信頼性学会誌 信頼性, 26(7), 627–635. https://doi.org/10.11348/reajshinrai.26.7_627

[W076] 中條武志; 久米均 (1984). *作業のフールプルーフ化に関する研究 —フールプルーフ化の原理— [A Study of Foolproofing Work: Principles of Foolproofing]*. https://doi.org/10.20684/quality.14.2_128

[W036] 于景元 (1992). *钱学森关于开放的复杂巨系统的研究 [Qian Xuesen's Research on Open Complex Giant Systems]*. https://doi.org/10.12011/1000-6788(1992)5-106514

[W011] 刘华平; 郭迪; 孙富春; 张新钰 (2023). *基于形态的具身智能研究: 历史回顾与前沿进展 [Morphology-Based Embodied Intelligence Research: Historical Review and Frontier Advances]*. https://doi.org/10.16383/j.aas.c220564

[SB-W016] 刘炳琪; 胡剑波; 刘畅; 李俊 (2020). *飞机差动刹车纠偏过程的 STAMP/STPA 安全性分析 [STAMP/STPA Safety Analysis of Aircraft Differential-Braking Direction Correction]*. https://hit.alljournals.cn/html/hitxb_cn/2020/4/20200410.html

[AFC-W021] 华为云计算技术有限公司 SRE 部 (2023). *确定性运维白皮书：稳定可靠篇 [Deterministic Operations and Maintenance White Paper: Stability and Reliability]*. https://res-static.hc-cdn.cn/cloudbu-site/china/zh-cn/SRE/1694663770441427235.pdf

[SG-W049] 吴学谋 / Xuemou Wu (n.d.). *泛系友人书（1）[Letter to Pansystems Friends, Part 1]*. https://www.survivor99.com/entropy/paper/p40.htm

[SB-W003] 周雪光; 赵伟 (2009). *英文文献中的中国组织现象研究 [Research on Chinese Organizational Phenomena in the English-Language Literature]*. https://stanford.edu/~xgzhou/review09_CH.pdf

[UX-W037] 日本プラントメンテナンス協会 [Japan Institute of Plant Maintenance] (2025). *からくり改善のメカニズム ～動力源とからくり機構～ [The Mechanisms of Karakuri Kaizen: Power Sources and Karakuri Mechanisms]*. 工場管理, 71(3), p. 16. https://pubdata.nikkan.co.jp/uploads/magazine_serial/202503/zkok/4.pdf

[UX-W036] 日本プラントメンテナンス協会 [Japan Institute of Plant Maintenance] (n.d.). *からくり改善とは [What is karakuri kaizen?]*. https://jipm-event.com/karakuri/about

[W010] 池俣吉人; 佐野明人 (2017). *受動歩行の力学的原理と応用 [Mechanical Principles and Applications of Passive Walking]*. https://doi.org/10.14243/jsaem.25.2

[AFC-W020] 火山引擎开发者社区 (n.d.). *火山引擎混沌工程之云原生场景实现 [Implementing Volcano Engine Chaos Engineering in Cloud-Native Scenarios]*. https://developer.volcengine.com/articles/7257050491912192019/

[GL-W008] 王威威 (2017). *“礼义化性”与“习俗化性”——论荀子思想中礼与俗的作用 [“Transforming Nature Through Ritual and Rightness” and “Transforming Nature Through Custom”: On the Roles of Ritual and Custom in Xunzi’s Thought]*. https://zxyj.cbpt.cnki.net/portal/journal/portal/client/paper/a8c46f882d4349587bbf95ab16ad3818

[TA-W027] 石庆馨; 张侃 (2004). *生态学界面设计研究综述 [A Review of Research on Ecological Interface Design]*. https://ir.psych.ac.cn/bitstream/311026/2244/3/%E7%94%9F%E6%80%81%E5%AD%A6%E7%95%8C%E9%9D%A2%E8%AE%BE%E8%AE%A1%E7%A0%94%E7%A9%B6%E7%BB%BC%E8%BF%B0.pdf

[W100] 科学技術振興機構 研究開発戦略センター (JST CRDS) (2026). *自律分散知能システム [Autonomous Distributed Intelligent Systems]*. JST CRDS report CRDS-FR-S203-202602. https://www.jst.go.jp/crds/pdf/CRDS-FR-S/CRDS-FR-S203-202602.pdf

[WME-W028] 科学时报社 (2010). *科学技术体系学研究的开创者 [A Pioneer in the Study of the System of Science and Technology]*. https://www.cas.cn/zt/rwzt/qxsssyzn/jnwz/201010/t20101031_3000052.html

[SB-W013] 赵博韬; 江泓 (2024). *走向都市认同的空间实践——亚历山大城市设计理论的价值、局限与启示 [Spatial Practices Toward Urban Identity: The Value, Limitations and Implications of Alexander’s Urban Design Theory]*. https://www.upi-planning.org.cn/Files/hjcsgh/MagazineHtml/202405/gwcg202405014.html

[WME-W027] 钱学森 (Qian Xuesen) (1986). *我对系统学认识的历程 [The Course of My Understanding of Systematology]*. Speech delivered 7 January 1986. https://sass.usst.edu.cn/2021/1213/c12540a261884/page.htm

[W035] 钱学森; 于景元; 戴汝为 (1990). *一个科学新领域 开放的复杂巨系统及其方法论 [A New Scientific Field: Open Complex Giant Systems and Their Methodology]*. https://sysengi.cjoe.ac.cn/CN/10.12011/1000-6788%281992%295-106514

[W009] 陈炜; 余跃庆; 张绪平; 苏丽颖 (2007). *欠驱动柔性机器人的振动可控性分析 [Analysis of Vibration Controllability in Underactuated Flexible Robots]*. https://doi.org/10.1360/aas-007-0391

[WME-W026] 顾基发 (Gu Jifa); 唐锡晋 (Tang Xijin) (2002). *物理—事理—人理系统方法论：一种东方的系统思考 [Wuli–Shili–Renli Systems Methodology: An Eastern Approach to Systems Thinking]*. https://www.ecsponline.com/yz/B37CE08123E4D4FBC887044740B535018000.pdf

[W078] 顾基发; 王浣尘; 唐锡晋 等 (2007). *综合集成方法体系与系统学研究 [Research on the Meta-synthesis Methodological System and Systematology]*. 科学出版社. ISBN 978-7-03-018469-6. https://www.ecsponline.com/yz/BFFEAD9A1C58E4D4580F2EC6EAE453EF2000.pdf

[W066] 문재성 (Jae-Sung Moon) (2021). *독립적 기능 제어를 이용한 몸통이 부착된 이족 로봇의 자연스럽고 유연한 평지 보행 [Natural and Flexible Level-Ground Walking of a Torso-Equipped Biped Robot Using Independent Function Control]*. https://doi.org/10.5302/J.ICROS.2021.21.0079

[GE-INTRO] Brad Pierce (2026). *What Is Gambits Engineering? Five Questions That Define This Emerging Discipline*. Zenodo. https://doi.org/10.5281/zenodo.23741487

[BC-W001] Hiroaki Kitano (2002). *Systems Biology: A Brief Overview*. Science, 295(5560), 1662–1664. https://doi.org/10.1126/science.1069492

[BC-W002] Uri Alon (2007). *Network motifs: theory and experimental approaches*. Nature Reviews Genetics, 8, 450–461. https://doi.org/10.1038/nrg2102

[BC-W003] Timothy S. Gardner; Charles R. Cantor; James J. Collins (2000). *Construction of a genetic toggle switch in Escherichia coli*. Nature, 403, 339–342. https://doi.org/10.1038/35002131

[BC-W004] Michael B. Elowitz; Stanislas Leibler (2000). *A synthetic oscillatory network of transcriptional regulators*. Nature, 403, 335–338. https://doi.org/10.1038/35002125

[BC-W005] Jeffrey D. Orth; Ines Thiele; Bernhard Ø. Palsson (2010). *What is flux balance analysis?*. Nature Biotechnology, 28, 245–248. https://doi.org/10.1038/nbt.1614

[BC-W006] Francesca Ceroni; Rhys Algar; Guy-Bart Stan; Tom Ellis (2015). *Quantifying cellular capacity identifies gene expression designs with reduced burden*. Nature Methods, 12, 415–418. https://doi.org/10.1038/nmeth.3339

[BC-W007] Jonathan R. Karr; Jayodita C. Sanghvi; Derek N. Macklin; Miriam V. Gutschow; Jared M. Jacobs; Benjamin Bolival Jr.; Nacyra Assad-Garcia; John I. Glass; Markus W. Covert (2012). *A Whole-Cell Computational Model Predicts Phenotype from Genotype*. Cell, 150, 389–401. https://doi.org/10.1016/j.cell.2012.05.044

[BC-W008] Krinos Li; Xianglu Xiao; Shenglong Deng; Lucas He; Zijun Zhong; Yuanjie Zou; Zhonghao Zhan; Zheng Hui; Weiye Bao; Guang Yang (2025). *Large Language Models Meet Virtual Cell: A Survey*. arXiv:2510.07706. https://doi.org/10.48550/arXiv.2510.07706

[BC-W009] Zihao Li; Zhichen Zeng; Xiao Lin; Feihao Fang; Yanru Qu; Zhe Xu; Zhining Liu; Xuying Ning; Tianxin Wei; Ge Liu; Hanghang Tong; Jingrui He (2025). *Flow Matching Meets Biology and Life Science: A Survey*. arXiv:2507.17731. Revised 2026. https://doi.org/10.48550/arXiv.2507.17731

[BC-W010] Sima Soltani; Mehrdad Jalali; Yahya Forghani; Reza Sheibani (2026). *Beyond Network Topology: Biological Evidence Integration and Reproducible Benchmarking for Protein Complex Detection*. arXiv:2606.03178. https://doi.org/10.48550/arXiv.2606.03178

[BC-W011] Pedro Zuidberg Dos Martires; Vincent Derkinderen; Luc De Raedt; Marcus Krantz (2024). *Automated Reasoning in Systems Biology: a Necessity for Precision Medicine*. arXiv:2410.13487. https://doi.org/10.48550/arXiv.2410.13487

[OI-W001] Elizabeth Wolfe Morrison; Frances J. Milliken (2000). *Organizational Silence: A Barrier to Change and Development in a Pluralistic World*. Academy of Management Review, 25(4), 706–725. https://doi.org/10.5465/amr.2000.3707697

[OI-W003] Chris Argyris (1976). *Single-Loop and Double-Loop Models in Research on Decision Making*. Administrative Science Quarterly, 21(3), 363–375. https://doi.org/10.2307/2391848

[OI-W004] Amy Edmondson (1999). *Psychological Safety and Learning Behavior in Work Teams*. Administrative Science Quarterly, 44(2), 350–383. https://doi.org/10.2307/2666999

[OI-W005] Barry M. Staw (1976). *Knee-Deep in the Big Muddy: A Study of Escalating Commitment to a Chosen Course of Action*. Organizational Behavior and Human Performance, 16(1), 27–44. https://doi.org/10.1016/0030-5073(76)90005-2

[OI-W007] Albert O. Hirschman (1970). *Exit, Voice, and Loyalty: Responses to Decline in Firms, Organizations, and States*. Harvard University Press.

[OI-W008] Charlan Jeanne Nemeth; Marie Personnaz; Bernard Personnaz; Jack A. Goncalo (2003). *The Liberating Role of Conflict in Group Creativity: A Cross Cultural Study*. Institute of Industrial Relations, University of California, Berkeley. Working Paper 90-03. https://irle.berkeley.edu/wp-content/uploads/2003/04/The-liberating-role-of-conflict-in-group-creativity.pdf

[OI-W009] Paul C. Nutt (1999). *Surprising but True: Half the Decisions in Organizations Fail*. Academy of Management Executive, 13(4), 75–90. https://doi.org/10.5465/ame.1999.2570556

[OI-W010] James G. March (1991). *Exploration and Exploitation in Organizational Learning*. Organization Science, 2(1), 71–87. https://doi.org/10.1287/orsc.2.1.71

[OI-W012] Michael Diehl; Wolfgang Stroebe (1987). *Productivity Loss in Brainstorming Groups: Toward the Solution of a Riddle*. Journal of Personality and Social Psychology, 53(3), 497–509. https://doi.org/10.1037/0022-3514.53.3.497

[OI-W013] Steven P. Dow; Alana Glassco; Jonathan Kass; Melissa Schwarz; Daniel L. Schwartz; Scott R. Klemmer (2010). *Parallel Prototyping Leads to Better Design Results, More Divergence, and Increased Self-Efficacy*. ACM Transactions on Computer-Human Interaction, 17(4), article 18, 1–24. https://doi.org/10.1145/1879831.1879836

[OI-W014] Tom Ritchey (2002). *General Morphological Analysis: A General Method for Non-Quantified Modelling*. Swedish Morphological Society. Revised 2013. https://www.swemorph.com/pdf/gma.pdf

[OI-W015] David G. Jansson; Steven M. Smith (1991). *Design Fixation*. Design Studies, 12(1), 3–11. https://doi.org/10.1016/0142-694X(91)90003-F

[OI-W017] Diane Vaughan (1997). *The Trickle-Down Effect: Policy Decisions, Risky Work, and the Challenger Tragedy*. California Management Review, 39(2), 80–102. https://doi.org/10.2307/41165888

[AB-W001] Wesley M. Cohen; Daniel A. Levinthal (1990). *Absorptive Capacity: A New Perspective on Learning and Innovation*. Administrative Science Quarterly, 35(1), 128–152. https://doi.org/10.2307/2393553

[AB-W002] Ingemar Dierickx; Karel Cool (1989). *Asset Stock Accumulation and Sustainability of Competitive Advantage*. Management Science, 35(12), 1504–1511. https://doi.org/10.1287/mnsc.35.12.1504

[AB-W003] P. N. Rosenstein-Rodan (1943). *Problems of Industrialisation of Eastern and South-Eastern Europe*. The Economic Journal, 53(210/211), 202–211. https://doi.org/10.2307/2226317

[AB-W004] Albert O. Hirschman (1958). *The Strategy of Economic Development*. Yale University Press. https://yalebooks.yale.edu/book/9780300001175/the-strategy-of-economic-development/

[AB-W005] Paul Collier; Frederick van der Ploeg; Michael Spence; Anthony J. Venables (2009). *Managing Resource Revenues in Developing Economies*. OxCarre Research Paper 15, University of Oxford. https://ora.ox.ac.uk/objects/uuid:b46dbfdf-569c-4bed-9d3d-da607b30dccf

[AB-W007] Jean-Pierre Noblet; Eric Simon (2008). *Capacité d'absorption : revue de littérature, opérationnalisation et exploration [Absorptive capacity: literature review, operationalization and exploration]*. XVIIe Conférence de l'Association internationale de management stratégique. https://www.strategie-aims.com/conferences/6-xviieme-conference-de-l-aims/communications/1486-capacite-dabsorption-revue-de-litterature-operationnalisation-et-exploration

[AB-W008] Florindo Rhaoni Picoli; Adriana Takahashi (2016). *Capacidade de Absorção, Aprendizagem Organizacional e Mecanismos de Integração Social [Absorptive capacity, organizational learning and social integration mechanisms]*. Revista de Administração Contemporânea, 20(1), 1–20. https://doi.org/10.1590/1982-7849rac2016140036

[AB-W009] 张杰 [Zhang Jie]; 陈志远 [Chen Zhiyuan]; 吴书凤 [Wu Shufeng]; 孙文浩 [Sun Wenhao] (2020). *对外技术引进与中国本土企业自主创新 [Foreign Technology Introduction and Local Firms' Innovation in China]*. 经济研究 [Economic Research Journal], no. 7, 92–105. https://www.rmbs.ruc.edu.cn/pub/www_rmbs/attachments/ee0c6fbebee64f3da5f15fe60f0b9fb2.pdf

[AB-W010] В. М. Полтерович [Victor Polterovich] (2009). *Проблема формирования национальной инновационной системы [The Problem of Creating a National Innovation System]*. Экономика и математические методы [Economics and Mathematical Methods], no. 2, 3–18. https://mpra.ub.uni-muenchen.de/22043/

[SM-W001] Laurence A. Walker (1997). *Flight Testing the X-36: The Test Pilot's Perspective*. NASA Contractor Report 198058. https://ntrs.nasa.gov/citations/19970031950

[SM-W002] Christian Gelzer (2014). *X-36 Tailless Fighter Agility Research Aircraft*. NASA. https://www.nasa.gov/aeronautics/x-36-tailless-fighter/

[SM-W003] Helen J. Huang; Alaa A. Ahmed (2011). *Tradeoff between Stability and Maneuverability during Whole-Body Movements*. PLoS ONE, 6(7), e21815. https://doi.org/10.1371/journal.pone.0021815

[SM-W004] John P. Scholz; Gregor Schöner (1999). *The uncontrolled manifold concept: identifying control variables for a functional task*. Experimental Brain Research, 126(3), 289–306. https://doi.org/10.1007/s002210050738

[SM-W006] Noël Bonneuil; Patrick Saint-Pierre (1998). *Domaine de victoire et stratégies viables chez les pêcheurs décrits par l'anthropologue Fredrik Barth [Winning domain and viable strategies among the fishermen described by anthropologist Fredrik Barth]*. Mathématiques et sciences humaines, 142, 43–66. https://www.numdam.org/item/MSH_1998__142__43_0/

[SM-W007] И. П. Кузнецов [I. P. Kuznetsov]; А. А. Паршин [A. A. Parshin]; Л. В. Халецкий [L. V. Khaletsky]; В. Ю. Шитов [V. Yu. Shitov] (2014). *Формирование требований к динамическим характеристикам и базовым параметрам контуров управления рулевого привода перспективного маневренного самолета [Developing requirements for the dynamic characteristics and basic parameters of flight-control actuator loops for a prospective maneuverable aircraft]*. Труды МАИ, no. 73, 1–24. https://mai.ru/upload/iblock/e23/e2395eaf1760a49d026f57acd88e2b18.pdf

[SM-W008] David D. Woods (2018). *The theory of graceful extensibility: basic rules that govern adaptive systems*. Environment Systems and Decisions, 38, 433–457. https://doi.org/10.1007/s10669-018-9708-3

[SM-W009] 中国民用航空局 [Civil Aviation Administration of China] (2020). *高风险货运固定翼无人机系统适航标准（试行）[Airworthiness standards for high-risk cargo fixed-wing unmanned aircraft systems (trial)]*. 民航适发〔2020〕1 号. 20 January 2020. https://www.caac.gov.cn/XXGK/XXGK/ZFGW/202005/t20200511_202468.html

[CG-W001] Alfredo Buttari; Jack Dongarra; Julie Langou; Julien Langou; Piotr Luszczek; Jakub Kurzak (2007). *Mixed Precision Iterative Refinement Techniques for the Solution of Dense Linear Systems*. The International Journal of High Performance Computing Applications 21(4):457–466. https://doi.org/10.1177/1094342007084026

[CG-W002] Erin Carson; Nicholas J. Higham (2018). *Accelerating the Solution of Linear Systems by Iterative Refinement in Three Precisions*. SIAM Journal on Scientific Computing 40(2):A817–A847. https://doi.org/10.1137/17M1140819

[CG-W003] Cindy Rubio-González; Cuong Nguyen; Hong Diep Nguyen; James Demmel; William Kahan; Koushik Sen; David H. Bailey; Costin Iancu; David Hough (2013). *Precimonious: Tuning Assistant for Floating-Point Precision*. Proceedings of SC '13, article 27, 12 pages. https://doi.org/10.1145/2503210.2503296

[CG-W004] Wei-Fan Chiang; Mark Baranowski; Ian Briggs; Alexey Solovyev; Ganesh Gopalakrishnan; Zvonimir Rakamarić (2017). *Rigorous Floating-Point Mixed-Precision Tuning*. Proceedings of POPL '17, 300–315. https://doi.org/10.1145/3009837.3009846

[CG-W005] Woongki Baek; Trishul M. Chilimbi (2010). *Green: A Framework for Supporting Energy-Conscious Programming using Controlled Approximation*. Proceedings of PLDI '10, 198–209. https://doi.org/10.1145/1806596.1806620

[CG-W006] Adrian Sampson; Werner Dietl; Emily Fortuna; Danushen Gnanapragasam; Luis Ceze; Dan Grossman (2011). *EnerJ: Approximate Data Types for Safe and General Low-Power Computation*. Proceedings of PLDI '11, 164–174. https://doi.org/10.1145/1993498.1993518

[CG-W025] Herbert H. Clark; Deanna Wilkes-Gibbs (1986). *Referring as a collaborative process*. Cognition, 22(1), 1–39. https://doi.org/10.1016/0010-0277(86)90010-7

[CG-W028] Daniel M. Wegner (1987). *Transactive Memory: A Contemporary Analysis of the Group Mind*. In Brian Mullen; George R. Goethals (eds.), Theories of Group Behavior, 185–208. Springer-Verlag. https://doi.org/10.1007/978-1-4612-4634-3_9

[CG-W009] Michael F. Schober; Herbert H. Clark (1989). *Understanding by addressees and overhearers*. Cognitive Psychology, 21(2), 211–232. https://doi.org/10.1016/0010-0285(89)90008-X

[CG-W007] Susan E. Brennan; Herbert H. Clark (1996). *Conceptual pacts and lexical choice in conversation*. Journal of Experimental Psychology: Learning, Memory, and Cognition, 22(6), 1482–1493. https://doi.org/10.1037/0278-7393.22.6.1482

[CG-W026] 光田航; 東中竜一郎; 大賀悠平; 杵渕哲也 (2021). *共同図形配置課題における対話の共通基盤構築過程の分析 [Analyzing common-ground construction in dialogue during a collaborative shape-arrangement task]*. 言語処理学会第 27 回年次大会発表論文集, 1697–1701. https://www.anlp.jp/proceedings/annual_meeting/2021/pdf_dir/B9-4.pdf

[CG-W027] 齋藤結; 光田航; 東中竜一郎; 南泰浩 (2023). *共通基盤の構築における名付けの有用性の分析 [An analysis of the usefulness of naming in constructing common ground]*. 言語処理学会第 29 回年次大会発表論文集, 1985–1989. https://www.anlp.jp/proceedings/annual_meeting/2023/pdf_dir/P8-7.pdf

[CG-W011] Daniel R. Nault; Rohit Voleti; Matthew Nicastro; Kevin G. Munhall (2023). *Investigating the influence of local and personal common ground on memory for conversation using an online referential communication task*. Journal of Experimental Psychology: General, 152(6), 1598–1621. https://doi.org/10.1037/xge0001341

[CG-W013] Jane W. S. Liu; Kwei-Jay Lin; Wei-Kuan Shih; Albert Chuang-shi Yu; Jen-Yao Chung; Wei Zhao (1991). *Algorithms for Scheduling Imprecise Computations*. Computer, 24(5), 58–68. https://doi.org/10.1109/2.76287

[CG-W014] Cristian Klein; Martina Maggio; Karl-Erik Årzén; Francisco Hernández-Rodriguez (2014). *Brownout: Building More Robust Cloud Applications*. Proceedings of the 36th International Conference on Software Engineering (ICSE 2014), 700–711. https://doi.org/10.1145/2568225.2568227

[CG-W015] Justin J. Meza; Thote Gowda; Ahmed Eid; Tomiwa Ijaware; Dmitry Chernyshev; Yi Yu; Md Nazim Uddin; Rohan Das; Chad Nachiappan; Sari Tran; Shuyang Shi; Tina Luo; David Ke Hong; Sankaralingam Panneerselvam; Hans Ragas; Svetlin Manavski; Weidong Wang; Francois Richard (2023). *Defcon: Preventing Overload with Graceful Feature Degradation*. 17th USENIX Symposium on Operating Systems Design and Implementation (OSDI 23), 607–622. https://www.usenix.org/conference/osdi23/presentation/meza

[CG-W016] Nadine B. Sarter; David D. Woods (1995). *How in the World Did We Ever Get into That Mode? Mode Error and Awareness in Supervisory Control*. Human Factors, 37(1), 5–19. https://doi.org/10.1518/001872095779049516

[CG-W017] Gabriel Szulanski (1996). *Exploring internal stickiness: Impediments to the transfer of best practice within the firm*. Strategic Management Journal, 17 Winter Special Issue, 27–43. https://doi.org/10.1002/smj.4250171105

[CG-W018] Sidney G. Winter; Gabriel Szulanski (2001). *Replication as Strategy*. Organization Science, 12(6), 730–743. https://doi.org/10.1287/orsc.12.6.730.10084

[CG-W019] Robert S. Huckman; Gary P. Pisano (2003). *The Effect of Organizational Context on Individual Performance*. NBER Working Paper 10027. https://doi.org/10.3386/w10027

[CG-W021] Glenn W. Brier (1950). *Verification of Forecasts Expressed in Terms of Probability*. Monthly Weather Review, 78(1), 1–3. https://doi.org/10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2

[CG-W023] Brian A. Nosek; Charles R. Ebersole; Alexander C. DeHaven; David T. Mellor (2018). *The preregistration revolution*. Proceedings of the National Academy of Sciences, 115(11), 2600–2606. https://doi.org/10.1073/pnas.1708274114

[CG-W024] Olmo R. van den Akker; Marcel A. L. M. van Assen; Marjan Bakker; Mahmoud Elsherif; Tsz Keung Wong; Jelte M. Wicherts (2023). *Preregistration in practice: A comparison of preregistered and non-preregistered studies in psychology*. Behavior Research Methods, 56, 5424–5433. First published online 2023; volume issued 2024. https://doi.org/10.3758/s13428-023-02277-0

[EF-W001] Philippe Bacchetta; Eric van Wincoop (2004). *A Scapegoat Model of Exchange-Rate Fluctuations*. American Economic Review, 94(2), 114–118. https://doi.org/10.1257/0002828041301849

[EF-W002] George Soros (2013). *Remarks delivered at the Global Economic Symposium*. Global Economic Symposium, Kiel, Germany, 1 October 2013. https://www.georgesoros.com/2013/10/01/the_future_of_europe/

[EF-W003] Michael Suk-Young Chwe (1998). *Culture, Circles, and Commercials: Publicity, Common Knowledge, and Social Coordination*. Rationality and Society, 10(1), 47–75. https://doi.org/10.1177/104346398010001002

[EF-W004] David S. Scharfstein; Jeremy C. Stein (1990). *Herd Behavior and Investment*. American Economic Review, 80(3), 465–479. https://www.jstor.org/stable/2006678

[EF-W005] Sushil Bikhchandani; David Hirshleifer; Ivo Welch (1992). *A Theory of Fads, Fashion, Custom, and Cultural Change as Informational Cascades*. Journal of Political Economy, 100(5), 992–1026. https://doi.org/10.1086/261849

[EF-W006] André Orléan (1988). *L'auto-référence dans la théorie keynésienne de la spéculation [Self-reference in the Keynesian theory of speculation]*. Cahiers d'Économie Politique, 14–15, 229–242. https://doi.org/10.3406/cep.1988.1072

[EF-W007] George Soros (2009). *Financial Markets*. Lecture delivered at Central European University, 27 October 2009. https://www.opensocietyfoundations.org/publications/george-soros-open-society-financial-crisis-and-way-ahead

[EF-W008] Stephen Morris; Hyun Song Shin (2002). *Social Value of Public Information*. American Economic Review, 92(5), 1521–1534. https://doi.org/10.1257/000282802762024610

[EF-W009] Marco Ottaviani; Peter Sørensen (2000). *Herd Behavior and Investment: Comment*. American Economic Review, 90(3), 695–704. https://doi.org/10.1257/aer.90.3.695

[EF-W010] David S. Scharfstein; Jeremy C. Stein (2000). *Herd Behavior and Investment: Reply*. American Economic Review, 90(3), 705–706. https://doi.org/10.1257/aer.90.3.705

[SW-W001] Marvin B. Lieberman; David B. Montgomery (1988). *First-mover advantages*. Strategic Management Journal, 9(S1), 41–58. https://doi.org/10.1002/smj.4250090706

[SW-W002] Marvin B. Lieberman; David B. Montgomery (1998). *First-mover (dis)advantages: Retrospective and link with the resource-based view*. Strategic Management Journal, 19(12), 1111–1125. https://doi.org/10.1002/(SICI)1097-0266(1998120)19:12<1111::AID-SMJ21>3.0.CO;2-W

[SW-W004] Robert S. Pindyck (1991). *Irreversibility, Uncertainty, and Investment*. Journal of Economic Literature, 29(3), 1110–1148. https://web.mit.edu/rpindyck/www/Papers/IrreverUncertInvestmentJEL1991.pdf

[SW-W005] Christophe Chamley; Douglas Gale (1994). *Information Revelation and Strategic Delay in a Model of Investment*. Econometrica, 62(5), 1065–1085. https://doi.org/10.2307/2951507

[SW-W006] Patrick Bolton; Christopher Harris (1999). *Strategic Experimentation*. Econometrica, 67(2), 349–374. https://doi.org/10.1111/1468-0262.00022

[SW-W007] 嘉本慎介 (Shinsuke Kamoto) (2006). *不確実性下における市場参入に対する先発の優位性の影響 [The Effect of First-Mover Advantage on Market Entry under Uncertainty]*. 現代ファイナンス, 20, 51–68. https://doi.org/10.24487/gendaifinance.20.0_51

[SW-W008] 川上桃子 (Momoko Kawakami) (2018). *「問題解決型コア部品」ベンダーとしての台湾企業の興隆過程と知識の獲得：液晶テレビ用SoC事業の事例分析 [The Rise of Taiwanese Firms as Suppliers of Problem-Solving Core Components and Their Acquisition of Knowledge: A Case Study of LCD Television SoCs]*. アジア経済, 59(4), 2–33. https://doi.org/10.24765/ajiakeizai.59.4_2

[SW-W009] 窦丽琛 (Dou Lichen); 李国平 (Li Guoping) (2004). *对"后发优势"的国内实证：基于技术创新扩散视角的分析 [An Empirical Test of Latecomer Advantage within China: An Analysis from the Perspective of Technological Innovation and Diffusion]*. 经济科学, 2004(4), 27–32. https://ccj.pku.edu.cn/Article/DownLoad?id=301274647&type=ArticleFile

[LC-W002] Rita Gunther McGrath; Ian C. MacMillan (1995). *Discovery-Driven Planning*. Harvard Business Review, 73(4), 44–54. https://hbr.org/1995/07/discovery-driven-planning

[LC-W003] Allen C. Ward; Jeffrey K. Liker; John J. Cristiano; Durward K. Sobek II (1995). *The Second Toyota Paradox: How Delaying Decisions Can Make Better Cars Faster*. Sloan Management Review, 36(3), 43–61. https://websites.umich.edu/~mrother/KATA_Files/Second_Toyota_Paradox.pdf

[LC-W004] Rita Gunther McGrath (1997). *A Real Options Logic for Initiating Technology Positioning Investments*. Academy of Management Review, 22(4), 974–996. https://doi.org/10.5465/amr.1997.9711022113

[LC-W005] Ian C. MacMillan; Alexander B. van Putten; Rita Gunther McGrath; James D. Thompson (2006). *Using Real Options Discipline for Highly Uncertain Technology Investments*. Research-Technology Management, 49(1), 29–37. https://business.columbia.edu/sites/default/files-efs/pubfiles/2833/Real%20Options%20DDP%20RTM%20Article.pdf

[LC-W006] Jeffrey K. Liker; James M. Morgan (2006). *The Toyota Way in Services: The Case of Lean Product Development*. Academy of Management Perspectives, 20(2), 5–20. https://doi.org/10.5465/amp.2006.20591002

[LC-W007] 石川晴雄 (Haruo Ishikawa); 佐々木直子 (Naoko Sasaki) (2018). *選好度を有する範囲概念に基づく多目的同時満足化設計（構造と制御の同時設計への適用）[Multi-Objective Simultaneous Satisficing Design Based on a Set-Based Concept with Preference: Application to Simultaneous Structural and Control Design]*. 日本機械学会論文集, 84(867), 18-00181. https://doi.org/10.1299/transjsme.18-00181

[LC-W008] Gaurav Tewari (2026). *Pilot Early, Commit Late: A Real-Options Model of Enterprise AI Adoption under Rapid Technological Progress*. arXiv preprint, 2609.15919. https://doi.org/10.48550/arXiv.2609.15919

[LC-W009] Sylvain Lenfle; Christophe Midler (2002). *Stratégies d'innovation et organisation de la conception dans les entreprises amont. Enseignements d'une recherche chez Usinor [Innovation Strategies and the Organization of Design in Upstream Firms: Lessons from Research at Usinor]*. Revue française de gestion, 28(140). https://sylvainlenfle.fr/images/Publications/article_RFG_2002.pdf

[LC-W010] Philip Weihs (2008). *Steuerung von Flexibilität. Management von Softwareinvestitionen mit Realoptionen [Managing Flexibility: Managing Software Investments with Real Options]*. Doctoral thesis, Wirtschaftsuniversität Wien. https://doi.org/10.57938/b9d98d1b-4613-4e90-98ae-91e95b68e886

[LC-W011] Brian M. Kennedy; Durward K. Sobek II; Michael N. Kennedy (2013). *Reducing Rework by Applying Set-Based Practices Early in the Systems Engineering Process*. First published online 2013; Systems Engineering, 17(3) (2014), 278–296. https://doi.org/10.1002/sys.21269

[CRE-W001] Senni Kirjavainen; Katja Hölttä-Otto (2020). *Deconstruction of Idea Generation Methods Into a Framework of Creativity Mechanisms*. First published 2020; journal volume 2021. https://doi.org/10.1115/1.4048539

[CRE-W002] Horst Geschka; Andrea Zirm (2011). *Kreativitätstechniken [Creativity Techniques]*. https://doi.org/10.1007/978-3-8349-6746-6_15

[CRE-W003] 姚威; 储昭卫 [Yao Wei; Chu Zhaowei] (2021). *系统化创新方法研究：理论进展与实践效果评价 [Systematic Innovation Methods: Theoretical Progress and Evaluation of Practical Effects]*. https://doi.org/10.12052/gdutxb.210034

[CRE-W004] 高橋誠; 日本創造学会 [Makoto Takahashi; Japan Creativity Society] (n.d.). *創造技法の分類とその種類 [Classification and Types of Creativity Techniques]*. https://www.japancreativity.jp/creativity/category/group/

[CRE-W005] Guy Aznar (2009). *Préciser le sens du mot « créativité » [Clarifying the Meaning of Creativity]*. https://gerflint.fr/Base/Europe4/aznar.pdf

[CRE-W006] Kleidson Daniel Medeiros Leopoldino; Mario Orestes Aguirre González; Paula de Oliveira Ferreira; José Raeudo Pereira; Marcus Eduardo Costa Souto (2016). *Creativity techniques: a systematic literature review*. https://doi.org/10.4322/pmd.2016.015

[CRE-W007] Г. Я. Буш [G. Ya. Bush] (1972). *Методы технического творчества [Methods of Technical Creativity]*. Riga: Liesma.

[CRE-W008] Henri Poincaré (1908). *L'invention mathématique [Mathematical Invention], in Science et méthode*. In Science et méthode, Livre I, chapter III. Paris: Ernest Flammarion.

[CRE-W009] Mary L. Gick; Keith J. Holyoak (1980). *Analogical Problem Solving*. https://doi.org/10.1016/0010-0285(80)90013-4

[CRE-W010] Kevin Dunbar; Isabelle Blanchette (2001). *The in vivo/in vitro approach to cognition: the case of analogy*. https://doi.org/10.1016/S1364-6613(00)01698-3

[CRE-W011] David Klahr; Herbert A. Simon (2001). *What Have Psychologists (And Others) Discovered About the Process of Scientific Discovery?*. https://www.psy.miami.edu/_assets/pdf/rpo-articles/klahr.pdf

[CRE-W012] Pat Langley; Gary L. Bradshaw; Herbert A. Simon (1981). *BACON.5: The Discovery of Conservation Laws*. https://www.ijcai.org/Proceedings/81-1/Papers/025.pdf

[CRE-W013] Nathalie Bonnardel; Todd Lubart (2019). *La créativité : approches et méthodes en psychologie et en ergonomie [Creativity: Approaches and Methods in Psychology and Ergonomics]*. https://doi.org/10.3917/rimhe.037.0079

[CRE-W014] Arthur Rimbaud (1912). *Lettre à Paul Demeny du 15 mai 1871 [Letter to Paul Demeny, 15 May 1871]*. Letter written 15 May 1871; first complete publication in La Nouvelle Revue française, October 1912.

[CRE-W015] André Breton (1924). *Manifeste du surréalisme [Manifesto of Surrealism]*. Paris: Éditions du Sagittaire.

[CRE-W016] Jean Lescure; Oulipo (n.d.). *S+7 [S+7 constraint]*. Oulipo. https://www.oulipo.net/fr/contraintes/s7

[CRE-W017] Gianni Rodari (1973). *Grammatica della fantasia: Introduzione all'arte di inventare storie [Grammar of Fantasy: An Introduction to the Art of Inventing Stories]*. Turin: Einaudi.

[CRE-W018] Maritza Isabel Álvarez Rodríguez (2009). *Escritura creativa. Aplicación de las técnicas de Gianni Rodari [Creative Writing: Applying Gianni Rodari's Techniques]*. http://ve.scielo.org/scielo.php?script=sci_arttext&pid=S1316-49102009000100010

[CRE-W019] Mauricio Pérez Jiménez (2002). *Técnicas y procedimientos. Un acercamiento al proceso creativo en el medio tecnográfico [Techniques and Procedures: An Approach to the Creative Process in the Technographic Medium]*. https://www.ull.es/revistas/index.php/artes/article/download/5539/3842

[CRE-W020] Viola Spolin (1963). *Improvisation for the Theater: A Handbook of Teaching and Directing Techniques*. Evanston: Northwestern University Press.

[CRE-W021] Elizabeth Waterhouse (2022). *Processing Choreography: Thinking with William Forsythe's Duo*. https://doi.org/10.14361/9783839455883

[CRE-W022] John Cage (1961). *Silence: Lectures and Writings*. Wesleyan University Press. Relevant essays first published in 1952 and 1957.

[CRE-W023] Nobuhiko Itani (2023). *Theater Games and Stanislavski's System: A Study on the Origins of Viola Spolin's Theatrical Education*. https://files.eric.ed.gov/fulltext/EJ1382327.pdf

[CRE-W024] Eeva Siljamäki; Panagiotis A. Kanellopoulos (2019). *Mapping visions of improvisation pedagogy in music education research*. First published 2019; journal volume 2020. https://doi.org/10.1177/1321103X19843003

[CRE-W025] Benjamin Baird; Jonathan Smallwood; Michael D. Mrazek; Julia W. Y. Kam; Michael S. Franklin; Jonathan W. Schooler (2012). *Inspired by Distraction: Mind Wandering Facilitates Creative Incubation*. https://doi.org/10.1177/0956797612446024

[CRE-W026] Ut Na Sio; Thomas C. Ormerod (2009). *Does Incubation Enhance Problem Solving? A Meta-Analytic Review*. https://doi.org/10.1037/a0014212

[CRE-W027] Simone M. Ritter; Ap Dijksterhuis (2014). *Creativity—the unconscious foundations of the incubation period*. https://doi.org/10.3389/fnhum.2014.00215

[CRE-W028] Marily Oppezzo; Daniel L. Schwartz (2014). *Give Your Ideas Some Legs: The Positive Effect of Walking on Creative Thinking*. https://doi.org/10.1037/a0036577

[CRE-W029] 田中博晃 [Hiroaki Tanaka] (2013). *KJ 法クイックマニュアル [KJ Method Quick Manual]*. https://www.jstage.jst.go.jp/article/methodologysig/3/0/3_3.8/_pdf

[CRE-W030] 川喜田二郎; 松沢哲郎; やまだようこ [Jiro Kawakita; Tetsuro Matsuzawa; Yoko Yamada] (2003). *KJ 法の原点と核心を語る：川喜田二郎さんインタビュー [Emergence and Essence of the KJ Method: An Interview with Jiro Kawakita]*. https://www.jstage.jst.go.jp/article/jaqp/2/1/2_6/_pdf

[CRE-W031] Michael Mose Biskjaer; Peter Dalsgaard; Kim Halskov (2017). *Understanding Creativity Methods in Design*. https://doi.org/10.1145/3064663.3064692

[CRE-W032] Dorian Peters; Lian Loke; Naseem Ahmadpour (2020). *Toolkits, cards and games – a review of analogue tools for collaborative ideation*. First published 2020; journal volume 2021. https://doi.org/10.1080/15710882.2020.1715444

[CRE-W033] Scarlett R. Herring; Brett R. Jones; Brian P. Bailey (2009). *Idea Generation Techniques among Creative Professionals*. https://doi.org/10.1109/HICSS.2009.241

[CRE-W034] Daniela Satomi Saito; Elisa Maria Pivetta; Vania Ribas Ulbricht; Claudia R. Batista; Rosane de Fatima Antunes Obregon (2015). *A web como espaço de criatividade e colaboração para o desenvolvimento de sites acessíveis [The Web as a Space for Creativity and Collaboration in Developing Accessible Websites]*. In Criatividade e Inovação na Educação, chapter 7, pp. 170–203. ISBN 9788566832280.

[CRE-W035] Ben Shneiderman (2007). *Creativity Support Tools: Accelerating Discovery and Innovation*. https://doi.org/10.1145/1323688.1323689

[CRE-W036] Hamit Coşkun (2011). *The Effects of Group Size, Memory Instruction, and Session Length on the Creative Performance in Electronic Brainstorming Groups*. https://files.eric.ed.gov/fulltext/EJ919891.pdf

[CRE-W037] Jeba Rezwana; Mary Lou Maher (2022). *Designing Creative AI Partners with COFI: A Framework for Modeling Interaction in Human-AI Co-Creative Systems*. First published 2022; journal volume 2023. https://doi.org/10.1145/3519026

[CRE-W038] Anil R. Doshi; Oliver P. Hauser (2024). *Generative AI enhances individual creativity but reduces the collective diversity of novel content*. https://doi.org/10.1126/sciadv.adn5290

[CRE-W039] Barrett R. Anderson; Jash Hemant Shah; Max Kreminski (2024). *Homogenization Effects of Large Language Models on Human Creative Ideation*. https://doi.org/10.1145/3635636.3656204

[CRE-W040] Karan Girotra; Lennart Meincke; Christian Terwiesch; Karl T. Ulrich (2023). *Ideas Are Dimes a Dozen: Large Language Models for Idea Generation in Innovation*. Working paper, 10 July 2023. https://mackinstitute.wharton.upenn.edu/wp-content/uploads/2023/08/LLM-Ideas-Working-Paper.pdf

[CRE-W041] Ut Na Sio; Hugues Lortie-Forgues (2024). *The impact of creativity training on creative performance: A meta-analytic review and critical evaluation of 5 decades of creativity training studies*. https://doi.org/10.1037/bul0000432

[CRE-W042] Patrícia Alves-Oliveira; Patrícia Arriaga; Carla Xavier; Guy Hoffman; Ana Paiva (2021). *Creativity Landscapes: Systematic Review Spanning 70 Years of Creativity Interventions for Children*. First published 2021; journal volume 2022. https://doi.org/10.1002/jocb.514

[CRE-W043] Viviane Kuntz; Tarcisio Vanzin; Vania Ribas Ulbricht; Claudia Regina Batista (2015). *Técnicas Criativas Aplicáveis em Cursos Online Abertos e Massivos (MOOCS) [Creative Techniques Applicable to MOOCs]*. In Criatividade e Inovação na Educação, chapter 8, pp. 204–229. ISBN 9788566832280.

[CRE-W044] Agustín Tristán López; Liliana Mendoza González (2016). *Taxonomías sobre creatividad [Taxonomies of Creativity]*. https://doi.org/10.18800/psico.201601.006

[CRE-W045] Oguz A. Acar; Murat Tarakci; Daan van Knippenberg (2018). *Creativity and Innovation Under Constraints: A Cross-Disciplinary Integrative Review*. First published 2018; journal volume 2019. https://doi.org/10.1177/0149206318805832

[CRE-W046] Elza Adamowicz (1998). *Surrealist Collage in Text and Image: Dissecting the Exquisite Corpse*. Cambridge University Press. ISBN 9780521592048.

[LP-W001] K. Anders Ericsson; Ralf Th. Krampe; Clemens Tesch-Römer (1993). *The Role of Deliberate Practice in the Acquisition of Expert Performance*. Psychological Review 100(3), 363–406. https://doi.org/10.1037/0033-295X.100.3.363

[LP-W002] Brooke N. Macnamara; David Z. Hambrick; Frederick L. Oswald (2014). *Deliberate Practice and Performance in Music, Games, Sports, Education, and Professions: A Meta-Analysis*. Psychological Science 25(8), 1608–1618. https://doi.org/10.1177/0956797614535810

[LP-W003] K. Anders Ericsson; Kyle W. Harwell (2019). *Deliberate Practice and Proposed Limits on the Effects of Practice on the Acquisition of Expert Performance: Why the Original Definition Matters and Recommendations for Future Research*. Frontiers in Psychology 10, 2396. https://doi.org/10.3389/fpsyg.2019.02396

[LP-W004] Brooke N. Macnamara; David Z. Hambrick (2020). *Toward a cumulative science of expertise: commentary on Moxley, Ericsson, and Tuffiash (2017)*. Psychological Research 85(3), 1108–1113. First published 2020; journal issue 2021. https://doi.org/10.1007/s00426-020-01307-2

[LP-W005] K. Anders Ericsson (2020). *Given that the detailed original criteria for deliberate practice have not changed, could the understanding of this complex concept have improved over time? A response to Macnamara and Hambrick (2020)*. Psychological Research 85(3), 1114–1120. First published 2020; journal issue 2021. https://doi.org/10.1007/s00426-020-01368-3

[LP-W006] Robert A. Duke; Amy L. Simmons; Carla Davis Cash (2009). *It's Not How Much; It's How: Characteristics of Practice Behavior and Retention of Performance Skills*. Journal of Research in Music Education 56(4), 310–321. https://doi.org/10.1177/0022429408328851

[LP-W007] Roger Chaffin; Gabriela Imreh (1997). *"Pulling Teeth and Torture": Musical Memory and Problem Solving*. Thinking & Reasoning 3(4), 315–336. https://doi.org/10.1080/135467897394310

[LP-W008] Friedrich Platz; Reinhard Kopiez; Andreas C. Lehmann; Anna Wolf (2014). *The influence of deliberate practice on musical achievement: a meta-analysis*. Frontiers in Psychology 5, 646. https://doi.org/10.3389/fpsyg.2014.00646

[LP-W009] П. Я. Гальперин [P. Ya. Galperin] (1959). Развитие исследований по формированию *умственных действий [Development of research on the formation of mental actions]*. Психологическая наука в СССР, т. 1.

[LP-W011] Geneviève Oléron (1955). *Récents travaux sur le transfert [Recent work on transfer]*. L'Année psychologique 55(2), 361–379. https://doi.org/10.3406/psy.1955.8799

[LP-W012] Ivan Wallan Tertuliano; Orlando Pereira de Souza Júnior; Antônio Sabino da Silva Filho; Umberto Cesar Corrêa (2008). *Estrutura de prática e freqüência de "feedback" extrínseco na aprendizagem de habilidades*

motoras [Practice structure and extrinsic feedback frequency in learning motor skills]. Revista Brasileira de Educação Física e Esporte 22(2), 103–118. https://doi.org/10.1590/S1807-55092008000200002

[LP-W013] Suzete Chiviacowsky; Camila Neves; Liliane Locatelli; Cibelly Oliveira (2005). *Aprendizagem motora em crianças: efeitos da freqüência autocontrolada de conhecimento de resultados [Motor learning in children: effects of self-controlled frequency of knowledge of results]*. Revista Brasileira de Ciências do Esporte 26(3), 177–190.

[LP-W014] 渋谷賢; 伊藤政展 [Satoshi Shibuya; Masanobu Ito] (1999). *運動学習における文脈干渉と要約フィードバックの相互作用 [Interaction of contextual interference and summary feedback in motor learning]*. 体育学研究 44(2), 100–111. https://doi.org/10.5432/jjpehss.KJ00003397504

[LP-W015] Judith Jiménez-Díaz; María Morera-Castro (2018). *¿Cómo diseñar la práctica para optimizar el desempeño y aprendizaje motor? Una revisión de literatura [How to design practice to optimize motor performance and learning? A literature review]*. Sportis 4(3), 587–603. https://doi.org/10.17979/sportis.2018.4.3.3103

[LP-W016] Julian Roelle; Tobias Richter (2025). *Üben aus Sicht der Lernpsychologie: Wie kann Üben das nachhaltige Lernen fördern? [Practice from the perspective of the psychology of learning: How can practice promote lasting learning?]*. Unterrichtswissenschaft 53, 179–204. https://doi.org/10.1007/s42010-025-00227-7

[LP-W017] 王植洵; 张锦坤 [Wang Zhixun; Zhang Jinkun] (2017). *提取练习效应的产生机制：编码、提取的整合演变 [The mechanism of the retrieval practice effect: From encoding and retrieval to their integration]*. 心理科学进展 25(9), 1512–1520. https://doi.org/10.3724/SP.J.1042.2017.01512

[LP-W018] Phillip G. Armour (2006). *The Learning Edge*. Communications of the ACM 49(6), 19–22.

[LP-W019] Nicholas C. Soderstrom; Robert A. Bjork (2015). *Learning Versus Performance: An Integrative Review*. Perspectives on Psychological Science 10(2), 176–199. https://doi.org/10.1177/1745691615569000

[LP-W020] John Dunlosky; Katherine A. Rawson; Elizabeth J. Marsh; Mitchell J. Nathan; Daniel T. Willingham (2013). *Improving Students' Learning With Effective Learning Techniques: Promising Directions From Cognitive and Educational Psychology*. Psychological Science in the Public Interest 14(1), 4–58. https://doi.org/10.1177/1529100612453266

[LP-W021] Mark A. Guadagnoli; Timothy D. Lee (2004). *Challenge Point: A Framework for Conceptualizing the Effects of Various Practice Conditions in Motor Learning*. Journal of Motor Behavior 36(2), 212–224. https://doi.org/10.3200/JMBR.36.2.212-224

[LP-W022] Henry L. Roediger III; Jeffrey D. Karpicke (2006). *Test-Enhanced Learning: Taking Memory Tests Improves Long-Term Retention*. Psychological Science 17(3), 249–255. https://doi.org/10.1111/j.1467-9280.2006.01693.x

[LP-W023] Nicholas J. Cepeda; Edward Vul; Doug Rohrer; John T. Wixted; Harold Pashler (2008). *Spacing Effects in Learning: A Temporal Ridgeline of Optimal Retention*. Psychological Science 19(11), 1095–1102. https://doi.org/10.1111/j.1467-9280.2008.02209.x

[LP-W024] Nate Kornell; Robert A. Bjork (2008). *Learning Concepts and Categories: Is Spacing the "Enemy of Induction"?* Psychological Science 19(6), 585–592. https://doi.org/10.1111/j.1467-9280.2008.02127.x

[LP-W025] Manu Kapur (2008). *Productive Failure*. Cognition and Instruction 26(3), 379–424. https://doi.org/10.1080/07370000802212669

[LP-W026] Tanmay Sinha; Manu Kapur (2021). *When Problem Solving Followed by Instruction Works: Evidence for Productive Failure*. Review of Educational Research 91(5), 761–798. https://doi.org/10.3102/00346543211019105

[LP-W027] Hermann Ebbinghaus (1885). *Über das Gedächtnis. Untersuchungen zur experimentellen Psychologie [On memory: Investigations in experimental psychology]*. Leipzig: Duncker & Humblot.

[LP-W028] Elizabeth Ligon Bjork; Robert A. Bjork (2011). *Making Things Hard on Yourself, But in a Good Way: Creating Desirable Difficulties to Enhance Learning*. In Psychology and the Real World: Essays Illustrating Fundamental Contributions to Society, 56–64.

[LP-W029] Katherine A. Rawson; John Dunlosky (2012). *When Is Practice Testing Most Effective for Improving the Durability and Efficiency of Student Learning?* Educational Psychology Review 24, 419–435. https://doi.org/10.1007/s10648-012-9203-1

[LP-W031] Mihaly Csikszentmihalyi; Judith LeFevre (1989). *Optimal Experience in Work and Leisure*. Journal of Personality and Social Psychology 56(5), 815–822. https://doi.org/10.1037/0022-3514.56.5.815

[LP-W032] Stanisław H. Czyż; Aleksandra M. Wójcik; Petra Solarská (2024). *The effect of contextual interference on transfer in motor learning: a systematic review and meta-analysis*. Frontiers in Psychology 15, 1377122. https://doi.org/10.3389/fpsyg.2024.1377122

[LP-W033] Sterre K. Ruitenburg; Mieke Goos; Guus Lambert; Paul A. Kirschner; Halszka Jarodzka; Gino Camp (2026). *A Meta-Analysis of the Effectiveness and Boundary Conditions of Retrieval Practice, Distributed Practice, and Interleaved Practice in Authentic School Classrooms*. Educational Psychology Review 38, article 137. https://doi.org/10.1007/s10648-026-10225-5

[LP-W034] Suzete Chiviacowsky; Gabriele Wulf (2002). *Self-Controlled Feedback: Does It Enhance Learning Because Performers Get Feedback When They Need It?* Research Quarterly for Exercise and Sport 73(4), 408–415. https://doi.org/10.1080/02701367.2002.10609040

[LP-W030] Richard A. Schmidt; Robert A. Bjork (1992). *New Conceptualizations of Practice: Common Principles in Three Paradigms Suggest New Concepts for Training*. Psychological Science 3(4), 207–217. https://doi.org/10.1111/j.1467-9280.1992.tb00029.x

[TD-W001] Michael J. Rohrbaugh; Varda Shoham (2001). *Brief Therapy Based on Interrupting Ironic Processes: The Palo Alto Model*. Clinical Psychology: Science and Practice 8(1):66–81. https://doi.org/10.1093/clipsy.8.1.66

[TD-W002] Steve de Shazer; Insoo Kim Berg; Eve Lipchik; Elam Nunnally; Alex Molnar; Wallace Gingerich; Michele Weiner-Davis (1986). *Brief Therapy: Focused Solution Development*. Family Process 25(2):207–221. https://doi.org/10.1111/j.1545-5300.1986.00207.x

[TD-W003] Solution Focused Brief Therapy Association Research Committee (2025). *Solution Focused Therapy Treatment Manual for Working with Individuals*. 3rd edition. Solution Focused Brief Therapy Association. https://www.sfbta.org/Solution%20Focused%20Therapy%20Treatment%20Manual%20for%20Working%20with%20Individuals%20-3rd%20Edition-%20final.pdf

[TD-W004] Wallace J. Gingerich; Lance T. Peterson (2013). *Effectiveness of Solution-Focused Brief Therapy: A Systematic Qualitative Review of Controlled Outcome Studies*. Research on Social Work Practice 23(3):266–283. https://doi.org/10.1177/1049731512470859

[TD-W005] P. Knekt; O. Lindfors; T. Härkänen; M. Välikoski; E. Virtala; M. A. Laaksonen; M. Marttunen; M. Kaipainen; C. Renlund; Helsinki Psychotherapy Study Group (2007). *Randomized trial on the effectiveness of long- and short-term psychodynamic psychotherapy and solution-focused therapy on psychiatric symptoms during a 3-year follow-up*. Psychological Medicine 38(5):689–703 (2008); first published 16 November 2007. https://doi.org/10.1017/S003329170700164X

[TD-W006] Zach W. Cooper; Orion Mowbray; Mohammed K. Ali; Leslie C. Johnson (2024). *Addressing depression and comorbid health conditions through solution-focused brief therapy in an integrated care setting: a randomized clinical trial*. BMC Primary Care 25:313. https://doi.org/10.1186/s12875-024-02561-8

[TD-W007] 邹晓伟; 田洋丽; 刘利丹; 万爱兰 (2021). *焦点解决短程治疗非自杀性自伤的理念分析 [An analysis of solution-focused brief therapy for nonsuicidal self-injury]*. 医学与哲学 42(24):50–54. https://doi.org/10.12014/j.issn.1002-0772.2021.24.11

[TD-W008] 黒沢幸子; 西野明樹; 鶴田芳映; 森俊夫 (2015). *事例とコンサルティを活かす解決志向ブリーフセラピーのコンサルテーション–11 ステップモデルの効果研究と実践への誘い– [Solution-focused brief-therapy consultation drawing on the case and consultee: an effectiveness study of an eleven-step model and an invitation to practice]*. コミュニティ心理学研究 18(2):186–204. https://doi.org/10.32236/jscpjournal.18.2_186

[TD-W009] Michele Orr; Sharlette Ryder; Julie Considine (2026). *Use of Solution-Focused Brief Therapy Within Adult Community Mental Health: A Scoping Review*. Clinical Psychology & Psychotherapy; first published 17 March 2026. https://doi.org/10.1002/cpp.70255

[TD-W010] William M. M. Kirby (1985). *[Citation Classic commentary on "Antibiotic susceptibility testing by a standardized single disk method"]*. Current Contents, no. 32, 12 August 1985, p. 18. https://garfield.library.upenn.edu/classics1985/A1985ANC2900001.pdf

[TD-W011] Comité de l'antibiogramme de la Société française de microbiologie (2025). *Recommandations 2025 [Recommendations 2025]*. CA-SFM/EUCAST, version 1.1, July 2025. https://www.sfm-microbiologie.org/wp-content/uploads/2025/07/CASFM2025_V1.1-JUILLET-2025.pdf

[TD-W012] Jochen W. L. Cals; Christopher C. Butler; Rogier M. Hopstaken; Kerenza Hood; Geert-Jan Dinant (2009). *Effect of point of care testing for C reactive protein and training in communication skills on antibiotic use in lower respiratory tract infections: cluster randomised trial*. BMJ 338: b1374. https://doi.org/10.1136/bmj.b1374

[TD-W013] Siri Aas Smedemark; Rune Aabenhus; Carl Llor; Anders Fournaise; Ole Olsen; Karsten Juhl Jørgensen (2022). *Biomarkers as point-of-care tests to guide prescription of antibiotics in people with acute respiratory infections in primary care*. Cochrane Database of Systematic Reviews, 2022, issue 10, CD010130. https://doi.org/10.1002/14651858.CD010130.pub3

[TD-W014] Rikke Vognbjerg Sydenham; Ulrik Stenz Justesen; Malene Plejdrup Hansen; Line Bjørnskov Pedersen; Rune Munck Aabenhus; Sonja Wehberg; Dorte Ejg Jarbøl (2021). *Prescribing antibiotics: the use of diagnostic tests in general practice. A register-based study*. Scandinavian Journal of Primary Health Care 39(4):466–475. https://doi.org/10.1080/02813432.2021.2004721

[TD-W015] Anna Maria Bonet-Esteve; Laia Font-Ribera; Judit Dorca-Vila; Antonia Retamal- Cañiz; Pere Roura-Poch; Josep Vidal-Alaball (2021). *Implantación de una prueba rápida de infección estreptocócica: ¿su uso también mejora la adherencia antibiótica? [Introducing a rapid test for streptococcal infection: does its use also improve antibiotic adherence?]*. Atención Primaria 53:102102. https://doi.org/10.1016/j.aprim.2021.102102

[TD-W016] Alastair D. Hay; Samantha Abbs; Matthew Ridd; Stephen Granier; J. Athene Lane; Peter Muir; Jodi Taylor; Grace Young; Kathy Eastwood; Hayley Dash; Lynne Bradshaw; Rebecca Clarke; Mandy Lui; Emma Bridgeman; Rachel C. M. Brierley; Emily Brown; Hannah V. Thornton; Paul M. Mitchell; Liang Zhu; Lucy Yardley; Chris Metcalfe (2026). *Rapid Respiratory Microbiological Point-of-Care Testing and Antibiotic Use in Primary Care*. JAMA Internal Medicine 186(7):817–826; first published 18 May 2026. https://doi.org/10.1001/jamainternmed.2026.1426

[TD-W017] Ritu Banerjee et al., for the Antibacterial Resistance Leadership Group (2020). *Randomized Trial Evaluating Clinical Impact of RAPid IDentification and Susceptibility Testing for Gram-negative Bacteremia: RAPIDS-GN*. Clinical Infectious Diseases 73(1): e39–e46 (2021); first published 6 May 2020. https://doi.org/10.1093/cid/ciaa528

[TD-W018] Ritu Banerjee; Abhigya Giri; Lauren Komarow; Maria Souli; Sarah B. Doernberg; Robin Patel (2024). *Impact of rapid antibiotic susceptibility testing for Gram-negative bacteremia varies by pathogen type and resistance: a secondary analysis of the RAPIDS GN trial*. Microbiology Spectrum 13(2) (2025); first published 16 December 2024. https://doi.org/10.1128/spectrum.01789-24

[TD-W019] Vanesa Anton-Vazquez; Paul Hine; Sanjeev Krishna; Marty Chaplin; Timothy Planche (2021). *Rapid versus standard antimicrobial susceptibility testing to guide treatment of bloodstream infection*. Cochrane Database of Systematic Reviews 2021, issue 5, CD013235. https://doi.org/10.1002/14651858.CD013235.pub2

[TD-W020] 西村翔 (2024). *抗菌薬適正使用に貢献する薬剤感受性検査とは [Susceptibility testing that contributes to appropriate antimicrobial use]*. 日本臨床微生物学会雑誌 34(1):1–17. https://www.jscm.org/journal/full/03401/034010001.pdf

[TD-W021] 王辉; 宁永忠; 陈宏斌; 等 (2016). *常见细菌药物敏感性试验报告规范中国专家共识 [Chinese expert consensus on standardized reporting of susceptibility tests for common bacteria]*. 中华检验医学杂志 39(1):18–22. https://doi.org/10.3760/cma.j.issn.1009-9158.2016.01.006

[TD-W022] Qian Chen; Yi-Ge Qu; Jing-Shu Li; Ji Yan (2026). *Rapid Diagnostics and Rapid Antimicrobial Susceptibility Testing for Bloodstream Infections: A Narrative Review and Proposed Clinical Actionability Framework*. International Journal of General Medicine 19:623052. https://doi.org/10.2147/IJGM.S623052

[TD-W023] Evdoxia Kyriazopoulou; Lydia Liaskou-Antoniou; George Adamis; Antonia Panagaki; Nikolaos Melachroinopoulos; Elina Drakou; Konstantinos Marousis; Georgios Chrysos; Andronikos Spyrou; Nikolaos Alexiou; Styliani Symbardi; Zoi Alexiou; Styliani Lagou; Virginia Kolonia; Theologia Gkavogianni; Miltiades Kyprianou; Ioannis Anagnostopoulos; Garyfallia Poulakou; Malvina Lada; Anna Makina; Efrosyni Roulia; Marina Koupetori; Vasileios Apostolopoulos; Dimitra Petrou; Thomas Nitsotolis; Anastasia Antoniadou; Evangelos J. Giamarellos-Bourboulis (2020). *Procalcitonin to Reduce Long-Term Infection-associated Adverse Events in Sepsis. A Randomized Trial*. American Journal of Respiratory and Critical Care Medicine 203(2):202–210 (2021); first published 6 August 2020. https://doi.org/10.1164/rccm.202004-1201OC

[TD-W024] Lars Kotthoff (2012). *Algorithm Selection for Combinatorial Search Problems: A survey*. arXiv:1210.7959 (2012); later journal publication 2014. https://doi.org/10.48550/arXiv.1210.7959

[TD-W025] Vladimir Estivill-Castro; Derick Wood (1992). *A Survey of Adaptive Sorting Algorithms*. ACM Computing Surveys 24(4):441–476. https://doi.org/10.1145/146370.146381

[TD-W026] Massachusetts Institute of Technology (2011). *Lecture 7: Linear-Time Sorting*. 6.006 Introduction to Algorithms, Fall 2011. MIT OpenCourseWare. https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-fall-2011/bf7d79105762bf79bbc0925438e1468a_MIT6_006F11_lec07.pdf

[TD-W027] Peter M. McIlroy; Keith Bostic; M. Douglas McIlroy (1993). *Engineering Radix Sort*. Computing Systems 6(1):5–27, Winter 1993. https://www.usenix.org/publications/compsystems/1993/win_mcilroy.pdf

[TD-W028] Jon L. Bentley; M. Douglas McIlroy (1993). *Engineering a Sort Function*. Software—Practice and Experience 23(11):1249–1265. https://doi.org/10.1002/spe.4380231105

[TD-W029] J. Ian Munro; Sebastian Wild (2018). *Nearly-Optimal Mergesorts: Fast, Practical Sorting Methods That Optimally Adapt to Existing Runs*. 26th European Symposium on Algorithms, LIPIcs 112, 63:1–63:16. https://doi.org/10.4230/LIPIcs.ESA.2018.63

[TD-W030] Vincent Jugé (2020). *L'algorithme de tri ShiversSort adaptatif [The adaptive ShiversSort sorting algorithm]*. 1024: Bulletin de la Société informatique de France 15:5–21, April 2020. https://www.societe-informatique-de-france.fr/wp-content/uploads/2020/04/1024_15_2020_5.pdf

[TD-W031] Python development team (2025). *CPython 3.14.0*. Software release 7 October 2025. https://github.com/python/cpython/tree/v3.14.0

[TD-W032] Manuel Blum; Robert W. Floyd; Vaughan Pratt; Ronald L. Rivest; Robert E. Tarjan (1973). *Time Bounds for Selection*. Journal of Computer and System Sciences 7:448–461. https://doi.org/10.1016/S0022-0000(73)80033-9

[SL-W001] Nawar N. Chaker; Rhett T. Epler; Gabriel Moreno; Dana Amiri; Elizabeth G. McDougal; Jay O'Toole (2025). The past, present, and future of adaptive selling: Toward an integrative framework. Journal of the Academy of Marketing Science 53, 1572–1599. https://doi.org/10.1007/s11747-025-01096-3

[SL-W002] William C. Moncrief; Greg W. Marshall (2004 online; 2005 issue). The evolution of the seven steps of selling. Industrial Marketing Management 34, 13–22. https://doi.org/10.1016/j.indmarman.2004.06.001

[SL-W003] Pramod Kumar Nayak (n.d.). Theories of Selling. Trinity Institute of Professional Studies, Sales Management course, Paper ID 17303.

[SL-W004] Uttarakhand Open University (n.d.). Sales and Distribution Management, MS 508. Unit 2, Personal Selling.

[SL-W005] Akinori Iwamoto (2023). The Origin of AIDA: Who Invented and Formulated the AIDA model? Proceedings of the 21st Biennial Conference on Historical Analysis & Research in Marketing. https://ojs.library.carleton.ca/index.php/pcharm/article/view/4340

[SL-W006] 安贺新（主编）(2009; 第 3 版 2019). 销售管理实务 [Sales Management Practice]. 清华大学出版社. ISBN 9787302530237.

[SL-W007] David Evers; Regina Schrank; Wolfgang Boos; Matthias Rief; Christian Schmitz; Timo Spatzier; Jan Wieseke (2024). Vom Produkt- zum Lösungsanbieter – Die Transformation zum Solution-Seller meistern [From product provider to solution provider: Mastering the transformation to solution seller]. FIR e. V. an der RWTH Aachen.

[SL-W008] Huthwaite International (n.d.). The SPIN Methodology. https://www.huthwaiteinternational.com/spin-methodology

[SL-W009] Huthwaite International (n.d.). SPIN Evidence of return on investment: Seven companies share their SPIN journeys and return on investment evidence.

[SL-W010] David Mattson (n.d.). How To Sell To the Modern Buyer ...The Sandler Way. Sandler white paper.

[SL-W011] Phillip Jesse McGowan (2020 online; 2021 issue). Sales failure: A review and future research directions. International Journal of Logistics Research and Applications. https://doi.org/10.1080/13675567.2020.1726306

[SL-W012] Enrique Bigné Alcañiz; Inés Küster Boluda; Francisco Torán Torres (2001). La venta adaptable y la orientación al mercado en el contexto de la venta personal [Adaptive selling and market orientation in personal selling]. Dirección y Organización 26, 152–168. https://www.revistadyo.es/DyO/index.php/dyo/article/view/209

[SL-W013] Valter Afonso Vieira; Deividi Pires; Ronie Galeano (2013). Determinantes do desempenho empresarial e das vendas cruzadas no varejo [Determinants of business performance and cross-selling in retail]. RAE 53(6), 565–579. https://doi.org/10.1590/S0034-759020130605

[SL-W014] Jean Lagueux; Lova Rajaobelina (2013). Proposition d'un modèle des antécédents et conséquences de la vente suggestive dans le domaine de la restauration [A proposed model of antecedents and consequences of suggestive selling in restaurants]. ACFAS research paper.

[SL-W015] 西川功一 [Koichi Nishikawa] (2010). 法人営業イノベーションとクラウド CRM システムの活用 [B2B sales innovation and the use of cloud CRM systems]. 日本生産管理学会論文誌 16(2), 147–152.

[SL-W016] И. В. Руденко [I. V. Rudenko] (2012). Управление продажами: истоки, сущность, подходы [Sales management: Origins, meaning and approaches]. Вестник Омского университета. Серия «Экономика» 4, 21–25.

[SL-W017] 粟島浩二 (2004). 卸売業における営業革新—生産財卸売業を事例として [Sales innovation in wholesaling: A case of industrial-goods wholesaling]. 流通 17, 72–79.

[SL-W018] Willem Verbeke; Bart Dietz; Ernst Verwaal (2010 online; 2011 issue). Drivers of sales performance: A contemporary meta-analysis. Have salespeople become knowledge brokers? Journal of the Academy of Marketing Science 39, 407–428. https://doi.org/10.1007/s11747-010-0211-8

[SL-W019] Harri Terho; Alexander Haas; Andreas Eggert; Wolfgang Ulaga (2012). "It's almost like taking the sales out of selling"—Towards a conceptualization of value-based selling in business markets. Industrial Marketing Management 41, 174–185. https://doi.org/10.1016/j.indmarman.2011.11.011

[SL-W020] Challenger (2024). What is the Challenger Sales Methodology? https://challengerinc.com/what-is-challenger-sales-methodology/

[SL-W021] Ben Taylor (2022). Defining Solution Selling. Richardson Sales Performance. https://www.richardson.com/sales-resources/solution-selling-sales-approach-methodology/

[SL-W022] Michael Cooper; Frank Winters (2000). Selling Services—Coordinating Sales and Project Management. Proceedings of the Project Management Institute Annual Seminars & Symposium, Houston, September 7–16.

[SL-W023] Frédéric Buchet (2018). Vendeurs : passez en mode solution [Salespeople: Switch to solution mode]. Dunod. ISBN 9782100769841.

[SL-W024] Ramon A. Avila; Scott A. Inks (2017). The Evolution of the Sales Process: Relationship Selling Versus the Challenger Sales. Atlantic Marketing Association Proceedings, 317–323.

[SL-W025] Matt Dixon; Ted McKenna (2024). What is the JOLT Effect? Everything You Need To Know. https://www.jolteffect.com/blog/what-is-the-jolt-effect

[SL-W026] Claus Ebster; Udo Wagner; Sabine Valis (2006). The effectiveness of verbal prompts on sales. Journal of Retailing and Consumer Services 13, 169–176. https://doi.org/10.1016/j.jretconser.2005.08.003

[SL-W027] Richardson Sales Performance (n.d.). Solution Selling® Training for Buyer Alignment and Sales Success.

[SL-W028] 山中正樹 [Masaki Yamanaka] (2019). 生産財における潜在的ニーズ獲得の論理 [A logic of latent-needs acquisition for industrial products]. 企業家研究 16, 1–23. https://doi.org/10.34418/fes.16.0_1

[SL-W029] Mack Hanan; James J. Cribbin; Herman C. Heiser (1970). Consultative Selling.

[NC-W001] Г. С. Альтшуллер (1985). Алгоритм решения изобретательских задач АРИЗ-85-В [Algorithm for solving inventive problems, ARIZ-85-V].

[NC-W002] Jacob Goldenberg and David Mazursky (2002). Creativity in Product Innovation. Cambridge University Press.

[NC-W003] 中川 徹 (2005). 創造的問題解決の新しいパラダイム—類比思考に頼らない USIT の 6 箱方式 [A new paradigm for creative problem solving: the six-box scheme of USIT without depending on analogical thinking]. 日本創造学会第 27 回研究大会.

[NC-W004] Kamal S. Birdi (2016). Creativity Training. In Paul Sparrow, Helen Shipton, Pawan Budhwar and Alan Brown (eds.), Human Resource Management, Innovation and Performance, 298–312. Palgrave Macmillan.

[NC-W005] Denis Dumas, Linda C. Schmidt and Patricia A. Alexander (2016). Predicting creative problem solving in engineering design. Thinking Skills and Creativity 21, 50–66. DOI 10.1016/j.tsc.2016.05.002.

[NC-W006] Christian Spreafico and Davide Russo (2016). TRIZ industrial case studies: a critical survey. Procedia CIRP 39, 51–56. DOI 10.1016/j.procir.2016.01.165.

[NC-W007] 姚威· 储昭卫· 胡顺顺 (2022). TRIZ 真的是创新“点金术”吗——对浙江省 TRIZ 应用效果的分析 [Is TRIZ really an innovative “Midas touch”? Application analysis based on Zhejiang Province]. 科技进步与对策 39(4), 10–19. DOI 10.6049/kjjbydc.2021060608.

[NC-W008] Arthur Koestler (1964). The Act of Creation.

[NC-W009] William J. J. Gordon (1961). Synectics: The Development of Creative Capacity.

[NC-W010] Sarnoff A. Mednick (1962). The associative basis of the creative process. Psychological Review 69(3), 220–232. DOI 10.1037/h0048850.

[NC-W011] Ching-Lin Wu, Shih-Yuan Huang, Pei-Zhen Chen and Hsueh-Chih Chen (2020). A Systematic Review of Creativity-Related Studies Applying the Remote Associates Test From 2000 to 2019. Frontiers in Psychology 11:573432. DOI 10.3389/fpsyg.2020.573432.

[NC-W012] 王硕阳· 林崇德· 孙汉银 (2020). 创造性问题解决中的联想能力及其作用 [The mechanism of associative abilities in creative problem solving]. 心理与行为研究 18(1), 8–16.

[NC-W013] 西田勇樹· 服部雅史· 織田涼 (2023). 遠隔連想テストは何を測るか [What does the Remote Associates Test measure?]. 心理学研究 94(5), 392–401. DOI 10.4992/jjpsy.94.22024.

[NC-W014] Margaret A. Boden (1997). Creativity and Artificial Intelligence. Proceedings of IJCAI 1997, 1563–1566.

[NC-W015] Ronald A. Finke (1992). Creative Inference in Imagery and Invention. AAAI Spring Symposium.

[NC-W016] Dean Keith Simonton (2021). Scientific Creativity: Discovery and Invention as Combinatorial. Frontiers in Psychology 12:721104. DOI 10.3389/fpsyg.2021.721104.

[NC-W017] Gilles Fauconnier and Mark Turner (1998). Conceptual Integration Networks. Cognitive Science 22(2), 133–187. DOI 10.1207/s15516709cog2202_1.

[NC-W018] Mehrnaz Azadyekta (2013). Comparison of the Effect Of Three Methods of Creativity Development in Elementary Students in Tehran City. The European Journal of Social & Behavioural Sciences 4(1), 88–96. DOI 10.15405/ejsbs.2013.1.11.

[NC-W019] Jacques Chabrillat, Marc Lecoutre and Pascal Lièvre (2023 online; 2024 issue). Le processus de bissociation au cœur de la fabrique des idées : le cas d’une innovation managériale [The process of bisociation at the heart of idea generation: the case of a managerial innovation]. Innovations 73(1), 185–214. DOI 10.3917/inno.pr2.0156.

[NC-W020] Don R. Swanson and Neil R. Smalheiser (1996). Undiscovered Public Knowledge: a Ten-Year Update. Proceedings of KDD 1996, 295–298.

[NC-W021] Tobias Kötter and Michael R. Berthold (2012). From Information Networks to Bisociative Information Networks. In Michael R. Berthold, ed., Bisociative Knowledge Discovery: An Introduction to Concept, Algorithms, Tools, and Applications. LNCS 7250, 33–50. DOI 10.1007/978-3-642-31830-6_3.

[NC-W022] Matjaž Juršič, Borut Sluban, Bojan Cestnik, Miha Grčar and Nada Lavrač (2012). Bridging Concept Identification for Constructing Information Networks from Text Documents. In Michael R. Berthold, ed., Bisociative Knowledge Discovery: An Introduction to Concept, Algorithms, Tools, and Applications. LNCS 7250, 66–90. DOI 10.1007/978-3-642-31830-6_6.

[NC-W023] Faez Ahmed and Mark Fuge (2018). Creative exploration using topic-based bisociative networks. Design Science 4, e12. DOI 10.1017/dsj.2018.5.

[NC-W024] Horace Walpole (1754). Letter to Sir Horace Mann, 28 January 1754.

[NC-W025] Robert K. Merton (1948). The Bearing of Empirical Research upon the Development of Social Theory. American Sociological Review 13(5), 505–515. DOI 10.2307/2087142. Expanded author account in Social Theory and Social Structure (1968), chapter V, 156–162.

[NC-W026] Christian Busch (2022 online; 2024 issue). Towards a Theory of Serendipity: A Systematic Review and Conceptualization. Journal of Management Studies 61(3), 1110–1151. DOI 10.1111/joms.12890.

[NC-W027] Stephann Makri and Ann Blandford (2012). Coming across information serendipitously, Part 1: A process model. Journal of Documentation 68(5), 684–705. DOI 10.1108/00220411211256030.

[NC-W028] Ohid Yaqub (2016; revised 2017). Serendipity: Towards a Taxonomy and a Theory. SPRU Working Paper 2016-17. Journal publication: Research Policy 47(1) (2018), 169–179. https://doi.org/10.1016/j.respol.2017.10.007

[NC-W029] Sylvie Catellin and Laurent Loty (2013). Sérendipité et indisciplinarité [Serendipity and indisciplinarity]. Hermès, La Revue 67(3), 32–40. DOI 10.4267/2042/51882.

[NC-W030] Stephann Makri, Ann Blandford, Mel Woods, Sarah Sharples and Deborah Maxwell (2014). “Making my own luck”: Serendipity strategies and how to support them in digital information environments. Journal of the Association for Information Science and Technology 65(11), 2179–2194. DOI 10.1002/asi.23200.

[NC-W031] Christian Busch and Harry G. Barkema (2020 online; 2022 issue). Planned Luck: How Incubators Can Facilitate Serendipity for Nascent Entrepreneurs Through Fostering Network Embeddedness. Entrepreneurship Theory and Practice 46(4), 884–919. DOI 10.1177/1042258720915798.

[NC-W032] Alessandra Stefane Cândido Elias da Trindade and Fernando Luiz Vechiato (2022). Criatividade e serendipidade na inovação de produtos e serviços em bibliotecas universitárias [Creativity and serendipity in university-library product and service innovation]. XXII Encontro Nacional de Pesquisa em Ciência da Informação.

[NC-W033] 张颖颖· 王进富 (2023). 机缘巧合实现：创业平台编配警觉性形成机理——组织间学习视角下的案例研究 [Seizing serendipity: the formation mechanism of entrepreneurial-platform orchestration alertness, a case study from interorganizational learning]. 科技进步与对策 40(19), 33–43. DOI 10.6049/kjjbydc.2023020030.

[NC-W034] 高橋菜奈子· 藤村聡· 真家美咲· 名倉早都季· 山崎裕子· 浅石卓真 (2026). VR 図書館内の 3D 書架におけるブラウジング行動：視線データに着目した探索的分析 [Browsing behavior on 3D bookshelves in a VR library: exploratory analysis focusing on eye-tracking data]. 日本図書館情報学会誌 72(1), 19–36. DOI 10.20651/jslis.72.1_19.

[XA-W001] Dawson R. Engler; M. Frans Kaashoek; James O’Toole Jr. (1995). Exokernel: An Operating System Architecture for Application-Level Resource Management. SOSP, 251–266.

[XA-W002] M. Frans Kaashoek; Dawson R. Engler; Gregory R. Ganger; Héctor M. Briceño; Russell Hunt; David Mazières; Thomas Pinckney; Robert Grimm; John Jannotti; Kenneth Mackenzie (1997). Application Performance and Flexibility on Exokernel Systems. SOSP, 52–65.

[XA-W003] Simon Kuenzer; Vlad-Andrei Bădoiu; Hugo Lefeuvre; Sharan Santhanam; Alexander Jung; Gaulthier Gain; Cyril Soldani; Costin Lupu; Ștefan Teodorescu; Costi Răducanu; Cristian Banu; Laurent Mathy; Răzvan Deaconescu; Costin Raiciu; Felipe Huici (2021). Unikraft: Fast, Specialized Unikernels the Easy Way. EuroSys.

[XA-W004] Ali Raza; Thomas Unger; Matthew Boyd; Eric B. Munson; Parul Sohal; Ulrich Drepper; Richard Jones; Daniel Bristot de Oliveira; Larry Woodman; Renato Mancuso; Jonathan Appavoo; Orran Krieger (2022; revised 2023). Unikernel Linux (UKL). arXiv:2206.00789.

[XA-W005] 田崎創 (2017). ライブラリ OS をとりまく現況 [The current landscape of library operating systems]. Internet Infrastructure Review 34, 28–31.

[XA-W006] Richard Jones (2023). Unikernel Linux (UKL) moves forward. Red Hat Research Quarterly, August.

[XA-W007] Hugo Lefeuvre; Gaulthier Gain; Daniel Dinca; Alexander Jung; Simon Kuenzer; Vlad Bădoiu; Răzvan Deaconescu; Laurent Mathy; Costin Raiciu; Pierre Olivier; Felipe Huici (2021). Unikraft and the Coming of Age of Unikernels. USENIX ;login: Online, 12 July.

[XA-W008] MIT PDOS exokernel project (n.d.). Exopc Exokernel Distribution.

[MG-W001] James E. Kelley Jr., Morgan R. Walker (1959). Critical-Path Planning and Scheduling. Proceedings of the Eastern Joint Computer Conference, pp. 160–173.

[MG-W002] Shailesh Sood (2003). Taming uncertainty. PM Network, 17(3), pp. 57–59.

[MG-W003] 諏訪晴彦, 森田大輔, 三道弘明 (2009). プロジェクト管理におけるダミーに基づくクリティカルチェーン・スケジューリング [Dummy-Based Critical Chain Scheduling in Project Management]. システム制御情報学会論文誌, 22(12), pp. 407–415. DOI: 10.5687/iscie.22.407.

[MG-W004] Bent Flyvbjerg (2006). From Nobel Prize to Project Management: Getting Risks Right. Project Management Journal, 37(3), pp. 5–15.

[MG-W005] Eric G. Too, Patrick Weaver (2013; journal issue 2014). The management of project management: A conceptual framework for project governance. International Journal of Project Management, 32, pp. 1382–1394. DOI: 10.1016/j.ijproman.2013.07.006.

[MG-W006] Sylvain Lenfle (2004). Peut-on gérer l’innovation par projet ? [Can innovation be managed through projects?]. In G. Garel, V. Giard and C. Midler (eds.), Faire de la recherche en management de projet. Vuibert.

[MG-W007] Eric von Hippel (1986). Lead Users: A Source of Novel Product Concepts. Management Science, 32(7), pp. 791–805.

[MG-W008] V. Krishnan, Karl T. Ulrich (2001). Product Development Decisions: A Review of the Literature. Management Science, 47(1), pp. 1–21.

[MG-W009] Tim Smolnik, Thorsten Bergmann (2020). Structuring and managing the new product development process – review on the evolution of the Stage-Gate® process. Journal of Business Chemistry, 2020(1). DOI: 10.17879/22139478907.

[MG-W010] Stefan Thomke, Donald Reinertsen (1998). Agile Product Development: Managing Development Flexibility in Uncertain Environments. California Management Review, 41(1), pp. 8–30.

[MG-W011] Antonie Jetter, Fatima Albar, Richard C. Sperry (2016). The Practice of Project Management in Product Development: Insights from the Literature and Cases in High-Tech. Project Management Institute.

[MG-W012] Thomas Bauch (2020). Produktmanagement Prozess-Navigator [Product-management process navigator]. PM1 Produktmanagement Insights.

[MG-W013] Olga Springer, Jakub Miler (2022). A comprehensive overview of software product management challenges. Empirical Software Engineering. DOI: 10.1007/s10664-022-10134-5.

[MG-W014] Harry Markowitz (1952). Portfolio Selection. The Journal of Finance, 7(1), pp. 77–91.

[MG-W015] F. M. Redington (1952). Review of the Principles of Life-office Valuations. Journal of the Institute of Actuaries, 78.

[MG-W016] J. L. Kelly Jr. (1956). A New Interpretation of Information Rate. The Bell System Technical Journal, 35, pp. 917–926.

[MG-W017] Scott Willenbrock (2011). Diversification Return, Portfolio Rebalancing, and the Commodity Return Puzzle. Financial Analysts Journal, 67(4), pp. 42–49.

[MG-W018] 陆靖昶, 朱坤 (2017). 广发 FOF 观察第 27 期丨资产配置中的再平衡策略 [FOF observation27: rebalancing strategies in asset allocation]. 广发基金, 13 March.

[MG-W019] Jonathan Yu-Meng Li (2023). Wasserstein-Kelly Portfolios: A Robust Data-Driven Solution to Optimize Portfolio Growth. arXiv:2302.13979.

[MG-W020] Thierry Roncalli, Guillaume Weisang (2012). Risk Parity Portfolios with Risk Factors. Working paper, September. MPRA Paper 44017.

[PWR-W001] Liam P. Maher; Aqsa Ejaz; Chi Lan Nguyen; Gerald R. Ferris (2021). Forty years of political skill and will in organizations: a review, meta-theoretical framework and directions for future research. Career Development International 27(1), 5–35 (issue 2022).

[PWR-W002] Michel Crozier (1960). Les relations de pouvoir dans un système d'organisation bureaucratique [Power relations in a bureaucratic organizational system]. Sociologie du travail 2(1), 61–75.

[PWR-W003] Jeffrey Pfeffer (2010). Power Play. Harvard Business Review, July–August, 84–92.

[PWR-W004] Robert Greene; Joost Elffers (producer) (1998). The 48 Laws of Power. Viking.

[PWR-W005] 大嶋玲未; 廣川佳子 (2020). 日本企業での新規提案場面における組織内政治の実態と機能 [Organizational politics in new-proposal situations in Japanese companies]. 産業・組織心理学研究 34(1), 59–70.

[PWR-W006] Fong Keng-Highberger; Zhiyu Feng; Kai Chi Yam; Xiao-Ping Chen; Hu Li (2024). Middle power plays: How and when Mach middle managers use downward abuse and upward guanxi to gain and maintain power. Journal of Organizational Behavior 45(7), 1088–1116.

[PWR-W007] Cameron Anderson; Daron L. Sharps; Christopher J. Soto; Oliver P. John (2020). People with disagreeable personalities (selfish, combative, and manipulative) do not have an advantage in pursuing power at work. Proceedings of the National Academy of Sciences 117(37), 22780–22786.

[PWR-W008] В. П. Пугачев (2006). Микрополитический аспект государственного управления человеческими ресурсами [The micropolitical aspect of public human-resource management]. Государственное управление. Электронный вестник.

[PWR-W009] Juliana Arcoverde Mansur; Filipe João Bera de Azevedo Sobral (2011). Política na terra do "jeitinho": consequências dos comportamentos políticos em organizações no Brasil [Politics in the land of jeitinho: consequences of political behavior in Brazilian organizations]. Revista de Administração Mackenzie 12(6).

[PWR-W010] Alexander Beresford (2015). Power, patronage, and gatekeeper politics in South Africa. African Affairs 114(455), 226–248.

[PWR-W011] John F. Padgett; Christopher K. Ansell (1993). Robust Action and the Rise of the Medici, 1400–1434. American Journal of Sociology 98(6), 1259–1319.

[PWR-W012] Ezequiel Alpuche de la Cruz; José Luis Bernal López; Cesaire Chiatchoua (2018). El poder y la organización: una mirada desde la Nueva Economía Institucional (NEI) [Power and organization: a view from New Institutional Economics]. Gestión y estrategia 54, 39–51.

[ST-W001] Henry Mintzberg; James A. Waters (1985). Of Strategies, Deliberate and Emergent. Strategic Management Journal 6(3), 257–272. DOI: 10.1002/smj.4250060306.

[ST-W002] Jane E. Dutton; Susan J. Ashford; Regina M. O'Neill; Katherine A. Lawrence (2001). Moves That Matter: Issue Selling and Organizational Change. Academy of Management Journal 44(4), 716–736. DOI: 10.2307/3069412.

[ST-W003] Linda Rouleau; Julia Balogun (2008). Exploring Middle Managers' Strategic Sensemaking Role Through Practical Knowledge. Les cahiers de recherche du GéPS 2(7), conference working paper, 23–25 September 2008.

[ST-W004] Saku Mantere; Eero Vaara (2008). On the Problem of Participation in Strategy: A Critical Discursive Perspective. Organization Science 19(2), 341–358. DOI: 10.1287/orsc.1070.0296.

[ST-W005] Christophe Torset (2005). La réflexion stratégique, objet et outil de recherche pour le management stratégique ? [Strategic reflection: object and research tool for strategic management?]. XIVe Conférence internationale de management stratégique, Angers.

[ST-W006] 李炜文; 高岩; 李晓彤 (2022). 战略过程中的中层管理者 [Middle managers in the strategy process]. 外国经济与管理 44(12), 69–86. DOI: 10.16538/j.cnki.fem.20220725.101.

[ST-W007] 许志行; 曹骞 (2019). 企业战略思维研究述评与展望 [Enterprise strategic thinking: a literature review and prospects]. 外国经济与管理 41(5), 58–73. DOI: 10.16538/j.cnki.fem.2019.05.005.

[ST-W008] 谷口正一郎 (2021). 組織メンバーの創発的行為形成に着目した組織変革のプロセスモデル：M-GTA を用いた X 社の事例分析 [A process model of organizational change focused on members' emergent actions: an M-GTA case analysis of Company X]. 組織開発研究 5, 22–41.

[ST-W009] Stefan Kühl, in conversation with Martin Claßen (2017). Die Dimension Macht: Funktion bei Veränderungen [The dimension of power: its function in change]. Working Paper 5/2017, 27 March 2017; scholar-authored interview.

[DC-W01] 孫子 (traditional attribution) (date uncertain). 孫子兵法 [The Art of War].

https://zh.wikisource.org/wiki/孫子兵法

[DC-W02] Lisa Raphals (2016). Sunzi versus Xunzi: Two Views of Deception and Indirection. Early China 39, 185–229. https://doi.org/10.1017/eac.2016.6

[DC-W03] Sextus Iulius Frontinus (date uncertain). Strategemata [Stratagems]. https://www.thelatinlibrary.com/frontinus/strat1.shtml

[DC-W04] Niccolò Machiavelli (1532). Il Principe [The Prince]. https://it.wikisource.org/wiki/Il_Principe/Capitolo_XVIII

[DC-W05] Marcel Meulder (1975). Marcel Détienne et Jean-Pierre Vernant, Les ruses de l'intelligence, la métis chez les Grecs [Review of The ruses of intelligence]. L'Antiquité classique 44(2), 771. Book review. https://www.persee.fr/doc/antiq_0770-2817_1975_num_44_2_1805_t1_0771_0000_1

[DC-W06] Olga Torres (2017). Los consejos de al-Harawi sobre estratagemas de guerra: un manual político-militar árabe del siglo XII [Al-Harawi's counsels on stratagems of war]. Revista de Estudios en Seguridad Internacional 3(2), 225–239. https://doi.org/10.18847/1.6.12

[DC-W07] Richards J. Heuer, Jr. (1981). Strategic Deception and Counterdeception: A Cognitive Process Approach. International Studies Quarterly 25(2), 294–327. https://doi.org/10.2307/2600359

[DC-W08] Георгий Львович Смолян (2013). Рефлексивное управление — технология принятия манипулятивных решений [Reflexive control: a technology of manipulative decision-making]. Труды Института системного анализа РАН 63(2), 54–61. https://gtmarket.ru/library/articles/7309

[DC-W09] João Ricardo da Cunha Croce Lopes (2021). Controle reflexivo russo: teoria militar e aplicações [Russian reflexive control: military theory and applications]. Coleção Meira Mattos 16, special issue, 15–41. https://doi.org/10.52781/cmm.e012

[DC-W10] Hayden B. Peake (2007). The Intelligence Officer's Bookshelf: review of STRATAGEM: Deception and Surprise in War. Studies in Intelligence 51(4). Book review. https://www.cia.gov/resources/csi/studies-in-intelligence/volume-51-no-4/the-intelligence-officers-bookshelf/

[DC-W11] Sarah Jenkins and Rick Delbridge (2020). Exploring Organizational Deception: Organizational Contexts, Social Relations and Types of Lying. Organization Theory 1(2). https://doi.org/10.1177/2631787720919436

[DC-W12] David Shulman (2008). More Lies than Meet the Eyes: Organizational Realities and Deceptions in Nonprofit Organizations. International Journal of Not-for-Profit Law 10(2). https://www.icnl.org/resources/research/ijnl/more-lies-than-meet-the-eyes-organizational-realities-and-deceptions-in-nonprofit-organizations

[DC-W13] David Arellano Gault (2016). Corrupción como proceso organizacional: comprendiendo la lógica de la desnormalización de la corrupción [Corruption as an organizational process]. Contaduría y Administración 62, 810–826 (2017 issue; first online 2016). https://doi.org/10.1016/j.cya.2016.01.005

[DC-W14] Joseph P. Gaspar and Maurice E. Schweitzer (2013). The Emotion Deception Model: A Review of Deception in Negotiation and the Role of Emotion in Deception. Negotiation and Conflict Management Research 6(3), 160–179. https://doi.org/10.1111/ncmr.12010

[DC-W15] Todd Rogers, Richard Zeckhauser, Francesca Gino, Michael I. Norton and Maurice E. Schweitzer (2016). Artful Paltering: The Risks and Rewards of Using Truthful Statements to Mislead Others. Journal of Personality and Social Psychology 112(3), 456–473 (2017 issue; first online 2016). https://doi.org/10.1037/pspi0000081

[DC-W16] Gustav Kuhn, Hugo A. Caffaratti, Robert Teszka and Ronald A. Rensink (2014). A psychologically-based taxonomy of misdirection. Frontiers in Psychology 5, 1392. https://doi.org/10.3389/fpsyg.2014.01392

[DC-W17] Mustafa Alhaj Ahmad Alaboud, Yvonne Steggemann, Timo Klein-Soetebier, Wilfried Kunde and Matthias Weigelt (2012). Täuschungshandlungen im Sport: Eine experimentelle Untersuchung zur Wirkung der Häufigkeitsverteilung auf die Blicktäuschung im Basketball [Deceptive actions in sport: frequency and the basketball head fake]. Zeitschrift für Sportpsychologie 19(3), 110–121. https://doi.org/10.1026/1612-5010/a000075

[DC-W18] Fatima Salahdine and Naima Kaabouch (2019). Social Engineering Attacks: A Survey. Future Internet 11(4), 89. https://doi.org/10.3390/fi11040089

[DC-W19] Diplomatic Conference, Geneva (1977). Protocol Additional to the Geneva Conventions of 12 August 1949, relating to the Protection of Victims of International Armed Conflicts (Protocol I). https://ihl-databases.icrc.org/en/ihl-treaties/api-1977/article-37

[DC-W20] 吉田靖之 (2026). 今日の海上における欺瞞規則：海戦法規上の背信行為と奇計に関する一考察 [Rules of deception at sea today: perfidy and ruses in naval warfare]. 高岡法学 44, 37–79. https://doi.org/10.24703/takahogaku.44.0_37

[DC-W21] Michel Pretalli (2022). Introduction, Ruse et magie de l'Antiquité à nos jours [Ruse and magic from antiquity to the present]. In Ruse et magie de l'Antiquité à nos jours, 9–17. https://pufc.univ-fcomte.fr/media/catalog/product/r/u/ruse-magie_introduction.pdf

[PM-W001] Stephanie Dick (2015). Of Models and Machines: Implementing Bounded Rationality. https://doi.org/10.1086/683527

[PM-W002] Richard Waldinger (1975). Achieving Several Goals Simultaneously. https://www.sri.com/wp-content/uploads/2021/12/763.pdf

[PM-W003] Jaco Quist; Philip Vergragt (2006). Past and Future of Backcasting: The Shift to Stakeholder Participation and a Proposal for a Methodological Framework. https://doi.org/10.1016/j.futures.2006.02.010

[PM-W004] Emilie J. K. Mutombo; Tom Bauler; Grégoire Wallenborn (2007). Méthodes participatives de prospective et de planification pour un développement durable : analyse d'approches et de réalisations [Participatory foresight and planning methods for sustainable development: approaches and applications]. https://www.belspo.be/belspo/organisation/Publ/pub_ostc/AP/rAP22_fr.pdf

[PM-W005] 和田春菜; 木下裕介; 水野有智; 福重真一; 梅田靖 (2013). 持続可能社会に向けたバックキャスティング型シナリオ作成手法の提案 [A design-support method for backcasting scenarios for a sustainable society]. https://doi.org/10.1299/kikaic.79.845

[PM-W006] Grant Wiggins; Jay McTighe (1998). Understanding by Design. https://www.acsouth.edu/wp-content/uploads/2023/07/Wiggins-and-McTighe-1998-What-is-backward-design.pdf

[PM-W007] Grant Wiggins; Jay McTighe (2011). The Understanding by Design Guide to Creating High-Quality Units. https://files.ascd.org/pdfs/publications/books/UBD_Guide_HighQU_downloads.pdf

[PM-W008] 王玲玲 (2023). 基于逆向教学设计视野下单元整体教学研究 [Integrated unit teaching from the perspective of backward instructional design]. https://doi.org/10.12238/er.v6i2.4890

[PM-W009] Jessica Wiese; Roger Buehler; Dale Griffin (2016). Backward Planning: Effects of Planning Direction on Predictions of Task Completion Time. https://doi.org/10.1017/S1930297500007269

[PM-W010] Jooyoung Park; Fang-Chi Lu; William M. Hedgcock (2017). Relative Effects of Forward and Backward Planning on Goal Pursuit. https://doi.org/10.1177/0956797617715510

[PM-W011] Allen Z. Ren; Brian Ichter; Anirudha Majumdar (2024). Thinking Forward and Backward: Effective Backward Planning with Large Language Models. https://arxiv.org/abs/2411.01790

[PM-W012] Sebastian Bollien (2021). Methoden zur Strategischen Vorausschau: Backcasting [Methods of strategic foresight: backcasting]. https://www.baks.bund.de/de/node/2274

[PM-W013] David N. Bengston; Lynne M. Westphal; Michael J. Dockry (2020). Back from the Future: The Backcasting Wheel for Mapping a Pathway to a Preferred Future. https://doi.org/10.1177/1946756720929724

[PM-W014] Yusuke Kishita; Mattias Höjer; Jaco Quist (2024). Consolidating Backcasting: A Design Framework towards a Users' Guide. https://doi.org/10.1016/j.techfore.2024.123285

[PM-W015] David McDonald; Gabriele Bammer; Peter Deane (2009). Research Integration Using Dialogue Methods. https://press-files.anu.edu.au/downloads/press/p60381/mobile/ch04s03.html

[PM-W016] The Pfeiffer Library (1998). The Robotics Decision: Solving Strategic Problems. https://home.snu.edu/~jsmith/library/body/v18.pdf

[PM-W017] Charles R. Schwenk (1990). Effects of Devil's Advocacy and Dialectical Inquiry on Decision Making: A Meta-analysis. https://doi.org/10.1016/0749-5978(90)90051-A

[PM-W018] Jerry Fjermestad (2019). Structured Conflict Approaches Used in Strategic Decision Making: From Mason's Initial Study to Virtual Teams. https://doi.org/10.1561/2900000013

[PM-W019] Charlan Nemeth; Keith Brown; John Rogers (2001). Devil's Advocate versus Authentic Dissent: Stimulating Quantity and Quality. https://doi.org/10.1002/ejsp.58

[PM-W020] Deborah J. Mitchell; J. Edward Russo; Nancy Pennington (1989). Back to the Future: Temporal Perspective in the Explanation of Events. https://doi.org/10.1002/bdm.3960020103

[PM-W021] Gary Klein (2007). Performing a Project Premortem. https://hbr.org/2007/09/performing-a-project-premortem

[PM-W022] Beth Veinott; Gary A. Klein; Sterling Wiggins (2010). Evaluating the Effectiveness of the PreMortem Technique on Plan Confidence. https://www.idl.iscram.org/files/veinott/2010/1049_Veinott_etal2010.pdf

[PM-W023] Guillermo M. Wippold; Kaylyn A. Garcia; Ariel Domlyn; Dylan Wong; Maria E. Fernandez (2025). Proactive Planning for Contextual Fit: The Role of the Implementation Premortem. https://doi.org/10.1186/s43058-025-00768-z

[PM-W024] UK Ministry of Defence, Development, Concepts and Doctrine Centre (2021). Red Teaming Handbook. https://www.gov.uk/government/publications/a-guide-to-red-teaming

[PM-W025] Héloïse Fayet; Amélie Férey (2023). « Imaginer au-delà de l'imaginaire ». Red teaming et serious games au service de l'anticipation et de la prospective [Imagining beyond the imaginable: red teaming and serious games for anticipation and foresight]. https://www.ifri.org/fr/briefings/imaginer-au-dela-de-limaginaire-red-teaming-et-serious-games-au-service-de-lanticipation

[PM-W026] 栗田真広 (2025). 国際政治学における分析手法としてのウォーゲームの活用–実験的ウォーゲームの台頭とその活用をめぐる議論を中心として– [Wargaming as an analytical method in international politics: the rise and debate over experimental wargaming]. https://www.nids.mod.go.jp/publication/security/pdf/2025/202503_02.pdf

[PM-W027] Giovanni Cunico; Nici Zimmermann; Nuno Videira (2023). Playing the New Devil's Advocate Role in Facilitated Modelling Processes to Address Group Homogeneity. https://doi.org/10.1080/01605682.2023.2263101

[PM-W028] George Wright; Gene Rowe (2011). Group-based judgmental forecasting: An integration of extant knowledge and the development of priorities for a new research agenda. https://doi.org/10.1016/j.ijforecast.2010.05.012

[PM-W029] Jooyoung Park; Fang-Chi Lu; William M. Hedgcock (2018). Corrigendum: Relative Effects of Forward and Backward Planning on Goal Pursuit. https://doi.org/10.1177/0956797617752922

[NN-W001] Fisher, Roger, and William Ury. 1981. Getting to Yes: Negotiating Agreement Without Giving In. Second edition with Bruce Patton, 1991.

[NN-W002] Organisation internationale du Travail. 2010. Processus et techniques de négociation : manuel de formation pour les acteurs du dialogue social en Afrique [Negotiation processes and techniques: A training manual for participants in social dialogue in Africa]. Dakar: OIT.

[NN-W003] Lipp, Wolfram Emanuel, Remigiusz Smolinski, and Peter Kesting. 2022. Toward a Process Model of First Offers and Anchoring in Negotiations. Negotiation and Conflict Management Research 16(1). DOI 10.34891/2022.574.

[NN-W004] Oesch, John, and Adam D. Galinsky. 2003. First Offers in Negotiations: Determinants and Effects. Working paper, SSRN 399722.

[NN-W005] Boothby, Erica J., Gus Cooney, and Maurice E. Schweitzer. 2022/2023. Embracing Complexity: A Review of Negotiation Research. Annual Review of Psychology 74:299–332. DOI 10.1146/annurev-psych-033020-014116.

[NN-W006] Schweitzer, Maurice E., Kori L. Krueger, Erica Boothby, and Gus Cooney. 2025. Negotiation. In The Handbook of Social Psychology, sixth edition. DOI 10.70400/NUZE7621.

[NN-W007] Стремовская, Алла Леонидовна. 2012. Возможные стратегии ведения переговорного процесса [Possible strategies for conducting negotiations]. Российский внешнеэкономический вестник 8:79–88.

[NN-W008] Sierralta Ríos, Aníbal. 2005. Negociaciones comerciales internacionales: texto y casos. Second, expanded edition, 2014. Fondo Editorial PUCP.

[NN-W009] White, James J. 1984. The Pros and Cons of Getting to YES. Journal of Legal Education 34:115–124.

[NN-W010] 苏勇, 程骏骏, 吴展. 2014. 合作式谈判研究述评与展望 [Review and prospects of cooperative negotiation research]. 外国经济与管理 36(3):65–73.

[NN-W011] 张真, 皇甫刚. 2007. 影响整合式谈判的情境因素和过程因素 [Contextual and process variables influencing integrative negotiation]. 心理科学进展 15(3):518–523.

[NN-W012] Levin, Daniel Z., Jorge Walter, and J. Keith Murnighan. 2010/2011. Dormant Ties: The Value of Reconnecting. Organization Science 22(4):923–939. DOI 10.1287/orsc.1100.0576.

[NN-W013] Hall, Jeffrey A. 2018/2019. How many hours does it take to make a friend? Journal of Social and Personal Relationships 36(4):1278–1296. DOI 10.1177/0265407518761225.

[NN-W014] Oswald, Debra L., Eddie M. Clark, and Cheryl M. Kelly. 2004. Friendship Maintenance: An Analysis of Individual and Dyad Behaviors. Journal of Social and Clinical Psychology 23(3):413–441. DOI 10.1521/jscp.23.3.413.35460.

[NN-W015] 本田周二. 2016. 友人関係における動機づけと友人とのコミュニケーションおよび精神的健康との関連 [Friendship motivation, communication and mental health]. 人間生活文化研究 26:572–582. DOI 10.9748/hcs.2016.572.

[NN-W016] 岡田涼. 2008. 親密な友人関係の形成・維持過程の動機づけモデルの構築 [A motivational model of the formation and maintenance of close friendships]. 教育心理学研究 56(4):575–588. DOI 10.5926/jjep1953.56.4_575.

[NN-W017] 李松, 刘峻君, 鲍秀琴, 陈旭. 2022. 人际感恩对关系维持的影响 [The impacts of interpersonal gratitude on relationship maintenance]. 心理科学进展 30(11):2586–2594. DOI 10.3724/SP.J.1042.2022.02586.

[NN-W018] Bidart, Claire. 2008. Étudier les réseaux. Apports et perspectives pour les sciences sociales [Studying networks: Contributions and perspectives for the social sciences]. Informations sociales 147:34–45. Reprinted as Étudier les réseaux sociaux [Studying social networks], Diversité 157 (2009):179–184. DOI 10.3406/diver.2009.3090.

[NN-W019] Wolff, Hans-Georg, and Klaus Moser. 2006. Entwicklung und Validierung einer Networkingskala [Development and validation of a networking scale]. Diagnostica 52(4):161–180. DOI 10.1026/0012-1924.52.4.161.

[NN-W020] Wolff, Hans-Georg, and Klaus Moser. 2009. Effects of Networking on Career Success: A Longitudinal Study. Journal of Applied Psychology 94(1):196–206. DOI 10.1037/a0013350.

[NN-W021] Casciaro, Tiziana, Francesca Gino, and Maryam Kouchaki. 2014. The Contaminating Effects of Building Instrumental Ties: How Networking Can Make Us Feel Dirty. Administrative Science Quarterly 59(4):705–735. DOI 10.1177/0001839214554990.

[NN-W022] Administrative Science Quarterly editors and authors. 2024. Corrigendum to The Contaminating Effects of Building Instrumental Ties: How Networking Can Make Us Feel Dirty. Administrative Science Quarterly 69(2):568. DOI 10.1177/00018392241243394.

[NN-W023] Eby, Lillian T., Tammy D. Allen, Sarah C. Evans, Thomas Ng, and David DuBois. 2008. Does Mentoring Matter? A Multidisciplinary Meta-Analysis Comparing Mentored and Non-Mentored Individuals. Journal of Vocational Behavior 72(2):254–267. DOI 10.1016/j.jvb.2007.04.005.

[NN-W024] Maesima, Giovania Mitie, Lívia Sanseverino Gomes, and Daniela Centenaro Levandowski. 2022. Internet e Relacionamentos Interpessoais: Uma Revisão Narrativa Crítica da Literatura Científica (2011–2020) [Internet and interpersonal relationships: A critical narrative review]. Interações 42:22–47. DOI 10.31211/interacoes.n42.2022.a2.

[NN-W025] Beersma, Bianca, and Carsten K. W. De Dreu. 2003. Social Motives in Integrative Negotiation: The Mediating Influence of Procedural Fairness. Social Justice Research 16:217–239.

[GS-W01] Edwin A. Locke; Gary P. Latham (2002). Building a Practically Useful Theory of Goal Setting and Task Motivation: A 35-Year Odyssey. American Psychologist 57(9), 705–717. https://doi.org/10.1037/0003-066X.57.9.705

[GS-W02] 桑原哲郎; 岸野哲 (1977). 目標設定理論の概念：目標設定理論（第 1 報） [Basic Concept of Creative Target-Setting: Theory of Creative Target-Setting, Part I]. 日本経営工学会誌 27(4), 343–348. https://doi.org/10.11221/jimapre.27.4_343

[GS-W03] 野上真; 古川久敬 (2010). 目標設定に関わる運用方略の効果性に関する研究の概括 [A Review of Research on the Effectiveness of Goal-Setting Management Strategies]. 産業・組織心理学研究 23(2), 129–144. https://doi.org/10.32222/jaiop.23.2_129

[GS-W04] Kennon M. Sheldon; Andrew J. Elliot (1999). Goal Striving, Need Satisfaction, and Longitudinal Well-Being: The Self-Concordance Model. Journal of Personality and Social Psychology 76(3), 482–497. https://doi.org/10.1037/0022-3514.76.3.482

[GS-W05] Harald Seelig (n.d.). Das Selbstkonkordanz-Modell [The Self-Concordance Model]. Institut für Sport und Sportwissenschaft, Universität Freiburg. https://www.sport.uni-freiburg.de/de/institut/psychologie/psych_proj/ssk/sktheorie

[GS-W06] Gabriele Oettingen; Hyeon-ju Pak; Karoline Schnetter (2001). Self-Regulation of Goal Setting: Turning Free Fantasies About the Future Into Binding Goals. Journal of Personality and Social Psychology 80(5), 736–753. https://doi.org/10.1037/0022-3514.80.5.736

[GS-W07] Peter M. Gollwitzer; Paschal Sheeran (n.d.). Implementation Intentions. National Cancer Institute, Health Behavior Constructs: Theory, Measurement, and Research. https://cancercontrol.cancer.gov/brp/research/constructs/implementation-intentions

[GS-W08] Guoxia Wang; Yi Wang; Xiaosong Gai (2021). A Meta-Analysis of the Effects of Mental Contrasting With Implementation Intentions on Goal Attainment. Frontiers in Psychology 12, 565202. https://doi.org/10.3389/fpsyg.2021.565202

[GS-W09] Carsten Wrosch; Michael F. Scheier; Gregory E. Miller; Richard Schulz; Charles S. Carver (2003). Adaptive Self-Regulation of Unattainable Goals: Goal Disengagement, Goal Reengagement, and Subjective Well-Being. Personality and Social Psychology Bulletin 29(12), 1494–1508. https://doi.org/10.1177/0146167203256921

[GS-W10] Д. А. Леонтьев (2024). Структура и нарушения саморегуляции деятельности: точка роста культурно-исторической деятельностной психологии [Structure and Disturbances of Activity Self-Regulation: A Growth Point for Cultural-Historical Activity Psychology]. Культурно-историческая психология 20(3), 27–35. https://doi.org/10.17759/chp.2024200303

[GS-W11] George T. Doran (1981). There's a S.M.A.R.T. Way to Write Management's Goals and Objectives. Management Review 70(11), 35–36. https://www.decisionskills.com/uploads/5/1/6/0/5160560/doran_1981_s.m.a.r.t-way-management-review.pdf

[GS-W12] Lisa D. Ordóñez; Maurice E. Schweitzer; Adam D. Galinsky; Max H. Bazerman (2009). Goals Gone Wild: The Systematic Side Effects of Over-Prescribing Goal Setting. Harvard Business School Working Paper 09-083. https://www.hbs.edu/ris/Publication%20Files/09-083.pdf

[GS-W13] Edwin A. Locke; Gary P. Latham (2009). Has Goal Setting Gone Wild, or Have Its Attackers Abandoned Good Scholarship? Academy of Management Perspectives 23(1), 17–23. https://doi.org/10.5465/AMP.2009.37008000

[GS-W14] Lisa D. Ordóñez; Maurice E. Schweitzer; Adam D. Galinsky; Max H. Bazerman (2009). On Good Scholarship, Goal Setting, and Scholars Gone Wild. Harvard Business School Working Paper 09-122. https://www.hbs.edu/ris/Publication%20Files/09-122.pdf

[GS-W15] 文鹏; 任晓雅; 陈诚 (2017). 目标设置对非伦理行为的影响：边界条件与理论基础 [The Impact of Goal Setting on Unethical Behavior: Boundary Conditions and Theoretical Bases]. 心理科学进展 25(8), 1401–1410. https://doi.org/10.3724/SP.J.1042.2017.01401

[GS-W16] Jaime Díaz-Ocejo; Juan Antonio Mora-Merida (2013). Revisión de algunas variables relevantes en el establecimiento de metas deportivas [Literature Review on Some Relevant Variables in Goal Setting in Sport]. Anales de Psicología 29(1), 233–242. https://doi.org/10.6018/analesps.29.1.137281

[GS-W17] Anderson Simas Frutuoso (2009). Estabelecimento de metas como estratégia motivacional. Artigo de revisão [Goal Setting as a Motivational Strategy: A Review Article]. EFDeportes, Revista Digital 14(137). https://www.efdeportes.com/efd137/estabelecimento-de-metas-como-estrategia-motivacional.htm

[T01-W001] Steel, P. (2007). The nature of procrastination. Psychological Bulletin 133, 65–94. DOI 10.1037/0033-2909.133.1.65. https://doi.org/10.1037/0033-2909.133.1.65

[T01-W002] Svartdal, F., et al. (2018). On the behavioral side of procrastination: Exploring behavioral delay in real-life settings. Frontiers in Psychology 9, 746. https://pmc.ncbi.nlm.nih.gov/articles/PMC5964561/

[T01-W003] 童廷婷、白幼玲、冯廷勇 (2024). 情绪调节改善拖延行为的认知机制：任务厌恶中介作用 [The cognitive mechanism of reducing procrastination by emotion regulation: The mediation role of task aversiveness]. 心理学报 56(4), 458 ff. https://journal.psych.ac.cn/xlxb/CN/10.3724/SP.J.1041.2024.00458

[T01-W004] Rozental, A., et al. (2018). Targeting procrastination using psychological treatments: A systematic review and meta-analysis. Frontiers in Psychology 9, 1588. https://pmc.ncbi.nlm.nih.gov/articles/PMC6125391/

[T02-W001] Cooper, A. C. (1971). The Founding of Technologically-Based Firms. Center for Venture Management. https://files.eric.ed.gov/fulltext/ED076773.pdf

[T02-W002] 榊原清則・古賀款久・本庄裕司・近藤一徳 (2000). 日本における技術系ベンチャー企業の経営実態と創業者に関する調査研究 [Survey research on technology-based start-ups and their entrepreneurs in Japan]. NISTEP 調査資料 73. https://nistep.repo.nii.ac.jp/record/4585/files/NISTEP-RM073-FullJ.pdf

[T02-W003] Sarasvathy, S. D. (2001). Causation and effectuation: Toward a theoretical shift from economic inevitability to entrepreneurial contingency. Academy of Management Review 26, 243–263. https://cdn.mises.org/sarasvathy_2001_causation_and_effectuation.pdf

[T02-W004] Arcuri, M. C., Russo, I., & Gandolfi, G. (2024, online). Productivity of innovation: The effect of innovativeness on start-up survival. Journal of Technology Transfer. DOI 10.1007/s10961-024-10069-7. https://doi.org/10.1007/s10961-024-10069-7

[T02-W005] Robb, A., Reedy, E. J., Ballou, J., DesRoches, D., Potter, F., & Zhao, Z. (2010). An Overview of the Kauffman Firm Survey: Results from the 2004–2008 Data. https://www.kauffman.org/wp-content/uploads/2019/12/kfs_2010_report.pdf

[T03-W001] Krypotos, A.-M., Effting, M., Kindt, M., & Beckers, T. (2015). Avoidance learning: A review of theoretical models and recent developments. Frontiers in Behavioral Neuroscience 9, 189. https://pmc.ncbi.nlm.nih.gov/articles/PMC4508580/

[T03-W002] Hofmann, S. G., & Hay, A. C. (2018). Rethinking avoidance: Toward a balanced approach to avoidance in treating anxiety disorders. Journal of Anxiety Disorders 55, 14–21. https://pmc.ncbi.nlm.nih.gov/articles/PMC5879019/

[T03-W003] Craske, M. G., et al. (2014). Maximizing exposure therapy: An inhibitory learning approach. Behaviour Research and Therapy 58, 10–23. https://pmc.ncbi.nlm.nih.gov/articles/PMC4114726/

[T03-W004] Sich, M. M. (2024). Hemmschwelle Expositionstherapie – virtuelle Realität (VR) als Lösungsansatz: Untersuchungen von Automatisierung und Akzeptanz von VR bei Patienten mit Angststörungen [Barriers to exposure therapy and virtual reality]. LMU München dissertation. https://edoc.ub.uni-muenchen.de/33741/1/Sich_Magdalena.pdf

[T03-W005] Shiban, Y., Pauli, P., & Mühlberger, A. (2013). Effect of multiple context exposure on renewal in spider phobia. Behaviour Research and Therapy 51, 68–74. https://www.sciencedirect.com/science/article/pii/S0005796712001611

[T04-W001] Smith, W. R. (1956). Product differentiation and market segmentation as alternative marketing strategies. Journal of Marketing 21, 3–8. https://journals.sagepub.com/doi/full/10.1177/002224295602100102

[T04-W002] 小原博 (2010). 歴史的視角からのマーケティング―販売管理，マーケティング管理，市場管理― [Marketing from a historical perspective: Sales management, marketing management and market management]. 同志社商学 61(6). https://doshisha.repo.nii.ac.jp/record/15136/files/017061060004.pdf

[T04-W003] Grönroos, C. (1994). From marketing mix to relationship marketing: Towards a paradigm shift in marketing. Management Decision 32(2), 4–20. https://markenmanagement.wordpress.com/wp-content/uploads/2012/01/groenroos_frommarketingmixtorelationshipmarketing.pdf

[T04-W005] Håkansson, H. (ed.) (1982). International Marketing and Purchasing of Industrial Goods: An Interaction Approach. John Wiley. Ch. 4, Robert Salle and Jean-Paul Valla, “Lyon Acier.” https://research.em-lyon.com/view/pdfCoverPage?download=true&filePid=136144430009453&instCode=33EML_INST

[T05-W001] Fondeur, Y., de Larquier, G., et al. (2012). Pratiques de recrutement et sélectivité sur le marché du travail [Recruitment practices and selectivity in the labor market]. Centre d’études de l’emploi, report 72. https://ceet.cnam.fr/medias/fichier/72-pratiques-recrutement-selectivite-marche-travail_1510322483969-pdf?ID_FICHE=1051257&INLINE=FALSE

[T05-W002] Wilhelmy, A., & Kleinmann, M. Selektion und Attraktion: Wie Organisationen und Bewerbende sich gegenseitig auswählen und beeinflussen [Selection and attraction: How organizations and applicants

select and influence one another]. https://www.psychologie.uzh.ch/dam/jcr:34649dfe-5a09-4b5b-b0d9-310da3f3930b/Wilhelmy&Kleinmann_Selektion&Attraktion_Final.pdf

[T05-W003] Schmidt, F. L., & Hunter, J. E. (1998). The validity and utility of selection methods in personnel psychology. Journal of Applied Psychology 83, 262–274. https://psycnet.apa.org/fulltext/1998-10661-006.pdf

[T05-W004] Sackett, P. R., Zhang, C., Berry, C. M., & Lievens, F. (2022). Revisiting meta-analytic estimates of validity in personnel selection: Addressing systematic overcorrection for restriction of range. Journal of Applied Psychology 107, 2040–2068. https://psycnet.apa.org/record/2022-17327-001

[T06-W001] Lambrechts, F., Grieten, S., Bouwen, R., & Corthouts, F. (2009). Process consultation revisited: Taking a relational practice perspective. Journal of Applied Behavioral Science 45, 39–58. https://b-m-institute.com/wp-content/uploads/2019/10/Lambrechts_2009.pdf

[T06-W002] Loizos Heracleous (2022). Helping at NASA: Guidelines for using process consultation to develop impactful research. Information and Organization 32(1), 100388. https://doi.org/10.1016/j.infoandorg.2022.100388

[T06-W003] Boni-Le Goff, I. (2012). Pratiques et identités professionnelles dans le conseil en management en France [Practices and professional identities of management consultants in France]. SociologieS. DOI 10.4000/sociologies.3072. https://doi.org/10.4000/sociologies.3072

[T06-W004] Ciampi, F. (2017). Knowing Through Consulting in Action: Meta-consulting Knowledge Creation Pathways. Firenze University Press. https://library.oapen.org/bitstream/handle/20.500.12657/55359/9788864536439.pdf?isAllowed=y&sequence=1

[T06-W005] Bloom, N., Eifert, B., Mahajan, A., McKenzie, D., & Roberts, J. (2013). Does management matter? Evidence from India. Quarterly Journal of Economics 128, 1–51. https://academic.oup.com/qje/article/128/1/1/1838606

[T07-W001] Bjork, R. A. (1972). Theoretical implications of directed forgetting. In Coding Processes in Human Memory, pp. 217 ff. https://bjorklab.psych.ucla.edu/wp-content/uploads/sites/13/2016/07/RBjork_1972.pdf

[T07-W002] Kluge, A., & Gronau, N. (2018). Intentional forgetting in organizations: The importance of eliminating retrieval cues for implementing new routines. Frontiers in Psychology 9, 51. https://pmc.ncbi.nlm.nih.gov/articles/PMC5799275/

[T07-W003] Martin de Holan, P., & Phillips, N. (2004). Remembrance of things past? The dynamics of organizational forgetting. Management Science 50(11), 1603–1613. DOI 10.1287/mnsc.1040.0273. https://doi.org/10.1287/mnsc.1040.0273

[T08-W001] Acas (2014). Challenging conversations and how to manage them. Advisory, Conciliation and Arbitration Service. https://www.acas.org.uk/acas-guide-to-challenging-conversations-and-how-to-manage-them

[T08-W002] King, C., Edlington, T., & Williams, B. (2020). Avoiding difficult conversations in the Australian health sector. SAGE Open Nursing 6. https://pmc.ncbi.nlm.nih.gov/articles/PMC7774411/

[T08-W003] 初鹿野・岩田 (2008). 選ばれていない参加者が発話するとき：もう一人の参加者について言及すること [Analysis of talk by an unselected participant]. 社会言語科学 10(2), 121–134. https://www.jstage.jst.go.jp/article/jajls/10/2/10_KJ00008440223/_pdf/-char/ja

[T09-W001] Gundersen, T. (2000). A Process Integration Primer. International Energy Agency / SINTEF. https://pubdb.bfe.admin.ch/en/publication/download/1401

[T09-W002] Linnhoff, B., & Flower, J. R. (1978). Synthesis of heat exchanger networks: II. Evolutionary generation of networks with various criteria of optimality. AIChE Journal 24(4), 642–654. DOI 10.1002/aic.690240412. https://doi.org/10.1002/aic.690240412

[T09-W003] Ryu, J., & Maravelias, C. T. (2019). Simultaneous Process and Heat Exchanger Network Synthesis Using a Discrete Temperature Grid. Industrial & Engineering Chemistry Research 58(15), 6002–6016. DOI 10.1021/acs.iecr.8b04083. https://doi.org/10.1021/acs.iecr.8b04083

[T09-W004] Li, B.-H., & Chang, C.-T. (2017). Retrofitting industrial heat exchanger network based on pinch analysis. AdCONIP, Taipei. https://skoge.folk.ntnu.no/prost/proceedings/adconip-2017/media/files/0078.pdf

[T09-W005] Linnhoff, B., & Flower, J. R. (1978). Synthesis of heat exchanger networks: I. Systematic generation of energy optimal networks. AIChE Journal 24(4), 633–642. DOI 10.1002/aic.690240411. https://doi.org/10.1002/aic.690240411

[T10-W001] National Bureau of Standards (1987). Investigation of L'Ambiance Plaza Building Collapse in Bridgeport, Connecticut. NBSIR 87-3640. https://nvlpubs.nist.gov/nistpubs/Legacy/IR/nbsir87-3640.pdf

[T10-W002] 刘征、罗志琪 (2012). 超大型深基坑半逆作法设计与实践 [Design and construction of ultra-large deep excavations]. 岩土工程学报, supplement, 507–511. https://www.cgejournal.com/cn/article/pdf/preview/14805.pdf

[T10-W003] Bruno Mazaré (2021). Renforcement de sol sous ouvrage existant et reprise en sous œuvre [Ground reinforcement below existing structures and underpinning]. Revue française de géotechnique 167, 1. https://doi.org/10.1051/geotech/2021008

[T10-W004] FHWA (2018). Bridge Preservation Guide: Maintaining a Resilient Infrastructure to Preserve Mobility. https://www.fhwa.dot.gov/bridge/preservation/guide/guide.pdf

[T11-W001] UCTE (2007). Final Report: System Disturbance on 4 November 2006. https://eepublicdownloads.entsoe.eu/clean-documents/pre2015/publications/ce/otherreports/Final-Report-20070130.pdf

[T11-W002] U.S.–Canada Power System Outage Task Force (2004). Final Report on the August 14, 2003 Blackout in the United States and Canada: Causes and Recommendations. https://eta-publications.lbl.gov/sites/default/files/2003-blackout-us-canada.pdf

[T11-W003] 顾雪平、白岩松、李少岩、刘艳 (2022). 电力系统黑启动恢复问题的研究评述 [Research review of power system black-start restoration]. 电工技术学报 37(13), 3183–3200. DOI 10.19595/j.cnki.1000-6753.tces.220027. https://doi.org/10.19595/j.cnki.1000-6753.tces.220027

[T11-W004] Balance Management Task Force (2003). Current State of Balance Management in Europe. https://eepublicdownloads.entsoe.eu/clean-documents/pre2015/ntc/archive/BalanceManagemeninEurope.pdf

[T12-W001] Griffith, A. A. (1921). The phenomena of rupture and flow in solids. Philosophical Transactions of the Royal Society A 221, 163–198. https://mbarkey.eng.ua.edu/courses/AEM644/Griffith1921fracture.pdf

[T12-W002] Da Chen, Xiao-Feng Zhang and Robert O. Ritchie (2000). Mechanisms of high-temperature fatigue and fracture in silicon carbide ceramics. In Fatigue and Fracture of High Temperature Materials, 1–8. TMS. https://imisweb.tms.org/iWeb/IWeb/Item_Detail.aspx?Category=PUBS_PDF&iProductCode=01-4461-1

[T12-W003] 中村眞実 (2017). 高圧水素ガス中におけるオーステナイト系ステンレス鋼溶接金属の SSRT 特性と疲労寿命特性に関する研究 [Slow strain rate and fatigue-life properties of stainless-steel weld metal under high-pressure hydrogen]. 九州大学博士論文. https://doi.org/10.15017/1931910

[T12-W004] Motooka, T., & Kiuchi, K. (2002). Corrosion fatigue of refractory materials in boiling nitric acid. Materials Transactions 43, 1220 ff. https://www.jstage.jst.go.jp/article/matertrans/43/5/43_5_1220/_article/-char/en

[T13-W001] Shewhart, W. A. (1931). Economic Control of Quality of Manufactured Product. D. Van Nostrand. https://archive.org/stream/in.ernet.dli.2015.150272/2015.150272.Economic-Control-Of-Quality-Of-Manufactured-Product_djvu.txt

[T13-W002] NIST/SEMATECH. e-Handbook of Statistical Methods, §6.1.6, What Is Process Capability? https://www.itl.nist.gov/div898/handbook/pmc/section1/pmc16.htm

[T13-W003] 仁科健 (n.d.). 品質のつくり込みは「ばらつき」への挑戦 [Building in quality is a challenge of variation]. 名古屋工業大学 repository. https://nitech.repo.nii.ac.jp/record/2477/files/FB50Kinen08.pdf

[T13-W004] Sharma, M., Sahni, S. P., & Sharma, S. (2019). Validating a destructive measurement system using Gauge R&R — a case study. Engineering Management in Production and Services 11(4). https://reference-global.com/download/article/10.2478/emj-2019-0031.pdf

[T13-W005] สุรศักดิ์ ชุบไธสง、ระพี กาญจนะ (2020). การลดของเสียในกระบวนการฉีดพลาสติกของชิ้นส่วนโทรศัพท์ โดยการออกแบบการทดลอง [Reducing defectives in plastic injection process of telephone part by design of experiment]. SWU Engineering Journal 15(3), 17–31. https://ph02.tci-thaijo.org/index.php/sej/article/view/239867

[T14-W001] Jeanneret Medina, M., Baudet, C., & Bonnal, P. (2019). Cinquante ans d'ingénierie des exigences : un concept suffisamment mûr pour être revisité [Fifty years of requirements engineering]. 4ème Conférence Internationale en Gestion de Projet, HAL 02143260. https://univ-lyon3.hal.science/hal-02143260/document

[T14-W002] NASA (2016). NASA Systems Engineering Handbook, rev. 2, §4.2, Technical Requirements Definition. https://www.nasa.gov/reference/4-2-technical-requirements-definition/

[T14-W003] Mars Climate Orbiter Mishap Investigation Board (1999). Phase I Report. NASA, 10 November. https://llis.nasa.gov/llis_lib/pdf/1009464main1_0641-mr.pdf

[T14-W004] Ågren, S. M., et al. (2019, online). The impact of requirements on systems development speed: A multiple-case study in automotive. Requirements Engineering. https://research.chalmers.se/publication/515041/file/515041_Fulltext.pdf

[T14-W005] Battelle Memorial Institute (2017). Sharing Data between Mobile Devices, Connected Vehicles and Infrastructure. Task 6: Prototype Acceptance Test Summary Report. FHWA-JPO-17-507. https://transops.s3.amazonaws.com/uploaded_files/Sharing%20Data%20Between%20Mobile%20Devices,%20Connected%20Vehicles,%20and%20Infrastructure.%20Prototype%20Acceptance%20Test%20Summary%20Report.pdf

[T15-W001] Триандафиллов, В. К. (1929). Характер операций современных армий [The character of operations of modern armies]. https://militera.lib.ru/science/triandafillov1/01.html

[T15-W002] U.S. Marine Corps (1997). Warfighting. Marine Corps Doctrinal Publication 1. https://www.marines.mil/Portals/1/Publications/MCDP%201%20Warfighting.pdf

[T15-W003] Desert Storm: Attrition or Maneuver? (1994–95 academic year). U.S. Army Command and General Staff College study. https://cgsc.contentdm.oclc.org/digital/api/collection/p4013coll3/id/1186/download

[T15-W004] Cole Petersen (2012). Clearing the air: Taking manoeuvre and attrition out of strategy. Infinity Journal 2(3), 15–19. https://doi.org/10.64148/msm.v2i3.3

[T15-W005] U.S. Department of Defense (1992). Conduct of the Persian Gulf War: Final Report to Congress. https://www.globalsecurity.org/military/library/report/1992/cpgw.pdf

[T16-W001] Ἀριστοτέλης (4th century BCE). Τέχνη ῥητορική [Rhetoric], I.2, 1355b–1356a. W. D. Ross Greek edition. https://beta.perseus.tufts.edu/urn%3Acts%3AgreekLit%3Atlg0086.tlg038.perseus-grc2%3A1.2/

[T16-W002] Quintilianus, M. F. (late 1st century CE). Institutio oratoria [The education of the orator], XI. https://www.thelatinlibrary.com/quintilian/quintilian.institutio11.shtml

[T16-W003] 《鬼谷子》 [Guiguzi], 捭闔第一 [Opening and closing, chapter 1]. Received text, traditional attribution and dating disputed. https://www.moshuyuan.com/chapter/zibu/gui-gu-zi/bai-he-di-yi.html

[T16-W004] Université de Genève. La mise en scène du discours [The staging of discourse]. Méthodes et problèmes. https://www.unige.ch/lettres/framo/enseignements/methodes/srhetorique/rdintegr.html

[T16-W005] Dalrymple, H. (1995). ‘Four Score and Seven Years Ago…’ Lincoln’s Words Still Inspire 130 Years Later. Library of Congress Information Bulletin. https://www.loc.gov/loc/lcib/9503/gettysburg.html

[T17-W001] Weber, M. (1922, posthumous publication). Wirtschaft und Gesellschaft [Economy and society], §12 Die Veralltäglichung des Charisma im Verwaltungsstab. https://www.textlog.de/7413.html

[T17-W002] Chazel, F. (2019). La sociologie wébérienne de la domination : l’apport comparé des deux versions [Weber’s sociology of domination: A comparison of the two versions]. Revue européenne des sciences sociales 57(1), 17–41. DOI 10.4000/ress.4883. https://doi.org/10.4000/ress.4883

[T17-W003] Jetter, C. (2021). Continuing revelation and institutionalization: Joseph Smith, Ralph Waldo Emerson and charismatic leadership in antebellum America. Studies in Church History. https://www.cambridge.org/core/journals/studies-in-church-history/article/continuing-revelation-and-institutionalization-joseph-smith-ralph-waldo-emerson-and-charismatic-leadership-in-antebellum-america/171C7017ACEA417929942D6ACBB31B00

[T17-W004] Antonakis, J., Fenley, M., & Liechti, S. (2011 issue, online 2012). Can charisma be taught? Tests of two interventions. Academy of Management Learning & Education 10, 374–396. https://journals.aom.org/doi/10.5465/amle.2010.0012

[T17-W005] Smith, J. F., Jr. (1909). Origin of the ‘Reorganized’ Church and the Question of Succession. https://www.gutenberg.org/files/50357/50357-h/50357-h.htm

[DP-W001] Stanley Reiter (1999). Coordinating Economic Activity: An Example. Conference paper abstract. https://econweb.ucsd.edu/~decent/99NYUConf/reiter.html

[T15-W006] Florian L. Waitl (2018). Crossing the Berm: Maneuver Support during Desert Shield and Desert Storm. In *Into the Breach: Historical Case Studies of Mobility Operations in Large-Scale Combat Operations*, ch. 9. Army University Press. https://home.army.mil/wood/7915/5751/8409/into-the-breach-lsco-volume-6.pdf

[T16-W006] Abraham Lincoln (1863). Gettysburg Address: Hay Copy. Abraham Lincoln Papers, Library of Congress. https://www.loc.gov/resource/mal.4356600/?st=text

[T16-W007] Edward Everett (1863). Edward Everett to Abraham Lincoln, 20 November 1863. Abraham Lincoln Papers, Library of Congress. https://www.loc.gov/resource/mal.2813300/?st=text

[T17-W006] Community of Christ (n.d.). History. The Era of Reorganization, Joseph Smith III and A Herald to the Saints. https://cofchrist.org/history/

[SMR-W001] David Fernandez Rivas et al. (2020). *Process intensification education contributes to sustainable development goals. Part 1*. Education for Chemical Engineers, 32, 1–14. https://doi.org/10.1016/j.ece.2020.04.003

[HCT-W001] George H. Heilmeier (2007). *Interview, January 16, 2007*. DARPA. https://www.esd.whs.mil/Portals/54/Documents/FOID/Reading%20Room/DARPA/15-F-0751_DARPA_Director_George_Heilmeier.pdf

[HCT-W002] Defense Advanced Research Projects Agency (n.d.). *The Heilmeier Catechism*. https://www.darpa.mil/about/heilmeier-catechism